Worked Solutions to Simmons' *Differential Equations with Applications and Historical Notes*

Third Edition

An independent, unofficial study companion
Chapters 1–14

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Solutions were prepared with AI assistance and checked by computer algebra and independent review;
see the Preface for scope and limitations.

Preface

What this is. A companion of worked solutions to the problems in George F. Simmons, *Differential Equations with Applications and Historical Notes*, Third Edition (CRC Press / Taylor & Francis). **You need the textbook to use it:** problems are cited by number only, in the form "Problem *s*–*n*" (problem *n* of Section *s*) or "Chapter *c*, Miscellaneous Problem *n*", following the textbook's own numbering. The problem statements themselves are the property of the copyright holder and are *not* reproduced here.

Coverage. Every problem set in the book is covered: the problems at the end of each numbered section, the miscellaneous sets for Chapters 1 and 2, and the exercises of the numerical-methods chapter. Chapter appendices, historical essays, and Sections 65 and 66 contain no problems; Section 72's single discussion exercise appears with Section 73.

Format of each solution. Every algebraic step, substitution, integration and choice of constant is shown, lost solutions and domains are noted, sketches are included where asked, and each problem ends with a boxed answer. Because statements are not reproduced, part labels (a), (b), ... refer to the textbook's own labels. log denotes the natural logarithm, as in the textbook.

How the solutions were checked, and the limits of that. Each section was (i) solved from the textbook's problem pages, (ii) verified by an independent computer-algebra/numerical script (in the `verify/` directory of the repository) that substitutes every answer back into its equation and redoes integrals, series coefficients and initial/boundary conditions, (iii) compared with the textbook's answer key, and (iv) re-derived by a second, independent reviewer. This is strong evidence of correctness but **not a guarantee**; proofs in particular are checked by reasoning rather than by script. Chapter 14 has no answer key to compare against. Corrections are welcome (see the repository's `CONTRIBUTING.md`).

AI assistance. The solutions and this typesetting were produced with the help of an AI system (Anthropic's Claude) and supervised by the repository owner.

Differences from the textbook's answer key. They are collected in the appendix *Remarks on the Textbook and Its Answer Key*.

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Chapter 1

The Nature of Differential Equations. Separable Equations

Section 2 General Remarks on Solutions

Problem 2-1 (Simmons)

Solution. In each case we compute the derivative(s) of the given function (differentiating implicitly when the function is given implicitly), substitute into the differential equation, and show that it reduces to an identity in x.

(a) $y = x^2 + c$ gives $y' = 2x$, which is the equation itself. ✓

(b) $y = cx^2$ gives $y' = 2cx$, so $xy' = 2cx^2 = 2(cx^2) = 2y$. ✓

(c) Differentiating $y^2 = e^{2x} + c$ implicitly with respect to x gives $2yy' = 2e^{2x}$, i.e. $yy' = e^{2x}$. ✓

(d) $y = ce^{kx}$ gives $y' = cke^{kx} = k(ce^{kx}) = ky$. ✓

(e) $y = c_1 \sin 2x + c_2 \cos 2x$ gives

$$y' = 2c_1 \cos 2x - 2c_2 \sin 2x, \qquad y'' = -4c_1 \sin 2x - 4c_2 \cos 2x = -4y,$$

so $y'' + 4y = -4y + 4y = 0$. ✓

(f) $y = c_1 e^{2x} + c_2 e^{-2x}$ gives $y' = 2c_1 e^{2x} - 2c_2 e^{-2x}$ and $y'' = 4c_1 e^{2x} + 4c_2 e^{-2x} = 4y$, so $y'' - 4y = 0$. ✓

(g) Since $(\sinh u)' = \cosh u$ and $(\cosh u)' = \sinh u$, $y = c_1 \sinh 2x + c_2 \cosh 2x$ gives $y' = 2c_1 \cosh 2x + 2c_2 \sinh 2x$ and $y'' = 4c_1 \sinh 2x + 4c_2 \cosh 2x = 4y$, so $y'' - 4y = 0$. ✓

(h) $y = \sin^{-1} xy$ means $\sin y = xy$ with $-\pi/2 \le y \le \pi/2$ (principal value). Differentiating $\sin y = xy$ implicitly,

$$(\cos y)\, y' = y + xy'.$$

On the principal branch $\cos y \ge 0$, so $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2y^2}$. Hence $y'\sqrt{1 - x^2y^2} = xy' + y$, which is the given equation. ✓

(i) $y = x \tan x$ gives $y' = \tan x + x \sec^2 x$, so

$$\begin{aligned} xy' &= x \tan x + x^2 \sec^2 x = x \tan x + x^2(1 + \tan^2 x) \qquad (\sec^2 x = 1 + \tan^2 x) \\ &= x \tan x + x^2 + (x \tan x)^2 = y + x^2 + y^2. \checkmark \end{aligned}$$

(j) Here $y > 0$ (so that $\log y$ is defined). Differentiating $x^2 = 2y^2 \log y$ implicitly,

$$2x = 4yy' \log y + 2y^2 \cdot \frac{y'}{y} = y'\,(4y \log y + 2y), \qquad \text{so} \qquad y' = \frac{x}{2y \log y + y}.$$

From the given relation, $2y \log y = x^2/y$; substituting,

$$y' = \frac{x}{x^2/y + y} = \frac{xy}{x^2 + y^2}. \checkmark$$

(k) Differentiating $y^2 = x^2 - cx$ gives $2yy' = 2x - c$. From the relation itself $c = (x^2 - y^2)/x$, so

$$2yy' = 2x - \frac{x^2 - y^2}{x} = \frac{x^2 + y^2}{x},$$

and multiplying by x gives $2xyy' = x^2 + y^2$. ✓

(l) $y = c^2 + c/x$ gives $y' = -c/x^2$. Then

$$y + xy' = c^2 + \frac{c}{x} - \frac{c}{x} = c^2, \qquad x^4(y')^2 = x^4 \cdot \frac{c^2}{x^4} = c^2,$$

so $y + xy' = x^4(y')^2$. ✓

(m) Differentiating $y = ce^{y/x}$ implicitly and using $ce^{y/x} = y$,

$$y' = ce^{y/x} \cdot \frac{xy' - y}{x^2} = \frac{y(xy' - y)}{x^2}.$$

Thus $x^2y' = xyy' - y^2$, i.e. $y'(xy - x^2) = y^2$, and so $y' = y^2/(xy - x^2)$ (wherever $xy - x^2 \neq 0$). ✓

(n) Differentiating $y + \sin y = x$ gives $y' + (\cos y)y' = 1$, so $y' = 1/(1 + \cos y)$. Using $x = y + \sin y$,

$$y\cos y - \sin y + x = y\cos y - \sin y + y + \sin y = y(1 + \cos y),$$

and therefore $(y\cos y - \sin y + x)y' = y(1 + \cos y) \cdot \dfrac{1}{1 + \cos y} = y$. ✓

(o) Differentiating $x + y = \tan^{-1} y$ gives $1 + y' = \dfrac{y'}{1 + y^2}$. Multiplying by $1 + y^2$: $(1 + y^2) + (1 + y^2)y' = y'$, i.e. $1 + y^2 + y^2y' = 0$. ✓

Answer. All fifteen functions (a)–(o) satisfy their differential equations identically, as shown.

Problem 2-2 (Simmons)

Solution. Parts (a)–(h) and (n) have the form $y' = f(x)$, solved by $y = \int f(x)\,dx + c$; the others are separable, solved by separating the variables and integrating. Throughout, c (or C) denotes an arbitrary constant, and we record any solutions lost when dividing.

(a) $y = \displaystyle\int (e^{3x} - x)\,dx = \tfrac{1}{3}e^{3x} - \tfrac{1}{2}x^2 + c.$

(b) For $x \neq 0$, $y' = 1/x$, so $y = \log|x| + c$ (that is, $y = \log x + c$ on $x > 0$).

(c) With the substitution $u = x^2$, $du = 2x\,dx$: $y = \displaystyle\int xe^{x^2}\,dx = \tfrac{1}{2}\int e^u\,du = \tfrac{1}{2}e^{x^2} + c.$

(d) Integrate by parts with $u = \sin^{-1} x$, $dv = dx$, so $du = dx/\sqrt{1 - x^2}$, $v = x$:

$$\begin{aligned} y = \int \sin^{-1} x\,dx &= x\sin^{-1} x - \int \frac{x}{\sqrt{1 - x^2}}\,dx \\ &= x\sin^{-1} x + \sqrt{1 - x^2} + c, \end{aligned}$$

since $\frac{d}{dx}\sqrt{1 - x^2} = -x/\sqrt{1 - x^2}$. (Here $|x| \leq 1$.)

(e) For $x \neq -1$, $y' = \dfrac{x}{1 + x} = 1 - \dfrac{1}{1 + x}$, so $y = x - \log|1 + x| + c$.

(f) $y = \displaystyle\int \frac{x}{1 + x^2}\,dx = \tfrac{1}{2}\log(1 + x^2) + c$ (substitute $u = 1 + x^2$, $du = 2x\,dx$).

(g) For $x \neq -1$ we need $\int \frac{x\,dx}{1+x^3}$, and $1 + x^3 = (1 + x)(x^2 - x + 1)$. Write

$$\frac{x}{(1 + x)(x^2 - x + 1)} = \frac{A}{1 + x} + \frac{Bx + C}{x^2 - x + 1}, \qquad x = A(x^2 - x + 1) + (Bx + C)(x + 1).$$

Putting $x = -1$ gives $-1 = 3A$, so $A = -\frac{1}{3}$; comparing coefficients of x^2 gives $0 = A + B$, so $B = \frac{1}{3}$; comparing constant terms gives $0 = A + C$, so $C = \frac{1}{3}$. Hence

$$\frac{x}{1+x^3} = -\frac{1}{3}\cdot\frac{1}{1+x} + \frac{1}{3}\cdot\frac{x+1}{x^2-x+1}.$$

For the second term write $x+1 = \frac{1}{2}(2x-1) + \frac{3}{2}$ and complete the square, $x^2 - x + 1 = (x - \frac{1}{2})^2 + \frac{3}{4}$:

$$\begin{aligned}\int \frac{x+1}{x^2-x+1}\,\mathrm{d}x &= \frac{1}{2}\int \frac{2x-1}{x^2-x+1}\,\mathrm{d}x + \frac{3}{2}\int \frac{dx}{(x-\frac{1}{2})^2+\frac{3}{4}} \\ &= \frac{1}{2}\log(x^2-x+1) + \frac{3}{2}\cdot\frac{2}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}} \\ &= \frac{1}{2}\log(x^2-x+1) + \sqrt{3}\,\tan^{-1}\frac{2x-1}{\sqrt{3}}.\end{aligned}$$

Therefore

$$\begin{aligned}y &= -\frac{1}{3}\log|1+x| + \frac{1}{6}\log(x^2-x+1) + \frac{\sqrt{3}}{3}\tan^{-1}\frac{2x-1}{\sqrt{3}} + c \\ &= \frac{1}{6}\log\frac{x^2-x+1}{(x+1)^2} + \frac{1}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}} + c.\end{aligned}$$

(h) With $u = \tan^{-1}x$, $du = dx/(1+x^2)$: $y = \displaystyle\int \frac{\tan^{-1}x}{1+x^2}\,\mathrm{d}x = \int u\,\mathrm{d}u = \tfrac{1}{2}(\tan^{-1}x)^2 + c$.

(i) The constant function $y = 1$ is a solution ($x\cdot 1\cdot 0 = 0 = 1-1$). For $y \neq 1$, $x \neq 0$, separate the variables:

$$\frac{y}{y-1}\,\mathrm{d}y = \frac{dx}{x}, \qquad \text{i.e.} \qquad \Big(1 + \frac{1}{y-1}\Big)dy = \frac{dx}{x}.$$

Integrating, $y + \log|y-1| = \log|x| + C$. Exponentiating, $e^y|y-1| = e^C|x|$, so $(y-1)e^y = kx$ with $k = \pm e^C \neq 0$. The value $k = 0$ gives the solution $y = 1$ found above, so all solutions are $(y-1)e^y = kx$, k arbitrary; for $k \neq 0$ this is the same as

$$x = c(y-1)e^y \qquad (c = 1/k).$$

(j) $y = 0$ is a solution. For $y \neq 0$ (and $x \neq 0$):

$$\frac{dy}{y^5} = -\frac{dx}{x^5} \implies -\frac{1}{4y^4} = \frac{1}{4x^4} + C \implies x^{-4} + y^{-4} = c \quad (c = -4C).$$

The solution $y = 0$ is not contained in this family (it was lost on dividing by y^5).

(k) The functions $y = n\pi$ (n an integer) are solutions, since then $y' = 0$ and $\tan y = 0$. Otherwise divide by $x\tan y$:

$$\cot y\,\mathrm{d}y = \frac{1-2x^2}{x}\,\mathrm{d}x = \Big(\frac{1}{x} - 2x\Big)dx \implies \log|\sin y| = \log|x| - x^2 + C.$$

Hence $|\sin y| = e^C|x|e^{-x^2}$, i.e. $\sin y = cxe^{-x^2}$ with $c = \pm e^C$; $c = 0$ recovers the solutions $y = n\pi$. So the general solution is $\sin y = cxe^{-x^2}$.

(l) $y = 0$ is a solution. For $y \neq 0$, $dy/y = 2x\,\mathrm{d}x$, so $\log|y| = x^2 + C$ and $y = \pm e^C e^{x^2}$. Including $y = 0$: $y = ce^{x^2}$, c arbitrary.

(m) $\sin y\,dy = x^2\,dx$, so $-\cos y = \frac{1}{3}x^3 + C$, i.e. $x^3 + 3\cos y = c$ $(c = -3C)$.

(n) $y' = \csc x$ (for $\sin x \neq 0$). Since

$$\frac{d}{dx}\log|\csc x + \cot x| = \frac{-\csc x\cot x - \csc^2 x}{\csc x + \cot x} = -\csc x,$$

we get $y = \int \csc x\,dx = -\log|\csc x + \cot x| + c$.

(o) $y = 0$ is a solution. For $y \neq 0$: $dy/y = -\tan x\,dx$, so $\log|y| = \log|\cos x| + C$ (because $\int \tan x\,dx = -\log|\cos x|$). Thus $y = c\cos x$, which includes $y = 0$ for $c = 0$.

(p) Similarly $dy/y = \tan x\,dx$ gives $\log|y| = -\log|\cos x| + C$, so $y = c/\cos x = c\sec x$ ($c = 0$ gives the solution $y = 0$).

(q) Divide by $(1+x^2)(1+y^2) > 0$:

$$\frac{dy}{1+y^2} + \frac{dx}{1+x^2} = 0 \implies \tan^{-1}x + \tan^{-1}y = C.$$

Taking the tangent of both sides and using $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$,

$$\frac{x+y}{1-xy} = \tan C =: c \implies x + y = c - cxy \implies y(1+cx) = c - x \implies y = \frac{c-x}{1+cx}.$$

(If C is an odd multiple of $\pi/2$, then $\tan C$ is undefined and the relation becomes $1 - xy = 0$, i.e. $y = 1/x$; this is the limiting case $c \to \infty$, and it also satisfies the equation, since $(1+x^2)(-1/x^2) + (1+1/x^2) = 0$.)

(r) We need $y > 0$ for $\log y$. The constant $y = 1$ is a solution ($\log 1 = 0$, $dy = 0$). For $y \neq 1$, $x \neq 0$:

$$\frac{dy}{y\log y} = \frac{dx}{x} \implies \log|\log y| = \log|x| + C \implies \log y = cx,$$

where we used the substitution $u = \log y$, $du = dy/y$, and $c = \pm e^C$. Hence $y = e^{cx}$; $c = 0$ gives the solution $y = 1$.

Answer.
(a) $y = \frac{1}{3}e^{3x} - \frac{1}{2}x^2 + c$
(b) $y = \log|x| + c$
(c) $y = \frac{1}{2}e^{x^2} + c$
(d) $y = x\sin^{-1}x + \sqrt{1-x^2} + c$
(e) $y = x - \log|1+x| + c$
(f) $y = \frac{1}{2}\log(1+x^2) + c$
(g) $y = \frac{1}{6}\log\frac{x^2-x+1}{(x+1)^2} + \frac{1}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}} + c$
(h) $y = \frac{1}{2}(\tan^{-1}x)^2 + c$
(i) $x = c(y-1)e^y$ (and $y = 1$)
(j) $x^{-4} + y^{-4} = c$ (and $y = 0$)
(k) $\sin y = cxe^{-x^2}$
(l) $y = ce^{x^2}$
(m) $x^3 + 3\cos y = c$
(n) $y = -\log|\csc x + \cot x| + c$
(o) $y = c\cos x$
(p) $y = c\sec x$
(q) $y = (c-x)/(1+cx)$
(r) $y = e^{cx}$

Problem 2-3 (Simmons)

Solution.

(a) By parts ($u = x$, $dv = e^x\,dx$): $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x$. So $y = xe^x - e^x + c$. At $x = 1$: $e - e + c = 3$, so $c = 3$ and $y = xe^x - e^x + 3$.

(b) With $u = \sin x$, $du = \cos x\,dx$: $y = \int 2\sin x\cos x\,dx = \sin^2 x + c$. At $x = 0$: $0 + c = 1$, so $y = \sin^2 x + 1$.

(c) By parts ($u = \log x$, $dv = dx$): $\int \log x\,dx = x\log x - \int x\cdot\frac{1}{x}\,dx = x\log x - x$. So $y = x\log x - x + c$; at $x = e$: $e\cdot 1 - e + c = 0$, so $c = 0$ and $y = x\log x - x$.

(d) $y' = \dfrac{1}{x^2-1} = \dfrac{1}{2}\Big(\dfrac{1}{x-1} - \dfrac{1}{x+1}\Big)$, so $y = \frac{1}{2}\log\left|\dfrac{x-1}{x+1}\right| + c$. The initial point $x = 2$ lies in the interval $x > 1$, where $\frac{x-1}{x+1} > 0$. At $x = 2$: $\frac{1}{2}\log\frac{1}{3} + c = 0$, so $c = \frac{1}{2}\log 3$ and

$$y = \frac{1}{2}\log\frac{x-1}{x+1} + \frac{1}{2}\log 3 = \frac{1}{2}\log\frac{3x-3}{x+1} \qquad (x > 1).$$

(e) Partial fractions:

$$\frac{1}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2},$$

$$A = \frac{1}{(-2)(2)} = -\frac{1}{4},\ B = \frac{1}{2\cdot 4} = \frac{1}{8},\ C = \frac{1}{(-2)(-4)} = \frac{1}{8}$$

(each found by the cover-up rule). Hence

$$y = -\frac{1}{4}\log|x| + \frac{1}{8}\log|x-2| + \frac{1}{8}\log|x+2| + c = \frac{1}{8}\log\frac{|x^2-4|}{x^2} + c.$$

The initial point $x = 1$ lies in $0 < x < 2$, where $|x^2-4| = 4 - x^2$. At $x = 1$: $\frac{1}{8}\log 3 + c = 0$, so $c = -\frac{1}{8}\log 3$ and

$$y = \frac{1}{8}\log\frac{4-x^2}{3x^2} \qquad (0 < x < 2).$$

(f) Partial fractions:

$$\frac{2x^2+x}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}, \qquad 2x^2 + x = A(x^2+1) + (Bx+C)(x+1).$$

$x = -1$: $1 = 2A$, $A = \frac{1}{2}$; coefficient of x^2: $2 = A + B$, $B = \frac{3}{2}$; constant term: $0 = A + C$, $C = -\frac{1}{2}$. Therefore

$$\begin{aligned} y &= \int\Big[\frac{1/2}{x+1} + \frac{\frac{3}{2}x}{x^2+1} - \frac{1/2}{x^2+1}\Big]dx = \frac{1}{2}\log|x+1| + \frac{3}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1}x + c \\ &= \frac{1}{4}\log\big[(x+1)^2(x^2+1)^3\big] - \frac{1}{2}\tan^{-1}x + c. \end{aligned}$$

At $x = 0$: $\frac{1}{4}\log 1 - 0 + c = 1$, so $c = 1$.

Answer. (a) $y = xe^x - e^x + 3$; (b) $y = \sin^2 x + 1$; (c) $y = x\log x - x$; (d) $y = \frac{1}{2}\log\frac{3x-3}{x+1}$; (e) $y = \frac{1}{8}\log\frac{4-x^2}{3x^2}$; (f) $y = \frac{1}{4}\log\big[(x+1)^2(x^2+1)^3\big] - \frac{1}{2}\tan^{-1}x + 1$.

Problem 2-4 (Simmons)

Solution.

(a) $y' = e^{3x}e^{-2y}$, so $e^{2y}\,dy = e^{3x}\,dx$ and $\frac{1}{2}e^{2y} = \frac{1}{3}e^{3x} + C$. Multiplying by 6: $3e^{2y} = 2e^{3x} + c$. At $(0,0)$: $3 = 2 + c$, $c = 1$. The integral curve is $3e^{2y} = 2e^{3x} + 1$.

(b) For $x \neq 0$, $dy = \left(2x + \frac{1}{x}\right)dx$, so $y = x^2 + \log|x| + c$. At $(1,1)$: $1 = 1 + 0 + c$, $c = 0$. Since the curve passes through a point with $x > 0$, it is $y = x^2 + \log x$.

(c) Dividing by $e^{-y}(1+x^2) > 0$: $\dfrac{dx}{1+x^2} + e^y\,dy = 0$, so $\tan^{-1} x + e^y = c$. At $(0,0)$: $0 + 1 = c$. The curve is $\tan^{-1} x + e^y = 1$.

(d) Near the given point $\sin 3x \neq 0$ and $\cos 2y \neq 0$, so we may divide by $\sin 3x \cos 2y$:

$$\frac{3\cos 3x}{\sin 3x}\,dx - \frac{2\sin 2y}{\cos 2y}\,dy = 0 \implies \log|\sin 3x| + \log|\cos 2y| = C \implies \sin 3x \cos 2y = c.$$

At $(\pi/12, \pi/8)$: $\sin\frac{\pi}{4}\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2} = \frac{1}{2}$, so $c = \frac{1}{2}$ and the curve is $2\sin 3x\cos 2y = 1$.

(e) Let $I = \int e^x \cos x\,dx$. Integrating by parts twice (with $u = \cos x$, then $u = \sin x$, and $dv = e^x\,dx$ each time):

$$I = e^x\cos x + \int e^x \sin x\,dx = e^x\cos x + e^x\sin x - I,$$

so $I = \frac{1}{2}e^x(\sin x + \cos x)$. Thus $y = \frac{1}{2}e^x(\sin x + \cos x) + c$; at $(0,0)$: $0 = \frac{1}{2} + c$, $c = -\frac{1}{2}$. The curve is $2y + 1 = e^x(\sin x + \cos x)$.

(f) Near $(1,0)$ we have $x > 0$ and $y + 1 > 0$. Separating,

$$\frac{y}{y+1}\,dy = \frac{x+1}{x}\,dx \implies \left(1 - \frac{1}{y+1}\right)dy = \left(1 + \frac{1}{x}\right)dx \implies y - \log(y+1) = x + \log x + C.$$

At $(1,0)$: $0 - 0 = 1 + 0 + C$, so $C = -1$. Rearranging, $\log x + \log(y+1) = y - x + 1$, i.e. $\log x(y+1) = y - x + 1$.

Answer. (a) $3e^{2y} = 2e^{3x} + 1$; (b) $y = x^2 + \log x$; (c) $\tan^{-1} x + e^y = 1$; (d) $2\sin 3x\cos 2y = 1$; (e) $2y + 1 = e^x(\sin x + \cos x)$; (f) $\log x(y+1) = y - x + 1$.

Problem 2-5 (Simmons)

Solution. Since the integrand e^{-t^2} is continuous, the fundamental theorem of calculus gives

$$\frac{d}{dx}\int_0^x e^{-t^2}\,dt = e^{-x^2}.$$

Hence, by the product rule,

$$y' = 2xe^{x^2}\int_0^x e^{-t^2}\,dt + e^{x^2}\cdot e^{-x^2} = 2x\,y + 1.$$

Answer. Shown: $y' = 2xy + 1$ identically.

Problem 2-6 (Simmons)

Solution.

(a) For $y = e^{2x}$: $y' = 2e^{2x}$, $y'' = 4e^{2x}$, so $y'' - 5y' + 6y = (4 - 10 + 6)e^{2x} = 0$. For $y = e^{3x}$: $y' = 3e^{3x}$, $y'' = 9e^{3x}$, so $y'' - 5y' + 6y = (9 - 15 + 6)e^{3x} = 0$.

(b) For $y = c_1e^{2x} + c_2e^{3x}$: $y' = 2c_1e^{2x} + 3c_2e^{3x}$, $y'' = 4c_1e^{2x} + 9c_2e^{3x}$, and

$$y'' - 5y' + 6y = c_1(4 - 10 + 6)e^{2x} + c_2(9 - 15 + 6)e^{3x} = 0$$

for all x and all constants c_1, c_2.

Answer. Shown: e^{2x}, e^{3x}, and every $c_1e^{2x} + c_2e^{3x}$ satisfy $y'' - 5y' + 6y = 0$.

Problem 2-7 (Simmons)

Solution. Here $y > 0$. Differentiating $xy = \log y + c$ implicitly with respect to x (the constant disappears):

$$y + x\frac{dy}{dx} = \frac{1}{y}\frac{dy}{dx} \implies \frac{dy}{dx}\left(\frac{1}{y} - x\right) = y \implies \frac{dy}{dx}\cdot\frac{1 - xy}{y} = y \implies \frac{dy}{dx} = \frac{y^2}{1 - xy},$$

valid wherever $1 - xy \neq 0$. This holds for every c.

Answer. Shown: every curve $xy = \log y + c$ satisfies $dy/dx = y^2/(1 - xy)$.

Problem 2-8 (Simmons)

Solution. For $y = e^{mx}$ we have $y' = me^{mx}$, $y'' = m^2e^{mx}$, $y''' = m^3e^{mx}$, so

$$2y''' + y'' - 5y' + 2y = (2m^3 + m^2 - 5m + 2)e^{mx}.$$

Since $e^{mx} \neq 0$, $y = e^{mx}$ is a solution exactly when $2m^3 + m^2 - 5m + 2 = 0$. Now $m = 1$ is a root ($2 + 1 - 5 + 2 = 0$), and dividing by $m - 1$,

$$2m^3 + m^2 - 5m + 2 = (m - 1)(2m^2 + 3m - 2) = (m - 1)(2m - 1)(m + 2).$$

So $m = 1$, $m = \frac{1}{2}$, $m = -2$.

Exactly as in Problem 6(b), let $y = c_1e^x + c_2e^{x/2} + c_3e^{-2x}$. Because the left side of the equation is a sum of constant multiples of y and its derivatives, it is linear in y:

$$\begin{aligned} 2y''' + y'' - 5y' + 2y = {} & c_1(2 + 1 - 5 + 2)e^x + c_2\left(\tfrac{2}{8} + \tfrac{1}{4} - \tfrac{5}{2} + 2\right)e^{x/2} \\ & + c_3(-16 + 4 + 10 + 2)e^{-2x} = 0. \end{aligned}$$

Answer. $m = 1,\ \frac{1}{2},\ -2$; $\quad y = c_1e^x + c_2e^{x/2} + c_3e^{-2x}$.

Section 3 Families of Curves. Orthogonal Trajectories

Notation. Throughout this section, solid (blue) curves in the sketches belong to the given family and dashed (green) curves are the orthogonal trajectories. Following the text, the differential equation of the orthogonal trajectories is obtained from that of the family by replacing dy/dx with $-dx/dy$ (in polar coordinates: replacing $r\,d\theta/dr$ with $-dr/(r\,d\theta)$).

Problem 3-1 (Simmons)

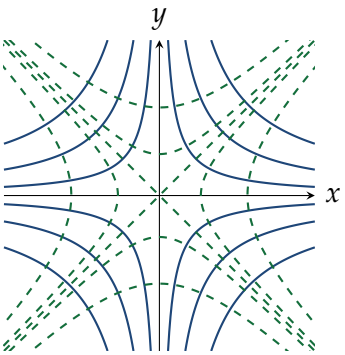
Solution.

(a) *Differential equation of the family.* Differentiating $xy = c$ with respect to x gives $y + xy' = 0$, i.e. $\frac{dy}{dx} = -\dfrac{y}{x}$ (c has already disappeared).

Orthogonal trajectories. Replacing dy/dx by $-dx/dy$:

$$-\frac{dx}{dy} = -\frac{y}{x} \implies x\,dx = y\,dy \implies \tfrac{1}{2}x^2 = \tfrac{1}{2}y^2 + C \implies x^2 - y^2 = c.$$

The family consists of rectangular hyperbolas with the coordinate axes as asymptotes (together with the axes themselves for $c = 0$); the trajectories are rectangular hyperbolas with asymptotes $y = \pm x$ (the lines $y = \pm x$ for $c = 0$).

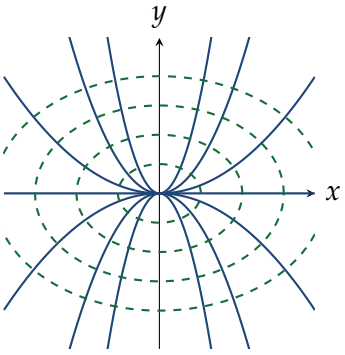


(b) *Family.* $y = cx^2$ gives $y' = 2cx$; eliminating $c = y/x^2$ gives $\frac{dy}{dx} = \dfrac{2y}{x}$.

Trajectories. Replace dy/dx by $-dx/dy$:

$$-\frac{dx}{dy} = \frac{2y}{x} \implies -x\,dx = 2y\,dy \implies -\tfrac{1}{2}x^2 = y^2 + C \implies x^2 + 2y^2 = c^2$$

(the constant $-2C$ must be positive and is written c^2). The family is the parabolas with vertex at the origin and axis the y-axis (and the line $y = 0$ for $c = 0$); the trajectories are ellipses centred at the origin with semi-axes c along the x-axis and $c/\sqrt{2}$ along the y-axis.



(c) *Family.* $r = c(1+\cos\theta)$ gives $\frac{dr}{d\theta} = -c\sin\theta$. Eliminating $c = r/(1+\cos\theta)$,

$$\frac{dr}{d\theta} = -\frac{r\sin\theta}{1+\cos\theta}, \qquad \text{i.e.} \qquad \frac{r\,d\theta}{dr} = -\frac{1+\cos\theta}{\sin\theta}.$$

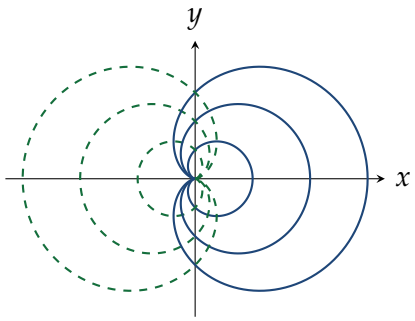
Trajectories. Replace $r\,d\theta/dr$ by $-dr/(r\,d\theta)$:

$$-\frac{dr}{r\,d\theta} = -\frac{1+\cos\theta}{\sin\theta} \implies \frac{dr}{r} = \frac{1+\cos\theta}{\sin\theta}\,\mathrm{d}\theta = \frac{\sin\theta}{1-\cos\theta}\,\mathrm{d}\theta,$$

where we multiplied numerator and denominator by $1-\cos\theta$ and used $(1+\cos\theta)(1-\cos\theta) = \sin^2\theta$. With $u = 1-\cos\theta$, $du = \sin\theta\,\mathrm{d}\theta$, the right side integrates to $\log(1-\cos\theta)$, so

$$\log r = \log(1-\cos\theta) + \log c \implies r = c(1-\cos\theta).$$

The family consists of cardioids symmetric about the x-axis with their cusp at the origin, lying to the right (they reach $r = 2c$ at $\theta = 0$); the trajectories are the mirror-image cardioids, with cusp at the origin and lying to the left (reaching $r = 2c$ at $\theta = \pi$).

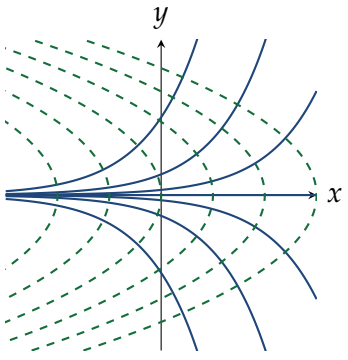


(d) *Family.* $y = ce^x$ gives $y' = ce^x = y$, so $\frac{dy}{dx} = y$.

Trajectories. Replace dy/dx by $-dx/dy$:

$$-\frac{dx}{dy} = y \implies dx = -y\,\mathrm{d}y \implies x = -\tfrac{1}{2}y^2 + C \implies y^2 = -2x + c.$$

The family is the exponential curves (and $y = 0$); the trajectories are congruent parabolas with axis the x-axis, opening to the left, with vertices at $(c/2, 0)$.



Answer. (a) $x^2 - y^2 = c$; (b) $x^2 + 2y^2 = c^2$; (c) $r = c(1-\cos\theta)$; (d) $y^2 = -2x + c$.

Problem 3-2 (Simmons)

Solution. We do (b) first and obtain (a) as the case $n = 4$.

(b) $y = cx^n$ gives $y' = ncx^{n-1}$; eliminating $c = y/x^n$ gives $\frac{dy}{dx} = \frac{ny}{x}$. Replacing dy/dx by $-dx/dy$,

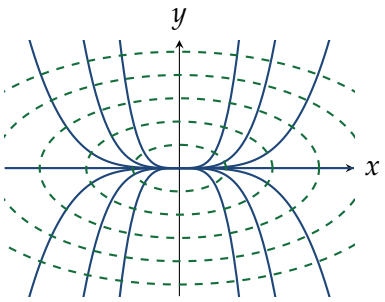
$$-\frac{dx}{dy} = \frac{ny}{x} \implies -x\,dx = ny\,dy \implies -\tfrac{1}{2}x^2 = \tfrac{n}{2}y^2 + C \implies x^2 + ny^2 = c^2.$$

These are ellipses centred at the origin with semi-axis c along the x-axis and $c/\sqrt{n}$ along the y-axis. (For $n = 1$ they are circles, the orthogonal trajectories of the lines $y = cx$, as in the text.)

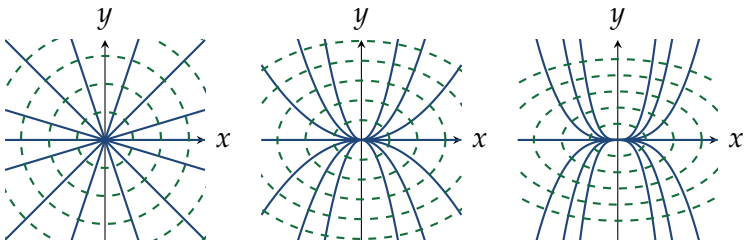
(a) With $n = 4$: the orthogonal trajectories of $y = cx^4$ are the ellipses $x^2 + 4y^2 = c^2$, with semi-axes c and $c/2$.

Sketches. For (a), and for (b) with $n = 1, 2, 3$:

(a) $n = 4$



(b) $n = 1$ (b) $n = 2$ (b) $n = 3$



Effect of increasing n. The ratio of the semi-axes of $x^2 + ny^2 = c^2$ is $c : c/\sqrt{n} = \sqrt{n} : 1$. As n increases, the ellipses become flatter: they are more and more elongated in the x-direction (squeezed toward the x-axis), reflecting the fact that the curves $y = cx^n$ hug the x-axis more and more closely near the origin.

Answer. (a) $x^2 + 4y^2 = c^2$; (b) $x^2 + ny^2 = c^2$. The trajectories are ellipses with axis ratio $\sqrt{n} : 1$, becoming more and more elongated in the x-direction as n increases.

Problem 3-3 (Simmons)

Solution. Let ψ be the angle from the polar radius to the tangent of a curve at a point; as shown in the text, $\tan\psi = r\,d\theta/dr$. For a curve of the given family,

$$\tan\psi = \frac{r}{dr/d\theta} = \frac{r}{F(r,\theta)}.$$

An orthogonal trajectory through the same point has tangent perpendicular to this one, so its angle ψ^* with the (same) polar radius is $\psi^* = \psi \pm \frac{\pi}{2}$, and hence $\tan\psi^* = -\cot\psi = -1/\tan\psi$.

This is exactly the text's rule: $r\,d\theta/dr$ is replaced by its negative reciprocal $-dr/(r\,d\theta)$. Along the trajectory, therefore,

$$\frac{r}{dr/d\theta} = \tan\psi^* = -\frac{F(r,\theta)}{r} \qquad\Longrightarrow\qquad \frac{dr}{d\theta} = -\frac{r^2}{F(r,\theta)},$$

which is the stated equation of the orthogonal trajectories.

Application. For $r = 2c\sin\theta$, $\frac{dr}{d\theta} = 2c\cos\theta$; eliminating $2c = r/\sin\theta$ gives

$$\frac{dr}{d\theta} = F(r,\theta) = \frac{r\cos\theta}{\sin\theta} = r\cot\theta.$$

The orthogonal trajectories therefore satisfy

$$\frac{dr}{d\theta} = -\frac{r^2}{r\cot\theta} = -r\tan\theta \;\Longrightarrow\; \frac{dr}{r} = -\tan\theta\,d\theta \;\Longrightarrow\; \log r = \log|\cos\theta| + \log 2c,$$

since $\int\tan\theta\,d\theta = -\log|\cos\theta|$. Hence $r = 2c\cos\theta$: the circles tangent to the y-axis at the origin (compare equation (13) of the text and Figure 3).

Answer. Shown: the trajectories satisfy $dr/d\theta = -r^2/F$. For $r = 2c\sin\theta$ the orthogonal trajectories are the circles $r = 2c\cos\theta$.

Problem 3-4 (Simmons)

Solution. Differentiating $r = c(1-\cos\theta)^{-1}$,

$$\frac{dr}{d\theta} = -\frac{c\sin\theta}{(1-\cos\theta)^2} = -\frac{r\sin\theta}{1-\cos\theta} =: F(r,\theta),$$

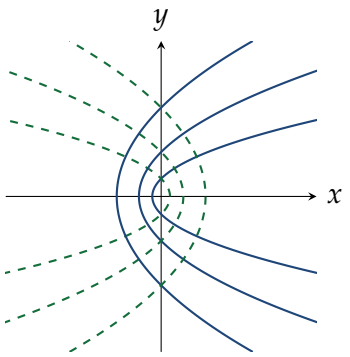
using $c/(1-\cos\theta) = r$. By Problem 3, the orthogonal trajectories satisfy

$$\frac{dr}{d\theta} = -\frac{r^2}{F} = \frac{r(1-\cos\theta)}{\sin\theta} \;\Longrightarrow\; \frac{dr}{r} = \frac{1-\cos\theta}{\sin\theta}\,d\theta = \frac{\sin\theta}{1+\cos\theta}\,d\theta,$$

where we multiplied numerator and denominator by $1+\cos\theta$. With $u = 1+\cos\theta$, $du = -\sin\theta\,d\theta$, the right side integrates to $-\log(1+\cos\theta)$, so

$$\log r = -\log(1+\cos\theta) + \log c \;\Longrightarrow\; r = \frac{c}{1+\cos\theta}.$$

Description. In rectangular coordinates, $r(1-\cos\theta) = c$ is $r = x + c$, so $x^2 + y^2 = x^2 + 2cx + c^2$, i.e. $y^2 = 2c\left(x + \frac{c}{2}\right)$: parabolas with focus at the origin, axis the x-axis, opening to the right, vertex $(-c/2, 0)$. Likewise $r(1+\cos\theta) = c$ is $r = c - x$, i.e. $y^2 = -2c\left(x - \frac{c}{2}\right)$: parabolas with focus at the origin opening to the left. The two families are confocal parabolas.



Answer. $r = c/(1+\cos\theta)$ (confocal parabolas $y^2 = c^2 - 2cx$ opening to the left).

Problem 3-5 (Simmons)

Solution. *Differential equation.* Differentiating $y^2 = 4c(x+c)$ gives $2yy' = 4c$, so $c = \frac{1}{2}yy'$. Substituting this into the equation of the family,

$$y^2 = 2yy'\left(x + \tfrac{1}{2}yy'\right) \qquad \Longrightarrow \qquad y^2 = 2xy\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2.$$

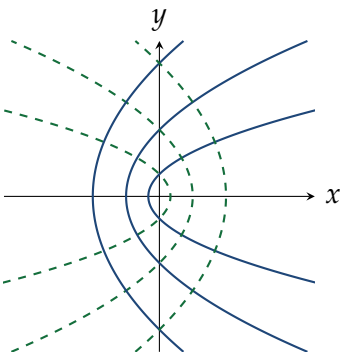
Invariance. Replacing dy/dx by $-dx/dy = -1/y'$:

$$y^2 = -\frac{2xy}{y'} + \frac{y^2}{(y')^2}.$$

Multiplying by $(y')^2$ gives $y^2(y')^2 = -2xyy' + y^2$, i.e. $y^2 = 2xyy' + y^2(y')^2$, which is the original equation. So the equation is unaltered.

Conclusion. The orthogonal trajectories of the family satisfy the same differential equation as the family itself, so they are curves of the same family: the family is *self-orthogonal*. Whenever two curves of the family meet, they cross at right angles. Concretely, through a point (x_0, y_0) with $y_0 \neq 0$, the condition $4c^2 + 4cx_0 - y_0^2 = 0$ is a quadratic in c with discriminant $16(x_0^2 + y_0^2) > 0$ and product of roots $-y_0^2/4 < 0$; so exactly two parabolas of the family pass through the point, one with $c > 0$ (opening right) and one with $c < 0$ (opening left), and these cross orthogonally.

Sketch. The parabola with parameter c has vertex $(-c, 0)$ and focus at the origin; it opens to the right for $c > 0$ (solid) and to the left for $c < 0$ (dashed, drawn in a different style only to distinguish the two halves of the one family).



Answer. $y^2 = 2xy\,y' + y^2(y')^2$. It is unchanged under $y' \to -1/y'$, so the family is self-orthogonal: any two of its curves that meet intersect at right angles.

Problem 3-6 (Simmons)

Solution. Let (x, y) be a point of the curve and $y' = dy/dx$ the slope there; (X, Y) denote running coordinates. The tangent line is $Y - y = y'(X - x)$ and the normal line is $Y - y = -\frac{1}{y'}(X - x)$. From these:

- the tangent meets the x-axis ($Y = 0$) at $X = x - y/y'$ and the y-axis ($X = 0$) at $Y = y - xy'$;
- the normal meets the x-axis at $X = x + yy'$ and the y-axis at $Y = y + x/y'$.

(a) The tangent meets the axes at $(x - y/y', 0)$ and $(0, y - xy')$. Their midpoint is (x, y) exactly when

$$\tfrac{1}{2}\big(x - y/y'\big) = x \quad \text{and} \quad \tfrac{1}{2}\big(y - xy'\big) = y,$$

and each of these is equivalent to $xy' = -y$. Separating, $\dfrac{dy}{y} = -\dfrac{dx}{x}$, so $\log|y| = -\log|x| + C$ and $xy = c$.

(b) The normal segment runs from (x, y) to $(x + yy', 0)$; its projection on the x-axis has length $|yy'|$ (the *subnormal*). So $yy' = \pm 1$, i.e. $y\,\mathrm{d}y = \pm dx$, and integrating, $\frac{1}{2}y^2 = \pm x + C$, or $y^2 = \pm 2x + c$.

(c) The tangent segment runs from (x, y) to $(x - y/y', 0)$; its projection has length $|y/y'|$ (the *subtangent*). So $y/y' = \pm 1$, i.e. $y' = \pm y$, and $\dfrac{dy}{y} = \pm dx$ gives $\log|y| = \pm x + C$, i.e. $y = ce^{\pm x}$.

(d) The tangent segment from (x, y) to $(x - y/y', 0)$ is bisected by the y-axis when its midpoint has abscissa 0:

$$\tfrac{1}{2}\Big(x + x - \frac{y}{y'}\Big) = 0 \implies \frac{y}{y'} = 2x \implies \frac{2\,dy}{y} = \frac{dx}{x} \implies 2\log|y| = \log|x| + C \implies y^2 = cx.$$

(e) The normal segment from (x, y) to $(0, y + x/y')$ is bisected by the x-axis when its midpoint has ordinate 0:

$$\tfrac{1}{2}\Big(y + y + \frac{x}{y'}\Big) = 0 \implies 2yy' = -x \implies 2y\,\mathrm{d}y = -x\,\mathrm{d}x,$$

and integrating, $y^2 = -\frac{1}{2}x^2 + C$, i.e. $x^2 + 2y^2 = c^2$ (with $c^2 = 2C$).

(f) The normal meets the x-axis at the point $(x + yy', 0)$. The distance from (x, y) to this point is $\sqrt{(yy')^2 + y^2}$, and the distance from (x, y) to the origin is $\sqrt{x^2 + y^2}$. Equating the squares,

$$y^2(y')^2 + y^2 = x^2 + y^2 \implies (yy')^2 = x^2 \implies yy' = \pm x \implies y\,\mathrm{d}y = \pm x\,\mathrm{d}x \implies y^2 = \pm x^2 + c.$$

(g) Since $\tan\psi = r\,d\theta/dr$, the condition $\psi = \theta$ gives $r\,d\theta/dr = \tan\theta$, which we write in differential form as

$$r\cos\theta\,\mathrm{d}\theta = \sin\theta\,\mathrm{d}r.$$

Rays. If θ is constant ($d\theta = 0$), this requires $\sin\theta = 0$: the line $\theta = 0$ (the x-axis; $\theta = \pi$ is the same line). Geometrically, along the x-axis the tangent coincides with the radius, so $\psi = 0 = \theta$, as required.

Other curves. Where $\sin\theta \neq 0$ and $r \neq 0$ we may separate:

$$\frac{dr}{r} = \frac{\cos\theta}{\sin\theta}\,\mathrm{d}\theta \implies \log r = \log|\sin\theta| + \log 2c \implies r = 2c\sin\theta,$$

the circles $x^2 + y^2 = 2cy$ tangent to the x-axis at the origin.

(h) Let $\psi = \alpha$ be the constant angle.

- If $\alpha = 0$ the tangent is along the radius, i.e. $r\,d\theta = 0$ along the curve, so $\theta = \theta_0$: the curves are the rays (straight lines through the origin).
- If $\alpha \neq 0$, put $k = \cot\alpha$. Then $r\,d\theta/dr = \tan\alpha$ gives $\dfrac{dr}{r} = \cot\alpha\,\mathrm{d}\theta = k\,\mathrm{d}\theta$, so $\log r = k\theta + C$ and
$$r = ce^{k\theta},$$
the logarithmic (equiangular) spirals. The case $\alpha = \pi/2$ ($k = 0$) gives the circles $r = c$ centred at the origin.

Answer. (a) $xy = c$; (b) $y^2 = \pm 2x + c$; (c) $y = ce^{\pm x}$; (d) $y^2 = cx$; (e) $x^2 + 2y^2 = c^2$; (f) $y^2 = \pm x^2 + c$; (g) $\theta = 0$ or $r = 2c\sin\theta$; (h) $\theta = \theta_0$ (if $\psi = 0$) or $r = ce^{k\theta}$ with $k = \cot\psi$.

Problem 3-7 (Simmons)

Solution. Let the curve be $y = y(x)$ with $y(0) = 0$. The condition is

$$\int_0^x y(t)\,dt = \frac{1}{3}xy(x) \qquad \text{for all } x > 0.$$

Differentiating both sides with respect to x (fundamental theorem of calculus on the left):

$$y = \frac{1}{3}(y + xy') \implies xy' = 2y \implies \frac{dy}{y} = \frac{2\,dx}{x} \implies \log y = 2\log x + C \implies y = cx^2,$$

with $c > 0$ since the curve rises into the first quadrant. Conversely, every such curve satisfies the condition: $\int_0^x ct^2\,dt = \frac{1}{3}cx^3 = \frac{1}{3}x \cdot cx^2$.

Answer. $y = cx^2$ $(c > 0)$.

Problem 3-8 (Simmons)

Solution. With the fourth vertex at (x, y), the vertices are $(0,0)$, $(x,0)$, $(0,y)$, (x,y), and $A = xy$. The condition is $dA/dx = kA$ for a constant k. Since $A > 0$, separate: $\dfrac{dA}{A} = k\,dx$, so $\log A = kx + C$ and $A = ce^{kx}$ $(c = e^C > 0)$. Hence the curve is

$$xy = ce^{kx}.$$

(Equivalently, $d(xy)/dx = kxy$ is $y + xy' = kxy$, i.e. $\dfrac{dy}{y} = \left(k - \dfrac{1}{x}\right)dx$, which integrates to $\log y = kx - \log x + C$, the same result.)

Answer. $xy = ce^{kx}$.

Problem 3-9 (Simmons)

Solution. A raindrop running slowly down a surface moves in the direction of steepest descent, i.e. (in horizontal projection) in the direction of $-\nabla z$; this direction is perpendicular to the level curves $z = $ const. Hence the projections of the paths onto the xy-plane are the orthogonal trajectories of the level curves

$$y^2 - x^2 = k.$$

Differential equation of the level curves. Differentiating, $2yy' - 2x = 0$, so $\frac{dy}{dx} = \dfrac{x}{y}$.

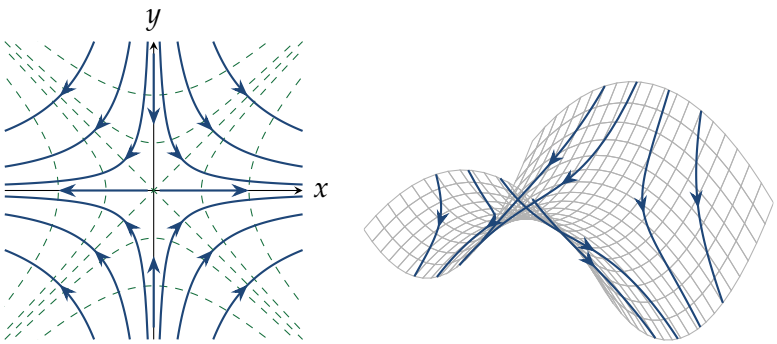
Orthogonal trajectories. Replacing dy/dx by $-dx/dy$:

$$-\frac{dx}{dy} = \frac{x}{y} \implies \frac{dx}{x} = -\frac{dy}{y} \implies \log|x| = -\log|y| + C \implies xy = c.$$

(Directly: $-\nabla z = (2x, -2y)$, so along a path $dy/dx = -2y/(2x) = -y/x$, the same equation.) So the raindrops run along the curves in which the vertical cylinders $xy = c$ cut the saddle

$z = y^2 - x^2$. On a path, (dx, dy) is a positive multiple of $(2x, -2y)$, so $|x|$ increases and $|y|$ decreases: a drop moves away from the yz-plane and toward the xz-plane, and runs off the saddle down the sides $x \to \pm\infty$ where the surface falls away ($z = -x^2$ along the x-axis). The special paths with $c = 0$ are the axes: a drop on the ridge line $x = 0$ (where $z = y^2$) slides toward the saddle point at the origin, and a drop on the line $y = 0$ (where $z = -x^2$) runs straight down and away from the origin.

Sketch. Left: contour map, level curves $y^2 - x^2 = k$ (dashed) and raindrop paths $xy = c$ (solid, arrows pointing downhill). Right: the saddle with a few paths.



The sketch is reasonable: the paths cross every contour at right angles, they descend from the high ridges along the y-direction (the front and back of the saddle) and pour off over the low sides along the x-direction, where a rider's legs would hang.

Answer. The paths are the curves in which the cylinders $xy = c$ intersect the saddle $z = y^2 - x^2$ (drops move with $|x|$ increasing, off the sides of the saddle).

Problem 3-10 (Simmons)

Solution. In each case we differentiate the equation of the family and eliminate the parameter.

(a) From $y = x\sin(x + c)$, $y' = \sin(x + c) + x\cos(x + c)$. Using $\sin(x + c) = y/x$,

$$xy' = x\sin(x + c) + x^2\cos(x + c) = y + x^2\cos(x + c), \qquad \text{so} \qquad xy' - y = x^2\cos(x + c).$$

Squaring and using $\cos^2 = 1 - \sin^2$:

$$(xy' - y)^2 = x^4\big(1 - \sin^2(x + c)\big) = x^4\left(1 - \frac{y^2}{x^2}\right) = x^2(x^2 - y^2).$$

(b) The centre of a circle through $(\pm 1, 0)$ is equidistant from these points, so it lies on the perpendicular bisector $x = 0$: centre $(0, k)$, radius$^2 = 1 + k^2$. The family is

$$x^2 + (y - k)^2 = 1 + k^2, \qquad \text{i.e.} \qquad x^2 + y^2 - 1 = 2ky.$$

Divide by y to isolate the parameter, $\dfrac{x^2 + y^2 - 1}{y} = 2k$, and differentiate (quotient rule):

$$\frac{(2x + 2yy')y - (x^2 + y^2 - 1)y'}{y^2} = 0$$

$$\implies \quad 2xy + 2y^2y' - (x^2 + y^2 - 1)y' = 0 \quad \implies \quad (x^2 - y^2 - 1)y' = 2xy.$$

(c) A circle with centre (a, a) on $y = x$ is tangent to both axes exactly when its radius is $|a|$. The family is $(x-a)^2 + (y-a)^2 = a^2$. Differentiating,

$$(x-a) + (y-a)y' = 0 \implies a = \frac{x + yy'}{1 + y'}.$$

Then

$$x - a = \frac{x + xy' - x - yy'}{1 + y'} = \frac{(x-y)y'}{1+y'}, \qquad y - a = \frac{y + yy' - x - yy'}{1+y'} = \frac{y - x}{1+y'}.$$

Substituting into $(x-a)^2 + (y-a)^2 = a^2$ and multiplying by $(1+y')^2$:

$$(x-y)^2(y')^2 + (x-y)^2 = (x+yy')^2, \qquad \text{i.e.} \qquad (x-y)^2\big(1 + (y')^2\big) = (x+yy')^2.$$

(At a point where $y' = -1$ the relation $(x-a) + (y-a)y' = 0$ forces $x = y$, and the final equation reads $0 = 0$ there as well, so it holds along every circle of the family.)

(d) The tangent to $x^2 = 4y$ (i.e. $y = x^2/4$, slope $x/2$) at $(2a, a^2)$ has slope a, so its equation is $y - a^2 = a(x - 2a)$, i.e.

$$y = ax - a^2.$$

Differentiating, $y' = a$. Eliminating a: $y = xy' - (y')^2$, i.e. $y + (y')^2 = xy'$.

(e) A non-vertical line $y = mx + b$ is tangent to the unit circle exactly when its distance from the origin is 1: $|b|/\sqrt{1+m^2} = 1$, i.e. $b^2 = 1 + m^2$. Along such a line $y' = m$ and $b = y - xy'$, so

$$(y - xy')^2 = 1 + (y')^2.$$

(The two vertical tangents $x = \pm 1$ are not of the form $y = mx + b$ and are not described by an equation in y'.)

Answer. (a) $(xy' - y)^2 = x^2(x^2 - y^2)$; (b) $(x^2 - y^2 - 1)y' = 2xy$; (c) $(x-y)^2(1+y'^2) = (x+yy')^2$; (d) $y + y'^2 = xy'$; (e) $(y - xy')^2 = 1 + y'^2$.

Problem 3-11 (Simmons)

Solution. *The parabola.* For $y = x^2/4$ we have $y' = x/2$, and

$$y + (y')^2 = \frac{x^2}{4} + \frac{x^2}{4} = \frac{x^2}{2} = x \cdot \frac{x}{2} = xy',$$

so the parabola satisfies $y + (y')^2 = xy'$: it is an integral curve. Now let $(2a, a^2)$ be any point of the parabola. The tangent line $y = ax - a^2$ at that point is also an integral curve (Problem 10(d)), and it passes through $(2a, a^2)$. It is a different curve from the parabola (a line is not a parabola; it touches the parabola only at that point, since $x^2/4 - (ax - a^2) = (x - 2a)^2/4$ vanishes only for $x = 2a$). So two integral curves pass through each point of the parabola: the parabola and its tangent line there. (Algebraically: solving $(y')^2 - xy' + y = 0$ gives $y' = \frac{1}{2}\big(x \pm \sqrt{x^2 - 4y}\big)$; on the parabola the discriminant $x^2 - 4y$ vanishes, so the right side fails to have a bounded $\partial/\partial y$ there, and the uniqueness part of Picard's theorem does not apply.)

The unit circle. On the upper or lower semicircle $y = \pm\sqrt{1 - x^2}$ ($|x| < 1$), implicit differentiation of $x^2 + y^2 = 1$ gives $y' = -x/y$. Then

$$(y - xy')^2 = \left(y + \frac{x^2}{y}\right)^2 = \left(\frac{x^2 + y^2}{y}\right)^2 = \frac{1}{y^2}, \qquad 1 + (y')^2 = 1 + \frac{x^2}{y^2} = \frac{x^2 + y^2}{y^2} = \frac{1}{y^2},$$

so the circle satisfies $(y - xy')^2 = 1 + (y')^2$: it is an integral curve. Through each of its points (with $|x| < 1$) passes also the tangent line to the circle at that point, which is an integral curve by Problem 10(e) and is distinct from the circle. Hence again two integral curves pass through each such point: the circle and its tangent line. (At the two points $(\pm 1, 0)$ the tangent line is vertical, so it is not a graph $y = y(x)$ and is not described by an equation in y'; there the circle is likewise not a graph over x, and these two points are excluded.)

Answer. Shown: $y = x^2/4$ satisfies $y + y'^2 = xy'$, and $x^2 + y^2 = 1$ satisfies $(y - xy')^2 = 1 + y'^2$; through each point of either curve pass two integral curves, the curve itself and its tangent line at that point.

Section 4 Growth, Decay, Chemical Reactions, and Mixing

Problem 4-1 (Simmons)

Solution. First, every function $y = ce^{kx}$ is a solution, since $y' = cke^{kx} = k(ce^{kx}) = ky$.
Conversely, let $f(x)$ be any solution on an interval I, so that $f'(x) = kf(x)$ for all $x \in I$. Put

$$g(x) = \frac{f(x)}{e^{kx}} = f(x)e^{-kx},$$

which is defined everywhere because $e^{kx} \neq 0$. By the product rule,

$$g'(x) = f'(x)e^{-kx} - kf(x)e^{-kx} = \big(f'(x) - kf(x)\big)e^{-kx} = 0 \cdot e^{-kx} = 0 \qquad (x \in I).$$

A function whose derivative vanishes at every point of an interval is constant there (mean value theorem), so $g(x) = c$ for some constant c. Hence $f(x) = ce^{kx}$ on I. Thus every solution has the form ce^{kx}; note in particular that this argument does not divide by y, so the solution $y \equiv 0$ (the case $c = 0$) is included and no solution is lost.

Answer. Shown: if $f' = kf$ on an interval then $(fe^{-kx})' = 0$, so $f = ce^{kx}$; these are all the solutions.

Problem 4-2 (Simmons)

Solution. An annual rate of r percent means $k = r/100$ in Example 1, so by formula (3) the amount after t years is

$$A = Pe^{rt/100} \qquad \Big(\text{the solution of } \frac{dA}{dt} = \tfrac{r}{100}A,\ A(0) = P\Big).$$

(a) The investment has doubled when $A = 2P$:

$$2P = Pe^{rT/100} \iff e^{rT/100} = 2 \iff \frac{rT}{100} = \log 2 \iff T = \frac{100\log 2}{r} \text{ years.}$$

(For example, at $r = 5$ percent, $T = 20\log 2 \approx 13.86$ years.)

(b) Setting $T = 10$ in part (a) and solving for r:

$$r = \frac{100\log 2}{10} = 10\log 2 \approx 10(0.693147) \approx 6.93 \text{ percent.}$$

Answer. (a) $T = \dfrac{100\log 2}{r}$ years; (b) $r = 10\log 2 \approx 6.93$ percent.

Problem 4-3 (Simmons)

Solution.

(a) The amount $A(t)$ grows for two reasons: interest, at the rate kA dollars per year, and new deposits, at the rate D dollars per year. Hence

$$\frac{dA}{dt} = kA + D, \qquad A(0) = 0 \quad \text{(no initial capital)}.$$

Since $kA + D > 0$ for $A \geq 0$, we may separate the variables:

$$\frac{dA}{kA+D} = dt \quad \Longrightarrow \quad \frac{1}{k}\log(kA+D) = t + c_0.$$

(Here we integrated with the substitution $u = kA + D$, $du = k\,dA$.) At $t = 0$, $A = 0$ gives $c_0 = \frac{1}{k}\log D$, so

$$\log\frac{kA+D}{D} = kt, \qquad kA + D = De^{kt}, \qquad A = \frac{D}{k}\left(e^{kt}-1\right) = D\left(\frac{e^{kt}-1}{k}\right).$$

(b) With $k = 0.06$, $t = 40$ and $A = 10^6$:

$$D = \frac{10^6 k}{e^{40k}-1} = \frac{60{,}000}{e^{2.4}-1} = \frac{60{,}000}{11.023176-1} = \frac{60{,}000}{10.023176} \approx 5986.13.$$

So $D \approx \$5986$ per year.

(c) With $k = 0.10$, $t = 40$:

$$D = \frac{10^6(0.1)}{e^4-1} = \frac{100{,}000}{54.598150-1} = \frac{100{,}000}{53.598150} \approx 1865.74,$$

so $D \approx \$1866$ per year. Indeed, $40 \times 1866 = 74{,}640 < 80{,}000$, confirming the parenthetical remark: without interest these deposits would total less than \$80,000.

Answer. (a) $A = \dfrac{D}{k}\left(e^{kt}-1\right)$; (b) $D \approx \$5986$ per year; (c) $D \approx \$1866$ per year.

Problem 4-4 (Simmons)

Solution.

(a) Interest adds kA dollars per year and withdrawals remove W dollars per year, so

$$\frac{dA}{dt} = kA - W, \qquad A(0) = P.$$

Write the equation as $\frac{d}{dt}\left(A - \frac{W}{k}\right) = k\left(A - \frac{W}{k}\right)$. By Problem 1 (applied to the function $A - W/k$), $A - W/k = Ce^{kt}$ for a constant C. At $t = 0$, $C = P - W/k$. Therefore

$$A = \frac{W}{k} + \left(P - \frac{W}{k}\right)e^{kt}.$$

(b) A is constant exactly when the coefficient of e^{kt} vanishes, i.e. $P = W/k$ (equivalently $dA/dt = kP - W = 0$ at $t = 0$). Thus

$$W_0 = kP,$$

which is just the interest earned on P.

(c) If $W > W_0 = kP$, then $P - W/k < 0$ and A decreases. It vanishes at the time T with

$$0 = \frac{W}{k} + \left(P - \frac{W}{k}\right)e^{kT} \iff e^{kT} = \frac{W/k}{W/k - P} = \frac{W}{W - kP} = \frac{W}{W - W_0}.$$

(The right side exceeds 1, so $T > 0$.) Hence

$$T = \frac{1}{k} \log \frac{W}{W - W_0} \text{ years.}$$

(d) With $k = 0.05$ and $W = 2W_0$: $\dfrac{W}{W - W_0} = \dfrac{2W_0}{W_0} = 2$, so

$$T = \frac{1}{0.05} \log 2 = 20 \log 2 \approx 13.86 \text{ years.}$$

Answer. (a) $A = \dfrac{W}{k} + \left(P - \dfrac{W}{k}\right) e^{kt}$; (b) $W_0 = kP$; (c) $T = \dfrac{1}{k} \log \dfrac{W}{W - W_0}$ years; (d) $20 \log 2 \approx 13.86$ years.

Problem 4-5 (Simmons)

Solution. Let $x = x(t)$ be his wealth in millions of dollars at time t (years), with $t = 0$ one year ago. Then

$$\frac{dx}{dt} = kx^2, \qquad x(0) = 10, \quad x(1) = 20.$$

Since $x > 0$ we may divide by x^2 and separate the variables:

$$\frac{dx}{x^2} = k\,dt \quad \Longrightarrow \quad -\frac{1}{x} = kt + c.$$

From $x(0) = 10$: $c = -\frac{1}{10}$. From $x(1) = 20$: $-\frac{1}{20} = k - \frac{1}{10}$, so $k = \frac{1}{20}$. Hence

$$-\frac{1}{x} = \frac{t}{20} - \frac{1}{10} = \frac{t - 2}{20}, \qquad \text{i.e.} \qquad x = \frac{20}{2 - t} \quad (t < 2).$$

Six months from now is $t = 1.5$: $x = 20/0.5 = 40$ million dollars.

One year from now is $t = 2$: as $t \to 2^-$, $x = 20/(2 - t) \to \infty$. The model predicts that his fortune becomes infinite at the end of the year (the solution "blows up" in finite time), which of course signals that this growth law cannot persist.

Answer. $x = 20/(2 - t)$ million dollars ($t = 0$ one year ago): 40 million dollars in 6 months; as $t \to 2$ (one year from now) $x \to \infty$.

Problem 4-6 (Simmons)

Solution. Measure t in hours from 6 P.M. By Example 2, $x = x_0 e^{kt}$, where x_0 is the population at 6 P.M. Tripling in one hour gives $3x_0 = x_0 e^k$, so $e^k = 3$, $k = \log 3$, and $x = x_0\, 3^t$. The population is $100x_0$ when

$$3^t = 100 \iff t = \frac{\log 100}{\log 3} = \frac{4.605170}{1.098612} \approx 4.1918 \text{ hours.}$$

Since $0.1918 \times 60 \approx 11.5$ minutes, this is about 4 hours $11\frac{1}{2}$ minutes after 6 P.M., i.e. at about 10:11 P.M.

Answer. $t = \log 100/\log 3 \approx 4.19$ hours after 6 P.M., i.e. at about 10:11 P.M.

Problem 4-7 (Simmons)

Solution. By Example 2, $x = x_0 e^{kt}$ with x_0 the original population. Doubling in 2 years gives $e^{2k} = 2$, so $e^k = 2^{1/2}$ and $x = x_0\, 2^{t/2}$. At $t = 3$,

$$10{,}000 = x_0\, 2^{3/2} \quad\Longrightarrow\quad x_0 = \frac{10{,}000}{2\sqrt{2}} = 2500\sqrt{2} \approx \frac{10{,}000}{2.828427} \approx 3535.5.$$

So the original population was about 3536 (in round numbers, about 3500).

Remark. The book's answer is 3531. That value results from using the rounded value $2\sqrt{2} \approx 2.832$ (indeed $10{,}000/2.832 \approx 3531$); with $2\sqrt{2} = 2.828427\ldots$ the exact value is $2500\sqrt{2} = 3535.53\ldots$ As a check, $3535.53 \times 2 = 7071.07$ after 2 years, and $7071.07 \times \sqrt{2} = 10{,}000.0$ after 3 years.

Answer. $x_0 = 10{,}000/2^{3/2} = 2500\sqrt{2} \approx 3536$.

Problem 4-8 (Simmons)

Solution. With t in years from the beginning of 1970 and x in billions, formula (5) gives $x = 3.6\, e^{t/50}$ ($k = 0.02 = 1/50$). The earth is full when $x = 30$:

$$e^{t/50} = \frac{30}{3.6} = \frac{25}{3} \quad\Longrightarrow\quad t = 50 \log \frac{25}{3} = 50(2.120264) \approx 106.0 \text{ years},$$

that is, in the year $1970 + 106 = 2076$.

In the year 2000, $t = 30$ and

$$x = 3.6\, e^{30/50} = 3.6\, e^{0.6} = 3.6(1.822119) \approx 6.56 \approx 6.6 \text{ billion}.$$

Answer. The earth will be full after $50\log(25/3) \approx 106$ years, in the year 2076; population in $2000 \approx 3.6e^{0.6} \approx 6.6$ billion.

Problem 4-9 (Simmons)

Solution.

(a) The hypothesis is $dx/dt = kx$, so by Problem 1 (or Example 2), $x = x_0 e^{kt}$, and $x_0 = x(0) = 2$. The condition $x(2) = 3$ gives

$$3 = 2e^{2k} \quad\Longrightarrow\quad e^{2k} = \tfrac{3}{2} \quad\Longrightarrow\quad e^k = \left(\tfrac{3}{2}\right)^{1/2}.$$

Therefore $x = 2\left(e^k\right)^t = 2\left(\tfrac{3}{2}\right)^{t/2}$, as was to be shown.

(b) At $t = 10$ days:

$$x(10) = 2\left(\frac{3}{2}\right)^5 = 2 \cdot \frac{243}{32} = \frac{243}{16} = 15.1875 \approx 15.2 \text{ grams}.$$

Answer. (a) Shown: $x = 2(3/2)^{t/2}$; (b) $x(10) = 243/16 \approx 15.2$ grams.

Problem 4-10 (Simmons)

Solution. As in Example 2, $x(0) = x_0$, and now

$$\frac{dx}{dt} = kx(x_1 - x), \qquad k > 0,$$

where we assume $0 < x_0 < x_1$ (a small colony placed in the environment). The constant solutions $x \equiv 0$ and $x \equiv x_1$ are excluded by $0 < x_0 < x_1$, and since solutions of this equation cannot cross these constant solutions, $0 < x < x_1$ for all t; so we may divide by $x(x_1 - x)$. By partial fractions,

$$\frac{1}{x(x_1 - x)} = \frac{1}{x_1}\left(\frac{1}{x} + \frac{1}{x_1 - x}\right) \qquad \left(\text{check: } \tfrac{(x_1-x)+x}{x_1 x(x_1-x)} = \tfrac{1}{x(x_1-x)}\right).$$

Separating the variables and integrating,

$$\frac{1}{x_1}\int\left(\frac{1}{x} + \frac{1}{x_1 - x}\right)dx = \int k\,\mathrm{d}t \quad \Longrightarrow \quad \frac{1}{x_1}\left(\log x - \log(x_1 - x)\right) = kt + c,$$

so $\log \dfrac{x}{x_1 - x} = kx_1 t + c'$. At $t = 0$, $c' = \log \dfrac{x_0}{x_1 - x_0}$, hence

$$\frac{x}{x_1 - x} = \frac{x_0}{x_1 - x_0}\, e^{kx_1 t}.$$

Solving for x: $x(x_1 - x_0) = x_0(x_1 - x)e^{kx_1 t}$, so $x\left[(x_1 - x_0) + x_0 e^{kx_1 t}\right] = x_0 x_1 e^{kx_1 t}$; dividing numerator and denominator by $e^{kx_1 t}$,

$$x = \frac{x_0 x_1}{x_0 + (x_1 - x_0)e^{-kx_1 t}}.$$

As $t \to \infty$, $e^{-kx_1 t} \to 0$ and $x \to x_1$, as required by the problem.

Most rapid increase. The growth rate $dx/dt = kx(x_1 - x) = k\left[\frac{x_1^2}{4} - (x - \frac{x_1}{2})^2\right]$ is a quadratic function of x that is largest when $x = \frac{1}{2}x_1$. Equivalently,

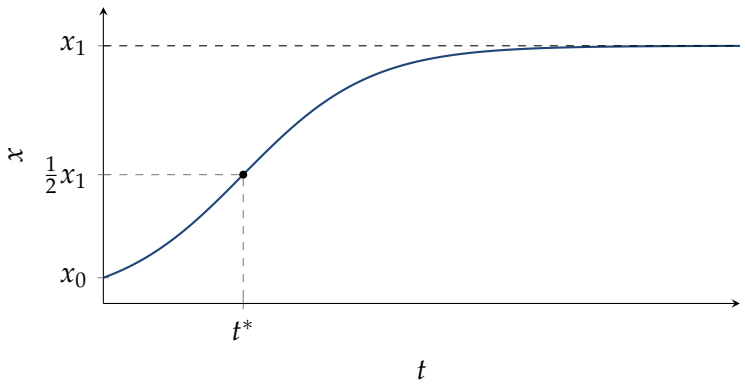
$$\frac{d^2x}{dt^2} = k(x_1 - 2x)\frac{dx}{dt},$$

which is positive for $x < x_1/2$ and negative for $x > x_1/2$: the graph has an inflection point when the population is half its limiting value. Setting $x = x_1/2$ in $\frac{x}{x_1 - x} = \frac{x_0}{x_1 - x_0}e^{kx_1 t}$ gives $1 = \frac{x_0}{x_1 - x_0}e^{kx_1 t}$, i.e. this happens at

$$t^* = \frac{1}{kx_1}\log\frac{x_1 - x_0}{x_0} \qquad (\text{if } x_0 < \tfrac{1}{2}x_1).$$

(If $x_0 \geq \frac{1}{2}x_1$, the growth rate decreases from the start, and the population increases most rapidly at $t = 0$.)

Sketch. The graph is the S-shaped "logistic curve": it starts at x_0, rises slowly, then steeply around t^* where it crosses $x = x_1/2$, and levels off, approaching the horizontal asymptote $x = x_1$. The figure shows the case $x_1 = 1$, $x_0 = 0.1$, $kx_1 = 1$ (so $t^* = \log 9 \approx 2.2$).



Answer. $x = \dfrac{x_0 x_1}{x_0 + (x_1 - x_0)e^{-kx_1 t}}$ (logistic curve, $x \to x_1$); the population increases most rapidly when $x = \frac{1}{2}x_1$, i.e. at $t = \frac{1}{kx_1}\log\frac{x_1 - x_0}{x_0}$.

Problem 4-11 (Simmons)

Solution. The hypothesis is $dn/dt = kn$ with $n(0) = n_0$, so by Example 2, $n = n_0 e^{kt}$. Hence

$$\frac{n_1}{n_0} = e^{kt_1}, \qquad \frac{n_2}{n_0} = e^{kt_2}.$$

Raising the first equation to the power t_2 and the second to the power t_1,

$$\left(\frac{n_1}{n_0}\right)^{t_2} = e^{kt_1 t_2} = \left(\frac{n_2}{n_0}\right)^{t_1}.$$

Answer. Shown: since $n_i/n_0 = e^{kt_i}$, both sides equal $e^{kt_1 t_2}$.

Problem 4-12 (Simmons)

Solution. By Example 3, $x = x_0 e^{-kt}$ with $kT = \log 2$ for the half-life $T = 1600$ years, so $e^{-k} = 2^{-1/1600}$ and

$$\frac{x}{x_0} = e^{-kt} = \left(\frac{1}{2}\right)^{t/1600}.$$

At $t = 2400$: $\dfrac{x}{x_0} = \left(\frac{1}{2}\right)^{3/2} = \dfrac{1}{2\sqrt{2}} = \dfrac{\sqrt{2}}{4} \approx 0.353553$, i.e. about 35.36 percent.

At $t = 8000$: $\dfrac{x}{x_0} = \left(\frac{1}{2}\right)^{5} = \dfrac{1}{32} = 0.03125$, i.e. 3.125 percent.

Remark. The book gives "about 35.35 percent"; the exact value $100\sqrt{2}/4 = 35.3553\ldots$ percent rounds to 35.36 (the book's figure is truncated rather than rounded).

Answer. After 2400 years: $100/(2\sqrt{2}) \approx 35.36$ percent; after 8000 years: $100/32 = 3.125$ percent.

Problem 4-13 (Simmons)

Solution. As in Problem 12, $x/x_0 = (1/2)^{t/20}$. When 99 percent has decayed, 1 percent remains:

$$\left(\frac{1}{2}\right)^{t/20} = \frac{1}{100} \iff \frac{t}{20}\log 2 = \log 100 \iff t = \frac{20\log 100}{\log 2} = \frac{20(4.605170)}{0.693147} \approx 132.9 \text{ days.}$$

Answer. $t = 20\log 100/\log 2 \approx 133$ days.

Problem 4-14 (Simmons)

Solution. A kill rate of 200 per 100 per hour means that grasshoppers die at the rate of 2 per grasshopper present per hour, i.e. the number x alive satisfies

$$-\frac{dx}{dt} = 2x, \qquad x(0) = x_0,$$

with t in hours. This is equation (6) with $k = 2$, so by (8), $x = x_0 e^{-2t}$. After 1 hour,

$$\frac{x}{x_0} = e^{-2} \approx 0.135335,$$

so about 13.53 percent are still alive. (Note that a naive reading "200 percent die per hour" would suggest none survive; the continuous model shows that the decay is exponential.)

Answer. $100e^{-2} \approx 13.53$ percent.

Problem 4-15 (Simmons)

Solution. By Example 3, $x = x_0 e^{-kt}$ for some constants x_0 and $k > 0$. Then

$$x_1 = x_0 e^{-kt_1}, \quad x_2 = x_0 e^{-kt_2} \quad\Longrightarrow\quad \frac{x_1}{x_2} = e^{k(t_2-t_1)} \quad\Longrightarrow\quad k = \frac{\log(x_1/x_2)}{t_2 - t_1}.$$

The half-life T satisfies $kT = \log 2$ (Example 3), so

$$T = \frac{\log 2}{k} = \frac{(t_2 - t_1)\log 2}{\log(x_1/x_2)}.$$

Answer. Shown: $k = \log(x_1/x_2)/(t_2 - t_1)$, hence $T = \log 2/k = (t_2 - t_1)\log 2/\log(x_1/x_2)$.

Problem 4-16 (Simmons)

Solution. When x grams of compound have formed, ax grams of the first substance and bx grams of the second have been used up, so the untransformed amounts are $aA - ax = a(A - x)$ and $bB - bx = b(B - x)$. The law of mass action gives

$$\frac{dx}{dt} = k\,a(A - x)\,b(B - x) = kab\,(A - x)(B - x), \qquad x(0) = 0, \quad k > 0.$$

The roles of the two substances are symmetric, so we may assume $B \le A$. Since $dx/dt > 0$ while $x < B$ and the constant function $x \equiv B$ is a solution, we have $0 \le x < B \le A$ for $t \ge 0$; the reaction stops only in the limit, when the second substance is exhausted.

Case $B = A$. The equation is $\dfrac{dx}{(A-x)^2} = kab\,dt$. Integrating,

$$\frac{1}{A-x} = kab\,t + c, \qquad c = \frac{1}{A} \ (\text{from } x(0)=0),$$

so

$$A - x = \frac{1}{kabt + 1/A} = \frac{A}{kAabt+1}, \qquad x = A - \frac{A}{kAabt+1} = \frac{kA^2abt}{kAabt+1}.$$

Case $B < A$. Partial fractions give

$$\frac{1}{(A-x)(B-x)} = \frac{1}{A-B}\left(\frac{1}{B-x} - \frac{1}{A-x}\right),$$

as is checked by combining the right side: $\frac{(A-x)-(B-x)}{(A-B)(A-x)(B-x)}$. Separating variables and integrating (with $0 \le x < B$, so both logarithms are defined),

$$\frac{1}{A-B}\big(-\log(B-x) + \log(A-x)\big) = kab\,t + c, \quad \text{i.e.} \quad \log\frac{A-x}{B-x} = k(A-B)ab\,t + c'.$$

At $t = 0$, $c' = \log(A/B)$, so with $E = e^{k(A-B)abt}$,

$$\frac{A-x}{B-x} = \frac{A}{B}E \implies B(A-x) = AE(B-x) \implies x(AE-B) = AB(E-1) \implies x = \frac{AB(E-1)}{AE-B}.$$

Dividing numerator and denominator by E,

$$x = \frac{AB\big(1 - e^{-k(A-B)abt}\big)}{A - Be^{-k(A-B)abt}}.$$

As $t \to \infty$, $x \to AB/A = B$: the reaction proceeds until the scarcer substance is used up.

The first formula is the limit of the second as $B \to A$. Fix $t > 0$ and put $s = kabt$. As functions of B, the numerator $N(B) = AB\big(1 - e^{-s(A-B)}\big)$ and denominator $D(B) = A - Be^{-s(A-B)}$ both tend to 0 as $B \to A$, so by l'Hospital's rule the limit is $N'(A)/D'(A)$. Using $\frac{d}{dB}e^{-s(A-B)} = se^{-s(A-B)}$,

$$\begin{aligned} N'(B) &= A\big(1 - e^{-s(A-B)}\big) - ABs\,e^{-s(A-B)} &&\implies N'(A) = -A^2 s,\\ D'(B) &= -e^{-s(A-B)} - Bs\,e^{-s(A-B)} &&\implies D'(A) = -1 - As. \end{aligned}$$

Hence

$$\lim_{B\to A} x = \frac{-A^2 s}{-1-As} = \frac{A^2 s}{As+1} = \frac{kA^2abt}{kAabt+1},$$

which is the formula of the case $B = A$.

Answer. If $B = A$: $x = \dfrac{kA^2abt}{kAabt+1}$; if $B < A$: $x = \dfrac{AB\big(1 - e^{-k(A-B)abt}\big)}{A - Be^{-k(A-B)abt}}$ (and the first is the limit of the second as $B \to A$).

Problem 4-17 (Simmons)

Solution. (The book's statement reads "$x =$ when $t = t_1$"; the symbol x_1 has evidently dropped out, and we take the condition to be $x = x_1$ when $t = t_1$, as in the book's answer.)

At time t the amount dissolved is $x_0 - x$ (initially nothing is dissolved), so the concentration of the actual solution is $(x_0 - x)/G$, while that of a saturated solution is S/G. The rate of dissolving is $-dx/dt$, so the law stated gives

$$-\frac{dx}{dt} = kx\left(\frac{S}{G} - \frac{x_0 - x}{G}\right) = \frac{k}{G}\,x(x + A), \qquad \text{where } A = S - x_0, \quad k > 0.$$

Assume first $A \neq 0$. Since $x + A = S - (\text{amount dissolved}) > 0$ until saturation and $x > 0$, we may divide by $x(x + A)$. Partial fractions:

$$\frac{1}{x(x+A)} = \frac{1}{A}\left(\frac{1}{x} - \frac{1}{x+A}\right).$$

Separating the variables and integrating,

$$\frac{1}{A}\big(\log x - \log(x + A)\big) = -\frac{k}{G}t + c \quad \Longrightarrow \quad \log\frac{x+A}{x} = \frac{kA}{G}t + c',$$

that is,

$$1 + \frac{A}{x} = \left(1 + \frac{A}{x_0}\right)e^{kAt/G} \qquad (\text{using } x = x_0 \text{ at } t = 0).$$

The condition $x = x_1$ at $t = t_1$ determines the unknown constant k: $e^{kAt_1/G} = \dfrac{1 + A/x_1}{1 + A/x_0}$, hence $e^{kAt/G} = \left(e^{kAt_1/G}\right)^{t/t_1}$ and

$$\frac{1 + (A/x)}{1 + (A/x_0)} = \left[\frac{1 + (A/x_1)}{1 + (A/x_0)}\right]^{t/t_1}, \qquad A = S - x_0.$$

Solving explicitly for x, with $R = \dfrac{1 + A/x_1}{1 + A/x_0}$,

$$x = \frac{A}{\left(1 + \dfrac{A}{x_0}\right)R^{t/t_1} - 1}.$$

(If $A > 0$, i.e. $S > x_0$, then $R > 1$ and $x \to 0$: everything eventually dissolves. If $A < 0$, then $x \to -A = x_0 - S$, the excess that can never dissolve.)

In the special case $A = 0$ ($S = x_0$) the equation is $-dx/dt = (k/G)x^2$, so $\frac{1}{x} = \frac{1}{x_0} + \frac{k}{G}t$, and the condition at t_1 gives $\dfrac{1}{x} - \dfrac{1}{x_0} = \dfrac{t}{t_1}\left(\dfrac{1}{x_1} - \dfrac{1}{x_0}\right)$, which is also the limit of the general formula as $A \to 0$.

Answer. $\dfrac{1 + (A/x)}{1 + (A/x_0)} = \left[\dfrac{1 + (A/x_1)}{1 + (A/x_0)}\right]^{t/t_1}$, where $A = S - x_0$;
i.e. $x = A \Big/ \left[\left(1 + \frac{A}{x_0}\right)R^{t/t_1} - 1\right]$ with $R = \frac{1 + A/x_1}{1 + A/x_0}$.

Problem 4-18 (Simmons)

Solution. By (2) and (3), $dx/dt = kxy$ with $k > 0$, and $y = 1 - x$, so

$$\frac{dx}{dt} = kx(1-x), \qquad x(0) = x_0, \quad 0 < x_0 < 1.$$

This is the equation of Problem 10 with $x_1 = 1$. Explicitly: for $0 < x < 1$, $\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$, so separating variables and integrating,

$$\log x - \log(1-x) = kt + c, \qquad \frac{x}{1-x} = \frac{x_0}{1-x_0} e^{kt} \quad \left(c = \log \tfrac{x_0}{1-x_0}\right).$$

Solving, $x(1-x_0) = x_0(1-x)e^{kt}$, so $x\left[(1-x_0) + x_0 e^{kt}\right] = x_0 e^{kt}$ and

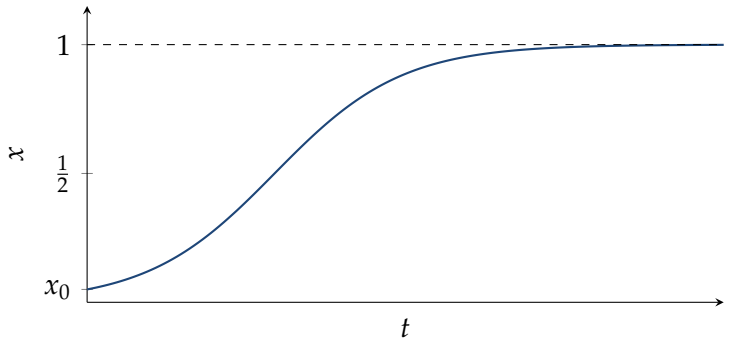
$$x = \frac{x_0}{x_0 + (1-x_0)e^{-kt}}.$$

Since $k > 0$, $e^{-kt} \to 0$ as $t \to \infty$, so

$$\lim_{t\to\infty} x = \frac{x_0}{x_0 + 0} = 1,$$

i.e. the proportion infected tends to 1: ultimately the disease spreads through the entire population (as long as $x_0 > 0$). Also $dx/dt > 0$ throughout, so x increases steadily; by Problem 10 the spread is fastest when half the population is infected, at $t = \frac{1}{k} \log \frac{1-x_0}{x_0}$ (if $x_0 < \frac{1}{2}$).

Sketch (case $x_0 = 0.05$, $k = 1$): an S-shaped curve rising from x_0 to the asymptote $x = 1$.



Answer. $x = \dfrac{x_0}{x_0 + (1-x_0)e^{-kt}}$; since $x \to 1$ as $t \to \infty$, the disease ultimately spreads through the entire population.

Problem 4-19 (Simmons)

Solution. Let x be the number of pounds of salt in the tank at time t (minutes). The volume stays at 100 gallons, so the concentration is $x/100$ lb/gal. No salt enters, and salt leaves at the rate $(x/100) \cdot 5 = x/20$ lb/min. Hence (as in Example 4)

$$\frac{dx}{dt} = -\frac{x}{20}, \qquad x(0) = 40 \quad \Longrightarrow \quad x = 40e^{-t/20}.$$

A concentration of 0.1 lb/gal means $x = 10$ pounds:

$$40e^{-t/20} = 10 \iff e^{t/20} = 4 \iff t = 20\log 4 = 40\log 2 \approx 27.73 \text{ minutes}.$$

Remark. The book gives $40\log 2 \cong 27.72$ minutes; this comes from using $\log 2 \approx 0.693$. With $\log 2 = 0.693147\ldots$ the value is $27.7259\ldots \approx 27.73$.

Answer. $t = 40\log 2 \approx 27.7$ minutes.

Problem 4-20 (Simmons)

Solution. Let x be the amount of pollutant at time t (hours), $x(0) = x_0$. The volume stays at 10 gallons, so the concentration is $x/10$, and pollutant is removed at the rate $(x/10)\cdot 5 = x/2$ per hour (the incoming water is pure). Thus

$$\frac{dx}{dt} = -\frac{x}{2} \quad\Longrightarrow\quad x = x_0 e^{-t/2}.$$

The pollution is half its initial level when $e^{-t/2} = \frac{1}{2}$, i.e.

$$t = 2\log 2 \approx 1.39 \text{ hours} \quad (\approx 1 \text{ hour } 23 \text{ minutes}).$$

Answer. $t = 2\log 2 \approx 1.39$ hours.

Problem 4-21 (Simmons)

Solution. Let x be the volume (ft^3) of carbon monoxide in the room at time t (minutes); $x(0) = 0$. The volume of air stays at 1800 ft^3, so the CO concentration is $x/1800$.

$$\text{rate of entering} = 0.06 \times 0.15 = 0.009 \text{ ft}^3/\text{min},$$
$$\text{rate of leaving} = \frac{x}{1800} \times 0.15 = \frac{x}{12{,}000} \text{ ft}^3/\text{min}.$$

Therefore

$$\frac{dx}{dt} = 0.009 - \frac{x}{12{,}000} = \frac{108 - x}{12{,}000}.$$

Since $x < 108$ (initially $x = 0$), separating the variables gives

$$\frac{dx}{108 - x} = \frac{dt}{12{,}000}, \qquad -\log(108 - x) = \frac{t}{12{,}000} + c,$$

and $x(0) = 0$ gives $c = -\log 108$. Hence $108 - x = 108e^{-t/12{,}000}$, i.e.

$$x = 108\left(1 - e^{-t/12{,}000}\right), \qquad \text{concentration} = \frac{x}{1800} = 0.06\left(1 - e^{-t/12{,}000}\right).$$

The concentration reaches 0.00018 when

$$1 - e^{-t/12{,}000} = \frac{0.00018}{0.06} = 0.003 \iff e^{-t/12{,}000} = 0.997$$
$$\iff t = -12{,}000\log(0.997)$$
$$= 12{,}000(0.0030045) \approx 36.05 \text{ minutes}.$$

A prudent person should leave no later than about 36 minutes after the smoking starts.

Answer. $t = -12{,}000\log 0.997 \approx 36$ minutes after the smoking starts.

Problem 4-22 (Simmons)

Solution. *By thinking.* Every 10-foot layer of water absorbs the same fraction of the light that falls on it, namely one-half (the water is uniform, so a layer at any depth behaves like the top layer). Thus the intensity is $\frac{1}{2}$ of the original at 10 ft, $\frac{1}{4}$ at 20 ft, $\frac{1}{8}$ at 30 ft and $\frac{1}{16}$ at 40 ft.

By a differential equation. Let $I(h)$ be the intensity at depth h. A thin layer from h to $h+\Delta h$ absorbs the fraction $-\Delta I/I$ of the light incident on it, and Lambert's law says $-\Delta I/I \approx k\,\Delta h$. Letting $\Delta h \to 0$,

$$\frac{dI}{dh} = -kI \quad \Longrightarrow \quad I = I_0 e^{-kh}.$$

From $I(10) = \frac{1}{2}I_0$ we get $e^{-10k} = \frac{1}{2}$, so $I = I_0(\frac{1}{2})^{h/10}$. Then $I = \frac{1}{16}I_0 = (\frac{1}{2})^4 I_0$ when $h/10 = 4$, i.e. $h = 40$ feet.

Answer. At a depth of 40 feet.

Problem 4-23 (Simmons)

Solution. As in Problem 22, $I = I_0 e^{-kh}$, and $I(15) = \frac{3}{5}I_0$ gives $e^{-15k} = \frac{3}{5}$, so

$$I = I_0 \left(\frac{3}{5}\right)^{h/15}.$$

Hence

$$I(30) = I_0\left(\tfrac{3}{5}\right)^2 = \tfrac{9}{25}I_0, \qquad I(60) = I_0\left(\tfrac{3}{5}\right)^4 = \tfrac{81}{625}I_0, \qquad I(50) = I_0\left(\tfrac{3}{5}\right)^{10/3} \approx 0.182\,I_0.$$

Answer. $I(30) = \frac{9}{25}I_0$, $I(60) = \frac{81}{625}I_0$, $I(50) = I_0(3/5)^{10/3} \approx 0.182 I_0$.

Problem 4-24 (Simmons)

Solution. Let $w(h)$ be the weight density of the air (weight per unit volume) at altitude h; it is the mass density times g, so by hypothesis $w(h) = c\,p(h)$ for a constant $c > 0$. The part of the column between altitudes h and $h+\Delta h$ has volume $1 \cdot \Delta h$ and weight approximately $w(h)\Delta h$. Since $p(h)$ is the weight of the air above altitude h,

$$p(h) - p(h+\Delta h) = \text{weight of air between } h \text{ and } h+\Delta h \approx w(h)\,\Delta h.$$

(Equivalently, $p(h) = \int_h^\infty w(s)\,ds$.) Dividing by Δh and letting $\Delta h \to 0$ (or differentiating the integral by the fundamental theorem of calculus),

$$\frac{dp}{dh} = -w(h) = -cp, \qquad c > 0.$$

This is equation (6) with h in place of t, so by (8),

$$p = p_0 e^{-ch}, \qquad p_0 = p(0) = \text{pressure at sea level.}$$

(Consistency check: $\int_h^\infty cp_0 e^{-cs}\,ds = p_0 e^{-ch}$, so this p is indeed the weight of the column above h.)

Answer. Shown: $dp/dh = -w = -cp$, and hence $p = p_0 e^{-ch}$.

Problem 4-25 (Simmons)

Solution. Let T be the body's temperature at time t (hours). Newton's law says

$$-\frac{dT}{dt} = k(T-10) \quad\Longrightarrow\quad \frac{d}{dt}(T-10) = -k(T-10) \quad\Longrightarrow\quad T-10 = (T_0-10)e^{-kt}$$

by (8), with $T_0 = 110$. So $T - 10 = 100e^{-kt}$. At $t = 1$: $50 = 100e^{-k}$, so $e^{-k} = \frac{1}{2}$ and $T - 10 = 100 \cdot 2^{-t}$. The temperature is 30°C when

$$2^{-t} = \frac{20}{100} = \frac{1}{5} \iff t = \frac{\log 5}{\log 2} \approx 2.322 \text{ hours.}$$

Since 1 hour has already elapsed, the additional time required is

$$\frac{\log 5}{\log 2} - 1 \approx 1.32 \text{ hours } (\approx 1 \text{ hour } 19 \text{ minutes}).$$

Answer. $\dfrac{\log 5}{\log 2} - 1 \approx 1.32$ hours.

Problem 4-26 (Simmons)

Solution. *By thinking.* By Newton's law of cooling the temperature difference from the surroundings (here simply the temperature itself, since the freezer is at 0°F) decays exponentially, so it is multiplied by the same factor in any two equal time intervals. From 15 to 30 minutes it went from 30 to 15, a factor $\frac{1}{2}$; so in the first 15 minutes it also halved, and the initial temperature was 60°F.

By a differential equation. $-dT/dt = k(T-0)$, so $T = T_0e^{-kt}$ (t in minutes). Then

$$T_0e^{-15k} = 30, \qquad T_0e^{-30k} = 15.$$

Dividing, $e^{-15k} = \frac{1}{2}$; substituting back, $T_0 = 30/e^{-15k} = 60$.

Answer. 60°F.

Problem 4-27 (Simmons)

Solution. By Newton's law of cooling with surroundings at 0°C, $T = 100e^{-kt}$. After 30 minutes, $T = 80$, so $e^{-30k} = 0.8$. After 60 minutes,

$$T = 100\left(e^{-30k}\right)^2 = 100(0.8)^2 = 64°\text{C}.$$

Thus during the next 30 minutes it cools $80 - 64 = 16°$.

Answer. 16°C.

Problem 4-28 (Simmons)

Solution. Let T be the temperature of the body and measure t in hours from 10 A.M. By Newton's law of cooling, $-dT/dt = k(T-5)$, so

$$T - 5 = 18e^{-kt} \qquad (\text{since } T-5 = 23-5 = 18 \text{ at } t = 0).$$

At noon ($t = 2$): $18.5 - 5 = 13.5 = 18e^{-2k}$, so $e^{-2k} = \frac{13.5}{18} = \frac{3}{4}$ and

$$T - 5 = 18\left(\tfrac{3}{4}\right)^{t/2}.$$

At the moment of death $T = 37$, i.e. $T - 5 = 32$:

$$18\left(\frac{3}{4}\right)^{t/2} = 32 \iff \left(\frac{3}{4}\right)^{t/2} = \frac{16}{9} = \left(\frac{3}{4}\right)^{-2} \iff t = -4.$$

So he was killed 4 hours before 10 A.M., at 6 A.M.

Answer. At 6 A.M.

Problem 4-29 (Simmons)

Solution. Let x be the amount of radiocarbon per gram of carbon in the specimen, t years after the death of the organism. By Example 3, $x = x_0e^{-kt}$ with $k = \log 2/5600$. The rate of disintegration $-dx/dt = kx$ is proportional to x, so the measured dpm R satisfies

$$\frac{R}{15.30} = \frac{x}{x_0} = e^{-kt} \quad \Longrightarrow \quad t = \frac{1}{k}\log\frac{15.30}{R} = \frac{5600}{\log 2}\log\frac{15.30}{R} \approx 8079.1\,\log\frac{15.30}{R} \text{ years.}$$

	Specimen	R (dpm)	$\log(15.30/R)$	Age t (years)
(a)	Tutankhamen chair leg	10.14	0.411365	3323
(b)	Hammurabi beam	9.52	0.474458	3833
(c)	Giant sloth dung	4.17	1.299937	10,502
(d)	Atlatl	6.42	0.868435	7016

Since the measurements were made in 1950: (a) about 3320 years old, i.e. about 1370 B.C.; (b) about 3830 years old, i.e. about 1880 B.C.; (c) about 10,500 years old; (d) about 7020 years old.

Remark. The book's answers are about 3330 (1380 B.C.), 3850 (1900 B.C.), 10,510 and 7010 years. The small differences (at most about 0.5 percent) presumably come from rounding in the book's intermediate computations (direct evaluation of the formula gives the values in the table); given that the half-life itself is known only approximately, the estimates agree.

Answer. (a) $\approx$ 3320 years ($\approx$ 1370 B.C.); (b) $\approx$ 3830 years ($\approx$ 1880 B.C.); (c) $\approx$ 10,500 years; (d) $\approx$ 7020 years. In general $t = \frac{5600}{\log 2}\log\frac{15.30}{R}$.

Section 5 Falling Bodies and Other Motion Problems

Throughout this section g denotes the (constant) acceleration of gravity at the surface of the earth; in numerical work we use $g = 32\,\text{ft/s}^2$, as Simmons does. The elliptic integrals are those of the text,

$$F(k,\phi) = \int_0^{\phi} \frac{d\phi}{\sqrt{1-k^2\sin^2\phi}}, \qquad E(k,\phi) = \int_0^{\phi} \sqrt{1-k^2\sin^2\phi}\,d\phi.$$

Problem 5-1 (Simmons)

Solution. Put $a = \sqrt{g/c}$, so that $g - cv^2 = c(a^2 - v^2)$. The constant function $v \equiv a$ is a solution of the equation (an equilibrium). Since our solution starts at $v(0) = 0 < a$ and solutions of this equation cannot cross the equilibrium solution (uniqueness), we have $0 \le v < a$ for all $t \ge 0$; in particular $a^2 - v^2 > 0$ and we may divide by it. Separating the variables,

$$\frac{dv}{a^2 - v^2} = c\,dt.$$

By partial fractions, $\dfrac{1}{a^2-v^2} = \dfrac{1}{2a}\left(\dfrac{1}{a+v} + \dfrac{1}{a-v}\right)$, so

$$\int \frac{dv}{a^2-v^2} = \frac{1}{2a}\big[\log(a+v) - \log(a-v)\big] = \frac{1}{2a}\log\frac{a+v}{a-v} \qquad (0 \le v < a).$$

Hence

$$\frac{1}{2a}\log\frac{a+v}{a-v} = ct + c_1.$$

The condition $v = 0$ at $t = 0$ gives $c_1 = \frac{1}{2a}\log 1 = 0$. Since $2ac = 2c\sqrt{g/c} = 2\sqrt{gc}$,

$$\log\frac{a+v}{a-v} = 2\sqrt{gc}\,t, \qquad \text{so} \qquad \frac{a+v}{a-v} = e^{2\sqrt{gc}\,t}.$$

Solving for v: $a + v = (a-v)e^{2\sqrt{gc}\,t}$, i.e. $v\big(1 + e^{2\sqrt{gc}\,t}\big) = a\big(e^{2\sqrt{gc}\,t} - 1\big)$, and multiplying numerator and denominator by $e^{-2\sqrt{gc}\,t}$,

$$v = \sqrt{\frac{g}{c}}\,\frac{1 - e^{-2\sqrt{gc}\,t}}{1 + e^{-2\sqrt{gc}\,t}} = \sqrt{\frac{g}{c}}\,\tanh\big(\sqrt{gc}\,t\big).$$

As $t \to \infty$, $e^{-2\sqrt{gc}\,t} \to 0$, so $v \to \sqrt{g/c}$. This is the terminal velocity. It can also be read off from the equation: the velocity stops changing when $g - cv^2 = 0$.

Check. $v(0) = 0$. With $u = \sqrt{gc}\,t$, $v = \sqrt{g/c}\tanh u$ gives $\frac{dv}{dt} = \sqrt{g/c}\,\sqrt{gc}\,\operatorname{sech}^2 u = g(1 - \tanh^2 u) = g - c\cdot\frac{g}{c}\tanh^2 u = g - cv^2$.

Answer. $v = \sqrt{\dfrac{g}{c}}\,\dfrac{1-e^{-2\sqrt{gc}\,t}}{1+e^{-2\sqrt{gc}\,t}} = \sqrt{\dfrac{g}{c}}\tanh\sqrt{gc}\,t$; the terminal velocity is $\sqrt{g/c}$.

Problem 5-2 (Simmons)

Solution. Let x (miles) be the distance traveled since the fuel ran out, t (hours) the time, and $v = dx/dt$ (miles/hour) the speed. The only horizontal force is the resistance $-kv$ ($k > 0$), so Newton's second law gives

$$m\frac{dv}{dt} = -kv, \qquad \text{i.e.} \qquad \frac{dv}{dt} = -cv, \quad c = \frac{k}{m} > 0.$$

With $v(0) = 60$ the solution is $v = 60e^{-ct}$. Integrating once more with $x(0) = 0$,

$$x = \int_0^t 60e^{-cs}\,ds = \frac{60}{c}\bigl(1 - e^{-ct}\bigr).$$

Since $e^{-ct} = v/60$, this says $x = (60 - v)/c$, that is,

$$v = 60 - cx.$$

(The same relation comes directly from $\frac{dv}{dt} = v\frac{dv}{dx}$: the equation becomes $v\,dv/dx = -cv$, so $dv/dx = -c$.) The speed therefore decreases linearly with distance. The data $v = 30$ when $x = 1$ give $30 = 60 - c$, so $c = 30$ (per hour). Thus $x = 2\bigl(1 - e^{-30t}\bigr)$. The speed $v = 60e^{-30t}$ is never exactly zero, but x increases steadily towards 2 as $t \to \infty$ and never exceeds it. So the torpedo coasts a total distance of 2 miles. (In terms of distance, $v = 60 - 30x$ drops to 0 exactly at $x = 2$.)

Answer. It coasts 2 miles (in all; $x(t) = 2(1 - e^{-30t})$ with t in hours).

Problem 5-3 (Simmons)

Solution. Measure y upward from the ground, so that the gravitational acceleration is $-g$:

$$\frac{d^2y}{dt^2} = -g, \qquad y(0) = 0, \quad y'(0) = v_0.$$

Integrating twice and using the initial conditions,

$$v = \frac{dy}{dt} = v_0 - gt, \qquad y = v_0 t - \tfrac{1}{2}gt^2.$$

Maximum height. The rock rises while $v > 0$ and reaches its highest point when $v = 0$, i.e. at $t = v_0/g$. There

$$y_{\max} = v_0 \cdot \frac{v_0}{g} - \frac{1}{2}g\frac{v_0^2}{g^2} = \frac{v_0^2}{2g}.$$

Return to the ground. $y = t\bigl(v_0 - \tfrac{1}{2}gt\bigr) = 0$ for $t = 0$ (the throw) and for $t = 2v_0/g$, when it hits the ground.

Numerical case. With $v_0 = 128\,\text{ft/s}$ and $g = 32\,\text{ft/s}^2$: the rock reaches its highest point at $t = 128/32 = 4\,\text{s}$, at height $128^2/64 = 16384/64 = 256\,\text{ft}$, and it hits the ground at $t = 8\,\text{s}$.

Answer. For $v_0 = 128$ ft/s: maximum height 256 ft, reached at $t = 4$ s; it hits the ground at $t = 8$ s. In general: maximum height $v_0^2/2g$ at $t = v_0/g$; it hits the ground at $t = 2v_0/g$.

Problem 5-4 (Simmons)

Solution. Measure y upward from the ground and let $v = dy/dt$. While the mass rises ($v > 0$), both gravity $-mg$ and the air resistance $-kv$ point downward, so

$$m\frac{dv}{dt} = -mg - kv, \qquad \text{i.e.} \qquad \frac{dv}{dt} = -(g + cv), \quad c = \frac{k}{m}.$$

Since $dv/dt \le -g < 0$ while $v > 0$, the velocity decreases and becomes 0 after a time at most v_0/g; the height at that moment is the maximum height. To get height as a function of velocity we use the chain rule $\frac{dv}{dt} = \frac{dv}{dy}\frac{dy}{dt} = v\frac{dv}{dy}$:

$$v\frac{dv}{dy} = -(g + cv), \qquad \text{so} \qquad dy = -\frac{v\,dv}{g + cv}.$$

As y goes from 0 to $y_{\max}$, v goes from v_0 to 0; hence

$$y_{\max} = -\int_{v_0}^{0} \frac{v\,dv}{g + cv} = \int_{0}^{v_0} \frac{v\,dv}{g + cv}.$$

Divide: $\dfrac{v}{g + cv} = \dfrac{1}{c} \cdot \dfrac{(g + cv) - g}{g + cv} = \dfrac{1}{c} - \dfrac{g}{c} \cdot \dfrac{1}{g + cv}$. Therefore

$$y_{\max} = \left[\frac{v}{c} - \frac{g}{c^2}\log(g + cv)\right]_0^{v_0} = \frac{v_0}{c} - \frac{g}{c^2}\log\frac{g + cv_0}{g} = \frac{v_0}{c} - \frac{g}{c^2}\log\left(1 + \frac{cv_0}{g}\right).$$

Substituting $c = k/m$ gives

$$y_{\max} = \frac{mv_0}{k} - \frac{m^2 g}{k^2}\log\left(1 + \frac{kv_0}{mg}\right),$$

as was to be shown.

Limit as $k \to 0$. Write $y_{\max} = N(k)/D(k)$ with

$$N(k) = mv_0 k - m^2 g \log\left(1 + \frac{kv_0}{mg}\right), \qquad D(k) = k^2.$$

As $k \to 0$, $N(k) \to 0$ and $D(k) \to 0$, so l'Hospital's rule applies:

$$N'(k) = mv_0 - m^2 g \cdot \frac{v_0/(mg)}{1 + kv_0/(mg)} = mv_0 - \frac{mv_0}{1 + kv_0/(mg)}, \qquad D'(k) = 2k.$$

Again $N'(k) \to mv_0 - mv_0 = 0$ and $D'(k) \to 0$, so we apply the rule once more:

$$N''(k) = mv_0 \cdot \frac{v_0/(mg)}{\big(1 + kv_0/(mg)\big)^2} = \frac{v_0^2/g}{\big(1 + kv_0/(mg)\big)^2}, \qquad D''(k) = 2.$$

Hence

$$\lim_{k\to 0} y_{\max} = \lim_{k\to 0} \frac{N''(k)}{D''(k)} = \frac{v_0^2/g}{2} = \frac{v_0^2}{2g},$$

which is the maximum height found in Problem 3 without air resistance.

Answer. Shown: $y_{\max} = \dfrac{mv_0}{k} - \dfrac{m^2 g}{k^2}\log\Big(1 + \dfrac{kv_0}{mg}\Big)$, and $y_{\max} \to v_0^2/2g$ as $k \to 0$.

Problem 5-5 (Simmons)

Solution. The gravitational force on the projectile is directed toward the center and has magnitude K/x^2 for some constant K. At the surface ($x = R$) it equals mg, so $K = mgR^2$. Newton's second law and the hint give

$$mv\frac{dv}{dx} = -\frac{mgR^2}{x^2}, \qquad \text{i.e.} \qquad v\,dv = -gR^2\frac{dx}{x^2}.$$

Integrating, $\frac{1}{2}v^2 = \dfrac{gR^2}{x} + C$. With $v = v_0$ at $x = R$ we get $C = \frac{1}{2}v_0^2 - gR$, so

$$v^2 = v_0^2 - 2gR + \frac{2gR^2}{x}. \qquad (*)$$

If $v_0^2 < 2gR$, the right side of $(*)$ is zero when $x = \dfrac{2gR^2}{2gR - v_0^2}$, a finite distance. The projectile cannot get past this point, since v^2 would be negative there. At this point its velocity is zero while the force on it is still directed toward the earth, so it stops and falls back. It does not travel indefinitely.

If $v_0^2 \geq 2gR$, then $(*)$ gives $v^2 \geq 2gR^2/x > 0$ for every x. The velocity stays positive, so x increases for all time. Moreover $x \to \infty$: if x stayed below some bound L, then $v \geq \sqrt{2gR^2/L} > 0$ for all t, and x would grow at least linearly in t, which is a contradiction. So the projectile keeps traveling indefinitely.

The projectile escapes exactly when $v_0 \geq \sqrt{2gR}$, so the escape velocity is $v_e = \sqrt{2gR}$.

Numerical value. $g = 32\,\text{ft/s}^2 = 32/5280\,\text{mi/s}^2 \approx 0.006061\,\text{mi/s}^2$ and $R = 4000\,\text{mi}$ give $2gR \approx 48.48\,\text{mi}^2/\text{s}^2$. So $v_e \approx 6.96\,\text{mi/s} \approx 7\,\text{miles/second}$, and $6.96 \times 3600 \approx 25{,}070 \approx 25{,}000\,\text{miles/hour}$.

Answer. Shown: the projectile escapes if and only if $v_0 \geq \sqrt{2gR} \approx 6.96$ mi/s (≈ 7 mi/s $\approx 25{,}000$ mi/h).

Problem 5-6 (Simmons)

Solution. By Problem 5, equation $(*)$, $v^2 = v_0^2 - 2gR + 2gR^2/x$, and $v_e^2 = 2gR$. Since $v_0 < v_e$, the velocity drops to zero, and the projectile turns back, at the distance $x_{\max}$ from the center where $v = 0$:

$$\frac{2gR^2}{x_{\max}} = 2gR - v_0^2 \qquad \Longrightarrow \qquad x_{\max} = \frac{2gR^2}{2gR - v_0^2} = \frac{v_e^2 R}{v_e^2 - v_0^2} = \frac{R}{1 - (v_0/v_e)^2}.$$

The height above the surface is

$$h = x_{\max} - R = R\left(\frac{1}{1 - (v_0/v_e)^2} - 1\right) = \frac{(v_0/v_e)^2}{1 - (v_0/v_e)^2}\,R.$$

Answer. Shown: $h = x_{\max} - R = \dfrac{(v_0/v_e)^2}{1 - (v_0/v_e)^2}\,R.$

Problem 5-7 (Simmons)

Solution. As in Problem 5 (with x the distance from the center and air resistance neglected), $v\,dv = -gR^2\,dx/x^2$, so $\frac{1}{2}v^2 = gR^2/x + C$. Suppose the body starts from rest at distance x_0 from the center. Then $C = -gR^2/x_0$ and

$$v^2 = 2gR^2\left(\frac{1}{x} - \frac{1}{x_0}\right).$$

Altitude $3R$, i.e. $x_0 = R + 3R = 4R$. At the surface ($x = R$),

$$v^2 = 2gR^2\left(\frac{1}{R} - \frac{1}{4R}\right) = 2gR \cdot \frac{3}{4} = \frac{3}{2}gR, \qquad |v| = \sqrt{1.5\,gR}.$$

Infinite height. Letting $x_0 \to \infty$, at $x = R$ we get $v^2 \to 2gR^2 \cdot \frac{1}{R} = 2gR$, so the speed at the surface is $\sqrt{2gR}$, which is the escape velocity of Problem 5. (The velocity is directed toward the center, so $v = dx/dt$ is negative; the values above are speeds.)

Answer. From altitude $3R$: speed $\sqrt{1.5\,gR}$ at the surface; from an infinite height: $\sqrt{2gR}$.

Problem 5-8 (Simmons)

Solution. Let x be the rock's distance from the center along the hole, with $v = dx/dt$. The force is directed toward the center with magnitude proportional to x. At the surface ($x = R$) it is mg, so the force is $-mgx/R$ (ignoring friction and air resistance). With the hint of Problem 5,

$$mv\frac{dv}{dx} = -\frac{mg}{R}\,x, \qquad \text{so} \qquad v\,dv = -\frac{g}{R}\,x\,dx.$$

Integrating, $\frac{1}{2}v^2 = -\frac{g}{2R}x^2 + C$. The rock is dropped from rest at the surface: $v = 0$ when $x = R$, so $C = gR/2$ and

$$v^2 = \frac{g}{R}\left(R^2 - x^2\right).$$

At the center, $x = 0$, this gives $v^2 = gR$, so the rock reaches the center with speed $\sqrt{gR}$.

Numerical value. With $g = 32/5280\,\text{mi/s}^2$ and $R = 4000\,\text{mi}$, $gR \approx 24.24\,\text{mi}^2/\text{s}^2$ and $\sqrt{gR} \approx 4.92 \approx 5\,\text{miles/second}$.

Answer. $\sqrt{gR}$, approximately $4.9 \approx 5$ miles/second.

Problem 5-9 (Simmons)

Solution.

(a) In the first quadrant the ellipse is the graph of $y = \frac{b}{a}\sqrt{a^2 - x^2}$, $0 \le x \le a$. For $0 \le x < a$,

$$y' = -\frac{b}{a}\cdot\frac{x}{\sqrt{a^2 - x^2}}, \qquad 1 + y'^2 = 1 + \frac{b^2x^2}{a^2(a^2 - x^2)} = \frac{a^4 - a^2x^2 + b^2x^2}{a^2(a^2 - x^2)} = \frac{a^4 - (a^2 - b^2)x^2}{a^2(a^2 - x^2)}.$$

The eccentricity of the ellipse ($a > b$) is $e = \sqrt{a^2 - b^2}/a$, so $a^2 - b^2 = a^2e^2$ and

$$1 + y'^2 = \frac{a^4 - a^2e^2x^2}{a^2(a^2 - x^2)} = \frac{a^2 - e^2x^2}{a^2 - x^2}.$$

The arc length formula $s = \int \sqrt{1 + y'^2}\,dx$ therefore gives the length of the first-quadrant arc as

$$\int_0^a \sqrt{\frac{a^2 - e^2x^2}{a^2 - x^2}}\,dx.$$

This is an improper integral, since the tangent is vertical at $(a, 0)$. It converges, because near $x = a$ the integrand is $\sqrt{a^2 - e^2x^2}/\left(\sqrt{a + x}\,\sqrt{a - x}\right) \le a/\sqrt{a(a - x)}$, which is integrable.

(b) Put $x = a\sin\phi$, $0 \le \phi \le \pi/2$. Then $dx = a\cos\phi\,d\phi$, and ϕ runs from 0 to $\pi/2$ as x runs from 0 to a. Also

$$a^2 - x^2 = a^2\cos^2\phi, \qquad a^2 - e^2x^2 = a^2\left(1 - e^2\sin^2\phi\right),$$

and $\cos\phi \ge 0$ on this interval, so $\sqrt{a^2 - x^2} = a\cos\phi$. Hence

$$\int_0^a \sqrt{\frac{a^2 - e^2x^2}{a^2 - x^2}}\,dx = \int_0^{\pi/2} \frac{a\sqrt{1 - e^2\sin^2\phi}}{a\cos\phi}\,a\cos\phi\,d\phi = a\int_0^{\pi/2} \sqrt{1 - e^2\sin^2\phi}\,d\phi = aE(e, \pi/2).$$

(The substitution also removes the singularity at the endpoint.) The ellipse is symmetric about both coordinate axes, so its four quadrant arcs have equal length. The complete circumference is therefore $4aE(e, \pi/2)$.

Check. For $a = 2$, $b = 1.2$ (so $e = 0.8$), the integral in (a), $aE(e, \pi/2)$, and the parametric arc length $\int_0^{\pi/2} \sqrt{a^2\sin^2 t + b^2\cos^2 t}\,dt$ all agree numerically to 12 digits.

Answer. Shown: the first-quadrant arc has length $\int_0^a \sqrt{(a^2 - e^2x^2)/(a^2 - x^2)}\,dx = aE(e, \pi/2)$, so the circumference is $4aE(e, \pi/2)$.

Problem 5-10 (Simmons)

Solution. One arch is $0 \le x \le \pi$. Since $y' = \cos x$,

$$L = \int_0^{\pi} \sqrt{1 + \cos^2 x}\,dx = 2\int_0^{\pi/2} \sqrt{1 + \cos^2 x}\,dx,$$

because $\cos^2(\pi - x) = \cos^2 x$, so the integrand is symmetric about $x = \pi/2$. Now

$$1 + \cos^2 x = 2 - \sin^2 x = 2\left(1 - \tfrac{1}{2}\sin^2 x\right),$$

so

$$L = 2\sqrt{2}\int_0^{\pi/2} \sqrt{1 - \left(\sqrt{1/2}\right)^2\sin^2 x}\,dx = 2\sqrt{2}\,E\left(\sqrt{1/2}, \pi/2\right).$$

Numerically, $E(\sqrt{1/2}, \pi/2) \approx 1.35064$, so $L \approx 3.8202$.

Answer. Shown: $L = \int_0^{\pi} \sqrt{1 + \cos^2 x}\,dx = 2\sqrt{2}\,E\left(\sqrt{1/2}, \pi/2\right) \approx 3.8202$.

Problem 5-11 (Simmons)

Solution. The curve exists where $\cos 2\theta \ge 0$, i.e. for $-\pi/4 \le \theta \le \pi/4$ (right loop) and $3\pi/4 \le \theta \le 5\pi/4$ (left loop). The equation is unchanged under $\theta \to -\theta$ and $\theta \to \pi - \theta$, so the lemniscate

is symmetric about both axes. Its total length is therefore four times the length of the part with $0 \le \theta \le \pi/4$.

In polar coordinates $ds = \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$. Differentiating $r^2 = a^2\cos 2\theta$ gives $r\dfrac{dr}{d\theta} = -a^2\sin 2\theta$, so for $0 \le \theta < \pi/4$ (where $r > 0$)

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = r^2 + \frac{a^4\sin^2 2\theta}{r^2} = \frac{a^4\cos^2 2\theta + a^4\sin^2 2\theta}{r^2} = \frac{a^4}{r^2} = \frac{a^2}{\cos 2\theta}.$$

Hence $ds = \dfrac{a\,d\theta}{\sqrt{\cos 2\theta}}$ and the total length is

$$L = 4a\int_0^{\pi/4} \frac{d\theta}{\sqrt{\cos 2\theta}}.$$

This integral is improper at $\theta = \pi/4$ but converges, since $\cos 2\theta = \sin(\pi/2 - 2\theta) \ge \frac{2}{\pi}(\pi/2 - 2\theta) = \frac{4}{\pi}(\pi/4 - \theta)$ there. Finally, $\cos 2\theta = 1 - 2\sin^2\theta = 1 - (\sqrt{2})^2\sin^2\theta$, so

$$L = 4a\int_0^{\pi/4} \frac{d\theta}{\sqrt{1 - (\sqrt{2})^2\sin^2\theta}} = 4aF\big(\sqrt{2}, \pi/4\big).$$

Here the modulus $k = \sqrt{2}$ exceeds 1, but the integral is still meaningful because $k\sin\theta \le 1$ on $0 \le \theta \le \pi/4$.

Check. Numerically $F(\sqrt{2}, \pi/4) \approx 1.311029$, so $L \approx 5.244115\,a$. Direct numerical integration of the Cartesian speed $\sqrt{x'(\theta)^2 + y'(\theta)^2}$, where $x = r\cos\theta$, $y = r\sin\theta$ gives the same value.

Answer. Shown: $L = 4a\int_0^{\pi/4} d\theta/\sqrt{\cos 2\theta} = 4aF(\sqrt{2}, \pi/4) \approx 5.2441\,a$.

Problem 5-12 (Simmons)

Solution. In Cartesian coordinates $r = a\sin\theta$ is $x^2 + y^2 = ay$, a circle of diameter a through the origin (center $(0, a/2)$). It is traced exactly once as θ runs over $0 \le \theta \le \pi$, where $r = a\sin\theta \ge 0$. The cylinder is the vertical cylinder on this circle. It lies within distance $a \le b$ of the z-axis, so it meets the sphere of radius b about the origin.

(a) The arc length element of the base circle $r = a\sin\theta$ is

$$ds = \sqrt{r^2 + (dr/d\theta)^2}\,d\theta = \sqrt{a^2\sin^2\theta + a^2\cos^2\theta}\,d\theta = a\,d\theta.$$

Above the point of the circle with parameter θ, the vertical line on the cylinder lies inside the sphere for $|z| \le \sqrt{b^2 - r^2} = \sqrt{b^2 - a^2\sin^2\theta}$. This is a segment of length $2\sqrt{b^2 - a^2\sin^2\theta}$. The cylinder is developable (it can be unrolled into a plane strip with s and z as Cartesian coordinates), so its area element is $ds\,dz$ and

$$A_{\text{cyl}} = \int_0^{\pi} 2\sqrt{b^2 - a^2\sin^2\theta}\;a\,d\theta = 2a\cdot 2\int_0^{\pi/2} \sqrt{b^2 - a^2\sin^2\theta}\,d\theta,$$

using the symmetry $\sin(\pi - \theta) = \sin\theta$. Factoring out b,

$$A_{\text{cyl}} = 4ab\int_0^{\pi/2} \sqrt{1 - (a/b)^2\sin^2\theta}\,d\theta = 4abE(a/b, \pi/2).$$

(b) The upper hemisphere is $z=\sqrt{b^2-r^2}$. For a surface $z=f(r)$ given in polar coordinates, the area element is $\sqrt{1+f'(r)^2}\,r\,dr\,d\theta$ (the surface is symmetric about the z-axis, so $\partial z/\partial\theta=0$). Here

$$f'(r)=-\frac{r}{\sqrt{b^2-r^2}},\qquad 1+f'(r)^2=\frac{b^2}{b^2-r^2},\qquad dS=\frac{b\,r}{\sqrt{b^2-r^2}}\,dr\,d\theta.$$

The part of the upper hemisphere inside the cylinder lies over the disk $0\le r\le a\sin\theta$, $0\le\theta\le\pi$. Its area is

$$\begin{aligned}\int_0^{\pi}\int_0^{a\sin\theta}\frac{b\,r\,dr}{\sqrt{b^2-r^2}}\,d\theta &= \int_0^{\pi}\Big[-b\sqrt{b^2-r^2}\Big]_{r=0}^{r=a\sin\theta}d\theta \qquad \Big(\tfrac{d}{dr}\big(-b\sqrt{b^2-r^2}\big)=\tfrac{br}{\sqrt{b^2-r^2}}\Big)\\ &= \int_0^{\pi}\Big(b^2-b\sqrt{b^2-a^2\sin^2\theta}\Big)d\theta\\ &= \pi b^2-2b\int_0^{\pi/2}\sqrt{b^2-a^2\sin^2\theta}\,d\theta \qquad \text{(symmetry about } \theta=\pi/2)\\ &= \pi b^2-2b^2E(a/b,\pi/2).\end{aligned}$$

(When $a=b$ the inner integral is improper at $r=b$ for $\theta=\pi/2$, but it converges and the computation is unchanged.) The lower hemisphere contributes the same amount by symmetry in the plane $z=0$, so

$$A_{\text{sph}}=2\big[\pi b^2-2b^2E(a/b,\pi/2)\big]=2b^2\big[\pi-2E(a/b,\pi/2)\big].$$

Remark. For $a=b$ we have $E(1,\pi/2)=\int_0^{\pi/2}\cos\theta\,d\theta=1$. This gives $A_{\text{cyl}}=4b^2$ and $A_{\text{sph}}=2b^2(\pi-2)$, the classical results for Viviani's window. Both formulas were also confirmed by direct numerical integration for several pairs (a,b).

Answer. Shown: (a) $A_{\text{cyl}}=4abE(a/b,\pi/2)$; (b) $A_{\text{sph}}=2b^2[\pi-2E(a/b,\pi/2)]$.

Problem 5-13 (Simmons)

Solution. *A preliminary substitution* (used in (a) and (b)). For $0\le u\le\pi/2$ define ϕ, $0\le\phi\le\pi/2$, by $\cos u=\cos^2\phi$. Then $\cos u$ runs from 1 to 0 as u runs from 0 to $\pi/2$, and correspondingly ϕ runs from 0 to $\pi/2$. Differentiating, $-\sin u\,du=-2\cos\phi\sin\phi\,d\phi$. Also, since $\sin u\ge 0$,

$$\sin u=\sqrt{1-\cos^4\phi}=\sqrt{(1-\cos^2\phi)(1+\cos^2\phi)}=\sin\phi\sqrt{1+\cos^2\phi}\qquad(\sin\phi\ge 0).$$

Hence, for $0<\phi<\pi/2$,

$$du=\frac{2\cos\phi\sin\phi}{\sin\phi\sqrt{1+\cos^2\phi}}\,d\phi=\frac{2\cos\phi\,d\phi}{\sqrt{1+\cos^2\phi}},\qquad \sqrt{\cos u}=\cos\phi.$$

We shall also use $1+\cos^2\phi=2-\sin^2\phi=2\big(1-\tfrac12\sin^2\phi\big)$, and we write $k=\sqrt{1/2}$, so that $1-\tfrac12\sin^2\phi=1-k^2\sin^2\phi$.

(a) With $x=\pi/2-y$ we have $dx=-dy$ and $\sin x=\cos y$; y runs from $\pi/2$ to 0. So

$$\int_0^{\pi/2}\frac{dx}{\sqrt{\sin x}}=\int_0^{\pi/2}\frac{dy}{\sqrt{\cos y}}.$$

Now put $\cos y = \cos^2\phi$ (the preliminary substitution with $u = y$):

$$\int_0^{\pi/2} \frac{dy}{\sqrt{\cos y}} = \int_0^{\pi/2} \frac{1}{\cos\phi} \cdot \frac{2\cos\phi\, d\phi}{\sqrt{1+\cos^2\phi}} = \int_0^{\pi/2} \frac{2\,d\phi}{\sqrt{2}\sqrt{1-k^2\sin^2\phi}} = \sqrt{2}\,F\big(\sqrt{1/2}, \pi/2\big).$$

(Both integrals are improper, at $x = 0$ and at $y = \pi/2$ respectively; the substitution turns them into a proper integral, which also shows they converge.)

(b) Put $\cos x = \cos^2\phi$ (the preliminary substitution with $u = x$):

$$\int_0^{\pi/2} \sqrt{\cos x}\, dx = \int_0^{\pi/2} \cos\phi \cdot \frac{2\cos\phi\, d\phi}{\sqrt{1+\cos^2\phi}} = \int_0^{\pi/2} \frac{2\cos^2\phi}{\sqrt{1+\cos^2\phi}}\, d\phi.$$

Write $\cos^2\phi = (1+\cos^2\phi) - 1$:

$$\frac{2\cos^2\phi}{\sqrt{1+\cos^2\phi}} = 2\sqrt{1+\cos^2\phi} - \frac{2}{\sqrt{1+\cos^2\phi}} = 2\sqrt{2}\sqrt{1-k^2\sin^2\phi} - \frac{\sqrt{2}}{\sqrt{1-k^2\sin^2\phi}}.$$

Integrating from 0 to $\pi/2$,

$$\int_0^{\pi/2} \sqrt{\cos x}\, dx = 2\sqrt{2}\,E\big(\sqrt{1/2}, \pi/2\big) - \sqrt{2}\,F\big(\sqrt{1/2}, \pi/2\big).$$

(c) Put $x = \pi/2 - \phi$: $dx = -d\phi$, $\sin x = \cos\phi$, and ϕ runs from $\pi/2$ to 0. Then

$$\begin{aligned}\int_0^{\pi/2} \sqrt{1+4\sin^2 x}\, dx &= \int_0^{\pi/2} \sqrt{1+4\cos^2\phi}\, d\phi = \int_0^{\pi/2} \sqrt{5-4\sin^2\phi}\, d\phi \quad (\cos^2\phi = 1-\sin^2\phi)\\ &= \sqrt{5}\int_0^{\pi/2} \sqrt{1-\tfrac{4}{5}\sin^2\phi}\, d\phi = \sqrt{5}\,E\big(\sqrt{4/5}, \pi/2\big).\end{aligned}$$

Check. Numerically, $F(\sqrt{1/2}, \pi/2) \approx 1.854075$ and $E(\sqrt{1/2}, \pi/2) \approx 1.350644$, so (a) $= 2.622058$ and (b) $= 1.198140$. Also $E(\sqrt{4/5}, \pi/2) \approx 1.178490$, so (c) $= 2.635184$. Each agrees with direct numerical quadrature of the left-hand side to at least 15 digits.

Answer. Shown: (a) $\sqrt{2}\,F(\sqrt{1/2}, \pi/2)$; (b) $2\sqrt{2}\,E(\sqrt{1/2}, \pi/2) - \sqrt{2}\,F(\sqrt{1/2}, \pi/2)$; (c) $\sqrt{5}\,E(\sqrt{4/5}, \pi/2)$.

Section 6 The Brachistochrone. Fermat and the Bernoullis

Throughout this section the cycloid (9) of the text is

$$x = a(\theta - \sin\theta), \qquad y = a(1 - \cos\theta),$$

generated by a point on a circle of radius a rolling along the x-axis; one arch corresponds to $0 \le \theta \le 2\pi$. For later use we compute its element of arc length once and for all:

$$\begin{aligned} \frac{dx}{d\theta} &= a(1-\cos\theta), \qquad \frac{dy}{d\theta} = a\sin\theta, \\ \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= a^2\left(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta\right) = 2a^2(1-\cos\theta) \\ &= 4a^2\sin^2\tfrac{\theta}{2} \qquad \left(1-\cos\theta = 2\sin^2\tfrac{\theta}{2}\right), \end{aligned}$$

so that, since $\sin\frac{\theta}{2} \ge 0$ for $0 \le \theta \le 2\pi$,

$$ds = 2a\sin\tfrac{\theta}{2}\,d\theta. \tag{$*$}$$

Problem 6-1 (Simmons)

Solution. By $(*)$, the length of one arch ($0 \le \theta \le 2\pi$) is

$$L = \int_0^{2\pi} 2a\sin\frac{\theta}{2}\,d\theta = \Big[-4a\cos\frac{\theta}{2}\Big]_0^{2\pi} = -4a\cos\pi + 4a\cos 0 = 4a + 4a = 8a.$$

The generating circle has diameter $2a$, and $8a = 4\cdot(2a)$.

Answer. Shown: one arch has length $8a = 4 \times$ (diameter $2a$).

Problem 6-2 (Simmons)

Solution. As θ runs from 0 to 2π, $x = a(\theta - \sin\theta)$ increases from 0 to $2\pi a$ (because $dx/d\theta = a(1-\cos\theta) \ge 0$), and $y \ge 0$. Hence the area under the arch is

$$\begin{aligned} A = \int_0^{2\pi a} y\,dx &= \int_0^{2\pi} a(1-\cos\theta)\cdot a(1-\cos\theta)\,d\theta = a^2\int_0^{2\pi}\left(1 - 2\cos\theta + \cos^2\theta\right)d\theta \\ &= a^2\int_0^{2\pi}\left(1 - 2\cos\theta + \frac{1+\cos 2\theta}{2}\right)d\theta = a^2\Big[\frac{3\theta}{2} - 2\sin\theta + \frac{\sin 2\theta}{4}\Big]_0^{2\pi} \\ &= a^2\cdot 3\pi = 3\pi a^2. \end{aligned}$$

The generating circle has area πa^2, so $A = 3\,(\pi a^2)$.

Answer. Shown: the area under one arch is $3\pi a^2 = 3 \times$ (area πa^2 of the circle).

Problem 6-3 (Simmons)

Solution. Here $0 \le y < c$, and we take $u = \sqrt{y/(c-y)} \ge 0$. Solving $u^2 = y/(c-y)$ for y:

$$u^2(c-y) = y \implies cu^2 = y(1+u^2) \implies y = \frac{cu^2}{1+u^2} = c - \frac{c}{1+u^2}, \qquad dy = \frac{2cu}{(1+u^2)^2}\,du.$$

Therefore

$$x = \int u \cdot \frac{2cu}{(1+u^2)^2}\,\mathrm{d}u = 2c\int \frac{u^2}{(1+u^2)^2}\,\mathrm{d}u.$$

The factor $1+u^2$ suggests the trigonometric substitution $u = \tan\phi$, $0 \le \phi < \pi/2$, for which $1+u^2 = \sec^2\phi$ and $\mathrm{d}u = \sec^2\phi\,\mathrm{d}\phi$:

$$x = 2c\int \frac{\tan^2\phi}{\sec^4\phi}\sec^2\phi\,\mathrm{d}\phi = 2c\int \tan^2\phi\cos^2\phi\,\mathrm{d}\phi = 2c\int \sin^2\phi\,\mathrm{d}\phi$$
$$= c\int (1-\cos 2\phi)\,\mathrm{d}\phi = \frac{c}{2}(2\phi - \sin 2\phi) + c_1.$$

In terms of ϕ the ordinate is

$$y = \frac{c\tan^2\phi}{1+\tan^2\phi} = c\,\frac{\tan^2\phi}{\sec^2\phi} = c\sin^2\phi = \frac{c}{2}(1-\cos 2\phi).$$

The curve passes through the origin A; $y = 0$ corresponds to $\phi = 0$, and $x = 0$ there forces $c_1 = 0$. Finally, putting $a = c/2$ and $\theta = 2\phi$,

$$x = a(\theta - \sin\theta), \qquad y = a(1-\cos\theta),$$

which are equations (9). (The substitutions are exactly those of the text combined into two steps: $u = \tan\phi$ is the text's relation (6).)

Answer. $x = \frac{c}{2}(2\phi - \sin 2\phi)$, $y = \frac{c}{2}(1-\cos 2\phi)$; with $a = c/2$, $\theta = 2\phi$: $x = a(\theta - \sin\theta)$, $y = a(1-\cos\theta)$.

Problem 6-4 (Simmons)

Solution. As in Figure 10 the y-axis points downward, so $y = a(1-\cos\theta)$ is the depth of the bead below its starting point. The point $(\pi a, 2a)$ corresponds to $\theta = \pi$. At the origin ($\theta = 0$) the tangent is vertical ($dy/dx = \sin\theta/(1-\cos\theta) = \cot\frac{\theta}{2} \to \infty$), so the released bead does start to move.

Since there is no friction, energy is conserved, and as in equation (2) the speed at depth y is

$$v = \frac{ds}{dt} = \sqrt{2gy} = \sqrt{2ga(1-\cos\theta)} = \sqrt{4ga\sin^2\tfrac{\theta}{2}} = 2\sqrt{ga}\,\sin\tfrac{\theta}{2} \qquad (0 \le \theta \le \pi).$$

Hence, using $(*)$,

$$\mathrm{d}t = \frac{\mathrm{d}s}{v} = \frac{2a\sin\frac{\theta}{2}\,\mathrm{d}\theta}{2\sqrt{ga}\,\sin\frac{\theta}{2}} = \sqrt{\frac{a}{g}}\,\mathrm{d}\theta \qquad (0 < \theta \le \pi),$$

and the time to reach the bottom is

$$T = \int_0^{\pi} \sqrt{\frac{a}{g}}\,\mathrm{d}\theta = \pi\sqrt{\frac{a}{g}}.$$

(The integrand ds/v looks singular at $\theta = 0$, where $v = 0$, but the factors $\sin\frac{\theta}{2}$ cancel, so there is no difficulty.)

Answer. Shown: the descent time from the origin to $(\pi a, 2a)$ is $T = \pi\sqrt{a/g}$.

Problem 6-5 (Simmons)

Solution. Let the bead be released from rest at the point with parameter θ_0, $0 < \theta_0 < \pi$, whose depth is $y_0 = a(1-\cos\theta_0)$. By conservation of energy the speed at a lower point θ ($\theta_0 < \theta \le \pi$) is determined by the drop $y - y_0$:

$$v = \sqrt{2g(y-y_0)} = \sqrt{2ga(\cos\theta_0 - \cos\theta)}.$$

Using $\cos\theta = 2\cos^2\frac{\theta}{2} - 1$ we get $\cos\theta_0 - \cos\theta = 2\big(\cos^2\frac{\theta_0}{2} - \cos^2\frac{\theta}{2}\big)$, so

$$v = 2\sqrt{ga}\,\sqrt{\cos^2\tfrac{\theta_0}{2} - \cos^2\tfrac{\theta}{2}}.$$

With $(*)$ the time of descent to $\theta = \pi$ is

$$T = \int_{\theta_0}^{\pi} \frac{ds}{v} = \int_{\theta_0}^{\pi} \frac{2a\sin\frac{\theta}{2}\,d\theta}{2\sqrt{ga}\sqrt{\cos^2\frac{\theta_0}{2} - \cos^2\frac{\theta}{2}}} = \sqrt{\frac{a}{g}}\int_{\theta_0}^{\pi} \frac{\sin\frac{\theta}{2}\,d\theta}{\sqrt{b^2 - \cos^2\frac{\theta}{2}}}, \qquad b = \cos\frac{\theta_0}{2} > 0.$$

Substitute $u = \cos\frac{\theta}{2}$, so $du = -\frac{1}{2}\sin\frac{\theta}{2}\,d\theta$, i.e. $\sin\frac{\theta}{2}\,d\theta = -2\,du$; when $\theta = \theta_0$, $u = b$, and when $\theta = \pi$, $u = 0$:

$$T = \sqrt{\frac{a}{g}}\int_b^0 \frac{-2\,du}{\sqrt{b^2-u^2}} = 2\sqrt{\frac{a}{g}}\int_0^b \frac{du}{\sqrt{b^2-u^2}} = 2\sqrt{\frac{a}{g}}\Big[\sin^{-1}\frac{u}{b}\Big]_0^b = 2\sqrt{\frac{a}{g}}\cdot\frac{\pi}{2} = \pi\sqrt{\frac{a}{g}}.$$

(The improper integral converges at $u = b$, since $\sin^{-1}(u/b)$ is continuous on $[0,b]$.) The result does not depend on θ_0; together with Problem 4 (the case $\theta_0 = 0$) this shows that from every starting point the bead reaches the bottom in the same time $\pi\sqrt{a/g}$.

Answer. Shown: from any starting point $\theta_0 \in [0,\pi)$ the descent time is $\pi\sqrt{a/g}$, independent of θ_0.

Problem 6-6 (Simmons)

Solution. At sunset the sun's rays are horizontal, so the shadow of the rock falls on the point of the hill's profile (the vertical section through the man in the direction of the sun) that is at the same height as the rock. Let h be the height of the hill, let y be height above the man's position and x horizontal distance, and let s be arc length along the profile measured from the base.

The rock. Its initial speed v_0 is such that it just reaches height h, so $v_0^2 = 2gh$. By conservation of energy, while rising it has speed

$$\frac{dy}{dt} = \sqrt{v_0^2 - 2gy} = \sqrt{2g(h-y)}.$$

The shadow. Its height is the rock's height y, and by hypothesis $ds/dt = k$, a constant. Therefore along the profile

$$\frac{ds}{dy} = \frac{ds/dt}{dy/dt} = \frac{k}{\sqrt{2g(h-y)}}.$$

It is convenient to measure depth below the top of the hill, $Y = h - y$ ($0 \le Y \le h$), so that $(ds/dY)^2 = k^2/(2gY)$. Since $ds^2 = dx^2 + dY^2$,

$$1 + \Big(\frac{dx}{dY}\Big)^2 = \frac{c}{Y}, \qquad c = \frac{k^2}{2g}, \qquad \text{i.e.} \qquad \Big(\frac{dx}{dY}\Big)^2 = \frac{c-Y}{Y}.$$

For this to be possible all the way down to the base ($Y = h$) we need $c \geq h$, i.e. $k \geq v_0$, which is reasonable: the shadow's path along the slope is at least as long as the rock's vertical path.

Solving. Let ξ be horizontal distance from the top of the hill, increasing toward the man, so $d\xi/dY = \sqrt{(c-Y)/Y} \geq 0$ and $\xi = 0$ when $Y = 0$. Substitute $Y = c\sin^2\phi$ ($0 \leq \phi \leq \pi/2$), $\mathrm{d}Y = 2c\sin\phi\cos\phi\,\mathrm{d}\phi$, $\sqrt{(c-Y)/Y} = \cos\phi/\sin\phi$:

$$\mathrm{d}\xi = \frac{\cos\phi}{\sin\phi}\,2c\sin\phi\cos\phi\,\mathrm{d}\phi = 2c\cos^2\phi\,\mathrm{d}\phi = c(1+\cos 2\phi)\,\mathrm{d}\phi, \qquad \xi = \frac{c}{2}(2\phi + \sin 2\phi),$$

the constant being 0 because $\xi = 0$ at $\phi = 0$. Also $Y = c\sin^2\phi = \frac{c}{2}(1-\cos 2\phi)$. With $a = c/2 = k^2/(4g)$ and $\theta = 2\phi$:

$$\xi = a(\theta + \sin\theta), \qquad Y = a(1-\cos\theta).$$

Now put $\theta = \psi - \pi$. Since $\sin(\psi-\pi) = -\sin\psi$ and $\cos(\psi-\pi) = -\cos\psi$,

$$\xi + \pi a = a(\psi - \sin\psi), \qquad 2a - Y = a(1-\cos\psi).$$

Thus, if we take a new horizontal coordinate $X = \xi + \pi a$ and measure height $H = 2a - Y$ above the horizontal line lying a distance $2a$ below the top of the hill, the profile is

$$X = a(\psi - \sin\psi), \qquad H = a(1-\cos\psi),$$

exactly the cycloid (9) generated by a circle of radius $a = k^2/(4g)$ rolling along that line. The top of the hill ($Y = 0$) is the top of the arch ($\psi = \pi$), where the tangent is horizontal, as it should be for a dome. (If $k = v_0$, then $2a = h$ and the face of the hill is a full half-arch, vertical at the base; if $k > v_0$, it is the upper portion of a half-arch.)

Answer. Shown: with $c = k^2/(2g)$ the profile satisfies $(dx/dY)^2 = (c-Y)/Y$ (Y = depth below the top), whose solution is the cycloid $X = a(\psi - \sin\psi)$, $H = a(1-\cos\psi)$, $a = k^2/(4g)$, with the hilltop at the top of the arch.

Miscellaneous Problems for Chapter 1

Chapter 1, Miscellaneous Problem 1 (Simmons)

Solution. Measure time t in hours from noon, and suppose it started snowing t_0 hours before noon ($t_0 > 0$). Since the snow falls steadily, its depth at time t is $D = \beta(t + t_0)$ for some constant $\beta > 0$. If the plow clears a strip of width w and has advanced x miles at time t, then it removes snow at the rate $wD\,dx/dt$, which by hypothesis is a constant V_0. Hence

$$\frac{dx}{dt} = \frac{C}{t + t_0}, \qquad C = \frac{V_0}{w\beta} > 0, \qquad x(0) = 0.$$

Integrating, $x = C\log(t + t_0) + c_1$, and $x(0) = 0$ gives $c_1 = -C\log t_0$, so

$$x = C\log\frac{t + t_0}{t_0}.$$

The data $x(2) = 2$ and $x(4) = 3$ give

$$2 = C\log\frac{2 + t_0}{t_0}, \qquad 3 = C\log\frac{4 + t_0}{t_0}.$$

Eliminating C: $3\log\frac{2+t_0}{t_0} = 2\log\frac{4+t_0}{t_0}$, i.e. $\left(\frac{2+t_0}{t_0}\right)^3 = \left(\frac{4+t_0}{t_0}\right)^2$, so

$$\begin{aligned} (2 + t_0)^3 &= t_0(4 + t_0)^2 && \text{(multiply by } t_0^3\text{)} \\ 8 + 12t_0 + 6t_0^2 + t_0^3 &= 16t_0 + 8t_0^2 + t_0^3 \\ 2t_0^2 + 4t_0 - 8 &= 0, \qquad \text{i.e.} \qquad t_0^2 + 2t_0 - 4 = 0. \end{aligned}$$

Thus $t_0 = -1 \pm \sqrt{5}$, and since $t_0 > 0$, $t_0 = \sqrt{5} - 1 \approx 1.236$ hours ≈ 1 hour 14.2 minutes. It started snowing at about 10:46 A.M.

Answer. $(\sqrt{5} - 1) \approx 1.236$ hours before noon, i.e. at about 10:46 A.M.

Chapter 1, Miscellaneous Problem 2 (Simmons)

Solution. Let $r(t)$ be the radius (inches) at time t (months). The volume is $V = \frac{4}{3}\pi r^3$ and the surface area is $4\pi r^2$, so the hypothesis reads

$$\frac{dV}{dt} = -k \cdot 4\pi r^2 \quad (k > 0).$$

Since $dV/dt = 4\pi r^2\,dr/dt$, dividing by $4\pi r^2$ (valid while $r > 0$) gives $dr/dt = -k$, so $r = r(0) - kt = \frac{1}{4} - kt$. The condition $r(1) = \frac{1}{8}$ gives $k = \frac{1}{8}$, hence

$$r = \frac{1}{4} - \frac{t}{8} = \frac{2 - t}{8} \qquad (0 \le t \le 2).$$

The radius is 0 when $t = 2$, i.e. one more month after the first measurement.

Answer. $r = (2 - t)/8$ inch (t in months); it disappears after one more month.

Chapter 1, Miscellaneous Problem 3 (Simmons)

Solution. The volume stays 100 gallons. If $x(t)$ is the number of pounds of salt at time t (minutes), salt enters at $1 \cdot 1 = 1$ lb/min and leaves at $1 \cdot \frac{x}{100}$ lb/min, so

$$\frac{dx}{dt} = 1 - \frac{x}{100} = \frac{100 - x}{100}, \qquad x(0) = 0.$$

Separating variables ($x < 100$ throughout),

$$\int_0^x \frac{\mathrm{d}x}{100 - x} = \int_0^t \frac{\mathrm{d}t}{100} \implies -\log\frac{100 - x}{100} = \frac{t}{100} \implies x = 100\big(1 - e^{-t/100}\big).$$

Setting $x = 50$: $e^{-t/100} = \frac{1}{2}$, so $t = 100\log 2 \approx 69.3$ minutes.

Answer. After $100\log 2 \approx 69.3$ minutes.

Chapter 1, Miscellaneous Problem 4 (Simmons)

Solution. The volume of liquid increases by $3 - 2 = 1$ gallon per minute, so at time t it is $100 + t$ gallons (the tank is "large", so it does not overflow). With $x(t)$ pounds of salt, the concentration is $x/(100 + t)$; no salt enters, and salt leaves at $2x/(100 + t)$ lb/min:

$$\frac{dx}{dt} = -\frac{2x}{100 + t}, \qquad x(0) = 200.$$

Separating variables ($x > 0$):

$$\int_{200}^x \frac{\mathrm{d}x}{x} = -2\int_0^t \frac{\mathrm{d}t}{100 + t} \implies \log\frac{x}{200} = -2\log\frac{100 + t}{100} \implies x = 200\Big(\frac{100}{100 + t}\Big)^2.$$

Setting $x = 100$: $(100 + t)^2 = 2 \cdot 100^2$, so $100 + t = 100\sqrt{2}$ and

$$t = 100(\sqrt{2} - 1) \approx 41.4 \text{ minutes.}$$

Answer. $100(\sqrt{2} - 1) \approx 41.4$ minutes.

Chapter 1, Miscellaneous Problem 5 (Simmons)

Solution. Place the center of the football at the origin with its long axis horizontal along the y-axis and the z-axis vertical. The semi-axes are 6 (along y) and 3 (along x and z), so the surface is

$$\frac{x^2}{9} + \frac{y^2}{36} + \frac{z^2}{9} = 1, \qquad \text{i.e.} \qquad 4x^2 + y^2 + 4z^2 = 36.$$

Water runs down a smooth surface along the paths of steepest descent. On a surface $z = f(x, y)$, with upward normal $\mathbf{n} \propto (-f_x, -f_y, 1)$, the component of the vertical vector $\mathbf{k}$ tangent to the surface is $\mathbf{k} - (\mathbf{k} \cdot \mathbf{n})\mathbf{n}$, whose horizontal part is a multiple of $(f_x, f_y) = \nabla f$. Hence the steepest-descent paths project onto the xy-plane as curves that cross the level curves $f = \text{const}$ at right angles: the projected paths are the orthogonal trajectories of the projected level curves.

The level curves $z = \text{const}$ project onto the family of ellipses

$$4x^2 + y^2 = K \qquad (K = 36 - 4z^2),$$

both for the upper half $z \geq 0$ and the lower half $z \leq 0$. Differentiating, $8x + 2yy' = 0$, so the family satisfies $y' = -4x/y$, and the orthogonal trajectories satisfy

$$\frac{dy}{dx} = \frac{y}{4x}.$$

Separating variables, $4\dfrac{\mathrm{d}y}{y} = \dfrac{\mathrm{d}x}{x}$, so $4\log|y| = \log|x| + \text{const}$, i.e.

$$x = cy^4.$$

(In separating we divided by x and y; the lines $x = 0$ (the case $c = 0$) and $y = 0$ (the limiting case $c \to \infty$) are also orthogonal trajectories: they are the axes of symmetry of the ellipses.) Therefore the water runs down along the curves in which the cylinders $x = cy^4$ cut the ellipsoid.

Answer. Along the intersections of the cylinders $x = cy^4$ with the ellipsoid $4x^2 + y^2 + 4z^2 = 36$ (long axis along the y-axis, z vertical).

Chapter 1, Miscellaneous Problem 6 (Simmons)

Solution. A family is self-orthogonal when its differential equation is unchanged if $y' = p$ is replaced by $-1/p$ (so that the orthogonal trajectories are again members of the family). This is the idea of Problem 5 of Section 3, where the parabolas $y^2 = 4c(x + c)$ were shown to be self-orthogonal; the book's cross-reference reads "Problem 3–2", but it is Problem 3–5 that deals with this property. We find the differential equation by eliminating a. Work at points with $x \neq 0$, $y \neq 0$. Differentiating the equation implicitly,

$$\frac{2x}{a^2} + \frac{2yp}{a^2 - c^2} = 0 \qquad \Longrightarrow \qquad x(a^2 - c^2) + a^2 yp = 0 \qquad \Longrightarrow \qquad a^2(x + yp) = c^2 x.$$

Here $x + yp \neq 0$ (otherwise $c^2 x = 0$), so

$$a^2 = \frac{c^2 x}{x + yp}, \qquad a^2 - c^2 = \frac{c^2 x - c^2(x + yp)}{x + yp} = -\frac{c^2 yp}{x + yp}.$$

Substituting into the equation of the family:

$$\frac{x^2(x + yp)}{c^2 x} - \frac{y^2(x + yp)}{c^2 yp} = 1 \implies (x + yp)\Big(x - \frac{y}{p}\Big) = c^2 \implies (x + yp)(xp - y) = c^2 p.$$

This is the differential equation of the family. Now replace p by $-1/p$:

$$\Big(x - \frac{y}{p}\Big)\Big(-\frac{x}{p} - y\Big) = -\frac{c^2}{p}.$$

Multiplying by $-p^2$ gives $(xp - y)(x + yp) = c^2 p$, the original equation. Hence the orthogonal trajectories of the family satisfy the same differential equation as the family itself, and so they are confocal conics of the same family: the family is self-orthogonal. (Concretely, through each point with $xy \neq 0$ the equation, quadratic in a^2, has exactly one root $a^2 > c^2$ and one root $0 < a^2 < c^2$, so exactly one ellipse and one hyperbola of the family pass through it, and they cross at right angles.)

Answer. Shown: the family satisfies $(x+yp)(xp-y)=c^2p$, $p=y'$, which is invariant under $p\to -1/p$; hence the confocal conics are self-orthogonal.

Chapter 1, Miscellaneous Problem 7 (Simmons)

Solution. Let $h(t)$ be the depth of the water above the hole. A body falling freely through height h acquires speed $\sqrt{2gh}$, so by Torricelli's law water leaves at the volume rate $\pi r^2\sqrt{2gh}$. The center of the hemisphere is at height R above the bottom, so at height η the horizontal cross section is a disk of radius ρ with $\rho^2 = R^2-(R-\eta)^2 = 2R\eta-\eta^2$. The volume of water is $V(h)=\int_0^h \pi(2R\eta-\eta^2)\,d\eta$, hence $dV/dt=\pi(2Rh-h^2)\,dh/dt$ and

$$\pi(2Rh-h^2)\frac{dh}{dt} = -\pi r^2\sqrt{2g}\,h^{1/2}.$$

Separating variables (for $h>0$):

$$\left(2Rh^{1/2}-h^{3/2}\right)dh = -r^2\sqrt{2g}\,dt.$$

The bowl empties at the time T when h has gone from R to 0:

$$r^2\sqrt{2g}\,T = \int_0^R \left(2Rh^{1/2}-h^{3/2}\right)dh = \left[\frac{4R}{3}h^{3/2}-\frac{2}{5}h^{5/2}\right]_0^R = \left(\frac{4}{3}-\frac{2}{5}\right)R^{5/2} = \frac{14}{15}R^{5/2}.$$

Therefore

$$T = \frac{14R^{5/2}}{15r^2\sqrt{2g}}.$$

Answer. $T=\dfrac{14R^{5/2}}{15r^2\sqrt{2g}}$ seconds (with R, r in feet, $g=32$ ft/s^2).

Chapter 1, Miscellaneous Problem 8 (Simmons)

Solution. Take the bowl to be a surface of revolution about the vertical y-axis, with the hole (area A_0) at the origin, and let $x=x(y)$ be the radius of the bowl at height y. When the water depth is y, the volume is $V=\int_0^y \pi x(\eta)^2\,d\eta$, so $dV/dt=\pi x^2\,dy/dt$, and by Torricelli's law (Problem 7)

$$\pi x^2\frac{dy}{dt} = -A_0\sqrt{2gy}.$$

We require $dy/dt=-k$, a constant ($k>0$). Then $\pi k x^2 = A_0\sqrt{2g}\,y^{1/2}$, so

$$y = \left(\frac{\pi k}{A_0\sqrt{2g}}\right)^2 x^4 = cx^4.$$

The bowl must have the shape of the surface obtained by revolving the curve $y=cx^4$ about the y-axis.

Answer. The surface obtained by revolving $y=cx^4$ about the (vertical) y-axis.

Chapter 1, Miscellaneous Problem 9 (Simmons)

Solution. Let the holes have area A_0, let A be the common area of the bases (the base of the cone being its top), and let H be the height of the cone. By Torricelli's law, a tank whose cross section at height y has area $S(y)$ satisfies $S(y)\,dy/dt = -A_0\sqrt{2gy}$, so its draining time from depth y_0 is

$$T = \frac{1}{A_0\sqrt{2g}}\int_0^{y_0} \frac{S(y)}{\sqrt{y}}\,dy.$$

Cylinder: $S = A$, $y_0 = h$:

$$T_1 = \frac{A}{A_0\sqrt{2g}}\int_0^{h} y^{-1/2}\,dy = \frac{2A\sqrt{h}}{A_0\sqrt{2g}}.$$

Cone: the radius of a cross section is proportional to the height above the vertex, so $S = A(y/H)^2$, $y_0 = H$:

$$T_2 = \frac{A}{H^2A_0\sqrt{2g}}\int_0^{H} y^{3/2}\,dy = \frac{A}{H^2A_0\sqrt{2g}}\cdot\frac{2}{5}H^{5/2} = \frac{2A\sqrt{H}}{5A_0\sqrt{2g}}.$$

Setting $T_1 = T_2$: $\sqrt{h} = \frac{1}{5}\sqrt{H}$, so $H = 25h$.

Answer. The cone has height $25h$.

Chapter 1, Miscellaneous Problem 10 (Simmons)

Solution. Once the water rotates rigidly with the can, the free surface is a surface of revolution about the axis. Take the y-axis along the axis of the can (upward), and let $y = y(x)$ be the profile of the surface in a vertical plane through the axis, $x \geq 0$ being distance from the axis. At the point (x, y) let α be the angle between the tangent and the horizontal, so that $\tan\alpha = dy/dx$. The reaction R is normal to the surface, so it makes the angle α with the vertical, pointing upward and toward the axis. The particle moves in a horizontal circle of radius x, so the resultant of R and mg is the horizontal force $mx\omega^2$ directed toward the axis. Resolving horizontally and vertically:

$$R\sin\alpha = mx\omega^2, \qquad R\cos\alpha - mg = 0.$$

Dividing, $\tan\alpha = x\omega^2/g$, i.e.

$$\frac{dy}{dx} = \frac{\omega^2}{g}x \qquad\Longrightarrow\qquad y = \frac{\omega^2}{2g}x^2 + y_0,$$

where y_0 is the height of the surface on the axis. This parabola, revolved about the axis, gives a paraboloid of revolution.

Answer. Shown: the profile satisfies $dy/dx = \omega^2 x/g$, so $y = \omega^2x^2/(2g) + y_0$, a paraboloid of revolution.

Chapter 1, Miscellaneous Problem 11 (Simmons)

Solution. Let the circle have radius a, highest point A and lowest point D, so that AD is a vertical diameter of length $2a$. Let the wire run from A to a lower point B of the circle, making an angle φ with the vertical AD ($0 \leq \varphi < \pi/2$). Since AD is a diameter, the angle ABD is a right angle, so the length of the wire is

$$AB = AD\cos\varphi = 2a\cos\varphi.$$

Let s be the distance travelled along the wire. The normal force of a frictionless wire is perpendicular to it, so the only force along the wire is the component $mg\cos\varphi$ of gravity:

$$\frac{d^2s}{dt^2} = g\cos\varphi, \qquad s(0) = 0, \quad \frac{ds}{dt}(0) = 0.$$

Integrating twice, $ds/dt = (g\cos\varphi)t$ and $s = \tfrac{1}{2}(g\cos\varphi)t^2$. The bead reaches B when $s = 2a\cos\varphi$:

$$\tfrac{1}{2}(g\cos\varphi)\,t^2 = 2a\cos\varphi \qquad\Longrightarrow\qquad t^2 = \frac{4a}{g}, \qquad t = 2\sqrt{\frac{a}{g}},$$

since $\cos\varphi > 0$ may be cancelled. This time is independent of φ, i.e. of the position of B; it is the time of free fall through the diameter ($\varphi = 0$).

Answer. Shown: the time is $2\sqrt{a/g}$ (a = radius) for every lower point.

Chapter 1, Miscellaneous Problem 12 (Simmons)

Solution. Let ρ be the mass per foot of the chain, so its mass is 4ρ, and let $x(t)$ be the length (ft) hanging over the edge at time t, so $x(0) = 1$, $x'(0) = 0$. All parts of the chain move with the same speed $v = dx/dt$. Neglecting friction, the net force on the chain is the weight $\rho g x$ of the overhanging part, so by Newton's second law

$$4\rho\frac{dv}{dt} = \rho g x, \qquad \text{i.e.} \qquad \frac{dv}{dt} = \frac{g}{4}x.$$

As in Section 5, write $dv/dt = (dv/dx)(dx/dt) = v\,dv/dx$:

$$v\frac{dv}{dx} = \frac{g}{4}x \;\Longrightarrow\; \int_0^v v\,\mathrm{d}v = \frac{g}{4}\int_1^x x\,\mathrm{d}x \;\Longrightarrow\; v^2 = \frac{g}{4}(x^2-1),$$

so (since $v > 0$ for $x > 1$)

$$\frac{dx}{dt} = \frac{\sqrt{g}}{2}\sqrt{x^2-1}.$$

Separating variables, the time for x to go from 1 to 4 is

$$T = \frac{2}{\sqrt{g}}\int_1^4 \frac{\mathrm{d}x}{\sqrt{x^2-1}} = \frac{2}{\sqrt{g}}\Big[\log\left(x+\sqrt{x^2-1}\right)\Big]_1^4 = \frac{2}{\sqrt{g}}\log\left(4+\sqrt{15}\right) = \sqrt{\frac{4}{g}}\log\left(4+\sqrt{15}\right),$$

the improper integral being convergent at $x = 1$. With $g = 32$ ft/s^2, $T = \frac{1}{2\sqrt{2}}\log(4+\sqrt{15}) \approx 0.7295$ seconds.

Check. $x = \cosh\left(\tfrac{1}{2}\sqrt{g}\,t\right)$ satisfies $x'' = \frac{g}{4}x$, $x(0) = 1$, $x'(0) = 0$, and $x(T) = \cosh\left(\log(4+\sqrt{15})\right) = \tfrac{1}{2}\left(4+\sqrt{15}+4-\sqrt{15}\right) = 4$.

Answer. $T = \sqrt{4/g}\,\log(4+\sqrt{15}) \approx 0.73$ seconds ($g = 32$ ft/s^2).

Chapter 1, Miscellaneous Problem 13 (Simmons)

Solution. From $\Delta T \approx \mu T\,\Delta\theta$ we have $\Delta T/\Delta\theta \approx \mu T$, with the approximation becoming exact as $\Delta\theta \to 0$. Hence

$$\frac{dT}{d\theta} = \mu T, \qquad T(0) = T_0,$$

where $\theta = 0$ is the point at which the rope leaves the post on the man's side (Figure 12). Since $T > 0$, separating variables gives

$$\int_{T_0}^{T} \frac{\mathrm{d}T}{T} = \mu \int_0^{\theta} \mathrm{d}\theta \implies \log \frac{T}{T_0} = \mu\theta \implies T = T_0 e^{\mu\theta}.$$

The restrained force grows exponentially with the angle of wrap; e.g. with $\mu = 0.3$ and two full turns ($\theta = 4\pi$), $T = e^{1.2\pi} T_0 \approx 43\, T_0$.

Answer. $\dfrac{dT}{d\theta} = \mu T; \quad T = T_0 e^{\mu\theta}.$

Chapter 1, Miscellaneous Problem 14 (Simmons)

Solution. We take the density a as weight per unit volume, L being a weight (if a is a mass density, replace a by ag below). The cross section at depth x bears the load L plus the weight of the part of the column above it:

$$W(x) = L + \int_0^x a\pi r(\xi)^2 \,\mathrm{d}\xi.$$

The hypothesis is $\pi r^2 = kW(x)$ for a constant k. At the top ($x = 0$), $\pi r_0^2 = kL$, so $k = \pi r_0^2 / L$. Differentiating with respect to x (fundamental theorem of calculus),

$$\frac{d}{dx}\left(\pi r^2\right) = k\, a \pi r^2.$$

Thus $S = \pi r^2$ satisfies $dS/dx = kaS$ with $S(0) = \pi r_0^2$, so $S = \pi r_0^2 e^{kax}$, i.e.

$$r^2 = r_0^2 e^{\pi r_0^2 ax/L}, \qquad r = r_0 e^{\pi r_0^2 ax/(2L)}.$$

Answer. $r = r_0 e^{\pi r_0^2 ax/2L}$.

Chapter 1, Miscellaneous Problem 15 (Simmons)

Solution. Let T_a be the room temperature, T_0 the initial temperature of the coffee, and T_c the temperature of the cream; "cool" cream means $T_c < T_a$. By Newton's law of cooling, $dT/dt = -k(T - T_a)$, a liquid starting at temperature T^* has temperature $T_a + (T^* - T_a)e^{-kt}$ after time t. Put $E = e^{-10k}$, so $0 < E < 1$. When cream is mixed in (same specific heat assumed), the temperature becomes the weighted average $\lambda T + (1-\lambda)T_c$, where $\lambda = m_{\text{coffee}}/(m_{\text{coffee}} + m_{\text{cream}})$, $0 < \lambda < 1$. Since the amount of cream is small, we use the same cooling constant k for both cups.

President: mixes first, then cools for 10 minutes:

$$T_P = T_a + \big(\lambda T_0 + (1-\lambda)T_c - T_a\big)E.$$

Prime Minister: cools for 10 minutes, then mixes:

$$T_M = \lambda\big(T_a + (T_0 - T_a)E\big) + (1-\lambda)T_c.$$

Subtracting, the terms $\lambda T_0 E$ cancel, and

$$T_P - T_M = T_a - \lambda T_a - T_a E + \lambda T_a E + (1-\lambda)T_c E - (1-\lambda)T_c$$

$$= (1-\lambda)T_a(1-E) - (1-\lambda)T_c(1-E) = (1-\lambda)(1-E)(T_a - T_c).$$

All three factors are positive, so $T_P > T_M$: the President drinks the hotter coffee.

Remark. The physical reason is that the President's coffee is closer to room temperature during the 10 minutes and therefore loses heat more slowly. (If the cream were exactly at room temperature the two would be equal in this model. A more refined model, in which the heat loss is through the same surface but the President's cup has a larger heat capacity and hence a smaller cooling constant, only increases T_P.)

Answer. The President.

Chapter 1, Miscellaneous Problem 16 (Simmons)

Solution. Let v be the speed of the submarine, so the destroyer's speed is $2v$, and let t be the time since the sighting. Whatever its direction, at time t the submarine is on the circle $r = vt$ about the origin. The strategy is to get onto this expanding circle and then stay on it while sweeping around through all directions.

Stage 1. The destroyer heads straight for the origin. After time t_1 it is at distance $3 - 2vt_1$ from the origin, and this equals the submarine's distance vt_1 when $3 = 3vt_1$, i.e. $vt_1 = 1$. So the destroyer goes 2 miles toward the origin, and is then 1 mile from it, exactly on the circle of possible submarine positions.

Stage 2. Choose the polar axis through this point, so the destroyer is at $r = 1$, $\theta = 0$. From now on it must keep $r = vt$, so $dr/dt = v$, while its speed is $2v$:

$$\left(\frac{dr}{dt}\right)^2 + r^2\left(\frac{d\theta}{dt}\right)^2 = 4v^2 \implies r^2\left(\frac{d\theta}{dt}\right)^2 = 3v^2 \implies r\frac{d\theta}{dt} = \pm\sqrt{3}\,v.$$

Dividing by $dr/dt = v$ gives $r\,d\theta/dr = \pm\sqrt{3}$, i.e. $\dfrac{\mathrm{d}r}{r} = \pm\dfrac{\mathrm{d}\theta}{\sqrt{3}}$, so $\log r = \pm\theta/\sqrt{3} + \text{const}$; the condition $r = 1$ at $\theta = 0$ makes the constant 0:

$$r = e^{\pm\theta/\sqrt{3}}.$$

Taking the $+$ sign (a counterclockwise sweep), suppose the submarine's course is the ray $\theta = \alpha$, $0 \le \alpha < 2\pi$. The destroyer reaches this ray when $\theta = \alpha$, at distance $r = e^{\alpha/\sqrt{3}}$ from the origin and at time $t = r/v$; at that moment the submarine is also at distance $vt = r$ on the same ray, i.e. directly below the destroyer. Hence the destroyer is certain to pass over the submarine before completing one revolution (at most $e^{2\pi/\sqrt{3}} \approx 37.6$ miles from the origin).

Answer. Go 2 miles straight toward the origin, then move outward along one of the spirals $r = e^{\pm\theta/\sqrt{3}}$ (polar axis through the point reached, r in miles).

Chapter 1, Miscellaneous Problem 17 (Simmons)

Solution. The initial configuration is unchanged by a rotation of 90° about the center, and so are the rules of motion; hence (by uniqueness of the motion) at every instant the four bugs are at the vertices of a square centered at the origin, which rotates and shrinks.

Let v be the common speed. The bug P starting on the polar axis is at (r, θ), with $r = a/\sqrt{2}$ (half the diagonal) and $\theta = 0$ initially. Facing the center from P, the bug on its right is the one at

polar angle $\theta + \pi/2$, at the same distance r. The vector from P to that bug is a side of the square, which makes an angle of 45° with the inward radius PO. Therefore the radial and transverse components of P's velocity are

$$\frac{dr}{dt} = -v\cos 45^\circ = -\frac{v}{\sqrt{2}}, \qquad r\frac{d\theta}{dt} = v\sin 45^\circ = \frac{v}{\sqrt{2}}.$$

Dividing, $\dfrac{1}{r}\dfrac{dr}{d\theta} = -1$, so $\log r = -\theta + \text{const}$, and $r = a/\sqrt{2}$ at $\theta = 0$ gives the path

$$r = \frac{a}{\sqrt{2}}\, e^{-\theta},$$

an equiangular spiral winding infinitely often around the center.

Distance walked. Since $dr/dt = -v/\sqrt{2}$ is constant, $r = \frac{a}{\sqrt{2}} - \frac{v}{\sqrt{2}}t$, which vanishes at $t = a/v$; the bugs meet at the center at time a/v and each has walked the distance $v \cdot (a/v) = a$. Equivalently, by arc length in polar coordinates,

$$s = \int_0^\infty \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, \mathrm{d}\theta = \int_0^\infty \sqrt{2}\cdot\frac{a}{\sqrt{2}}e^{-\theta}\, \mathrm{d}\theta = a.$$

Answer. $r = \dfrac{a}{\sqrt{2}}e^{-\theta}$; total distance $= a$ (the side of the table).

Chapter 2

First Order Equations

Section 7 Homogeneous Equations

Throughout this section we use the method of the text: if $y' = f(x,y)$ with f homogeneous of degree 0, then $f(x,y) = f(1, y/x)$, and the substitution

$$y = zx, \qquad \frac{dy}{dx} = z + x\frac{dz}{dx}$$

turns the equation into $z + xz' = f(1,z)$, whose variables separate: $\dfrac{dz}{f(1,z)-z} = \dfrac{dx}{x}$. As in the text, arbitrary constants are renamed freely (e.g. $\log|x| + c = \log C|x|$), and we point out solutions that are lost when we divide by a factor that may vanish.

Problem 7-1 (Simmons)

Solution.

(a) $M = x^2 - 2y^2$ and $N = xy$ are both homogeneous of degree 2 ($M(tx,ty) = t^2M$, $N(tx,ty) = t^2N$), so the equation is homogeneous. It reads

$$\frac{dy}{dx} = \frac{2y^2 - x^2}{xy} = \frac{2z^2 - 1}{z} \qquad (z = y/x).$$

With $y = zx$:

$$z + x\frac{dz}{dx} = \frac{2z^2-1}{z} \quad\Longrightarrow\quad x\frac{dz}{dx} = \frac{2z^2 - 1 - z^2}{z} = \frac{z^2-1}{z} \quad\Longrightarrow\quad \frac{z\,dz}{z^2-1} = \frac{dx}{x}.$$

Integrating, $\frac{1}{2}\log|z^2-1| = \log|x| + c$, i.e. $z^2 - 1 = cx^2$. Replacing z by y/x: $y^2/x^2 - 1 = cx^2$, so

$$y^2 = x^2 + cx^4.$$

(The solutions $y = \pm x$ lost in dividing by $z^2 - 1$ are the case $c = 0$.)

(b) $x^2y' = 3xy + 2y^2$, and $3xy + 2y^2$, x^2 are both homogeneous of degree 2. Thus $y' = 3z + 2z^2$, and $y = zx$ gives

$$z + xz' = 3z + 2z^2 \implies xz' = 2z(z+1) \implies \frac{dz}{z(z+1)} = \frac{2\,dx}{x}.$$

Since $\dfrac{1}{z(z+1)} = \dfrac{1}{z} - \dfrac{1}{z+1}$, integration gives $\log\left|\dfrac{z}{z+1}\right| = 2\log|x| + c$, i.e. $\dfrac{z}{z+1} = cx^2$. With $z = y/x$, $\dfrac{z}{z+1} = \dfrac{y}{x+y}$, so

$$y = cx^2(x+y), \qquad \text{equivalently} \qquad y = \frac{cx^3}{1-cx^2}.$$

(The lost solution $y = 0$ is $c = 0$; the other lost solution $y = -x$ (i.e. $z = -1$) also satisfies the equation, since $x^2(-1) - 3x(-x) - 2x^2 = 0$; it corresponds to "$c = \infty$".)

(c) Dividing by x^2: $y' = 3\left(1 + \frac{y^2}{x^2}\right)\tan^{-1}\frac{y}{x} + \frac{y}{x}$, a function of y/x alone, so the equation is homogeneous (the right side of the original is homogeneous of degree 2, like x^2). With $y = zx$:

$$z + xz' = 3(1+z^2)\tan^{-1}z + z \implies \frac{dz}{(1+z^2)\tan^{-1}z} = \frac{3\,dx}{x}.$$

Since $\frac{d}{dz}\tan^{-1} z = \frac{1}{1+z^2}$, the left side is $d(\tan^{-1} z)/\tan^{-1} z$, and integration gives $\log|\tan^{-1} z| = 3\log|x| + c$, i.e. $\tan^{-1} z = cx^3$. Hence $y/x = \tan cx^3$:

$$y = x\tan cx^3.$$

(The solution $y = 0$, lost when dividing by $\tan^{-1} z$, is $c = 0$.)

(d) Dividing by $x\sin(y/x)$: $y' = \dfrac{y}{x} + \dfrac{1}{\sin(y/x)}$, a function of y/x, so the equation is homogeneous. With $y = zx$:

$$z + xz' = z + \frac{1}{\sin z} \implies \sin z\,dz = \frac{dx}{x} \implies -\cos z = \log|x| + c.$$

Writing $c = \log C$ ($C > 0$) and replacing z by y/x:

$$\cos\frac{y}{x} + \log cx = 0$$

(for $x > 0$; for $x < 0$ read $\log c|x|$).

(e) $y' = \dfrac{y}{x} + 2e^{-y/x}$ is a function of y/x, so the equation is homogeneous. With $y = zx$:

$$z + xz' = z + 2e^{-z} \implies e^z\,dz = \frac{2\,dx}{x} \implies e^z = 2\log|x| + c = \log cx^2.$$

Hence $z = \log(\log cx^2)$ and

$$y = x\log(\log cx^2).$$

(f) $M = x - y$ and $N = -(x + y)$ are homogeneous of degree 1. The equation is $y' = \dfrac{x-y}{x+y} = \dfrac{1-z}{1+z}$, so with $y = zx$:

$$xz' = \frac{1-z}{1+z} - z = \frac{1-2z-z^2}{1+z} \implies \frac{(1+z)\,dz}{1-2z-z^2} = \frac{dx}{x}.$$

Because $d(1-2z-z^2) = -2(1+z)\,dz$, integration gives $-\frac{1}{2}\log|1-2z-z^2| = \log|x| + c$, i.e. $x^2(1-2z-z^2) = c$. With $z = y/x$:

$$x^2 - 2xy - y^2 = c.$$

(g) Dividing by x, $y' = 2 + 3\dfrac{y}{x}$ is a function of y/x (both x and $2x + 3y$ are homogeneous of degree 1), so the equation is homogeneous. With $y = zx$:

$$z + xz' = 2 + 3z \implies xz' = 2(1+z) \implies \frac{dz}{1+z} = \frac{2\,dx}{x} \implies 1 + z = cx^2.$$

Thus $y/x = cx^2 - 1$, i.e.

$$y = cx^3 - x.$$

(The lost solution $z = -1$, i.e. $y = -x$, is $c = 0$.)

(h) $\sqrt{x^2+y^2}$ and x are homogeneous of degree 1 (for $t > 0$), so the equation is homogeneous. We first take $x > 0$; then $\sqrt{x^2+y^2} = x\sqrt{1+z^2}$, and $y = zx$ gives

$$z + xz' = \sqrt{1+z^2} \implies \frac{dz}{\sqrt{1+z^2} - z} = \frac{dx}{x}.$$

Rationalizing,

$$\frac{1}{\sqrt{1+z^2}-z} = \frac{\sqrt{1+z^2}+z}{(1+z^2)-z^2} = \sqrt{1+z^2}+z.$$

Using the standard integral $\int \sqrt{1+z^2}\,dz = \frac{1}{2}\left[z\sqrt{1+z^2} + \log\left(z+\sqrt{1+z^2}\right)\right]$,

$$\tfrac{1}{2}z\sqrt{1+z^2} + \tfrac{1}{2}\log\left(z+\sqrt{1+z^2}\right) + \tfrac{1}{2}z^2 = \log x + c.$$

Multiply by 2 and put $z = y/x$, so that $\sqrt{1+z^2} = \sqrt{x^2+y^2}/x$:

$$\frac{y\sqrt{x^2+y^2}}{x^2} + \log\frac{y+\sqrt{x^2+y^2}}{x} + \frac{y^2}{x^2} = 2\log x + c.$$

Multiplying by x^2 and using $\log\frac{y+\sqrt{x^2+y^2}}{x} = \log\left(y+\sqrt{x^2+y^2}\right) - \log x$:

$$y\sqrt{x^2+y^2} + x^2\log\left(y+\sqrt{x^2+y^2}\right) - 3x^2\log x + y^2 = cx^2.$$

For $x < 0$: the equation is unchanged by the reflection $x \mapsto -x$ (if $X = -x$, then $X\,dy/dX = (-x)(-y') = xy'$), so the solutions for $x < 0$ are the mirror images of those for $x > 0$, and are given by the same formula with $\log x$ replaced by $\log|x|$.

(i) $y' = \dfrac{y^2}{x^2} + 2\dfrac{y}{x}$ is a function of y/x (degree 2 throughout). With $y = zx$:

$$z + xz' = z^2 + 2z \implies xz' = z(z+1) \implies \frac{dz}{z(z+1)} = \frac{dx}{x} \implies \log\left|\frac{z}{z+1}\right| = \log|x| + c,$$

so $\dfrac{z}{z+1} = cx$, i.e. $z = cx(z+1)$, $z = \dfrac{cx}{1-cx}$, and

$$y = \frac{cx^2}{1-cx}.$$

(Lost solutions: $y = 0$ is $c = 0$; $y = -x$ also satisfies the equation, since $x^2(-1) = x^2 - 2x^2$; it corresponds to "$c = \infty$".)

(j) $M = x^3 + y^3$, $N = -xy^2$ are homogeneous of degree 3. Here $y' = \dfrac{x^3+y^3}{xy^2} = \dfrac{1+z^3}{z^2}$, and $y = zx$ gives

$$z + xz' = \frac{1}{z^2} + z \implies z^2\,dz = \frac{dx}{x} \implies \frac{z^3}{3} = \log|x| + c.$$

So $y^3/x^3 = 3\log|x| + 3c = \log c|x|^3$, i.e.

$$y^3 = x^3\log cx^3.$$

Answer. (a) $y^2 = x^2 + cx^4$; (b) $y = cx^2(x+y)$; (c) $y = x\tan cx^3$; (d) $\cos(y/x) + \log cx = 0$; (e) $y = x\log(\log cx^2)$; (f) $x^2 - 2xy - y^2 = c$; (g) $y = cx^3 - x$; (h) $y\sqrt{x^2+y^2} + x^2\log\left(y + \sqrt{x^2+y^2}\right) - 3x^2\log x + y^2 = cx^2$; (i) $y = cx^2/(1-cx)$; (j) $y^3 = x^3\log cx^3$.

Problem 7-2 (Simmons)

Solution. A circle tangent to the y-axis at the origin has its center on the x-axis, at $(a,0)$, and radius $|a|$:

$$(x-a)^2+y^2=a^2 \quad\Longleftrightarrow\quad x^2+y^2=2ax.$$

Differentiating, $2x+2yy'=2a$; eliminating $a=(x^2+y^2)/(2x)$ gives

$$2yy'=\frac{x^2+y^2}{x}-2x=\frac{y^2-x^2}{x}, \qquad \text{so} \qquad y'=\frac{y^2-x^2}{2xy}$$

is the differential equation of the family. The orthogonal trajectories satisfy $y'=-1/(\text{old slope})$:

$$\frac{dy}{dx}=\frac{2xy}{x^2-y^2}=\frac{2z}{1-z^2}, \qquad z=\frac{y}{x},$$

which is homogeneous. With $y=zx$:

$$x\frac{dz}{dx}=\frac{2z}{1-z^2}-z=\frac{z+z^3}{1-z^2} \quad\Longrightarrow\quad \frac{(1-z^2)\,dz}{z(1+z^2)}=\frac{dx}{x}.$$

Partial fractions: $\dfrac{1-z^2}{z(1+z^2)}=\dfrac{1}{z}-\dfrac{2z}{1+z^2}$ (check: $\frac{(1+z^2)-2z^2}{z(1+z^2)}$). Integrating,

$$\log|z|-\log(1+z^2)=\log|x|+c \quad\Longrightarrow\quad \frac{z}{1+z^2}=cx.$$

Since $\dfrac{z}{1+z^2}=\dfrac{y/x}{(x^2+y^2)/x^2}=\dfrac{xy}{x^2+y^2}$, this is $xy=cx(x^2+y^2)$, i.e. $y=c(x^2+y^2)$, or with a new constant

$$x^2+y^2=cy.$$

These are the circles tangent to the x-axis at the origin (center $(0,c/2)$). The solution $z=0$, i.e. the x-axis $y=0$, lost in the division, is also a trajectory (it passes through all the centers and so cuts every circle of the family at right angles); it is the limiting case $c\to\infty$.

Answer. $x^2+y^2=cy$: the circles tangent to the x-axis at the origin.

Problem 7-3 (Simmons)

Solution. If $z=ax+by+c$, then $\dfrac{dz}{dx}=a+b\dfrac{dy}{dx}=a+bf(z)$. This equation does not contain x explicitly, so its variables separate:

$$\frac{dz}{a+bf(z)}=dx, \qquad x=\int\frac{dz}{a+bf(z)}+c'.$$

(If $b=0$ the original equation $y'=f(ax+c)$ is already separated. Any root z_0 of $a+bf(z_0)=0$ gives the additional solution $ax+by+c=z_0$.)

(a) Put $z=x+y$. Then $z'=1+y'=1+z^2$, so

$$\frac{dz}{1+z^2}=dx \;\Longrightarrow\; \tan^{-1}z=x+c \;\Longrightarrow\; z=\tan(x+c),$$

i.e. $x+y=\tan(x+c)$, or $y=\tan(x+c)-x$.

(b) Put $z = x - y + 1$. Then $z' = 1 - y' = 1 - \sin^2 z = \cos^2 z$, so

$$\sec^2 z \, dz = dx \implies \tan z = x + c,$$

i.e. $\tan(x - y + 1) = x + c$. In the division by $\cos^2 z$ we lose the solutions $\cos z = 0$, i.e. $x - y + 1 = \frac{\pi}{2} + n\pi$, the straight lines $y = x + 1 - \frac{\pi}{2} - n\pi$ (n an integer); these indeed satisfy the equation, since then $y' = 1$ and $\sin^2(\frac{\pi}{2} + n\pi) = 1$.

Answer. (a) $x + y = \tan(x + c)$; (b) $\tan(x - y + 1) = x + c$ (plus the lines $y = x + 1 - \frac{\pi}{2} - n\pi$).

Problem 7-4 (Simmons)

Solution.

(a) With $x = z - h$, $y = w - k$ we have $dx = dz$, $dy = dw$, so $dy/dx = dw/dz$, and

$$ax + by + c = az + bw + (c - ah - bk), \qquad dx + ey + f = dz + ew + (f - dh - ek).$$

Choose h, k so that both constant terms vanish:

$$ah + bk = c, \qquad dh + ek = f.$$

This linear system has determinant $ae - bd \neq 0$, so it has the unique solution (Cramer's rule)

$$h = \frac{ce - bf}{ae - bd}, \qquad k = \frac{af - cd}{ae - bd}.$$

(Geometrically, $(-h, -k)$ is the point where the lines $ax + by + c = 0$ and $dx + ey + f = 0$ meet.) With this choice the equation becomes

$$\frac{dw}{dz} = F\left(\frac{az + bw}{dz + ew}\right) = F\left(\frac{a + b(w/z)}{d + e(w/z)}\right),$$

whose right side is homogeneous of degree 0 in z, w. So the equation is homogeneous.

(b) Suppose $ae = bd$.

Case $(d, e) \neq (0, 0)$. The vectors (a, b) and (d, e) are linearly dependent, so $(a, b) = \lambda(d, e)$ for some constant λ, and $ax + by = \lambda(dx + ey)$. Put

$$z = dx + ey.$$

Then $ax + by + c = \lambda z + c$ and

$$\frac{dz}{dx} = d + e\frac{dy}{dx} = d + eF\left(\frac{\lambda z + c}{z + f}\right),$$

an equation in which x does not appear, so its variables separate: $\dfrac{dz}{d + eF\big((\lambda z + c)/(z + f)\big)} = dx$.

(If here $e = 0$, then $d \neq 0$ and $b = \lambda e = 0$, and the original equation is simply $y' = F\big((ax + c)/(dx + f)\big)$, which is already separated: $y = \int F \, dx$.)

Case $d = e = 0$. The equation is $y' = F\big((ax + by + c)/f\big)$, a function of $ax + by$; by Problem 3 the substitution $z = ax + by$ separates the variables: $z' = a + bF\big((z + c)/f\big)$.

In the main case, therefore, the substitution is $z = dx + ey$ (equivalently $z = ax + by$ when $(a, b) \neq (0, 0)$, since then the two differ by a constant factor).

Answer. (a) Take $h = \dfrac{ce-bf}{ae-bd}$, $k = \dfrac{af-cd}{ae-bd}$; then $\dfrac{dw}{dz} = F\left(\dfrac{az+bw}{dz+ew}\right)$, which is homogeneous. (b) $z = dx + ey$ (then $ax+by = \lambda z$ and $z' = d + eF\big((\lambda z + c)/(z+f)\big)$ is separable).

Problem 7-5 (Simmons)

Solution. We use Problem 4. When $ae \neq bd$ we translate the origin to the intersection point (x_0, y_0) of the two lines, i.e. we put $x = z + x_0$, $y = w + y_0$ (so $dy/dx = dw/dz$), which makes the equation homogeneous in z, w; then we put $w = uz$.

(a) $ae - bd = 1\cdot(-1) - 1\cdot 1 = -2 \neq 0$. The lines $x + y + 4 = 0$ and $x - y - 6 = 0$ meet where $2x - 2 = 0$: $x_0 = 1$, $y_0 = -5$. With $x = z + 1$, $y = w - 5$: $x + y + 4 = z + w$, $x - y - 6 = z - w$, so

$$\frac{dw}{dz} = \frac{z+w}{z-w} = \frac{1+u}{1-u} \qquad (w = uz).$$

This is the equation of Example 1 in the variables z, w. Explicitly, $z\,u' = \dfrac{1+u}{1-u} - u = \dfrac{1+u^2}{1-u}$, so

$$\frac{(1-u)\,du}{1+u^2} = \frac{dz}{z} \implies \tan^{-1}u - \tfrac{1}{2}\log(1+u^2) = \log|z| + c \implies \tan^{-1}\frac{w}{z} = \log\sqrt{z^2+w^2} + c,$$

since $\frac{1}{2}\log(1+u^2) + \log|z| = \log\sqrt{z^2+w^2}$. Returning to x, y ($z = x-1$, $w = y+5$):

$$\tan^{-1}\frac{y+5}{x-1} = \log\sqrt{(x-1)^2 + (y+5)^2} + c.$$

(b) Here $ae = bd$ ($1\cdot 1 = 1\cdot 1$), so by Problem 4(b) we put $z = x + y$. Then

$$\frac{dz}{dx} = 1 + \frac{z+4}{z-6} = \frac{2z-2}{z-6} \quad\implies\quad \frac{(z-6)\,dz}{2(z-1)} = dx.$$

Since $\dfrac{z-6}{z-1} = 1 - \dfrac{5}{z-1}$, integration gives $z - 5\log|z-1| = 2x + c$, i.e. $x + y - 5\log|x+y-1| = 2x + c$:

$$y - x = 5\log|x+y-1| + c.$$

(The line $x + y = 1$, i.e. $z = 1$, lost in the division, is also a solution: there $y' = -1$ and $\frac{x+y+4}{x+y-6} = \frac{5}{-5} = -1$.)

(c) The equation is $\dfrac{dy}{dx} = \dfrac{-2x+2y}{y-1}$, with $a = -2$, $b = 2$, $c = 0$, $d = 0$, $e = 1$, $f = -1$; $ae - bd = -2 \neq 0$. The lines $x - y = 0$ and $y - 1 = 0$ meet at $(1, 1)$. With $x = z + 1$, $y = w + 1$: $2x - 2y = 2z - 2w$ and $y - 1 = w$, so the equation becomes

$$(2z - 2w)\,dz + w\,dw = 0, \qquad \frac{dw}{dz} = \frac{2w-2z}{w} = \frac{2u-2}{u} \quad (w = uz).$$

Then

$$z\frac{du}{dz} = \frac{2u-2}{u} - u = -\frac{u^2-2u+2}{u} \quad\implies\quad \frac{u\,du}{u^2-2u+2} = -\frac{dz}{z}.$$

Write $\dfrac{u}{u^2-2u+2} = \dfrac{1}{2}\cdot\dfrac{2u-2}{u^2-2u+2} + \dfrac{1}{(u-1)^2+1}$. Integrating,

$$\tfrac{1}{2}\log(u^2-2u+2) + \tan^{-1}(u-1) = -\log|z| + c.$$

Multiplying by 2 and combining logarithms, $\log\left[z^2(u^2-2u+2)\right] = \log\left[(w-z)^2+z^2\right]$, so

$$\log\left[(w-z)^2+z^2\right] + 2\tan^{-1}\frac{w-z}{z} = c.$$

Since $w-z = y-x$ and $z = x-1$:

$$\log\left[(y-x)^2+(x-1)^2\right] + 2\tan^{-1}\frac{y-x}{x-1} = c.$$

(d) $ae-bd = 1\cdot 4 - 1\cdot 1 = 3 \neq 0$. Solving $x+y=1$, $x+4y=-2$ gives $3y=-3$: $y_0=-1$, $x_0=2$. With $x=z+2$, $y=w-1$: $x+y-1 = z+w$, $x+4y+2 = z+4w$, so

$$\frac{dw}{dz} = \frac{z+w}{z+4w} = \frac{1+u}{1+4u}, \qquad z\frac{du}{dz} = \frac{1+u}{1+4u} - u = \frac{1-4u^2}{1+4u}.$$

Separating, $\dfrac{(1+4u)\,du}{(1-2u)(1+2u)} = \dfrac{dz}{z}$. Partial fractions: $1+4u = A(1+2u)+B(1-2u)$; $u=\frac{1}{2}$ gives $A=\frac{3}{2}$, $u=-\frac{1}{2}$ gives $B=-\frac{1}{2}$. So

$$\int\left(\frac{3/2}{1-2u} - \frac{1/2}{1+2u}\right)du = -\tfrac{3}{4}\log|1-2u| - \tfrac{1}{4}\log|1+2u| = \log|z| + c.$$

Multiplying by -4: $3\log|1-2u| + \log|1+2u| + 4\log|z| = c$, i.e. $(1-2u)^3(1+2u)z^4 = c$, i.e. $(z-2w)^3(z+2w) = c$. Now $z-2w = x-2y-4$ and $z+2w = x+2y$, so

$$(x+2y)(x-2y-4)^3 = c.$$

(The lost solutions $u=\pm\frac{1}{2}$ are included as $c=0$.)

(e) The equation is $\dfrac{dy}{dx} = \dfrac{2x+3y-1}{4x+4}$; $ae-bd = 2\cdot 0 - 3\cdot 4 = -12 \neq 0$. The lines meet where $x=-1$ and $-2+3y-1=0$, i.e. at $(-1,1)$. With $x=z-1$, $y=w+1$: $2x+3y-1 = 2z+3w$, $4x+4 = 4z$, so

$$\frac{dw}{dz} = \frac{2z+3w}{4z} = \frac{2+3u}{4}, \qquad z\frac{du}{dz} = \frac{2+3u}{4} - u = \frac{2-u}{4}.$$

Thus $\dfrac{du}{2-u} = \dfrac{dz}{4z}$, so $-\log|2-u| = \frac{1}{4}\log|z| + c$, i.e. $-4\log|2-u| - \log|z| = 4c$, i.e. $(2-u)^4 z = C$. Since $(2-u)^4 z = (2z-w)^4/z^3$: $(2z-w)^4 = Cz^3$. With $2z-w = 2x+2-y+1 = 2x-y+3$ and $z=x+1$:

$$(2x-y+3)^4 = c(x+1)^3.$$

Answer. (a) $\tan^{-1}\dfrac{y+5}{x-1} = \log\sqrt{(x-1)^2+(y+5)^2} + c$; (b) $y-x = 5\log|x+y-1| + c$; (c) $\log\left[(y-x)^2+(x-1)^2\right] + 2\tan^{-1}\dfrac{y-x}{x-1} = c$; (d) $(x+2y)(x-2y-4)^3 = c$; (e) $(2x-y+3)^4 = c(x+1)^3$.

Problem 7-6 (Simmons)

Solution. With $y = zx^n$ we have $y' = x^n z' + nx^{n-1}z$. We take $x>0$ when fractional powers occur.

(a) The awkward term is $xy^2 = z^2x^{2n+1}$; it becomes a function of z alone if $n = -\frac{1}{2}$. So put $y = zx^{-1/2}$ (i.e. $z^2 = xy^2$). Then $y' = x^{-1/2}z' - \frac{1}{2}x^{-3/2}z$ and the right side is $\dfrac{1-z^2}{2x^{3/2}z}$. Multiplying the equation by $x^{3/2}$:

$$xz' - \frac{z}{2} = \frac{1-z^2}{2z} \implies xz' = \frac{1-z^2+z^2}{2z} = \frac{1}{2z} \implies 2z\,dz = \frac{dx}{x}.$$

Hence $z^2 = \log x + c$, i.e. $xy^2 = \log x + c$, or $x = ce^{xy^2}$. (For $x < 0$, implicit differentiation shows that $xy^2 = \log|x| + c$ satisfies the equation as well: $y^2 + 2xyy' = 1/x$ is the given equation; and this is again $x = ce^{xy^2}$, now with $c < 0$.)

(b) *With* $n = -\frac{1}{2}$. Again $z^2 = xy^2$, $y = zx^{-1/2}$; the right side is $\dfrac{2+3z^2}{4x^{3/2}z}$, and multiplying by $x^{3/2}$:

$$xz' - \frac{z}{2} = \frac{2+3z^2}{4z} \implies xz' = \frac{2+5z^2}{4z} \implies \frac{4z\,dz}{2+5z^2} = \frac{dx}{x}.$$

Integrating, $\frac{2}{5}\log(2+5z^2) = \log x + c$, so $2 + 5z^2 = cx^{5/2}$:

$$2 + 5xy^2 = cx^{5/2}.$$

With $n = \frac{3}{4}$ (the value given in the answer key) the equation separates even more simply. Put $y = zx^{3/4}$; then $y' = x^{3/4}z' + \frac{3}{4}x^{-1/4}z$, while

$$\frac{2+3xy^2}{4x^2y} = \frac{2+3x^{5/2}z^2}{4x^{11/4}z} = \frac{1}{2x^{11/4}z} + \frac{3}{4}x^{-1/4}z.$$

The terms $\frac{3}{4}x^{-1/4}z$ cancel, leaving $x^{3/4}z' = \dfrac{1}{2x^{11/4}z}$, i.e.

$$z\,dz = \tfrac{1}{2}x^{-7/2}\,dx \implies \frac{z^2}{2} = -\frac{1}{5}x^{-5/2} + c.$$

Since $z^2 = y^2x^{-3/2}$, multiplying by $10x^{5/2}$ gives $5xy^2 = -2 + cx^{5/2}$, the same result. So both $n = -\frac{1}{2}$ and $n = \frac{3}{4}$ work.

(c) Try $n = -1$, i.e. $z = xy$, $y = z/x$, $y' = \dfrac{z'}{x} - \dfrac{z}{x^2}$. The right side becomes

$$\frac{z/x - z^2/x}{x + xz} = \frac{z(1-z)}{x^2(1+z)}.$$

Hence

$$\frac{z'}{x} = \frac{z}{x^2} + \frac{z(1-z)}{x^2(1+z)} = \frac{z(1+z)+z(1-z)}{x^2(1+z)} = \frac{2z}{x^2(1+z)}, \qquad \text{so} \qquad \frac{(1+z)\,dz}{z} = \frac{2\,dx}{x}.$$

Integrating, $\log|z| + z = 2\log|x| + c$, i.e. $ze^z = Cx^2$, i.e. $xye^{xy} = Cx^2$, or $ye^{xy} = Cx$. Writing $c = 1/C$:

$$x = cye^{xy}.$$

(The solution $y = 0$, lost in dividing by z, corresponds to $C = 0$ in $ye^{xy} = Cx$.)

Answer. (a) $n = -\frac{1}{2}$: $x = ce^{xy^2}$; (b) $n = -\frac{1}{2}$ or $n = \frac{3}{4}$: $2 + 5xy^2 = cx^{5/2}$; (c) $n = -1$: $x = cye^{xy}$.

Problem 7-7 (Simmons)

Solution. Let the equation be $y' = f(x,y)$ with $f(tx,ty) = f(x,y)$. Consider a non-vertical line $y = mx$ through the origin. At any point (x, mx) of it with $x \neq 0$, taking $t = 1/x$,

$$f(x, mx) = f\big(1, m\big).$$

So every integral curve that crosses the line does so with the *same* slope $\mu = f(1,m)$, independent of the point. If $\beta = \tan^{-1} m$ is the inclination of the line and $\varphi = \tan^{-1}\mu$ that of the integral curve at the crossing, the angle between them is $\varphi - \beta$, the same at every crossing. (Equivalently, $\tan(\varphi - \beta) = \dfrac{\mu - m}{1 + \mu m}$ is constant.)

For the vertical line $x = 0$: at $(0,y)$, $y \neq 0$, take $t = 1/y$ to get $f(0,y) = f(0,1)$, again a constant slope, so again a constant angle.

Remark. If f is homogeneous of degree 0 only for $t > 0$ (as with $\sqrt{x^2+y^2}/x$ in Problem 1(h)), the argument applies to each of the two rays from the origin separately: the angle is constant along each ray.

Answer. Shown: on $y = mx$ every integral curve has slope $f(1,m)$, a constant, so all crossings are at the same angle.

Problem 7-8 (Simmons)

Solution. The described map is $(x,y) \mapsto (x_1, y_1) = (kx, ky)$ with $k > 0$ (same direction, distance multiplied by k). Let $C\colon y = g(x)$ be an integral curve, so $g'(x) = f\big(x, g(x)\big)$. Its image C_1 is given parametrically by

$$x_1 = kx, \qquad y_1 = kg(x).$$

Along C_1, by the chain rule,

$$\frac{dy_1}{dx_1} = \frac{dy_1/dx}{dx_1/dx} = \frac{kg'(x)}{k} = g'(x) = f\big(x, g(x)\big).$$

Since f is homogeneous of degree 0, $f(x, g(x)) = f\big(kx, kg(x)\big) = f(x_1, y_1)$. Hence $dy_1/dx_1 = f(x_1, y_1)$ along C_1, i.e. C_1 is an integral curve. (Geometrically: the stretching maps each point to a point on the same ray from the origin and preserves slopes; by Problem 7 the slope field is constant along rays, so integral curves go to integral curves.)

Answer. Shown: $dy_1/dx_1 = g'(x) = f(x,y) = f(kx,ky) = f(x_1,y_1)$, so C_1 is an integral curve.

Problem 7-9 (Simmons)

Solution. Assume (as throughout the chapter) that f is continuous, so that through every point of its domain there passes an integral curve. Fix $k > 0$ and a point (x_0, y_0), and let $C\colon y = g(x)$ be an integral curve through it; then $g(x_0) = y_0$ and its slope there is $g'(x_0) = f(x_0, y_0)$.

By hypothesis the stretched curve $C_1\colon x_1 = kx$, $y_1 = kg(x)$ is an integral curve. It passes through (kx_0, ky_0), and, exactly as in Problem 8, its slope there is

$$\frac{dy_1}{dx_1} = \frac{kg'(x_0)}{k} = g'(x_0) = f(x_0, y_0).$$

But C_1 is an integral curve, so its slope at (kx_0, ky_0) must equal $f(kx_0, ky_0)$. Therefore

$$f(kx_0, ky_0) = f(x_0, y_0)$$

for every point (x_0, y_0) and every $k > 0$, i.e. $f(tx, ty) = t^0 f(x, y)$: f is homogeneous of degree 0, and the equation $y' = f(x, y)$ is homogeneous (it can be written as $y' = f(1, y/x)$ for $x > 0$).

Answer. Shown: invariance under $(x, y) \mapsto (kx, ky)$ forces $f(kx, ky) = f(x, y)$, so the equation is homogeneous.

Problem 7-10 (Simmons)

Solution. At P let θ_1 be the inclination of the tangent to the curve of the first family, so $\tan\theta_1 = f(x, y)$, and let θ_2 be the inclination of the tangent to the curve of the second family. The angle from the first curve to the second is α, so $\theta_2 = \theta_1 + \alpha$. Hence the slope of the second curve is, by the addition formula for the tangent,

$$y' = \tan\theta_2 = \tan(\theta_1 + \alpha) = \frac{\tan\theta_1 + \tan\alpha}{1 - \tan\theta_1 \tan\alpha} = \frac{f(x, y) + \tan\alpha}{1 - f(x, y)\tan\alpha},$$

which is the stated equation (valid wherever $f(x, y)\tan\alpha \neq 1$, i.e. wherever the second curve has a non-vertical tangent).

Answer. Shown: $y' = \tan(\theta_1 + \alpha)$ with $\tan\theta_1 = f$ gives the stated equation.

Problem 7-11 (Simmons)

Solution. With $\alpha = \pi/4$, $\tan\alpha = 1$, the curves sought satisfy $y' = \dfrac{f + 1}{1 - f}$.

(a) The lines $y = cx$ satisfy $y' = c = y/x$, so $f = y/x$ and

$$y' = \frac{y/x + 1}{1 - y/x} = \frac{x + y}{x - y}.$$

This is the equation of Example 1, whose solution is $\tan^{-1}\dfrac{y}{x} = \log\sqrt{x^2 + y^2} + c$ (derived there by $y = zx$: $\frac{(1-z)\,dz}{1+z^2} = \frac{dx}{x}$). In polar coordinates $\sqrt{x^2 + y^2} = r$ and $\tan^{-1}(y/x)$ differs from θ by a constant multiple of π, which can be absorbed into c. So $\theta = \log r + c$, i.e.

$$r = ce^{\theta},$$

a family of logarithmic spirals.

(b) Differentiating $x^2 + y^2 = c^2$: $2x + 2yy' = 0$, so $f = -x/y$. Then

$$y' = \frac{-x/y + 1}{1 + x/y} = \frac{y - x}{y + x} = \frac{z - 1}{z + 1} \qquad (y = zx).$$

So $xz' = \dfrac{z - 1}{z + 1} - z = -\dfrac{1 + z^2}{1 + z}$, and

$$\frac{(1 + z)\,dz}{1 + z^2} = -\frac{dx}{x} \implies \tan^{-1} z + \tfrac{1}{2}\log(1 + z^2) = -\log|x| + c,$$

and since $\frac{1}{2}\log(1+z^2)+\log|x| = \log\sqrt{x^2+y^2}$,

$$\tan^{-1}\frac{y}{x} + \log\sqrt{x^2+y^2} = c.$$

In polar coordinates, $\theta + \log r = c$, i.e.

$$r = ce^{-\theta}.$$

(c) Differentiating $x^2 - 2xy - y^2 = c$: $2x - 2y - 2xy' - 2yy' = 0$, so $f = \dfrac{x-y}{x+y}$. Then

$$y' = \frac{\frac{x-y}{x+y}+1}{1-\frac{x-y}{x+y}} = \frac{(x-y)+(x+y)}{(x+y)-(x-y)} = \frac{2x}{2y} = \frac{x}{y}.$$

Separating, $y\,dy = x\,dx$, so $y^2 = x^2 - c$:

$$x^2 - y^2 = c.$$

Answer. (a) $r = ce^{\theta}$ (polar coordinates); (b) $r = ce^{-\theta}$; (c) $x^2 - y^2 = c$.

Section 8 Exact Equations

For Problems 1–19 the instruction is: *Determine which of the following equations are exact, and solve the ones that are.* In each case we write the equation as $M\,dx + N\,dy = 0$ and apply the test $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ of the text. When the equation is exact we find f with $\frac{\partial f}{\partial x} = M$, $\frac{\partial f}{\partial y} = N$ by the method of equations (5) and (6): integrate one of the two equations partially, then determine the "constant" of integration (a function of the other variable) from the remaining equation. The general solution is $f(x, y) = c$.

Problem 8-1 (Simmons)

Solution. $M = y$, $N = x + \dfrac{2}{y}$; $\frac{\partial M}{\partial y} = 1 = \frac{\partial N}{\partial x}$, so the equation is *exact*. Integrating $\frac{\partial f}{\partial x} = y$ with respect to x: $f = xy + g(y)$. Then $\frac{\partial f}{\partial y} = x + g'(y) = x + \dfrac{2}{y}$ gives $g'(y) = \dfrac{2}{y}$, $g(y) = 2\log|y| = \log y^2$. Hence

$$xy + \log y^2 = c.$$

Answer. Exact; $xy + \log y^2 = c$.

Problem 8-2 (Simmons)

Solution. $M = \sin x \tan y + 1$, $N = \cos x \sec^2 y$:

$$\frac{\partial M}{\partial y} = \sin x \sec^2 y, \qquad \frac{\partial N}{\partial x} = -\sin x \sec^2 y.$$

These are not identically equal, so the equation is *not exact*.

Answer. Not exact.

Problem 8-3 (Simmons)

Solution. $\frac{\partial M}{\partial y} = 1 = \frac{\partial N}{\partial x}$: *exact*. From $\frac{\partial f}{\partial x} = y - x^3$, $f = xy - \frac{1}{4}x^4 + g(y)$; then $\frac{\partial f}{\partial y} = x + g'(y) = x + y^3$, so $g'(y) = y^3$, $g(y) = \frac{1}{4}y^4$. Thus $xy - \frac{1}{4}x^4 + \frac{1}{4}y^4 = c$, or, multiplying by 4,

$$4xy - x^4 + y^4 = c.$$

Answer. Exact; $4xy - x^4 + y^4 = c$.

Problem 8-4 (Simmons)

Solution. In the form $M\,dx + N\,dy = 0$: $M = 2y^2 - 4x + 5$, $N = -4 + 2y - 4xy$. Then

$$\frac{\partial M}{\partial y} = 4y, \qquad \frac{\partial N}{\partial x} = -4y,$$

which are not identically equal: *not exact*.

Answer. Not exact.

Problem 8-5 (Simmons)

Solution.

$$\frac{\partial M}{\partial y} = 1 + \cos xy - xy \sin xy, \qquad \frac{\partial N}{\partial x} = 1 + \cos xy - xy \sin xy,$$

so the equation is *exact*. Since $\frac{\partial}{\partial x} \sin xy = y \cos xy$, integrating M with respect to x gives $f = xy + \sin xy + g(y)$. Then $\frac{\partial f}{\partial y} = x + x \cos xy + g'(y) = N$ forces $g'(y) = 0$; take $g = 0$:

$$xy + \sin xy = c.$$

Answer. Exact; $xy + \sin xy = c$.

Problem 8-6 (Simmons)

Solution.

$$\frac{\partial M}{\partial y} = \cos x \cdot 2 \cos y(-\sin y) = -2 \cos x \sin y \cos y, \qquad \frac{\partial N}{\partial x} = 2 \cos x \sin y \cos y.$$

These differ (they have opposite signs), so the equation is *not exact*.

Answer. Not exact.

Problem 8-7 (Simmons)

Solution. Moving everything to one side and changing signs (which does not affect exactness):

$$(e^y + \cos x \cos y)\,dx + (xe^y - \sin x \sin y)\,dy = 0.$$

Here $\frac{\partial M}{\partial y} = e^y - \cos x \sin y$ and $\frac{\partial N}{\partial x} = e^y - \cos x \sin y$: *exact*. Integrating M with respect to x: $f = xe^y + \sin x \cos y + g(y)$. Then $\frac{\partial f}{\partial y} = xe^y - \sin x \sin y + g'(y) = N$ gives $g'(y) = 0$. Hence

$$xe^y + \sin x \cos y = c.$$

Answer. Exact; $xe^y + \sin x \cos y = c$.

Problem 8-8 (Simmons)

Solution. $M = -\frac{1}{y} \sin \frac{x}{y}$, $N = \frac{x}{y^2} \sin \frac{x}{y}$. Using $\frac{\partial}{\partial y} \frac{x}{y} = -\frac{x}{y^2}$ and $\frac{\partial}{\partial x} \frac{x}{y} = \frac{1}{y}$,

$$\frac{\partial M}{\partial y} = \frac{1}{y^2} \sin \frac{x}{y} - \frac{1}{y} \cos \frac{x}{y} \cdot \left(-\frac{x}{y^2}\right) = \frac{1}{y^2} \sin \frac{x}{y} + \frac{x}{y^3} \cos \frac{x}{y},$$
$$\frac{\partial N}{\partial x} = \frac{1}{y^2} \sin \frac{x}{y} + \frac{x}{y^2} \cos \frac{x}{y} \cdot \frac{1}{y} = \frac{1}{y^2} \sin \frac{x}{y} + \frac{x}{y^3} \cos \frac{x}{y}.$$

They agree: *exact*. Since $\frac{\partial}{\partial x}\cos\frac{x}{y} = -\frac{1}{y}\sin\frac{x}{y} = M$, we have $f = \cos\frac{x}{y} + g(y)$; then $\frac{\partial f}{\partial y} = \frac{x}{y^2}\sin\frac{x}{y} + g'(y) = N$ gives $g' = 0$. So

$$\cos\frac{x}{y} = c,$$

which (on any interval where the cosine is one-to-one) is equivalent to $\dfrac{x}{y} = c$.

Answer. Exact; $\cos\dfrac{x}{y} = c$, or $\dfrac{x}{y} = c$.

Problem 8-9 (Simmons)

Solution. $\frac{\partial M}{\partial y} = 1$, $\frac{\partial N}{\partial x} = -1$: *not exact*.

Answer. Not exact.

Problem 8-10 (Simmons)

Solution. $\frac{\partial M}{\partial y} = 6xy^2 + \cos x = \frac{\partial N}{\partial x}$: *exact*. Integrating M with respect to x: $f = x^2y^3 + y\sin x + g(y)$; then $\frac{\partial f}{\partial y} = 3x^2y^2 + \sin x + g'(y) = N$ gives $g' = 0$. Hence

$$x^2y^3 + y\sin x = c.$$

Answer. Exact; $x^2y^3 + y\sin x = c$.

Problem 8-11 (Simmons)

Solution. Bring all terms to one side:

$$\left(\frac{y}{1-x^2y^2} - 1\right)dx + \frac{x}{1-x^2y^2}\,\mathrm{d}y = 0.$$

By the quotient rule,

$$\frac{\partial M}{\partial y} = \frac{(1-x^2y^2) + y\cdot 2x^2y}{(1-x^2y^2)^2} = \frac{1+x^2y^2}{(1-x^2y^2)^2}, \qquad \frac{\partial N}{\partial x} = \frac{(1-x^2y^2) + x\cdot 2xy^2}{(1-x^2y^2)^2} = \frac{1+x^2y^2}{(1-x^2y^2)^2},$$

so the equation is *exact*. Here it is easier to integrate N with respect to y: with $u = xy$, $du = x\,\mathrm{d}y$,

$$\int \frac{x\,\mathrm{d}y}{1-x^2y^2} = \int \frac{du}{1-u^2} = \frac{1}{2}\log\left|\frac{1+u}{1-u}\right|, \qquad \text{so} \qquad f = \frac{1}{2}\log\left|\frac{1+xy}{1-xy}\right| + h(x).$$

Then $\frac{\partial f}{\partial x} = \frac{1}{2}\left(\frac{y}{1+xy} + \frac{y}{1-xy}\right) + h'(x) = \frac{y}{1-x^2y^2} + h'(x)$, and equating this to M gives $h'(x) = -1$, $h(x) = -x$. Multiplying $f = c$ by 2:

$$\log\frac{1+xy}{1-xy} - 2x = c.$$

Answer. Exact; $\log\dfrac{1+xy}{1-xy} - 2x = c$.

Problem 8-12 (Simmons)

Solution. $\frac{\partial M}{\partial y} = 8xy^3 + \cos y = \frac{\partial N}{\partial x}$: *exact*. Integrating M with respect to x: $f = x^2y^4 + x\sin y + g(y)$; then $\frac{\partial f}{\partial y} = 4x^2y^3 + x\cos y + g'(y) = N$ gives $g' = 0$:

$$x^2y^4 + x\sin y = c.$$

Answer. Exact; $x^2y^4 + x\sin y = c$.

Problem 8-13 (Simmons)

Solution. $M = \dfrac{y}{1-x^2y^2} + x$, $N = \dfrac{x}{1-x^2y^2}$. Exactly as in Problem 11 (the added term x does not depend on y), $\frac{\partial M}{\partial y} = \dfrac{1+x^2y^2}{(1-x^2y^2)^2} = \frac{\partial N}{\partial x}$: *exact*. As in Problem 11, $\int N\,dy = \frac{1}{2}\log\left|\frac{1+xy}{1-xy}\right|$, so $f = \frac{1}{2}\log\left|\frac{1+xy}{1-xy}\right| + h(x)$ and $\frac{\partial f}{\partial x} = \dfrac{y}{1-x^2y^2} + h'(x) = M$ gives $h'(x) = x$, $h = \frac{1}{2}x^2$. (In other words, the equation is $d\left(\frac{1}{2}\log\frac{1+xy}{1-xy}\right) + d\left(\frac{1}{2}x^2\right) = 0$, because $y\,dx + x\,dy = d(xy)$.) Multiplying by 2:

$$\log\left(\frac{1+xy}{1-xy}\right) + x^2 = c.$$

Answer. Exact; $\log\dfrac{1+xy}{1-xy} + x^2 = c$.

Problem 8-14 (Simmons)

Solution. $M = 2x + 2x\sqrt{x^2-y}$, $N = -\sqrt{x^2-y}$. Then

$$\frac{\partial M}{\partial y} = 2x\cdot\frac{-1}{2\sqrt{x^2-y}} = -\frac{x}{\sqrt{x^2-y}}, \qquad \frac{\partial N}{\partial x} = -\frac{2x}{2\sqrt{x^2-y}} = -\frac{x}{\sqrt{x^2-y}},$$

so the equation is *exact*. Integrating N with respect to y: $f = -\int(x^2-y)^{1/2}\,dy = \frac{2}{3}(x^2-y)^{3/2} + h(x)$. Then

$$\frac{\partial f}{\partial x} = \tfrac{2}{3}\cdot\tfrac{3}{2}(x^2-y)^{1/2}\cdot 2x + h'(x) = 2x\sqrt{x^2-y} + h'(x) = M$$

gives $h'(x) = 2x$, $h = x^2$. Thus $x^2 + \frac{2}{3}(x^2-y)^{3/2} = c$, or

$$3x^2 + 2(x^2-y)^{3/2} = c.$$

Answer. Exact; $3x^2 + 2(x^2-y)^{3/2} = c$.

Problem 8-15 (Simmons)

Solution. $\frac{\partial M}{\partial y} = \frac{x}{y} + x$, $\frac{\partial N}{\partial x} = \frac{y}{x} + y$. These are not identically equal (e.g. at $(x, y) = (1, 2)$ they are $\frac{3}{2}$ and 4): *not exact*.

Answer. Not exact.

Problem 8-16 (Simmons)

Solution. Using $\frac{d}{dy}\csc y = -\csc y \cot y$ and $\frac{d}{dx}\cot x = -\csc^2 x$,

$$\frac{\partial M}{\partial y} = 2ye^{y^2} + \csc y \cot y \csc^2 x, \qquad \frac{\partial N}{\partial x} = 2ye^{y^2} + \csc y \cot y \csc^2 x,$$

so the equation is *exact*. Integrating M with respect to x (and using $\int \csc^2 x\,dx = -\cot x$): $f = xe^{y^2} + \csc y \cot x + g(y)$. Then $\frac{\partial f}{\partial y} = 2xye^{y^2} - \csc y \cot y \cot x + g'(y) = N$ gives $g' = 0$. Hence

$$xe^{y^2} + \csc y \cot x = c.$$

Answer. Exact; $xe^{y^2} + \csc y \cot x = c$.

Problem 8-17 (Simmons)

Solution. $\frac{\partial M}{\partial y} = 2y \sin 2x$ and $\frac{\partial N}{\partial x} = -2y \cdot 2\cos x(-\sin x) = 4y \sin x \cos x = 2y \sin 2x$: *exact*. Integrating N with respect to y: $f = -y^2 \cos^2 x + h(x)$; then $\frac{\partial f}{\partial x} = 2y^2 \cos x \sin x + h'(x) = y^2 \sin 2x + h'(x) = M$ gives $h'(x) = 1$, $h = x$. So

$$x - y^2 \cos^2 x = c.$$

Answer. Exact; $x - y^2 \cos^2 x = c$.

Problem 8-18 (Simmons)

Solution.

$$\frac{\partial M}{\partial y} = x \cdot \left(-\tfrac{3}{2}\right)(x^2 + y^2)^{-5/2} \cdot 2y = -\frac{3xy}{(x^2 + y^2)^{5/2}} = \frac{\partial N}{\partial x}$$

(by symmetry), so the equation is *exact*. With $u = x^2 + y^2$, $\int x(x^2 + y^2)^{-3/2}\,dx = \frac{1}{2}\int u^{-3/2}\,du = -(x^2 + y^2)^{-1/2}$, so $f = -(x^2 + y^2)^{-1/2} + g(y)$, and $\frac{\partial f}{\partial y} = y(x^2 + y^2)^{-3/2} + g'(y) = N$ gives $g' = 0$. Thus $-1/\sqrt{x^2 + y^2} = \text{const}$, i.e.

$$x^2 + y^2 = c^2,$$

the circles centered at the origin.

Answer. Exact; $x^2 + y^2 = c^2$.

Problem 8-19 (Simmons)

Solution. $\frac{\partial M}{\partial y} = \dfrac{3x^2}{y} = \frac{\partial N}{\partial x}$: *exact*. Integrating M with respect to x: $f = x^3(1+\log y) + g(y)$; then $\frac{\partial f}{\partial y} = \dfrac{x^3}{y} + g'(y) = \dfrac{x^3}{y} - 2y$ gives $g'(y) = -2y$, $g = -y^2$. So

$$x^3(1+\log y) - y^2 = c.$$

Answer. Exact; $x^3(1+\log y) - y^2 = c$.

Problem 8-20 (Simmons)

Solution. Write the equation as

$$\left(\frac{y}{(x+y)^2} - 1\right) dx + \left(1 - \frac{x}{(x+y)^2}\right) dy = 0.$$

Exactness.

$$\frac{\partial M}{\partial y} = \frac{(x+y)^2 - y\cdot 2(x+y)}{(x+y)^4} = \frac{x-y}{(x+y)^3},$$
$$\frac{\partial N}{\partial x} = -\frac{(x+y)^2 - x\cdot 2(x+y)}{(x+y)^4} = -\frac{y-x}{(x+y)^3} = \frac{x-y}{(x+y)^3},$$

so the equation is exact. *First way: integrate M with respect to x.* Since $\int \frac{y\,dx}{(x+y)^2} = -\frac{y}{x+y}$,

$$f_1 = -\frac{y}{x+y} - x + g(y), \qquad \frac{\partial f_1}{\partial y} = -\frac{(x+y)-y}{(x+y)^2} + g'(y) = -\frac{x}{(x+y)^2} + g'(y).$$

Equating to N gives $g'(y) = 1$, $g = y$. So $-\dfrac{y}{x+y} - x + y = c$; multiplying by $x+y$:

$$-y + (y-x)(y+x) = c(x+y), \qquad \text{i.e.} \qquad -y + y^2 - x^2 = c(x+y).$$

Second way: integrate N with respect to y. Since $\int \frac{-x\,dy}{(x+y)^2} = \frac{x}{x+y}$,

$$f_2 = \frac{x}{x+y} + y + h(x), \qquad \frac{\partial f_2}{\partial x} = \frac{(x+y)-x}{(x+y)^2} + h'(x) = \frac{y}{(x+y)^2} + h'(x).$$

Equating to M gives $h'(x) = -1$, $h = -x$. So $\dfrac{x}{x+y} + y - x = c$; multiplying by $x+y$:

$$x + y^2 - x^2 = c(x+y).$$

Reconciliation. Since $\dfrac{x}{x+y} = 1 - \dfrac{y}{x+y}$, we have $f_2 = f_1 + 1$: the two potential functions differ only by a constant, so $f_1 = c$ and $f_2 = c+1$ describe the same family of curves. In the cleared forms: $x + y^2 - x^2 = c(x+y)$ is the same as $-y + y^2 - x^2 = (c-1)(x+y)$, since the left sides differ by $x+y$.

Answer. $-y + y^2 - x^2 = c(x+y)$, or equivalently $x + y^2 - x^2 = c(x+y)$; the potentials differ by the constant 1.

Problem 8-21 (Simmons)

Solution. The denominators factor as $x(4y^2 - x^2)$ and $y(4y^2 - x^2)$, and the coefficients split as

$$M = \frac{4y^2 - 2x^2}{x(4y^2 - x^2)} = \frac{1}{x} - \frac{x}{4y^2 - x^2}, \qquad N = \frac{8y^2 - x^2}{y(4y^2 - x^2)} = \frac{1}{y} + \frac{4y}{4y^2 - x^2}$$

(check: $\frac{(4y^2-x^2)-x^2}{x(4y^2-x^2)}$ and $\frac{(4y^2-x^2)+4y^2}{y(4y^2-x^2)}$).

(a) *Exactness.*

$$\frac{\partial M}{\partial y} = -x \cdot \frac{-8y}{(4y^2 - x^2)^2} = \frac{8xy}{(4y^2 - x^2)^2}, \qquad \frac{\partial N}{\partial x} = 4y \cdot \frac{-(-2x)}{(4y^2 - x^2)^2} = \frac{8xy}{(4y^2 - x^2)^2},$$

so the equation is exact. Integrating M with respect to x (with $\int \frac{-x\,dx}{4y^2 - x^2} = \frac{1}{2}\log|4y^2 - x^2|$):

$$f = \log|x| + \tfrac{1}{2}\log|4y^2 - x^2| + g(y), \qquad \frac{\partial f}{\partial y} = \frac{4y}{4y^2 - x^2} + g'(y).$$

Equating to N: $g'(y) = 1/y$, $g = \log|y|$. So $\log|x| + \log|y| + \frac{1}{2}\log|4y^2 - x^2| = c$; doubling and exponentiating,

$$x^2y^2(4y^2 - x^2) = c.$$

(b) *As a homogeneous equation.*

$$\frac{dy}{dx} = -\frac{M}{N} = -\frac{(4y^2 - 2x^2)\,y(4y^2 - x^2)}{x(4y^2 - x^2)(8y^2 - x^2)} = -\frac{y(4y^2 - 2x^2)}{x(8y^2 - x^2)},$$

which is homogeneous of degree 0. With $y = zx$:

$$z + x\frac{dz}{dx} = -\frac{z(4z^2 - 2)}{8z^2 - 1} \implies x\frac{dz}{dx} = \frac{-4z^3 + 2z - 8z^3 + z}{8z^2 - 1} = \frac{3z(1 - 4z^2)}{8z^2 - 1},$$

so

$$\frac{(8z^2 - 1)\,dz}{z(1 - 4z^2)} = \frac{3\,dx}{x}.$$

Partial fractions: $8z^2 - 1 = A(1 - 4z^2) + Bz(1 + 2z) + Cz(1 - 2z)$; $z = 0$ gives $A = -1$, $z = \frac{1}{2}$ gives $1 = B$, $z = -\frac{1}{2}$ gives $1 = -C$. Hence

$$\int \left(-\frac{1}{z} + \frac{1}{1 - 2z} - \frac{1}{1 + 2z}\right) dz = -\log|z| - \tfrac{1}{2}\log|1 - 2z| - \tfrac{1}{2}\log|1 + 2z| = 3\log|x| + c.$$

Multiplying by -2: $2\log|z| + \log|1 - 4z^2| + 6\log|x| = c$, i.e. $z^2(1 - 4z^2)x^6 = c$. Since $z^2(1 - 4z^2)x^6 = \frac{y^2}{x^2} \cdot \frac{x^2 - 4y^2}{x^2} \cdot x^6 = x^2y^2(x^2 - 4y^2)$, this is (changing the sign of c)

$$x^2y^2(4y^2 - x^2) = c,$$

the same as in (a).

Answer. $x^2y^2(4y^2 - x^2) = c$ (by both methods).

Problem 8-22 (Simmons)

Solution.

(a) $\frac{\partial M}{\partial y} = 2xy + nx^2$ and $\frac{\partial N}{\partial x} = 3x^2 + 2xy$; these are identically equal iff $n = 3$. Then $M = xy^2 + 3x^2y$, so $f = \frac{1}{2}x^2y^2 + x^3y + g(y)$, and $\frac{\partial f}{\partial y} = x^2y + x^3 + g'(y) = N$ gives $g' = 0$. Multiplying $f = c$ by 2:

$$x^2y^2 + 2x^3y = c.$$

(b) $\frac{\partial M}{\partial y} = e^{2xy} + 2xye^{2xy} = (1 + 2xy)e^{2xy}$ and $\frac{\partial N}{\partial x} = ne^{2xy} + 2nxye^{2xy} = n(1 + 2xy)e^{2xy}$. Since $(1 + 2xy)e^{2xy}$ is not identically zero, these agree iff $n = 1$. Then integrating $N = xe^{2xy}$ with respect to y: $f = \frac{1}{2}e^{2xy} + h(x)$, and $\frac{\partial f}{\partial x} = ye^{2xy} + h'(x) = M$ gives $h'(x) = x$, $h = \frac{1}{2}x^2$. Multiplying $f = c$ by 2:

$$x^2 + e^{2xy} = c.$$

Answer. (a) $n = 3$, $x^2y^2 + 2x^3y = c$; (b) $n = 1$, $x^2 + e^{2xy} = c$.

Section 9 Integrating Factors

Throughout, equation (2) of the section is $M(x,y)\,dx + N(x,y)\,dy = 0$, and the condition that μ be an integrating factor for (2) is equation (5):

$$\frac{1}{\mu}\left(N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y}\right) = \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}. \tag{5}$$

We also use the two tests of the text: if $(\partial M/\partial y - \partial N/\partial x)/N = g(x)$ then $\mu = e^{\int g(x)\,dx}$ is an integrating factor (formula (7)); if $(\partial M/\partial y - \partial N/\partial x)/(-M) = h(y)$ then $\mu = e^{\int h(y)\,dy}$ is one (formula (9)).

Problem 9-1 (Simmons)

Solution. Let $\mu = \mu(z)$ with $z = xy$, where $\mu(z) = e^{\int g(z)\,dz}$, so that $\mu'(z) = g(z)\mu(z)$ and $\mu > 0$. By the chain rule,

$$\frac{\partial \mu}{\partial x} = \mu'(z)\frac{\partial z}{\partial x} = y\,\mu'(z), \qquad \frac{\partial \mu}{\partial y} = \mu'(z)\frac{\partial z}{\partial y} = x\,\mu'(z).$$

Hence the left side of (5) is

$$\frac{1}{\mu}\left(N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y}\right) = \frac{\mu'(z)}{\mu(z)}\,(Ny - Mx) = g(z)\,(Ny - Mx).$$

By hypothesis $\partial M/\partial y - \partial N/\partial x = g(z)(Ny - Mx)$, so the left side of (5) equals the right side, i.e. (5) holds. Since (5) is just the expanded form of $\partial(\mu M)/\partial y = \partial(\mu N)/\partial x$, the equation $\mu M\,dx + \mu N\,dy = 0$ is exact, and μ is an integrating factor for (2).

Answer. Shown: if $(M_y - N_x)/(Ny - Mx) = g(xy)$, then $\mu = e^{\int g(z)\,dz}$, $z = xy$, satisfies (5), so $\mu(M\,dx + N\,dy)$ is exact.

Problem 9-2 (Simmons)

Solution. In each part we write the equation as $M\,dx + N\,dy = 0$, compute $M_y - N_x$ ($M_y = \partial M/\partial y$, $N_x = \partial N/\partial x$), and test whether $(M_y - N_x)/N$ depends on x only, $(M_y - N_x)/(-M)$ on y only, or $(M_y - N_x)/(Ny - Mx)$ on xy only (Problem 1). Having found μ, we solve the exact equation $\mu M\,dx + \mu N\,dy = 0$ by finding f with $f_x = \mu M$, $f_y = \mu N$; the solution is $f(x,y) = c$.

(a) Here $M = -2xy$, $N = 3x^2 - y^2$, so $M_y = -2x$, $N_x = 6x$, and $M_y - N_x = -8x$. Now

$$\frac{M_y - N_x}{-M} = \frac{-8x}{2xy} = -\frac{4}{y} = h(y), \qquad \mu = e^{\int -4\,dy/y} = e^{-4\log|y|} = y^{-4}.$$

Multiplying by y^{-4}:

$$-\frac{2x}{y^3}\,dx + \left(\frac{3x^2}{y^4} - \frac{1}{y^2}\right)dy = 0.$$

From $f_x = -2xy^{-3}$ we get $f = -x^2y^{-3} + \phi(y)$; then $f_y = 3x^2y^{-4} + \phi'(y)$ must equal $3x^2y^{-4} - y^{-2}$, so $\phi'(y) = -y^{-2}$ and $\phi(y) = y^{-1}$. Thus

$$-\frac{x^2}{y^3} + \frac{1}{y} = c_1 \quad\Longrightarrow\quad -x^2 + y^2 = c_1y^3 \quad\Longrightarrow\quad x^2 - y^2 = cy^3 \quad (c = -c_1).$$

Since μ is undefined at $y = 0$, we check that line separately: $y = 0$ ($dy = 0$) makes the equation $0 = 0$, so $y = 0$ is also a solution.

(b) $M = xy - 1$, $N = x^2 - xy$; $M_y = x$, $N_x = 2x - y$, $M_y - N_x = y - x$.

$$\frac{M_y - N_x}{N} = \frac{y - x}{x(x - y)} = -\frac{1}{x} = g(x), \qquad \mu = e^{-\log|x|} = \frac{1}{|x|},$$

and we may use $\mu = 1/x$ (a constant multiple of an integrating factor is again one). Then

$$\left(y - \frac{1}{x}\right) dx + (x - y)\, dy = 0.$$

$f_x = y - 1/x$ gives $f = xy - \log|x| + \phi(y)$, and $f_y = x + \phi'(y) = x - y$ gives $\phi = -\frac{1}{2}y^2$. So $xy - \log|x| - \frac{1}{2}y^2 = c_1$, or, multiplying by 2 and writing $2\log|x| = \log x^2$,

$$2xy - \log x^2 - y^2 = c.$$

(The line $x = 0$, where μ fails, also satisfies the equation, since there $dx = 0$ and $N = 0$.)

(c) $M = y$, $N = x + 3x^3y^4$; $M_y = 1$, $N_x = 1 + 9x^2y^4$, so $M_y - N_x = -9x^2y^4$. Neither $(M_y - N_x)/N$ nor $(M_y - N_x)/(-M)$ is a function of one variable, but

$$Ny - Mx = xy + 3x^3y^5 - xy = 3x^3y^5, \qquad \frac{M_y - N_x}{Ny - Mx} = \frac{-9x^2y^4}{3x^3y^5} = -\frac{3}{xy} = -\frac{3}{z},$$

so by Problem 1, $\mu = e^{\int -3\,dz/z} = z^{-3} = (xy)^{-3}$. Multiplying the equation $(x\,dy + y\,dx) + 3x^3y^4\,dy = 0$ by $(xy)^{-3}$ and using $x\,dy + y\,dx = d(xy)$:

$$\frac{d(xy)}{(xy)^3} + 3y\,dy = 0 \quad \Longrightarrow \quad -\frac{1}{2(xy)^2} + \frac{3}{2}y^2 = c_1.$$

Multiplying by $2x^2y^2$: $-1 + 3x^2y^4 = 2c_1x^2y^2$, i.e.

$$3x^2y^4 = 1 + cx^2y^2.$$

The lines $x = 0$ and $y = 0$ (where μ is undefined) also satisfy the equation.

(d) $M = e^x$, $N = e^x \cot y + 2y \csc y$; $M_y = 0$, $N_x = e^x \cot y$.

$$\frac{M_y - N_x}{-M} = \frac{-e^x \cot y}{-e^x} = \cot y = h(y), \qquad \mu = e^{\int \cot y\,dy} = e^{\log|\sin y|} = |\sin y|;$$

we take $\mu = \sin y$. Then

$$e^x \sin y\,dx + (e^x \cos y + 2y)\,dy = 0,$$

since $\sin y \cot y = \cos y$ and $\sin y \csc y = 1$. From $f_x = e^x \sin y$, $f = e^x \sin y + \phi(y)$; $f_y = e^x \cos y + \phi' = e^x \cos y + 2y$ gives $\phi = y^2$. Hence

$$e^x \sin y + y^2 = c.$$

(e) $M = (x + 2)\sin y$, $N = x \cos y$; $M_y = (x + 2)\cos y$, $N_x = \cos y$, $M_y - N_x = (x + 1)\cos y$.

$$\frac{M_y - N_x}{N} = \frac{x + 1}{x} = 1 + \frac{1}{x} = g(x), \qquad \mu = e^{x + \log|x|} = |x|e^x;$$

we take $\mu = xe^x$. The equation becomes $x(x+2)e^x \sin y\,dx + x^2e^x \cos y\,dy = 0$. From $f_y = x^2e^x \cos y$, $f = x^2e^x \sin y + \phi(x)$, and $f_x = (2x+x^2)e^x \sin y + \phi'(x)$, which equals $x(x+2)e^x \sin y$ when $\phi' = 0$. Hence

$$x^2e^x \sin y = c.$$

(f) $M = y$, $N = x - 2x^2y^3$; $M_y = 1$, $N_x = 1 - 4xy^3$, $M_y - N_x = 4xy^3$. Also $Ny - Mx = xy - 2x^2y^4 - xy = -2x^2y^4$, so

$$\frac{M_y - N_x}{Ny - Mx} = \frac{4xy^3}{-2x^2y^4} = -\frac{2}{xy} = -\frac{2}{z}, \qquad \mu = e^{\int -2\,dz/z} = z^{-2} = (xy)^{-2}$$

by Problem 1. Writing the equation as $(y\,dx + x\,dy) - 2x^2y^3\,dy = 0$ and multiplying by $(xy)^{-2}$:

$$\frac{d(xy)}{(xy)^2} - 2y\,dy = 0 \quad \Longrightarrow \quad -\frac{1}{xy} - y^2 = c_1.$$

Multiplying by $-xy$: $1 + xy^3 = -c_1xy$, i.e.

$$1 + xy^3 = cxy.$$

The lines $x = 0$ and $y = 0$ are also solutions.

(g) $M = x + 3y^2$, $N = 2xy$; $M_y = 6y$, $N_x = 2y$.

$$\frac{M_y - N_x}{N} = \frac{4y}{2xy} = \frac{2}{x}, \qquad \mu = e^{2\log|x|} = x^2.$$

The equation becomes $(x^3 + 3x^2y^2)\,dx + 2x^3y\,dy = 0$. From $f_y = 2x^3y$, $f = x^3y^2 + \phi(x)$; $f_x = 3x^2y^2 + \phi' = x^3 + 3x^2y^2$ gives $\phi = \frac{1}{4}x^4$. So $\frac{1}{4}x^4 + x^3y^2 = c_1$, i.e.

$$x^4 + 4x^3y^2 = c.$$

(h) $M = y$, $N = 2x - ye^y$; $M_y = 1$, $N_x = 2$.

$$\frac{M_y - N_x}{-M} = \frac{-1}{-y} = \frac{1}{y}, \qquad \mu = e^{\log|y|} \to \mu = y.$$

The equation becomes $y^2\,dx + (2xy - y^2e^y)\,dy = 0$. From $f_x = y^2$, $f = xy^2 + \phi(y)$; $f_y = 2xy + \phi'(y) = 2xy - y^2e^y$, so $\phi'(y) = -y^2e^y$. Integrating by parts twice,

$$\int y^2e^y\,dy = y^2e^y - 2\int ye^y\,dy = y^2e^y - 2(ye^y - e^y) = e^y(y^2 - 2y + 2),$$

so $\phi = -e^y(y^2 - 2y + 2)$ and

$$xy^2 - e^y(y^2 - 2y + 2) = c.$$

(i) $M = y\log y - 2xy$, $N = x + y$ (with $y > 0$); $M_y = \log y + 1 - 2x$, $N_x = 1$, so $M_y - N_x = \log y - 2x$ and

$$\frac{M_y - N_x}{-M} = \frac{\log y - 2x}{-y(\log y - 2x)} = -\frac{1}{y}, \qquad \mu = e^{-\log y} = \frac{1}{y}.$$

The equation becomes $(\log y - 2x)\,dx + \left(\frac{x}{y} + 1\right)\,dy = 0$. From $f_x = \log y - 2x$, $f = x\log y - x^2 + \phi(y)$; $f_y = x/y + \phi' = x/y + 1$ gives $\phi = y$. Hence

$$x\log y - x^2 + y = c.$$

(j) $M = y^2 + xy + 1$, $N = x^2 + xy + 1$; $M_y = 2y + x$, $N_x = 2x + y$, so $M_y - N_x = y - x$. Also

$$Ny - Mx = (x^2y + xy^2 + y) - (xy^2 + x^2y + x) = y - x, \qquad \frac{M_y - N_x}{Ny - Mx} = 1 = g(z),$$

so by Problem 1, $\mu = e^{\int 1\,dz} = e^z = e^{xy}$. The equation $e^{xy}(y^2 + xy + 1)\,dx + e^{xy}(x^2 + xy + 1)\,dy = 0$ is exact; we try $f = e^{xy}(x + y)$:

$$f_x = ye^{xy}(x + y) + e^{xy} = e^{xy}(xy + y^2 + 1), \qquad f_y = xe^{xy}(x + y) + e^{xy} = e^{xy}(x^2 + xy + 1),$$

exactly μM and μN. (Systematically: $f_y = e^{xy}(x^2 + xy + 1)$ integrates in y to $e^{xy}(x + y) + \phi(x)$, and comparing f_x forces $\phi' = 0$.) Hence

$$e^{xy}(x + y) = c.$$

(k) $M = x^3 + xy^3$, $N = 3y^2$; $M_y = 3xy^2$, $N_x = 0$.

$$\frac{M_y - N_x}{N} = \frac{3xy^2}{3y^2} = x, \qquad \mu = e^{x^2/2}.$$

The equation becomes $e^{x^2/2}(x^3 + xy^3)\,dx + 3y^2e^{x^2/2}\,dy = 0$. From f_y, $f = y^3e^{x^2/2} + \phi(x)$; $f_x = xy^3e^{x^2/2} + \phi'(x)$, so $\phi'(x) = x^3e^{x^2/2}$. With $u = x^2/2$, $du = x\,dx$, $x^2 = 2u$:

$$\int x^3e^{x^2/2}\,dx = \int 2ue^u\,du = 2(u - 1)e^u = (x^2 - 2)e^{x^2/2}.$$

Hence $f = e^{x^2/2}(y^3 + x^2 - 2)$ and

$$e^{x^2/2}(y^3 + x^2 - 2) = c.$$

Answer. (a) $\mu = y^{-4}$, $x^2 - y^2 = cy^3$ (and $y = 0$); (b) $\mu = 1/x$, $2xy - \log x^2 - y^2 = c$; (c) $\mu = (xy)^{-3}$, $3x^2y^4 = 1 + cx^2y^2$; (d) $\mu = \sin y$, $e^x \sin y + y^2 = c$; (e) $\mu = xe^x$, $x^2e^x \sin y = c$; (f) $\mu = (xy)^{-2}$, $1 + xy^3 = cxy$; (g) $\mu = x^2$, $x^4 + 4x^3y^2 = c$; (h) $\mu = y$, $xy^2 - e^y(y^2 - 2y + 2) = c$; (i) $\mu = 1/y$, $x \log y - x^2 + y = c$; (j) $\mu = e^{xy}$, $e^{xy}(x + y) = c$; (k) $\mu = e^{x^2/2}$, $e^{x^2/2}(y^3 + x^2 - 2) = c$.

Problem 9-3 (Simmons)

Solution. Suppose $\mu = \mu(z)$ with $z = x + y$. Then $\partial\mu/\partial x = \mu'(z)$ and $\partial\mu/\partial y = \mu'(z)$, so condition (5) becomes

$$\frac{\mu'(z)}{\mu(z)}(N - M) = \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}, \qquad \text{i.e.} \qquad \frac{\mu'(z)}{\mu(z)} = \frac{\partial M/\partial y - \partial N/\partial x}{N - M}.$$

The left side is a function of $z = x + y$ only; hence such a μ can exist only if the right side is a function of $x + y$ alone. Conversely, if

$$\frac{\partial M/\partial y - \partial N/\partial x}{N - M} = g(z), \qquad z = x + y,$$

then $\mu(z) = e^{\int g(z)\,dz}$ satisfies $\mu'/\mu = g$, so by the computation above it satisfies (5) and is an integrating factor for (2).

Answer. Exactly when $(\partial M/\partial y - \partial N/\partial x)/(N - M)$ is a function $g(z)$ of $z = x + y$; then $\mu = e^{\int g(z)\,dz}$.

Problem 9-4 (Simmons)

Solution. The formulas of the text are

$$d\left(\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{x^2} \quad (11), \qquad d\left(\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{y^2} \quad (12),$$

$$d(xy) = x\,dy + y\,dx \quad (13), \qquad d(x^2 + y^2) = 2(x\,dx + y\,dy) \quad (14),$$

$$d\left(\tan^{-1}\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{x^2 + y^2} \quad (15), \qquad d\left(\log\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{xy} \quad (16).$$

Whenever we divide by an expression that can vanish we note the corresponding lines separately.

(a) Dividing $x\,dy - y\,dx = (1 + y^2)\,dy$ by y^2 and using (12):

$$\frac{x\,dy - y\,dx}{y^2} = -d\left(\frac{x}{y}\right) = \left(\frac{1}{y^2} + 1\right)dy.$$

Integrating, $-\dfrac{x}{y} = -\dfrac{1}{y} + y + c$. (The line $y = 0$ also satisfies the equation.)

(b) Dividing by xy and using (16): $d\left(\log\dfrac{x}{y}\right) = \dfrac{y\,dx - x\,dy}{xy} = y^2\,dy$, so

$$\log\frac{x}{y} = \frac{1}{3}y^3 + c.$$

(The lines $x = 0$ and $y = 0$ also satisfy the equation.)

(c) Rearranged, $x\,dy - y\,dx = (x^5 + x^3y^2)\,dx = x^3(x^2 + y^2)\,dx$. Dividing by $x^2 + y^2$ and using (15),

$$\frac{x\,dy - y\,dx}{x^2 + y^2} = -d\left(\tan^{-1}\frac{x}{y}\right) = x^3\,dx, \qquad \text{so} \qquad \tan^{-1}\frac{x}{y} = -\frac{1}{4}x^4 + c.$$

(d) Expanding, $y\,dy + x\,dy = y\,dx - x\,dx$, i.e. $x\,dx + y\,dy = y\,dx - x\,dy$. Dividing by $x^2 + y^2$ and using (14) and (15):

$$\frac{d(x^2 + y^2)}{2(x^2 + y^2)} = d\left(\tan^{-1}\frac{x}{y}\right) \quad \Longrightarrow \quad \tfrac{1}{2}\log(x^2 + y^2) = \tan^{-1}\frac{x}{y} + c,$$

that is, $\log\sqrt{x^2 + y^2} = \tan^{-1}(x/y) + c$.

(e) Rearranged, $x\,dy - y\,dx = (x^2 + 9y^2)\,dx$. The analogue of (15) for $3y/x$ is

$$d\left(\tan^{-1}\frac{3y}{x}\right) = \frac{1}{1 + 9y^2/x^2}\cdot\frac{3(x\,dy - y\,dx)}{x^2} = \frac{3(x\,dy - y\,dx)}{x^2 + 9y^2}.$$

Dividing the equation by $x^2 + 9y^2$ therefore gives $\frac{1}{3}d\left(\tan^{-1}(3y/x)\right) = dx$, whence

$$\tan^{-1}\frac{3y}{x} = 3x + c.$$

(f) Rearranged, $y^2\,dx + (x\,dy - y\,dx) = 0$. Dividing by y^2 and using (12):

$$dx - d\left(\frac{x}{y}\right) = 0 \quad\Longrightarrow\quad x - \frac{x}{y} = c_1 \quad\Longrightarrow\quad \frac{x}{y} = x - c_1 \quad\Longrightarrow\quad y = \frac{x}{x+c}$$

with $c = -c_1$. (The line $y = 0$ is also a solution.)

(g) Dividing by x^2 and using (11): $d\left(\frac{y}{x}\right) = \left(2 - \frac{3}{x^2}\right)\,dx$, so $\frac{y}{x} = 2x + \frac{3}{x} + c$, i.e.

$$y = 2x^2 + 3 + cx.$$

(h) By (13) the equation reads $d(xy) = \sqrt{xy}\,dy$. In the region $xy > 0$ we divide by $\sqrt{xy}$:

$$\frac{d(xy)}{\sqrt{xy}} = dy \quad\Longrightarrow\quad 2\sqrt{xy} = y + c.$$

(The lines $x = 0$ and $y = 0$, where $\sqrt{xy} = 0$, also satisfy the equation.)

(i) Regrouping, $(y\,dx + x\,dy) - xy^2\,dx + x^2y^2\,dy = 0$, i.e. $d(xy) + x^2y^2\left(dy - \frac{dx}{x}\right) = 0$. Dividing by x^2y^2:

$$\frac{d(xy)}{(xy)^2} + dy - \frac{dx}{x} = 0 \quad\Longrightarrow\quad -\frac{1}{xy} + y - \log x = c,$$

i.e. $-\frac{1}{xy} - \log x + y = c$. (The lines $x = 0$, $y = 0$ also satisfy the equation.)

(j) Dividing by y^2 and using (12) and (13):

$$\frac{x\,dy - y\,dx}{y^2} = x^2y^2(x\,dy + y\,dx) \quad\Longleftrightarrow\quad -d\left(\frac{x}{y}\right) = (xy)^2\,d(xy).$$

Integrating, $-\frac{x}{y} = \frac{(xy)^3}{3} + c_1$; multiplying by $3y$ gives $-3x = x^3y^4 + 3c_1y$, i.e.

$$3x + x^3y^4 + cy = 0 \qquad (c = 3c_1).$$

(The line $y = 0$ is also a solution.)

(k) By (13), $d(xy) + x^2y^5\,dy = 0$. Dividing by x^2y^2:

$$\frac{d(xy)}{(xy)^2} + y^3\,dy = 0 \quad\Longrightarrow\quad -\frac{1}{xy} + \frac{y^4}{4} = c_1.$$

Multiplying by $4xy$: $-4 + xy^5 = 4c_1xy$, i.e. $x(y^5 + cy) = 4$ with $c = -4c_1$. (The lines $x = 0$, $y = 0$ also satisfy the equation.)

(l) Rearranged, $2xy^2\,dx - (y\,dx - x\,dy) = 0$. Dividing by y^2 and using (12):

$$2x\,dx - d\left(\frac{x}{y}\right) = 0 \quad\Longrightarrow\quad x^2 - \frac{x}{y} = c_1 \quad\Longrightarrow\quad y = \frac{x}{x^2 - c_1} = \frac{x}{x^2 + c}.$$

(The line $y = 0$ is also a solution.)

(m) Multiplying by x and using (13): $x\,dy + y\,dx = d(xy) = x\sin x\,dx$. Integrating by parts, $\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x$, so

$$xy = \sin x - x\cos x + c, \qquad \text{i.e.} \qquad xy + x\cos x = \sin x + c.$$

Answer. (a) $-x/y = -1/y + y + c$; (b) $\log(x/y) = \frac{1}{3}y^3 + c$; (c) $\tan^{-1}(x/y) = -\frac{1}{4}x^4 + c$; (d) $\log\sqrt{x^2+y^2} = \tan^{-1}(x/y) + c$; (e) $\tan^{-1}(3y/x) = 3x + c$; (f) $y = x/(x+c)$; (g) $y = 2x^2 + 3 + cx$; (h) $2\sqrt{xy} = y + c$; (i) $-1/(xy) - \log x + y = c$; (j) $3x + x^3y^4 + cy = 0$; (k) $x(y^5 + cy) = 4$; (l) $y = x/(x^2 + c)$; (m) $xy + x\cos x = \sin x + c$.

Problem 9-5 (Simmons)

Solution. The argument of the tangent is y/x^2, so we choose $n = 2$: $y = x^2 z$, which gives $\frac{dy}{dx} = 2xz + x^2\frac{dz}{dx}$. The right side becomes

$$\frac{2x^2 z}{x} + \frac{x^3}{x^2 z} + x\tan z = 2xz + \frac{x}{z} + x\tan z.$$

Equating and cancelling $2xz$:

$$x^2\frac{dz}{dx} = \frac{x}{z} + x\tan z \quad\Longrightarrow\quad x\frac{dz}{dx} = \frac{1}{z} + \frac{\sin z}{\cos z} = \frac{\cos z + z\sin z}{z\cos z}.$$

This is separable:

$$\frac{z\cos z}{\cos z + z\sin z}\,\mathrm{d}z = \frac{\mathrm{d}x}{x}.$$

Since $d(\cos z + z\sin z) = (-\sin z + \sin z + z\cos z)\,\mathrm{d}z = z\cos z\,\mathrm{d}z$, the left side is $d\log|\cos z + z\sin z|$, and integration gives

$$\log|\cos z + z\sin z| = \log|x| + \text{const}, \qquad \cos z + z\sin z = cx.$$

(The value $c = 0$ gives the constant solutions $z = z_0$ with $\cos z_0 + z_0\sin z_0 = 0$, which were divided out in the separation; they are thus included.) Substituting $z = y/x^2$ and multiplying by x^2:

$$x^2\cos\frac{y}{x^2} + y\sin\frac{y}{x^2} = cx^3.$$

Answer. With $n = 2$: $x^2\cos(y/x^2) + y\sin(y/x^2) = cx^3$.

Problem 9-6 (Simmons)

Solution. Place the light source at the pole O and let $P = (r,\theta)$ be a point of the curve, as in Figure 19. Let ψ be the angle from the radius vector OP (extended beyond P) to the tangent line at P. Writing the curve as $x = r\cos\theta$, $y = r\sin\theta$ with $r = r(\theta)$, its tangent vector is

$$\left(\frac{dx}{d\theta}, \frac{dy}{d\theta}\right) = \frac{dr}{d\theta}(\cos\theta, \sin\theta) + r(-\sin\theta, \cos\theta),$$

whose component along the radial unit vector $(\cos\theta, \sin\theta)$ is $dr/d\theta$ and whose component along the perpendicular unit vector $(-\sin\theta, \cos\theta)$ is r. Hence

$$\tan\psi = \frac{r}{dr/d\theta} = r\frac{d\theta}{dr}.$$

The angle ψ. In Example 2 we found $\alpha = \beta$ (law of reflection) and $\theta = \alpha + \phi = 2\beta$ (geometry, with $\phi = \beta$), so $\alpha = \theta/2$. The angle α between the tangent and the segment PO and the angle

ψ between the tangent and the extension of OP beyond P are supplementary, which is the hint: $\psi = \pi - \alpha = \pi - \theta/2$. (Equivalently: the tangent line makes the angle $\phi = \theta + \psi$ with the x-axis; reflection in a line of inclination ϕ turns the ray direction θ into direction $2\phi - \theta$, and requiring this to be $0 \pmod{2\pi}$ gives $\psi = -\theta/2 \pmod{\pi}$, the same condition.)

The differential equation. Therefore

$$r\frac{d\theta}{dr} = \tan\left(\pi - \frac{\theta}{2}\right) = -\tan\frac{\theta}{2}, \qquad \text{so} \qquad \frac{dr}{r} = -\cot\frac{\theta}{2}\,d\theta.$$

Since $\int \cot(\theta/2)\,d\theta = 2\log\sin(\theta/2)$ (for $0 < \theta < 2\pi$),

$$\log r = -2\log\sin\frac{\theta}{2} + \text{const}, \qquad r = \frac{c_1}{\sin^2(\theta/2)} = \frac{2c_1}{1-\cos\theta},$$

using $\sin^2(\theta/2) = \frac{1}{2}(1-\cos\theta)$. With $c = 2c_1 > 0$,

$$r = \frac{c}{1-\cos\theta}.$$

This is the polar equation of a parabola with focus at the origin and axis along the x-axis (eccentricity 1); its vertex A ($\theta = \pi$) is at $(-c/2, 0)$, as in the figure. Indeed, $r - r\cos\theta = c$ means $\sqrt{x^2+y^2} = x + c$, i.e. $y^2 = 2cx + c^2$, the result of Example 2.

Answer. $r = c/(1-\cos\theta)$, a parabola with focus at the origin (equivalently $y^2 = 2cx + c^2$).

Section 10 Linear Equations

Throughout, equation (1) is the linear equation in standard form $\dfrac{dy}{dx} + P(x)y = Q(x)$, whose integrating factor is $e^{\int P\,dx}$; equation (3) is $\dfrac{d}{dx}\left(e^{\int P\,dx}y\right) = Qe^{\int P\,dx}$, and (4) is the general solution $y = e^{-\int P\,dx}\left(\int Qe^{\int P\,dx}\,dx + c\right)$. Following the text, in each problem we *multiply by* $e^{\int P\,dx}$ *and integrate*, working on an interval on which the coefficients are continuous.

Problem 10-1 (Simmons)

Solution. Multiplying (1) by dx and rearranging gives

$$\big(P(x)y - Q(x)\big)\,dx + 1\cdot\,dy = 0, \qquad M = P(x)y - Q(x), \quad N = 1.$$

Then $\partial M/\partial y = P(x)$ and $\partial N/\partial x = 0$, so

$$\frac{\partial M/\partial y - \partial N/\partial x}{N} = P(x),$$

a function of x alone. By formula (7) of Section 9,

$$\mu = e^{\int P(x)\,dx}$$

is an integrating factor depending only on x. Note $\mu' = P\mu$ and $\mu > 0$.
Check of exactness: $\partial(\mu M)/\partial y = \mu P$ and $\partial(\mu N)/\partial x = \mu' = \mu P$.
Now we find f with $f_x = \mu M = \mu P y - \mu Q$ and $f_y = \mu N = \mu$. From $f_y = \mu(x)$,

$$f = \mu(x)\,y + g(x).$$

Then $f_x = \mu' y + g'(x) = \mu P y + g'(x)$, and comparing with $\mu P y - \mu Q$ gives $g'(x) = -\mu Q$, i.e. $g(x) = -\int Q\mu\,dx$. The general solution $f = c$ is therefore

$$e^{\int P\,dx}y - \int Qe^{\int P\,dx}\,dx = c, \qquad \text{i.e.} \qquad y = e^{-\int P\,dx}\left(\int Qe^{\int P\,dx}\,dx + c\right),$$

which is (4).

Answer. $\mu = e^{\int P\,dx}$; solving the exact equation gives $y = e^{-\int P\,dx}\left(\int Qe^{\int P\,dx}\,dx + c\right)$, formula (4).

Problem 10-2 (Simmons)

Solution.

(a) For $x \neq 0$, dividing by x: $y' - \dfrac{3}{x}y = x^3$, so $P = -3/x$, $\int P\,dx = -3\log|x|$, and we may take $e^{\int P\,dx} = x^{-3}$ (the sign of $|x|^{-3}$ versus x^{-3} is a constant factor, which does not matter). Multiplying by x^{-3}:

$$\frac{d}{dx}\left(\frac{y}{x^3}\right) = 1 \quad\Longrightarrow\quad \frac{y}{x^3} = x + c \quad\Longrightarrow\quad y = x^4 + cx^3.$$

(b) $P = 1$, integrating factor e^x:

$$\frac{d}{dx}(e^x y) = \frac{e^x}{1 + e^{2x}}.$$

With $u = e^x$, $du = e^x\,dx$: $\displaystyle\int \frac{e^x\,dx}{1+e^{2x}} = \int \frac{du}{1+u^2} = \tan^{-1} u = \tan^{-1} e^x$. Hence $e^x y = \tan^{-1} e^x + c$, i.e.

$$y = e^{-x}\tan^{-1} e^x + ce^{-x}.$$

(c) Dividing by $(1+x^2)\,dx$: $\displaystyle y' + \frac{2x}{1+x^2}y = \frac{\cot x}{1+x^2}$. Here $\int P\,dx = \log(1+x^2)$, so the integrating factor is $1+x^2$, and multiplying back we recover the original form, whose left side is the exact derivative $\dfrac{d}{dx}\big((1+x^2)y\big)$:

$$\frac{d}{dx}\big((1+x^2)y\big) = \cot x \quad\Longrightarrow\quad (1+x^2)y = \log|\sin x| + c,$$

since $\int \cot x\,dx = \log|\sin x|$. Thus

$$y = (1+x^2)^{-1}\log|\sin x| + c(1+x^2)^{-1}$$

(on an interval where $\sin x > 0$ this is $\log(\sin x)$, as in the book).

(d) $P = 1$, integrating factor e^x:

$$\frac{d}{dx}(e^x y) = 2x + x^2 e^x.$$

Integrating by parts twice, $\int x^2 e^x\,dx = x^2e^x - 2\int xe^x\,dx = x^2e^x - 2xe^x + 2e^x$. Hence $e^x y = x^2 + (x^2 - 2x + 2)e^x + c$, and

$$y = x^2e^{-x} + x^2 - 2x + 2 + ce^{-x}.$$

(e) $P = \cot x$, $\int P\,dx = \log|\sin x|$, integrating factor $\sin x$:

$$\frac{d}{dx}(y\sin x) = 2x\csc x\sin x = 2x \quad\Longrightarrow\quad y\sin x = x^2 + c \quad\Longrightarrow\quad y = x^2\csc x + c\csc x.$$

(f) Dividing by $x\,dx$ $(x \neq 0)$: $\displaystyle y' = \frac{2y}{x} - x^2$, i.e. $\displaystyle y' - \frac{2}{x}y = -x^2$. Integrating factor $e^{-2\log|x|} = x^{-2}$:

$$\frac{d}{dx}\left(\frac{y}{x^2}\right) = -1 \quad\Longrightarrow\quad \frac{y}{x^2} = -x + c \quad\Longrightarrow\quad y = -x^3 + cx^2.$$

(g) Dividing by x: $\displaystyle y' + \left(\frac{1}{x} + \cot x\right)y = 1$. Then $\int P\,dx = \log|x| + \log|\sin x|$, integrating factor $x\sin x$:

$$\frac{d}{dx}(xy\sin x) = x\sin x.$$

By parts, $\int x\sin x\,dx = -x\cos x + \sin x$. Hence

$$xy\sin x = \sin x - x\cos x + c.$$

(h) $P = -2x$, integrating factor e^{-x^2}:

$$\frac{d}{dx}\left(e^{-x^2}y\right) = 6x \quad\Longrightarrow\quad e^{-x^2}y = 3x^2 + c \quad\Longrightarrow\quad y = 3x^2e^{x^2} + ce^{x^2}.$$

(i) For $x > 0$, $x \neq 1$: $y' + \dfrac{1}{x \log x} y = \dfrac{3x^2}{\log x}$. With $u = \log x$, $du = dx/x$, $\displaystyle\int \frac{dx}{x \log x} = \log|\log x|$, so the integrating factor is $\log x$ (up to sign):

$$\frac{d}{dx}(y \log x) = 3x^2 \quad \Longrightarrow \quad y \log x = x^3 + c \quad \Longrightarrow \quad y = \frac{x^3 + c}{\log x}.$$

(j) Dividing by $x^2\,dx$: $y' + \dfrac{1 - 2x}{x^2} y = 1$. Now

$$\int \frac{1 - 2x}{x^2}\,dx = \int \left(\frac{1}{x^2} - \frac{2}{x}\right) dx = -\frac{1}{x} - 2\log|x|, \qquad e^{\int P\,dx} = x^{-2} e^{-1/x}.$$

Multiplying through:

$$\frac{d}{dx}\left(x^{-2} e^{-1/x} y\right) = x^{-2} e^{-1/x}.$$

With $u = -1/x$, $du = x^{-2}\,dx$: $\int x^{-2} e^{-1/x}\,dx = e^{-1/x}$. So $x^{-2} e^{-1/x} y = e^{-1/x} + c$, i.e.

$$y = x^2 + c x^2 e^{1/x} = x^2\left(1 + c e^{1/x}\right).$$

Answer. (a) $y = x^4 + cx^3$; (b) $y = e^{-x} \tan^{-1} e^x + c e^{-x}$; (c) $y = (1 + x^2)^{-1} \log(\sin x) + c(1 + x^2)^{-1}$; (d) $y = x^2 e^{-x} + x^2 - 2x + 2 + c e^{-x}$; (e) $y = x^2 \csc x + c \csc x$; (f) $y = -x^3 + cx^2$; (g) $xy \sin x = \sin x - x \cos x + c$; (h) $y = 3x^2 e^{x^2} + c e^{x^2}$; (i) $y = (x^3 + c)/\log x$; (j) $y = x^2(1 + c e^{1/x})$.

Problem 10-3 (Simmons)

Solution. *Reduction.* Let $n \neq 0, 1$ and $z = y^{1-n}$. Then $\dfrac{dz}{dx} = (1 - n) y^{-n} \dfrac{dy}{dx}$. Multiplying Bernoulli's equation by $(1 - n)y^{-n}$ (where $y \neq 0$) gives

$$(1 - n) y^{-n} \frac{dy}{dx} + (1 - n) P(x) y^{1-n} = (1 - n) Q(x), \qquad \text{i.e.} \qquad \frac{dz}{dx} + (1 - n) P(x)\, z = (1 - n) Q(x),$$

which is linear in z. (When $n > 0$, $y \equiv 0$ is also a solution, lost in this division.)

(a) Dividing by x: $y' + \dfrac{1}{x} y = x^3 y^3$, so $n = 3$ and $z = y^{-2}$. The linear equation is

$$z' - \frac{2}{x} z = -2x^3.$$

Integrating factor x^{-2}: $\dfrac{d}{dx}\left(\dfrac{z}{x^2}\right) = -2x$, so $\dfrac{z}{x^2} = -x^2 + c$, $z = -x^4 + cx^2$:

$$\frac{1}{y^2} = -x^4 + cx^2 \qquad (\text{and also } y = 0).$$

(b) Dividing by xy^2: $y' + \dfrac{1}{x} y = (\cos x)\, y^{-2}$, so $n = -2$ and $z = y^3$. The linear equation is

$$z' + \frac{3}{x} z = 3 \cos x.$$

Integrating factor $e^{3\log|x|} \to x^3$: $\dfrac{d}{dx}(x^3 z) = 3x^3\cos x$. Integrating by parts repeatedly,

$$\begin{aligned}\int x^3\cos x\,dx &= x^3\sin x - 3\int x^2\sin x\,dx = x^3\sin x - 3\left(-x^2\cos x + 2\int x\cos x\,dx\right)\\ &= x^3\sin x + 3x^2\cos x - 6(x\sin x + \cos x).\end{aligned}$$

Hence

$$x^3y^3 = 3x^3\sin x + 9x^2\cos x - 18x\sin x - 18\cos x + c,$$
$$y^3 = 3\sin x + 9x^{-1}\cos x - 18x^{-2}\sin x - 18x^{-3}\cos x + cx^{-3}.$$

(c) Dividing by $x\,dx$: $y' + \dfrac{1}{x}y = y^2$, so $n = 2$ and $z = y^{-1}$. The linear equation is

$$z' - \frac{1}{x}z = -1.$$

Integrating factor x^{-1}: $\dfrac{d}{dx}\left(\dfrac{z}{x}\right) = -\dfrac{1}{x}$, so $\dfrac{z}{x} = -\log x + c$ (for $x > 0$), i.e. $\dfrac{1}{y} = cx - x\log x$. Multiplying by y:

$$1 = cxy - xy\log x, \qquad \text{i.e.} \qquad 1 + xy\log x = cxy \qquad (\text{and also } y = 0).$$

Answer. Shown: $z = y^{1-n}$ gives $z' + (1-n)Pz = (1-n)Q$. (a) $1/y^2 = -x^4 + cx^2$; (b) $y^3 = 3\sin x + 9x^{-1}\cos x - 18x^{-2}\sin x - 18x^{-3}\cos x + cx^{-3}$; (c) $1 + xy\log x = cxy$.

Problem 10-4 (Simmons)

Solution. In each case we regard x as a function of y, using $dx/dy = 1/y'$.

(a) Since $y' = dy/dx$, the equation is $(e^y - 2xy)\,dy = y^2\,dx$, i.e.

$$\frac{dx}{dy} = \frac{e^y - 2xy}{y^2} \quad\Longrightarrow\quad \frac{dx}{dy} + \frac{2}{y}x = \frac{e^y}{y^2},$$

linear in x. Integrating factor $e^{2\log|y|} = y^2$: $\dfrac{d}{dy}(y^2x) = e^y$, so

$$xy^2 = e^y + c.$$

(The line $y = 0$ is also a solution of the original equation: $y' = 0$ gives $0 = 0$.)

(b) The equation is $y = y'(x + y^2e^y)$, i.e. $\dfrac{dx}{dy} = \dfrac{x + y^2e^y}{y}$, or

$$\frac{dx}{dy} - \frac{1}{y}x = ye^y.$$

Integrating factor $1/y$: $\dfrac{d}{dy}\left(\dfrac{x}{y}\right) = e^y$, so $\dfrac{x}{y} = e^y + c$, i.e.

$$x = ye^y + cy.$$

(Again $y = 0$ is also a solution.)

(c) Collecting the y' terms, $y'\big(x - x^3(y-1)\big) = -2$, so

$$\frac{dx}{dy} = \frac{x - x^3(y-1)}{-2} \quad\Longrightarrow\quad \frac{dx}{dy} + \frac{1}{2}x = \frac{y-1}{2}x^3.$$

This is a Bernoulli equation in x with $n = 3$; by Problem 3 put $z = x^{-2}$, so that $dz/dy = -2x^{-3}\,dx/dy$. Multiplying the equation by $-2x^{-3}$:

$$\frac{dz}{dy} - z = 1 - y.$$

Integrating factor e^{-y}: $\dfrac{d}{dy}\left(e^{-y}z\right) = (1-y)e^{-y}$. By parts, $\int (1-y)e^{-y}\,dy = -(1-y)e^{-y} - \int e^{-y}\,dy = -(1-y)e^{-y} + e^{-y} = ye^{-y}$. Hence $e^{-y}z = ye^{-y} + c$, $z = y + ce^{y}$, and since $z = 1/x^2$,

$$1 = x^2(y + ce^{y}).$$

($x = 0$ is not a solution, since it would give $2 = 0$.)

(d) The equation is already written with x as dependent variable. For $f(y) \neq 0$, dividing by $f(y)^2$:

$$\frac{dx}{dy} + 3\frac{f'(y)}{f(y)}x = \frac{f'(y)}{f(y)^2}.$$

Since $\int 3f'/f\,dy = 3\log|f|$, the integrating factor is $f(y)^3$:

$$\frac{d}{dy}\left(f(y)^3 x\right) = f(y)f'(y) \quad\Longrightarrow\quad f(y)^3 x = \tfrac{1}{2}f(y)^2 + c_1,$$

i.e.

$$2xf(y)^3 = f(y)^2 + c.$$

Answer. (a) $xy^2 = e^y + c$; (b) $x = ye^y + cy$; (c) $1 = x^2(y + ce^y)$; (d) $2xf(y)^3 = f(y)^2 + c$.

Problem 10-5 (Simmons)

Solution. As in Section 3, we find the differential equation of the family by eliminating c, replace y' by $-1/y'$, and solve the new equation.

(a) Differentiating, $y' = 1 - ce^{-x}$; since $ce^{-x} = y - x$, the family satisfies $y' = 1 + x - y$. The orthogonal trajectories satisfy

$$\frac{dy}{dx} = -\frac{1}{1+x-y} \quad\Longleftrightarrow\quad \frac{dx}{dy} = -(1+x-y) \quad\Longleftrightarrow\quad \frac{dx}{dy} + x = y - 1,$$

which is linear in x (as in Problem 4). Integrating factor e^{y}: $\dfrac{d}{dy}(e^{y}x) = (y-1)e^{y}$. By parts, $\int (y-1)e^{y}\,dy = (y-1)e^{y} - e^{y} = (y-2)e^{y}$. Hence $e^{y}x = (y-2)e^{y} + c$:

$$x = y - 2 + ce^{-y}.$$

(b) Differentiating, $2yy' = ce^{x} + 1$; since $ce^{x} = y^2 - x - 1$, the family satisfies $2yy' = y^2 - x$, i.e. $y' = (y^2 - x)/(2y)$. The orthogonal trajectories satisfy

$$\frac{dy}{dx} = -\frac{2y}{y^2 - x} \quad\Longleftrightarrow\quad \frac{dx}{dy} = -\frac{y^2 - x}{2y} \quad\Longleftrightarrow\quad \frac{dx}{dy} - \frac{1}{2y}x = -\frac{y}{2}.$$

For $y > 0$ the integrating factor is $e^{-\frac{1}{2}\log y} = y^{-1/2}$:

$$\frac{d}{dy}\left(y^{-1/2}x\right) = -\frac{1}{2}y^{1/2} \quad \Longrightarrow \quad y^{-1/2}x = -\frac{1}{3}y^{3/2} + c_1.$$

Multiplying by $3y^{1/2}$: $3x = -y^2 + 3c_1\sqrt{y}$, i.e.

$$3x + y^2 = c\sqrt{y}.$$

(For $y < 0$ the same computation gives $3x + y^2 = c\sqrt{-y}$, the mirror images. The line $y = 0$, which the family crosses vertically, is also an orthogonal trajectory.)

Answer. (a) $x = y - 2 + ce^{-y}$; (b) $3x + y^2 = c\sqrt{y}$.

Problem 10-6 (Simmons)

Solution. Let f and g be differentiable on an interval on which $f(x) \neq 0$ (so that the family is a genuine one-parameter family). Differentiating $y = cf(x) + g(x)$ gives

$$y' = cf'(x) + g'(x).$$

To obtain the differential equation of the family we eliminate c: from the family itself, $c = \dfrac{y - g(x)}{f(x)}$. Substituting,

$$y' = \frac{f'(x)}{f(x)}\big(y - g(x)\big) + g'(x),$$

that is,

$$\frac{dy}{dx} + \left(-\frac{f'(x)}{f(x)}\right)y = g'(x) - \frac{f'(x)g(x)}{f(x)}.$$

This has the form $y' + P(x)y = Q(x)$ with $P = -f'/f$ and $Q = g' - f'g/f$, functions of x alone; hence it is linear.

Answer. Shown: eliminating c gives $y' - \dfrac{f'}{f}y = g' - \dfrac{f'g}{f}$, which is linear.

Problem 10-7 (Simmons)

Solution. Here $y > 0$ (for $\log y$ to be defined). With $z = \log y$, $z' = y'/y$. Dividing the equation by y:

$$\frac{y'}{y} + P = Q\log y \quad \Longrightarrow \quad z' + P = Qz \quad \Longrightarrow \quad z' - Q(x)z = -P(x),$$

a linear equation in z, which can be solved by (4); then $y = e^z$.

For $xy' = 2x^2y + y\log y$, divide by x: $y' - 2xy = \dfrac{1}{x}\,y\log y$, so $P = -2x$, $Q = 1/x$. The equation for $z = \log y$ is

$$z' - \frac{1}{x}z = 2x.$$

Integrating factor $1/x$: $\dfrac{d}{dx}\left(\dfrac{z}{x}\right) = 2$, so $\dfrac{z}{x} = 2x + c$, and

$$\log y = 2x^2 + cx.$$

Answer. Shown: $z = \log y$ gives the linear equation $z' - Qz = -P$; the example gives $\log y = 2x^2 + cx$.

Problem 10-8 (Simmons)

Solution. On $0 < x < \pi/2$ (and similarly on $\pi/2 < x < \pi$), $\sin 2x \neq 0$, and dividing gives

$$y' - \frac{2}{\sin 2x} y = \frac{2\cos x}{\sin 2x} = \frac{2\cos x}{2\sin x \cos x} = \csc x.$$

Now, with $u = 2x$ and $\int \csc u \, du = \log|\tan(u/2)|$,

$$\int \frac{2\,dx}{\sin 2x} = \int \csc u \, du = \log|\tan x|,$$

so $\int P\,dx = -\log|\tan x|$ and the integrating factor is $1/|\tan x|$; we take $\cot x$ (a constant multiple). Then

$$\frac{d}{dx}(y\cot x) = \cot x \csc x = \frac{\cos x}{\sin^2 x} \quad \Longrightarrow \quad y\cot x = -\frac{1}{\sin x} + c,$$

and multiplying by $\tan x$ (using $\tan x / \sin x = \sec x$),

$$y = c\tan x - \sec x = \frac{c\sin x - 1}{\cos x}.$$

As $x \to \pi/2$, $\cos x \to 0$ while $c\sin x - 1 \to c - 1$. If $c \neq 1$, $|y| \to \infty$. If $c = 1$,

$$y = \frac{\sin x - 1}{\cos x} = \frac{(\sin x - 1)(\sin x + 1)}{\cos x(1 + \sin x)} = \frac{-\cos^2 x}{\cos x(1+\sin x)} = -\frac{\cos x}{1+\sin x} \longrightarrow 0$$

as $x \to \pi/2$ from either side. So exactly one solution remains bounded, namely $c = 1$:

$$y = \tan x - \sec x \ \big(= -\cos x/(1+\sin x)\big).$$

Answer. $y = \tan x - \sec x$.

Problem 10-9 (Simmons)

Solution. The volume decreases at $4 - 3 = 1$ gal/min, so at time t the tank holds $10 - t$ gallons ($0 \le t \le 10$), with concentration $x/(10-t)$ lb/gal. Salt enters at $3 \cdot 1 = 3$ lb/min and leaves at $4x/(10-t)$ lb/min, so

$$\frac{dx}{dt} = 3 - \frac{4x}{10-t}, \qquad x(0) = 2, \qquad \text{i.e.} \qquad \frac{dx}{dt} + \frac{4}{10-t}x = 3.$$

Since $\int \frac{4\,dt}{10-t} = -4\log(10-t)$, the integrating factor is $(10-t)^{-4}$:

$$\frac{d}{dt}\Big[(10-t)^{-4}x\Big] = 3(10-t)^{-4} \quad \Longrightarrow \quad (10-t)^{-4}x = (10-t)^{-3} + c,$$

because $\frac{d}{dt}(10-t)^{-3} = 3(10-t)^{-4}$. Thus $x = (10-t) + c(10-t)^4$. At $t = 0$: $2 = 10 + 10^4 c$, so $c = -8/10^4$, and

$$x(t) = (10-t) - \frac{8}{10^4}(10-t)^4, \qquad 0 \le t \le 10.$$

(The tank is empty at $t = 10$, where $x = 0$; for $0 \le t < 10$, $x = (10-t)[1 - 0.0008(10-t)^3] \ge 0.2(10-t) > 0$.)

Remark. The book's answer prints the coefficient as $\frac{8}{10}$; this is a misprint for $\frac{8}{10^4}$, since with $\frac{8}{10}$ the initial condition would give $x(0) = 10 - 8000 \neq 2$.

Answer. $x(t) = (10-t) - \dfrac{8}{10^4}(10-t)^4$ lb, $0 \le t \le 10$.

Problem 10-10 (Simmons)

Solution. The volume at time t is $40 - t$ gallons ($0 \le t \le 40$). With $x(t)$ the pounds of salt, salt enters at $2 \cdot 3 = 6$ lb/min and leaves at $3x/(40-t)$ lb/min:

$$\frac{dx}{dt} + \frac{3}{40-t}x = 6, \qquad x(0) = 0.$$

Since $\int \frac{3\,dt}{40-t} = -3\log(40-t)$, the integrating factor is $(40-t)^{-3}$:

$$\frac{d}{dt}\left[(40-t)^{-3}x\right] = 6(40-t)^{-3} \quad \Longrightarrow \quad (40-t)^{-3}x = 3(40-t)^{-2} + c,$$

because $\frac{d}{dt}3(40-t)^{-2} = 6(40-t)^{-3}$. So $x = 3(40-t) + c(40-t)^3$; $x(0) = 0$ gives $120 + 64000c = 0$, $c = -3/1600$:

$$x(t) = 3(40-t) - \frac{3}{1600}(40-t)^3.$$

(a) The volume is 20 gallons when $t = 20$:

$$x(20) = 3 \cdot 20 - \frac{3}{1600} \cdot 8000 = 60 - 15 = 45 \text{ lb}.$$

(b) $x'(t) = -3 + \dfrac{9}{1600}(40-t)^2$, which vanishes when $(40-t)^2 = \dfrac{1600}{3}$, i.e. $40 - t = \dfrac{40}{\sqrt{3}}$ (taking $40 - t > 0$):

$$t = 40 - \frac{40}{\sqrt{3}} = \frac{40}{3}\left(3 - \sqrt{3}\right) \approx 16.9 \text{ min}.$$

Since $x''(t) = -\frac{18}{1600}(40-t) < 0$ for $t < 40$, and $x(0) = x(40) = 0$, this is the maximum. The maximum amount is $x = 3 \cdot \frac{40}{\sqrt{3}} - \frac{3}{1600} \cdot \frac{64000}{3\sqrt{3}} = \frac{120-40}{\sqrt{3}} = \frac{80}{\sqrt{3}} \approx 46.2$ lb.

Answer. (a) 45 lb; (b) after $\frac{40}{3}(3 - \sqrt{3}) \approx 16.9$ minutes (when the tank holds $80/\sqrt{3} \approx 46.2$ lb).

Problem 10-11 (Simmons)

Solution.

(a) A decays at the rate $k_1 x$, so $dx/dt = -k_1 x$, $x(0) = x_0$, and $x = x_0 e^{-k_1 t}$. B is produced at the rate A decays and itself decays at the rate $k_2 y$; we assume no B is present initially, $y(0) = 0$. Thus

$$\frac{dy}{dt} = k_1 x - k_2 y \qquad \text{i.e.} \qquad \frac{dy}{dt} + k_2 y = k_1 x_0 e^{-k_1 t}.$$

Integrating factor $e^{k_2 t}$:

$$\frac{d}{dt}\left(e^{k_2 t} y\right) = k_1 x_0 e^{(k_2 - k_1)t}.$$

Case $k_2 \neq k_1$. Integrating, $e^{k_2 t} y = \dfrac{k_1 x_0}{k_2 - k_1} e^{(k_2 - k_1)t} + c$; $y(0) = 0$ gives $c = -\dfrac{k_1 x_0}{k_2 - k_1}$. Multiplying by $e^{-k_2 t}$:

$$y = \frac{k_1 x_0}{k_2 - k_1}\left(e^{-k_1 t} - e^{-k_2 t}\right).$$

Case $k_2 = k_1$. The right side is the constant $k_1 x_0$, so $e^{k_1 t} y = k_1 x_0 t + c$ with $c = 0$:

$$y = k_1 x_0 t e^{-k_1 t}.$$

(This is also the limit of the first formula as $k_2 \to k_1$.)

(b) Here A is radium and B is radon, with $k_1 = 0.00043$ and $k_2 = 66$ per year; at the completion of the basement we take the trapped radon to be $y(0) = 0$. Since $k_2 \neq k_1$,

$$y'(t) = \frac{k_1 x_0}{k_2 - k_1}\left(-k_1 e^{-k_1 t} + k_2 e^{-k_2 t}\right) = 0 \iff k_2 e^{-k_2 t} = k_1 e^{-k_1 t} \iff e^{(k_2 - k_1)t} = \frac{k_2}{k_1},$$

so the only critical point is

$$t_{\max} = \frac{\log(k_2/k_1)}{k_2 - k_1}.$$

Because $y(0) = 0$, $y > 0$ for $t > 0$ (as $k_2 > k_1$ makes $e^{-k_1 t} > e^{-k_2 t}$), and $y \to 0$ as $t \to \infty$, this single critical point is the maximum. Numerically,

$$t_{\max} = \frac{\log(66/0.00043)}{65.99957} = \frac{\log 153488.4}{65.99957} \approx \frac{11.9413}{65.99957} \approx 0.18093 \text{ yr} \approx 0.18093 \times 365 \approx 66 \text{ days}.$$

(Check of the data: $\log 2/1600 \approx 0.00043$ and $\log 2/(3.8/365) \approx 66.6$ per year, consistent with the given rate constants.)

Answer. (a) $y = \dfrac{k_1 x_0}{k_2 - k_1}\left(e^{-k_1 t} - e^{-k_2 t}\right)$ if $k_2 \neq k_1$; $y = k_1 x_0 t e^{-k_1 t}$ if $k_2 = k_1$. (b) $t = \dfrac{\log(k_2/k_1)}{k_2 - k_1} \approx 0.181$ yr, about 66 days.

Section 11 Reduction of Order

Throughout this section we use the two substitutions of the text:

- *dependent variable y missing*: put $y' = p$, $y'' = \dfrac{dp}{dx}$;
- *independent variable x missing*: put $y' = p$, $y'' = \dfrac{dp}{dx} = \dfrac{dp}{dy}\dfrac{dy}{dx} = p\,\dfrac{dp}{dy}$.

c, c_1, c_2, C, K, … denote arbitrary constants.

Problem 11-1 (Simmons)

Solution.

(a) The variable x is missing, so put $y' = p$, $y'' = p\,dp/dy$:

$$yp\frac{dp}{dy} + p^2 = 0, \qquad \text{i.e.} \qquad p\left(y\frac{dp}{dy} + p\right) = 0.$$

If $p = 0$ then y is constant; this is a solution (case $c_1 = 0$ below). Otherwise $y\,dp/dy + p = 0$, i.e. $\dfrac{d}{dy}(yp) = 0$, so $yp = c$. Then

$$y\frac{dy}{dx} = c \implies y\,\mathrm{d}y = c\,\mathrm{d}x \implies \tfrac{1}{2}y^2 = cx + C \implies y^2 = c_1x + c_2 \quad (c_1 = 2c,\ c_2 = 2C).$$

(Alternatively, $yy'' + (y')^2 = (yy')'$, which leads to the same result at once.)

(b) The variable y is missing; with $y' = p$, $y'' = dp/dx$ the equation becomes the separable first order equation

$$x\frac{dp}{dx} = p + p^3 = p(1 + p^2).$$

The constant $p = 0$ (that is, $y = $ const) is a solution, lost if we divide by p; it is the degenerate case noted at the end. For $p \neq 0$ we separate variables and use the partial fraction decomposition $\dfrac{1}{p(1+p^2)} = \dfrac{1}{p} - \dfrac{p}{1+p^2}$:

$$\int\left(\frac{1}{p} - \frac{p}{1+p^2}\right)\mathrm{d}p = \int\frac{dx}{x} \implies \log|p| - \tfrac{1}{2}\log(1+p^2) = \log|x| + C \implies \frac{p^2}{1+p^2} = Kx^2,$$

with $K = e^{2C} > 0$. Writing $K = 1/c_1^2$ and solving for p^2,

$$p^2(1 - Kx^2) = Kx^2 \implies p^2 = \frac{x^2}{c_1^2 - x^2} \implies \frac{dy}{dx} = \frac{\pm x}{\sqrt{c_1^2 - x^2}}.$$

Integrating (substitute $u = c_1^2 - x^2$, $du = -2x\,\mathrm{d}x$),

$$y = \mp\sqrt{c_1^2 - x^2} + c_2, \qquad \text{so} \qquad (y - c_2)^2 = c_1^2 - x^2,$$

that is, $x^2 + (y - c_2)^2 = c_1^2$: circles with centres on the y-axis. In addition there are the constant solutions $y = c$ (from $p = 0$).

(c) We assume $k > 0$ (for $k = 0$ the equation is $y'' = 0$, with solution $y = c_1x + c_2$). The variable x is missing, so put $y' = p$, $y'' = p\,dp/dy$:

$$p\frac{dp}{dy} = k^2y \implies p\,\mathrm{d}p = k^2y\,\mathrm{d}y \implies p^2 = k^2y^2 + k^2A,$$

where we have written the constant of integration as k^2A. Thus $\dfrac{dy}{dx} = \pm k\sqrt{y^2 + A}$, and we distinguish three cases.

Case $A = 0$. Then $dy/dx = \pm ky$, so $y = ce^{kx}$ or $y = ce^{-kx}$.

Case $A = a^2 > 0$. Separating, $\displaystyle\int \frac{dy}{\sqrt{y^2 + a^2}} = \pm k \int dx$ gives $\sinh^{-1}(y/a) = \pm kx + b$, so $y = a\sinh(\pm kx + b)$. Since

$$a\sinh(kx + b) = \frac{ae^b}{2}e^{kx} - \frac{ae^{-b}}{2}e^{-kx},$$

and $a\sinh(-kx + b) = -a\sinh(kx - b)$ is of the same form, this is $c_1e^{kx} + c_2e^{-kx}$ with $c_1c_2 < 0$.

Case $A = -a^2 < 0$ (so $|y| \geq a$). Separating, $\displaystyle\int \frac{dy}{\sqrt{y^2 - a^2}} = \pm k \int dx$ gives $\cosh^{-1}(|y|/a) = \pm kx + b$, so $y = \pm a\cosh(kx \pm b)$ (cosh is even), i.e.

$$y = \pm\Big(\frac{ae^{\pm b}}{2}e^{kx} + \frac{ae^{\mp b}}{2}e^{-kx}\Big),$$

which is $c_1e^{kx} + c_2e^{-kx}$ with $c_1c_2 > 0$.

Conversely, every pair (c_1, c_2) arises in one of these cases ($c_1c_2 < 0$, $c_1c_2 > 0$, or $c_1c_2 = 0$, the last giving the exponentials and $y = 0$). Hence the general solution is

$$y = c_1e^{kx} + c_2e^{-kx}.$$

Direct check: $y'' = k^2c_1e^{kx} + k^2c_2e^{-kx} = k^2y$.

(d) The variable y is missing; with $y' = p$,

$$x^2\frac{dp}{dx} = 2xp + p^2, \qquad \text{i.e.} \qquad \frac{dp}{dx} - \frac{2}{x}\,p = \frac{p^2}{x^2},$$

a Bernoulli equation. The constant $p = 0$ ($y = \text{const}$) is a solution. For $p \neq 0$ put $w = 1/p$, so $dw/dx = -p^{-2}\,dp/dx$; dividing the equation by $-p^2$ gives the linear equation

$$\frac{dw}{dx} + \frac{2}{x}\,w = -\frac{1}{x^2}.$$

An integrating factor is $e^{\int 2\,dx/x} = x^2$:

$$\frac{d}{dx}(x^2w) = -1 \implies x^2w = -x + c_1 \implies p = \frac{1}{w} = \frac{x^2}{c_1 - x}.$$

To integrate, divide: $\dfrac{x^2}{x - c_1} = x + c_1 + \dfrac{c_1^2}{x - c_1}$, so

$$\frac{dy}{dx} = -x - c_1 - \frac{c_1^2}{x - c_1} \implies y = -\tfrac{1}{2}x^2 - c_1x - c_1^2\log|x - c_1| + c_2.$$

(Together with the constant solutions $y = c$.)

(e) The variable x is missing; with $y' = p$, $y'' = p\,dp/dy$,

$$2yp\frac{dp}{dy} = 1 + p^2 \implies \frac{2p\,dp}{1+p^2} = \frac{dy}{y} \implies \log(1+p^2) = \log|y| + C \implies 1 + p^2 = c_1 y,$$

with $c_1 \neq 0$ (no solution is lost: $y \neq 0$ is forced by the equation, and $1 + p^2$ never vanishes). Hence $p = \pm\sqrt{c_1 y - 1}$ and

$$\frac{dy}{\sqrt{c_1 y - 1}} = \pm\,dx \implies \frac{2}{c_1}\sqrt{c_1 y - 1} = \pm x + C \implies 2\sqrt{c_1 y - 1} = \pm c_1 x + c_2$$

($c_2 = c_1 C$). Squaring shows these are the parabolas $c_1 y = 1 + \frac{1}{4}(c_1 x + c_2)^2$.

(f) The variable x is missing; with $y' = p$, $y'' = p\,dp/dy$,

$$yp\frac{dp}{dy} = p^2.$$

If $p = 0$, y is constant. If $p \neq 0$: $\dfrac{dp}{p} = \dfrac{dy}{y}$, so $\log|p| = \log|y| + C$, $p = c_1 y$. Then $dy/dx = c_1 y$ gives $y = c_2 e^{c_1 x}$. This family contains the constants ($c_1 = 0$) and $y = 0$ ($c_2 = 0$), so the general solution is

$$y = c_2 e^{c_1 x}.$$

(g) The variable y is missing; with $y' = p$, $y'' = dp/dx$,

$$x\frac{dp}{dx} + p = 4x \iff \frac{d}{dx}(xp) = 4x \implies xp = 2x^2 + c_1 \implies \frac{dy}{dx} = 2x + \frac{c_1}{x},$$

and so $y = x^2 + c_1 \log|x| + c_2$.

Answer. (a) $y^2 = c_1 x + c_2$; (b) $x^2 + (y - c_2)^2 = c_1^2$ (and $y = c$); (c) $y = c_1 e^{kx} + c_2 e^{-kx}$; (d) $y = -\frac{1}{2}x^2 - c_1 x - c_1^2 \log(x - c_1) + c_2$ (and $y = c$); (e) $2\sqrt{c_1 y - 1} = \pm c_1 x + c_2$; (f) $y = c_2 e^{c_1 x}$; (g) $y = x^2 + c_1 \log x + c_2$.

Problem 11-2 (Simmons)

Solution.

(a) The variable y is missing; with $y' = p$, $y'' = dp/dx$,

$$(x^2 + 2p)\frac{dp}{dx} + 2xp = 0, \qquad \text{or} \qquad 2xp\,dx + (x^2 + 2p)\,dp = 0.$$

This is exact: $\frac{\partial}{\partial p}(2xp) = 2x = \frac{\partial}{\partial x}(x^2 + 2p)$. A function with $F_x = 2xp$, $F_p = x^2 + 2p$ is $F = x^2 p + p^2$, so the solutions satisfy

$$x^2 p + p^2 = c.$$

The condition $p = y' = 0$ at $x = 0$ gives $c = 0$, hence $p(x^2 + p) = 0$, so $p = 0$ or $p = -x^2$. Integrating with $y(0) = 1$:

$$y = 1 \qquad \text{or} \qquad y = 1 - \tfrac{1}{3}x^3, \text{ i.e. } 3y + x^3 = 3.$$

Both satisfy the equation and the initial conditions (for $y = 1 - x^3/3$: $(x^2 - 2x^2)(-2x) + 2x(-x^2) = 2x^3 - 2x^3 = 0$).

Remark. The problem has two solutions because the coefficient $x^2 + 2y'$ of y'' vanishes at the initial point, so the equation cannot be solved for y'' there and the usual uniqueness theorem does not apply.

(b) The variable x is missing; with $y' = p$, $y'' = p\,dp/dy$,

$$yp\frac{dp}{dy} = y^2p + p^2.$$

Since $p = 1 \neq 0$ at the initial point, we may divide by p near $x = 0$:

$$y\frac{dp}{dy} = y^2 + p \qquad \text{or} \qquad \frac{dp}{dy} - \frac{1}{y}\,p = y,$$

which is linear in p with integrating factor $e^{-\int dy/y} = 1/y$:

$$\frac{d}{dy}\Big(\frac{p}{y}\Big) = 1 \implies \frac{p}{y} = y + c \implies p = y^2 + cy.$$

At $x = 0$, $y = -\frac{1}{2}$ and $p = 1$: $1 = \frac{1}{4} - \frac{c}{2}$, so $c = -\frac{3}{2}$ and

$$\frac{dy}{dx} = y^2 - \tfrac{3}{2}y = \tfrac{1}{2}y(2y - 3).$$

Separating and using $\dfrac{1}{y(2y-3)} = -\dfrac{1}{3y} + \dfrac{2/3}{2y-3}$,

$$\int \Big(-\frac{1}{3y} + \frac{2/3}{2y-3}\Big)\,\mathrm{d}y = \int \frac{dx}{2} \implies \frac{1}{3}\log\Big|\frac{2y-3}{y}\Big| = \frac{x}{2} + C \implies \frac{2y-3}{y} = Ke^{3x/2}.$$

At $x = 0$, $y = -\frac{1}{2}$: $K = \dfrac{-1-3}{-1/2} = 8$. Hence

$$2y - 3 = 8ye^{3x/2}, \qquad \text{i.e.} \qquad y = \frac{3}{2 - 8e^{3x/2}}.$$

Check: $y(0) = 3/(-6) = -\frac{1}{2}$, and $y' = \dfrac{36e^{3x/2}}{(2-8e^{3x/2})^2}$ gives $y'(0) = 36/36 = 1$.

(c) The variable x is missing; with $y' = p$, $y'' = p\,dp/dy$,

$$p\frac{dp}{dy} = pe^y.$$

Since $p = 2 \neq 0$ initially, divide by p: $dp/dy = e^y$, so $p = e^y + c$; at $y = 0$, $p = 2$, so $c = 1$. (Equivalently: $y'' = (e^y)'$ integrates directly to $y' = e^y + c$.) Now

$$\frac{dy}{dx} = e^y + 1 \implies \int \frac{dy}{e^y + 1} = \int dx.$$

Writing $\dfrac{1}{e^y+1} = \dfrac{e^{-y}}{1+e^{-y}}$ (substitute $u = 1 + e^{-y}$, $du = -e^{-y}dy$), the integral is $-\log(1 + e^{-y})$, so

$$-\log(1 + e^{-y}) = x + C.$$

At $x = 0$, $y = 0$: $C = -\log 2$. Hence $1 + e^{-y} = 2e^{-x}$, $e^{-y} = 2e^{-x} - 1$, and

$$y = -\log(2e^{-x} - 1), \qquad x < \log 2.$$

Answer. (a) $y = 1$ or $3y + x^3 = 3$; (b) $2y - 3 = 8ye^{3x/2}$, i.e. $y = 3/(2 - 8e^{3x/2})$; (c) $y = -\log(2e^{-x} - 1)$.

Problem 11-3 (Simmons)

Solution.

(a) *Method 1 (y missing).* With $y' = p$, $y'' = dp/dx$:

$$\frac{dp}{dx} = 1 + p^2 \implies \tan^{-1} p = x + c_1 \implies p = \tan(x + c_1),$$

and since $\int \tan u \, du = -\log|\cos u|$,

$$y = -\log|\cos(x + c_1)| + c_2.$$

Method 2 (x missing). With $y' = p$, $y'' = p\,dp/dy$:

$$p\frac{dp}{dy} = 1 + p^2 \implies \frac{p\,dp}{1 + p^2} = dy \implies \tfrac{1}{2}\log(1 + p^2) = y + C \implies 1 + p^2 = Ae^{2y},$$

with $A = e^{2C} > 0$. Thus $dy/dx = \pm\sqrt{Ae^{2y} - 1}$, and $\displaystyle\int \frac{dy}{\sqrt{Ae^{2y} - 1}} = \pm \int dx$. To evaluate the integral put $\sqrt{A}\,e^y = \sec\theta$ with $0 < \theta < \pi/2$; then $\sqrt{A}\,e^y\,dy = \sec\theta\tan\theta\,d\theta$, i.e. $dy = \tan\theta\,d\theta$, and $\sqrt{Ae^{2y} - 1} = \sqrt{\sec^2\theta - 1} = \tan\theta$. So the integral is $\int d\theta = \theta$, and

$$\theta = \pm x + C' \implies \sqrt{A}\,e^y = \sec(\pm x + C') \implies y = -\log\cos(x + c_1) - \tfrac{1}{2}\log A,$$

where $c_1 = \pm C'$ (note $\cos(-x + C') = \cos(x - C')$).

Reconciliation. Method 2 gives the same family with $c_2 = -\frac{1}{2}\log A$; the absolute value in Method 1 merely allows intervals on which $\cos(x + c_1) < 0$, which are obtained from the others by replacing c_1 by $c_1 + \pi$. Conversely, from Method 1, $1 + p^2 = \sec^2(x + c_1)$ and $e^{2y} = e^{2c_2}\sec^2(x + c_1)$, so $1 + p^2 = Ae^{2y}$ with $A = e^{-2c_2}$: the first integral of Method 2.

(b) *Method 1 (y missing).* With $y' = p$:

$$\frac{dp}{dx} = 1 - p^2.$$

The constants $p = \pm 1$ are solutions, giving $y = \pm x + c$. For $p \neq \pm 1$,

$$\int \frac{dp}{1 - p^2} = \int dx \implies \frac{1}{2}\log\left|\frac{1 + p}{1 - p}\right| = x + C \implies \frac{1 + p}{1 - p} = Ke^{2x}\ (K \neq 0) \implies p = \frac{Ke^{2x} - 1}{Ke^{2x} + 1}.$$

Multiplying numerator and denominator by e^{-x}, $p = \dfrac{Ke^x - e^{-x}}{Ke^x + e^{-x}}$, which is exactly $\dfrac{d}{dx}\log|Ke^x + e^{-x}|$. Hence

$$y = \log|Ke^x + e^{-x}| + c_2.$$

(For $K = e^{2c} > 0$ this is $y = \log\cosh(x + c) + \text{const}$; for $K = -e^{2c} < 0$ it is $y = \log|\sinh(x + c)| + \text{const}$.)

Method 2 (x missing). With $y' = p$, $y'' = p\,dp/dy$:

$$p\frac{dp}{dy} + p^2 = 1.$$

Put $w = p^2$, so $p\,dp/dy = \tfrac{1}{2}dw/dy$; then $dw/dy + 2w = 2$, a linear equation with integrating factor e^{2y}: $(e^{2y}w)' = 2e^{2y}$, so $e^{2y}w = e^{2y} + C$, i.e.

$$p^2 = 1 + Ce^{-2y}.$$

Then $dy/dx = \pm\sqrt{1 + Ce^{-2y}} = \pm e^{-y}\sqrt{e^{2y} + C}$. With $u = e^y$ ($du = e^y\,dy$):

$$\int \frac{du}{\sqrt{u^2 + C}} = \pm \int dx \implies \log\left|u + \sqrt{u^2 + C}\right| = \pm x + c \implies u + \sqrt{u^2 + C} = Ke^{\pm x} \quad (K > 0).$$

Solving for u: $\sqrt{u^2 + C} = Ke^{\pm x} - u$; squaring, $u^2 + C = K^2e^{\pm 2x} - 2uKe^{\pm x} + u^2$, hence

$$e^y = u = \frac{K}{2}e^{\pm x} - \frac{C}{2K}e^{\mp x}.$$

Reconciliation. Both methods give

$$e^y = Ae^x + Be^{-x}, \qquad \text{i.e.} \qquad y = \log(Ae^x + Be^{-x}),$$

(A, B constants, not both zero, on an interval where $Ae^x + Be^{-x} > 0$). In Method 1, $e^y = e^{c_2}|Ke^x + e^{-x}|$, so $A = \sigma Ke^{c_2}$, $B = \sigma e^{c_2}$ with $\sigma = \pm 1$ the sign of $Ke^x + e^{-x}$, while $y = x + c$ is $B = 0$; in Method 2, $C = -4AB$ (one checks $p^2 = 1 + Ce^{-2y}$ directly: $(Ae^x - Be^{-x})^2 = (Ae^x + Be^{-x})^2 - 4AB$). That this is the general solution is also clear from the substitution $u = e^y$: $u'' = (y'' + y'^2)e^y = u$.

Remark. The book's answer $y = \log(c_1e^x + e^{-x}) + c_2$ is the case $B > 0$ of this family ($c_1 = A/B$, $c_2 = \log B$). The solutions with $B \le 0$, for example $y = x$ and $y = \log\sinh x$ ($x > 0$), are not of that form; they are included in $y = \log(Ae^x + Be^{-x})$.

Answer. (a) $y = -\log\cos(x + c_1) + c_2$ by both methods (first integral $1 + (y')^2 = e^{-2c_2}e^{2y}$); (b) $y = \log(Ae^x + Be^{-x})$, i.e. $y = \log(c_1e^x + e^{-x}) + c_2$ together with $y = \log(c_1e^x - e^{-x}) + c_2$ and $y = x + c_2$.

Problem 11-4 (Simmons)

Solution. Let x be the rock's displacement from the centre of the earth along the hole, and R the radius of the earth. Inside the earth the gravitational force is proportional to the distance from the centre (Problem 5–8), $F = -\lambda x$; at the surface its magnitude is mg, so $\lambda R = mg$ and $\lambda = mg/R$. Newton's second law gives

$$\frac{d^2x}{dt^2} = -\frac{g}{R}x, \qquad x = R,\ \frac{dx}{dt} = 0 \text{ when } t = 0.$$

The independent variable t is missing, so put $v = dx/dt$, $d^2x/dt^2 = v\,dv/dx$:

$$v\frac{dv}{dx} = -\frac{g}{R}x \implies v^2 = \frac{g}{R}(R^2 - x^2)$$

(using $v = 0$ at $x = R$). During the fall x decreases, so

$$\frac{dx}{\sqrt{R^2 - x^2}} = -\sqrt{\frac{g}{R}}\,dt \implies \cos^{-1}\frac{x}{R} = \sqrt{\frac{g}{R}}\,t \implies x = R\cos\sqrt{\frac{g}{R}}\,t$$

(the constant is 0 since $x = R$ at $t = 0$). This formula satisfies the differential equation for all t and therefore describes the whole motion: the rock reaches the other end $x = -R$ (with $v = 0$,

the "pause") when $\sqrt{g/R}\,t = \pi$, and first returns to $x = R$ when $\sqrt{g/R}\,t = 2\pi$. The round trip therefore takes

$$T = 2\pi\sqrt{\frac{R}{g}}.$$

Numerically, with $R = 4000$ miles $= 2.112 \times 10^7$ ft and $g = 32$ ft/s^2,

$$T = 2\pi\sqrt{660{,}000}\ \text{s} \approx 2\pi(812.4)\ \text{s} \approx 5104\ \text{s} \approx 85\ \text{minutes}.$$

(With $R = 6.371 \times 10^6$ m and $g = 9.81$ m/s^2 one gets $T \approx 5063$ s ≈ 84 minutes.)

Remark. The book's answer is $T = 2\pi\sqrt{R/g} \cong 89$ minutes. The formula agrees with ours, but the numerical value it yields with the book's own data ($R \approx 4000$ miles, $g = 32$ ft/s^2) is about 85 minutes, not 89. (A value of 89 minutes would require $R \approx 4400$ miles.)

Answer. $T = 2\pi\sqrt{R/g} \approx 85$ minutes (the book states $\cong 89$ minutes; see the remark).

Problem 11-5 (Simmons)

Solution. As in Figure 10 the positive y-axis points downward, so y measures depth below the cusp at the origin; one arch ($0 \le \theta \le 2\pi$) has its lowest point at $\theta = \pi$, i.e. at $(\pi a, 2a)$.

Arc length. We have $dx/d\theta = a(1-\cos\theta)$, $dy/d\theta = a\sin\theta$, so

$$\left(\frac{ds}{d\theta}\right)^2 = a^2\left[(1-\cos\theta)^2 + \sin^2\theta\right] = 2a^2(1-\cos\theta) = 4a^2\sin^2\frac{\theta}{2}.$$

Measure arc length s from the lowest point, positive on the side $0 \le \theta < \pi$ and negative on the side $\pi < \theta \le 2\pi$. Then

$$s = \int_\theta^\pi 2a\sin\frac{\phi}{2}\,d\phi = \Big[-4a\cos\frac{\phi}{2}\Big]_\theta^\pi = 4a\cos\frac{\theta}{2}, \qquad 0 \le \theta \le 2\pi$$

(for $\theta > \pi$ the same formula gives the negative of the arc length, as required by our sign convention). Consequently

$$y = a(1-\cos\theta) = 2a\sin^2\frac{\theta}{2} = 2a\Big(1-\cos^2\frac{\theta}{2}\Big) = 2a - \frac{s^2}{8a}.$$

Energy. Let the bead be released from rest at the point with arc length s_0 and depth $y_0 = 2a - s_0^2/(8a)$. With no friction, the gain in kinetic energy equals the loss of potential energy: $\frac{1}{2}mv^2 = mg(y - y_0)$. Hence

$$v^2 = 2g(y-y_0) = 2g\Big(\frac{s_0^2}{8a} - \frac{s^2}{8a}\Big) = \frac{g}{4a}(s_0^2 - s^2), \qquad \text{i.e.} \qquad 4av^2 = g(s_0^2 - s^2).$$

Differentiation. Since $v = ds/dt$ up to sign, $v^2 = (ds/dt)^2$, and differentiating $4a(ds/dt)^2 = g(s_0^2 - s^2)$ with respect to t gives

$$8a\,\frac{ds}{dt}\,\frac{d^2s}{dt^2} = -2gs\,\frac{ds}{dt}.$$

Dividing by $8a\,ds/dt$ (which vanishes only at isolated instants, the turning points; by continuity the result holds there too),

$$\frac{d^2s}{dt^2} + \frac{g}{4a}\,s = 0.$$

Solution. This is the equation of Example 2 with $k = \sqrt{g/4a}$, whose general solution is $s = c_1 \sin kt + c_2 \cos kt$. The bead starts from rest at $s = s_0$: $s(0) = s_0$ gives $c_2 = s_0$, and $s'(0) = kc_1 = 0$ gives $c_1 = 0$. Therefore

$$s = s_0 \cos \sqrt{\frac{g}{4a}}\, t.$$

(It indeed satisfies $4a(ds/dt)^2 = 4as_0^2 \frac{g}{4a} \sin^2 kt = g(s_0^2 - s^2)$.) The motion is periodic, with period

$$P = \frac{2\pi}{k} = 2\pi\sqrt{\frac{4a}{g}} = 4\pi\sqrt{\frac{a}{g}}.$$

Tautochrone property. The bead first reaches the lowest point $s = 0$ when $kt = \pi/2$, i.e. after time

$$t = \frac{\pi}{2k} = \frac{P}{4} = \pi\sqrt{\frac{a}{g}},$$

which does not depend on the starting position s_0. This is precisely the statement of Problem 6–5: the bead reaches the bottom in the same time $\pi\sqrt{a/g}$ wherever it is released.

Answer. $4av^2 = g(s_0^2 - s^2)$ and $s'' + (g/4a)s = 0$ shown; $s = s_0 \cos \sqrt{g/4a}\, t$; period $4\pi\sqrt{a/g}$ (time to the bottom $\pi\sqrt{a/g}$, independent of s_0).

Section 12 The Hanging Chain. Pursuit Curves

We use the notation of the text: in Example 1 the lowest point of the chain lies on the y-axis, T_0 is the (horizontal) tension there, T the tension at (x, y) acting along the tangent at angle θ, s the arc length from the lowest point, and $w(s)$ the weight per unit length, so that

$$T\cos\theta = T_0, \qquad T\sin\theta = \int_0^s w(s)\,ds, \qquad T_0 y'' = w(s)\sqrt{1+(y')^2}. \tag{1}$$

Problem 12-1 (Simmons)

Solution. For the uniform chain, $w(s) = w_0$ and the curve is $y = \dfrac{1}{a}\cosh ax$ with $a = w_0/T_0$. Squaring and adding the two equilibrium equations in (1),

$$T^2 = T^2\cos^2\theta + T^2\sin^2\theta = T_0^2 + (w_0 s)^2.$$

The arc length from the lowest point $(0, 1/a)$ to (x, y) is

$$s = \int_0^x \sqrt{1+\sinh^2 ax}\,dx = \int_0^x \cosh ax\,dx = \frac{1}{a}\sinh ax.$$

Therefore, using $w_0 = aT_0$,

$$T^2 = T_0^2 + \frac{w_0^2}{a^2}\sinh^2 ax = T_0^2\left(1+\sinh^2 ax\right) = T_0^2\cosh^2 ax,$$

so (as $T > 0$)

$$T = T_0\cosh ax = aT_0\cdot\frac{1}{a}\cosh ax = w_0 y.$$

Answer. Shown: $T = T_0\cosh ax = w_0 y$.

Problem 12-2 (Simmons)

Solution. The portion of the chain between the lowest point (at $x = 0$) and (x, y) now carries, besides its own weight $\int_0^s w(s)\,ds$, the load resting on the horizontal interval $[0, x]$, whose weight is $\int_0^x L(x)\,dx$. The horizontal balance $T\cos\theta = T_0$ is unchanged, and the vertical balance becomes

$$T\sin\theta = \int_0^s w(s)\,ds + \int_0^x L(x)\,dx.$$

Dividing by $T\cos\theta = T_0$ and using $\tan\theta = y'$,

$$T_0 y' = \int_0^s w(s)\,ds + \int_0^x L(x)\,dx.$$

Differentiating with respect to x (by the fundamental theorem of calculus and the chain rule, $\frac{d}{dx}\int_0^s w\,ds = w(s)\,\frac{ds}{dx} = w(s)\sqrt{1+(y')^2}$):

$$T_0 y'' = w(s)\sqrt{1+(y')^2} + L(x).$$

Answer. $T_0 y'' = w(s)\sqrt{1+(y')^2} + L(x)$.

Problem 12-3 (Simmons)

Solution. With $w(s) = 0$ and $L(x) = L_0$ the equation of Problem 2 becomes

$$T_0 y'' = L_0, \qquad \text{i.e.} \qquad y'' = \frac{L_0}{T_0}.$$

Integrating twice, $y' = \dfrac{L_0}{T_0}x + c_1$ and $y = \dfrac{L_0}{2T_0}x^2 + c_1 x + c_2$. Since the lowest point is on the y-axis, $y'(0) = 0$, so $c_1 = 0$, and

$$y = \frac{L_0}{2T_0}x^2 + y_0,$$

where y_0 is the height of the lowest point. The cable hangs in a parabola.

Answer. A parabola, $y = \dfrac{L_0}{2T_0}x^2 + y_0$.

Problem 12-4 (Simmons)

Solution. "Uniform density" means that equal amounts of material have equal weights; the hypothesis says that a small piece of the cable containing weight ΔW of material is stretched to a length Δs proportional to the tension T in it:

$$\Delta s = kT\,\Delta W \qquad (k > 0 \text{ constant}).$$

Hence the weight per unit (stretched) length is

$$w(s) = \lim \frac{\Delta W}{\Delta s} = \frac{1}{kT}.$$

The derivation of (1) used only the equilibrium of the portion of cable between the lowest point and (x, y), so it remains valid: $T_0 y'' = w(s)\sqrt{1+(y')^2}$. Moreover, from $T\cos\theta = T_0$ and $\tan\theta = y'$,

$$T = \frac{T_0}{\cos\theta} = T_0 \sec\theta = T_0\sqrt{1+\tan^2\theta} = T_0\sqrt{1+(y')^2}.$$

Substituting,

$$T_0 y'' = \frac{\sqrt{1+(y')^2}}{kT_0\sqrt{1+(y')^2}} = \frac{1}{kT_0}, \qquad \text{so} \qquad y'' = \frac{1}{kT_0^2} = \text{const.}$$

Integrating twice with $y'(0) = 0$ (lowest point on the y-axis),

$$y = \frac{x^2}{2kT_0^2} + y_0,$$

a parabola.

Answer. Shown: $y'' = 1/(kT_0^2)$ is constant, so $y = x^2/(2kT_0^2) + y_0$, a parabola.

Problem 12-5 (Simmons)

Solution. Put the bottom of the curtain on the x-axis and the lowest point of the cord on the y-axis, at height $y_0 > 0$. The rod hanging from the point (x, y) of the cord reaches down to the x-axis, so its length is y. If each rod has weight ρ per unit length and there are n rods per unit horizontal distance, the load carried per unit horizontal length is

$$L(x) = n\rho\, y = \kappa y, \qquad \kappa = n\rho > 0.$$

The cord itself has $w(s) = 0$, so by Problem 2

$$T_0 y'' = \kappa y, \qquad \text{i.e.} \qquad y'' = a^2 y, \quad a = \sqrt{\kappa / T_0}.$$

The variable x is missing; with $y' = p$, $y'' = p\, dp/dy$:

$$p\frac{dp}{dy} = a^2 y \implies p^2 = a^2(y^2 - y_0^2),$$

the constant being fixed by $p = 0$ at the lowest point $y = y_0$. For $x > 0$ the cord rises, so

$$\frac{dy}{\sqrt{y^2 - y_0^2}} = a\,dx \implies \cosh^{-1}\frac{y}{y_0} = ax \implies y = y_0 \cosh ax$$

(constant zero since $y = y_0$ at $x = 0$); by symmetry ($y'' = a^2 y$ is unchanged under $x \to -x$) the same formula holds for $x < 0$. With $c = y_0/2$,

$$y = c\,(e^{ax} + e^{-ax}),$$

a catenary.

Answer. $y = c(e^{ax} + e^{-ax})$ (a catenary), with the bottom of the curtain on the x-axis and the lowest point of the cord on the y-axis.

Problem 12-6 (Simmons)

Solution. Let the curve be $y = y(x) > 0$ and write the property as

$$\int_{x_1}^{x} y\,dx = a\int_{x_1}^{x} \sqrt{1 + (y')^2}\,dx \qquad \text{for all } x_1, x,$$

with a constant $a > 0$. Differentiating with respect to x gives

$$y = a\sqrt{1 + (y')^2}; \tag{$*$}$$

conversely, if $(*)$ holds then the two integrals have equal derivatives and both vanish at $x = x_1$, so they are equal. Thus we must solve $(*)$. It gives $y \geq a$ and

$$(y')^2 = \frac{y^2 - a^2}{a^2}.$$

Case $y' = 0$: then $y = a$, a horizontal straight line (indeed the area under a segment of length ℓ is $a\ell$).

Case $y' \neq 0$: separating variables,

$$\frac{dy}{\sqrt{y^2-a^2}} = \pm\frac{dx}{a} \implies \cosh^{-1}\frac{y}{a} = \pm\frac{x-b}{a} \implies y = a\cosh\frac{x-b}{a},$$

since cosh is even. This is a catenary; check: $a\sqrt{1+\sinh^2\frac{x-b}{a}} = a\cosh\frac{x-b}{a} = y$.

Remark. The same conclusion follows by eliminating the constant a: differentiating $y^2 = a^2(1+y'^2)$ gives $y'(y-a^2y'') = 0$, so on any interval either $y' = 0$ (the line) or $y'' = y/a^2$ (whose solutions satisfying $(*)$ are the catenaries above). A twice differentiable curve cannot switch from one alternative to the other, since at a junction y'' would have to jump from 0 to $1/a$; if only a continuous tangent is required, a horizontal segment $y = a$ may be joined tangentially at its end to half of the catenary $y = a\cosh\frac{x-b}{a}$, and this composite curve also has the property.

Answer. A horizontal straight line $y = a$ or a catenary $y = a\cosh\dfrac{x-b}{a}$.

Problem 12-7 (Simmons)

Solution. The tractrix of Example 2 is the curve

$$y = a\log\frac{a+\sqrt{a^2-x^2}}{x} - \sqrt{a^2-x^2}, \qquad 0 < x \leq a, \qquad \text{with slope} \qquad m_1 = \frac{dy}{dx} = -\frac{\sqrt{a^2-x^2}}{x}.$$

Consider the circle $x^2+(y-h)^2 = a^2$ with centre $(0,h)$, $h > 0$. Its lower half is $y = h - \sqrt{a^2-x^2}$, and differentiating implicitly, $2x + 2(y-h)y' = 0$, so on the lower half the slope is

$$m_2 = -\frac{x}{y-h} = \frac{x}{\sqrt{a^2-x^2}}.$$

At any point (x,y), $0 < x < a$, where the tractrix meets the lower half of this circle,

$$m_1m_2 = -\frac{\sqrt{a^2-x^2}}{x}\cdot\frac{x}{\sqrt{a^2-x^2}} = -1,$$

so the curves cross at right angles.

It remains to see that the two curves do meet. A point (x,y) of the tractrix lies on the lower half of the circle exactly when $h = H(x)$, where

$$H(x) = y(x) + \sqrt{a^2-x^2} = a\log\frac{a+\sqrt{a^2-x^2}}{x}$$

is the height of the point T in which the tangent at $P = (x,y)$ meets the y-axis (recall $PT = a$; geometrically, the circle of radius a about T passes through P, and its radius TP lies along the tangent of the tractrix, which is why the curves are orthogonal). H is continuous on $(0,a]$ with $H(a) = 0$ and $H(x) \to \infty$ as $x \to 0^+$, and

$$H'(x) = -\frac{\sqrt{a^2-x^2}}{x} - \frac{x}{\sqrt{a^2-x^2}} = -\frac{a^2}{x\sqrt{a^2-x^2}} < 0.$$

Hence for each $h > 0$ there is exactly one point of the tractrix on the lower half of the circle with centre $(0,h)$, and there the curves are orthogonal.

Answer. Shown: at each intersection the slopes are $-\sqrt{a^2-x^2}/x$ and $x/\sqrt{a^2-x^2}$, whose product is -1.

Problem 12-8 (Simmons)

Solution. From Example 3, the dog's path (starting at $(c,0)$, $0<x\le c$) satisfies

$$\frac{dy}{dx}=p=\frac{1}{2}\Big[\Big(\frac{x}{c}\Big)^k-\Big(\frac{c}{x}\Big)^k\Big],\qquad y=0 \text{ at } x=c,$$

and the rabbit's height at time t is, by (4), $at=y-xy'$. Note also

$$1+p^2=\frac{1}{4}\Big[\Big(\frac{x}{c}\Big)^k+\Big(\frac{c}{x}\Big)^k\Big]^2,$$

since $(u-v)^2+4=(u+v)^2$ when $uv=1$.

(a) For $k<1$, integrate term by term:

$$\int\Big(\frac{x}{c}\Big)^k dx=\frac{c}{1+k}\Big(\frac{x}{c}\Big)^{1+k},\qquad \int\Big(\frac{c}{x}\Big)^k dx=c^k\frac{x^{1-k}}{1-k}=\frac{c}{1-k}\Big(\frac{x}{c}\Big)^{1-k}.$$

Hence

$$y=\frac{c}{2}\Big[\frac{1}{1+k}\Big(\frac{x}{c}\Big)^{1+k}-\frac{1}{1-k}\Big(\frac{x}{c}\Big)^{1-k}\Big]+C.$$

At $x=c, y=0$: $C=-\dfrac{c}{2}\Big[\dfrac{1}{1+k}-\dfrac{1}{1-k}\Big]=\dfrac{ck}{1-k^2}$. So

$$y=\frac{c}{2}\Big[\frac{1}{1+k}\Big(\frac{x}{c}\Big)^{1+k}-\frac{1}{1-k}\Big(\frac{x}{c}\Big)^{1-k}\Big]+\frac{ck}{1-k^2}.$$

Because $1-k>0$, both powers of x tend to 0 as $x\to 0^+$, so the path ends on the y-axis at the point $\big(0,ck/(1-k^2)\big)$. The dog gets there after running the finite distance

$$\int_0^c\sqrt{1+p^2}\,dx=\frac{1}{2}\int_0^c\Big[\Big(\frac{x}{c}\Big)^k+\Big(\frac{c}{x}\Big)^k\Big]dx=\frac{c}{2}\Big[\frac{1}{1+k}+\frac{1}{1-k}\Big]=\frac{c}{1-k^2},$$

and at that moment the rabbit is at the same point, since its height is $at=y-xy'$ and $xy'=\frac{1}{2}\big[c^{-k}x^{1+k}-c^kx^{1-k}\big]\to 0$ as $x\to 0^+$. Thus the dog catches the rabbit at $\big(0,ck/(1-k^2)\big)$, and the rabbit has run

$$\frac{ck}{1-k^2}.$$

(Consistency check: the times agree, since the rabbit's distance divided by a equals the dog's distance divided by b: $\dfrac{ck}{a(1-k^2)}=\dfrac{c}{b(1-k^2)}$ because $k=a/b$.)

(b) For $k=1$,

$$\frac{dy}{dx}=\frac{1}{2}\Big(\frac{x}{c}-\frac{c}{x}\Big)\implies y=\frac{1}{2}\Big(\frac{x^2}{2c}-c\log x\Big)+C.$$

At $x=c, y=0$: $C=-\dfrac{c}{4}+\dfrac{c}{2}\log c$, so

$$y=\frac{1}{2}\Big[\frac{x^2-c^2}{2c}-c\log\frac{x}{c}\Big].$$

Now $y \to +\infty$ as $x \to 0^+$, so the dog's path has infinite length and the dog never reaches the y-axis; since $dx/ds = -1/\sqrt{1+p^2} < 0$, x decreases steadily and, the arc from $x = c$ to any $x > 0$ being finite, $x \to 0$ as $t \to \infty$.

Distance. The rabbit is at $R = (0, at)$ with $at = y - xy'$, and the dog at $D = (x, y)$. Hence $at - y = -xy'$ and

$$|DR|^2 = x^2 + (at-y)^2 = x^2\bigl(1 + (y')^2\bigr) = x^2 \cdot \frac{1}{4}\Bigl(\frac{x}{c} + \frac{c}{x}\Bigr)^2, \qquad \text{so} \qquad |DR| = \frac{x^2}{2c} + \frac{c}{2}.$$

(This is just the length of the tangent segment DR.) As x decreases to 0, $|DR|$ decreases strictly and tends to $c/2$, but $x > 0$ at every time. Therefore the dog comes closer than $c/2 + \epsilon$ for every $\epsilon > 0$, but never as close as $c/2$: the distance approaches $c/2$.

Answer. (a) $y = \dfrac{c}{2}\Bigl[\dfrac{1}{1+k}\Bigl(\dfrac{x}{c}\Bigr)^{1+k} - \dfrac{1}{1-k}\Bigl(\dfrac{x}{c}\Bigr)^{1-k}\Bigr] + \dfrac{ck}{1-k^2}$; the rabbit runs $ck/(1-k^2)$. (b) $y = \dfrac{1}{2}\Bigl[\dfrac{x^2-c^2}{2c} - c\log\dfrac{x}{c}\Bigr]$; the distance $x^2/(2c) + c/2$ decreases to $c/2$: the dog gets within $c/2 + \epsilon$ for any $\epsilon > 0$ but never as close as $c/2$.

Problem 12-9 (Simmons)

Solution. *The path.* For completeness we first rederive the solution quoted in the text. The equation of the path is

$$\frac{dy}{dx} = \frac{a\sqrt{x^2+y^2} + by}{bx} = k\sqrt{1 + (y/x)^2} + \frac{y}{x}, \qquad k = \frac{a}{b}, \quad 0 < x \le c.$$

It is homogeneous; put $y = vx$, $y' = v + xv'$:

$$v + x\frac{dv}{dx} = k\sqrt{1+v^2} + v \implies \frac{dv}{\sqrt{1+v^2}} = k\frac{dx}{x} \implies \log\bigl(v + \sqrt{1+v^2}\bigr) = k\log x + C.$$

At $x = c$, $y = 0$, so $v = 0$ and $C = -k\log c$. Hence $v + \sqrt{1+v^2} = (x/c)^k$, and multiplying by x,

$$c^k\bigl(y + \sqrt{x^2+y^2}\bigr) = x^{k+1}.$$

Solving for y. Write $X = x^{k+1}/c^k$. Then $\sqrt{x^2+y^2} = X - y$; squaring, $x^2 + y^2 = X^2 - 2Xy + y^2$, so

$$y = \frac{X^2 - x^2}{2X} = \frac{1}{2}\Bigl(X - \frac{x^2}{X}\Bigr) = \frac{1}{2}\Bigl(\frac{x^{k+1}}{c^k} - \frac{c^k}{x^{k-1}}\Bigr).$$

(Squaring introduced nothing extraneous, since $X - y = \frac{1}{2}(X + x^2/X) > 0$.)

Fate of the boat. The boat heads toward the origin, so $dx/dt = -b\cos\theta < 0$ while $x > 0$: it moves steadily toward the bank $x = 0$ along the path above. As $x \to 0^+$, $x^{k+1}/c^k \to 0$, and the second term behaves as follows.

- $a > b$ $(k > 1)$: $c^k/x^{k-1} \to \infty$, so $y \to -\infty$. The boat is carried downstream without limit and never lands.
- $a = b$ $(k = 1)$: $y = \dfrac{1}{2}\Bigl(\dfrac{x^2}{c} - c\Bigr) \to -\dfrac{c}{2}$. The boat lands at $(0, -c/2)$, at a distance $c/2$ downstream from the point toward which it aims.

- $a < b$ $(k < 1)$: $c^k x^{1-k} \to 0$, so $y \to 0$. The boat lands at the origin, the point it is aiming at.

Thus the boat reaches the opposite bank if and only if $a \le b$ (current no faster than the boat).

Remark. In the borderline case $a = b$ the landing at $(0, -c/2)$ is approached only in the limit: near that point $\cos\theta = x/\sqrt{x^2+y^2} \approx 2x/c$, so $dx/dt \approx -2bx/c$ and $x \approx \text{const} \cdot e^{-2bt/c}$; the boat's speed tends to zero as it comes in, and it gets as close to $(0, -c/2)$ as we please. For $a < b$, by contrast, near the origin $|y| \approx \frac{1}{2}c^k x^{1-k}$, so $dx/dt \approx -2b(x/c)^k$, and x^{1-k} decreases at a nearly constant rate: the boat reaches the origin in finite time.

Answer. $y = \frac{1}{2}\left(\frac{x^{k+1}}{c^k} - \frac{c^k}{x^{k-1}}\right)$, $k = a/b$. If $a > b$, $y \to -\infty$ as $x \to 0$: the boat never lands. If $a = b$, $y \to -c/2$: it lands at $(0, -c/2)$. If $a < b$, $y \to 0$: it lands at the origin.

Section 13 Simple Electric Circuits

Throughout this section we use the notation of the text: L, R, C are positive constants (inductance, resistance, capacitance), $E = E(t)$ is the impressed emf, Q is the charge on the capacitor and $I = dQ/dt$ is the current. Kirchhoff's law gives

$$L\frac{dI}{dt} + RI + \frac{1}{C}Q = E, \tag{1}$$

and in the absence of a capacitor

$$L\frac{dI}{dt} + RI = E. \tag{4}$$

Equation (4) is first order linear; dividing by L and multiplying by the integrating factor $e^{\int (R/L)\,dt} = e^{Rt/L}$ turns it into

$$\frac{d}{dt}\left(Ie^{Rt/L}\right) = \frac{1}{L}E(t)\,e^{Rt/L}. \tag{$*$}$$

Problem 13-1 (Simmons)

Solution. In Example 1 the solution of $L\,dI/dt + RI = E_0$ with $I(0) = I_0$ is $I = E_0/R + (I_0 - E_0/R)e^{-Rt/L}$. With $I_0 = 0$ this becomes

$$I(t) = \frac{E_0}{R}\left(1 - e^{-Rt/L}\right).$$

Since $R/L > 0$, $e^{-Rt/L}$ decreases strictly from 1 to 0 as t goes from 0 to ∞, so $|I(t)|$ increases strictly towards (but never reaches) the value $|E_0|/R$. The theoretical maximum of the current is therefore the steady-state value E_0/R (the sign of E_0 merely fixes the direction of the current). The current equals half of this when

$$\begin{aligned} \frac{E_0}{R}\left(1 - e^{-Rt/L}\right) = \frac{1}{2}\cdot\frac{E_0}{R} &\iff 1 - e^{-Rt/L} = \frac{1}{2} \qquad (E_0 \neq 0) \\ &\iff e^{-Rt/L} = \frac{1}{2} \iff -\frac{R}{L}t = -\log 2 \iff t = \frac{L\log 2}{R}. \end{aligned}$$

Because $|I|$ is strictly increasing, this is the unique instant at which half the maximum is reached.

Answer. $I = (E_0/R)(1 - e^{-Rt/L})$ reaches $\frac{1}{2}(E_0/R)$ exactly when $e^{-Rt/L} = \frac{1}{2}$, i.e. at $t = (L\log 2)/R$ seconds.

Problem 13-2 (Simmons)

Solution. We use the integrated form $(*)$ of (4) and the initial condition $I(0) = I_0$.

(a) With $E = E_0e^{-kt}$, $(*)$ reads

$$\frac{d}{dt}\left(Ie^{Rt/L}\right) = \frac{E_0}{L}e^{(R/L-k)t}.$$

Case $R \neq kL$ (i.e. $R/L - k \neq 0$). Integrating,

$$Ie^{Rt/L} = \frac{E_0}{L}\cdot\frac{e^{(R/L-k)t}}{R/L-k} + c = \frac{E_0}{R-kL}e^{(R/L-k)t} + c, \qquad \text{so} \qquad I = \frac{E_0}{R-kL}e^{-kt} + ce^{-Rt/L}.$$

Putting $t=0$: $I_0 = \dfrac{E_0}{R-kL} + c$, so $c = I_0 - \dfrac{E_0}{R-kL}$ and

$$I = \frac{E_0}{R-kL}e^{-kt} + \Big(I_0 - \frac{E_0}{R-kL}\Big)e^{-Rt/L}.$$

Case $R = kL$. Then the right side of $(*)$ is the constant E_0/L, so $Ie^{Rt/L} = E_0 t/L + c$; $t=0$ gives $c = I_0$, hence

$$I = \Big(I_0 + \frac{E_0}{L}t\Big)e^{-Rt/L}.$$

(b) With $E = E_0 \sin \omega t$, $(*)$ reads $\dfrac{d}{dt}\big(Ie^{Rt/L}\big) = \dfrac{E_0}{L}e^{Rt/L}\sin\omega t$. We need the standard integral (two integrations by parts, or check by differentiation), with $a = R/L$:

$$\int e^{at}\sin\omega t\,dt = \frac{e^{at}(a\sin\omega t - \omega\cos\omega t)}{a^2+\omega^2} + c.$$

Indeed, differentiating the right side gives

$$\frac{e^{at}}{a^2+\omega^2}\big[a(a\sin\omega t - \omega\cos\omega t) + (a\omega\cos\omega t + \omega^2\sin\omega t)\big] = e^{at}\sin\omega t.$$

Hence

$$Ie^{Rt/L} = \frac{E_0}{L}\cdot\frac{e^{Rt/L}\big(\frac{R}{L}\sin\omega t - \omega\cos\omega t\big)}{\frac{R^2}{L^2}+\omega^2} + c.$$

Multiplying numerator and denominator by L^2 and then by $e^{-Rt/L}$,

$$I = \frac{E_0\,(R\sin\omega t - L\omega\cos\omega t)}{R^2+L^2\omega^2} + ce^{-Rt/L}.$$

At $t=0$: $I_0 = -\dfrac{E_0 L\omega}{R^2+L^2\omega^2} + c$, so $c = I_0 + \dfrac{E_0 L\omega}{R^2+L^2\omega^2}$.

The steady-state part can be written as a single sine. Define the angle α $(0<\alpha<\pi/2)$ by

$$\cos\alpha = \frac{R}{\sqrt{R^2+L^2\omega^2}}, \qquad \sin\alpha = \frac{L\omega}{\sqrt{R^2+L^2\omega^2}}, \qquad \text{i.e.} \quad \tan\alpha = \frac{L\omega}{R}.$$

Then $R\sin\omega t - L\omega\cos\omega t = \sqrt{R^2+L^2\omega^2}\,(\sin\omega t\cos\alpha - \cos\omega t\sin\alpha) = \sqrt{R^2+L^2\omega^2}\,\sin(\omega t - \alpha)$, and therefore

$$I = \frac{E_0}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t-\alpha) + \Big(I_0 + \frac{E_0 L\omega}{R^2+L^2\omega^2}\Big)e^{-Rt/L}.$$

Answer. (a) $I = \dfrac{E_0}{R-kL}e^{-kt} + \Big(I_0 - \dfrac{E_0}{R-kL}\Big)e^{-Rt/L}$ if $R \neq kL$; $I = (I_0 + E_0 t/L)e^{-Rt/L}$ if $R = kL$.
(b) $I = \dfrac{E_0}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t-\alpha) + \Big(I_0 + \dfrac{E_0 L\omega}{R^2+L^2\omega^2}\Big)e^{-Rt/L}$, where $\tan\alpha = L\omega/R$.

Problem 13-3 (Simmons)

Solution. We assume, as the problem implicitly does, that $E(t)$ is differentiable, so that by (4) $I' = (E - RI)/L$ is differentiable too.

(a) At an (interior) maximum or minimum of the differentiable function $I(t)$ we have $dI/dt = 0$. Putting this in (4) gives $RI = E$, i.e. $E = RI$, which is Ohm's law.

(b) Differentiating (4) with respect to t gives

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} = \frac{dE}{dt}.$$

At an extremum of I, $dI/dt = 0$, so there

$$\frac{dE}{dt} = L\frac{d^2I}{dt^2}.$$

At a minimum of I the second derivative satisfies $d^2I/dt^2 > 0$ (for a nondegenerate minimum, the usual second-derivative-test situation; in general $d^2I/dt^2 \geq 0$), and since $L > 0$ we get $dE/dt > 0$: the emf is increasing. At a maximum $d^2I/dt^2 < 0$ (in general ≤ 0), so $dE/dt < 0$: the emf is decreasing.

Remark. In the degenerate case $d^2I/dt^2 = 0$ at the extremum the argument only gives $dE/dt = 0$; the statement of the problem tacitly refers to the generic situation of a nondegenerate extremum.

Answer. Shown: at an extremum of I, $I' = 0$, so (4) gives $E = RI$; differentiating (4), $E' = LI''$ there, which is positive at a minimum and negative at a maximum.

Problem 13-4 (Simmons)

Solution. With $L = 0$, equation (3) $L\,d^2Q/dt^2 + R\,dQ/dt + Q/C = E$ becomes the first order linear equation

$$R\frac{dQ}{dt} + \frac{1}{C}Q = E, \qquad \text{i.e.} \qquad \frac{dQ}{dt} + \frac{1}{RC}Q = \frac{E}{R}.$$

The integrating factor is $e^{\int dt/(RC)} = e^{t/RC}$, so

$$\frac{d}{dt}\left(Qe^{t/RC}\right) = \frac{E(t)}{R}e^{t/RC}, \qquad Q(t) = \frac{e^{-t/RC}}{R}\int_0^t E(s)e^{s/RC}\,ds, \tag{$**$}$$

where the lower limit 0 builds in the condition $Q(0) = 0$.

(a) $E = E_0$:

$$Q = \frac{E_0e^{-t/RC}}{R}\int_0^t e^{s/RC}\,ds = \frac{E_0e^{-t/RC}}{R}\cdot RC\left(e^{t/RC} - 1\right) = E_0C\left(1 - e^{-t/RC}\right).$$

(b) $E = E_0e^{-t}$: the integrand in $(**)$ is $E_0e^{(1/RC-1)s}$.

Case 1, $RC = 1$. The exponent vanishes, and $Q = \dfrac{e^{-t}}{R}E_0t = E_0Cte^{-t}$ (since $1/R = C$).

Case 2, $RC \neq 1$. Then $\dfrac{1}{RC} - 1 = \dfrac{1 - RC}{RC} \neq 0$ and

$$Q = \frac{E_0e^{-t/RC}}{R}\cdot\frac{RC}{1-RC}\left(e^{(1-RC)t/RC} - 1\right) = \frac{E_0C}{1-RC}\left(e^{-t} - e^{-t/RC}\right) = \frac{E_0C}{RC-1}\left(e^{-t/RC} - e^{-t}\right).$$

(c) $E = E_0\cos\omega t$. Write $a = 1/(RC)$. We need

$$\int e^{as}\cos\omega s\,ds = \frac{e^{as}(a\cos\omega s + \omega\sin\omega s)}{a^2 + \omega^2},$$

which is verified by differentiation: the derivative of the right side is $\frac{e^{as}}{a^2+\omega^2}[a^2\cos\omega s + a\omega\sin\omega s - a\omega\sin\omega s + \omega^2\cos\omega s] = e^{as}\cos\omega s$. Hence

$$\int_0^t e^{as}\cos\omega s\,\mathrm{d}s = \frac{e^{at}(a\cos\omega t + \omega\sin\omega t) - a}{a^2+\omega^2},$$

and by $(**)$

$$Q = \frac{E_0}{R(a^2+\omega^2)}\big(a\cos\omega t + \omega\sin\omega t - ae^{-at}\big).$$

Now $a^2+\omega^2 = \dfrac{1+R^2C^2\omega^2}{R^2C^2}$, so $\dfrac{1}{R(a^2+\omega^2)} = \dfrac{RC^2}{R^2C^2\omega^2+1}$, and multiplying in $a = 1/(RC)$:

$$\begin{aligned} Q &= \frac{RC^2E_0}{R^2C^2\omega^2+1}\cdot\frac{1}{RC}\big(\cos\omega t + RC\omega\sin\omega t - e^{-t/RC}\big) \\ &= \frac{E_0C}{R^2C^2\omega^2+1}\big(RC\omega\sin\omega t + \cos\omega t - e^{-t/RC}\big). \end{aligned}$$

In every case $Q(0) = 0$, as is evident by inspection.

Answer. (a) $Q = E_0C(1-e^{-t/RC})$. (b) If $RC = 1$: $Q = E_0Cte^{-t}$; if $RC \neq 1$: $Q = \dfrac{E_0C}{RC-1}\big(e^{-t/RC} - e^{-t}\big)$.
(c) $Q = \dfrac{E_0C}{R^2C^2\omega^2+1}\big(RC\omega\sin\omega t + \cos\omega t - e^{-t/RC}\big)$.

Problem 13-5 (Simmons)

Solution. With $R = 0$ and $E = 0$, equation (1) becomes

$$L\frac{dI}{dt} + \frac{1}{C}Q = 0, \qquad I = \frac{dQ}{dt}, \qquad \text{i.e.} \qquad \frac{d^2Q}{dt^2} = -\omega_0^2Q, \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

This is a second order equation with the independent variable t missing, so we reduce its order as in Section 11, taking Q as the independent variable: $\dfrac{dI}{dt} = \dfrac{dI}{dQ}\dfrac{dQ}{dt} = I\dfrac{dI}{dQ}$. Then

$$LI\frac{dI}{dQ} = -\frac{Q}{C} \implies \frac{L}{2}I^2 = -\frac{Q^2}{2C} + \text{const.}$$

At $t = 0$, $Q = Q_0$ and $I = 0$, so the constant is $Q_0^2/(2C)$, and

$$I^2 = \frac{Q_0^2 - Q^2}{LC} = \omega_0^2(Q_0^2 - Q^2). \tag{i}$$

(This is conservation of energy: $\frac{1}{2}LI^2 + Q^2/(2C) = Q_0^2/(2C)$.)
Suppose first $Q_0 > 0$. At $t = 0$, $dI/dt = -Q_0/(LC) < 0$, so for small $t > 0$ the current is negative and Q decreases from Q_0; by (i), as long as $I < 0$,

$$\frac{dQ}{dt} = -\omega_0\sqrt{Q_0^2 - Q^2} \implies \frac{-dQ}{\sqrt{Q_0^2-Q^2}} = \omega_0\,\mathrm{d}t \implies \cos^{-1}\frac{Q}{Q_0} = \omega_0 t + c.$$

At $t = 0$, $\cos^{-1}1 = 0$, so $c = 0$ and $Q = Q_0\cos\omega_0 t$ (for $0 \le \omega_0 t \le \pi$, the range of $\cos^{-1}$).

The formula holds for all t, and for every Q_0. The function $Q_0 \cos \omega_0 t$ satisfies $Q'' = -\omega_0^2 Q$ for all t, with $Q(0) = Q_0$, $Q'(0) = 0$. It is the only such function: if Q is any solution with these initial values, then $D = Q - Q_0 \cos \omega_0 t$ satisfies $D'' + \omega_0^2 D = 0$, $D(0) = D'(0) = 0$, and

$$\frac{d}{dt}\left[(D')^2 + \omega_0^2 D^2\right] = 2D'\left(D'' + \omega_0^2 D\right) = 0,$$

so $(D')^2 + \omega_0^2 D^2$ is constant, equal to its value 0 at $t = 0$; hence $D \equiv 0$. This argument does not use the sign of Q_0, so the result holds for $Q_0 \leq 0$ as well. Therefore

$$Q = Q_0 \cos \frac{t}{\sqrt{LC}}, \qquad I = \frac{dQ}{dt} = -\frac{Q_0}{\sqrt{LC}} \sin \frac{t}{\sqrt{LC}}.$$

The charge and current oscillate with period $2\pi\sqrt{LC}$ and never die out, since there is no resistance to dissipate energy.

Answer. $Q = Q_0 \cos\left(t/\sqrt{LC}\right)$, $\quad I = -\dfrac{Q_0}{\sqrt{LC}} \sin\left(t/\sqrt{LC}\right)$.

Miscellaneous Problems for Chapter 2

The text introduces this set as follows (paraphrased): the differential equations below are representatives of all the types discussed in Chapter 2, given in random order, and many of them can be solved by more than one method. They are meant as practice in *recognizing* which method or methods apply to a given equation, without the hint provided by the title of the section in which an equation appears. Accordingly, for each equation we first identify its type and the method used, and then solve it completely.

Throughout, c, c_1, c_2 denote arbitrary constants, which may be renamed from line to line (e.g. $2c$ or e^c is again written c). For an equation written as $M\,dx + N\,dy = 0$, *exact* means $\partial M/\partial y = \partial N/\partial x$ (Section 8), and we use the integrating-factor tests of Section 9: if $(\partial M/\partial y - \partial N/\partial x)/N = g(x)$ then $\mu = e^{\int g(x)\,dx}$ is an integrating factor, and if $(\partial M/\partial y - \partial N/\partial x)/(-M) = h(y)$ then $\mu = e^{\int h(y)\,dy}$ is one.

Chapter 2, Miscellaneous Problem 1 (Simmons)

Solution. *Type:* second order with the independent variable x missing; reduction of order (Section 11).

Put $y' = p$; then $y'' = \dfrac{dp}{dx} = \dfrac{dp}{dy}\dfrac{dy}{dx} = p\dfrac{dp}{dy}$, and the equation becomes

$$yp\frac{dp}{dy} = p^2.$$

If $p \equiv 0$, then y is constant; this is included below ($c_1 = 0$). Otherwise, for $p \neq 0$, $y \neq 0$, separate variables:

$$\frac{dp}{p} = \frac{dy}{y} \implies \log|p| = \log|y| + \text{const} \implies p = c_1 y,$$

that is, $\dfrac{dy}{dx} = c_1 y$. Separating again, $\dfrac{dy}{y} = c_1\,dx$, so $\log|y| = c_1 x + \text{const}$ and

$$y = c_2 e^{c_1 x}.$$

The solution $y \equiv 0$ (lost when dividing by y) is the case $c_2 = 0$, and the constants are the case $c_1 = 0$.

Check. $y' = c_1 c_2 e^{c_1 x}$, $y'' = c_1^2 c_2 e^{c_1 x}$, so $yy'' = c_1^2 c_2^2 e^{2c_1 x} = (y')^2$.

Answer. $y = c_2 e^{c_1 x}$.

Chapter 2, Miscellaneous Problem 2 (Simmons)

Solution. *Type:* not linear in y, but *linear in* x when x is regarded as a function of y (Section 10).

Clearly $y \equiv 0$ is a solution (both sides vanish). For $y \neq 0$, invert:

$$\frac{dx}{dy} = \frac{1 - xy}{y^2} = \frac{1}{y^2} - \frac{x}{y} \qquad \implies \qquad \frac{dx}{dy} + \frac{1}{y}x = \frac{1}{y^2}.$$

An integrating factor is $e^{\int dy/y} = |y|$; we may use y itself. Multiplying by y,

$$y\frac{dx}{dy} + x = \frac{1}{y}, \qquad \text{i.e.} \qquad \frac{d}{dy}(xy) = \frac{1}{y},$$

so

$$xy = \log|y| + c.$$

(On the half-plane $y > 0$ considered in the book this reads $xy = \log y + c$.) The singular solution $y \equiv 0$ is not contained in this family.

Check. Differentiating $xy = \log y + c$ implicitly: $y + xy' = y'/y$, so $y'(1/y - x) = y$, i.e. $y'(1 - xy) = y^2$.

Answer. $xy = \log y + c$ (with $\log|y|$ for $y < 0$); also $y \equiv 0$.

Chapter 2, Miscellaneous Problem 3 (Simmons)

Solution. *Type:* coefficients linear in x, y with nonparallel lines; reduces to a homogeneous equation by translating the origin (Section 7, Problem 4(a) there). (It is not exact: $\partial M/\partial y = 3 \neq -3 = \partial N/\partial x$.)

The lines $2x + 3y + 1 = 0$ and $-3x + 2y + 5 = 0$ intersect where (multiplying the first by 2, the second by 3, and subtracting) $13x - 13 = 0$, i.e. $x = 1$, and then $3y = -3$, $y = -1$. Put $x = X + 1$, $y = Y - 1$ (so $dx = dX$, $dy = dY$):

$$2x + 3y + 1 = 2X + 3Y, \qquad 2y - 3x + 5 = 2Y - 3X,$$

and the equation becomes the homogeneous equation

$$(2X + 3Y)\,dX + (2Y - 3X)\,dY = 0.$$

Put $Y = vX$, $dY = v\,dX + X\,dv$:

$$\begin{aligned}
(2 + 3v)\,dX + (2v - 3)(v\,dX + X\,dv) &= 0 \qquad\qquad (\text{divide by } X)\\
(2 + 3v + 2v^2 - 3v)\,dX + X(2v - 3)\,dv &= 0\\
2(1 + v^2)\,dX + X(2v - 3)\,dv &= 0\\
\frac{dX}{X} + \frac{2v - 3}{2(1 + v^2)}\,dv &= 0.
\end{aligned}$$

Since $\displaystyle\int \frac{2v - 3}{2(1 + v^2)}\,dv = \frac{1}{2}\int \frac{2v}{1 + v^2}\,dv - \frac{3}{2}\int \frac{dv}{1 + v^2} = \frac{1}{2}\log(1 + v^2) - \frac{3}{2}\tan^{-1} v$, integration gives

$$\log|X| + \tfrac{1}{2}\log(1 + v^2) - \tfrac{3}{2}\tan^{-1} v = c.$$

Multiplying by 2 and using $X^2(1 + v^2) = X^2 + Y^2$:

$$\log(X^2 + Y^2) - 3\tan^{-1}\frac{Y}{X} = c.$$

Returning to $X = x - 1$, $Y = y + 1$:

$$3\tan^{-1}\frac{y + 1}{x - 1} = \log\left[(y + 1)^2 + (x - 1)^2\right] + c.$$

Answer. $\displaystyle 3\tan^{-1}\frac{y + 1}{x - 1} = \log\left[(y + 1)^2 + (x - 1)^2\right] + c.$

Chapter 2, Miscellaneous Problem 4 (Simmons)

Solution. *Type:* homogeneous (Section 7): the right side of $y' = \sqrt{x^2+y^2}/x$ is unchanged when x, y are replaced by tx, ty ($t > 0$).

First let $x > 0$. Then $\sqrt{x^2+y^2} = x\sqrt{1+(y/x)^2}$ and the equation is $y' = \sqrt{1+(y/x)^2}$. Put $y = vx$, $y' = v + xv'$:

$$v + x\frac{dv}{dx} = \sqrt{1+v^2} \implies \frac{dv}{\sqrt{1+v^2}-v} = \frac{dx}{x}.$$

(Note $\sqrt{1+v^2} - v > 0$ for all v, so no division by zero occurs.) Rationalizing, $\dfrac{1}{\sqrt{1+v^2}-v} = \dfrac{\sqrt{1+v^2}+v}{(1+v^2)-v^2} = \sqrt{1+v^2}+v$, so

$$\int \left(\sqrt{1+v^2}+v\right) dv = \log x + c.$$

Using the standard integral (verified by differentiation)

$$\int \sqrt{1+v^2}\,dv = \tfrac{1}{2}\left[v\sqrt{1+v^2} + \log\left(v+\sqrt{1+v^2}\right)\right],$$

we get

$$\tfrac{1}{2}v\sqrt{1+v^2} + \tfrac{1}{2}\log\left(v+\sqrt{1+v^2}\right) + \tfrac{1}{2}v^2 = \log x + c,$$

or, after multiplying by 2,

$$v\sqrt{1+v^2} + \log\left(v+\sqrt{1+v^2}\right) + v^2 = 2\log x + c.$$

Now substitute $v = y/x$. For $x > 0$ we have $\sqrt{1+v^2} = \sqrt{x^2+y^2}/x$ and

$$v + \sqrt{1+v^2} = \frac{y+\sqrt{x^2+y^2}}{x},$$

so

$$\frac{y\sqrt{x^2+y^2}}{x^2} + \log\left(y+\sqrt{x^2+y^2}\right) - \log x + \frac{y^2}{x^2} = 2\log x + c.$$

Multiplying by x^2 and collecting the logarithms of x:

$$y\sqrt{x^2+y^2} + x^2\log\left(y+\sqrt{x^2+y^2}\right) + y^2 = 3x^2\log x + cx^2.$$

For $x < 0$: if $y(x)$ is a solution there, then $w(s) = y(-s)$, $s = -x > 0$, satisfies $s\,dw/ds = (-x)(-y') = xy' = \sqrt{s^2+w^2}$, the same equation; so the same formula holds with $\log x$ replaced by $\log|x|$ (note $x^2 = s^2$). In all cases $y + \sqrt{x^2+y^2} > 0$ for $x \neq 0$, so the logarithm is defined.

Answer. $y\sqrt{x^2+y^2} + x^2\log\left(y+\sqrt{x^2+y^2}\right) + y^2 = 3x^2\log x + cx^2$ ($\log|x|$ if $x < 0$).

Chapter 2, Miscellaneous Problem 5 (Simmons)

Solution. *Type:* Bernoulli equation in x as a function of y (Section 10).

Both $x \equiv 0$ and $y \equiv 0$ are solutions (each makes both sides vanish). For $y \neq 0$,

$$\frac{dx}{dy} = \frac{x^3 - xy}{y^2} \qquad \Longrightarrow \qquad \frac{dx}{dy} + \frac{1}{y}x = \frac{1}{y^2}x^3,$$

a Bernoulli equation with $n = 3$. For $x \neq 0$ put $z = x^{1-3} = x^{-2}$, so $dz/dy = -2x^{-3}\,dx/dy$. Multiplying the equation by $-2x^{-3}$:

$$-2x^{-3}\frac{dx}{dy} - \frac{2}{y}x^{-2} = -\frac{2}{y^2} \qquad \Longrightarrow \qquad \frac{dz}{dy} - \frac{2}{y}z = -\frac{2}{y^2}.$$

This is linear; an integrating factor is $e^{-\int 2\,dy/y} = y^{-2}$:

$$\frac{d}{dy}\left(\frac{z}{y^2}\right) = -\frac{2}{y^4} \implies \frac{z}{y^2} = \frac{2}{3y^3} + c \implies \frac{1}{x^2} = z = \frac{2}{3y} + cy^2.$$

Multiplying by $3x^2y$ and renaming $3c$ as c:

$$3y = 2x^2 + cx^2y^3.$$

Neither line $x \equiv 0$ nor $y \equiv 0$ belongs to this family (for $x = 0$ the relation forces $y = 0$, and for $y = 0$ it forces $x = 0$), so both are additional solutions, lost when we divided by y^2 and by x^3.

Check. Differentiate $3y = 2x^2 + cx^2y^3$ with respect to y (with $x = x(y)$) and eliminate $c = (3y - 2x^2)/(x^2y^3)$: $3 = 4xx' + 2cxy^3x' + 3cx^2y^2$ gives, after simplification, $y^2x' = x^3 - xy$. (Verified symbolically in the script.)

Answer. $3y = 2x^2 + cx^2y^3$; also $x \equiv 0$ and $y \equiv 0$.

Chapter 2, Miscellaneous Problem 6 (Simmons)

Solution. *Type:* integrating factor found by grouping terms (Section 9).

Here $M = x^2y^3 + y$, $N = -(x^3y^2 - x)$, and $\partial M/\partial y = 3x^2y^2 + 1 \neq -3x^2y^2 + 1 = \partial N/\partial x$, so the equation is not exact. Regroup it as

$$x^2y^2(y\,dx - x\,dy) + (y\,dx + x\,dy) = 0.$$

Dividing by x^3y^3 (for $x \neq 0$, $y \neq 0$):

$$\frac{y\,dx - x\,dy}{xy} + \frac{y\,dx + x\,dy}{(xy)^3} = 0.$$

Now $\dfrac{y\,dx - x\,dy}{xy} = \dfrac{dx}{x} - \dfrac{dy}{y} = d\log\left|\dfrac{x}{y}\right|$ and $\dfrac{y\,dx + x\,dy}{(xy)^3} = \dfrac{d(xy)}{(xy)^3} = d\left(-\dfrac{1}{2x^2y^2}\right)$. Hence the equation is $d\left[\log|x/y| - 1/(2x^2y^2)\right] = 0$ and

$$\log\left|\frac{x}{y}\right| - \frac{1}{2x^2y^2} = c, \qquad \text{i.e.} \qquad -\frac{1}{2x^2y^2} = \log\left|\frac{y}{x}\right| + c.$$

(One checks that $\mu = 1/(xy)^3$ makes the equation exact: $\partial(\mu M)/\partial y = -2/(x^3y^3) = \partial(\mu N)/\partial x$.) The lines $x \equiv 0$ and $y \equiv 0$, excluded by the division, are also solutions of the original equation.

Answer. $-\dfrac{1}{2x^2y^2} = \log\dfrac{y}{x} + c$; also $x \equiv 0$, $y \equiv 0$.

Chapter 2, Miscellaneous Problem 7 (Simmons)

Solution. *Type:* second order with x missing, so reduction of order applies (Section 11); the quickest route, however, is to observe that the left side is an exact derivative.

Since $(yy')' = yy'' + (y')^2$ and $(y^2)' = 2yy'$, the equation reads

$$\frac{d}{dx}(yy' - y^2) = 0 \qquad \Longrightarrow \qquad yy' - y^2 = k$$

for a constant k. With $u = y^2$ we have $yy' = \frac{1}{2}u'$, so $u' - 2u = 2k$, a linear equation. With integrating factor e^{-2x}:

$$\left(ue^{-2x}\right)' = 2ke^{-2x} \implies ue^{-2x} = -ke^{-2x} + c_2 \implies y^2 = c_2e^{2x} + c_1 \quad (c_1 = -k).$$

(Reduction of order gives the same: with $y' = p$, $y'' = p\,dp/dy$, the equation is $yp\,dp/dy + p^2 - 2yp = 0$; for $p \neq 0$, $dp/dy + p/y = 2$, i.e. $(yp)' = 2y$ with respect to y, so $yp = y^2 + k$, which is the first integral above.)

Check. From $y^2 = c_2e^{2x} + c_1$: $yy' = c_2e^{2x}$, $(yy')' = yy'' + (y')^2 = 2c_2e^{2x}$, and $2yy' = 2c_2e^{2x}$; the difference is 0.

Answer. $y^2 = c_2e^{2x} + c_1$.

Chapter 2, Miscellaneous Problem 8 (Simmons)

Solution. *Type:* exact (Section 8); equivalently linear, $xy' + y = x\cos x$.

The left side is $d(xy)$, so $d(xy) = x\cos x\,dx$. Integrating by parts ($u = x$, $dv = \cos x\,dx$): $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x$. Hence

$$xy = x\sin x + \cos x + c.$$

Answer. $xy = x\sin x + \cos x + c$.

Chapter 2, Miscellaneous Problem 9 (Simmons)

Solution. *Type:* homogeneous (Section 7): written as $y^2\,dx + (x^2 - xy)\,dy = 0$, both coefficients are homogeneous of degree 2. (It is also a Bernoulli equation for x as a function of y.)

The lines $y \equiv 0$ and $x \equiv 0$ are solutions (each makes both sides vanish). Since the answer is most simply expressed with x in terms of y, put $x = uy$, $dx = u\,dy + y\,du$:

$$\begin{aligned} y^2(u\,dy + y\,du) + (u^2y^2 - uy^2)\,dy &= 0 && \text{(divide by } y^2\text{)} \\ u\,dy + y\,du + (u^2 - u)\,dy &= 0 \\ u^2\,dy + y\,du &= 0 \\ \frac{du}{u^2} + \frac{dy}{y} &= 0 && (u \neq 0,\ y \neq 0). \end{aligned}$$

Integrating, $-\dfrac{1}{u} + \log|y| = c$, i.e. $\dfrac{y}{x} = \log|y| - c$. Multiplying by x and renaming $-c$ as c:

$$y = x\log|y| + cx.$$

Check. Differentiate $y = x\log y + cx$: $y' = \log y + xy'/y + c$. Eliminating $c = (y - x\log y)/x$ gives $y' = y/x + xy'/y$, i.e. $y'(xy - x^2) = y^2$, which is $xy\,y' = x^2y' + y^2$.

Answer. $y = x\log y + cx$; also $x \equiv 0$, $y \equiv 0$.

Chapter 2, Miscellaneous Problem 10 (Simmons)

Solution. *Type:* exact (Section 8).

In differential form: $(ye^x - 2xy^3)\,dx + (e^x - 3x^2y^2)\,dy = 0$, with

$$\frac{\partial M}{\partial y} = e^x - 6xy^2 = \frac{\partial N}{\partial x}.$$

We seek f with $f_x = ye^x - 2xy^3$: $f = ye^x - x^2y^3 + g(y)$. Then $f_y = e^x - 3x^2y^2 + g'(y)$ must equal $N = e^x - 3x^2y^2$, so $g' = 0$, and g may be taken 0. The solution is $f = c$:

$$ye^x - x^2y^3 = c.$$

Answer. $ye^x - x^2y^3 = c$.

Chapter 2, Miscellaneous Problem 11 (Simmons)

Solution. *Type:* second order with the dependent variable y missing; reduction of order (Section 11).

Put $p = y'$, $y'' = dp/dx$: $\dfrac{dp}{dx} = -2xp^2$. If $p \equiv 0$, then $y = c_2$ (constant). Otherwise separate variables:

$$-\frac{dp}{p^2} = 2x\,dx \implies \frac{1}{p} = x^2 + k \implies \frac{dy}{dx} = \frac{1}{x^2+k},$$

k a constant. The integration depends on the sign of k.

$k > 0$: write $k = 1/c_1^2$ ($c_1 \neq 0$). Then $\dfrac{dy}{dx} = \dfrac{c_1^2}{1 + c_1^2x^2}$, and with $u = c_1x$,

$$y = c_1 \tan^{-1}(c_1x) - c_2, \qquad \text{i.e.} \qquad c_1 \tan^{-1} c_1x = y + c_2.$$

$k = 0$: $y' = 1/x^2$, $y = -\dfrac{1}{x} + c_2$.

$k = -a^2 < 0$ ($a > 0$): by partial fractions $\dfrac{1}{x^2 - a^2} = \dfrac{1}{2a}\left(\dfrac{1}{x-a} - \dfrac{1}{x+a}\right)$, so $y = \dfrac{1}{2a}\log\left|\dfrac{x-a}{x+a}\right| + c_2$.

Remark. The book's answer $c_1 \tan^{-1} c_1x = y + c_2$ is the case $k > 0$ (the one in which the solution is defined for all x); the other two families and the constant solutions also satisfy the equation, as is checked by direct substitution in the verification script.

Answer. $c_1 \tan^{-1} c_1x = y + c_2$; also $y = c_2 - 1/x$, $y = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c_2$, and $y = c_2$.

Chapter 2, Miscellaneous Problem 12 (Simmons)

Solution. *Type:* linear (Section 10).

For $x \neq 0$: $\dfrac{dy}{dx} = x + \dfrac{y}{x}$, i.e. $y' - \dfrac{1}{x}y = x$. An integrating factor is $e^{-\int dx/x} = 1/x$ (up to sign), and

$$\frac{1}{x}y' - \frac{1}{x^2}y = 1 \implies \frac{d}{dx}\left(\frac{y}{x}\right) = 1 \implies \frac{y}{x} = x + c \implies y = x^2 + cx.$$

(The line $x \equiv 0$ is also an integral curve of the differential form: $dx = 0$ and the right side $x\,dy$ vanishes there.)

Answer. $y = x^2 + cx$.

Chapter 2, Miscellaneous Problem 13 (Simmons)

Solution. *Type:* linear (Section 10); the left side is already the exact derivative $(xy)'$.

$(xy)' = x^2 \cos x$. Integrating by parts twice,

$$\begin{aligned}
\int x^2 \cos x\,dx &= x^2 \sin x - \int 2x \sin x\,dx && (u = x^2,\ dv = \cos x\,dx)\\
&= x^2 \sin x - \left(-2x\cos x + \int 2\cos x\,dx\right) && (u = 2x,\ dv = \sin x\,dx)\\
&= x^2 \sin x + 2x\cos x - 2\sin x.
\end{aligned}$$

Hence $xy = x^2 \sin x + 2x\cos x - 2\sin x + c$, and dividing by x,

$$y = x\sin x + 2\cos x - 2x^{-1}\sin x + cx^{-1}.$$

Answer. $y = x\sin x + 2\cos x - 2x^{-1}\sin x + cx^{-1}$.

Chapter 2, Miscellaneous Problem 14 (Simmons)

Solution. *Type:* coefficients linear in x, y with *parallel* lines ($6x + 4y = 2(3x + 2y)$); substitution $z = 3x + 2y$ (Section 7, Problem 4(b) there). It is not exact ($4 \neq 3$).

Put $z = 3x + 2y$, so $dz = 3\,dx + 2\,dy$, $dy = \frac{1}{2}(dz - 3\,dx)$:

$$\begin{aligned}
(2z + 3)\,dx + \tfrac{1}{2}(z + 2)(dz - 3\,dx) &= 0 && (\times 2)\\
(4z + 6)\,dx + (z + 2)\,dz - (3z + 6)\,dx &= 0\\
z\,dx + (z + 2)\,dz &= 0\\
dx + \left(1 + \frac{2}{z}\right)dz &= 0 && (z \neq 0).
\end{aligned}$$

Integrating, $x + z + 2\log|z| = c$, i.e.

$$(3x + 2y) + \log(3x + 2y)^2 + x = c.$$

Lost solution. We divided by z. The line $z = 0$, i.e. $y = -\frac{3}{2}x$, does satisfy the equation: with $dy = -\frac{3}{2}\,dx$, $(6x - 6x + 3)\,dx + (3x - 3x + 2)(-\frac{3}{2})\,dx = 3\,dx - 3\,dx = 0$. So $3x + 2y = 0$ is an additional solution, not in the family.

Answer. $(3x + 2y) + \log(3x + 2y)^2 + x = c$; also the line $3x + 2y = 0$.

Chapter 2, Miscellaneous Problem 15 (Simmons)

Solution. *Type:* exact (Section 8).

Write it as $[\cos(x+y) - x\sin(x+y)]\,dx - x\sin(x+y)\,dy = 0$. Then

$$\frac{\partial M}{\partial y} = -\sin(x+y) - x\cos(x+y) = \frac{\partial N}{\partial x},$$

so it is exact. In fact $d\big[x\cos(x+y)\big] = \cos(x+y)\,dx - x\sin(x+y)(dx+dy)$, which is precisely the left side. (Systematically: $f_y = -x\sin(x+y)$ gives $f = x\cos(x+y) + g(x)$; then $f_x = \cos(x+y) - x\sin(x+y) + g'(x) = M$ forces $g' = 0$.) Hence

$$x\cos(x+y) = c.$$

Answer. $x\cos(x+y) = c$.

Chapter 2, Miscellaneous Problem 16 (Simmons)

Solution. *Type:* second order with y missing; reduction of order (Section 11), leading to a linear first order equation.

Put $p = y'$: $x^2p' + xp = 1$. For $x \neq 0$ divide by x: $xp' + p = 1/x$, i.e. $(xp)' = 1/x$. Hence $xp = \log|x| + c_1$, and

$$y' = \frac{\log|x|}{x} + \frac{c_1}{x} \implies y = \tfrac{1}{2}(\log|x|)^2 + c_1\log|x| + c_2,$$

using $\int \frac{\log x}{x}\,dx = \frac{1}{2}(\log x)^2$ (substitute $u = \log x$). On $x > 0$ this is $y = \frac{1}{2}(\log x)^2 + c_1\log x + c_2$.

Answer. $y = \frac{1}{2}(\log x)^2 + c_1\log x + c_2$.

Chapter 2, Miscellaneous Problem 17 (Simmons)

Solution. *Type:* exact (Section 8).

$\frac{\partial M}{\partial y} = 2ye^{xy} + xy^2e^{xy}$ and $\frac{\partial N}{\partial x} = ye^{xy} + (ye^{xy} + xy^2e^{xy}) = 2ye^{xy} + xy^2e^{xy}$: exact. From $f_y = e^{xy} + xye^{xy} = \frac{\partial}{\partial y}(ye^{xy})$ we get $f = ye^{xy} + g(x)$. Then $f_x = y^2e^{xy} + g'(x) = M$ requires $g'(x) = \cos x$, $g = \sin x$. Hence

$$ye^{xy} + \sin x = c.$$

Answer. $ye^{xy} + \sin x = c$.

Chapter 2, Miscellaneous Problem 18 (Simmons)

Solution. *Type:* reducible to a separable equation by the substitution $z = x - y$.

The equation requires $x - y > 0$. Note $\log(x-y) \neq 0$ along any solution, since where $\log(x-y) = 0$ the equation would read $0 = 1$. So $y' = 1 + \dfrac{1}{\log(x-y)}$. With $z = x - y > 0$, $z' = 1 - y' =$

$-\dfrac{1}{\log z}$, so

$$\log z\,dz = -dx \implies z\log z - z = -x + c,$$

using $\int \log z\,dz = z\log z - z$ (integration by parts). Substituting $z = x - y$: $(x-y)\log(x-y) - x + y = -x + c$, i.e.

$$(x-y)\log(x-y) = c - y.$$

Answer. $(x-y)\log(x-y) = c - y$.

Chapter 2, Miscellaneous Problem 19 (Simmons)

Solution. *Type:* linear (Section 10).

An integrating factor is $e^{\int 2x\,dx} = e^{x^2}$:

$$e^{x^2}y' + 2xe^{x^2}y = 1 \implies \left(ye^{x^2}\right)' = 1 \implies ye^{x^2} = x + c \implies y = xe^{-x^2} + ce^{-x^2}.$$

Answer. $y = xe^{-x^2} + ce^{-x^2}$.

Chapter 2, Miscellaneous Problem 20 (Simmons)

Solution. *Type:* homogeneous (both coefficients are homogeneous of degree 2); it also has an integrating factor depending on x alone (Section 9), which is quicker.

Write $M = y^2 - 3xy - 2x^2$, $N = -(x^2 - xy) = xy - x^2$. Then $\frac{\partial M}{\partial y} = 2y - 3x$, $\frac{\partial N}{\partial x} = y - 2x$, and

$$\frac{\partial M/\partial y - \partial N/\partial x}{N} = \frac{y-x}{x(y-x)} = \frac{1}{x} = g(x), \qquad \mu = e^{\int dx/x} = x.$$

Multiplying by x: $(xy^2 - 3x^2y - 2x^3)\,dx + (x^2y - x^3)\,dy = 0$, which is exact (both mixed partials equal $2xy - 3x^2$). From $f_y = x^2y - x^3$: $f = \frac{1}{2}x^2y^2 - x^3y + g(x)$; then $f_x = xy^2 - 3x^2y + g'(x)$ must equal $xy^2 - 3x^2y - 2x^3$, so $g' = -2x^3$, $g = -\frac{1}{2}x^4$. Thus $\frac{1}{2}x^2y^2 - x^3y - \frac{1}{2}x^4 = \text{const}$, or

$$x^2y^2 - 2x^3y - x^4 = c.$$

(Multiplying by $\mu = x$ introduces no spurious solutions; the line $x \equiv 0$, which is the case $c = 0$ of the family, is indeed an integral curve of the original equation.)

Answer. $x^2y^2 - 2x^3y - x^4 = c$.

Chapter 2, Miscellaneous Problem 21 (Simmons)

Solution. *Type:* linear (Section 10); the left side is already the derivative $[(1+x^2)y]'$.

$\left[(1+x^2)y\right]' = 4x^3$, so $(1+x^2)y = x^4 + c$ and

$$y = \frac{x^4}{1+x^2} + \frac{c}{1+x^2} = x^4(1+x^2)^{-1} + c(1+x^2)^{-1}.$$

Answer. $y = x^4(1+x^2)^{-1} + c(1+x^2)^{-1}$.

Chapter 2, Miscellaneous Problem 22 (Simmons)

Solution. *Type:* exact (Section 8).

Write it as $(e^x \sin y - y \sin xy)\,dx + (e^x \cos y - x \sin xy)\,dy = 0$. Then

$$\frac{\partial M}{\partial y} = e^x \cos y - \sin xy - xy \cos xy = \frac{\partial N}{\partial x}.$$

Note that $d(e^x \sin y) = e^x \sin y\,dx + e^x \cos y\,dy$ and $d(\cos xy) = -\sin xy\,(y\,dx + x\,dy)$, so the equation is $d(e^x \sin y + \cos xy) = 0$:

$$e^x \sin y + \cos xy = c.$$

Answer. $e^x \sin y + \cos xy = c$.

Chapter 2, Miscellaneous Problem 23 (Simmons)

Solution. *Type:* second order with y missing; reduction of order (Section 11).

Put $p = y'$: $(1+x^2)\dfrac{dp}{dx} = -xp$. If $p \equiv 0$, y is constant (the case $c_1 = 0$ below). Otherwise

$$\frac{dp}{p} = -\frac{x\,dx}{1+x^2} \implies \log|p| = -\tfrac{1}{2}\log(1+x^2) + \text{const} \implies p = \frac{c_1}{\sqrt{1+x^2}}.$$

Since $\dfrac{d}{dx}\log\left(x+\sqrt{1+x^2}\right) = \dfrac{1 + x/\sqrt{1+x^2}}{x+\sqrt{1+x^2}} = \dfrac{1}{\sqrt{1+x^2}}$, integration gives

$$y = c_1 \log\left(x+\sqrt{1+x^2}\right) + c_2 \qquad \left(= c_1 \sinh^{-1} x + c_2\right).$$

Answer. $y = c_1 \log\left(x+\sqrt{1+x^2}\right) + c_2$.

Chapter 2, Miscellaneous Problem 24 (Simmons)

Solution. *Type:* exact (Section 8).

Write it as $(2xy - e^y - x)\,dx + (-xe^y - y + x^2)\,dy = 0$:

$$\frac{\partial M}{\partial y} = 2x - e^y = \frac{\partial N}{\partial x}.$$

From $f_x = 2xy - e^y - x$: $f = x^2y - xe^y - \frac{1}{2}x^2 + g(y)$. Then $f_y = x^2 - xe^y + g'(y)$ must equal $-xe^y - y + x^2$, so $g' = -y$, $g = -\frac{1}{2}y^2$. The solution is $x^2y - xe^y - \frac{1}{2}x^2 - \frac{1}{2}y^2 = \text{const}$; multiplying by -2,

$$2xe^y + x^2 + y^2 - 2x^2y = c.$$

Answer. $2xe^y + x^2 + y^2 - 2x^2y = c$.

Chapter 2, Miscellaneous Problem 25 (Simmons)

Solution. *Type:* integrating factor depending on y alone (Section 9).

Write $M = e^x(1+x)$, $N = -(xe^x - ye^y) = -xe^x + ye^y$. Then $\frac{\partial M}{\partial y} = 0$, $\frac{\partial N}{\partial x} = -e^x - xe^x = -e^x(1+x)$, so the equation is not exact, but

$$\frac{\partial M/\partial y - \partial N/\partial x}{-M} = \frac{e^x(1+x)}{-e^x(1+x)} = -1 = h(y), \qquad \mu = e^{\int(-1)\,dy} = e^{-y}.$$

Multiplying by e^{-y}:

$$(1+x)e^{x-y}\,dx + \left(-xe^{x-y} + y\right)dy = 0,$$

which is exact: both mixed partials equal $-(1+x)e^{x-y}$. From $f_y = -xe^{x-y} + y$: $f = xe^{x-y} + \frac{1}{2}y^2 + g(x)$; then $f_x = (1+x)e^{x-y} + g'(x) = (1+x)e^{x-y}$, so $g' = 0$. Hence $xe^{x-y} + \frac{1}{2}y^2 = \text{const}$, or

$$2xe^xe^{-y} + y^2 = c.$$

(Since $e^{-y} \neq 0$, no solutions are gained or lost by multiplying by μ.)

Answer. $2xe^xe^{-y} + y^2 = c$.

Chapter 2, Miscellaneous Problem 26 (Simmons)

Solution. *Type.* Both coefficients are homogeneous of degree 6, so the equation is homogeneous; but it is quicker to treat it as a Bernoulli equation. Solving for y' ($x \neq 0$, $y \neq 0$),

$$y' = \frac{x^2y^4 + x^6}{x^3y^3} = \frac{y}{x} + \frac{x^3}{y^3},$$

which is Bernoulli with $n = -3$. Put $u = y^4$ (i.e. $u = y^{1-n}$). Then $u' = 4y^3y'$, and multiplying the equation by $4y^3$,

$$u' = \frac{4y^4}{x} + 4x^3 = \frac{4}{x}u + 4x^3, \qquad \text{i.e.} \qquad u' - \frac{4}{x}u = 4x^3.$$

This is linear, with integrating factor $e^{-\int 4\,dx/x} = x^{-4}$:

$$\left(\frac{u}{x^4}\right)' = \frac{4x^3}{x^4} = \frac{4}{x},$$
$$\frac{u}{x^4} = 4\log|x| + c = \log x^4 + c.$$

Hence $y^4 = x^4\log x^4 + cx^4$.

Answer. $y^4 - x^4\log x^4 = cx^4$.

Chapter 2, Miscellaneous Problem 27 (Simmons)

Solution. *Type.* Linear first order: $y' - (3\tan x)\,y = 1$. An integrating factor is

$$e^{-\int 3\tan x\,dx} = e^{3\log|\cos x|} = |\cos x|^3,$$

and we may use $\cos^3 x$ (a constant sign change does not matter). Then

$$(y\cos^3 x)' = \cos^3 x.$$

Writing $\cos^3 x = (1-\sin^2 x)\cos x$ and substituting $s = \sin x$, $ds = \cos x\,dx$,

$$\int \cos^3 x\,dx = \int (1-s^2)\,ds = \sin x - \tfrac{1}{3}\sin^3 x + c.$$

Therefore $y\cos^3 x = \sin x - \frac{1}{3}\sin^3 x + c$, or, multiplying by 3 and renaming $3c$ as c,

$$3y\cos^3 x = 3\sin x - \sin^3 x + c.$$

Answer. $3y\cos^3 x = 3\sin x - \sin^3 x + c$.

Chapter 2, Miscellaneous Problem 28 (Simmons)

Solution. *Type.* Homogeneous (the right side is a function of y/x). Put $y = vx$, $y' = v + xv'$:

$$v + xv' = 1 + v - v^2 \quad\Longrightarrow\quad x\frac{dv}{dx} = 1 - v^2.$$

The constant solutions $v = \pm 1$ give $y = x$ and $y = -x$, which are solutions (for $y = \pm x$ both sides equal ± 1). For $v \neq \pm 1$ separate variables and use partial fractions $\frac{1}{1-v^2} = \frac{1}{2}\left(\frac{1}{1+v} + \frac{1}{1-v}\right)$:

$$\begin{aligned}
\int \frac{dv}{1-v^2} &= \int \frac{dx}{x},\\
\tfrac{1}{2}\log\left|\frac{1+v}{1-v}\right| &= \log|x| + c_0,\\
\frac{1+v}{1-v} &= cx^2 \qquad (c = \pm e^{2c_0}).
\end{aligned}$$

Solving for v: $1 + v = cx^2 - cx^2 v$, so $v(1+cx^2) = cx^2 - 1$ and

$$y = xv = \frac{x(cx^2-1)}{cx^2+1}.$$

The value $c = 0$ gives $y = -x$; the solution $y = x$ is the limit $c \to \infty$.

Answer. $y = \dfrac{x(cx^2-1)}{cx^2+1}$ (together with $y = x$, the case $c = \infty$).

Chapter 2, Miscellaneous Problem 29 (Simmons)

Solution. *Type.* Homogeneous (numerator and denominator are both homogeneous of degree 2). Because the exponent is x/y, it is convenient to regard x as the unknown function of y and put

$x = vy$, so that $\frac{dx}{dy} = v + y\frac{dv}{dy}$. Inverting the equation and dividing numerator and denominator by y^2,

$$\frac{dx}{dy} = \frac{y^2 + y^2e^{v^2} + 2x^2e^{v^2}}{2xye^{v^2}} = \frac{1 + e^{v^2} + 2v^2e^{v^2}}{2ve^{v^2}}.$$

Hence

$$y\frac{dv}{dy} = \frac{1 + e^{v^2} + 2v^2e^{v^2}}{2ve^{v^2}} - v = \frac{1 + e^{v^2}}{2ve^{v^2}}.$$

Separating the variables,

$$\int \frac{2ve^{v^2}}{1 + e^{v^2}}\,dv = \int \frac{dy}{y}.$$

On the left the numerator is the derivative of $1 + e^{v^2}$, so

$$\log\left(1 + e^{v^2}\right) = \log|y| + c_0, \qquad \text{i.e.} \qquad 1 + e^{v^2} = cy.$$

With $v = x/y$ this is $1 + e^{(x/y)^2} = cy$.

Answer. $1 + e^{(x/y)^2} = cy$.

Chapter 2, Miscellaneous Problem 30 (Simmons)

Solution. *Type.* This has the form $dy/dx = (a_1x + b_1y + c_1)/(a_2x + b_2y + c_2)$ and could be reduced to a homogeneous equation by translating to the intersection point $(-\frac{2}{5}, -\frac{4}{5})$ of the two lines (Section 7, Problem 4). However, written in differential form,

$$(x + 2y + 2)\,dx + (2x - y)\,dy = 0,$$

it is *exact*: $M = x + 2y + 2$, $N = 2x - y$, and $\frac{\partial M}{\partial y} = 2 = \frac{\partial N}{\partial x}$. We seek F with $F_x = M$, $F_y = N$:

$$F = \int (x + 2y + 2)\,dx = \tfrac{1}{2}x^2 + 2xy + 2x + h(y), \qquad F_y = 2x + h'(y) = 2x - y \;\Rightarrow\; h(y) = -\tfrac{1}{2}y^2.$$

So $\frac{1}{2}x^2 + 2xy + 2x - \frac{1}{2}y^2 = c$, i.e.

$$x^2 + 4xy + 4x - y^2 = c.$$

Remark. The book's answer $(5y + 4)^2 - 4(5y + 4)(5x + 2) - (5x + 2)^2 = c$ (obtained by the translation $x = X - \frac{2}{5}$, $y = Y - \frac{4}{5}$) is the same family: expanding, $(5y + 4)^2 - 4(5y + 4)(5x + 2) - (5x + 2)^2 = -25(x^2 + 4xy + 4x - y^2) - 20$.

Answer. $x^2 + 4xy + 4x - y^2 = c$, equivalently $(5y + 4)^2 - 4(5y + 4)(5x + 2) - (5x + 2)^2 = c$.

Chapter 2, Miscellaneous Problem 31 (Simmons)

Solution. *Type.* Exact: $M = 3x^2 \log y$, $N = x^3/y$ (with $y > 0$), and $\frac{\partial M}{\partial y} = 3x^2/y = \frac{\partial N}{\partial x}$. Indeed the left side is

$$\log y\; d(x^3) + x^3\; d(\log y) = d(x^3 \log y),$$

so the solutions are $x^3 \log y = c$.

Answer. $x^3 \log y = c$.

Chapter 2, Miscellaneous Problem 32 (Simmons)

Solution. *Type.* Exact. Here (in a region where $x/(x+3) > 0$)

$$\frac{\partial M}{\partial y} = \frac{6y}{x^2+3x}, \qquad \frac{\partial N}{\partial x} = 2y\left(\frac{1}{x} - \frac{1}{x+3}\right) = 2y \cdot \frac{3}{x(x+3)} = \frac{6y}{x^2+3x},$$

since $\log \frac{5x}{x+3} = \log 5 + \log x - \log(x+3)$ (up to the signs inside absolute values, which do not affect derivatives). Integrate M in x, using $\frac{1}{x(x+3)} = \frac{1}{3}\left(\frac{1}{x} - \frac{1}{x+3}\right)$:

$$F = 3y^2 \int \frac{dx}{x(x+3)} = y^2 \log \frac{x}{x+3} + h(y).$$

Then

$$F_y = 2y \log \frac{x}{x+3} + h'(y) = 2y \log \frac{5x}{x+3} + 3 \sin y,$$

so $h'(y) = 2y \log 5 + 3 \sin y$ and $h(y) = y^2 \log 5 - 3 \cos y$. Thus $F = y^2 \log \frac{5x}{x+3} - 3 \cos y$, and the solutions are $F = c$.

Answer. $y^2 \log \dfrac{5x}{x+3} - 3 \cos y = c$.

Chapter 2, Miscellaneous Problem 33 (Simmons)

Solution. *Type.* Exact (and also homogeneous). With $M = (y-x)(x+y)^{-3}$, $N = -2x(x+y)^{-3}$,

$$\frac{\partial M}{\partial y} = (x+y)^{-3} - 3(y-x)(x+y)^{-4} = \frac{(x+y) - 3y + 3x}{(x+y)^4} = \frac{4x-2y}{(x+y)^4},$$
$$\frac{\partial N}{\partial x} = -2(x+y)^{-3} + 6x(x+y)^{-4} = \frac{-2x-2y+6x}{(x+y)^4} = \frac{4x-2y}{(x+y)^4}.$$

Integrate N with respect to y:

$$F = -2x \int (x+y)^{-3}\, dy = \frac{x}{(x+y)^2} + g(x).$$

Then $F_x = \dfrac{1}{(x+y)^2} - \dfrac{2x}{(x+y)^3} + g'(x) = \dfrac{y-x}{(x+y)^3} + g'(x)$, which equals M exactly when $g' = 0$. Hence $x/(x+y)^2 = C$, i.e. (with $c = 1/C$, and $c = 0$ giving the solution $x = 0$)

$$x = c(x+y)^2,$$

valid away from the line $x + y = 0$, where the equation is undefined.

Answer. $x = c(x+y)^2$.

Chapter 2, Miscellaneous Problem 34 (Simmons)

Solution. *Type.* Integrating factor by grouping (it is also Bernoulli: $y' + y/x = -y^2$). Regroup as

$$xy^2\,dx + (y\,dx + x\,dy) = 0, \qquad \text{i.e.} \qquad xy^2\,dx + d(xy) = 0.$$

The function $y = 0$ is a solution. For $xy \neq 0$ divide by $x^2y^2 = (xy)^2$:

$$\frac{dx}{x} + \frac{d(xy)}{(xy)^2} = 0 \quad \Longrightarrow \quad d\left(\log|x| - \frac{1}{xy}\right) = 0.$$

Hence $\log|x| - \dfrac{1}{xy} = c$.

Answer. $\log x - \dfrac{1}{xy} = c$ (and the solution $y = 0$).

Chapter 2, Miscellaneous Problem 35 (Simmons)

Solution. *Type.* Second order with the dependent variable y missing: reduction of order. Put $p = y'$, $y'' = p'$:

$$x^2p' = 3xp - 2p^2, \qquad \text{i.e.} \qquad p' - \frac{3}{x}p = -\frac{2}{x^2}p^2,$$

a Bernoulli equation in p ($n = 2$). The solution $p = 0$ gives $y = c$ (constant). For $p \neq 0$ put $w = 1/p$, $w' = -p'/p^2$; dividing the equation by $-p^2$,

$$w' + \frac{3}{x}w = \frac{2}{x^2}.$$

The integrating factor is $e^{\int 3\,dx/x} = x^3$:

$$(x^3w)' = 2x \implies x^3w = x^2 + c_1 \implies p = \frac{x^3}{x^2 + c_1}.$$

Integrate once more, dividing first: $\dfrac{x^3}{x^2 + c_1} = x - \dfrac{c_1x}{x^2 + c_1}$, so (substituting $u = x^2 + c_1$ in the second term)

$$y = \int \left(x - \frac{c_1x}{x^2 + c_1}\right) dx = \frac{1}{2}x^2 - \frac{c_1}{2}\log\left|x^2 + c_1\right| + c_2.$$

Answer. $y = \frac{1}{2}x^2 - \frac{c_1}{2}\log(x^2 + c_1) + c_2$ (and the constant solutions $y = c$).

Chapter 2, Miscellaneous Problem 36 (Simmons)

Solution. *Type.* Exact (also homogeneous). $M = 3x^2y - y^3$, $N = x^3 - 3xy^2$, and $\frac{\partial M}{\partial y} = 3x^2 - 3y^2 = \frac{\partial N}{\partial x}$. Then

$$F = \int (3x^2y - y^3)\,dx = x^3y - xy^3 + h(y), \qquad F_y = x^3 - 3xy^2 + h'(y) = N \;\Rightarrow\; h' = 0.$$

Answer. $x^3y - xy^3 = c$.

Chapter 2, Miscellaneous Problem 37 (Simmons)

Solution. *Type.* Linear. In standard form ($x \neq 0$)

$$y' + \frac{2}{x(x^2+1)}\,y = \frac{(x^2+1)^2}{x}.$$

By partial fractions $\dfrac{2}{x(x^2+1)} = \dfrac{2}{x} - \dfrac{2x}{x^2+1}$, so

$$\int \frac{2\,dx}{x(x^2+1)} = \log x^2 - \log(x^2+1), \qquad \text{integrating factor } \frac{x^2}{x^2+1}.$$

Multiplying through,

$$\left(\frac{x^2}{x^2+1}\,y\right)' = \frac{x^2}{x^2+1}\cdot\frac{(x^2+1)^2}{x} = x(x^2+1), \qquad \frac{x^2 y}{x^2+1} = \frac{(x^2+1)^2}{4} + c,$$

where we integrated $x(x^2+1)$ by the substitution $u = x^2+1$. Multiplying by $4(x^2+1)$ and renaming $4c$ as c,

$$4x^2 y = (x^2+1)^3 + c(x^2+1).$$

Answer. $4x^2y = (x^2+1)^3 + c(x^2+1)$.

Chapter 2, Miscellaneous Problem 38 (Simmons)

Solution. *Type.* Exact (it is also of the type reducible to homogeneous by translation to the intersection $(-1,1)$ of the two lines). In differential form

$$(3x+2y+1)\,dx + (2x+3y-1)\,dy = 0, \qquad \frac{\partial M}{\partial y} = 2 = \frac{\partial N}{\partial x}.$$

Then $F = \frac{3}{2}x^2 + 2xy + x + h(y)$ and $F_y = 2x + h'(y) = 2x + 3y - 1$ gives $h = \frac{3}{2}y^2 - y$. Thus (multiplying by 2)

$$3x^2 + 4xy + 3y^2 + 2x - 2y = c.$$

Remark. The book writes this as $3(y-1)^2 + 4(y-1)(x+1) + 3(x+1)^2 = c$; expanding, that left side equals $3x^2 + 4xy + 3y^2 + 2x - 2y + 2$, so the two forms agree.

Answer. $3x^2 + 4xy + 3y^2 + 2x - 2y = c$, i.e. $3(y-1)^2 + 4(y-1)(x+1) + 3(x+1)^2 = c$.

Chapter 2, Miscellaneous Problem 39 (Simmons)

Solution. *Type.* Exact:

$$\frac{\partial M}{\partial y} = e^{x^2y}\left[x^2(1+2x^2y) + 2x^2\right] = e^{x^2y}(3x^2 + 2x^4y), \qquad \frac{\partial N}{\partial x} = 3x^2 e^{x^2y} + x^3\cdot 2xy\,e^{x^2y},$$

which agree. Integrating N in y: $F = xe^{x^2y} + g(x)$, and $F_x = e^{x^2y} + 2x^2y\,e^{x^2y} + g'(x) = M$ forces $g' = 0$.

Answer. $xe^{x^2y} = c$.

Chapter 2, Miscellaneous Problem 40 (Simmons)

Solution. *Type.* Exact: $\frac{\partial M}{\partial y} = 3x^2e^y = \frac{\partial N}{\partial x}$. Then $F = x^3e^y - x^2 + h(y)$, and $F_y = x^3e^y + h'(y) = x^3e^y - \sin y$ gives $h = \cos y$.

Answer. $x^3e^y - x^2 + \cos y = c$.

Chapter 2, Miscellaneous Problem 41 (Simmons)

Solution. *Type.* Second order with x missing: reduction of order with $p = y'$, $y'' = p\,\dfrac{dp}{dy}$. The equation becomes

$$y^2 p\frac{dp}{dy} + p^3 = 0.$$

If $p = 0$, then $y = c$ (constant), which is a solution. Otherwise $y^2\,dp/dy = -p^2$, and separating,

$$\frac{dp}{p^2} = -\frac{dy}{y^2} \implies -\frac{1}{p} = \frac{1}{y} - c_1 \implies \frac{dx}{dy} = \frac{1}{p} = c_1 - \frac{1}{y}.$$

Integrating with respect to y,

$$x = c_1 y - \log|y| + c_2',$$

which, writing $c_2' = -\log c_2$, is $x = c_1 y - \log c_2 y$.

Answer. $x = c_1 y - \log c_2 y$ (and the constant solutions $y = c$).

Chapter 2, Miscellaneous Problem 42 (Simmons)

Solution. *Type.* Homogeneous, not exact, solved by an integrating factor. Here $M = 3xy + y^2$, $N = 3xy + x^2$,

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = (3x + 2y) - (3y + 2x) = x - y \neq 0.$$

Neither $(M_y - N_x)/N$ nor $(M_y - N_x)/(-M)$ is a function of one variable, but

$$\frac{M_y - N_x}{N - M} = \frac{x - y}{x^2 - y^2} = \frac{1}{x + y}$$

is a function of $z = x + y$ alone. This means there is an integrating factor $\mu(z)$, $z = x + y$: the condition $(\mu M)_y = (\mu N)_x$ reads $\mu'(z)M + \mu M_y = \mu'(z)N + \mu N_x$, i.e.

$$\frac{\mu'(z)}{\mu} = \frac{M_y - N_x}{N - M} = \frac{1}{z}, \qquad \text{so} \qquad \mu = z = x + y.$$

Multiplying the equation by $x + y$,

$$(3x^2y + 4xy^2 + y^3)\,dx + (x^3 + 4x^2y + 3xy^2)\,dy = 0,$$

which is exact ($\partial/\partial y$ of the first and $\partial/\partial x$ of the second coefficient are both $3x^2+8xy+3y^2$). Then

$$F=\int (3x^2y+4xy^2+y^3)\,dx = x^3y+2x^2y^2+xy^3+h(y), \qquad F_y = x^3+4x^2y+3xy^2+h'(y),$$

so $h'=0$ and $F=x^3y+2x^2y^2+xy^3=xy(x+y)^2$.

Remark. Since the equation is homogeneous, $1/(xM+yN)=1/[4xy(x+y)]$ is also an integrating factor (Section 9); it differs from $x+y$ by the factor $1/[4xy(x+y)^2]=1/(4F)$, a function of F, and leads to $\log|xy(x+y)^2|=\text{const}$, the same solutions.

Answer. $xy(x+y)^2=c$.

Chapter 2, Miscellaneous Problem 43 (Simmons)

Solution. *Type.* Homogeneous (also Riccati). With $y=vx$:

$$x^2(v+xv')=x^2+vx^2+v^2x^2 \implies xv'=1+v^2 \implies \int\frac{dv}{1+v^2}=\int\frac{dx}{x}.$$

So $\tan^{-1}v=\log|x|+\log c=\log c|x|$, i.e. $v=\tan(\log cx)$ and $y=x\tan(\log cx)$ (for $x>0$; for $x<0$ replace x by $|x|$ inside the logarithm).

Answer. $y=x\tan(\log cx)$.

Chapter 2, Miscellaneous Problem 44 (Simmons)

Solution. *Type.* Bernoulli with $n=2$ (here $x>0$). Clearly $y=0$ is a solution. For $y\neq 0$ put $w=1/y$, $w'=-y'/y^2$. Dividing the equation by y^2,

$$\frac{xy'}{y^2}+\frac{1}{y}=\log x \implies -xw'+w=\log x \implies w'-\frac{1}{x}w=-\frac{\log x}{x}.$$

The integrating factor is $e^{-\int dx/x}=1/x$:

$$\left(\frac{w}{x}\right)'=-\frac{\log x}{x^2}.$$

Integrating by parts ($u=\log x$, $dv=x^{-2}dx$, $v=-1/x$): $\int\frac{\log x}{x^2}\,dx=-\frac{\log x}{x}+\int\frac{dx}{x^2}=-\frac{\log x}{x}-\frac{1}{x}$. Hence

$$\frac{w}{x}=\frac{\log x}{x}+\frac{1}{x}+c, \qquad w=1+\log x+cx.$$

Answer. $\dfrac{1}{y}=1+\log x+cx$ (and $y=0$).

Chapter 2, Miscellaneous Problem 45 (Simmons)

Solution. *Type.* Exact ($x>-3$, $y>0$). With $M=\cos y/(x+3)$, $N=-\sin y\log(5x+15)+1/y$:

$$\frac{\partial M}{\partial y}=-\frac{\sin y}{x+3}, \qquad \frac{\partial N}{\partial x}=-\sin y\cdot\frac{5}{5x+15}=-\frac{\sin y}{x+3}.$$

Since $\frac{d}{dx}\log(5x+15) = \frac{1}{x+3}$, we may take $F = \cos y\,\log(5x+15) + h(y)$. Then

$$F_y = -\sin y\log(5x+15) + h'(y) = N \implies h'(y) = \frac{1}{y}, \quad h = \log y.$$

Answer. $\cos y\,\log(5x+15) + \log y = c$.

Chapter 2, Miscellaneous Problem 46 (Simmons)

Solution. *Type.* Second order with y missing: reduction of order, $p = y'$, $y'' = p'$:

$$x^2\frac{dp}{dx} = -p^2.$$

The solution $p = 0$ gives $y = c$. For $p \neq 0$,

$$\frac{dp}{p^2} = -\frac{dx}{x^2} \implies -\frac{1}{p} = \frac{1}{x} - c_1 \implies \frac{1}{p} = c_1 - \frac{1}{x} = \frac{c_1x-1}{x}, \qquad p = \frac{x}{c_1x-1}.$$

Case $c_1 = 0$: $p = -x$, so $y = -\frac{1}{2}x^2 + c_2$.

Case $c_1 \neq 0$: dividing, $\dfrac{x}{c_1x-1} = \dfrac{1}{c_1} + \dfrac{1}{c_1(c_1x-1)}$, hence

$$y = \frac{x}{c_1} + \frac{1}{c_1^2}\log|c_1x-1| + c_2', \qquad \text{i.e.} \qquad c_1^2y = c_1x + \log(c_1x-1) + c_2.$$

Remark. The book's answer is printed as $c_1y^2 = c_1x + \log(c_1x-1) + c_2$. This is a misprint for $c_1^2y = \cdots$: for instance with $c_1 = 2$, $c_2 = 5$ the printed relation gives a function that does not satisfy the equation at $x = 3$, whereas $y = [c_1x + \log(c_1x-1) + c_2]/c_1^2$ gives $y' = x/(c_1x-1)$, $y'' = -1/(c_1x-1)^2$ and $x^2y'' + (y')^2 = 0$ identically.

Answer. $c_1^2y = c_1x + \log(c_1x-1) + c_2$; also $y = -\frac{1}{2}x^2 + c_2$ and $y = c$.

Chapter 2, Miscellaneous Problem 47 (Simmons)

Solution. *Type.* Linear in y: $xy' + (x+1)y = 1$, i.e.

$$y' + \left(1 + \frac{1}{x}\right)y = \frac{1}{x}.$$

Integrating factor: $e^{\int(1+1/x)\,dx} = e^{x+\log|x|} = |x|e^x$; we use xe^x. Then

$$(xe^xy)' = xe^x\cdot\frac{1}{x} = e^x \implies xye^x = e^x + c.$$

Answer. $xye^x - e^x = c$.

Chapter 2, Miscellaneous Problem 48 (Simmons)

Solution. *Type.* Bernoulli (also homogeneous): $y' - \frac{2}{x}y = \frac{1}{x^2}y^2$. The function $y = 0$ is a solution. For $y \neq 0$ put $w = 1/y$, $w' = -y'/y^2$; dividing by $-y^2$,

$$w' + \frac{2}{x}w = -\frac{1}{x^2}.$$

With integrating factor x^2: $(x^2 w)' = -1$, so $x^2 w = c - x$ and

$$y = \frac{x^2}{c - x}.$$

Answer. $y = \dfrac{x^2}{c - x}$ (and $y = 0$).

Chapter 2, Miscellaneous Problem 49 (Simmons)

Solution. *Type.* Second order with x missing: $p = y'$, $y'' = p\,dp/dy$:

$$p\frac{dp}{dy} = 2yp^3.$$

The case $p = 0$ gives the constant solutions $y = c$. Otherwise $dp/dy = 2yp^2$, and

$$\frac{dp}{p^2} = 2y\,dy \implies -\frac{1}{p} = y^2 + c_1 \implies \frac{dx}{dy} = \frac{1}{p} = -y^2 - c_1.$$

Integrating, $x = -\frac{1}{3}y^3 - c_1 y + c_2$, i.e.

$$y^3 = 3(c_2 - x - c_1 y).$$

Answer. $y^3 = 3(c_2 - x - c_1 y)$ (and $y = c$).

Chapter 2, Miscellaneous Problem 50 (Simmons)

Solution. *Type.* Linear in x as a function of y. Integrating factor $e^{\int \cot y\,dy} = e^{\log|\sin y|}$; we use $\sin y$:

$$\frac{d}{dy}(x \sin y) = \sec y \sin y = \tan y \implies x \sin y = \log|\sec y| + c.$$

Answer. $x = \csc y\,\big[\log(\sec y) + c\big]$.

Chapter 2, Miscellaneous Problem 51 (Simmons)

Solution. Let $x_1(t)$ and $x_2(t)$ be the pounds of salt in the first and second tanks at time t (minutes). Both tanks keep a constant volume of 50 gal, because inflow and outflow are each 2 gal/min, so the concentrations are $x_1/50$ and $x_2/50$ lb/gal.

First tank. Pure water enters; brine leaves at $2 \cdot \frac{x_1}{50}$ lb/min:

$$\frac{dx_1}{dt} = -\frac{x_1}{25}, \quad x_1(0) = 25 \implies x_1 = 25e^{-t/25}.$$

Second tank. Salt enters at the rate at which it leaves the first tank and leaves at $2 \cdot \frac{x_2}{50}$ lb/min:

$$\frac{dx_2}{dt} = \frac{x_1}{25} - \frac{x_2}{25} = e^{-t/25} - \frac{x_2}{25}, \qquad x_2(0) = 0.$$

This is linear; multiplying by $e^{t/25}$ gives $(x_2 e^{t/25})' = 1$, so $x_2 e^{t/25} = t + c$, and $x_2(0) = 0$ gives $c = 0$:

$$x_2(t) = te^{-t/25}.$$

Then $x_2'(t) = e^{-t/25}\left(1 - \frac{t}{25}\right)$, which is positive for $t < 25$ and negative for $t > 25$. Hence x_2 is greatest at $t = 25$ min, when it contains $x_2(25) = 25/e \approx 9.20$ lb of salt.

Answer. At $t = 25$ minutes (the second tank then holds $25/e \approx 9.20$ lb of salt).

Chapter 2, Miscellaneous Problem 52 (Simmons)

Solution. *Proof.* Let y_1 satisfy $y_1' = p + qy_1 + ry_1^2$. For any differentiable y write $z = y - y_1$. Then

$$\begin{aligned} y' - (p + qy + ry^2) &= y_1' + z' - p - q(y_1 + z) - r(y_1^2 + 2y_1 z + z^2) \\ &= \left[y_1' - p - qy_1 - ry_1^2\right] + z' - qz - 2ry_1 z - rz^2 \\ &= z' - (q + 2ry_1)z - rz^2. \end{aligned}$$

So y is a solution of the Riccati equation if and only if $z = y - y_1$ is a solution of the Bernoulli equation $z' - (q + 2ry_1)z = rz^2$. Consequently every solution of the Riccati equation is $y_1 + z$ for some solution z of the Bernoulli equation, and conversely; i.e. the general solution is $y = y_1 + z$ with z the general solution of the Bernoulli equation.

The example. Here $p = -x^5$, $q = 1/x$, $r = x^3$, and $y_1 = x$ is a solution because $y_1' = 1$ and $\frac{x}{x} + x^3 \cdot x^2 - x^5 = 1$. The Bernoulli equation is

$$z' - \left(\frac{1}{x} + 2x^4\right) z = x^3 z^2.$$

Besides $z = 0$, put $w = 1/z$, $w' = -z'/z^2$; dividing by $-z^2$,

$$w' + \left(\frac{1}{x} + 2x^4\right) w = -x^3.$$

The integrating factor is $e^{\int (1/x + 2x^4)\,dx} = xe^{2x^5/5}$, so

$$\left(xe^{2x^5/5} w\right)' = -x^4 e^{2x^5/5} \implies xe^{2x^5/5} w = -\tfrac{1}{2} e^{2x^5/5} + C,$$

(the integral by the substitution $u = \frac{2}{5}x^5$, $du = 2x^4\,dx$). Hence

$$\frac{1}{z} = w = \frac{2Ce^{-2x^5/5} - 1}{2x}, \qquad y = x + \frac{2x}{Ke^{-2x^5/5} - 1} = x\,\frac{Ke^{-2x^5/5} + 1}{Ke^{-2x^5/5} - 1} \quad (K = 2C).$$

To put this in a neater implicit form, let $A = Ke^{-2x^5/5}$, so $y/x = (A+1)/(A-1)$; then

$$\frac{y - x}{y + x} = \frac{y/x - 1}{y/x + 1} = \frac{(A+1) - (A-1)}{(A+1) + (A-1)} = \frac{1}{A} = \frac{1}{K} e^{2x^5/5}.$$

Writing $c = 1/K$, the general solution is $ce^{2x^5/5} = \dfrac{y-x}{y+x}$. The value $c = 0$ gives the particular solution $y = x$ (the case $z = 0$), and the solution $y = -x$ corresponds to $c = \infty$ ($K = 0$).

Answer. Proof as above; the general solution is $ce^{2x^5/5} = \dfrac{y-x}{y+x}$, i.e. $y = x\,\dfrac{1+ce^{2x^5/5}}{1-ce^{2x^5/5}}$.

Chapter 2, Miscellaneous Problem 53 (Simmons)

Solution.

(a) Expanding,

$$\frac{dy}{dt} = s + (I-s)y - Iy^2,$$

a Riccati equation with $p = s(t)$, $q = I - s(t)$, $r = -I$. The constant $y_1 = 1$ is a solution, since then both sides of $(*)$ vanish. By Problem 52, $y = 1 + z$, where

$$z' - (q + 2ry_1)z = rz^2, \qquad q + 2ry_1 = I - s - 2I = -(s+I),$$

that is,

$$\frac{dz}{dt} + [s(t) + I]z = -Iz^2.$$

(Directly: substituting $y = 1 + z$ in $(*)$ gives $z' = -z(s + I + Iz)$.)

(b) With $s = at$ the Bernoulli equation is $z' + (at + I)z = -Iz^2$. Apart from $z = 0$ (i.e. $y = 1$), put $w = 1/z$, $w' = -z'/z^2$; dividing by $-z^2$,

$$w' - (at + I)w = I.$$

An integrating factor is $e^{-(\frac{1}{2}at^2+It)}$:

$$\frac{d}{dt}\left[e^{-(\frac{1}{2}at^2+It)}w\right] = Ie^{-(\frac{1}{2}at^2+It)}, \qquad w = e^{\frac{1}{2}at^2+It}\left[I\int e^{-(\frac{1}{2}at^2+It)}\,dt + c\right].$$

Therefore

$$y = 1 + z, \qquad \frac{1}{z} = e^{\frac{1}{2}at^2+It}\left[I\int e^{-(\frac{1}{2}at^2+It)}\,dt + c\right].$$

If the initial proportion is $y(0) = y_0 < 1$, then $1/z(0) = 1/(y_0 - 1)$, and taking the integral from 0 to t fixes the constant:

$$y(t) = 1 + \frac{e^{-(\frac{1}{2}at^2+It)}}{\dfrac{1}{y_0-1} + I\displaystyle\int_0^t e^{-(\frac{1}{2}a\tau^2+I\tau)}\,d\tau}.$$

Answer. (a) $\dfrac{dz}{dt} + [s(t) + I]z = -Iz^2$. (b) $y = 1 + z$, where $\dfrac{1}{z} = e^{\frac{1}{2}at^2+It}\left[I\displaystyle\int e^{-(\frac{1}{2}at^2+It)}\,dt + c\right]$.

Chapter 2, Miscellaneous Problem 54 (Simmons)

Solution.

(a) By Problem 52, $y = y_1 + z$ where z solves $z' - (q + 2ry_1)z = rz^2$. Besides $z = 0$, put $w = 1/z$; as in Problem 52, w satisfies the *linear* equation

$$w' + (q + 2ry_1)w = -r.$$

The general solution of a first order linear equation has the form $w = cu(x) + v(x)$, where $u = e^{-\int (q+2ry_1)\,dx}$ solves the homogeneous equation and v is any particular solution (explicitly, $v = u\int(-r/u)\,\mathrm{d}x$). Hence

$$y = y_1 + \frac{1}{cu + v} = \frac{c\,uy_1 + (vy_1 + 1)}{c\,u + v},$$

which is of the stated form with $f = uy_1$, $g = vy_1 + 1$, $F = u$, $G = v$. (The solution $y = y_1$, i.e. $z = 0$, is the limiting member $c \to \infty$, $y = f/F$.)

(b) Let $y = \dfrac{cf + g}{cF + G}$ with $fG - Fg \neq 0$ (if $fG - Fg \equiv 0$, then $f/F = g/G$ and y does not depend on c, so there is no genuine one-parameter family). To eliminate c, solve for it:

$$y(cF + G) = cf + g \implies c = \frac{g - yG}{yF - f}.$$

Differentiating this identity with respect to x (c is constant) and clearing the denominator,

$$(g' - y'G - yG')(yF - f) - (g - yG)(y'F + yF' - f') = 0.$$

The terms containing y' are

$$y'\left[-G(yF - f) - F(g - yG)\right] = y'(fG - Fg),$$

and the remaining terms are

$$\begin{aligned}&(g' - yG')(yF - f) - (g - yG)(yF' - f')\\ &\quad = y^2(GF' - FG') + y(Fg' + fG' - gF' - Gf') + (gf' - fg').\end{aligned}$$

Hence

$$y' = \frac{(FG' - GF')\,y^2 + (gF' + Gf' - Fg' - fG')\,y + (fg' - gf')}{fG - Fg},$$

which is of the form $y' = p(x) + q(x)y + r(x)y^2$, a Riccati equation, with

$$p = \frac{fg' - gf'}{fG - Fg}, \quad q = \frac{gF' + Gf' - Fg' - fG'}{fG - Fg}, \quad r = \frac{FG' - GF'}{fG - Fg}.$$

Answer. (a) $y = y_1 + 1/(cu + v) = \dfrac{c\,uy_1 + vy_1 + 1}{cu + v}$ with $w = cu + v$ the general solution of $w' + (q + 2ry_1)w = -r$. (b) Eliminating c gives $y' = \dfrac{(FG' - GF')y^2 + (gF' + Gf' - Fg' - fG')y + (fg' - gf')}{fG - Fg}$, a Riccati equation.

Chapter 2, Miscellaneous Problem 55 (Simmons)

Solution. Take the upward direction as positive, and let $v(t)$ and $h(t)$ be the velocity and height of the rocket, with $v(0) = h(0) = 0$. The total mass is

$$m(t) = m_1 + m_2 - at, \qquad 0 \le t \le t_b = \frac{m_2}{a},$$

where t_b is the burnout time (all the fuel m_2 is gone). We use the variable-mass law stated before Problem 55,

$$w\frac{dm}{dt} + F = m\frac{dv}{dt},$$

where w is the velocity, relative to the rocket, of the mass leaving it. The exhaust moves backward (downward) at speed b relative to the rocket, so $w = -b$; also $dm/dt = -a$ and $F = -mg$. Hence

$$(-b)(-a) - mg = m\frac{dv}{dt}, \qquad \text{i.e.} \qquad \frac{dv}{dt} = \frac{ab}{m_1 + m_2 - at} - g.$$

(For the rocket to leave the ground we need $ab > (m_1 + m_2)g$.) Integrating from 0 with $v(0) = 0$,

$$v(t) = -b\log(m_1 + m_2 - at) + b\log(m_1 + m_2) - gt = b\log\frac{m_1 + m_2}{m_1 + m_2 - at} - gt.$$

Burnout velocity. At $t_b = m_2/a$ the mass is m_1, so

$$v_b = b\log\frac{m_1 + m_2}{m_1} - \frac{gm_2}{a} = b\log\left(1 + \frac{m_2}{m_1}\right) - \frac{gm_2}{a}.$$

Burnout height. Write $M = m_1 + m_2$. Then

$$h_b = \int_0^{t_b} v\,\mathrm{d}t = -b\int_0^{t_b} \log\left(1 - \frac{at}{M}\right)dt - \frac{1}{2}gt_b^2.$$

Substitute $u = 1 - at/M$, $dt = -\frac{M}{a}\,\mathrm{d}u$; $t = 0 \mapsto u = 1$, $t = t_b \mapsto u = m_1/M$:

$$\begin{aligned} -\int_0^{t_b} \log\left(1 - \frac{at}{M}\right)dt &= \frac{M}{a}\int_1^{m_1/M} \log u\,\mathrm{d}u = \frac{M}{a}\Big[u\log u - u\Big]_1^{m_1/M} \\ &= \frac{M}{a}\left[\frac{m_1}{M}\log\frac{m_1}{M} - \frac{m_1}{M} + 1\right] = \frac{1}{a}\left[m_1\log\frac{m_1}{M} + m_2\right]. \end{aligned}$$

Therefore, with $\frac{1}{2}gt_b^2 = \frac{gm_2^2}{2a^2}$,

$$h_b = -\frac{gm_2^2}{2a^2} + \frac{bm_2}{a} + \frac{bm_1}{a}\log\frac{m_1}{m_1 + m_2}.$$

Answer. Burnout velocity $v_b = b\log\left(1 + \dfrac{m_2}{m_1}\right) - \dfrac{gm_2}{a}$; burnout height $h_b = -\dfrac{gm_2^2}{2a^2} + \dfrac{bm_2}{a} + \dfrac{bm_1}{a}\log\dfrac{m_1}{m_1 + m_2}$.

Chapter 2, Miscellaneous Problem 56 (Simmons)

Solution. Let $r(t)$, $m(t)$, $v(t)$ be the radius, mass and (downward) velocity of the drop, and ρ the density of water, so $m = \frac{4}{3}\pi\rho r^3$. The drop gains mass at a rate proportional to its surface area $4\pi r^2$:

$$\frac{dm}{dt} = 4\pi\rho r^2\frac{dr}{dt} = \lambda\cdot 4\pi r^2 \implies \frac{dr}{dt} = \frac{\lambda}{\rho} = k\ (\text{constant}).$$

With $r(0) = 0$ this gives $r = kt$. The vapor gathered is at rest, so its velocity relative to the drop is $w = -v$, and the variable-mass law $w\,dm/dt + F = m\,dv/dt$ with $F = mg$ becomes

$$-v\frac{dm}{dt} + mg = m\frac{dv}{dt}, \qquad \text{i.e.} \qquad \frac{d}{dt}(mv) = mg.$$

Since $m = \frac{4}{3}\pi\rho k^3 t^3$, cancelling the constant $\frac{4}{3}\pi\rho k^3$:

$$\frac{d}{dt}(t^3 v) = gt^3 \implies t^3 v = \frac{gt^4}{4} + C.$$

At $t = 0$ the left side is 0 (as v is bounded), so $C = 0$. Hence $v = \frac{1}{4}gt$ and $dv/dt = g/4$: the drop falls with constant acceleration $g/4$.

Answer. Shown: $r = kt$, $(t^3 v)' = gt^3$, so $v = gt/4$ and the acceleration is $g/4$.

Chapter 2, Miscellaneous Problem 57 (Simmons)

Solution. As in Problem 56, $dr/dt = k$ (constant, $k > 0$), so now $r = r_0 + kt$, and $\frac{d}{dt}(mv) = mg$ with $m = \frac{4}{3}\pi\rho r^3$ gives

$$\frac{d}{dt}(r^3 v) = gr^3.$$

Change the independent variable to r: since $dr = k\,\mathrm{d}t$,

$$\frac{d}{dr}(r^3 v) = \frac{g}{k}r^3 \implies r^3 v = \frac{g}{4k}\left(r^4 - r_0^4\right),$$

where the constant was chosen so that $v = 0$ when $r = r_0$ (start from rest). Thus

$$v = \frac{g}{4k}\left(r - \frac{r_0^4}{r^3}\right), \qquad \frac{dv}{dt} = k\frac{dv}{dr} = \frac{g}{4}\left(1 + \frac{3r_0^4}{r^4}\right).$$

If $r_0 = 0$ the acceleration is the constant $g/4$. If $r_0 > 0$, then $r = r_0 + kt$ increases with t, so $3r_0^4/r^4$ strictly decreases (from 3 toward 0) and the acceleration is not constant. Hence it is constant if and only if $r_0 = 0$.

Answer. Shown: $a = \frac{g}{4}\left(1 + \frac{3r_0^4}{r^4}\right)$, constant $(= g/4)$ iff $r_0 = 0$.

Chapter 2, Miscellaneous Problem 58 (Simmons)

Solution. Let x be the distance fallen, $v = dx/dt$, and $m = \frac{4}{3}\pi\rho r^3$. In time dt the drop sweeps out a cylinder of cross-section πr^2 and length $v\,\mathrm{d}t$, and collects the water in it; with σ the (uniform) mass of water per unit volume of mist,

$$\frac{dm}{dt} = \sigma\pi r^2 v, \qquad \text{i.e.} \qquad 4\pi\rho r^2\frac{dr}{dt} = \sigma\pi r^2\frac{dx}{dt} \implies \frac{dr}{dt} = \kappa\frac{dx}{dt}, \quad \kappa = \frac{\sigma}{4\rho}.$$

With $r = 0$ when $x = 0$ this gives $r = \kappa x$, so $m = \frac{4}{3}\pi\rho\kappa^3 x^3$. As in Problem 56 the collected water is at rest, so $\frac{d}{dt}(mv) = mg$, i.e. (cancelling constants)

$$\frac{d}{dt}(x^3 v) = gx^3.$$

Let $u = x^3 v$ and use x as independent variable, $\frac{d}{dt} = v\frac{d}{dx} = \frac{u}{x^3}\frac{d}{dx}$:

$$\frac{u}{x^3}\frac{du}{dx} = gx^3 \implies u\,\mathrm{d}u = gx^6\,\mathrm{d}x \implies \frac{u^2}{2} = \frac{gx^7}{7} + C.$$

At the start $x = 0$ and $u = 0$, so $C = 0$. Then $x^6 v^2 = \frac{2}{7} g x^7$, i.e.

$$v^2 = \frac{2g}{7}\, x.$$

Differentiating with respect to t: $2v\frac{dv}{dt} = \frac{2g}{7} v$, and since $v > 0$ once the drop is moving, $\frac{dv}{dt} = \frac{g}{7}$. (Equivalently, $v^2 = 2 \cdot \frac{g}{7} \cdot x$ is exactly the velocity–distance relation of uniform acceleration $g/7$ from rest.)

Answer. Shown: $r = \kappa x$, $(x^3 v)' = g x^3$ gives $v^2 = \frac{2}{7} g x$, so the acceleration is $g/7$.

Chapter 2, Miscellaneous Problem 59 (Simmons)

Solution. First a computation used in all three parts. From $(*)$,

$$\frac{dm}{dv} = m_0 \cdot \frac{v}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2},$$

and therefore

$$\frac{d(mv)}{dv} = m + v\frac{dm}{dv} = m_0 \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \left[\left(1 - \frac{v^2}{c^2}\right) + \frac{v^2}{c^2}\right] = \frac{m_0}{(1 - v^2/c^2)^{3/2}}. \tag{1}$$

(a) Let A be the acceleration due to the gravitational field, so the force on the particle is $F = mA$ (gravitational force is proportional to the mass). With $w = -v$ the variable-mass law $w\, dm/dt + F = m\, dv/dt$ becomes $F = \frac{d}{dt}(mv)$, i.e.

$$\frac{d}{dt}(mv) = mA.$$

By (1), $\frac{d}{dt}(mv) = \frac{m_0}{(1 - v^2/c^2)^{3/2}} \frac{dv}{dt}$, so

$$\frac{m_0}{(1 - v^2/c^2)^{3/2}} \frac{dv}{dt} = \frac{m_0 A}{(1 - v^2/c^2)^{1/2}} \implies \frac{dv}{1 - v^2/c^2} = A\, \mathrm{d}t.$$

Using $\dfrac{1}{1 - v^2/c^2} = \dfrac{1}{2}\left(\dfrac{1}{1 - v/c} + \dfrac{1}{1 + v/c}\right)$ and $v(0) = 0$,

$$\frac{c}{2} \log \frac{1 + v/c}{1 - v/c} = At \implies \frac{c + v}{c - v} = e^{2At/c} \implies v = c\, \frac{e^{2At/c} - 1}{e^{2At/c} + 1} = c\, \frac{1 - e^{-2At/c}}{1 + e^{-2At/c}} = c \tanh \frac{At}{c}.$$

As $t \to \infty$, $e^{-2At/c} \to 0$, so $v \to c$ (and $v < c$ for all t).

(b) Since $dx = v\, \mathrm{d}t$, $F\, \mathrm{d}x = \frac{d}{dt}(mv)\, v\, \mathrm{d}t = v\, \mathrm{d}(mv)$. By (1), $d(mv) = m_0 (1 - v^2/c^2)^{-3/2}\, \mathrm{d}v$, so (substituting $s = 1 - v^2/c^2$, $ds = -2v\, dv/c^2$)

$$E = m_0 \int_0^v \frac{v\, \mathrm{d}v}{(1 - v^2/c^2)^{3/2}} = m_0 c^2 \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2}\right]_0^v$$
$$= c^2 \left(\frac{m_0}{\sqrt{1 - v^2/c^2}} - m_0\right) = c^2 (m - m_0) = Mc^2.$$

(c) Now suppose only that $E = Mc^2$, i.e. $\int_0^v v\,\mathrm{d}(mv) = (m - m_0)c^2$ for every v, with $m = m(v)$ unknown and $m(0) = m_0$. Differentiating with respect to v,

$$v\,\frac{d(mv)}{dv} = c^2\frac{dm}{dv} \implies v\left(m + v\frac{dm}{dv}\right) = c^2\frac{dm}{dv} \implies (c^2 - v^2)\frac{dm}{dv} = mv.$$

Separating the variables ($0 \le v < c$),

$$\frac{dm}{m} = \frac{v\,\mathrm{d}v}{c^2 - v^2} \implies \log m = -\tfrac{1}{2}\log(c^2 - v^2) + K.$$

At $v = 0$: $\log m_0 = -\log c + K$, so $K = \log(m_0 c)$ and

$$m = \frac{m_0 c}{\sqrt{c^2 - v^2}} = \frac{m_0}{\sqrt{1 - v^2/c^2}},$$

which is $(*)$.

Answer. (a) $v = c\,\dfrac{1 - e^{-2At/c}}{1 + e^{-2At/c}} = c\tanh(At/c) \to c$ as $t \to \infty$ (A = acceleration of the field). (b) $E = m_0c^2\left[(1 - v^2/c^2)^{-1/2} - 1\right] = Mc^2$. (c) $E = Mc^2$ implies $(c^2 - v^2)m' = mv$, whose solution with $m(0) = m_0$ is $(*)$.

Chapter 3

Second Order Linear Equations

Section 14 Introduction

As instructed in the text, throughout these problems we assume the fact proved in Section 15: if $y_1(x)$ and $y_2(x)$ are two solutions of the homogeneous equation

$$y'' + P(x)y' + Q(x)y = 0 \tag{2}$$

and neither is a constant multiple of the other, then $c_1y_1(x) + c_2y_2(x)$ is its general solution. Together with Theorem B (general solution of the complete equation $= y_g + y_p$) this lets us write down general solutions as soon as suitable particular solutions are known. When an equation is written with a leading coefficient $a(x)$ in front of y'', the theory applies (after division by $a(x)$) on any interval on which $a(x) \neq 0$ and the resulting coefficients are continuous.

Problem 14-1 (Simmons)

Solution.

(a) For $y_1 = 1$ we have $y_1' = y_1'' = 0$, so $xy_1'' - y_1' = 0$. For $y_2 = x^2$ we have $y_2' = 2x$, $y_2'' = 2$, so

$$xy_2'' - y_2' = 2x - 2x = 0.$$

Both are solutions, and $y_2/y_1 = x^2$ is not constant, so neither is a constant multiple of the other. Hence the general solution of $xy'' - y' = 0$ (on any interval not containing 0, where $P = -1/x$ is continuous) is

$$y = c_1 + c_2x^2.$$

(b) For $y_p = ax^3$: $y_p' = 3ax^2$, $y_p'' = 6ax$, so

$$xy_p'' - y_p' = 6ax^2 - 3ax^2 = 3ax^2.$$

This equals $3x^2$ for all x if and only if $3a = 3$, i.e. $a = 1$. Thus $y_p = x^3$, and by Theorem B the general solution of the complete equation is $y_g + y_p$:

$$y = c_1 + c_2x^2 + x^3.$$

This agrees with Example 1 of Section 11, where reduction of order gave $y = x^3 + \frac{1}{2}c_1x^2 + c_2$; the two forms differ only in the names of the constants ($\frac{1}{2}c_1 \leftrightarrow c_2$, $c_2 \leftrightarrow c_1$).

(c) Yes. A constant obviously has $y' = y'' = 0$, so $y_1 = 1$ is a solution of the reduced equation. Since differentiation lowers the exponent of a power by one and xy'', y' are then both multiples of x^{n-1}, it is natural to try $y = x^n$:

$$x \cdot n(n-1)x^{n-2} - nx^{n-1} = n(n-2)x^{n-1},$$

which vanishes for $n = 0$ and $n = 2$, giving $y_1 = 1$ and $y_2 = x^2$. (Equivalently, one can notice that $xy'' - y' = x^2\,(y'/x)'$, so the reduced equation says y'/x is constant, i.e. $y' = 2c_2x$ and $y = c_1 + c_2x^2$.) For y_p, the same computation with $y = x^n$ gives $n(n-2)x^{n-1}$, which must equal $3x^2$: the exponent forces $n = 3$, and then $n(n-2) = 3$ holds exactly, so $y_p = x^3$.

Answer. (a) $y = c_1 + c_2x^2$; (b) $a = 1$, $y = c_1 + c_2x^2 + x^3$; (c) yes: try constants and powers x^n, which give $n(n-2)x^{n-1}$; $n = 0, 2$ for the reduced and $n = 3$ for the complete equation.

Problem 14-2 (Simmons)

Solution. For $y_1 = 1$: $y_1' = y_1'' = 0$, so $xy_1'' + y_1' = 0$. For $y_2 = \log x$ ($x > 0$): $y_2' = 1/x$, $y_2'' = -1/x^2$, so

$$xy_2'' + y_2' = x\left(-\frac{1}{x^2}\right) + \frac{1}{x} = 0.$$

Since $y_2/y_1 = \log x$ is not constant, the general solution (for $x > 0$) is

$$y = c_1 + c_2 \log x.$$

Inspection: a constant is clearly a solution. Moreover, the left side is an exact derivative, $xy'' + y' = (xy')'$, so the equation says $xy' = c_2$, i.e. $y' = c_2/x$, and integration gives $y = c_2 \log x + c_1$; this exhibits both $y_1 = 1$ and $y_2 = \log x$.

Answer. $y = c_1 + c_2 \log x$; yes, since $xy'' + y' = (xy')'$.

Problem 14-3 (Simmons)

Solution.

(a) For $y_1 = e^{-x}$: $y_1' = -e^{-x}$, $y_1'' = e^{-x}$, so $y_1'' - y_1' - 2y_1 = (1 + 1 - 2)e^{-x} = 0$. For $y_2 = e^{2x}$: $y_2' = 2e^{2x}$, $y_2'' = 4e^{2x}$, so $y_2'' - y_2' - 2y_2 = (4 - 2 - 2)e^{2x} = 0$. The ratio $y_2/y_1 = e^{3x}$ is not constant, hence

$$y = c_1 e^{-x} + c_2 e^{2x}.$$

(b) With $y_p = ax + b$ we have $y_p' = a$, $y_p'' = 0$, and

$$y_p'' - y_p' - 2y_p = -a - 2ax - 2b = -2ax - (a + 2b).$$

For this to equal $4x$ identically we need $-2a = 4$ and $a + 2b = 0$, so $a = -2$ and $b = 1$. Thus $y_p = -2x + 1$, and by Theorem B

$$y = c_1 e^{-x} + c_2 e^{2x} - 2x + 1.$$

Answer. (a) $y = c_1 e^{-x} + c_2 e^{2x}$; (b) $a = -2$, $b = 1$; $y = c_1 e^{-x} + c_2 e^{2x} - 2x + 1$.

Problem 14-4 (Simmons)

Solution.

(a) Each term on the left is a power of x times a derivative of y, with the power exactly compensating for the order of the derivative; so for $y = ax^n$ every term is a multiple of x^{n+1}:

$$x^3 \cdot an(n-1)x^{n-2} + x^2 \cdot anx^{n-1} + x \cdot ax^n = a\big[n(n-1) + n + 1\big]x^{n+1} = a(n^2+1)x^{n+1}.$$

To obtain the constant 1 we need $n + 1 = 0$, i.e. $n = -1$, and then $a(1 + 1) = 1$, $a = \frac{1}{2}$. So $y = \dfrac{1}{2x}$.
Check: $y' = -\dfrac{1}{2x^2}$, $y'' = \dfrac{1}{x^3}$, and $x^3 \cdot \dfrac{1}{x^3} + x^2\left(-\dfrac{1}{2x^2}\right) + x \cdot \dfrac{1}{2x} = 1 - \frac{1}{2} + \frac{1}{2} = 1$.

(b) The right side is constant and y itself does not appear, so try $y = ax$: then $y' = a$, $y'' = 0$ and $y'' - 2y' = -2a = 6$, so $a = -3$: $y = -3x$.

(c) Since $(\sin x)'' = -\sin x$, try $y = a \sin x$: then $y'' - 2y = -a \sin x - 2a \sin x = -3a \sin x$, which equals $\sin x$ when $a = -\frac{1}{3}$: $y = -\frac{1}{3} \sin x$.

Answer. (a) $y = 1/(2x)$; (b) $y = -3x$; (c) $y = -\frac{1}{3}\sin x$.

Problem 14-5 (Simmons)

Solution. In (a), (b), (c), (e) the coefficients are constants, so the theory applies on every interval; in each case the two solutions of the reduced equation found below have a nonconstant ratio.

(a) Reduced equation $y'' = 0$: $y_1 = 1$ and $y_2 = x$ (both have zero second derivative; ratio x). Complete equation: $(e^x)'' = e^x$, so $y_p = e^x$. Hence

$$y = c_1 x + c_2 + e^x.$$

(b) Reduced equation $y'' - 2y' = 0$: $y_1 = 1$ is a solution; trying $y = e^{mx}$ gives $(m^2 - 2m)e^{mx} = 0$, so $m = 0$ or $m = 2$, and $y_2 = e^{2x}$. Complete equation: try $y = ax$; then $y'' - 2y' = -2a = 4$, $a = -2$, so $y_p = -2x$. Hence

$$y = c_1 + c_2 e^{2x} - 2x.$$

(c) Reduced equation $y'' - y = 0$: $y = e^{mx}$ gives $m^2 - 1 = 0$, so $y_1 = e^x$, $y_2 = e^{-x}$ (ratio e^{-2x}). Complete equation: try $y = a\sin x$; then $y'' - y = -2a\sin x = \sin x$, so $a = -\frac{1}{2}$ and $y_p = -\frac{1}{2}\sin x$. Hence

$$y = c_1 e^x + c_2 e^{-x} - \tfrac{1}{2}\sin x.$$

(d) Here the equation is itself homogeneous (its right side is 0), so the reduced and complete equations coincide and we may take $y_p = 0$. Trying $y = x$: $y' = 1$, $y'' = 0$ and $(x-1)\cdot 0 - x\cdot 1 + x = 0$, so $y_1 = x$ is a solution. Trying $y = e^x$:

$$(x-1)e^x - xe^x + e^x = (x - 1 - x + 1)e^x = 0,$$

so $y_2 = e^x$ is a solution. Their ratio e^x/x is not constant, so on any interval not containing $x = 1$ (where the leading coefficient vanishes; dividing by $x - 1$ gives continuous P, Q elsewhere)

$$y = c_1 x + c_2 e^x.$$

(e) Reduced equation $y'' + 2y' = 0$: $y = e^{mx}$ gives $m^2 + 2m = 0$, $m = 0, -2$, so $y_1 = 1$, $y_2 = e^{-2x}$. Complete equation: try $y = ae^x$; then $y'' + 2y' = 3ae^x = 6e^x$, $a = 2$, so $y_p = 2e^x$. Hence

$$y = c_1 + c_2 e^{-2x} + 2e^x.$$

Answer. (a) $y = c_1 x + c_2 + e^x$; (b) $y = c_1 + c_2 e^{2x} - 2x$; (c) $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2}\sin x$; (d) $y = c_1 x + c_2 e^x$; (e) $y = c_1 + c_2 e^{-2x} + 2e^x$.

Problem 14-6 (Simmons)

Solution. In each case we differentiate twice and eliminate c_1, c_2 from the three equations for y, y', y''.

(a) $y = c_1 x + c_2 x^2$, $y' = c_1 + 2c_2 x$, $y'' = 2c_2$. Thus $c_2 = \frac{1}{2}y''$ and $c_1 = y' - 2c_2 x = y' - xy''$. Substituting in y:

$$y = (y' - xy'')x + \tfrac{1}{2}y''x^2 = xy' - \tfrac{1}{2}x^2 y'' \quad\Longrightarrow\quad x^2 y'' - 2xy' + 2y = 0.$$

(b) $y' = kc_1e^{kx} - kc_2e^{-kx}$, $y'' = k^2c_1e^{kx} + k^2c_2e^{-kx} = k^2y$. Hence $y'' - k^2y = 0$.

(c) $y' = kc_1\cos kx - kc_2\sin kx$, $y'' = -k^2c_1\sin kx - k^2c_2\cos kx = -k^2y$. Hence $y'' + k^2y = 0$.

(d) $y' = -2c_2e^{-2x}$, $y'' = 4c_2e^{-2x} = -2y'$. Hence $y'' + 2y' = 0$ (c_1 disappears on the first differentiation).

(e) $y' = c_1 + c_2\cos x$, $y'' = -c_2\sin x$. Where $\sin x \neq 0$, $c_2 = -y''/\sin x$ and $c_1 = y' - c_2\cos x = y' + y''\cot x$. Substituting in y:

$$y = x(y' + y''\cot x) - y'' = xy' + (x\cot x - 1)y'' \quad\Longrightarrow\quad (1 - x\cot x)y'' - xy' + y = 0.$$

Multiplying by $\sin x$ gives the equivalent form $(\sin x - x\cos x)y'' - x\sin x\,y' + \sin x\,y = 0$, which makes sense also where $\sin x = 0$.

(f) Write the family as $e^{-x}y = c_1 + c_2x$. Differentiating twice eliminates both constants: $(e^{-x}y)'' = 0$. Since $(e^{-x}y)' = e^{-x}(y' - y)$ and $(e^{-x}y)'' = e^{-x}(y'' - 2y' + y)$, and $e^{-x} \neq 0$,

$$y'' - 2y' + y = 0.$$

(g) $y' = c_1e^x - 3c_2e^{-3x}$, $y'' = c_1e^x + 9c_2e^{-3x}$. Subtracting, $y' - y = -4c_2e^{-3x}$ and $y'' - y' = 12c_2e^{-3x}$, so $y'' - y' = -3(y' - y)$, i.e.

$$y'' + 2y' - 3y = 0.$$

(h) $y' = c_1 - c_2x^{-2}$, $y'' = 2c_2x^{-3}$. Thus $c_2 = \frac{1}{2}x^3y''$ and $c_1 = y' + c_2x^{-2} = y' + \frac{1}{2}xy''$. Substituting in y:

$$y = x\left(y' + \tfrac{1}{2}xy''\right) + \tfrac{1}{2}x^2y'' = xy' + x^2y'' \quad\Longrightarrow\quad x^2y'' + xy' - y = 0.$$

Answer. (a) $x^2y'' - 2xy' + 2y = 0$; (b) $y'' - k^2y = 0$; (c) $y'' + k^2y = 0$; (d) $y'' + 2y' = 0$; (e) $(1 - x\cot x)y'' - xy' + y = 0$; (f) $y'' - 2y' + y = 0$; (g) $y'' + 2y' - 3y = 0$; (h) $x^2y'' + xy' - y = 0$.

Problem 14-7 (Simmons)

Solution. For $x \neq 0$,

$$y' = -c_1x^{-2} + 5c_2x^4, \qquad y'' = 2c_1x^{-3} + 20c_2x^3,$$

so

$$\begin{aligned} x^2y'' - 3xy' - 5y &= (2c_1x^{-1} + 20c_2x^5) + (3c_1x^{-1} - 15c_2x^5) - (5c_1x^{-1} + 5c_2x^5) \\ &= (2 + 3 - 5)c_1x^{-1} + (20 - 15 - 5)c_2x^5 = 0. \end{aligned}$$

Thus y is a solution on any interval not containing 0 (where x^{-1} is defined).

The initial conditions at $x_0 \neq 0$ are the linear system

$$\frac{1}{x_0}\,c_1 + x_0^5\,c_2 = y_0, \qquad -\frac{1}{x_0^2}\,c_1 + 5x_0^4\,c_2 = y_0'.$$

Its determinant is

$$\begin{vmatrix} x_0^{-1} & x_0^5 \\ -x_0^{-2} & 5x_0^4 \end{vmatrix} = 5x_0^3 + x_0^3 = 6x_0^3 \neq 0,$$

so the system has exactly one solution. By Cramer's rule,

$$c_1 = \frac{5x_0^4 y_0 - x_0^5 y_0'}{6x_0^3} = \frac{x_0(5y_0 - x_0 y_0')}{6}, \qquad c_2 = \frac{x_0^{-1} y_0' + x_0^{-2} y_0}{6x_0^3} = \frac{y_0 + x_0 y_0'}{6x_0^5}.$$

(Check: $c_1/x_0 + c_2 x_0^5 = \frac{1}{6}(5y_0 - x_0 y_0') + \frac{1}{6}(y_0 + x_0 y_0') = y_0$, and $-c_1/x_0^2 + 5c_2 x_0^4 = -\frac{5y_0 - x_0 y_0'}{6x_0} + \frac{5(y_0 + x_0 y_0')}{6x_0} = y_0'$.)

Answer. Verified; the constants are uniquely $c_1 = \frac{1}{6}x_0(5y_0 - x_0 y_0')$, $c_2 = (y_0 + x_0 y_0')/(6x_0^5)$, since the determinant is $6x_0^3 \neq 0$.

Problem 14-8 (Simmons)

Solution. For $y = x^2 \sin x$:

$$y' = 2x \sin x + x^2 \cos x, \qquad y'' = 2\sin x + 4x \cos x - x^2 \sin x.$$

Hence

$$\begin{aligned} x^2 y'' &= 2x^2 \sin x + 4x^3 \cos x - x^4 \sin x, \\ -4xy' &= -8x^2 \sin x - 4x^3 \cos x, \\ (x^2 + 6)y &= x^4 \sin x + 6x^2 \sin x, \end{aligned}$$

and adding, the coefficients of $x^2 \sin x$ $(2 - 8 + 6)$, of $x^3 \cos x$ $(4 - 4)$ and of $x^4 \sin x$ $(-1 + 1)$ all vanish. So $y = x^2 \sin x$ is a solution; $y = 0$ is trivially a solution. Moreover $y(0) = 0 \cdot \sin 0 = 0$ and $y'(0) = 0 + 0 = 0$, and obviously the zero function satisfies the same conditions.

There is no contradiction. To apply Theorem A the equation must be in the form (1), so we divide by x^2:

$$y'' - \frac{4}{x} y' + \left(1 + \frac{6}{x^2}\right) y = 0.$$

Here $P(x) = -4/x$ and $Q(x) = 1 + 6/x^2$ are not continuous (not even defined) at $x = 0$, so the hypotheses of Theorem A fail on every interval $[a, b]$ containing the point $x_0 = 0$ at which the initial conditions are imposed. Theorem A therefore says nothing about uniqueness of solutions with prescribed $y(0), y'(0)$. (On any interval not containing 0 the two solutions do have different initial data.)

Answer. Both are solutions with $y(0) = y'(0) = 0$; no contradiction, because in normalized form $P = -4/x$, $Q = 1 + 6/x^2$ are discontinuous at $x_0 = 0$, so Theorem A does not apply there.

Problem 14-9 (Simmons)

Solution. Let $y(x)$ be a solution of (2) on $[a, b]$ whose graph is tangent to the x-axis at $x_0 \in [a, b]$. Then the curve meets the axis there, $y(x_0) = 0$, and its tangent line is the axis, so $y'(x_0) = 0$. The function identically equal to 0 is also a solution of the homogeneous equation (2) (the trivial solution), and it satisfies the same conditions $y(x_0) = 0$, $y'(x_0) = 0$. By the uniqueness part of Theorem A (with $R(x) = 0$, $y_0 = y_0' = 0$) there is only one solution on $[a, b]$ with these initial values, so $y(x) \equiv 0$ on $[a, b]$.

Answer. Tangency at x_0 means $y(x_0) = y'(x_0) = 0$; the zero solution has the same initial data, so by uniqueness (Theorem A) $y \equiv 0$.

Problem 14-10 (Simmons)

Solution. Let $x_0 \in [a, b]$ with $y_1(x_0) = y_2(x_0) = 0$.

Case 1: $y_1'(x_0) = 0$. Then $y_1(x_0) = y_1'(x_0) = 0$, so by Problem 9 (i.e. by the uniqueness in Theorem A) $y_1 \equiv 0$ on $[a, b]$, and $y_1 = 0 \cdot y_2$ is a constant multiple of y_2.

Case 2: $y_1'(x_0) \neq 0$. Put $k = y_2'(x_0)/y_1'(x_0)$ and consider

$$y(x) = y_2(x) - k\, y_1(x).$$

By Theorem C (with constants 1 and $-k$), y is a solution of (2) on $[a, b]$, and

$$y(x_0) = y_2(x_0) - ky_1(x_0) = 0, \qquad y'(x_0) = y_2'(x_0) - ky_1'(x_0) = 0.$$

Again by uniqueness (the zero function has the same initial data), $y \equiv 0$, i.e. $y_2(x) = ky_1(x)$ for all x in $[a, b]$.

In either case one of the functions is a constant multiple of the other.

Answer. Shown: if $y_1'(x_0) = 0$ then $y_1 \equiv 0 = 0 \cdot y_2$; otherwise $y_2 - ky_1$ with $k = y_2'(x_0)/y_1'(x_0)$ has zero initial data at x_0, so $y_2 = ky_1$.

Section 15 The General Solution of the Homogeneous Equation

Throughout we use the results of the text. For the homogeneous equation

$$y'' + P(x)y' + Q(x)y = 0 \tag{1}$$

with P, Q continuous on $[a, b]$:

- **Theorem A.** If y_1, y_2 are linearly independent solutions of (1) on $[a, b]$, then $c_1y_1 + c_2y_2$ is the general solution on $[a, b]$.
- **Lemma 1.** The Wronskian $W = W(y_1, y_2) = y_1y_2' - y_2y_1'$ of two solutions is either identically zero or never zero on $[a, b]$ (indeed $W = ce^{-\int P\,dx}$, Abel's formula).
- **Lemma 2.** Two solutions are linearly dependent if and only if $W \equiv 0$.

Hence two solutions of (1) are linearly independent exactly when $W \neq 0$ at one (hence every) point. Theorem 14-A (existence and uniqueness) is also used freely.

Problem 15-1 (Simmons)

Solution. Here $P = 0$ and $Q = -1$ are continuous everywhere. For $y_1 = e^x$, $y_1'' - y_1 = e^x - e^x = 0$; for $y_2 = e^{-x}$, $y_2'' - y_2 = e^{-x} - e^{-x} = 0$. The Wronskian is

$$W(y_1, y_2) = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^xe^{-x} - e^{-x}e^x = -2 \neq 0$$

for all x. By Lemma 2 the solutions are linearly independent on every interval.

Answer. Shown: $W(e^x, e^{-x}) = -2 \neq 0$, so they are linearly independent on every interval.

Problem 15-2 (Simmons)

Solution. Dividing by x^2 gives $y'' - \frac{2}{x}y' + \frac{2}{x^2}y = 0$, with $P = -2/x$ and $Q = 2/x^2$ continuous on every interval not containing 0. For $y_1 = x$: $x^2 \cdot 0 - 2x \cdot 1 + 2x = 0$. For $y_2 = x^2$: $x^2 \cdot 2 - 2x \cdot 2x + 2x^2 = 2x^2 - 4x^2 + 2x^2 = 0$. The Wronskian is

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2 \neq 0 \quad (x \neq 0),$$

so y_1, y_2 are linearly independent and by Theorem A $y = c_1x + c_2x^2$ is the general solution.

For the initial conditions, $y' = c_1 + 2c_2x$, so

$$y(1) = c_1 + c_2 = 3, \qquad y'(1) = c_1 + 2c_2 = 5.$$

Subtracting, $c_2 = 2$, hence $c_1 = 1$, and $y = x + 2x^2$. (The interval must contain $x = 1$, so it is a subinterval of $(0, \infty)$.)

Answer. $W = x^2 \neq 0$, so $y = c_1x + c_2x^2$ is general; the particular solution is $y = x + 2x^2$.

Problem 15-3 (Simmons)

Solution. The coefficients are constants, hence continuous everywhere. For $y_1 = e^x$: $e^x - 3e^x + 2e^x = 0$; for $y_2 = e^{2x}$: $4e^{2x} - 6e^{2x} + 2e^{2x} = 0$. Also

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0,$$

so by Theorem A the general solution is $y = c_1 e^x + c_2 e^{2x}$ on any interval.
With $y' = c_1 e^x + 2c_2 e^{2x}$:

$$c_1 + c_2 = -1, \qquad c_1 + 2c_2 = 1.$$

Subtracting, $c_2 = 2$ and $c_1 = -3$, so $y = -3e^x + 2e^{2x}$.

Answer. $W = e^{3x} \neq 0$; the particular solution is $y = -3e^x + 2e^{2x}$.

Problem 15-4 (Simmons)

Solution. The coefficients are constants. For $y_1 = e^{2x}$: $4e^{2x} - 8e^{2x} + 4e^{2x} = 0$. For $y_2 = xe^{2x}$: $y_2' = (1+2x)e^{2x}$, $y_2'' = (4+4x)e^{2x}$, so

$$y_2'' - 4y_2' + 4y_2 = \big[(4+4x) - 4(1+2x) + 4x\big]e^{2x} = 0.$$

The Wronskian is

$$W = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & (1+2x)e^{2x} \end{vmatrix} = (1+2x)e^{4x} - 2xe^{4x} = e^{4x} \neq 0.$$

By Theorem A, $y = c_1 e^{2x} + c_2 xe^{2x}$ is the general solution on any interval.

Answer. Shown: $W = e^{4x} \neq 0$, so $c_1 e^{2x} + c_2 xe^{2x}$ is the general solution.

Problem 15-5 (Simmons)

Solution. Try $y = x^m$: $x^2 \cdot m(m-1)x^{m-2} - 2x^m = (m^2 - m - 2)x^m = (m-2)(m+1)x^m$, which vanishes for $m = 2$ and $m = -1$. So $y_1 = x^2$ and $y_2 = x^{-1}$ are solutions. On $[1,2]$, $P = 0$ and $Q = -2/x^2$ are continuous, and

$$W = \begin{vmatrix} x^2 & x^{-1} \\ 2x & -x^{-2} \end{vmatrix} = -1 - 2 = -3 \neq 0,$$

so they are linearly independent and the general solution is $y = c_1 x^2 + c_2 x^{-1}$. From $y' = 2c_1 x - c_2 x^{-2}$:

$$c_1 + c_2 = 1, \qquad 2c_1 - c_2 = 8.$$

Adding, $3c_1 = 9$, $c_1 = 3$, $c_2 = -2$. Hence $y = 3x^2 - 2x^{-1}$.

Answer. $y_1 = x^2$, $y_2 = x^{-1}$; the particular solution is $y = 3x^2 - 2x^{-1}$.

Problem 15-6 (Simmons)

Solution. All coefficients are constants, so Theorem A applies on $[0,2]$ in every part.

(a) e^x: $e^x + e^x - 2e^x = 0$. e^{-2x}: $4e^{-2x} - 2e^{-2x} - 2e^{-2x} = 0$. $W = e^x(-2e^{-2x}) - e^{-2x}e^x = -3e^{-x} \neq 0$. So $y = c_1e^x + c_2e^{-2x}$ and $y' = c_1e^x - 2c_2e^{-2x}$. Then

$$c_1 + c_2 = 8, \qquad c_1 - 2c_2 = 2 \implies 3c_2 = 6,\ c_2 = 2,\ c_1 = 6.$$

Thus $y = 6e^x + 2e^{-2x}$.

(b) The same functions and Wronskian as in (a). Now

$$c_1e + c_2e^{-2} = 0, \qquad c_1e - 2c_2e^{-2} = 0.$$

Subtracting, $3c_2e^{-2} = 0$, so $c_2 = 0$ and then $c_1 = 0$. Hence $y = 0$ (consistent with uniqueness: the zero function has these initial data).

(c) e^{-2x}: $4e^{-2x} - 10e^{-2x} + 6e^{-2x} = 0$. e^{-3x}: $9e^{-3x} - 15e^{-3x} + 6e^{-3x} = 0$. $W = e^{-2x}(-3e^{-3x}) - e^{-3x}(-2e^{-2x}) = -e^{-5x} \neq 0$. With $y = c_1e^{-2x} + c_2e^{-3x}$, $y' = -2c_1e^{-2x} - 3c_2e^{-3x}$:

$$c_1 + c_2 = 1, \qquad -2c_1 - 3c_2 = 1.$$

Substituting $c_2 = 1 - c_1$: $-2c_1 - 3 + 3c_1 = 1$, so $c_1 = 4$, $c_2 = -3$, and $y = 4e^{-2x} - 3e^{-3x}$.

(d) 1: $0 + 0 = 0$. e^{-x}: $e^{-x} - e^{-x} = 0$. $W = \begin{vmatrix} 1 & e^{-x} \\ 0 & -e^{-x} \end{vmatrix} = -e^{-x} \neq 0$. With $y = c_1 + c_2e^{-x}$, $y' = -c_2e^{-x}$:

$$y'(2) = -c_2e^{-2} = e^{-2} \Rightarrow c_2 = -1, \qquad y(2) = c_1 - e^{-2} = 0 \Rightarrow c_1 = e^{-2}.$$

Thus $y = e^{-2} - e^{-x}$.

Answer. (a) $y = 6e^x + 2e^{-2x}$; (b) $y = 0$; (c) $y = 4e^{-2x} - 3e^{-3x}$; (d) $y = e^{-2} - e^{-x}$.

Problem 15-7 (Simmons)

Solution.

(a) The variable y is missing, so put $p = y'$, $y'' = p'$ (Section 11, Method 1): $p' + p^2 = 0$. The function $p \equiv 0$ is a solution, and by uniqueness for this first order equation any solution that vanishes at one point is $\equiv 0$. Otherwise $p \neq 0$ and

$$\frac{dp}{p^2} = -dx \implies -\frac{1}{p} = -x - c_1 \implies p = \frac{1}{x + c_1}.$$

Then $y' = 0$ gives $y = c_2$, while $y' = 1/(x + c_1)$ gives, by integrating, $y = \log|x + c_1| + c_2$ (on an interval not containing $-c_1$). Hence all solutions are

$$y = c_2 \text{ (a constant)} \qquad \text{or} \qquad y = \log|x + c_1| + c_2.$$

(Check: $y' = 1/(x + c_1)$, $y'' = -1/(x + c_1)^2$, and $y'' + y'^2 = 0$.)

(b) For $y_1 = 1$: $y_1'' + y_1'^2 = 0$. For $y_2 = \log x$ $(x > 0)$: $y_2' = 1/x$, $y_2'' = -1/x^2$, so $y_2'' + y_2'^2 = -1/x^2 + 1/x^2 = 0$. Their Wronskian is

$$W = \begin{vmatrix} 1 & \log x \\ 0 & 1/x \end{vmatrix} = \frac{1}{x} \neq 0$$

for $x > 0$, so they are linearly independent. However, $y = c_1 + c_2 \log x$ is *not* the general solution: for it, $y' = c_2/x$, $y'' = -c_2/x^2$ and

$$y'' + y'^2 = \frac{c_2(c_2 - 1)}{x^2},$$

which is 0 only for $c_2 = 0$ or $c_2 = 1$. For instance $2\log x$ is not a solution. The reason is that the equation $y'' + (y')^2 = 0$ is not linear (the term $(y')^2$), so the superposition principle (Theorem 14-C) fails and Theorem A, which concerns linear equations (1), does not apply. (Indeed the true general solution $\log|x + c_1| + c_2$ with $c_1 \neq 0$ is not even of the form $c_1 + c_2 \log x$.)

Answer. (a) $y = c_2$ or $y = \log|x + c_1| + c_2$; (b) 1 and $\log x$ are independent solutions, but $c_1 + c_2 \log x$ is not the general solution (indeed generally not a solution) because the equation is nonlinear.

Problem 15-8 (Simmons)

Solution. Let y_1, y_2 be solutions of (1) on $[a, b]$. By Lemmas 1 and 2 they are linearly dependent if and only if $W(y_1, y_2)(x_0) = 0$ at a single point x_0 of the interval (since W is then $\equiv 0$ by Lemma 1, and Lemma 2 gives dependence).

(a) If $y_1(x_0) = y_2(x_0) = 0$, then

$$W(x_0) = y_1(x_0)y_2'(x_0) - y_2(x_0)y_1'(x_0) = 0 \cdot y_2'(x_0) - 0 \cdot y_1'(x_0) = 0.$$

(b) If y_1 and y_2 both have a maximum or minimum at x_0 (an interior point, or an endpoint at which the derivatives vanish), then $y_1'(x_0) = y_2'(x_0) = 0$ (Fermat's rule for a differentiable function at an interior extremum), so

$$W(x_0) = y_1(x_0) \cdot 0 - y_2(x_0) \cdot 0 = 0.$$

In both cases $W(x_0) = 0$, hence $W \equiv 0$ and the solutions are linearly dependent.

Answer. Shown: in each case $W(x_0) = 0$ (common zero, or both derivatives zero at x_0), so by Lemmas 1 and 2 the solutions are linearly dependent.

Problem 15-9 (Simmons)

Solution. Note $g(x) = x^3$ for $x \geq 0$ and $g(x) = -x^3$ for $x \leq 0$. It is differentiable with $g'(x) = 3x^2$ for $x \geq 0$, $g'(x) = -3x^2$ for $x \leq 0$ (both give $g'(0) = 0$), i.e. $g' = 3x|x|$.

(a) For $x \geq 0$: $W = x^3 \cdot 3x^2 - x^3 \cdot 3x^2 = 0$. For $x \leq 0$: $W = x^3(-3x^2) - (-x^3)(3x^2) = -3x^5 + 3x^5 = 0$. So $W(f, g) \equiv 0$ on $[-1, 1]$.

(b) Suppose $g = cf$ on $[-1, 1]$ for a constant c. At $x = 1$: $1 = c$; at $x = -1$: $g(-1) = 1$ and $f(-1) = -1$, so $1 = -c$. These are incompatible. Suppose $f = cg$: at $x = 1$, $1 = c$; at $x = -1$, $-1 = c \cdot 1$, again incompatible. Neither function is identically zero, so neither is a constant multiple of the other: f and g are linearly independent.

(c) There is no contradiction. Lemma 2 is a statement about two *solutions of an equation (1)* with P and Q continuous on the interval. Here f and g cannot both be solutions of such an equation on $[-1, 1]$: both satisfy $y(0) = y'(0) = 0$, so by the uniqueness theorem (Problem 14-9) each

would be identically zero, which is false. (They are solutions of $x^2y'' - 4xy' + 6y = 0$, check: $x^2 \cdot 6x - 4x \cdot 3x^2 + 6x^3 = 0$ for f, and likewise for g since $g'' = 6|x|$; but dividing by x^2 gives $P = -4/x$, $Q = 6/x^2$, which are not continuous at 0.) So Lemma 2 does not apply to f and g.

Answer. (a) $W \equiv 0$; (b) $g = \pm f$ with different signs on the two halves, so they are independent; (c) no contradiction, since f, g are not solutions of any equation (1) with continuous P, Q on $[-1, 1]$.

Problem 15-10 (Simmons)

Solution.

(a) Since y_1 and y_2 solve the equation,

$$y_1' P + y_1 Q = -y_1'', \qquad y_2' P + y_2 Q = -y_2'',$$

a linear system for P and Q at each x. Its determinant is $y_1'y_2 - y_1y_2' = -W(y_1, y_2)$, which is nonzero because y_1, y_2 are independent (Lemmas 1, 2). Cramer's rule gives

$$P = \frac{\begin{vmatrix} -y_1'' & y_1 \\ -y_2'' & y_2 \end{vmatrix}}{-W} = \frac{-y_1''y_2 + y_1y_2''}{-W} = -\frac{y_1y_2'' - y_2y_1''}{W},$$

$$Q = \frac{\begin{vmatrix} y_1' & -y_1'' \\ y_2' & -y_2'' \end{vmatrix}}{-W} = \frac{-y_1'y_2'' + y_2'y_1''}{-W} = \frac{y_1'y_2'' - y_2'y_1''}{W}.$$

Thus P and Q, hence the equation, are determined by y_1, y_2.

(b) *Pair* $\sin x$, $\cos x$: $W = -\sin^2 x - \cos^2 x = -1$, $y_1'' = -\sin x$, $y_2'' = -\cos x$.

$$P = -\frac{\sin x(-\cos x) - \cos x(-\sin x)}{-1} = 0,$$

$$Q = \frac{\cos x(-\cos x) - (-\sin x)(-\sin x)}{-1} = \frac{-\cos^2 x - \sin^2 x}{-1} = 1.$$

Pair $\sin x$, $\sin x - \cos x$: $y_1 = \sin x$, $y_2 = \sin x - \cos x$, $y_2' = \cos x + \sin x$, $y_2'' = \cos x - \sin x$, $y_1' = \cos x$, $y_1'' = -\sin x$. Then $W = \sin x(\cos x + \sin x) - (\sin x - \cos x)\cos x = \sin^2 x + \cos^2 x = 1$ and

$$\begin{aligned} P &= -\big[\sin x(\cos x - \sin x) - (\sin x - \cos x)(-\sin x)\big] \\ &= -\big[\sin x \cos x - \sin^2 x + \sin^2 x - \sin x \cos x\big] = 0, \\ Q &= \cos x(\cos x - \sin x) - (\cos x + \sin x)(-\sin x) \\ &= \cos^2 x - \sin x \cos x + \sin x \cos x + \sin^2 x = 1. \end{aligned}$$

Both pairs give $y'' + y = 0$.

(c) Let $y_1 = e^{2x}$, $y_2 = xe^{2x}$. Then $y_1' = 2e^{2x}$, $y_1'' = 4e^{2x}$, $y_2' = (1 + 2x)e^{2x}$, $y_2'' = (4 + 4x)e^{2x}$, and $W = e^{4x}$ (Problem 4). So

$$P = -\frac{e^{2x}(4 + 4x)e^{2x} - xe^{2x} \cdot 4e^{2x}}{e^{4x}} = -(4 + 4x - 4x) = -4,$$

$$Q = \frac{2e^{2x}(4 + 4x)e^{2x} - (1 + 2x)e^{2x} \cdot 4e^{2x}}{e^{4x}} = 8 + 8x - 4 - 8x = 4.$$

The equation is $y'' - 4y' + 4y = 0$, the equation of Problem 4.

Answer. (a) shown by Cramer's rule; (b) both pairs give $y'' + y = 0$; (c) $y'' - 4y' + 4y = 0$.

Problem 15-11 (Simmons)

Solution.

(a) Assume P is differentiable. With $y = uv$: $y' = u'v + uv'$, $y'' = u''v + 2u'v' + uv''$, so (1) becomes

$$uv'' + (2u' + Pu)v' + (u'' + Pu' + Qu)v = 0.$$

The v' term disappears if $2u' + Pu = 0$; take the nonzero solution

$$u = e^{-\frac{1}{2}\int P\,dx}, \qquad u' = -\tfrac{1}{2}Pu, \qquad u'' = -\tfrac{1}{2}P'u - \tfrac{1}{2}Pu' = -\tfrac{1}{2}P'u + \tfrac{1}{4}P^2u.$$

Then

$$u'' + Pu' + Qu = u\left(-\tfrac{1}{2}P' + \tfrac{1}{4}P^2 - \tfrac{1}{2}P^2 + Q\right) = u\left(Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2\right).$$

Dividing by $u \neq 0$:

$$v'' + \left(Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2\right)v = 0.$$

(b) Here $P = 2x$, $Q = 1 + x^2$, so $u = e^{-\frac{1}{2}\int 2x\,dx} = e^{-x^2/2}$ and

$$Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2 = 1 + x^2 - 1 - x^2 = 0.$$

So $v'' = 0$, $v = c_1x + c_2$, and

$$y = uv = (c_1x + c_2)e^{-x^2/2}.$$

This is the general solution, since $e^{-x^2/2}$ and $xe^{-x^2/2}$ have Wronskian $e^{-x^2} \neq 0$ (and they solve the equation by the construction above).

Answer. (a) $u = e^{-\frac{1}{2}\int P\,dx}$ and $v'' + \left(Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2\right)v = 0$; (b) $y = (c_1x + c_2)e^{-x^2/2}$.

Section 16 The Use of a Known Solution to Find Another

Throughout this section, equation (1) is $y'' + P(x)y' + Q(x)y = 0$, and the method of the text is used: if y_1 is a known solution that does not vanish on the interval under consideration, then $y_2 = vy_1$ is a second, linearly independent solution, where

$$v = \int \frac{1}{y_1^2}\, e^{-\int P\,dx}\,dx. \tag{3}$$

Before applying (3) every equation must first be divided by its leading coefficient so that it has the form (1). Constants of integration in $\int P\,dx$ and in (3) may be dropped, since they only multiply y_2 by a nonzero constant or add a multiple of y_1 to it, and either change is absorbed into the arbitrary constants of the general solution $y = c_1y_1 + c_2y_2$. When P has a term k/x, we work on an interval with $x > 0$ (so that $\int dx/x = \log x$); on an interval with $x < 0$ one gets $\log|x| = \log(-x)$ instead, which changes $e^{-\int P\,dx}$ only by a constant factor ± 1 and hence leads to the same general solution.

Problem 16-1 (Simmons)

Solution. By construction $v' = \dfrac{1}{y_1^2}e^{-\int P\,dx}$ (formula (3) differentiated), on an interval where $y_1 \neq 0$. Since $y_2 = vy_1$, we have $y_2' = v'y_1 + vy_1'$, and so

$$\begin{aligned} W(y_1, y_2) &= y_1y_2' - y_2y_1' = y_1(v'y_1 + vy_1') - vy_1y_1' \\ &= v'y_1^2 + vy_1y_1' - vy_1y_1' = v'y_1^2 \\ &= y_1^2 \cdot \frac{1}{y_1^2}e^{-\int P\,dx} = e^{-\int P\,dx}. \end{aligned}$$

An exponential is never zero, so $W(y_1, y_2) \neq 0$ at every point of the interval. By the theory of Section 15 (two solutions of (1) are linearly independent if and only if their Wronskian does not vanish; or directly: if $c_1y_1 + c_2y_2 \equiv 0$ then also $c_1y_1' + c_2y_2' \equiv 0$, and this homogeneous linear system for c_1, c_2 has determinant $W \neq 0$, so $c_1 = c_2 = 0$), the solutions y_1 and y_2 are linearly independent.

Answer. Shown: $W(y_1, y_2) = v'y_1^2 = e^{-\int P\,dx} \neq 0$, so y_1 and $y_2 = vy_1$ are linearly independent.

Problem 16-2 (Simmons)

Solution.

(a) Here $P(x) = 0$, so $e^{-\int P\,dx} = 1$. On any interval where $\sin x \neq 0$, formula (3) gives

$$v = \int \frac{dx}{\sin^2 x} = \int \csc^2 x\,dx = -\cot x, \qquad y_2 = vy_1 = -\cot x\,\sin x = -\cos x.$$

The function $y_2 = -\cos x$ satisfies $y_2'' + y_2 = \cos x - \cos x = 0$ on the whole line, and

$$W(\sin x, -\cos x) = \sin x \cdot \sin x - (-\cos x)\cos x = 1 \neq 0,$$

so the two solutions are independent everywhere. Absorbing the sign into the constant, the general solution is

$$y = c_1 \sin x + c_2 \cos x.$$

(b) Again $P(x) = 0$ and $e^{-\int P\,dx} = 1$; $y_1 = e^x$ never vanishes, so

$$v = \int \frac{dx}{e^{2x}} = \int e^{-2x}\,dx = -\tfrac{1}{2}e^{-2x}, \qquad y_2 = vy_1 = -\tfrac{1}{2}e^{-2x}e^x = -\tfrac{1}{2}e^{-x}.$$

Check: $y_2'' - y_2 = -\tfrac{1}{2}e^{-x} + \tfrac{1}{2}e^{-x} = 0$. The general solution is

$$y = c_1e^x + c_2e^{-x}.$$

Answer. (a) $y_2 = -\cos x$, $y = c_1\sin x + c_2\cos x$; (b) $y_2 = -\tfrac{1}{2}e^{-x}$, $y = c_1e^x + c_2e^{-x}$.

Problem 16-3 (Simmons)

Solution. $y_1 = 1$ is a solution because $y_1' = y_1'' = 0$. Dividing by x puts the equation in the form (1): $y'' + \dfrac{3}{x}y' = 0$, so $P(x) = 3/x$ and (for $x > 0$)

$$e^{-\int P\,dx} = e^{-3\log x} = x^{-3}.$$

Then formula (3) gives

$$v = \int \frac{1}{1^2}\,x^{-3}\,dx = \frac{x^{-2}}{-2} = -\frac{1}{2}x^{-2}, \qquad y_2 = vy_1 = -\frac{1}{2}x^{-2}.$$

Check: $y_2' = x^{-3}$, $y_2'' = -3x^{-4}$, so $xy_2'' + 3y_2' = -3x^{-3} + 3x^{-3} = 0$. The general solution is

$$y = c_1 + c_2x^{-2}.$$

Answer. $y_2 = -\tfrac{1}{2}x^{-2}$, $y = c_1 + c_2x^{-2}$.

Problem 16-4 (Simmons)

Solution. With $y_1 = x^2$: $y_1' = 2x$, $y_1'' = 2$, so $x^2(2) + x(2x) - 4x^2 = 2x^2 + 2x^2 - 4x^2 = 0$; thus y_1 is a solution.

Dividing by x^2: $y'' + \dfrac{1}{x}y' - \dfrac{4}{x^2}y = 0$, so $P(x) = 1/x$ and, for $x > 0$, $e^{-\int P\,dx} = e^{-\log x} = x^{-1}$. Formula (3) gives

$$v = \int \frac{1}{x^4}\,x^{-1}\,dx = \int x^{-5}\,dx = -\frac{1}{4}x^{-4}, \qquad y_2 = vy_1 = -\frac{1}{4}x^{-4}\cdot x^2 = -\frac{1}{4}x^{-2}.$$

Check: $y_2' = \tfrac{1}{2}x^{-3}$, $y_2'' = -\tfrac{3}{2}x^{-4}$, and $x^2y_2'' + xy_2' - 4y_2 = -\tfrac{3}{2}x^{-2} + \tfrac{1}{2}x^{-2} + x^{-2} = 0$. The general solution is

$$y = c_1x^2 + c_2x^{-2}.$$

Answer. $y_2 = -\tfrac{1}{4}x^{-2}$, $y = c_1x^2 + c_2x^{-2}$.

Problem 16-5 (Simmons)

Solution. Indeed $y_1 = x$ gives $(1-x^2)\cdot 0 - 2x\cdot 1 + 2x = 0$. We work on the interval $-1 < x < 1$, where $1 - x^2 > 0$. Dividing by $1 - x^2$:

$$y'' - \frac{2x}{1-x^2}y' + \frac{2}{1-x^2}y = 0, \qquad P(x) = -\frac{2x}{1-x^2}.$$

With the substitution $u = 1 - x^2$, $du = -2x\,dx$,

$$-\int P\,dx = \int \frac{2x}{1-x^2}\,dx = -\int \frac{du}{u} = -\log(1-x^2), \qquad e^{-\int P\,dx} = \frac{1}{1-x^2}.$$

Formula (3), on an interval where $x \neq 0$ (so that $y_1 \neq 0$), gives

$$v = \int \frac{1}{x^2}\cdot\frac{1}{1-x^2}\,dx.$$

By partial fractions, $\dfrac{1}{x^2(1-x^2)} = \dfrac{1}{x^2} + \dfrac{1}{1-x^2}$ (indeed $\frac{(1-x^2)+x^2}{x^2(1-x^2)} = \frac{1}{x^2(1-x^2)}$), and $\dfrac{1}{1-x^2} = \dfrac{1}{2}\left(\dfrac{1}{1+x} + \dfrac{1}{1-x}\right)$. Hence

$$v = -\frac{1}{x} + \frac{1}{2}\big[\log(1+x) - \log(1-x)\big] = -\frac{1}{x} + \frac{1}{2}\log\frac{1+x}{1-x},$$

and

$$y_2 = vy_1 = \frac{x}{2}\log\frac{1+x}{1-x} - 1.$$

This function is defined and twice differentiable on all of $-1 < x < 1$ (including $x = 0$), and a direct substitution (carried out in the verification script) shows that it satisfies the equation there; for $x \neq 0$ its Wronskian with y_1 is $e^{-\int P\,dx} = 1/(1-x^2) \neq 0$ (Problem 1); since y_1, y_2 are smooth, $W = y_1y_2' - y_2y_1'$ is continuous, so $W = 1/(1-x^2)$ also at $x = 0$. The general solution on $-1 < x < 1$ is therefore

$$y = c_1x + c_2\left[\frac{x}{2}\log\left(\frac{1+x}{1-x}\right) - 1\right].$$

Answer. $y = c_1x + c_2\left[\dfrac{x}{2}\log\left(\dfrac{1+x}{1-x}\right) - 1\right], \quad -1 < x < 1.$

Problem 16-6 (Simmons)

Solution. *Verification.* For $x > 0$, with $y_1 = x^{-1/2}\sin x$:

$$\begin{aligned} y_1' &= -\tfrac{1}{2}x^{-3/2}\sin x + x^{-1/2}\cos x, \\ y_1'' &= \tfrac{3}{4}x^{-5/2}\sin x - \tfrac{1}{2}x^{-3/2}\cos x - \tfrac{1}{2}x^{-3/2}\cos x - x^{-1/2}\sin x \\ &= \tfrac{3}{4}x^{-5/2}\sin x - x^{-3/2}\cos x - x^{-1/2}\sin x. \end{aligned}$$

Hence

$$\begin{aligned} x^2y_1'' &= \tfrac{3}{4}x^{-1/2}\sin x - x^{1/2}\cos x - x^{3/2}\sin x, \\ xy_1' &= -\tfrac{1}{2}x^{-1/2}\sin x + x^{1/2}\cos x, \\ \left(x^2 - \tfrac{1}{4}\right)y_1 &= x^{3/2}\sin x - \tfrac{1}{4}x^{-1/2}\sin x, \end{aligned}$$

and the sum is $\left(\frac{3}{4}-\frac{1}{2}-\frac{1}{4}\right)x^{-1/2}\sin x+(-1+1)x^{1/2}\cos x+(-1+1)x^{3/2}\sin x=0$.

Second solution. Dividing by x^2: $y''+\frac{1}{x}y'+\left(1-\frac{1}{4x^2}\right)y=0$, so $P=1/x$ and $e^{-\int P\,dx}=e^{-\log x}=x^{-1}$ $(x>0)$. On an interval where $\sin x\neq 0$, formula (3) gives

$$v=\int\frac{1}{x^{-1}\sin^2 x}\,x^{-1}\,dx=\int\csc^2 x\,dx=-\cot x,$$
$$y_2=vy_1=-\cot x\cdot x^{-1/2}\sin x=-x^{-1/2}\cos x.$$

To see that $x^{-1/2}\cos x$ is a solution on the whole half-line $x>0$, substitute $y=x^{-1/2}u$:

$$y'=x^{-1/2}u'-\tfrac{1}{2}x^{-3/2}u,\qquad y''=x^{-1/2}u''-x^{-3/2}u'+\tfrac{3}{4}x^{-5/2}u,$$
$$\begin{aligned}x^2y''+xy'+\left(x^2-\tfrac{1}{4}\right)y&=x^{3/2}u''-x^{1/2}u'+\tfrac{3}{4}x^{-1/2}u+x^{1/2}u'-\tfrac{1}{2}x^{-1/2}u+x^{3/2}u-\tfrac{1}{4}x^{-1/2}u\\&=x^{3/2}\,(u''+u).\end{aligned}$$

Thus $x^{-1/2}u$ solves Bessel's equation (for $p=\frac{1}{2}$) exactly when $u''+u=0$; both $u=\sin x$ (which re-verifies y_1) and $u=\cos x$ qualify. Moreover $W(x^{-1/2}\sin x,\,-x^{-1/2}\cos x)=x^{-1}\neq 0$ for all $x>0$ (Problem 1), so y_1,y_2 are independent on every interval of positive x. The general solution there is

$$y=c_1x^{-1/2}\sin x+c_2x^{-1/2}\cos x.$$

Answer. $y_1=x^{-1/2}\sin x$ verified; $\quad y=c_1x^{-1/2}\sin x+c_2x^{-1/2}\cos x\ \ (x>0)$.

Problem 16-7 (Simmons)

Solution. In each case $y_1=x$, $y_1'=1$, $y_1''=0$, and we work on intervals where $x>0$ and all coefficients are defined (in (a) we take $x>1$).

(a) Check: $0-\frac{x}{x-1}+\frac{x}{x-1}=0$. Here $P=-\dfrac{x}{x-1}=-1-\dfrac{1}{x-1}$, so on an interval with $x>1$,

$$-\int P\,dx=x+\log(x-1),\qquad e^{-\int P\,dx}=(x-1)e^x.$$

Then

$$v=\int\frac{(x-1)e^x}{x^2}\,dx=\frac{e^x}{x},$$

because $\dfrac{d}{dx}\dfrac{e^x}{x}=\dfrac{xe^x-e^x}{x^2}=\dfrac{(x-1)e^x}{x^2}$. Hence $y_2=vy_1=e^x$. (On $0<x<1$ the factor $|x-1|=1-x$ only changes the sign of v.) Check: $e^x-\frac{x}{x-1}e^x+\frac{1}{x-1}e^x=e^x\left(1-\frac{x-1}{x-1}\right)=0$. The general solution is $y=c_1x+c_2e^x$.

(b) Check: $0+2x-2x=0$. Dividing by x^2: $P=2/x$, $e^{-\int P\,dx}=x^{-2}$, and

$$v=\int\frac{1}{x^2}\,x^{-2}\,dx=\int x^{-4}\,dx=-\frac{1}{3}x^{-3},\qquad y_2=-\frac{1}{3}x^{-2}.$$

Check: for $y=x^{-2}$, $x^2(6x^{-4})+2x(-2x^{-3})-2x^{-2}=(6-4-2)x^{-2}=0$. The general solution is $y=c_1x+c_2x^{-2}$.

(c) Check: $0-x(x+2)+(x+2)x=0$. Dividing by x^2: $P=-\dfrac{x+2}{x}=-1-\dfrac{2}{x}$, so

$$-\int P\,dx=x+2\log x,\qquad e^{-\int P\,dx}=x^2e^x,$$

and

$$v = \int \frac{1}{x^2}\, x^2 e^x \, dx = e^x, \qquad y_2 = xe^x.$$

Check: $(xe^x)' = (x+1)e^x$, $(xe^x)'' = (x+2)e^x$, and $x^2(x+2)e^x - x(x+2)(x+1)e^x + (x+2)xe^x = x(x+2)e^x\,[x-(x+1)+1] = 0$. The general solution is $y = c_1 x + c_2 x e^x$.

Answer. (a) $y = c_1 x + c_2 e^x$; (b) $y = c_1 x + c_2 x^{-2}$; (c) $y = c_1 x + c_2 x e^x$.

Problem 16-8 (Simmons)

Solution. The function $y_1 = x$ is an obvious solution: $y_1'' - xf(x)y_1' + f(x)y_1 = 0 - xf(x) + xf(x) = 0$. The equation is already in the form (1) with $P(x) = -xf(x)$, so

$$e^{-\int P\,dx} = e^{\int xf(x)\,dx},$$

and on an interval where $x \neq 0$ formula (3) gives

$$v = \int \frac{1}{x^2}\, e^{\int xf(x)\,dx}\,dx = \int x^{-2} e^{\int xf(x)\,dx}\,dx, \qquad y_2 = x\int x^{-2} e^{\int xf(x)\,dx}\,dx.$$

(By Problem 1, $W(y_1, y_2) = e^{\int xf\,dx} \neq 0$, so y_1, y_2 are independent.) The general solution is

$$y = c_1 x + c_2 x \int x^{-2} e^{\int xf(x)\,dx}\,dx.$$

Answer. $y = c_1 x + c_2 x \displaystyle\int x^{-2} e^{\int xf(x)\,dx}\,dx.$

Problem 16-9 (Simmons)

Solution. With $y_1 = y_1' = y_1'' = e^x$: $xe^x - (2x+1)e^x + (x+1)e^x = (x - 2x - 1 + x + 1)e^x = 0$.

Dividing by x: $P = -\dfrac{2x+1}{x} = -2 - \dfrac{1}{x}$, so for $x > 0$

$$-\int P\,dx = 2x + \log x, \qquad e^{-\int P\,dx} = xe^{2x},$$

and

$$v = \int \frac{1}{e^{2x}}\, xe^{2x}\,dx = \int x\,dx = \frac{x^2}{2}, \qquad y_2 = \frac{1}{2}x^2 e^x.$$

Check, with $y = x^2 e^x$: $y' = (x^2+2x)e^x$, $y'' = (x^2+4x+2)e^x$, and

$$\begin{aligned} &x(x^2+4x+2) - (2x+1)(x^2+2x) + (x+1)x^2 \\ &\quad = (x^3+4x^2+2x) - (2x^3+5x^2+2x) + (x^3+x^2) = 0. \end{aligned}$$

The general solution is

$$y = c_1 e^x + c_2 x^2 e^x.$$

Answer. $y = c_1 e^x + c_2 x^2 e^x$.

Problem 16-10 (Simmons)

Solution.

(a) Since the coefficients of y'', y', y add up to $x-(x+n)+n=0$, the function $y_1=e^x$ (for which $y_1=y_1'=y_1''$) is a solution: $xe^x-(x+n)e^x+ne^x=0$.

Dividing by x: $P=-\dfrac{x+n}{x}=-1-\dfrac{n}{x}$, so for $x>0$

$$-\int P\,dx = x+n\log x, \qquad e^{-\int P\,dx}=x^n e^x,$$

and formula (3) gives

$$v=\int \frac{1}{e^{2x}}\,x^n e^x\,dx=\int x^n e^{-x}\,dx, \qquad y_2=e^x\int x^n e^{-x}\,dx.$$

This integral can be evaluated explicitly. Repeated integration by parts gives

$$\int x^n e^{-x}\,dx=-e^{-x}\left(x^n+nx^{n-1}+n(n-1)x^{n-2}+\cdots+n!\right)=-n!\,e^{-x}\sum_{k=0}^{n}\frac{x^k}{k!},$$

which is confirmed by differentiation: if $S(x)=\sum_{k=0}^{n}x^k/k!$ then $S'=S-x^n/n!$, so

$$\frac{d}{dx}\left[-n!\,e^{-x}S\right]=n!\,e^{-x}(S-S')=n!\,e^{-x}\frac{x^n}{n!}=x^n e^{-x}.$$

Therefore $y_2=-n!\,S(x)$, and dropping the constant factor $-n!$ we may take

$$y_2=\sum_{k=0}^{n}\frac{x^k}{k!}=1+x+\frac{x^2}{2!}+\cdots+\frac{x^n}{n!}.$$

We check directly that this polynomial solves the equation for all x: with $S'=S-\frac{x^n}{n!}$ and $S''=S'-\frac{x^{n-1}}{(n-1)!}=S-\frac{x^n}{n!}-\frac{x^{n-1}}{(n-1)!}$,

$$\begin{aligned} xS''-(x+n)S'+nS &= xS-\frac{x^{n+1}}{n!}-\frac{x^n}{(n-1)!}-(x+n)S+\frac{(x+n)x^n}{n!}+nS \\ &= \left(-\frac{x^{n+1}}{n!}+\frac{x^{n+1}}{n!}\right)+\left(-\frac{x^n}{(n-1)!}+\frac{n\,x^n}{n!}\right)=0. \end{aligned}$$

The solutions e^x and $S(x)$ are linearly independent, since their ratio $S(x)e^{-x}$ is not constant (it tends to 0 as $x\to\infty$ but equals 1 at $x=0$). Thus two linearly independent solutions are

$$y_1=e^x, \qquad y_2=e^x\int x^n e^{-x}\,dx \quad (\text{equivalently } y_2=\textstyle\sum_{k=0}^{n}x^k/k!).$$

(b) Using $y_2=-e^x\int x^n e^{-x}\,dx=n!\sum_{k=0}^{n}x^k/k!$:

- $n=1$: $\int xe^{-x}\,dx=-(x+1)e^{-x}$, so $y_2=x+1$ and $y=c_1e^x+c_2(x+1)$.
- $n=2$: $\int x^2e^{-x}\,dx=-(x^2+2x+2)e^{-x}$, so $y=c_1e^x+c_2(x^2+2x+2)$.
- $n=3$: $\int x^3e^{-x}\,dx=-(x^3+3x^2+6x+6)e^{-x}$, so $y=c_1e^x+c_2(x^3+3x^2+6x+6)$.

(Each integral is obtained by integrating by parts, e.g. $\int x^2e^{-x}\,dx=-x^2e^{-x}+2\int xe^{-x}\,dx$, and is checked by differentiation.)

Answer. (a) $y_1 = e^x$, $y_2 = e^x \int x^n e^{-x}\,dx$ (a constant multiple of $\sum_{k=0}^{n} x^k/k!$);
(b) $n=1$: $y = c_1e^x + c_2(x+1)$; $n=2$: $y = c_1e^x + c_2(x^2+2x+2)$; $n=3$: $y = c_1e^x + c_2(x^3+3x^2+6x+6)$.

Problem 16-11 (Simmons)

Solution. The coefficients add up to $1 - f(x) + f(x) - 1 = 0$, so $y_1 = e^x$ is a solution: $e^x - f(x)e^x + [f(x)-1]e^x = 0$. Here $P = -f(x)$ and $e^{-\int P\,dx} = e^{\int f(x)\,dx}$, so formula (3) gives

$$v = \int \frac{1}{e^{2x}}\, e^{\int f(x)\,dx}\,dx = \int e^{-2x+\int f(x)\,dx}\,dx, \qquad y_2 = e^x \int e^{-2x+\int f(x)\,dx}\,dx.$$

By Problem 1 these are linearly independent, and the general solution is

$$y = c_1e^x + c_2e^x \int e^{-2x+\int f(x)\,dx}\,dx.$$

Answer. $y = c_1e^x + c_2e^x \int e^{-2x+\int f(x)\,dx}\,dx$.

Problem 16-12 (Simmons)

Solution. Let y_1 be a solution of (1) that does not vanish on the interval, and suppose y_2 is any other solution; write $v = y_2/y_1$, so $y_2 = vy_1$. By the quotient rule,

$$v' = \left(\frac{y_2}{y_1}\right)' = \frac{y_1y_2' - y_2y_1'}{y_1^2} = \frac{W(y_1,y_2)}{y_1^2}.$$

By Abel's formula (Section 15, formula (3)), the Wronskian of two solutions of (1) satisfies $W' + PW = 0$, so $W(y_1,y_2) = c\,e^{-\int P\,dx}$ for some constant c. Hence

$$v' = \frac{c}{y_1^2}\,e^{-\int P\,dx}, \qquad v = c\int \frac{1}{y_1^2}\,e^{-\int P\,dx}\,dx + c'.$$

For $y_2 = vy_1$ to be linearly independent of y_1 we need $W \neq 0$, i.e. $c \neq 0$; and the constant c' merely adds $c'y_1$ to y_2. Choosing $c = 1$, $c' = 0$ we obtain exactly formula (3):

$$v = \int \frac{1}{y_1^2}\,e^{-\int P\,dx}\,dx.$$

Conversely, for this v the function vy_1 is indeed a solution, as shown in the text (or: its Wronskian with y_1 is $e^{-\int P\,dx}$, matching Abel's formula).

Answer. Shown: $v' = W(y_1,y_2)/y_1^2$ and, by Abel's formula $W = ce^{-\int P\,dx}$ (take $c=1$), $v = \int y_1^{-2}e^{-\int P\,dx}\,dx$, which is formula (3).

Section 17 The Homogeneous Equation with Constant Coefficients

Throughout this section we consider $y'' + py' + qy = 0$ with constants p, q (equation (1)). Substituting $y = e^{mx}$ gives $(m^2 + pm + q)e^{mx} = 0$, so m must satisfy the auxiliary equation $m^2 + pm + q = 0$ with roots $m_{1,2} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$. The general solution is

$$\begin{aligned} &y = c_1 e^{m_1 x} + c_2 e^{m_2 x} && (m_1 \neq m_2 \text{ real}),\\ &y = e^{ax}(c_1 \cos bx + c_2 \sin bx) && (m_{1,2} = a \pm ib,\ b > 0),\\ &y = c_1 e^{mx} + c_2 x e^{mx} && (m_1 = m_2 = m). \end{aligned}$$

If the equation is not monic, we simply keep its leading coefficient in the auxiliary equation.

Problem 17-1 (Simmons)

Solution. In each part we write the auxiliary equation, find its roots, and use the corresponding form of the general solution.

(a) $m^2 + m - 6 = (m+3)(m-2) = 0$, so $m = 2, -3$ (distinct real): $y = c_1 e^{2x} + c_2 e^{-3x}$.

(b) $m^2 + 2m + 1 = (m+1)^2 = 0$, so $m = -1$ (double): $y = c_1 e^{-x} + c_2 x e^{-x}$.

(c) $m^2 + 8 = 0$, so $m = \pm i\sqrt{8} = \pm 2\sqrt{2}\, i$ ($a = 0$, $b = 2\sqrt{2}$): $y = c_1 \cos 2\sqrt{2}\, x + c_2 \sin 2\sqrt{2}\, x$.

(d) Dividing by 2 (or directly) $2m^2 - 4m + 8 = 0$, i.e. $m^2 - 2m + 4 = 0$: $m = \dfrac{2 \pm \sqrt{4 - 16}}{2} = 1 \pm i\sqrt{3}$. Hence $y = e^x \left(c_1 \cos \sqrt{3}\, x + c_2 \sin \sqrt{3}\, x\right)$.

(e) $m^2 - 4m + 4 = (m-2)^2 = 0$, so $m = 2$ (double): $y = c_1 e^{2x} + c_2 x e^{2x}$.

(f) $m^2 - 9m + 20 = (m-4)(m-5) = 0$, so $m = 4, 5$: $y = c_1 e^{5x} + c_2 e^{4x}$ (the order of the constants is immaterial).

(g) $2m^2 + 2m + 3 = 0$, so $m = \dfrac{-2 \pm \sqrt{4 - 24}}{4} = \dfrac{-2 \pm 2i\sqrt{5}}{4} = -\dfrac{1}{2} \pm \dfrac{\sqrt{5}}{2} i$. Hence $y = e^{-x/2}\left(c_1 \cos \dfrac{\sqrt{5}}{2} x + c_2 \sin \dfrac{\sqrt{5}}{2} x\right)$.

(h) $4m^2 - 12m + 9 = (2m-3)^2 = 0$, so $m = \frac{3}{2}$ (double): $y = c_1 e^{3x/2} + c_2 x e^{3x/2}$.

(i) $m^2 + m = m(m+1) = 0$, so $m = 0, -1$: $y = c_1 + c_2 e^{-x}$.

(j) $m^2 - 6m + 25 = 0$, so $m = \dfrac{6 \pm \sqrt{36 - 100}}{2} = 3 \pm 4i$: $y = e^{3x}(c_1 \cos 4x + c_2 \sin 4x)$.

(k) $4m^2 + 20m + 25 = (2m+5)^2 = 0$, so $m = -\frac{5}{2}$ (double): $y = c_1 e^{-5x/2} + c_2 x e^{-5x/2}$.

(l) $m^2 + 2m + 3 = 0$, so $m = \dfrac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm i\sqrt{2}$: $y = e^{-x}\left(c_1 \cos \sqrt{2}\, x + c_2 \sin \sqrt{2}\, x\right)$.

(m) $y'' - 4y = 0$ gives $m^2 - 4 = 0$, so $m = \pm 2$: $y = c_1 e^{2x} + c_2 e^{-2x}$.

(n) $4m^2 - 8m + 7 = 0$, so

$$m = \frac{8 \pm \sqrt{64 - 112}}{8} = \frac{8 \pm 4\sqrt{3}\, i}{8} = 1 \pm \frac{\sqrt{3}}{2} i, \qquad y = e^x \left(c_1 \cos \frac{\sqrt{3}}{2} x + c_2 \sin \frac{\sqrt{3}}{2} x\right).$$

(o) $2m^2 + m - 1 = (2m-1)(m+1) = 0$, so $m = \frac{1}{2}, -1$: $y = c_1 e^{x/2} + c_2 e^{-x}$.

(p) $16m^2 - 8m + 1 = (4m-1)^2 = 0$, so $m = \frac{1}{4}$ (double): $y = c_1 e^{x/4} + c_2 x e^{x/4}$.

(q) $m^2 + 4m + 5 = 0$, so $m = \dfrac{-4 \pm \sqrt{16-20}}{2} = -2 \pm i$: $y = e^{-2x}(c_1 \cos x + c_2 \sin x)$.

(r) $m^2 + 4m - 5 = (m+5)(m-1) = 0$, so $m = 1, -5$: $y = c_1 e^{x} + c_2 e^{-5x}$.

Answer. (a) $c_1 e^{2x} + c_2 e^{-3x}$; (b) $c_1 e^{-x} + c_2 x e^{-x}$; (c) $c_1 \cos 2\sqrt{2}x + c_2 \sin 2\sqrt{2}x$; (d) $e^x(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x)$; (e) $c_1 e^{2x} + c_2 x e^{2x}$; (f) $c_1 e^{5x} + c_2 e^{4x}$; (g) $e^{-x/2}\big(c_1 \cos \frac{\sqrt{5}}{2}x + c_2 \sin \frac{\sqrt{5}}{2}x\big)$; (h) $c_1 e^{3x/2} + c_2 x e^{3x/2}$; (i) $c_1 + c_2 e^{-x}$; (j) $e^{3x}(c_1 \cos 4x + c_2 \sin 4x)$; (k) $c_1 e^{-5x/2} + c_2 x e^{-5x/2}$; (l) $e^{-x}(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x)$; (m) $c_1 e^{2x} + c_2 e^{-2x}$; (n) $e^x\big(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x\big)$; (o) $c_1 e^{x/2} + c_2 e^{-x}$; (p) $c_1 e^{x/4} + c_2 x e^{x/4}$; (q) $e^{-2x}(c_1 \cos x + c_2 \sin x)$; (r) $c_1 e^{x} + c_2 e^{-5x}$.

Problem 17-2 (Simmons)

Solution.

(a) $m^2 - 5m + 6 = (m-2)(m-3) = 0$, so $y = c_1 e^{2x} + c_2 e^{3x}$ and $y' = 2c_1 e^{2x} + 3c_2 e^{3x}$. The conditions at $x = 1$ give

$$c_1 e^2 + c_2 e^3 = e^2, \qquad 2c_1 e^2 + 3c_2 e^3 = 3e^2.$$

Subtracting twice the first equation from the second: $c_2 e^3 = e^2$, so $c_2 = e^{-1}$, and then $c_1 e^2 = e^2 - e^2 = 0$, so $c_1 = 0$. Hence $y = e^{-1} e^{3x} = e^{3x-1}$.

(b) $m^2 - 6m + 5 = (m-1)(m-5) = 0$: $y = c_1 e^x + c_2 e^{5x}$, $y' = c_1 e^x + 5c_2 e^{5x}$. Then $c_1 + c_2 = 3$ and $c_1 + 5c_2 = 11$; subtracting, $4c_2 = 8$, so $c_2 = 2$, $c_1 = 1$: $y = e^x + 2e^{5x}$.

(c) $m^2 - 6m + 9 = (m-3)^2 = 0$: $y = (c_1 + c_2 x)e^{3x}$, $y' = \big(c_2 + 3c_1 + 3c_2 x\big)e^{3x}$. Then $y(0) = c_1 = 0$ and $y'(0) = c_2 + 3c_1 = c_2 = 5$. So $y = 5x e^{3x}$.

(d) $m^2 + 4m + 5 = 0$ gives $m = -2 \pm i$, so $y = e^{-2x}(c_1 \cos x + c_2 \sin x)$ and

$$y' = e^{-2x}\big[(-2c_1 + c_2)\cos x + (-2c_2 - c_1)\sin x\big].$$

Then $y(0) = c_1 = 1$ and $y'(0) = -2c_1 + c_2 = 0$, so $c_2 = 2$: $y = e^{-2x}(\cos x + 2\sin x)$.

(e) $m^2 + 4m + 2 = 0$ gives $m = \dfrac{-4 \pm \sqrt{8}}{2} = -2 \pm \sqrt{2}$, so $y = c_1 e^{(-2+\sqrt{2})x} + c_2 e^{(-2-\sqrt{2})x}$. The conditions are

$$c_1 + c_2 = -1, \qquad (-2+\sqrt{2})c_1 + (-2-\sqrt{2})c_2 = 2 + 3\sqrt{2}.$$

Substituting $c_1 = -1 - c_2$ in the second equation: $(2-\sqrt{2}) + c_2\big[(2-\sqrt{2}) + (-2-\sqrt{2})\big] = 2 + 3\sqrt{2}$, i.e. $2 - \sqrt{2} - 2\sqrt{2}\,c_2 = 2 + 3\sqrt{2}$, so $c_2 = \dfrac{-4\sqrt{2}}{2\sqrt{2}} = -2$ and $c_1 = 1$:

$$y = e^{(-2+\sqrt{2})x} - 2e^{(-2-\sqrt{2})x}.$$

(f) $m^2 + 8m - 9 = (m+9)(m-1) = 0$: $y = c_1 e^x + c_2 e^{-9x}$, $y' = c_1 e^x - 9c_2 e^{-9x}$. The conditions at $x = 1$:

$$c_1 e + c_2 e^{-9} = 2, \qquad c_1 e - 9c_2 e^{-9} = 0.$$

Subtracting: $10c_2 e^{-9} = 2$, so $c_2 = \frac{e^9}{5}$; then $c_1 e = 9c_2 e^{-9} = \frac{9}{5}$, so $c_1 = \frac{9}{5e}$. Therefore

$$y = \frac{9}{5}e^{x-1} + \frac{1}{5}e^{9-9x} = \frac{9}{5}e^{x-1} + \frac{1}{5}e^{-9(x-1)}.$$

Answer. (a) $y = e^{3x-1}$; (b) $y = e^x + 2e^{5x}$; (c) $y = 5xe^{3x}$; (d) $y = e^{-2x}(\cos x + 2\sin x)$; (e) $y = e^{(-2+\sqrt{2})x} - 2e^{(-2-\sqrt{2})x}$; (f) $y = \frac{9}{5}e^{x-1} + \frac{1}{5}e^{-9(x-1)}$.

Problem 17-3 (Simmons)

Solution. "The general solution approaches 0" means that every solution does, i.e. $c_1y_1 + c_2y_2 \to 0$ for all choices of c_1, c_2, where y_1, y_2 are the basic solutions. Let m_1, m_2 be the roots of $m^2 + pm + q = 0$; then $m_1 + m_2 = -p$ and $m_1m_2 = q$.

Step 1: every solution tends to 0 if and only if both roots have negative real part.

($\Leftarrow$) Suppose both roots have negative real part. The basic solutions are e^{m_1x}, e^{m_2x} (distinct real), or $e^{ax}\cos bx$, $e^{ax}\sin bx$ with $a < 0$ (complex roots), or e^{mx}, xe^{mx} with $m < 0$ (equal roots). Each tends to 0 as $x \to \infty$: $e^{kx} \to 0$ for $k < 0$; $|e^{ax}\cos bx| \le e^{ax} \to 0$ and likewise for the sine; and $xe^{mx} = x/e^{-mx} \to 0$ for $m < 0$ (l'Hôpital's rule). Hence all solutions tend to 0.

($\Rightarrow$) Suppose some root m_1 has $\operatorname{Re} m_1 \ge 0$; we exhibit a solution that does not tend to 0. If m_1 is real, then $y = e^{m_1x}$ is a solution with $y(x) \ge 1$ for all $x \ge 0$. If $m_1 = a + ib$ with $b > 0$, $a \ge 0$, then $y = e^{ax}\cos bx$ is a solution, and at the points $x_k = 2\pi k/b \to \infty$ we have $y(x_k) = e^{ax_k} \ge 1$. In either case $y \not\to 0$.

Step 2: both roots have negative real part if and only if $p > 0$ and $q > 0$.

Complex roots ($p^2 < 4q$, $m_{1,2} = a \pm ib$, $b \neq 0$): here $a = -p/2$ and $q = a^2 + b^2 > 0$ automatically. So $a < 0 \iff p > 0 \iff (p > 0 \text{ and } q > 0)$.

Real roots ($p^2 \ge 4q$, including equal roots): if both roots are negative, then $p = -(m_1 + m_2) > 0$ and $q = m_1m_2 > 0$. Conversely, if $p > 0$ and $q > 0$, then $m_1m_2 = q > 0$ shows that m_1, m_2 have the same sign, and $m_1 + m_2 = -p < 0$ shows that this common sign is negative.

Combining Steps 1 and 2: every solution of (1) tends to 0 as $x \to \infty$ if and only if $p > 0$ and $q > 0$.

Answer. Shown: all solutions of $y'' + py' + qy = 0$ tend to 0 as $x \to \infty \iff$ both roots of $m^2 + pm + q = 0$ have negative real part $\iff p > 0$ and $q > 0$.

Problem 17-4 (Simmons)

Solution. Let y be a solution of $y'' + py' + qy = 0$ on an interval, with p, q constants. First, y has derivatives of all orders: y is twice differentiable, and from the equation

$$y'' = -py' - qy$$

the right side is differentiable (since y and y' are), hence y'' is differentiable, so y''' exists and $y''' = -py'' - qy'$. Now differentiate the identity $y'' + py' + qy = 0$ term by term; since p and q are constants,

$$y''' + py'' + qy' = 0, \qquad \text{i.e.} \qquad (y')'' + p\,(y')' + q\,(y') = 0.$$

This says exactly that $u = y'$ satisfies $u'' + pu' + qu = 0$, so y' is again a solution of (1). (The constancy of p and q is essential: for variable coefficients we would get the extra terms $P'y' + Q'y$.)

Answer. Shown: differentiating $y'' + py' + qy = 0$ gives $(y')'' + p(y')' + q(y') = 0$, so y' is a solution.

Problem 17-5 (Simmons)

Solution. *The transformation.* Take $x > 0$ and put $x = e^z$, i.e. $z = \log x$, so $\mathrm{d}z/\,\mathrm{d}x = 1/x$. Regard y as a function of z. By the chain rule,

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz},$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\cdot\frac{1}{x}\frac{d^2y}{dz^2} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right).$$

Substituting in $x^2y'' + pxy' + qy = 0$:

$$\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) + p\frac{dy}{dz} + qy = 0, \qquad \text{i.e.} \qquad \frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = 0, \tag{$*$}$$

a linear equation with the constant coefficients $p-1$ and q. Its auxiliary equation is $m^2 + (p-1)m + q = 0$, and then $e^{mz} = x^m$. (For $x < 0$ put $x = -e^z$, $z = \log|x|$; again $\mathrm{d}z/\,\mathrm{d}x = 1/x$ and the same computation applies, so the solutions below hold for $x < 0$ with x^m replaced by $|x|^m$ and $\log x$ by $\log|x|$.) We work with $x > 0$. Note that the solutions in z translate as: $e^{m_1 z}, e^{m_2 z} \to x^{m_1}, x^{m_2}$; $e^{az}\cos bz,\ e^{az}\sin bz \to x^a\cos(b\log x),\ x^a\sin(b\log x)$; $e^{mz}, ze^{mz} \to x^m, x^m\log x$.

(a) Here $p = 3$, $q = 10$: $(*)$ is $y_{zz} + 2y_z + 10y = 0$, with $m^2 + 2m + 10 = 0$, $m = -1 \pm 3i$. Thus $y = e^{-z}(c_1\cos 3z + c_2\sin 3z)$, i.e.

$$y = x^{-1}\left[c_1\cos(3\log x) + c_2\sin(3\log x)\right] = x^{-1}\left[c_1\cos(\log x^3) + c_2\sin(\log x^3)\right].$$

(b) Dividing by 2: $x^2y'' + 5xy' + 4y = 0$, $p = 5$, $q = 4$: $y_{zz} + 4y_z + 4y = 0$, $(m+2)^2 = 0$, so $y = (c_1 + c_2z)e^{-2z}$:

$$y = c_1x^{-2} + c_2x^{-2}\log x.$$

(c) $p = 2$, $q = -12$: $y_{zz} + y_z - 12y = 0$, $m^2 + m - 12 = (m+4)(m-3) = 0$, so $y = c_1e^{3z} + c_2e^{-4z}$:

$$y = c_1x^3 + c_2x^{-4}.$$

(d) Dividing by 4: $x^2y'' - \frac{3}{4}y = 0$, $p = 0$, $q = -\frac{3}{4}$: $y_{zz} - y_z - \frac{3}{4}y = 0$, $m^2 - m - \frac{3}{4} = (m - \frac{3}{2})(m + \frac{1}{2}) = 0$:

$$y = c_1x^{3/2} + c_2x^{-1/2}.$$

(e) $p = -3$, $q = 4$: $y_{zz} - 4y_z + 4y = 0$, $(m-2)^2 = 0$, so $y = (c_1 + c_2z)e^{2z}$:

$$y = c_1x^2 + c_2x^2\log x.$$

(f) $p = 2$, $q = -6$: $y_{zz} + y_z - 6y = 0$, $m^2 + m - 6 = (m+3)(m-2) = 0$:

$$y = c_1x^2 + c_2x^{-3}.$$

(g) $p = 2$, $q = 3$: $y_{zz} + y_z + 3y = 0$, $m = \dfrac{-1 \pm \sqrt{1-12}}{2} = -\dfrac{1}{2} \pm \dfrac{\sqrt{11}}{2}i$, so $y = e^{-z/2}\left(c_1\cos\frac{\sqrt{11}}{2}z + c_2\sin\frac{\sqrt{11}}{2}z\right)$:

$$y = x^{-1/2}\left[c_1\cos\left(\tfrac{1}{2}\sqrt{11}\log x\right) + c_2\sin\left(\tfrac{1}{2}\sqrt{11}\log x\right)\right].$$

(h) $p = 1, q = -2$: $y_{zz} - 2y = 0$, $m = \pm\sqrt{2}$:

$$y = c_1 x^{\sqrt{2}} + c_2 x^{-\sqrt{2}}.$$

(i) $p = 1, q = -16$: $y_{zz} - 16y = 0$, $m = \pm 4$:

$$y = c_1 x^4 + c_2 x^{-4}.$$

Answer. With $x = e^z$ the equation becomes $y_{zz} + (p-1)y_z + qy = 0$. (a) $x^{-1}[c_1 \cos(\log x^3) + c_2 \sin(\log x^3)]$; (b) $c_1 x^{-2} + c_2 x^{-2} \log x$; (c) $c_1 x^3 + c_2 x^{-4}$; (d) $c_1 x^{3/2} + c_2 x^{-1/2}$; (e) $c_1 x^2 + c_2 x^2 \log x$; (f) $c_1 x^2 + c_2 x^{-3}$; (g) $x^{-1/2}[c_1 \cos(\frac{1}{2}\sqrt{11}\log x) + c_2 \sin(\frac{1}{2}\sqrt{11}\log x)]$; (h) $c_1 x^{\sqrt{2}} + c_2 x^{-\sqrt{2}}$; (i) $c_1 x^4 + c_2 x^{-4}$.

Problem 17-6 (Simmons)

Solution. As in the statement, we assume $Q > 0$ on the interval considered (so that $\sqrt{Q}$ is real), and $z = z(x)$ has $z'(x) \neq 0$ (so that z is a legitimate new independent variable). Regard y as a function of z. Then

$$y' = \frac{dy}{dz}\, z', \qquad y'' = \frac{d^2y}{dz^2}\, z'^2 + \frac{dy}{dz}\, z''.$$

Substituting in $(*)$ and dividing by z'^2:

$$\frac{d^2y}{dz^2} + \frac{z'' + Pz'}{z'^2}\frac{dy}{dz} + \frac{Q}{z'^2}\, y = 0. \tag{1}$$

Only if. Suppose (1) has constant coefficients: $Q/z'^2 = q$ and $(z'' + Pz')/z'^2 = \tilde{p}$ constants. Since $Q > 0$ and $z'^2 > 0$, $q > 0$, and $z' = \pm\sqrt{Q/q}$, so $z' = k\sqrt{Q}$ with the constant $k = \pm 1/\sqrt{q} \neq 0$ (the sign cannot change, since z' is continuous and never 0). Then $z'' = \dfrac{kQ'}{2\sqrt{Q}}$ and

$$\tilde{p} = \frac{z'' + Pz'}{z'^2} = \frac{\dfrac{kQ'}{2\sqrt{Q}} + Pk\sqrt{Q}}{k^2 Q} = \frac{Q' + 2PQ}{2k\, Q^{3/2}}, \qquad \text{so} \qquad \frac{Q' + 2PQ}{Q^{3/2}} = 2k\tilde{p}$$

is constant.

If. Suppose $(Q' + 2PQ)/Q^{3/2} = C$ is constant, and choose $z = \int \sqrt{Q}\, dx$, so $z' = \sqrt{Q} > 0$ and $z'' = Q'/(2\sqrt{Q})$. Then by the computation above with $k = 1$, the coefficient of dy/dz in (1) is

$$\frac{z'' + Pz'}{z'^2} = \frac{Q' + 2PQ}{2Q^{3/2}} = \frac{C}{2},$$

and the coefficient of y is $Q/z'^2 = Q/Q = 1$. Thus $(*)$ becomes

$$\frac{d^2y}{dz^2} + \frac{C}{2}\frac{dy}{dz} + y = 0,$$

an equation with constant coefficients.

Remark. For Euler's equation of Problem 5 we have $P = p/x$, $Q = q/x^2$ (with $q > 0$ here), and $(Q' + 2PQ)/Q^{3/2} = (-2q + 2pq)x^{-3}/(q^{3/2}x^{-3}) = 2(p-1)/\sqrt{q}$, a constant, and $z = \int \sqrt{q}\, dx/x = \sqrt{q}\log x$, consistent with the substitution $z = \log x$ up to the constant factor. (If $Q < 0$ the same argument works with $\sqrt{-Q}$ in place of $\sqrt{Q}$.)

Answer. Shown: with $z' = k\sqrt{Q}$ the coefficients in (1) are $Q/z'^2 = 1/k^2$ and $(Q' + 2PQ)/(2kQ^{3/2})$; both are constant iff $(Q' + 2PQ)/Q^{3/2}$ is constant, and then $z = \int \sqrt{Q}\,dx$ works.

Problem 17-7 (Simmons)

Solution.

(a) Dividing by x (take $x > 0$): $y'' + \dfrac{x^2-1}{x}y' + x^2 y = 0$, so $P = \dfrac{x^2-1}{x}$ and $Q = x^2$. Then $Q' = 2x$ and $2PQ = 2x(x^2-1)$, so

$$\frac{Q' + 2PQ}{Q^{3/2}} = \frac{2x + 2x^3 - 2x}{x^3} = 2,$$

a constant. Hence the transformation is possible, with $z = \int \sqrt{Q}\,dx = \int x\,dx = \frac{x^2}{2}$. Then $z' = x$, $z'' = 1$, and equation (1) of Problem 6 has coefficients

$$\frac{z'' + Pz'}{z'^2} = \frac{1 + (x^2-1)}{x^2} = 1, \qquad \frac{Q}{z'^2} = 1,$$

i.e.

$$\frac{d^2y}{dz^2} + \frac{dy}{dz} + y = 0.$$

The auxiliary equation $m^2 + m + 1 = 0$ has roots $m = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, so $y = e^{-z/2}\big(c_1 \cos\frac{\sqrt{3}}{2}z + c_2 \sin\frac{\sqrt{3}}{2}z\big)$. With $z = x^2/2$:

$$y = e^{-x^2/4}\left(c_1 \cos\frac{\sqrt{3}\,x^2}{4} + c_2 \sin\frac{\sqrt{3}\,x^2}{4}\right).$$

(Direct substitution, carried out in the verification script, shows that this function satisfies the original equation for every x, so the formula is valid on $x < 0$ as well.)

(b) Here $P = 3x$, $Q = x^2$, so $Q' + 2PQ = 2x + 6x^3$. For $x > 0$,

$$\frac{Q' + 2PQ}{Q^{3/2}} = \frac{2x + 6x^3}{x^3} = \frac{2}{x^2} + 6,$$

which is not constant; for $x < 0$, $Q^{3/2} = |x|^3 = -x^3$ and the quotient is $-\frac{2}{x^2} - 6$, again not constant (and $Q = 0$ at $x = 0$ anyway, so $\sqrt{Q}$ would not give a legitimate change of variable there). By Problem 6, equation (b) cannot be transformed into an equation with constant coefficients by a change of independent variable.

Answer. (a) Possible, with $z = x^2/2$: $y = e^{-x^2/4}\big(c_1 \cos\frac{\sqrt{3}}{4}x^2 + c_2 \sin\frac{\sqrt{3}}{4}x^2\big)$; (b) not possible.

Problem 17-8 (Simmons)

Solution.

(a) For any constant m, $y = e^{mx}$ gives $y'' - (m_1 + m_2)y' + m_1 m_2 y = \big[m^2 - (m_1 + m_2)m + m_1 m_2\big]e^{mx} = (m - m_1)(m - m_2)e^{mx}$, which is 0 for $m = m_1$ and for $m = m_2$. So $e^{m_1 x}$ and $e^{m_2 x}$ are

solutions; the equation is linear and homogeneous, so the linear combination $\dfrac{1}{m_1 - m_2}e^{m_1 x} - \dfrac{1}{m_1 - m_2}e^{m_2 x}$ (the constant $1/(m_1 - m_2)$ exists because $m_1 \neq m_2$) is a solution.

(b) With m_2 and x fixed, the quotient $\dfrac{e^{m_1 x} - e^{m_2 x}}{m_1 - m_2}$ has the form $0/0$ as $m_1 \to m_2$. Differentiating numerator and denominator with respect to m_1 (l'Hospital's rule):

$$\lim_{m_1 \to m_2} \frac{e^{m_1 x} - e^{m_2 x}}{m_1 - m_2} = \lim_{m_1 \to m_2} \frac{x e^{m_1 x}}{1} = x e^{m_2 x}.$$

(c) Replacing m_1 by m_2 in the equation of (a) gives $y'' - 2m_2 y' + m_2^2 y = 0$. For $y = x e^{m_2 x}$:

$$y' = (1 + m_2 x)e^{m_2 x}, \qquad y'' = (2m_2 + m_2^2 x)e^{m_2 x},$$

so

$$\begin{aligned} y'' - 2m_2 y' + m_2^2 y &= \big[(2m_2 + m_2^2 x) - 2m_2(1 + m_2 x) + m_2^2 x\big]e^{m_2 x} \\ &= \big[(2m_2 - 2m_2) + (m_2^2 - 2m_2^2 + m_2^2)x\big]e^{m_2 x} = 0. \end{aligned}$$

Thus $x e^{m_2 x}$ is a solution of the equation with equal roots, independent of $e^{m_2 x}$ (their ratio is x), which is the second solution of formula (12) in the text.

Answer. Shown: (a) the quotient is a solution; (b) its limit is $x e^{m_2 x}$; (c) $x e^{m_2 x}$ solves $y'' - 2m_2 y' + m_2^2 y = 0$.

Section 18 The Method of Undetermined Coefficients

Throughout this section we use the method of the text. For $y'' + py' + qy = R(x)$ with constant p, q, the general solution is $y = y_g + y_p$, where y_g is the general solution of the reduced homogeneous equation (found from the auxiliary equation $m^2 + pm + q = 0$) and y_p is any particular solution. The trial form of y_p is chosen from the form of $R(x)$: Ae^{ax} for e^{ax}; $A\sin bx + B\cos bx$ for $\sin bx$ or $\cos bx$; a polynomial of the same degree for a polynomial. If the trial form already solves the reduced homogeneous equation, it is multiplied by x (by x^2 for a double root); if $q = 0$ a polynomial trial form is multiplied by x. In each case the coefficients are found by substituting into the equation and comparing coefficients.

Problem 18-1 (Simmons)

Solution.

(a) The auxiliary equation $m^2 + 3m - 10 = (m+5)(m-2) = 0$ gives $y_g = c_1e^{2x} + c_2e^{-5x}$. Since 4 is not a root, try $y_p = Ae^{4x}$:

$$y_p'' + 3y_p' - 10y_p = A(16 + 12 - 10)e^{4x} = 18Ae^{4x} = 6e^{4x} \implies A = \tfrac{1}{3}.$$

$$y = c_1e^{2x} + c_2e^{-5x} + \tfrac{1}{3}e^{4x}.$$

(b) $m^2 + 4 = 0$ gives $m = \pm 2i$, so $y_g = c_1\sin 2x + c_2\cos 2x$; $\sin x$ is not a solution of the reduced equation, so try $y_p = A\sin x + B\cos x$. Then $y_p'' = -A\sin x - B\cos x$ and

$$y_p'' + 4y_p = 3A\sin x + 3B\cos x = 3\sin x \implies A = 1,\ B = 0.$$

$$y = c_1\sin 2x + c_2\cos 2x + \sin x.$$

(c) $m^2 + 10m + 25 = (m+5)^2 = 0$: double root -5, so $y_g = c_1e^{-5x} + c_2xe^{-5x}$. Since e^{-5x} and xe^{-5x} both solve the reduced equation, try $y_p = Ax^2e^{-5x}$. Then

$$\begin{aligned} y_p' &= A(2x - 5x^2)e^{-5x}, \\ y_p'' &= A(2 - 20x + 25x^2)e^{-5x}, \\ y_p'' + 10y_p' + 25y_p &= Ae^{-5x}\left[(2 - 20x + 25x^2) + (20x - 50x^2) + 25x^2\right] = 2Ae^{-5x}. \end{aligned}$$

Setting $2A = 14$ gives $A = 7$.

$$y = c_1e^{-5x} + c_2xe^{-5x} + 7x^2e^{-5x}.$$

(d) $m^2 - 2m + 5 = 0$ gives $m = 1 \pm 2i$, so $y_g = e^x(c_1\cos 2x + c_2\sin 2x)$. The right side is a quadratic polynomial; try $y_p = A + Bx + Cx^2$. Then $y_p' = B + 2Cx$, $y_p'' = 2C$ and

$$y_p'' - 2y_p' + 5y_p = 5Cx^2 + (5B - 4C)x + (2C - 2B + 5A) = 25x^2 + 12.$$

Equating coefficients:

$$5C = 25, \qquad 5B - 4C = 0, \qquad 2C - 2B + 5A = 12$$

so $C = 5$, $B = 4$, and $5A = 12 - 10 + 8 = 10$, $A = 2$. Hence $y_p = 2 + 4x + 5x^2$ and

$$y = e^x(c_1\cos 2x + c_2\sin 2x) + 2 + 4x + 5x^2.$$

(e) $m^2 - m - 6 = (m-3)(m+2) = 0$, so $y_g = c_1 e^{3x} + c_2 e^{-2x}$. Now -2 is a simple root, so Ae^{-2x} fails; use $y_p = Axe^{-2x}$:

$$y_p' = A(1-2x)e^{-2x}, \qquad y_p'' = A(-4+4x)e^{-2x},$$
$$y_p'' - y_p' - 6y_p = Ae^{-2x}\big[(-4+4x) - (1-2x) - 6x\big] = -5Ae^{-2x} = 20e^{-2x},$$

so $A = -4$.

$$y = c_1 e^{3x} + c_2 e^{-2x} - 4xe^{-2x}.$$

(f) $m^2 - 3m + 2 = (m-1)(m-2) = 0$, so $y_g = c_1 e^x + c_2 e^{2x}$. Try $y_p = A\sin 2x + B\cos 2x$; then $y_p' = 2A\cos 2x - 2B\sin 2x$, $y_p'' = -4A\sin 2x - 4B\cos 2x$ and

$$\begin{aligned} y_p'' - 3y_p' + 2y_p &= (-4A + 6B + 2A)\sin 2x + (-4B - 6A + 2B)\cos 2x \\ &= (-2A + 6B)\sin 2x + (-6A - 2B)\cos 2x. \end{aligned}$$

Matching with $14\sin 2x - 18\cos 2x$:

$$-2A + 6B = 14, \qquad -6A - 2B = -18.$$

The second gives $B = 9 - 3A$; then $-2A + 54 - 18A = 14$, so $A = 2$ and $B = 3$.

$$y = c_1 e^x + c_2 e^{2x} + 2\sin 2x + 3\cos 2x.$$

(g) $y_g = c_1 \sin x + c_2 \cos x$, and $\cos x$ solves the reduced equation, so use (as in Example 1 of the text) $y_p = x(A\sin x + B\cos x)$. Then

$$y_p'' + y_p = 2A\cos x - 2B\sin x = 2\cos x \implies A = 1,\ B = 0,$$

so $y_p = x\sin x$.

$$y = c_1 \sin x + c_2 \cos x + x\sin x.$$

(h) The auxiliary equation $m^2 - 2m = m(m-2) = 0$ gives $y_g = c_1 + c_2 e^{2x}$. Here $q = 0$, so the polynomial trial form is multiplied by x: $y_p = Ax + Bx^2$. Then $y_p' = A + 2Bx$, $y_p'' = 2B$ and

$$y_p'' - 2y_p' = 2B - 2A - 4Bx = 12x - 10.$$

Thus $-4B = 12$, so $B = -3$, and $2B - 2A = -10$ gives $A = 2$. Hence $y_p = 2x - 3x^2$ and

$$y = c_1 + c_2 e^{2x} + 2x - 3x^2.$$

(i) $(m-1)^2 = 0$, so $y_g = c_1 e^x + c_2 x e^x$. Both e^x and xe^x solve the reduced equation, so try $y_p = Ax^2 e^x$. By the computation of the text (here $a = 1$ is a double root), $y_p'' - 2y_p' + y_p = 2Ae^x$; directly, $y_p' = A(2x + x^2)e^x$, $y_p'' = A(2 + 4x + x^2)e^x$ and $y_p'' - 2y_p' + y_p = Ae^x[(2 + 4x + x^2) - (4x + 2x^2) + x^2] = 2Ae^x$. Setting $2A = 6$ gives $A = 3$.

$$y = c_1 e^x + c_2 x e^x + 3x^2 e^x.$$

(j) $m^2 - 2m + 2 = 0$ gives $m = 1 \pm i$, so $y_g = e^x(c_1\cos x + c_2 \sin x)$. The right side $e^x \sin x$ is part of y_g, so the trial form is $y_p = xe^x(A\sin x + B\cos x)$. To keep the algebra short, put $y = e^x u$. Then $y' = e^x(u + u')$, $y'' = e^x(u + 2u' + u'')$ and

$$y'' - 2y' + 2y = e^x\big[(u + 2u' + u'') - 2(u + u') + 2u\big] = e^x(u'' + u),$$

so the equation becomes $u'' + u = \sin x$. This is Example 1 of the text, with particular solution $u = x(A\sin x + B\cos x)$, $u'' + u = 2A\cos x - 2B\sin x$, so $A = 0$, $B = -\frac{1}{2}$, i.e. $u = -\frac{1}{2}x\cos x$. Hence $y_p = -\frac{1}{2}xe^x\cos x$ and

$$y = e^x(c_1\cos x + c_2\sin x) - \tfrac{1}{2}xe^x\cos x.$$

(k) $m^2 + m = m(m+1) = 0$, so $y_g = c_1 + c_2e^{-x}$. Since $q = 0$, the polynomial trial form of degree 4 is multiplied by x (the constant term would be a solution of the reduced equation, so it is omitted):

$$y_p = Bx^5 + Cx^4 + Dx^3 + Ex^2 + Fx.$$

Then

$$\begin{aligned} y_p' &= 5Bx^4 + 4Cx^3 + 3Dx^2 + 2Ex + F, \\ y_p'' &= 20Bx^3 + 12Cx^2 + 6Dx + 2E, \\ y_p'' + y_p' &= 5Bx^4 + (20B + 4C)x^3 + (12C + 3D)x^2 + (6D + 2E)x + (2E + F). \end{aligned}$$

Equating with $10x^4 + 2$, from the top down:

$$\begin{aligned} 5B = 10 &\qquad \Rightarrow B = 2, \\ 20B + 4C = 0 &\qquad \Rightarrow C = -10, \\ 12C + 3D = 0 &\qquad \Rightarrow D = 40, \\ 6D + 2E = 0 &\qquad \Rightarrow E = -120, \\ 2E + F = 2 &\qquad \Rightarrow F = 242. \end{aligned}$$

$$y = c_1 + c_2e^{-x} + 2x^5 - 10x^4 + 40x^3 - 120x^2 + 242x.$$

Answer. (a) $y = c_1e^{2x} + c_2e^{-5x} + \frac{1}{3}e^{4x}$; (b) $y = c_1\sin 2x + c_2\cos 2x + \sin x$; (c) $y = c_1e^{-5x} + c_2xe^{-5x} + 7x^2e^{-5x}$; (d) $y = e^x(c_1\cos 2x + c_2\sin 2x) + 2 + 4x + 5x^2$; (e) $y = c_1e^{3x} + c_2e^{-2x} - 4xe^{-2x}$; (f) $y = c_1e^x + c_2e^{2x} + 2\sin 2x + 3\cos 2x$; (g) $y = c_1\sin x + c_2\cos x + x\sin x$; (h) $y = c_1 + c_2e^{2x} + 2x - 3x^2$; (i) $y = c_1e^x + c_2xe^x + 3x^2e^x$; (j) $y = e^x(c_1\cos x + c_2\sin x) - \frac{1}{2}xe^x\cos x$; (k) $y = c_1 + c_2e^{-x} + 2x^5 - 10x^4 + 40x^3 - 120x^2 + 242x$.

Problem 18-2 (Simmons)

Solution. The reduced equation has auxiliary equation $m^2 + k^2 = 0$, $m = \pm ik$, so $y_g = c_1\sin kx + c_2\cos kx$. Two cases arise.

Case $b \neq k$. Then $\sin bx$ does not solve the reduced equation; try $y_p = A\sin bx + B\cos bx$. Since $y_p'' = -b^2y_p$,

$$y_p'' + k^2y_p = (k^2 - b^2)(A\sin bx + B\cos bx) = \sin bx \implies A = \frac{1}{k^2 - b^2},\ B = 0.$$

Case $b = k$. Now $\sin kx$ solves the reduced equation, so try $y_p = x(A\sin kx + B\cos kx)$:

$$y_p'' = 2kA\cos kx - 2kB\sin kx - k^2x(A\sin kx + B\cos kx),$$

so $y_p'' + k^2y_p = 2kA\cos kx - 2kB\sin kx = \sin kx$. Hence $A = 0$ and $B = -\dfrac{1}{2k}$, giving $y_p = -\dfrac{x\cos kx}{2k}$.

Answer. $y = c_1 \sin kx + c_2 \cos kx + \dfrac{\sin bx}{k^2 - b^2}$ if $b \neq k$; $y = c_1 \sin kx + c_2 \cos kx - \dfrac{x \cos kx}{2k}$ if $b = k$.

Problem 18-3 (Simmons)

Solution. *Proof of superposition.* Write $L[y] = y'' + Py' + Qy$. Differentiation is linear, so $L[y_1 + y_2] = (y_1 + y_2)'' + P(y_1 + y_2)' + Q(y_1 + y_2) = L[y_1] + L[y_2] = R_1 + R_2$. □

(a) The reduced equation $y'' + 4y = 0$ has $y_g = c_1 \sin 2x + c_2 \cos 2x$. By superposition a particular solution is the sum of particular solutions u_1, u_2, u_3 of

$$y'' + 4y = 4\cos 2x, \qquad y'' + 4y = 6\cos x, \qquad y'' + 4y = 8x^2 - 4x.$$

- $4\cos 2x$ solves the reduced equation's family, so try $u_1 = x(A\sin 2x + B\cos 2x)$. Then $u_1'' + 4u_1 = 4A\cos 2x - 4B\sin 2x = 4\cos 2x$, so $A = 1$, $B = 0$: $u_1 = x\sin 2x$.
- Try $u_2 = A\cos x + B\sin x$; $u_2'' + 4u_2 = 3A\cos x + 3B\sin x = 6\cos x$, so $A = 2$, $B = 0$: $u_2 = 2\cos x$.
- Try $u_3 = A + Bx + Cx^2$; $u_3'' + 4u_3 = (2C + 4A) + 4Bx + 4Cx^2 = 8x^2 - 4x$, so $C = 2$, $B = -1$, $2C + 4A = 0 \Rightarrow A = -1$: $u_3 = -1 - x + 2x^2$.

$$y = c_1 \sin 2x + c_2 \cos 2x + x\sin 2x + 2\cos x - 1 - x + 2x^2.$$

(b) $y_g = c_1 \sin 3x + c_2 \cos 3x$. Split the right side into four pieces:

- $2\sin 3x$: resonant, try $u_1 = x(A\sin 3x + B\cos 3x)$; $u_1'' + 9u_1 = 6A\cos 3x - 6B\sin 3x = 2\sin 3x$, so $A = 0$, $B = -\frac{1}{3}$: $u_1 = -\frac{1}{3}x\cos 3x$.
- $4\sin x$: try $u_2 = A\sin x + B\cos x$; $u_2'' + 9u_2 = 8A\sin x + 8B\cos x = 4\sin x$, so $A = \frac{1}{2}$, $B = 0$: $u_2 = \frac{1}{2}\sin x$.
- $-26e^{-2x}$: try $u_3 = Ae^{-2x}$; $u_3'' + 9u_3 = 13Ae^{-2x} = -26e^{-2x}$, so $A = -2$: $u_3 = -2e^{-2x}$.
- $27x^3$: try $u_4 = A + Bx + Cx^2 + Dx^3$; $u_4'' + 9u_4 = (2C + 9A) + (6D + 9B)x + 9Cx^2 + 9Dx^3 = 27x^3$, so $D = 3$, $C = 0$, $6D + 9B = 0 \Rightarrow B = -2$, $2C + 9A = 0 \Rightarrow A = 0$: $u_4 = 3x^3 - 2x$.

$$y = c_1 \sin 3x + c_2 \cos 3x - \tfrac{1}{3}x\cos 3x + \tfrac{1}{2}\sin x - 2e^{-2x} + 3x^3 - 2x.$$

Answer. Superposition shown: $L[y_1 + y_2] = R_1 + R_2$. (a) $y = c_1 \sin 2x + c_2 \cos 2x + x\sin 2x + 2\cos x - 1 - x + 2x^2$; (b) $y = c_1 \sin 3x + c_2 \cos 3x - \frac{1}{3}x\cos 3x + \frac{1}{2}\sin x - 2e^{-2x} + 3x^3 - 2x$.

Section 19 The Method of Variation of Parameters

Throughout this section we use the method of the text. For $y'' + P(x)y' + Q(x)y = R(x)$, if y_1, y_2 are independent solutions of the reduced equation with Wronskian $W = W(y_1, y_2) = y_1y_2' - y_2y_1'$, then

$$y_p = v_1y_1 + v_2y_2, \qquad v_1' = \frac{-y_2R}{W}, \quad v_2' = \frac{y_1R}{W},$$

(formulas (11)–(13) of the text), where R is the right side of the equation *in the standard form with leading coefficient* 1. Constants of integration in v_1, v_2 may be omitted, since they only add solutions of the homogeneous equation. We write log for the natural logarithm and use $\int \sec x\,dx = \log(\sec x + \tan x)$ and $\int \csc x\,dx = -\log(\csc x + \cot x)$.

Problem 19-1 (Simmons)

Solution. *Inspection.* Try $y_p = ax + b$: $y_p'' - 2y_p' + y_p = -2a + ax + b = 2x$ requires $a = 2$ and $b = 2a = 4$. So $y_p = 2x + 4$.

Variation of parameters. $(m-1)^2 = 0$, so $y_1 = e^x$, $y_2 = xe^x$, and

$$W = e^x(e^x + xe^x) - xe^x \cdot e^x = e^{2x}.$$

With $R = 2x$:

$$v_1' = \frac{-xe^x \cdot 2x}{e^{2x}} = -2x^2e^{-x}, \qquad v_2' = \frac{e^x \cdot 2x}{e^{2x}} = 2xe^{-x}.$$

Integrating by parts (twice for v_1):

$$v_1 = -2\int x^2e^{-x}\,dx = 2e^{-x}(x^2 + 2x + 2), \qquad v_2 = 2\int xe^{-x}\,dx = -2e^{-x}(x+1).$$

(Check: $\frac{d}{dx}\left[2e^{-x}(x^2+2x+2)\right] = 2e^{-x}(-x^2 - 2x - 2 + 2x + 2) = -2x^2e^{-x}$.) Therefore

$$y_p = v_1y_1 + v_2y_2 = 2(x^2 + 2x + 2) - 2x(x+1) = 2x + 4.$$

Both methods give the same function.

Answer. $y_p = 2x + 4$.

Problem 19-2 (Simmons)

Solution. *Undetermined coefficients.* $m^2 - m - 6 = (m-3)(m+2)$; -1 is not a root, so try $y_p = Ae^{-x}$: $A(1+1-6)e^{-x} = -4Ae^{-x} = e^{-x}$, so $A = -\frac{1}{4}$ and $y_p = -\frac{1}{4}e^{-x}$.

Variation of parameters. $y_1 = e^{3x}$, $y_2 = e^{-2x}$,

$$W = e^{3x}(-2e^{-2x}) - e^{-2x}(3e^{3x}) = -5e^x.$$

$$v_1' = \frac{-e^{-2x}e^{-x}}{-5e^x} = \tfrac{1}{5}e^{-4x}, \qquad v_2' = \frac{e^{3x}e^{-x}}{-5e^x} = -\tfrac{1}{5}e^x,$$

so $v_1 = -\frac{1}{20}e^{-4x}$ and $v_2 = -\frac{1}{5}e^x$. Then

$$y_p = -\tfrac{1}{20}e^{-4x}e^{3x} - \tfrac{1}{5}e^xe^{-2x} = -\left(\tfrac{1}{20} + \tfrac{1}{5}\right)e^{-x} = -\tfrac{1}{4}e^{-x}.$$

Answer. $y_p = -\frac{1}{4}e^{-x}$ (by both methods).

Problem 19-3 (Simmons)

Solution.

(a) $y_1 = \sin 2x$, $y_2 = \cos 2x$, $W = \sin 2x(-2\sin 2x) - \cos 2x(2\cos 2x) = -2$. Then

$$v_1' = \frac{-\cos 2x \tan 2x}{-2} = \tfrac{1}{2}\sin 2x, \qquad v_2' = \frac{\sin 2x \tan 2x}{-2} = -\frac{1-\cos^2 2x}{2\cos 2x} = -\tfrac{1}{2}(\sec 2x - \cos 2x).$$

Hence $v_1 = -\frac{1}{4}\cos 2x$ and, using $\int \sec 2x\,dx = \frac{1}{2}\log(\sec 2x + \tan 2x)$,

$$v_2 = -\tfrac{1}{4}\log(\sec 2x + \tan 2x) + \tfrac{1}{4}\sin 2x.$$

$$\begin{aligned} y_p &= -\tfrac{1}{4}\sin 2x\cos 2x + \cos 2x\left[-\tfrac{1}{4}\log(\sec 2x + \tan 2x) + \tfrac{1}{4}\sin 2x\right] \\ &= -\tfrac{1}{4}\cos 2x\,\log(\sec 2x + \tan 2x). \end{aligned}$$

(valid on $-\pi/4 < x < \pi/4$.)

(b) $(m+1)^2 = 0$: $y_1 = e^{-x}$, $y_2 = xe^{-x}$, $W = e^{-2x}$ (as in Problem 1 with $x \to -x$ in the exponent: $W = e^{-x}(1-x)e^{-x} - xe^{-x}(-e^{-x}) = e^{-2x}$). With $R = e^{-x}\log x$ $(x > 0)$:

$$v_1' = \frac{-xe^{-x}e^{-x}\log x}{e^{-2x}} = -x\log x, \qquad v_2' = \frac{e^{-x}e^{-x}\log x}{e^{-2x}} = \log x.$$

By parts, $\int x\log x\,dx = \frac{1}{2}x^2\log x - \frac{1}{4}x^2$ and $\int \log x\,dx = x\log x - x$, so

$$v_1 = -\tfrac{1}{2}x^2\log x + \tfrac{1}{4}x^2, \qquad v_2 = x\log x - x.$$

$$y_p = e^{-x}\left[-\tfrac{1}{2}x^2\log x + \tfrac{1}{4}x^2 + x^2\log x - x^2\right] = e^{-x}\left[\tfrac{1}{2}x^2\log x - \tfrac{3}{4}x^2\right].$$

(c) $m^2 - 2m - 3 = (m-3)(m+1)$: $y_1 = e^{3x}$, $y_2 = e^{-x}$, $W = e^{3x}(-e^{-x}) - e^{-x}(3e^{3x}) = -4e^{2x}$. Then

$$v_1' = \frac{-e^{-x}\cdot 64xe^{-x}}{-4e^{2x}} = 16xe^{-4x}, \qquad v_2' = \frac{e^{3x}\cdot 64xe^{-x}}{-4e^{2x}} = -16x.$$

Integrating by parts, $\int xe^{-4x}\,dx = -\frac{1}{4}xe^{-4x} - \frac{1}{16}e^{-4x}$, so

$$v_1 = -4xe^{-4x} - e^{-4x}, \qquad v_2 = -8x^2.$$

$$y_p = e^{3x}(-4x-1)e^{-4x} - 8x^2e^{-x} = -e^{-x}(8x^2 + 4x + 1).$$

(d) $m^2 + 2m + 5 = 0$, $m = -1 \pm 2i$: $y_1 = e^{-x}\cos 2x$, $y_2 = e^{-x}\sin 2x$. Since $y_1' = e^{-x}(-\cos 2x - 2\sin 2x)$ and $y_2' = e^{-x}(-\sin 2x + 2\cos 2x)$,

$$W = e^{-2x}\left[\cos 2x(-\sin 2x + 2\cos 2x) - \sin 2x(-\cos 2x - 2\sin 2x)\right] = 2e^{-2x}.$$

With $R = e^{-x}\sec 2x$:

$$v_1' = \frac{-e^{-x}\sin 2x\cdot e^{-x}\sec 2x}{2e^{-2x}} = -\tfrac{1}{2}\tan 2x, \qquad v_2' = \frac{e^{-x}\cos 2x\cdot e^{-x}\sec 2x}{2e^{-2x}} = \tfrac{1}{2}.$$

So $v_1 = \frac{1}{4}\log(\cos 2x)$ (since $\frac{d}{dx}\log\cos 2x = -2\tan 2x$) and $v_2 = \frac{1}{2}x$.

$$y_p = \tfrac{1}{4}e^{-x}\cos 2x\,\log(\cos 2x) + \tfrac{1}{2}xe^{-x}\sin 2x.$$

(e) The equation must first be put in standard form: $y'' + \frac{3}{2}y' + \frac{1}{2}y = \frac{1}{2}e^{-3x}$, so $R = \frac{1}{2}e^{-3x}$. The auxiliary equation $2m^2 + 3m + 1 = (2m+1)(m+1) = 0$ gives $y_1 = e^{-x/2}$, $y_2 = e^{-x}$ and

$$W = e^{-x/2}(-e^{-x}) - e^{-x}\left(-\tfrac{1}{2}e^{-x/2}\right) = -\tfrac{1}{2}e^{-3x/2}.$$

$$v_1' = \frac{-e^{-x}\cdot\frac{1}{2}e^{-3x}}{-\frac{1}{2}e^{-3x/2}} = e^{-5x/2}, \qquad v_2' = \frac{e^{-x/2}\cdot\frac{1}{2}e^{-3x}}{-\frac{1}{2}e^{-3x/2}} = -e^{-2x},$$

so $v_1 = -\frac{2}{5}e^{-5x/2}$ and $v_2 = \frac{1}{2}e^{-2x}$. Then

$$y_p = -\tfrac{2}{5}e^{-5x/2}e^{-x/2} + \tfrac{1}{2}e^{-2x}e^{-x} = \left(-\tfrac{2}{5} + \tfrac{1}{2}\right)e^{-3x} = \tfrac{1}{10}e^{-3x}.$$

(Check by undetermined coefficients: Ae^{-3x} gives $(18 - 9 + 1)A = 10A = 1$.)

(f) $m^2 - 3m + 2 = (m-1)(m-2)$: $y_1 = e^x$, $y_2 = e^{2x}$, $W = e^x \cdot 2e^{2x} - e^{2x}e^x = e^{3x}$. With $R = (1+e^{-x})^{-1}$:

$$v_1' = \frac{-e^{2x}}{(1+e^{-x})e^{3x}} = -\frac{e^{-x}}{1+e^{-x}}, \qquad v_2' = \frac{e^x}{(1+e^{-x})e^{3x}} = \frac{e^{-2x}}{1+e^{-x}}.$$

Since $\frac{d}{dx}\log(1+e^{-x}) = -\frac{e^{-x}}{1+e^{-x}}$, we get $v_1 = \log(1+e^{-x})$. For v_2 put $u = e^{-x}$, $dx = -du/u$:

$$v_2 = \int \frac{u^2}{1+u}\left(-\frac{du}{u}\right) = -\int \frac{u}{1+u}\,\mathrm{d}u = -\int\left(1 - \frac{1}{1+u}\right)\mathrm{d}u = -u + \log(1+u),$$

i.e. $v_2 = -e^{-x} + \log(1+e^{-x})$. Then

$$y_p = e^x\log(1+e^{-x}) + e^{2x}\left[-e^{-x} + \log(1+e^{-x})\right] = e^x\log(1+e^{-x}) - e^x + e^{2x}\log(1+e^{-x}).$$

Answer. (a) $y_p = -\frac{1}{4}\cos 2x\log(\sec 2x + \tan 2x)$; (b) $y_p = \frac{1}{2}x^2e^{-x}\log x - \frac{3}{4}x^2e^{-x}$; (c) $y_p = -e^{-x}(8x^2 + 4x + 1)$; (d) $y_p = \frac{1}{2}xe^{-x}\sin 2x + \frac{1}{4}e^{-x}\cos 2x\log(\cos 2x)$; (e) $y_p = \frac{1}{10}e^{-3x}$; (f) $y_p = e^x\log(1+e^{-x}) - e^x + e^{2x}\log(1+e^{-x})$.

Problem 19-4 (Simmons)

Solution. In all parts $y_1 = \sin x$, $y_2 = \cos x$, $W = -\sin^2 x - \cos^2 x = -1$, so

$$v_1' = \cos x\,R(x), \qquad v_2' = -\sin x\,R(x), \qquad y_p = v_1\sin x + v_2\cos x.$$

(a) $R = \sec x$: $v_1' = 1$, $v_2' = -\tan x$, so $v_1 = x$, $v_2 = \log(\cos x)$ and

$$y_p = x\sin x + \cos x\log(\cos x).$$

(b) $R = \cot^2 x$: $v_1' = \dfrac{\cos^3 x}{\sin^2 x} = \dfrac{\cos x(1-\sin^2 x)}{\sin^2 x} = \dfrac{\cos x}{\sin^2 x} - \cos x$, so $v_1 = -\csc x - \sin x$. Also $v_2' = -\dfrac{\cos^2 x}{\sin x} = -\dfrac{1-\sin^2 x}{\sin x} = -\csc x + \sin x$, so $v_2 = \log(\csc x + \cot x) - \cos x$. Hence

$$\begin{aligned} y_p &= -1 - \sin^2 x + \cos x\log(\csc x + \cot x) - \cos^2 x \\ &= \cos x\log(\csc x + \cot x) - 2. \end{aligned}$$

(c) $R = \cot 2x = \dfrac{\cos 2x}{2\sin x\cos x}$. Then

$$v_1' = \cos x\cot 2x = \frac{\cos 2x}{2\sin x} = \frac{1-2\sin^2 x}{2\sin x} = \tfrac{1}{2}\csc x - \sin x,$$

$$v_2' = -\sin x\cot 2x = -\frac{\cos 2x}{2\cos x} = -\frac{2\cos^2 x - 1}{2\cos x} = -\cos x + \tfrac{1}{2}\sec x.$$

Thus $v_1 = -\frac{1}{2}\log(\csc x + \cot x) + \cos x$ and $v_2 = -\sin x + \frac{1}{2}\log(\sec x + \tan x)$, and

$$\begin{aligned} y_p &= \sin x\left[-\tfrac{1}{2}\log(\csc x+\cot x) + \cos x\right] + \cos x\left[-\sin x + \tfrac{1}{2}\log(\sec x+\tan x)\right] \\ &= \tfrac{1}{2}\cos x\log(\sec x+\tan x) - \tfrac{1}{2}\sin x\log(\csc x+\cot x). \end{aligned}$$

(d) $R = x\cos x$: $v_1' = x\cos^2 x = \frac{1}{2}x(1+\cos 2x)$ and $v_2' = -x\sin x\cos x = -\frac{1}{2}x\sin 2x$. Integrating by parts, $\int x\cos 2x\,dx = \frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x$ and $\int x\sin 2x\,dx = -\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x$, so (choosing the constant in v_1 suitably)

$$v_1 = \tfrac{1}{4}x^2 + \tfrac{1}{4}x\sin 2x - \tfrac{1}{4}\sin^2 x, \qquad v_2 = \tfrac{1}{4}x\cos 2x - \tfrac{1}{8}\sin 2x.$$

(Indeed $\frac{1}{2}\int x\cos 2x\,dx = \frac{1}{4}x\sin 2x + \frac{1}{8}\cos 2x$ and $\frac{1}{8}\cos 2x - \frac{1}{8} = -\frac{1}{4}\sin^2 x$.) Then

$$\begin{aligned} y_p &= \tfrac{1}{4}x^2\sin x + \tfrac{1}{4}x(\sin x\sin 2x + \cos x\cos 2x) - \tfrac{1}{4}\sin^3 x - \tfrac{1}{8}\cos x\sin 2x \\ &= \tfrac{1}{4}x^2\sin x + \tfrac{1}{4}x\cos x - \tfrac{1}{4}\sin x(\sin^2 x + \cos^2 x), \end{aligned}$$

using $\sin x\sin 2x + \cos x\cos 2x = \cos(2x-x) = \cos x$ and $\frac{1}{8}\cos x\sin 2x = \frac{1}{4}\sin x\cos^2 x$. Hence

$$y_p = \tfrac{1}{4}\left(x^2\sin x + x\cos x - \sin x\right).$$

(e) $R = \tan x$: $v_1' = \sin x$, $v_2' = -\sin x\tan x = -\dfrac{1-\cos^2 x}{\cos x} = -\sec x + \cos x$, so $v_1 = -\cos x$, $v_2 = -\log(\sec x+\tan x) + \sin x$ and

$$y_p = -\sin x\cos x - \cos x\log(\sec x+\tan x) + \sin x\cos x = -\cos x\log(\sec x+\tan x).$$

(f) $R = \sec x\tan x$: $v_1' = \tan x$, $v_2' = -\sin x\sec x\tan x = -\tan^2 x = 1-\sec^2 x$, so $v_1 = -\log(\cos x)$, $v_2 = x - \tan x$, and

$$y_p = -\sin x\log(\cos x) + x\cos x - \sin x.$$

(g) $R = \sec x\csc x$: $v_1' = \cos x\sec x\csc x = \csc x$, $v_2' = -\sin x\sec x\csc x = -\sec x$, so $v_1 = -\log(\csc x+\cot x)$, $v_2 = -\log(\sec x+\tan x)$, and

$$y_p = -\sin x\log(\csc x+\cot x) - \cos x\log(\sec x+\tan x).$$

Answer. (a) $y_p = x\sin x + \cos x\log(\cos x)$; (b) $y_p = \cos x\log(\csc x+\cot x) - 2$; (c) $y_p = \frac{1}{2}\cos x\log(\sec x+\tan x) - \frac{1}{2}\sin x\log(\csc x+\cot x)$; (d) $y_p = \frac{1}{4}(x^2\sin x + x\cos x - \sin x)$; (e) $y_p = -\cos x\log(\sec x+\tan x)$; (f) $y_p = x\cos x - \sin x - \sin x\log(\cos x)$; (g) $y_p = -\sin x\log(\csc x+\cot x) - \cos x\log(\sec x+\tan x)$.

Problem 19-5 (Simmons)

Solution.

(a) Take $y_1 = \sin x$, $y_2 = \cos x$, $W = -1$. Then $v_1' = \cos x\, f(x)$ and $v_2' = -\sin x\, f(x)$, and we may choose the antiderivatives that vanish at $x = 0$ (writing the variable of integration as t):

$$v_1(x) = \int_0^x f(t)\cos t\,dt, \qquad v_2(x) = -\int_0^x f(t)\sin t\,dt.$$

Since $\sin x$ and $\cos x$ do not depend on t, they can be moved inside the integrals:

$$\begin{aligned} y_p &= \sin x \int_0^x f(t)\cos t\,dt - \cos x \int_0^x f(t)\sin t\,dt \\ &= \int_0^x f(t)\left[\sin x\cos t - \cos x\sin t\right]\,dt = \int_0^x f(t)\sin(x-t)\,dt, \end{aligned}$$

by the subtraction formula $\sin(x-t) = \sin x\cos t - \cos x\sin t$.

(b) Now $y_1 = \sin kx$, $y_2 = \cos kx$ and $W = \sin kx(-k\sin kx) - \cos kx(k\cos kx) = -k$. Then $v_1' = \dfrac{\cos kx\, f(x)}{k}$, $v_2' = -\dfrac{\sin kx\, f(x)}{k}$; taking the antiderivatives vanishing at 0,

$$y_p = \frac{1}{k}\int_0^x f(t)\left[\sin kx\cos kt - \cos kx\sin kt\right]\,dt = \frac{1}{k}\int_0^x f(t)\sin k(x-t)\,dt.$$

Answer. (a) Shown: $y_p = \int_0^x f(t)\sin(x-t)\,dt$; (b) $y_p(x) = \dfrac{1}{k}\displaystyle\int_0^x f(t)\sin k(x-t)\,dt$.

Problem 19-6 (Simmons)

Solution. In each part the leading coefficient is not 1; we divide by it to get R in standard form. The two solutions of the reduced equation are found by inspection and, if necessary, by the reduction of order of Section 16: if y_1 is a solution of $y'' + Py' + Qy = 0$, then $y_2 = vy_1$ with $v' = e^{-\int P\,dx}/y_1^2$.

(a) Standard form: $y'' - \dfrac{2x}{x^2-1}y' + \dfrac{2}{x^2-1}y = x^2 - 1$, so $R = x^2 - 1$. By inspection $y_1 = x$ is a solution of the reduced equation ($-2x + 2x = 0$). Reduction of order: $\int P\,dx = -\log|x^2-1|$, so $v' = \dfrac{|x^2-1|}{x^2}$, i.e. (taking the sign positive) $v' = 1 - x^{-2}$ and $v = x + x^{-1}$, giving $y_2 = x\left(x + \frac{1}{x}\right) = x^2 + 1$. Direct check: $(x^2-1)\cdot 2 - 2x\cdot 2x + 2(x^2+1) = 0$. Then

$$W = x(2x) - (x^2+1)(1) = x^2 - 1,$$

$$v_1' = \frac{-(x^2+1)(x^2-1)}{x^2-1} = -(x^2+1), \qquad v_2' = \frac{x(x^2-1)}{x^2-1} = x,$$

so $v_1 = -\frac{1}{3}x^3 - x$, $v_2 = \frac{1}{2}x^2$, and

$$y_p = x\left(-\tfrac{1}{3}x^3 - x\right) + (x^2+1)\tfrac{1}{2}x^2 = \tfrac{1}{6}x^4 - \tfrac{1}{2}x^2.$$

$$y = c_1 x + c_2(x^2+1) + \tfrac{1}{6}x^4 - \tfrac{1}{2}x^2.$$

(b) Standard form: $P = \dfrac{2-x^2}{x(x+1)}$, $Q = -\dfrac{2+x}{x(x+1)}$, $R = \dfrac{x(x+1)^2}{x(x+1)} = x+1$. By inspection $y_1 = e^x$ solves the reduced equation: $e^x[(x^2+x) + (2-x^2) - (2+x)] = 0$. Reduction of

order: by partial fractions, $\dfrac{2-x^2}{x(x+1)} = -1 + \dfrac{2+x}{x(x+1)} = -1 + \dfrac{2}{x} - \dfrac{1}{x+1}$, whence $\int P\,dx = -x + 2\log x - \log(x+1)$ and $e^{-\int P} = \dfrac{e^x(x+1)}{x^2}$. Then $v' = \dfrac{e^{-\int P}}{y_1^2} = \dfrac{e^{-x}(x+1)}{x^2}$, and $v = -\dfrac{e^{-x}}{x}$ (indeed $\frac{d}{dx}\left(-e^{-x}/x\right) = e^{-x}/x + e^{-x}/x^2$), giving $y_2 = -1/x$; we use $y_2 = 1/x$. Direct check: with $y = 1/x$, $y' = -x^{-2}$, $y'' = 2x^{-3}$,

$$(x^2+x)\frac{2}{x^3} - \frac{2-x^2}{x^2} - \frac{2+x}{x} = \frac{2}{x} + \frac{2}{x^2} - \frac{2}{x^2} + 1 - \frac{2}{x} - 1 = 0.$$

Now

$$W = e^x\left(-\frac{1}{x^2}\right) - \frac{1}{x}e^x = -\frac{e^x(x+1)}{x^2},$$

$$v_1' = \frac{-\frac{1}{x}(x+1)}{-e^x(x+1)/x^2} = xe^{-x}, \qquad v_2' = \frac{e^x(x+1)}{-e^x(x+1)/x^2} = -x^2,$$

so $v_1 = -(x+1)e^{-x}$ (by parts) and $v_2 = -\frac{1}{3}x^3$. Then

$$y_p = e^x\left[-(x+1)e^{-x}\right] + \frac{1}{x}\left(-\tfrac{1}{3}x^3\right) = -x - 1 - \tfrac{1}{3}x^2.$$

$$y = c_1e^x + c_2x^{-1} - x - 1 - \tfrac{1}{3}x^2.$$

(c) Standard form: $y'' + \dfrac{x}{1-x}y' - \dfrac{1}{1-x}y = 1 - x$, so $R = 1 - x$. By inspection $y_1 = x$ ($0 + x - x = 0$) and $y_2 = e^x$ ($(1-x)e^x + xe^x - e^x = 0$) solve the reduced equation, and they are independent. Then

$$W = xe^x - e^x = (x-1)e^x,$$

$$v_1' = \frac{-e^x(1-x)}{(x-1)e^x} = 1, \qquad v_2' = \frac{x(1-x)}{(x-1)e^x} = -xe^{-x},$$

so $v_1 = x$ and $v_2 = (x+1)e^{-x}$ (by parts: $-\int xe^{-x}\,dx = xe^{-x} + e^{-x}$). Hence

$$y_p = x\cdot x + e^x(x+1)e^{-x} = x^2 + x + 1.$$

The term x is a solution of the reduced equation and can be absorbed into c_1x:

$$y = c_1x + c_2e^x + x^2 + 1.$$

(Equivalently $y = c_1x + c_2e^x + x^2 + x + 1$ with a different c_1.)

(d) Standard form: $y'' - \dfrac{1+x}{x}y' + \dfrac{1}{x}y = xe^{2x}$, so $R = xe^{2x}$. By inspection $y_1 = e^x$ ($xe^x - (1+x)e^x + e^x = 0$) and $y_2 = 1 + x$ ($0 - (1+x) + (1+x) = 0$). Then

$$W = e^x\cdot 1 - (1+x)e^x = -xe^x,$$

$$v_1' = \frac{-(1+x)xe^{2x}}{-xe^x} = (1+x)e^x, \qquad v_2' = \frac{e^x\cdot xe^{2x}}{-xe^x} = -e^{2x},$$

so $v_1 = xe^x$ (since $\frac{d}{dx}(xe^x) = (1+x)e^x$) and $v_2 = -\frac{1}{2}e^{2x}$. Then

$$y_p = xe^x\cdot e^x - \tfrac{1}{2}e^{2x}(1+x) = \tfrac{1}{2}e^{2x}(x-1).$$

$$y = c_1e^x + c_2(x+1) + \tfrac{1}{2}e^{2x}(x-1).$$

(e) This is an Euler equation: $y = x$ and $y = x^2$ solve the reduced equation ($x^2 \cdot 0 - 2x + 2x = 0$ and $2x^2 - 4x^2 + 2x^2 = 0$). Standard form: $R = \dfrac{xe^{-x}}{x^2} = \dfrac{e^{-x}}{x}$ ($x > 0$). Then $y_1 = x$, $y_2 = x^2$, $W = x \cdot 2x - x^2 = x^2$ and

$$v_1' = \frac{-x^2 \cdot e^{-x}/x}{x^2} = -\frac{e^{-x}}{x}, \qquad v_2' = \frac{x \cdot e^{-x}/x}{x^2} = \frac{e^{-x}}{x^2}.$$

The integral $\int e^{-x}x^{-1}\,\mathrm{d}x$ is not an elementary function, so we leave it as such: $v_1 = -\int \frac{e^{-x}}{x}\,\mathrm{d}x$. For v_2, integrate by parts with $u = e^{-x}$, $dw = x^{-2}dx$:

$$v_2 = \int \frac{e^{-x}}{x^2}\,\mathrm{d}x = -\frac{e^{-x}}{x} - \int \frac{e^{-x}}{x}\,\mathrm{d}x.$$

Hence, with $I(x) = \displaystyle\int \frac{e^{-x}}{x}\,\mathrm{d}x$,

$$y_p = xv_1 + x^2v_2 = -xI - xe^{-x} - x^2I = -xe^{-x} - (x^2 + x)I(x).$$

$$y = c_1x + c_2x^2 - xe^{-x} - (x^2 + x)\int \frac{e^{-x}}{x}\,\mathrm{d}x.$$

Answer. (a) $y = c_1x + c_2(x^2 + 1) + \frac{1}{6}x^4 - \frac{1}{2}x^2$; (b) $y = c_1e^x + c_2x^{-1} - x - 1 - \frac{1}{3}x^2$; (c) $y = c_1x + c_2e^x + x^2 + 1$; (d) $y = c_1e^x + c_2(x + 1) + \frac{1}{2}e^{2x}(x - 1)$; (e) $y = c_1x + c_2x^2 - xe^{-x} - (x^2 + x)\displaystyle\int \frac{e^{-x}}{x}\,\mathrm{d}x$ (the integral is not elementary).

Section 20 Vibrations in Mechanical and Electrical Systems

Throughout this section we use the notation of the text. For the cart of Figure 25 (mass M, spring constant k, damping constant c) the free motion satisfies

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + a^2x = 0, \qquad b = \frac{c}{2M}, \quad a = \sqrt{\frac{k}{M}}, \tag{11}$$

and in the underdamped case $b < a$ we put $\alpha = \sqrt{a^2 - b^2}$. Under the impressed force $F_0 \cos \omega t$ the steady-state motion is

$$x = \frac{F_0}{\sqrt{(k - \omega^2 M)^2 + \omega^2 c^2}} \cos(\omega t - \phi), \tag{27}$$

with amplitude

$$\frac{F_0}{\sqrt{(k - \omega^2 M)^2 + \omega^2 c^2}}. \tag{29}$$

As in the text, $g = 32$ ft/s^2 in numerical work with British units.

Problem 20-1 (Simmons)

Solution. The impressed frequency is $\omega/2\pi$ (with $\omega \geq 0$), so we must find the value of ω that maximizes (29). Since $F_0 > 0$ is fixed, the amplitude is largest exactly when the positive quantity under the square root,

$$G = (k - \omega^2 M)^2 + \omega^2 c^2,$$

is smallest. G depends on ω only through $u = \omega^2 \geq 0$:

$$G(u) = (k - uM)^2 + uc^2 = M^2u^2 - (2kM - c^2)u + k^2.$$

This is a quadratic in u with positive leading coefficient, so it is strictly convex ($G''(u) = 2M^2 > 0$), and its only critical point is given by

$$G'(u) = -2M(k - uM) + c^2 = 0 \quad \Longleftrightarrow \quad u = u^* = \frac{k}{M} - \frac{c^2}{2M^2}.$$

Two cases arise.

Case $u^* > 0$, that is, $c^2 < 2kM$. Then G decreases on $[0, u^*]$ and increases on $[u^*, \infty)$, so G has its minimum on $u \geq 0$ at $u = u^*$, and the amplitude is maximal at

$$\omega = \omega^* = \sqrt{\frac{k}{M} - \frac{c^2}{2M^2}}, \qquad \text{i.e. at the impressed frequency} \qquad \frac{\omega^*}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{M} - \frac{c^2}{2M^2}}.$$

The minimum value of G is

$$G(u^*) = \left(\frac{c^2}{2M}\right)^2 + \left(\frac{k}{M} - \frac{c^2}{2M^2}\right)c^2 = \frac{kc^2}{M} - \frac{c^4}{4M^2} = c^2\left(\frac{k}{M} - \frac{c^2}{4M^2}\right) = c^2\alpha^2,$$

so the maximum amplitude is $F_0/(c\alpha)$, which is very large when c is small, in agreement with the discussion of resonance in the text.

Case $u^* \leq 0$, that is, $c^2 \geq 2kM$. Then $G'(u) > 0$ for all $u > 0$ (as G' is increasing and $G'(u^*) = 0$ with $u^* \leq 0$), so G increases and the amplitude (29) strictly *decreases* as ω increases from 0. There is no positive impressed frequency at which the amplitude is a maximum (the supremum is approached only as $\omega \to 0$, where there is no vibration at all).

Does a resonance frequency necessarily exist? No. The motion is underdamped when $b < a$, i.e. $c^2/4M^2 < k/M$, i.e. $c^2 < 4kM$. A resonance frequency exists if and only if $c^2 < 2kM$, i.e. $\dfrac{k}{M} - \dfrac{c^2}{2M^2} > 0$, which is more restrictive. Whenever $2kM \le c^2 < 4kM$ the free motion is underdamped (so a natural frequency exists) but there is no resonance frequency.

Comparison with the natural frequency. By (22) the natural frequency is $\dfrac{1}{2\pi}\sqrt{\dfrac{k}{M} - \dfrac{c^2}{4M^2}}$. When the resonance frequency exists,

$$\left(\frac{k}{M} - \frac{c^2}{4M^2}\right) - \left(\frac{k}{M} - \frac{c^2}{2M^2}\right) = \frac{c^2}{4M^2} > 0$$

(since $c > 0$), and because the square root is increasing,

$$\frac{1}{2\pi}\sqrt{\frac{k}{M} - \frac{c^2}{2M^2}} < \frac{1}{2\pi}\sqrt{\frac{k}{M} - \frac{c^2}{4M^2}},$$

so the resonance frequency is always less than the natural frequency (which in turn is less than the undamped frequency $\frac{1}{2\pi}\sqrt{k/M}$).

Answer. The amplitude is maximal at the impressed frequency $\dfrac{1}{2\pi}\sqrt{\dfrac{k}{M} - \dfrac{c^2}{2M^2}}$ (maximum amplitude $F_0/(c\alpha)$), provided $\dfrac{k}{M} - \dfrac{c^2}{2M^2} > 0$ ($c^2 < 2kM$). It need not exist: this condition is more restrictive than the underdamping condition $\dfrac{k}{M} - \dfrac{c^2}{4M^2} > 0$; for $2kM \le c^2 < 4kM$ the amplitude decreases steadily with ω. When it exists it is less than the natural frequency $\dfrac{1}{2\pi}\sqrt{\dfrac{k}{M} - \dfrac{c^2}{4M^2}}$, since $\dfrac{c^2}{2M^2} > \dfrac{c^2}{4M^2}$.

Problem 20-2 (Simmons)

Solution. Formula (20) is the same function as (19),

$$x = \frac{x_0}{\alpha} e^{-bt}(\alpha\cos\alpha t + b\sin\alpha t),$$

because with $\theta = \tan^{-1}(b/\alpha)$ we have $\cos\theta = \alpha/\sqrt{\alpha^2+b^2}$, $\sin\theta = b/\sqrt{\alpha^2+b^2}$, hence

$$\sqrt{\alpha^2+b^2}\cos(\alpha t - \theta) = \alpha\cos\alpha t + b\sin\alpha t.$$

We use (19) to differentiate, and take $x_0 > 0$ (the cart is pulled out to a positive position, as in the text). By the product rule,

$$\begin{aligned}\frac{dx}{dt} &= \frac{x_0}{\alpha} e^{-bt}\Big[-b(\alpha\cos\alpha t + b\sin\alpha t) + (-\alpha^2\sin\alpha t + \alpha b\cos\alpha t)\Big] \\ &= \frac{x_0}{\alpha} e^{-bt}\Big[-\alpha b\cos\alpha t - b^2\sin\alpha t - \alpha^2\sin\alpha t + \alpha b\cos\alpha t\Big] \\ &= -\frac{x_0(\alpha^2+b^2)}{\alpha} e^{-bt}\sin\alpha t.\end{aligned}$$

Since $x_0(\alpha^2+b^2)e^{-bt}/\alpha > 0$, the sign of dx/dt is the sign of $-\sin\alpha t$. Hence $dx/dt = 0$ exactly when $\sin\alpha t = 0$, i.e. at $t = n\pi/\alpha$ ($n = 0, 1, 2, \dots$), and:

- at $t = 2n\pi/\alpha = nT$ (recall $T = 2\pi/\alpha$ by (21)), $\sin \alpha t$ changes from negative to positive, so dx/dt changes from positive to negative: x has a (local) maximum. At $t = 0$ we have $dx/dt < 0$ on $(0, T/2)$, so x decreases from its initial value: $t = 0$ is also a maximum point (one-sided, at the start of the motion);
- at $t = (2n+1)\pi/\alpha$, $\sin \alpha t$ changes from positive to negative and x has a minimum.

Thus x assumes its maximum values precisely at $t = 0, T, 2T, \dots$. (Alternatively, $d^2x/dt^2 = -x_0(\alpha^2 + b^2)e^{-bnT} < 0$ at $t = nT$, as one finds by differentiating once more and using $\sin 2n\pi = 0$, $\cos 2n\pi = 1$.)

The maximum values are, from (19) with $\cos(\alpha nT) = \cos 2n\pi = 1$ and $\sin(\alpha nT) = 0$,

$$x(nT) = \frac{x_0}{\alpha} e^{-bnT} \cdot \alpha = x_0 e^{-bnT}.$$

If $x_1 = x(nT)$ and $x_2 = x((n+1)T)$ are two successive maximum values, then

$$\frac{x_1}{x_2} = \frac{x_0 e^{-bnT}}{x_0 e^{-b(n+1)T}} = e^{bT},$$

independently of n. Hence $\log(x_1/x_2) = bT$, the logarithmic decrement.

Remark. The maxima do *not* occur where $\cos(\alpha t - \theta) = 1$ in (20) (that would be at $t = (\theta + 2n\pi)/\alpha$), because the factor e^{-bt} is also decreasing; the curve touches the envelope $\pm \frac{x_0\sqrt{\alpha^2+b^2}}{\alpha} e^{-bt}$ slightly after each maximum. If $x_0 < 0$ the roles of maxima and minima are interchanged, and the same ratio e^{bT} holds for successive minima.

Answer. Shown: $dx/dt = -\dfrac{x_0(\alpha^2+b^2)}{\alpha} e^{-bt} \sin \alpha t$ changes from $+$ to $-$ at $t = nT$, so the maxima occur at $t = 0, T, 2T, \dots$ with values $x(nT) = x_0 e^{-bnT}$; hence $x_1/x_2 = e^{bT}$ and the logarithmic decrement is bT.

Problem 20-3 (Simmons)

Solution. Let ρ be the (mass) density of water. By Archimedes' principle, in equilibrium the weight of the buoy equals the weight of the water displaced by its lower hemisphere:

$$Mg = \rho g \cdot \tfrac{2}{3}\pi r^3 \quad \Longrightarrow \quad M = \tfrac{2}{3}\pi\rho r^3.$$

Let x be the downward displacement of the buoy from equilibrium. The water line then lies a height x above the equator of the sphere, and the extra submerged volume is the slab of the sphere between the equator and height x. Measuring z vertically from the centre, the horizontal cross section at height z is a disk of area $\pi(r^2 - z^2)$, so

$$\text{extra volume} = \int_0^x \pi(r^2 - z^2)\,\mathrm{d}z = \pi\Big(r^2 x - \frac{x^3}{3}\Big)$$

(the same formula with $x < 0$ gives the volume lost when the buoy rises). The net (restoring) force is the weight of this extra displaced water, directed against x:

$$M\frac{d^2x}{dt^2} = -\rho g \pi\Big(r^2 x - \frac{x^3}{3}\Big).$$

For a slight depression ($|x| \ll r$) the term $x^3/3$ is negligible compared with $r^2 x$; this amounts to replacing the slab by a cylinder of base πr^2, the area of the water-line cross section. So

$$\tfrac{2}{3}\pi\rho r^3 \frac{d^2x}{dt^2} = -\rho g \pi r^2 x \quad\Longrightarrow\quad \frac{d^2x}{dt^2} + \frac{3g}{2r}\,x = 0.$$

This is equation (3) with $a^2 = 3g/2r$, so the motion is simple harmonic and by (7) its period is

$$T = \frac{2\pi}{a} = 2\pi\sqrt{\frac{2r}{3g}}.$$

Note that the density of water cancels out: since the buoy floats half-submerged, its mean density is necessarily $\rho/2$.

Answer. $T = 2\pi\sqrt{2r/3g}$ seconds.

Problem 20-4 (Simmons)

Solution. Let W be the weight of the buoy (in pounds), so its mass is $M = W/g$ slugs with $g = 32$. Its radius is 1 ft, so its horizontal cross section has area $\pi \cdot 1^2 = \pi$ ft^2. In equilibrium the buoyant force balances the weight. If the buoy is pushed down a further distance x (ft), it displaces an extra volume πx ft^3 (exactly, since the sides are vertical) of water weighing $62.4\pi x$ lb, and this is the net upward force. Hence

$$\frac{W}{g}\frac{d^2x}{dt^2} = -62.4\pi x \quad\Longrightarrow\quad \frac{d^2x}{dt^2} + \frac{62.4\pi g}{W}\,x = 0,$$

simple harmonic motion with $a^2 = 62.4\pi g/W$ and period

$$T = \frac{2\pi}{a} = 2\pi\sqrt{\frac{W}{62.4\pi g}}.$$

Solving for W and substituting $T = 1.9$, $g = 32$:

$$W = \frac{62.4\pi g\,T^2}{4\pi^2} = \frac{62.4\,g\,T^2}{4\pi} = \frac{62.4\cdot 32\cdot (1.9)^2}{4\pi} = \frac{7208.448}{4\pi} \approx 573.6 \text{ lb}.$$

Remark. With $g = 32.2$ ft/s^2 the same formula gives about 577 lb; the answer is proportional to g.

Answer. $W = \dfrac{62.4\,g\,T^2}{4\pi} = \dfrac{62.4\cdot 32\cdot 1.9^2}{4\pi} \approx 574$ pounds.

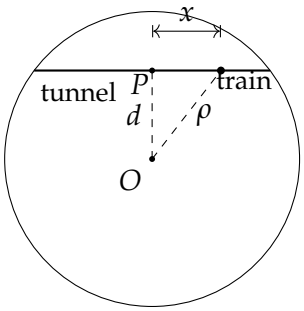
Problem 20-5 (Simmons)

Solution. *Gravity inside the earth.* We regard the earth as a homogeneous ball of radius R and mass M_E. By Newton's shell theorem, at a point at distance $\rho \le R$ from the centre the spherical shell outside radius ρ exerts no net force, while the ball of radius ρ attracts as if its mass $M_E\rho^3/R^3$ were concentrated at the centre. So a particle of mass m is pulled toward the centre with a force of magnitude

$$F = \frac{Gm\,M_E\rho^3/R^3}{\rho^2} = \frac{GM_E m}{R^3}\,\rho = \frac{mg}{R}\,\rho,$$

because at the surface $mg = GM_E m/R^2$. In vector form, $\mathbf{F} = -\dfrac{mg}{R}\,\mathbf{r}$, where $\mathbf{r}$ is the position vector from the centre.

Equation of motion. Let the tunnel be a chord at distance d ($0 \le d < R$) from the centre O, let P be the midpoint of the chord, and let x be the train's coordinate along the tunnel measured from P. Then $\mathbf{r} = x\,\mathbf{e} + d\,\mathbf{n}$, where $\mathbf{e}$ is a unit vector along the tunnel and $\mathbf{n}$ a unit vector from O to P perpendicular to it.



The component of $\mathbf{F}$ along the tunnel is $\mathbf{F}\cdot\mathbf{e} = -\dfrac{mg}{R}x$; the component $-\dfrac{mg}{R}d$ perpendicular to the tunnel is balanced by the reaction of the (frictionless) tracks and does not affect the motion along the tunnel. Newton's second law gives

$$m\frac{d^2x}{dt^2} = -\frac{mg}{R}\,x \quad\Longrightarrow\quad \frac{d^2x}{dt^2} + \frac{g}{R}\,x = 0.$$

This is (3) with $a = \sqrt{g/R}$. If the tunnel has length $2L$ and the train starts from rest at the end $x = L$, then (6) gives

$$x = L\cos\sqrt{\frac{g}{R}}\,t, \qquad \frac{dx}{dt} = -L\sqrt{\frac{g}{R}}\,\sin\sqrt{\frac{g}{R}}\,t.$$

The train reaches the other end $x = -L$ with zero velocity at $t = \pi\sqrt{R/g}$ and is back at the start at

$$T = 2\pi\sqrt{\frac{R}{g}},$$

which involves neither d nor L: the round-trip time is the same for every such tunnel.

Estimate. With $R \approx 4000$ miles $= 4000\cdot 5280 = 21{,}120{,}000$ ft and $g = 32$ ft/s^2,

$$\sqrt{\frac{R}{g}} = \sqrt{660{,}000}\ \text{s} \approx 812.4\ \text{s}, \qquad T = 2\pi\sqrt{\frac{R}{g}} \approx 5104\ \text{s} \approx 85\ \text{minutes},$$

i.e. roughly an hour and a half. (With $R = 3960$ miles one gets 84.6 minutes.)

Greatest speed. The speed $|dx/dt| = L\sqrt{g/R}\,\left|\sin\sqrt{g/R}\,t\right|$ is greatest when $|\sin| = 1$, that is, when $x = 0$, at the midpoint of the tunnel (where the train is closest to the centre), and its value is

$$v_{\max} = L\sqrt{\frac{g}{R}}.$$

Numerically, $\sqrt{g/R} \approx 1/812.4\ \text{s}^{-1} \approx 0.001231\ \text{s}^{-1}$; with L in miles,

$$v_{\max} \approx 0.001231\,L\ \text{miles/s} \approx 0.074\,L\ \text{miles/min} \approx 4.43\,L\ \text{miles/hr}.$$

Answer. The motion along the tunnel obeys $x'' + (g/R)x = 0$, so the round-trip time is $2\pi\sqrt{R/g}$ for every tunnel, about 85 minutes (roughly $1\frac{1}{2}$ hours) with $R = 4000$ miles. The greatest speed, reached at the midpoint, is $L\sqrt{g/R} \approx 0.074L$ miles/min $\approx 4.43L$ miles/hr.

Remark. The book's answer gives "approximately 90 minutes" as a rough estimate; the computation with $R = 4000$ miles gives 85 minutes, consistent with the book's speeds $0.074L$ miles/min and $4.43L$ miles/hr (which correspond to the same value of R).

Problem 20-6 (Simmons)

Solution. The mass is $M = 128/32 = 4$ slugs. With $c = 0$ (no resistance), equation (24) becomes

$$4\frac{d^2x}{dt^2} + 64x = 32\sin 4t \quad\Longrightarrow\quad \frac{d^2x}{dt^2} + 16x = 8\sin 4t,$$

with x measured (in feet) away from the wall as in Figure 25, and initial conditions $x(0) = 6$ in $= \frac{1}{2}$ ft, $x'(0) = 0$.

Homogeneous equation. The auxiliary equation $m^2 + 16 = 0$ has roots $m = \pm 4i$, so $x_h = c_1\cos 4t + c_2\sin 4t$. The free vibrations have $a = 4$ and period $2\pi/4 = \pi/2$, the same as the period of $\sin 4t$.

Particular solution. Since $\sin 4t$ solves the homogeneous equation, the trial function $A\cos 4t + B\sin 4t$ fails, and by the method of undetermined coefficients we try $x_p = t(A\cos 4t + B\sin 4t)$. Then

$$x_p' = (A\cos 4t + B\sin 4t) + t(-4A\sin 4t + 4B\cos 4t),$$
$$x_p'' = 2(-4A\sin 4t + 4B\cos 4t) + t(-16A\cos 4t - 16B\sin 4t) = -8A\sin 4t + 8B\cos 4t - 16x_p,$$

so $x_p'' + 16x_p = -8A\sin 4t + 8B\cos 4t$. Equating this to $8\sin 4t$ gives $-8A = 8$ and $8B = 0$, i.e. $A = -1$, $B = 0$:

$$x_p = -t\cos 4t.$$

Initial conditions. The general solution is $x = c_1\cos 4t + c_2\sin 4t - t\cos 4t$, with

$$x' = -4c_1\sin 4t + 4c_2\cos 4t - \cos 4t + 4t\sin 4t.$$

$x(0) = c_1 = \frac{1}{2}$, and $x'(0) = 4c_2 - 1 = 0$ gives $c_2 = \frac{1}{4}$. Therefore

$$x(t) = \tfrac{1}{2}\cos 4t + \tfrac{1}{4}\sin 4t - t\cos 4t.$$

Pure resonance. At $t = n\pi/2$ ($n = 1, 2, \dots$) we have $\cos 4t = 1$, $\sin 4t = 0$, so $x(n\pi/2) = \frac{1}{2} - \frac{n\pi}{2}$, whose absolute value tends to ∞ as $n \to \infty$. The oscillation $-t\cos 4t$ has an amplitude growing linearly in t, while the other two terms stay bounded by $\frac{1}{2} + \frac{1}{4}$; so $|x(t)|$ takes arbitrarily large values, as asserted. This happens because the forcing has exactly the natural period $\pi/2$ of the undamped system: each push reinforces the motion.

Answer. $x(t) = \frac{1}{2}\cos 4t + \frac{1}{4}\sin 4t - t\cos 4t$ (ft); e.g. $x(n\pi/2) = \frac{1}{2} - n\pi/2$, so $|x(t)|$ is unbounded.

Problem 20-7 (Simmons)

Solution. Here (30) is $LQ'' + RQ' + \frac{1}{C}Q = E_0\cos\omega t$, with L, R, C, E_0, ω positive constants.

Why the real part works. If a complex-valued function $Q = Q_1 + iQ_2$ (Q_1, Q_2 real) satisfies the complex equation, then, because L, R, $1/C$ are real and differentiation commutes with taking real parts,

$$\Big(LQ_1'' + RQ_1' + \tfrac{1}{C}Q_1\Big) + i\Big(LQ_2'' + RQ_2' + \tfrac{1}{C}Q_2\Big) = E_0\cos\omega t + iE_0\sin\omega t.$$

Equating real parts, $Q_1 = \operatorname{Re} Q$ satisfies (30).

Undetermined coefficients. Try $Q = Ae^{i\omega t}$ with a complex constant A. Then $Q' = i\omega Ae^{i\omega t}$ and $Q'' = -\omega^2 Ae^{i\omega t}$, and substitution gives

$$\Big(-\omega^2 L + i\omega R + \frac{1}{C}\Big)Ae^{i\omega t} = E_0 e^{i\omega t} \quad\Longrightarrow\quad A = \frac{E_0}{(1/C - \omega^2 L) + i\omega R}.$$

The denominator is not zero since its imaginary part $\omega R > 0$. Write

$$D = \sqrt{(1/C - \omega^2 L)^2 + \omega^2 R^2}$$

for its modulus. Multiplying numerator and denominator by the complex conjugate,

$$A = \frac{E_0\,[(1/C - \omega^2 L) - i\omega R]}{D^2}.$$

Real part. Using $e^{i\omega t} = \cos\omega t + i\sin\omega t$,

$$\begin{aligned} Ae^{i\omega t} &= \frac{E_0}{D^2}\,[(1/C - \omega^2 L) - i\omega R]\,[\cos\omega t + i\sin\omega t] \\ &= \frac{E_0}{D^2}\Big\{\big[(1/C - \omega^2 L)\cos\omega t + \omega R\sin\omega t\big] + i\big[(1/C - \omega^2 L)\sin\omega t - \omega R\cos\omega t\big]\Big\}, \end{aligned}$$

so

$$\operatorname{Re}\big(Ae^{i\omega t}\big) = \frac{E_0}{D}\Big[\frac{1/C - \omega^2 L}{D}\cos\omega t + \frac{\omega R}{D}\sin\omega t\Big].$$

Now define the angle ϕ ($0 < \phi < \pi$) by

$$\cos\phi = \frac{1/C - \omega^2 L}{D}, \qquad \sin\phi = \frac{\omega R}{D};$$

this is possible because the squares of these two numbers add up to 1 and the second is positive, and then $\tan\phi = \omega R/(1/C - \omega^2 L)$ as required (when $1/C \neq \omega^2 L$; if $1/C = \omega^2 L$ then $\phi = \pi/2$). By the subtraction formula for the cosine,

$$Q = \operatorname{Re}\big(Ae^{i\omega t}\big) = \frac{E_0}{D}\big(\cos\phi\cos\omega t + \sin\phi\sin\omega t\big) = \frac{E_0}{\sqrt{(1/C - \omega^2 L)^2 + \omega^2 R^2}}\cos(\omega t - \phi),$$

which is $(*)$, and by the first paragraph it is a solution of (30).

Equivalently, in polar form the denominator is $(1/C - \omega^2 L) + i\omega R = De^{i\phi}$, so $Ae^{i\omega t} = \dfrac{E_0}{D}e^{i(\omega t - \phi)}$ and its real part is visibly $\dfrac{E_0}{D}\cos(\omega t - \phi)$; the quantity $(1/C - \omega^2 L) + i\omega R$ equals $i\omega Z$, where $Z = R + i(\omega L - 1/\omega C)$ is the complex impedance of the circuit.

Remark. The condition $\tan\phi = \omega R/(1/C - \omega^2 L)$ alone determines ϕ only up to a multiple of π; the calculation shows that the correct choice is the one with $0 < \phi < \pi$ (in the first quadrant if $1/C > \omega^2 L$, in the second if $1/C < \omega^2 L$). The same remark applies to ϕ in (27).

Answer. $Q = \dfrac{E_0}{(1/C - \omega^2 L) + i\omega R}\, e^{i\omega t}$ is a particular solution of the complex equation; its real part is $Q = \dfrac{E_0}{\sqrt{(1/C - \omega^2 L)^2 + \omega^2 R^2}} \cos(\omega t - \phi)$ with $\cos\phi = (1/C - \omega^2 L)/D$, $\sin\phi = \omega R/D$ (so $\tan\phi = \omega R/(1/C - \omega^2 L)$), which is $(*)$ and solves (30).

Section 21 Newton's Law of Gravitation and the Motion of the Planets

Throughout we use the notation and equations of the text. A particle m moves in the plane with polar coordinates (r,θ) centred at the fixed mass M, and

$$m\left(r\frac{d^2\theta}{dt^2}+2\frac{dr}{dt}\frac{d\theta}{dt}\right)=F_\theta, \qquad m\left[\frac{d^2r}{dt^2}-r\left(\frac{d\theta}{dt}\right)^2\right]=F_r, \tag{7}$$

$$r^2\frac{d\theta}{dt}=h, \tag{8}$$

$$r=\frac{h^2/k}{1+e\cos\theta}, \tag{15}$$

$$b^2=a^2(1-e^2), \qquad b^2=\frac{h^2a}{k}, \qquad T^2=\left(\frac{4\pi^2}{k}\right)a^3, \tag{20),(21),(22}$$

where $k=GM$, a and b are the semimajor and semiminor axes of an elliptic orbit and T is its period. The speed satisfies, by (4), $v^2=r^2(d\theta/dt)^2+(dr/dt)^2$.

Problem 21-1 (Simmons)

Solution. First we see why the constant drops out. By (22), $T^2=(4\pi^2/k)a^3$ holds for every planet with the same constant $k=GM$. For the earth, $T=1$ year and $a=1$ astronomical unit, so in these units $1=(4\pi^2/k)\cdot 1$, i.e. $4\pi^2/k=1$, and (22) becomes $T^2=a^3$. Since $T>0$,

$$T=a^{3/2}=a\sqrt{a} \qquad (T \text{ in years, } a \text{ in astronomical units}).$$

(a) $a=2$: $T=2^{3/2}=2\sqrt{2}\approx 2.83$ years.

(b) $a=3$: $T=3^{3/2}=3\sqrt{3}\approx 5.20$ years.

(c) $a=25$: $T=25^{3/2}=\left(\sqrt{25}\right)^3=5^3=125$ years.

Answer. (a) $T=2\sqrt{2}\approx 2.83$ years; (b) $T=3\sqrt{3}\approx 5.20$ years; (c) $T=125$ years.

Problem 21-2 (Simmons)

Solution. We use the form $T^2=a^3$ of Kepler's third law from Problem 1 (T in years, a in astronomical units).

(a) With 1 year $=365$ days, Mercury's period is $T=88/365\approx 0.24110$ year. Solving $T^2=a^3$ for $a>0$,

$$a=T^{2/3}=\left(\frac{88}{365}\right)^{2/3}.$$

Numerically, $\log(88/365)\approx -1.42256$, so $\log a\approx \frac{2}{3}(-1.42256)=-0.94837$ and

$$a\approx e^{-0.94837}\approx 0.387\approx 0.39 \text{ astronomical units.}$$

(Using 365.25 days per year changes only the fourth decimal: $a\approx 0.3872$.) In miles, $a\approx 0.387\times 93{,}000{,}000\approx 36{,}000{,}000$ miles (about 58,000,000 km).

(b) With $a=9.54$,

$$T=a^{3/2}=9.54\sqrt{9.54}\approx 9.54\times 3.0887\approx 29.47\approx 29.5 \text{ years.}$$

Answer. (a) $a = (88/365)^{2/3} \approx 0.39$ astronomical units $\approx$ 36,000,000 miles; (b) $T = 9.54^{3/2} \approx 29.5$ years.

Problem 21-3 (Simmons)

Solution. Here we assume only the two kinematic laws (8) and (15), in which $h > 0$, k and $e \geq 0$ are constants; nothing is assumed about the force. Note that $k > 0$: at $\theta = 0$, (15) gives $r = (h^2/k)/(1+e) > 0$, so $h^2/k > 0$. The components F_θ, F_r of the force are then determined by Newton's second law in the form (7).

(a) Multiply the first of equations (7) by r and recognise a derivative:

$$rF_\theta = m\left(r^2\frac{d^2\theta}{dt^2} + 2r\frac{dr}{dt}\frac{d\theta}{dt}\right) = m\,\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right) = m\,\frac{dh}{dt} = 0,$$

because $r^2\,d\theta/dt = h$ is constant by (8). Since $r > 0$, $F_\theta = 0$: the force is central.

(b) Write (15) as

$$\frac{1}{r} = \frac{k}{h^2}(1 + e\cos\theta).$$

Differentiating with respect to t (chain rule, $\theta = \theta(t)$):

$$-\frac{1}{r^2}\frac{dr}{dt} = -\frac{ke}{h^2}\sin\theta\,\frac{d\theta}{dt}.$$

Multiplying by $-r^2$ and using (8), $r^2\,d\theta/dt = h$:

$$\frac{dr}{dt} = \frac{ke}{h^2}\sin\theta\left(r^2\frac{d\theta}{dt}\right) = \frac{ke}{h^2}\,h\sin\theta = \frac{ke}{h}\sin\theta.$$

(c) Differentiate (b) with respect to t; k, e, h are constants, and $d\theta/dt = h/r^2$ by (8):

$$\frac{d^2r}{dt^2} = \frac{ke}{h}\cos\theta\,\frac{d\theta}{dt} = \frac{ke}{h}\cos\theta\cdot\frac{h}{r^2} = \frac{ke\cos\theta}{r^2}.$$

(d) By (8), $r(d\theta/dt)^2 = r\,(h/r^2)^2 = h^2/r^3$. By (15), $h^2 = kr(1+e\cos\theta)$, so

$$r\left(\frac{d\theta}{dt}\right)^2 = \frac{h^2}{r^3} = \frac{kr(1+e\cos\theta)}{r^3} = \frac{k(1+e\cos\theta)}{r^2}.$$

Substituting this and (c) into the second of equations (7):

$$F_r = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right] = m\left[\frac{ke\cos\theta}{r^2} - \frac{k + ke\cos\theta}{r^2}\right] = -\frac{mk}{r^2}.$$

Thus $\mathbf{F} = F_\theta\mathbf{u}_\theta + F_r\mathbf{u}_r = -(mk/r^2)\,\mathbf{u}_r$: since $k > 0$, the force points toward the origin (opposite to $\mathbf{u}_r$) and has magnitude mk/r^2, which is proportional to m and inversely proportional to r^2. Writing the constant k as GM gives $F_r = -GMm/r^2$, which is Newton's law of gravitation (11).

Answer. Shown: (8) and (15) give $F_\theta = 0$, $\dfrac{dr}{dt} = \dfrac{ke}{h}\sin\theta$, $\dfrac{d^2r}{dt^2} = \dfrac{ke\cos\theta}{r^2}$, and $F_r = -\dfrac{mk}{r^2} = -G\dfrac{Mm}{r^2}$, an attraction toward the origin with magnitude inversely proportional to r^2.

Problem 21-4 (Simmons)

Solution. The orbit is an ellipse (15) with $0 \le e < 1$ and semimajor axis a. By (4) the velocity has components $r\,d\theta/dt$ along $\mathbf{u}_\theta$ and dr/dt along $\mathbf{u}_r$, so

$$v^2 = r^2\left(\frac{d\theta}{dt}\right)^2 + \left(\frac{dr}{dt}\right)^2.$$

By (8), $r^2(d\theta/dt)^2 = (r^2\,d\theta/dt)^2/r^2 = h^2/r^2$, and by Problem 3(b), $dr/dt = (ke/h)\sin\theta$. Hence

$$v^2 = \frac{h^2}{r^2} + \frac{k^2}{h^2}e^2\sin^2\theta.$$

We now eliminate θ. From (15), $1 + e\cos\theta = h^2/(kr)$, so $e\cos\theta = \dfrac{h^2}{kr} - 1$, and

$$e^2\sin^2\theta = e^2 - e^2\cos^2\theta = e^2 - \left(\frac{h^2}{kr} - 1\right)^2 = e^2 - \frac{h^4}{k^2r^2} + \frac{2h^2}{kr} - 1.$$

Therefore

$$v^2 = \frac{h^2}{r^2} + \frac{k^2}{h^2}\left(e^2 - 1\right) - \frac{h^2}{r^2} + \frac{2k}{r} = \frac{2k}{r} - \frac{k^2(1-e^2)}{h^2}.$$

Finally, the text showed that $a = \dfrac{h^2}{k(1-e^2)}$ (half the sum of the least and greatest values of r), so $\dfrac{k^2(1-e^2)}{h^2} = \dfrac{k}{a}$, and

$$v^2 = \frac{2k}{r} - \frac{k}{a} = k\left(\frac{2}{r} - \frac{1}{a}\right).$$

Remark. The same result follows from conservation of energy (18): $\frac{1}{2}mv^2 - km/r = E$, together with $e^2 = 1 + E(2h^2/mk^2)$, which gives $E = -mk^2(1-e^2)/(2h^2) = -mk/(2a)$; hence $v^2 = 2k/r - k/a$. In particular the total energy of an elliptic orbit depends only on its semimajor axis.

Answer. Shown: $v^2 = \dfrac{h^2}{r^2} + \dfrac{k^2e^2}{h^2}\sin^2\theta = \dfrac{2k}{r} - \dfrac{k^2(1-e^2)}{h^2} = k\left(\dfrac{2}{r} - \dfrac{1}{a}\right)$.

Problem 21-5 (Simmons)

Solution. As in the text, we treat each fragment as a particle moving under the attraction of the fixed sun alone (ignoring the attractions of the fragments on one another and of the other planets). Let the explosion occur at time t_0 at the point P_0, at distance r_0 from the sun, and let v_0 be the common speed of the fragments just after the explosion.

Which fragments are in elliptic orbits. A fragment that neither falls into the sun nor escapes stays at distances bounded away from 0 and ∞. A fragment with $h = 0$ moves on a straight line through

the sun ($d\theta/dt = 0$), and so ends in the sun; hence the fragments considered have $h \neq 0$, and the analysis of the text (with θ replaced by $-\theta$ if $h < 0$) shows that each moves on a conic (15). Parabolic and hyperbolic orbits ($e \geq 1$) are unbounded and lead out of the solar system, so each fragment considered moves on an ellipse with the sun at a focus, having some semimajor axis a; different fragments have, in general, different ellipses and different eccentricities.

All these ellipses have the same semimajor axis. Apply Problem 4 to a fragment at the instant t_0, when it is at distance r_0 with speed v_0:

$$v_0^2 = k\left(\frac{2}{r_0} - \frac{1}{a}\right) \qquad \Longrightarrow \qquad \frac{1}{a} = \frac{2}{r_0} - \frac{v_0^2}{k}.$$

The right side involves only r_0, v_0 and $k = GM$, which are the same for every fragment (the direction of motion and the mass of the fragment do not enter). Hence every such fragment has the same semimajor axis

$$a = \frac{kr_0}{2k - r_0 v_0^2}$$

(necessarily $r_0 v_0^2 < 2k$, since $a > 0$; if $r_0 v_0^2 \geq 2k$ no fragment would have an elliptic orbit at all).

All have the same period. By Kepler's third law (22), $T^2 = (4\pi^2/k)a^3$, so every such fragment has the same period

$$T = 2\pi\sqrt{\frac{a^3}{k}}.$$

(For $h < 0$ the area swept is $\frac{1}{2}|h|(t_2 - t_1)$, and the derivation of (22) is unchanged with h replaced by $|h|$.)

Reunion. The motion on an elliptic orbit is periodic with period T: after time T the radius vector has swept out the whole area πab of the ellipse exactly once, so the fragment is back at its starting point P_0 (and every fragment's orbit passes through P_0, where it began). Hence at the time $t_0 + T$ (and again at $t_0 + 2T$, $t_0 + 3T, \ldots$) every fragment that has neither fallen into the sun nor escaped is again at P_0: all of them reunite at the point where they began to diverge.

Answer. Shown: by Problem 4, $1/a = 2/r_0 - v_0^2/k$ is the same for all fragments in elliptic orbits, so by (22) they all have the same period $T = 2\pi\sqrt{a^3/k}$, and all are back at the explosion point at time $t_0 + T$.

Section 22 Higher Order Linear Equations. Coupled Harmonic Oscillators

Throughout Problems 1–15 we use the method of the text. For $y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0$ we substitute $y = e^{rx}$ and obtain the auxiliary equation $r^n + a_1 r^{n-1} + \cdots + a_n = 0$. Each root contributes to the general solution as follows:

- a real root r of multiplicity k contributes $(c_1 + c_2 x + \cdots + c_k x^{k-1}) e^{rx}$;
- a pair of complex roots $\alpha \pm i\beta$ of multiplicity k contributes $e^{\alpha x}\left[(A_1 + \cdots + A_k x^{k-1}) \cos \beta x + (B_1 + \cdots + B_k x^{k-1}) \sin \beta x\right]$.

Altogether this gives n linearly independent solutions, and their general linear combination is the general solution. To factor the auxiliary polynomial we look for integer roots among the divisors of the constant term, and divide out the corresponding linear factors (synthetic division).

Problems 22-1 to 22-15 (Simmons)

We treat each equation as a separate problem below.

Problem 22-1 (Simmons)

Solution. The auxiliary equation is

$$r^3 - 3r^2 + 2r = r(r^2 - 3r + 2) = r(r-1)(r-2) = 0,$$

with the three distinct real roots $r = 0, 1, 2$. They give the solutions $e^{0x} = 1$, e^x, e^{2x}, so the general solution is $y = c_1 + c_2 e^x + c_3 e^{2x}$.

Answer. $y = c_1 + c_2 e^x + c_3 e^{2x}$.

Problem 22-2 (Simmons)

Solution. The auxiliary polynomial is $p(r) = r^3 - 3r^2 + 4r - 2$. Since $p(1) = 1 - 3 + 4 - 2 = 0$, $r - 1$ is a factor, and division gives

$$r^3 - 3r^2 + 4r - 2 = (r-1)(r^2 - 2r + 2).$$

The quadratic factor has roots $r = \dfrac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$. Hence the roots are 1 and $1 \pm i$ (so $\alpha = 1$, $\beta = 1$), giving the solutions e^x, $e^x \cos x$, $e^x \sin x$:

$$y = c_1 e^x + e^x (c_2 \cos x + c_3 \sin x).$$

Answer. $y = c_1 e^x + e^x (c_2 \cos x + c_3 \sin x)$.

Problem 22-3 (Simmons)

Solution. The auxiliary equation is $r^3 - 1 = (r-1)(r^2+r+1) = 0$. The quadratic factor has roots

$$r = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i.$$

Thus $r = 1$ gives e^x, and the pair $\alpha \pm i\beta$ with $\alpha = -\frac{1}{2}$, $\beta = \frac{1}{2}\sqrt{3}$ gives $e^{-x/2}\cos\frac{1}{2}\sqrt{3}x$ and $e^{-x/2}\sin\frac{1}{2}\sqrt{3}x$.

Answer. $y = c_1 e^x + e^{-x/2}\left(c_2 \cos\frac{1}{2}\sqrt{3}\,x + c_3 \sin\frac{1}{2}\sqrt{3}\,x\right)$.

Problem 22-4 (Simmons)

Solution. The auxiliary equation is $r^3 + 1 = (r+1)(r^2-r+1) = 0$. The quadratic factor has roots

$$r = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i.$$

So $r = -1$ gives e^{-x}, and $\alpha = \frac{1}{2}$, $\beta = \frac{1}{2}\sqrt{3}$ give $e^{x/2}\cos\frac{1}{2}\sqrt{3}x$, $e^{x/2}\sin\frac{1}{2}\sqrt{3}x$.

Answer. $y = c_1 e^{-x} + e^{x/2}\left(c_2 \cos\frac{1}{2}\sqrt{3}\,x + c_3 \sin\frac{1}{2}\sqrt{3}\,x\right)$.

Problem 22-5 (Simmons)

Solution. By the binomial theorem the auxiliary polynomial is $r^3 + 3r^2 + 3r + 1 = (r+1)^3$, so $r = -1$ is a root of multiplicity 3. It contributes $(c_1 + c_2 x + c_3 x^2)e^{-x}$, which is the whole general solution.

Answer. $y = (c_1 + c_2 x + c_3 x^2)e^{-x}$.

Problem 22-6 (Simmons)

Solution. The coefficients $1, 4, 6, 4, 1$ are binomial coefficients, so the auxiliary polynomial is $r^4 + 4r^3 + 6r^2 + 4r + 1 = (r+1)^4$: the root $r = -1$ has multiplicity 4, giving $(c_1 + c_2 x + c_3 x^2 + c_4 x^3)e^{-x}$.

Answer. $y = (c_1 + c_2 x + c_3 x^2 + c_4 x^3)e^{-x}$.

Problem 22-7 (Simmons)

Solution. The auxiliary equation is $r^4 - 1 = (r^2-1)(r^2+1) = (r-1)(r+1)(r^2+1) = 0$, with roots $1, -1, \pm i$. These give e^x, e^{-x}, $\cos x$, $\sin x$.

Answer. $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$.

Problem 22-8 (Simmons)

Solution. The auxiliary equation is a quadratic in r^2: $r^4 + 5r^2 + 4 = (r^2+1)(r^2+4) = 0$, with roots $\pm i$ and $\pm 2i$. These give $\cos x$, $\sin x$, $\cos 2x$, $\sin 2x$.

Answer. $y = c_1 \cos x + c_2 \sin x + c_3 \cos 2x + c_4 \sin 2x$.

Problem 22-9 (Simmons)

Solution. Here a is a real constant. The auxiliary polynomial is a perfect square:

$$r^4 - 2a^2r^2 + a^4 = (r^2 - a^2)^2 = (r-a)^2(r+a)^2.$$

If $a \neq 0$, the roots a and $-a$ are distinct and each has multiplicity 2, so they contribute $(c_1 + c_2x)e^{ax}$ and $(c_3 + c_4x)e^{-ax}$:

$$y = (c_1 + c_2x)e^{ax} + (c_3 + c_4x)e^{-ax}.$$

(In the degenerate case $a = 0$ the equation is $y^{(4)} = 0$, the root $r = 0$ has multiplicity 4, and $y = c_1 + c_2x + c_3x^2 + c_4x^3$.)

Answer. $y = (c_1 + c_2x)e^{ax} + (c_3 + c_4x)e^{-ax}$ $(a \neq 0)$.

Problem 22-10 (Simmons)

Solution. With a real and $a \neq 0$, the auxiliary polynomial is

$$r^4 + 2a^2r^2 + a^4 = (r^2 + a^2)^2 = (r - ia)^2(r + ia)^2,$$

so the complex pair $\pm ia$ ($\alpha = 0$, $\beta = a$) has multiplicity 2. By the rule for repeated complex roots it contributes $(c_1 + c_2x)\cos ax + (c_3 + c_4x)\sin ax$, which is the general solution. (If $a = 0$ the equation is $y^{(4)} = 0$ with general solution $c_1 + c_2x + c_3x^2 + c_4x^3$.)

Remark. The answer printed in the book for Problem 10, $y = (c_1 + c_2x)e^{-x} + c_3\cos x + c_4 \sin x$, is the answer to Problem 11 repeated by mistake: it does not contain a, and, for example, $y = e^{-x}$ gives $y^{(4)} + 2a^2y'' + a^4y = (1 + a^2)^2e^{-x} \neq 0$.

Answer. $y = (c_1 + c_2x)\cos ax + (c_3 + c_4x)\sin ax$ $(a \neq 0)$.

Problem 22-11 (Simmons)

Solution. The auxiliary polynomial $p(r) = r^4 + 2r^3 + 2r^2 + 2r + 1$ satisfies $p(-1) = 1 - 2 + 2 - 2 + 1 = 0$. Grouping the terms,

$$p(r) = (r^4 + 2r^3 + r^2) + (r^2 + 2r + 1) = r^2(r+1)^2 + (r+1)^2 = (r+1)^2(r^2+1).$$

So $r = -1$ is a double root and $\pm i$ are simple roots, giving $(c_1 + c_2x)e^{-x}$ and $c_3\cos x + c_4 \sin x$.

Answer. $y = (c_1 + c_2x)e^{-x} + c_3\cos x + c_4 \sin x$.

Problem 22-12 (Simmons)

Solution. Let $p(r) = r^4 + 2r^3 - 2r^2 - 6r + 5$. Since $p(1) = 1 + 2 - 2 - 6 + 5 = 0$, division by $r - 1$ gives $p(r) = (r-1)(r^3 + 3r^2 + r - 5)$. The cubic again vanishes at $r = 1$ ($1 + 3 + 1 - 5 = 0$), and $r^3 + 3r^2 + r - 5 = (r-1)(r^2 + 4r + 5)$. Hence

$$p(r) = (r-1)^2(r^2 + 4r + 5), \qquad r^2 + 4r + 5 = 0 \iff r = \frac{-4 \pm \sqrt{16 - 20}}{2} = -2 \pm i.$$

The double root 1 gives $(c_1 + c_2 x)e^x$; the pair $-2 \pm i$ gives $e^{-2x}(c_3 \cos x + c_4 \sin x)$.

Answer. $y = (c_1 + c_2 x)e^x + e^{-2x}(c_3 \cos x + c_4 \sin x)$.

Problem 22-13 (Simmons)

Solution. Let $p(r) = r^3 - 6r^2 + 11r - 6$. Then $p(1) = 1 - 6 + 11 - 6 = 0$ and $p(r) = (r-1)(r^2 - 5r + 6) = (r-1)(r-2)(r-3)$. The distinct roots $1, 2, 3$ give e^x, e^{2x}, e^{3x}.

Answer. $y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$.

Problem 22-14 (Simmons)

Solution. Let $p(r) = r^4 + r^3 - 3r^2 - 5r - 2$. Testing divisors of -2: $p(-1) = 1 - 1 - 3 + 5 - 2 = 0$ and $p(2) = 16 + 8 - 12 - 10 - 2 = 0$. Dividing by $(r-2)(r+1) = r^2 - r - 2$:

$$r^4 + r^3 - 3r^2 - 5r - 2 = (r^2 - r - 2)(r^2 + 2r + 1) = (r-2)(r+1)(r+1)^2 = (r-2)(r+1)^3.$$

(Check: $(r^2 - r - 2)(r^2 + 2r + 1) = r^4 + 2r^3 + r^2 - r^3 - 2r^2 - r - 2r^2 - 4r - 2 = r^4 + r^3 - 3r^2 - 5r - 2$.) Thus $r = 2$ is simple and $r = -1$ is triple, giving $c_1 e^{2x}$ and $(c_2 + c_3 x + c_4 x^2)e^{-x}$.

Answer. $y = c_1 e^{2x} + (c_2 + c_3 x + c_4 x^2)e^{-x}$.

Problem 22-15 (Simmons)

Solution. Group the auxiliary polynomial in pairs of terms:

$$\begin{aligned} r^5 - 6r^4 - 8r^3 + 48r^2 + 16r - 96 &= r^4(r-6) - 8r^2(r-6) + 16(r-6) \\ &= (r-6)(r^4 - 8r^2 + 16) = (r-6)(r^2 - 4)^2 \\ &= (r-6)(r-2)^2(r+2)^2. \end{aligned}$$

The roots are 2 (double), -2 (double) and 6 (simple), giving $(c_1 + c_2 x)e^{2x}$, $(c_3 + c_4 x)e^{-2x}$ and $c_5 e^{6x}$.

Answer. $y = (c_1 + c_2 x)e^{2x} + (c_3 + c_4 x)e^{-2x} + c_5 e^{6x}$.

Check. For each of Problems 1–15 the verification script substitutes every basis function into the differential equation (all residuals vanish), confirms the factorization of the auxiliary polynomial, and checks that the Wronskian of the basis is not identically zero, so the basis is linearly independent.

Problem 22-16 (Simmons)

Solution. Let x_1, x_2 be the displacements of the carts from their equilibrium positions, measured positive to the right. As in Section 20 there is no friction and the springs obey Hooke's law: a spring whose length differs from its equilibrium length by δ exerts a restoring force of magnitude $k\delta$ at each end, directed so as to return it to that length. (If the springs are under some tension already at equilibrium, those equilibrium forces balance one another on each cart, and since Hooke's law is linear only the changes computed below are left over.)

Take the configuration of the figure: $x_1 > 0$, $x_2 > 0$, $x_2 > x_1$.

- *Left spring* (k_1): its left end is fixed and its right end has moved x_1 to the right, so it is stretched by x_1. It pulls m_1 toward the wall, i.e. to the left: force $-k_1 x_1$ on m_1.
- *Middle spring* (k_3): its left end has moved x_1 and its right end x_2, so it is stretched by $x_2 - x_1 > 0$. A stretched spring pulls its two ends toward each other: it pulls m_1 to the right with force $+k_3(x_2 - x_1)$ and pulls m_2 to the left with force $-k_3(x_2 - x_1)$.
- *Right spring* (k_2): its right end is fixed and its left end has moved x_2 to the right, so it is compressed by x_2. It pushes m_2 back to the left: force $-k_2 x_2$ on m_2.

Newton's second law (mass $\times$ acceleration $=$ total force) for each cart gives

$$m_1 \frac{d^2x_1}{dt^2} = -k_1 x_1 + k_3(x_2 - x_1) = -(k_1 + k_3)x_1 + k_3 x_2,$$
$$m_2 \frac{d^2x_2}{dt^2} = -k_3(x_2 - x_1) - k_2 x_2 = k_3 x_1 - (k_2 + k_3)x_2,$$

which are equations (6). Although we used a particular configuration to decide the directions, the resulting formulas are valid for all positions: if, say, $x_1 < 0$, the left spring is compressed by $-x_1$ and pushes m_1 to the right with force $k_1(-x_1) = -k_1 x_1$; similarly for the other springs. In every case the signed force is given by the same linear expression.

Answer. Shown: $m_1 x_1'' = -k_1 x_1 + k_3(x_2 - x_1) = -(k_1 + k_3)x_1 + k_3 x_2$ and $m_2 x_2'' = -k_3(x_2 - x_1) - k_2 x_2 = k_3 x_1 - (k_2 + k_3)x_2$.

Problem 22-17 (Simmons)

Solution. Solve the first equation of (6) for x_2 (possible since $k_3 > 0$):

$$x_2 = \frac{1}{k_3}\left(m_1 \frac{d^2x_1}{dt^2} + (k_1 + k_3)x_1\right), \qquad \text{so} \qquad \frac{d^2x_2}{dt^2} = \frac{1}{k_3}\left(m_1 \frac{d^4x_1}{dt^4} + (k_1 + k_3)\frac{d^2x_1}{dt^2}\right).$$

Substitute both into the second equation $m_2 x_2'' = k_3 x_1 - (k_2 + k_3)x_2$:

$$\frac{m_2}{k_3}\left(m_1 x_1^{(4)} + (k_1 + k_3)x_1''\right) = k_3 x_1 - \frac{k_2 + k_3}{k_3}\left(m_1 x_1'' + (k_1 + k_3)x_1\right).$$

Multiply by k_3 and bring everything to the left:

$$m_1 m_2 x_1^{(4)} + \left[m_2(k_1 + k_3) + m_1(k_2 + k_3)\right]x_1'' + \left[(k_1 + k_3)(k_2 + k_3) - k_3^2\right]x_1 = 0.$$

Finally divide by $m_1 m_2$:

$$\frac{d^4x_1}{dt^4} + \left[\frac{k_1 + k_3}{m_1} + \frac{k_2 + k_3}{m_2}\right]\frac{d^2x_1}{dt^2} + \left[\left(\frac{k_1 + k_3}{m_1}\right)\left(\frac{k_2 + k_3}{m_2}\right) - \frac{k_3^2}{m_1 m_2}\right]x_1 = 0.$$

Answer. $\dfrac{d^4x_1}{dt^4} + \left[\dfrac{k_1+k_3}{m_1} + \dfrac{k_2+k_3}{m_2}\right]\dfrac{d^2x_1}{dt^2} + \left[\dfrac{(k_1+k_3)(k_2+k_3)}{m_1m_2} - \dfrac{k_3^2}{m_1m_2}\right]x_1 = 0.$

Problem 22-18 (Simmons)

Solution. *The equation for x_1.* With $m_1 = m_2 = m$, $k_1 = k_2 = k_3 = k$ the coefficients in Problem 17 become

$$\frac{2k}{m} + \frac{2k}{m} = \frac{4k}{m}, \qquad \frac{2k}{m}\cdot\frac{2k}{m} - \frac{k^2}{m^2} = \frac{3k^2}{m^2},$$

so

$$\frac{d^4x_1}{dt^4} + \frac{4k}{m}\frac{d^2x_1}{dt^2} + \frac{3k^2}{m^2}x_1 = 0.$$

Its auxiliary equation is a quadratic in r^2:

$$r^4 + \frac{4k}{m}r^2 + \frac{3k^2}{m^2} = \left(r^2 + \frac{k}{m}\right)\left(r^2 + \frac{3k}{m}\right) = 0,$$

with the purely imaginary roots $r = \pm i\sqrt{k/m}$ and $r = \pm i\sqrt{3k/m}$, all simple. Hence

$$x_1 = c_1\cos\sqrt{\frac{k}{m}}\,t + c_2\sin\sqrt{\frac{k}{m}}\,t + c_3\cos\sqrt{\frac{3k}{m}}\,t + c_4\sin\sqrt{\frac{3k}{m}}\,t.$$

The equation for x_2, directly. In the special case, equations (6) read

$$mx_1'' = -2kx_1 + kx_2, \qquad mx_2'' = kx_1 - 2kx_2.$$

This time solve the *second* equation for x_1: $x_1 = \dfrac{m}{k}x_2'' + 2x_2$, so $x_1'' = \dfrac{m}{k}x_2^{(4)} + 2x_2''$. Substituting in the first equation:

$$m\left(\frac{m}{k}x_2^{(4)} + 2x_2''\right) = -2k\left(\frac{m}{k}x_2'' + 2x_2\right) + kx_2, \quad \text{i.e.} \quad \frac{m^2}{k}x_2^{(4)} + 4mx_2'' + 3kx_2 = 0.$$

Multiplying by k/m^2,

$$\frac{d^4x_2}{dt^4} + \frac{4k}{m}\frac{d^2x_2}{dt^2} + \frac{3k^2}{m^2}x_2 = 0,$$

which is the same equation as for x_1. (Consistently, $x_2 = (mx_1'' + 2kx_1)/k$ applied to the solution above gives $x_2 = c_1\cos\sqrt{k/m}\,t + c_2\sin\sqrt{k/m}\,t - c_3\cos\sqrt{3k/m}\,t - c_4\sin\sqrt{3k/m}\,t$: in the slow mode the carts move together, in the fast mode in opposite directions.)

Normal frequencies. The solutions are combinations of oscillations with circular frequencies $\omega_1 = \sqrt{k/m}$ and $\omega_2 = \sqrt{3k/m}$; the corresponding frequencies (cycles per unit time, $\omega/2\pi$, as in Section 20) are

$$\frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad \text{and} \qquad \frac{1}{2\pi}\sqrt{\frac{3k}{m}}.$$

Answer. $x_1 = c_1\cos\sqrt{k/m}\,t + c_2\sin\sqrt{k/m}\,t + c_3\cos\sqrt{3k/m}\,t + c_4\sin\sqrt{3k/m}\,t$; x_2 satisfies the same equation $x^{(4)} + (4k/m)x'' + (3k^2/m^2)x = 0$; normal frequencies $\dfrac{1}{2\pi}\sqrt{\dfrac{k}{m}}$ and $\dfrac{1}{2\pi}\sqrt{\dfrac{3k}{m}}$.

Problem 22-19 (Simmons)

Solution. The auxiliary equation of $y^{(4)} = 0$ is $r^4 = 0$: the root $r = 0$ has multiplicity 4, so $y_g = (c_1 + c_2x + c_3x^2 + c_4x^3)e^{0x}$. (Equivalently, integrate four times.) We write it as $y = c_1x^3 + c_2x^2 + c_3x + c_4$.

For $y^{(4)} = \sin x + 24$ we need a particular solution. Since $\frac{d^4}{dx^4}\sin x = \sin x$ and $\frac{d^4}{dx^4}x^4 = 24$, the function $y_p = \sin x + x^4$ works. (Undetermined coefficients gives the same: $A\sin x + B\cos x$ yields $A = 1$, $B = 0$; for the constant 24, since 0 is a quadruple root, the trial function is Cx^4, and $24C = 24$.) Therefore $y = y_g + y_p$.

Answer. $y^{(4)} = 0$: $y = c_1x^3 + c_2x^2 + c_3x + c_4$; $\quad y^{(4)} = \sin x + 24$: $y = c_1x^3 + c_2x^2 + c_3x + c_4 + \sin x + x^4$.

Problem 22-20 (Simmons)

Solution. By Problem 1 the reduced equation has $y_g = c_1 + c_2e^x + c_3e^{2x}$ (roots $0, 1, 2$).

Term 10. A constant trial function would be a solution of the reduced equation ($r = 0$ is a simple root), so we try $y = Ax$: $y' = A$, $y'' = y''' = 0$, and $2A = 10$, $A = 5$.

Term $42e^{3x}$. Since 3 is not a root, try $y = Be^{3x}$: $y''' - 3y'' + 2y' = (27 - 27 + 6)Be^{3x} = 6Be^{3x}$, so $6B = 42$, $B = 7$.

Thus $y_p = 5x + 7e^{3x}$ and $y = y_g + y_p$.

Answer. $y = c_1 + c_2e^x + c_3e^{2x} + 5x + 7e^{3x}$.

Problem 22-21 (Simmons)

Solution. The auxiliary equation $r^3 - r = r(r-1)(r+1) = 0$ has roots $0, 1, -1$, so $y_g = c_1 + c_2e^x + c_3e^{-x}$. Since $r = 0$ is a simple root, the trial particular solution for the constant 1 is $y = Ax$, giving $-A = 1$, $A = -1$. Hence

$$y = c_1 + c_2e^x + c_3e^{-x} - x, \qquad y' = c_2e^x - c_3e^{-x} - 1, \qquad y'' = c_2e^x + c_3e^{-x}.$$

The initial conditions give

$$c_1 + c_2 + c_3 = 4, \qquad c_2 - c_3 - 1 = 4, \qquad c_2 + c_3 = 4.$$

From the last two, $c_2 - c_3 = 5$ and $c_2 + c_3 = 4$, so $c_2 = \frac{9}{2}$, $c_3 = -\frac{1}{2}$; then the first gives $c_1 = 4 - 4 = 0$.

Check. $y = \frac{9}{2}e^x - \frac{1}{2}e^{-x} - x$: $y''' - y' = (\frac{9}{2}e^x + \frac{1}{2}e^{-x}) - (\frac{9}{2}e^x + \frac{1}{2}e^{-x} - 1) = 1$; $y(0) = \frac{9}{2} - \frac{1}{2} = 4$, $y'(0) = \frac{9}{2} + \frac{1}{2} - 1 = 4$, $y''(0) = \frac{9}{2} - \frac{1}{2} = 4$.

Answer. $y = \frac{9}{2}e^x - \frac{1}{2}e^{-x} - x$.

Problem 22-22 (Simmons)

Solution. Work on $x > 0$ and put $x = e^z$, $z = \log x$, so $dz/dx = 1/x$. Write $\dot{y} = dy/dz$, $\ddot{y} = d^2y/dz^2$, $\dddot{y} = d^3y/dz^3$. By the chain rule, $\dfrac{d}{dx}(\cdot) = \dfrac{1}{x}\dfrac{d}{dz}(\cdot)$, so

$$\begin{aligned} y' &= \frac{\dot{y}}{x}, && \Longrightarrow \quad xy' = \dot{y};\\ y'' &= \frac{d}{dx}\left(\frac{\dot{y}}{x}\right) = \frac{\ddot{y}}{x^2} - \frac{\dot{y}}{x^2}, && \Longrightarrow \quad x^2y'' = \ddot{y} - \dot{y};\\ y''' &= \frac{d}{dx}\left(\frac{\ddot{y} - \dot{y}}{x^2}\right) = \frac{\dddot{y} - \ddot{y}}{x^3} - \frac{2(\ddot{y} - \dot{y})}{x^3}, && \Longrightarrow \quad x^3y''' = \dddot{y} - 3\ddot{y} + 2\dot{y}. \end{aligned}$$

(In the second and third lines we used the product rule together with $\frac{d}{dx}\dot{y} = \ddot{y}/x$, $\frac{d}{dx}\ddot{y} = \dddot{y}/x$, $\frac{d}{dx}x^{-1} = -x^{-2}$, $\frac{d}{dx}x^{-2} = -2x^{-3}$.) Substituting into the Euler equation:

$$(\dddot{y} - 3\ddot{y} + 2\dot{y}) + a_1(\ddot{y} - \dot{y}) + a_2\dot{y} + a_3y = 0,$$

that is,

$$\frac{d^3y}{dz^3} + (a_1 - 3)\frac{d^2y}{dz^2} + (2 - a_1 + a_2)\frac{dy}{dz} + a_3y = 0,$$

a linear equation with constant coefficients. (In general $x^k y^{(k)} = D(D-1)\cdots(D-k+1)y$ with $D = d/dz$, which is why the method works for every order. Note that the auxiliary equation is obtained directly by trying $y = x^r = e^{rz}$.)

On $x < 0$ one uses $x = -e^z$ instead; since each term $x^k y^{(k)}$ is unchanged by $x \to -x$, the same constant-coefficient equation results, and the solutions below hold with $\log x$ replaced by $\log|x|$ (and x^r by $|x|^r$, which for integer r just changes the sign of the constant).

(a) $a_1 = 3$, $a_2 = a_3 = 0$: the transformed equation is $\dddot{y} + 0\cdot\ddot{y} + (2-3)\dot{y} = \dddot{y} - \dot{y} = 0$, with auxiliary equation $r^3 - r = r(r-1)(r+1) = 0$. Hence $y = c_1 + c_2e^z + c_3e^{-z}$, and since $e^z = x$:

$$y = c_1 + c_2x + c_3x^{-1}.$$

(b) $a_1 = 1$, $a_2 = -2$, $a_3 = 2$: the transformed equation is $\dddot{y} - 2\ddot{y} + (2-1-2)\dot{y} + 2y = \dddot{y} - 2\ddot{y} - \dot{y} + 2y = 0$, with

$$r^3 - 2r^2 - r + 2 = r^2(r-2) - (r-2) = (r-2)(r-1)(r+1) = 0.$$

Hence $y = c_1e^z + c_2e^{2z} + c_3e^{-z}$, i.e.

$$y = c_1x + c_2x^2 + c_3x^{-1}.$$

(c) $a_1 = 2$, $a_2 = 1$, $a_3 = -1$: the transformed equation is $\dddot{y} - \ddot{y} + (2-2+1)\dot{y} - y = \dddot{y} - \ddot{y} + \dot{y} - y = 0$, with

$$r^3 - r^2 + r - 1 = r^2(r-1) + (r-1) = (r-1)(r^2+1) = 0,$$

roots $1, \pm i$. Hence $y = c_1e^z + c_2\cos z + c_3\sin z$, and with $z = \log x$:

$$y = c_1x + c_2\cos(\log x) + c_3\sin(\log x).$$

Answer. Shown: $x = e^z$ gives $y_{zzz} + (a_1-3)y_{zz} + (2-a_1+a_2)y_z + a_3y = 0$. (a) $y = c_1 + c_2x + c_3x^{-1}$; (b) $y = c_1x + c_2x^2 + c_3x^{-1}$; (c) $y = c_1x + c_2\cos(\log x) + c_3\sin(\log x)$ $(x > 0$; use $\log|x|$ for $x < 0)$.

Problem 22-23 (Simmons)

Solution. Put $u = y'$. Then $u' = y''$, $u'' = y'''$, $u''' = y^{(4)}$, and the equation becomes

$$x^3u''' + 8x^2u'' + 8xu' - 8u = 0,$$

a third order Euler equation with $a_1 = 8$, $a_2 = 8$, $a_3 = -8$. By Problem 22, $x = e^z$ transforms it into

$$\frac{d^3u}{dz^3} + (8-3)\frac{d^2u}{dz^2} + (2-8+8)\frac{du}{dz} - 8u = \frac{d^3u}{dz^3} + 5\frac{d^2u}{dz^2} + 2\frac{du}{dz} - 8u = 0.$$

The auxiliary polynomial $r^3 + 5r^2 + 2r - 8$ vanishes at $r = 1$ ($1 + 5 + 2 - 8 = 0$), and

$$r^3 + 5r^2 + 2r - 8 = (r-1)(r^2 + 6r + 8) = (r-1)(r+2)(r+4),$$

so $u = Ae^z + Be^{-2z} + Ce^{-4z} = Ax + Bx^{-2} + Cx^{-4}$. Integrating $y' = u$ once:

$$y = \frac{A}{2}x^2 - Bx^{-1} - \frac{C}{3}x^{-3} + c_4.$$

Since A, B, C are arbitrary, so are $c_1 = A/2$, $c_2 = -B$, $c_3 = -C/3$, and $y = c_1x^2 + c_2x^{-1} + c_3x^{-3} + c_4$. It contains four arbitrary constants multiplying the four linearly independent functions $x^2, x^{-1}, x^{-3}, 1$, each of which satisfies the equation, so it is the general solution of this fourth order equation (on any interval not containing 0).

Answer. Shown: with $u = y'$ and $x = e^z$, $u = Ax + Bx^{-2} + Cx^{-4}$, hence $y = c_1x^2 + c_2x^{-1} + c_3x^{-3} + c_4$.

Section 23 Operator Methods for Finding Particular Solutions

Write $D = d/dx$, so that $y'' - 3y' + 2y = f$ becomes $p(D)y = f$ with $p(D) = D^2 - 3D + 2$. Polynomials in D with constant coefficients multiply and factor like ordinary polynomials, and in particular commute with one another. We use the tools of the text:

- formula (9): $\dfrac{1}{D-r}f(x) = e^{rx}\displaystyle\int e^{-rx}f(x)\,dx$ (constants of integration are suppressed, since only particular solutions are wanted), which is the solution of $y' - ry = f$;
- *Method 1* (successive integrations): factor $p(D) = (D-r_1)\cdots(D-r_n)$ and apply $1/(D-r_j)$ one after another, each time by (9);
- *Method 2* (partial fractions): decompose $1/p(D)$ into terms $A/(D-r)$ and apply (9) to each;
- *Method 3* (series): for polynomial f expand $1/p(D)$ in powers of D (high derivatives of a polynomial vanish, so the series terminates when applied to f);
- *Method 4* (exponential shift rule): $p(D)\left[e^{kx}g(x)\right] = e^{kx}p(D+k)g(x)$ (10), and consequently $\dfrac{1}{p(D)}\left[e^{kx}g(x)\right] = e^{kx}\dfrac{1}{p(D+k)}g(x)$ (11).

The rule (10) follows from $(D-r)\left(e^{kx}g\right) = ke^{kx}g + e^{kx}g' - re^{kx}g = e^{kx}(D+k-r)g$ applied to each factor of $p(D)$. A particular solution is only determined up to a solution of the reduced equation, so the answers obtained by different methods may differ from the book's by such terms; in every such case we say so, and the verification script substitutes each answer back into the equation.

Problem 23-1 (Simmons)

Solution. The equation is $(D^2-4)y = e^{2x}$, i.e. $(D-2)(D+2)y = e^{2x}$.

Method 1. Apply $1/(D+2)$ first and then $1/(D-2)$. By (9) with $r=-2$,

$$\frac{1}{D+2}e^{2x} = e^{-2x}\int e^{2x}e^{2x}\,dx = e^{-2x}\cdot\frac{e^{4x}}{4} = \frac{e^{2x}}{4},$$

and then by (9) with $r=2$,

$$y = \frac{1}{D-2}\,\frac{e^{2x}}{4} = e^{2x}\int e^{-2x}\,\frac{e^{2x}}{4}\,dx = e^{2x}\int\frac{1}{4}\,dx = \frac{x}{4}e^{2x}.$$

If instead we apply the factors in the other order,

$$\frac{1}{D-2}e^{2x} = e^{2x}\int e^{-2x}e^{2x}\,dx = xe^{2x}, \qquad y = \frac{1}{D+2}\,xe^{2x} = e^{-2x}\int xe^{4x}\,dx.$$

By parts, $\int xe^{4x}\,dx = \frac{x}{4}e^{4x} - \frac{1}{4}\int e^{4x}\,dx = \left(\frac{x}{4} - \frac{1}{16}\right)e^{4x}$, so

$$y = \left(\frac{x}{4} - \frac{1}{16}\right)e^{2x}.$$

The two results differ by $\frac{1}{16}e^{2x}$, which is a solution of $y'' - 4y = 0$; both are particular solutions.

Method 2. Since $\dfrac{1}{(D-2)(D+2)} = \dfrac{1}{4}\left[\dfrac{1}{D-2} - \dfrac{1}{D+2}\right]$ (check: $\frac{(D+2)-(D-2)}{(D-2)(D+2)} = \frac{4}{(D-2)(D+2)}$),

$$y = \frac{1}{4}\left[\frac{1}{D-2}e^{2x} - \frac{1}{D+2}e^{2x}\right] = \frac{1}{4}\left[xe^{2x} - \frac{e^{2x}}{4}\right] = \left(\frac{x}{4} - \frac{1}{16}\right)e^{2x},$$

using the two values computed above.

Answer. $y_p = \left(\dfrac{x}{4} - \dfrac{1}{16}\right)e^{2x}$ (Method 2, and Method 1 in the order $(D-2)$ then $(D+2)$); Method 1 in the other order gives $y_p = \dfrac{x}{4}e^{2x}$, which differs by the homogeneous solution $\frac{1}{16}e^{2x}$.

Problem 23-2 (Simmons)

Solution. Here $(D-1)(D+1)y = x^2e^{2x}$.

Method 1. First, by (9) with $r = -1$,

$$\frac{1}{D+1}x^2e^{2x} = e^{-x}\int x^2e^{3x}\,dx.$$

Integrating by parts twice, $\int x^2e^{3x}\,dx = e^{3x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right)$ (differentiating the right side returns x^2e^{3x}), so $\frac{1}{D+1}x^2e^{2x} = e^{2x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right)$. Next, with $r = 1$,

$$y = \frac{1}{D-1}e^{2x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right) = e^x\int e^x\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right)dx.$$

Using $\int e^x(ax^2+bx+c)\,dx = e^x\left[ax^2 + (b-2a)x + (c-b+2a)\right]$ with $a = \frac{1}{3}$, $b = -\frac{2}{9}$, $c = \frac{2}{27}$: $b - 2a = -\frac{8}{9}$, $c - b + 2a = \frac{2+6+18}{27} = \frac{26}{27}$. Hence

$$y = e^{2x}\left(\frac{x^2}{3} - \frac{8x}{9} + \frac{26}{27}\right) = \frac{1}{27}\left(9x^2 - 24x + 26\right)e^{2x}.$$

Method 2. $\dfrac{1}{(D-1)(D+1)} = \dfrac{1}{2}\left[\dfrac{1}{D-1} - \dfrac{1}{D+1}\right]$. We have

$$\frac{1}{D-1}x^2e^{2x} = e^x\int x^2e^x\,dx = e^x\cdot e^x(x^2-2x+2) = e^{2x}(x^2-2x+2),$$

and the second term was found above. Therefore

$$y = \frac{1}{2}e^{2x}\left[(x^2-2x+2) - \left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right)\right] = \frac{1}{2}e^{2x}\left[\frac{2}{3}x^2 - \frac{16}{9}x + \frac{52}{27}\right] = e^{2x}\left(\frac{x^2}{3} - \frac{8x}{9} + \frac{26}{27}\right),$$

the same function as before.

Method 4. By (11) with $k = 2$ (the exponential factor is e^{2x}, $g = x^2$):

$$y = \frac{1}{D^2-1}e^{2x}x^2 = e^{2x}\frac{1}{(D+2)^2-1}x^2 = e^{2x}\frac{1}{D^2+4D+3}x^2.$$

Expand in powers of D through D^2 (higher powers annihilate x^2):

$$\frac{1}{3+4D+D^2} = \frac{1}{3}\left[1 + \frac{4D+D^2}{3}\right]^{-1} = \frac{1}{3}\left[1 - \frac{4D+D^2}{3} + \frac{(4D+D^2)^2}{9} - \cdots\right]$$

$$= \frac{1}{3}\left[1 - \frac{4D}{3} - \frac{D^2}{3} + \frac{16D^2}{9} + \cdots\right] = \frac{1}{3} - \frac{4}{9}D + \frac{13}{27}D^2 + \cdots.$$

Applying it to x^2 (with $Dx^2 = 2x$, $D^2x^2 = 2$): $\frac{x^2}{3} - \frac{8x}{9} + \frac{26}{27}$, so again

$$y = e^{2x}\left(\frac{x^2}{3} - \frac{8x}{9} + \frac{26}{27}\right).$$

Answer. $y_p = \dfrac{1}{27}\left(9x^2 - 24x + 26\right)e^{2x}$ by all three methods.

Problem 23-3 (Simmons)

Solution. $(D+2)^2 y = 10x^3 e^{-2x}$. By (9) with $r = -2$, applied twice:

$$\frac{1}{D+2}\,10x^3e^{-2x} = e^{-2x}\int e^{2x}\cdot 10x^3e^{-2x}\,dx = e^{-2x}\cdot\frac{10x^4}{4} = \frac{5}{2}x^4e^{-2x},$$

$$y = \frac{1}{D+2}\,\frac{5}{2}x^4e^{-2x} = e^{-2x}\int \frac{5}{2}x^4\,dx = e^{-2x}\cdot\frac{x^5}{2} = \frac{1}{2}x^5e^{-2x}.$$

Answer. $y_p = \dfrac{1}{2}x^5e^{-2x}$.

Problem 23-4 (Simmons)

Solution. $(D-1)^2 y = e^x$. By (9) with $r = 1$:

$$\frac{1}{D-1}e^x = e^x\int e^{-x}e^x\,dx = xe^x, \qquad y = \frac{1}{D-1}xe^x = e^x\int e^{-x}xe^x\,dx = e^x\int x\,dx = \frac{1}{2}x^2e^x.$$

Answer. $y_p = \dfrac{1}{2}x^2e^x$.

Problem 23-5 (Simmons)

Solution. $(D-1)(D+1)y = e^{-x}$; apply $1/(D+1)$ first, then $1/(D-1)$.

$$\frac{1}{D+1}e^{-x} = e^{-x}\int e^xe^{-x}\,dx = xe^{-x}, \qquad y = \frac{1}{D-1}xe^{-x} = e^x\int e^{-x}xe^{-x}\,dx = e^x\int xe^{-2x}\,dx.$$

By parts, $\int xe^{-2x}\,dx = -\frac{x}{2}e^{-2x} + \frac{1}{2}\int e^{-2x}\,dx = \left(-\frac{x}{2} - \frac{1}{4}\right)e^{-2x}$, so

$$y = \left(-\frac{x}{2} - \frac{1}{4}\right)e^{-x}.$$

Answer. $y_p = \left(-\frac{1}{2}x - \frac{1}{4}\right)e^{-x}$.

Problem 23-6 (Simmons)

Solution. $D^2 - 2D - 3 = (D-3)(D+1)$, so $y = \dfrac{1}{D-3}\dfrac{1}{D+1}6e^{5x}$.

$$\frac{1}{D+1}6e^{5x} = e^{-x}\int e^{x}\cdot 6e^{5x}\,dx = e^{-x}e^{6x} = e^{5x},$$

$$y = \frac{1}{D-3}e^{5x} = e^{3x}\int e^{-3x}e^{5x}\,dx = e^{3x}\cdot\frac{e^{2x}}{2} = \frac{1}{2}e^{5x}.$$

Answer. $y_p = \dfrac{1}{2}e^{5x}$.

Problem 23-7 (Simmons)

Solution. $y = \dfrac{1}{1-D+D^2}(x^3 - 3x^2 + 1)$. Put $u = D - D^2$ and use $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots$, keeping terms through D^3 (since D^4 annihilates a cubic): $u = D - D^2$, $u^2 = D^2 - 2D^3 + \cdots$, $u^3 = D^3 + \cdots$, so

$$\frac{1}{1-D+D^2} = 1 + D + (-1+1)D^2 + (-2+1)D^3 + \cdots = 1 + D - D^3 + \cdots.$$

With $f = x^3 - 3x^2 + 1$, $f' = 3x^2 - 6x$, $f''' = 6$:

$$y = f + f' - f''' = x^3 - 3x^2 + 1 + 3x^2 - 6x - 6 = x^3 - 6x - 5.$$

Answer. $y_p = x^3 - 6x - 5$.

Problem 23-8 (Simmons)

Solution. $y = \dfrac{1}{1-2D+D^3}f$, $f = 2x^3 - 3x^2 + 4x + 5$. With $u = 2D - D^3$, $1 + u + u^2 + u^3 + \cdots$ and $u^2 = 4D^2 + \cdots$ (next term $-4D^4$), $u^3 = 8D^3 + \cdots$:

$$\frac{1}{1-2D+D^3} = 1 + 2D + 4D^2 + (-1+8)D^3 + \cdots = 1 + 2D + 4D^2 + 7D^3 + \cdots.$$

Now $f' = 6x^2 - 6x + 4$, $f'' = 12x - 6$, $f''' = 12$, so

$$\begin{aligned} y &= f + 2f' + 4f'' + 7f''' \\ &= (2x^3 - 3x^2 + 4x + 5) + (12x^2 - 12x + 8) + (48x - 24) + 84 \\ &= 2x^3 + 9x^2 + 40x + 73. \end{aligned}$$

Answer. $y_p = 2x^3 + 9x^2 + 40x + 73.$

Problem 23-9 (Simmons)

Solution. $(1+4D^2)y = x^4$, so $y = \dfrac{1}{1+4D^2}x^4 = (1 - 4D^2 + 16D^4 - \cdots)x^4$, using $\frac{1}{1+r} = 1 - r + r^2 - \cdots$ with $r = 4D^2$ ($D^6x^4 = 0$). Since $D^2x^4 = 12x^2$ and $D^4x^4 = 24$,

$$y = x^4 - 48x^2 + 16 \cdot 24 = x^4 - 48x^2 + 384.$$

Answer. $y_p = x^4 - 48x^2 + 384.$

Problem 23-10 (Simmons)

Solution. $(D^5 - D^3)y = D^3(D^2-1)y = x^2$, so $y = \dfrac{1}{D^3}\dfrac{1}{D^2-1}x^2$. Here $\dfrac{1}{D^2-1} = -\dfrac{1}{1-D^2} = -(1 + D^2 + D^4 + \cdots)$. Since $1/D^3$ means three integrations, we integrate x^2 three times first:

$$\frac{1}{D^3}x^2 = \frac{x^5}{60} \qquad (\tfrac{x^3}{3},\ \tfrac{x^4}{12},\ \tfrac{x^5}{60}).$$

Then apply $-(1 + D^2 + D^4)$ (the term D^6 and beyond kill x^5), using $D^2\frac{x^5}{60} = \frac{x^3}{3}$, $D^4\frac{x^5}{60} = 2x$:

$$y = -\left(\frac{x^5}{60} + \frac{x^3}{3} + 2x\right) = -\frac{x^5}{60} - \frac{x^3}{3} - 2x.$$

Remark. If $-(1+D^2)$ is applied first, giving $-(x^2+2)$, and then three integrations are performed, one gets $-\frac{x^5}{60} - \frac{x^3}{3}$; this differs from the result above by $2x$, a solution of the reduced equation $y^{(5)} - y''' = 0$ (whose general solution contains $c_1 + c_2x + c_3x^2$). The result above is the book's.

Answer. $y_p = -\dfrac{1}{60}x^5 - \dfrac{1}{3}x^3 - 2x.$

Problem 23-11 (Simmons)

Solution. $y = \dfrac{1}{D^6-1}x^{10} = -\dfrac{1}{1-D^6}x^{10} = -(1 + D^6 + D^{12} + \cdots)x^{10}$, and $D^{12}x^{10} = 0$. Since $D^6x^{10} = 10\cdot 9\cdot 8\cdot 7\cdot 6\cdot 5\,x^4 = \dfrac{10!}{4!}x^4 = 151{,}200\,x^4$,

$$y = -x^{10} - 151{,}200\,x^4.$$

Answer. $y_p = -x^{10} - 151{,}200\,x^4.$

Problem 23-12 (Simmons)

Solution. $(D^2+D-1)y = f$, $f = 3x - x^4$, so $y = -\dfrac{1}{1-D-D^2}f$. Expanding by long division (or from $\frac{1}{1-u} = \sum u^n$, $u = D + D^2$):

$$\frac{1}{1-D-D^2} = 1 + D + 2D^2 + 3D^3 + 5D^4 + \cdots$$

(the coefficients c_n satisfy $c_n = c_{n-1} + c_{n-2}$, the Fibonacci numbers: indeed $(1-D-D^2)(1+D+2D^2+3D^3+5D^4) = 1 + O(D^5)$). The derivatives of f are $f' = 3 - 4x^3$, $f'' = -12x^2$, $f''' = -24x$, $f^{(4)} = -24$. Thus

$$\begin{aligned}\frac{1}{1-D-D^2}f &= (3x - x^4) + (3 - 4x^3) + 2(-12x^2) + 3(-24x) + 5(-24)\\ &= -x^4 - 4x^3 - 24x^2 - 69x - 117,\end{aligned}$$

and $y = -(\cdots) = x^4 + 4x^3 + 24x^2 + 69x + 117$.

Answer. $y_p = x^4 + 4x^3 + 24x^2 + 69x + 117$.

Problem 23-13 (Simmons)

Solution. $y = \dfrac{1}{1+D^2}x^4 = (1 - D^2 + D^4 - \cdots)x^4 = x^4 - 12x^2 + 24$, since $D^2x^4 = 12x^2$, $D^4x^4 = 24$.

Answer. $y_p = x^4 - 12x^2 + 24$.

Problem 23-14 (Simmons)

Solution. $D^2(D-1)y = 12x - 2$, so $y = \dfrac{1}{D^2}\dfrac{1}{D-1}(12x-2)$ with $\dfrac{1}{D-1} = -(1 + D + D^2 + \cdots)$. Integrating twice (suppressing constants):

$$\int (12x-2)\,dx = 6x^2 - 2x, \qquad \int (6x^2 - 2x)\,dx = 2x^3 - x^2,$$

so $\dfrac{1}{D^2}(12x-2) = 2x^3 - x^2 =: g$. Then $g' = 6x^2 - 2x$, $g'' = 12x - 2$, $g''' = 12$ and

$$y = -(g + g' + g'' + g''') = -\big(2x^3 - x^2 + 6x^2 - 2x + 12x - 2 + 12\big) = -2x^3 - 5x^2 - 10x - 10.$$

Answer. $y_p = -2x^3 - 5x^2 - 10x - 10$.

Problem 23-15 (Simmons)

Solution. $D^2(D+1)y = 9x^2 - 2x + 1$, so $y = \dfrac{1}{D+1}\dfrac{1}{D^2}(9x^2 - 2x + 1)$ with $\dfrac{1}{1+D} = 1 - D + D^2 - D^3 + D^4 - \cdots$. Integrate twice:

$$\int (9x^2 - 2x + 1)\,dx = 3x^3 - x^2 + x, \qquad g = \int (3x^3 - x^2 + x)\,dx = \frac{3}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2}x^2.$$

Then $g' = 3x^3 - x^2 + x$, $g'' = 9x^2 - 2x + 1$, $g''' = 18x - 2$, $g^{(4)} = 18$, and

$$\begin{aligned} y &= g - g' + g'' - g''' + g^{(4)} \\ &= \tfrac{3}{4}x^4 - \tfrac{1}{3}x^3 + \tfrac{1}{2}x^2 - 3x^3 + x^2 - x + 9x^2 - 2x + 1 - 18x + 2 + 18 \\ &= \tfrac{3}{4}x^4 - \tfrac{10}{3}x^3 + \tfrac{21}{2}x^2 - 21x + 21. \end{aligned}$$

Answer. $y_p = \dfrac{3}{4}x^4 - \dfrac{10}{3}x^3 + \dfrac{21}{2}x^2 - 21x + 21.$

Problem 23-16 (Simmons)

Solution. With $p(D) = D^2 - 4D + 3$ and $k = 2$, $p(D+2) = (D+2)^2 - 4(D+2) + 3 = D^2 + 4D + 4 - 4D - 8 + 3 = D^2 - 1$. By (11),

$$y = e^{2x}\frac{1}{D^2 - 1}x^3 = -e^{2x}(1 + D^2 + D^4 + \cdots)x^3 = -e^{2x}(x^3 + 6x),$$

since $D^2x^3 = 6x$ and $D^4x^3 = 0$.

Answer. $y_p = -e^{2x}(x^3 + 6x)$.

Problem 23-17 (Simmons)

Solution. With $k = 2$: $p(D+2) = (D+2)^2 - 7(D+2) + 12 = D^2 + 4D + 4 - 7D - 14 + 12 = D^2 - 3D + 2$, so

$$y = e^{2x}\frac{1}{2 - 3D + D^2}(x^3 - 5x^2).$$

Write $\dfrac{1}{2 - 3D + D^2} = \dfrac{1}{2}\cdot\dfrac{1}{1-u}$ with $u = \tfrac{3}{2}D - \tfrac{1}{2}D^2$. Then $u^2 = \tfrac{9}{4}D^2 - \tfrac{3}{2}D^3 + \cdots$, $u^3 = \tfrac{27}{8}D^3 + \cdots$, and

$$\frac{1}{1-u} = 1 + \frac{3}{2}D + \left(-\frac{1}{2} + \frac{9}{4}\right)D^2 + \left(-\frac{3}{2} + \frac{27}{8}\right)D^3 + \cdots = 1 + \frac{3}{2}D + \frac{7}{4}D^2 + \frac{15}{8}D^3 + \cdots.$$

For $f = x^3 - 5x^2$: $f' = 3x^2 - 10x$, $f'' = 6x - 10$, $f''' = 6$, so

$$\begin{aligned} \left(1 + \tfrac{3}{2}D + \tfrac{7}{4}D^2 + \tfrac{15}{8}D^3\right) f &= x^3 - 5x^2 + \tfrac{9}{2}x^2 - 15x + \tfrac{21}{2}x - \tfrac{35}{2} + \tfrac{45}{4} \\ &= x^3 - \tfrac{1}{2}x^2 - \tfrac{9}{2}x - \tfrac{25}{4}. \end{aligned}$$

Therefore

$$y = \frac{1}{2}e^{2x}\left(x^3 - \frac{1}{2}x^2 - \frac{9}{2}x - \frac{25}{4}\right) = \frac{1}{8}e^{2x}\left(4x^3 - 2x^2 - 18x - 25\right).$$

Answer. $y_p = \dfrac{1}{8}e^{2x}\left(4x^3 - 2x^2 - 18x - 25\right).$

Problem 23-18 (Simmons)

Solution. Here $p(D) = (D+1)^2$. By linearity treat the two terms separately.
Term $2x^2e^{-2x}$ ($k=-2$): $p(D-2) = (D-1)^2$, so

$$\frac{1}{(D+1)^2}e^{-2x}\,2x^2 = e^{-2x}\frac{1}{(1-D)^2}\,2x^2 = e^{-2x}(1+2D+3D^2+\cdots)2x^2,$$

using $(1-D)^{-2} = \sum_{n\geq 0}(n+1)D^n$. With $D(2x^2) = 4x$, $D^2(2x^2) = 4$: $e^{-2x}(2x^2+8x+12) = 2e^{-2x}(x^2+4x+6)$.

Term $3e^{2x}$ ($k=2$): $p(D+2) = (D+3)^2$, and for the constant $g=3$, $\dfrac{1}{(D+3)^2}\,3 = \dfrac{3}{9} = \dfrac{1}{3}$ (all derivatives of a constant vanish). So this term gives $\frac{1}{3}e^{2x}$.
Adding,

$$y = 2e^{-2x}(x^2+4x+6) + \tfrac{1}{3}e^{2x}.$$

Answer. $y_p = 2e^{-2x}(x^2+4x+6) + \dfrac{1}{3}e^{2x}$.

Problem 23-19 (Simmons)

Solution. $(D^3-8)y = 16x^2$, so $y = -\dfrac{1}{8}\dfrac{1}{1-D^3/8}16x^2 = -\dfrac{1}{8}\left(1+\dfrac{D^3}{8}+\cdots\right)16x^2$. Since $D^3x^2 = 0$, $y = -\dfrac{1}{8}\cdot 16x^2 = -2x^2$. (Direct check: $y''' = 0$, $-8y = 16x^2$.)

Answer. $y_p = -2x^2$.

Problem 23-20 (Simmons)

Solution. $y = \dfrac{1}{D^4-1}(1-x^3) = -(1+D^4+\cdots)(1-x^3) = -(1-x^3) = x^3-1$, because D^4 annihilates $1-x^3$. (Direct check: $y^{(4)} = 0$ and $-y = 1-x^3$.)

Answer. $y_p = x^3-1$.

Problem 23-21 (Simmons)

Solution. $D\left(D^2-\frac{1}{4}\right)y = x$. Use $\dfrac{1}{D^2-\frac{1}{4}} = -4\dfrac{1}{1-4D^2} = -4(1+4D^2+\cdots)$ and apply it to x first: $-4x$ (since $D^2x = 0$). Then integrate once for $1/D$:

$$y = \frac{1}{D}(-4x) = -2x^2.$$

(Check: $y' = -4x$, $y''' = 0$, and $y''' - \frac{1}{4}y' = x$.)

Remark. Applying $1/D$ first would give $x^2/2$ and then $-4\left(\frac{x^2}{2}+4\right) = -2x^2-16$; this differs from the above by the constant -16, a solution of the reduced equation.

Answer. $y_p = -2x^2$.

Problem 23-22 (Simmons)

Solution. $D^4 y = x^{-3}$ means $y = \dfrac{1}{D^4} x^{-3}$: integrate four times (constants suppressed), for $x \neq 0$:

$$\int x^{-3}\,dx = -\frac{1}{2x^2}, \qquad \int \left(-\frac{1}{2x^2}\right) dx = \frac{1}{2x},$$
$$\int \frac{1}{2x}\,dx = \frac{1}{2}\log|x|, \qquad \int \frac{1}{2}\log|x|\,dx = \frac{1}{2}\left(x\log|x| - x\right),$$

the last by parts ($\int \log|x|\,dx = x\log|x| - x$). So $y = \frac{1}{2}x(\log|x| - 1)$, which also holds for $x < 0$ since $\frac{d}{dx}\log|x| = \frac{1}{x}$ there as well.

Answer. $y_p = \dfrac{1}{2}x(\log|x| - 1)$ (i.e. $\frac{1}{2}x(\log x - 1)$ for $x > 0$).

Problem 23-23 (Simmons)

Solution. $D(D^2 - D + 1)y = x + 1$, so $y = \dfrac{1}{1 - D + D^2}\dfrac{1}{D}(x+1)$. First $\dfrac{1}{D}(x+1) = \dfrac{x^2}{2} + x =: g$. Next, with $u = D - D^2$: $\dfrac{1}{1-u} = 1 + u + u^2 + \cdots = 1 + D + (-1+1)D^2 + \cdots = 1 + D + 0 \cdot D^2 + \cdots$ (through D^2, enough since $D^3 g = 0$). Thus

$$y = g + g' = \frac{x^2}{2} + x + (x+1) = \frac{x^2}{2} + 2x + 1.$$

Answer. $y_p = \dfrac{1}{2}x^2 + 2x + 1$.

Problem 23-24 (Simmons)

Solution. $D^2(D+2)y = x$, so $y = \dfrac{1}{D+2}\dfrac{1}{D^2}x$. First $\dfrac{1}{D^2}x = \dfrac{x^3}{6} =: g$. Next

$$\frac{1}{D+2} = \frac{1}{2}\cdot\frac{1}{1 + D/2} = \frac{1}{2}\left(1 - \frac{D}{2} + \frac{D^2}{4} - \frac{D^3}{8} + \cdots\right),$$

and $g' = \frac{x^2}{2}$, $g'' = x$, $g''' = 1$, so

$$y = \frac{1}{2}\left(\frac{x^3}{6} - \frac{x^2}{4} + \frac{x}{4} - \frac{1}{8}\right) = \frac{x^3}{12} - \frac{x^2}{8} + \frac{x}{8} - \frac{1}{16} = \frac{1}{48}\left(4x^3 - 6x^2 + 6x - 3\right).$$

Answer. $y_p = \dfrac{1}{48}\left(4x^3 - 6x^2 + 6x - 3\right)$.

Problem 23-25 (Simmons)

Solution. Rule (10) with $p(D) = (D-r)^m$ and $k = r$ gives $(D-r)^m\left[e^{rx}u\right] = e^{rx}D^m u$. So substituting $y = e^{rx}u(x)$ (equivalently multiplying the equation by e^{-rx}) reduces each equation to $D^m u =$ a given function, to be integrated m times with m arbitrary constants; the result is the general solution because it has m constants for an order m equation and every solution has this form (any solution y gives $u = e^{-rx}y$).

(a) $r = 2$, $m = 3$. $y = e^{2x}u$: $e^{2x}D^3u = e^{2x}$, i.e. $u''' = 1$, so $u = \frac{x^3}{6} + c_1x^2 + c_2x + c_3$, and

$$y = \left(\frac{x^3}{6} + c_1x^2 + c_2x + c_3\right)e^{2x}.$$

(b) $r = -1$, $m = 3$. $y = e^{-x}u$: $e^{-x}u''' = 12e^{-x}$, so $u''' = 12$ and $u = 2x^3 + c_1x^2 + c_2x + c_3$:

$$y = \left(2x^3 + c_1x^2 + c_2x + c_3\right)e^{-x}.$$

(c) $r = 2$, $m = 2$. $y = e^{2x}u$: $e^{2x}u'' = e^{2x}\sin x$, so $u'' = \sin x$, $u' = -\cos x + c_1$, $u = -\sin x + c_1x + c_2$:

$$y = \left(-\sin x + c_1x + c_2\right)e^{2x}.$$

Answer. (a) $y = \left(\frac{1}{6}x^3 + c_1x^2 + c_2x + c_3\right)e^{2x}$; (b) $y = \left(2x^3 + c_1x^2 + c_2x + c_3\right)e^{-x}$; (c) $y = \left(-\sin x + c_1x + c_2\right)e^{2x}$.

Problem 23-26 (Simmons)

Solution.

(a) Since $q(r)$ divides $p(r)$, $p(r) = s(r)q(r)$ for some polynomial $s(r)$, and therefore $p(D) = s(D)q(D)$ as operators (operator polynomials with constant coefficients multiply exactly as ordinary polynomials do). If $q(D)y = 0$, then $p(D)y = s(D)\left[q(D)y\right] = s(D)0 = 0$, because a linear differential operator with constant coefficients maps the zero function to zero.

(b) If r_1 is a root of multiplicity k, then $p(r) = (r-r_1)^k s(r)$ for a polynomial s, so $(r-r_1)^k$ is a factor of $p(r)$. Part (a) with $q(r) = (r-r_1)^k$ shows that every solution of $(D-r_1)^k y = 0$ solves $p(D)y = 0$.

(c) Let y be any function with k derivatives and set $u = e^{-r_1x}y$, so that $y = e^{r_1x}u$. The exponential shift rule (10) with $p(D)$ replaced by $(D-r_1)^k$ and k replaced by r_1 reads

$$(D-r_1)^k\left[e^{r_1x}u\right] = e^{r_1x}\,(D+r_1-r_1)^k u = e^{r_1x}D^k u.$$

Hence $(D-r_1)^k y = 0$ is equivalent to $e^{r_1x}D^k u = 0$, that is (as $e^{r_1x} \neq 0$) to $D^k u = \dfrac{d^k u}{dx^k} = 0$. Integrating k times, each integration adding one arbitrary constant, shows that the solutions of $u^{(k)} = 0$ are exactly the polynomials $u = c_1 + c_2x + \cdots + c_kx^{k-1}$ (the x^{k-1} coefficient is $c_k = u^{(k-1)}/(k-1)!$, etc.), and so

$$y = e^{r_1x}u = (c_1 + c_2x + \cdots + c_kx^{k-1})e^{r_1x},$$

which has k arbitrary constants and contains every solution, i.e. is the general solution of the kth order equation $(D-r_1)^k y = 0$.

Answer. Shown: (a) $p(D) = s(D)q(D)$, so $q(D)y = 0 \Rightarrow p(D)y = 0$; (b) follows from (a) with $q = (r - r_1)^k$; (c) $(D - r_1)^k(e^{r_1 x}u) = e^{r_1 x}D^k u$, so $u^{(k)} = 0$ and $y = (c_1 + c_2 x + \cdots + c_k x^{k-1})e^{r_1 x}$.

Chapter 4

Qualitative Properties of Solutions

Section 24 Oscillations and the Sturm Separation Theorem

Throughout, $s(x)$ and $c(x)$ denote the solutions of $y''+y=0$ (2) determined by $s(0)=0,\ s'(0)=1$ and $c(0)=1,\ c'(0)=0$, exactly as in the text, and we may use the facts proved there:

$$s'=c, \qquad c'=-s, \qquad s^2+c^2=1, \qquad W[s,c]=-1,$$

together with the uniqueness theorem (Theorem 14-A): a solution of a second order linear homogeneous equation is determined by its value and its derivative at one point. π denotes the first positive zero of s, as in the text.

Problem 24-1 (Simmons)

Solution. The tool is Theorem 14-A: two solutions of $y''+y=0$ that have the same value and the same derivative at one point are identical. Note also that the equation $y''+y=0$ has constant coefficients, so if $y(x)$ is a solution then so is $y(x+a)$ for every constant a (the chain rule gives $\frac{d^2}{dx^2}y(x+a)=y''(x+a)=-y(x+a)$), and so is every linear combination $\alpha y_1(x)+\beta y_2(x)$ of solutions.

(5) Fix a and put $f(x)=s(x+a)$ and $g(x)=s(x)c(a)+c(x)s(a)$. Both are solutions of $y''+y=0$ (translate of a solution; linear combination of the solutions s,c with the constant coefficients $c(a),s(a)$). At $x=0$:

$$\begin{aligned} f(0)&=s(a), & f'(0)&=s'(a)=c(a),\\ g(0)&=s(0)c(a)+c(0)s(a)=s(a), & g'(0)&=s'(0)c(a)+c'(0)s(a)=c(a). \end{aligned}$$

The initial data coincide, so $f\equiv g$, which is (5).

(6) Put $f(x)=c(x+a)$ and $g(x)=c(x)c(a)-s(x)s(a)$; again both solve $y''+y=0$. Then

$$\begin{aligned} f(0)&=c(a), & f'(0)&=c'(a)=-s(a),\\ g(0)&=c(0)c(a)-s(0)s(a)=c(a), & g'(0)&=c'(0)c(a)-s'(0)s(a)=-s(a), \end{aligned}$$

so $f\equiv g$, which is (6).

(7, 8) Put $a=x$ in (5) and (6) (they hold for every a and every x):

$$s(2x)=s(x)c(x)+c(x)s(x)=2s(x)c(x), \qquad c(2x)=c(x)c(x)-s(x)s(x)=c(x)^2-s(x)^2.$$

(9, 10) First we need the value of $c(\pi)$. Since $s(\pi)=0$, formula (4) gives $c(\pi)^2=1-0=1$, so $c(\pi)=\pm1$. By definition π is the first positive zero of s and $s(0)=0,\ s'(0)=1>0$, so $s>0$ on $(0,\pi)$. Hence for $x<\pi$,

$$\frac{s(x)-s(\pi)}{x-\pi}=\frac{s(x)}{x-\pi}<0,$$

and letting $x\to\pi^-$ we get $s'(\pi)\le 0$. But $s'(\pi)=c(\pi)=\pm1$, so $c(\pi)=-1$.

Now (5) and (6) with $a=\pi$ give

$$s(x+\pi)=s(x)c(\pi)+c(x)s(\pi)=-s(x), \qquad c(x+\pi)=c(x)c(\pi)-s(x)s(\pi)=-c(x).$$

Applying this twice,

$$s(x+2\pi)=-s(x+\pi)=s(x), \qquad c(x+2\pi)=-c(x+\pi)=c(x),$$

which are (9) and (10).

Remark. The existence of the zero π, which the text obtains from the convexity picture, can also be made rigorous at once. Suppose $s > 0$ for all $x > 0$. Then $s'' = -s < 0$ on $(0,\infty)$, so s' is decreasing. If $s'(x_1) < 0$ at some x_1, then $s'(x) \le s'(x_1) < 0$ for $x \ge x_1$ and s would become negative; so $s' \ge 0$ and s is nondecreasing. Then $s(x) \ge s(1) > 0$ for $x \ge 1$, so $s'' = -s \le -s(1)$ there and $s'(x) \le s'(1) - s(1)(x-1) \to -\infty$, contradicting $s' \ge 0$. Hence s has a positive zero, and the first one exists because zeros of a nontrivial solution are isolated.

Answer. Shown: (5) and (6) by uniqueness (same value and slope at $x = 0$); (7), (8) by putting $a = x$; $c(\pi) = -1$ gives $s(x+\pi) = -s(x)$, $c(x+\pi) = -c(x)$, hence (9) and (10).

Problem 24-2 (Simmons)

Solution. Let $y_1 = a\sin x + b\cos x$ and $y_2 = c\sin x + d\cos x$. Both are solutions of $y'' + y = 0$ (linear combinations of the solutions $\sin x$, $\cos x$). Their Wronskian is

$$\begin{aligned} W(y_1,y_2) &= y_1y_2' - y_2y_1' \\ &= (a\sin x + b\cos x)(c\cos x - d\sin x) - (c\sin x + d\cos x)(a\cos x - b\sin x) \\ &= \big[ac\sin x\cos x - ad\sin^2 x + bc\cos^2 x - bd\sin x\cos x\big] \\ &\quad - \big[ac\sin x\cos x - bc\sin^2 x + ad\cos^2 x - bd\sin x\cos x\big] \\ &= -ad\sin^2 x + bc\cos^2 x + bc\sin^2 x - ad\cos^2 x \\ &= -(ad-bc)(\sin^2 x + \cos^2 x) = -(ad-bc) \neq 0. \end{aligned}$$

A nonvanishing Wronskian means y_1, y_2 are linearly independent (Section 15); in particular neither is the trivial solution (if, say, $a = b = 0$ then $ad - bc = 0$). So y_1, y_2 are two linearly independent solutions of $y'' + P(x)y' + Q(x)y = 0$ with $P = 0$, $Q = 1$, and the Sturm separation theorem (Theorem 24-A) applies: their zeros are distinct (no common zero, since $W \neq 0$) and y_1 vanishes exactly once between any two successive zeros of y_2, and conversely. That is, the zeros occur alternately.

Answer. Shown: $W(y_1,y_2) = bc - ad \neq 0$, so the functions are independent solutions of $y'' + y = 0$ and the Sturm separation theorem gives distinct, alternating zeros.

Problem 24-3 (Simmons)

Solution. **Normal form.** For $x > 0$ divide by x^2:

$$y'' + \frac{1}{x}y' + \Big(1 - \frac{p^2}{x^2}\Big)y = 0, \qquad P(x) = \frac{1}{x}, \quad Q(x) = 1 - \frac{p^2}{x^2}.$$

With $y = uv$ the first derivative term disappears for $v = e^{-\frac{1}{2}\int P\,dx} = e^{-\frac{1}{2}\log x} = x^{-1/2}$ (equation (14)), and the new coefficient is given by (15):

$$\begin{aligned} q(x) &= Q - \tfrac{1}{4}P^2 - \tfrac{1}{2}P' = 1 - \frac{p^2}{x^2} - \frac{1}{4x^2} + \frac{1}{2x^2} \\ &= 1 + \frac{1-4p^2}{4x^2}. \end{aligned}$$

So with $y = x^{-1/2}u$, i.e. $u = \sqrt{x}\,y$, Bessel's equation becomes

$$u'' + \Big(1 + \frac{1-4p^2}{4x^2}\Big)u = 0.$$

(Direct check: $y = x^{-1/2}u$ gives $y' = x^{-1/2}u' - \frac{1}{2}x^{-3/2}u$ and $y'' = x^{-1/2}u'' - x^{-3/2}u' + \frac{3}{4}x^{-5/2}u$; substituting, $x^2y'' + xy' + (x^2 - p^2)y = x^{3/2}\big[u'' + (1 + \frac{1-4p^2}{4x^2})u\big]$.)

Zeros. Since $x^{-1/2}$ never vanishes on $x > 0$, y and u have exactly the same zeros there. Let u be a nontrivial solution. Theorem 24-C cannot be quoted verbatim because for $p > \frac{1}{2}$ the coefficient q is negative for small x; but $q(x) = \dfrac{4x^2 + 1 - 4p^2}{4x^2} > 0$ for all $x \geq p$, $x > 0$. Put $X = p + 1$ and $w(t) = u(t + X)$ for $t > 0$. Then $w'' + \tilde{q}(t)w = 0$ with $\tilde{q}(t) = q(t + X) > 0$ for all $t > 0$, w is nontrivial, and

$$\int_1^T \tilde{q}(t)\,\mathrm{d}t = \int_{1+X}^{T+X} \Big(1 + \frac{1-4p^2}{4\xi^2}\Big)\,\mathrm{d}\xi = (T-1) + \frac{1-4p^2}{4}\Big(\frac{1}{1+X} - \frac{1}{T+X}\Big) \xrightarrow[T\to\infty]{} \infty.$$

Hence $\int_1^\infty \tilde{q} = \infty$, and by Theorem 24-C w has infinitely many positive zeros, so u has infinitely many zeros in (X, ∞), and so does $y = x^{-1/2}u$. Thus every nontrivial solution of Bessel's equation has infinitely many positive zeros. (For $0 \leq p \leq \frac{1}{2}$ one can apply Theorem 24-C to u directly, since then $q > 0$ for all $x > 0$.)

Answer. Normal form: $u'' + \Big(1 + \dfrac{1-4p^2}{4x^2}\Big)u = 0$ with $u = \sqrt{x}\,y$; since $q \to 1$ with $\int^\infty q = \infty$, Theorem 24-C gives infinitely many positive zeros.

Problem 24-4 (Simmons)

Solution. First, the hypothesis of Theorem C fails: $\int_1^\infty \dfrac{k}{x^2}\,\mathrm{d}x = k < \infty$.

For $x > 0$ the Euler equation $x^2y'' + ky = 0$ has solutions x^m with $m(m-1) + k = 0$, i.e. $m^2 - m + k = 0$, $m = \dfrac{1 \pm \sqrt{1-4k}}{2}$. Three cases.

($k > 1/4$) $m = \frac{1}{2} \pm i\beta$ with $\beta = \frac{1}{2}\sqrt{4k-1} > 0$, and $x^{1/2 \pm i\beta} = \sqrt{x}\,e^{\pm i\beta \log x}$, so the real general solution is

$$y = \sqrt{x}\,\big[A\cos(\beta\log x) + B\sin(\beta\log x)\big] = R\sqrt{x}\,\cos(\beta\log x - \varphi),$$

where $A = R\cos\varphi$, $B = R\sin\varphi$ and $R = \sqrt{A^2 + B^2} > 0$ for a nontrivial solution. Since $\sqrt{x} > 0$, $y(x) = 0$ iff $\beta\log x - \varphi = \frac{\pi}{2} + n\pi$, i.e.

$$x_n = \exp\Big(\frac{\varphi + \pi/2 + n\pi}{\beta}\Big), \qquad n = 0, \pm1, \pm2, \ldots$$

These are positive and pairwise distinct (the exponent is strictly increasing in n), and $x_n \to \infty$ as $n \to \infty$. So there are infinitely many positive zeros.

($k = 1/4$) $m = \frac{1}{2}$ is a double root, so $y = \sqrt{x}\,(A + B\log x)$. If $B \neq 0$ the factor $A + B\log x$ is strictly monotonic and vanishes only at $x = e^{-A/B}$; if $B = 0$ then $A \neq 0$ and there is no zero. So there is at most one positive zero.

($0 < k < 1/4$) Now $m_1 = \frac{1+\sqrt{1-4k}}{2} \neq m_2 = \frac{1-\sqrt{1-4k}}{2}$ are real, and $y = Ax^{m_1} + Bx^{m_2}$. (Both are positive since $m_1 + m_2 = 1$, $m_1m_2 = k > 0$.) If $A = 0$ or $B = 0$ there are no zeros. Otherwise $y(x) = 0 \iff x^{m_1 - m_2} = -B/A$, and $x \mapsto x^{m_1-m_2}$ with $m_1 - m_2 = \sqrt{1-4k} > 0$ is strictly increasing on $(0, \infty)$, so there is at most one solution. Again at most one positive zero.

In all cases with $k \leq 1/4$ every nontrivial solution has at most one (hence finitely many) positive zeros, while for $k > 1/4$ it has infinitely many.

Answer. Shown: for $k > \frac{1}{4}$, $y = R\sqrt{x}\cos(\beta \log x - \varphi)$, $\beta = \frac{1}{2}\sqrt{4k-1}$, has the zeros $x_n = e^{(\varphi+\pi/2+n\pi)/\beta}$ (infinitely many); for $k \le \frac{1}{4}$ every nontrivial solution has at most one positive zero.

Section 25 The Sturm Comparison Theorem

We use the following observation about the proof of Theorem 25-B (Sturm comparison theorem). In that proof one only works on the interval $[x_1, x_2]$ between two successive zeros of z, and the only facts used are $z'' = -rz$, $y'' = -qy$ and $q > r$ on (x_1, x_2) (the positivity of q and r is never used). Hence:

(B′) *If $z'' + rz = 0$ has successive zeros $x_1 < x_2$, $y'' + qy = 0$ with y nontrivial, and $q(x) > r(x)$ for $x_1 < x < x_2$ (with q, r continuous on $[x_1, x_2]$), then y vanishes at least once in the open interval (x_1, x_2).*

(Indeed, if $y, z > 0$ on (x_1, x_2) then $W = yz' - zy'$ satisfies $W' = (q - r)yz > 0$, so $W(x_2) > W(x_1)$, while $W(x_1) = y(x_1)z'(x_1) \geq 0$ and $W(x_2) = y(x_2)z'(x_2) \leq 0$; a contradiction. The cases of other signs reduce to this by replacing y or z by its negative.) In each problem below, a "comparison with $v'' + \lambda v = 0$" means this statement with $v = \sin\sqrt{\lambda}\,(x - x_0)$, whose successive zeros are $x_0 + n\pi/\sqrt{\lambda}$.

Problem 25-1 (Simmons)

Solution. By Problem 24-3, $u = \sqrt{x}\,y_p$ solves

$$u'' + q(x)u = 0, \qquad q(x) = 1 + \frac{1 - 4p^2}{4x^2},$$

and u has the same positive zeros as y_p. So $x_1 < x_2$ are successive zeros of u (successive zeros exist, as zeros of a nontrivial solution are isolated on any closed interval, by the proof of Theorem 25-A, which uses only Theorem 14-A, and there are infinitely many by Problem 24-3).

(a) Let $0 \leq p < \frac{1}{2}$. Then $1 - 4p^2 > 0$, so $q(x) > 1$ for all $x > 0$.

$x_2 - x_1 < \pi$. Compare u (coefficient q) with $v'' + v = 0$, $v = \sin(x - x_1)$, whose successive zeros are x_1 and $x_1 + \pi$. Since $q > 1$, Theorem 25-B (applied with v in the role of "z" and u in the role of "y") says that u vanishes in the open interval $(x_1, x_1 + \pi)$. As x_2 is the first zero of u after x_1, $x_2 < x_1 + \pi$, i.e. $x_2 - x_1 < \pi$.

Limit. Let $0 < \delta < 1$. Because $0 < \dfrac{1 - 4p^2}{4x^2} \leq \dfrac{1}{4x^2} < \delta$ for $x > \dfrac{1}{2\sqrt{\delta}} =: X_\delta$, we have $q(x) < 1 + \delta$ for $x > X_\delta$. Take $x_1 > X_\delta$ and let $w = \sin\big(\sqrt{1 + \delta}\,(x - x_1)\big)$, a solution of $w'' + (1 + \delta)w = 0$. Now u has the successive zeros x_1, x_2 and on (x_1, x_2) the coefficient of w is larger than q. By (B′) (with u in the role of "z", coefficient $r = q$, and w in the role of "y", coefficient $1 + \delta$), w vanishes in (x_1, x_2). The first zero of w after x_1 is $x_1 + \pi/\sqrt{1 + \delta}$, hence $x_1 + \pi/\sqrt{1 + \delta} < x_2$. Combining,

$$\frac{\pi}{\sqrt{1 + \delta}} < x_2 - x_1 < \pi \qquad (x_1 > X_\delta).$$

Given $\varepsilon > 0$ choose δ so small that $\pi/\sqrt{1 + \delta} > \pi - \varepsilon$; then $\pi - \varepsilon < x_2 - x_1 < \pi$ whenever $x_1 > X_\delta$. Hence $x_2 - x_1 \to \pi$ as $x_1 \to \infty$.

(b) Let $p > \frac{1}{2}$. Then $q(x) = 1 - \dfrac{4p^2 - 1}{4x^2} < 1$ for all $x > 0$.

$x_2 - x_1 > \pi$. Take $z = u$ (coefficient q), with successive zeros x_1, x_2, and $y = \sin(x - x_1)$, which solves $y'' + y = 0$ with $1 > q$ on (x_1, x_2). By (B′), y vanishes in (x_1, x_2); its zeros are $x_1 + n\pi$, so $x_1 + \pi < x_2$, i.e. $x_2 - x_1 > \pi$.

Limit. Let $0 < \delta < 1$. For $x > X_\delta := \dfrac{\sqrt{4p^2 - 1}}{2\sqrt{\delta}}$ we have $\dfrac{4p^2 - 1}{4x^2} < \delta$, so $q(x) > 1 - \delta > 0$. Take $x_1 > X_\delta$ and $v = \sin\big(\sqrt{1 - \delta}\,(x - x_1)\big)$, a solution of $v'' + (1 - \delta)v = 0$ with successive zeros x_1

and $x_1 + \pi/\sqrt{1-\delta}$. Since $q > 1-\delta$ on this interval (all its points exceed $x_1 > X_\delta$), Theorem 25-B shows that u vanishes in $\left(x_1,\, x_1 + \pi/\sqrt{1-\delta}\right)$, so $x_2 < x_1 + \pi/\sqrt{1-\delta}$. Thus

$$\pi < x_2 - x_1 < \frac{\pi}{\sqrt{1-\delta}} \qquad (x_1 > X_\delta),$$

and since $\pi/\sqrt{1-\delta} \to \pi$ as $\delta \to 0$, we get $x_2 - x_1 \to \pi$ as $x_1 \to \infty$.

Remark. For $p = \frac{1}{2}$, $q \equiv 1$, $u = A\sin x + B\cos x$ and $y_{1/2} = x^{-1/2}u$, so the spacing is exactly π (Theorem 25-C).

Answer. (a) $\dfrac{\pi}{\sqrt{1+\delta}} < x_2 - x_1 < \pi$ for $x_1 > \dfrac{1}{2\sqrt{\delta}}$; (b) $\pi < x_2 - x_1 < \dfrac{\pi}{\sqrt{1-\delta}}$ for large x_1; in both cases $x_2 - x_1 \to \pi$ as $x_1 \to \infty$.

Problem 25-2 (Simmons)

Solution. Here q is continuous on $x > 0$, and the inequalities are understood for all $x > 0$.

($q > k/x^2$, $k > 1/4$) Let $z'' + \dfrac{k}{x^2}z = 0$. By Problem 24-4 every nontrivial solution z has the infinitely many positive zeros $x_n = e^{(\varphi+\pi/2+n\pi)/\beta}$ $(n \in \mathbb{Z})$, with $x_n < x_{n+1}$ and $x_n \to \infty$, and x_n, x_{n+1} are successive zeros. On each (x_n, x_{n+1}) we have $q(x) > k/x^2 > 0$, so by Theorem 25-B y has at least one zero in (x_n, x_{n+1}). These intervals are pairwise disjoint, so y has infinitely many positive zeros.

($q < 1/(4x^2)$) Let $z(x) = \sqrt{x}$. Then $z'' = -\frac{1}{4}x^{-3/2}$, so $z'' + \dfrac{1}{4x^2}z = -\frac{1}{4}x^{-3/2} + \frac{1}{4}x^{-3/2} = 0$, and $z > 0$ has no zero on $(0,\infty)$.

Suppose y had two positive zeros $a < b$. By Theorem 25-A (valid for any continuous q; its proof only uses Theorem 14-A) y has only finitely many zeros in $[a,b]$, so there are successive zeros $x_1 < x_2$ of y in $[a,b]$. On (x_1,x_2) the coefficient of z is $\dfrac{1}{4x^2} > q(x)$. Applying (B′) (with y in the role of "z", coefficient q, and $\sqrt{x}$ in the role of "y", coefficient $1/4x^2$), $\sqrt{x}$ must vanish somewhere in $(x_1,x_2) \subset (0,\infty)$, which is false. Hence y has at most one positive zero, in particular only finitely many.

Answer. Shown: comparison with $z'' + (k/x^2)z = 0$ ($k > \frac{1}{4}$, infinitely many zeros, Problem 24-4) gives infinitely many zeros of y; comparison with the zero-free solution $\sqrt{x}$ of $z'' + z/(4x^2) = 0$ shows y has at most one positive zero when $q < 1/4x^2$.

Problem 25-3 (Simmons)

Solution. **Theorem.** Let q be continuous on $[a,\infty)$ and suppose $q(x) \geq k$ for all $x \geq a$, where $k > 0$ is a constant. Then every nontrivial solution of $y'' + q(x)y = 0$ has infinitely many zeros in $[a,\infty)$. More precisely, every interval $[x_0,\, x_0 + \pi/\sqrt{k'})$ with $x_0 \geq a$ and $0 < k' < k$ contains a zero, and if $x_1 < x_2$ are successive zeros with $x_1 \geq a$ then $x_2 - x_1 \leq \pi/\sqrt{k}$.

Proof. Let $0 < k' < k$ and $L = \pi/\sqrt{k'}$. Let $x_0 \geq a$, and $v(x) = \sin\sqrt{k'}\,(x - x_0)$, which solves $v'' + k'v = 0$ and has the successive zeros x_0 and $x_0 + L$. On $(x_0, x_0 + L)$ we have $q(x) \geq k > k'$. By Theorem 25-B (or (B′)), y has a zero in $(x_0, x_0 + L)$.

Infinitely many zeros. Apply this with $x_0 = a + nL$, $n = 0, 1, 2, \ldots$: each of the pairwise disjoint intervals $(a + nL,\ a + (n+1)L)$ contains a zero of y, so there are infinitely many zeros, and they are unbounded.

Spacing. Let $x_1 < x_2$ be successive zeros of y with $x_1 \geq a$. Taking $x_0 = x_1$ above, y has a zero in $(x_1, x_1 + L)$ (this is a zero different from x_1), so $x_2 < x_1 + L$, i.e. $x_2 - x_1 < \pi/\sqrt{k'}$ for every $k' < k$. Letting $k' \to k$ gives $x_2 - x_1 \leq \pi/\sqrt{k}$. ■

Special cases.

- $q(x) = \sin^2 x + 1 \geq 1$ for all x, so the Theorem applies with $a = 0$, $k = 1$: every nontrivial solution of $y'' + (\sin^2 x + 1)y = 0$ has infinitely many positive zeros, with successive zeros at most π apart.
- $q(x) = f(x) + k$ with $f \geq 0$ continuous and $k > 0$ (the answer in the book): then $q \geq k$, so every solution of $y'' + [f(x) + k]y = 0$ has infinitely many positive zeros. (The sine example is $f = \sin^2 x$, $k = 1$.)

Answer. Theorem: if q is continuous with $q(x) \geq k > 0$ on $[a, \infty)$, every nontrivial solution of $y'' + q(x)y = 0$ has infinitely many zeros, successive ones at most $\pi/\sqrt{k}$ apart. In particular this holds for $y'' + (\sin^2 x + 1)y = 0$ and, as in the book, for $y'' + [f(x) + k]y = 0$ with $f \geq 0$, $k > 0$.

Chapter 5

Power Series Solutions and Special Functions

Section 26 Introduction. A Review of Power Series

Throughout, the *ratio test* is used in the form of the text: if $u_n \neq 0$ and $\lim_{n\to\infty} |u_{n+1}/u_n| = L$, then $\sum u_n$ converges when $L < 1$ and diverges when $L > 1$. For a power series $\sum a_n x^n$ with all $a_n \neq 0$ this gives the radius of convergence $R = \lim |a_n/a_{n+1}|$ whenever the limit exists (with $R = \infty$ if $|a_n/a_{n+1}| \to \infty$).

Problem 26-1 (Simmons)

Solution.

((5)) Here $a_n = n! \neq 0$. Fix $x \neq 0$. Then

$$\left|\frac{a_{n+1}x^{n+1}}{a_n x^n}\right| = \frac{(n+1)!}{n!}\,|x| = (n+1)|x| \longrightarrow \infty,$$

so $L = \infty > 1$ and the series diverges for every $x \neq 0$. Equivalently $|a_n/a_{n+1}| = 1/(n+1) \to 0$. Hence $R = 0$ (the series converges only at $x = 0$).

((6)) Here $a_n = 1/n! \neq 0$, and for any fixed x

$$\left|\frac{a_{n+1}x^{n+1}}{a_n x^n}\right| = \frac{n!}{(n+1)!}\,|x| = \frac{|x|}{n+1} \longrightarrow 0 < 1,$$

so the series converges for every x. Equivalently $|a_n/a_{n+1}| = n+1 \to \infty$. Hence $R = \infty$.

((7)) Here $a_n = 1$, so $|a_n/a_{n+1}| = 1 \to 1$; directly, $\left|\dfrac{x^{n+1}}{x^n}\right| = |x|$, so $L = |x|$: convergence for $|x| < 1$ and divergence for $|x| > 1$. Hence $R = 1$.

Answer. (5) $R = 0$; (6) $R = \infty$; (7) $R = 1$.

Problem 26-2 (Simmons)

Solution. Let $a_n = \dfrac{p(p-1)\cdots(p-n+1)}{n!}$. Because p is not 0 or a positive integer, none of the factors $p - k$ ($k = 0, 1, 2, \dots$) vanishes, so every $a_n \neq 0$ and the ratio test applies. We have

$$\frac{a_{n+1}}{a_n} = \frac{p(p-1)\cdots(p-n+1)(p-n)}{(n+1)!} \cdot \frac{n!}{p(p-1)\cdots(p-n+1)} = \frac{p-n}{n+1}.$$

Therefore

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{|p-n|}{n+1} = \frac{|\,n-p\,|}{n+1} = \left|\frac{1-p/n}{1+1/n}\right| \longrightarrow 1 \qquad (n \to \infty).$$

For fixed $x \neq 0$, the ratio of successive terms satisfies $\left|\dfrac{a_{n+1}x^{n+1}}{a_n x^n}\right| \to |x| = L$. By the ratio test the series converges if $|x| < 1$ and diverges if $|x| > 1$. (Equivalently, $R = \lim |a_n/a_{n+1}| = 1$.) At $x = 0$ the series trivially converges.

Answer. Shown: the ratio of consecutive terms tends to $|x|$, so $R = 1$.

Problem 26-3 (Simmons)

Solution. Regarded as power series in x, each of these has every second coefficient equal to 0, so the quotient a_n/a_{n+1} is not defined; we therefore apply the ratio test directly to the nonzero terms u_n, for a fixed $x \neq 0$.

((13)) $u_n = (-1)^n \dfrac{x^{2n+1}}{(2n+1)!}$, so

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{|x|^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{|x|^{2n+1}} = \frac{x^2}{(2n+2)(2n+3)} \longrightarrow 0 < 1.$$

((14)) $u_n = (-1)^n \dfrac{x^{2n}}{(2n)!}$, so

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{x^{2n+2}}{(2n+2)!} \cdot \frac{(2n)!}{x^{2n}} = \frac{x^2}{(2n+1)(2n+2)} \longrightarrow 0 < 1.$$

In both cases $L = 0 < 1$ for every real x, so both series converge for all x: $R = \infty$.

Answer. Shown: $|u_{n+1}/u_n| \to 0$ for every x, hence $R = \infty$ for both series.

Problem 26-4 (Simmons)

Solution.

(Preliminary: $a^n/n! \to 0$) For any constant a the series $\sum_{n\geq 0} a^n/n!$ converges (ratio test: $|a|/(n+1) \to 0 < 1$; see Problem 1(6)). The terms of a convergent series tend to 0, so $a^n/n! \to 0$.

(General setup) By Taylor's formula, for a function f with derivatives of all orders,

$$f(x) = \sum_{k=0}^{N} \frac{f^{(k)}(0)}{k!} x^k + R_N(x), \qquad R_N(x) = \frac{f^{(N+1)}(\bar{x})}{(N+1)!}\, x^{N+1},$$

with $\bar{x}$ between 0 and x. The expansion holds at x exactly when $R_N(x) \to 0$ as $N \to \infty$. Fix an arbitrary x.

((12) $f = e^x$) Every derivative is $f^{(k)} = e^x$, so $f^{(k)}(0) = 1$, and since $\bar{x}$ lies between 0 and x, $e^{\bar{x}} \leq e^{|x|}$. Hence

$$|R_N(x)| = \frac{e^{\bar{x}}}{(N+1)!}|x|^{N+1} \leq e^{|x|}\,\frac{|x|^{N+1}}{(N+1)!} \longrightarrow 0$$

by the preliminary remark with $a = |x|$ (the factor $e^{|x|}$ is a constant). So $e^x = \sum_{k\geq 0} x^k/k!$ for all x.

((13) $f = \sin x$) The derivatives cycle: $\sin x,\ \cos x,\ -\sin x,\ -\cos x,\ \sin x, \ldots$, so $f^{(k)}(0) = 0, 1, 0, -1, 0, 1, \ldots$ for $k = 0, 1, 2, 3, 4, 5, \ldots$. The Taylor polynomial therefore contains only the odd powers, x^{2n+1} with coefficient $(-1)^n/(2n+1)!$. Every derivative is $\pm\sin\bar{x}$ or $\pm\cos\bar{x}$, hence of absolute value ≤ 1:

$$|R_N(x)| \leq \frac{|x|^{N+1}}{(N+1)!} \longrightarrow 0.$$

So $\sin x = \sum_{n\geq 0} (-1)^n x^{2n+1}/(2n+1)!$ for all x.

((14) $f = \cos x$) Here $f^{(k)}(0) = 1, 0, -1, 0, 1, 0, \ldots$, so only even powers occur with coefficient $(-1)^n/(2n)!$ for x^{2n}, and again $|f^{(N+1)}(\bar{x})| \leq 1$ gives $|R_N(x)| \leq |x|^{N+1}/(N+1)! \to 0$. Thus $\cos x = \sum_{n\geq 0} (-1)^n x^{2n}/(2n)!$ for all x.

Answer. Shown: in each case $R_N(x) \to 0$ for every x because $|R_N(x)| \le e^{|x|}|x|^{N+1}/(N+1)! \to 0$.

Problem 26-5 (Simmons)

Solution.

(Expansion of $1/(1-x)$) Let $|x|<1$ and $s_n = 1 + x + \cdots + x^n$. Then

$$\frac{1}{1-x} - s_n = \frac{1}{1-x} - \frac{1-x^{n+1}}{1-x} = \frac{x^{n+1}}{1-x}, \qquad \left|\frac{x^{n+1}}{1-x}\right| = \frac{|x|^{n+1}}{|1-x|} \longrightarrow 0$$

since $|x|<1$. Thus $s_n \to 1/(1-x)$, i.e. $\dfrac{1}{1-x} = \sum_{n\ge 0} x^n$ for $|x|<1$.

(Expansion of $1/(1+x)$) If $|x|<1$ then $|-x|<1$, so we may replace x by $-x$ in the preceding result:

$$\frac{1}{1+x} = \sum_{n=0}^{\infty}(-x)^n = 1 - x + x^2 - x^3 + \cdots, \qquad |x|<1.$$

($\log(1+x)$) The series for $1/(1+t)$ has radius of convergence 1, so it may be integrated termwise between limits inside $(-1,1)$. For $|x|<1$,

$$\log(1+x) = \int_0^x \frac{dt}{1+t} = \int_0^x \sum_{n=0}^{\infty}(-1)^n t^n\,dt = \sum_{n=0}^{\infty}(-1)^n\frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

($\tan^{-1} x$) For $|t|<1$ we have $|t^2|<1$, so replacing x by t^2 in the series for $1/(1+x)$,

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty}(-1)^n t^{2n} = 1 - t^2 + t^4 - t^6 + \cdots, \qquad |t|<1.$$

This series (in powers of t) has radius of convergence 1, so termwise integration from 0 to x, $|x|<1$, is allowed:

$$\tan^{-1} x = \int_0^x \frac{dt}{1+t^2} = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots.$$

Answer. Shown: both geometric expansions hold for $|x|<1$, and termwise integration gives the series for $\log(1+x)$ and $\tan^{-1} x$ for $|x|<1$.

Problem 26-6 (Simmons)

Solution.

(a) For $|x|<1$, $\dfrac{1}{1-x} = \sum_{n\ge 0} x^n$, so by the rule for multiplying series (all $a_n = b_n = 1$)

$$\frac{1}{(1-x)^2} = \left(\sum_{n=0}^{\infty} x^n\right)^2 = \sum_{n=0}^{\infty} c_n x^n, \qquad c_n = \sum_{k=0}^{n} a_k b_{n-k} = \sum_{k=0}^{n} 1\cdot 1 = n+1.$$

Hence $\dfrac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \cdots = \sum_{n=0}^{\infty}(n+1)x^n$, $|x|<1$.

(b) Since $\dfrac{d}{dx}\dfrac{1}{1-x} = \dfrac{1}{(1-x)^2}$ and a power series may be differentiated termwise inside its interval of convergence,

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} nx^{n-1} = 1 + 2x + 3x^2 + \cdots, \qquad |x| < 1.$$

Shifting the index ($n \to n+1$) shows this is the same series as in (a).

Answer. $\dfrac{1}{(1-x)^2} = \displaystyle\sum_{n=1}^{\infty} nx^{n-1} = \sum_{n=0}^{\infty}(n+1)x^n = 1 + 2x + 3x^2 + \cdots$ for $|x| < 1$ (agrees with the book).

Problem 26-7 (Simmons)

Solution.

(a) Since $1 \cdot 2 \cdots (2n) = (2n)!$, the series is $y = \sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n}}{(2n)!}$, which converges for all x (Problem 3), so it may be differentiated termwise twice:

$$y' = \sum_{n=1}^{\infty}(-1)^n \frac{x^{2n-1}}{(2n-1)!}, \qquad y'' = \sum_{n=1}^{\infty}(-1)^n \frac{x^{2n-2}}{(2n-2)!}.$$

Put $m = n-1$ in the last sum:

$$y'' = \sum_{m=0}^{\infty}(-1)^{m+1}\frac{x^{2m}}{(2m)!} = -\sum_{m=0}^{\infty}(-1)^m \frac{x^{2m}}{(2m)!} = -y.$$

So $y'' + y = 0$.

(b) The factors $2 \cdot 4 \cdots (2n) = 2^n n!$, so $2^2 \cdot 4^2 \cdots (2n)^2 = 2^{2n}(n!)^2$ and

$$y = \sum_{n=0}^{\infty} a_n x^{2n}, \qquad a_n = \frac{(-1)^n}{2^{2n}(n!)^2}.$$

Convergence. For fixed $x \neq 0$, the ratio of consecutive terms is

$$\left|\frac{a_{n+1}x^{2n+2}}{a_n x^{2n}}\right| = \frac{2^{2n}(n!)^2}{2^{2n+2}((n+1)!)^2}x^2 = \frac{x^2}{4(n+1)^2} \longrightarrow 0 < 1,$$

so the series converges for all x and can be differentiated termwise as often as we like.

Substitution. Termwise,

$$y' = \sum_{n=1}^{\infty} 2na_n x^{2n-1}, \qquad y'' = \sum_{n=1}^{\infty} 2n(2n-1)a_n x^{2n-2},$$

$$xy'' + y' = \sum_{n=1}^{\infty}\big[2n(2n-1) + 2n\big]a_n x^{2n-1} = \sum_{n=1}^{\infty} 4n^2 a_n x^{2n-1},$$

$$xy = \sum_{m=0}^{\infty} a_m x^{2m+1} = \sum_{n=1}^{\infty} a_{n-1}x^{2n-1} \qquad (m = n-1).$$

Therefore

$$xy'' + y' + xy = \sum_{n=1}^{\infty}\big(4n^2 a_n + a_{n-1}\big)x^{2n-1}.$$

Now $\dfrac{a_n}{a_{n-1}} = \dfrac{(-1)^n 2^{2n-2}((n-1)!)^2}{(-1)^{n-1}2^{2n}(n!)^2} = -\dfrac{1}{4n^2}$, i.e. $4n^2 a_n + a_{n-1} = 0$ for every $n \geq 1$. Every coefficient vanishes, so $xy'' + y' + xy = 0$. (This y is the Bessel function $J_0(x)$.)

Answer. (a) $y'' = -y$ for the cosine series. (b) The series converges for all x (ratio $\to x^2/(4(n+1)^2) \to 0$) and satisfies $xy'' + y' + xy = 0$ since $4n^2 a_n + a_{n-1} = 0$.

Section 27 Series Solutions of First Order Equations

In every problem we substitute $y=\sum_{n\ge 0}a_nx^n$, use $y'=\sum_{n\ge 0}(n+1)a_{n+1}x^n$, and equate coefficients of like powers of x (legitimate because a power series that is identically zero has all coefficients zero). As in the text, this shows only that *if* a power-series solution exists, it must be the one found; in each case where we recognise the sum as an elementary function we check directly that it is a solution.

Problem 27-1 (Simmons)

Solution.

(a) We have $y'=\sum_{n\ge 0}(n+1)a_{n+1}x^n$ and $2xy=\sum_{m\ge 0}2a_mx^{m+1}=\sum_{n\ge 1}2a_{n-1}x^n$. Equating coefficients:

$$x^0:\ a_1=0; \qquad x^n\ (n\ge 1):\ (n+1)a_{n+1}=2a_{n-1}.$$

Since $a_1=0$, the recurrence $a_{n+1}=\dfrac{2a_{n-1}}{n+1}$ gives $a_3=\frac{2}{3}a_1=0$, and in general all odd coefficients vanish. For the even ones put $n=2k-1$: $a_{2k}=\dfrac{2a_{2k-2}}{2k}=\dfrac{a_{2k-2}}{k}$, so

$$a_2=a_0,\quad a_4=\frac{a_2}{2}=\frac{a_0}{2!},\quad a_6=\frac{a_4}{3}=\frac{a_0}{3!},\ \dots,\quad a_{2k}=\frac{a_0}{k!},$$

with a_0 arbitrary. Hence

$$y=a_0\Big(1+x^2+\frac{x^4}{2!}+\frac{x^6}{3!}+\frac{x^8}{4!}+\cdots\Big)=a_0\sum_{k=0}^{\infty}\frac{(x^2)^k}{k!}=a_0e^{x^2}.$$

The series converges for all x (ratio test), so termwise differentiation was justified.

Direct solution. The equation is separable: $y\equiv 0$ is a solution, and otherwise $\dfrac{dy}{y}=2x\,dx$ gives $\log|y|=x^2+c$, i.e. $y=Ce^{x^2}$ (the case $C=0$ is $y=0$). This agrees with the series with $C=a_0$.

(b) Now $y'+y=\sum_{n\ge 0}\big[(n+1)a_{n+1}+a_n\big]x^n$ must equal 1:

$$x^0:\ a_1+a_0=1; \qquad x^n\ (n\ge 1):\ (n+1)a_{n+1}+a_n=0.$$

So $a_1=1-a_0=-(a_0-1)$ and $a_{n+1}=-\dfrac{a_n}{n+1}$ for $n\ge 1$, whence

$$a_2=\frac{a_0-1}{2},\quad a_3=-\frac{a_0-1}{2\cdot 3},\quad \dots,\quad a_n=(-1)^n\frac{a_0-1}{n!}\ (n\ge 1)$$

(by induction: $a_{n+1}=-a_n/(n+1)=(-1)^{n+1}(a_0-1)/(n+1)!$). Therefore

$$y=a_0+(a_0-1)\sum_{n=1}^{\infty}\frac{(-x)^n}{n!}=1+(a_0-1)\Big(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\cdots\Big)=1+(a_0-1)e^{-x},$$

using $a_0=1+(a_0-1)$.

Direct solution. $y'=1-y$ gives $\dfrac{dy}{1-y}=dx$ (and $y=1$ is the constant solution), so $-\log|1-y|=x+c$, i.e. $y=1+Ce^{-x}$. This agrees with the series, $C=a_0-1$ (and $C=0$, $a_0=1$ gives $y=1$).

Answer. (a) $y = a_0\left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots\right) = a_0 e^{x^2}$; (b) $y = a_0 - (a_0 - 1)x + \frac{a_0 - 1}{2!}x^2 - \cdots = 1 + (a_0 - 1)e^{-x}$.

Problem 27-2 (Simmons)

Solution.

(a) $xy' = \sum_{n\geq 0} n a_n x^n$ (the $n = 0$ term is 0), and $y = \sum_{n\geq 0} a_n x^n$. Equating coefficients of x^n:

$$n a_n = a_n \iff (n-1)a_n = 0 \qquad (n = 0, 1, 2, \ldots).$$

For $n = 0$: $-a_0 = 0$, so $a_0 = 0$; for $n = 1$: $0 \cdot a_1 = 0$, so a_1 is arbitrary; for $n \geq 2$: $a_n = 0$. Hence

$$y = a_1 x.$$

Direct solution. On $x > 0$ or on $x < 0$, separation gives $\dfrac{dy}{y} = \dfrac{dx}{x}$, $\log|y| = \log|x| + c$, so $y = Cx$ (including $C = 0$). On an interval through 0 the equation at $x = 0$ forces $y(0) = 0$; a solution could a priori be $C_1 x$ for $x > 0$ and $C_2 x$ for $x < 0$, but $y'(0)$ must exist, which forces $C_1 = C_2$. So the general solution is $y = Cx$, the same as the series answer: *no discrepancy*.

(b) $x^2 y' = \sum_{m\geq 0}(m+1)a_{m+1}x^{m+2} = \sum_{n\geq 2}(n-1)a_{n-1}x^n$, and this must equal $y = \sum_{n\geq 0} a_n x^n$:

$$x^0 :\ a_0 = 0; \qquad x^1 :\ a_1 = 0; \qquad x^n\ (n \geq 2) :\ a_n = (n-1)a_{n-1}.$$

Starting from $a_1 = 0$ the recurrence gives $a_2 = a_3 = \cdots = 0$. Thus the only power-series solution is $y \equiv 0$.

Direct solution. $y = 0$ is a solution; otherwise $\dfrac{dy}{y} = \dfrac{dx}{x^2}$ gives $\log|y| = -\dfrac{1}{x} + c$, so

$$y = Ce^{-1/x} \qquad (x \neq 0).$$

Explanation of the discrepancy. The power-series method finds only solutions analytic at $x = 0$. The solutions $Ce^{-1/x}$ are not analytic at 0 unless $C = 0$: as $x \to 0^-$ we have $e^{-1/x} \to \infty$, so such a solution cannot even be continued through 0 from the left; and for $x > 0$, $e^{-1/x} \to 0$ as $x \to 0^+$ together with all its derivatives (each derivative has the form (polynomial in $1/x$)$\cdot e^{-1/x}$), so its Taylor series at 0 is identically zero while the function itself is not. (For instance $y = Ce^{-1/x}$ for $x > 0$, $y = 0$ for $x \leq 0$ is a differentiable solution on the whole line which is not analytic at 0.) The equation $x^2 y' = y$ has a singular point at $x = 0$ (writing $y' = y/x^2$), and the theory of Section 27 (existence of analytic solutions) does not apply there. Thus the only solution analytic at 0 is $y = 0$, the case $C = 0$.

Answer. (a) $y = a_1 x$; the series and direct solutions agree. (b) The series gives only $y = 0$, whereas the direct solution is $y = 0$ or $y = ce^{-1/x}$; the latter is analytic at $x = 0$ only when $c = 0$.

Problem 27-3 (Simmons)

Solution. The function $y = \sin^{-1} x$ on $(-1, 1)$ is the solution of $y' = (1 - x^2)^{-1/2}$ with $y(0) = 0$. (Any two solutions differ by a constant, and the constant is fixed by $y(0) = 0$.)

(Way 1: binomial series, then equate coefficients) By the binomial series (9) with $p = -\frac{1}{2}$ and x replaced by $-x^2$ (valid for $|x^2| < 1$):

$$(1-x^2)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-x^2)^n,$$

where

$$\binom{-1/2}{n}(-1)^n = \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2n-1}{2})}{n!}(-1)^n = \frac{1\cdot 3\cdots(2n-1)}{2^n n!} = \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)},$$

because $2\cdot 4\cdots(2n) = 2^n n!$. So

$$(1-x^2)^{-1/2} = 1 + \sum_{n=1}^{\infty} \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}\, x^{2n}, \qquad |x| < 1.$$

Now put $y = \sum a_k x^k$, so $y' = \sum (k+1)a_{k+1}x^k$ and compare coefficients. The series for the right side contains only even powers, so $(k+1)a_{k+1} = 0$ for odd k, i.e. $a_2 = a_4 = \cdots = 0$; and $y(0) = 0$ gives $a_0 = 0$. For $k = 2n$:

$$(2n+1)a_{2n+1} = \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}, \quad a_{2n+1} = \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}\cdot\frac{1}{2n+1} \quad (n \geq 1), \qquad a_1 = 1.$$

(The series has radius of convergence 1, the same as that of y', so termwise differentiation is justified for $|x| < 1$.)

(Way 2: Taylor coefficients $a_n = y^{(n)}(0)/n!$) From $y' = (1-x^2)^{-1/2}$ we get $y'' = x(1-x^2)^{-3/2} = \dfrac{xy'}{1-x^2}$, i.e.

$$(1-x^2)y'' = xy'.$$

Differentiate this n times, using Leibniz's rule on the left, $\frac{d^n}{dx^n}[(1-x^2)y''] = (1-x^2)y^{(n+2)} - 2nxy^{(n+1)} - n(n-1)y^{(n)}$, and on the right $\frac{d^n}{dx^n}[xy'] = xy^{(n+1)} + ny^{(n)}$. Setting $x = 0$:

$$y^{(n+2)}(0) - n(n-1)y^{(n)}(0) = ny^{(n)}(0) \implies y^{(n+2)}(0) = n^2\, y^{(n)}(0).$$

With $y(0) = 0$ and $y'(0) = 1$ we get $y^{(2)}(0) = 0 \cdot y(0) = 0$, hence all even derivatives vanish, and

$$y^{(2n+1)}(0) = (2n-1)^2\, y^{(2n-1)}(0) = \big[1\cdot 3\cdots(2n-1)\big]^2.$$

Using $(2n+1)! = (2n+1)\cdot[1\cdot 3\cdots(2n-1)]\cdot[2\cdot 4\cdots(2n)]$,

$$a_{2n+1} = \frac{[1\cdot 3\cdots(2n-1)]^2}{(2n+1)!} = \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}\cdot\frac{1}{2n+1},$$

the same as in Way 1.

(Result) Both methods give

$$\sin^{-1} x = x + \sum_{n=1}^{\infty} \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}\,\frac{x^{2n+1}}{2n+1} = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \cdots, \qquad |x| < 1.$$

(The formula for $\pi/6$) Since $\frac{1}{2}$ lies in $(-1,1)$ and $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, setting $x = \frac{1}{2}$:

$$\begin{aligned}\frac{\pi}{6} &= \frac{1}{2} + \frac{1}{2}\cdot\frac{(1/2)^3}{3} + \frac{1\cdot 3}{2\cdot 4}\cdot\frac{(1/2)^5}{5} + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\cdot\frac{(1/2)^7}{7} + \cdots \\ &= \frac{1}{2} + \frac{1}{2}\cdot\frac{1}{3\cdot 2^3} + \frac{1\cdot 3}{2\cdot 4}\cdot\frac{1}{5\cdot 2^5} + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\cdot\frac{1}{7\cdot 2^7} + \cdots.\end{aligned}$$

Answer. $\sin^{-1} x = x + \displaystyle\sum_{n=1}^{\infty} \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)} \frac{x^{2n+1}}{2n+1}$ ($|x| < 1$), and putting $x = \frac{1}{2}$ gives the stated series for $\pi/6$.

Problem 27-4 (Simmons)

Solution. Since $\tan x$ is analytic at 0 and solves $(*)$ with $y(0) = 0$, a power-series solution $y = \sum a_n x^n$ with $a_0 = y(0) = 0$ exists, and the computations below determine its coefficients uniquely.

(a) Write $y^2 = \sum_{n\geq 0} c_n x^n$ with $c_n = \sum_{k=0}^{n} a_k a_{n-k}$ (the product rule for series). This is the complication due to nonlinearity: the coefficient a_{n+1} depends on *all* the earlier a_k through a convolution. Equating coefficients in $\sum (n+1)a_{n+1}x^n = 1 + \sum c_n x^n$:

$$a_1 = 1 + c_0, \qquad (n+1)a_{n+1} = c_n \quad (n \geq 1).$$

Here $a_0 = 0$, so $c_0 = a_0^2 = 0$ and $a_1 = 1$.

All even coefficients vanish. Suppose $a_0 = a_2 = \cdots = a_{2j} = 0$ for all even indices up to n, where n is odd. In $c_n = \sum_{k=0}^{n} a_k a_{n-k}$, k and $n-k$ have opposite parity, so one factor in every product is an even-indexed coefficient, which is 0. Thus $c_n = 0$ and $a_{n+1} = 0$. By induction every $a_{2j} = 0$.

The odd coefficients. With $a_0 = a_2 = a_4 = 0$ and $a_1 = 1$:

$$\begin{aligned}
2a_2 &= c_1 = 2a_0a_1 = 0, \\
3a_3 &= c_2 = 2a_0a_2 + a_1^2 = 1 && \Longrightarrow\ a_3 = \tfrac{1}{3}, \\
4a_4 &= c_3 = 2a_0a_3 + 2a_1a_2 = 0, \\
5a_5 &= c_4 = 2a_0a_4 + 2a_1a_3 + a_2^2 = \tfrac{2}{3} && \Longrightarrow\ a_5 = \tfrac{2}{15}.
\end{aligned}$$

Hence

$$y = x + \tfrac{1}{3}x^3 + \tfrac{2}{15}x^5 + \cdots = \tan x.$$

(b) From $y' = 1 + y^2$:

$$\begin{aligned}
y'' &= 2yy', \\
y''' &= 2(y')^2 + 2yy'', \\
y^{(4)} &= 4y'y'' + 2y'y'' + 2yy''' = 6y'y'' + 2yy''', \\
y^{(5)} &= 6(y'')^2 + 6y'y''' + 2y'y''' + 2yy^{(4)} = 6(y'')^2 + 8y'y''' + 2yy^{(4)}.
\end{aligned}$$

With $y(0) = 0$ and $y'(0) = 1 + 0 = 1$:

$$y''(0) = 0, \quad y'''(0) = 2\cdot 1 + 0 = 2, \quad y^{(4)}(0) = 6\cdot 1\cdot 0 + 0 = 0, \quad y^{(5)}(0) = 0 + 8\cdot 1\cdot 2 + 0 = 16.$$

Then $a_n = y^{(n)}(0)/n!$ gives

$$a_1 = 1, \quad a_2 = 0, \quad a_3 = \frac{2}{3!} = \frac{1}{3}, \quad a_4 = 0, \quad a_5 = \frac{16}{5!} = \frac{16}{120} = \frac{2}{15},$$

the same coefficients as in (a): $\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots$.

Answer. Both methods give $y = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots = \tan x$ (shown).

Problem 27-5 (Simmons)

Solution.

(Method (a): equate coefficients) With $y=\sum a_n x^n$, $y(0)=0$ gives $a_0=0$. Then $y'+y=x$ reads $\sum_{n\ge 0}\big[(n+1)a_{n+1}+a_n\big]x^n = x$, so

$$x^0:\ a_1+a_0=0\Rightarrow a_1=0; \qquad x^1:\ 2a_2+a_1=1\Rightarrow a_2=\tfrac{1}{2}; \qquad x^n\ (n\ge 2):\ a_{n+1}=-\frac{a_n}{n+1}.$$

Therefore $a_3=-\frac{1}{2\cdot 3}=-\frac{1}{3!}$, $a_4=\frac{1}{4!}$, and in general (induction) $a_n=\dfrac{(-1)^n}{n!}$ for $n\ge 2$. Hence

$$y=\frac{x^2}{2!}-\frac{x^3}{3!}+\frac{x^4}{4!}-\cdots=\sum_{n=2}^{\infty}\frac{(-x)^n}{n!}.$$

(Method (b): differentiate repeatedly) From $y'=x-y$: $y''=1-y'$ and $y^{(k)}=-y^{(k-1)}$ for $k\ge 3$. At 0: $y(0)=0$, $y'(0)=0-0=0$, $y''(0)=1-0=1$, $y'''(0)=-1$, $y^{(4)}(0)=1$, …, i.e. $y^{(n)}(0)=(-1)^n$ for $n\ge 2$. So $a_n=y^{(n)}(0)/n!=(-1)^n/n!$ for $n\ge 2$, $a_0=a_1=0$: the same series.

(Recognition) Since $e^{-x}=\sum_{n\ge 0}(-x)^n/n!=1-x+\sum_{n\ge 2}(-x)^n/n!$,

$$y=\sum_{n=2}^{\infty}\frac{(-x)^n}{n!}=e^{-x}-1+x.$$

(Direct solution) $y'+y=x$ is linear with integrating factor e^x: $(ye^x)'=xe^x$, and by parts $\int xe^x\,dx=(x-1)e^x+C$, so $y=x-1+Ce^{-x}$. The condition $y(0)=0$ gives $-1+C=0$, $C=1$:

$$y=x-1+e^{-x},$$

which agrees with the series.

Answer. $y=\dfrac{x^2}{2!}-\dfrac{x^3}{3!}+\dfrac{x^4}{4!}-\cdots=e^{-x}+x-1$ (agrees with the book).

Section 28 Second Order Linear Equations. Ordinary Points

Throughout we seek solutions $y=\sum_{n=0}^{\infty}a_nx^n$ about the ordinary point $x=0$. We use the shifted forms

$$y'=\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n,\qquad xy'=\sum_{n=0}^{\infty}na_nx^n,$$
$$y''=\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}x^n,\qquad x^2y''=\sum_{n=0}^{\infty}n(n-1)a_nx^n,$$

and set the coefficient of every power x^n in the left side of the equation equal to zero. By Theorem A of the section, a power series so obtained converges at least on the interval on which the power series of $P(x)$ and $Q(x)$ converge, and $a_0=y(0)$, $a_1=y'(0)$ are arbitrary.

Problem 28-1 (Simmons)

Solution. Here $P=2x/(1+x^2)$ and $Q=-2/(1+x^2)$ are analytic at 0 with radius of convergence 1, so $x=0$ is an ordinary point and solutions $\sum a_nx^n$ are valid at least for $|x|<1$. Substituting,

$$\sum_n(n+1)(n+2)a_{n+2}x^n+\sum_n n(n-1)a_nx^n+2\sum_n na_nx^n-2\sum_n a_nx^n=0,$$

so the coefficient of x^n gives

$$(n+1)(n+2)a_{n+2}+\big[n(n-1)+2n-2\big]a_n=0.$$

Since $n(n-1)+2n-2=n^2+n-2=(n+2)(n-1)$, this is

$$(n+1)(n+2)a_{n+2}=-(n+2)(n-1)a_n\qquad\Longrightarrow\qquad\boxed{a_{n+2}=-\frac{n-1}{n+1}\,a_n}\qquad(n=0,1,2,\dots).$$

Odd coefficients. For $n=1$ the factor $n-1$ vanishes, so $a_3=0$, and then $a_5=a_7=\cdots=0$. Only a_1 survives: the odd part is a_1x.

Even coefficients. $a_2=-\frac{-1}{1}a_0=a_0$, $a_4=-\frac{1}{3}a_2=-\frac{1}{3}a_0$, $a_6=-\frac{3}{5}a_4=\frac{1}{5}a_0$, $a_8=-\frac{5}{7}a_6=-\frac{1}{7}a_0$, $\dots$. Indeed, multiplying the ratios $a_{2k+2}/a_{2k}=-(2k-1)/(2k+1)$ gives the general term

$$a_{2k}=\frac{(-1)^{k-1}}{2k-1}\,a_0\qquad(k=0,1,2,\dots)$$

(for $k=0$: $(-1)^{-1}/(-1)=1$). Hence the general solution is

$$y=a_0\Big(1+x^2-\tfrac{1}{3}x^4+\tfrac{1}{5}x^6-\tfrac{1}{7}x^8+\cdots\Big)+a_1x=a_0\Big(1+\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{2k-1}x^{2k}\Big)+a_1x,\qquad|x|<1.$$

The ratio $|a_{2k+2}x^{2k+2}/a_{2k}x^{2k}|=\frac{2k-1}{2k+1}x^2\to x^2$ confirms the radius of convergence 1.

Elementary form. The series in a_0 is

$$1+x\sum_{k=1}^{\infty}\frac{(-1)^{k-1}x^{2k-1}}{2k-1}=1+x\left(x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots\right)=1+x\tan^{-1}x,$$

because $\tan^{-1}x=x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots$ for $|x|<1$. Therefore

$$y=a_0(1+x\tan^{-1}x)+a_1x.$$

Check. $y_1 = 1 + x\tan^{-1}x$ has $y_1' = \tan^{-1}x + \frac{x}{1+x^2}$ and $y_1'' = \frac{1}{1+x^2} + \frac{1-x^2}{(1+x^2)^2} = \frac{2}{(1+x^2)^2}$. Then $(1+x^2)y_1'' + 2xy_1' - 2y_1 = \frac{2}{1+x^2} + 2x\tan^{-1}x + \frac{2x^2}{1+x^2} - 2 - 2x\tan^{-1}x = \frac{2+2x^2}{1+x^2} - 2 = 0$, and $y_2 = x$ obviously satisfies the equation ($0 + 2x - 2x = 0$). The closed form is analytic for all real x, even though the series stops converging at $|x| = 1$.

Answer. $y = a_0\left(1 + x^2 - \frac{x^4}{3} + \frac{x^6}{5} - \frac{x^8}{7} + \cdots\right) + a_1 x = a_0(1 + x\tan^{-1}x) + a_1 x$, with $a_{n+2} = -\frac{n-1}{n+1}a_n$. Yes, it is elementary.

Problem 28-2 (Simmons)

Solution.

(a) Substituting $y = \sum a_n x^n$:

$$\sum_n (n+1)(n+2)a_{n+2}x^n + \sum_n n a_n x^n + \sum_n a_n x^n = 0 \implies (n+1)(n+2)a_{n+2} + (n+1)a_n = 0,$$

so (dividing by $n+1 \neq 0$)

$$a_{n+2} = -\frac{a_n}{n+2} \qquad (n \geq 0).$$

Even index: $a_2 = -\frac{a_0}{2}$, $a_4 = -\frac{a_2}{4} = \frac{a_0}{2\cdot 4}$, $a_6 = -\frac{a_0}{2\cdot 4\cdot 6}$, and in general

$$a_{2k} = \frac{(-1)^k a_0}{2\cdot 4\cdots(2k)} = \frac{(-1)^k a_0}{2^k k!}.$$

Odd index: $a_3 = -\frac{a_1}{3}$, $a_5 = \frac{a_1}{3\cdot 5}$, $a_7 = -\frac{a_1}{3\cdot 5\cdot 7}$, and in general

$$a_{2k+1} = \frac{(-1)^k a_1}{3\cdot 5\cdots(2k+1)}.$$

Hence $y = a_0 y_1(x) + a_1 y_2(x)$ with

$$y_1(x) = 1 - \frac{x^2}{2} + \frac{x^4}{2\cdot 4} - \frac{x^6}{2\cdot 4\cdot 6} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^k k!},$$

$$y_2(x) = x - \frac{x^3}{3} + \frac{x^5}{3\cdot 5} - \frac{x^7}{3\cdot 5\cdot 7} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{3\cdot 5\cdots(2k+1)}.$$

Here $y_1(0) = 1$, $y_1'(0) = 0$ and $y_2(0) = 0$, $y_2'(0) = 1$, so they are independent.

(b) For both series the ratio of consecutive nonzero terms is, by the recurrence,

$$\left|\frac{a_{n+2}x^{n+2}}{a_n x^n}\right| = \frac{|x|^2}{n+2} \longrightarrow 0 \qquad (n \to \infty)$$

for every fixed x. Since the limit is $0 < 1$, the ratio test shows that both series converge absolutely for all x.

(c) The series y_1 is $\sum_k \frac{1}{k!}\left(-\frac{x^2}{2}\right)^k = e^{-x^2/2}$, the exponential series with $u = -x^2/2$. (Direct check: $y_1' = -xe^{-x^2/2}$, $y_1'' = (x^2-1)e^{-x^2/2}$, so $y_1'' + xy_1' + y_1 = (x^2 - 1 - x^2 + 1)e^{-x^2/2} = 0$.)

Second solution (Section 16). For $y'' + Py' + Qy = 0$ with a known solution y_1, a second solution is $y_2 = y_1 \int \frac{e^{-\int P\,dx}}{y_1^2}\,dx$. Here $P = x$, so $e^{-\int P\,dx} = e^{-x^2/2}$ and

$$\frac{e^{-x^2/2}}{y_1^2} = \frac{e^{-x^2/2}}{e^{-x^2}} = e^{x^2/2}, \qquad y_2^*(x) = e^{-x^2/2}\int_0^x e^{t^2/2}\,dt$$

(the lower limit 0 is a choice of the constant of integration; any other choice adds a multiple of y_1).

Identification with y_2. The function y_2^* satisfies $y_2^*(0) = 0$ and

$$y_2^{*\prime}(x) = -xe^{-x^2/2}\int_0^x e^{t^2/2}\,dt + e^{-x^2/2}e^{x^2/2} = 1 - x\,y_2^*(x),$$

so $y_2^{*\prime}(0) = 1$. The series y_2 also has $y_2(0) = 0$, $y_2'(0) = 1$. By the uniqueness part of Theorem A (the solution with prescribed $y(0), y'(0)$ is unique), $y_2^* = y_2$. Explicitly,

$$y_2(x) = e^{-x^2/2}\int_0^x e^{t^2/2}\,dt = \left(1 - \tfrac{x^2}{2} + \tfrac{x^4}{8} - \cdots\right)\left(x + \tfrac{x^3}{6} + \tfrac{x^5}{40} + \cdots\right) = x - \frac{x^3}{3} + \frac{x^5}{15} - \cdots,$$

and indeed the coefficient of x^3 is $\frac{1}{6} - \frac{1}{2} = -\frac{1}{3}$ and that of x^5 is $\frac{1}{40} - \frac{1}{12} + \frac{1}{8} = \frac{3-10+15}{120} = \frac{1}{15}$, in agreement with $-\frac{1}{3}$ and $\frac{1}{3\cdot 5}$ of (a).

Answer. (a) $a_{n+2} = -\dfrac{a_n}{n+2}$; $y_1 = \displaystyle\sum_{k\ge 0}\frac{(-1)^k x^{2k}}{2^k k!} = 1 - \frac{x^2}{2} + \frac{x^4}{2\cdot 4} - \cdots$, $y_2 = \displaystyle\sum_{k\ge 0}\frac{(-1)^k x^{2k+1}}{3\cdot 5\cdots(2k+1)} = x - \frac{x^3}{3} + \frac{x^5}{3\cdot 5} - \cdots$. (b) ratio $\to x^2/(n+2) \to 0$. (c) $y_1 = e^{-x^2/2}$ and $y_2 = e^{-x^2/2}\displaystyle\int_0^x e^{t^2/2}\,dt$.

Problem 28-3 (Simmons)

Solution. Substituting $y = \sum a_n x^n$, with $y' = \sum (n+1)a_{n+1}x^n$ and $xy = \sum a_{n-1}x^n$ (where $a_{-1} = 0$):

$$\sum_n (n+1)(n+2)a_{n+2}x^n + \sum_n (n+1)a_{n+1}x^n - \sum_n a_{n-1}x^n = 0.$$

Equating the coefficient of x^n to zero,

$$(n+1)(n+2)a_{n+2} + (n+1)a_{n+1} - a_{n-1} = 0,$$

that is,

$$\boxed{a_{n+2} = -\frac{(n+1)a_{n+1} - a_{n-1}}{(n+1)(n+2)}} \qquad (n \ge 0,\ a_{-1} = 0).$$

This involves $a_{n+2}, a_{n+1}, a_{n-1}$: a three-term recursion, which does not allow a_n to be written in a closed form. Here $P = 1$, $Q = -x$ are entire, so by Theorem A the series converge for all x. Explicitly for the first values:

$$a_2 = -\tfrac{1}{2}a_1, \quad a_3 = -\frac{2a_2 - a_0}{6}, \quad a_4 = -\frac{3a_3 - a_1}{12}, \quad a_5 = -\frac{4a_4 - a_2}{20}, \quad a_6 = -\frac{5a_5 - a_3}{30}, \ \ldots$$

(a) $a_0 = 1$, $a_1 = 0$: $a_2 = 0$; $a_3 = -\frac{0-1}{6} = \frac{1}{6}$; $a_4 = -\frac{3\cdot\frac{1}{6} - 0}{12} = -\frac{1}{24}$; $a_5 = -\frac{4\cdot(-\frac{1}{24}) - 0}{20} = \frac{1}{120}$; $a_6 = -\frac{5\cdot\frac{1}{120} - \frac{1}{6}}{30} = -\frac{\frac{1}{24} - \frac{1}{6}}{30} = \frac{1}{240}$; $a_7 = -\frac{6a_6 - a_4}{42} = -\frac{\frac{1}{40} + \frac{1}{24}}{42} = -\frac{1}{630}$. Thus

$$y_1(x) = 1 + \frac{x^3}{2\cdot 3} - \frac{x^4}{2\cdot 3\cdot 4} + \frac{x^5}{2\cdot 3\cdot 4\cdot 5} + \frac{x^6}{240} - \frac{x^7}{630} + \cdots.$$

(b) $a_0 = 0$, $a_1 = 1$: $a_2 = -\frac{1}{2}$; $a_3 = -\frac{2(-\frac{1}{2})-0}{6} = \frac{1}{6}$; $a_4 = -\frac{3\cdot\frac{1}{6}-1}{12} = \frac{1}{24}$; $a_5 = -\frac{4\cdot\frac{1}{24}-(-\frac{1}{2})}{20} = -\frac{\frac{1}{6}+\frac{1}{2}}{20} = -\frac{1}{30}$; $a_6 = -\frac{5(-\frac{1}{30})-\frac{1}{6}}{30} = \frac{\frac{1}{6}+\frac{1}{6}}{30} = \frac{1}{90}$; $a_7 = -\frac{6a_6-a_4}{42} = -\frac{\frac{1}{15}-\frac{1}{24}}{42} = -\frac{1}{1680}$. Thus

$$y_2(x) = x - \frac{x^2}{2} + \frac{x^3}{2\cdot 3} + \frac{x^4}{2\cdot 3\cdot 4} - \frac{4x^5}{2\cdot 3\cdot 4\cdot 5} + \frac{x^6}{90} - \frac{x^7}{1680} + \cdots$$

(note $\frac{4}{2\cdot 3\cdot 4\cdot 5} = \frac{1}{30}$).

Answer. $a_{n+2} = -\dfrac{(n+1)a_{n+1} - a_{n-1}}{(n+1)(n+2)}$; $y_1 = 1 + \dfrac{x^3}{2\cdot 3} - \dfrac{x^4}{2\cdot 3\cdot 4} + \dfrac{x^5}{2\cdot 3\cdot 4\cdot 5} + \cdots$, $y_2 = x - \dfrac{x^2}{2} + \dfrac{x^3}{2\cdot 3} + \dfrac{x^4}{2\cdot 3\cdot 4} - \dfrac{4x^5}{2\cdot 3\cdot 4\cdot 5} + \cdots$.

Problem 28-4 (Simmons)

Solution.

(a) With $y = \sum a_n x^n$ we have $y'' = \sum (n+1)(n+2)a_{n+2}x^n$, $(p+\frac{1}{2})y = \sum (p+\frac{1}{2})a_n x^n$ and $x^2 y = \sum_n a_{n-2}x^n$ (with $a_{-1} = a_{-2} = 0$). The coefficient of x^n in the equation is

$$(n+1)(n+2)a_{n+2} + \left(p+\tfrac{1}{2}\right)a_n - \tfrac{1}{4}a_{n-2} = 0,$$

which is the stated recursion (it relates three coefficients).

(b) Let $y = we^{-x^2/4}$. Then

$$y' = \left(w' - \tfrac{x}{2}w\right)e^{-x^2/4},$$
$$y'' = \left(w'' - \tfrac{1}{2}w - \tfrac{x}{2}w'\right)e^{-x^2/4} - \tfrac{x}{2}\left(w' - \tfrac{x}{2}w\right)e^{-x^2/4} = \left(w'' - xw' - \tfrac{1}{2}w + \tfrac{x^2}{4}w\right)e^{-x^2/4}.$$

Hence

$$y'' + \left(p + \tfrac{1}{2} - \tfrac{1}{4}x^2\right)y = e^{-x^2/4}\left[w'' - xw' - \tfrac{1}{2}w + \tfrac{x^2}{4}w + pw + \tfrac{1}{2}w - \tfrac{x^2}{4}w\right] = e^{-x^2/4}\left[w'' - xw' + pw\right].$$

Since $e^{-x^2/4} \neq 0$, the equation is equivalent to $w'' - xw' + pw = 0$.

(c) Put $w = \sum a_n x^n$: $\sum (n+1)(n+2)a_{n+2}x^n - \sum n a_n x^n + p\sum a_n x^n = 0$, so

$$(n+1)(n+2)a_{n+2} = (n-p)a_n \quad\Longleftrightarrow\quad \boxed{a_{n+2} = -\frac{p-n}{(n+1)(n+2)}\,a_n} \qquad (n \geq 0),$$

a two-term recursion. Then

$$a_2 = -\frac{p}{2!}a_0, \quad a_4 = -\frac{p-2}{3\cdot 4}a_2 = \frac{p(p-2)}{4!}a_0, \quad a_6 = -\frac{p-4}{5\cdot 6}a_4 = -\frac{p(p-2)(p-4)}{6!}a_0, \ \ldots,$$
$$a_3 = -\frac{p-1}{3!}a_1, \quad a_5 = -\frac{p-3}{4\cdot 5}a_3 = \frac{(p-1)(p-3)}{5!}a_1, \quad a_7 = -\frac{(p-1)(p-3)(p-5)}{7!}a_1, \ \ldots,$$

and in general

$$a_{2k} = (-1)^k \frac{p(p-2)\cdots(p-2k+2)}{(2k)!}a_0, \qquad a_{2k+1} = (-1)^k \frac{(p-1)(p-3)\cdots(p-2k+1)}{(2k+1)!}a_1.$$

Therefore

$$w(x) = a_0\left[1 - \frac{p}{2!}x^2 + \frac{p(p-2)}{4!}x^4 - \cdots\right] + a_1\left[x - \frac{p-1}{3!}x^3 + \frac{(p-1)(p-3)}{5!}x^5 - \cdots\right].$$

Because $|a_{n+2}x^{n+2}/(a_n x^n)| = \frac{|p-n|}{(n+1)(n+2)}x^2 \to 0$, both series converge for all x. The general solution of the original equation of the problem is $y = we^{-x^2/4}$, i.e. the same expression multiplied by $e^{-x^2/4}$.

Answer. (a) shown; (b) $w'' - xw' + pw = 0$; (c) $a_{n+2} = -\frac{p-n}{(n+1)(n+2)}a_n$ and $w = a_0\big[1 - \frac{p}{2!}x^2 + \frac{p(p-2)}{4!}x^4 - \cdots\big] + a_1\big[x - \frac{(p-1)}{3!}x^3 + \frac{(p-1)(p-3)}{5!}x^5 - \cdots\big]$ (and $y = we^{-x^2/4}$).

Problem 28-5 (Simmons)

Solution.

(a) The equation is $u'' + q(x)u = 0$ with $q(x) = x$.

Positive zeros. $q(x) = x > 0$ for all $x > 0$ and $\int_1^\infty x\,dx = \infty$. By Theorem C of Section 24, every nontrivial solution has infinitely many zeros on the positive x-axis.

Negative zeros. On the interval $x < 0$ we have $q(x) = x < 0$. By Theorem B of Section 24, a nontrivial solution of $u'' + q(x)u = 0$ with $q < 0$ has at most one zero. (Reason: if $u(x_0) = 0$ and, say, $u'(x_0) > 0$, then $u > 0$ just to the right of x_0 and $u'' = -xu > 0$ there, so u' increases and u can never return to 0 to the right; similarly to the left, and the case $u'(x_0) < 0$ is analogous.) Hence there is at most one negative zero.

(b) Here $P = 0$, $Q = x$ are entire, so all solutions are given by power series valid for all x. Substituting $y = \sum a_n x^n$ and using $xy = \sum_{n\ge1} a_{n-1}x^n$:

$$2a_2 + \sum_{n\ge1}\big[(n+1)(n+2)a_{n+2} + a_{n-1}\big]x^n = 0.$$

So $a_2 = 0$ and $(n+1)(n+2)a_{n+2} = -a_{n-1}$ for $n \ge 1$; shifting the index $n \to n-1$,

$$\boxed{a_{n+3} = -\frac{a_n}{(n+2)(n+3)}} \qquad (n \ge 0).$$

The recursion links $a_0, a_3, a_6, \ldots$ (a_0 free), $a_1, a_4, a_7, \ldots$ (a_1 free), and $a_2 = a_5 = a_8 = \cdots = 0$.

Indices $3n$:

$$a_{3n} = \frac{(-1)^n a_0}{(2\cdot3)(5\cdot6)\cdots\big((3n-1)3n\big)} = \frac{(-1)^n a_0}{[2\cdot5\cdots(3n-1)]\,[3\cdot6\cdots3n]} = \frac{(-1)^n a_0}{2\cdot5\cdots(3n-1)\,3^n n!}.$$

Indices $3n+1$:

$$a_{3n+1} = \frac{(-1)^n a_1}{(3\cdot4)(6\cdot7)\cdots(3n)(3n+1)} = \frac{(-1)^n a_1}{[3\cdot6\cdots3n][4\cdot7\cdots(3n+1)]} = \frac{(-1)^n a_1}{4\cdot7\cdots(3n+1)\,3^n n!}.$$

Therefore the general solution of Airy's equation is

$$y = a_0\Big[1 + \sum_{n=1}^{\infty}\frac{(-1)^n x^{3n}}{2\cdot5\cdots(3n-1)\,3^n n!}\Big] + a_1\Big[x + \sum_{n=1}^{\infty}\frac{(-1)^n x^{3n+1}}{4\cdot7\cdots(3n+1)\,3^n n!}\Big],$$

that is, $y = a_0\big(1 - \frac{x^3}{6} + \frac{x^6}{180} - \frac{x^9}{12960} + \cdots\big) + a_1\big(x - \frac{x^4}{12} + \frac{x^7}{504} - \frac{x^{10}}{45360} + \cdots\big)$.

Convergence. In the first series the ratio of consecutive terms is, from the recursion with $n \to 3n$,

$$\left|\frac{a_{3n+3}x^{3n+3}}{a_{3n}x^{3n}}\right| = \frac{|x|^3}{(3n+2)(3n+3)} \longrightarrow 0,$$

and in the second $\left|\dfrac{a_{3n+4}x^{3n+4}}{a_{3n+1}x^{3n+1}}\right| = \dfrac{|x|^3}{(3n+3)(3n+4)} \to 0$. Hence both series converge for every x (infinite radius of convergence).

(c) If $y(x)$ solves $y''(x) = -xy(x)$ for all x, put $z(x) = y(-x)$. Then $z''(x) = y''(-x) = -(-x)y(-x) = xz(x)$, so $z'' - xz = 0$; conversely every solution of $z'' - xz = 0$ arises this way. Replacing x by $-x$ in the general solution of (b) (note $(-x)^{3n} = (-1)^n x^{3n}$ and $(-x)^{3n+1} = -(-1)^n x^{3n+1}$, so the factors $(-1)^n$ cancel):

$$y = a_0\Big[1 + \sum_{n=1}^{\infty} \frac{x^{3n}}{2\cdot 5\cdots(3n-1)\,3^n n!}\Big] + a_1\Big[-x - \sum_{n=1}^{\infty} \frac{x^{3n+1}}{4\cdot 7\cdots(3n+1)\,3^n n!}\Big].$$

Since a_1 is arbitrary, the sign in the second bracket may be absorbed into it, which gives the equivalent form with $+x + \sum\cdots$ (all terms positive).

Answer. (a) shown via Theorems 24-C and 24-B; (b) $a_{n+3} = -\dfrac{a_n}{(n+2)(n+3)}$, $a_2 = 0$, $y = a_0\Big[1 + \sum_{n\ge1} \dfrac{(-1)^n x^{3n}}{2\cdot 5\cdots(3n-1)3^n n!}\Big] + a_1\Big[x + \sum_{n\ge1} \dfrac{(-1)^n x^{3n+1}}{4\cdot 7\cdots(3n+1)3^n n!}\Big]$; (c) $y = a_0\Big[1 + \sum_{n\ge1} \dfrac{x^{3n}}{2\cdot 5\cdots(3n-1)3^n n!}\Big] + a_1\Big[-x - \sum_{n\ge1} \dfrac{x^{3n+1}}{4\cdot 7\cdots(3n+1)3^n n!}\Big]$.

Problem 28-6 (Simmons)

Solution.

(a) In the form $y'' - \frac{x}{1-x^2}y' + \frac{p^2}{1-x^2}y = 0$ the coefficients are analytic at 0 with power series (in x^2) of radius 1 (since $\frac{1}{1-x^2} = \sum x^{2n}$). So $x = 0$ is an ordinary point and the series solutions are valid for $|x| < 1$ (Theorem A). Substituting $y = \sum a_n x^n$:

$$\sum (n+1)(n+2)a_{n+2}x^n - \sum n(n-1)a_n x^n - \sum n a_n x^n + p^2\sum a_n x^n = 0,$$

whence $(n+1)(n+2)a_{n+2} - \big[n(n-1) + n - p^2\big]a_n = 0$, and since $n(n-1)+n = n^2$,

$$\boxed{a_{n+2} = \frac{n^2 - p^2}{(n+1)(n+2)}\,a_n} \qquad (n \ge 0),$$

which can also be written $a_{n+2} = -\dfrac{(p-n)(p+n)}{(n+1)(n+2)}a_n$.

Even part:

$$a_2 = -\frac{p^2}{2!}a_0, \qquad a_4 = \frac{2^2 - p^2}{3\cdot 4}a_2 = \frac{p^2(p^2-4)}{4!}a_0 = \frac{p\cdot p(p-2)(p+2)}{4!}a_0,$$
$$a_6 = \frac{4^2 - p^2}{5\cdot 6}a_4 = -\frac{p^2(p^2-4)(p^2-16)}{6!}a_0,$$

and generally $a_{2k} = \dfrac{\prod_{j=0}^{k-1}\big[(2j)^2 - p^2\big]}{(2k)!}a_0$.

Odd part: $a_3 = \frac{1-p^2}{2\cdot 3}a_1 = -\frac{(p-1)(p+1)}{3!}a_1$, $a_5 = \frac{3^2-p^2}{4\cdot 5}a_3 = \frac{(p-1)(p-3)(p+1)(p+3)}{5!}a_1$, and generally $a_{2k+1} = \dfrac{\prod_{j=0}^{k-1}\big[(2j+1)^2 - p^2\big]}{(2k+1)!}a_1$. So two independent solutions on $|x| < 1$ are

$$y_1(x) = 1 - \frac{p\cdot p}{2!}x^2 + \frac{p(p-2)\,p(p+2)}{4!}x^4 - \frac{p(p-2)(p-4)\,p(p+2)(p+4)}{6!}x^6 + \cdots,$$

$$y_2(x) = x - \frac{(p-1)(p+1)}{3!}x^3 + \frac{(p-1)(p-3)(p+1)(p+3)}{5!}x^5 - \cdots.$$

They are independent since $y_1(0) = 1,\ y_1'(0) = 0, y_2(0) = 0,\ y_2'(0) = 1$.

(b) Let $p = n \geq 0$ be an integer. In the recursion, the coefficient $n^2 - p^2$ at index n vanishes: $a_{n+2} = 0$, hence $a_{n+4} = a_{n+6} = \cdots = 0$ (each later coefficient of the same parity is a multiple of the previous one).

- If n is even, the even series y_1 stops: y_1 is a polynomial of degree at most n. Its coefficient of x^n is $a_n = \prod_{j=0}^{n/2-1}[(2j)^2 - n^2]/n! \cdot a_0 \neq 0$, since each factor $(2j)^2 - n^2$ with $2j < n$ is nonzero. (For $n = 0$, $y_1 = 1$.) So $\deg y_1 = n$.
- If n is odd, the odd series y_2 stops, and its coefficient of x^n is $\prod_{j=0}^{(n-1)/2-1}[(2j+1)^2 - n^2]/n! \cdot a_1 \neq 0$, so $\deg y_2 = n$.

Thus for every integer $n \geq 0$ there is a polynomial solution of degree exactly n. Examples: $p = 0$: 1; $p = 1$: x; $p = 2$: $1 - 2x^2 = -T_2(x)$; $p = 3$: $x - \frac{4}{3}x^3 = -\frac{1}{3}T_3(x)$, where T_n are the Chebyshev polynomials $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$.

Answer. (a) $a_{n+2} = \dfrac{n^2 - p^2}{(n+1)(n+2)}a_n$; $y_1 = 1 - \dfrac{p^2}{2!}x^2 + \dfrac{p^2(p^2-4)}{4!}x^4 - \cdots$, $y_2 = x - \dfrac{(p^2-1)}{3!}x^3 + \dfrac{(p^2-1)(p^2-9)}{5!}x^5 - \cdots$. (b) For $p = n$ the recursion gives $a_{n+2} = 0$, so y_1 (n even) or y_2 (n odd) is a polynomial of degree n.

Problem 28-7 (Simmons)

Solution.

(a) Since $P = -2x$ and $Q = 2p$ are polynomials, every point is ordinary. With $y = \sum a_n x^n$,

$$\sum (n+1)(n+2)a_{n+2}x^n - 2\sum n a_n x^n + 2p\sum a_n x^n = 0 \implies (n+1)(n+2)a_{n+2} = 2(n-p)a_n,$$

$$\boxed{a_{n+2} = -\frac{2(p-n)}{(n+1)(n+2)}\,a_n} \qquad (n \geq 0).$$

Even index: $a_2 = -\frac{2p}{2!}a_0$, $a_4 = -\frac{2(p-2)}{3\cdot 4}a_2 = \frac{2^2 p(p-2)}{4!}a_0$, $a_6 = -\frac{2(p-4)}{5\cdot 6}a_4 = -\frac{2^3 p(p-2)(p-4)}{6!}a_0$, and in general

$$a_{2k} = (-2)^k \frac{p(p-2)\cdots(p-2k+2)}{(2k)!}a_0.$$

Odd index: $a_3 = -\frac{2(p-1)}{3!}a_1$, $a_5 = -\frac{2(p-3)}{4\cdot 5}a_3 = \frac{2^2(p-1)(p-3)}{5!}a_1$, $a_7 = -\frac{2^3(p-1)(p-3)(p-5)}{7!}a_1$, and in general

$$a_{2k+1} = (-2)^k \frac{(p-1)(p-3)\cdots(p-2k+1)}{(2k+1)!}a_1.$$

These are exactly the series y_1, y_2 of the statement, with $y_1(0) = 1, y_1'(0) = 0, y_2(0) = 0, y_2'(0) = 1$, hence independent, and $y = a_0 y_1 + a_1 y_2$ is the general solution.

Direct convergence check. For both series (fixed x) the ratio of successive terms is

$$\left|\frac{a_{n+2}x^{n+2}}{a_n x^n}\right| = \frac{2|p-n|}{(n+1)(n+2)}\,x^2 \longrightarrow 0 \qquad (n \to \infty),$$

because the numerator grows linearly and the denominator quadratically. By the ratio test both series converge for all x.

(b) If $p = n$ is a nonnegative integer, the recursion gives $a_{n+2} = 0$ and so all later coefficients of that parity vanish. The terminating series is y_1 for n even and y_2 for n odd. Using the formulas of (a):

$$
\begin{aligned}
&p = 0 : y_1 = 1 \quad (\text{since } a_2 = -\tfrac{2\cdot 0}{2} = 0);\\
&p = 1 : y_2 = x \quad (\text{since } a_3 \propto (p-1) = 0);\\
&p = 2 : y_1 = 1 - \tfrac{2\cdot 2}{2}x^2 = 1 - 2x^2 \quad (a_4 \propto p - 2 = 0);\\
&p = 3 : y_2 = x - \tfrac{2\cdot 2}{3!}x^3 = x - \tfrac{2}{3}x^3 \quad (a_5 \propto p - 3 = 0);\\
&p = 4 : y_1 = 1 - \tfrac{2\cdot 4}{2}x^2 + \tfrac{2^2\cdot 4\cdot 2}{4!}x^4 = 1 - 4x^2 + \tfrac{4}{3}x^4 \quad (a_6 \propto p - 4 = 0);\\
&p = 5 : y_2 = x - \tfrac{2\cdot 4}{3!}x^3 + \tfrac{2^2\cdot 4\cdot 2}{5!}x^5 = x - \tfrac{4}{3}x^3 + \tfrac{4}{15}x^5 \quad (a_7 \propto p - 5 = 0).
\end{aligned}
$$

(c) Each polynomial in (b) has leading term $c\,x^p$; multiplying by $2^p/c$ makes the leading term $2^p x^p$:

$$
\begin{aligned}
&H_0 = 1, \qquad H_1 = 2\cdot x = 2x, \qquad H_2 = -2(1 - 2x^2) = 4x^2 - 2 \quad (c = -2),\\
&H_3 = -12\big(x - \tfrac{2}{3}x^3\big) = 8x^3 - 12x \quad (c = -\tfrac{2}{3},\ 8/c = -12),\\
&H_4 = 12\big(1 - 4x^2 + \tfrac{4}{3}x^4\big) = 16x^4 - 48x^2 + 12 \quad (c = \tfrac{4}{3},\ 16/c = 12),\\
&H_5 = 120\big(x - \tfrac{4}{3}x^3 + \tfrac{4}{15}x^5\big) = 32x^5 - 160x^3 + 120x \quad (c = \tfrac{4}{15},\ 32/c = 120).
\end{aligned}
$$

(d) Let $h_n(x) = (-1)^n e^{x^2}\dfrac{d^n}{dx^n}e^{-x^2}$, so that $u_n := \dfrac{d^n}{dx^n}e^{-x^2} = (-1)^n h_n e^{-x^2}$. Then

$$
u_{n+1} = u_n' = (-1)^n\big(h_n' - 2xh_n\big)e^{-x^2} \quad \text{and} \quad u_{n+1} = (-1)^{n+1}h_{n+1}e^{-x^2},
$$

so

$$
\boxed{h_{n+1} = 2xh_n - h_n'}.
$$

Starting with $h_0 = e^{x^2}e^{-x^2} = 1$:

$$
\begin{aligned}
h_1 &= 2x\cdot 1 - 0 = 2x,\\
h_2 &= 2x(2x) - 2 = 4x^2 - 2,\\
h_3 &= 2x(4x^2 - 2) - 8x = 8x^3 - 4x - 8x = 8x^3 - 12x,\\
h_4 &= 2x(8x^3 - 12x) - (24x^2 - 12) = 16x^4 - 24x^2 - 24x^2 + 12 = 16x^4 - 48x^2 + 12,\\
h_5 &= 2x(16x^4 - 48x^2 + 12) - (64x^3 - 96x)\\
&= 32x^5 - 96x^3 + 24x - 64x^3 + 96x = 32x^5 - 160x^3 + 120x.
\end{aligned}
$$

These coincide with $H_0, \dots, H_5$ of (c).

Answer. (a) $a_{n+2} = -\dfrac{2(p-n)}{(n+1)(n+2)}a_n$, giving y_1, y_2 as stated; ratio $\to 0$, so $R = \infty$. (b), (c) $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$, $H_3 = 8x^3 - 12x$, $H_4 = 16x^4 - 48x^2 + 12$, $H_5 = 32x^5 - 160x^3 + 120x$. (d) The recursion $h_{n+1} = 2xh_n - h_n'$ for $h_n = (-1)^n e^{x^2}D^n e^{-x^2}$ reproduces all of them.

Section 29 Regular Singular Points

Terminology (text of the section): a singular point x_0 of $y'' + P(x)y' + Q(x)y = 0$ is *regular* if $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are both analytic at x_0, and *irregular* otherwise. For a regular singular point at 0 we write $xP(x) = p_0 + p_1 x + \cdots$, $x^2 Q(x) = q_0 + q_1 x + \cdots$; the Frobenius series $y = x^m \sum_{n\ge 0} a_n x^n$ ($a_0 \neq 0$) then leads to the *indicial equation* $m(m-1) + p_0 m + q_0 = 0$. Each Frobenius computation below substitutes $y = \sum a_n x^{n+m}$, $y' = \sum (n+m) a_n x^{n+m-1}$, $y'' = \sum (n+m)(n+m-1) a_n x^{n+m-2}$ and equates the coefficient of each power of x to zero.

Problem 29-1 (Simmons)

Solution. In each case we divide by the coefficient of y'' to get P and Q; singular points are the points where the leading coefficient vanishes (and P or Q genuinely fails to be analytic).

(a) $P = \dfrac{-2(x-1)}{x^3(x-1)} = -\dfrac{2}{x^3}$, $Q = \dfrac{3x}{x^3(x-1)} = \dfrac{3}{x^2(x-1)}$. Singular points: $x = 0$ (P and Q have poles) and $x = 1$ (Q has a pole).

- $x = 0$: $xP = -2/x^2$ is not analytic at 0. *Irregular.*
- $x = 1$: $(x-1)P = -\dfrac{2(x-1)}{x^3}$ and $(x-1)^2 Q = \dfrac{3(x-1)}{x^2}$ are analytic at 1. *Regular.*

(b) The leading coefficient is $x^2(x^2-1)^2$, which vanishes at $0, 1, -1$. Since $(x-1)^2 = (1-x)^2$,

$$P = \frac{-x(1-x)}{x^2(x-1)^2(x+1)^2} = -\frac{1}{x(1-x)(1+x)^2}, \qquad Q = \frac{2}{x^2(x-1)^2(x+1)^2}.$$

- $x = 0$: $xP = -\dfrac{1}{(1-x)(1+x)^2}$ and $x^2 Q = \dfrac{2}{(x-1)^2(x+1)^2}$ are analytic at 0. *Regular.*
- $x = 1$: $(x-1)P = \dfrac{1}{x(1+x)^2}$ (as $(x-1)/(1-x) = -1$) and $(x-1)^2 Q = \dfrac{2}{x^2(x+1)^2}$ are analytic at 1. *Regular.*
- $x = -1$: $(x+1)P = -\dfrac{1}{x(1-x)(1+x)}$ still has the factor $1 + x$ in the denominator, so it is not analytic at -1. *Irregular.* (Here $(x+1)^2 Q = \frac{2}{x^2(x-1)^2}$ is analytic, but one failure suffices.)

(c) $P = \dfrac{2-x}{x^2}$, $Q = 0$; the only singular point is $x = 0$, and $xP = \dfrac{2-x}{x} = \dfrac{2}{x} - 1$ is not analytic there. *Irregular.*

(d) $P = -\dfrac{x+1}{x(3x+1)}$, $Q = \dfrac{2}{x(3x+1)}$; singular points $x = 0$ and $x = -\frac{1}{3}$.

- $x = 0$: $xP = -\dfrac{x+1}{3x+1}$ and $x^2 Q = \dfrac{2x}{3x+1}$ are analytic at 0. *Regular.*
- $x = -\frac{1}{3}$: since $x + \frac{1}{3} = \frac{3x+1}{3}$, $(x+\frac{1}{3})P = -\dfrac{x+1}{3x}$ and $(x+\frac{1}{3})^2 Q = \dfrac{2(3x+1)}{9x}$ are analytic at $-\frac{1}{3}$. *Regular.*

Answer. (a) $x = 0$ irregular, $x = 1$ regular; (b) $x = 0$ and $x = 1$ regular, $x = -1$ irregular; (c) $x = 0$ irregular; (d) $x = 0$ and $x = -\frac{1}{3}$ regular.

Problem 29-2 (Simmons)

Solution. In every case $P = 0$ and $Q = \dfrac{\sin x}{x^k}$ ($k = 0, 1, 2, 3, 4$ in (a)–(e)), where $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$.

(a) $Q = \sin x$ is analytic at 0: *ordinary point.*

(b) $Q = \dfrac{\sin x}{x} = 1 - \dfrac{x^2}{3!} + \dfrac{x^4}{5!} - \cdots$ is analytic at 0 (the singularity is removable): *ordinary point.*

(c) $Q = \dfrac{\sin x}{x^2} = \dfrac{1}{x} - \dfrac{x}{3!} + \cdots$ is not analytic, so 0 is singular. But $xP = 0$ and $x^2Q = \sin x$ are analytic: *regular singular point.*

(d) $Q = \dfrac{\sin x}{x^3} = \dfrac{1}{x^2} - \dfrac{1}{3!} + \cdots$ is not analytic. $xP = 0$ and $x^2Q = \dfrac{\sin x}{x} = 1 - \frac{x^2}{3!} + \cdots$ are analytic: *regular singular point.*

(e) $Q = \dfrac{\sin x}{x^4} = \dfrac{1}{x^3} - \dfrac{1}{6x} + \cdots$ is not analytic, and $x^2Q = \dfrac{\sin x}{x^2} = \dfrac{1}{x} - \dfrac{x}{6} + \cdots$ is not analytic either: *irregular singular point.*

Answer. (a) ordinary point; (b) ordinary point; (c) regular singular point; (d) regular singular point; (e) irregular singular point.

Problem 29-3 (Simmons)

Solution.

(a) Dividing by x^3: $P = \dfrac{\cos 2x - 1}{x^3}$, $Q = \dfrac{2}{x^2}$. Since $\cos 2x - 1 = -2x^2 + \frac{2}{3}x^4 - \cdots$ (because $\cos 2x = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots$),

$$xP = \frac{\cos 2x - 1}{x^2} = -2 + \frac{2}{3}x^2 - \cdots, \qquad x^2Q = 2,$$

both analytic at 0, so $x = 0$ is a regular singular point with $p_0 = -2$, $q_0 = 2$. The indicial equation is

$$m(m-1) - 2m + 2 = 0 \iff m^2 - 3m + 2 = (m-1)(m-2) = 0, \qquad m_1 = 2,\ m_2 = 1.$$

(b) Dividing by $4x^2$: $P = \dfrac{2x^4 - 5x}{4x^2} = \dfrac{2x^3 - 5}{4x}$, $Q = \dfrac{3x^2 + 2}{4x^2}$. Then $xP = \dfrac{2x^3 - 5}{4} = -\dfrac{5}{4} + \dfrac{x^3}{2}$, $x^2Q = \dfrac{3x^2 + 2}{4} = \dfrac{1}{2} + \dfrac{3}{4}x^2$, so $p_0 = -\frac{5}{4}$, $q_0 = \frac{1}{2}$ and the indicial equation is

$$m(m-1) - \tfrac{5}{4}m + \tfrac{1}{2} = 0 \iff m^2 - \tfrac{9}{4}m + \tfrac{1}{2} = 0 \iff 4m^2 - 9m + 2 = (4m-1)(m-2) = 0,$$

with roots $m_1 = 2$, $m_2 = \frac{1}{4}$.

Answer. (a) $m(m-1)-2m+2=0$, $m_1=2$, $m_2=1$; (b) $m(m-1)-\frac{5}{4}m+\frac{1}{2}=0$, $m_1=2$, $m_2=\frac{1}{4}$.

Problem 29-4 (Simmons)

Solution. In each part we work for $x>0$.

(a) $y''+\dfrac{1}{2x}y'+\dfrac{1}{4x}y=0$: $xP=\frac{1}{2}$ and $x^2Q=\frac{x}{4}$ are analytic, so 0 is a regular singular point. Substituting $y=\sum a_n x^{n+m}$,

$$\sum 4(n+m)(n+m-1)a_n x^{n+m-1}+\sum 2(n+m)a_n x^{n+m-1}+\sum a_n x^{n+m}=0.$$

The coefficient of x^{n+m-1} (with $a_{-1}=0$) is

$$2(n+m)(2n+2m-1)a_n+a_{n-1}=0,$$

since $4(n+m)(n+m-1)+2(n+m)=2(n+m)\big(2n+2m-2+1\big)$. For $n=0$ (with $a_0\neq 0$): $2m(2m-1)=0$, the indicial equation, with roots $m=\frac{1}{2}$ and $m=0$.

$m=\frac{1}{2}$: $2(n+\frac{1}{2})(2n)a_n=2n(2n+1)a_n=-a_{n-1}$, i.e. $a_n=-\dfrac{a_{n-1}}{2n(2n+1)}$; so $a_1=-\frac{1}{2\cdot 3}$, $a_2=\frac{1}{2\cdot 3\cdot 4\cdot 5}$, and with $a_0=1$, $a_n=\dfrac{(-1)^n}{(2n+1)!}$. Hence

$$y_1=x^{1/2}\Big(1-\frac{x}{3!}+\frac{x^2}{5!}-\cdots\Big)=\sum_{n=0}^{\infty}\frac{(-1)^n x^{n+1/2}}{(2n+1)!}=\sin\sqrt{x}.$$

$m=0$: $2n(2n-1)a_n=-a_{n-1}$, i.e. $a_n=-\dfrac{a_{n-1}}{2n(2n-1)}$, so with $a_0=1$, $a_n=\dfrac{(-1)^n}{(2n)!}$ and

$$y_2=1-\frac{x}{2!}+\frac{x^2}{4!}-\cdots=\sum_{n=0}^{\infty}\frac{(-1)^n x^n}{(2n)!}=\cos\sqrt{x}.$$

Both series converge for all x (ratio $\to 0$) and the solutions are independent (their ratio $\tan\sqrt{x}$ is not constant; also they have different leading exponents).

Check. With $y=\sin\sqrt{x}$: $y'=\frac{\cos\sqrt{x}}{2\sqrt{x}}$, $y''=-\frac{\sin\sqrt{x}}{4x}-\frac{\cos\sqrt{x}}{4x^{3/2}}$, so $4xy''+2y'+y=-\sin\sqrt{x}-\frac{\cos\sqrt{x}}{\sqrt{x}}+\frac{\cos\sqrt{x}}{\sqrt{x}}+\sin\sqrt{x}=0$.

(b) $y''+\dfrac{3-x}{2x}y'-\dfrac{1}{2x}y=0$: $xP=\frac{3-x}{2}$, $x^2Q=-\frac{x}{2}$ are analytic: regular singular point. Substituting:

$$\sum 2(n+m)(n+m-1)a_n x^{n+m-1}+\sum 3(n+m)a_n x^{n+m-1}-\sum (n+m)a_n x^{n+m}-\sum a_n x^{n+m}=0.$$

Coefficient of x^{n+m-1}: $(n+m)(2n+2m+1)a_n-\big[(n+m-1)+1\big]a_{n-1}=0$, i.e.

$$(n+m)\big[(2n+2m+1)a_n-a_{n-1}\big]=0.$$

$n=0$: $m(2m+1)=0$, so $m=0$ or $m=-\frac{1}{2}$.

$m=0$: $n\big[(2n+1)a_n-a_{n-1}\big]=0$, so for $n\geq 1$, $a_n=\dfrac{a_{n-1}}{2n+1}$, giving $a_n=\dfrac{1}{3\cdot 5\cdots(2n+1)}$ ($a_0=1$): $a_1=\frac{1}{3}$, $a_2=\frac{1}{15}$, …

$$y_1=\sum_{n=0}^{\infty}\frac{x^n}{1\cdot 3\cdot 5\cdots(2n+1)}=1+\frac{x}{3}+\frac{x^2}{15}+\cdots.$$

$m=-\frac{1}{2}$: here $n+m=n-\frac{1}{2}\neq 0$ for $n\geq 1$, so $(2n)a_n=a_{n-1}$, $a_n=\dfrac{1}{2^n n!}$ and

$$y_2=x^{-1/2}\sum_{n=0}^{\infty}\frac{(x/2)^n}{n!}=x^{-1/2}e^{x/2}.$$

The series converge for all x (by the ratio test, or directly for y_2 as an exponential series). The solutions are independent (different leading behaviour: $y_1\to 1$, $y_2\to\infty$ as $x\to 0^+$).

(c) $y''+\dfrac{x+1}{2x}y'+\dfrac{3}{2x}y=0$: $xP=\frac{x+1}{2}$, $x^2Q=\frac{3x}{2}$ analytic: regular singular point. Substituting:

$$\sum 2(n+m)(n+m-1)a_n x^{n+m-1}+\sum(n+m)a_n x^{n+m}+\sum(n+m)a_n x^{n+m-1}+3\sum a_n x^{n+m}=0.$$

Coefficient of x^{n+m-1}: $(n+m)(2n+2m-1)a_n+\big[(n+m-1)+3\big]a_{n-1}=0$, i.e.

$$(n+m)(2n+2m-1)a_n=-(n+m+2)\,a_{n-1}.$$

$n=0$: $m(2m-1)=0$, $m=\frac{1}{2}$ or $m=0$.

$m=\frac{1}{2}$: $(n+\frac{1}{2})(2n)a_n=-(n+\frac{5}{2})a_{n-1}$, i.e. $a_n=-\dfrac{2n+5}{2n(2n+1)}\,a_{n-1}$. With $a_0=1$: $a_1=-\frac{7}{2\cdot 3}=-\frac{7}{6}$, $a_2=-\frac{9}{4\cdot 5}a_1=\frac{21}{40}$, $a_3=-\frac{11}{6\cdot 7}a_2=-\frac{11}{80}$. Multiplying the ratios,

$$a_n=(-1)^n\frac{7\cdot 9\cdots(2n+5)}{(2\cdot 4\cdots 2n)(3\cdot 5\cdots(2n+1))}=(-1)^n\frac{(2n+3)(2n+5)}{15\cdot 2^n n!},$$

because $\frac{7\cdot 9\cdots(2n+5)}{3\cdot 5\cdots(2n+1)}=\frac{(2n+3)(2n+5)}{3\cdot 5}$. Hence

$$y_1=x^{1/2}\Big(1-\frac{7}{6}x+\frac{21}{40}x^2-\frac{11}{80}x^3+\cdots\Big)=\frac{x^{1/2}}{15}\sum_{n=0}^{\infty}\frac{(2n+3)(2n+5)}{n!}\Big(-\frac{x}{2}\Big)^n.$$

The sum can be done in closed form: $(2n+3)(2n+5)=4n(n-1)+20n+15$ and $\sum\frac{n(n-1)}{n!}t^n=t^2e^t$, $\sum\frac{n}{n!}t^n=te^t$, so with $t=-\frac{x}{2}$ the sum is $(4t^2+20t+15)e^t=(x^2-10x+15)e^{-x/2}$:

$$y_1=\frac{1}{15}\,x^{1/2}\,(x^2-10x+15)\,e^{-x/2}.$$

$m=0$: $n(2n-1)a_n=-(n+2)a_{n-1}$, i.e. $a_n=-\dfrac{n+2}{n(2n-1)}a_{n-1}$. With $a_0=1$: $a_1=-\frac{3}{1\cdot 1}=-3$, $a_2=-\frac{4}{2\cdot 3}a_1=2$, $a_3=-\frac{5}{3\cdot 5}a_2=-\frac{2}{3}$, $a_4=-\frac{6}{4\cdot 7}a_3=\frac{1}{7}$. The general term is

$$a_n=(-1)^n\frac{(n+1)(n+2)}{2\,[1\cdot 3\cdots(2n-1)]}=(-1)^n\frac{(n+1)(n+2)\,2^n n!}{2\,(2n)!},$$

so

$$y_2=1-3x+2x^2-\frac{2}{3}x^3+\frac{1}{7}x^4-\cdots.$$

Both series converge for all x (the ratios $|a_n/a_{n-1}|\sim\frac{1}{2n}\to 0$).

Check of y_1: for $y=x^{1/2}(x^2-10x+15)e^{-x/2}/15$ SymPy confirms $2xy''+(x+1)y'+3y\equiv 0$ (see the verification script), and the first coefficients $-\frac{7}{6}$, $\frac{21}{40}$, $-\frac{11}{80}$ agree with the expansion of $(x^2-10x+15)e^{-x/2}/15$.

(d) $y''+\dfrac{1}{2x}y'-\dfrac{x+1}{2x^2}y=0$: $xP=\frac{1}{2}$, $x^2Q=-\frac{x+1}{2}$ analytic: regular singular point. Substituting:

$$\sum 2(n+m)(n+m-1)a_n x^{n+m}+\sum(n+m)a_n x^{n+m}-\sum a_n x^{n+m}-\sum a_n x^{n+m+1}=0.$$

Coefficient of x^{n+m}: $\big[2(n+m)(n+m-1)+(n+m)-1\big]a_n = a_{n-1}$. With $r=n+m$, $2r(r-1)+r-1 = 2r^2-r-1 = (2r+1)(r-1)$, so

$$(2n+2m+1)(n+m-1)\,a_n = a_{n-1}.$$

$n=0$: $(2m+1)(m-1)=0$, $m=1$ or $m=-\tfrac12$.

$m=1$: $(2n+3)n\,a_n = a_{n-1}$, so $a_n = \dfrac{a_{n-1}}{n(2n+3)}$; $a_1=\tfrac15$, $a_2=\tfrac{1}{2\cdot7}\cdot\tfrac15=\tfrac{1}{70}$, $a_3=\tfrac{1}{3\cdot9}\cdot\tfrac{1}{70}=\tfrac{1}{1890}$, and in general $a_n = \dfrac{1}{n!\cdot 5\cdot 7\cdots(2n+3)}$:

$$y_1 = x\Big(1+\frac{x}{5}+\frac{x^2}{70}+\frac{x^3}{1890}+\cdots\Big) = x\sum_{n=0}^{\infty}\frac{x^n}{n!\,5\cdot7\cdots(2n+3)}.$$

$m=-\tfrac12$: $(2n)(n-\tfrac32)a_n = n(2n-3)a_n = a_{n-1}$, so $a_n = \dfrac{a_{n-1}}{n(2n-3)}$: $a_1=\tfrac{1}{1\cdot(-1)}=-1$, $a_2=\tfrac{-1}{2\cdot1}=-\tfrac12$, $a_3=\tfrac{-1/2}{3\cdot3}=-\tfrac{1}{18}$, and in general $a_n = -\dfrac{1}{n!\cdot 1\cdot3\cdots(2n-3)}$ for $n\ge1$ (empty product $=1$ when $n=1$):

$$y_2 = x^{-1/2}\Big(1-x-\frac{x^2}{2}-\frac{x^3}{18}-\cdots\Big) = x^{-1/2}\Big(1-\sum_{n=1}^{\infty}\frac{x^n}{n!\,1\cdot3\cdots(2n-3)}\Big).$$

Both converge for all x (ratios $\to 0$). The two roots differ by $\tfrac32$, not an integer, so no complication arises, and the solutions are independent (different leading exponents).

Answer. (a) $y_1 = x^{1/2}(1-\tfrac{x}{3!}+\tfrac{x^2}{5!}-\cdots) = \sin\sqrt{x}$, $y_2 = 1-\tfrac{x}{2!}+\tfrac{x^2}{4!}-\cdots = \cos\sqrt{x}$.
(b) $y_1 = \sum_{n\ge0}\frac{x^n}{1\cdot3\cdot5\cdots(2n+1)}$, $y_2 = x^{-1/2}\sum_{n\ge0}\frac{x^n}{2^n n!} = x^{-1/2}e^{x/2}$.
(c) $y_1 = x^{1/2}(1-\tfrac76 x+\tfrac{21}{40}x^2-\cdots) = \tfrac{1}{15}x^{1/2}(x^2-10x+15)e^{-x/2}$, $y_2 = 1-3x+2x^2-\cdots$.
(d) $y_1 = x(1+\tfrac15 x+\tfrac{1}{70}x^2+\cdots)$, $y_2 = x^{-1/2}(1-x-\tfrac12 x^2-\cdots)$.

Problem 29-5 (Simmons)

Solution. Here $xP=1$ and $x^2Q=x^2$, so $p_0=1$, $q_0=0$ and the origin is a regular singular point. Substituting $y=\sum a_n x^{n+m}$:

$$\sum(n+m)(n+m-1)a_n x^{n+m}+\sum(n+m)a_n x^{n+m}+\sum a_n x^{n+m+2}=0,$$

and the coefficient of x^{n+m} is

$$(n+m)^2 a_n + a_{n-2} = 0 \qquad (a_{-1}=a_{-2}=0),$$

because $(n+m)(n+m-1)+(n+m)=(n+m)^2$. For $n=0$ (with $a_0\neq0$) the indicial equation is $m(m-1)+m=m^2=0$, which has only the (double) root $m=0$.

With $m=0$: $n=1$ gives $a_1=0$, and for $n\ge2$

$$a_n = -\frac{a_{n-2}}{n^2}.$$

Hence all odd coefficients vanish ($a_1=0\Rightarrow a_3=a_5=\cdots=0$), and for even indices $n=2k$, $a_{2k} = -\dfrac{a_{2k-2}}{4k^2}$, so

$$a_{2k} = \frac{(-1)^k a_0}{(4\cdot1^2)(4\cdot2^2)\cdots(4k^2)} = \frac{(-1)^k a_0}{4^k(k!)^2} = \frac{(-1)^k a_0}{2^{2k}(k!)^2}.$$

Taking $a_0 = 1$,

$$y = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k}(k!)^2} x^{2k} = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \cdots,$$

which is $J_0(x)$. The ratio of consecutive terms is $\frac{x^2}{4(k+1)^2} \to 0$, so it converges for all x. Since the indicial equation has a single root, the method yields only this one Frobenius series; a second independent solution is of a different type (it contains a logarithm), as discussed in the text and in Section 30.

Answer. Indicial equation $m^2 = 0$ (only the root $m = 0$); recursion $a_n = -a_{n-2}/n^2$ gives $y = \sum_{n\ge 0} \dfrac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$.

Problem 29-6 (Simmons)

Solution.

(a) $P = \dfrac{1}{x^2}$, $Q = -\dfrac{1}{x^3}$. Then $xP = \dfrac{1}{x}$ is not analytic at 0 (and neither is $x^2Q = -\frac{1}{x}$). So $x = 0$ is a singular point which is not regular: *irregular*.

(b) $y_1 = x$ is a solution: $y_1'' + \frac{1}{x^2}y_1' - \frac{1}{x^3}y_1 = 0 + \frac{1}{x^2} - \frac{1}{x^2} = 0$. By Section 16, $y_2 = y_1 \displaystyle\int \frac{e^{-\int P\,dx}}{y_1^2}\,dx$. Here $\int P\,dx = -\frac{1}{x}$, so $e^{-\int P\,dx} = e^{1/x}$ and

$$y_2 = x \int \frac{e^{1/x}}{x^2}\,dx.$$

Since $\dfrac{d}{dx}e^{1/x} = -\dfrac{e^{1/x}}{x^2}$, the integral is $-e^{1/x}$ (taking the constant 0), so

$$y_2(x) = -x\,e^{1/x} \qquad (x \neq 0).$$

Check. $y_2' = -e^{1/x} + \dfrac{e^{1/x}}{x}$, $y_2'' = \dfrac{e^{1/x}}{x^2} - \dfrac{e^{1/x}}{x^2} - \dfrac{e^{1/x}}{x^3} = -\dfrac{e^{1/x}}{x^3}$, so $y_2'' + \frac{1}{x^2}y_2' - \frac{1}{x^3}y_2 = e^{1/x}\big[-\frac{1}{x^3} - \frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^2}\big] = 0$. The Wronskian $y_1y_2' - y_2y_1' = x\big(-e^{1/x} + \frac{e^{1/x}}{x}\big) + xe^{1/x} = e^{1/x} \neq 0$, so y_1, y_2 are independent.

(c) Suppose, for $x > 0$, that $-xe^{1/x} = x^m \sum_{n=0}^{\infty} a_n x^n$ with $a_0 \neq 0$, the series having a positive radius of convergence. Put $g(x) = \sum a_n x^n$; then $g(x) \to a_0$ as $x \to 0^+$ and

$$e^{1/x} = -x^{m-1} g(x).$$

Choose an integer $N \ge 1 - m$ and multiply by x^N: the right side becomes $-x^{m-1+N}g(x)$ with $m - 1 + N \ge 0$, which stays bounded as $x \to 0^+$. But the left side satisfies, from $e^u \ge \dfrac{u^{N+1}}{(N+1)!}$ with $u = 1/x$,

$$x^N e^{1/x} \ge \frac{x^N}{(N+1)!\,x^{N+1}} = \frac{1}{(N+1)!\,x} \longrightarrow \infty \qquad (x \to 0^+).$$

This is a contradiction. Hence y_2 is not a Frobenius series (for any exponent m, even a negative integer or irrational one). The reason: $e^{1/x}$ has an essential singularity at 0, far worse than any pole or power x^m; this is what the irregular singular point does.

Answer. (a) $xP = 1/x$ is not analytic; (b) $y_2 = -xe^{1/x}$; (c) shown: $x^N e^{1/x} \to \infty$ while $x^{N+m-1}g(x)$ stays bounded.

Problem 29-7 (Simmons)

Solution. Insert $y = \sum_{n\ge 0} a_n x^{n+m}$ with $a_0 \neq 0$. The three terms of the equation produce

$$y'' = \sum (n+m)(n+m-1)a_n x^{n+m-2}, \quad \frac{p}{x^b}y' = \sum p(n+m)a_n x^{n+m-1-b}, \quad \frac{q}{x^c}y = \sum q a_n x^{n+m-c}.$$

The lowest power of x that can occur is x^{m-d}, where $d = \max\{2,\, b+1,\, c\}$, and it is produced only by the $n = 0$ terms (higher n give higher powers). The coefficient of x^{m-d} must vanish; as $a_0 \neq 0$ this gives the indicial equation, which collects those of the three terms whose depth (2, $b+1$, c respectively) equals d:

$$\big[\, m(m-1) \text{ if } d = 2 \,\big] + \big[\, pm \text{ if } d = b+1 \,\big] + \big[\, q \text{ if } d = c \,\big] = 0. \tag{$*$}$$

(a) $b = 2$, $c = 3$: $d = \max\{2,3,3\} = 3$, attained by the second and third terms, but not by y'' (depth 2). So $(*)$ reads

$$pm + q = 0 \qquad \Longrightarrow \qquad m = -\frac{q}{p},$$

(allowed since $p \neq 0$). Thus the only possible exponent is $m = -q/p$, and at most one Frobenius solution $x^m \sum a_n x^n$ can exist (and in general, as in Problem 8, it need not converge).

(b) The equation $(*)$ is quadratic in m exactly when the term $m(m-1)$ is present, i.e. when $d = 2$. Since $d = \max\{2, b+1, c\}$ and b, c are positive integers, $d = 2$ holds if and only if $b+1 \le 2$ and $c \le 2$, i.e. $b = 1$ and $c \le 2$.

Conversely, if $b > 1$ or $c > 2$, then $d > 2$ and the $m(m-1)$ term is absent. Then $(*)$ is at most linear in m: it is $pm = 0$ (if $d = b+1 > c$), $pm + q = 0$ (if $d = b+1 = c$) or $q = 0$ (if $d = c > b+1$). The last is impossible since $q \neq 0$, so no Frobenius series exists at all in that case, and in the other two cases there is a single admissible m ($m = 0$ or $m = -q/p$). Hence we cannot hope for two Frobenius solutions.

Explicitly, for $b = 1$: if $c = 2$ then $(*)$ is $m(m-1) + pm + q = 0$, and if $c = 1$ then $(*)$ is $m(m-1) + pm = 0$ (roots 0 and $1-p$); in both cases there are two roots to work with, in agreement with the regularity conditions ($xP = p$ and $x^2Q = qx^{2-c}$ analytic exactly for $b = 1$, $c \le 2$).

Answer. (a) The lowest-power terms give $pm + q = 0$, so $m = -q/p$ is the only possible exponent. (b) The indicial condition is quadratic iff $d = \max\{2, b+1, c\} = 2$, i.e. iff $b = 1$ and $c \le 2$.

Problem 29-8 (Simmons)

Solution. Here $P = \dfrac{3x-1}{x^2}$, so $xP = 3 - \dfrac{1}{x}$ is not analytic: $x = 0$ is irregular. Insert $y = \sum a_n x^{n+m}$ ($a_0 \neq 0$):

$$\sum (n+m)(n+m-1)a_n x^{n+m} + \sum 3(n+m)a_n x^{n+m} - \sum (n+m)a_n x^{n+m-1} + \sum a_n x^{n+m} = 0.$$

The lowest power is x^{m-1}, coming only from $-y'$ with $n = 0$; its coefficient is $-ma_0 = 0$. As $a_0 \neq 0$, $m = 0$.

For $m = 0$ the coefficient of x^{n-1} ($n \geq 1$) collects the term $-na_n$ from $-y'$ and the terms of $x^2y'' + 3xy' + y$ with index $n-1$:

$$-na_n + \big[(n-1)(n-2) + 3(n-1) + 1\big]a_{n-1} = 0.$$

Since $(n-1)(n-2) + 3(n-1) + 1 = (n-1)(n+1) + 1 = n^2$, this is $-na_n + n^2 a_{n-1} = 0$, i.e.

$$a_n = n\, a_{n-1} \qquad (n \geq 1).$$

With $a_0 = 1$: $a_1 = 1$, $a_2 = 2 \cdot 1$, $a_3 = 3 \cdot 2 \cdot 1$, $\dots$, $a_n = n!$. So the formal series is

$$y = \sum_{n=0}^{\infty} n!\, x^n = 1 + x + 2x^2 + 6x^3 + 24x^4 + \cdots .$$

Divergence. For fixed $x \neq 0$ the ratio of consecutive terms is $\left|\dfrac{(n+1)!\,x^{n+1}}{n!\,x^n}\right| = (n+1)|x| \to \infty$, so by the ratio test the series diverges for every $x \neq 0$; its radius of convergence is 0. It converges only at $x = 0$ and therefore does not define a function that solves the equation on any interval.

Answer. $m = 0$ and $a_n = na_{n-1}$, so $a_n = n!$ and $y = \sum n!\, x^n$, with radius of convergence 0 (ratio $(n+1)|x| \to \infty$).

Section 30 Regular Singular Points (Continued)

Throughout we use the Frobenius method of the text. For $y''+P(x)y'+Q(x)y=0$ with a regular singular point at $x=0$ we write $xP(x)=\sum p_n x^n$, $x^2Q(x)=\sum q_n x^n$, and substitute $y=\sum a_n x^{m+n}$. The coefficient of the lowest power gives the indicial equation $f(m)=m(m-1)+p_0 m+q_0=0$, and the remaining coefficients give $a_n f(m+n)=(\text{terms in } a_0,\dots,a_{n-1})$. We work on $x>0$.

Problem 30-1 (Simmons)

Solution. Here $P=-3/x$, $Q=(4x+4)/x^2$, so $xP=-3$, $x^2Q=4+4x$: $p_0=-3$, $q_0=4$, and $x=0$ is a regular singular point. The indicial equation is

$$m(m-1)-3m+4=m^2-4m+4=(m-2)^2=0,$$

so $m_1=m_2=2$. Equal exponents give only one Frobenius solution (Case A), namely the one with $m=2$.

Put $y=\sum_{n\ge 0}a_n x^{n+2}$. Then $y'=\sum(n+2)a_n x^{n+1}$, $y''=\sum(n+2)(n+1)a_n x^n$, and

$$\begin{aligned} x^2y''-3xy'+4y+4xy &= \sum_{n\ge0}\big[(n+2)(n+1)-3(n+2)+4\big]a_n x^{n+2}+4\sum_{n\ge0}a_n x^{n+3}\\ &= \sum_{n\ge0}n^2a_n x^{n+2}+4\sum_{n\ge1}a_{n-1}x^{n+2}, \end{aligned}$$

because $(n+2)(n+1)-3(n+2)+4=(n+2)(n-2)+4=n^2$. Setting the coefficient of x^{n+2} equal to zero:

$$n^2a_n+4a_{n-1}=0\quad(n\ge1)\qquad\Longrightarrow\qquad a_n=-\frac{4}{n^2}a_{n-1}.$$

(For $n=0$ we get $0\cdot a_0=0$, so a_0 is arbitrary; take $a_0=1$.) Hence

$$a_n=\frac{(-4)^n}{(n!)^2},\qquad a_1=-4,\quad a_2=\frac{16}{4}=4,\quad a_3=-\frac{64}{36}=-\frac{16}{9},\ \dots$$

and

$$y=x^2\sum_{n=0}^{\infty}\frac{(-4)^n}{(n!)^2}x^n=x^2\left(1-4x+4x^2-\tfrac{16}{9}x^3+\cdots\right).$$

The ratio test gives $|a_nx^n/a_{n-1}x^{n-1}|=4|x|/n^2\to0$, so the series converges for all x.

Remark. Since $J_0(z)=\sum(-1)^n(z/2)^{2n}/(n!)^2$, this solution is $x^2J_0(4\sqrt{x})$.

Answer. $y=x^2\displaystyle\sum_{n=0}^{\infty}\frac{(-4)^n}{(n!)^2}x^n=x^2(1-4x+4x^2-\cdots)$ (times an arbitrary constant).

Problem 30-2 (Simmons)

Solution. Dividing by $4x^2$: $y''-2y'+\left(1+\frac{1}{4x^2}\right)y=0$, so $xP=-2x$, $x^2Q=\frac14+x^2$, giving $p_0=0$, $q_0=\frac14$ and the indicial equation

$$m(m-1)+\tfrac14=\left(m-\tfrac12\right)^2=0,\qquad m_1=m_2=\tfrac12.$$

Equal exponents: there is only one Frobenius solution. Put $y=\sum a_n x^{n+1/2}$. Collecting the coefficient of $x^{n+1/2}$ in $4x^2y''-8x^2y'+(4x^2+1)y$:

$$4x^2y''\to4\left(n+\tfrac12\right)\left(n-\tfrac12\right)a_n,\qquad -8x^2y'\to-8\left(n-\tfrac12\right)a_{n-1},$$

$$4x^2y \to 4a_{n-2}, \qquad\qquad y \to a_n.$$

Since $4(n^2 - \frac{1}{4}) + 1 = 4n^2$, the recursion is

$$4n^2a_n - 4(2n-1)a_{n-1} + 4a_{n-2} = 0, \qquad \text{i.e.} \qquad n^2a_n = (2n-1)a_{n-1} - a_{n-2} \quad (a_{-1} = 0).$$

With $a_0 = 1$: $a_1 = 1$, $a_2 = (3-1)/4 = \frac{1}{2}$, $a_3 = (\frac{5}{2} - 1)/9 = \frac{1}{6}$. This suggests $a_n = 1/n!$, and indeed this satisfies the recursion:

$$\frac{(2n-1)}{(n-1)!} - \frac{1}{(n-2)!} = \frac{(2n-1)-(n-1)}{(n-1)!} = \frac{n}{(n-1)!} = \frac{n^2}{n!}.$$

Hence

$$y_1 = x^{1/2}\sum_{n=0}^{\infty}\frac{x^n}{n!} = x^{1/2}e^x.$$

For a second solution use the method of Section 16: $P = -2$, so $e^{-\int P\,dx} = e^{2x}$ and

$$v' = \frac{e^{2x}}{y_1^2} = \frac{e^{2x}}{xe^{2x}} = \frac{1}{x}, \qquad v = \log x, \qquad y_2 = y_1 v = x^{1/2}e^x \log x.$$

(This has the form $y_1 \log x + x^{1/2}\sum c_n x^n$ predicted by the text, with all $c_n = 0$.) These are independent since $y_2/y_1 = \log x$ is not constant.

Check. $y_1' = (\frac{1}{2}x^{-1/2} + x^{1/2})e^x$, $y_1'' = (-\frac{1}{4}x^{-3/2} + x^{-1/2} + x^{1/2})e^x$. Then $4x^2y_1'' - 8x^2y_1' + (4x^2 + 1)y_1 = e^x[-x^{1/2} + 4x^{3/2} + 4x^{5/2} - 4x^{3/2} - 8x^{5/2} + 4x^{5/2} + x^{1/2}] = 0$.

Answer. $y = c_1x^{1/2}e^x + c_2x^{1/2}e^x\log x$.

Problem 30-3 (Simmons)

Solution.

(a) Multiply by x: $x^2y'' + 2xy' + x^2y = 0$, so $p_0 = 2$, $q_0 = 0$ (and $p_n = 0$ for $n \geq 1$, $q_2 = 1$, other $q_n = 0$). Indicial equation: $m(m-1) + 2m = m(m+1) = 0$, so $m_1 = 0$, $m_2 = -1$ (they differ by the integer 1). Substituting $y = \sum a_n x^{n+m}$ into $xy'' + 2y' + xy = 0$:

$$\sum_{n\geq 0}(n+m)(n+m+1)a_nx^{n+m-1} + \sum_{n\geq 0}a_nx^{n+m+1} = 0.$$

The coefficient of x^{n+m-1} is

$$(n+m)(n+m+1)a_n + a_{n-2} = 0 \qquad (a_{-1} = a_{-2} = 0).$$

Case $m = 0$: $n(n+1)a_n = -a_{n-2}$. For $n = 1$: $2a_1 = 0$, so $a_1 = 0$ and then all odd $a_n = 0$. For $n = 2k$: $(2k)(2k+1)a_{2k} = -a_{2k-2}$, so with $a_0 = 1$, $a_{2k} = (-1)^k/(2k+1)!$ and

$$y_1 = \sum_{k=0}^{\infty}\frac{(-1)^kx^{2k}}{(2k+1)!} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots = \frac{\sin x}{x}.$$

Case $m = -1$: $n(n-1)a_n = -a_{n-2}$. Now $n = 0$ is trivial (a_0 arbitrary), and for $n = 1$ the equation reads $0 \cdot a_1 = -a_{-1} = 0$, which holds for every a_1 (the right side vanishes: Case C of the text), so we may choose $a_1 = 0$; then all odd a_n vanish. For $n = 2k$: $(2k)(2k-1)a_{2k} = -a_{2k-2}$, so $a_{2k} = (-1)^k/(2k)!$ and

$$y_2 = x^{-1}\sum_{k=0}^{\infty}\frac{(-1)^kx^{2k}}{(2k)!} = x^{-1}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) = \frac{\cos x}{x}.$$

(A nonzero a_1 would only add a multiple of y_1.) The solutions are independent: $y_2/y_1 = \cot x$ is not constant.

(b) Here $P = -1$, $Q = (x^2-2)/x^2$: $xP = -x$, $x^2Q = -2 + x^2$, so $p_0 = 0$, $q_0 = -2$ and

$$m(m-1) - 2 = (m-2)(m+1) = 0, \qquad m_1 = 2,\ m_2 = -1 \quad (m_1 - m_2 = 3).$$

Substituting $y = \sum a_n x^{n+m}$, the coefficient of x^{n+m} in $x^2y'' - x^2y' + x^2y - 2y$ is

$$\big[(n+m)(n+m-1) - 2\big]a_n - (n+m-1)a_{n-1} + a_{n-2} = 0.$$

Since $(s)(s-1) - 2 = (s-2)(s+1)$ with $s = n+m$, this reads

$$(n+m-2)(n+m+1)\,a_n = (n+m-1)a_{n-1} - a_{n-2}.$$

Case $m = 2$: $n(n+3)a_n = (n+1)a_{n-1} - a_{n-2}$. With $a_0 = 1$:

$$a_1 = \tfrac{2}{4} = \tfrac{1}{2},$$
$$a_2 = \tfrac{3\cdot\frac{1}{2} - 1}{2\cdot 5} = \tfrac{1}{20},$$
$$a_3 = \tfrac{4\cdot\frac{1}{20} - \frac{1}{2}}{3\cdot 6} = \tfrac{-3/10}{18} = -\tfrac{1}{60}.$$

Thus

$$y_1 = x^2\left(1 + \tfrac{1}{2}x + \tfrac{1}{20}x^2 - \tfrac{1}{60}x^3 + \cdots\right).$$

Case $m = -1$: $(n-3)n\,a_n = (n-2)a_{n-1} - a_{n-2}$. With $a_0 = 1$:

$$n = 1: \ -2a_1 = -a_0 \Rightarrow a_1 = \tfrac{1}{2},$$
$$n = 2: \ -2a_2 = 0\cdot a_1 - a_0 \Rightarrow a_2 = \tfrac{1}{2},$$
$$n = 3: 0\cdot a_3 = 1\cdot a_2 - a_1 = \tfrac{1}{2} - \tfrac{1}{2} = 0.$$

At $n = 3$, where $f(m+n) = 0$, the right side is also zero (Case C), so a_3 is unrestricted; we choose $a_3 = 0$ (any other choice adds a multiple of y_1). Then

$$n = 4: \ 4a_4 = 2a_3 - a_2 = -\tfrac{1}{2} \Rightarrow a_4 = -\tfrac{1}{8},$$

so

$$y_2 = x^{-1}\left(1 + \tfrac{1}{2}x + \tfrac{1}{2}x^2 + 0\cdot x^3 - \tfrac{1}{8}x^4 + \cdots\right).$$

The two solutions are independent because their leading powers x^2 and x^{-1} differ.

(c) Multiply by x: $x^2y'' - xy' + 4x^4y = 0$, so $p_0 = -1$, $q_0 = 0$ and $q_4 = 4$ is the only other nonzero coefficient. The indicial equation is $m(m-1) - m = m(m-2) = 0$: $m_1 = 2$, $m_2 = 0$. Substituting $y = \sum a_n x^{n+m}$ into $xy'' - y' + 4x^3y = 0$:

$$\sum_{n\ge 0}(n+m)(n+m-2)a_n x^{n+m-1} + 4\sum_{n\ge 0} a_n x^{n+m+3} = 0,$$

and the coefficient of x^{n+m-1} gives

$$(n+m)(n+m-2)a_n = -4a_{n-4} \qquad (a_{-1} = a_{-2} = a_{-3} = a_{-4} = 0).$$

Case $m = 2$: $n(n+2)a_n = -4a_{n-4}$. For $n = 1, 2, 3$ this forces $a_1 = a_2 = a_3 = 0$, hence $a_n = 0$ unless $4 \mid n$. For $n = 4k$: $4k(4k+2)a_{4k} = -4a_{4k-4}$, i.e. $a_{4k} = -\dfrac{a_{4k-4}}{2k(2k+1)}$. With $a_0 = 1$, $a_{4k} = \dfrac{(-1)^k}{(2k+1)!}$ (induction: $\frac{(-1)^k}{(2k+1)!}\cdot 2k(2k+1) = -\frac{(-1)^{k-1}}{(2k-1)!}$), so

$$y_1 = x^2\sum_{k=0}^{\infty}\frac{(-1)^k x^{4k}}{(2k+1)!} = \sum_{k=0}^{\infty}\frac{(-1)^k (x^2)^{2k+1}}{(2k+1)!} = \sin x^2.$$

Case $m = 0$: $n(n-2)a_n = -4a_{n-4}$. For $n = 1$: $-a_1 = 0$. For $n = 2$: $0 \cdot a_2 = -4a_{-2} = 0$, so a_2 is unrestricted (Case C); we choose $a_2 = 0$ (a nonzero a_2 would only add a multiple of y_1). For $n = 3$: $3a_3 = 0$. Hence $a_n = 0$ unless $4 \mid n$, and for $n = 4k$, $4k(4k-2)a_{4k} = -4a_{4k-4}$, i.e. $a_{4k} = -\dfrac{a_{4k-4}}{2k(2k-1)}$, so $a_{4k} = (-1)^k/(2k)!$ and

$$y_2 = \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k}}{(2k)!} = \cos x^2.$$

These are independent ($y_2/y_1 = \cot x^2$).

Answer. (a) $y_1 = \dfrac{\sin x}{x} = 1 - \dfrac{x^2}{3!} + \dfrac{x^4}{5!} - \cdots$, $\quad y_2 = \dfrac{\cos x}{x} = x^{-1}\Big(1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \cdots\Big)$;
(b) $y_1 = x^2\big(1 + \frac{1}{2}x + \frac{1}{20}x^2 - \frac{1}{60}x^3 + \cdots\big)$, $\quad y_2 = x^{-1}\big(1 + \frac{1}{2}x + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \cdots\big)$;
(c) $y_1 = x^2\big(1 - \frac{x^4}{3!} + \frac{x^8}{5!} - \cdots\big) = \sin x^2$, $\quad y_2 = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \cdots = \cos x^2$.

Problem 30-4 (Simmons)

Solution. $xP = 1$, $x^2Q = x^2 - 1$, so $p_0 = 1$, $q_0 = -1$ and the indicial equation is $m(m-1) + m - 1 = m^2 - 1 = 0$: $m_1 = 1$, $m_2 = -1$, and $m_1 - m_2 = 2$.

Substituting $y = \sum a_n x^{n+m}$, the coefficient of x^{n+m} is

$$\big[(n+m)^2 - 1\big]a_n + a_{n-2} = 0 \qquad (a_{-1} = a_{-2} = 0).$$

Exponent $m = -1$: $[(n-1)^2 - 1]a_n = n(n-2)a_n = -a_{n-2}$. For $n = 1$: $-a_1 = 0$. For $n = 2$:

$$0 \cdot a_2 = -a_0 \neq 0$$

(as $a_0 \neq 0$). This is impossible, so no Frobenius series with $m = -1$ exists (Case B). Hence there is only one Frobenius solution, that with $m = m_1 = 1$.

Exponent $m = 1$: $[(n+1)^2 - 1]a_n = n(n+2)a_n = -a_{n-2}$. For $n = 1$: $3a_1 = 0$, so $a_1 = 0$ and all odd coefficients vanish. For $n = 2k$: $4k(k+1)a_{2k} = -a_{2k-2}$, so, with $a_0 = 1$,

$$a_{2k} = \frac{(-1)^k}{4^k k!\,(k+1)!}$$

(induction: $a_{2k} = -\dfrac{a_{2k-2}}{4k(k+1)} = \dfrac{(-1)^k}{4^{k-1}(k-1)!\,k! \cdot 4k(k+1)} = \dfrac{(-1)^k}{4^k\,k!\,(k+1)!}$, since $k\,(k-1)! = k!$ and $(k+1)\,k! = (k+1)!$). Thus

$$y = x\left(1 - \frac{x^2}{2^2\,2!} + \frac{x^4}{2^4\,2!\,3!} - \cdots\right) = x\sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k}k!(k+1)!}x^{2k}.$$

The ratio of successive terms is $x^2/(4(k+1)(k+2)) \to 0$, so the series converges for all x.

Remark. This is $2J_1(x)$.

Answer. $m_1 = 1$, $m_2 = -1$; no Frobenius series for $m = -1$ (the recursion demands $0 \cdot a_2 = -a_0 \neq 0$). The only Frobenius solution is $y = x\Big(1 - \dfrac{x^2}{2^2\,2!} + \dfrac{x^4}{2^4\,2!\,3!} - \cdots\Big)$.

Problem 30-5 (Simmons)

Solution. $p_0 = 1$, $q_0 = -\tfrac14$, so the indicial equation is $m^2 - \tfrac14 = 0$: $m_1 = \tfrac12$, $m_2 = -\tfrac12$, $m_1 - m_2 = 1$. The coefficient of x^{n+m} in the substituted equation is

$$\left[(n+m)^2 - \tfrac14\right]a_n + a_{n-2} = 0.$$

Exponent $m = -\tfrac12$: $(n-\tfrac12)^2 - \tfrac14 = n(n-1)$, so $n(n-1)a_n = -a_{n-2}$. For $n = 1$ the equation reads $0 \cdot a_1 = -a_{-1} = 0$: the right side is zero, so a_1 is unrestricted (Case C) and the process does not break down. Choose $a_1 = 0$; then all odd coefficients vanish. For $n = 2k$: $2k(2k-1)a_{2k} = -a_{2k-2}$, so $a_{2k} = (-1)^k/(2k)!$ and

$$y_2 = x^{-1/2}\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k}}{(2k)!} = x^{-1/2}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) = \frac{\cos x}{\sqrt{x}}.$$

Exponent $m = \tfrac12$: $(n+\tfrac12)^2 - \tfrac14 = n(n+1)$, so $n(n+1)a_n = -a_{n-2}$. Then $a_1 = 0$, and $a_{2k} = (-1)^k/(2k+1)!$, giving

$$y_1 = x^{1/2}\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k}}{(2k+1)!} = x^{1/2}\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots\right) = \frac{\sin x}{\sqrt{x}}.$$

They are independent since $y_2/y_1 = \cot x$ is not constant. (Had we kept $a_1 \neq 0$ in the case $m = -\tfrac12$, the odd coefficients would give $a_1 x^{-1/2}\sum(-1)^k x^{2k+1}/(2k+1)! = a_1 y_1$, only a multiple of y_1.)

Answer. $m_1 - m_2 = 1$; $y_1 = x^{1/2}\left(1 - \dfrac{x^2}{3!} + \dfrac{x^4}{5!} - \cdots\right) = \dfrac{\sin x}{\sqrt{x}}$ and $y_2 = x^{-1/2}\left(1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \cdots\right) = \dfrac{\cos x}{\sqrt{x}}$.

Problem 30-6 (Simmons)

Solution. Bessel's equation of order 0 is $x^2y'' + xy' + x^2y = 0$; write $L[y] = x^2y'' + xy' + x^2y$. Its exponents are $m_1 = m_2 = 0$ ($p_0 = 1$, $q_0 = 0$, $m^2 = 0$), and the Frobenius solution (Problem 29-5) is

$$y_1 = \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}(n!)^2}x^{2n} = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \cdots.$$

(It satisfies $4n^2a_{2n} = -a_{2n-2}$, from the coefficient of x^{2n} in $L[\sum a_k x^k]$, namely $k^2a_k + a_{k-2}$ with $k = 2n$.)

Following (11) we seek $y_2 = y_1\log x + u$, $u = \sum_{k\geq 0} c_k x^k$. We have

$$(y_1\log x)' = y_1'\log x + \frac{y_1}{x}, \qquad (y_1\log x)'' = y_1''\log x + \frac{2y_1'}{x} - \frac{y_1}{x^2},$$

so

$$L[y_1\log x] = \log x\, L[y_1] + 2xy_1' - y_1 + y_1 = 2xy_1',$$

because $L[y_1] = 0$. Therefore $L[y_2] = 0$ is equivalent to

$$L[u] = -2xy_1'.$$

Compute the right side:

$$y_1' = \sum_{n\geq 1} \frac{(-1)^n 2n}{2^{2n}(n!)^2} x^{2n-1}, \qquad -2xy_1' = \sum_{n\geq 1} \frac{(-1)^{n+1} 4n}{2^{2n}(n!)^2} x^{2n}.$$

For the left side, $L[\sum c_k x^k] = \sum [k^2 c_k + c_{k-2}] x^k$ (with $c_{-1} = c_{-2} = 0$). Matching coefficients of x^k:

- k odd: $k^2 c_k + c_{k-2} = 0$ with $k = 1$ gives $c_1 = 0$, hence all odd $c_k = 0$.
- $k = 0$: $0 \cdot c_0 = 0$, so c_0 is arbitrary; we take $c_0 = 0$ (a nonzero c_0 only adds $c_0 y_1$).
- $k = 2n \geq 2$:

$$4n^2 c_{2n} + c_{2n-2} = \frac{(-1)^{n+1} 4n}{2^{2n}(n!)^2}. \tag{5.1}$$

For $n = 1$: $4c_2 + 0 = \dfrac{4}{4} = 1$, so $c_2 = \frac{1}{4} = \dfrac{(-1)^2}{2^2(1!)^2} \cdot 1$. We claim

$$c_{2n} = \frac{(-1)^{n+1}}{2^{2n}(n!)^2} H_n, \qquad H_n = 1 + \tfrac{1}{2} + \cdots + \tfrac{1}{n}, \quad H_0 = 0.$$

This holds for $n = 1$. If it holds for $n - 1$, then $c_{2n-2} = \dfrac{(-1)^n H_{n-1}}{2^{2n-2}((n-1)!)^2} = \dfrac{(-1)^n 4n^2 H_{n-1}}{2^{2n}(n!)^2}$, and

$$4n^2 c_{2n} + c_{2n-2} = \frac{(-1)^{n+1} 4n^2}{2^{2n}(n!)^2}\big(H_n - H_{n-1}\big) = \frac{(-1)^{n+1} 4n^2}{2^{2n}(n!)^2} \cdot \frac{1}{n} = \frac{(-1)^{n+1} 4n}{2^{2n}(n!)^2},$$

which is (5.1). Hence

$$y_2 = y_1 \log x + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2^{2n}(n!)^2} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) x^{2n} = y_1 \log x + \frac{x^2}{4} - \frac{3x^4}{128} + \cdots.$$

It is independent of y_1 because of the term $\log x$.

Answer. Shown: $y_2 = y_1 \log x + \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{2^{2n}(n!)^2}\big(1 + \frac{1}{2} + \cdots + \frac{1}{n}\big) x^{2n}$.

Section 31 Gauss's Hypergeometric Equation

Notation: $F(a,b,c,x)=\sum_{n\ge 0}\dfrac{(a)_n(b)_n}{n!\,(c)_n}x^n$ with the Pochhammer symbol $(a)_n=a(a+1)\cdots(a+n-1)$, $(a)_0=1$; this is the hypergeometric series (4) of the text. The equation

$$x(1-x)y''+[c-(a+b+1)x]y'-aby=0 \tag{5.2}$$

has, near $x=0$, the general solution (when c is not an integer)

$$y=c_1F(a,b,c,x)+c_2x^{1-c}F(a-c+1,\,b-c+1,\,2-c,\,x), \tag{5.3}$$

and, near $x=1$ (when $c-a-b$ is not an integer),

$$y=c_1F(a,b,a+b-c+1,1-x)+c_2(1-x)^{c-a-b}F(c-b,\,c-a,\,c-a-b+1,\,1-x). \tag{5.4}$$

In each problem we bring the equation to the form (5.2) by a linear change of the independent variable and read off a, b, c from $c=$ (constant term of the y' coefficient), $a+b+1=-$(coefficient of t in the y' coefficient), $ab=-$(coefficient of y).

Problem 31-1 (Simmons)

Solution. We use two facts: $(1)_n=n!$, and for the half-integers

$$\left(\tfrac12\right)_n=\frac{1\cdot3\cdots(2n-1)}{2^n}=\frac{(2n)!}{4^n n!},\qquad \left(\tfrac32\right)_n=\frac{3\cdot5\cdots(2n+1)}{2^n}=\frac{(2n+1)!}{4^n n!}, \tag{5.5}$$

so $\left(\frac32\right)_n=(2n+1)\left(\frac12\right)_n$. (Check: $(2n)!=2\cdot4\cdots2n\cdot1\cdot3\cdots(2n-1)=2^n n!\cdot2^n(\tfrac12)_n$.)

(a) The coefficient of $(-x)^n$ in $F(-p,b,b,-x)$ is $\dfrac{(-p)_n(b)_n}{n!(b)_n}=\dfrac{(-p)_n}{n!}$ (the $(b)_n$ cancel), and $(-p)_n(-x)^n=(-p)(-p+1)\cdots(-p+n-1)(-1)^nx^n=p(p-1)\cdots(p-n+1)x^n$. Hence

$$F(-p,b,b,-x)=\sum_{n=0}^{\infty}\frac{p(p-1)\cdots(p-n+1)}{n!}x^n=\sum_{n=0}^{\infty}\binom{p}{n}x^n=(1+x)^p,$$

the binomial series.

(b) $\dfrac{(1)_n(1)_n}{n!(2)_n}=\dfrac{n!\,n!}{n!\,(n+1)!}=\dfrac{1}{n+1}$, since $(2)_n=2\cdot3\cdots(n+1)=(n+1)!$. Thus

$$xF(1,1,2,-x)=x\sum_{n=0}^{\infty}\frac{(-x)^n}{n+1}=\sum_{n=0}^{\infty}\frac{(-1)^nx^{n+1}}{n+1}=x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots=\log(1+x).$$

(c) By (5.5), $\dfrac{(\frac12)_n(\frac12)_n}{n!(\frac32)_n}=\dfrac{(\frac12)_n}{n!\,(2n+1)}=\dfrac{(2n)!}{4^n(n!)^2(2n+1)}$, so

$$xF\left(\tfrac12,\tfrac12,\tfrac32,x^2\right)=\sum_{n=0}^{\infty}\frac{(2n)!}{4^n(n!)^2}\frac{x^{2n+1}}{2n+1}=x+\frac12\frac{x^3}{3}+\frac{1\cdot3}{2\cdot4}\frac{x^5}{5}+\cdots,$$

because $\dfrac{(2n)!}{4^n(n!)^2}=\dfrac{1\cdot3\cdots(2n-1)}{2\cdot4\cdots2n}$. This is the Maclaurin series of $\sin^{-1}x$ (obtained by integrating $(1-x^2)^{-1/2}=\sum\binom{-1/2}{n}(-x^2)^n=\sum\frac{(1/2)_n}{n!}x^{2n}$ term by term).

(d) $\dfrac{(\frac12)_n(1)_n}{n!(\frac32)_n} = \dfrac{(\frac12)_n}{(\frac12)_n(2n+1)} = \dfrac{1}{2n+1}$, so

$$xF\left(\tfrac12,1,\tfrac32,-x^2\right) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \tan^{-1}x.$$

(e) The coefficient of x^n in $F(a,b,a,x/b)$ is $\dfrac{(a)_n(b)_n}{n!(a)_n}b^{-n} = \dfrac{1}{n!}\cdot\dfrac{b(b+1)\cdots(b+n-1)}{b^n}$ (note $(a)_n$ cancels). For fixed n, $\dfrac{b(b+1)\cdots(b+n-1)}{b^n} = \prod_{j=0}^{n-1}\left(1+\frac{j}{b}\right) \to 1$ as $b\to\infty$, so term by term the series tends to $\sum x^n/n! = e^x$.

(f) The coefficient of x^{2n} in $F\left(a,a,\frac32,\frac{-x^2}{4a^2}\right)$ is

$$\frac{(a)_n^2}{n!(\frac32)_n}\cdot\frac{(-1)^n}{4^n a^{2n}} = \left[\prod_{j=0}^{n-1}\left(1+\frac{j}{a}\right)\right]^2\frac{(-1)^n}{4^n n!\,(\frac32)_n} \xrightarrow[a\to\infty]{} \frac{(-1)^n}{4^n n!(\frac32)_n} = \frac{(-1)^n}{(2n+1)!},$$

using $4^n n!(\frac32)_n = (2n+1)!$ from (5.5). So the limit is $\sum(-1)^n x^{2n}/(2n+1)! = (\sin x)/x$, and multiplying by x gives $\sin x$.

(g) Likewise with $c=\frac12$: the coefficient tends to $\dfrac{(-1)^n}{4^n n!(\frac12)_n} = \dfrac{(-1)^n}{(2n)!}$ (since $4^n n!(\frac12)_n = (2n)!$), giving $\sum(-1)^n x^{2n}/(2n)! = \cos x$.

Answer. Shown: (a) the binomial series; (b) $\log(1+x)$ series; (c) $\sin^{-1}x$ series; (d) $\tan^{-1}x$ series; (e) $\to e^x$; (f) $\to \sin x$; (g) $\to \cos x$, by comparing coefficients.

Problem 31-2 (Simmons)

Solution.

(a) The equation is already of the form (5.2): $c=\frac32$, $a+b+1=2$, $ab=-2$. So $a+b=1$, $ab=-2$, and a,b are the roots of $s^2-s-2=(s-2)(s+1)=0$: $a=2$, $b=-1$. Since $c=\frac32$ is not an integer, (5.3) applies:

$$F(2,-1,\tfrac32,x), \qquad x^{1-c}F(a-c+1,b-c+1,2-c,x) = x^{-1/2}F\left(\tfrac32,-\tfrac32,\tfrac12,x\right).$$

Because $(-1)_n = 0$ for $n\ge 2$, the first series terminates: $F(2,-1,\frac32,x) = 1+\dfrac{2\cdot(-1)}{\frac32}x = 1-\dfrac{4}{3}x$.

$$y = c_1\left(1-\frac{4}{3}x\right) + c_2 x^{-1/2}F\left(\frac32,-\frac32,\frac12,x\right) \qquad (0<x<1).$$

(b) Put $t=-x$ (so $x=0\leftrightarrow t=0$). Then $\dfrac{dy}{dx} = -\dfrac{dy}{dt}$, $\dfrac{d^2y}{dx^2} = \dfrac{d^2y}{dt^2}$, and $2x^2+2x = 2t^2-2t = -2t(1-t)$, $1+5x = 1-5t$:

$$-2t(1-t)y_{tt} - (1-5t)y_t + y = 0 \quad\Longrightarrow\quad t(1-t)y_{tt} + \left(\tfrac12 - \tfrac52 t\right)y_t - \tfrac12 y = 0.$$

This is (5.2) with $c=\frac12$, $a+b+1=\frac52$, $ab=\frac12$; thus $a+b=\frac32$, $ab=\frac12$ and $s^2-\frac32 s+\frac12 = (s-1)(s-\frac12)=0$: $\{a,b\}=\{1,\frac12\}$. By (5.3) ($c=\frac12$ is not an integer):

$$y = c_1F\left(\tfrac12,1,\tfrac12,t\right) + c_2 t^{1/2}F\left(1,\tfrac32,\tfrac32,t\right),$$

since $a-c+1=1$, $b-c+1=\tfrac{3}{2}$, $2-c=\tfrac{3}{2}$. Now $F(a,b,b,t)=\sum \frac{(a)_n}{n!}t^n=(1-t)^{-a}$ (the $(b)_n$ cancel and the binomial series applies), so $F(\tfrac12,1,\tfrac12,t)=F(1,\tfrac12,\tfrac12,t)=\dfrac{1}{1-t}$ and $F(1,\tfrac32,\tfrac32,t)=\dfrac{1}{1-t}$. With $t=-x$:

$$y=c_1F\big(\tfrac12,1,\tfrac12,-x\big)+c_2(-x)^{1/2}F\big(1,\tfrac32,\tfrac32,-x\big)=\frac{c_1}{1+x}+c_2\frac{(-x)^{1/2}}{1+x}\qquad(|x|<1).$$

(The second solution is written with $(-x)^{1/2}$, which is real for $x<0$ ($t>0$); for $x>0$ the real solution is $x^{1/2}/(1+x)$, and in both cases it is $|x|^{1/2}/(1+x)$, as direct substitution confirms.)

Check. For $y=(1+x)^{-1}$: $y'=-(1+x)^{-2}$, $y''=2(1+x)^{-3}$, and $(2x^2+2x)y''+(1+5x)y'+y=\dfrac{4x}{(1+x)^2}-\dfrac{1+5x}{(1+x)^2}+\dfrac{1+x}{(1+x)^2}=0$.

(c) The singular points are $x=\pm1$. Put $t=\dfrac{x+1}{2}$, so $x=2t-1$ (then $x=-1\leftrightarrow t=0$ and $x=1\leftrightarrow t=1$), $\dfrac{dy}{dx}=\tfrac12 y_t$, $\dfrac{d^2y}{dx^2}=\tfrac14 y_{tt}$. Now $x^2-1=(x-1)(x+1)=(2t-2)(2t)=-4t(1-t)$ and $5x+4=10t-1$. The equation becomes

$$-4t(1-t)\cdot\tfrac14 y_{tt}+(10t-1)\cdot\tfrac12 y_t+4y=0\implies t(1-t)y_{tt}+\big(\tfrac12-5t\big)y_t-4y=0.$$

So $c=\tfrac12$, $a+b+1=5$, $ab=4$: $a+b=4$, $ab=4$, hence $a=b=2$. By (5.3):

$$y=c_1F\big(2,2,\tfrac12,t\big)+c_2t^{1/2}F\big(\tfrac52,\tfrac52,\tfrac32,t\big),\qquad t=\frac{x+1}{2},$$

because $a-c+1=\tfrac52=b-c+1$, $2-c=\tfrac32$. This converges for $|t|<1$, i.e. $-3<x<1$, and we use $x>-1$ for the real power.

(d) Here $x^2-x-6=(x-3)(x+2)$, so the singular points are $x=3$ and $x=-2$. Put $t=\dfrac{3-x}{5}$, so $x=3-5t$, $x=3\leftrightarrow t=0$, $x=-2\leftrightarrow t=1$, $\dfrac{dy}{dx}=-\tfrac15 y_t$, $\dfrac{d^2y}{dx^2}=\tfrac{1}{25}y_{tt}$. Then $x-3=-5t$, $x+2=5(1-t)$, $x^2-x-6=-25t(1-t)$, and $5+3x=14-15t$:

$$-25t(1-t)\cdot\tfrac{1}{25}y_{tt}-\tfrac15(14-15t)y_t+y=0\implies t(1-t)y_{tt}+\big(\tfrac{14}{5}-3t\big)y_t-y=0.$$

So $c=\tfrac{14}{5}$, $a+b+1=3$, $ab=1$: $a+b=2$, $ab=1$, hence $a=b=1$. Since c is not an integer, (5.3) gives (with $a-c+1=b-c+1=-\tfrac45$, $2-c=-\tfrac45$, $1-c=-\tfrac95$)

$$y=c_1F\big(1,1,\tfrac{14}{5},t\big)+c_2t^{-9/5}F\big(-\tfrac45,-\tfrac45,-\tfrac45,t\big),\qquad t=\frac{3-x}{5}.$$

The second function is elementary: $F(a,b,b,t)=(1-t)^{-a}$ gives $F(-\tfrac45,-\tfrac45,-\tfrac45,t)=(1-t)^{4/5}$, so $y_2=t^{-9/5}(1-t)^{4/5}$. (Valid for $0<t<1$, i.e. $-2<x<3$.)

Answer. (a) $y=c_1\big(1-\tfrac43x\big)+c_2x^{-1/2}F\big(\tfrac32,-\tfrac32,\tfrac12,x\big)$;
(b) $y=c_1F\big(\tfrac12,1,\tfrac12,-x\big)+c_2(-x)^{1/2}F\big(1,\tfrac32,\tfrac32,-x\big)=\dfrac{c_1}{1+x}+c_2\dfrac{(-x)^{1/2}}{1+x}$;
(c) $y=c_1F\big(2,2,\tfrac12,\tfrac{x+1}{2}\big)+c_2\big(\tfrac{x+1}{2}\big)^{1/2}F\big(\tfrac52,\tfrac52,\tfrac32,\tfrac{x+1}{2}\big)$;
(d) $y=c_1F\big(1,1,\tfrac{14}{5},\tfrac{3-x}{5}\big)+c_2\big(\tfrac{3-x}{5}\big)^{-9/5}F\big(-\tfrac45,-\tfrac45,-\tfrac45,\tfrac{3-x}{5}\big)$.

Problem 31-3 (Simmons)

Solution. With $t = \frac{1}{2}(1-x)$ we have $x = 1-2t$, $\dfrac{dt}{dx} = -\frac{1}{2}$, so

$$\frac{dy}{dx} = -\frac{1}{2}\frac{dy}{dt}, \qquad \frac{d^2y}{dx^2} = \frac{1}{4}\frac{d^2y}{dt^2},$$

and $1-x^2 = 1-(1-2t)^2 = 4t-4t^2 = 4t(1-t)$. Therefore

$$\begin{aligned}(1-x^2)y'' - xy' + p^2y &= 4t(1-t)\cdot\tfrac{1}{4}y_{tt} - (1-2t)\left(-\tfrac{1}{2}y_t\right) + p^2y \\ &= t(1-t)y_{tt} + \left(\tfrac{1}{2}-t\right)y_t + p^2y = 0.\end{aligned}$$

This is the hypergeometric equation (5.2) in t with $c = \frac{1}{2}$, $a+b+1 = 1$, $ab = -p^2$; so $a+b = 0$, $ab = -p^2$, giving $a = p$, $b = -p$. Since $c = \frac{1}{2}$ is not an integer, (5.3) gives the general solution near $t = 0$ (that is, near $x = 1$):

$$\begin{aligned}y &= c_1F\left(p,-p,\tfrac{1}{2},t\right) + c_2t^{1/2}F\left(p-\tfrac{1}{2}+1,\,-p-\tfrac{1}{2}+1,\,2-\tfrac{1}{2},\,t\right) \\ &= c_1F\left(p,-p,\tfrac{1}{2},t\right) + c_2t^{1/2}F\left(p+\tfrac{1}{2},\,-p+\tfrac{1}{2},\,\tfrac{3}{2},\,t\right),\end{aligned}$$

with $t = \frac{1}{2}(1-x)$, which is the stated result (for $0 < t < 1$, i.e. $-1 < x < 1$).

Answer. Shown: $t(1-t)y_{tt} + \left(\frac{1}{2}-t\right)y_t + p^2y = 0$ with $a = p$, $b = -p$, $c = \frac{1}{2}$, and the stated general solution follows from (5.3).

Problem 31-4 (Simmons)

Solution.

(a) The equation is (5.2) with $c = p$, $a+b+1 = p+2$, $ab = p$. Thus $a+b = p+1$, $ab = p$, and $s^2-(p+1)s+p = (s-p)(s-1) = 0$: $a = p$, $b = 1$. Since $c = p$ is not an integer, (5.3) gives

$$\begin{aligned}y &= c_1F(p,1,p,x) + c_2x^{1-p}F(p-p+1,\,1-p+1,\,2-p,\,x) \\ &= c_1F(p,1,p,x) + c_2x^{1-p}F(1,\,2-p,\,2-p,\,x).\end{aligned}$$

(b) Since the coefficient of x^n in $F(\alpha,\beta,\beta,x)$ is $\dfrac{(\alpha)_n(\beta)_n}{n!(\beta)_n} = \dfrac{(\alpha)_n}{n!}$, we get

$$F(\alpha,\beta,\beta,x) = \sum\frac{(\alpha)_n}{n!}x^n = (1-x)^{-\alpha} \qquad \text{(binomial series, } |x|<1\text{)}.$$

Hence $F(p,1,p,x) = F(1,p,p,x) = (1-x)^{-1}$ (using the symmetry $F(a,b,c,x) = F(b,a,c,x)$) and $F(1,2-p,2-p,x) = (1-x)^{-1}$, so

$$y = \frac{c_1}{1-x} + c_2\frac{x^{1-p}}{1-x}.$$

Check. For $y = 1/(1-x)$: $y' = (1-x)^{-2}$, $y'' = 2(1-x)^{-3}$, so $x(1-x)y'' + [p-(p+2)x]y' - py = \dfrac{2x+p-(p+2)x-p(1-x)}{(1-x)^2} = \dfrac{0}{(1-x)^2} = 0$.

For $y = x^s/(1-x)$ with $s = 1-p$, multiply the left side of the equation by $(1-x)^2/x^{s-1}$. Using $y' = \dfrac{sx^{s-1}}{1-x} + \dfrac{x^s}{(1-x)^2}$ and $y'' = \dfrac{s(s-1)x^{s-2}}{1-x} + \dfrac{2sx^{s-1}}{(1-x)^2} + \dfrac{2x^s}{(1-x)^3}$, we obtain the polynomial

$$s(s-1)(1-x)^2 + 2sx(1-x) + 2x^2 + [p-(p+2)x][s+(1-s)x] - px(1-x).$$

With $s = 1-p$ (so $s(s-1) = p^2-p$, $1-s = p$) its coefficients are: constant term $(p^2-p) + p(1-p) = 0$; coefficient of x: $-2(p^2-p) + 2(1-p) + (2p^2+p-2) - p = 0$; coefficient of x^2: $(p^2-p) - 2(1-p) + 2 - p(p+2) + p = 0$.

(c) For $p = 1$, (b) gives $y_1 = \dfrac{1}{1-x}$ and the second solution degenerates to $x^0/(1-x) = y_1$, so we need a new second solution. Write the equation as $y'' + Py' + Qy = 0$ with

$$P = \frac{1-3x}{x(1-x)} = \frac{1}{x} - \frac{2}{1-x}$$

(partial fractions: $\frac{1-3x}{x(1-x)} = \frac{A}{x} + \frac{B}{1-x}$ with $A = 1$ and, evaluating $1-3x$ at $x = 1$, $B = -2$). For $0 < x < 1$,

$$\int P\,dx = \log x + 2\log(1-x), \qquad e^{-\int P\,dx} = \frac{1}{x(1-x)^2}.$$

Section 16: $y_2 = vy_1$ with

$$v' = \frac{e^{-\int P\,dx}}{y_1^2} = \frac{1}{x(1-x)^2}\cdot(1-x)^2 = \frac{1}{x}, \qquad v = \log x, \qquad y_2 = \frac{\log x}{1-x}.$$

They are independent ($y_2/y_1 = \log x$), so $y = c_1\dfrac{1}{1-x} + c_2\dfrac{\log x}{1-x}$.

Answer. (a) $y = c_1F(p,1,p,x) + c_2x^{1-p}F(1,2-p,2-p,x)$; (b) $y = \dfrac{c_1}{1-x} + c_2\dfrac{x^{1-p}}{1-x}$; (c) $y = \dfrac{c_1}{1-x} + c_2\dfrac{\log x}{1-x}$.

Problem 31-5 (Simmons)

Solution. With $t = e^x$: $\dfrac{dy}{dx} = t\dfrac{dy}{dt}$ and $\dfrac{d^2y}{dx^2} = t\dfrac{d}{dt}\Big(t\dfrac{dy}{dt}\Big) = t^2y_{tt} + ty_t$. The equation becomes

$$(1-t)(t^2y_{tt} + ty_t) + \tfrac{1}{2}ty_t + ty = 0.$$

Dividing by t (note $t = e^x > 0$):

$$t(1-t)y_{tt} + (1-t)y_t + \tfrac{1}{2}y_t + y = 0 \quad\Longrightarrow\quad t(1-t)y_{tt} + \big(\tfrac{3}{2}-t\big)y_t + y = 0.$$

This is hypergeometric with singular points $t = 0$ and $t = 1$; the point $x = 0$ corresponds to $t = 1$. Put $s = 1-t = 1-e^x$ (so $x = 0 \leftrightarrow s = 0$): $t(1-t) = s(1-s)$, $y_t = -y_s$, $y_{tt} = y_{ss}$, $\frac{3}{2} - t = \frac{1}{2} + s$, so

$$s(1-s)y_{ss} - \big(\tfrac{1}{2}+s\big)y_s + y = 0 \quad\Longleftrightarrow\quad s(1-s)y_{ss} + \big[-\tfrac{1}{2}-s\big]y_s + y = 0.$$

This is (5.2) with $c = -\frac{1}{2}$, $a+b+1 = 1$, $ab = -1$: $a+b = 0$, $ab = -1$ gives $a = 1$, $b = -1$. Since $c = -\frac{1}{2}$ is not an integer, (5.3) applies in the variable s:

$$y = c_1F\big(1,-1,-\tfrac{1}{2},s\big) + c_2s^{1-c}F\big(a-c+1,\,b-c+1,\,2-c,\,s\big)$$

$$= c_1 F\big(1,-1,-\tfrac12,s\big) + c_2 s^{3/2} F\big(\tfrac52,\tfrac12,\tfrac52,s\big),$$

with $s = 1-e^x$. (This is the same as formula (5.4) for t near 1, with $a+b-c+1$ and $c-a-b$ evaluated at $a=1, b=-1, c=\frac32$.)

Both functions can be simplified. Since $(-1)_n = 0$ for $n \ge 2$,

$$F\big(1,-1,-\tfrac12,s\big) = 1 + \frac{1\cdot(-1)}{-\frac12}s = 1+2s = 3-2e^x,$$

and $F(\frac52,\frac12,\frac52,s) = (1-s)^{-1/2} = e^{-x/2}$, so $y_2 = (1-e^x)^{3/2}e^{-x/2}$. Thus

$$y = c_1 F\big(1,-1,-\tfrac12,1-e^x\big) + c_2(1-e^x)^{3/2}F\big(\tfrac52,\tfrac12,\tfrac52,1-e^x\big) = c_1(3-2e^x) + c_2(1-e^x)^{3/2}e^{-x/2}.$$

(For $x<0$, $1-e^x>0$ and the power is real; for $x>0$ replace $(1-e^x)^{3/2}$ by $(e^x-1)^{3/2}$, which is the same solution up to a constant factor. The series converge for $|1-e^x|<1$.)

Answer. $y = c_1 F\big(1,-1,-\tfrac12,1-e^x\big) + c_2(1-e^x)^{3/2}F\big(\tfrac52,\tfrac12,\tfrac52,1-e^x\big) = c_1(3-2e^x) + c_2(1-e^x)^{3/2}e^{-x/2}$.

Problem 31-6 (Simmons)

Solution.

(a) Differentiate the series term by term (legitimate inside the radius of convergence $|x|<1$); the $n=0$ term is constant:

$$F'(a,b,c,x) = \sum_{n=1}^{\infty} \frac{(a)_n (b)_n}{n!\,(c)_n}\, n x^{n-1} = \sum_{m=0}^{\infty} \frac{(a)_{m+1}(b)_{m+1}}{m!\,(c)_{m+1}} x^m \qquad (n = m+1).$$

Since $(a)_{m+1} = a(a+1)_m$ (the factors are $a, a+1, \dots, a+m$) and similarly $(b)_{m+1} = b(b+1)_m$, $(c)_{m+1} = c(c+1)_m$,

$$F'(a,b,c,x) = \frac{ab}{c}\sum_{m=0}^{\infty} \frac{(a+1)_m (b+1)_m}{m!\,(c+1)_m} x^m = \frac{ab}{c} F(a+1,b+1,c+1,x).$$

(b) By Problem 3, with $t = \frac12(1-x) \in (0,1)$ (that is, $-1<x<1$), every solution is $y = c_1 F(p,-p,\frac12,t) + c_2 t^{1/2} G(t)$, where $G(t) = F(p+\frac12,-p+\frac12,\frac32,t)$. By the chain rule $\dfrac{dy}{dx} = -\dfrac12 \dfrac{dy}{dt}$, and

$$\frac{d}{dt}F\big(p,-p,\tfrac12,t\big) = \frac{p(-p)}{1/2}F\big(p+1,1-p,\tfrac32,t\big) = -2p^2 F\big(p+1,1-p,\tfrac32,t\big) \quad \text{by (a)},$$
$$\frac{d}{dt}\big[t^{1/2}G(t)\big] = \frac12 t^{-1/2}G(t) + t^{1/2}G'(t).$$

Hence

$$y'(x) = -\frac12\Big[-2c_1 p^2 F\big(p+1,1-p,\tfrac32,t\big) + c_2\Big(\tfrac12 t^{-1/2}G(t) + t^{1/2}G'(t)\Big)\Big].$$

As $x \to 1^-$, i.e. $t \to 0^+$: the functions $F(p+1,1-p,\frac32,t)$, $G(t)$, $G'(t)$ are analytic at $t=0$, hence bounded near 0, with $G(0)=1$, and $t^{1/2}G'(t) \to 0$, while $\frac12 t^{-1/2}G(t) \to +\infty$. Thus the first term and $t^{1/2}G'$ stay bounded, and

$$y'(x) \text{ is bounded near } x=1 \iff c_2 = 0,$$

because if $c_2 \neq 0$, then $|y'| \geq \frac{|c_2|}{4} t^{-1/2} |G(t)| - (\text{bounded}) \to \infty$. So the solutions with bounded derivative are exactly $y = c_1 F\left(p, -p, \frac{1}{2}, \frac{1-x}{2}\right)$.

Polynomial solutions. A polynomial has a bounded derivative near $x = 1$, so a polynomial solution must be $c_1 F(p, -p, \frac{1}{2}, t)$ for some constant c_1 (on $-1 < x < 1$, hence everywhere, as polynomials). Suppose $c_1 \neq 0$. If p is not an integer, no coefficient $\dfrac{(p)_k (-p)_k}{k! (\frac{1}{2})_k}$ vanishes (the factors $p + j$ and $-p + j$ are nonzero for every integer $j \geq 0$), so $F(p, -p, \frac{1}{2}, t)$ is a genuine infinite power series in $t = \frac{1}{2}(1 - x)$, i.e. in $x - 1$; it cannot equal a polynomial, whose Taylor series at $x = 1$ is finite. Hence p must be an integer, $p = n \in \{0, 1, 2, \dots\}$. Conversely, for $p = n$ we have $(-n)_k = 0$ for $k > n$, so $F(n, -n, \frac{1}{2}, t)$ terminates and is a polynomial of degree n in t (the coefficient of t^n is $\frac{(n)_n (-n)_n}{n! (\frac{1}{2})_n} \neq 0$), hence a polynomial of degree n in x. Therefore the polynomial solutions occur only when $p = n$ is a non-negative integer, and are then exactly the constant multiples of $F\left(n, -n, \frac{1}{2}, \frac{1-x}{2}\right)$. (For $n = 0$ this is the constant 1; for $n = 1$, $F = 1 + \frac{1 \cdot (-1)}{1/2} \cdot \frac{1-x}{2} = x$; for $n = 2$, $F = 1 + \frac{2(-2)}{1/2} t + \frac{(2 \cdot 3)(-2)(-1)}{2! \cdot \frac{1}{2} \cdot \frac{3}{2}} t^2 = 1 - 8t + 8t^2 = 2x^2 - 1$, since $t = \frac{1-x}{2}$.)

Answer. (a) shown by shifting the index in the series; (b) y' is bounded near $x = 1$ iff $c_2 = 0$, giving $y = c_1 F\left(p, -p, \frac{1}{2}, \frac{1-x}{2}\right)$, and polynomial solutions occur exactly for $p = n = 0, 1, 2, \dots$ and are multiples of $F\left(n, -n, \frac{1}{2}, \frac{1-x}{2}\right)$.

Section 32 The Point at Infinity

For $y'' + P(x)y' + Q(x)y = 0$, the substitution $t = 1/x$ gives $y' = -t^2\dfrac{dy}{dt}$ and $y'' = t^4\dfrac{d^2y}{dt^2} + 2t^3\dfrac{dy}{dt}$ (formulas (3), (4) of the text), and the transformed equation is

$$\frac{d^2y}{dt^2} + \left[\frac{2}{t} - \frac{P(1/t)}{t^2}\right]\frac{dy}{dt} + \frac{Q(1/t)}{t^4}\,y = 0. \tag{5.6}$$

The point $x = \infty$ is ordinary, regular singular (with the exponents of the indicial equation at $t = 0$), or irregular singular according as $t = 0$ is for (5.6). For a regular singular point at $t = 0$, both $t\tilde{P}(t)$ and $t^2\tilde{Q}(t)$ must be analytic at $t = 0$, where $\tilde{P} = \frac{2}{t} - \frac{P(1/t)}{t^2}$ and $\tilde{Q} = \frac{Q(1/t)}{t^4}$.

Problem 32-1 (Simmons)

Solution.

(a) Dividing by $1 - x^2$: $P(x) = \dfrac{-2x}{1-x^2}$, $Q(x) = \dfrac{p(p+1)}{1-x^2}$. Then

$$P(1/t) = \frac{-2/t}{1-1/t^2} = \frac{-2t}{t^2-1}, \qquad Q(1/t) = \frac{p(p+1)}{1-1/t^2} = \frac{p(p+1)t^2}{t^2-1}.$$

Therefore

$$\tilde{P} = \frac{2}{t} - \frac{P(1/t)}{t^2} = \frac{2}{t} + \frac{2}{t(t^2-1)} = \frac{2(t^2-1)+2}{t(t^2-1)} = \frac{2t}{t^2-1}, \qquad \tilde{Q} = \frac{Q(1/t)}{t^4} = \frac{p(p+1)}{t^2(t^2-1)}.$$

(Equivalently, direct substitution of (3) and (4) into the equation, using $1 - x^2 = (t^2-1)/t^2$, gives $y_{tt} + \frac{2t}{t^2-1}y_t + \frac{p(p+1)}{t^2(t^2-1)}y = 0$.) Now

$$t\tilde{P} = \frac{2t^2}{t^2-1}, \qquad t^2\tilde{Q} = \frac{p(p+1)}{t^2-1}$$

are analytic at $t = 0$ (the only singularities of these rational functions are $t = \pm 1$), so $t = 0$ is a regular singular point, with $p_0 = t\tilde{P}\big|_{t=0} = 0$ and $q_0 = t^2\tilde{Q}\big|_{t=0} = -p(p+1)$. The indicial equation is

$$m(m-1) - p(p+1) = 0 \iff m^2 - m - p(p+1) = (m-p-1)(m+p) = 0,$$

so the exponents are $m = p+1$ and $m = -p$.

(b) $P(x) = 1/x$, $Q(x) = (x^2 - p^2)/x^2 = 1 - p^2/x^2$. Then $P(1/t) = t$, $Q(1/t) = 1 - p^2t^2$, and

$$\tilde{P} = \frac{2}{t} - \frac{t}{t^2} = \frac{1}{t}, \qquad \tilde{Q} = \frac{1-p^2t^2}{t^4},$$

so the equation at t is $y_{tt} + \dfrac{1}{t}y_t + \dfrac{1-p^2t^2}{t^4}y = 0$. Here $t\tilde{P} = 1$ is analytic, but

$$t^2\tilde{Q} = \frac{1-p^2t^2}{t^2} = \frac{1}{t^2} - p^2$$

is not analytic at $t = 0$ (it becomes infinite as $t \to 0$, for every value of p). So $t = 0$ is not a regular singular point; since it is a singular point ($\tilde{Q}$ is unbounded there), it is an irregular singular point.

Answer. (a) $x=\infty$ is a regular singular point with exponents $p+1$ and $-p$. (b) $x=\infty$ is an irregular singular point.

Problem 32-2 (Simmons)

Solution. Equation (9), the transform at $t=1/x$ of $x(1-x)y''+[c-(a+b+1)x]y'-aby=0$, is

$$y''+\frac{(1-a-b)-(2-c)t}{t(1-t)}y'+\frac{ab}{t^2(1-t)}y=0,$$

(primes are now d/dt). Multiplying by $t(1-t)$:

$$t(1-t)y''+\big[(1-a-b)-(2-c)t\big]y'+\frac{ab}{t}y=0. \tag{5.7}$$

Put $y=t^a w$. Then

$$y'=t^a\Big(w'+\frac{a}{t}w\Big),\qquad y''=t^a\Big(w''+\frac{2a}{t}w'+\frac{a(a-1)}{t^2}w\Big).$$

Substituting in (5.7) and dividing by t^a:

$$t(1-t)\Big[w''+\frac{2a}{t}w'+\frac{a(a-1)}{t^2}w\Big]+\big[(1-a-b)-(2-c)t\big]\Big[w'+\frac{a}{t}w\Big]+\frac{ab}{t}w=0.$$

Collect terms.

- w'': $t(1-t)$.
- w': $2a(1-t)+(1-a-b)-(2-c)t=(1+a-b)-(2a+2-c)t$, and $2a+2-c=a+(1+a-c)+1$.
- w:
$$\frac{(1-t)a(a-1)}{t}+\frac{a[(1-a-b)-(2-c)t]}{t}+\frac{ab}{t}$$
$$=\frac{1}{t}\big[a(a-1)+a(1-a-b)+ab\big]-a(a-1)-a(2-c).$$
The bracket is $a^2-a+a-a^2-ab+ab=0$, and $-a(a-1)-a(2-c)=-a(a+1-c)$.

Hence

$$t(1-t)w''+\big\{(1+a-b)-[a+(1+a-c)+1]t\big\}w'-a(1+a-c)w=0,$$

which is the hypergeometric equation with parameters

$$A=a,\qquad B=1+a-c,\qquad C=1+a-b$$

(indeed $A+B+1=2a+2-c$, $AB=a(1+a-c)$).

Solutions. If $a-b$ is not an integer (in particular $a\neq b$), then $C=1+a-b$ is not an integer, so by (6) of Section 31 the general solution near $t=0$ is

$$w=c_1F(A,B,C,t)+c_2t^{1-C}F(A-C+1,\,B-C+1,\,2-C,\,t).$$

Here $1-C=b-a$, and

$$A-C+1=a-(1+a-b)+1=b,$$

$$B - C + 1 = (1 + a - c) - (1 + a - b) + 1 = 1 + b - c,$$
$$2 - C = 1 - a + b = 1 + b - a.$$

Since $y = t^a w$,

$$y = c_1 t^a F\big(a, 1 + a - c, 1 + a - b, t\big) + c_2 t^b F\big(b, 1 + b - c, 1 + b - a, t\big)$$

(because $t^a \cdot t^{b-a} = t^b$). Replacing t by $1/x$ gives the two solutions

$$y_1 = x^{-a} F\big(a, 1 + a - c, 1 + a - b, \tfrac{1}{x}\big), \qquad y_2 = x^{-b} F\big(b, 1 + b - c, 1 + b - a, \tfrac{1}{x}\big),$$

valid for $|1/x| < 1$, i.e. $|x| > 1$. They are independent: $y_1/y_2 = x^{b-a} \cdot \big[1 + O(1/x)\big]$, which is not constant because $a \neq b$ (near $t = 0$ the two solutions have the different exponents a and b, and each series starts with the term 1).

Answer. Shown: $y = t^a w$ turns (9) into the hypergeometric equation with $A = a$, $B = 1 + a - c$, $C = 1 + a - b$, and the two independent solutions for large x are $y_1 = x^{-a} F(a, 1 + a - c, 1 + a - b, 1/x)$, $y_2 = x^{-b} F(b, 1 + b - c, 1 + b - a, 1/x)$.

Problem 32-3 (Simmons)

Solution. Write the equation as $y'' + Py' + Qy = 0$ with $P(x) = \dfrac{c - x}{x}$, $Q(x) = -\dfrac{a}{x}$. Then $P(1/t) = (c - 1/t)t = ct - 1$ and $Q(1/t) = -at$, so by (5.6)

$$\tilde{P} = \frac{2}{t} - \frac{ct - 1}{t^2} = \frac{2t - ct + 1}{t^2} = \frac{1 + (2 - c)t}{t^2}, \qquad \tilde{Q} = \frac{-at}{t^4} = -\frac{a}{t^3},$$

so the transformed equation is

$$y_{tt} + \frac{1 + (2 - c)t}{t^2} y_t - \frac{a}{t^3} y = 0.$$

For $t = 0$ to be a regular singular point we would need $t\tilde{P}$ and $t^2\tilde{Q}$ to be analytic at $t = 0$. But

$$t\tilde{P} = \frac{1 + (2 - c)t}{t} = \frac{1}{t} + (2 - c)$$

is unbounded as $t \to 0$, hence not analytic there (and likewise $t^2\tilde{Q} = -a/t$ is unbounded when $a \neq 0$). Thus $t = 0$ is a singular point of the transformed equation that is not regular, i.e. an irregular singular point, and $x = \infty$ is an irregular singular point of (10).

Answer. Verified: $t\tilde{P} = 1/t + (2 - c)$ is not analytic at $t = 0$, so $x = \infty$ is an irregular singular point of the confluent hypergeometric equation.

Problem 32-4 (Simmons)

Solution. Dividing $xy'' + (c - x)y' - ay = 0$ by x: $P = \dfrac{c - x}{x}$, $Q = -\dfrac{a}{x}$. So

$$xP = c - x, \qquad x^2 Q = -ax,$$

both polynomials, hence analytic at 0: $x=0$ is a regular singular point, with $p_0=c$, $p_1=-1$, $q_0=0$, $q_1=-a$. The indicial equation is

$$m(m-1)+cm+0=m(m+c-1)=0,$$

with exponents $m_1=0$ and $m_2=1-c$.

Exponent 0: put $y=\sum_{n\ge0}u_nx^n$ (we write u_n for the coefficients so as not to confuse them with the constant a). Then

$$xy''=\sum_{n\ge1}n(n-1)u_nx^{n-1},\quad cy'=\sum_{n\ge1}cnu_nx^{n-1},\quad -xy'=-\sum_{n\ge1}nu_nx^n,\quad -ay=-\sum_{n\ge0}au_nx^n.$$

Shifting the first two sums by one index, the coefficient of x^n ($n\ge0$) in the equation is

$$(n+1)n\,u_{n+1}+c(n+1)u_{n+1}-nu_n-au_n=0,$$

that is,

$$(n+1)(n+c)\,u_{n+1}=(n+a)\,u_n,\qquad u_{n+1}=\frac{n+a}{(n+1)(n+c)}u_n.$$

Because c is not zero or a negative integer, $n+c\neq0$ for all $n\ge0$, so the recursion never breaks down. Taking $u_0=1$,

$$u_1=\frac{a}{1\cdot c},\quad u_2=\frac{a(a+1)}{1\cdot2\cdot c(c+1)},\quad\dots,\quad u_n=\frac{a(a+1)\cdots(a+n-1)}{n!\,c(c+1)\cdots(c+n-1)}$$

(by induction: multiplying u_n by $\frac{a+n}{(n+1)(c+n)}$ gives the formula for $n+1$). Hence the solution is

$$y=1+\sum_{n=1}^{\infty}\frac{a(a+1)\cdots(a+n-1)}{n!\,c(c+1)\cdots(c+n-1)}x^n=F(a,c,x).$$

The ratio $|u_{n+1}x^{n+1}/(u_nx^n)|=\frac{|n+a|}{(n+1)|n+c|}|x|\to0$, so it converges for all x.

Answer. Exponents 0 and $1-c$; the exponent-0 solution is $F(a,c,x)=1+\sum_{n\ge1}\dfrac{a(a+1)\cdots(a+n-1)}{n!\,c(c+1)\cdots(c+n-1)}x^n$, from $u_{n+1}=\dfrac{n+a}{(n+1)(n+c)}u_n$.

Problem 32-5 (Simmons)

Solution. Laguerre's equation is the confluent hypergeometric equation (10) with $c=1$ and $a=-p$. Since $c=1$ is not zero or a negative integer, Problem 4 shows that

$$y_1=F(-p,1,x)=1+\sum_{n=1}^{\infty}\frac{(-p)(-p+1)\cdots(-p+n-1)}{n!\,1\cdot2\cdots n}x^n=\sum_{n=0}^{\infty}\frac{(-p)_n}{(n!)^2}x^n$$

is a solution; it is analytic (entire) with $y_1(0)=1$.

A second solution is unbounded. The exponents are 0 and $1-c=0$, a double root, so we find a second solution by reduction of order (Section 16). Here $P=\dfrac{1-x}{x}=\dfrac{1}{x}-1$, $\int P\,dx=\log x-x$, and $e^{-\int P\,dx}=\dfrac{e^x}{x}$. On an interval $(0,\delta)$ on which $y_1\neq0$ (possible since $y_1(0)=1$ and y_1 is continuous),

$$y_2=y_1v,\qquad v'=\frac{e^x}{x\,y_1^2}.$$

Near $x = 0$, $e^x/y_1^2 \to 1$, so $v'(x) \sim 1/x$ and $v(x) \to -\infty$ as $x \to 0^+$ (more precisely, $v' \geq \frac{1}{2x}$ on some $(0, \delta')$, so $v(x) = v(\delta') - \int_x^{\delta'} v' \leq v(\delta') - \frac{1}{2}\log(\delta'/x) \to -\infty$). The general solution on $(0, \delta')$ is $y = c_1 y_1 + c_2 y_1 v = y_1(c_1 + c_2 v)$. Since $y_1 \to 1$ and $v \to -\infty$,

$$y \text{ is bounded near } 0 \iff c_2 = 0,$$

for if $c_2 \neq 0$ then $|y| = |y_1|\,|c_1 + c_2 v| \to \infty$. Hence the only solutions bounded near the origin are $y = c_1 F(-p, 1, x)$.

Polynomials. If $p = n \in \{0, 1, 2, \dots\}$ then $(-n)_k = (-n)(-n+1)\cdots(-n+k-1) = 0$ for $k > n$ (the factor $-n + n = 0$ occurs), so the series terminates:

$$L_n(x) = F(-n, 1, x) = \sum_{k=0}^{n} \frac{(-n)_k}{(k!)^2} x^k = \sum_{k=0}^{n} \frac{(-1)^k\, n!}{(n-k)!\,(k!)^2} x^k,$$

a polynomial of degree n (the coefficient of x^n is $(-1)^n/n! \neq 0$), with $L_n(0) = 1$. For example $L_0 = 1$, $L_1 = 1 - x$, $L_2 = 1 - 2x + \frac{1}{2}x^2$.

Answer. Shown: $F(-p, 1, x)$ is the only solution (up to a constant factor) bounded near 0, since any second solution contains $\log x$ and is unbounded; for $p = n$ the series terminates and $L_n(x) = \sum_{k=0}^{n} \frac{(-1)^k n!}{(n-k)!\,(k!)^2} x^k$ is a polynomial of degree n.

Chapter 6

Fourier Series and Orthogonal Functions

Section 33 The Fourier Coefficients

Throughout this section f is integrable on $-\pi \le x \le \pi$, and its Fourier series is

$$f(x) \sim \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \ (n \ge 0), \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \ (n \ge 1),$$

formulas (7) and (9) of the text. We use repeatedly that the value of f at finitely many points does not affect the integrals, and the orthogonality relations (2), (5), (6), (8) of the text together with $\int_{-\pi}^{\pi}\cos^2 nx\,dx = \int_{-\pi}^{\pi}\sin^2 nx\,dx = \pi$ $(n \ge 1)$. Also $\cos n\pi = (-1)^n$, $\sin n\pi = 0$, and $\sin\frac{(2k-1)\pi}{2} = (-1)^{k+1}$, $\cos\frac{(2k-1)\pi}{2} = 0$, $\sin k\pi = 0$, $\cos k\pi = (-1)^k$ for integers k. All coefficient formulas below were re-derived with SymPy (exact integration for $n = 0, \dots, 16$) and the series were compared numerically with the functions; see `verify/sec33.py`.

Problem 33-1 (Simmons)

Solution. Only the interval $[-\pi, \pi/2]$ contributes to the integrals.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi/2} \pi\,dx = \frac{3\pi}{2},$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi/2} \pi\cos nx\,dx = \left[\frac{\sin nx}{n}\right]_{-\pi}^{\pi/2} = \frac{\sin(n\pi/2) - \sin(-n\pi)}{n} = \frac{\sin(n\pi/2)}{n},$$
$$b_n = \int_{-\pi}^{\pi/2} \sin nx\,dx = \left[-\frac{\cos nx}{n}\right]_{-\pi}^{\pi/2} = \frac{\cos n\pi - \cos(n\pi/2)}{n}.$$

Now separate odd and even n.

Odd $n = 2k-1$ $(k \ge 1)$: $\sin\frac{n\pi}{2} = (-1)^{k+1}$, $\cos n\pi = -1$, $\cos\frac{n\pi}{2} = 0$, so

$$a_{2k-1} = \frac{(-1)^{k+1}}{2k-1}, \qquad b_{2k-1} = -\frac{1}{2k-1}.$$

Even $n = 2m$ $(m \ge 1)$: $\sin m\pi = 0$, $\cos n\pi = 1$, $\cos\frac{n\pi}{2} = \cos m\pi = (-1)^m$, so

$$a_{2m} = 0, \qquad b_{2m} = \frac{1 - (-1)^m}{2m} = \begin{cases} 0, & m \text{ even}, \\ \dfrac{1}{m}, & m \text{ odd}. \end{cases}$$

Thus the nonzero even-index coefficients occur for $m = 2k-1$, i.e. $n = 2(2k-1)$, with $b_{2(2k-1)} = \dfrac{1}{2k-1}$. Substituting,

$$f(x) \sim \frac{3\pi}{4} + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x - \sin(2k-1)x + \sin 2(2k-1)x}{2k-1}.$$

Answer. $\dfrac{3\pi}{4} + \displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x - \sin(2k-1)x + \sin 2(2k-1)x}{2k-1}$ (agrees with the book).

Problem 33-2 (Simmons)

Solution. Only $[0, \pi/2]$ contributes:

$$a_0 = \frac{1}{\pi}\int_0^{\pi/2} \mathrm{d}x = \frac{1}{2},$$
$$a_n = \frac{1}{\pi}\int_0^{\pi/2} \cos nx\,\mathrm{d}x = \frac{\sin(n\pi/2)}{n\pi},$$
$$b_n = \frac{1}{\pi}\int_0^{\pi/2} \sin nx\,\mathrm{d}x = \frac{1-\cos(n\pi/2)}{n\pi}.$$

For odd $n = 2k-1$: $a_n = \dfrac{(-1)^{k+1}}{\pi(2k-1)}$, $b_n = \dfrac{1}{\pi(2k-1)}$. For even $n = 2m$: $a_{2m} = 0$ and $b_{2m} = \dfrac{1-(-1)^m}{2m\pi}$, which is 0 for m even and $\dfrac{1}{m\pi}$ for m odd; i.e. $b_{2(2k-1)} = \dfrac{1}{\pi(2k-1)}$. Hence

$$f(x) \sim \frac{1}{4} + \frac{1}{\pi}\sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x + \sin(2k-1)x + \sin 2(2k-1)x}{2k-1}.$$

Answer. $\dfrac{1}{4} + \dfrac{1}{\pi}\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x + \sin(2k-1)x + \sin 2(2k-1)x}{2k-1}$ (agrees with the book).

Problem 33-3 (Simmons)

Solution.

$$a_0 = \frac{1}{\pi}\int_0^{\pi} \sin x\,\mathrm{d}x = \frac{1}{\pi}\big[-\cos x\big]_0^{\pi} = \frac{2}{\pi}.$$

For a_n and b_n we use the product formulas $\sin x\cos nx = \frac{1}{2}[\sin(n+1)x - \sin(n-1)x]$ and $\sin x \sin nx = \frac{1}{2}[\cos(n-1)x - \cos(n+1)x]$.

The case $n = 1$. $a_1 = \frac{1}{\pi}\int_0^{\pi}\sin x\cos x\,\mathrm{d}x = \frac{1}{2\pi}\int_0^{\pi}\sin 2x\,\mathrm{d}x = 0$ and $b_1 = \frac{1}{\pi}\int_0^{\pi}\sin^2 x\,\mathrm{d}x = \frac{1}{\pi}\cdot\frac{\pi}{2} = \frac{1}{2}$.

The case $n \geq 2$. Then $n \pm 1 \neq 0$ and

$$a_n = \frac{1}{2\pi}\left[\frac{1-\cos(n+1)\pi}{n+1} - \frac{1-\cos(n-1)\pi}{n-1}\right],$$
$$b_n = \frac{1}{2\pi}\left[\frac{\sin(n-1)\pi}{n-1} - \frac{\sin(n+1)\pi}{n+1}\right] = 0.$$

If n is odd, $n \pm 1$ are even, $\cos(n\pm1)\pi = 1$, and $a_n = 0$. If $n = 2m$ is even, $\cos(n\pm1)\pi = -1$ and

$$a_{2m} = \frac{1}{2\pi}\left[\frac{2}{2m+1} - \frac{2}{2m-1}\right] = \frac{1}{\pi}\cdot\frac{(2m-1)-(2m+1)}{4m^2-1} = -\frac{2}{\pi(4m^2-1)}.$$

Therefore

$$f(x) \sim \frac{1}{\pi} + \frac{1}{2}\sin x - \frac{2}{\pi}\sum_{m=1}^{\infty}\frac{\cos 2mx}{4m^2-1}.$$

Answer. $\dfrac{1}{\pi} - \dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty} \dfrac{\cos 2nx}{4n^2-1} + \dfrac{1}{2}\sin x$ (agrees with the book).

Problem 33-4 (Simmons)

Solution. Now $f(x) = 0$ on $[-\pi, 0)$ and $f(x) = \cos x$ on $[0, \pi]$. Using $\cos x \cos nx = \frac{1}{2}[\cos(n+1)x + \cos(n-1)x]$ and $\cos x \sin nx = \frac{1}{2}[\sin(n+1)x + \sin(n-1)x]$:

Cosine coefficients. $a_0 = \frac{1}{\pi}\int_0^\pi \cos x\,dx = 0$ and $a_1 = \frac{1}{\pi}\int_0^\pi \cos^2 x\,dx = \frac{1}{2}$; for $n \geq 2$,

$$a_n = \frac{1}{2\pi}\left[\frac{\sin(n+1)\pi}{n+1} + \frac{\sin(n-1)\pi}{n-1}\right] = 0.$$

Sine coefficients. $b_1 = \frac{1}{\pi}\int_0^\pi \cos x \sin x\,dx = \frac{1}{2\pi}\int_0^\pi \sin 2x\,dx = 0$; for $n \geq 2$,

$$b_n = \frac{1}{2\pi}\left[\frac{1-\cos(n+1)\pi}{n+1} + \frac{1-\cos(n-1)\pi}{n-1}\right].$$

For n odd this is 0. For $n = 2m$ it equals

$$b_{2m} = \frac{1}{2\pi}\left[\frac{2}{2m+1} + \frac{2}{2m-1}\right] = \frac{1}{\pi}\cdot\frac{4m}{4m^2-1} = \frac{4m}{\pi(4m^2-1)}.$$

Hence

$$f(x) \sim \frac{1}{2}\cos x + \frac{4}{\pi}\sum_{m=1}^{\infty}\frac{m\sin 2mx}{4m^2-1}.$$

Answer. $\dfrac{1}{2}\cos x + \dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty}\dfrac{n\sin 2nx}{4n^2-1}$ (agrees with the book).

Problem 33-5 (Simmons)

Solution.

(a) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\pi\,dx = 2\pi$, and for $n \geq 1$, by (2), $a_n = \int_{-\pi}^{\pi}\cos nx\,dx = 0$, $b_n = \int_{-\pi}^{\pi}\sin nx\,dx = 0$. So the series is $\frac{1}{2}a_0 = \pi$.

(b) For $n \geq 0$, $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\sin x\cos nx\,dx = 0$ by (5) for $n \geq 1$, and for $n = 0$ because $\sin x$ is odd (or by (2)). For b_n: if $n \neq 1$, (8) gives $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\sin x\sin nx\,dx = 0$, and for $n = 1$, $b_1 = \frac{1}{\pi}\int_{-\pi}^{\pi}\sin^2 x\,dx = \frac{1}{\pi}\cdot\pi = 1$. So the series is just $\sin x$.

(c) For $n \geq 1$, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos x\sin nx\,dx = 0$ by (5), which says $\int_{-\pi}^{\pi}\sin nx\cos mx\,dx = 0$ for all integers $m, n \geq 1$ (here $m = 1$). $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos x\,dx = 0$; $a_n = 0$ for $n \neq 1$ by (6), and $a_1 = \frac{1}{\pi}\int_{-\pi}^{\pi}\cos^2 x\,dx = 1$. So the series is just $\cos x$.

(d) The Fourier coefficients are linear in f (the integrals (7), (9) are linear), so the Fourier series of $\pi + \sin x + \cos x$ is the sum of the three series above: $\pi + \sin x + \cos x$.

Alternate line of thought. Suppose $f(x)$ is itself a finite trigonometric sum

$$f(x) = \tfrac{1}{2}\alpha_0 + \sum_{n=1}^{N}(\alpha_n\cos nx + \beta_n\sin nx).$$

A finite sum is trivially uniformly convergent, so the term-by-term integration used to derive (3), (7), (9) is legitimate and yields $a_n = \alpha_n$, $b_n = \beta_n$. Thus the Fourier series of any finite trigonometric series is that series itself, which gives (a)–(d) at once (with $\alpha_0 = 2\pi$ in (a)). In particular these Fourier series converge to f at *every* x.

Answer. (a) π; (b) $\sin x$; (c) $\cos x$; (d) $\pi + \sin x + \cos x$. Any finite trigonometric series is its own Fourier series.

Problem 33-6 (Simmons)

Solution. Example 3 of the text gives, for $f_0(x) = -\frac{\pi}{2}$ on $[-\pi, 0)$, $\frac{\pi}{2}$ on $[0, \pi]$,

$$f_0(x) \sim 2\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

We use the linearity of Fourier coefficients: the series of $cf + g$ is c times that of f plus that of g, and the series of a constant C is C itself. Put $S(x) = \sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots$, so $f_0 \sim 2S$ and the square wave $q(x) = \mp 1$ (i.e. -1 on $[-\pi, 0)$, $+1$ on $[0, \pi]$) $= \frac{2}{\pi} f_0$ has series $\frac{4}{\pi} S$.

(a) $f = a\,q = \frac{2a}{\pi} f_0 \sim \dfrac{4a}{\pi} S$.

(b) Take $a = 1$ in (a): $f = q \sim \dfrac{4}{\pi} S$.

(c) $f = \frac{\pi}{4}\, q = \frac{1}{2} f_0 \sim S$.

(d) $f = \frac{1}{2} + \frac{3}{2} q$ (check: $x < 0$: $\frac{1}{2} - \frac{3}{2} = -1$; $x > 0$: $\frac{1}{2} + \frac{3}{2} = 2$), so $f \sim \dfrac{1}{2} + \dfrac{3}{2} \cdot \dfrac{4}{\pi} S = \dfrac{1}{2} + \dfrac{6}{\pi} S$.

(e) $f = \frac{3}{2} + \frac{1}{2} q$ (check: $\frac{3}{2} - \frac{1}{2} = 1$, $\frac{3}{2} + \frac{1}{2} = 2$), so $f \sim \dfrac{3}{2} + \dfrac{1}{2} \cdot \dfrac{4}{\pi} S = \dfrac{3}{2} + \dfrac{2}{\pi} S$.

Answer. With $S = \sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots$: (a) $\frac{4a}{\pi} S$; (b) $\frac{4}{\pi} S$; (c) S; (d) $\frac{1}{2} + \frac{6}{\pi} S$; (e) $\frac{3}{2} + \frac{2}{\pi} S$ (agrees with the book).

Problem 33-7 (Simmons)

Solution. Let f_1 be the function of Problem 1 (π on $[-\pi, \frac{\pi}{2}]$, 0 on $(\frac{\pi}{2}, \pi]$), f_2 that of Problem 2 (1 on $[0, \frac{\pi}{2}]$, 0 elsewhere), and E that of Example 2 (0 on $[-\pi, 0)$, π on $[0, \pi]$). Then $\pi - E$ equals π on $[-\pi, 0)$ and 0 on $[0, \pi]$. Compare with f_1, which is π on $[-\pi, 0)$ and also π on $[0, \pi/2]$:

$$f_1 - (\pi - E) = \begin{cases} 0, & -\pi \le x < 0, \\ \pi, & 0 \le x \le \frac{\pi}{2}, \\ 0, & \frac{\pi}{2} < x \le \pi \end{cases} \quad = \ \pi f_2$$

(up to values at isolated points, which do not matter). So $f_2 = \dfrac{1}{\pi}\left[f_1 - (\pi - E)\right]$. By (13) of the text, $E \sim \frac{\pi}{2} + 2\sum_{k \ge 1} \frac{\sin(2k-1)x}{2k-1}$, hence

$$\pi - E \sim \frac{\pi}{2} - 2\sum_{k=1}^{\infty} \frac{\sin(2k-1)x}{2k-1}.$$

Subtract this from the result of Problem 1:

$$\begin{aligned} f_1 - (\pi - E) &\sim \left(\frac{3\pi}{4} - \frac{\pi}{2}\right) \\ &\quad + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x - \sin(2k-1)x + \sin 2(2k-1)x + 2\sin(2k-1)x}{2k-1} \\ &= \frac{\pi}{4} + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x + \sin(2k-1)x + \sin 2(2k-1)x}{2k-1}. \end{aligned}$$

Dividing by π gives

$$f_2(x) \sim \frac{1}{4} + \frac{1}{\pi}\sum_{k=1}^{\infty} \frac{(-1)^{k+1}\cos(2k-1)x + \sin(2k-1)x + \sin 2(2k-1)x}{2k-1},$$

which is exactly the series found directly in Problem 2.

Answer. $f_2 = \frac{1}{\pi}[f_1 - (\pi - E)]$, giving

$$\frac{1}{4} + \frac{1}{\pi}\sum_{k\geq 1} \frac{(-1)^{k+1}\cos(2k-1)x + \sin(2k-1)x + \sin 2(2k-1)x}{2k-1},$$

the same as in Problem 2.

Problem 33-8 (Simmons)

Solution. (The reference "Figure 33" in the book is a misprint: the sawtooth wave meant is that of Figure 39, the graph of the sum of the series (11), namely $y = x$ on $-\pi < x < \pi$ extended with period 2π.)

The idea: the sawtooth $y = x$ rises with slope 1 from $-\pi$ to π and drops by 2π at each odd multiple of π. Split the rise into two halves, one on each half period, and move each half to a common level. Let the square wave be

$$q(x) = -\tfrac{\pi}{2} \ \ (-\pi < x < 0), \qquad q(x) = +\tfrac{\pi}{2} \ \ (0 < x < \pi), \qquad q(x + 2\pi) = q(x),$$

and subtract it from $y = x$:

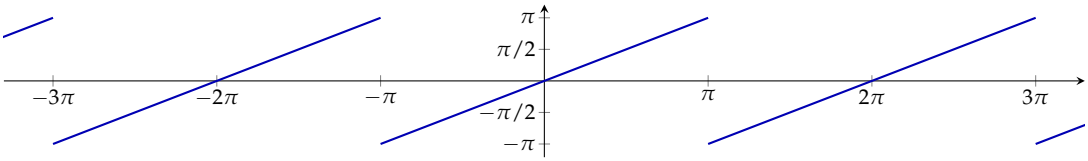
$$s(x) = x - q(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0, \\ x - \frac{\pi}{2}, & 0 < x < \pi. \end{cases}$$

On $(-\pi, 0)$ the function s rises from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$; on $(0, \pi)$ it again rises from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. Hence s is a sawtooth with slope 1, going from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ on each interval of length π and jumping down by π at each multiple of π; its period is π (indeed $s(x + \pi) = s(x)$: for $-\pi < x < 0$, $s(x + \pi) = (x + \pi) - \frac{\pi}{2} = x + \frac{\pi}{2} = s(x)$). Therefore

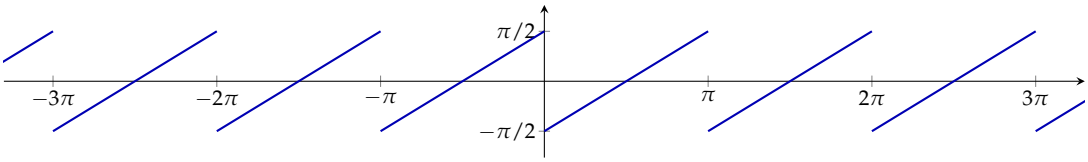
$$x \ = \ s(x) + q(x) \qquad (-\pi < x < \pi),$$

and, both sides having period 2π, the identity holds for the whole periodic sawtooth. Graphically, in the three sketches below the first graph is the vertical sum of the second and the third: on $(0, \pi)$ we add the level $+\frac{\pi}{2}$ to the rising segment from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ and get the segment from 0 to π; on $(-\pi, 0)$ we add $-\frac{\pi}{2}$ and get the segment from $-\pi$ to 0. The jumps are consistent: at $x = 0$ the square wave jumps up by π and s jumps down by π, so their sum is continuous (as is $y = x$); at $x = \pi$ the square wave jumps down by π and s jumps down by π, giving the total drop 2π of the big sawtooth.

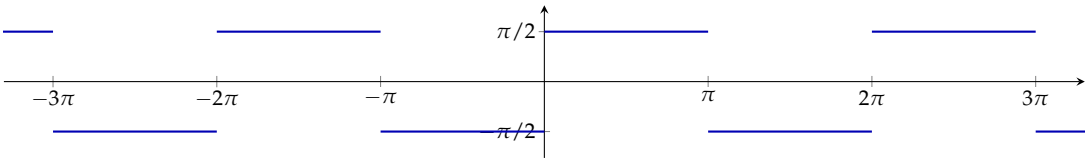
$y = x$ (sawtooth, period 2π)



$y = s(x)$ (sawtooth, period π)



$y = q(x)$ (square wave, period 2π)



Answer. Shown: $x = s(x) + q(x)$ on $(-\pi, \pi)$, where $s = x \mp \frac{\pi}{2}$ is the sawtooth of period π (from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ on each half period) and $q = \mp\frac{\pi}{2}$ is the square wave of period 2π.

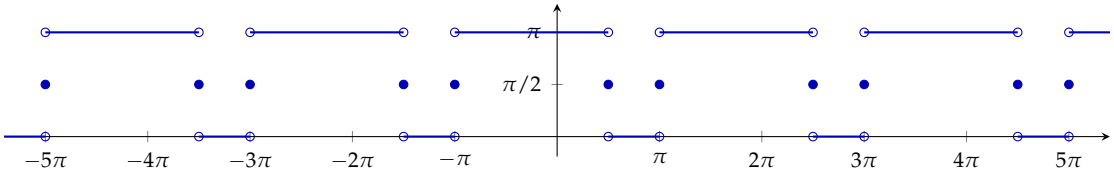
Section 34 The Problem of Convergence

Throughout, f is extended periodically, $f(x+2\pi)=f(x)$. By Dirichlet's theorem (text, Section 34), if f satisfies the Dirichlet conditions, its Fourier series converges at every x to $\tfrac12[f(x-)+f(x+)]$, which is $f(x)$ at points of continuity. At the points $x=\pm\pi,\pm3\pi,\dots$ the one-sided limits are $f(\pi-)$ and $f(-\pi+)$ (the periodic extension). In the sketches, a filled dot marks the value of the sum at a jump and an open circle marks a one-sided limit that is not attained. In the graphs the horizontal axis is marked in multiples of π. The coefficient formulas and the numerical sums were checked with SymPy and by numerical partial sums (`verify/sec34.py`).

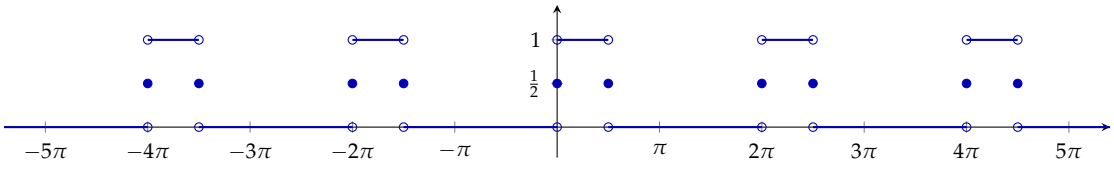
Problem 34-1 (Simmons)

Solution. In each case f is bounded, piecewise monotone with finitely many jumps, so Dirichlet's theorem applies.

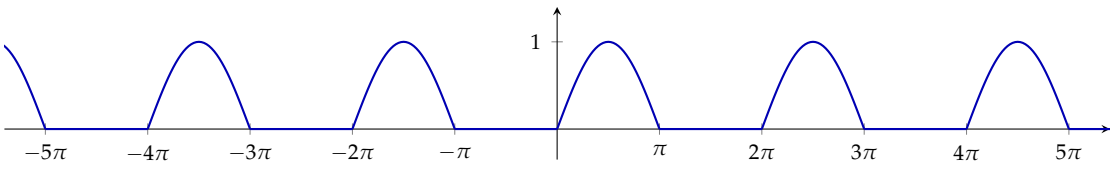
(33.1) $f=\pi$ on $[-\pi,\tfrac{\pi}{2}]$, $f=0$ on $(\tfrac{\pi}{2},\pi]$. Within one period the only discontinuity is the jump at $x=\tfrac{\pi}{2}$ ($f(\tfrac{\pi}{2}-)=\pi$, $f(\tfrac{\pi}{2}+)=0$) and, because of periodicity, at $x=\pi$ ($f(\pi-)=0$, $f(\pi+)=f(-\pi+)=\pi$). The sum equals $\pi/2$ at both kinds of points, i.e. at $x=\tfrac{\pi}{2}+2k\pi$ and $x=\pi+2k\pi$; elsewhere the sum equals π on $(-\pi+2k\pi,\tfrac{\pi}{2}+2k\pi)$ and 0 on $(\tfrac{\pi}{2}+2k\pi,\pi+2k\pi)$.



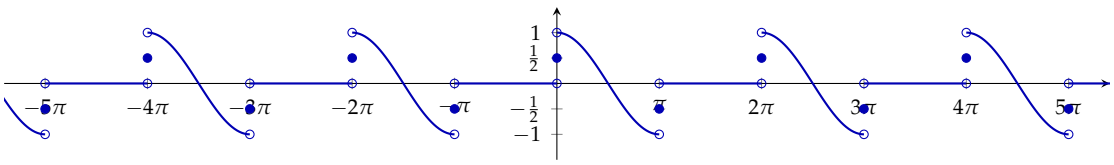
(33.2) $f=1$ on $[0,\tfrac{\pi}{2}]$, 0 elsewhere in $[-\pi,\pi]$. Jumps at $x=2k\pi$ (from 0 to 1) and $x=\tfrac{\pi}{2}+2k\pi$ (from 1 to 0); the sum is $\tfrac12$ at these points, 1 on $(2k\pi,\tfrac{\pi}{2}+2k\pi)$ and 0 elsewhere. At $x=\pm\pi$ the function is 0 on both sides, so it is continuous there.



(33.3) $f=0$ on $[-\pi,0)$, $\sin x$ on $[0,\pi]$. This function is continuous everywhere (the pieces match: $f(0)=\sin 0=0$, $f(\pi)=\sin\pi=0=f(-\pi)$), so the sum is f itself: the "half-wave rectified" sine, equal to $\sin x$ on $[2k\pi,(2k+1)\pi]$ and 0 on $[(2k+1)\pi,(2k+2)\pi]$.



(33.4) $f=0$ on $[-\pi,0)$, $\cos x$ on $[0,\pi]$. At $x=2k\pi$: $f(0-)=0$, $f(0+)=\cos 0=1$, so the sum is $\tfrac12$. At $x=(2k+1)\pi$: $f(\pi-)=\cos\pi=-1$, $f(\pi+)=f(-\pi+)=0$, so the sum is $-\tfrac12$. Elsewhere the sum is $\cos x$ on $(2k\pi,(2k+1)\pi)$ and 0 on $((2k-1)\pi,2k\pi)$.



Answer. Graphs above: periodic extensions of the given functions, with the value at each jump equal to the average of the two one-sided limits ($\frac{\pi}{2}$; $\frac{1}{2}$; none, since continuous; $\frac{1}{2}$ at $2k\pi$ and $-\frac{1}{2}$ at $(2k+1)\pi$).

Problem 34-2 (Simmons)

Solution. The example in the text treats $g(x) = 0$ for $-\pi \le x < 0$ and $g(x) = x$ for $0 \le x < \pi$, and finds

$$g(x) \sim \frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2} + \sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n}.$$

Our function is $f(x) = g(-x)$: for $-\pi < x \le 0$, $-x \in [0,\pi)$ and $g(-x) = -x$; for $0 < x \le \pi$, $-x \in [-\pi, 0)$ and $g(-x) = 0$. (Values at the single points $\pm\pi$ do not affect coefficients.) If g has coefficients a_n, b_n, the substitution $u = -x$ gives

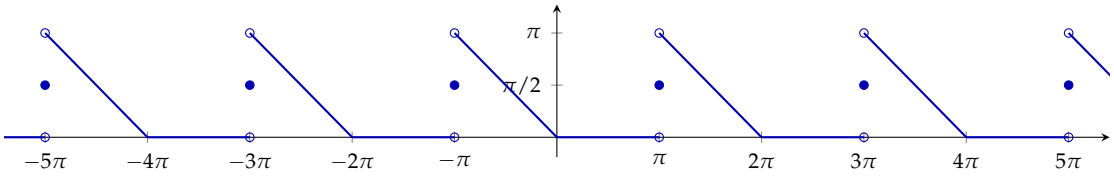
$$\frac{1}{\pi}\int_{-\pi}^{\pi} g(-x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} g(u)\cos nu\,du = a_n,$$
$$\frac{1}{\pi}\int_{-\pi}^{\pi} g(-x)\sin nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} g(u)\sin nu\,du = -b_n.$$

So the cosine coefficients are unchanged and the sine coefficients change sign (equivalently, replace x by $-x$ in the series, since cos is even and sin is odd):

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2} - \sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n}$$

at every point of continuity.

Graph of the sum. f is continuous at $x = 0$ ($f(0-) = 0 = f(0+)$) and at all other interior points of $(-\pi, \pi)$. At $x = \pm\pi$: $f(\pi-) = 0$ and $f(\pi+) = f(-\pi+) = \pi$, so by Dirichlet's theorem the sum is $\frac{\pi}{2}$ at all odd multiples of π. The graph is a periodic "reverse sawtooth": the line $y = -x$ falling from π to 0 on $(-\pi, 0]$, then 0 on $(0, \pi)$, repeated every 2π.



Answer. $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2} - \sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n}$ (agrees with the book), with sum $\frac{\pi}{2}$ at $x = (2k+1)\pi$; graph above.

Problem 34-3 (Simmons)

Solution. *Coefficients.*

$$a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-\pi)\,dx + \int_{0}^{\pi} x\,dx\right] = \frac{1}{\pi}\left[-\pi^2 + \frac{\pi^2}{2}\right] = -\frac{\pi}{2},$$

$$a_n = \frac{1}{\pi}\left[-\pi\int_{-\pi}^{0}\cos nx\,dx + \int_{0}^{\pi} x\cos nx\,dx\right] = \frac{1}{\pi}\left[-\pi\cdot 0 + \left(\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right)\Big|_0^{\pi}\right]$$
$$= \frac{(-1)^n - 1}{\pi n^2},$$

$$b_n = \frac{1}{\pi}\left[-\pi\int_{-\pi}^{0}\sin nx\,dx + \int_{0}^{\pi} x\sin nx\,dx\right].$$

Here $\int_{-\pi}^{0}\sin nx\,dx = \left[-\frac{\cos nx}{n}\right]_{-\pi}^{0} = \frac{-1+(-1)^n}{n}$ and $\int_0^{\pi} x\sin nx\,dx = \left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = -\frac{\pi(-1)^n}{n}$ (integration by parts with $u = x$). Hence

$$b_n = \frac{1}{\pi}\left[\frac{\pi(1-(-1)^n)}{n} - \frac{\pi(-1)^n}{n}\right] = \frac{1-2(-1)^n}{n}.$$

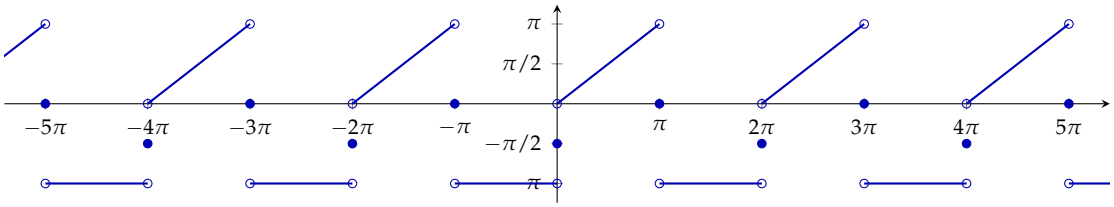
For $n = 2k-1$ odd: $a_n = -\dfrac{2}{\pi(2k-1)^2}$, $b_n = \dfrac{3}{2k-1}$. For $n = 2k$ even: $a_n = 0$, $b_n = -\dfrac{1}{2k}$. Therefore

$$f(x) = -\frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2} + 3\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{2n-1} - \sum_{n=1}^{\infty}\frac{\sin 2nx}{2n}$$

at all points of continuity (and the series converges to the average of the one-sided limits at jumps).

Cross-check by linearity. $f = g + E - \pi$, where g is the function of the text's example and $E = \pi$ on $[0,\pi)$, 0 on $[-\pi,0)$ (Example 2 of Section 33): indeed for $x < 0$, $0 + 0 - \pi = -\pi$; for $x \ge 0$, $x + \pi - \pi = x$. Adding the series $\frac{\pi}{4} + \cdots$ of g, $\frac{\pi}{2} + 2\sum\frac{\sin(2n-1)x}{2n-1}$ of E and the constant $-\pi$ gives constant $-\frac{\pi}{4}$, sine coefficients $\frac{1}{n} + \frac{2}{n} = \frac{3}{n}$ for odd n and $-\frac{1}{n}$ for even n, the same as above.

Graph of the sum. At $x = 0$: $f(0-) = -\pi$, $f(0+) = 0$, so the sum is $-\frac{\pi}{2}$ (at all $x = 2k\pi$). At $x = \pi$: $f(\pi-) = \pi$ and $f(\pi+) = f(-\pi+) = -\pi$, so the sum is 0 (at all $x = (2k+1)\pi$). Elsewhere the sum is f: constant $-\pi$ on $(2k\pi - \pi, 2k\pi)$ and the line $y = x - 2k\pi$ on $(2k\pi, (2k+1)\pi)$.



Numerical sums. At $x = 0$ every sine vanishes and $\cos 0 = 1$, so

$$-\frac{\pi}{2} = -\frac{\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} \quad\Longrightarrow\quad \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi}{2}\cdot\frac{\pi}{4} = \frac{\pi^2}{8}.$$

At $x = \pi$ every sine vanishes and $\cos(2n-1)\pi = -1$, so

$$0 = -\frac{\pi}{4} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} \quad\Longrightarrow\quad \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Answer. $f(x) = -\dfrac{\pi}{4} - \dfrac{2}{\pi}\displaystyle\sum_{n\ge1}\frac{\cos(2n-1)x}{(2n-1)^2} + 3\sum_{n\ge1}\frac{\sin(2n-1)x}{2n-1} - \sum_{n\ge1}\frac{\sin 2nx}{2n}$; the sum is $-\frac{\pi}{2}$ at $x = 2k\pi$ and 0 at $x = (2k+1)\pi$. Both points give $\displaystyle\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$ (agrees with the book).

Problem 34-4 (Simmons)

Solution.

(a) Integrate by parts twice. With $u = x^2$:

$$\int x^2\cos nx\,dx = \frac{x^2\sin nx}{n} - \frac{2}{n}\int x\sin nx\,dx = \frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3},$$
$$\int x^2\sin nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2}{n}\int x\cos nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3},$$

using $\int x\sin nx\,dx = -\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}$ and $\int x\cos nx\,dx = \frac{x\sin nx}{n} + \frac{\cos nx}{n^2}$. Since $f = 0$ on $[-\pi, 0)$,

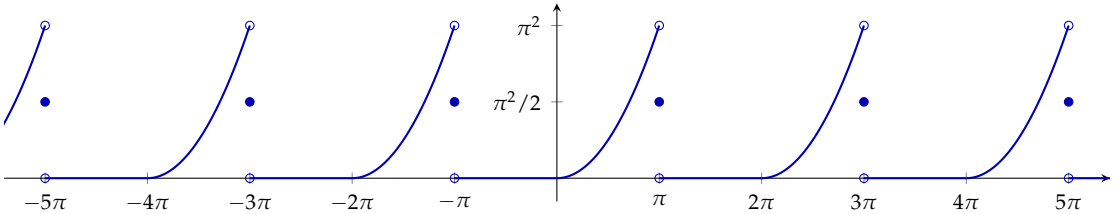
$$a_0 = \frac{1}{\pi}\int_0^{\pi} x^2\,dx = \frac{\pi^2}{3}, \qquad \text{so } \tfrac{1}{2}a_0 = \frac{\pi^2}{6},$$
$$a_n = \frac{1}{\pi}\left[\frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}\right]_0^{\pi} = \frac{1}{\pi}\cdot\frac{2\pi(-1)^n}{n^2} = \frac{2(-1)^n}{n^2},$$
$$b_n = \frac{1}{\pi}\left[-\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}\right]_0^{\pi} = \frac{1}{\pi}\left[-\frac{\pi^2(-1)^n}{n} + \frac{2(-1)^n - 2}{n^3}\right]$$
$$= \frac{\pi(-1)^{n+1}}{n} + \frac{2\big((-1)^n - 1\big)}{\pi n^3}.$$

The second part of b_n vanishes for even n and equals $-\dfrac{4}{\pi n^3}$ for odd n. Thus

$$\sum_{n=1}^{\infty} b_n\sin nx = \pi\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{(2n-1)^3},$$

and adding $\frac{1}{2}a_0 + \sum a_n\cos nx$ gives the stated series. By Dirichlet's theorem it converges to $f(x)$ at all points of continuity.

(b) f is continuous at $x = 0$ ($f(0-) = 0$, $f(0+) = 0$), and at $x = \pi$: $f(\pi-) = \pi^2$, $f(\pi+) = f(-\pi+) = 0$, so the sum is $\frac{\pi^2}{2}$ at all odd multiples of π; elsewhere it is x^2 on $[2k\pi, (2k+1)\pi)$ shifted ($y = (x - 2k\pi)^2$) and 0 on $((2k-1)\pi, 2k\pi]$.



(c) At $x = 0$ the sum is $f(0) = 0$; all sines vanish and $\cos 0 = 1$:

$$0 = \frac{\pi^2}{6} + 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} \quad\Longrightarrow\quad \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}, \quad \text{i.e.} \quad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \cdots = \frac{\pi^2}{12}.$$

At $x=\pi$ the sum is $\frac{1}{2}\big[f(\pi-)+f(\pi+)\big]=\frac{\pi^2}{2}$; all sines vanish and $(-1)^n\cos n\pi=(-1)^{2n}=1$:

$$\frac{\pi^2}{2}=\frac{\pi^2}{6}+2\sum_{n=1}^{\infty}\frac{1}{n^2}\quad\Longrightarrow\quad\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{1}{2}\cdot\frac{\pi^2}{3}=\frac{\pi^2}{6}.$$

(d) Let $S=\sum_{n\ge1}1/n^2$ (a convergent series). Adding $2\sum_{n\ge1}\left(\frac{1}{2n}\right)^2=\frac{1}{2}S$ to both sides of $\sum_{n\ge1}\frac{(-1)^{n+1}}{n^2}=\frac{\pi^2}{12}$: on the left, odd n keep the term $\frac{1}{n^2}$, and for even $n=2m$ the term $-\frac{1}{(2m)^2}+\frac{2}{(2m)^2}=+\frac{1}{(2m)^2}$, so the left side becomes $\sum_{n\ge1}\frac{1}{n^2}=S$. Hence

$$S=\frac{\pi^2}{12}+\frac{1}{2}S\quad\Longrightarrow\quad\frac{1}{2}S=\frac{\pi^2}{12}\quad\Longrightarrow\quad S=\frac{\pi^2}{6}.$$

Answer. (a) shown ($a_0=\frac{\pi^2}{3}$, $a_n=\frac{2(-1)^n}{n^2}$, $b_n=\frac{\pi(-1)^{n+1}}{n}+\frac{2((-1)^n-1)}{\pi n^3}$); (b) graph above (sum $\frac{\pi^2}{2}$ at odd multiples of π); (c) $x=0$ gives $\frac{\pi^2}{12}$, $x=\pi$ gives $\frac{\pi^2}{6}$; (d) $S=\frac{\pi^2}{12}+\frac{S}{2}$, so $S=\frac{\pi^2}{6}$.

Problem 34-5 (Simmons)

Solution.

(a) Differentiating,

$$\frac{d}{dx}\left[\frac{e^x(\cos nx+n\sin nx)}{1+n^2}\right]=\frac{e^x[(\cos nx+n\sin nx)+(-n\sin nx+n^2\cos nx)]}{1+n^2}=e^x\cos nx,$$
$$\frac{d}{dx}\left[\frac{e^x(\sin nx-n\cos nx)}{1+n^2}\right]=\frac{e^x[(\sin nx-n\cos nx)+(n\cos nx+n^2\sin nx)]}{1+n^2}=e^x\sin nx,$$

so

$$\int e^x\cos nx\,dx=\frac{e^x(\cos nx+n\sin nx)}{1+n^2},\quad\int e^x\sin nx\,dx=\frac{e^x(\sin nx-n\cos nx)}{1+n^2}\qquad(n\ge0).$$

At $x=\pm\pi$ we have $\sin n\pi=0$ and $\cos(\pm n\pi)=(-1)^n$, and $e^{\pi}-e^{-\pi}=2\sinh\pi$. Therefore

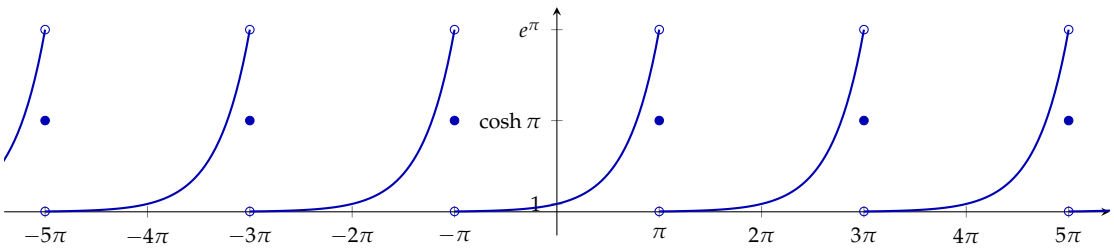
$$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}e^x\,dx=\frac{e^{\pi}-e^{-\pi}}{\pi}=\frac{2\sinh\pi}{\pi},$$
$$a_n=\frac{1}{\pi}\cdot\frac{(-1)^n(e^{\pi}-e^{-\pi})}{1+n^2}=\frac{2(-1)^n\sinh\pi}{\pi(1+n^2)},$$
$$b_n=\frac{1}{\pi}\cdot\frac{-n(-1)^n(e^{\pi}-e^{-\pi})}{1+n^2}=-\frac{2n(-1)^n\sinh\pi}{\pi(1+n^2)}.$$

Hence

$$f(x)=\frac{\sinh\pi}{\pi}\left[1+2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+1}\cos nx-2\sum_{n=1}^{\infty}\frac{(-1)^n n}{n^2+1}\sin nx\right]$$

at every point of continuity, i.e. for $-\pi<x<\pi$ (and periodically).

(b) f is continuous except at $x=(2k+1)\pi$, where $f(\pi-)=e^{\pi}$ and $f(\pi+)=f(-\pi+)=e^{-\pi}$; the sum there is $\frac{1}{2}(e^{\pi}+e^{-\pi})=\cosh\pi\approx11.592$. Elsewhere the sum is $e^{x-2k\pi}$ on $((2k-1)\pi,(2k+1)\pi)$.



(c) At $x = 0$ the function is continuous with value $e^0 = 1$; sines vanish, cosines equal 1:

$$1 = \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+1}\right] \implies \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+1} = \frac{1}{2}\left[\frac{\pi}{\sinh\pi} - 1\right].$$

At $x = \pi$ the sum is $\cosh\pi$; $\sin n\pi = 0$ and $(-1)^n\cos n\pi = 1$:

$$\cosh\pi = \frac{\sinh\pi}{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{1}{n^2+1}\right] \implies \sum_{n=1}^{\infty}\frac{1}{n^2+1} = \frac{1}{2}\left[\frac{\pi\cosh\pi}{\sinh\pi} - 1\right] = \frac{1}{2}\left[\frac{\pi}{\tanh\pi} - 1\right].$$

Answer. (a) $f(x) = \dfrac{\sinh\pi}{\pi}\left[1 + 2\displaystyle\sum_{n\geq 1}\frac{(-1)^n}{n^2+1}\cos nx - 2\sum_{n\geq 1}\frac{(-1)^n n}{n^2+1}\sin nx\right]$ (agrees with the book); (b) graph above (sum $\cosh\pi$ at odd multiples of π); (c) both sums obtained from $x = \pi$ and $x = 0$.

Problem 34-6 (Simmons)

Solution. The Dirichlet conditions: bounded, finitely many discontinuities, finitely many maxima and minima. We show that f and g satisfy them but $f + g$ does not.

g satisfies them. $g = -2x$ is continuous, bounded by 2π and strictly decreasing: no discontinuities and no maxima or minima in the open interval.

f satisfies them. For $x \neq 0$, $|f(x)| \leq x^2 + 2|x| \leq \pi^2 + 2\pi$, so f is bounded. It is continuous at $x \neq 0$, and at 0 since $|x^2\sin\frac{1}{x}| \leq x^2 \to 0$; so there are no discontinuities. For $x \neq 0$, $f'(x) = 2x\sin\frac{1}{x} - \cos\frac{1}{x} + 2$, and at 0: $f'(0) = \lim_{h\to 0}\left(h\sin\frac{1}{h} + 2\right) = 2$.

- If $|x| < \frac{1}{2}$ ($x \neq 0$), then $|2x\sin\frac{1}{x}| \leq 2|x| < 1$ and $\cos\frac{1}{x} \leq 1$, so $f'(x) \geq 2 - 2|x| - 1 = 1 - 2|x| > 0$; also $f'(0) = 2 > 0$. Hence f is strictly increasing on $(-\frac{1}{2}, \frac{1}{2})$ and has no extremum there.
- On each of the compact intervals $[\frac{1}{2}, \pi]$ and $[-\pi, -\frac{1}{2}]$, f' is a real-analytic function of x (built from x, $\sin\frac{1}{x}$, $\cos\frac{1}{x}$ with $x \neq 0$) and is not identically zero, since $f'(\pm 1) = 2 + 2\sin 1 - \cos 1 > 0$ (note $f'(-1) = 2 + (-2)\sin(-1) - \cos(-1)$ is the same number). A non-zero analytic function has isolated zeros, so only finitely many in a compact interval. A local extremum at an interior point requires $f' = 0$ there, so f has only finitely many extrema in $|x| \geq \frac{1}{2}$.

Altogether f has finitely many maxima and minima in $[-\pi, \pi)$, so f satisfies the Dirichlet conditions.

$f + g$ does not. $(f+g)(x) = x^2\sin\frac{1}{x} =: h(x)$ for $x \neq 0$ and $h(0) = 0$. It is bounded and continuous, but for $x \neq 0$, $h'(x) = 2x\sin\frac{1}{x} - \cos\frac{1}{x}$. Take

$$x_k = \frac{1}{2k\pi} : \ h'(x_k) = -1 < 0, \qquad y_k = \frac{1}{(2k+1)\pi} : \ h'(y_k) = +1 > 0 \qquad (k = 1, 2, \dots),$$

(because $\sin\frac{1}{x_k} = \sin 2k\pi = 0$, $\cos 2k\pi = 1$, and $\sin(2k+1)\pi = 0$, $\cos(2k+1)\pi = -1$). Then $y_k < x_k$. On the closed interval $[y_k, x_k]$ the continuous function h attains its maximum. It is not attained at y_k (since $h'(y_k) > 0$, h takes larger values just to the right of y_k) nor at x_k (since $h'(x_k) < 0$, h takes larger values just to the left of x_k). So h has a local maximum at an interior point $m_k \in (y_k, x_k)$. The intervals (y_k, x_k) are pairwise disjoint (they are ordered as $x_{k+1} < y_k < x_k$: indeed $\frac{1}{2(k+1)\pi} < \frac{1}{(2k+1)\pi}$), and $m_k \to 0$. Thus h has infinitely many local maxima (and likewise minima) in $(0, \frac{1}{2\pi})$, so $f+g$ violates the Dirichlet conditions.

Hence f and g belong to the class, but $f+g$ does not: the class is not closed under addition, so it is not a linear space.

Answer. f and $g = -2x$ satisfy the Dirichlet conditions, but $f+g = x^2 \sin\frac{1}{x}$ ($= 0$ at $x = 0$) has infinitely many maxima and minima near $x = 0$; so the class is not closed under addition.

Problem 34-7 (Simmons)

Solution. *Conventions.* Dirichlet conditions: $|f| \le M$; f has finitely many discontinuities; f has finitely many points of local maximum or minimum (local maximum at c: $f(c) \ge f(x)$ for all x near c; similarly for minimum). The limits are $f(x-) = \lim_{t\to 0^+} f(x-t)$ and $f(x+) = \lim_{t\to 0^+} f(x+t)$; at the right endpoint we mean $f(\pi-)$, and "exist" means the limit exists as a finite number.

Step 1: the exceptional points. Let D be the set consisting of $-\pi$, π, the discontinuities of f, and the points of local extremum of f in $(-\pi, \pi)$. By hypothesis D is finite; write $D = \{p_0 < p_1 < \cdots < p_r\}$, $p_0 = -\pi$, $p_r = \pi$. The open intervals $I_j = (p_{j-1}, p_j)$ are the complement of D in $(-\pi, \pi)$.

Step 2: on each I_j, f is monotonic. On I_j, f is continuous (no point of I_j is a discontinuity) and has no local extremum. We claim f is injective on I_j. Suppose $f(a) = f(b)$ with $a < b$ in I_j. By continuity on $[a, b]$, f attains a maximum value μ and a minimum value ν there. If $\mu > f(a)$, the maximum is attained at some interior point $c \in (a, b)$, and then $f(c) \ge f(x)$ for all $x \in (a, b)$, so c is a local maximum in I_j: contradiction. Similarly if $\nu < f(a)$ we get a local minimum. Otherwise $\mu = \nu = f(a)$, so f is constant on $[a, b]$ and every point of (a, b) is a local maximum: contradiction. Hence f is injective. A continuous injective function on an interval is strictly monotonic (if it were not, there would be $a < b < c$ with $f(b)$ not between $f(a)$ and $f(c)$; the intermediate value theorem would then produce two points with equal values). So f is strictly increasing or strictly decreasing on each I_j.

Step 3: limits at the ends of I_j. f is monotonic and bounded ($|f| \le M$) on I_j. A bounded monotonic function has finite limits at both ends of its interval: e.g. if f is increasing, $\lim_{x\to p_j^-} f(x) = \sup_{I_j} f$ and $\lim_{x\to p_{j-1}^+} f(x) = \inf_{I_j} f$ (the standard monotone-limit argument: given $\varepsilon > 0$ choose $x_0 \in I_j$ with $f(x_0) > \sup f - \varepsilon$; then $\sup f - \varepsilon < f(x) \le \sup f$ for $x_0 < x < p_j$); the decreasing case is the same with sup and inf exchanged.

Step 4: conclusion. Let $x_0 \in (-\pi, \pi)$. If $x_0 \notin D$, then x_0 lies in some I_j where f is continuous, so $f(x_0-) = f(x_0+) = f(x_0)$. If $x_0 = p_j \in D$ with $0 < j < r$, then x_0 is the right end of I_j and the left end of I_{j+1}; by Step 3, $f(x_0-) = \lim_{x\to p_j^-} f(x)$ exists and $f(x_0+) = \lim_{x\to p_j^+} f(x)$ exists. At the left endpoint $-\pi = p_0$, which is the left end of I_1, Step 3 gives that $f(-\pi+)$ exists; at the right endpoint $\pi = p_r$, the right end of I_r, $f(\pi-)$ exists. ■

Remark on the hint. Steps 1–3 make the hint precise: between consecutive points of the finite set D the function is continuous and does not oscillate (monotone), so near each point it behaves monotonically on each side, which forces the one-sided limits to exist (they are finite because f is bounded; e.g. $\sin\frac{1}{x}$ is bounded but fails the finiteness of extrema, and $\frac{1}{\pi - x}$ is monotonic but

unbounded: the one-sided limit fails to be finite).

Answer. Shown: D (endpoints, discontinuities, extrema) is finite; on each gap f is continuous without extrema, hence monotonic and bounded, hence has finite one-sided limits at both ends of the gap; so $f(x\pm)$ exist at every point, $f(-\pi+)$ and $f(\pi-)$ at the endpoints.

Section 35 Even and Odd Functions. Cosine and Sine Series

Throughout, the relevant facts are the Euler formulas

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$$

the reductions (5), (6) of the text (for f even: $a_n = \frac{2}{\pi}\int_0^{\pi} f\cos nx\,dx$, $b_n = 0$; for f odd: $a_n = 0$, $b_n = \frac{2}{\pi}\int_0^{\pi} f\sin nx\,dx$), and the convergence theorem of Section 34: at every point the series converges to the average of the one-sided limits of the periodic extension of f (and to the value of f at points of continuity). The elementary integrals

$$\int_0^{\pi} x\cos nx\,dx = \Big[\frac{x\sin nx}{n}\Big]_0^{\pi} - \frac{1}{n}\int_0^{\pi}\sin nx\,dx = \frac{(-1)^n - 1}{n^2}, \tag{I1}$$

$$\int_0^{\pi} x\sin nx\,dx = \Big[-\frac{x\cos nx}{n}\Big]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx = -\frac{\pi(-1)^n}{n}, \tag{I2}$$

$$\int_0^{\pi}\sin nx\,dx = \frac{1-(-1)^n}{n} \tag{I3}$$

(for $n \geq 1$) will be used repeatedly.

Problem 35-1 (Simmons)

Solution. We use the rules (even)(even) = even, (even)(odd) = odd, (odd)(odd) = even, and check $f(-x)$ directly.

(1) $f(x) = x^5\sin x$: x^5 is odd and $\sin x$ is odd, so the product is *even*. Directly, $f(-x) = (-x)^5\sin(-x) = (-x^5)(-\sin x) = f(x)$.

(2) $f(x) = x^2\sin 2x$: x^2 is even, $\sin 2x$ is odd, so the product is *odd*: $f(-x) = x^2\sin(-2x) = -f(x)$.

(3) $f(x) = e^x$: $f(-x) = e^{-x}$. Since $e^{-x} \neq e^x$ and $e^{-x} \neq -e^x$ (for instance at $x = 1$: $e^{-1} \approx 0.37$, $e^1 \approx 2.72$), e^x is *neither*.

(4) $f(x) = (\sin x)^3$: $f(-x) = (-\sin x)^3 = -\sin^3 x$, *odd*.

(5) $f(x) = \sin x^2$: $f(-x) = \sin((-x)^2) = \sin x^2$, *even*.

(6) $f(x) = \cos(x + x^3)$: $f(-x) = \cos(-x - x^3) = \cos(x + x^3)$ (cosine is even), so *even*.

(7) $f(x) = x + x^2 + x^3$: $f(-x) = -x + x^2 - x^3$. This equals neither $f(x)$ (it differs in the odd terms) nor $-f(x) = -x - x^2 - x^3$ (it differs in the x^2 term; e.g. $f(1) = 3$, $f(-1) = -1$). *Neither.*

(8) $f(x) = \log\frac{1+x}{1-x}$ (defined for $|x| < 1$): $f(-x) = \log\frac{1-x}{1+x} = \log\Big(\frac{1+x}{1-x}\Big)^{-1} = -f(x)$, *odd*.

Answer. even, odd, neither, odd, even, even, neither, odd (in the order listed).

Problem 35-2 (Simmons)

Solution. Let f be defined on $-a \le x \le a$ (so $f(-x)$ is defined whenever $f(x)$ is) and put

$$f_e(x) = \tfrac{1}{2}[f(x) + f(-x)], \qquad f_o(x) = \tfrac{1}{2}[f(x) - f(-x)].$$

f_e is even: $f_e(-x) = \tfrac{1}{2}[f(-x) + f(x)] = f_e(x)$.
f_o is odd: $f_o(-x) = \tfrac{1}{2}[f(-x) - f(x)] = -\tfrac{1}{2}[f(x) - f(-x)] = -f_o(x)$.
Their sum is f: $f_e + f_o = \tfrac{1}{2}[f(x) + f(-x) + f(x) - f(-x)] = f(x)$.

Hence $f = f_e + f_o$ as required. (The decomposition is unique: if $f = g_e + g_o$ is another one, then $f(-x) = g_e(x) - g_o(x)$, so $g_e = f_e$ and $g_o = f_o$ by adding and subtracting.) For example, $e^x = \cosh x + \sinh x$.

Answer. Shown: $f = f_e + f_o$ with $f_e = \tfrac{1}{2}[f(x) + f(-x)]$ even and $f_o = \tfrac{1}{2}[f(x) - f(-x)]$ odd.

Problem 35-3 (Simmons)

Solution. In the substitution $x = -t$ we have $dx = -dt$, and $x = -a \mapsto t = a$, $x = 0 \mapsto t = 0$. Hence

$$\int_{-a}^{0} f(x)\,dx = \int_{a}^{0} f(-t)(-dt) = \int_{0}^{a} f(-t)\,dt = \int_{0}^{a} f(-x)\,dx$$

(the name of the variable of integration is immaterial). Therefore

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} \big[f(-x) + f(x)\big]\,dx.$$

If f is even, $f(-x) = f(x)$ by (1), so the integrand is $2f(x)$ and $\int_{-a}^{a} f\,dx = 2\int_{0}^{a} f\,dx$, which is (3). If f is odd, $f(-x) = -f(x)$ by (2), so the integrand is 0 and $\int_{-a}^{a} f\,dx = 0$, which is (4).

Answer. Shown: $\int_{-a}^{a} f = \int_{0}^{a}[f(x) + f(-x)]\,dx$, which is $2\int_{0}^{a} f$ for even f and 0 for odd f.

Problem 35-4 (Simmons)

Solution. *Sine series.* By (6), $a_n = 0$ and, using (I3),

$$b_n = \frac{2}{\pi}\int_0^{\pi} \frac{\pi}{4}\sin nx\,dx = \frac{1}{2}\int_0^{\pi} \sin nx\,dx = \frac{1-(-1)^n}{2n} = \begin{cases} 1/n, & n \text{ odd},\\ 0, & n \text{ even}. \end{cases}$$

Thus the sine series is $\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots = \sum_{n\ge 1} \frac{\sin(2n-1)x}{2n-1}$. The odd extension of $\pi/4$ is the square wave which is $\pi/4$ on $(0,\pi)$, $-\pi/4$ on $(-\pi,0)$; it is piecewise smooth and continuous on $0 < x < \pi$, so the series converges to $\pi/4$ there. (At $x = 0, \pi$ the sum is 0.)

Putting $x = \pi/2$: $\sin\frac{(2n-1)\pi}{2} = (-1)^{n-1}$, so

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \qquad \text{(Leibniz–Gregory series)}.$$

Cosine series. By (5), $b_n = 0$, $a_0 = \frac{2}{\pi}\int_0^{\pi}\frac{\pi}{4}\,dx = \frac{\pi}{2}$, and for $n \ge 1$ $a_n = \frac{2}{\pi}\cdot\frac{\pi}{4}\int_0^{\pi}\cos nx\,dx = \frac{1}{2}\cdot\frac{\sin n\pi}{n} = 0$. So the cosine series is $\frac{1}{2}a_0 = \frac{\pi}{4}$, i.e. the constant itself (as it must be: a finite trigonometric series is its own Fourier series).

Answer. Sine series as stated; $x=\pi/2$ gives $1-\frac13+\frac15-\cdots=\frac{\pi}{4}$; the cosine series of $\pi/4$ is just $\pi/4$.

Problem 35-5 (Simmons)

Solution. Since $\cos\frac12(-x)=\cos\frac12 x$, f is even, so $b_n=0$ and $a_n=\frac{2}{\pi}\int_0^{\pi}\cos\frac{x}{2}\cos nx\,dx$. For $n=0$:

$$a_0=\frac{2}{\pi}\int_0^{\pi}\cos\frac{x}{2}\,dx=\frac{2}{\pi}\Big[2\sin\frac{x}{2}\Big]_0^{\pi}=\frac{4}{\pi}.$$

For $n\ge 1$ use $\cos A\cos B=\frac12[\cos(A+B)+\cos(A-B)]$:

$$a_n=\frac{1}{\pi}\int_0^{\pi}\Big[\cos\big(n+\tfrac12\big)x+\cos\big(n-\tfrac12\big)x\Big]\,dx=\frac{1}{\pi}\left[\frac{\sin(n+\frac12)\pi}{n+\frac12}+\frac{\sin(n-\frac12)\pi}{n-\frac12}\right].$$

Now $\sin(n\pi\pm\frac{\pi}{2})=\pm\cos n\pi=\pm(-1)^n$, so

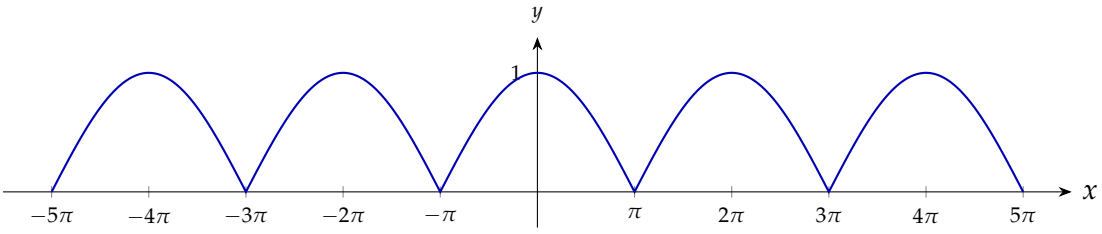
$$a_n=\frac{(-1)^n}{\pi}\left[\frac{1}{n+\frac12}-\frac{1}{n-\frac12}\right]=\frac{(-1)^n}{\pi}\cdot\frac{-1}{n^2-\frac14}=-\frac{4(-1)^n}{\pi(4n^2-1)}.$$

Hence

$$\cos\frac{x}{2}=\frac{2}{\pi}-\frac{4}{\pi}\sum_{n=1}^{\infty}(-1)^n\frac{\cos nx}{4n^2-1},\qquad -\pi\le x\le\pi,$$

and the series is the Fourier series of the period-2π function.

Graph. The periodic extension is continuous, because $f(-\pi)=f(\pi)=\cos\frac{\pi}{2}=0$, and it is piecewise smooth, so the series converges to it at *every* x (Section 34), with corners at the odd multiples of π. On $[-\pi,\pi]$ the extension is $\cos\frac{x}{2}\ge 0$; on the next period it repeats, and in fact the sum equals $|\cos\frac{x}{2}|$ for all x.



Answer. $f(x)=\dfrac{2}{\pi}-\dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\cos nx}{4n^2-1}$; the sum is the continuous wave $|\cos\frac{x}{2}|$.

Problem 35-6 (Simmons)

Solution. We take $\sin x$ on $0\le x\le\pi$.

Sine series. By (6), $b_n=\frac{2}{\pi}\int_0^{\pi}\sin x\sin nx\,dx$. Since $\sin x\sin nx=\frac12[\cos(n-1)x-\cos(n+1)x]$,

$$b_1=\frac{2}{\pi}\int_0^{\pi}\sin^2 x\,dx=\frac{2}{\pi}\cdot\frac{\pi}{2}=1,\qquad b_n=\frac{1}{\pi}\Big[\frac{\sin(n-1)x}{n-1}-\frac{\sin(n+1)x}{n+1}\Big]_0^{\pi}=0\quad(n\ge 2).$$

So the sine series is simply $\sin x=\sin x$ (a finite trigonometric series is its own Fourier series).

Cosine series. By (5), $b_n = 0$ and $a_n = \frac{2}{\pi}\int_0^\pi \sin x \cos nx\,dx$. Here $a_0 = \frac{2}{\pi}[-\cos x]_0^\pi = \frac{4}{\pi}$, and $a_1 = \frac{2}{\pi}\int_0^\pi \sin x\cos x\,dx = \frac{1}{\pi}[\sin^2 x]_0^\pi = 0$. For $n \geq 2$, with $\sin x\cos nx = \frac{1}{2}[\sin(n+1)x - \sin(n-1)x]$ and (I3) (valid for $n \pm 1 \neq 0$),

$$a_n = \frac{1}{\pi}\left[\frac{1-(-1)^{n+1}}{n+1} - \frac{1-(-1)^{n-1}}{n-1}\right] = \frac{1+(-1)^n}{\pi}\left[\frac{1}{n+1} - \frac{1}{n-1}\right] = -\frac{2\,(1+(-1)^n)}{\pi(n^2-1)}.$$

So $a_n = 0$ for odd n, and for $n = 2m$: $a_{2m} = -\dfrac{4}{\pi(4m^2-1)}$. Therefore

$$\sin x = \frac{2}{\pi} - \frac{4}{\pi}\sum_{m=1}^{\infty}\frac{\cos 2mx}{4m^2-1}, \qquad 0 \leq x \leq \pi,$$

valid on the closed interval because the even extension $|\sin x|$ is continuous and piecewise smooth.

Answer. Sine series: $\sin x$ itself. Cosine series: $\sin x = \dfrac{2}{\pi} - \dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{\cos 2nx}{4n^2-1}$, $0 \leq x \leq \pi$.

Problem 35-7 (Simmons)

Solution. In both parts note that $f(x) = \frac{\pi}{2} - |x|$ on $[-\pi, \pi]$ (for $x < 0$, $|x| = -x$), so f is even and $b_n = 0$.

(a) $a_n = \dfrac{2}{\pi}\displaystyle\int_0^\pi \left(\frac{\pi}{2} - x\right)\cos nx\,dx$. For $n = 0$: $a_0 = \frac{2}{\pi}\left[\frac{\pi x}{2} - \frac{x^2}{2}\right]_0^\pi = 0$. For $n \geq 1$, $\int_0^\pi \cos nx\,dx = 0$, so by (I1)

$$a_n = -\frac{2}{\pi}\int_0^\pi x\cos nx\,dx = -\frac{2}{\pi}\cdot\frac{(-1)^n - 1}{n^2} = \frac{2\,[1-(-1)^n]}{\pi n^2} = \begin{cases} \dfrac{4}{\pi n^2}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

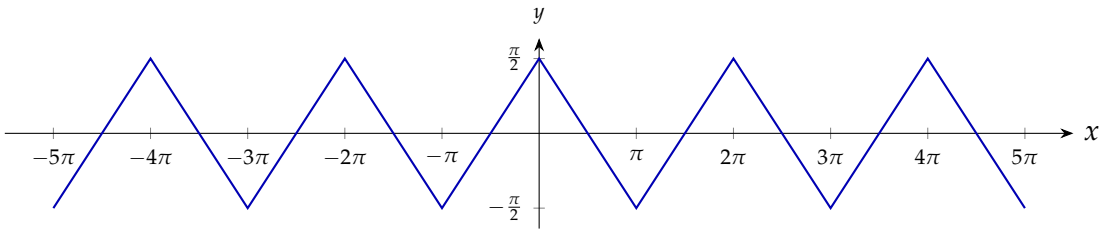
Hence $f(x) = \dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2}$.

(b) From (8), $|x| = \dfrac{\pi}{2} - \dfrac{4}{\pi}\left(\cos x + \dfrac{\cos 3x}{3^2} + \dfrac{\cos 5x}{5^2} + \cdots\right)$ on $[-\pi, \pi]$. Therefore

$$f(x) = \frac{\pi}{2} - |x| = \frac{4}{\pi}\left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right),$$

the same result, with no integration.

Graph. The periodic extension is continuous (since $f(\pi) = f(-\pi) = -\frac{\pi}{2}$), hence the series converges to it everywhere: a triangular wave of period 2π with peaks $\frac{\pi}{2}$ at $x = 0, \pm 2\pi, \pm 4\pi$ and troughs $-\frac{\pi}{2}$ at the odd multiples of π.



Answer. $f(x) = \dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty} \dfrac{\cos(2n-1)x}{(2n-1)^2}$ (triangular wave between $\pm\frac{\pi}{2}$).

Problem 35-8 (Simmons)

Solution.

(a) On $(-\pi,\pi)$: $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\,dx = \frac{1}{\pi}\cdot 2\pi^2 = 2\pi$ (the term $-x$ is odd and integrates to 0). For $n \geq 1$, $\pi\cos nx$ integrates to 0 over a full period and $x\cos nx$ is odd, so

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\cos nx\,dx = 0.$$

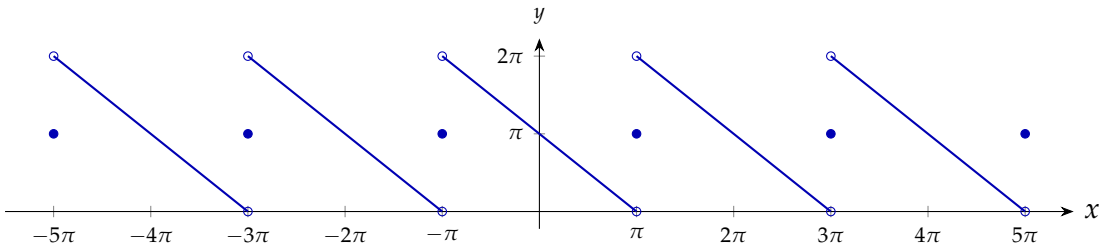
For b_n: $\pi\sin nx$ is odd, and $x\sin nx$ is even, so by (3) and (I2)

$$b_n = -\frac{1}{\pi}\int_{-\pi}^{\pi} x\sin nx\,dx = -\frac{2}{\pi}\int_{0}^{\pi} x\sin nx\,dx = -\frac{2}{\pi}\left(-\frac{\pi(-1)^n}{n}\right) = \frac{2(-1)^n}{n}.$$

Hence

$$\pi - x = \pi + 2\sum_{n=1}^{\infty}(-1)^n\frac{\sin nx}{n}, \qquad -\pi < x < \pi.$$

Graph: the periodic extension is $\pi - (x - 2k\pi)$ on $(2k\pi - \pi, 2k\pi + \pi)$, a sawtooth falling from 2π to 0 on each period, with jumps at $x = \pm\pi, \pm 3\pi, \pm 5\pi$; at each jump the series converges to the average of the one-sided limits 2π and 0, namely π (indeed every term $\sin n(2k+1)\pi$ vanishes and the sum is π).



(b) The cosine series on $[0,\pi]$ uses the even extension $\pi - |x|$. By (5): $a_0 = \frac{2}{\pi}\int_0^{\pi}(\pi - x)\,dx = \frac{2}{\pi}\cdot\frac{\pi^2}{2} = \pi$, and for $n \geq 1$, using $\int_0^{\pi}\cos nx\,dx = 0$ and (I1),

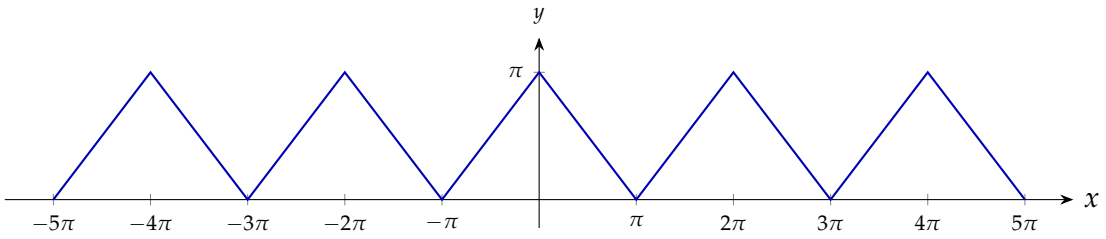
$$a_n = -\frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{2[1-(-1)^n]}{\pi n^2} = \begin{cases} \dfrac{4}{\pi n^2}, & n \text{ odd},\\ 0, & n \text{ even}. \end{cases}$$

So

$$\pi - x = \frac{\pi}{2} + \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2}, \qquad 0 \leq x \leq \pi.$$

(This is also $\frac{\pi}{2} + f(x)$ for the function f of Problem 7, since $\pi - x = \frac{\pi}{2} + (\frac{\pi}{2} - x)$ for $x \geq 0$.)

Graph: the sum is the even extension $\pi - |x|$ made periodic: a continuous triangular wave with peaks π at $x = 0, \pm 2\pi, \pm 4\pi$ and zeros at the odd multiples of π.



(c) The sine series uses the odd extension. By (6) and (I3), (I2): for $n \geq 1$,

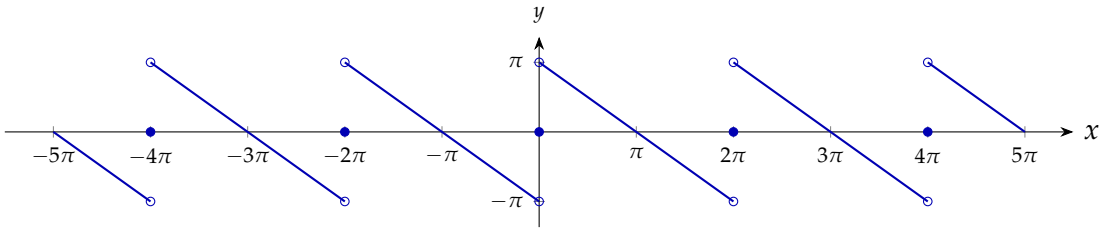
$$b_n = \frac{2}{\pi}\int_0^{\pi} (\pi - x)\sin nx\,dx = \frac{2}{\pi}\left[\pi \cdot \frac{1-(-1)^n}{n} + \frac{\pi(-1)^n}{n}\right] = \frac{2}{n}.$$

So

$$\pi - x = 2\sum_{n=1}^{\infty} \frac{\sin nx}{n}, \qquad 0 < x \leq \pi.$$

At $x = \pi$ every term vanishes and the series sums to $0 = \pi - \pi$, so the closed endpoint $x = \pi$ is included; at $x = 0$ the sum is $0 \neq \pi$.

Graph: the odd extension equals $\pi - x$ on $(0, \pi)$ and $-\pi - x$ on $(-\pi, 0)$; extended with period 2π this is $\pi - (x - 2k\pi)$ on $(2k\pi, 2k\pi + 2\pi)$, a sawtooth falling from π to $-\pi$ with jumps at $x = 0, \pm 2\pi, \pm 4\pi$, where the sum is the average value 0.



Answer. (a) $\pi - x = \pi + 2\sum(-1)^n \frac{\sin nx}{n}$; (b) $\pi - x = \frac{\pi}{2} + \frac{4}{\pi}\sum \frac{\cos(2n-1)x}{(2n-1)^2}$; (c) $\pi - x = 2\sum \frac{\sin nx}{n}$ (sums over $n \geq 1$), with the graphs drawn above.

Problem 35-9 (Simmons)

Solution. By (5), $b_n = 0$ and $a_n = \dfrac{2}{\pi}\left[\displaystyle\int_0^{\pi/2} x\cos nx\,dx + \int_{\pi/2}^{\pi} (\pi - x)\cos nx\,dx\right]$.

Constant term. $a_0 = \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi/2} + \frac{2}{\pi}\int_{\pi/2}^{\pi}(\pi - x)\,dx = \frac{2}{\pi}\left(\frac{\pi^2}{8} + \frac{\pi^2}{8}\right) = \frac{\pi}{2}$, so $\frac{1}{2}a_0 = \frac{\pi}{4}$.

General $n \geq 1$. Let $I_n = \int_0^{\pi/2} x\cos nx\,dx$. In the second integral put $u = \pi - x$: $\cos nx = \cos(n\pi - nu) = (-1)^n \cos nu$, so $\int_{\pi/2}^{\pi}(\pi - x)\cos nx\,dx = (-1)^n \int_0^{\pi/2} u\cos nu\,du = (-1)^n I_n$. Thus

$$a_n = \frac{2}{\pi}\left[1 + (-1)^n\right] I_n,$$

which is 0 for odd n. For even $n = 2m$, integrating by parts,

$$I_{2m} = \left[\frac{x\sin 2mx}{2m}\right]_0^{\pi/2} - \frac{1}{2m}\int_0^{\pi/2} \sin 2mx\,dx = 0 + \left[\frac{\cos 2mx}{4m^2}\right]_0^{\pi/2} = \frac{\cos m\pi - 1}{4m^2} = \frac{(-1)^m - 1}{4m^2},$$

so $a_{2m} = \dfrac{4}{\pi} I_{2m} = \dfrac{(-1)^m - 1}{\pi m^2}$, which is 0 for even m and $-\dfrac{2}{\pi m^2}$ for odd m. Writing the odd m as $m = 2n-1$ ($n = 1, 2, \dots$), the only nonzero coefficients are

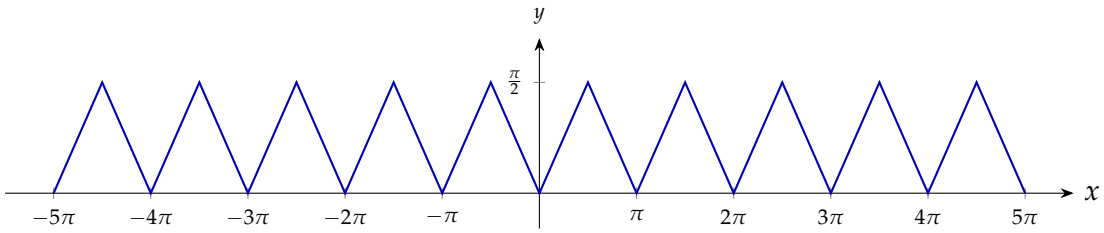
$$a_{2(2n-1)} = -\frac{2}{\pi(2n-1)^2}, \qquad \text{multiplying } \cos 2(2n-1)x,$$

and therefore

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2(2n-1)x}{(2n-1)^2}, \qquad 0 \le x \le \pi,$$

valid on the whole closed interval because the even periodic extension is continuous and piecewise smooth.

Graph. The even extension equals $|x|$ for $|x| \le \frac{\pi}{2}$ and $\pi - |x|$ for $\frac{\pi}{2} \le |x| \le \pi$. Extended with period 2π, the sum is $f(x) = \operatorname{dist}(x, \pi\mathbb{Z})$, a zigzag of period π with zeros at $x = k\pi$ and peaks $\frac{\pi}{2}$ at $x = (k + \frac{1}{2})\pi$.



Answer. Shown: $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\sum_{n\ge1} \frac{\cos 2(2n-1)x}{(2n-1)^2}$; the sum is the zigzag $\operatorname{dist}(x, \pi\mathbb{Z})$.

Problem 35-10 (Simmons)

Solution.

(a) x^2 is even, so $b_n = 0$ and $a_n = \frac{2}{\pi}\int_0^\pi x^2 \cos nx\,dx$. Now $a_0 = \frac{2}{\pi}\cdot\frac{\pi^3}{3} = \frac{2\pi^2}{3}$, so $\frac{1}{2}a_0 = \frac{\pi^2}{3}$. For $n \ge 1$, integrating by parts twice and using (I2),

$$\int_0^\pi x^2 \cos nx\,dx = \Big[\frac{x^2 \sin nx}{n}\Big]_0^\pi - \frac{2}{n}\int_0^\pi x \sin nx\,dx = -\frac{2}{n}\Big(-\frac{\pi(-1)^n}{n}\Big) = \frac{2\pi(-1)^n}{n^2},$$

hence $a_n = \dfrac{4(-1)^n}{n^2}$, giving the stated series. The periodic extension of x^2 from $[-\pi, \pi]$ is continuous (the values at $\pm\pi$ agree) and piecewise smooth, so the series converges to x^2 on the closed interval $[-\pi, \pi]$.

(b) Sine series: $a_n = 0$, $b_n = \frac{2}{\pi}\int_0^\pi x^2 \sin nx\,dx$. Integrate by parts:

$$\int_0^\pi x^2 \sin nx\,dx = \Big[-\frac{x^2\cos nx}{n}\Big]_0^\pi + \frac{2}{n}\int_0^\pi x\cos nx\,dx = -\frac{\pi^2(-1)^n}{n} + \frac{2[(-1)^n - 1]}{n^3}$$

by (I1). Therefore

$$b_n = \frac{2\pi(-1)^{n+1}}{n} - \frac{4[1-(-1)^n]}{\pi n^3} = \frac{2\pi(-1)^{n+1}}{n} - \begin{cases} \dfrac{8}{\pi n^3}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases}$$

that is,

$$x^2 = 2\pi\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n} - \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{(2n-1)^3}, \qquad 0 \le x < \pi.$$

(The odd extension of x^2 jumps at $x = \pm\pi$, so the series is valid on $0 \le x < \pi$ but not at $x = \pi$, where it sums to $0 \neq \pi^2$.)

By (7), $x = 2\sum(-1)^{n+1}\frac{\sin nx}{n}$ for $-\pi < x < \pi$, so the first sum equals $2\pi \cdot \frac{x}{2} = \pi x$ and

$$x^2 = \pi x - \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{(2n-1)^3}, \qquad 0 \le x < \pi.$$

Put $x = \pi/2$: $\sin\frac{(2n-1)\pi}{2} = (-1)^{n+1}$, so

$$\frac{\pi^2}{4} = \frac{\pi^2}{2} - \frac{8}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{(2n-1)^3} \quad\Longrightarrow\quad \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi}{8}\cdot\frac{\pi^2}{4} = \frac{\pi^3}{32},$$

i.e. $1 - \frac{1}{3^3} + \frac{1}{5^3} - \cdots = \frac{\pi^3}{32}$.

(c) Let $Z = \sum_{n\ge1} n^{-3}$; this series converges absolutely, so its terms may be regrouped into odd and even n:

$$Z = \underbrace{\sum_{n\ge1}\frac{1}{(2n-1)^3}}_{s} + \sum_{n\ge1}\frac{1}{(2n)^3} = s + \frac{1}{8}Z.$$

Hence $\frac{7}{8}Z = s$, i.e. $Z = \frac{8}{7}s$.

Answer. (a) $a_0/2 = \pi^2/3$, $a_n = 4(-1)^n/n^2$; (b) $x^2 = 2\pi\sum(-1)^{n+1}\frac{\sin nx}{n} - \frac{8}{\pi}\sum\frac{\sin(2n-1)x}{(2n-1)^3}$ on $0 \le x < \pi$ and $1 - \frac{1}{3^3} + \frac{1}{5^3} - \cdots = \frac{\pi^3}{32}$; (c) $\sum n^{-3} = \frac{8}{7}s$.

Problem 35-11 (Simmons)

Solution.

(a) Cosine series on $[0,\pi]$: $b_n = 0$, $a_n = \frac{2}{\pi}\int_0^\pi x^3\cos nx\,dx$. For $n = 0$: $a_0 = \frac{2}{\pi}\cdot\frac{\pi^4}{4} = \frac{\pi^3}{2}$, so $\frac{1}{2}a_0 = \frac{\pi^3}{4}$. For $n \ge 1$, repeated integration by parts gives the antiderivative

$$\int x^3\cos nx\,dx = \frac{x^3\sin nx}{n} + \frac{3x^2\cos nx}{n^2} - \frac{6x\sin nx}{n^3} - \frac{6\cos nx}{n^4}$$

(differentiate to check: the terms telescope to $x^3\cos nx$). Evaluating between 0 and π (all sine terms vanish):

$$\int_0^\pi x^3\cos nx\,dx = \frac{3\pi^2(-1)^n}{n^2} - \frac{6(-1)^n}{n^4} + \frac{6}{n^4} = \frac{3\pi^2(-1)^n}{n^2} + \frac{6[1-(-1)^n]}{n^4}.$$

Thus

$$a_n = \frac{6\pi(-1)^n}{n^2} + \frac{12[1-(-1)^n]}{\pi n^4} = \frac{6\pi(-1)^n}{n^2} + \begin{cases}\dfrac{24}{\pi n^4}, & n \text{ odd},\\ 0, & n \text{ even},\end{cases}$$

which is the stated expansion. It holds on the closed interval $0 \le x \le \pi$, since the even extension $|x|^3$ has a continuous periodic extension ($|\pm\pi|^3$ agree) and is piecewise smooth.

(b) We need two auxiliary sums obtained from Problem 10(a): at $x=0$, $0=\frac{\pi^2}{3}+4\sum\frac{(-1)^n}{n^2}$, so

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}=-\frac{\pi^2}{12};\qquad \text{and at } x=\pi:\ \pi^2=\frac{\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{1}{n^2}\ \Rightarrow\ \sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}.$$

Now put $x=0$ in (a): $0=\dfrac{\pi^3}{4}+6\pi\Big(-\dfrac{\pi^2}{12}\Big)+\dfrac{24}{\pi}\sum\dfrac{1}{(2n-1)^4}$, i.e.

$$\frac{24}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^4}=\frac{\pi^3}{2}-\frac{\pi^3}{4}=\frac{\pi^3}{4}\quad\Longrightarrow\quad\sum_{n=1}^{\infty}\frac{1}{(2n-1)^4}=\frac{\pi^4}{96}.$$

Then, with $Z_4=\sum_{n\ge1}n^{-4}$ (absolutely convergent, so we may split into odd and even n),

$$Z_4=\frac{\pi^4}{96}+\frac{1}{16}Z_4\ \Longrightarrow\ \frac{15}{16}Z_4=\frac{\pi^4}{96}\ \Longrightarrow\ Z_4=\frac{16\pi^4}{15\cdot96}=\frac{\pi^4}{90}.$$

Answer. (a) $a_0/2=\pi^3/4$, $a_n=\frac{6\pi(-1)^n}{n^2}+\frac{12[1-(-1)^n]}{\pi n^4}$; (b) $\sum\frac{1}{(2n-1)^4}=\frac{\pi^4}{96}$, $\sum\frac{1}{n^4}=\frac{\pi^4}{90}$.

Problem 35-12 (Simmons)

Solution.

(a) x^4 is even, so $b_n=0$ and $a_n=\frac{2}{\pi}\int_0^{\pi}x^4\cos nx\,dx$. First $a_0=\frac{2}{\pi}\cdot\frac{\pi^5}{5}=\frac{2\pi^4}{5}$, so $\frac12a_0=\frac{\pi^4}{5}$. For $n\ge1$, repeated integration by parts gives

$$\int x^4\cos nx\,dx=\frac{x^4\sin nx}{n}+\frac{4x^3\cos nx}{n^2}-\frac{12x^2\sin nx}{n^3}-\frac{24x\cos nx}{n^4}+\frac{24\sin nx}{n^5}$$

(check by differentiating). Between 0 and π all sine terms vanish, and the cosine terms vanish at $x=0$ because of the factors x^3, x:

$$\int_0^{\pi}x^4\cos nx\,dx=(-1)^n\Big[\frac{4\pi^3}{n^2}-\frac{24\pi}{n^4}\Big],\qquad a_n=\frac{2}{\pi}\cdot4\pi(-1)^n\frac{\pi^2n^2-6}{n^4}=8(-1)^n\frac{\pi^2n^2-6}{n^4}.$$

This is the stated series; it is valid on $[-\pi,\pi]$ since the periodic extension of x^4 is continuous and piecewise smooth.

(b) Put $x=\pi$ ($\cos n\pi=(-1)^n$, so $(-1)^n\cos n\pi=1$):

$$\pi^4=\frac{\pi^4}{5}+8\sum_{n=1}^{\infty}\frac{\pi^2n^2-6}{n^4}=\frac{\pi^4}{5}+8\pi^2\sum_{n=1}^{\infty}\frac{1}{n^2}-48\sum_{n=1}^{\infty}\frac{1}{n^4}.$$

Using $\sum n^{-2}=\pi^2/6$ (Problem 11(b)),

$$48\sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{5}+\frac{8\pi^4}{6}-\pi^4=\pi^4\cdot\frac{3+20-15}{15}=\frac{8\pi^4}{15}\quad\Longrightarrow\quad\sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{8\pi^4}{15\cdot48}=\frac{\pi^4}{90}.$$

Answer. (a) $a_n=8(-1)^n(\pi^2n^2-6)/n^4$, $\frac12a_0=\pi^4/5$; (b) $\sum_{n\ge1}n^{-4}=\pi^4/90$.

Problem 35-13 (Simmons)

Solution.

(a) $\cos \alpha x$ is even in x, so $b_n = 0$ and $a_n = \frac{2}{\pi}\int_0^\pi \cos \alpha x \cos nx \, dx$. With $\cos \alpha x \cos nx = \frac{1}{2}[\cos(\alpha + n)x + \cos(\alpha - n)x]$ (note $\alpha \pm n \neq 0$ because α is not an integer),

$$a_n = \frac{1}{\pi}\left[\frac{\sin(\alpha+n)\pi}{\alpha+n} + \frac{\sin(\alpha-n)\pi}{\alpha-n}\right], \qquad n \geq 0.$$

Since $\sin(\alpha \pm n)\pi = \sin \alpha\pi \cos n\pi \pm \cos \alpha\pi \sin n\pi = (-1)^n \sin \alpha\pi$,

$$a_n = \frac{(-1)^n \sin \alpha\pi}{\pi}\Big[\frac{1}{\alpha+n} + \frac{1}{\alpha-n}\Big] = \frac{(-1)^n \, 2\alpha \sin \alpha\pi}{\pi(\alpha^2 - n^2)},$$

and for $n = 0$ the same computation gives $a_0 = \dfrac{2\sin \alpha\pi}{\alpha\pi}$, so $\frac{1}{2}a_0 = \dfrac{\sin \alpha\pi}{\alpha\pi}$. This is the stated series. Because the even periodic extension of $\cos \alpha x$ is continuous (its values at $\pm\pi$ agree) and piecewise smooth, the expansion holds on the closed interval $[-\pi, \pi]$.

(b) Put $x = \pi$ in (a); $(-1)^n \cos n\pi = 1$:

$$\cos \alpha\pi = \frac{\sin \alpha\pi}{\alpha\pi} + \frac{2\alpha \sin \alpha\pi}{\pi}\sum_{n=1}^{\infty} \frac{1}{\alpha^2 - n^2}.$$

Since α is not an integer, $\sin \alpha\pi \neq 0$; multiplying by $\pi / \sin \alpha\pi$ gives

$$\pi \cot \alpha\pi = \frac{1}{\alpha} + 2\alpha \sum_{n=1}^{\infty} \frac{1}{\alpha^2 - n^2}.$$

(c) Rename $\alpha = t$ with $0 < t < 1$ (so t is not an integer) and move $\frac{1}{t}$ to the left:

$$\pi \cot \pi t - \frac{1}{t} = \sum_{n=1}^{\infty} \frac{2t}{t^2 - n^2} = \sum_{n=1}^{\infty} \frac{-2t}{n^2 - t^2}.$$

Integrate from 0 to x, $0 < x < 1$. Term by term this is legitimate: for $0 \leq t \leq x$ the nth term is bounded in absolute value by $\frac{2x}{n^2 - x^2}$, and $\sum \frac{2x}{n^2 - x^2}$ converges (compare with $\sum n^{-2}$), so the series converges uniformly on $[0, x]$ by the Weierstrass M-test.

Left side. An antiderivative of $\pi \cot \pi t - \frac{1}{t}$ is $\log \sin \pi t - \log t = \log \dfrac{\sin \pi t}{t}$ (positive argument for $0 < t < 1$), and $\dfrac{\sin \pi t}{t} \to \pi$ as $t \to 0^+$. Thus

$$\int_0^x \Big(\pi \cot \pi t - \frac{1}{t}\Big)\, dt = \log \frac{\sin \pi x}{x} - \log \pi = \log \frac{\sin \pi x}{\pi x}.$$

Right side. Since $\dfrac{d}{dt}\log(n^2 - t^2) = \dfrac{-2t}{n^2 - t^2}$,

$$\int_0^x \frac{-2t}{n^2 - t^2}\, dt = \log(n^2 - x^2) - \log n^2 = \log\Big(1 - \frac{x^2}{n^2}\Big).$$

Therefore

$$\log\Big(\frac{\sin \pi x}{\pi x}\Big) = \sum_{n=1}^{\infty} \log\Big(1 - \frac{x^2}{n^2}\Big), \qquad 0 < x < 1.$$

The partial sums of the right side equal $\log \prod_{n=1}^{N}(1 - x^2/n^2)$; since they converge, and the exponential function is continuous, the partial products converge to $\exp(\text{sum})$:

$$\frac{\sin \pi x}{\pi x} = \prod_{n=1}^{\infty}\Big(1 - \frac{x^2}{n^2}\Big) = \Big(1 - \frac{x^2}{1^2}\Big)\Big(1 - \frac{x^2}{2^2}\Big)\Big(1 - \frac{x^2}{3^2}\Big)\cdots.$$

Finally, replace x by x/π (so $0<x<\pi$): $\dfrac{\sin x}{x}=\prod_{n\ge1}\Big(1-\dfrac{x^2}{n^2\pi^2}\Big)=\Big(1-\dfrac{x^2}{\pi^2}\Big)\Big(1-\dfrac{x^2}{4\pi^2}\Big)\Big(1-\dfrac{x^2}{9\pi^2}\Big)\cdots$. The factor $1-\frac{x^2}{n^2\pi^2}=\big(1-\frac{x}{n\pi}\big)\big(1+\frac{x}{n\pi}\big)$ vanishes exactly at $x=\pm n\pi$, which are the nonzero roots of $\sin x=0$.

Answer. (a) $a_n=\frac{2\alpha(-1)^n\sin\alpha\pi}{\pi(\alpha^2-n^2)}$, $\frac12a_0=\frac{\sin\alpha\pi}{\alpha\pi}$ (series as stated); (b) $\pi\cot\alpha\pi=\frac{1}{\alpha}+2\alpha\sum\frac{1}{\alpha^2-n^2}$; (c) $\log\frac{\sin\pi x}{\pi x}=\sum\log(1-\frac{x^2}{n^2})$ and Euler's product $\frac{\sin x}{x}=\prod(1-\frac{x^2}{n^2\pi^2})$.

Problem 35-14 (Simmons)

Solution. The identities themselves follow from $\cos2x=\cos^2x-\sin^2x$ and $\sin^2x+\cos^2x=1$: adding and subtracting give $2\cos^2x=1+\cos2x$ and $2\sin^2x=1-\cos2x$.

The right-hand sides are finite trigonometric series, $\frac12a_0+\sum(a_n\cos nx+b_n\sin nx)$ with only a_0 and a_2 nonzero. By the orthogonality relations of Section 33 (equivalently, the remark in Section 33 that any finite trigonometric series is automatically the Fourier series of its sum), integrating a finite trigonometric series term by term against $\cos nx$ or $\sin nx$ over $[-\pi,\pi]$ returns exactly its own coefficients. Hence the right-hand sides are the Fourier series of $\sin^2x$ and $\cos^2x$. Consistently with the evenness of these functions, only cosine terms (no sine terms) appear. Explicitly: $\sin^2x$ has $\frac12a_0=\frac12$, $a_2=-\frac12$; $\cos^2x$ has $\frac12a_0=\frac12$, $a_2=\frac12$; all other coefficients vanish.

Answer. Shown: each right side is a finite cosine polynomial, hence equal to the Fourier series of its sum (by uniqueness/orthogonality).

Problem 35-15 (Simmons)

Solution. On $0<x<\pi$ we compute $b_n=\frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx$ for $f=\sin^2x$ and $f=\cos^2x$, using $\sin^2x=\frac12(1-\cos2x)$, $\cos^2x=\frac12(1+\cos2x)$:

$$b_n^{(\sin^2)}=\frac{1}{\pi}\Big[\int_0^\pi\sin nx\,dx-J_n\Big],\qquad b_n^{(\cos^2)}=\frac{1}{\pi}\Big[\int_0^\pi\sin nx\,dx+J_n\Big],$$

where $J_n=\int_0^\pi\cos2x\sin nx\,dx$. By (I3), $\int_0^\pi\sin nx\,dx=\frac{1-(-1)^n}{n}$, i.e. $\frac{2}{n}$ for odd n and 0 for even n. Using $\cos2x\sin nx=\frac12[\sin(n+2)x+\sin(n-2)x]$ and the fact that $\int_0^\pi\sin kx\,dx=\frac{1-(-1)^k}{k}$ for every integer $k\neq0$ (for negative k this is also valid since the integrand is odd in k and so is the right side; and $\int_0^\pi\sin0\,dx=0$):

n even: $n\pm2$ are even, every integral vanishes, so $J_n=0$ and $b_n=0$ for both functions.

n odd: $n\pm2$ are odd (nonzero), so

$$J_n=\frac12\Big[\frac{2}{n+2}+\frac{2}{n-2}\Big]=\frac{1}{n+2}+\frac{1}{n-2}.$$

Writing $n=2m-1$ ($m=1,2,\dots$), $n+2=2m+1$ and $n-2=2m-3$:

$$b_{2m-1}^{(\sin^2)}=\frac{1}{\pi}\Big[\frac{2}{2m-1}-\frac{1}{2m+1}-\frac{1}{2m-3}\Big],\qquad b_{2m-1}^{(\cos^2)}=\frac{1}{\pi}\Big[\frac{2}{2m-1}+\frac{1}{2m+1}+\frac{1}{2m-3}\Big].$$

Thus

$$\sin^2x=\frac{1}{\pi}\sum_{m=1}^{\infty}\Big[\frac{2}{2m-1}-\frac{1}{2m+1}-\frac{1}{2m-3}\Big]\sin(2m-1)x,\qquad 0\le x\le\pi,$$

$$\cos^2 x = \frac{1}{\pi}\sum_{m=1}^{\infty}\Big[\frac{2}{2m-1}+\frac{1}{2m+1}+\frac{1}{2m-3}\Big]\sin(2m-1)x, \qquad 0<x<\pi.$$

The first is valid on the closed interval because $\sin^2 x$ vanishes at $x=0$ and $x=\pi$ (its odd extension is continuous and piecewise smooth); the second only on the open interval, since $\cos^2 0=\cos^2\pi=1$ while every sine series vanishes there.

Check of $\sin^2 x+\cos^2 x=1$. Adding the coefficients, the terms $\mp\frac{1}{2m+1}\mp\frac{1}{2m-3}$ cancel and

$$\sin^2 x+\cos^2 x=\frac{1}{\pi}\sum_{m=1}^{\infty}\frac{4}{2m-1}\sin(2m-1)x=\frac{4}{\pi}\Big(\sin x+\frac{\sin 3x}{3}+\cdots\Big)=1, \quad 0<x<\pi,$$

by Problem 4. (This is the sine series of the constant 1, as it should be.)

Answer. $\sin^2 x=\frac{1}{\pi}\sum_{n\ge1}\big[\frac{2}{2n-1}-\frac{1}{2n+1}-\frac{1}{2n-3}\big]\sin(2n-1)x$ on $0\le x\le\pi$; $\cos^2 x=\frac{1}{\pi}\sum_{n\ge1}\big[\frac{2}{2n-1}+\frac{1}{2n+1}+\frac{1}{2n-3}\big]\sin(2n-1)x$ on $0<x<\pi$; their sum is $\frac{4}{\pi}\sum\frac{\sin(2n-1)x}{2n-1}=1$.

Remark. The printed answer to the second series in the book has minus signs in front of $\frac{1}{2n+1}$ and $\frac{1}{2n-3}$, i.e. it repeats the $\sin^2 x$ coefficients. That cannot be right for $\cos^2 x$: for example $b_1=\frac{2}{\pi}\int_0^\pi\cos^2 x\sin x\,dx=\frac{2}{\pi}\cdot\frac{2}{3}=\frac{4}{3\pi}$, which matches the plus-sign formula ($\frac{1}{\pi}[2+\frac{1}{3}-1]=\frac{4}{3\pi}$), whereas the printed form would give $\frac{1}{\pi}[2-\frac{1}{3}+1]=\frac{8}{3\pi}$ (which is b_1 of $\sin^2 x$). The plus signs are also the ones that make $\sin^2 x+\cos^2 x=1$ work.

Problem 35-16 (Simmons)

Solution. (The book prints the second identity with a misprint, "$\frac{3}{4}x$"; the intended term is $\frac{3}{4}\cos x$.)

Proof of the identities. By the addition formulas,

$$\begin{aligned}\sin 3x &= \sin(2x+x)=\sin 2x\cos x+\cos 2x\sin x=2\sin x\cos^2 x+(1-2\sin^2 x)\sin x\\ &=2\sin x(1-\sin^2 x)+\sin x-2\sin^3 x=3\sin x-4\sin^3 x,\\ \cos 3x &= \cos(2x+x)=\cos 2x\cos x-\sin 2x\sin x=(2\cos^2 x-1)\cos x-2\sin^2 x\cos x\\ &=(2\cos^2 x-1)\cos x-2(1-\cos^2 x)\cos x=4\cos^3 x-3\cos x.\end{aligned}$$

Solving for the cubes gives

$$\sin^3 x=\tfrac{1}{4}(3\sin x-\sin 3x)=\tfrac{3}{4}\sin x-\tfrac{1}{4}\sin 3x,$$
$$\cos^3 x=\tfrac{1}{4}(3\cos x+\cos 3x)=\tfrac{3}{4}\cos x+\tfrac{1}{4}\cos 3x.$$

Fourier series. The right-hand sides are finite trigonometric series. By the orthogonality of $1,\cos nx,\sin nx$ on $[-\pi,\pi]$, a finite trigonometric series is its own Fourier series, i.e. its coefficients are exactly those given by the Euler formulas applied to its sum (Section 33). Hence the right sides are the Fourier series of $\sin^3 x$ and $\cos^3 x$. This is consistent with parity: $\sin^3 x$ is odd and contains only sines, $\cos^3 x$ is even and contains only cosines.

Answer. Proved: $\sin 3x=3\sin x-4\sin^3 x$, $\cos 3x=4\cos^3 x-3\cos x$; the right-hand sides are finite trigonometric series and therefore are the Fourier series of the left-hand sides.

Section 36 Extension to Arbitrary Intervals

For a function f on $-L \le x < L$ we use the substitution $t = \pi x/L$ of the text, i.e. $g(t) = f(Lt/\pi)$, and the formulas

$$f(x) = \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty}\Big(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\Big),$$
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx, \qquad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx.$$

For even (odd) f the integrals reduce to $\frac{2}{L}\int_0^L$ with $b_n = 0$ ($a_n = 0$). A function given on $0 \le x \le L$ has a cosine (sine) series obtained from its even (odd) extension to $[-L, L]$. Convergence is as in Section 34: to $f(x)$ at points of continuity of the periodic extension, and to the average of the one-sided limits at jumps. We use $\int_0^\pi t\cos nt\,dt = \dfrac{(-1)^n - 1}{n^2}$, $\int_0^\pi t\sin nt\,dt = -\dfrac{\pi(-1)^n}{n}$, $\int_0^\pi \sin nt\,dt = \dfrac{1-(-1)^n}{n}$ $(n \ge 1)$.

Problem 36-1 (Simmons)

Solution. In Example 1 of the text, the function f_1 equal to 0 on $-2 \le x < 0$ and 1 on $0 \le x < 2$ has the expansion

$$f_1(x) = \frac{1}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\frac{(2n-1)\pi x}{2}.$$

Our function takes the values -3 and 3, so $f = 6f_1 - 3$ (indeed $6\cdot 0 - 3 = -3$ and $6\cdot 1 - 3 = 3$). A Fourier series is linear in the function (the coefficients are integrals of f), so

$$f(x) = 6\Big[\frac{1}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\frac{(2n-1)\pi x}{2}\Big] - 3 = \frac{12}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\frac{(2n-1)\pi x}{2}.$$

This holds at every point except the jumps $x = 0, \pm 2, \pm 4, \ldots$ (where the sum is 0, the average of -3 and 3).

Answer. $f(x) = \dfrac{12}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\frac{(2n-1)\pi x}{2}.$

Problem 36-2 (Simmons)

Solution.

(a) Here $L = 1$ and $f(x) = 1 - |x|$ is even, so $b_n = 0$ and $a_n = 2\int_0^1 (1-x)\cos n\pi x\,dx$. For $n = 0$: $a_0 = 2\int_0^1 (1-x)\,dx = 1$. For $n \ge 1$, $\int_0^1 \cos n\pi x\,dx = 0$, and with $t = n\pi x$,

$$\int_0^1 x\cos n\pi x\,dx = \frac{1}{n^2\pi^2}\int_0^{n\pi} t\cos t\,dt = \frac{1}{n^2\pi^2}\big[t\sin t + \cos t\big]_0^{n\pi} = \frac{(-1)^n - 1}{n^2\pi^2}.$$

So $a_n = -2\cdot\dfrac{(-1)^n-1}{n^2\pi^2} = \dfrac{2[1-(-1)^n]}{n^2\pi^2}$, i.e. $\dfrac{4}{n^2\pi^2}$ for odd n and 0 for even n. Hence

$$f(x) = \frac{1}{2} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{\cos(2n-1)\pi x}{(2n-1)^2}, \qquad -1 \le x \le 1.$$

(Cross-check with (8) of Section 35, $|t| = \frac{\pi}{2} - \frac{4}{\pi}\sum \frac{\cos(2n-1)t}{(2n-1)^2}$ with $t = \pi x$: $1 - |x| = 1 - \frac{|t|}{\pi} = \frac{1}{2} + \frac{4}{\pi^2}\sum\cdots$.) The periodic extension is continuous, so the expansion holds on the whole closed interval.

(b) Here $L = 2$ and $|x|$ is even: $b_n = 0$ and $a_n = \int_0^2 x\cos\frac{n\pi x}{2}\,dx$. Then $a_0 = \int_0^2 x\,dx = 2$. For $n \geq 1$ substitute $t = \frac{n\pi x}{2}$, $x = \frac{2t}{n\pi}$:

$$a_n = \frac{4}{n^2\pi^2}\int_0^{n\pi} t\cos t\,dt = \frac{4}{n^2\pi^2}\left[(-1)^n - 1\right] = \begin{cases} -\dfrac{8}{n^2\pi^2}, & n \text{ odd},\\ 0, & n \text{ even}. \end{cases}$$

Therefore

$$|x| = 1 - \frac{8}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{2}, \qquad -2 \leq x \leq 2,$$

again valid on the closed interval since the periodic extension is continuous.

Answer. (a) $f(x) = \dfrac{1}{2} + \dfrac{4}{\pi^2}\displaystyle\sum_{n=1}^{\infty}\frac{\cos(2n-1)\pi x}{(2n-1)^2}$; (b) $|x| = 1 - \dfrac{8}{\pi^2}\displaystyle\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{2}$.

Problem 36-3 (Simmons)

Solution. The series has period L in x (each term $\sin\frac{2n\pi x}{L}$ has period L), so we expand on the symmetric interval $-\ell \leq x < \ell$ of length $2\ell = L$, $\ell = L/2$. Let F be the function with period L that equals $\frac{L}{2} - x$ on $0 < x < L$. On $-\ell < x < 0$ the point $x + L$ lies in $(0, L)$, so

$$F(x) = \frac{L}{2} - (x + L) = -\frac{L}{2} - x \quad (-\ell < x < 0), \qquad F(x) = \frac{L}{2} - x \quad (0 < x < \ell),$$

i.e. $F(x) = \ell - x$ on $(0, \ell)$ and $F(-x) = -\ell + x = -F(x)$: F is *odd*. Thus $a_n = 0$ and, with the half-length ℓ,

$$b_n = \frac{2}{\ell}\int_0^{\ell}(\ell - x)\sin\frac{n\pi x}{\ell}\,dx.$$

Substitute $t = \pi x/\ell$, $x = \ell t/\pi$, $dx = \ell\,dt/\pi$:

$$\begin{aligned} b_n &= \frac{2}{\ell}\cdot\frac{\ell}{\pi}\int_0^{\pi}\ell\Big(1 - \frac{t}{\pi}\Big)\sin nt\,dt = \frac{2\ell}{\pi}\Big[\int_0^{\pi}\sin nt\,dt - \frac{1}{\pi}\int_0^{\pi} t\sin nt\,dt\Big]\\ &= \frac{2\ell}{\pi}\Big[\frac{1 - (-1)^n}{n} + \frac{(-1)^n}{n}\Big] = \frac{2\ell}{\pi n} = \frac{L}{\pi n}. \end{aligned}$$

Since $\sin\dfrac{n\pi x}{\ell} = \sin\dfrac{2n\pi x}{L}$, the Fourier series of F is $\dfrac{L}{\pi}\displaystyle\sum_{n\geq 1}\frac{1}{n}\sin\frac{2n\pi x}{L}$. The periodic function F is continuous and smooth on $0 < x < L$ (its only jumps are at $x = 0, \pm L, \pm 2L, \ldots$; at $x = \pm\ell$ the two branches agree, both having the value 0), so the series converges to $F(x) = \frac{L}{2} - x$ for $0 < x < L$.

Answer. Shown: the Fourier series of the period-L sawtooth $\frac{L}{2} - x$ has $a_n = 0$, $b_n = \frac{L}{n\pi}$, so $\frac{L}{2} - x = \frac{L}{\pi}\sum\frac{1}{n}\sin\frac{2n\pi x}{L}$ on $0 < x < L$.

Problem 36-4 (Simmons)

Solution. Take $L=1$ and extend f evenly to $[-1,1]$: $b_n=0$, $a_n=2\int_0^1 (x^2-x+\tfrac16)\cos n\pi x\,dx$. For $n=0$: $a_0=2\big[\tfrac13-\tfrac12+\tfrac16\big]=0$. For $n\ge 1$ let $m=n$ and put $t=m\pi x$. Then $\int_0^1 \cos m\pi x\,dx=0$ and

$$\int_0^1 x\cos m\pi x\,dx=\frac{(-1)^m-1}{m^2\pi^2}\quad\text{(from Problem 2(a))},$$
$$\int_0^1 x^2\cos m\pi x\,dx=\Big[\frac{x^2\sin m\pi x}{m\pi}\Big]_0^1-\frac{2}{m\pi}\int_0^1 x\sin m\pi x\,dx,$$
$$\int_0^1 x\sin m\pi x\,dx=\Big[-\frac{x\cos m\pi x}{m\pi}\Big]_0^1+\frac{1}{m\pi}\int_0^1\cos m\pi x\,dx=-\frac{(-1)^m}{m\pi},$$

so $\int_0^1 x^2\cos m\pi x\,dx=\dfrac{2(-1)^m}{m^2\pi^2}$. Therefore

$$a_m=2\Big[\frac{2(-1)^m}{m^2\pi^2}-\frac{(-1)^m-1}{m^2\pi^2}\Big]=\frac{2[1+(-1)^m]}{m^2\pi^2}=\begin{cases}\dfrac{4}{m^2\pi^2}, & m\text{ even},\\ 0, & m\text{ odd}.\end{cases}$$

Putting $m=2n$ gives $a_{2n}=\dfrac{1}{n^2\pi^2}$, so

$$x^2-x+\frac16=\frac{1}{\pi^2}\sum_{n=1}^{\infty}\frac{\cos 2n\pi x}{n^2},\qquad 0\le x\le 1,$$

valid on the closed interval since $f(0)=f(1)=\tfrac16$ makes the even periodic extension continuous.

Answer. $f(x)=\dfrac{1}{\pi^2}\displaystyle\sum_{n=1}^{\infty}\frac{\cos 2n\pi x}{n^2}$.

Problem 36-5 (Simmons)

Solution. Here $L=2$, and $b_n=0$ for the even extension, with $a_n=\dfrac{2}{L}\displaystyle\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx=\int_0^1 2\cos\frac{n\pi x}{2}\,dx$. For $n=0$: $a_0=\int_0^1 2\,dx=2$, so $\tfrac12 a_0=1$. For $n\ge1$:

$$a_n=2\Big[\frac{2}{n\pi}\sin\frac{n\pi x}{2}\Big]_0^1=\frac{4}{n\pi}\sin\frac{n\pi}{2}.$$

This is 0 for even n; for odd $n=2m-1$, $\sin\frac{(2m-1)\pi}{2}=(-1)^{m+1}$. Hence

$$f(x)=1+\frac{4}{\pi}\sum_{m=1}^{\infty}\frac{(-1)^{m+1}}{2m-1}\cos\frac{(2m-1)\pi x}{2},\qquad 0\le x\le 2,\ x\ne 1.$$

At the jump $x=1$ the series sums to the average 1 (every cosine term vanishes there).

Answer. $f(x)=1+\dfrac{4}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{2n-1}\cos\frac{(2n-1)\pi x}{2}$.

Problem 36-6 (Simmons)

Solution. Here $L = 1$, so $t = \pi x$ runs over $[-\pi, \pi]$ and $g(t) = \cos t$. Directly, by orthogonality of $\{1, \cos nt, \sin nt\}$ on $[-\pi, \pi]$: $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos t \cos nt \, dt$ equals 1 for $n = 1$ and 0 for $n \neq 1$ (for $n = 0$ as well), and $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos t \sin nt \, dt = 0$ for all n (odd integrand). So the Fourier series of $\cos t$ is just $\cos t$, and in terms of x, $\cos \pi x = \cos \pi x$. (This is the remark of Section 33: a finite trigonometric series is its own Fourier series.) The expansion is valid for all x.

Answer. $\cos \pi x = \cos \pi x$ (only $a_1 = 1$ is nonzero).

Problem 36-7 (Simmons)

Solution. Here $L = 1$, $b_n = 0$ and $a_n = 2\int_0^1 f(x) \cos n\pi x \, dx$. Note that $f(1-x) = f(x)$: for $0 \le x < \frac{1}{2}$, $1 - x \in (\frac{1}{2}, 1]$ and $f(1-x) = (1-x) - \frac{3}{4} = \frac{1}{4} - x$. Hence in $\int_{1/2}^1 f(x)\cos n\pi x \, dx$, the substitution $x \to 1 - x$ gives $\int_0^{1/2} f(x) \cos n\pi(1-x)\, dx = (-1)^n \int_0^{1/2} f(x)\cos n\pi x \, dx$, so

$$a_n = 2\big[1 + (-1)^n\big] \int_0^{1/2} \Big(\frac{1}{4} - x\Big) \cos n\pi x \, dx,$$

which vanishes for odd n. Also $a_0 = 2\int_0^1 f \, dx = 4\int_0^{1/2}(\frac{1}{4} - x)\, dx = 4\big[\frac{1}{8} - \frac{1}{8}\big] = 0$.

For even $n = 2m$, $m \ge 1$: $\displaystyle\int_0^{1/2} \cos 2m\pi x \, dx = \frac{\sin m\pi}{2m\pi} = 0$ and

$$\begin{aligned}\int_0^{1/2} x \cos 2m\pi x \, dx &= \Big[\frac{x \sin 2m\pi x}{2m\pi}\Big]_0^{1/2} - \frac{1}{2m\pi}\int_0^{1/2} \sin 2m\pi x \, dx \\ &= 0 - \frac{1}{2m\pi}\cdot\frac{1 - \cos m\pi}{2m\pi} = -\frac{1 - (-1)^m}{4m^2\pi^2}.\end{aligned}$$

Therefore

$$a_{2m} = 4\Big[0 + \frac{1 - (-1)^m}{4m^2\pi^2}\Big] = \frac{1 - (-1)^m}{m^2\pi^2} = \begin{cases} \dfrac{2}{m^2\pi^2}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases}$$

The only nonzero coefficients are a_{2m} with $m = 2n - 1$ odd, multiplying $\cos 2(2n-1)\pi x$:

$$f(x) = \frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{\cos 2(2n-1)\pi x}{(2n-1)^2}, \qquad 0 \le x \le 1,$$

valid on the closed interval because f is continuous with $f(0) = \frac{1}{4}$, $f(1) = \frac{1}{4}$, $f(\frac{1}{2}) = -\frac{1}{4}$ and piecewise smooth, so its even periodic extension is continuous.

Answer. $f(x) = \dfrac{2}{\pi^2}\displaystyle\sum_{n=1}^{\infty} \frac{\cos 2(2n-1)\pi x}{(2n-1)^2}$.

Problem 36-8 (Simmons)

Solution.

(a) First, $\dfrac{e^x+1}{e^x-1}-1=\dfrac{e^x+1-(e^x-1)}{e^x-1}=\dfrac{2}{e^x-1}$, so $\dfrac{x}{2}\Big(\dfrac{e^x+1}{e^x-1}-1\Big)=\dfrac{x}{e^x-1}$, confirming the given identity. Let $h(x)=\dfrac{x}{2}\cdot\dfrac{e^x+1}{e^x-1}$. Then

$$h(-x)=\frac{-x}{2}\cdot\frac{e^{-x}+1}{e^{-x}-1}=\frac{-x}{2}\cdot\frac{1+e^x}{1-e^x}=\frac{x}{2}\cdot\frac{e^x+1}{e^x-1}=h(x)$$

(multiply numerator and denominator by e^x), so h is even and its power series contains only even powers of x. Hence

$$\sum_{n\ge0}\frac{B_n}{n!}x^n=-\frac{x}{2}+h(x)$$

has x-coefficient $B_1=-\frac12$ (the term $-\frac{x}{2}$ is the only odd-power contribution), and the coefficient of every odd power x^n, $n\ge3$, is 0. By uniqueness of power series coefficients, $B_n=0$ for odd $n>1$.

(b) Multiply $(*)$ by $\dfrac{e^x-1}{x}=\sum_{j\ge0}\dfrac{x^j}{(j+1)!}$: the product equals 1. In the Cauchy product the coefficient of x^{n-1} $(n\ge1)$ is

$$\sum_{k=0}^{n-1}\frac{B_k}{k!}\cdot\frac{1}{(n-1-k+1)!}=\sum_{k=0}^{n-1}\frac{B_k}{k!\,(n-k)!}.$$

For $n=1$ this is the constant term, $B_0=1$ (consistent with the right side 1). For $n\ge2$ the coefficient of x^{n-1} on the right side is 0, so $\sum_{k=0}^{n-1}\frac{B_k}{k!(n-k)!}=0$. Multiplying by $n!$ and using $\binom{n}{k}=\frac{n!}{k!(n-k)!}$ gives

$$\sum_{k=0}^{n-1}\binom{n}{k}B_k=0\qquad(n\ge2),$$

which is $(**)$.

(c) We use $B_0=1$, $B_1=-\frac12$ and $B_3=B_5=\cdots=0$ from (a), and solve $(**)$ successively for the unknown even-index number B_{n-1}, whose coefficient is $\binom{n}{n-1}=n$.

$n=3$: $\binom30B_0+\binom31B_1+\binom32B_2=0$, i.e. $1-\frac32+3B_2=0$, so $B_2=\frac16$.

$n=5$: $B_0+5B_1+10B_2+10B_3+5B_4=0$: $1-\frac52+\frac{10}{6}+0+5B_4=0$. Since $1-\frac52+\frac53=\frac16$, $5B_4=-\frac16$ and $B_4=-\frac{1}{30}$.

$n=7$: $B_0+7B_1+21B_2+35B_3+35B_4+21B_5+7B_6=0$: $1-\frac72+\frac{21}{6}+0-\frac{35}{30}+0+7B_6=0$. Here $1-\frac72+\frac72-\frac76=-\frac16$, so $7B_6=\frac16$ and $B_6=\frac{1}{42}$.

$n=9$: with $\binom9k=1,9,36,84,126,126,84,36,9$: $1+9\big(-\frac12\big)+36\cdot\frac16+0+126\big(-\frac{1}{30}\big)+0+84\cdot\frac{1}{42}+0+9B_8=0$, i.e. $1-\frac92+6-\frac{21}{5}+2+9B_8=0$. The numbers add to $\frac{3}{10}$, so $9B_8=-\frac{3}{10}$ and $B_8=-\frac{1}{30}$.

$n=11$: with $\binom{11}{k}=1,11,55,165,330,462,462,330,165,55,11$ and $B_3=B_5=B_7=B_9=0$:

$$1+11\Big(-\frac12\Big)+55\cdot\frac16+330\Big(-\frac{1}{30}\Big)+462\cdot\frac{1}{42}+165\Big(-\frac{1}{30}\Big)+11B_{10}=0,$$

i.e. $1-\frac{11}{2}+\frac{55}{6}-11+11-\frac{11}{2}+11B_{10}=0$. The numbers add to $-\frac56$, so $11B_{10}=\frac56$ and $B_{10}=\frac{5}{66}$.

Answer. (a) $B_1=-\frac12$, $B_{2k+1}=0$ for $k\ge1$; (b) $\sum_{k=0}^{n-1}\binom{n}{k}B_k=0$ for $n\ge2$; (c) $B_2=\frac16$, $B_4=-\frac{1}{30}$, $B_6=\frac{1}{42}$, $B_8=-\frac{1}{30}$, $B_{10}=\frac{5}{66}$.

Problem 36-9 (Simmons)

Solution.

(a) Multiply the two power series in t (Cauchy product). The coefficient of t^n in the product is

$$\sum_{k=0}^{n} \frac{B_k}{k!} \cdot \frac{x^{n-k}}{(n-k)!}.$$

Equating it with $\dfrac{B_n(x)}{n!}$ and multiplying by $n!$:

$$B_n(x) = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} B_k x^{n-k} = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k},$$

which is the stated formula. The highest power is x^n (from $k=0$) with coefficient $\binom{n}{0}B_0 = 1 \neq 0$; so $B_n(x)$ is a polynomial of degree exactly n (monic).

(b) At $x=0$ all terms with $n-k \geq 1$ vanish and only $k=n$ survives: $B_n(0) = \binom{n}{n}B_n = B_n$. At $x=1$: $B_n(1) = \sum_{k=0}^{n}\binom{n}{k}B_k = \sum_{k=0}^{n-1}\binom{n}{k}B_k + B_n = 0 + B_n = B_n$ for $n \geq 2$, by $(**)$. (For $n=1$: $B_1(1) = 1 + B_1 = \frac{1}{2} \neq B_1$, so the restriction $n \geq 2$ is needed; $B_0(1) = B_0 = 1$ trivially.)

(c) Differentiating the formula in (a) for $n+1$,

$$B'_{n+1}(x) = \sum_{k=0}^{n} \binom{n+1}{k} (n+1-k) B_k x^{n-k}.$$

(The term $k = n+1$ is the constant B_{n+1}, with derivative 0.) Now $(n+1-k)\binom{n+1}{k} = (n+1-k)\dfrac{(n+1)!}{k!(n+1-k)!} = \dfrac{(n+1)\,n!}{k!(n-k)!} = (n+1)\binom{n}{k}$, so

$$B'_{n+1}(x) = (n+1)\sum_{k=0}^{n}\binom{n}{k}B_k x^{n-k} = (n+1)B_n(x).$$

Integrating from 0 to x and using $B_{n+1}(0) = B_{n+1}$ from (b):

$$B_{n+1}(x) - B_{n+1} = (n+1)\int_0^x B_n(t)\,dt, \qquad \text{i.e.} \qquad B_{n+1}(x) = B_{n+1} + (n+1)\int_0^x B_n(t)\,dt.$$

Putting $x=1$ and using $B_{n+1}(1) = B_{n+1}$, which holds when $n+1 \geq 2$, i.e. $n \geq 1$: $B_{n+1} = B_{n+1} + (n+1)\int_0^1 B_n(t)\,dt$, hence $\int_0^1 B_n(x)\,dx = 0$ for $n \geq 1$.

(d) Use (a) with $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$, $B_4 = -\frac{1}{30}$:

$$\begin{aligned}
B_0(x) &= 1,\\
B_1(x) &= x + B_1 = x - \tfrac{1}{2},\\
B_2(x) &= x^2 + 2B_1x + B_2 = x^2 - x + \tfrac{1}{6},\\
B_3(x) &= x^3 + 3B_1x^2 + 3B_2x + B_3 = x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x,\\
B_4(x) &= x^4 + 4B_1x^3 + 6B_2x^2 + 4B_3x + B_4 = x^4 - 2x^3 + x^2 - \tfrac{1}{30}.
\end{aligned}$$

Answer. (a) $B_n(x) = \sum_{k=0}^{n}\binom{n}{k}B_k x^{n-k}$; (b) $B_n(0) = B_n$, $B_n(1) = B_n$ $(n \geq 2)$; (c) $B'_{n+1} = (n+1)B_n$, $B_{n+1}(x) = B_{n+1} + (n+1)\int_0^x B_n$, $\int_0^1 B_n = 0$ $(n \geq 1)$; (d) the listed polynomials.

Problem 36-10 (Simmons)

Solution. Throughout $n \geq 1$. We use the facts from Problems 8 and 9: $B_m' = mB_{m-1}$, $B_m(0) = B_m(1)$ for $m \geq 2$, and $B_1(x) = x - \frac{1}{2}$ (so $B_1(1) - B_1(0) = 1$).

Step 1: symmetry. From the generating function, replacing x by $1-x$ and t by $-t$,

$$\sum_m \frac{B_m(1-x)}{m!}t^m = \frac{te^{(1-x)t}}{e^t - 1} = \frac{te^{-xt}}{1-e^{-t}} = \frac{(-t)e^{x(-t)}}{e^{-t}-1} = \sum_m \frac{B_m(x)}{m!}(-t)^m,$$

so $B_m(1-x) = (-1)^m B_m(x)$. In particular $B_{2n}(1-x) = B_{2n}(x)$.

Step 2: the cosine series on $[0,1]$. With $L = 1$, the cosine series is $\frac{1}{2}a_0 + \sum_{m\geq 1} a_m \cos m\pi x$ with $a_m = 2\int_0^1 B_{2n}(x)\cos m\pi x\,\mathrm{d}x$. For odd m, $\cos m\pi(1-x) = -\cos m\pi x$, so by the symmetry of Step 1 (substituting $x \to 1-x$ in the integral over $[0,1]$) $\int_0^1 B_{2n}\cos m\pi x\,\mathrm{d}x = -\int_0^1 B_{2n}\cos m\pi x\,\mathrm{d}x$, hence $a_m = 0$. Also $a_0 = 2\int_0^1 B_{2n}\,\mathrm{d}x = 0$ by Problem 9(c). Remaining are the coefficients with $m = 2k$, $k \geq 1$, namely $a_{2k} = 2\,C_{2n}$, where for fixed integer $k \geq 1$ we define

$$C_r = \int_0^1 B_r(x)\cos 2\pi kx\,\mathrm{d}x, \qquad S_r = \int_0^1 B_r(x)\sin 2\pi kx\,\mathrm{d}x.$$

Step 3: recursion. For $r \geq 1$ integrate by parts, using $B_r' = rB_{r-1}$, $\sin 2\pi k = \sin 0 = 0$ and $\cos 2\pi k = \cos 0 = 1$:

$$C_r = \Big[B_r\frac{\sin 2\pi kx}{2\pi k}\Big]_0^1 - \frac{r}{2\pi k}\int_0^1 B_{r-1}\sin 2\pi kx\,\mathrm{d}x = -\frac{r}{2\pi k}S_{r-1},$$
$$S_r = \Big[-B_r\frac{\cos 2\pi kx}{2\pi k}\Big]_0^1 + \frac{r}{2\pi k}C_{r-1} = -\frac{B_r(1)-B_r(0)}{2\pi k} + \frac{r}{2\pi k}C_{r-1}.$$

For $r \geq 2$, $B_r(1) = B_r(0)$, so $S_r = \dfrac{r}{2\pi k}C_{r-1}$. For $r = 1$: $B_1(1) - B_1(0) = 1$ and $C_0 = \int_0^1 \cos 2\pi kx\,\mathrm{d}x = 0$, so $S_1 = -\dfrac{1}{2\pi k}$.

Step 4: induction. Write $\omega = 2\pi k$. We claim

$$S_{2j-1} = (-1)^j\frac{(2j-1)!}{\omega^{2j-1}}, \qquad C_{2j} = (-1)^{j+1}\frac{(2j)!}{\omega^{2j}} \qquad (j \geq 1).$$

For $j = 1$: $S_1 = -\frac{1}{\omega}$, and $C_2 = -\frac{2}{\omega}S_1 = \frac{2}{\omega^2} = (-1)^2\frac{2!}{\omega^2}$. If S_{2j-1} is as claimed, then $C_{2j} = -\frac{2j}{\omega}S_{2j-1} = (-1)^{j+1}\frac{(2j)!}{\omega^{2j}}$, and then $S_{2j+1} = \frac{2j+1}{\omega}C_{2j} = (-1)^{j+1}\frac{(2j+1)!}{\omega^{2j+1}}$, which is the claim for $j+1$.

Step 5: assembly. Thus $C_{2n} = (-1)^{n+1}\dfrac{(2n)!}{(2\pi k)^{2n}}$, and

$$a_{2k} = 2C_{2n} = (-1)^{n+1}\frac{2\,(2n)!}{(2\pi)^{2n}}\cdot\frac{1}{k^{2n}},$$

which gives the series

$$B_{2n}(x) = (-1)^{n+1}\frac{2(2n)!}{(2\pi)^{2n}}\sum_{k=1}^{\infty}\frac{\cos 2k\pi x}{k^{2n}}.$$

Convergence. B_{2n} is a polynomial, and $B_{2n}(0) = B_{2n}(1)$ (Problem 9(b), as $2n \geq 2$), so its even periodic extension is continuous and piecewise smooth; its cosine series therefore converges to $B_{2n}(x)$ at every x of $[0,1]$ (indeed uniformly, as the coefficients are $O(k^{-2})$).

Answer. Shown: $a_{2k} = (-1)^{n+1}\dfrac{2(2n)!}{(2\pi)^{2n}k^{2n}}$, all other $a_m = 0$, which is the stated series.

Problem 36-11 (Simmons)

Solution. In Problem 10 (with n replaced by p) put $x = 0$. Then $\cos 2k\pi \cdot 0 = 1$ and $B_{2p}(0) = B_{2p}$ (Problem 9(b)), so

$$B_{2p} = (-1)^{p+1}\frac{2(2p)!}{(2\pi)^{2p}}\sum_{k=1}^{\infty}\frac{1}{k^{2p}}.$$

Solving for the sum (and $(2\pi)^{2p} = 2^{2p}\pi^{2p}$):

$$\sum_{k=1}^{\infty}\frac{1}{k^{2p}} = (-1)^{p+1}\frac{(2\pi)^{2p}B_{2p}}{2(2p)!} = (-1)^{p+1}\frac{2^{2p}B_{2p}}{2(2p)!}\pi^{2p}$$

(using $(-1)^{p+1}\cdot(-1)^{p+1} = 1$ to move the sign). Now insert $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, $B_8 = -\frac{1}{30}$, $B_{10} = \frac{5}{66}$ from Problem 8(c):

$$\begin{aligned}
p=1:&\quad (+1)\frac{2^2\cdot\frac{1}{6}}{2\cdot 2!}\pi^2 = \frac{4/6}{4}\pi^2 = \frac{\pi^2}{6},\\
p=2:&\quad (-1)\frac{2^4\cdot\left(-\frac{1}{30}\right)}{2\cdot 4!}\pi^4 = \frac{16/30}{48}\pi^4 = \frac{\pi^4}{90},\\
p=3:&\quad (+1)\frac{2^6\cdot\frac{1}{42}}{2\cdot 6!}\pi^6 = \frac{64/42}{1440}\pi^6 = \frac{64}{60480}\pi^6 = \frac{\pi^6}{945},\\
p=4:&\quad (-1)\frac{2^8\cdot\left(-\frac{1}{30}\right)}{2\cdot 8!}\pi^8 = \frac{256/30}{80640}\pi^8 = \frac{256}{2419200}\pi^8 = \frac{\pi^8}{9450},\\
p=5:&\quad (+1)\frac{2^{10}\cdot\frac{5}{66}}{2\cdot 10!}\pi^{10} = \frac{5120/66}{7257600}\pi^{10} = \frac{5120}{479001600}\pi^{10} = \frac{\pi^{10}}{93555}.
\end{aligned}$$

(For the last: $479001600/5120 = 93555$.)

Answer. $\sum k^{-2p} = (-1)^{p+1}\frac{2^{2p}B_{2p}}{2(2p)!}\pi^{2p}$, giving $\frac{\pi^2}{6}$, $\frac{\pi^4}{90}$, $\frac{\pi^6}{945}$, $\frac{\pi^8}{9450}$, $\frac{\pi^{10}}{93555}$ for $p = 1,\dots,5$.

Section 37 Orthogonal Functions

Throughout, R is the space of functions that are Riemann integrable on the interval in question; the product of two such functions is again integrable, so $(f,g)=\int_a^b fg\,dx$ and $\|f\|=\sqrt{(f,f)}$ always exist. For the trigonometric system on $[-\pi,\pi]$ we use the elementary orthogonality relations (for integers $j,k\ge 0$)

$$\int_{-\pi}^{\pi}\cos jx\,\sin kx\,dx=0, \tag{O1}$$

$$\int_{-\pi}^{\pi}\cos jx\,\cos kx\,dx=\int_{-\pi}^{\pi}\sin jx\,\sin kx\,dx=0\quad(j\neq k), \tag{O2}$$

$$\int_{-\pi}^{\pi}\cos^2 kx\,dx=\int_{-\pi}^{\pi}\sin^2 kx\,dx=\pi\quad(k\ge 1),\qquad\int_{-\pi}^{\pi}1\,dx=2\pi, \tag{O3}$$

which are equations (4) of this section together with the orthogonality of the sequence (2) shown in Section 33. (For (O1), the integrand is odd when $k\ge 1$ and equals 0 when $k=0$.)

Problem 37-1 (Simmons)

Solution. Recall the definitions of the ordinary Fourier coefficients,

$$a_k=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos kx\,dx\quad(k\ge 0),\qquad b_k=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin kx\,dx\quad(k\ge 1).$$

Equivalently,

$$\int_{-\pi}^{\pi}f\cos kx\,dx=\pi a_k,\qquad\int_{-\pi}^{\pi}f\sin kx\,dx=\pi b_k,\qquad\int_{-\pi}^{\pi}f\,dx=\pi a_0. \tag{6.1}$$

(a) Multiply s_n by f and integrate term by term (a finite sum, so this is always legitimate):

$$\begin{aligned}\int_{-\pi}^{\pi}fs_n\,dx&=\frac{a_0}{2}\int_{-\pi}^{\pi}f\,dx+\sum_{k=1}^{n}\Big(a_k\int_{-\pi}^{\pi}f\cos kx\,dx+b_k\int_{-\pi}^{\pi}f\sin kx\,dx\Big)\\&=\frac{a_0}{2}(\pi a_0)+\sum_{k=1}^{n}\big(a_k\cdot\pi a_k+b_k\cdot\pi b_k\big)\qquad\text{by (6.1)}\\&=\pi\Big[\tfrac12 a_0^2+\sum_{k=1}^{n}(a_k^2+b_k^2)\Big].\end{aligned}$$

Dividing by π gives

$$\frac{1}{\pi}\int_{-\pi}^{\pi}fs_n\,dx=\tfrac12 a_0^2+\sum_{k=1}^{n}(a_k^2+b_k^2). \tag{6.2}$$

(b) Write $s_n=\frac{a_0}{2}+\sum_{k=1}^{n}u_k$ with $u_k=a_k\cos kx+b_k\sin kx$. Then

$$s_n^2=\frac{a_0^2}{4}+a_0\sum_{k=1}^{n}u_k+\sum_{j=1}^{n}\sum_{k=1}^{n}u_ju_k.$$

We integrate each group of products over $[-\pi,\pi]$.

- Constant term: $\int_{-\pi}^{\pi}\frac{a_0^2}{4}\,dx=\frac{a_0^2}{4}\cdot 2\pi=\pi\cdot\frac{a_0^2}{2}$ by (O3).

- Cross terms with the constant: $\int_{-\pi}^{\pi} u_k\,dx = a_k\int\cos kx\,dx + b_k\int\sin kx\,dx = 0$ for $k \geq 1$, because $\int_{-\pi}^{\pi}\cos kx\,dx = \frac{\sin kx}{k}\Big|_{-\pi}^{\pi} = 0$ and $\int_{-\pi}^{\pi}\sin kx\,dx = -\frac{\cos kx}{k}\Big|_{-\pi}^{\pi} = 0$.

- Products $u_j u_k$:

$$u_j u_k = a_j a_k \cos jx\cos kx + a_j b_k \cos jx\sin kx + b_j a_k \sin jx\cos kx + b_j b_k \sin jx\sin kx.$$

  By (O1) the two mixed terms integrate to 0 for all j, k. By (O2) the other two integrate to 0 if $j \neq k$, and by (O3) for $j = k$ we get $a_k^2\pi + b_k^2\pi$. Hence $\int_{-\pi}^{\pi} u_j u_k\,dx = \pi(a_k^2 + b_k^2)\,\delta_{jk}$ (Kronecker delta).

Adding the three contributions,

$$\int_{-\pi}^{\pi} s_n^2\,dx = \pi\Big[\tfrac{1}{2}a_0^2 + \sum_{k=1}^{n}(a_k^2 + b_k^2)\Big],$$

and dividing by π,

$$\frac{1}{\pi}\int_{-\pi}^{\pi} s_n^2\,dx = \tfrac{1}{2}a_0^2 + \sum_{k=1}^{n}(a_k^2 + b_k^2). \tag{6.3}$$

(c) Expanding the square, $(f - s_n)^2 = f^2 - 2fs_n + s_n^2$, and integrating,

$$\frac{1}{\pi}\int_{-\pi}^{\pi}(f - s_n)^2\,dx = \frac{1}{\pi}\int f^2\,dx - 2\cdot\frac{1}{\pi}\int fs_n\,dx + \frac{1}{\pi}\int s_n^2\,dx.$$

Writing $T_n = \frac{1}{2}a_0^2 + \sum_{k=1}^{n}(a_k^2 + b_k^2)$, (6.2) and (6.3) say that the last two integrals are T_n each, so

$$\frac{1}{\pi}\int_{-\pi}^{\pi}(f - s_n)^2\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx - 2T_n + T_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx - T_n.$$

The integrand $(f - s_n)^2$ is nonnegative, so the left side is ≥ 0. Therefore

$$T_n = \tfrac{1}{2}a_0^2 + \sum_{k=1}^{n}(a_k^2 + b_k^2) \leq \frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2\,dx \qquad \text{for every } n.$$

Now the right side is a fixed finite number M (since f is integrable, so is f^2). The partial sums T_n of the series $\frac{1}{2}a_0^2 + \sum(a_k^2 + b_k^2)$, whose terms are all ≥ 0, are therefore nondecreasing and bounded above by M; hence they converge, and letting $n \to \infty$ in $T_n \leq M$ gives

$$\tfrac{1}{2}a_0^2 + \sum_{k=1}^{\infty}(a_k^2 + b_k^2) \leq \frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2\,dx,$$

which is $(*)$.

Corollary. Since the series $\sum(a_k^2 + b_k^2)$ converges, its general term tends to 0: $a_k^2 + b_k^2 \to 0$, hence $a_k \to 0$ and $b_k \to 0$ as $k \to \infty$.

Answer. Shown: $T_n = \frac{1}{\pi}\int fs_n = \frac{1}{\pi}\int s_n^2$, so $0 \leq \frac{1}{\pi}\int(f - s_n)^2 = \frac{1}{\pi}\int f^2 - T_n$; letting $n \to \infty$ gives Bessel's inequality $(*)$, and consequently $a_n, b_n \to 0$.

Problem 37-2 (Simmons)

Solution. The orthonormal sequence (5) is, in the order of the text,

$$\phi_1 = \frac{1}{\sqrt{2\pi}}, \qquad \phi_{2k} = \frac{\cos kx}{\sqrt{\pi}}, \qquad \phi_{2k+1} = \frac{\sin kx}{\sqrt{\pi}} \qquad (k = 1, 2, \dots).$$

(It is orthonormal on $[-\pi, \pi]$: by (O3), $\int \phi_1^2 = 2\pi/(2\pi) = 1$, $\int \phi_{2k}^2 = \int \phi_{2k+1}^2 = \pi/\pi = 1$, and distinct members are orthogonal by (O1), (O2) together with $\int_{-\pi}^{\pi} \cos kx\,dx = \int_{-\pi}^{\pi} \sin kx\,dx = 0$.) Let $f \in R$ on $[-\pi, \pi]$, with ordinary coefficients a_k, b_k as in (6.1). The coefficients (8), $\alpha_m = \int_{-\pi}^{\pi} f\phi_m\,dx$, are

$$\begin{aligned} \alpha_1 &= \int_{-\pi}^{\pi} \frac{f}{\sqrt{2\pi}}\,dx = \frac{\pi a_0}{\sqrt{2\pi}} = \sqrt{\frac{\pi}{2}}\,a_0, \\ \alpha_{2k} &= \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f\cos kx\,dx = \frac{\pi a_k}{\sqrt{\pi}} = \sqrt{\pi}\,a_k, \\ \alpha_{2k+1} &= \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} f\sin kx\,dx = \frac{\pi b_k}{\sqrt{\pi}} = \sqrt{\pi}\,b_k. \end{aligned}$$

So the coefficients (8) differ from the ordinary ones only by the constant factors $\sqrt{\pi/2}$ (for the constant term) and $\sqrt{\pi}$ (for the others). The Fourier series (6) is $\sum \alpha_m \phi_m$, whose terms are

$$\begin{aligned} \alpha_1\phi_1 &= \sqrt{\frac{\pi}{2}}\,a_0 \cdot \frac{1}{\sqrt{2\pi}} = a_0\sqrt{\frac{\pi}{2\cdot 2\pi}} = \frac{a_0}{2}, \\ \alpha_{2k}\phi_{2k} &= \sqrt{\pi}\,a_k \cdot \frac{\cos kx}{\sqrt{\pi}} = a_k \cos kx, \\ \alpha_{2k+1}\phi_{2k+1} &= \sqrt{\pi}\,b_k \cdot \frac{\sin kx}{\sqrt{\pi}} = b_k \sin kx. \end{aligned}$$

Hence $\sum_m \alpha_m\phi_m = \frac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos kx + b_k \sin kx)$, term for term the ordinary Fourier series.

Answer. $\alpha_1 = \sqrt{\pi/2}\,a_0$, $\alpha_{2k} = \sqrt{\pi}\,a_k$, $\alpha_{2k+1} = \sqrt{\pi}\,b_k$, but $\alpha_m\phi_m$ equals $\frac{1}{2}a_0$, $a_k \cos kx$, $b_k \sin kx$ respectively, so the two series coincide.

Problem 37-3 (Simmons)

Solution. Inequality (19) is $|(f,g)| \le \|f\|\,\|g\|$ for $f, g \in R$. For every real α, using the bilinearity and symmetry of the inner product,

$$F(\alpha) = \|f + \alpha g\|^2 = (f + \alpha g, f + \alpha g) = (f,f) + 2\alpha(f,g) + \alpha^2(g,g) = \|g\|^2\alpha^2 + 2(f,g)\,\alpha + \|f\|^2,$$

and $F(\alpha) = \int_a^b (f + \alpha g)^2\,dx \ge 0$ for all real α, since the integrand is a square.

Case 1: $\|g\| \neq 0$. Then F is a genuine quadratic polynomial in α with positive leading coefficient $\|g\|^2$ and with no negative values. If its discriminant were positive it would have two distinct real roots and would take negative values between them. Hence the discriminant is ≤ 0:

$$4(f,g)^2 - 4\|g\|^2\|f\|^2 \le 0 \quad \Longrightarrow \quad (f,g)^2 \le \|f\|^2\|g\|^2.$$

Taking square roots gives $|(f,g)| \le \|f\|\,\|g\|$.

Case 2: $\|g\| = 0$ (g a null function, which need not vanish identically). Now $F(\alpha) = 2(f,g)\alpha + \|f\|^2 \ge 0$ for all real α. If $(f,g) \neq 0$, choosing $\alpha = -\left(\|f\|^2 + 1\right)/\left(2(f,g)\right)$ gives $F(\alpha) = -1 < 0$, a contradiction. Hence $(f,g) = 0$ and (19) reads $0 \le 0$, which is true.

In both cases $|(f,g)| \le \|f\|\|g\|$, equivalently $(f,g)^2 \le \|f\|^2\|g\|^2$.

Answer. Shown: $F(\alpha) = \|g\|^2\alpha^2 + 2(f,g)\alpha + \|f\|^2 \geq 0$ forces discriminant ≤ 0, i.e. $|(f,g)| \leq \|f\|\|g\|$ (and $(f,g) = 0$ trivially if $\|g\| = 0$).

Problem 37-4 (Simmons)

Solution. By bilinearity and symmetry of the inner product, $(f,g) = (g,f)$,

$$\|f+g\|^2 = (f+g, f+g) = (f,f) + 2(f,g) + (g,g) = \|f\|^2 + 2(f,g) + \|g\|^2,$$
$$\|f-g\|^2 = (f-g, f-g) = (f,f) - 2(f,g) + (g,g) = \|f\|^2 - 2(f,g) + \|g\|^2.$$

Adding, the terms $\pm 2(f,g)$ cancel:

$$\|f+g\|^2 + \|f-g\|^2 = 2\|f\|^2 + 2\|g\|^2.$$

Answer. Shown: $\|f+g\|^2 + \|f-g\|^2 = 2\|f\|^2 + 2\|g\|^2$.

Problem 37-5 (Simmons)

Solution. From the expansion in Problem 4,

$$\|f-g\|^2 = \|f\|^2 - 2(f,g) + \|g\|^2.$$

Hence $\|f-g\|^2 = \|f\|^2 + \|g\|^2$ holds if and only if $-2(f,g) = 0$, that is, if and only if $(f,g) = 0$, which is by definition the statement that f and g are orthogonal. Both implications are contained in this chain of equivalences:

- if $(f,g) = 0$ then $\|f-g\|^2 = \|f\|^2 + \|g\|^2$ (Pythagorean theorem);
- if $\|f-g\|^2 = \|f\|^2 + \|g\|^2$ then $-2(f,g) = 0$, so $(f,g) = 0$ (converse).

(The same computation with $f+g$ gives $\|f+g\|^2 = \|f\|^2 + \|g\|^2 \iff (f,g) = 0$.)

Answer. Shown: $\|f-g\|^2 = \|f\|^2 + \|g\|^2 - 2(f,g)$, so $\|f-g\|^2 = \|f\|^2 + \|g\|^2 \iff (f,g) = 0$.

Problem 37-6 (Simmons)

Solution. Let $f \in R$ on $[a,b]$ be a null function, $\int_a^b f^2\,dx = 0$, and suppose f is continuous at a point $x_0 \in [a,b]$. We argue by contradiction: assume $f(x_0) \neq 0$ and put $\varepsilon = \frac{1}{2}|f(x_0)| > 0$.

By continuity at x_0 there is $\delta > 0$ such that $|f(x) - f(x_0)| < \varepsilon$ whenever $x \in [a,b]$ and $|x - x_0| < \delta$. For such x,

$$|f(x)| \geq |f(x_0)| - |f(x) - f(x_0)| > 2\varepsilon - \varepsilon = \varepsilon.$$

The set $J = [a,b] \cap (x_0 - \delta, x_0 + \delta)$ is an interval containing x_0 and has positive length, because $a < b$ and $(x_0 - \delta, x_0 + \delta)$ is an open interval around x_0 (at an endpoint of $[a,b]$ it still contains a one-sided neighborhood of x_0 inside $[a,b]$). So J contains a closed interval $[c,d]$ with $d - c = \ell > 0$. On $[c,d]$ we have $f^2 > \varepsilon^2$, and since $f^2 \geq 0$ everywhere,

$$0 = \int_a^b f^2\,dx \;\geq\; \int_c^d f^2\,dx \;\geq\; \varepsilon^2(d-c) = \varepsilon^2\ell > 0,$$

a contradiction. Hence $f(x_0) = 0$.

If f is continuous at every point of $[a, b]$ (a continuous null function), the argument applies at every x_0, so $f(x) = 0$ for all $x \in [a, b]$, i.e. f is identically zero.

Answer. Shown: if $f(x_0) \neq 0$ at a point of continuity then $f^2 > \varepsilon^2$ on an interval of positive length, so $\int f^2 > 0$; hence null functions vanish at points of continuity, and continuous null functions vanish identically.

Section 38 The Mean Convergence of Fourier Series

We use the notation of the text: $E_n = \int_a^b [f-p_n]^2\,dx = \|f-p_n\|^2$ is the mean square error, a sequence p_n converges in the mean to f if $E_n \to 0$, and we write $S_k = \sum_{n\ge 1} n^{-k}$ and $O_k = \sum_{n \text{ odd}} n^{-k}$. In Problems 7–9 the values $S_2 = \pi^2/6$, $S_4 = \pi^4/90$, $S_6 = \pi^6/945$, $S_8 = \pi^8/9450$ are *derived* one after the other, each from Parseval's equation and the preceding ones.

Problem 38-1 (Simmons)

Solution.

(a) Fix $x \in [0,1]$.

- If $x = 0$, then $x \le 1/n$ for every n, so $f_n(0) = 0$ for all n.
- If $0 < x \le 1$, choose an integer $N \ge 2/x$. For every $n \ge N$ we have $2/n \le 2/N \le x$, so $x \ge 2/n$ and therefore $f_n(x) = 0$ (the third line of the definition; note $x \le 1$).

Thus for each x the numerical sequence $f_n(x)$ is 0 from some index on, so $f_n(x) \to 0$: the sequence converges pointwise to the zero function.

(b) For $n \ge 2$ we have $2/n \le 1$, so the interval $(1/n, 2/n)$ lies in $[0,1]$ and $f_n^2 = n$ there, while $f_n^2 = 0$ elsewhere (the values at the points $1/n$, $2/n$ do not affect the integral). Hence

$$E_n = \int_0^1 [f_n(x)-0]^2\,dx = \int_{1/n}^{2/n} n\,dx = n\Big(\frac{2}{n} - \frac{1}{n}\Big) = 1 \qquad (n \ge 2).$$

So $E_n = 1$ for all $n \ge 2$ and $E_n \not\to 0$; the sequence does not converge in the mean to 0. (Indeed $\|f_n - 0\| = 1$ for all $n \ge 2$.)

Answer. (a) $f_n(x) = 0$ for all $n \ge \max(2/x, 1)$ (and always if $x = 0$), so $f_n \to 0$ pointwise; (b) $E_n = \int_0^1 f_n^2\,dx = 1$ for $n \ge 2$, so no mean convergence to 0.

Problem 38-2 (Simmons)

Solution. *Structure of the sequence.* The intervals come in "rounds": round k ($k = 1, 2, 3, \dots$) consists of the 2^k intervals $\big[\frac{j}{2^k}, \frac{j+1}{2^k}\big]$, $j = 0, 1, \dots, 2^k - 1$, listed from left to right, each of length 2^{-k}. (Round 1 is $[0, \frac12], [\frac12, 1]$; round 2 is $[0, \frac14], \dots, [\frac34, 1]$; and so on.) The rounds $1, \dots, k-1$ together contain $2 + 4 + \cdots + 2^{k-1} = 2^k - 2$ intervals, so I_n belongs to round k exactly when

$$2^k - 2 < n \le (2^k - 2) + 2^k = 2^{k+1} - 2.$$

(a) Let I_n belong to round k. Since $n \le 2^{k+1} - 2$, we get $2^k \ge (n+2)/2$, hence the length of I_n is

$$|I_n| = 2^{-k} \le \frac{2}{n+2}.$$

Because $f_n^2 = f_n$ is the indicator function of I_n,

$$E_n = \int_0^1 [f_n(x)-0]^2\,dx = \int_0^1 f_n\,dx = |I_n| \le \frac{2}{n+2} \longrightarrow 0 \quad (n \to \infty).$$

(As $n \to \infty$ the round index $k \to \infty$ as well, since each round is finite.) Hence $f_n \to 0$ in the mean.

(b) Fix any $x \in [0,1]$. In each round k the closed intervals cover $[0,1]$, so x lies in at least one of them: therefore $f_n(x) = 1$ for at least one index n in every round, i.e. for infinitely many n. On the other hand, x belongs to at most two intervals of round k (a point lies in two adjacent closed intervals only if it is a common endpoint $j/2^k$, and it cannot lie in three because intervals separated by one other are disjoint). For $k \geq 2$ round k has $2^k \geq 4 > 2$ intervals, so some interval of round k misses x, and $f_n(x) = 0$ for that n. Thus $f_n(x) = 0$ for infinitely many n as well. A numerical sequence taking both the values 0 and 1 infinitely often is not convergent (a limit L would have to satisfy $|L-0| < \frac{1}{2}$ and $|L-1| < \frac{1}{2}$ simultaneously for large n). Hence $\{f_n(x)\}$ diverges at every $x \in [0,1]$.

Answer. (a) $E_n = |I_n| \leq 2/(n+2) \to 0$, so $f_n \to 0$ in the mean; (b) every $x \in [0,1]$ has $f_n(x) = 1$ and $f_n(x) = 0$ infinitely often, so $f_n(x)$ converges at no point.

Problem 38-3 (Simmons)

Solution. In the text, $E_n = E_n(b_1, \ldots, b_n) = \int_a^b \left[f - (b_1\phi_1 + \cdots + b_n\phi_n)\right]^2 \mathrm{d}x$ is a function of the n variables $b_1, \ldots, b_n$, with $a_k = \int_a^b f\phi_k \,\mathrm{d}x$ fixed numbers (the Fourier coefficients) and $\int_a^b f^2 \,\mathrm{d}x$ a fixed number. At an interior minimum every partial derivative vanishes, so $\partial E_n/\partial b_k = 0$ $(k = 1, \ldots, n)$.

From (8):

$$E_n = \int_a^b f^2 \,\mathrm{d}x - 2\sum_{j=1}^{n} a_j b_j + \sum_{j=1}^{n} b_j^2.$$

Only the $j = k$ terms depend on b_k, so

$$\frac{\partial E_n}{\partial b_k} = -2a_k + 2b_k = 0 \quad \Longrightarrow \quad b_k = a_k.$$

From (9):

$$E_n = \int_a^b f^2 \,\mathrm{d}x - \sum_{j=1}^{n} a_j^2 + \sum_{j=1}^{n} (b_j - a_j)^2.$$

Again only the $j = k$ term of the last sum depends on b_k:

$$\frac{\partial E_n}{\partial b_k} = 2(b_k - a_k) = 0 \quad \Longrightarrow \quad b_k = a_k.$$

Both formulas give the same condition. Since $\partial^2 E_n/\partial b_k^2 = 2 > 0$ (and the mixed second derivatives are 0), the stationary point $b_k = a_k$ $(k = 1, \ldots, n)$ is indeed a minimum, in agreement with Theorem 1. (Formulas (8) and (9) are the same function: $-2a_jb_j + b_j^2 = -a_j^2 + (b_j - a_j)^2$.)

Answer. Both $\partial E_n/\partial b_k = -2a_k + 2b_k$ (from (8)) and $\partial E_n/\partial b_k = 2(b_k - a_k)$ (from (9)) vanish exactly when $b_k = a_k$.

Problem 38-4 (Simmons)

Solution. The functions $\sin kx$ are orthogonal on $[0,\pi]$ with $\int_0^\pi \sin^2 kx\,dx = \pi/2$ (from the text: $\int_0^\pi \sin mx \sin nx\,dx = 0$ for $m \neq n$, $= \pi/2$ for $m = n$). So $\phi_k(x) = \sqrt{2/\pi}\sin kx$ form an orthonormal sequence on $[0,\pi]$, and we may write

$$p(x) = \sum_{k=1}^{5} b_k \sin kx = \sum_{k=1}^{5} c_k \phi_k(x), \qquad c_k = b_k\sqrt{\pi/2}.$$

By Theorem 1, $\int_0^\pi [1-p]^2\,dx$ is minimized exactly when c_k equals the Fourier coefficient $\alpha_k = \int_0^\pi 1\cdot\phi_k\,dx$, i.e. when

$$b_k = \frac{\alpha_k}{\sqrt{\pi/2}} = \sqrt{\frac{2}{\pi}}\cdot\sqrt{\frac{2}{\pi}}\int_0^\pi \sin kx\,dx = \frac{2}{\pi}\int_0^\pi \sin kx\,dx.$$

Now

$$\int_0^\pi \sin kx\,dx = \Big[-\frac{\cos kx}{k}\Big]_0^\pi = \frac{1-\cos k\pi}{k} = \frac{1-(-1)^k}{k} = \begin{cases} 2/k, & k \text{ odd},\\ 0, & k \text{ even}.\end{cases}$$

Therefore

$$b_1 = \frac{2}{\pi}\cdot 2 = \frac{4}{\pi}, \quad b_2 = 0, \quad b_3 = \frac{2}{\pi}\cdot\frac{2}{3} = \frac{4}{3\pi}, \quad b_4 = 0, \quad b_5 = \frac{2}{\pi}\cdot\frac{2}{5} = \frac{4}{5\pi}.$$

(Direct check with Problem 3: $E = \int_0^\pi (1-\sum b_k\sin kx)^2\,dx = \pi - 2\sum b_k\frac{1-(-1)^k}{k} + \frac{\pi}{2}\sum b_k^2$, and $\partial E/\partial b_k = -2\frac{1-(-1)^k}{k} + \pi b_k = 0$ gives the same $b_k = \frac{2(1-(-1)^k)}{\pi k}$.) These are the first five terms of the sine series $1 = \frac{4}{\pi}\big(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\big)$ on $(0,\pi)$.

Answer. $b_1 = \dfrac{4}{\pi}$, $b_2 = 0$, $b_3 = \dfrac{4}{3\pi}$, $b_4 = 0$, $b_5 = \dfrac{4}{5\pi}$ (agrees with the book).

Problem 38-5 (Simmons)

Solution. As in Problem 4, with orthonormal $\phi_k = \sqrt{2/\pi}\sin kx$ on $[0,\pi]$, Theorem 1 gives

$$b_k = \frac{2}{\pi}\int_0^\pi x\sin kx\,dx.$$

Integrating by parts ($u = x$, $dv = \sin kx\,dx$, $v = -\cos kx/k$):

$$\int_0^\pi x\sin kx\,dx = \Big[-\frac{x\cos kx}{k}\Big]_0^\pi + \frac{1}{k}\int_0^\pi \cos kx\,dx = -\frac{\pi\cos k\pi}{k} + \frac{1}{k^2}\big[\sin kx\big]_0^\pi = \frac{\pi(-1)^{k+1}}{k}.$$

Hence $b_k = \dfrac{2(-1)^{k+1}}{k}$:

$$b_1 = 2, \qquad b_2 = -\frac{2}{2} = -1, \qquad b_3 = \frac{2}{3}.$$

Answer. $b_1 = 2$, $b_2 = -1$, $b_3 = \dfrac{2}{3}$ (agrees with the book).

Problem 38-6 (Simmons)

Solution. Take $[a,b]=[-\pi,\pi]$ and $\{\phi_m\}=\{\frac{1}{\sqrt{2\pi}},\frac{\cos x}{\sqrt{\pi}},\frac{\sin x}{\sqrt{\pi}},\frac{\cos 2x}{\sqrt{\pi}},\frac{\sin 2x}{\sqrt{\pi}},\dots\}$. By Problem 37–2 the Fourier coefficients $\alpha_m=\int_{-\pi}^{\pi} f\phi_m\,dx$ are

$$\alpha_1=\sqrt{\frac{\pi}{2}}\,a_0,\qquad \alpha_{2k}=\sqrt{\pi}\,a_k,\qquad \alpha_{2k+1}=\sqrt{\pi}\,b_k\qquad (k\ge 1),$$

where a_k,b_k are the ordinary Fourier coefficients of f. Then

$$\alpha_1^2=\frac{\pi}{2}a_0^2,\qquad \alpha_{2k}^2+\alpha_{2k+1}^2=\pi(a_k^2+b_k^2).$$

Parseval's equation (14) is $\sum_{m=1}^{\infty}\alpha_m^2=\int_{-\pi}^{\pi}f^2\,dx$. The series has nonnegative terms, so it may be grouped without changing its sum; grouping the terms in the pairs $(\alpha_{2k},\alpha_{2k+1})$, it becomes

$$\frac{\pi}{2}a_0^2+\pi\sum_{k=1}^{\infty}(a_k^2+b_k^2)=\int_{-\pi}^{\pi}[f(x)]^2\,dx.$$

Dividing by π:

$$\frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2\,dx=\tfrac{1}{2}a_0^2+\sum_{n=1}^{\infty}(a_n^2+b_n^2),$$

which is (17).

Answer. Shown: with $\alpha_1^2=\frac{\pi}{2}a_0^2$ and $\alpha_{2k}^2+\alpha_{2k+1}^2=\pi(a_k^2+b_k^2)$, (14) becomes (17) after division by π.

Problem 38-7 (Simmons)

Solution. Both series are Fourier series on $[-\pi,\pi]$, so the coefficients can be read off and (17) applied.

First series. For $f(x)=x$ (odd), $a_n=0$ for all $n\ge 0$, and, integrating by parts,

$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}x\sin nx\,dx=\frac{2}{\pi}\int_0^{\pi}x\sin nx\,dx=\frac{2}{\pi}\cdot\frac{\pi(-1)^{n+1}}{n}=\frac{2(-1)^{n+1}}{n}$$

(using the integral computed in Problem 5), which agrees with the displayed series. The left side of (17) is

$$\frac{1}{\pi}\int_{-\pi}^{\pi}x^2\,dx=\frac{1}{\pi}\cdot\frac{2\pi^3}{3}=\frac{2\pi^2}{3},$$

and the right side is $\sum_{n\ge1}b_n^2=\sum 4/n^2=4S_2$. Thus

$$4S_2=\frac{2\pi^2}{3}\quad\Longrightarrow\quad\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}.$$

Second series. For $f(x)=x^2$ (even), $b_n=0$, and the displayed series gives $\frac{1}{2}a_0=\pi^2/3$, i.e. $a_0=2\pi^2/3$, and $a_n=4(-1)^n/n^2$ for $n\ge1$. (These are indeed the coefficients: $a_0=\frac{2}{\pi}\int_0^{\pi}x^2\,dx=\frac{2\pi^2}{3}$ and, by two integrations by parts, $a_n=\frac{2}{\pi}\int_0^{\pi}x^2\cos nx\,dx=\frac{2}{\pi}\cdot\frac{2\pi(-1)^n}{n^2}=\frac{4(-1)^n}{n^2}$.) The left side of (17) is

$$\frac{1}{\pi}\int_{-\pi}^{\pi}x^4\,dx=\frac{1}{\pi}\cdot\frac{2\pi^5}{5}=\frac{2\pi^4}{5},$$

and the right side is

$$\tfrac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2 = \frac{1}{2}\cdot\frac{4\pi^4}{9} + \sum_{n=1}^{\infty}\frac{16}{n^4} = \frac{2\pi^4}{9} + 16S_4.$$

Therefore

$$16S_4 = \frac{2\pi^4}{5} - \frac{2\pi^4}{9} = \frac{2\pi^4(9-5)}{45} = \frac{8\pi^4}{45} \quad\Longrightarrow\quad S_4 = \frac{8\pi^4}{45\cdot 16} = \frac{\pi^4}{90}.$$

Answer. $\displaystyle\sum_1^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$ (from $4S_2 = \frac{2\pi^2}{3}$) and $\displaystyle\sum_1^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$ (from $\frac{2\pi^4}{9} + 16S_4 = \frac{2\pi^4}{5}$).

Problem 38-8 (Simmons)

Solution. *The sine series of x^2.* On $(0,\pi)$ it is $x^2 = \sum b_n \sin nx$ with $b_n = \frac{2}{\pi}\int_0^{\pi} x^2\sin nx\,dx$. Integrating by parts twice ($\int x^2\sin nx\,dx = -\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}$),

$$\int_0^{\pi} x^2\sin nx\,dx = -\frac{\pi^2(-1)^n}{n} + 0 + \frac{2\big((-1)^n-1\big)}{n^3},$$

so

$$b_n = \frac{2\pi(-1)^{n+1}}{n} + \frac{4\big((-1)^n-1\big)}{\pi n^3} = \begin{cases} \dfrac{2\pi}{n} - \dfrac{8}{\pi n^3}, & n \text{ odd},\\[2mm] -\dfrac{2\pi}{n}, & n \text{ even}. \end{cases}$$

Parseval's equation. The sine series is the Fourier series on $[-\pi,\pi]$ of the odd function F with $F(x) = x^2$ on $[0,\pi]$ ($F(x) = -x^2$ on $[-\pi,0)$), for which $a_n = 0$ and $b_n = \frac{2}{\pi}\int_0^{\pi} x^2\sin nx\,dx$ as above. Since $F^2 = x^4$, (17) gives

$$\sum_{n=1}^{\infty} b_n^2 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^4\,dx = \frac{2\pi^4}{5}.$$

Squaring $b_n = \dfrac{2\pi(-1)^{n+1}}{n} + \dfrac{4((-1)^n-1)}{\pi n^3}$ and summing, three sums appear.

- Square of the first term: $\sum \dfrac{4\pi^2}{n^2} = 4\pi^2 S_2 = 4\pi^2\cdot\dfrac{\pi^2}{6} = \dfrac{2\pi^4}{3}$.
- Twice the product:

  $$2\cdot\frac{2\pi(-1)^{n+1}}{n}\cdot\frac{4((-1)^n-1)}{\pi n^3} = \frac{16\,(-1)^{n+1}((-1)^n-1)}{n^4} = \frac{16(-1+(-1)^n)}{n^4},$$

  which is $-\dfrac{32}{n^4}$ for odd n and 0 for even n. Hence the sum is $-32O_4$, where $O_k = \sum_{n\text{ odd}} n^{-k}$.
- Square of the second term: $\dfrac{16((-1)^n-1)^2}{\pi^2 n^6} = \dfrac{64}{\pi^2 n^6}$ for odd n, 0 for even n; the sum is $\dfrac{64}{\pi^2}O_6$.

Thus

$$\frac{2\pi^4}{5} = \frac{2\pi^4}{3} - 32\,O_4 + \frac{64}{\pi^2}\,O_6.$$

The odd sums. Splitting S_k into odd and even n (even $n = 2m$): $S_k = O_k + 2^{-k}S_k$, so $O_k = (1-2^{-k})S_k$. Using $S_4 = \pi^4/90$ from Problem 7,

$$O_4 = \frac{15}{16}\cdot\frac{\pi^4}{90} = \frac{\pi^4}{96}.$$

Then $32O_4 = \pi^4/3$ and the equation becomes

$$\frac{2\pi^4}{5} = \frac{2\pi^4}{3} - \frac{\pi^4}{3} + \frac{64}{\pi^2}O_6 \quad\Longrightarrow\quad \frac{64}{\pi^2}O_6 = \frac{2\pi^4}{5} - \frac{\pi^4}{3} = \frac{\pi^4}{15} \quad\Longrightarrow\quad O_6 = \frac{\pi^6}{960}.$$

Finally $S_6 = \dfrac{O_6}{1-2^{-6}} = \dfrac{64}{63}\cdot\dfrac{\pi^6}{960} = \dfrac{\pi^6}{63\cdot 15} = \dfrac{\pi^6}{945}$.

Answer. $\displaystyle\sum_1^\infty \frac{1}{n^6} = \frac{\pi^6}{945}.$

Problem 38-9 (Simmons)

Solution. *The cosine series of x^4.* It is the Fourier series on $[-\pi,\pi]$ of the even function x^4: $b_n = 0$ and $a_n = \frac{2}{\pi}\int_0^\pi x^4\cos nx\,\mathrm{d}x$. With

$$\int x^4\cos nx\,\mathrm{d}x = \frac{x^4\sin nx}{n} + \frac{4x^3\cos nx}{n^2} - \frac{12x^2\sin nx}{n^3} - \frac{24x\cos nx}{n^4} + \frac{24\sin nx}{n^5}$$

(repeated integration by parts; differentiate to check), evaluating between 0 and π where $\sin n\pi = 0$, $\cos n\pi = (-1)^n$:

$$\int_0^\pi x^4\cos nx\,\mathrm{d}x = (-1)^n\Big(\frac{4\pi^3}{n^2} - \frac{24\pi}{n^4}\Big),$$

so for $n \ge 1$,

$$a_n = \frac{2}{\pi}\cdot(-1)^n\Big(\frac{4\pi^3}{n^2} - \frac{24\pi}{n^4}\Big) = (-1)^n\Big(\frac{8\pi^2}{n^2} - \frac{48}{n^4}\Big) = \frac{8(-1)^n(\pi^2n^2-6)}{n^4},$$

and $a_0 = \frac{2}{\pi}\int_0^\pi x^4\,\mathrm{d}x = \frac{2\pi^4}{5}$ (so $\frac{1}{2}a_0 = \frac{\pi^4}{5}$), matching the series in 35–12(a).

Parseval's equation (17). The left side is $\frac{1}{\pi}\int_{-\pi}^{\pi} x^8\,\mathrm{d}x = \frac{1}{\pi}\cdot\frac{2\pi^9}{9} = \frac{2\pi^8}{9}$. The right side is

$$\tfrac{1}{2}a_0^2 + \sum_{n=1}^\infty a_n^2 = \frac{1}{2}\cdot\frac{4\pi^8}{25} + 64\sum_{n=1}^\infty\frac{(\pi^2n^2-6)^2}{n^8} = \frac{2\pi^8}{25} + 64\sum_{n=1}^\infty\frac{\pi^4n^4 - 12\pi^2n^2 + 36}{n^8}$$
$$= \frac{2\pi^8}{25} + 64\big(\pi^4S_4 - 12\pi^2S_6 + 36S_8\big).$$

By Problems 7 and 8, $\pi^4S_4 = \dfrac{\pi^8}{90}$ and $12\pi^2S_6 = \dfrac{12\pi^8}{945} = \dfrac{4\pi^8}{315}$, so

$$\pi^4S_4 - 12\pi^2S_6 = \pi^8\Big(\frac{1}{90} - \frac{4}{315}\Big) = \pi^8\cdot\frac{7-8}{630} = -\frac{\pi^8}{630}.$$

Equating the two sides,

$$\frac{2\pi^8}{9} = \frac{2\pi^8}{25} + 64\Big(-\frac{\pi^8}{630} + 36S_8\Big) \implies 64\Big(36S_8 - \frac{\pi^8}{630}\Big) = \pi^8\Big(\frac{2}{9} - \frac{2}{25}\Big) = \frac{32\pi^8}{225}.$$

Dividing by 64: $36S_8 - \dfrac{\pi^8}{630} = \dfrac{\pi^8}{450}$, so

$$36S_8 = \pi^8\Big(\frac{1}{450} + \frac{1}{630}\Big) = \pi^8\,\frac{7+5}{3150} = \frac{2\pi^8}{525}, \qquad S_8 = \frac{2\pi^8}{525\cdot 36} = \frac{\pi^8}{9450}.$$

Answer. $\displaystyle\sum_{1}^{\infty} \frac{1}{n^8} = \frac{\pi^8}{9450}.$

Chapter 7

Partial Differential Equations and Boundary Value Problems

Section 40 Eigenvalues, Eigenfunctions, and the Vibrating String

Throughout, the wave equation (8) of the text is $a^2\,\partial^2y/\partial x^2 = \partial^2y/\partial t^2$, with boundary conditions $y(0,t) = y(\pi,t) = 0$ and initial conditions $y_t(x,0) = 0$, $y(x,0) = f(x)$ (the "plucked string"); its formal solution is Bernoulli's series

$$y(x,t) = \sum_{n=1}^{\infty} b_n \sin nx\, \cos nat, \qquad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx. \tag{$***$}$$

Eigenvalue parameters λ are real, as in the text.

Problem 40-1 (Simmons)

Solution. All six are special cases of (f), so we solve (f) first and then specialize. Let $\ell = b - a > 0$ and consider $y'' + \lambda y = 0$, $y(a) = y(b) = 0$. It is convenient to measure x from a, i.e. to write solutions in terms of $x - a$. We treat the three signs of λ.

Case $\lambda < 0$. Write $\lambda = -\mu^2$, $\mu > 0$. The general solution is $y = c_1\cosh\mu(x-a) + c_2\sinh\mu(x-a)$. Then $y(a) = c_1 = 0$, and $y(b) = c_2\sinh\mu\ell = 0$. Since $\sinh\mu\ell \neq 0$ for $\mu\ell > 0$, $c_2 = 0$. Only $y \equiv 0$: no eigenvalues.

Case $\lambda = 0$. $y = c_1 + c_2(x-a)$; $y(a) = c_1 = 0$ and $y(b) = c_2\ell = 0$ give $c_2 = 0$. Only the trivial solution: 0 is not an eigenvalue.

Case $\lambda > 0$. Write $\lambda = \mu^2$, $\mu > 0$. The general solution is $y = c_1\sin\mu(x-a) + c_2\cos\mu(x-a)$. The condition $y(a) = 0$ gives $c_2 = 0$, so $y = c_1\sin\mu(x-a)$. Then $y(b) = c_1\sin\mu\ell = 0$. For a nontrivial solution we need $c_1 \neq 0$, hence $\sin\mu\ell = 0$, i.e. $\mu\ell = n\pi$ with $n = 1,2,3,\ldots$ (positive since $\mu > 0$). Thus

$$\lambda_n = \mu^2 = \frac{n^2\pi^2}{\ell^2} = \frac{n^2\pi^2}{(b-a)^2}, \qquad y_n(x) = \sin\frac{n\pi(x-a)}{b-a}, \quad n = 1,2,3,\ldots$$

(any nonzero constant multiple is also an eigenfunction).

(f) By the computation above, $\lambda_n = \dfrac{n^2\pi^2}{(b-a)^2}$, $y_n(x) = \sin\dfrac{n\pi(x-a)}{b-a}$.

(a) Here $a = 0$, $b = \pi/2$, $\ell = \pi/2$: $\lambda_n = \dfrac{n^2\pi^2}{\pi^2/4} = 4n^2$, $y_n(x) = \sin\dfrac{n\pi x}{\pi/2} = \sin 2nx$. Direct check: $y_n'' = -4n^2\sin 2nx = -\lambda_n y_n$, $y_n(0) = 0$, $y_n(\pi/2) = \sin n\pi = 0$.

(b) $a = 0$, $b = 2\pi$, $\ell = 2\pi$: $\lambda_n = \dfrac{n^2\pi^2}{4\pi^2} = \dfrac{n^2}{4}$, $y_n(x) = \sin\dfrac{n\pi x}{2\pi} = \sin\dfrac{nx}{2}$. Check: $y_n(2\pi) = \sin n\pi = 0$.

(c) $a = 0$, $b = 1$, $\ell = 1$: $\lambda_n = n^2\pi^2$, $y_n(x) = \sin n\pi x$.

(d) $a = 0$, $b = L$, $\ell = L$: $\lambda_n = \dfrac{n^2\pi^2}{L^2}$, $y_n(x) = \sin\dfrac{n\pi x}{L}$.

(e) $a = -L$, $b = L$, $\ell = 2L$: $\lambda_n = \dfrac{n^2\pi^2}{4L^2}$, $y_n(x) = \sin\dfrac{n\pi(x+L)}{2L}$. Check: $y_n(-L) = \sin 0 = 0$, $y_n(L) = \sin n\pi = 0$.

Answer. (a) $\lambda_n = 4n^2$, $y_n = \sin 2nx$; (b) $\lambda_n = n^2/4$, $y_n = \sin\frac{1}{2}nx$; (c) $\lambda_n = n^2\pi^2$, $y_n = \sin n\pi x$; (d) $\lambda_n = n^2\pi^2/L^2$, $y_n = \sin(n\pi x/L)$; (e) $\lambda_n = n^2\pi^2/4L^2$, $y_n = \sin\frac{n\pi(x+L)}{2L}$; (f)

$\lambda_n = n^2\pi^2/(b-a)^2$, $y_n = \sin\frac{n\pi(x-a)}{b-a}$; $n = 1, 2, 3, \ldots$ in every case.

Problem 40-2 (Simmons)

Solution.

(a) Let F and G be twice differentiable and put $y = F(x+at) + G(x-at)$. By the chain rule (using $\partial(x \pm at)/\partial x = 1$ and $\partial(x \pm at)/\partial t = \pm a$):

$$\frac{\partial y}{\partial x} = F'(x+at) + G'(x-at), \qquad \frac{\partial^2 y}{\partial x^2} = F''(x+at) + G''(x-at),$$
$$\frac{\partial y}{\partial t} = aF'(x+at) - aG'(x-at), \qquad \frac{\partial^2 y}{\partial t^2} = a^2F''(x+at) + a^2G''(x-at).$$

Hence $\partial^2 y/\partial t^2 = a^2\,\partial^2 y/\partial x^2$, which is (8).

(b) Regard y as a function $y(\alpha, \beta)$ of $\alpha = x + at$, $\beta = x - at$. Then $\alpha_x = \beta_x = 1$, $\alpha_t = a$, $\beta_t = -a$, and the chain rule gives

$$y_x = y_\alpha + y_\beta, \qquad y_t = a\,(y_\alpha - y_\beta).$$

Applying the same rules again (the operators are $\partial_x = \partial_\alpha + \partial_\beta$ and $\partial_t = a(\partial_\alpha - \partial_\beta)$, and $y_{\alpha\beta} = y_{\beta\alpha}$):

$$y_{xx} = (\partial_\alpha + \partial_\beta)(y_\alpha + y_\beta) = y_{\alpha\alpha} + 2y_{\alpha\beta} + y_{\beta\beta},$$
$$y_{tt} = a^2(\partial_\alpha - \partial_\beta)(y_\alpha - y_\beta) = a^2\left(y_{\alpha\alpha} - 2y_{\alpha\beta} + y_{\beta\beta}\right).$$

Therefore

$$a^2 y_{xx} - y_{tt} = a^2\left(2y_{\alpha\beta} + 2y_{\alpha\beta}\right) = 4a^2\, y_{\alpha\beta},$$

and the wave equation $a^2y_{xx} - y_{tt} = 0$ becomes (as $a \neq 0$)

$$\frac{\partial^2 y}{\partial\alpha\,\partial\beta} = 0.$$

Integrate with respect to β (holding α fixed): $\partial y/\partial\alpha$ is independent of β, so $\partial y/\partial\alpha = \varphi(\alpha)$ for an arbitrary function φ. Integrating with respect to α (holding β fixed) gives

$$y = \int \varphi(\alpha)\,d\alpha + G(\beta) = F(\alpha) + G(\beta),$$

where $F(\alpha) = \int \varphi(\alpha)\,d\alpha$ and G is an arbitrary function (the "constant" of integration may depend on β). Returning to x, t:

$$y(x,t) = F(x+at) + G(x-at),$$

which is $(*)$.

Answer. (a) $y_{xx} = F'' + G''$, $y_{tt} = a^2(F'' + G'')$, so (8) holds. (b) $a^2y_{xx} - y_{tt} = 4a^2y_{\alpha\beta}$, so $y_{\alpha\beta} = 0$, whence $y = F(\alpha) + G(\beta) = F(x+at) + G(x-at)$.

Problem 40-3 (Simmons)

Solution. Here (11) is $y_t(x,0) = 0$ (released from rest) and (12) is $y(x,0) = f(x)$, valid for all real x.

(a) By $(*)$, $y = F(x+at) + G(x-at)$, so $y_t = aF'(x+at) - aG'(x-at)$. The initial conditions give

$$y_t(x,0) = 0: \quad aF'(x) - aG'(x) = 0 \implies F'(x) = G'(x) \implies F(x) - G(x) = K \text{ (a constant)},$$
$$y(x,0) = f(x): \quad F(x) + G(x) = f(x).$$

Adding and subtracting, $F(x) = \frac{1}{2}[f(x) + K]$ and $G(x) = \frac{1}{2}[f(x) - K]$. Then

$$y(x,t) = \tfrac{1}{2}\left[f(x+at) + K\right] + \tfrac{1}{2}\left[f(x-at) - K\right] = \tfrac{1}{2}\left[f(x+at) + f(x-at)\right],$$

the constants cancelling. This is $(**)$.

(b) *Oddness.* $y(0,t) = 0$ for all $t \geq 0$ gives, by $(**)$, $f(at) + f(-at) = 0$. As t runs over $[0,\infty)$, $s = at$ runs over $[0,\infty)$, so $f(-s) = -f(s)$ for all $s \geq 0$. If $s < 0$, then $-s > 0$ and the relation just proved reads $f(s) = -f(-s)$, i.e. again $f(-s) = -f(s)$. Hence f is odd on all of $\mathbb{R}$.

Periodicity. $y(\pi,t) = 0$ gives $f(\pi + at) + f(\pi - at) = 0$, i.e. with $s = at \geq 0$,

$$f(\pi + s) = -f(\pi - s) = f(s - \pi),$$

where the last step uses oddness: $-f(\pi - s) = f(-(\pi - s)) = f(s - \pi)$. For $s < 0$, put $s = -r$, $r > 0$: the identity $f(\pi - r) = -f(\pi + r)$ is the same statement, so $f(\pi + s) = f(s - \pi)$ holds for all real s. Now let $u = s - \pi$ range over $\mathbb{R}$: $f(u + 2\pi) = f(u)$ for all u. Thus f has period 2π.

(c) Oddness at $x = 0$: $f(0) = -f(0)$, so $f(0) = 0$. Periodicity then oddness at π: $f(\pi) = f(-\pi + 2\pi) = f(-\pi) = -f(\pi)$, so $f(\pi) = 0$. (Indeed also $f(k\pi) = 0$ for every integer k.)

(d) Bernoulli's series is $y = \sum_{n=1}^{\infty} b_n \sin nx \cos nat$ with $f(x) = \sum b_n \sin nx$ (equation (18)). By the product formula $2\sin A \cos B = \sin(A+B) + \sin(A-B)$ with $A = nx$, $B = nat$,

$$\sin nx \cos nat = \tfrac{1}{2}\left[\sin n(x+at) + \sin n(x-at)\right].$$

Therefore, formally,

$$y(x,t) = \tfrac{1}{2}\sum_{n=1}^{\infty} b_n \sin n(x+at) + \tfrac{1}{2}\sum_{n=1}^{\infty} b_n \sin n(x-at) = \tfrac{1}{2}\left[f(x+at) + f(x-at)\right],$$

where $f(s) = \sum b_n \sin ns$ is understood for all real s (the sum of the sine series, which is automatically odd and 2π-periodic, consistent with (b)). This is $(**)$.

Answer. Shown: (a) $F - G = K$, $F + G = f$ give $y = \frac{1}{2}[f(x+at) + f(x-at)]$; (b) $y(0,t) = 0 \Rightarrow f$ odd, $y(\pi,t) = 0 \Rightarrow f(u + 2\pi) = f(u)$; (c) $f(0) = f(\pi) = 0$; (d) $\sin nx \cos nat = \frac{1}{2}[\sin n(x+at) + \sin n(x-at)]$.

Problem 40-4 (Simmons)

Solution.

(a) Substitute $y = u(x)v(t)$. Since v does not depend on x and u not on t,

$$v(t)\,\frac{d}{dx}\big(gx\,u'(x)\big) = u(x)\,v''(t).$$

Dividing by $u(x)v(t)$ (at points where neither vanishes):

$$\frac{(gx\,u')'}{u(x)} = \frac{v''(t)}{v(t)}.$$

The left side depends only on x, the right side only on t; for them to be equal for all x and t both must equal one constant, which we call $-\lambda$. The left side gives $(gx\,u')' = -\lambda u$, i.e.

$$\frac{d}{dx}\left(gx\,\frac{du}{dx}\right) + \lambda u = 0, \qquad (***)$$

and the right side gives $v'' + \lambda v = 0$.

(b) Take $\lambda > 0$ so that $z = 2\sqrt{\lambda x/g}$ is real and positive for $x > 0$. Then

$$\frac{dz}{dx} = 2\sqrt{\lambda/g}\cdot\frac{1}{2\sqrt{x}} = \sqrt{\frac{\lambda}{gx}}, \qquad x = \frac{gz^2}{4\lambda}.$$

With u regarded as a function of z, by the chain rule

$$gx\,\frac{du}{dx} = gx\sqrt{\frac{\lambda}{gx}}\,\frac{du}{dz} = \sqrt{\lambda g x}\,\frac{du}{dz}.$$

Using $x = gz^2/4\lambda$, $\lambda g x = g^2 z^2/4$, so $\sqrt{\lambda g x} = gz/2$. Hence

$$gx\,\frac{du}{dx} = \frac{gz}{2}\,\frac{du}{dz}.$$

Also $\dfrac{dz}{dx} = \sqrt{\dfrac{\lambda}{gx}} = \dfrac{2\lambda}{gz}$ (since $\sqrt{\lambda/(gx)} = \sqrt{4\lambda^2/(g^2z^2)}$). Differentiating once more with respect to x:

$$\frac{d}{dx}\left(gx\,\frac{du}{dx}\right) = \frac{dz}{dx}\,\frac{d}{dz}\left(\frac{gz}{2}\,\frac{du}{dz}\right) = \frac{2\lambda}{gz}\cdot\frac{g}{2}\left(z\,\frac{d^2u}{dz^2} + \frac{du}{dz}\right) = \frac{\lambda}{z}\left(z\,\frac{d^2u}{dz^2} + \frac{du}{dz}\right).$$

Substituting into $(***)$:

$$\frac{\lambda}{z}\left(z\,u_{zz} + u_z\right) + \lambda u = 0.$$

Multiplying by z/λ ($\lambda \neq 0$, $z \neq 0$):

$$z\,\frac{d^2u}{dz^2} + \frac{du}{dz} + zu = 0.$$

Bessel's equation $x^2y'' + xy' + (x^2 - p^2)y = 0$ with $p = 0$ is $x^2y'' + xy' + x^2y = 0$; dividing by x gives $xy'' + y' + xy = 0$, which is the equation above with z in place of x and u in place of y. Its solution that stays bounded at the free end $z = 0$ ($x = 0$) is $u = J_0(z) = J_0\big(2\sqrt{\lambda x/g}\big)$.

Answer. (a) Separation gives $(gxu')' + \lambda u = 0$ and $v'' + \lambda v = 0$. (b) With $z = 2\sqrt{\lambda x/g}$: $\frac{d}{dx}(gxu') = \frac{\lambda}{z}(zu'' + u')$, so $(***)$ becomes $zu_{zz} + u_z + zu = 0$, Bessel's equation of order 0.

Problem 40-5 (Simmons)

Solution. The solution is $(***)$: $y = \sum b_n \sin nx \cos nat$ with $b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx$. In all three cases f is continuous, $f(0) = f(\pi) = 0$, and f is piecewise smooth, so integration by parts (with $f(0) = f(\pi) = 0$ killing the boundary terms) is convenient:

$$\int_0^\pi f(x)\sin nx\,dx = \left[-\frac{f(x)\cos nx}{n}\right]_0^\pi + \frac{1}{n}\int_0^\pi f'(x)\cos nx\,dx = \frac{1}{n}\int_0^\pi f'(x)\cos nx\,dx. \qquad (1)$$

(a) Here $f' = 2c/\pi$ on $(0,\pi/2)$ and $f' = -2c/\pi$ on $(\pi/2,\pi)$. By (1),

$$\int_0^\pi f\sin nx\,dx = \frac{1}{n}\left[\frac{2c}{\pi}\int_0^{\pi/2}\cos nx\,dx - \frac{2c}{\pi}\int_{\pi/2}^{\pi}\cos nx\,dx\right]$$
$$= \frac{2c}{\pi n}\left[\frac{\sin nx}{n}\Big|_0^{\pi/2} - \frac{\sin nx}{n}\Big|_{\pi/2}^{\pi}\right] = \frac{2c}{\pi n^2}\left[\sin\frac{n\pi}{2} - \left(0 - \sin\frac{n\pi}{2}\right)\right] = \frac{4c}{\pi n^2}\sin\frac{n\pi}{2}.$$

Hence $b_n = \dfrac{2}{\pi}\cdot\dfrac{4c}{\pi n^2}\sin\dfrac{n\pi}{2} = \dfrac{8c}{\pi^2 n^2}\sin\dfrac{n\pi}{2}$. This vanishes for even n; for $n = 2k-1$, $\sin\frac{(2k-1)\pi}{2} = (-1)^{k+1}$. Therefore

$$y(x,t) = \frac{8c}{\pi^2}\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin(2n-1)x\,\cos(2n-1)at}{(2n-1)^2}.$$

(b) $f(x) = x(\pi - x)/\pi$, so $f' = (\pi - 2x)/\pi$, $f'' = -2/\pi$. Integrating (1) by parts once more (the boundary term $[f'\sin nx/n]_0^\pi$ vanishes because $\sin 0 = \sin n\pi = 0$):

$$\int_0^\pi f\sin nx\,dx = \frac{1}{n}\left\{\left[\frac{f'\sin nx}{n}\right]_0^\pi - \frac{1}{n}\int_0^\pi f''\sin nx\,dx\right\} = \frac{2}{\pi n^2}\int_0^\pi \sin nx\,dx = \frac{2}{\pi n^2}\cdot\frac{1-(-1)^n}{n}.$$

So $b_n = \dfrac{2}{\pi}\cdot\dfrac{2(1-(-1)^n)}{\pi n^3} = \dfrac{4\,(1-(-1)^n)}{\pi^2 n^3}$, which is 0 for even n and $\dfrac{8}{\pi^2 n^3}$ for odd n. With $n \to 2n-1$:

$$y(x,t) = \frac{8}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin(2n-1)x\,\cos(2n-1)at}{(2n-1)^3}.$$

(c) Here $f' = 1$ on $(0,\pi/4)$, $f' = 0$ on $(\pi/4,3\pi/4)$, $f' = -1$ on $(3\pi/4,\pi)$. By (1),

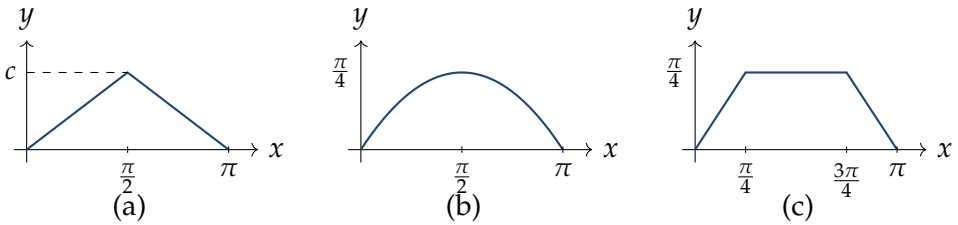
$$\int_0^\pi f\sin nx\,dx = \frac{1}{n}\left[\int_0^{\pi/4}\cos nx\,dx - \int_{3\pi/4}^{\pi}\cos nx\,dx\right] = \frac{1}{n^2}\left[\sin\frac{n\pi}{4} - \left(\sin n\pi - \sin\frac{3n\pi}{4}\right)\right]$$
$$= \frac{1}{n^2}\left[\sin\frac{n\pi}{4} + \sin\frac{3n\pi}{4}\right].$$

Hence $b_n = \dfrac{2}{\pi n^2}\left[\sin\dfrac{n\pi}{4} + \sin\dfrac{3n\pi}{4}\right]$ and

$$y(x,t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^2}\left[\sin\frac{n\pi}{4} + \sin\frac{3n\pi}{4}\right]\sin nx\,\cos nat.$$

(Using $\sin\frac{n\pi}{4} + \sin\frac{3n\pi}{4} = 2\sin\frac{n\pi}{2}\cos\frac{n\pi}{4}$, the terms with n even vanish; e.g. $b_1 = \frac{2\sqrt{2}}{\pi}$.)

Sketches of the initial shapes. (a) is a triangle with peak height c at $x = \pi/2$ (drawn for $c = 1$), (b) is a parabolic arch of height $\pi/4$ at $x = \pi/2$ (since $f(\pi/2) = \frac{1}{\pi}\cdot\frac{\pi}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}$), and (c) is a trapezoid of height $\pi/4$.



Answer. (a) $y = \dfrac{8c}{\pi^2}\sum_1^\infty (-1)^{n+1}\dfrac{\sin(2n-1)x\cos(2n-1)at}{(2n-1)^2}$;

(b) $y = \dfrac{8}{\pi^2}\sum_1^\infty \dfrac{\sin(2n-1)x\cos(2n-1)at}{(2n-1)^3}$;

(c) $y = \dfrac{2}{\pi}\sum_1^\infty \dfrac{1}{n^2}\left[\sin\dfrac{n\pi}{4} + \sin\dfrac{3n\pi}{4}\right]\sin nx\cos nat.$

Problem 40-6 (Simmons)

Solution. Take $f(x) = c\sin mx$ with $m = 1, 2, \ldots$ ($m = 1$ is the single arch). By $(***)$,

$$b_n = \frac{2}{\pi}\int_0^\pi c\sin mx\sin nx\,dx.$$

By the orthogonality relation (20), this is 0 for $n \neq m$, and for $n = m$

$$b_m = \frac{2c}{\pi}\int_0^\pi \sin^2 mx\,dx = \frac{2c}{\pi}\cdot\frac{\pi}{2} = c.$$

Thus the series has the single term $b_m \sin mx\cos mat$:

$$y(x,t) = c\ \sin mx\ \cos mat.$$

In particular for $m = 1$: $y(x,t) = c\ \sin x\cos at$.

(One checks directly that this satisfies $a^2y_{xx} = y_{tt}$ [both equal $-a^2m^2c\sin mx\cos mat$], $y(0,t) = y(\pi,t) = 0$, $y_t(x,0) = 0$, $y(x,0) = c\sin mx$.)

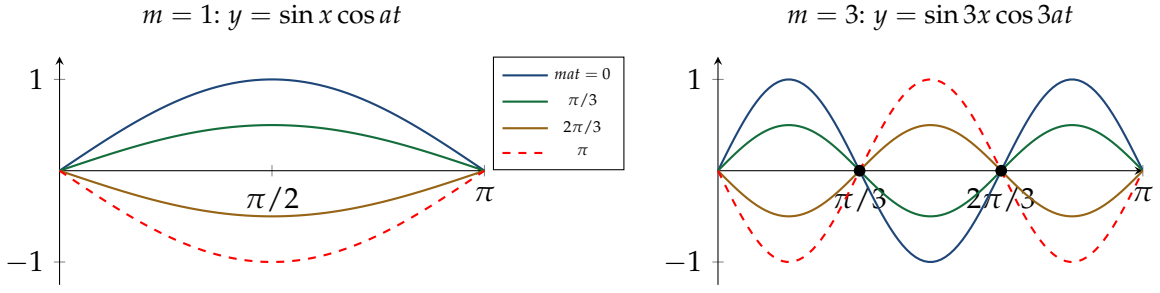
Shape. At each fixed time t the displacement is $y = [c\cos mat]\cdot\sin mx$: the same curve $\sin mx$ multiplied by the number $c\cos mat$, which oscillates between c and $-c$ with period $2\pi/(ma)$. So the string always has the shape of the original curve $c\sin mx$, with amplitude $c\cos mat$ varying harmonically; for $m = 1$ it is one arch that rises and falls (and for $\cos at < 0$ lies below the axis).

Nodes. The points where the string never moves are those with $\sin mx = 0$ in $0 < x < \pi$, since there $y = c\sin mx\cos mat = 0$ for all t. Now $\sin mx = 0$ iff $x = k\pi/m$, k an integer, and $0 < k\pi/m < \pi$ iff $k = 1, 2, \ldots, m-1$. So there are exactly $m-1$ nodes,

$$x = \frac{\pi}{m},\ \frac{2\pi}{m},\ \ldots,\ \frac{(m-1)\pi}{m},$$

and these divide the string into m segments, each vibrating as a single arch. At any other $x \in (0,\pi)$ we have $\sin mx \neq 0$, so the string at that point does move (it is zero only when $\cos mat = 0$). (For $m = 1$, there are no nodes besides the ends.) Between consecutive nodes neighbouring arches have opposite signs of $\sin mx$, so adjacent segments move in opposite directions.

Sketches. With $c = 1$, the figure shows $y = \sin mx\cos mat$ for $m = 1$ and $m = 3$ at the times with $\cos mat = 1, \frac{1}{2}, 0, -\frac{1}{2}, -1$, i.e. $mat = 0, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \pi$ (the dots mark the $m-1$ nodes, i.e. none for $m = 1$ and two for $m = 3$).



The string oscillates between the solid blue curve ($mat = 0$) and the dashed curve ($mat = \pi$), passing through the flat position at $mat = \pi/2$; the nodes stay fixed.

Answer. $y(x,t) = c\sin x\cos at$ for $f = c\sin x$, and $y = c\sin nx\cos nat$ for $f = c\sin nx$. At each instant the string is the curve $\sin nx$ scaled by $c\cos nat$ (a standing wave); it has exactly $n-1$ nodes $x = k\pi/n$, $k = 1,\dots,n-1$, in $(0,\pi)$.

Problem 40-7 (Simmons)

Solution. Seek solutions of $a^2y_{xx} = y_{tt}$ of the form $y = u(x)v(t)$. Substitution gives $a^2u''v = uv''$, i.e.

$$\frac{u''(x)}{u(x)} = \frac{1}{a^2}\frac{v''(t)}{v(t)} = -\lambda$$

(each side depends on a different variable, so both equal a constant $-\lambda$). Hence

$$u'' + \lambda u = 0, \qquad v'' + \lambda a^2 v = 0.$$

Space part. The boundary conditions $y(0,t) = y(\pi,t) = 0$ require $u(0) = u(\pi) = 0$ (for nontrivial v). By Problem 1(d) with $L = \pi$ (or the text), nontrivial solutions exist only for $\lambda = n^2$, $n = 1,2,3,\dots$, with eigenfunctions $u_n(x) = \sin nx$.

Time part. For $\lambda = n^2$: $v'' + n^2a^2v = 0$, whose general solution is $v = A\sin nat + B\cos nat$. The initial condition $y(x,0) = 0$ requires $u_n(x)v(0) = 0$, i.e. $v(0) = B = 0$. So $v_n(t) = \sin nat$ and we have the product solutions

$$y_n(x,t) = \sin nx\,\sin nat,$$

each satisfying (8), (9), (10) and $y_n(x,0) = 0$. Any linear combination, and (formally) any series

$$y(x,t) = \sum_{n=1}^{\infty} c_n \sin nx\,\sin nat$$

does too.

Determining c_n. Differentiating term by term (formally),

$$\left.\frac{\partial y}{\partial t}\right|_{t=0} = \sum_{n=1}^{\infty} c_n\, na\,\sin nx\,\cos nat\Big|_{t=0} = \sum_{n=1}^{\infty}(c_n na)\sin nx.$$

The requirement $y_t(x,0) = g(x)$ thus says $g(x) = \sum(c_nna)\sin nx$, a Fourier sine series. By the formula (21) of the text (derived from the orthogonality relation (20) and $\int_0^\pi \sin^2 nx\,dx = \pi/2$), its coefficients are

$$c_nna = \frac{2}{\pi}\int_0^\pi g(x)\sin nx\,dx \quad\Longrightarrow\quad c_n = \frac{2}{\pi na}\int_0^\pi g(x)\sin nx\,dx.$$

This gives the required solution.

Answer. $y(x,t) = \sum_{n=1}^{\infty} c_n \sin nx\sin nat$ with $c_n = \dfrac{2}{\pi na}\displaystyle\int_0^\pi g(x)\sin nx\,dx$ (obtained from $u_n = \sin nx$, $v_n = \sin nat$ and $y_t(x,0) = \sum c_nna\sin nx = g(x)$).

Section 41 The Heat Equation

In this section $w = w(x,t)$ is temperature, the one-dimensional heat equation is $a^2\,\partial^2 w/\partial x^2 = \partial w/\partial t$ with $a^2 = k/c\rho$ (equation (4)), and the basic solution of the problem $w(x,0) = f(x)$, $w(0,t) = w(\pi,t) = 0$ is

$$w(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx, \qquad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx. \tag{†}$$

Problem 41-1 (Simmons)

Solution. Equation (1) is $a^2\left(w_{xx}+w_{yy}+w_{zz}\right) = w_t$ with $a^2 = k/c\rho$. We use the three physical principles (a), (b), (c) of the text: heat flows from hot to cold; the rate of flow across an area S perpendicular to the direction of flow is $-kS\times$(rate of change of temperature in that direction); and the heat stored when the temperature changes by Δw in mass Δm is $c\,\Delta m\,\Delta w$.

Consider a box B with one corner at (x,y,z) and edges $\Delta x, \Delta y, \Delta z$ along the axes. Let ρ be the density.

Heat stored. The mass is $\Delta m = \rho\,\Delta x\,\Delta y\,\Delta z$. If the temperature changes by Δw in the time Δt, the heat stored is $\Delta H = c\,\Delta m\,\Delta w = c\rho\,\Delta x\,\Delta y\,\Delta z\,\Delta w$, so the rate of storage is

$$\frac{\Delta H}{\Delta t} = c\rho\,\Delta x\,\Delta y\,\Delta z\,\frac{\Delta w}{\Delta t}. \tag{1}$$

Heat flowing in. We assume no heat is generated inside the box, so heat enters only through the six faces.

Faces perpendicular to the x-axis. Each has area $\Delta y\,\Delta z$. By (b), the rate of flow into the box through the face at x is $-k\,\Delta y\,\Delta z\left.\dfrac{\partial w}{\partial x}\right|_x$ (positive when $w_x < 0$, in accordance with (a)), and through the opposite face at $x+\Delta x$ it is $+k\,\Delta y\,\Delta z\left.\dfrac{\partial w}{\partial x}\right|_{x+\Delta x}$. Their sum is

$$k\,\Delta y\,\Delta z\left[\left.\frac{\partial w}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial w}{\partial x}\right|_x\right].$$

Faces perpendicular to the y-axis (area $\Delta x\,\Delta z$) and *to the z-axis* (area $\Delta x\,\Delta y$) give in the same way

$$k\,\Delta x\,\Delta z\left[\left.\frac{\partial w}{\partial y}\right|_{y+\Delta y} - \left.\frac{\partial w}{\partial y}\right|_y\right], \qquad k\,\Delta x\,\Delta y\left[\left.\frac{\partial w}{\partial z}\right|_{z+\Delta z} - \left.\frac{\partial w}{\partial z}\right|_z\right].$$

Balance. Equating the rate of storage (1) with the total rate of inflow:

$$c\rho\,\Delta x\,\Delta y\,\Delta z\,\frac{\Delta w}{\Delta t} = k\,\Delta y\,\Delta z\,\Delta_x w_x + k\,\Delta x\,\Delta z\,\Delta_y w_y + k\,\Delta x\,\Delta y\,\Delta_z w_z,$$

where $\Delta_x w_x = w_x|_{x+\Delta x} - w_x|_x$, etc. Dividing by $c\rho\,\Delta x\,\Delta y\,\Delta z$:

$$\frac{\Delta w}{\Delta t} = \frac{k}{c\rho}\left[\frac{w_x|_{x+\Delta x} - w_x|_x}{\Delta x} + \frac{w_y|_{y+\Delta y} - w_y|_y}{\Delta y} + \frac{w_z|_{z+\Delta z} - w_z|_z}{\Delta z}\right].$$

Now let $\Delta x, \Delta y, \Delta z, \Delta t \to 0$. The difference quotients tend to $\partial w/\partial t$, $\partial^2 w/\partial x^2$, $\partial^2 w/\partial y^2$, $\partial^2 w/\partial z^2$ respectively, so with $a^2 = k/c\rho$,

$$a^2\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) = \frac{\partial w}{\partial t},$$

which is (1).

Answer. Shown: storage rate $c\rho\,\Delta x\Delta y\Delta z\,\Delta w/\Delta t$ equals net inflow through the three pairs of faces; dividing by $c\rho\,\Delta x\Delta y\Delta z$ and passing to the limit gives $a^2(w_{xx}+w_{yy}+w_{zz}) = w_t$, $a^2 = k/c\rho$.

Problem 41-2 (Simmons)

Solution. We want $w_t = a^2 w_{xx}$ for $0 < x < \pi$, $t > 0$, with $w(x,0) = f(x)$, $w(0,t) = w_1$, $w(\pi,t) = w_2$. Following the hint, let $g(x)$ be the steady-state temperature of Example 2, that is, the solution of $g'' = 0$ with $g(0) = w_1$, $g(\pi) = w_2$:

$$g(x) = w_1 + \frac{w_2 - w_1}{\pi}\,x.$$

Put $w = W(x,t) + g(x)$. Since $g_{xx} = 0$ and $g_t = 0$,

$$a^2 w_{xx} - w_t = a^2 W_{xx} + a^2 g'' - W_t = a^2 W_{xx} - W_t,$$

so W satisfies the same heat equation $a^2 W_{xx} = W_t$. The boundary and initial conditions become

$$W(0,t) = w(0,t) - g(0) = w_1 - w_1 = 0, \qquad W(\pi,t) = w_2 - g(\pi) = 0,$$
$$W(x,0) = f(x) - g(x).$$

This is exactly the problem of the text with initial temperature $f(x) - g(x)$ and zero end temperatures, solved by $(\dagger)$:

$$W(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx, \qquad b_n = \frac{2}{\pi}\int_0^{\pi}[f(x) - g(x)]\sin nx\,dx.$$

Therefore

$$w(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx + g(x),$$

where $g(x) = w_1 + \frac{1}{\pi}(w_2 - w_1)x$ and $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi}[f(x) - g(x)]\sin nx\,dx$. *Remark on b_n.* Since $\int_0^{\pi}\sin nx\,dx = \dfrac{1-(-1)^n}{n}$ and $\int_0^{\pi} x\sin nx\,dx = -\dfrac{\pi(-1)^n}{n}$,

$$\int_0^{\pi} g\sin nx\,dx = w_1\,\frac{1-(-1)^n}{n} - (w_2 - w_1)\frac{(-1)^n}{n} = \frac{w_1 - (-1)^n w_2}{n},$$

so equivalently $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi} f(x)\sin nx\,dx - \frac{2\,[w_1 - (-1)^n w_2]}{n\pi}$. As $t \to \infty$ every term of the series tends to 0, so $w(x,t) \to g(x)$, the steady state.

Answer. $w(x,t) = \displaystyle\sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx + g(x)$, where $g(x) = w_1 + \frac{1}{\pi}(w_2 - w_1)x$ and $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi}[f(x) - g(x)]\sin nx\,dx$.

Problem 41-3 (Simmons)

Solution. Use the set-up of the text: rod of cross-sectional area A, density ρ, thermal conductivity k, and specific heat which, to avoid confusion with the constant c of the statement, we denote by c_0; so $a^2 = k/c_0\rho$. Let P be the perimeter of a cross-section, so that the lateral surface area of the slice between x and $x+\Delta x$ is $P\,\Delta x$.

Newton's law of cooling: the rate at which a surface loses heat to its surroundings is proportional to the area of the surface and to the difference between the temperature w of the surface and the temperature w_0 of the surroundings. Hence the slice loses heat through its lateral surface at the rate

$$h\,P\,\Delta x\,(w - w_0),$$

where $h > 0$ is a constant (it is positive: when $w > w_0$ heat is lost, and when $w < w_0$ heat is gained).

The heat inflow through the two ends of the slice is as in the text, $kA\,[w_x|_{x+\Delta x} - w_x|_x]$, and the rate of storage is $c_0\rho A\,\Delta x\,\dfrac{\Delta w}{\Delta t}$. Balancing:

$$c_0\rho A\,\Delta x\,\frac{\Delta w}{\Delta t} = kA\left[\left.\frac{\partial w}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial w}{\partial x}\right|_{x}\right] - hP\,\Delta x\,(w - w_0).$$

Divide by $c_0\rho A\,\Delta x$:

$$\frac{\Delta w}{\Delta t} = \frac{k}{c_0\rho}\cdot\frac{w_x|_{x+\Delta x} - w_x|_x}{\Delta x} - \frac{hP}{c_0\rho A}\,(w - w_0).$$

Let $\Delta x, \Delta t \to 0$ and write $a^2 = k/c_0\rho$ and $c = hP/(c_0\rho A) > 0$:

$$\frac{\partial w}{\partial t} = a^2\frac{\partial^2 w}{\partial x^2} - c\,(w - w_0), \qquad \text{i.e.} \qquad a^2\frac{\partial^2 w}{\partial x^2} = \frac{\partial w}{\partial t} + c\,(w - w_0).$$

Answer. Shown: lateral loss $hP\,\Delta x\,(w - w_0)$ added to the slice balance gives $a^2 w_{xx} = w_t + c(w - w_0)$ with $c = hP/c_0\rho A > 0$.

Problem 41-4 (Simmons)

Solution. The problem is

$$a^2 w_{xx} = w_t + cw \quad (0 < x < \pi,\ t > 0), \qquad w(0,t) = w(\pi,t) = 0, \qquad w(x,0) = f(x).$$

Reduction to the heat equation. Put $w = e^{-ct}u(x,t)$. Then $w_t = e^{-ct}(u_t - cu)$ and $w_{xx} = e^{-ct}u_{xx}$, so

$$a^2 w_{xx} - w_t - cw = e^{-ct}\left[a^2 u_{xx} - u_t + cu - cu\right] = e^{-ct}\left[a^2 u_{xx} - u_t\right].$$

Thus the equation holds iff $a^2 u_{xx} = u_t$. The conditions become $u(0,t) = u(\pi,t) = 0$, $u(x,0) = w(x,0) = f(x)$: exactly the basic problem of the text, whose solution is (†):

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx, \qquad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,\mathrm{d}x.$$

(Equivalently, separating variables directly in $a^2 w_{xx} = w_t + cw$ with $w = X(x)T(t)$ gives $X''/X = (T' + cT)/(a^2T) = -\lambda$, so $X'' + \lambda X = 0$, $X(0) = X(\pi) = 0$, hence $\lambda = n^2$, $X_n = \sin nx$, and $T' + (c + n^2a^2)T = 0$, whence $T_n = e^{-(c+n^2a^2)t}$: the same result.) Therefore

$$w(x,t) = e^{-ct}\sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx.$$

Answer. $w(x,t) = e^{-ct}\sum_{n=1}^{\infty} b_n e^{-n^2a^2t}\sin nx$, where $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi} f(x)\sin nx\,dx$.

Problem 41-5 (Simmons)

Solution. *Boundary conditions.* Insulating an end means that no heat flows through it. By principle (b), the rate of flow across the end is proportional to $\partial w/\partial x$ there, so no flow means $\partial w/\partial x = 0$ at the end:

$$\left.\frac{\partial w}{\partial x}\right|_{x=0} = 0, \qquad \left.\frac{\partial w}{\partial x}\right|_{x=\pi} = 0 \qquad (t \geq 0).$$

Temperature. In Example 1 the rod is at the uniform temperature 100°C at $t = 0$, and its lateral surface is insulated. With the ends insulated as well, no heat can enter or leave the rod, and since the temperature is already uniform there is no temperature difference to drive any flow of heat inside the rod. Common sense says that nothing changes:

$$w(x,t) = 100 \quad \text{for all } x,\ t.$$

Check. $w \equiv 100$ satisfies $a^2w_{xx} = w_t$ (both sides are 0), the initial condition $w(x,0) = 100$, and the boundary conditions $w_x = 0$ at $x = 0, \pi$. (This solution is unique: for two solutions the difference D satisfies the heat equation, $D_x = 0$ at the ends, and $D(x,0) = 0$; then $\frac{d}{dt}\int_0^{\pi} D^2\,dx = 2\int_0^{\pi} DD_t\,dx = 2a^2\int_0^{\pi} DD_{xx}\,dx = 2a^2[DD_x]_0^{\pi} - 2a^2\int_0^{\pi} D_x^2\,dx \leq 0$, so $\int D^2$ cannot grow from 0, and $D \equiv 0$.) It also agrees with the result of Problem 6 below: with $f \equiv 100$ one finds $a_0 = 200$ and $a_n = 0$ for $n \geq 1$, so $w = \frac{1}{2}a_0 = 100$.

Answer. Boundary conditions: $w_x(0,t) = 0$, $w_x(\pi,t) = 0$. Temperature: $w(x,t) = 100$ (constant).

Problem 41-6 (Simmons)

Solution. The problem is

$$a^2w_{xx} = w_t, \qquad w_x(0,t) = w_x(\pi,t) = 0, \qquad w(x,0) = f(x).$$

Separation of variables. Let $w = u(x)v(t)$. Then $a^2u''v = uv'$, so

$$\frac{u''}{u} = \frac{1}{a^2}\frac{v'}{v} = -\lambda, \qquad u'' + \lambda u = 0, \qquad v' + \lambda a^2 v = 0.$$

The boundary conditions require $u'(0) = u'(\pi) = 0$.

Eigenvalues.

- $\lambda = -\mu^2 < 0$: $u = c_1\cosh\mu x + c_2\sinh\mu x$, $u' = \mu(c_1\sinh\mu x + c_2\cosh\mu x)$. $u'(0) = \mu c_2 = 0$ gives $c_2 = 0$, then $u'(\pi) = \mu c_1\sinh\mu\pi = 0$ gives $c_1 = 0$. No eigenvalues.
- $\lambda = 0$: $u = c_1 + c_2x$, $u' = c_2$, so $u'(0) = 0$ forces $c_2 = 0$ and then $u'(\pi) = 0$ holds. Thus $\lambda_0 = 0$ *is* an eigenvalue, with eigenfunction $u_0 = 1$.
- $\lambda = \mu^2 > 0$: $u = c_1\cos\mu x + c_2\sin\mu x$, $u' = \mu(-c_1\sin\mu x + c_2\cos\mu x)$. $u'(0) = \mu c_2 = 0$ gives $c_2 = 0$; then $u'(\pi) = -\mu c_1\sin\mu\pi = 0$ requires $\sin\mu\pi = 0$, so $\mu = n$, $n = 1,2,3,\ldots$

So the eigenvalues are $\lambda_n = n^2$, $n = 0,1,2,\ldots$, with eigenfunctions $u_n(x) = \cos nx$ (for $n = 0$, $u_0 = 1$).

Time factor. $v' + n^2a^2v = 0$ gives $v_n(t) = e^{-n^2a^2t}$ (and $v_0 = 1$). The product solutions are $e^{-n^2a^2t}\cos nx$, $n = 0,1,2,\ldots$, each satisfying the PDE and the boundary conditions ($\partial_x\cos nx =$

$-n\sin nx$ vanishes at 0 and π). Forming the series, with the constant term written as $\frac{1}{2}a_0$ for convenience:

$$w(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n e^{-n^2a^2t}\cos nx.$$

Initial condition. At $t=0$: $f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx$, the Fourier cosine series of f on $[0,\pi]$. The functions $\cos nx$ are orthogonal on $[0,\pi]$: for $m \neq n$, $\int_0^\pi \cos mx \cos nx\,dx = \frac{1}{2}\int_0^\pi [\cos(m-n)x + \cos(m+n)x]\,dx = 0$, and $\int_0^\pi \cos^2 nx\,dx = \pi/2$ ($n \geq 1$), $\int_0^\pi 1\,dx = \pi$. Multiplying the series by $\cos nx$ and integrating term by term over $[0,\pi]$ gives $\int_0^\pi f\cos nx\,dx = a_n\pi/2$ for $n \geq 1$, and multiplying by 1 gives $\int_0^\pi f\,dx = \frac{1}{2}a_0\pi$. Hence for $n = 0,1,2,\ldots$

$$a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx,$$

(the factor $\frac{1}{2}$ in front of a_0 makes the single formula valid for $n=0$). The solution is

$$w(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n e^{-n^2a^2t}\cos nx, \qquad a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx, \quad n = 0,1,2,\ldots$$

Note that $\frac{1}{2}a_0 = \frac{1}{\pi}\int_0^\pi f\,dx$ is the mean initial temperature, which is the limiting temperature as $t\to\infty$ (no heat is lost). For $f \equiv 100$ this gives $w \equiv 100$, in agreement with Problem 5.

Answer. $w(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n e^{-n^2a^2t}\cos nx$, where $a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx$ for $n = 0,1,2,\ldots$

Problem 41-7 (Simmons)

Solution. Steady state means $\partial w/\partial t = 0$, so the equation reduces to Laplace's equation $w_{xx} + w_{yy} = 0$ in $0 < x < \pi$, $y > 0$. Let $w = u(x)v(y)$: $u''v + uv'' = 0$, so

$$\frac{u''}{u} = -\frac{v''}{v} = -\lambda, \qquad u'' + \lambda u = 0, \qquad v'' - \lambda v = 0.$$

The conditions $w(0,y) = w(\pi,y) = 0$ give $u(0) = u(\pi) = 0$. This is the eigenvalue problem of the text: for $\lambda < 0$ or $\lambda = 0$ only $u \equiv 0$ is possible (Problem 40-1), and $\lambda_n = n^2$, $u_n = \sin nx$ for $n = 1,2,3,\ldots$

The y-equation. For $\lambda = n^2$, $v'' - n^2v = 0$ has general solution $v = Ae^{ny} + Be^{-ny}$. The condition $w \to 0$ as $y\to\infty$ (for every x) cannot hold if $A \neq 0$, since $e^{ny}\to\infty$; so $A = 0$ and $v_n(y) = e^{-ny}$. The product solutions are $w_n(x,y) = e^{-ny}\sin nx$, each satisfying Laplace's equation ($w_{xx} + w_{yy} = -n^2w_n + n^2w_n = 0$), both side conditions, and $w_n \to 0$ as $y\to\infty$. Form

$$w(x,y) = \sum_{n=1}^{\infty} b_n e^{-ny}\sin nx.$$

Condition at $y = 0$. $f(x) = w(x,0) = \sum_{n=1}^{\infty} b_n \sin nx$, the Fourier sine series of f on $[0,\pi]$, so by (15) (or formula (21) of Section 40)

$$b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx.$$

Hence

$$w(x,y) = \sum_{n=1}^{\infty} b_n e^{-ny}\sin nx, \qquad b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx.$$

(Since $|b_n| \le C$ is bounded and $e^{-ny} \le e^{-y}$ for $n \ge 1$, $y \ge 0$, the series is dominated by $\frac{C e^{-y}}{1-e^{-y}} \to 0$ (with $C = \frac{2}{\pi}\int_0^\pi |f|\,dx$) as $y \to \infty$, consistent with the last condition.)

Answer. $w(x,y) = \displaystyle\sum_{n=1}^{\infty} b_n e^{-ny} \sin nx$, where $b_n = \dfrac{2}{\pi}\displaystyle\int_0^\pi f(x)\sin nx\,dx$.

Section 42 The Dirichlet Problem for a Circle. Poisson's Integral

Throughout this section $w = w(r,\theta)$ is harmonic in the unit disc, i.e. it satisfies Laplace's equation

$$w_{rr} + \frac{1}{r}w_r + \frac{1}{r^2}w_{\theta\theta} = 0,$$

and the solution of the Dirichlet problem with boundary function f is, by the text,

$$w(r,\theta) = \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty} r^n(a_n\cos n\theta + b_n \sin n\theta),$$

where

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)\cos n\phi\,d\phi, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)\sin n\phi\,d\phi.$$

Each term $r^n\cos n\theta$, $r^n \sin n\theta$ is harmonic (it is the real or imaginary part of $(x+iy)^n$).

Problem 42-1 (Simmons)

Solution. Assume F has continuous second partial derivatives, so $w_{xy} = w_{yx}$. Write w_x, w_y, w_{xx}, w_{xy}, w_{yy} for the partial derivatives of F, evaluated at $(r\cos\theta, r\sin\theta)$. Since

$$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial y}{\partial r} = \sin\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta,$$

the chain rule gives the formulas of the hint:

$$w_r = w_x\cos\theta + w_y\sin\theta, \tag{7.1}$$
$$w_\theta = -r\sin\theta\, w_x + r\cos\theta\, w_y. \tag{7.2}$$

The same chain rule applies to every function of (x,y), in particular to w_x and w_y:

$$\frac{\partial w_x}{\partial r} = w_{xx}\cos\theta + w_{xy}\sin\theta, \quad \frac{\partial w_y}{\partial r} = w_{xy}\cos\theta + w_{yy}\sin\theta,$$

$$\frac{\partial w_x}{\partial \theta} = -r\sin\theta\, w_{xx} + r\cos\theta\, w_{xy}, \quad \frac{\partial w_y}{\partial \theta} = -r\sin\theta\, w_{xy} + r\cos\theta\, w_{yy}.$$

Computation of $\partial(rw_r)/\partial r$. From (7.1), $rw_r = r\cos\theta\, w_x + r\sin\theta\, w_y$, so by the product rule

$$\begin{aligned}\frac{\partial}{\partial r}(rw_r) &= \cos\theta\, w_x + r\cos\theta\,(w_{xx}\cos\theta + w_{xy}\sin\theta) + \sin\theta\, w_y + r\sin\theta\,(w_{xy}\cos\theta + w_{yy}\sin\theta)\\ &= (w_x\cos\theta + w_y\sin\theta) + r\big(w_{xx}\cos^2\theta + 2w_{xy}\sin\theta\cos\theta + w_{yy}\sin^2\theta\big)\\ &= w_r + r\big(w_{xx}\cos^2\theta + 2w_{xy}\sin\theta\cos\theta + w_{yy}\sin^2\theta\big).\end{aligned} \tag{A}$$

Computation of $\partial^2 w/\partial\theta^2$. Differentiating (7.2) with respect to θ,

$$\begin{aligned}w_{\theta\theta} &= -r\cos\theta\, w_x - r\sin\theta\,(-r\sin\theta\, w_{xx} + r\cos\theta\, w_{xy})\\ &\quad - r\sin\theta\, w_y + r\cos\theta\,(-r\sin\theta\, w_{xy} + r\cos\theta\, w_{yy})\\ &= -r(w_x\cos\theta + w_y\sin\theta) + r^2\big(w_{xx}\sin^2\theta - 2w_{xy}\sin\theta\cos\theta + w_{yy}\cos^2\theta\big)\\ &= -r\,w_r + r^2\big(w_{xx}\sin^2\theta - 2w_{xy}\sin\theta\cos\theta + w_{yy}\cos^2\theta\big).\end{aligned} \tag{B}$$

Combination. Divide (A) by r and (B) by r^2 and add:

$$\begin{aligned}\frac{1}{r}\frac{\partial}{\partial r}(rw_r)+\frac{1}{r^2}w_{\theta\theta} &= \frac{w_r}{r}+\left(w_{xx}\cos^2\theta+2w_{xy}\sin\theta\cos\theta+w_{yy}\sin^2\theta\right)\\ &\quad -\frac{w_r}{r}+\left(w_{xx}\sin^2\theta-2w_{xy}\sin\theta\cos\theta+w_{yy}\cos^2\theta\right)\\ &= w_{xx}(\cos^2\theta+\sin^2\theta)+w_{yy}(\sin^2\theta+\cos^2\theta)=w_{xx}+w_{yy}.\end{aligned}$$

The left side equals $\dfrac{1}{r}\left[\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)+\frac{1}{r}\frac{\partial^2 w}{\partial\theta^2}\right]$, which is the first equality of the problem. Finally $\frac{1}{r}\frac{\partial}{\partial r}(rw_r)=\frac{1}{r}(w_r+rw_{rr})=w_{rr}+\frac{1}{r}w_r$ gives the second form.

Answer. Shown: $w_{xx}+w_{yy}=\dfrac{1}{r}\Big[(rw_r)_r+\dfrac{1}{r}w_{\theta\theta}\Big]=w_{rr}+\dfrac{1}{r}w_r+\dfrac{1}{r^2}w_{\theta\theta}$, where $(rw_r)_r=w_r+r(w_{xx}\cos^2\theta+2w_{xy}\sin\theta\cos\theta+w_{yy}\sin^2\theta)$ and $w_{\theta\theta}=-rw_r+r^2(w_{xx}\sin^2\theta-2w_{xy}\sin\theta\cos\theta+w_{yy}\cos^2\theta)$.

Problem 42-2 (Simmons)

Solution. In each case we compute the Fourier coefficients a_n, b_n of f on $[-\pi,\pi]$ and insert the factor r^n as in (6) of the text.

(a) $f(\theta)=\cos\frac{\theta}{2}$ is even, so $b_n=0$ and $a_n=\dfrac{2}{\pi}\displaystyle\int_0^{\pi}\cos\frac{\theta}{2}\cos n\theta\,d\theta$. Using $2\cos A\cos B=\cos(A+B)+\cos(A-B)$,

$$a_n=\frac{1}{\pi}\int_0^{\pi}\Big[\cos\big(n+\tfrac12\big)\theta+\cos\big(n-\tfrac12\big)\theta\Big]\,d\theta=\frac{1}{\pi}\left[\frac{\sin(n+\frac12)\pi}{n+\frac12}+\frac{\sin(n-\frac12)\pi}{n-\frac12}\right].$$

Now $\sin(n\pi+\frac{\pi}{2})=\cos n\pi=(-1)^n$ and $\sin(n\pi-\frac{\pi}{2})=-\cos n\pi=-(-1)^n$, hence

$$a_n=\frac{(-1)^n}{\pi}\left[\frac{2}{2n+1}-\frac{2}{2n-1}\right]=\frac{(-1)^n}{\pi}\cdot\frac{2(2n-1)-2(2n+1)}{4n^2-1}=-\frac{4}{\pi}\,\frac{(-1)^n}{4n^2-1},$$

for $n=0,1,2,\dots$ (this formula is valid also for $n=0$, where it gives $a_0=4/\pi$, in agreement with the direct computation $a_0=\frac{1}{\pi}\big[2\sin\frac{\theta}{2}\big]_{-\pi}^{\pi}=\frac{4}{\pi}$). Thus $\frac12 a_0=\frac{2}{\pi}$ and

$$w(r,\theta)=\frac{2}{\pi}-\frac{4}{\pi}\sum_{n=1}^{\infty}(-1)^n\frac{r^n\cos n\theta}{4n^2-1}.$$

(Here $f(-\pi)=f(\pi)=0$, so the periodic extension of f is continuous.)

(b) As printed, $f(\theta)=0$ on $(-\pi,\pi)$. Then all $a_n=b_n=0$ and

$$w(r,\theta)\equiv 0.$$

Remark. The printed answer in the book, $w=2\big(r\sin\theta-\frac12 r^2\sin2\theta+\frac13 r^3\sin3\theta-\cdots\big)$, is the solution for $f(\theta)=\theta$, $-\pi<\theta<\pi$; evidently the "θ" was misprinted as "0" in the statement. We therefore also solve that (clearly intended) problem. With $f(\theta)=\theta$ (odd), $a_n=0$ and

$$\begin{aligned}b_n&=\frac{2}{\pi}\int_0^{\pi}\theta\sin n\theta\,d\theta=\frac{2}{\pi}\left\{\Big[-\frac{\theta\cos n\theta}{n}\Big]_0^{\pi}+\frac{1}{n}\int_0^{\pi}\cos n\theta\,d\theta\right\}\\ &=\frac{2}{\pi}\left(-\frac{\pi(-1)^n}{n}+0\right)=\frac{2(-1)^{n+1}}{n}\end{aligned}$$

(integration by parts with $u=\theta$, $dv=\sin n\theta\,d\theta$). Hence

$$w(r,\theta)=2\sum_{n=1}^{\infty}(-1)^{n+1}\frac{r^n\sin n\theta}{n}=2\Big(r\sin\theta-\frac{r^2}{2}\sin 2\theta+\frac{r^3}{3}\sin 3\theta-\cdots\Big),$$

which agrees with the book's answer. The series can be summed: with $z=re^{i\theta}$ we have $\sum_{n\ge1}(-1)^{n+1}z^n/n=\log(1+z)$, and $r^n\sin n\theta=\operatorname{Im}z^n$, so

$$w=2\operatorname{Im}\log(1+z)=2\arg(1+z)=2\tan^{-1}\frac{r\sin\theta}{1+r\cos\theta},$$

which tends to θ as $r\to1$ for $|\theta|<\pi$.

(c) $f(\theta)=\sin\theta$ for $0\le\theta\le\pi$ and 0 for $-\pi\le\theta<0$; f is continuous ($\sin0=\sin\pi=0$). The coefficients are

$$a_n=\frac{1}{\pi}\int_0^{\pi}\sin\theta\cos n\theta\,d\theta,\qquad b_n=\frac{1}{\pi}\int_0^{\pi}\sin\theta\sin n\theta\,d\theta.$$

$n=0$: $a_0=\frac{1}{\pi}\int_0^{\pi}\sin\theta\,d\theta=\frac{2}{\pi}$, so $\frac12a_0=\frac1\pi$.
$n=1$: $a_1=\frac{1}{\pi}\int_0^{\pi}\sin\theta\cos\theta\,d\theta=\frac{1}{2\pi}\big[\sin^2\theta\big]_0^{\pi}=0$, and $b_1=\frac{1}{\pi}\int_0^{\pi}\sin^2\theta\,d\theta=\frac1\pi\cdot\frac\pi2=\frac12$.
$n\ge2$: Use $2\sin\theta\cos n\theta=\sin(n+1)\theta-\sin(n-1)\theta$ and $\int_0^{\pi}\sin k\theta\,d\theta=\dfrac{1-(-1)^k}{k}$ ($k\ne0$):

$$a_n=\frac{1}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1}-\frac{1-(-1)^{n-1}}{n-1}\right]=\frac{1+(-1)^n}{2\pi}\left[\frac{1}{n+1}-\frac{1}{n-1}\right]=-\frac{1+(-1)^n}{\pi(n^2-1)}.$$

So $a_n=0$ for odd $n\ge3$, and for $n=2m$, $a_{2m}=-\dfrac{2}{\pi(4m^2-1)}$. Also, from $2\sin\theta\sin n\theta=\cos(n-1)\theta-\cos(n+1)\theta$ and $\int_0^{\pi}\cos k\theta\,d\theta=0$ for $k\ge1$, we get $b_n=0$ for $n\ge2$. Therefore

$$w(r,\theta)=\frac1\pi+\frac12 r\sin\theta-\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{r^{2n}\cos2n\theta}{4n^2-1}.$$

(d) $f=1$ on $(0,\pi]$, $f=0$ on $[-\pi,0]$ (a jump at $\theta=0$ and at $\pm\pi$). Then

$$a_0=\frac{1}{\pi}\int_0^{\pi}d\theta=1,\qquad a_n=\frac{1}{\pi}\int_0^{\pi}\cos n\theta\,d\theta=\frac{\sin n\pi}{n\pi}=0,$$

$$b_n=\frac{1}{\pi}\int_0^{\pi}\sin n\theta\,d\theta=\frac{1-\cos n\pi}{n\pi}=\frac{1-(-1)^n}{n\pi}=\begin{cases}\dfrac{2}{n\pi}, & n\text{ odd},\\ 0, & n\text{ even}.\end{cases}$$

Writing $n=2k-1$,

$$w(r,\theta)=\frac12+\frac{2}{\pi}\sum_{k=1}^{\infty}\frac{r^{2k-1}\sin(2k-1)\theta}{2k-1}=\frac12+\frac{2}{\pi}\Big(r\sin\theta+\frac13r^3\sin3\theta+\frac15r^5\sin5\theta+\cdots\Big).$$

(e) $f(\theta)=\frac14\theta^2$ is even, so $b_n=0$, and $a_n=\dfrac{2}{\pi}\displaystyle\int_0^{\pi}\frac{\theta^2}{4}\cos n\theta\,d\theta=\frac{1}{2\pi}\int_0^{\pi}\theta^2\cos n\theta\,d\theta$. For $n=0$: $a_0=\frac{1}{2\pi}\cdot\frac{\pi^3}{3}=\frac{\pi^2}{6}$, so $\frac12a_0=\frac{\pi^2}{12}$. For $n\ge1$ integrate by parts twice:

$$\int_0^{\pi}\theta^2\cos n\theta\,d\theta=\Big[\frac{\theta^2\sin n\theta}{n}\Big]_0^{\pi}-\frac{2}{n}\int_0^{\pi}\theta\sin n\theta\,d\theta=-\frac{2}{n}\int_0^{\pi}\theta\sin n\theta\,d\theta,$$

$$\int_0^{\pi} \theta \sin n\theta \, d\theta = \Big[-\frac{\theta\cos n\theta}{n}\Big]_0^{\pi} + \frac{1}{n}\int_0^{\pi} \cos n\theta \, d\theta = -\frac{\pi(-1)^n}{n},$$

so $\int_0^{\pi} \theta^2 \cos n\theta \, d\theta = \dfrac{2\pi(-1)^n}{n^2}$ and $a_n = \dfrac{(-1)^n}{n^2}$. Hence

$$w(r,\theta) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} (-1)^n \frac{r^n \cos n\theta}{n^2}.$$

Answer. (a) $w = \dfrac{2}{\pi} - \dfrac{4}{\pi}\displaystyle\sum_{n\ge 1}(-1)^n \frac{r^n\cos n\theta}{4n^2-1}$; (b) $w \equiv 0$ as printed (the book's answer is for $f = \theta$: $w = 2\displaystyle\sum_{n\ge 1}(-1)^{n+1}\frac{r^n \sin n\theta}{n}$); (c) $w = \dfrac{1}{\pi} + \dfrac{1}{2} r\sin\theta - \dfrac{2}{\pi}\displaystyle\sum_{n\ge 1}\frac{r^{2n}\cos 2n\theta}{4n^2-1}$; (d) $w = \dfrac{1}{2} + \dfrac{2}{\pi}\Big(r\sin\theta + \dfrac{r^3}{3}\sin 3\theta + \dfrac{r^5}{5}\sin 5\theta + \cdots\Big)$; (e) $w = \dfrac{\pi^2}{12} + \displaystyle\sum_{n\ge 1}(-1)^n \frac{r^n\cos n\theta}{n^2}$.

Problem 42-3 (Simmons)

Solution. *The series.* Laplace's equation in polar coordinates, $w_{rr} + \frac{1}{r}w_r + \frac{1}{r^2}w_{\theta\theta} = 0$, is the same in the disc $0 \le r \le R$ as in the unit disc, so separation of variables $w = u(r)v(\theta)$ proceeds exactly as in the text: periodicity in θ forces $v_n = \alpha_n \cos n\theta + \beta_n \sin n\theta$ ($n = 0, 1, 2, \dots$), the radial equation $r^2u'' + ru' - n^2u = 0$ has solutions $A + B\log r$ ($n = 0$) and $Ar^n + Br^{-n}$ ($n \ge 1$), and boundedness (continuity) at $r = 0$ forces $B = 0$. Superposition gives

$$w(r,\theta) = \tfrac{1}{2}\alpha_0 + \sum_{n=1}^{\infty} r^n(\alpha_n \cos n\theta + \beta_n \sin n\theta).$$

The boundary condition is now imposed at $r = R$:

$$f(\theta) = w(R,\theta) = \tfrac{1}{2}\alpha_0 + \sum_{n=1}^{\infty} R^n(\alpha_n \cos n\theta + \beta_n \sin n\theta).$$

This is the Fourier series of f, so by uniqueness of Fourier coefficients $\alpha_0 = a_0$ and $R^n\alpha_n = a_n$, $R^n\beta_n = b_n$, i.e. $\alpha_n = a_n/R^n$, $\beta_n = b_n/R^n$, where $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f\cos n\phi \, d\phi$, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f \sin n\phi \, d\phi$. Therefore

$$w(r,\theta) = \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty}\Big(\frac{r}{R}\Big)^n (a_n \cos n\theta + b_n \sin n\theta).$$

Each term is harmonic, because $(r/R)^n \cos n\theta = R^{-n}\,\mathrm{Re}(x+iy)^n$ (similarly for sin), and the term-by-term sum converges for $r < R$ (see below), while at $r = R$ it reduces to the Fourier series of f.

The Poisson integral. Put $\rho = r/R < 1$. Substituting the integral formulas for a_n, b_n and using $\cos n\theta\cos n\phi + \sin n\theta \sin n\phi = \cos n(\theta - \phi)$,

$$w = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)\Big[\frac{1}{2} + \sum_{n=1}^{\infty}\rho^n \cos n(\theta-\phi)\Big]\, d\phi.$$

The interchange of sum and integral is justified: if $|f| \le M$ then $|a_n|, |b_n| \le 2M$, so the series in n is dominated by $\sum 4M\rho^n < \infty$, and the convergence is uniform in ϕ (Weierstrass test). To sum the

bracket let $\alpha = \theta - \phi$ and $z = \rho e^{i\alpha}$, $|z| = \rho < 1$. Since $\rho^n \cos n\alpha = \operatorname{Re} z^n$ and $\sum_{n\ge1} z^n = \dfrac{z}{1-z}$,

$$\frac{1}{2} + \sum_{n=1}^{\infty} \rho^n \cos n\alpha = \operatorname{Re}\Big[\frac{1}{2} + \frac{z}{1-z}\Big] = \operatorname{Re}\frac{1+z}{2(1-z)} = \operatorname{Re}\frac{(1+z)(1-\bar z)}{2|1-z|^2}$$
$$= \operatorname{Re}\frac{1-|z|^2+(z-\bar z)}{2|1-z|^2} = \frac{1-\rho^2}{2(1-2\rho\cos\alpha+\rho^2)},$$

because $z - \bar z = 2i \operatorname{Im} z$ is purely imaginary and $|1-z|^2 = 1 - 2\rho\cos\alpha + \rho^2$. Hence

$$w(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1-\rho^2}{1-2\rho\cos(\theta-\phi)+\rho^2} f(\phi)\,d\phi.$$

Multiplying numerator and denominator by R^2 and using $\rho = r/R$:

$$\frac{1-\rho^2}{1-2\rho\cos\alpha+\rho^2} = \frac{R^2-r^2}{R^2-2Rr\cos\alpha+r^2},$$

which gives

$$w(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{R^2-r^2}{R^2-2Rr\cos(\theta-\phi)+r^2}\, f(\phi)\,d\phi.$$

Answer. Shown: $w = \frac{1}{2}a_0 + \sum_{n\ge1}(r/R)^n(a_n\cos n\theta + b_n \sin n\theta)$, and $w(r,\theta) = \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} \frac{R^2-r^2}{R^2-2Rr\cos(\theta-\phi)+r^2} f(\phi)\,d\phi$.

Problem 42-4 (Simmons)

Solution. We understand, as is necessary, that the closed disc bounded by C lies in the region (otherwise the theorem fails or is meaningless: $\log r$ is harmonic in the region $r > 0$, and the circle $r = 1$ lies in it, but the center of the circle is not in the region, so w is not defined there). Let C have center P_0 and radius R_0. Place the origin of polar coordinates (ρ,θ) at P_0 and let $w = w(\rho,\theta)$ in these coordinates; w is of class C^2 on the closed disc and satisfies, for $0 < \rho \le R_0$,

$$w_{\rho\rho} + \frac{1}{\rho}w_\rho + \frac{1}{\rho^2}w_{\theta\theta} = 0 \quad\Longleftrightarrow\quad \frac{\partial}{\partial\rho}(\rho w_\rho) = -\frac{1}{\rho}w_{\theta\theta}, \tag{1}$$

by Laplace's equation in polar form (Problem 1). For $0 \le \rho \le R_0$ define the average of w over the circle of radius ρ about P_0:

$$m(\rho) = \frac{1}{2\pi}\int_0^{2\pi} w(\rho,\theta)\,d\theta.$$

Then m is continuous on $[0, R_0]$ and $m(0) = w(P_0)$ (as $\rho \to 0$, $w(\rho,\theta) \to w(P_0)$ uniformly in θ by continuity of w at P_0). We must show $m(R_0) = m(0)$.

Since w is C^1, we may differentiate under the integral sign: $m'(\rho) = \dfrac{1}{2\pi}\displaystyle\int_0^{2\pi} w_\rho\,d\theta$, and then by (1)

$$\frac{d}{d\rho}(\rho m'(\rho)) = \frac{1}{2\pi}\int_0^{2\pi}\frac{\partial}{\partial\rho}(\rho w_\rho)\,d\theta = -\frac{1}{2\pi\rho}\int_0^{2\pi} w_{\theta\theta}\,d\theta = -\frac{1}{2\pi\rho}\big[w_\theta(\rho,2\pi) - w_\theta(\rho,0)\big] = 0,$$

because w_θ has period 2π in θ. Hence $\rho m'(\rho) = c$ is constant on $(0, R_0]$. But w_ρ is bounded near P_0 (it is the directional derivative $w_x\cos\theta + w_y\sin\theta$ of a C^1 function), so $|\rho m'(\rho)| \le \rho \sup|w_\rho| \to 0$

as $\rho \to 0$; thus $c = 0$, so $m'(\rho) = 0$ for $0 < \rho \le R_0$. Therefore m is constant on $(0, R_0]$, and by continuity at 0,

$$m(R_0) = m(0) = w(P_0), \qquad \text{i.e.} \qquad w(P_0) = \frac{1}{2\pi}\int_0^{2\pi} w(R_0, \theta)\,\mathrm{d}\theta,$$

the average of the values of w on the circumference of C.

Remark. Equivalently: w restricted to the closed disc solves the Dirichlet problem with boundary function $f(\theta) = w(R_0, \theta)$, whose solution is the Poisson integral of Problem 3 (the solution is unique by the maximum principle); putting $r = 0$ in that formula gives the same average, as noted at the end of the text.

Answer. Shown: $w(P_0) = \dfrac{1}{2\pi}\displaystyle\int_0^{2\pi} w(R_0, \theta)\,\mathrm{d}\theta$, because the circle average $m(\rho)$ satisfies $m' \equiv 0$ and $m(0) = w(P_0)$.

Section 43 Sturm–Liouville Problems

Throughout, P, Q, R denote functions of x with as many continuous derivatives as the formulas require; primes denote d/dx. For the equation $Py'' + Qy' + Ry = 0$ the *adjoint equation* (Problem 3) is $P\mu'' + (2P' - Q)\mu' + (P'' - Q' + R)\mu = 0$.

Problem 43-1 (Simmons)

Solution. Expand the exact form with the product rule:

$$[Py']' + [Sy]' = Py'' + P'y' + Sy' + S'y = Py'' + (P' + S)y' + S'y.$$

This coincides with $Py'' + Qy' + Ry$ for every function y if and only if the coefficients of y' and of y agree:

$$Q = P' + S, \qquad R = S'.$$

Necessity. If the equation is exact, the first relation gives $S = Q - P'$; differentiating, $S' = Q' - P''$, and the second relation $R = S'$ gives $R = Q' - P''$, i.e.

$$P'' - Q' + R = 0.$$

Sufficiency. Suppose $P'' - Q' + R = 0$ and define $S = Q - P'$. Then $Q = P' + S$ holds by definition, and $S' = Q' - P'' = R$ by the hypothesis. Both coefficient relations hold, so $Py'' + Qy' + Ry = [Py']' + [Sy]'$, i.e. the equation is exact (with $S = Q - P'$).

Answer. Shown: the equation is exact iff $P'' - Q' + R = 0$, and then $S = Q - P'$.

Problem 43-2 (Simmons)

Solution. Here $P = x^2$, $Q = x$, $R = -n^2$, so $P'' = 2$, $Q' = 1$ and

$$P'' - Q' + R = 2 - 1 - n^2 = 1 - n^2.$$

By Problem 1 the equation is exact iff $1 - n^2 = 0$, i.e. $n = \pm 1$; for positive integers n this means $n = 1$ only.

For $n = 1$: $S = Q - P' = x - 2x = -x$, and the equation is $[x^2y']' + [-xy]' = 0$ (indeed $[x^2y']' - [xy]' = x^2y'' + 2xy' - xy' - y = x^2y'' + xy' - y$). Integrating,

$$x^2y' - xy = c_1 \quad\Longrightarrow\quad y' - \frac{1}{x}y = \frac{c_1}{x^2} \qquad (x \neq 0).$$

This is linear with integrating factor $e^{-\int dx/x} = \dfrac{1}{x}$ (for $x > 0$ or $x < 0$ separately): $\left(\dfrac{y}{x}\right)' = \dfrac{c_1}{x^3}$, so

$$\frac{y}{x} = -\frac{c_1}{2x^2} + c_2, \qquad y = -\frac{c_1}{2x} + c_2 x.$$

Check: $y = x$ and $y = 1/x$ both satisfy $x^2y'' + xy' - y = 0$ (for $y = 1/x$: $x^2 \cdot 2x^{-3} - x \cdot x^{-2} - x^{-1} = 0$).

Answer. Exact only for $n = 1$; then $y = -\dfrac{c_1}{2x} + c_2 x$ (equivalently $y = C_1 x^{-1} + C_2 x$).

Problem 43-3 (Simmons)

Solution. *The adjoint equation.* The equation $\mu P y'' + \mu Q y' + \mu R y = 0$ has coefficients $\tilde{P} = \mu P$, $\tilde{Q} = \mu Q$, $\tilde{R} = \mu R$. By Problem 1 it is exact iff $\tilde{P}'' - \tilde{Q}' + \tilde{R} = 0$, i.e.

$$(\mu P)'' - (\mu Q)' + \mu R = 0.$$

Expanding with the product rule, $(\mu P)'' = P\mu'' + 2P'\mu' + P''\mu$ and $(\mu Q)' = Q\mu' + Q'\mu$, so the condition becomes

$$P\mu'' + (2P' - Q)\mu' + (P'' - Q' + R)\mu = 0,$$

the adjoint equation. For each case below we compute P, Q, R, then $2P' - Q$ and $P'' - Q' + R$.

(a) $P = 1 - x^2$, $Q = -2x$, $R = p(p+1)$; $P' = -2x$, $P'' = -2$, $Q' = -2$. $2P' - Q = -4x + 2x = -2x$; $P'' - Q' + R = -2 + 2 + p(p+1) = p(p+1)$. Adjoint:

$$(1 - x^2)\mu'' - 2x\mu' + p(p+1)\mu = 0$$

(the Legendre equation again).

(b) $P = x^2$, $Q = x$, $R = x^2 - p^2$; $P' = 2x$, $P'' = 2$, $Q' = 1$. $2P' - Q = 4x - x = 3x$; $P'' - Q' + R = 2 - 1 + x^2 - p^2 = 1 + x^2 - p^2$. Adjoint:

$$x^2\mu'' + 3x\mu' + (1 + x^2 - p^2)\mu = 0.$$

(c) $P = 1 - x^2$, $Q = -x$, $R = p^2$; $P' = -2x$, $P'' = -2$, $Q' = -1$. $2P' - Q = -4x + x = -3x$; $P'' - Q' + R = -2 + 1 + p^2 = p^2 - 1$. Adjoint:

$$(1 - x^2)\mu'' - 3x\mu' + (p^2 - 1)\mu = 0.$$

(d) $P = 1$, $Q = -2x$, $R = 2p$; $P' = P'' = 0$, $Q' = -2$. $2P' - Q = 2x$; $P'' - Q' + R = 2 + 2p$. Adjoint:

$$\mu'' + 2x\mu' + (2 + 2p)\mu = 0.$$

(e) $P = 1$, $Q = 0$, $R = x$; $P' = P'' = Q' = 0$. $2P' - Q = 0$; $P'' - Q' + R = x$. Adjoint:

$$\mu'' + x\mu = 0$$

(Airy's equation is its own adjoint).

(f) $P = x$, $Q = 1 - x$, $R = p$; $P' = 1$, $P'' = 0$, $Q' = -1$. $2P' - Q = 2 - 1 + x = 1 + x$; $P'' - Q' + R = 0 + 1 + p = 1 + p$. Adjoint:

$$x\mu'' + (1 + x)\mu' + (1 + p)\mu = 0.$$

Answer. (a) $(1 - x^2)\mu'' - 2x\mu' + p(p+1)\mu = 0$; (b) $x^2\mu'' + 3x\mu' + (1 + x^2 - p^2)\mu = 0$; (c) $(1 - x^2)\mu'' - 3x\mu' + (p^2 - 1)\mu = 0$; (d) $\mu'' + 2x\mu' + (2 + 2p)\mu = 0$; (e) $\mu'' + x\mu = 0$; (f) $x\mu'' + (1 + x)\mu' + (1 + p)\mu = 0$.

Problem 43-4 (Simmons)

Solution. Here $P = 1$, $Q = -\big(2x + \frac{3}{x}\big)$, $R = -4$, on $x > 0$ (or on $x < 0$; the computation is the same). Then $P' = P'' = 0$ and $Q' = -\big(2 - \frac{3}{x^2}\big)$, so

$$2P' - Q = 2x + \frac{3}{x}, \qquad P'' - Q' + R = 2 - \frac{3}{x^2} - 4 = -2 - \frac{3}{x^2},$$

and the adjoint equation is

$$\mu'' + \Big(2x + \frac{3}{x}\Big)\mu' - \Big(2 + \frac{3}{x^2}\Big)\mu = 0.$$

By inspection try $\mu = x$: $\mu' = 1$, $\mu'' = 0$, and

$$0 + \Big(2x + \frac{3}{x}\Big) - \Big(2 + \frac{3}{x^2}\Big)x = 2x + \frac{3}{x} - 2x - \frac{3}{x} = 0.$$

So $\mu = x$ is an integrating factor. Multiplying the equation by x:

$$xy'' - (2x^2 + 3)y' - 4xy = 0.$$

This is exact: with $\tilde{P} = x$, $\tilde{Q} = -(2x^2 + 3)$, $\tilde{R} = -4x$, $\tilde{P}'' - \tilde{Q}' + \tilde{R} = 0 + 4x - 4x = 0$. Then $S = \tilde{Q} - \tilde{P}' = -(2x^2 + 3) - 1 = -(2x^2 + 4)$ (and $S' = -4x = \tilde{R}$), so the equation is

$$\big[xy'\big]' - \big[(2x^2 + 4)y\big]' = 0 \quad \Longrightarrow \quad xy' - (2x^2 + 4)y = c_1.$$

(Check of the expansion: $[xy']' - [(2x^2 + 4)y]' = xy'' + y' - (2x^2 + 4)y' - 4xy = xy'' - (2x^2 + 3)y' - 4xy$.) Dividing by x gives the linear equation

$$y' - \Big(2x + \frac{4}{x}\Big)y = \frac{c_1}{x}.$$

Its integrating factor is $\exp\Big[-\int \Big(2x + \frac{4}{x}\Big)dx\Big] = e^{-x^2 - 4\log|x|} = \dfrac{e^{-x^2}}{x^4}$, so

$$\frac{d}{dx}\Big[\frac{e^{-x^2}}{x^4}\,y\Big] = c_1\frac{e^{-x^2}}{x^5}, \qquad \frac{e^{-x^2}}{x^4}\,y = c_1\int \frac{e^{-x^2}}{x^5}\,\mathrm{d}x + c_2.$$

Hence

$$y = x^4 e^{x^2}\left[c_1 \int \frac{e^{-x^2}}{x^5}\,\mathrm{d}x + c_2\right].$$

(The integral is not elementary, but $x^4e^{x^2}$ is an elementary solution: $y = x^4e^{x^2}$ is the case $c_1 = 0$.)

Answer. $\mu = x$; $y = x^4 e^{x^2}\left[c_1 \int \dfrac{e^{-x^2}}{x^5}\,\mathrm{d}x + c_2\right]$.

Problem 43-5 (Simmons)

Solution. The adjoint of $Py'' + Qy' + Ry = 0$ is $P_1\mu'' + Q_1\mu' + R_1\mu = 0$ with

$$P_1 = P, \qquad Q_1 = 2P' - Q, \qquad R_1 = P'' - Q' + R.$$

Applying the same rule to this equation, the adjoint of the adjoint has coefficients

$$\begin{aligned} P_2 &= P_1 = P, \\ Q_2 &= 2P_1' - Q_1 = 2P' - (2P' - Q) = Q, \\ R_2 &= P_1'' - Q_1' + R_1 = P'' - (2P'' - Q') + (P'' - Q' + R) = P'' - 2P'' + Q' + P'' - Q' + R = R. \end{aligned}$$

Thus the adjoint of the adjoint is $Py'' + Qy' + Ry = 0$, the original equation.

Answer. Shown: $(P_2, Q_2, R_2) = (P, Q, R)$.

Problem 43-6 (Simmons)

Solution.

(a) The adjoint is $P\mu'' + (2P' - Q)\mu' + (P'' - Q' + R)\mu = 0$. It is the same equation as $Py'' + Qy' + Ry = 0$ iff the corresponding coefficients are equal:

$$P = P, \qquad 2P' - Q = Q, \qquad P'' - Q' + R = R.$$

The first is automatic. The second says $2P' = 2Q$, i.e. $Q = P'$. If $Q = P'$, then $Q' = P''$ and the third condition $P'' - Q' = 0$ holds as well. Hence the equation is self-adjoint iff $P' = Q$. (The "if" direction: $Q = P'$ implies both nontrivial coefficient conditions; the "only if" direction: the second coefficient condition alone forces $Q = P'$.) In that case the equation reads $Py'' + P'y' + Ry = 0$, and since $[Py']' = Py'' + P'y'$ this is

$$[Py']' + Ry = 0.$$

(The book's displayed form omits the factor y in the last term, which is a misprint.)

(b) We test $P' = Q$ for each equation of Problem 3:

	P	P'	Q	self-adjoint?
(a) Legendre	$1 - x^2$	$-2x$	$-2x$	yes
(b) Bessel	x^2	$2x$	x	no
(c) Chebyshev	$1 - x^2$	$-2x$	$-x$	no
(d) Hermite	1	0	$-2x$	no
(e) Airy	1	0	0	yes
(f) Laguerre	x	1	$1 - x$	no

This agrees with Problem 3: the adjoints of (a) and (e) are the equations themselves, while those of (b), (c), (d), (f) are different equations.

Answer. (a) Self-adjoint $\iff$ $P' = Q$, and then the equation is $[Py']' + Ry = 0$. (b) Legendre's and Airy's equations.

Problem 43-7 (Simmons)

Solution. Let $E(x) = e^{\int (Q/P)\,dx}$ (on an interval where $P \neq 0$), so that $E' = (Q/P)E$, and $h = E/P$, which is nowhere zero. Multiplying the equation by h gives an equation with coefficients

$$\tilde{P} = hP = E, \qquad \tilde{Q} = hQ = \frac{Q}{P}E = E', \qquad \tilde{R} = hR.$$

Then $\tilde{Q} = E' = \tilde{P}'$, so by Problem 6(a) the new equation is self-adjoint; explicitly it is $[Ey']' + \dfrac{ER}{P}y = 0$. Multiplication by the nowhere-vanishing function h does not change the solutions.

Answer. Shown: $\tilde{P} = e^{\int (Q/P)dx}$, $\tilde{Q} = \tilde{P}'$, so the equation becomes $[\tilde{P}y']' + \tilde{R}y = 0$.

Problem 43-8 (Simmons)

Solution. Standard form means $[Py']' + Ry = 0$. For an equation already satisfying $Q = P'$ nothing is needed; otherwise multiply by $h = \frac{1}{P}e^{\int (Q/P)dx}$.

(a) Legendre: $Q = P'$ (Problem 6(b)), so directly

$$\left[(1-x^2)y'\right]' + p(p+1)y = 0.$$

(b) Bessel (take $x > 0$): $\dfrac{Q}{P} = \dfrac{1}{x}$, $e^{\int Q/P} = e^{\log x} = x$, so $h = \dfrac{x}{x^2} = \dfrac{1}{x}$. Multiplying,

$$xy'' + y' + \left(x - \frac{p^2}{x}\right)y = 0 \quad \Longleftrightarrow \quad \left[xy'\right]' + \left(x - \frac{p^2}{x}\right)y = 0.$$

(c) Chebyshev (take $|x| < 1$): $\dfrac{Q}{P} = \dfrac{-x}{1-x^2}$, $\displaystyle\int \frac{-x\,dx}{1-x^2} = \tfrac{1}{2}\log(1-x^2)$, so $e^{\int Q/P} = \sqrt{1-x^2}$ and $h = \dfrac{\sqrt{1-x^2}}{1-x^2} = \dfrac{1}{\sqrt{1-x^2}}$. Multiplying,

$$\sqrt{1-x^2}\,y'' - \frac{x}{\sqrt{1-x^2}}\,y' + \frac{p^2}{\sqrt{1-x^2}}\,y = 0.$$

Since $\dfrac{d}{dx}\sqrt{1-x^2} = -\dfrac{x}{\sqrt{1-x^2}}$, this is

$$\left[\sqrt{1-x^2}\,y'\right]' + \frac{p^2}{\sqrt{1-x^2}}\,y = 0.$$

(d) Hermite: $\dfrac{Q}{P} = -2x$, $e^{\int Q/P} = e^{-x^2}$, $h = e^{-x^2}$. Multiplying: $e^{-x^2}y'' - 2xe^{-x^2}y' + 2pe^{-x^2}y = 0$, and $\dfrac{d}{dx}e^{-x^2} = -2xe^{-x^2}$, so

$$\left[e^{-x^2}y'\right]' + 2pe^{-x^2}y = 0.$$

(e) Airy: $Q = 0 = P'$, so already

$$[y']' + xy = 0.$$

(f) Laguerre (take $x > 0$): $\dfrac{Q}{P} = \dfrac{1-x}{x} = \dfrac{1}{x} - 1$, $\displaystyle\int \frac{Q}{P}\,dx = \log x - x$, $e^{\int Q/P} = xe^{-x}$, $h = \dfrac{xe^{-x}}{x} = e^{-x}$. Multiplying: $xe^{-x}y'' + (1-x)e^{-x}y' + pe^{-x}y = 0$, and $\dfrac{d}{dx}(xe^{-x}) = (1-x)e^{-x}$, so

$$\left[xe^{-x}y'\right]' + pe^{-x}y = 0.$$

Answer. (a) $[(1-x^2)y']' + p(p+1)y = 0$; (b) $[xy']' + \left(x - \frac{p^2}{x}\right)y = 0$; (c) $\left[\sqrt{1-x^2}\,y'\right]' + \dfrac{p^2}{\sqrt{1-x^2}}y = 0$; (d) $\left[e^{-x^2}y'\right]' + 2pe^{-x^2}y = 0$; (e) $[y']' + xy = 0$; (f) $\left[xe^{-x}y'\right]' + pe^{-x}y = 0$.

Problem 43-9 (Simmons)

Solution. Recall (3): $[p(x)y']' + [\lambda q(x) + r(x)]y = 0$, with p, p', q, r continuous and $p > 0$ on $[a,b]$; equivalently $y'' + \dfrac{p'}{p}y' + \dfrac{\lambda q + r}{p}y = 0$, a linear equation with continuous coefficients. Hence (Theorem A of Section 14) a solution is determined uniquely by $y(a)$ and $y'(a)$; in particular a solution with $y(a) = y'(a) = 0$ is identically zero. The boundary condition at a is

$$c_1 y(a) + c_2 y'(a) = 0, \qquad (c_1, c_2) \neq (0,0).$$

Let u and v be eigenfunctions for the same λ (so both are nontrivial solutions of (3) satisfying (9)). Their Wronskian is $W(x) = u(x)v'(x) - v(x)u'(x)$.

Step 1: $W(a) = 0$. Both u and v satisfy the condition at a:

$$c_1 u(a) + c_2 u'(a) = 0, \qquad c_1 v(a) + c_2 v'(a) = 0.$$

Regard this as a homogeneous linear system for the unknowns (c_1, c_2), with coefficient matrix $\begin{pmatrix} u(a) & u'(a) \\ v(a) & v'(a) \end{pmatrix}$. It has the nontrivial solution (c_1, c_2), so the determinant vanishes: $u(a)v'(a) - v(a)u'(a) = W(a) = 0$.

Step 2: u, v are linearly dependent. Since $W(a) = 0$, the system $\alpha u(a) + \beta v(a) = 0$, $\alpha u'(a) + \beta v'(a) = 0$ for the unknowns (α, β) has determinant $W(a) = 0$ and hence a nontrivial solution $(\alpha, \beta) \neq (0,0)$. Then $z = \alpha u + \beta v$ is a solution of the linear equation (3) (superposition) with $z(a) = z'(a) = 0$, so $z \equiv 0$ on $[a,b]$ by the uniqueness theorem. Thus $\alpha u + \beta v \equiv 0$ with $(\alpha, \beta) \neq (0,0)$, i.e. u and v are linearly dependent. (Equivalently, by Abel's formula $W(x) = W(a)p(a)/p(x) \equiv 0$.)

Step 3: conclusion. Since $u \not\equiv 0$ is an eigenfunction, in $\alpha u + \beta v = 0$ we cannot have $\beta = 0$ (that would force $\alpha u \equiv 0$ with $\alpha \neq 0$, contradicting $u \not\equiv 0$). So $v = -(\alpha/\beta)u$ is a constant multiple of u. Hence every eigenfunction belonging to a given eigenvalue is determined up to a constant factor. (Only the condition at one endpoint was used.)

Answer. Shown: two eigenfunctions for the same λ have $W(a) = 0$, hence are linearly dependent, so $v = ku$.

Problem 43-10 (Simmons)

Solution.

(a) Let y_m, y_n be eigenfunctions for eigenvalues $\lambda_m \neq \lambda_n$. Calculation (6) of the text (which uses only the differential equation, multiplication by y_n, y_m, subtraction, and integration by parts) gives

$$(\lambda_m - \lambda_n)\int_a^b q\, y_m y_n \, dx = p(b)\big[y_m(b)y_n'(b) - y_n(b)y_m'(b)\big] - p(a)\big[y_m(a)y_n'(a) - y_n(a)y_m'(a)\big].$$

By the periodic conditions $y_m(b) = y_m(a)$, $y_n(b) = y_n(a)$, $y_m'(b) = y_m'(a)$, $y_n'(b) = y_n'(a)$, and $p(b) = p(a)$, so the two brackets are equal and the right side is $\big(p(b) - p(a)\big)\big[y_m(a)y_n'(a) - y_n(a)y_m'(a)\big] = 0$. Thus $(\lambda_m - \lambda_n)\int_a^b q y_m y_n \, dx = 0$, and since $\lambda_m \neq \lambda_n$,

$$\int_a^b q(x) y_m(x) y_n(x)\, dx = 0.$$

(b) Here $p = q = 1$, $r = 0$, $[a,b] = [-\pi, \pi]$. We consider $y'' + \lambda y = 0$ with $y(-\pi) = y(\pi)$, $y'(-\pi) = y'(\pi)$ in the three cases for λ.

$\lambda < 0$: write $\lambda = -k^2$, $k > 0$; $y = Ae^{kx} + Be^{-kx}$. Then

$$y(\pi) - y(-\pi) = A(e^{k\pi} - e^{-k\pi}) + B(e^{-k\pi} - e^{k\pi}) = 2\sinh(k\pi)\,(A - B),$$
$$y'(\pi) - y'(-\pi) = kA(e^{k\pi} - e^{-k\pi}) - kB(e^{-k\pi} - e^{k\pi}) = 2k\sinh(k\pi)\,(A + B).$$

Since $\sinh k\pi \neq 0$, both conditions give $A = B$ and $A = -B$, so $A = B = 0$: no eigenvalues.

$\lambda = 0$: $y = A + Bx$; $y(\pi) - y(-\pi) = 2\pi B = 0$ gives $B = 0$, and then $y' = 0$ satisfies the second condition. So $\lambda_0 = 0$ is an eigenvalue with the single (up to a constant factor) eigenfunction $y_0 = 1$.

$\lambda > 0$: write $\lambda = k^2$, $k > 0$; $y = A\cos kx + B\sin kx$, $y' = -Ak\sin kx + Bk\cos kx$. Using that cos is even and sin is odd,

$$y(\pi) - y(-\pi) = 2B\sin k\pi,$$
$$y'(\pi) - y'(-\pi) = -2Ak\sin k\pi.$$

Both vanish iff $A\sin k\pi = 0$ and $B\sin k\pi = 0$. A nontrivial solution therefore requires $\sin k\pi = 0$, i.e. $k = n$, a positive integer; and then *both* A and B are arbitrary. So

$$\lambda_n = n^2 \ (n = 1, 2, 3, \dots), \qquad y = A\cos nx + B\sin nx,$$

with two linearly independent eigenfunctions $\cos nx$ and $\sin nx$ for each λ_n. For $\lambda_0 = 0$ there is only one. This verifies that each eigenvalue has one or two independent eigenfunctions (consistent with the orthogonality in part (a): $\cos nx, \sin nx$ for different n are orthogonal on $[-\pi, \pi]$, while $\cos nx$ and $\sin nx$ for the same n are orthogonal as well).

(c) Equation (3) is a linear homogeneous second order differential equation with continuous coefficients and $p > 0$; by the existence and uniqueness theorem (Theorem A of Section 14) its solutions are determined by the two numbers $y(a), y'(a)$, so the solution space is two-dimensional: every solution is $c_1y_1 + c_2y_2$ for any fixed pair of independent solutions (e.g. those with $(y, y')(a) = (1, 0)$ and $(0, 1)$). The eigenfunctions for a given λ are solutions of (3), so they all lie in this two-dimensional space, and no three of them can be linearly independent. (In contrast, for the separated conditions (9) the space is at most one-dimensional, by Problem 9.)

Answer. (a) Right side of (6) is $(p(b) - p(a))[\cdots] = 0$, so $\int_a^b qy_my_n\,dx = 0$ for $\lambda_m \neq \lambda_n$. (b) $\lambda_0 = 0$, $y_0 = 1$; $\lambda_n = n^2$ $(n \geq 1)$ with eigenfunctions $\cos nx$ and $\sin nx$. (c) The solution space of a second order linear equation is two-dimensional.

Chapter 8

Some Special Functions of Mathematical Physics

Section 44 Legendre Polynomials

Throughout, $P_n(x)$ is the Legendre polynomial of the text, defined by formula (8),

$$P_n(x) = \sum_{k=0}^{[n/2]} (-1)^k \frac{(2n-2k)!}{2^n\, k!\,(n-k)!\,(n-2k)!}\, x^{n-2k},$$

which is the same polynomial as the hypergeometric form (5) and as Rodrigues' formula (9), $P_n(x) = \dfrac{1}{2^n n!}\dfrac{d^n}{dx^n}(x^2-1)^n$. We shall also use the identity

$$1\cdot 3\cdots(2m-1) = \frac{(2m)!}{2^m\, m!}, \tag{$**$}$$

which follows from $(2m)! = [1\cdot 3\cdots(2m-1)]\,[2\cdot 4\cdots 2m] = [1\cdot 3\cdots(2m-1)]\,2^m m!$.

Problem 44-1 (Simmons)

Solution. The relation holds for $|t|$ small, and the coefficients of a convergent power series in t are unique; so we may compare coefficients of t^n after expanding the left side in powers of t for a fixed x.

(a) Put $x = 1$. Then $1 - 2t + t^2 = (1-t)^2$ and, for $|t| < 1$,

$$\frac{1}{\sqrt{(1-t)^2}} = \frac{1}{1-t} = 1 + t + t^2 + \cdots + t^n + \cdots \qquad \text{(geometric series)},$$

(note $\sqrt{(1-t)^2} = |1-t| = 1-t$ for $|t|<1$). Comparing with $\sum P_n(1)t^n$ gives $P_n(1) = 1$ for all n. Put $x = -1$. Then $1 + 2t + t^2 = (1+t)^2$ and, for $|t| < 1$,

$$\frac{1}{1+t} = \sum_{n=0}^{\infty} (-1)^n t^n,$$

so $P_n(-1) = (-1)^n$.

(b) Put $x = 0$: the generating function becomes $(1+t^2)^{-1/2}$. By the binomial series $(1+u)^{-1/2} = \sum_{n\ge 0}\binom{-1/2}{n}u^n$ ($|u|<1$) with $u = t^2$,

$$(1+t^2)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} t^{2n}.$$

Only even powers of t occur, so the coefficient of t^{2n+1} is 0:

$$P_{2n+1}(0) = 0.$$

The coefficient of t^{2n} is

$$P_{2n}(0) = \binom{-1/2}{n} = \frac{\left(-\frac12\right)\left(-\frac32\right)\left(-\frac52\right)\cdots\left(-\frac{2n-1}{2}\right)}{n!} \qquad (n \text{ factors})$$
$$= \frac{(-1)^n\, 1\cdot 3\cdot 5\cdots(2n-1)}{2^n\, n!}.$$

(For $n = 0$ this is $P_0(0) = 1$.)

Answer. Shown: $P_n(1) = 1$, $P_n(-1) = (-1)^n$; $P_{2n+1}(0) = 0$ and $P_{2n}(0) = (-1)^n \dfrac{1\cdot 3\cdots(2n-1)}{2^n n!}$.

Problem 44-2 (Simmons)

Solution.

(a) Differentiate the left side with respect to t (chain rule):

$$\frac{d}{dt}(1-2xt+t^2)^{-1/2} = -\tfrac{1}{2}(1-2xt+t^2)^{-3/2}(-2x+2t) = \frac{x-t}{(1-2xt+t^2)^{3/2}}.$$

A power series may be differentiated term by term inside its interval of convergence, so the derivative of the right side is $\sum_{n=1}^{\infty} nP_n(x)t^{n-1}$ (the $n=0$ term is constant). Hence

$$\frac{x-t}{(1-2xt+t^2)^{3/2}} = \sum_{n=1}^{\infty} nP_n(x)t^{n-1}.$$

Multiply both sides by $1-2xt+t^2$. Since $\dfrac{1-2xt+t^2}{(1-2xt+t^2)^{3/2}} = (1-2xt+t^2)^{-1/2}$ is the generating function,

$$(x-t)\,(1-2xt+t^2)^{-1/2} = (1-2xt+t^2)\sum_{n=1}^{\infty} nP_n(x)t^{n-1},$$

and replacing $(1-2xt+t^2)^{-1/2}$ by $\sum_{n=0}^{\infty} P_n(x)t^n$ gives the required identity

$$(x-t)\sum_{n=0}^{\infty} P_n(x)t^n = (1-2xt+t^2)\sum_{n=1}^{\infty} nP_n(x)t^{n-1}.$$

(b) Fix $n \geq 1$ and find the coefficient of t^n on each side.

Left side. $x\sum P_k t^k$ contributes xP_n; $-t\sum P_k t^k = -\sum P_k t^{k+1}$ contributes $-P_{n-1}$ (take $k = n-1$). Total: $xP_n - P_{n-1}$.

Right side. Write $1-2xt+t^2$ times $\sum_{k\geq 1} kP_k t^{k-1}$:

- the term $1\cdot\sum kP_k t^{k-1}$ gives t^n when $k-1=n$: coefficient $(n+1)P_{n+1}$;
- the term $-2xt\sum kP_k t^{k-1} = -2x\sum kP_k t^k$ gives, for $k=n$: $-2xnP_n$;
- the term $t^2\sum kP_k t^{k-1} = \sum kP_k t^{k+1}$ gives, for $k+1=n$: $(n-1)P_{n-1}$.

Total: $(n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1}$.

Equating,

$$\begin{aligned} xP_n - P_{n-1} &= (n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1} \\ \Longrightarrow\quad (n+1)P_{n+1} &= xP_n + 2nxP_n - P_{n-1} - (n-1)P_{n-1} \\ &= (2n+1)xP_n - nP_{n-1}. \end{aligned}$$

(For $n=0$ the coefficient of t^0 gives $xP_0 = P_1$, which is the recursion with the term $0\cdot P_{-1}$ dropped.)

(c) With $P_0 = 1$, $P_1 = x$:

$n=1$: $2P_2 = 3xP_1 - P_0 = 3x^2-1$, so

$$P_2(x) = \tfrac{1}{2}(3x^2-1).$$

$n=2$: $3P_3 = 5xP_2 - 2P_1 = \tfrac{5}{2}x(3x^2-1) - 2x = \tfrac{1}{2}(15x^3-5x-4x) = \tfrac{1}{2}(15x^3-9x)$, so

$$P_3(x) = \tfrac{1}{2}(5x^3-3x).$$

$n=3$: $4P_4 = 7xP_3 - 3P_2 = \tfrac{7}{2}x(5x^3-3x) - \tfrac{3}{2}(3x^2-1) = \tfrac{1}{2}(35x^4-21x^2-9x^2+3) = \tfrac{1}{2}(35x^4-30x^2+3)$, so

$$P_4(x) = \tfrac{1}{8}(35x^4-30x^2+3).$$

$n=4$: $5P_5 = 9xP_4 - 4P_3 = \tfrac{9}{8}x(35x^4-30x^2+3) - 2(5x^3-3x) = \tfrac{1}{8}(315x^5-270x^3+27x-80x^3+48x) = \tfrac{1}{8}(315x^5-350x^3+75x)$, so

$$P_5(x) = \tfrac{1}{8}(63x^5-70x^3+15x).$$

Answer. (c) $P_2 = \tfrac{1}{2}(3x^2-1)$, $P_3 = \tfrac{1}{2}(5x^3-3x)$, $P_4 = \tfrac{1}{8}(35x^4-30x^2+3)$, $P_5 = \tfrac{1}{8}(63x^5-70x^3+15x)$; this agrees with the book's key.

Problem 44-3 (Simmons)

Solution.

(a) Write $c_0 = 1$ and $c_j = \dfrac{1\cdot 3\cdots(2j-1)}{2^j\, j!}$ for $j \ge 1$. The binomial series is

$$(1-u)^{-1/2} = \sum_{j=0}^{\infty} c_j u^j \qquad (|u|<1),$$

because $\binom{-1/2}{j}(-1)^j = \dfrac{\frac{1}{2}\cdot\frac{3}{2}\cdots\frac{2j-1}{2}}{j!} = c_j$. Since $1-2xt+t^2 = 1-t(2x-t)$, we take $u = t(2x-t)$ and get

$$(1-2xt+t^2)^{-1/2} = \sum_{j=0}^{\infty} c_j\, t^j (2x-t)^j,$$

which is exactly the displayed series (the terms $j = 1, 2, \dots, n-1, n$ have coefficients $\frac{1}{2}, \frac{1\cdot 3}{2^2 2!}, \dots, c_{n-1}, c_n$). It converges when $|u| = |t||2x-t| < 1$; for $|x| \le 1$ this holds if $|t|(2+|t|) < 1$, i.e. $|t| < \sqrt{2}-1$. Moreover, for such t and $|x| \le 1$ the series obtained by expanding every $(2x-t)^j$ by the binomial theorem converges absolutely, since $\sum_j c_j |t|^j (2|x|+|t|)^j$ converges (it is dominated by the binomial series at $u = |t|(2+|t|) < 1$). Hence the terms may be rearranged by powers of t.

(b) The term $c_j t^j (2x-t)^j$ is divisible by t^j and so contains no power of t below t^j; therefore a term with $j > n$ cannot contribute to t^n, and only $j = 0, 1, \dots, n$ matter. (Also, j must satisfy $j \ge n/2$, as we now see.) By the binomial theorem,

$$(2x-t)^j = \sum_{i=0}^{j} \binom{j}{i} (2x)^{j-i}(-t)^i,$$

so $t^j(2x-t)^j$ contains t^n exactly when $j+i = n$, i.e. $i = n-j$, with $0 \le i \le j$. Put $k = n-j$ (so $j = n-k$, $0 \le k \le n-k$, i.e. $0 \le k \le [n/2]$). The contribution is

$$c_{n-k}\binom{n-k}{k}(2x)^{(n-k)-k}(-1)^k = (-1)^k\, c_{n-k}\binom{n-k}{k}(2x)^{n-2k}.$$

Thus the coefficient of t^n is

$$\sum_{k=0}^{[n/2]} (-1)^k c_{n-k} \binom{n-k}{k} (2x)^{n-2k}. \tag{$*$}$$

Now list the first terms:

- $k=0$: $c_n(2x)^n = \dfrac{1\cdot 3\cdots(2n-1)}{2^n n!}(2x)^n$;
- $k=1$: $-c_{n-1}\binom{n-1}{1}(2x)^{n-2} = -\dfrac{1\cdot 3\cdots(2n-3)}{2^{n-1}(n-1)!}\,\dfrac{n-1}{1!}(2x)^{n-2}$;
- $k=2$: $+c_{n-2}\binom{n-2}{2}(2x)^{n-4} = \dfrac{1\cdot 3\cdots(2n-5)}{2^{n-2}(n-2)!}\,\dfrac{(n-2)(n-3)}{2!}(2x)^{n-4}$;

and so on, which is the expression in (b).

(c) By $(**)$, $c_m = \dfrac{(2m)!}{2^m m!\cdot 2^m m!} = \dfrac{(2m)!}{2^{2m}(m!)^2}$, so $c_{n-k} = \dfrac{(2n-2k)!}{2^{2n-2k}\,[(n-k)!]^2}$. Also $\binom{n-k}{k} = \dfrac{(n-k)!}{k!\,(n-2k)!}$. Hence the kth term of $(*)$ is

$$(-1)^k \frac{(2n-2k)!}{2^{2n-2k}\,[(n-k)!]^2}\cdot\frac{(n-k)!}{k!\,(n-2k)!}\cdot 2^{n-2k}x^{n-2k}$$
$$= (-1)^k \frac{(2n-2k)!}{2^{2n-2k-(n-2k)}\,k!\,(n-k)!\,(n-2k)!}\,x^{n-2k} = (-1)^k\frac{(2n-2k)!}{2^n\,k!\,(n-k)!\,(n-2k)!}\,x^{n-2k},$$

because $(2n-2k)-(n-2k)=n$. Summing over $k=0,\dots,[n/2]$,

$$\text{coefficient of } t^n = \sum_{k=0}^{[n/2]}(-1)^k\frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!}x^{n-2k} = P_n(x)$$

by formula (8). Therefore $(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x)t^n$. (For $n=0$ the coefficient is $c_0 = 1 = P_0$.)

Answer. Shown: the coefficient of t^n in $(1-2xt+t^2)^{-1/2}$ is $\sum_{k=0}^{[n/2]}(-1)^k\frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!}x^{n-2k} = P_n(x)$.

Problem 44-4 (Simmons)

Solution. Equation (1) is $(1-x^2)y'' - 2xy' + n(n+1)y = 0$.

(a) Since $w = (x^2-1)^n$, $w' = n(x^2-1)^{n-1}\cdot 2x$, so $(x^2-1)w' = 2nx(x^2-1)^n = 2nxw$, i.e. $(x^2-1)w' - 2nxw = 0$.

Differentiate this $k+1$ times, using Leibniz' rule $(fg)^{(m)} = \sum_i \binom{m}{i} f^{(i)} g^{(m-i)}$.

For $(x^2-1)w'$: with $f = x^2-1$, $f' = 2x$, $f'' = 2$, $f''' = 0$, and $g = w'$ ($g^{(m)} = w^{(m+1)}$),

$$\left[(x^2-1)w'\right]^{(k+1)} = (x^2-1)w^{(k+2)} + (k+1)(2x)w^{(k+1)} + \binom{k+1}{2}2\,w^{(k)}$$
$$= (x^2-1)w^{(k+2)} + 2(k+1)xw^{(k+1)} + (k+1)k\,w^{(k)}.$$

For $2nxw$: with $f = x$, $f' = 1$, $f'' = 0$,

$$\big[2nxw\big]^{(k+1)} = 2n\left[xw^{(k+1)} + (k+1)w^{(k)}\right] = 2nxw^{(k+1)} + 2(k+1)nw^{(k)}.$$

Subtracting,

$$(x^2-1)w^{(k+2)} + 2(k+1)xw^{(k+1)} + (k+1)kw^{(k)} - 2nxw^{(k+1)} - 2(k+1)nw^{(k)} = 0,$$

which is the stated identity. Now choose $k = n$, so that $w^{(k)} = w^{(n)} = y$, $w^{(k+1)} = y'$, $w^{(k+2)} = y''$:

$$\begin{aligned} 0 &= (x^2-1)y'' + 2(n+1)xy' + (n+1)n\,y - 2nxy' - 2(n+1)n\,y \\ &= (x^2-1)y'' + 2xy' + \big[n(n+1) - 2n(n+1)\big]y \\ &= (x^2-1)y'' + 2xy' - n(n+1)y. \end{aligned}$$

Multiplying by -1,

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0,$$

which is Legendre's equation (1). So $y = \dfrac{d^n}{dx^n}(x^2-1)^n$ is a solution.

(b) Since $x^2 - 1 = (x-1)(x+1)$, we have $w = (x-1)^n(x+1)^n = uv$, and

$$y = (uv)^{(n)} = \sum_{j=0}^{n} \binom{n}{j} u^{(n-j)} v^{(j)}.$$

For $0 \le m \le n$,

$$u^{(m)}(x) = n(n-1)\cdots(n-m+1)(x-1)^{n-m},$$

which vanishes at $x = 1$ unless $m = n$, in which case $u^{(n)}(x) = n!$. Therefore at $x = 1$ every term in the sum with $n - j < n$, i.e. $j \ge 1$, is zero, and only the term $j = 0$ remains:

$$y(1) = u^{(n)}(1)\,v(1) = n! \cdot (1+1)^n = n!\,2^n.$$

Remark. Hence $y/(2^n n!)$ is a solution of Legendre's equation with value 1 at $x = 1$; this is $P_n(x)$ by Rodrigues' formula (9), consistent with $P_n(1) = 1$.

Answer. (a) $y = \dfrac{d^n}{dx^n}(x^2-1)^n$ satisfies $(1-x^2)y'' - 2xy' + n(n+1)y = 0$; (b) $y(1) = n!\,2^n$.

Section 45 Properties of Legendre Polynomials

Throughout, $P_n(x)$ is the Legendre polynomial of Section 44, a polynomial of degree n (with leading coefficient $\dfrac{(2n)!}{2^n(n!)^2}$) satisfying Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ and $P_n(\pm 1) = (\pm 1)^n$.

Problem 45-1 (Simmons)

Solution. Here $n \geq 1$ (for $n = 0$ the formula reads $\int \cos\theta\, d\theta = \sin\theta$, which is true). We verify the formula by showing that the derivative of the right side is $\cos^{2n+1}\theta$. By the product and chain rule, and $\sin^2\theta = 1 - \cos^2\theta$ in the second step,

$$\begin{aligned}
\frac{d}{d\theta}\left[\frac{1}{2n+1}\cos^{2n}\theta\sin\theta\right] &= \frac{1}{2n+1}\left[-2n\cos^{2n-1}\theta\sin^2\theta + \cos^{2n+1}\theta\right] \\
&= \frac{1}{2n+1}\left[-2n\cos^{2n-1}\theta(1-\cos^2\theta) + \cos^{2n+1}\theta\right] \\
&= \frac{1}{2n+1}\left[(2n+1)\cos^{2n+1}\theta - 2n\cos^{2n-1}\theta\right] \\
&= \cos^{2n+1}\theta - \frac{2n}{2n+1}\cos^{2n-1}\theta.
\end{aligned}$$

The derivative of the second term on the right of (4) is $\dfrac{2n}{2n+1}\cos^{2n-1}\theta$. Adding,

$$\frac{d}{d\theta}\,[\text{right side}] = \cos^{2n+1}\theta - \frac{2n}{2n+1}\cos^{2n-1}\theta + \frac{2n}{2n+1}\cos^{2n-1}\theta = \cos^{2n+1}\theta,$$

the integrand on the left. Hence (4) holds (up to the constant of integration, which is absorbed in the indefinite integrals).

Remark. The formula arises from integration by parts: $\int \cos^{2n+1}\theta\, d\theta = \int \cos^{2n}\theta\, d(\sin\theta) = \cos^{2n}\theta\sin\theta + 2n\int \cos^{2n-1}\theta\sin^2\theta\, d\theta = \cos^{2n}\theta\sin\theta + 2n\int \cos^{2n-1}\theta\, d\theta - 2n\int \cos^{2n+1}\theta\, d\theta$; solving for $\int \cos^{2n+1}\theta\, d\theta$ gives (4).

Answer. Shown: the derivative of the right side of (4) equals $\cos^{2n+1}\theta$.

Problem 45-2 (Simmons)

Solution. (The two forms agree because $\frac{d}{dx}[(1-x^2)y'] = (1-x^2)y'' - 2xy'$.) Let $m \neq n$ be non-negative integers. Multiply the equation for P_m by P_n and the equation for P_n by P_m:

$$\begin{aligned}
P_n\frac{d}{dx}\left[(1-x^2)P_m'\right] + m(m+1)P_mP_n &= 0, \\
P_m\frac{d}{dx}\left[(1-x^2)P_n'\right] + n(n+1)P_nP_m &= 0.
\end{aligned}$$

Subtract the second from the first:

$$P_n\frac{d}{dx}\left[(1-x^2)P_m'\right] - P_m\frac{d}{dx}\left[(1-x^2)P_n'\right] + \left[m(m+1) - n(n+1)\right]P_mP_n = 0.$$

The first two terms form an exact derivative:

$$\frac{d}{dx}\Big\{(1-x^2)\left[P_nP_m' - P_mP_n'\right]\Big\} = P_n\frac{d}{dx}\left[(1-x^2)P_m'\right] + P_n'(1-x^2)P_m'$$

$$- P_m \frac{d}{dx}\left[(1-x^2)P_n'\right] - P_m'(1-x^2)P_n',$$

and the two middle terms cancel. Also $m(m+1) - n(n+1) = m^2 - n^2 + m - n = (m-n)(m+n+1)$. Hence

$$\frac{d}{dx}\Big\{(1-x^2)\left[P_nP_m' - P_mP_n'\right]\Big\} + (m-n)(m+n+1)P_mP_n = 0.$$

Integrate from -1 to 1:

$$\Big[(1-x^2)\left(P_nP_m' - P_mP_n'\right)\Big]_{-1}^{1} + (m-n)(m+n+1)\int_{-1}^{1} P_mP_n\,dx = 0.$$

The bracketed term vanishes at both limits because the polynomial factor $P_nP_m' - P_mP_n'$ is finite there while $1-x^2 = 0$ at $x = \pm 1$. Thus

$$(m-n)(m+n+1)\int_{-1}^{1} P_m(x)P_n(x)\,dx = 0.$$

Since $m \neq n$ and $m+n+1 \geq 1 > 0$, the product $(m-n)(m+n+1) \neq 0$, so

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = 0 \qquad (m \neq n).$$

Answer. Shown: $\displaystyle\int_{-1}^{1} P_mP_n\,dx = 0$ for $m \neq n$, using only Legendre's equation.

Problem 45-3 (Simmons)

Solution. *Evaluation of the integral.* Let $0 < |t| < 1$. Then $1 - 2xt + t^2 \geq (1-|t|)^2 > 0$ for $|x| \leq 1$, so the integrand is continuous, and treating t as a constant,

$$\int \frac{dx}{1-2xt+t^2} = -\frac{1}{2t}\log(1-2xt+t^2)$$

(since $\frac{d}{dx}(1-2xt+t^2) = -2t$). Evaluating between -1 and 1, where $1-2t+t^2 = (1-t)^2$ and $1+2t+t^2 = (1+t)^2$:

$$\int_{-1}^{1} \frac{dx}{1-2xt+t^2} = -\frac{1}{2t}\left[\log(1-t)^2 - \log(1+t)^2\right] = \frac{1}{t}\left[\log(1+t) - \log(1-t)\right]$$

(here $1 \pm t > 0$, so $\log(1\pm t)^2 = 2\log(1\pm t)$). With the Maclaurin series, valid for $|t| < 1$,

$$\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \cdots, \qquad \log(1-t) = -t - \frac{t^2}{2} - \frac{t^3}{3} - \cdots,$$

we get $\log(1+t) - \log(1-t) = 2\left(t + \dfrac{t^3}{3} + \dfrac{t^5}{5} + \cdots\right) = 2\sum_{n=0}^{\infty} \dfrac{t^{2n+1}}{2n+1}$, and therefore

$$\int_{-1}^{1} \frac{dx}{1-2xt+t^2} = \sum_{n=0}^{\infty} \frac{2}{2n+1}\,t^{2n}.$$

(At $t = 0$ both sides equal 2.)

Conclusion. The two power series in t,

$$\sum_{n=0}^{\infty} \left(\int_{-1}^{1} P_n^2 \, dx \right) t^{2n} \qquad \text{and} \qquad \sum_{n=0}^{\infty} \frac{2}{2n+1} t^{2n},$$

have the same sum for $|t|$ small, so their coefficients agree (uniqueness of power series). Comparing the coefficients of t^{2n},

$$\int_{-1}^{1} P_n(x)^2 \, dx = \frac{2}{2n+1}.$$

Justification of the squaring and term-by-term integration. For $|x| \le 1$ and $|t| < \sqrt{2} - 1$, the series $\sum P_n(x) t^n$ converges absolutely and uniformly in x (see the estimate in Problem 3(a) of Section 44), so its square is the Cauchy product, which may be integrated term by term; the cross terms $2 \int P_m P_n \, dx$, $m \neq n$, vanish by the first part of (2), leaving $\sum_n \left(\int P_n^2 \, dx \right) t^{2n}$ (only powers t^{m+n}, $m \neq n$, come from the cross terms, and they all have zero coefficient).

Answer. Shown: $\displaystyle\int_{-1}^{1} \frac{dx}{1 - 2xt + t^2} = \sum_{n \ge 0} \frac{2}{2n+1} t^{2n}$, hence $\displaystyle\int_{-1}^{1} P_n^2 \, dx = \frac{2}{2n+1}$.

Problem 45-4 (Simmons)

Solution. By formula (6), $f(x) = \sum a_n P_n(x)$ with $a_n = \left(n + \frac{1}{2}\right) \displaystyle\int_{-1}^{1} f(x) P_n(x) \, dx$. We need $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$.

(a) Since $f = 0$ on $[-1, 0)$, each integral runs over $[0, 1]$ with $f(x) = x$.

$$a_0 = \tfrac{1}{2} \int_0^1 x \, dx = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4},$$

$$a_1 = \tfrac{3}{2} \int_0^1 x \cdot x \, dx = \tfrac{3}{2} \cdot \tfrac{1}{3} = \tfrac{1}{2},$$

$$a_2 = \tfrac{5}{2} \int_0^1 x \cdot \tfrac{1}{2}(3x^2 - 1) \, dx = \tfrac{5}{4} \int_0^1 (3x^3 - x) \, dx = \tfrac{5}{4} \left[\tfrac{3}{4} - \tfrac{1}{2} \right] = \tfrac{5}{4} \cdot \tfrac{1}{4} = \tfrac{5}{16}.$$

Hence

$$f(x) = \tfrac{1}{4} P_0(x) + \tfrac{1}{2} P_1(x) + \tfrac{5}{16} P_2(x) + \cdots.$$

(b) Use $\int e^x \, dx = e^x$, $\int x e^x \, dx = (x - 1)e^x$, $\int x^2 e^x \, dx = (x^2 - 2x + 2)e^x$ (each checked by differentiation).

$$a_0 = \tfrac{1}{2} \int_{-1}^{1} e^x \, dx = \tfrac{1}{2}(e - e^{-1}),$$

$$a_1 = \tfrac{3}{2} \int_{-1}^{1} x e^x \, dx = \tfrac{3}{2} \Big[(x - 1)e^x \Big]_{-1}^{1} = \tfrac{3}{2} \left[0 - (-2)e^{-1} \right] = 3e^{-1},$$

$$a_2 = \tfrac{5}{2} \int_{-1}^{1} \tfrac{1}{2}(3x^2 - 1) e^x \, dx = \tfrac{5}{4} \left[3 \int_{-1}^{1} x^2 e^x \, dx - \int_{-1}^{1} e^x \, dx \right].$$

Here $\int_{-1}^{1} x^2 e^x \, dx = (1 - 2 + 2)e - (1 + 2 + 2)e^{-1} = e - 5e^{-1}$, and $\int_{-1}^{1} e^x \, dx = e - e^{-1}$, so

$$a_2 = \tfrac{5}{4} \left[3e - 15e^{-1} - e + e^{-1} \right] = \tfrac{5}{4} \left(2e - 14e^{-1} \right) = \tfrac{1}{2} \left(5e - 35e^{-1} \right).$$

Hence

$$e^x = \tfrac{1}{2}(e - e^{-1})P_0(x) + 3e^{-1}P_1(x) + \tfrac{1}{2}(5e - 35e^{-1})P_2(x) + \cdots.$$

Answer. (a) $f(x) = \frac{1}{4}P_0 + \frac{1}{2}P_1 + \frac{5}{16}P_2 + \cdots$; (b) $e^x = \frac{1}{2}(e - e^{-1})P_0 + 3e^{-1}P_1 + \frac{1}{2}(5e - 35e^{-1})P_2 + \cdots$ (agrees with the key).

Problem 45-5 (Simmons)

Solution. Since $P_0, \ldots, P_n$ have degrees $0, \ldots, n$, every polynomial of degree $\le n$ is a linear combination of them (the text's argument: x^j is a combination of $P_0, \ldots, P_j$ for each j). So

$$p(x) = \sum_{j=0}^{n} a_j P_j(x)$$

for some constants a_j. Fix m with $0 \le m \le n-1$. The polynomial P_m has degree $m \le n-1$, so $P_m(x) = \sum_{k=0}^{m} \beta_k x^k$ for some constants β_k, and by hypothesis

$$\int_{-1}^{1} P_m(x)p(x)\,dx = \sum_{k=0}^{m} \beta_k \int_{-1}^{1} x^k p(x)\,dx = 0.$$

On the other hand, by the orthogonality relations (2),

$$\int_{-1}^{1} P_m(x)p(x)\,dx = \sum_{j=0}^{n} a_j \int_{-1}^{1} P_m P_j\,dx = a_m \int_{-1}^{1} P_m^2\,dx = \frac{2a_m}{2m+1}.$$

Hence $\dfrac{2a_m}{2m+1} = 0$, i.e. $a_m = 0$ for $m = 0, 1, \ldots, n-1$. Therefore $p(x) = a_n P_n(x)$, and $c = a_n$ (indeed $c \neq 0$, since p has degree n).

Answer. Shown: $p(x) = cP_n(x)$ with $c = a_n \neq 0$.

Problem 45-6 (Simmons)

Solution. "Deviation from zero in the sense of least squares" of a function q means the value of $I(q) = \displaystyle\int_{-1}^{1} [q(x) - 0]^2\,dx = \int_{-1}^{1} q(x)^2\,dx$; we must show $I(p) \ge I(rP_n)$ for every polynomial p of degree n with leading coefficient 1.

The leading coefficient of P_n is $A_n = \dfrac{(2n)!}{2^n (n!)^2}$, so $r = 1/A_n$ and rP_n is monic. Let p be monic of degree n. Write

$$p(x) = \sum_{k=0}^{n} b_k P_k(x).$$

The coefficient of x^n on the right comes only from $b_n P_n$ (since P_k has degree $k < n$ for $k < n$), and equals $b_n A_n$; as p is monic, $b_n A_n = 1$, so

$$b_n = \frac{1}{A_n} = r.$$

Now by orthogonality (2),

$$I(p) = \int_{-1}^{1} \Big(\sum_{k=0}^{n} b_k P_k \Big)^2 \,\mathrm{d}x = \sum_{j,k} b_j b_k \int_{-1}^{1} P_j P_k \,\mathrm{d}x = \sum_{k=0}^{n} \frac{2}{2k+1}\, b_k^2.$$

Therefore

$$I(p) = \sum_{k=0}^{n-1} \frac{2}{2k+1} b_k^2 + \frac{2}{2n+1} r^2 \;\geq\; \frac{2}{2n+1} r^2,$$

with equality if and only if $b_0 = b_1 = \cdots = b_{n-1} = 0$, i.e. $p = rP_n$. For $p = rP_n$ itself, $b_n = r$ and all other $b_k = 0$, so $I(rP_n) = \dfrac{2r^2}{2n+1}$, which is the minimum. Hence among monic polynomials of degree n, rP_n deviates least from zero in the least-squares sense (and it is the only minimizer).

Remark. Explicitly, $r = \dfrac{2^n (n!)^2}{(2n)!}$ and the minimum value is $I(rP_n) = \dfrac{2}{2n+1} \left[\dfrac{2^n (n!)^2}{(2n)!} \right]^2$.

Answer. Shown: for monic p of degree n, $\int_{-1}^{1} p^2 \,\mathrm{d}x \geq \dfrac{2r^2}{2n+1} = \int_{-1}^{1} (rP_n)^2 \,\mathrm{d}x$, with equality only for $p = rP_n$.

Section 46 Bessel Functions. The Gamma Function

Throughout, $J_p(x)$ is given by the series (6) of the text,

$$J_p(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(x/2)^{2n+p}}{n!\,(p+n)!}, \qquad (p+n)! = \Gamma(p+n+1),$$

which is a power series (times x^p) with infinite radius of convergence, so it may be differentiated and rearranged term by term for $x > 0$.

Problem 46-1 (Simmons)

Solution. Recall (7) and (8):

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{x}{2}\right)^{2n}, \qquad J_1(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,(n+1)!}\left(\frac{x}{2}\right)^{2n+1}.$$

(a) Differentiate J_0 term by term. The constant term ($n = 0$) has derivative 0, and for $n \geq 1$

$$\frac{d}{dx}\left(\frac{x}{2}\right)^{2n} = 2n \cdot \frac{1}{2}\left(\frac{x}{2}\right)^{2n-1} = n\left(\frac{x}{2}\right)^{2n-1}.$$

Hence, since $\dfrac{n}{(n!)^2} = \dfrac{1}{(n-1)!\,n!}$,

$$J_0'(x) = \sum_{n=1}^{\infty} \frac{(-1)^n}{(n-1)!\,n!}\left(\frac{x}{2}\right)^{2n-1}.$$

Put $n = k+1$ (so k runs from 0 to ∞):

$$J_0'(x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{k!\,(k+1)!}\left(\frac{x}{2}\right)^{2k+1} = -\sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(k+1)!}\left(\frac{x}{2}\right)^{2k+1} = -J_1(x).$$

(b) Write $xJ_1(x) = 2 \cdot \dfrac{x}{2} J_1(x)$, so

$$xJ_1(x) = 2\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,(n+1)!}\left(\frac{x}{2}\right)^{2n+2}.$$

Differentiating term by term, $\frac{d}{dx}\left(\frac{x}{2}\right)^{2n+2} = (n+1)\left(\frac{x}{2}\right)^{2n+1}$, so

$$\begin{aligned}\frac{d}{dx}\left[xJ_1(x)\right] &= 2\sum_{n=0}^{\infty} \frac{(-1)^n(n+1)}{n!\,(n+1)!}\left(\frac{x}{2}\right)^{2n+1} = 2\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{x}{2}\right)^{2n+1} \qquad \left(\tfrac{n+1}{(n+1)!} = \tfrac{1}{n!}\right)\\ &= x\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{x}{2}\right)^{2n} = xJ_0(x).\end{aligned}$$

Answer. (a) $J_0' = -J_1$; (b) $(xJ_1)' = xJ_0$, both shown by termwise differentiation of the series (7), (8).

Problem 46-2 (Simmons)

Solution. Rolle's theorem: if f is differentiable on $[\alpha,\beta]$ and $f(\alpha)=f(\beta)=0$, then $f'(\xi)=0$ for some ξ with $\alpha<\xi<\beta$. All functions below are power series, hence differentiable everywhere.

(a) Let $0<\alpha<\beta$ with $J_0(\alpha)=J_0(\beta)=0$. Apply Rolle's theorem to $f=J_0$ on $[\alpha,\beta]$: there is $\xi\in(\alpha,\beta)$ with $J_0'(\xi)=0$. By Problem 1(a), $J_0'(\xi)=-J_1(\xi)$, so $J_1(\xi)=0$ with $\alpha<\xi<\beta$.

(b) Let $0<\alpha<\beta$ with $J_1(\alpha)=J_1(\beta)=0$. Then $f(x)=xJ_1(x)$ satisfies $f(\alpha)=\alpha J_1(\alpha)=0$ and $f(\beta)=0$. By Rolle's theorem there is $\xi\in(\alpha,\beta)$ with $f'(\xi)=0$, and by Problem 1(b), $f'(\xi)=\xi J_0(\xi)$. Since $\xi>\alpha>0$, we may divide by ξ and get $J_0(\xi)=0$.

Answer. Shown: J_1 vanishes between consecutive zeros of J_0 (since $J_0'=-J_1$), and J_0 vanishes between zeros of J_1 (since $(xJ_1)'=xJ_0$).

Problem 46-3 (Simmons)

Solution.

(a) Let $t=s^2$ with $s\ge 0$. Then $dt=2s\,ds$, $t^{-1/2}=1/s$, and $t:0\to\infty$ corresponds to $s:0\to\infty$. Hence

$$\Gamma\Big(\frac12\Big)=\int_0^\infty \frac{1}{s}\,e^{-s^2}\,2s\,ds=2\int_0^\infty e^{-s^2}\,ds.$$

(b) Squaring the result of (a) and writing the second factor with the dummy variable y gives the displayed product of two integrals. The product of the two (absolutely convergent) one-variable integrals equals the double integral of $e^{-x^2}e^{-y^2}=e^{-(x^2+y^2)}$ over the first quadrant $x\ge0,\ y\ge0$ (Fubini; the integrand is positive).

In polar coordinates $x=r\cos\theta$, $y=r\sin\theta$, the quadrant is $0\le r<\infty$, $0\le\theta\le\pi/2$, $x^2+y^2=r^2$, and $dx\,dy=r\,dr\,d\theta$. Therefore

$$\Gamma\Big(\frac12\Big)^2=4\int_0^{\pi/2}\int_0^\infty e^{-r^2}\,r\,dr\,d\theta=4\Big(\int_0^{\pi/2}d\theta\Big)\Big(\int_0^\infty re^{-r^2}\,dr\Big).$$

Now $\int_0^{\pi/2}d\theta=\pi/2$ and $\int_0^\infty re^{-r^2}\,dr=\big[-\tfrac12 e^{-r^2}\big]_0^\infty=\tfrac12$. So

$$\Gamma\Big(\frac12\Big)^2=4\cdot\frac{\pi}{2}\cdot\frac12=\pi.$$

Since $\Gamma(\tfrac12)=\int_0^\infty t^{-1/2}e^{-t}\,dt>0$, we take the positive square root: $\Gamma(\tfrac12)=\sqrt{\pi}$.

Answer. Shown: $\Gamma(\tfrac12)=2\int_0^\infty e^{-s^2}ds$, $\Gamma(\tfrac12)^2=\pi$, hence $\Gamma(\tfrac12)=\sqrt{\pi}$.

Problem 46-4 (Simmons)

Solution. By (10), $\Gamma(p+1)=p\,\Gamma(p)$, i.e. $(p)!=p\,(p-1)!$.

Specific values.

$$\Big(\frac12\Big)!=\frac12\Big(-\frac12\Big)!=\frac{\sqrt{\pi}}{2},\qquad \Big(\frac32\Big)!=\frac32\Big(\frac12\Big)!=\frac32\cdot\frac{\sqrt{\pi}}{2}=\frac{3\sqrt{\pi}}{4}.$$

General formulas. We prove the second formula by induction on n and then deduce the first.

For $n = 0$ the second formula says $(-\frac{1}{2})! = \frac{0!}{2^0\,0!}\sqrt{\pi} = \sqrt{\pi}$, which is (14). Suppose it holds for some $n \geq 0$. Then by (10) applied with $p = n + \frac{1}{2}$,

$$\left(n+\frac{1}{2}\right)! = \left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)! = \frac{2n+1}{2}\cdot\frac{(2n)!}{2^{2n}\,n!}\sqrt{\pi} = \frac{(2n+1)!}{2^{2n+1}\,n!}\sqrt{\pi}. \qquad (*)$$

This is the first formula for the index n. Since $(n+1)-\frac{1}{2} = n+\frac{1}{2}$, $(*)$ also gives the second formula for the index $n+1$ once we check that

$$\frac{(2n+1)!}{2^{2n+1}\,n!} = \frac{(2n+2)!}{2^{2n+2}(n+1)!}.$$

Indeed, $(2n+2)! = (2n+2)(2n+1)! = 2(n+1)(2n+1)!$ and $(n+1)! = (n+1)\,n!$, so the right side equals $\dfrac{2(n+1)(2n+1)!}{2^{2n+2}(n+1)\,n!} = \dfrac{(2n+1)!}{2^{2n+1}\,n!}$. This completes the induction: the second formula holds for all $n \geq 0$, and $(*)$ gives the first formula for all $n \geq 0$.

Answer. $(\frac{1}{2})! = \dfrac{\sqrt{\pi}}{2}$, $(\frac{3}{2})! = \dfrac{3\sqrt{\pi}}{4}$; and $(n+\frac{1}{2})! = \dfrac{(2n+1)!}{2^{2n+1}n!}\sqrt{\pi}$, $(n-\frac{1}{2})! = \dfrac{(2n)!}{2^{2n}n!}\sqrt{\pi}$.

Problem 46-5 (Simmons)

Solution. We use the leading behavior of the series (6) near $x = 0$. Since the series (6) is x^p times a power series in x^2,

$$J_p(x) = \frac{(x/2)^p}{p!}\left(1 + O(x^2)\right) \qquad (x \to 0^+).$$

$J_{1/2}$. With $p = \frac{1}{2}$ and $(\frac{1}{2})! = \sqrt{\pi}/2$ (Problem 4),

$$\sqrt{x}\,J_{1/2}(x) = \sqrt{x}\,\frac{(x/2)^{1/2}}{\sqrt{\pi}/2}\left(1+O(x^2)\right) = \frac{2x}{\sqrt{2\pi}}\left(1+O(x^2)\right) = \sqrt{\frac{2}{\pi}}\,x\left(1+O(x^2)\right).$$

On the other hand $a\cos x + b\sin x = a + bx + O(x^2)$. Comparing the constant terms gives $a = 0$, and comparing the coefficients of x gives $b = \sqrt{2/\pi}$. Hence $\sqrt{x}J_{1/2}(x) = \sqrt{2/\pi}\,\sin x$, i.e.

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\,\sin x.$$

$J_{-1/2}$. With $p = -\frac{1}{2}$ and $(-\frac{1}{2})! = \Gamma(\frac{1}{2}) = \sqrt{\pi}$,

$$\sqrt{x}\,J_{-1/2}(x) = \sqrt{x}\,\frac{(x/2)^{-1/2}}{\sqrt{\pi}}\left(1+O(x^2)\right) = \sqrt{\frac{2}{\pi}}\left(1+O(x^2)\right).$$

This has constant term $\sqrt{2/\pi}$ and *no* term in x, while $c\cos x + d\sin x = c + dx + O(x^2)$. Hence $c = \sqrt{2/\pi}$ and $d = 0$, so

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\,\cos x.$$

Answer. $a = 0$, $b = \sqrt{2/\pi}$, $c = \sqrt{2/\pi}$, $d = 0$; hence $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\sin x$ and $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\cos x$.

Problem 46-6 (Simmons)

Solution. By (6), $J_{1/2}(x) = \sum_{n=0}^{\infty} \dfrac{(-1)^n (x/2)^{2n+1/2}}{n!\,\Gamma(n+\frac{3}{2})}$ and $J_{-1/2}(x) = \sum_{n=0}^{\infty} \dfrac{(-1)^n (x/2)^{2n-1/2}}{n!\,\Gamma(n+\frac{1}{2})}$. From Problem 4 (with $\Gamma(m+1) = m!$),

$$\Gamma\Big(n+\frac{3}{2}\Big) = \Big(n+\frac{1}{2}\Big)! = \frac{(2n+1)!}{2^{2n+1}n!}\sqrt{\pi}, \qquad \Gamma\Big(n+\frac{1}{2}\Big) = \Big(n-\frac{1}{2}\Big)! = \frac{(2n)!}{2^{2n}n!}\sqrt{\pi}.$$

For $J_{1/2}$: the nth term is

$$\frac{(-1)^n\, x^{2n+1/2}\, 2^{-2n-1/2}}{n!} \cdot \frac{2^{2n+1}\, n!}{(2n+1)!\sqrt{\pi}} = (-1)^n \frac{x^{2n+1/2}\, 2^{1/2}}{(2n+1)!\sqrt{\pi}}$$
$$= \sqrt{\frac{2}{\pi x}}\, \frac{(-1)^n x^{2n+1}}{(2n+1)!} \qquad (x^{2n+1/2} = x^{-1/2} x^{2n+1}).$$

Summing over n and recognizing the Maclaurin series of $\sin x$,

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sqrt{\frac{2}{\pi x}} \sin x.$$

For $J_{-1/2}$: the nth term is

$$\frac{(-1)^n\, x^{2n-1/2}\, 2^{-2n+1/2}}{n!} \cdot \frac{2^{2n}\, n!}{(2n)!\sqrt{\pi}} = (-1)^n \frac{x^{2n-1/2}\, 2^{1/2}}{(2n)!\sqrt{\pi}} = \sqrt{\frac{2}{\pi x}}\, \frac{(-1)^n x^{2n}}{(2n)!},$$

so, by the Maclaurin series of $\cos x$,

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \sqrt{\frac{2}{\pi x}} \cos x.$$

Answer. Shown: the series for $J_{\pm 1/2}$ reduce, term by term, to $\sqrt{2/(\pi x)}$ times the series for $\sin x$ and $\cos x$.

Problem 46-7 (Simmons)

Solution. Take $x > 0$, $a \neq 0$, $b \neq 0$ (with $a > 0$ so that $z = ax^b > 0$). We do the change of variables in two stages: first $z = ax^b$ (with w regarded as a function of x), then $w = yx^c$.

Stage 1: $z = ax^b$. Here $\frac{dz}{dx} = abx^{b-1}$, so by the chain rule

$$\frac{dw}{dz} = \frac{1}{abx^{b-1}} \frac{dw}{dx},$$
$$\frac{d^2w}{dz^2} = \frac{1}{abx^{b-1}} \frac{d}{dx}\Big(\frac{1}{abx^{b-1}} \frac{dw}{dx}\Big) = \frac{1}{a^2b^2x^{2b-2}}\Big(\frac{d^2w}{dx^2} - \frac{b-1}{x}\frac{dw}{dx}\Big),$$

because $\frac{d}{dx} x^{1-b} = (1-b)x^{-b}$. Therefore

$$z^2 \frac{d^2w}{dz^2} = a^2x^{2b} \cdot \frac{1}{a^2b^2x^{2b-2}}\Big(w_{xx} - \frac{b-1}{x} w_x\Big) = \frac{1}{b^2}\Big(x^2 w_{xx} - (b-1)\, x\, w_x\Big),$$
$$z \frac{dw}{dz} = ax^b \frac{w_x}{abx^{b-1}} = \frac{1}{b}\, x\, w_x.$$

Substituting in Bessel's equation and multiplying by b^2:

$$\begin{gathered} x^2 w_{xx} - (b-1)x w_x + b\,x w_x + (a^2b^2x^{2b} - p^2b^2)w = 0, \\ \text{i.e.} \quad x^2 w_{xx} + x w_x + (a^2b^2x^{2b} - p^2b^2)w = 0. \end{gathered} \tag{1}$$

Stage 2: $w = yx^c$. Then

$$w_x = x^c y' + c x^{c-1} y, \qquad w_{xx} = x^c y'' + 2c x^{c-1} y' + c(c-1)x^{c-2} y.$$

Insert in (1):

$$x^c \Big[x^2 y'' + 2c\,xy' + c(c-1)y + xy' + c\,y + (a^2b^2x^{2b} - p^2b^2)y \Big] = 0.$$

Divide by $x^c \neq 0$ and collect: $c(c-1) + c = c^2$, so

$$x^2 y'' + (2c+1)xy' + \big[a^2b^2x^{2b} + c^2 - p^2b^2\big]y = 0,$$

as claimed.

General solution. Conversely, y solves this equation iff $w = yx^c$, regarded as a function of $z = ax^b$, solves Bessel's equation of order p. By (16) and (18)–(20) of the text, the general solution of Bessel's equation is $w = c_1 J_p(z) + c_2 J_{-p}(z)$ if p is not an integer, and $w = c_1 J_p(z) + c_2 Y_p(z)$ in all cases. Since $y = x^{-c}w$ and $z = ax^b$,

$$y = x^{-c}\big[c_1 J_p(ax^b) + c_2 J_{-p}(ax^b)\big] \ \ (p \text{ not an integer}), \qquad y = x^{-c}\big[c_1 J_p(ax^b) + c_2 Y_p(ax^b)\big] \ \ (\text{all } p).$$

Answer. $y = x^{-c}\big[c_1 J_p(ax^b) + c_2 J_{-p}(ax^b)\big]$ if p is not an integer; $y = x^{-c}\big[c_1 J_p(ax^b) + c_2 Y_p(ax^b)\big]$ in all cases.

Problem 46-8 (Simmons)

Solution. Take $x > 0$. Multiplying $y'' + xy = 0$ by x^2 gives

$$x^2 y'' + 0 \cdot xy' + x^3 y = 0.$$

Match this with the equation of Problem 7, $x^2y'' + (2c+1)xy' + [a^2b^2x^{2b} + c^2 - p^2b^2]y = 0$:

$$\begin{aligned} 2c+1 = 0 &\implies c = -\tfrac{1}{2}, \\ 2b = 3 &\implies b = \tfrac{3}{2}, \\ a^2b^2 = 1 &\implies a = \tfrac{1}{b} = \tfrac{2}{3} \quad (a > 0), \\ c^2 - p^2b^2 = 0 &\implies p^2 = \frac{c^2}{b^2} = \frac{1/4}{9/4} = \frac{1}{9} \implies p = \tfrac{1}{3}. \end{aligned}$$

Since $p = \frac{1}{3}$ is not an integer, Problem 7 gives the general solution $y = x^{-c}[c_1 J_p(ax^b) + c_2 J_{-p}(ax^b)]$ with $x^{-c} = x^{1/2}$, $ax^b = \frac{2}{3}x^{3/2}$:

$$y = x^{1/2}\Big[c_1 J_{1/3}\Big(\frac{2}{3}x^{3/2}\Big) + c_2 J_{-1/3}\Big(\frac{2}{3}x^{3/2}\Big)\Big].$$

Answer. $y = x^{1/2}\big[c_1 J_{1/3}(\tfrac{2}{3}x^{3/2}) + c_2 J_{-1/3}(\tfrac{2}{3}x^{3/2})\big]$ (for $x > 0$), with $c = -\tfrac{1}{2}$, $b = \tfrac{3}{2}$, $a = \tfrac{2}{3}$, $p = \tfrac{1}{3}$.

Problem 46-9 (Simmons)

Solution. By (19) and (17), for an integer $m \geq 0$ and fixed $x > 0$,

$$Y_m(x) = \lim_{p\to m} Y_p(x) = \lim_{p\to m} \frac{J_p(x)\cos p\pi - J_{-p}(x)}{\sin p\pi}.$$

Both numerator and denominator are differentiable functions of p (the series (6) and (15) are series in which $1/\Gamma$ is an entire function of p, so $J_{\pm p}(x)$ is differentiable in p). At $p = m$:

- the denominator is $\sin m\pi = 0$;
- the numerator is $J_m(x)\cos m\pi - J_{-m}(x) = (-1)^m J_m(x) - (-1)^m J_m(x) = 0$, using $J_{-m} = (-1)^m J_m$ (shown in the text).

So the limit has the indeterminate form $0/0$ and l'Hospital's rule applies (differentiating with respect to p):

$$Y_m(x) = \lim_{p\to m} \frac{\dfrac{\partial J_p}{\partial p}\cos p\pi - \pi J_p \sin p\pi - \dfrac{\partial}{\partial p}J_{-p}}{\pi\cos p\pi}.$$

At $p = m$: $\cos m\pi = (-1)^m$, $\sin m\pi = 0$, so

$$Y_m(x) = \frac{(-1)^m \dfrac{\partial J_p}{\partial p}\Big|_{p=m} - \dfrac{\partial}{\partial p}J_{-p}\Big|_{p=m}}{\pi(-1)^m} = \frac{1}{\pi}\Big[\frac{\partial J_p}{\partial p} - (-1)^m \frac{\partial}{\partial p}J_{-p}\Big]_{p=m},$$

where in the last step we divided numerator and denominator by $(-1)^m$ and used $1/(-1)^m = (-1)^m$ and $(-1)^m(-1)^m = 1$ (so the first term loses its sign factor and the second acquires $(-1)^m$).

Answer. Shown: by l'Hospital's rule, $Y_m(x) = \dfrac{1}{\pi}\Big[\dfrac{\partial J_p}{\partial p} - (-1)^m \dfrac{\partial J_{-p}}{\partial p}\Big]_{p=m}$.

Section 47 Properties of Bessel Functions

Throughout, $J_p(x) = \sum_{n=0}^{\infty} (-1)^n \dfrac{(x/2)^{2n+p}}{n!\,\Gamma(p+n+1)}$ as in (1), and we work on $x > 0$. We use the identities (2)–(7) of the text.

Remark. As printed in the book, formula (5) reads $J_p'(x) - \frac{p}{x} J_p(x) = J_{p+1}(x)$; this is a misprint. Carrying out the differentiation in (3), $\frac{d}{dx}[x^{-p} J_p] = x^{-p} J_p' - p x^{-p-1} J_p = -x^{-p} J_{p+1}$, gives

$$J_p'(x) - \frac{p}{x} J_p(x) = -J_{p+1}(x), \tag{5′}$$

which is the form consistent with (3), with (6) (= (4)+(5′)) and with (7) (= (4)−(5′)). We use (5′) below.

Problem 47-1 (Simmons)

Solution. Multiply the series (1) by x^{-p}; since $(x/2)^{2n+p} = x^{2n+p}/2^{2n+p}$,

$$x^{-p} J_p(x) = \sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n}}{2^{2n+p}\, n!\, \Gamma(p+n+1)}.$$

Differentiate term by term (a power series in x). The term $n = 0$ is constant. For $n \geq 1$ the derivative of x^{2n} is $2n\, x^{2n-1}$, and $\dfrac{2n}{2^{2n+p}\, n!} = \dfrac{1}{2^{2n+p-1}(n-1)!}$, so

$$\frac{d}{dx}\left[x^{-p} J_p(x)\right] = \sum_{n=1}^{\infty} \frac{(-1)^n\, x^{2n-1}}{2^{2n+p-1}\,(n-1)!\,\Gamma(p+n+1)}.$$

Put $n = k+1$, $k = 0, 1, 2, \ldots$ (so $2n-1 = 2k+1$, $2n+p-1 = 2k+p+1$, $\Gamma(p+n+1) = \Gamma(p+k+2)$):

$$\begin{aligned}\frac{d}{dx}\left[x^{-p} J_p(x)\right] &= \sum_{k=0}^{\infty} \frac{(-1)^{k+1}\, x^{2k+1}}{2^{2k+p+1}\, k!\, \Gamma(p+k+2)} = -x^{-p} \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+p+1}}{2^{2k+p+1}\, k!\, \Gamma\big((p+1)+k+1\big)} \\ &= -x^{-p} \sum_{k=0}^{\infty} (-1)^k \frac{(x/2)^{2k+(p+1)}}{k!\,\big((p+1)+k\big)!} = -x^{-p} J_{p+1}(x),\end{aligned}$$

where we used $x^{2k+1} = x^{-p}\, x^{2k+p+1}$.

Answer. Verified: $\dfrac{d}{dx}[x^{-p} J_p] = -x^{-p} J_{p+1}$.

Problem 47-2 (Simmons)

Solution. Let $p \geq 0$. We use (2) and (3):

$$\frac{d}{dx}\left[x^{p+1} J_{p+1}(x)\right] = x^{p+1} J_p(x), \qquad \frac{d}{dx}\left[x^{-p} J_p(x)\right] = -x^{-p} J_{p+1}(x). \tag{$*$}$$

(The first is (2) with p replaced by $p+1$.)

Preliminaries. J_p is a nontrivial solution of Bessel's equation; if $J_p(x_0) = 0 = J_p'(x_0)$ at some $x_0 > 0$ the uniqueness theorem for the linear equation (whose coefficients are continuous for $x > 0$)

would force $J_p \equiv 0$. Hence the zeros of J_p in $(0,\infty)$ are simple, and in particular isolated, so "consecutive positive zeros" makes sense. The same holds for J_{p+1}. Also J_p and J_{p+1} have no common positive zero: indeed, if $J_p(x_0)=0$, then $(5')$ gives $J_p'(x_0) = -J_{p+1}(x_0)$, so $J_{p+1}(x_0)=0$ would give $J_p(x_0) = J_p'(x_0) = 0$, which is impossible.

(i) A zero of J_{p+1} lies between consecutive zeros of J_p. Let $0<\alpha<\beta$ be zeros of J_p. The function $f(x) = x^{-p}J_p(x)$ is differentiable on $[\alpha,\beta]$ (as $\alpha>0$) and $f(\alpha)=f(\beta)=0$. By Rolle's theorem $f'(\xi)=0$ for some $\xi\in(\alpha,\beta)$, and by $(*)$, $-\xi^{-p}J_{p+1}(\xi)=0$, i.e. $J_{p+1}(\xi)=0$.

(ii) A zero of J_p lies between consecutive zeros of J_{p+1}. Let $0<\alpha<\beta$ be zeros of J_{p+1} and $g(x) = x^{p+1}J_{p+1}(x)$. Then $g(\alpha)=g(\beta)=0$, so $g'(\xi)=0$ for some $\xi\in(\alpha,\beta)$, and by $(*)$, $\xi^{p+1}J_p(\xi)=0$, i.e. $J_p(\xi)=0$.

(iii) Exactly one. Let $\alpha<\beta$ be consecutive positive zeros of J_p. By (i) there is at least one zero of J_{p+1} in (α,β). If there were two, $\xi_1<\xi_2$, in (α,β), then (ii) applied to them would give a zero of J_p in $(\xi_1,\xi_2)\subset(\alpha,\beta)$, contradicting the consecutiveness of α,β. So there is exactly one. Symmetrically, if $\alpha<\beta$ are consecutive positive zeros of J_{p+1}, (ii) gives at least one zero of J_p between them, and two such zeros $\xi_1<\xi_2$ would by (i) enclose a zero of J_{p+1} strictly between α and β, a contradiction. Hence exactly one.

Answer. Shown: via Rolle's theorem applied to $x^{-p}J_p$ and $x^{p+1}J_{p+1}$ (formulas (2), (3)) the positive zeros of J_p and J_{p+1} interlace.

Problem 47-3 (Simmons)

Solution. Formula (7) with p replaced by n reads $\dfrac{2n}{x}J_n = J_{n-1}+J_{n+1}$, i.e.

$$J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x). \tag{$*$}$$

$n=1$: $\quad J_2 = \dfrac{2}{x}J_1 - J_0.$

$n=2$: $\quad J_3 = \dfrac{4}{x}J_2 - J_1 = \dfrac{4}{x}\Big(\dfrac{2}{x}J_1 - J_0\Big) - J_1 = \Big(\dfrac{8}{x^2}-1\Big)J_1 - \dfrac{4}{x}J_0.$

$n=3$:

$$\begin{aligned} J_4 &= \frac{6}{x}J_3 - J_2 = \frac{6}{x}\Big[\Big(\frac{8}{x^2}-1\Big)J_1 - \frac{4}{x}J_0\Big] - \Big(\frac{2}{x}J_1 - J_0\Big) \\ &= \Big(\frac{48}{x^3} - \frac{6}{x} - \frac{2}{x}\Big)J_1 + \Big(-\frac{24}{x^2}+1\Big)J_0 = \Big(\frac{48}{x^3} - \frac{8}{x}\Big)J_1 - \Big(\frac{24}{x^2}-1\Big)J_0. \end{aligned}$$

Answer. $J_2 = \dfrac{2}{x}J_1 - J_0$; $\quad J_3 = \Big(\dfrac{8}{x^2}-1\Big)J_1 - \dfrac{4}{x}J_0$; $\quad J_4 = \Big(\dfrac{48}{x^3}-\dfrac{8}{x}\Big)J_1 - \Big(\dfrac{24}{x^2}-1\Big)J_0$. (Agrees with the book.)

Problem 47-4 (Simmons)

Solution. We expand f in the Bessel series (11) with $p=0$: $f(x) = \sum a_n J_0(\lambda_n x)$, where by (13)

$$a_n = \frac{2}{J_1(\lambda_n)^2}\int_0^1 x f(x) J_0(\lambda_n x)\,dx = \frac{2}{J_1(\lambda_n)^2}\int_0^{1/2} x J_0(\lambda_n x)\,dx,$$

because $f = 0$ on $(\frac{1}{2}, 1]$ and $f = 1$ on $[0, \frac{1}{2})$ (the single point $x = \frac{1}{2}$ does not affect the integral). (Here $J_{p+1} = J_1$ since $p = 0$.)

The integral. Substitute $t = \lambda_n x$ ($dx = dt/\lambda_n$) and use (10), $\int tJ_0(t)\,dt = tJ_1(t)$:

$$\int_0^{1/2} xJ_0(\lambda_n x)\,dx = \frac{1}{\lambda_n^2}\int_0^{\lambda_n/2} tJ_0(t)\,dt = \frac{1}{\lambda_n^2}\Big[tJ_1(t)\Big]_0^{\lambda_n/2} = \frac{1}{\lambda_n^2}\cdot\frac{\lambda_n}{2}J_1\Big(\frac{\lambda_n}{2}\Big) = \frac{J_1(\lambda_n/2)}{2\lambda_n}.$$

Therefore

$$a_n = \frac{2}{J_1(\lambda_n)^2}\cdot\frac{J_1(\lambda_n/2)}{2\lambda_n} = \frac{J_1(\lambda_n/2)}{\lambda_n J_1(\lambda_n)^2}.$$

Convergence to f. f is piecewise constant, so f and f' have only a finite number of jump discontinuities on $0 \le x \le 1$ (the single jump is at $x = \frac{1}{2}$). By the Bessel expansion theorem (Theorem A), for $0 < x < 1$ the series converges to $f(x)$ at points of continuity, i.e. to 1 for $0 < x < \frac{1}{2}$ and to 0 for $\frac{1}{2} < x < 1$, and at the jump $x = \frac{1}{2}$ to $\frac{1}{2}[f(\frac{1}{2}-) + f(\frac{1}{2}+)] = \frac{1}{2}(1+0) = \frac{1}{2} = f(\frac{1}{2})$, which is the prescribed value. At the endpoints, again by the text, the series converges at $x = 1$ to $0 = f(1)$ (each $J_0(\lambda_n) = 0$) and at $x = 0$ (case $p = 0$) to $f(0+) = 1 = f(0)$. Hence the series equals $f(x)$ on the whole interval $0 \le x \le 1$.

Answer. Shown: $a_n = \dfrac{J_1(\lambda_n/2)}{\lambda_n J_1(\lambda_n)^2}$ and, by Theorem A, the series converges to $f(x)$ for every $x \in [0,1]$ (including $x = \frac{1}{2}$, where it gives $\frac{1}{2}$).

Problem 47-5 (Simmons)

Solution. Here $p \ge 0$. By (13),

$$a_n = \frac{2}{J_{p+1}(\lambda_n)^2}\int_0^1 x\cdot x^p J_p(\lambda_n x)\,dx = \frac{2}{J_{p+1}(\lambda_n)^2}\int_0^1 x^{p+1}J_p(\lambda_n x)\,dx.$$

Substitute $t = \lambda_n x$, $x = t/\lambda_n$, $dx = dt/\lambda_n$:

$$\int_0^1 x^{p+1}J_p(\lambda_n x)\,dx = \frac{1}{\lambda_n^{p+2}}\int_0^{\lambda_n} t^{p+1}J_p(t)\,dt.$$

By (8) with p replaced by $p+1$, $\int t^{p+1}J_p(t)\,dt = t^{p+1}J_{p+1}(t) + \text{const}$, and the integrand is integrable at 0; the antiderivative vanishes at $t = 0$ since $p + 1 > 0$. Hence

$$\int_0^{\lambda_n} t^{p+1}J_p(t)\,dt = \lambda_n^{p+1}J_{p+1}(\lambda_n), \qquad \int_0^1 x^{p+1}J_p(\lambda_n x)\,dx = \frac{\lambda_n^{p+1}J_{p+1}(\lambda_n)}{\lambda_n^{p+2}} = \frac{J_{p+1}(\lambda_n)}{\lambda_n}. \quad (*)$$

Therefore

$$a_n = \frac{2}{J_{p+1}(\lambda_n)^2}\cdot\frac{J_{p+1}(\lambda_n)}{\lambda_n} = \frac{2}{\lambda_n J_{p+1}(\lambda_n)}$$

(note $J_{p+1}(\lambda_n) \neq 0$, since J_p and J_{p+1} have no common positive zero, Problem 2), which is the stated series.

Answer. $a_n = \dfrac{2}{\lambda_n J_{p+1}(\lambda_n)}$, so $x^p = \displaystyle\sum_{n=1}^{\infty}\frac{2}{\lambda_n J_{p+1}(\lambda_n)}J_p(\lambda_n x)$.

Problem 47-6 (Simmons)

Solution. *The identity.* By Problem 5, $x^p = \sum_n \dfrac{2}{\lambda_n J_{p+1}(\lambda_n)} J_p(\lambda_n x)$. Multiply both sides by $x\,g(x)$ and integrate from 0 to 1, integrating the series term by term (this is the formal step):

$$\int_0^1 x^{p+1} g(x)\,dx = \sum_{n=1}^{\infty} \frac{2}{\lambda_n J_{p+1}(\lambda_n)} \int_0^1 x g(x) J_p(\lambda_n x)\,dx.$$

Dividing by 2 gives the stated formula. Call it $(\dagger)$.

$g(x) = x^p$. Left side of $(\dagger)$:

$$\frac{1}{2}\int_0^1 x^{2p+1}\,dx = \frac{1}{2}\cdot\frac{1}{2p+2} = \frac{1}{4(p+1)}.$$

Right side: $\displaystyle\int_0^1 x^{p+1} J_p(\lambda_n x)\,dx = \frac{J_{p+1}(\lambda_n)}{\lambda_n}$ by $(*)$ of Problem 5, so the nth term is $\dfrac{1}{\lambda_n J_{p+1}(\lambda_n)}\cdot \dfrac{J_{p+1}(\lambda_n)}{\lambda_n} = \dfrac{1}{\lambda_n^2}$. Thus

$$\sum_{n=1}^{\infty} \frac{1}{\lambda_n^2} = \frac{1}{4(p+1)}.$$

$g(x) = x^{p+2}$. Left side of $(\dagger)$: $\dfrac{1}{2}\displaystyle\int_0^1 x^{2p+3}\,dx = \frac{1}{2}\cdot\frac{1}{2p+4} = \frac{1}{4(p+2)}$.

Right side: we need $I_n = \displaystyle\int_0^1 x^{p+3} J_p(\lambda_n x)\,dx = \frac{1}{\lambda_n^{p+4}}\int_0^{\lambda_n} t^{p+3} J_p(t)\,dt$ (again $t = \lambda_n x$). Write $t^{p+3} = t^2 \cdot t^{p+1}$ and integrate by parts with $u = t^2$, $dv = t^{p+1}J_p(t)\,dt$, $v = t^{p+1}J_{p+1}(t)$ (by (8)):

$$\int_0^{\lambda} t^{p+3} J_p(t)\,dt = \Big[t^{p+3} J_{p+1}(t)\Big]_0^{\lambda} - 2\int_0^{\lambda} t^{p+2} J_{p+1}(t)\,dt.$$

By (8) with $p+2$ in place of p, $\int t^{p+2}J_{p+1}(t)\,dt = t^{p+2}J_{p+2}(t)$. So, with $\lambda = \lambda_n$ (the boundary terms at 0 vanish),

$$\int_0^{\lambda} t^{p+3} J_p(t)\,dt = \lambda^{p+3} J_{p+1}(\lambda) - 2\lambda^{p+2} J_{p+2}(\lambda).$$

Now (7) with $p+1$ in place of p gives $\dfrac{2(p+1)}{\lambda} J_{p+1}(\lambda) = J_p(\lambda) + J_{p+2}(\lambda)$, and $J_p(\lambda_n) = 0$, so $J_{p+2}(\lambda_n) = \dfrac{2(p+1)}{\lambda_n} J_{p+1}(\lambda_n)$. Hence

$$\int_0^{\lambda_n} t^{p+3} J_p(t)\,dt = \lambda_n^{p+3} J_{p+1}(\lambda_n) - 4(p+1)\lambda_n^{p+1} J_{p+1}(\lambda_n), \qquad I_n = J_{p+1}(\lambda_n)\Big[\frac{1}{\lambda_n} - \frac{4(p+1)}{\lambda_n^3}\Big].$$

So the nth term on the right of $(\dagger)$ is $\dfrac{1}{\lambda_n J_{p+1}(\lambda_n)} I_n = \dfrac{1}{\lambda_n^2} - \dfrac{4(p+1)}{\lambda_n^4}$, and $(\dagger)$ becomes

$$\frac{1}{4(p+2)} = \sum_{n=1}^{\infty}\frac{1}{\lambda_n^2} - 4(p+1)\sum_{n=1}^{\infty}\frac{1}{\lambda_n^4} = \frac{1}{4(p+1)} - 4(p+1)\sum_{n=1}^{\infty}\frac{1}{\lambda_n^4},$$

using the first result. Solving,

$$\sum_{n=1}^{\infty}\frac{1}{\lambda_n^4} = \frac{1}{4(p+1)}\Big[\frac{1}{4(p+1)} - \frac{1}{4(p+2)}\Big] = \frac{1}{4(p+1)}\cdot\frac{(p+2)-(p+1)}{4(p+1)(p+2)} = \frac{1}{16(p+1)^2(p+2)}.$$

Answer. $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\lambda_n^2}=\frac{1}{4(p+1)}$ and $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\lambda_n^4}=\frac{1}{16(p+1)^2(p+2)}$.

Problem 47-7 (Simmons)

Solution. Take $p=\frac{1}{2}$. By Problem 46–5, $J_{1/2}(x)=\sqrt{2/(\pi x)}\,\sin x$, and since $\sqrt{2/(\pi x)}\neq 0$ for $x>0$, the positive zeros of $J_{1/2}$ are exactly the positive zeros of $\sin x$: $\lambda_n=n\pi$, $n=1,2,3,\dots$ Problem 6 gives

$$\sum_{n=1}^{\infty}\frac{1}{n^2\pi^2}=\frac{1}{4(\frac{1}{2}+1)}=\frac{1}{6},\qquad \sum_{n=1}^{\infty}\frac{1}{n^4\pi^4}=\frac{1}{16(\frac{3}{2})^2(\frac{5}{2})}=\frac{1}{16\cdot\frac{9}{4}\cdot\frac{5}{2}}=\frac{1}{90}.$$

(Indeed $16\cdot\frac{9}{4}=36$ and $36\cdot\frac{5}{2}=90$.) Multiplying by π^2 and π^4 respectively:

$$\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6},\qquad \sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{90}.$$

Answer. Shown: with $p=\frac{1}{2}$, $\lambda_n=n\pi$, giving $\sum 1/n^2=\pi^2/6$ and $\sum 1/n^4=\pi^4/90$.

Problem 47-8 (Simmons)

Solution. Here $B\neq 0$ and C are constants and $x>0$.

Part 1: the transformation. Let $y=\dfrac{u'}{Bu}$. Then

$$y'=\frac{1}{B}\Big(\frac{u''}{u}-\frac{u'^2}{u^2}\Big),\qquad By^2=\frac{u'^2}{Bu^2}.$$

Therefore

$$y'+By^2=\frac{1}{B}\frac{u''}{u}-\frac{u'^2}{Bu^2}+\frac{u'^2}{Bu^2}=\frac{u''}{Bu},$$

and the Riccati equation $y'+By^2=Cx^m$ becomes $\dfrac{u''}{Bu}=Cx^m$, i.e.

$$u''-BCx^m u=0\qquad (u\neq 0).$$

Conversely, if u is a nonvanishing solution of this linear equation, $y=u'/(Bu)$ solves the Riccati equation; the general solution of the Riccati equation is $y=u'/(Bu)$ with $u=c_1u_1+c_2u_2$ the general solution of the linear equation.

Part 2: reduction to Bessel's equation. Let $m\neq -2$. Multiplying the linear equation by x^2,

$$x^2u''+0\cdot xu'-BC\,x^{m+2}u=0.$$

Compare with Problem 46–7: $x^2y''+(2c+1)xy'+[a^2b^2x^{2b}+c^2-p^2b^2]y=0$ (with $y\to u$):

$$\begin{aligned}2c+1=0&\Rightarrow c=-\tfrac{1}{2},\\ 2b=m+2&\Rightarrow b=\tfrac{m+2}{2}\ (\neq 0),\\ a^2b^2=-BC&\Rightarrow a^2=-\frac{4BC}{(m+2)^2},\end{aligned}$$

$$c^2 - p^2b^2 = 0 \Rightarrow p^2 = \frac{c^2}{b^2} = \frac{1}{(m+2)^2} \;\Rightarrow\; p = \frac{1}{|m+2|}.$$

(The equation involves only p^2, so either sign of p may be used.) Hence, by Problem 46–7, if $BC < 0$ (so $a = 2\sqrt{-BC}/|m+2|$ is real) the general solution is

$$u = x^{1/2}\Big[c_1 J_p\big(ax^{(m+2)/2}\big) + c_2 J_{-p}\big(ax^{(m+2)/2}\big)\Big], \qquad p = \frac{1}{|m+2|},$$

when p is not an integer (and with J_{-p} replaced by Y_p in general). If $BC > 0$, a is imaginary, $a = i\,a_0$ with $a_0 = 2\sqrt{BC}/|m+2|$, and the same formula holds with $J_{\pm p}(ia_0 s)$, which is $i^{\pm p}$ times the modified Bessel function $I_{\pm p}(a_0 s)$, $s = x^{(m+2)/2}$. In either case the nature of the solution is governed by the order $p = 1/|m+2|$.

Part 3: when is the solution elementary? As recalled in the text (Liouville's theorem, footnote 7), J_p is an elementary function exactly when p is half an odd integer, $p = j + \frac{1}{2}$ with j an integer (then $J_{\pm p}$ are combinations of sin, cos and powers, by repeated use of (7) starting from $J_{\pm 1/2}$; for imaginary argument sin, cos become sinh, cosh, i.e. exponentials). Thus the Riccati equation is solvable in elementary terms iff

$$\frac{1}{|m+2|} = \frac{2j+1}{2} \quad \text{for some integer } j \geq 0, \qquad \text{i.e.} \qquad |m+2| = \frac{2}{2j+1}.$$

- If $m+2 = +\dfrac{2}{2j+1}$ $(j = 0, 1, 2, \dots)$, then $m = \dfrac{2}{2j+1} - 2 = \dfrac{2-4j-2}{2j+1} = \dfrac{-4j}{2j+1}$, which is $-4k/(2k+1)$ with $k = j \geq 0$.
- If $m+2 = -\dfrac{2}{2j+1}$, then $m = -2 - \dfrac{2}{2j+1} = \dfrac{-4j-4}{2j+1}$. Put $k = -(j+1) \leq -1$; then $2k+1 = -(2j+1)$ and $\dfrac{-4k}{2k+1} = \dfrac{4(j+1)}{-(2j+1)} = \dfrac{-4(j+1)}{2j+1}$, the same value.

Therefore m has the required form with k ranging over all integers, $k \geq 0$ for $m+2 > 0$ and $k \leq -1$ for $m+2 < 0$; and each such m arises, since for $m = -4k/(2k+1)$ we have $m+2 = \dfrac{2}{2k+1}$, so $p = \dfrac{|2k+1|}{2}$ is half an odd integer. (Examples: $k = 0$, $m = 0$; $k = 1$, $m = -\frac{4}{3}$; $k = -1$, $m = -4$.) Conversely, if $m \neq -2$ is not of this form, $p = 1/|m+2|$ is not half an odd integer and $J_{\pm p}$ (or Y_p) are not elementary. ■

Answer. Shown: $By = u'/u$ gives $u'' - BCx^m u = 0$; with $p = 1/|m+2|$ this is a Bessel equation (Problem 46–7 with $c = -\frac{1}{2}$, $b = \frac{m+2}{2}$), whose solutions are elementary iff p is a half odd integer, i.e. iff $m = -4k/(2k+1)$, $k \in \mathbb{Z}$.

Problem 47-9 (Simmons)

Solution. *Reduction.* The equation $y' = x^2 + y^2$ is $y' - y^2 = x^2$, which is the special Riccati equation $y' + By^2 = Cx^m$ with $B = -1$, $C = 1$, $m = 2$. By Problem 8, $y = \dfrac{u'}{Bu} = -\dfrac{u'}{u}$ where

$$u'' - BCx^m u = u'' + x^2 u = 0.$$

Solving for u. Multiply by x^2: $x^2u'' + x^4u = 0$. In the notation of Problem 46–7: $2c + 1 = 0$ so $c = -\frac{1}{2}$; $2b = 4$ so $b = 2$; $a^2b^2 = 1$ so $a = \frac{1}{2}$; $c^2 - p^2b^2 = 0$ so $p^2 = \frac{1/4}{4} = \frac{1}{16}$, $p = \frac{1}{4}$ (not an

integer). So

$$u = x^{1/2} f\Big(\frac{x^2}{2}\Big), \qquad f(z) = AJ_{1/4}(z) + BJ_{-1/4}(z), \qquad z = \frac{x^2}{2}$$

(here A, B are the arbitrary constants; B is not to be confused with the earlier constant $B = -1$).

Computing $y = -u'/u$. Since $dz/dx = x$,

$$u' = \tfrac{1}{2}x^{-1/2} f(z) + x^{1/2}\, x\, f'(z), \qquad y = -\frac{u'}{u} = -\frac{1}{2x} - x\,\frac{f'(z)}{f(z)}.$$

Now we use (4) with $p = \frac{1}{4}$ and (5′) with $p = -\frac{1}{4}$:

$$J'_{1/4}(z) + \frac{1}{4z} J_{1/4}(z) = J_{-3/4}(z), \qquad J'_{-1/4}(z) + \frac{1}{4z} J_{-1/4}(z) = -J_{3/4}(z).$$

(For the second: (5′) says $J'_p - \frac{p}{z} J_p = -J_{p+1}$; with $p = -\frac{1}{4}$ this is $J'_{-1/4} + \frac{1}{4z} J_{-1/4} = -J_{3/4}$.) Hence

$$f'(z) = AJ_{-3/4}(z) - BJ_{3/4}(z) - \frac{1}{4z} f(z).$$

Since $4z = 2x^2$, $\dfrac{x}{4z} = \dfrac{1}{2x}$, so

$$y = -\frac{1}{2x} - \frac{x\big(AJ_{-3/4} - BJ_{3/4}\big)}{f} + \frac{x}{4z} = -\frac{1}{2x} + \frac{1}{2x} - x\,\frac{AJ_{-3/4}(z) - BJ_{3/4}(z)}{AJ_{1/4}(z) + BJ_{-1/4}(z)}.$$

The terms $\mp\frac{1}{2x}$ cancel, leaving

$$y = -x\,\frac{AJ_{-3/4}(z) - BJ_{3/4}(z)}{AJ_{1/4}(z) + BJ_{-1/4}(z)}, \qquad z = \tfrac{1}{2}x^2.$$

Normalizing the constants. Only the ratio B/A matters (if $A \neq 0$; divide numerator and denominator by A). Put $c = -B/A$, so $B/A = -c$:

$$y = -x\,\frac{J_{-3/4}(z) + cJ_{3/4}(z)}{J_{1/4}(z) - cJ_{-1/4}(z)} = x\,\frac{J_{-3/4}(z) + cJ_{3/4}(z)}{cJ_{-1/4}(z) - J_{1/4}(z)},$$

which is the stated form. (The remaining choice $A = 0$ gives $y = xJ_{3/4}(z)/J_{-1/4}(z)$, the limit of the family as $c \to \infty$.)

Answer. $y = x\,\dfrac{J_{-3/4}(\frac{1}{2}x^2) + cJ_{3/4}(\frac{1}{2}x^2)}{cJ_{-1/4}(\frac{1}{2}x^2) - J_{1/4}(\frac{1}{2}x^2)}$, obtained from $y = -u'/u$, $u'' + x^2 u = 0$, $u = x^{1/2}[AJ_{1/4}(\frac{x^2}{2}) + BJ_{-1/4}(\frac{x^2}{2})]$, $c = -B/A$.

Chapter 9

Laplace Transforms

Section 48 Introduction

Throughout,

$$\mathcal{L}[f(x)] = F(p) = \int_0^\infty e^{-px} f(x)\,dx = \lim_{b\to\infty} \int_0^b e^{-px} f(x)\,dx$$

as in (6), (7) of the text, and $\mathcal{L}$ is linear: $\mathcal{L}[\alpha f + \beta g] = \alpha\mathcal{L}[f] + \beta\mathcal{L}[g]$.

Problem 48-1 (Simmons)

Solution. In every case we compute $\int_0^b$ explicitly and let $b \to \infty$.

((8)) $f = 1$. For $p \neq 0$,

$$\int_0^b e^{-px}\,dx = \Big[-\frac{e^{-px}}{p}\Big]_0^b = \frac{1 - e^{-pb}}{p}.$$

If $p > 0$, then $e^{-pb} \to 0$ as $b \to \infty$, so the limit is $1/p$. If $p < 0$, then $e^{-pb} \to \infty$ and the limit does not exist; if $p = 0$ the integral is $\int_0^b dx = b \to \infty$. Hence $\mathcal{L}[1] = \dfrac{1}{p}$ for $p > 0$ (and only then).

((9)) $f = x$. Integrate by parts with $u = x$, $dv = e^{-px}dx$ ($du = dx$, $v = -e^{-px}/p$), $p > 0$:

$$\int_0^b xe^{-px}\,dx = \Big[-\frac{xe^{-px}}{p}\Big]_0^b + \frac{1}{p}\int_0^b e^{-px}\,dx = -\frac{be^{-pb}}{p} + \frac{1}{p}\cdot\frac{1 - e^{-pb}}{p}.$$

For $p > 0$, $be^{-pb} = b/e^{pb} \to 0$ as $b \to \infty$ (exponential beats a power), and $e^{-pb} \to 0$. So the limit is $\dfrac{1}{p^2}$:

$$\mathcal{L}[x] = \frac{1}{p^2}, \qquad p > 0.$$

(For $p \leq 0$ the integrand $xe^{-px} \geq x$ does not tend to 0 and the integral diverges.)

((11)) $f = e^{ax}$. Here $e^{-px}e^{ax} = e^{-(p-a)x}$, so by the computation in (8) with p replaced by $p - a$,

$$\int_0^b e^{-(p-a)x}\,dx = \frac{1 - e^{-(p-a)b}}{p-a} \longrightarrow \frac{1}{p-a} \qquad (p > a),$$

and the integral diverges for $p \leq a$. Thus $\mathcal{L}[e^{ax}] = \dfrac{1}{p-a}$, $p > a$.

((12), (13)) Let $p > 0$ and put

$$S = \int_0^\infty e^{-px}\sin ax\,dx, \qquad C = \int_0^\infty e^{-px}\cos ax\,dx.$$

Both integrals converge (absolutely) by the comparison test, since $|e^{-px}\sin ax| \leq e^{-px}$ and $|e^{-px}\cos ax| \leq e^{-px}$ and $\int_0^\infty e^{-px}dx = 1/p$ converges. Integrate by parts with $dv = e^{-px}dx$, $v = -e^{-px}/p$. For S, take $u = \sin ax$, $du = a\cos ax\,dx$:

$$S = \Big[-\frac{e^{-px}\sin ax}{p}\Big]_0^\infty + \frac{a}{p}\int_0^\infty e^{-px}\cos ax\,dx = \frac{a}{p}\,C,$$

because $e^{-px}\sin ax \to 0$ as $x \to \infty$ and $\sin 0 = 0$. For C, take $u = \cos ax$, $du = -a\sin ax\,dx$:

$$C = \Big[-\frac{e^{-px}\cos ax}{p}\Big]_0^\infty - \frac{a}{p}\int_0^\infty e^{-px}\sin ax\,dx = \frac{1}{p} - \frac{a}{p}\,S,$$

because the boundary term at $x=0$ is $-(-1/p)=1/p$. Substituting $C=\frac{1}{p}-\frac{a}{p}S$ into $S=\frac{a}{p}C$:

$$S=\frac{a}{p}\Big(\frac{1}{p}-\frac{a}{p}S\Big)=\frac{a}{p^2}-\frac{a^2}{p^2}S \implies S\Big(1+\frac{a^2}{p^2}\Big)=\frac{a}{p^2} \implies S=\frac{a}{p^2+a^2}.$$

Then

$$C=\frac{1}{p}-\frac{a}{p}\cdot\frac{a}{p^2+a^2}=\frac{(p^2+a^2)-a^2}{p(p^2+a^2)}=\frac{p}{p^2+a^2}.$$

Hence $\mathcal{L}[\sin ax]=\dfrac{a}{p^2+a^2}$ and $\mathcal{L}[\cos ax]=\dfrac{p}{p^2+a^2}$ for $p>0$.

Answer. $\mathcal{L}[1]=\dfrac{1}{p}$, $\mathcal{L}[x]=\dfrac{1}{p^2}$, $\mathcal{L}[e^{ax}]=\dfrac{1}{p-a}$ $(p>a)$, $\mathcal{L}[\sin ax]=\dfrac{a}{p^2+a^2}$, $\mathcal{L}[\cos ax]=\dfrac{p}{p^2+a^2}$ (all for $p>0$ except (11), which needs $p>a$).

Problem 48-2 (Simmons)

Solution. Use $\sinh ax=\frac{1}{2}(e^{ax}-e^{-ax})$, $\cosh ax=\frac{1}{2}(e^{ax}+e^{-ax})$, linearity, and formula (11), $\mathcal{L}[e^{kx}]=1/(p-k)$ for $p>k$. Both $\mathcal{L}[e^{ax}]$ (needs $p>a$) and $\mathcal{L}[e^{-ax}]$ (needs $p>-a$) exist exactly when $p>|a|$.

(a)

$$\mathcal{L}[\sinh ax]=\tfrac{1}{2}\Big(\frac{1}{p-a}-\frac{1}{p+a}\Big)=\tfrac{1}{2}\cdot\frac{(p+a)-(p-a)}{(p-a)(p+a)}=\tfrac{1}{2}\cdot\frac{2a}{p^2-a^2}=\frac{a}{p^2-a^2}.$$

(b)

$$\mathcal{L}[\cosh ax]=\tfrac{1}{2}\Big(\frac{1}{p-a}+\frac{1}{p+a}\Big)=\tfrac{1}{2}\cdot\frac{(p+a)+(p-a)}{p^2-a^2}=\frac{p}{p^2-a^2}.$$

Answer. Shown: $\mathcal{L}[\sinh ax]=\dfrac{a}{p^2-a^2}$ and $\mathcal{L}[\cosh ax]=\dfrac{p}{p^2-a^2}$ for $p>|a|$.

Problem 48-3 (Simmons)

Solution. By the double-angle identities $\sin^2 ax=\frac{1}{2}(1-\cos 2ax)$ and $\cos^2 ax=\frac{1}{2}(1+\cos 2ax)$. Using linearity, $\mathcal{L}[1]=1/p$ and $\mathcal{L}[\cos 2ax]=\dfrac{p}{p^2+(2a)^2}=\dfrac{p}{p^2+4a^2}$ (formula (13) with a replaced by $2a$), for $p>0$:

$$\mathcal{L}[\sin^2 ax]=\frac{1}{2}\Big(\frac{1}{p}-\frac{p}{p^2+4a^2}\Big)=\frac{1}{2}\cdot\frac{(p^2+4a^2)-p^2}{p(p^2+4a^2)}=\frac{2a^2}{p(p^2+4a^2)},$$

$$\mathcal{L}[\cos^2 ax]=\frac{1}{2}\Big(\frac{1}{p}+\frac{p}{p^2+4a^2}\Big)=\frac{1}{2}\cdot\frac{2p^2+4a^2}{p(p^2+4a^2)}=\frac{p^2+2a^2}{p(p^2+4a^2)}.$$

Relation. The two transforms add up to

$$\mathcal{L}[\sin^2 ax]+\mathcal{L}[\cos^2 ax]=\frac{1}{p}=\mathcal{L}[1],$$

which is just $\mathcal{L}[\sin^2 ax+\cos^2 ax]=\mathcal{L}[1]$; and their difference is $\mathcal{L}[\cos^2 ax]-\mathcal{L}[\sin^2 ax]=\dfrac{p}{p^2+4a^2}=\mathcal{L}[\cos 2ax]$.

Answer. $\mathcal{L}[\sin^2 ax] = \frac{1}{2}\left(\frac{1}{p} - \frac{p}{p^2+4a^2}\right)$, $\mathcal{L}[\cos^2 ax] = \frac{1}{2}\left(\frac{1}{p} + \frac{p}{p^2+4a^2}\right)$; their sum is $\frac{1}{p} = \mathcal{L}[1]$.

Problem 48-4 (Simmons)

Solution. We use linearity with (8)–(13): $\mathcal{L}[1] = 1/p$, $\mathcal{L}[x^n] = n!/p^{n+1}$, $\mathcal{L}[e^{ax}] = 1/(p-a)$, $\mathcal{L}[\sin ax] = a/(p^2+a^2)$, $\mathcal{L}[\cos ax] = p/(p^2+a^2)$.

(a) $\mathcal{L}[10] = 10\,\mathcal{L}[1] = \frac{10}{p}$ $(p > 0)$.

(b) $\mathcal{L}[x^5] = \frac{5!}{p^6}$ and $\mathcal{L}[\cos 2x] = \frac{p}{p^2+4}$, so $\mathcal{L}[x^5 + \cos 2x] = \frac{5!}{p^6} + \frac{p}{p^2+4}$ $(= \frac{120}{p^6} + \frac{p}{p^2+4})$, $p > 0$.

(c) $\mathcal{L}[2e^{3x} - \sin 5x] = \frac{2}{p-3} - \frac{5}{p^2+25}$, $p > 3$.

(d) $4\sin x \cos x = 2\sin 2x$ (double-angle formula), so

$$\mathcal{L}[4\sin x\cos x + 2e^{-x}] = 2\cdot\frac{2}{p^2+4} + \frac{2}{p+1} = \frac{4}{p^2+4} + \frac{2}{p+1}, \qquad p > 0.$$

(e) $x^6\sin^2 3x + x^6\cos^2 3x = x^6(\sin^2 3x + \cos^2 3x) = x^6$, so $\mathcal{L} = \frac{6!}{p^7} = \frac{720}{p^7}$, $p > 0$.

Answer. (a) $\frac{10}{p}$; (b) $\frac{5!}{p^6} + \frac{p}{p^2+4}$; (c) $\frac{2}{p-3} - \frac{5}{p^2+25}$; (d) $\frac{4}{p^2+4} + \frac{2}{p+1}$; (e) $\frac{6!}{p^7}$.

Problem 48-5 (Simmons)

Solution. We match the given expressions against the table (8)–(13) and use linearity.

(a) $\mathcal{L}[x^3] = 3!/p^4 = 6/p^4$, so $\frac{30}{p^4} = 5\cdot\frac{6}{p^4} = \mathcal{L}[5x^3]$. Thus $f = 5x^3$.

(b) $\frac{2}{p+3} = 2\cdot\frac{1}{p-(-3)} = \mathcal{L}[2e^{-3x}]$. Thus $f = 2e^{-3x}$.

(c) $\mathcal{L}[x^2] = 2!/p^3 = 2/p^3$, so $\frac{4}{p^3} = \mathcal{L}[2x^2]$; and $\mathcal{L}[\sin 2x] = \frac{2}{p^2+4}$, so $\frac{6}{p^2+4} = 3\cdot\frac{2}{p^2+4} = \mathcal{L}[3\sin 2x]$. Thus $f = 2x^2 + 3\sin 2x$.

(d) Partial fractions: $\frac{1}{p^2+p} = \frac{1}{p(p+1)} = \frac{1}{p} - \frac{1}{p+1}$ (check: $\frac{(p+1)-p}{p(p+1)} = \frac{1}{p(p+1)}$). Since $\mathcal{L}[1] = 1/p$ and $\mathcal{L}[e^{-x}] = 1/(p+1)$, $f = 1 - e^{-x}$.

(e) $\frac{1}{p^4+p^2} = \frac{1}{p^2(p^2+1)} = \frac{1}{p^2} - \frac{1}{p^2+1}$ (check: $\frac{(p^2+1)-p^2}{p^2(p^2+1)} = \frac{1}{p^2(p^2+1)}$). Since $\mathcal{L}[x] = 1/p^2$ and $\mathcal{L}[\sin x] = 1/(p^2+1)$, $f = x - \sin x$.

Answer. (a) $5x^3$; (b) $2e^{-3x}$; (c) $2x^2 + 3\sin 2x$; (d) $1 - e^{-x}$; (e) $x - \sin x$.

Problem 48-6 (Simmons)

Solution. Formula (10) of the text says $\mathcal{L}[x^n] = \int_0^\infty e^{-px} x^n \, dx = n!/p^{n+1}$ for positive integers n (it is also true for $n = 0$, by (8)). Taking $p = 1$,

$$n! = \int_0^\infty e^{-x} x^n \, dx.$$

The right side makes sense for non-integer $n > -1$ as well (it converges at $x = 0$ because x^n is integrable there when $n > -1$, and at ∞ because $x^n e^{-x/2}$ is bounded). So the natural definition is

$$\tfrac{1}{2}! \;:=\; \int_0^\infty e^{-x} x^{1/2} \, dx.$$

This is consistent with the characteristic property of the factorial, $n! = n\,(n-1)!$: indeed integrating by parts ($u = x^{1/2}$, $dv = e^{-x}dx$),

$$\int_0^\infty e^{-x} x^{1/2} \, dx = \Big[-x^{1/2} e^{-x}\Big]_0^\infty + \frac{1}{2}\int_0^\infty e^{-x} x^{-1/2} \, dx = \frac{1}{2}\int_0^\infty e^{-x} x^{-1/2} \, dx,$$

i.e. $\frac{1}{2}! = \frac{1}{2}\,(-\frac{1}{2})!$ with $(-\frac{1}{2})! := \int_0^\infty e^{-x} x^{-1/2} dx$. By formula (4) of Section 49 with $p = 1$ and (5) there, $(-\frac{1}{2})! = \mathcal{L}[x^{-1/2}]\big|_{p=1} = \sqrt{\pi}$ (equivalently, substituting $x = s^2$ gives $2\int_0^\infty e^{-s^2} ds = 2 \cdot \frac{\sqrt{\pi}}{2}$). Therefore

$$\tfrac{1}{2}! = \tfrac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{2} \approx 0.886227.$$

With this definition $\mathcal{L}[x^{1/2}] = \dfrac{\frac{1}{2}!}{p^{3/2}} = \dfrac{\sqrt{\pi}}{2p^{3/2}}$, in line with formula (10).

Answer. $\frac{1}{2}! := \displaystyle\int_0^\infty e^{-x} x^{1/2} \, dx = \frac{\sqrt{\pi}}{2} \approx 0.8862.$

Section 49 A Few Remarks on the Theory

Problem 49-1 (Simmons)

Solution. Here $I = \int_0^\infty e^{-s^2} ds$; it converges since $e^{-s^2} \le e^{-s}$ for $s \ge 1$ and $\int_1^\infty e^{-s} ds < \infty$, so I is a finite positive number. Since the integrand of the double integral is a product, the product of the two single integrals equals the iterated (equivalently, double) integral over the first quadrant $Q = \{x \ge 0,\ y \ge 0\}$ (the integrand is positive, so the order of integration is immaterial).

In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, $dx\,dy = r\,\mathrm{d}r\,\mathrm{d}\theta$, and $x^2 + y^2 = r^2$; the quadrant Q corresponds to $0 \le r < \infty$, $0 \le \theta \le \pi/2$. Therefore

$$I^2 = \int_0^{\pi/2} \int_0^\infty e^{-r^2}\, r\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{\pi/2} \mathrm{d}\theta \cdot \int_0^\infty r e^{-r^2}\,\mathrm{d}r.$$

The inner integral is evaluated by the substitution $u = r^2$, $du = 2r\,\mathrm{d}r$:

$$\int_0^\infty r e^{-r^2}\,\mathrm{d}r = \lim_{b\to\infty} \Big[-\tfrac{1}{2} e^{-r^2}\Big]_0^b = \lim_{b\to\infty} \tfrac{1}{2}\left(1 - e^{-b^2}\right) = \tfrac{1}{2},$$

so

$$I^2 = \frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}.$$

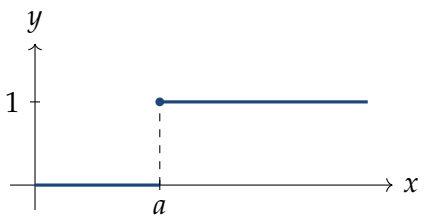
Since $I > 0$, $I = \sqrt{\pi/4} = \sqrt{\pi}/2$.

Answer. Shown: $I^2 = \pi/4$, hence $\displaystyle\int_0^\infty e^{-s^2} ds = \frac{\sqrt{\pi}}{2}$.

Problem 49-2 (Simmons)

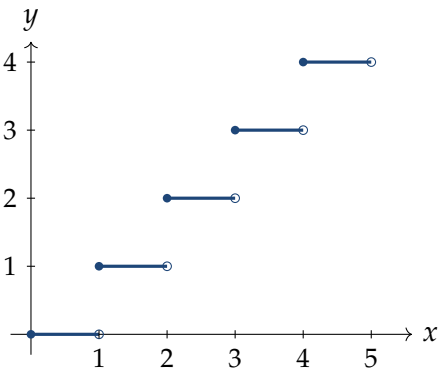
Solution. All four functions are piecewise continuous and bounded (hence of exponential order with $c = 0$), so each transform exists for $p > 0$. Throughout, $p > 0$.

(a) $u(x-a) = 0$ for $x < a$ and $= 1$ for $x \ge a$: a step of height 1 at $x = a$.



$$\mathcal{L}[u(x-a)] = \int_0^a e^{-px}\cdot 0\,\mathrm{d}x + \int_a^\infty e^{-px}\,\mathrm{d}x = \lim_{b\to\infty}\Big[-\frac{e^{-px}}{p}\Big]_a^b = \frac{e^{-ap}}{p}.$$

(b) $[x] = n$ for $n \le x < n+1$ ($n = 0, 1, 2, \ldots$): a staircase with unit jumps at the positive integers.



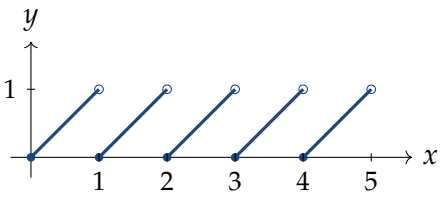
For $p > 0$ put $q = e^{-p} \in (0,1)$. On $[n, n+1)$ the integrand is ne^{-px} and $\int_n^{n+1} e^{-px}dx = \dfrac{e^{-pn} - e^{-p(n+1)}}{p} = \dfrac{q^n(1-q)}{p}$. Hence

$$\mathcal{L}[[x]] = \sum_{n=0}^{\infty} n \int_n^{n+1} e^{-px}\,\mathrm{d}x = \frac{1-q}{p}\sum_{n=1}^{\infty} nq^n.$$

(Termwise integration is legitimate: $\int_0^b$ is a finite sum of such pieces, and the series below converges.) Using $\sum_{n=1}^{\infty} nq^n = q\dfrac{d}{dq}\sum_{n=0}^{\infty} q^n = q\dfrac{d}{dq}\dfrac{1}{1-q} = \dfrac{q}{(1-q)^2}$ for $|q| < 1$,

$$\mathcal{L}[[x]] = \frac{1-q}{p}\cdot\frac{q}{(1-q)^2} = \frac{q}{p(1-q)} = \frac{e^{-p}}{p(1-e^{-p})} = \frac{1}{p\,(e^{p}-1)}.$$

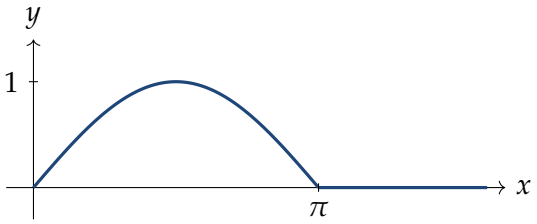
(c) $x - [x] = x - n$ on $[n, n+1)$: a sawtooth rising linearly from 0 to 1 on each unit interval and dropping back to 0 at each integer.



Both x and $[x]$ have transforms for $p > 0$, so by linearity, $\mathcal{L}[x] = 1/p^2$ (formula (9) of Section 48) and part (b):

$$\mathcal{L}[x - [x]] = \frac{1}{p^2} - \frac{1}{p(e^{p}-1)} = \frac{(e^{p}-1)-p}{p^2(e^{p}-1)} = \frac{e^{p}-1-p}{p^2(e^{p}-1)}.$$

(d) $f(x) = \sin x$ on $[0, \pi]$ (one arch of the sine) and 0 for $x > \pi$.



Only $[0,\pi]$ contributes: $\mathcal{L}[f]=\int_0^\pi e^{-px}\sin x\,dx$. Let J be this integral. Integrating by parts twice (first $u=\sin x$, $dv=e^{-px}dx$; then $u=\cos x$, $dv=e^{-px}dx$):

$$J=\Big[-\frac{e^{-px}\sin x}{p}\Big]_0^\pi+\frac{1}{p}\int_0^\pi e^{-px}\cos x\,dx=\frac{1}{p}\int_0^\pi e^{-px}\cos x\,dx,$$

$$\int_0^\pi e^{-px}\cos x\,dx=\Big[-\frac{e^{-px}\cos x}{p}\Big]_0^\pi-\frac{1}{p}\int_0^\pi e^{-px}\sin x\,dx=\frac{e^{-p\pi}+1}{p}-\frac{J}{p},$$

since $-\dfrac{e^{-p\pi}\cos\pi}{p}+\dfrac{\cos 0}{p}=\dfrac{e^{-p\pi}+1}{p}$. Substituting,

$$J=\frac{1+e^{-p\pi}}{p^2}-\frac{J}{p^2}\implies J\Big(1+\frac{1}{p^2}\Big)=\frac{1+e^{-p\pi}}{p^2}\implies J=\frac{1+e^{-\pi p}}{p^2+1}.$$

Answer. (a) $\dfrac{e^{-ap}}{p}$; (b) $\dfrac{1}{p(e^p-1)}$; (c) $\dfrac{e^p-1-p}{p^2(e^p-1)}$; (d) $\dfrac{1+e^{-\pi p}}{p^2+1}$ (all for $p>0$; graphs above).

Problem 49-3 (Simmons)

Solution. We must show that $\int_0^\infty e^{-px}e^{x^2}dx$ diverges for *every* real p. By the hint,

$$e^{-px}e^{x^2}=e^{x^2-px}=e^{(x-p/2)^2}\,e^{-p^2/4}.$$

Let $x_0=\max\{0,\ p/2+1\}$. For $x\ge x_0$ we have $x-p/2\ge 1$, so $(x-p/2)^2\ge 1$, and therefore

$$e^{-px}e^{x^2}\ \ge\ e^{1-p^2/4}=:m>0\qquad(x\ge x_0).$$

Since the integrand is positive (so the integral over $[0,x_0]$ is ≥ 0),

$$\int_0^b e^{-px}e^{x^2}\,dx\ \ge\ \int_{x_0}^b m\,dx=m\,(b-x_0)\longrightarrow\infty\quad(b\to\infty).$$

Hence the limit in the definition does not exist (it is $+\infty$) for any p, and $\mathcal{L}[e^{x^2}]$ does not exist for any value of p. (Consistent with the text: e^{x^2} is not of exponential order.)

Answer. Shown: $\int_0^\infty e^{-px+x^2}dx$ diverges for every p, because the integrand is bounded below by the positive constant $e^{1-p^2/4}$ for large x.

Problem 49-4 (Simmons)

Solution. $\mathcal{L}[x^{-1}]=\int_0^\infty e^{-px}x^{-1}dx$. The trouble is at $x=0$, where $1/x$ is not integrable, and this cannot be cured by the choice of p. Fix any real p. For $0<x\le 1$ we have $-px\ge -|p|$, hence $e^{-px}\ge e^{-|p|}$, and so for $0<\varepsilon<1$

$$\int_\varepsilon^1\frac{e^{-px}}{x}\,dx\ \ge\ e^{-|p|}\int_\varepsilon^1\frac{dx}{x}=e^{-|p|}\log\frac{1}{\varepsilon}\longrightarrow\infty\qquad(\varepsilon\to 0^+).$$

Thus the integral already diverges on $(0,1]$ (the integrand is positive on $(0,\infty)$, so the rest of the range cannot cancel this), for every p. Hence $\mathcal{L}[x^{-1}]$ does not exist for any p.

Answer. Shown: $\int_0^1 e^{-px}x^{-1}dx = \infty$ for every real p, so $\mathcal{L}[x^{-1}]$ does not exist.

Problem 49-5 (Simmons)

Solution. *Transform.* Since $f_\epsilon = 0$ for $x > \epsilon$, for $p \neq 0$

$$\mathcal{L}[f_\epsilon] = \int_0^\epsilon e^{-px} \cdot \frac{1}{\epsilon}\,\mathrm{d}x = \frac{1}{\epsilon}\Big[-\frac{e^{-px}}{p}\Big]_0^\epsilon = \frac{1-e^{-p\epsilon}}{p\epsilon},$$

valid for every $p \neq 0$ (the integral is over a finite interval, so no restriction is needed for existence).

Limit. Note first that the printed limit "$\epsilon \to \infty$" is a misprint for $\epsilon \to 0$, which is what the discussion that follows the problem ("$\delta(x) = \lim_{\epsilon\to 0} f_\epsilon(x)$", $\mathcal{L}[\delta] = 1$) requires. Indeed, for fixed $p > 0$, as $\epsilon \to \infty$ we have $e^{-p\epsilon} \to 0$ and $\mathcal{L}[f_\epsilon] \sim 1/(p\epsilon) \to 0$, not 1. So we prove

$$\lim_{\epsilon\to 0} \mathcal{L}[f_\epsilon] = 1.$$

Fix $p \neq 0$ and put $u = p\epsilon$, so $u \to 0$ as $\epsilon \to 0$ and $\mathcal{L}[f_\epsilon] = (1-e^{-u})/u$. This is $\dfrac{g(u)-g(0)}{u-0}$ with $g(u) = 1-e^{-u}$, $g(0) = 0$, so its limit is $g'(0) = e^{-0} = 1$ (equivalently, by L'Hôpital's rule, $\lim_{u\to 0} \dfrac{e^{-u}}{1} = 1$). Thus

$$\lim_{\epsilon\to 0} \frac{1-e^{-p\epsilon}}{p\epsilon} = 1.$$

Answer. Shown: $\mathcal{L}[f_\epsilon] = \dfrac{1-e^{-p\epsilon}}{p\epsilon}$, and this tends to 1 as $\epsilon \to 0$ (the printed "$\epsilon \to \infty$" should read $\epsilon \to 0$; as $\epsilon \to \infty$ it tends to 0 for $p > 0$).

Section 50 Applications to Differential Equations

Throughout we use the rules of the text: the shifting formula $\mathcal{L}[e^{ax}f(x)] = F(p-a)$ (formula (9)), the table of transform pairs, and, for $Y = \mathcal{L}[y]$,

$$\mathcal{L}[y'] = pY - y(0), \qquad \mathcal{L}[y''] = p^2Y - py(0) - y'(0).$$

Inverse transforms are taken with the standard table (uniqueness holds among continuous functions, which every solution of the differential equation is).

Problem 50-1 (Simmons)

Solution.

(a) Since $\mathcal{L}[x^5] = 5!/p^6$, the shift $p \to p+2$ (here $a = -2$) gives

$$\mathcal{L}[x^5e^{-2x}] = \frac{5!}{(p+2)^6} = \frac{120}{(p+2)^6}.$$

(b) Write $(1-x^2)e^{-x} = e^{-x} - x^2e^{-x}$. By linearity and the shift $p \to p+1$, using $\mathcal{L}[1] = 1/p$ and $\mathcal{L}[x^2] = 2!/p^3$,

$$\mathcal{L}[(1-x^2)e^{-x}] = \frac{1}{p+1} - \frac{2!}{(p+1)^3}.$$

(c) $\mathcal{L}[\cos 2x] = \dfrac{p}{p^2+4}$, and the shift $p \to p-3$ ($a = 3$) gives

$$\mathcal{L}[e^{3x}\cos 2x] = \frac{p-3}{(p-3)^2+4}.$$

Answer. (a) $\dfrac{5!}{(p+2)^6}$; (b) $\dfrac{1}{p+1} - \dfrac{2!}{(p+1)^3}$; (c) $\dfrac{p-3}{(p-3)^2+4}$.

Problem 50-2 (Simmons)

Solution.

(a) Since $\mathcal{L}[\sin 3x] = \dfrac{3}{p^2+9}$, the shift gives $\mathcal{L}[e^{-2x}\sin 3x] = \dfrac{3}{(p+2)^2+9}$. Therefore

$$\mathcal{L}^{-1}\left[\frac{6}{(p+2)^2+9}\right] = 2e^{-2x}\sin 3x.$$

(b) $\mathcal{L}[x^3] = 3!/p^4 = 6/p^4$, so $\mathcal{L}[e^{-3x}x^3] = 6/(p+3)^4$ and

$$\mathcal{L}^{-1}\left[\frac{12}{(p+3)^4}\right] = 2x^3e^{-3x}.$$

(c) Complete the square: $p^2+2p+5 = (p+1)^2+4$, and write the numerator as $p+3 = (p+1)+2$:

$$\frac{p+3}{p^2+2p+5} = \frac{p+1}{(p+1)^2+2^2} + \frac{2}{(p+1)^2+2^2}.$$

By the shift ($a=-1$) applied to $\mathcal{L}[\cos 2x] = p/(p^2+4)$ and $\mathcal{L}[\sin 2x] = 2/(p^2+4)$,

$$\mathcal{L}^{-1}\left[\frac{p+3}{p^2+2p+5}\right] = e^{-x}\cos 2x + e^{-x}\sin 2x.$$

Answer. (a) $2e^{-2x}\sin 3x$; (b) $2x^3e^{-3x}$; (c) $e^{-x}\cos 2x + e^{-x}\sin 2x$.

Problem 50-3 (Simmons)

Solution. Let $Y = \mathcal{L}[y]$ in each part.

(a) Transforming, $pY - y(0) + Y = \dfrac{3}{p-2}$, and $y(0)=0$ gives

$$(p+1)Y = \frac{3}{p-2} \quad\Longrightarrow\quad Y = \frac{3}{(p-2)(p+1)}.$$

Partial fractions: $\dfrac{3}{(p-2)(p+1)} = \dfrac{A}{p-2} + \dfrac{B}{p+1}$ with $3 = A(p+1)+B(p-2)$; $p=2$ gives $A=1$, $p=-1$ gives $B=-1$. So $Y = \dfrac{1}{p-2} - \dfrac{1}{p+1}$ and

$$y = e^{2x} - e^{-x}.$$

(b) Transforming with $y(0)=0$, $y'(0)=3$:

$$\big(p^2Y - 3\big) - 4pY + 4Y = 0 \quad\Longrightarrow\quad (p-2)^2Y = 3 \quad\Longrightarrow\quad Y = \frac{3}{(p-2)^2}.$$

Since $\mathcal{L}[x] = 1/p^2$, the shift gives $\mathcal{L}[xe^{2x}] = 1/(p-2)^2$, hence

$$y = 3xe^{2x}.$$

(c) Transforming, with $\mathcal{L}[2] = 2/p$:

$$\big(p^2Y - 1\big) + 2pY + 2Y = \frac{2}{p} \quad\Longrightarrow\quad (p^2+2p+2)Y = 1 + \frac{2}{p} = \frac{p+2}{p}.$$

Thus $Y = \dfrac{p+2}{p(p^2+2p+2)}$. Write $\dfrac{p+2}{p(p^2+2p+2)} = \dfrac{A}{p} + \dfrac{Bp+C}{p^2+2p+2}$, so

$$p+2 = A(p^2+2p+2) + (Bp+C)p.$$

Comparing coefficients: p^2: $A+B=0$; p^1: $2A+C=1$; p^0: $2A=2$. Hence $A=1$, $B=-1$, $C=-1$:

$$Y = \frac{1}{p} - \frac{p+1}{(p+1)^2+1}.$$

By the shift, $\mathcal{L}^{-1}[(p+1)/((p+1)^2+1)] = e^{-x}\cos x$, so

$$y = 1 - e^{-x}\cos x.$$

(d) Using $\mathcal{L}[3x^2] = 6/p^3$:

$$(p^2Y - 1) + pY = \frac{6}{p^3} \quad\Longrightarrow\quad p(p+1)Y = 1 + \frac{6}{p^3} \quad\Longrightarrow\quad Y = \frac{1}{p(p+1)} + \frac{6}{p^4(p+1)}.$$

Repeatedly use $\dfrac{1}{p^k(p+1)} = \dfrac{1}{p^k} - \dfrac{1}{p^{k-1}(p+1)}$ (valid because $\frac{1}{p^k} - \frac{1}{p^{k-1}(p+1)} = \frac{(p+1)-p}{p^k(p+1)}$):

$$\begin{aligned}
\frac{1}{p(p+1)} &= \frac{1}{p} - \frac{1}{p+1},\\
\frac{1}{p^2(p+1)} &= \frac{1}{p^2} - \frac{1}{p} + \frac{1}{p+1},\\
\frac{1}{p^3(p+1)} &= \frac{1}{p^3} - \frac{1}{p^2} + \frac{1}{p} - \frac{1}{p+1},\\
\frac{1}{p^4(p+1)} &= \frac{1}{p^4} - \frac{1}{p^3} + \frac{1}{p^2} - \frac{1}{p} + \frac{1}{p+1}.
\end{aligned}$$

Therefore

$$Y = \Big(\frac{1}{p} - \frac{1}{p+1}\Big) + 6\Big(\frac{1}{p^4} - \frac{1}{p^3} + \frac{1}{p^2} - \frac{1}{p} + \frac{1}{p+1}\Big) = -\frac{5}{p} + \frac{6}{p^2} - \frac{6}{p^3} + \frac{6}{p^4} + \frac{5}{p+1}.$$

Using $\mathcal{L}[x^n] = n!/p^{n+1}$ (so $6/p^2 \leftrightarrow 6x$, $6/p^3 \leftrightarrow 3x^2$, $6/p^4 \leftrightarrow x^3$):

$$y = -5 + 6x - 3x^2 + x^3 + 5e^{-x}.$$

(e) By the shift, $\mathcal{L}[3e^{-x}\sin x] = \dfrac{3}{(p+1)^2+1} = \dfrac{3}{p^2+2p+2}$. Transforming, with $y(0) = 0$, $y'(0) = 3$:

$$(p^2Y - 3) + 2pY + 5Y = \frac{3}{p^2+2p+2} \quad\Longrightarrow\quad (p^2+2p+5)Y = 3 + \frac{3}{p^2+2p+2}.$$

Put $u = (p+1)^2$, so $p^2+2p+2 = u+1$ and $p^2+2p+5 = u+4$. Then

$$Y = \frac{3}{u+4} + \frac{3}{(u+1)(u+4)}, \qquad \frac{3}{(u+1)(u+4)} = \frac{1}{u+1} - \frac{1}{u+4},$$

the latter because $\frac{1}{u+1} - \frac{1}{u+4} = \frac{3}{(u+1)(u+4)}$. Hence

$$Y = \frac{2}{(p+1)^2+4} + \frac{1}{(p+1)^2+1}.$$

Since $\mathcal{L}[e^{-x}\sin 2x] = 2/((p+1)^2+4)$ and $\mathcal{L}[e^{-x}\sin x] = 1/((p+1)^2+1)$,

$$y = e^{-x}\sin 2x + e^{-x}\sin x.$$

Check. Each solution was substituted into its differential equation and initial conditions by SymPy (see `verify/sec50.py`); e.g. in (a), $y' + y = 2e^{2x} + e^{-x} + e^{2x} - e^{-x} = 3e^{2x}$ and $y(0) = 0$.

Answer. (a) $y = e^{2x} - e^{-x}$; (b) $y = 3xe^{2x}$; (c) $y = 1 - e^{-x}\cos x$; (d) $y = -5 + 6x - 3x^2 + x^3 + 5e^{-x}$; (e) $y = e^{-x}\sin 2x + e^{-x}\sin x$.

Problem 50-4 (Simmons)

Solution. Let $Y = \mathcal{L}[y]$. Then

$$(p^2Y - py_0 - y_0') - 2a(pY - y_0) + a^2Y = 0,$$

so

$$(p^2 - 2ap + a^2)Y = py_0 + y_0' - 2ay_0.$$

Since $p^2 - 2ap + a^2 = (p-a)^2$ and $py_0 + y_0' - 2ay_0 = y_0(p-a) + (y_0' - ay_0)$,

$$Y = \frac{y_0}{p-a} + \frac{y_0' - ay_0}{(p-a)^2}.$$

Now $\mathcal{L}[e^{ax}] = 1/(p-a)$ and, by the shift applied to $\mathcal{L}[x] = 1/p^2$, $\mathcal{L}[xe^{ax}] = 1/(p-a)^2$. Hence

$$y = y_0e^{ax} + (y_0' - ay_0)xe^{ax} = \bigl[y_0 + (y_0' - ay_0)x\bigr]e^{ax}.$$

With $c_1 = y_0$ and $c_2 = y_0' - ay_0$ arbitrary, this is the general solution $y = (c_1 + c_2x)e^{ax}$ of Section 17 for a double root a.

Answer. $y(x) = y_0e^{ax} + (y_0' - ay_0)\,xe^{ax}$.

Problem 50-5 (Simmons)

Solution. Here (3) is the formula $\mathcal{L}[y'] = p\mathcal{L}[y] - y(0)$. (In the integral the dummy variable is renamed: $\int_0^x f(t)\,dt$.)

Proof of the formula. Let f be piecewise continuous and of exponential order, and put

$$g(x) = \int_0^x f(t)\,dt.$$

Then g is continuous, $g(0) = 0$, $g'(x) = f(x)$ (at every point of continuity of f), and g is of exponential order. By (3),

$$\mathcal{L}[g'] = p\mathcal{L}[g] - g(0), \qquad \text{i.e.} \qquad F(p) = p\mathcal{L}[g] - 0.$$

Dividing by p ($p > 0$) gives $\mathcal{L}[g] = F(p)/p$:

$$\mathcal{L}\left[\int_0^x f(t)\,dt\right] = \frac{F(p)}{p}. \qquad \blacksquare$$

Inverse transform of $1/(p(p+1))$, first way (the integral formula). Write $\dfrac{1}{p(p+1)} = \dfrac{1}{p}\cdot F(p)$ with $F(p) = \dfrac{1}{p+1} = \mathcal{L}[e^{-x}]$. By the formula just proved,

$$\mathcal{L}^{-1}\left[\frac{1}{p(p+1)}\right] = \int_0^x e^{-t}\,dt = \Bigl[-e^{-t}\Bigr]_0^x = 1 - e^{-x}.$$

Second way (partial fractions). $\dfrac{1}{p(p+1)} = \dfrac{1}{p} - \dfrac{1}{p+1}$ (since $\frac{(p+1)-p}{p(p+1)} = \frac{1}{p(p+1)}$), so

$$\mathcal{L}^{-1}\left[\frac{1}{p(p+1)}\right] = \mathcal{L}^{-1}\left[\frac{1}{p}\right] - \mathcal{L}^{-1}\left[\frac{1}{p+1}\right] = 1 - e^{-x}.$$

The two methods agree.

Answer. Shown: $\mathcal{L}\left[\int_0^x f\right] = F(p)/p$; $\mathcal{L}^{-1}\left[\dfrac{1}{p(p+1)}\right] = 1 - e^{-x}$ by both methods.

Problem 50-6 (Simmons)

Solution. Let $Y = \mathcal{L}[y]$. By Problem 5, $\mathcal{L}\left[\int_0^x y\,dt\right] = Y/p$, and $\mathcal{L}[y'] = pY - y(0) = pY$. Transforming the equation:

$$pY + 4Y + \frac{5}{p}Y = \frac{1}{p+1} \quad\Longrightarrow\quad \frac{p^2+4p+5}{p}\,Y = \frac{1}{p+1} \quad\Longrightarrow\quad Y = \frac{p}{(p+1)(p^2+4p+5)}.$$

Partial fractions: $\dfrac{p}{(p+1)(p^2+4p+5)} = \dfrac{A}{p+1} + \dfrac{Bp+C}{p^2+4p+5}$, so

$$p = A(p^2+4p+5) + (Bp+C)(p+1).$$

Coefficients: p^2: $A + B = 0$; p^0: $5A + C = 0$; p^1: $4A + B + C = 1$. Putting $p = -1$ in the identity gives $-1 = 2A$, so $A = -\frac{1}{2}$, $B = \frac{1}{2}$, $C = \frac{5}{2}$ (and indeed $4A + B + C = -2 + \frac{1}{2} + \frac{5}{2} = 1$). Since $p^2 + 4p + 5 = (p+2)^2 + 1$ and $\frac{1}{2}p + \frac{5}{2} = \frac{1}{2}(p+2) + \frac{3}{2}$,

$$Y = -\frac{1}{2}\cdot\frac{1}{p+1} + \frac{1}{2}\cdot\frac{p+2}{(p+2)^2+1} + \frac{3}{2}\cdot\frac{1}{(p+2)^2+1}.$$

By the shift ($a = -2$) applied to $\cos x$ and $\sin x$, and $\mathcal{L}[e^{-x}] = 1/(p+1)$:

$$y = -\frac{1}{2}e^{-x} + \frac{1}{2}e^{-2x}\cos x + \frac{3}{2}e^{-2x}\sin x.$$

Check. SymPy confirms $y(0) = 0$ and that $y' + 4y + 5\int_0^x y\,dt - e^{-x} \equiv 0$ for this y.

Answer. $y = \dfrac{3}{2}e^{-2x}\sin x + \dfrac{1}{2}e^{-2x}\cos x - \dfrac{1}{2}e^{-x}$.

Section 51 Derivatives and Integrals of Laplace Transforms

We use the formulas of the text, with $F(p)=\mathcal{L}[f(x)]$ and $Y=\mathcal{L}[y]$:

$$\mathcal{L}[(-1)^n x^n f(x)] = F^{(n)}(p) \quad \text{(formula (4))},$$
$$\mathcal{L}[xy] = -Y', \qquad \mathcal{L}[xy'] = -\frac{d}{dp}[pY - y(0)], \qquad \mathcal{L}[xy''] = -\frac{d}{dp}[p^2Y - py(0) - y'(0)],$$
$$\mathcal{L}\left[\frac{f(x)}{x}\right] = \int_p^\infty F(s)\,ds, \qquad \int_0^\infty \frac{f(x)}{x}\,dx = \int_0^\infty F(p)\,dp.$$

Problem 51-1 (Simmons)

Solution. Since $\mathcal{L}[\cos ax] = p/(p^2+a^2)$, formula (2), $\mathcal{L}[-xf(x)] = F'(p)$, gives $\mathcal{L}[x\cos ax] = -\dfrac{d}{dp}\dfrac{p}{p^2+a^2}$. By the quotient rule,

$$\mathcal{L}[x\cos ax] = -\frac{(p^2+a^2)\cdot 1 - p\cdot 2p}{(p^2+a^2)^2} = -\frac{a^2-p^2}{(p^2+a^2)^2} = \frac{p^2-a^2}{(p^2+a^2)^2}. \qquad \blacksquare$$

Now write the numerator as $p^2-a^2 = (p^2+a^2) - 2a^2$:

$$\frac{p^2-a^2}{(p^2+a^2)^2} = \frac{1}{p^2+a^2} - \frac{2a^2}{(p^2+a^2)^2} \quad\Longrightarrow\quad \frac{1}{(p^2+a^2)^2} = \frac{1}{2a^2}\left[\frac{1}{p^2+a^2} - \frac{p^2-a^2}{(p^2+a^2)^2}\right].$$

Using $\mathcal{L}^{-1}\left[\dfrac{1}{p^2+a^2}\right] = \dfrac{\sin ax}{a}$ and the result just proved,

$$\mathcal{L}^{-1}\left[\frac{1}{(p^2+a^2)^2}\right] = \frac{1}{2a^2}\left(\frac{\sin ax}{a} - x\cos ax\right).$$

Answer. $\mathcal{L}[x\cos ax] = \dfrac{p^2-a^2}{(p^2+a^2)^2}$; $\mathcal{L}^{-1}\left[\dfrac{1}{(p^2+a^2)^2}\right] = \dfrac{1}{2a^2}\left(\dfrac{\sin ax}{a} - x\cos ax\right)$.

Problem 51-2 (Simmons)

Solution.

(a) By (3), $\mathcal{L}[x^2 f(x)] = F''(p)$, with $F(p) = \dfrac{a}{p^2+a^2}$. Then

$$F'(p) = -\frac{2ap}{(p^2+a^2)^2},$$

$$F''(p) = -\frac{2a(p^2+a^2)^2 - 2ap\cdot 2(p^2+a^2)\cdot 2p}{(p^2+a^2)^4} = -\frac{2a(p^2+a^2) - 8ap^2}{(p^2+a^2)^3} = \frac{6ap^2-2a^3}{(p^2+a^2)^3}.$$

Hence

$$\mathcal{L}[x^2\sin ax] = \frac{6ap^2-2a^3}{(p^2+a^2)^3}.$$

(b) Write $x^{3/2} = x \cdot x^{1/2}$ and use Example 2 of the text, $\mathcal{L}[x^{1/2}] = \dfrac{\sqrt{\pi}}{2p^{3/2}}$. By (2),

$$\mathcal{L}[x^{3/2}] = -\frac{d}{dp}\left(\frac{\sqrt{\pi}}{2}\,p^{-3/2}\right) = -\frac{\sqrt{\pi}}{2}\cdot\left(-\frac{3}{2}\right)p^{-5/2} = \frac{3\sqrt{\pi}}{4\,p^{5/2}} = \frac{3}{4p^2}\sqrt{\frac{\pi}{p}}.$$

(This agrees with $\Gamma(\tfrac{5}{2})/p^{5/2}$, as $\Gamma(\tfrac{5}{2}) = \tfrac{3}{2}\cdot\tfrac{1}{2}\sqrt{\pi} = \tfrac{3}{4}\sqrt{\pi}$.)

Answer. (a) $\dfrac{6ap^2 - 2a^3}{(p^2+a^2)^3}$; (b) $\dfrac{3}{4p^2}\sqrt{\dfrac{\pi}{p}}$.

Problem 51-3 (Simmons)

Solution. In both parts the coefficients are first-degree polynomials, so the transformed equation is of first order in Y. Throughout, $y(0) = 0$ while $y'(0)$ is an unspecified constant; it disappears on differentiation with respect to p.

(a) Term by term (with $y(0) = 0$):

$$\begin{aligned}
\mathcal{L}[xy''] &= -\frac{d}{dp}\left[p^2Y - y'(0)\right] = -2pY - p^2Y',\\
\mathcal{L}[3xy'] &= -3\frac{d}{dp}\left[pY\right] = -3Y - 3pY',\\
\mathcal{L}[-y'] &= -pY,\\
\mathcal{L}[-4xy] &= 4Y',\\
\mathcal{L}[-9y] &= -9Y.
\end{aligned}$$

Adding and setting the sum equal to 0:

$$(-p^2 - 3p + 4)Y' + (-2p - 3 - p - 9)Y = 0 \quad\Longrightarrow\quad -(p^2 + 3p - 4)Y' = (3p + 12)Y.$$

Since $p^2 + 3p - 4 = (p+4)(p-1)$ and $3p + 12 = 3(p+4)$, we may cancel $p+4$ (we consider $p > 1$) and get

$$-(p-1)Y' = 3Y \quad\Longrightarrow\quad \frac{dY}{Y} = -\frac{3\,dp}{p-1} \quad\Longrightarrow\quad Y = \frac{C}{(p-1)^3}.$$

(The solution $Y \equiv 0$ is the case $C = 0$.) Since $\mathcal{L}[x^2] = 2/p^3$, the shift gives $\mathcal{L}[x^2e^x] = 2/(p-1)^3$, so

$$y = cx^2e^x \qquad (c = C/2).$$

Check. For $y = x^2e^x$: $y' = (x^2+2x)e^x$, $y'' = (x^2+4x+2)e^x$, and $xy'' + (3x-1)y' - (4x+9)y = e^x\left[x^3 + 4x^2 + 2x + 3x^3 + 6x^2 - x^2 - 2x - 4x^3 - 9x^2\right] = 0$; also $y(0) = 0$.

(b) Now the right side is $\mathcal{L}[3e^{-x}] = 3/(p+1)$, and (with $y(0) = 0$)

$$\begin{aligned}
\mathcal{L}[xy''] &= -2pY - p^2Y',\\
\mathcal{L}[2xy'] &= -2\frac{d}{dp}\left[pY\right] = -2Y - 2pY',\\
\mathcal{L}[3y'] &= 3pY,\\
\mathcal{L}[xy] &= -Y',
\end{aligned}$$

$$\mathcal{L}[3y] = 3Y.$$

Adding:

$$-(p^2+2p+1)Y' + (-2p-2+3p+3)Y = \frac{3}{p+1} \quad\Longrightarrow\quad -(p+1)^2Y' + (p+1)Y = \frac{3}{p+1}.$$

Divide by $-(p+1)^2$:

$$Y' - \frac{1}{p+1}Y = -\frac{3}{(p+1)^3}.$$

This is linear with integrating factor $e^{-\int dp/(p+1)} = \dfrac{1}{p+1}$:

$$\frac{d}{dp}\left[\frac{Y}{p+1}\right] = -\frac{3}{(p+1)^4} \quad\Longrightarrow\quad \frac{Y}{p+1} = \frac{1}{(p+1)^3} + C, \qquad Y = \frac{1}{(p+1)^2} + C(p+1).$$

The transform of a piecewise continuous function of exponential order tends to 0 as $p \to \infty$; $C(p+1)$ does not, so $C = 0$ and

$$Y = \frac{1}{(p+1)^2}.$$

Since $\mathcal{L}[x] = 1/p^2$, the shift gives $\mathcal{L}[xe^{-x}] = 1/(p+1)^2$, hence

$$y = xe^{-x}.$$

Check. $y' = (1-x)e^{-x}$, $y'' = (x-2)e^{-x}$, so $xy'' + (2x+3)y' + (x+3)y = e^{-x}\left[x^2 - 2x + (2x+3)(1-x) + x^2 + 3x\right] = e^{-x}\left[x^2 - 2x + 3 - x - 2x^2 + x^2 + 3x\right] = 3e^{-x}$; and $y(0) = 0$.

Answer. (a) $y = cx^2e^x$; (b) $y = xe^{-x}$.

Problem 51-4 (Simmons)

Solution. Transform the equation. By formula 50-(4), $\mathcal{L}[y''] = p^2Y - py_0 - y_0'$, and by formula (3) of this section, with $f = y$, $\mathcal{L}[x^2y] = Y''(p)$. Therefore

$$(p^2Y - py_0 - y_0') + Y'' = 0, \qquad \text{i.e.} \qquad Y'' + p^2Y = py_0 + y_0'. \qquad \blacksquare$$

This has exactly the form of the original equation (second order, with the coefficient x^2 replaced by p^2, now with a right-hand side), so transforming has not simplified the problem. By contrast, in Example 3 and Problem 3 the coefficients of y'', y', y have degree at most 1, so each term $x\,y^{(k)}$ produces only a first derivative of a transform; the term xy'' then contributes $-p^2Y'$ and the second derivative Y'' never appears, which is why those equations reduce to first order in Y.

Answer. Shown: $Y'' + p^2Y = py_0 + y_0'$.

Problem 51-5 (Simmons)

Solution. We use (12): $\displaystyle\int_0^\infty \frac{f(x)}{x}\,dx = \int_0^\infty F(p)\,dp$, valid when the left side exists. In both parts $f(x)/x$ has a finite limit as $x \to 0^+$ (namely $b-a$, resp. b) and decays exponentially, so the integral exists.

(a) Here $f = e^{-ax} - e^{-bx}$ and $F(p) = \dfrac{1}{p+a} - \dfrac{1}{p+b}$. Thus

$$\int_0^\infty F(p)\,dp = \Big[\log(p+a) - \log(p+b)\Big]_0^\infty = \lim_{p\to\infty} \log\frac{p+a}{p+b} - \log\frac{a}{b} = 0 - \log\frac{a}{b} = \log\frac{b}{a}.$$

(b) Here $f = e^{-ax}\sin bx$ and, by the shift, $F(p) = \dfrac{b}{(p+a)^2+b^2}$. Thus

$$\int_0^\infty \frac{b\,dp}{(p+a)^2+b^2} = \left[\tan^{-1}\frac{p+a}{b}\right]_0^\infty = \frac{\pi}{2} - \tan^{-1}\frac{a}{b}.$$

For $a, b > 0$, $\tan^{-1}(a/b) + \tan^{-1}(b/a) = \pi/2$ (the angles of a right triangle with legs a, b), so $\frac{\pi}{2} - \tan^{-1}\frac{a}{b} = \tan^{-1}\frac{b}{a}$.

Answer. (a) $\log\dfrac{b}{a}$; (b) $\tan^{-1}\dfrac{b}{a}$.

Problem 51-6 (Simmons)

Solution. From Example 3 of the text, $J_0(x) = \sum_{n\ge 0} \dfrac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$ and

$$\mathcal{L}[J_0(x)] = \frac{1}{\sqrt{p^2+1}} \qquad (p > 0).$$

(a) By definition, $\mathcal{L}[J_0] = \int_0^\infty e^{-px} J_0(x)\,dx$. Formally letting $p \to 0^+$ on both sides,

$$\int_0^\infty J_0(x)\,dx = \lim_{p\to 0^+} \frac{1}{\sqrt{p^2+1}} = 1. \qquad \blacksquare$$

(b) Let $g(x) = \dfrac{1}{\pi}\displaystyle\int_0^\pi \cos(x\cos t)\,dt$. Transform g formally, interchanging $\mathcal{L}$ with the t-integral, and use $\mathcal{L}[\cos kx] = p/(p^2+k^2)$ with $k = \cos t$:

$$\mathcal{L}[g] = \frac{1}{\pi}\int_0^\pi \frac{p\,dt}{p^2+\cos^2 t}.$$

Since the integrand has period π and is symmetric about $\pi/2$ ($\cos^2(\pi - t) = \cos^2 t$), $\int_0^\pi = 2\int_0^{\pi/2}$. On $0 \le t < \pi/2$ put $u = \tan t$, so that $dt = \dfrac{du}{1+u^2}$ and $\cos^2 t = \dfrac{1}{1+u^2}$:

$$\int_0^{\pi/2} \frac{dt}{p^2+\cos^2 t} = \int_0^\infty \frac{du}{(1+u^2)\left(p^2 + \frac{1}{1+u^2}\right)} = \int_0^\infty \frac{du}{p^2(1+u^2)+1} = \int_0^\infty \frac{du}{p^2u^2 + (p^2+1)}.$$

With $v = pu/\sqrt{p^2+1}$ ($du = \dfrac{\sqrt{p^2+1}}{p}\,dv$) this is

$$\frac{\sqrt{p^2+1}}{p}\cdot\frac{1}{p^2+1}\int_0^\infty \frac{dv}{v^2+1} = \frac{1}{p\sqrt{p^2+1}}\cdot\frac{\pi}{2}.$$

Therefore $\displaystyle\int_0^{\pi} \frac{dt}{p^2+\cos^2 t} = \frac{\pi}{p\sqrt{p^2+1}}$ and

$$\mathcal{L}[g] = \frac{1}{\pi}\cdot p \cdot \frac{\pi}{p\sqrt{p^2+1}} = \frac{1}{\sqrt{p^2+1}} = \mathcal{L}[J_0].$$

Both functions are continuous with the same transform, so $g = J_0$:

$$J_0(x) = \frac{1}{\pi}\int_0^{\pi} \cos(x\cos t)\,\mathrm{d}t. \qquad \blacksquare$$

Answer. Shown formally: (a) $\int_0^{\infty} J_0\,\mathrm{d}x = \lim_{p\to 0} 1/\sqrt{p^2+1} = 1$; (b) both sides have transform $1/\sqrt{p^2+1}$.

Problem 51-7 (Simmons)

Solution. In each part we take the Laplace transform with respect to x (with t a parameter), interchange $\mathcal{L}$ and the t-integral formally, and recognize the result.

(a) Since $\mathcal{L}_x[\sin xt] = \dfrac{t}{p^2+t^2}$,

$$\mathcal{L}[f] = \int_0^{\infty} \frac{1}{t}\cdot\frac{t}{p^2+t^2}\,\mathrm{d}t = \int_0^{\infty}\frac{dt}{p^2+t^2} = \frac{1}{p}\left[\tan^{-1}\frac{t}{p}\right]_0^{\infty} = \frac{\pi}{2p} = \mathcal{L}\left[\frac{\pi}{2}\right].$$

Hence $f(x) = \pi/2$.

(b) Since $\mathcal{L}_x[\cos xt] = \dfrac{p}{p^2+t^2}$,

$$\mathcal{L}[f] = \int_0^{\infty} \frac{p}{(p^2+t^2)(1+t^2)}\,\mathrm{d}t.$$

For $p \neq 1$, $\dfrac{1}{(p^2+t^2)(1+t^2)} = \dfrac{1}{p^2-1}\left[\dfrac{1}{1+t^2} - \dfrac{1}{p^2+t^2}\right]$ (since the bracket equals $\frac{p^2-1}{(1+t^2)(p^2+t^2)}$), so

$$\mathcal{L}[f] = \frac{p}{p^2-1}\left[\frac{\pi}{2} - \frac{\pi}{2p}\right] = \frac{p}{p^2-1}\cdot\frac{\pi(p-1)}{2p} = \frac{\pi}{2(p+1)}.$$

(For $p = 1$ directly, $\int_0^{\infty}\frac{dt}{(1+t^2)^2} = \frac{\pi}{4} = \frac{\pi}{2(1+1)}$, so the formula holds for all $p > 0$.) Since $\mathcal{L}\left[\frac{\pi}{2}e^{-x}\right] = \dfrac{\pi}{2(p+1)}$, we conclude

$$f(x) = \frac{\pi}{2}e^{-x}.$$

Answer. Shown formally: (a) $\mathcal{L}[f] = \pi/(2p)$, so $f = \pi/2$; (b) $\mathcal{L}[f] = \pi/(2(p+1))$, so $f = \frac{\pi}{2}e^{-x}$.

Problem 51-8 (Simmons)

Solution.

(a) Split the range of integration into periods:

$$F(p)=\int_0^{\infty} e^{-px}f(x)\,\mathrm{d}x=\sum_{n=0}^{\infty}\int_{na}^{(n+1)a} e^{-px}f(x)\,\mathrm{d}x.$$

In the nth integral put $x=u+na$ $(0\le u\le a)$. Then $f(u+na)=f(u)$ by periodicity and $e^{-px}=e^{-pna}e^{-pu}$, so

$$\int_{na}^{(n+1)a} e^{-px}f(x)\,\mathrm{d}x=e^{-npa}\int_0^{a} e^{-pu}f(u)\,\mathrm{d}u.$$

Hence, with $I=\int_0^a e^{-pu}f(u)\,\mathrm{d}u$ independent of n,

$$F(p)=I\sum_{n=0}^{\infty}\left(e^{-ap}\right)^n=\frac{I}{1-e^{-ap}}\qquad (p>0,\ \text{since } e^{-ap}<1),$$

which is the stated formula. ∎

(b) The function has period $a=2$ and equals 1 on $[0,1)$, 0 on $[1,2)$. By (a),

$$F(p)=\frac{1}{1-e^{-2p}}\int_0^1 e^{-px}\,\mathrm{d}x=\frac{1}{1-e^{-2p}}\cdot\frac{1-e^{-p}}{p}.$$

As $1-e^{-2p}=(1-e^{-p})(1+e^{-p})$, the factor $1-e^{-p}$ cancels:

$$F(p)=\frac{1}{p(1+e^{-p})}.$$

Remark. In the book's answer key this result appears under the label "8. (a)"; it is the answer to part (b), as computed here (part (a) is a proof).

Answer. (a) Shown: $F(p)=\dfrac{1}{1-e^{-ap}}\displaystyle\int_0^a e^{-px}f(x)\,\mathrm{d}x$; (b) $F(p)=\dfrac{1}{p\,(1+e^{-p})}$.

Section 52 Convolutions and Abel's Mechanical Problem

Throughout this section $\mathcal{L}[f] = F(p)$, $\mathcal{L}[g] = G(p)$, and we use the convolution theorem

$$\mathcal{L}\left[\int_0^x f(x-t)\,g(t)\,dt\right] = F(p)\,G(p),$$

together with the standard transforms $\mathcal{L}[x^n] = n!/p^{n+1}$, $\mathcal{L}[e^{ax}] = 1/(p-a)$, $\mathcal{L}[\sin ax] = a/(p^2+a^2)$, $\mathcal{L}[\cos ax] = p/(p^2+a^2)$ and the shift rule $\mathcal{L}[e^{-x}\varphi(x)] = \Phi(p+1)$.

Problem 52-1 (Simmons)

Solution. Take $a > 0$. Since $\mathcal{L}[\sin ax] = \dfrac{a}{p^2+a^2}$, we have $\dfrac{1}{(p^2+a^2)^2} = \mathcal{L}\left[\dfrac{\sin ax}{a}\right]\mathcal{L}\left[\dfrac{\sin ax}{a}\right]$, so by the convolution theorem

$$\mathcal{L}^{-1}\left[\frac{1}{(p^2+a^2)^2}\right] = \frac{1}{a^2}\int_0^x \sin a(x-t)\,\sin at\,dt.$$

Use $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ with $A = a(x-t)$, $B = at$, so $A - B = a(x-2t)$ and $A + B = ax$:

$$\begin{aligned}\int_0^x \sin a(x-t)\sin at\,dt &= \frac{1}{2}\int_0^x \cos a(x-2t)\,dt - \frac{1}{2}\cos ax\int_0^x dt\\ &= \frac{1}{2}\left[-\frac{\sin a(x-2t)}{2a}\right]_0^x - \frac{x}{2}\cos ax\\ &= \frac{1}{2}\cdot\frac{-\sin(-ax)+\sin ax}{2a} - \frac{x}{2}\cos ax = \frac{\sin ax}{2a} - \frac{x\cos ax}{2}.\end{aligned}$$

(here $\frac{d}{dt}\sin a(x-2t) = -2a\cos a(x-2t)$ gives the antiderivative $-\sin a(x-2t)/(2a)$.) Therefore

$$\mathcal{L}^{-1}\left[\frac{1}{(p^2+a^2)^2}\right] = \frac{1}{a^2}\left(\frac{\sin ax}{2a} - \frac{x\cos ax}{2}\right) = \frac{\sin ax - ax\cos ax}{2a^3}.$$

Answer. $\mathcal{L}^{-1}\left[\dfrac{1}{(p^2+a^2)^2}\right] = \dfrac{\sin ax - ax\cos ax}{2a^3}.$

Problem 52-2 (Simmons)

Solution. Each integral is a convolution $k * y$, so we apply $\mathcal{L}$ and write $Y = \mathcal{L}[y]$.

(a) The kernel is $k(x) = x$, $\mathcal{L}[x] = 1/p^2$, so the equation is $y = 1 - x * y$. Then

$$Y = \frac{1}{p} - \frac{1}{p^2}Y \implies \frac{p^2+1}{p^2}Y = \frac{1}{p} \implies Y = \frac{p}{p^2+1},$$

hence $y(x) = \cos x$.

(b) Since $e^x\displaystyle\int_0^x e^{-t}y(t)\,dt = \int_0^x e^{x-t}y(t)\,dt = e^x * y$, the equation is $y = e^x + e^x * y$, so

$$Y = \frac{1}{p-1} + \frac{1}{p-1}Y \implies \frac{p-2}{p-1}Y = \frac{1}{p-1} \implies Y = \frac{1}{p-2},$$

hence $y(x) = e^{2x}$.

(c) Here $k(x) = 2\cos x$, $\mathcal{L}[2\cos x] = 2p/(p^2+1)$. Then

$$\frac{1}{p+1} = Y\left(1 + \frac{2p}{p^2+1}\right) = Y\,\frac{p^2+2p+1}{p^2+1} = Y\,\frac{(p+1)^2}{p^2+1} \;\Longrightarrow\; Y = \frac{p^2+1}{(p+1)^3}.$$

Write $p^2+1 = (p+1)^2 - 2(p+1) + 2$ (indeed $(p+1)^2 - 2(p+1) + 2 = p^2+2p+1-2p-2+2 = p^2+1$). Then

$$Y = \frac{1}{p+1} - \frac{2}{(p+1)^2} + \frac{2}{(p+1)^3}.$$

Because $\mathcal{L}[e^{-x}x^n] = n!/(p+1)^{n+1}$, we have $\mathcal{L}^{-1}\!\left[1/(p+1)^2\right] = xe^{-x}$ and $\mathcal{L}^{-1}\!\left[2/(p+1)^3\right] = x^2e^{-x}$. Therefore

$$y(x) = e^{-x}\left(1 - 2x + x^2\right) = e^{-x}(x-1)^2.$$

(d) Now $k(x) = x$ and $\mathcal{L}[3\sin 2x] = 6/(p^2+4)$:

$$\frac{6}{p^2+4} = Y\left(1 + \frac{1}{p^2}\right) = Y\,\frac{p^2+1}{p^2} \;\Longrightarrow\; Y = \frac{6p^2}{(p^2+4)(p^2+1)}.$$

Partial fractions in $u = p^2$: $\dfrac{u}{(u+4)(u+1)} = \dfrac{A}{u+4} + \dfrac{B}{u+1}$ with $A = \dfrac{-4}{-4+1} = \dfrac{4}{3}$, $B = \dfrac{-1}{-1+4} = -\dfrac{1}{3}$. So

$$Y = 6\left[\frac{4/3}{p^2+4} - \frac{1/3}{p^2+1}\right] = \frac{8}{p^2+4} - \frac{2}{p^2+1}.$$

Since $\mathcal{L}[\sin 2x] = 2/(p^2+4)$, $\mathcal{L}[\sin x] = 1/(p^2+1)$: $y(x) = 4\sin 2x - 2\sin x$.

Check. Direct substitution is confirmed symbolically in `verify/sec52.py`; e.g. in (a), $1 - \int_0^x (x-t)\cos t\,dt = 1 - (1-\cos x) = \cos x$, since $\int_0^x (x-t)\cos t\,dt = [(x-t)\sin t]_0^x + \int_0^x \sin t\,dt = 1 - \cos x$.

Answer. (a) $y = \cos x$; (b) $y = e^{2x}$; (c) $y = e^{-x}(x-1)^2$; (d) $y = 4\sin 2x - 2\sin x$.

Problem 52-3 (Simmons)

Solution. Equation (7) says $T(y) = \dfrac{1}{\sqrt{2g}}\displaystyle\int_0^y \frac{f(v)}{\sqrt{y-v}}\,dv$, i.e. $T = \dfrac{1}{\sqrt{2g}}\,y^{-1/2} * f$. With $\mathcal{L}[y^{-1/2}] = \sqrt{\pi}/\sqrt{p}$ this gives $\mathcal{L}[T] = \dfrac{1}{\sqrt{2g}}\sqrt{\dfrac{\pi}{p}}\,\mathcal{L}[f]$, so equation (8) reads

$$\mathcal{L}[f] = \sqrt{\frac{2g}{\pi}}\;p^{1/2}\,\mathcal{L}[T].$$

We now invert this. Write $p^{1/2} = p\cdot p^{-1/2}$ and note that $p^{-1/2} = \mathcal{L}\!\left[\dfrac{y^{-1/2}}{\sqrt{\pi}}\right]$. By the convolution theorem

$$p^{-1/2}\,\mathcal{L}[T] = \mathcal{L}[G], \qquad G(y) = \frac{1}{\sqrt{\pi}}\int_0^y \frac{T(t)}{\sqrt{y-t}}\,dt.$$

(the convolution of $y^{-1/2}/\sqrt{\pi}$ with T evaluated at y). Since $G(0)=0$ (the integral over an interval of length 0; more precisely $|G(y)| \le \frac{2}{\sqrt{\pi}}\sqrt{y}\max_{[0,y]}|T| \to 0$), the rule $\mathcal{L}[G'] = p\mathcal{L}[G] - G(0)$ gives $p\,\mathcal{L}[G] = \mathcal{L}[G']$. Hence

$$\mathcal{L}[f] = \sqrt{\frac{2g}{\pi}}\,\mathcal{L}[G'] \quad\Longrightarrow\quad f(y) = \sqrt{\frac{2g}{\pi}}\cdot\frac{1}{\sqrt{\pi}}\frac{d}{dy}\int_0^y \frac{T(t)\,dt}{\sqrt{y-t}} = \frac{\sqrt{2g}}{\pi}\frac{d}{dy}\int_0^y \frac{T(t)\,dt}{\sqrt{y-t}},$$

which is the asserted formula.

Constant $T(y)=T_0$. Substituting $u = y-t$, $du = -dt$:

$$\int_0^y \frac{T_0\,dt}{\sqrt{y-t}} = T_0\int_0^y u^{-1/2}\,du = 2T_0\sqrt{y}, \qquad \frac{d}{dy}\left(2T_0\sqrt{y}\right) = \frac{T_0}{\sqrt{y}}.$$

Therefore

$$f(y) = \frac{\sqrt{2g}\,T_0}{\pi}\cdot\frac{1}{\sqrt{y}} = \frac{\sqrt{b}}{\sqrt{y}}, \qquad b = \frac{2gT_0^2}{\pi^2},$$

which is (9), $f(y)=\sqrt{b/y}$, with $b = 2gT_0^2/\pi^2$ as in the text.

Check. Putting this f back into (7): $\dfrac{1}{\sqrt{2g}}\displaystyle\int_0^y \frac{\sqrt{2g}\,T_0}{\pi\sqrt{v}\sqrt{y-v}}\,dv = \frac{T_0}{\pi}\int_0^y \frac{dv}{\sqrt{v(y-v)}} = \frac{T_0}{\pi}\cdot\pi = T_0$, using $\int_0^y \frac{dv}{\sqrt{v(y-v)}} = \pi$ (put $v = y\sin^2\phi$).

Answer. Shown: $f(y) = \dfrac{\sqrt{2g}}{\pi}\dfrac{d}{dy}\displaystyle\int_0^y \frac{T(t)\,dt}{\sqrt{y-t}}$; for $T\equiv T_0$ this gives $f(y) = \dfrac{\sqrt{2g}\,T_0}{\pi\sqrt{y}} = \sqrt{b/y}$ with $b = 2gT_0^2/\pi^2$.

Problem 52-4 (Simmons)

Solution. By Problem 3, $f(y) = \dfrac{\sqrt{2g}}{\pi}\dfrac{d}{dy}\displaystyle\int_0^y \frac{k\sqrt{t}}{\sqrt{y-t}}\,dt$. Put $t = yu$, $dt = y\,du$:

$$\int_0^y \frac{\sqrt{t}}{\sqrt{y-t}}\,dt = y\int_0^1 \frac{\sqrt{u}}{\sqrt{1-u}}\,du = y\cdot 2\int_0^{\pi/2}\sin^2\phi\,d\phi = y\cdot\frac{\pi}{2},$$

using $u = \sin^2\phi$, $du = 2\sin\phi\cos\phi\,d\phi$. (Indeed $\sqrt{u}/\sqrt{1-u}\,du = \frac{\sin\phi}{\cos\phi}\,2\sin\phi\cos\phi\,d\phi = 2\sin^2\phi\,d\phi$ and $\int_0^{\pi/2}\sin^2\phi\,d\phi = \pi/4$.) Hence

$$f(y) = \frac{\sqrt{2g}}{\pi}\cdot\frac{d}{dy}\left(\frac{\pi k y}{2}\right) = \frac{k\sqrt{2g}}{2} = \frac{k\sqrt{g}}{\sqrt{2}} =: c,$$

a constant. (Check with transforms: $\mathcal{L}[k\sqrt{y}] = \dfrac{k\sqrt{\pi}}{2p^{3/2}}$, so by (8) $\mathcal{L}[f] = \sqrt{\frac{2g}{\pi}}\,p^{1/2}\dfrac{k\sqrt{\pi}}{2p^{3/2}} = \dfrac{k\sqrt{2g}}{2p}$.)

Equation (6) now says $\sqrt{1+(dx/dy)^2} = c$, that is,

$$\left(\frac{dx}{dy}\right)^2 = c^2 - 1 = \frac{gk^2}{2} - 1.$$

A solution exists only if $c \ge 1$, i.e. $gk^2 \ge 2$ (a descent time $k\sqrt{y}$ cannot be shorter than the free-fall time $\sqrt{2y/g}$ along the vertical, which is the case $c=1$, $x\equiv$ const). For $c>1$, $dx/dy = \pm\sqrt{c^2-1}$

is constant, so $x = \pm\sqrt{c^2-1}\,y + \text{const}$; since the curve passes through the origin the constant is 0, and taking the branch with $x \geq 0$,

$$y = \frac{x}{\sqrt{c^2-1}} = \frac{x}{\sqrt{gk^2/2-1}},$$

a straight line through the origin: $y = cx$ in the book's notation, with the slope constant fixed by k (and g).

Check. A bead sliding from rest down a frictionless incline of inclination θ from height y travels the length $y/\sin\theta$ with acceleration $g\sin\theta$, so $T = \sqrt{2(y/\sin\theta)/(g\sin\theta)} = \sqrt{2/g}\,\sqrt{y}/\sin\theta$. This equals $k\sqrt{y}$ exactly when $\sin\theta = \sqrt{2/g}/k = 1/c$; then $\tan\theta = 1/\sqrt{c^2-1}$ is the slope found above.

Answer. A straight line through the origin, $y = \dfrac{x}{\sqrt{gk^2/2-1}}$ (the book writes $y = cx$); it exists when $gk^2 \geq 2$ with $f(y) \equiv k\sqrt{g/2}$.

Problem 52-5 (Simmons)

Solution. **By Laplace transforms.** Apply $\mathcal{L}$ with $y(0) = y'(0) = 0$: $p^2Y + a^2Y = F(p)$, so

$$Y = F(p)\cdot\frac{1}{p^2+a^2} = \mathcal{L}[f]\cdot\mathcal{L}\left[\frac{\sin ax}{a}\right].$$

By the convolution theorem $y(x) = \displaystyle\int_0^x f(t)\,\frac{\sin a(x-t)}{a}\,\mathrm{d}t$, the claimed formula.

Direct verification (for continuous f). Expand $\sin a(x-t) = \sin ax\cos at - \cos ax\sin at$:

$$y(x) = \frac{1}{a}\left[\sin ax\,C(x) - \cos ax\,S(x)\right], \quad C(x) = \int_0^x f(t)\cos at\,\mathrm{d}t, \quad S(x) = \int_0^x f(t)\sin at\,\mathrm{d}t,$$

where $C' = f(x)\cos ax$, $S' = f(x)\sin ax$ (fundamental theorem of calculus). Then

$$\begin{aligned} y' &= \frac{1}{a}\Big[a\cos ax\,C + \sin ax\,f\cos ax + a\sin ax\,S - \cos ax\,f\sin ax\Big] = \cos ax\,C + \sin ax\,S, \\ y'' &= -a\sin ax\,C + \cos ax\,f\cos ax + a\cos ax\,S + \sin ax\,f\sin ax \\ &= -a\left[\sin ax\,C - \cos ax\,S\right] + f(\cos^2 ax + \sin^2 ax) = -a^2y + f(x). \end{aligned}$$

So $y'' + a^2y = f(x)$. Also $C(0) = S(0) = 0$ gives $y(0) = 0$ and $y'(0) = 0$. By uniqueness of solutions of linear initial value problems, this is the solution.

Answer. Shown: $y(x) = \frac{1}{a}\int_0^x f(t)\sin a(x-t)\,dt$ satisfies $y'' + a^2y = f(x)$, $y(0) = y'(0) = 0$.

Section 53 More about Convolutions. The Unit Step and Impulse Functions

Notation: $f * g(t) = \int_0^t f(t-\tau)g(\tau)\,d\tau$; for $y'' + ay' + by = f(t)$ with $y(0) = y'(0) = 0$ put $z(p) = p^2 + ap + b$. The indicial response A and the impulsive response h satisfy $\mathcal{L}[A] = \dfrac{1}{p\,z(p)}$, $\mathcal{L}[h] = \dfrac{1}{z(p)}$, $A' = h$, $A(0) = 0$, and the solution is given by

$$y(t) = \int_0^t A(t-\tau)f'(\tau)\,d\tau + f(0)A(t) \tag{13}$$

$$y(t) = \int_0^t h(t-\tau)f(\tau)\,d\tau \tag{15}$$

Problem 53-1 (Simmons)

Solution. The problem asks us to show commutativity; we also prove the associative and distributive laws that are stated. (Functions are assumed piecewise continuous on $[0,\infty)$.)

Commutativity. In $f * g(t) = \int_0^t f(t-\tau)g(\tau)\,d\tau$ put $\sigma = t - \tau$. Then $\tau = t - \sigma$, $d\tau = -d\sigma$; $\tau = 0 \leftrightarrow \sigma = t$ and $\tau = t \leftrightarrow \sigma = 0$. So

$$f * g(t) = \int_{\sigma=t}^{0} f(\sigma)\,g(t-\sigma)(-d\sigma) = \int_0^t g(t-\sigma)f(\sigma)\,d\sigma = g * f(t).$$

Distributivity. By linearity of the integral,

$$\begin{aligned} f * (g+h)(t) &= \int_0^t f(t-\tau)[g(\tau)+h(\tau)]\,d\tau \\ &= \int_0^t f(t-\tau)g(\tau)\,d\tau + \int_0^t f(t-\tau)h(\tau)\,d\tau = f*g(t) + f*h(t). \end{aligned}$$

The second law $(f+g)*h = f*h + g*h$ follows in the same way (or from the first and commutativity: $(f+g)*h = h*(f+g) = h*f + h*g = f*h + g*h$).

Associativity. By definition

$$f * (g*h)(t) = \int_0^t f(t-\tau)\left[\int_0^\tau g(\tau-\sigma)h(\sigma)\,d\sigma\right]d\tau.$$

The region of integration is the triangle $0 \le \sigma \le \tau \le t$. Reversing the order (for fixed $\sigma \in [0,t]$, τ runs from σ to t):

$$f * (g*h)(t) = \int_0^t h(\sigma)\left[\int_\sigma^t f(t-\tau)\,g(\tau-\sigma)\,d\tau\right]d\sigma.$$

In the inner integral put $u = \tau - \sigma$ (u runs from 0 to $t-\sigma$, $t - \tau = (t-\sigma) - u$):

$$\int_\sigma^t f(t-\tau)g(\tau-\sigma)\,d\tau = \int_0^{t-\sigma} f\big((t-\sigma)-u\big)g(u)\,du = (f*g)(t-\sigma).$$

Hence

$$f * (g*h)(t) = \int_0^t (f*g)(t-\sigma)\,h(\sigma)\,d\sigma = [(f*g)*h](t).$$

(Alternatively, all three laws follow from $\mathcal{L}[f*g] = \mathcal{L}[f]\mathcal{L}[g]$ and the corresponding laws of ordinary multiplication, together with uniqueness of the inverse transform.)

Answer. Shown: $f*g = g*f$ (via $\sigma = t-\tau$); also $f*(g*h) = (f*g)*h$ and $f*(g+h) = f*g + f*h$, $(f+g)*h = f*h + g*h$.

Problem 53-2 (Simmons)

Solution. We use $f*g(t) = \int_0^t f(t-\tau)g(\tau)\,d\tau$ (and $f*g = g*f$ by Problem 1, so the order is at our disposal). Assume $a \neq 0$ where we divide by a.

(a) Take $f = 1$, $g = \sin a\tau$:

$$1 * \sin at = \int_0^t \sin a\tau\,d\tau = \Big[-\frac{\cos a\tau}{a}\Big]_0^t = \frac{1-\cos at}{a}.$$

(b)

$$e^{at} * e^{bt} = \int_0^t e^{a(t-\tau)}e^{b\tau}\,d\tau = e^{at}\int_0^t e^{(b-a)\tau}\,d\tau = e^{at}\,\frac{e^{(b-a)t}-1}{b-a} = \frac{e^{bt}-e^{at}}{b-a} = \frac{e^{at}-e^{bt}}{a-b}.$$

(division by $b-a$ is legitimate because $a \neq b$).

(c) Take $f = t$, $g = e^{a\tau}$, so $f(t-\tau) = t-\tau$; integrate by parts with $u = t-\tau$, $dv = e^{a\tau}d\tau$:

$$\begin{aligned} t * e^{at} = \int_0^t (t-\tau)e^{a\tau}\,d\tau &= \Big[(t-\tau)\frac{e^{a\tau}}{a}\Big]_0^t + \frac{1}{a}\int_0^t e^{a\tau}\,d\tau \\ &= -\frac{t}{a} + \frac{e^{at}-1}{a^2} = \frac{e^{at}-1-at}{a^2}. \end{aligned}$$

(d) By $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ with $A = a(t-\tau)$, $B = b\tau$:

$$\sin a(t-\tau)\sin b\tau = \tfrac{1}{2}\big[\cos\big(at-(a+b)\tau\big) - \cos\big(at+(b-a)\tau\big)\big].$$

Since $a \neq b$ we have $b-a \neq 0$; assume also $a+b \neq 0$. Then

$$\begin{aligned} \int_0^t \cos\big(at-(a+b)\tau\big)\,d\tau &= \Big[-\frac{\sin(at-(a+b)\tau)}{a+b}\Big]_0^t = \frac{\sin at + \sin bt}{a+b}, \\ \int_0^t \cos\big(at+(b-a)\tau\big)\,d\tau &= \Big[\frac{\sin(at+(b-a)\tau)}{b-a}\Big]_0^t = \frac{\sin bt - \sin at}{b-a}. \end{aligned}$$

(at $\tau = t$: $at-(a+b)t = -bt$, whose sine is $-\sin bt$; and $at+(b-a)t = bt$.) Therefore

$$\begin{aligned} \sin at * \sin bt &= \frac{1}{2}\left[\frac{\sin at + \sin bt}{a+b} + \frac{\sin bt - \sin at}{a-b}\right] \\ &= \frac{(\sin at + \sin bt)(a-b) + (\sin bt - \sin at)(a+b)}{2(a^2-b^2)} \\ &= \frac{2a\sin bt - 2b\sin at}{2(a^2-b^2)} = \frac{a\sin bt - b\sin at}{a^2-b^2}. \end{aligned}$$

Remark. If $b = -a \neq 0$ the above formula is $0/0$, but the result is still obtained from the formula for $b = a$: $\sin at * \sin(-at) = -\sin at * \sin at = -\dfrac{\sin at - at\cos at}{2a}$ (Section 52, Problem 1). Both sides of the displayed formula are odd in b, and the case $a = b$ is the limit, consistent with this.

Answer. (a) $\dfrac{1-\cos at}{a}$; (b) $\dfrac{e^{at}-e^{bt}}{a-b}$; (c) $\dfrac{e^{at}-1-at}{a^2}$; (d) $\dfrac{a\sin bt - b\sin at}{a^2-b^2}$.

Problem 53-3 (Simmons)

Solution. We must check $\mathcal{L}[f*g] = \mathcal{L}[f]\mathcal{L}[g]$ using the results of Problem 2.

(a) $\mathcal{L}[1]\mathcal{L}[\sin at] = \dfrac{1}{p}\cdot\dfrac{a}{p^2+a^2}$. On the other side

$$\mathcal{L}\left[\frac{1-\cos at}{a}\right] = \frac{1}{a}\left[\frac{1}{p} - \frac{p}{p^2+a^2}\right] = \frac{1}{a}\cdot\frac{p^2+a^2-p^2}{p(p^2+a^2)} = \frac{a}{p(p^2+a^2)}.$$

(b) $\mathcal{L}[e^{at}]\mathcal{L}[e^{bt}] = \dfrac{1}{(p-a)(p-b)}$, and

$$\mathcal{L}\left[\frac{e^{at}-e^{bt}}{a-b}\right] = \frac{1}{a-b}\left[\frac{1}{p-a} - \frac{1}{p-b}\right] = \frac{1}{a-b}\cdot\frac{(p-b)-(p-a)}{(p-a)(p-b)} = \frac{1}{(p-a)(p-b)}.$$

(c) $\mathcal{L}[t]\mathcal{L}[e^{at}] = \dfrac{1}{p^2(p-a)}$, and

$$\begin{aligned}\mathcal{L}\left[\frac{e^{at}-1-at}{a^2}\right] &= \frac{1}{a^2}\left[\frac{1}{p-a} - \frac{1}{p} - \frac{a}{p^2}\right] = \frac{1}{a^2}\cdot\frac{p^2 - p(p-a) - a(p-a)}{p^2(p-a)} \\ &= \frac{1}{a^2}\cdot\frac{a^2}{p^2(p-a)} = \frac{1}{p^2(p-a)},\end{aligned}$$

since $p^2 - p^2 + ap - ap + a^2 = a^2$.

(d) $\mathcal{L}[\sin at]\mathcal{L}[\sin bt] = \dfrac{ab}{(p^2+a^2)(p^2+b^2)}$, and

$$\begin{aligned}\mathcal{L}\left[\frac{a\sin bt - b\sin at}{a^2-b^2}\right] &= \frac{1}{a^2-b^2}\left[\frac{ab}{p^2+b^2} - \frac{ab}{p^2+a^2}\right] = \frac{ab}{a^2-b^2}\cdot\frac{(p^2+a^2)-(p^2+b^2)}{(p^2+a^2)(p^2+b^2)} \\ &= \frac{ab}{(p^2+a^2)(p^2+b^2)}.\end{aligned}$$

In every case the two sides agree.

Answer. Verified in all four cases (a)–(d): $\mathcal{L}[f*g] = \mathcal{L}[f]\,\mathcal{L}[g]$.

Problem 53-4 (Simmons)

Solution. Method 1 is formula (13) with the indicial response A, method 2 is formula (15) with the impulsive response $h = A'$.

(a) Here $z(p) = p^2+5p+6 = (p+2)(p+3)$. Partial fractions:

$$\mathcal{L}[A] = \frac{1}{p(p+2)(p+3)} = \frac{1/6}{p} - \frac{1/2}{p+2} + \frac{1/3}{p+3}$$

(residues: at 0: $\frac{1}{2\cdot 3}$; at -2: $\frac{1}{(-2)(1)}$; at -3: $\frac{1}{(-3)(-1)}$). So

$$A(t) = \tfrac{1}{6} - \tfrac{1}{2}e^{-2t} + \tfrac{1}{3}e^{-3t}, \qquad h(t) = A'(t) = e^{-2t} - e^{-3t}$$

(consistent with $\mathcal{L}[h] = 1/((p+2)(p+3)) = 1/(p+2) - 1/(p+3)$).
Method 1. $f = 5e^{3t}$, $f' = 15e^{3t}$, $f(0) = 5$:

$$\begin{aligned}\int_0^t A(t-\tau)\,15e^{3\tau}\,\mathrm{d}\tau &= 15\left[\frac{1}{6}\cdot\frac{e^{3t}-1}{3} - \frac{1}{2}e^{-2t}\frac{e^{5t}-1}{5} + \frac{1}{3}e^{-3t}\frac{e^{6t}-1}{6}\right]\\ &= \tfrac{5}{6}(e^{3t}-1) - \tfrac{3}{2}(e^{3t}-e^{-2t}) + \tfrac{5}{6}(e^{3t}-e^{-3t}),\end{aligned}$$

using $\int_0^t e^{-2(t-\tau)}e^{3\tau}d\tau = e^{-2t}\frac{e^{5t}-1}{5}$ etc. Adding $f(0)A(t) = \frac{5}{6} - \frac{5}{2}e^{-2t} + \frac{5}{3}e^{-3t}$:

$$y = \left(\tfrac{5}{6} - \tfrac{3}{2} + \tfrac{5}{6}\right)e^{3t} + \left(-\tfrac{5}{6} + \tfrac{5}{6}\right) + \left(\tfrac{3}{2} - \tfrac{5}{2}\right)e^{-2t} + \left(-\tfrac{5}{6} + \tfrac{5}{3}\right)e^{-3t} = \tfrac{1}{6}e^{3t} - e^{-2t} + \tfrac{5}{6}e^{-3t}.$$

Method 2.

$$\begin{aligned}y &= \int_0^t \left[e^{-2(t-\tau)} - e^{-3(t-\tau)}\right]5e^{3\tau}\,\mathrm{d}\tau = 5\left[e^{-2t}\frac{e^{5t}-1}{5} - e^{-3t}\frac{e^{6t}-1}{6}\right]\\ &= e^{3t} - e^{-2t} - \tfrac{5}{6}\left(e^{3t} - e^{-3t}\right) = \tfrac{1}{6}e^{3t} - e^{-2t} + \tfrac{5}{6}e^{-3t}.\end{aligned}$$

Both methods agree. (Check: $y(0) = \frac{1}{6} - 1 + \frac{5}{6} = 0$, $y'(0) = \frac{1}{2} + 2 - \frac{5}{2} = 0$.)

(b) $z(p) = p^2 + p - 6 = (p+3)(p-2)$ and

$$\mathcal{L}[A] = \frac{1}{p(p-2)(p+3)} = -\frac{1/6}{p} + \frac{1/10}{p-2} + \frac{1/15}{p+3}$$

(residues: at 0: $\frac{1}{(-2)(3)}$; at 2: $\frac{1}{2\cdot 5}$; at -3: $\frac{1}{(-3)(-5)}$), so

$$A = -\tfrac{1}{6} + \tfrac{1}{10}e^{2t} + \tfrac{1}{15}e^{-3t}, \qquad h = A' = \tfrac{1}{5}\left(e^{2t} - e^{-3t}\right).$$

Method 1. $f = t$, $f' = 1$, $f(0) = 0$, so $y = \int_0^t A(t-\tau)\,\mathrm{d}\tau = \int_0^t A(s)\,\mathrm{d}s$ ($s = t - \tau$):

$$\begin{aligned}y &= -\frac{t}{6} + \frac{e^{2t}-1}{20} - \frac{e^{-3t}-1}{45}\\ &= -\frac{t}{6} + \frac{1}{20}e^{2t} - \frac{1}{45}e^{-3t} + \left(-\frac{1}{20} + \frac{1}{45}\right)\\ &= -\frac{t}{6} - \frac{1}{36} - \frac{1}{45}e^{-3t} + \frac{1}{20}e^{2t}.\end{aligned}$$

Method 2. $y = \dfrac{1}{5}\displaystyle\int_0^t \tau\left[e^{2(t-\tau)} - e^{-3(t-\tau)}\right]d\tau$. By parts,

$$\int_0^t \tau e^{-2\tau}d\tau = -\frac{te^{-2t}}{2} + \frac{1-e^{-2t}}{4}, \qquad \int_0^t \tau e^{3\tau}d\tau = \frac{te^{3t}}{3} - \frac{e^{3t}-1}{9},$$

so $e^{2t}\int_0^t \tau e^{-2\tau}d\tau = -\frac{t}{2} + \frac{e^{2t}-1}{4}$ and $e^{-3t}\int_0^t \tau e^{3\tau}d\tau = \frac{t}{3} - \frac{1-e^{-3t}}{9}$. Thus

$$y = \frac{1}{5}\left[-\frac{t}{2} + \frac{e^{2t}-1}{4} - \frac{t}{3} + \frac{1-e^{-3t}}{9}\right] = \frac{1}{5}\left[-\frac{5t}{6} + \frac{e^{2t}}{4} - \frac{e^{-3t}}{9} - \frac{5}{36}\right] = -\frac{t}{6} - \frac{1}{36} + \frac{e^{2t}}{20} - \frac{e^{-3t}}{45}.$$

Same result.

(c) $z(p) = p^2 - p = p(p-1)$, $\mathcal{L}[A] = \dfrac{1}{p^2(p-1)} = \dfrac{1}{p-1} - \dfrac{1}{p} - \dfrac{1}{p^2}$ (the numerator check: $p^2 - p(p-1) - (p-1) = 1$), so

$$A = e^t - 1 - t, \qquad h = A' = e^t - 1.$$

Method 1. $f = t^2$, $f' = 2t$, $f(0) = 0$:

$$y = \int_0^t \left[e^{t-\tau} - 1 - (t-\tau)\right] 2\tau \, d\tau.$$

Term by term: $2e^t \int_0^t \tau e^{-\tau} d\tau = 2e^t \left[1 - (1+t)e^{-t}\right] = 2e^t - 2 - 2t$; $\int_0^t 2\tau \, d\tau = t^2$; $\int_0^t (t-\tau) 2\tau \, d\tau = t \cdot t^2 - \frac{2}{3}t^3 = \frac{1}{3}t^3$. Hence

$$y = 2e^t - 2 - 2t - t^2 - \tfrac{1}{3}t^3.$$

Method 2. $y = \displaystyle\int_0^t (e^{t-\tau} - 1)\tau^2 d\tau = e^t \int_0^t \tau^2 e^{-\tau} d\tau - \frac{t^3}{3}$. Integrating by parts twice, $\int_0^t \tau^2 e^{-\tau} d\tau = 2 - (t^2 + 2t + 2)e^{-t}$, so

$$y = 2e^t - (t^2 + 2t + 2) - \tfrac{1}{3}t^3,$$

the same function.

Answer. (a) $y = \frac{1}{6}e^{3t} + \frac{5}{6}e^{-3t} - e^{-2t}$; (b) $y = -\frac{t}{6} - \frac{1}{36} - \frac{1}{45}e^{-3t} + \frac{1}{20}e^{2t}$; (c) $y = 2e^t - \frac{1}{3}t^3 - t^2 - 2t - 2$.

Problem 53-5 (Simmons)

Solution.

(a) Here $z(p) = p^2 + 3p + 2 = (p+1)(p+2)$, so $a = -1$, $b = -2$ and $\dfrac{1}{(p+1)(p+2)} = \dfrac{1}{p+1} - \dfrac{1}{p+2}$, i.e. $A = 1$, $B = -1$ ($A = \frac{1}{-1+2}$, $B = \frac{1}{-2+1}$). Hence $h(t) = e^{-t} - e^{-2t}$ and

$$y(t) = \int_0^t f(\tau) \left[e^{-(t-\tau)} - e^{-2(t-\tau)}\right] d\tau.$$

(b) *Case* $f = e^{3t}$:

$$\begin{aligned} y &= e^{-t} \int_0^t e^{4\tau} d\tau - e^{-2t} \int_0^t e^{5\tau} d\tau = e^{-t} \frac{e^{4t} - 1}{4} - e^{-2t} \frac{e^{5t} - 1}{5} \\ &= \tfrac{1}{4}e^{3t} - \tfrac{1}{4}e^{-t} - \tfrac{1}{5}e^{3t} + \tfrac{1}{5}e^{-2t} = \tfrac{1}{20}e^{3t} - \tfrac{1}{4}e^{-t} + \tfrac{1}{5}e^{-2t}. \end{aligned}$$

Case $f = t$: using $\int_0^t \tau e^{\tau} d\tau = (t-1)e^t + 1$ and $\int_0^t \tau e^{2\tau} d\tau = \left(\frac{t}{2} - \frac{1}{4}\right) e^{2t} + \frac{1}{4}$,

$$\begin{aligned} y &= e^{-t} \left[(t-1)e^t + 1\right] - e^{-2t} \left[\left(\tfrac{t}{2} - \tfrac{1}{4}\right) e^{2t} + \tfrac{1}{4}\right] \\ &= t - 1 + e^{-t} - \tfrac{t}{2} + \tfrac{1}{4} - \tfrac{1}{4}e^{-2t} = \tfrac{1}{2}t - \tfrac{3}{4} + e^{-t} - \tfrac{1}{4}e^{-2t}. \end{aligned}$$

(c) From (a), $\mathcal{L}[A] = \dfrac{1}{p(p+1)(p+2)} = \dfrac{1/2}{p} - \dfrac{1}{p+1} + \dfrac{1/2}{p+2}$ (residues $\frac{1}{1\cdot 2}$, $\frac{1}{(-1)(1)}$, $\frac{1}{(-2)(-1)}$), so

$$A(t) = \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t}.$$

Case $f = e^{3t}$: $f' = 3e^{3t}$, $f(0) = 1$:

$$\begin{aligned} y &= \int_0^t \left[\tfrac{1}{2} - e^{-(t-\tau)} + \tfrac{1}{2}e^{-2(t-\tau)}\right] 3e^{3\tau}\,d\tau + A(t) \\ &= 3\left[\tfrac{1}{2}\cdot\frac{e^{3t}-1}{3} - e^{-t}\frac{e^{4t}-1}{4} + \tfrac{1}{2}e^{-2t}\frac{e^{5t}-1}{5}\right] + \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t} \\ &= \tfrac{1}{2}(e^{3t}-1) - \tfrac{3}{4}(e^{3t}-e^{-t}) + \tfrac{3}{10}(e^{3t}-e^{-2t}) + \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t} \\ &= \left(\tfrac{1}{2} - \tfrac{3}{4} + \tfrac{3}{10}\right)e^{3t} + \left(\tfrac{3}{4} - 1\right)e^{-t} + \left(-\tfrac{3}{10} + \tfrac{1}{2}\right)e^{-2t} = \tfrac{1}{20}e^{3t} - \tfrac{1}{4}e^{-t} + \tfrac{1}{5}e^{-2t}. \end{aligned}$$

Case $f = t$: $f' = 1$, $f(0) = 0$, so $y = \int_0^t A(s)ds = \frac{t}{2} - (1 - e^{-t}) + \frac{1}{4}(1 - e^{-2t}) = \frac{t}{2} - \frac{3}{4} + e^{-t} - \frac{1}{4}e^{-2t}$. Both cases agree with (b).

Answer. (a) $y = \displaystyle\int_0^t f(\tau)\left[e^{-(t-\tau)} - e^{-2(t-\tau)}\right]d\tau$; (b), (c) $f = e^{3t}$: $y = \frac{1}{20}e^{3t} - \frac{1}{4}e^{-t} + \frac{1}{5}e^{-2t}$; $f = t$: $y = \frac{1}{2}t - \frac{3}{4} + e^{-t} - \frac{1}{4}e^{-2t}$.

Problem 53-6 (Simmons)

Solution. Assume f is continuous with f' piecewise continuous and both of exponential order, so that $\mathcal{L}[f']$ exists. We justify each line.

1. $\mathcal{L}[y] = \mathcal{L}[f]/z(p)$ is equation (4): transform $y'' + ay' + by = f$ with $y(0) = y'(0) = 0$ to get $z(p)\mathcal{L}[y] = \mathcal{L}[f]$, with $z(p) = p^2 + ap + b$.
2. $\dfrac{\mathcal{L}[f]}{z(p)} = \dfrac{1}{p\,z(p)} \cdot p\,\mathcal{L}[f]$: multiply and divide by $p \neq 0$.
3. $\dfrac{1}{p\,z(p)} = \mathcal{L}[A(t)]$ by (7), where A is the response to $u(t)$ ($\mathcal{L}[u] = 1/p$) with $A(0) = A'(0) = 0$.
4. The rule for transforming a derivative gives $\mathcal{L}[f'] = p\mathcal{L}[f] - f(0)$, i.e. $p\mathcal{L}[f] = \mathcal{L}[f'] + f(0)$.
5. Distribute: $\mathcal{L}[A]\{\mathcal{L}[f'] + f(0)\} = \mathcal{L}[A]\mathcal{L}[f'] + f(0)\mathcal{L}[A]$. The convolution theorem gives $\mathcal{L}[A]\mathcal{L}[f'] = \mathcal{L}[A * f']$, and $f(0)$ is a constant, so $f(0)\mathcal{L}[A]$ is the transform of $f(0)A$.
6. By linearity of $\mathcal{L}$, $\mathcal{L}[A * f'] + f(0)\mathcal{L}[A] = \mathcal{L}[A * f' + f(0)A]$, and $A * f'(t) = \int_0^t A(t - \tau)f'(\tau)\,\mathrm{d}\tau$ by definition (2).

Since a function is determined by its transform (uniqueness of inversion for continuous functions),

$$y(t) = \int_0^t A(t-\tau)f'(\tau)\,\mathrm{d}\tau + f(0)A(t),$$

which is (13). All steps are valid.

Answer. Checked: each line follows from (4), (7), $\mathcal{L}[f'] = p\mathcal{L}[f] - f(0)$, the convolution theorem, and linearity; this gives (13).

Problem 53-7 (Simmons)

Solution. Here the leading coefficient is M rather than 1, so $z(p) = Mp^2 + k$ (transforming $Mx'' + kx = f$ with zero initial data gives $z(p)\mathcal{L}[x] = \mathcal{L}[f]$). The derivation of (13) and (15) uses only this relation, so the formulas hold with this $z(p)$ (A is the response to $u(t)$, h the response to $\delta(t)$). Put $\omega = \sqrt{k/M}$, so $Mp^2 + k = M(p^2 + \omega^2)$ and $k = M\omega^2$.

Indicial response. $\mathcal{L}[A] = \dfrac{1}{p(Mp^2+k)} = \dfrac{1}{k}\left[\dfrac{1}{p} - \dfrac{Mp}{Mp^2+k}\right] = \dfrac{1}{k}\left[\dfrac{1}{p} - \dfrac{p}{p^2+\omega^2}\right]$ (check: $\frac{1}{p} - \frac{Mp}{Mp^2+k} = \frac{Mp^2+k-Mp^2}{p(Mp^2+k)} = \frac{k}{p(Mp^2+k)}$). So

$$A(t) = \frac{1}{k}\left(1 - \cos\sqrt{\frac{k}{M}}\,t\right).$$

Impulsive response.

$$h(t) = A'(t) = \frac{\omega}{k}\sin\omega t = \frac{1}{\sqrt{Mk}}\sin\sqrt{\frac{k}{M}}\,t,$$

since $\omega/k = \sqrt{k/M}/k = 1/\sqrt{Mk}$. (Check: $\mathcal{L}[h] = \dfrac{\omega}{\sqrt{Mk}\,(p^2+\omega^2)} = \dfrac{1}{M(p^2+\omega^2)} = \dfrac{1}{Mp^2+k}$ because $\omega/\sqrt{Mk} = 1/M$.)

Solution. By (15),

$$x(t) = \int_0^t h(t-\tau)f(\tau)\,d\tau = \frac{1}{\sqrt{Mk}}\int_0^t f(\tau)\sin\sqrt{\frac{k}{M}}\,(t-\tau)\,d\tau.$$

(Equivalently (13): $x = \int_0^t A(t-\tau)f'(\tau)d\tau + f(0)A(t)$; this agrees with Problem 5 of Section 52 with $a = \omega$ and f/M as input, since $\frac{1}{\omega M} = \frac{1}{\sqrt{Mk}}$.)

Answer. $A(t) = \dfrac{1}{k}\left(1 - \cos\sqrt{k/M}\,t\right)$, $h(t) = \dfrac{1}{\sqrt{Mk}}\sin\sqrt{k/M}\,t$, $x(t) = \dfrac{1}{\sqrt{Mk}}\displaystyle\int_0^t f(\tau)\sin\sqrt{\tfrac{k}{M}}\,(t-\tau)\,d\tau$.

Problem 53-8 (Simmons)

Solution. Transforming with $I(0) = 0$: $(Lp + R)\mathcal{L}[I] = \mathcal{L}[E]$, so $z(p) = Lp + R$ and (4) reads $\mathcal{L}[I] = \mathcal{L}[E]/(Lp+R)$. Put $r = R/L$.

Indicial response (input $u(t)$, $\mathcal{L}[u] = 1/p$):

$$\mathcal{L}[A] = \frac{1}{p(Lp+R)} = \frac{1}{R}\left[\frac{1}{p} - \frac{L}{Lp+R}\right] = \frac{1}{R}\left[\frac{1}{p} - \frac{1}{p+R/L}\right], \qquad A(t) = \frac{1-e^{-Rt/L}}{R}.$$

Impulsive response (input $\delta(t)$, $\mathcal{L}[\delta] = 1$):

$$\mathcal{L}[h] = \frac{1}{Lp+R} = \frac{1}{L}\cdot\frac{1}{p+R/L}, \qquad h(t) = \frac{1}{L}e^{-Rt/L} = A'(t).$$

(a) $E = E_0u(t)$ is E_0 times the unit step, so the output is E_0 times the indicial response:

$$I(t) = E_0A(t) = \frac{E_0}{R}\left(1 - e^{-Rt/L}\right).$$

(b) $E = E_0\delta(t)$, so the output is E_0 times the impulsive response:

$$I(t) = E_0 h(t) = \frac{E_0}{L} e^{-Rt/L}.$$

(Check: $\mathcal{L}[I] = \dfrac{E_0}{Lp + R}$ and δ produces the instantaneous jump $LI(0^+) = E_0$.)

(c) By (15),

$$I(t) = \int_0^t h(t-\tau)E(\tau)\,d\tau = \frac{E_0}{L}\int_0^t e^{-r(t-\tau)}\sin\omega\tau\,d\tau = \frac{E_0}{L}e^{-rt}\int_0^t e^{r\tau}\sin\omega\tau\,d\tau.$$

Since

$$\begin{aligned}\frac{d}{d\tau}\left[e^{r\tau}(r\sin\omega\tau - \omega\cos\omega\tau)\right] &= e^{r\tau}\left[r^2\sin\omega\tau + r\omega\cos\omega\tau - r\omega\cos\omega\tau + \omega^2\sin\omega\tau\right] \\ &= (r^2+\omega^2)e^{r\tau}\sin\omega\tau,\end{aligned}$$

$$\int_0^t e^{r\tau}\sin\omega\tau\,d\tau = \frac{e^{rt}(r\sin\omega t - \omega\cos\omega t) + \omega}{r^2+\omega^2}.$$

Therefore

$$I(t) = \frac{E_0}{L(r^2+\omega^2)}\left[r\sin\omega t - \omega\cos\omega t + \omega e^{-rt}\right].$$

Now $L(r^2+\omega^2) = \dfrac{R^2+L^2\omega^2}{L}$ and $r = R/L$, so

$$I(t) = \frac{E_0}{R^2+L^2\omega^2}\left[R\sin\omega t - L\omega\cos\omega t\right] + \frac{E_0 L\omega}{R^2+L^2\omega^2}e^{-Rt/L}.$$

Write $R\sin\omega t - L\omega\cos\omega t = \sqrt{R^2+L^2\omega^2}\,\sin(\omega t - \alpha)$ with $\cos\alpha = R/\sqrt{R^2+L^2\omega^2}$, $\sin\alpha = L\omega/\sqrt{R^2+L^2\omega^2}$, i.e. $\tan\alpha = L\omega/R$ (as $\sin(\omega t-\alpha) = \sin\omega t\cos\alpha - \cos\omega t\sin\alpha$). Then

$$I(t) = \frac{E_0}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t - \alpha) + \frac{E_0 L\omega}{R^2+L^2\omega^2}e^{-Rt/L}.$$

Check. At $t = 0$: $\dfrac{E_0(-L\omega)}{R^2+L^2\omega^2} + \dfrac{E_0 L\omega}{R^2+L^2\omega^2} = 0$, as required. Also, by (13) with $E(0) = 0$,

$$I(t) = \int_0^t A(t-\tau)\,E_0\omega\cos\omega\tau\,d\tau,$$

which gives the same function (verified symbolically in `verify/sec53.py`).

Answer. (a) $I = \dfrac{E_0}{R}\left(1 - e^{-Rt/L}\right)$; (b) $I = \dfrac{E_0}{L}e^{-Rt/L}$; (c) $I = \dfrac{E_0}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t - \alpha) + \dfrac{E_0 L\omega}{R^2+L^2\omega^2}e^{-Rt/L}$, $\tan\alpha = \dfrac{L\omega}{R}$.

Chapter 10

Systems of First Order Equations

Section 54 General Remarks on Systems

The device used throughout this section is the one in the text: an nth order equation $y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$ is turned into a first order system by introducing the new unknowns $y_1 = y$, $y_2 = y'$, $\dots$, $y_n = y^{(n-1)}$. Then $y_k' = y_{k+1}$ for $k = 1, \dots, n-1$ (this is just the definition of the new unknowns) and the equation itself supplies y_n'.

Problem 54-1 (Simmons)

Solution.

(a) The equation is of second order, so we need two unknown functions. Put

$$y_1 = y = y(x), \qquad y_2 = y'.$$

(In the book's notation y and $z = y'$.) Then by definition $y_1' = y' = y_2$. Solving the equation for the highest derivative gives $y'' = x^2 y' + xy$, i.e.

$$y_2' = y'' = x^2 y_2 + x y_1.$$

Hence the equivalent system is

$$\begin{cases} y_1' = y_2, \\ y_2' = x y_1 + x^2 y_2, \end{cases} \qquad \text{or, with } y,\ z = y' : \qquad \begin{cases} \dfrac{dy}{dx} = z, \\ \dfrac{dz}{dx} = xy + x^2 z. \end{cases}$$

Equivalence. If $y(x)$ solves the second order equation, then $y_1 = y$, $y_2 = y'$ satisfy both equations above, as just shown. Conversely, if (y_1, y_2) solves the system, then $y_2 = y_1'$ by the first equation, so $y_2' = y_1''$, and the second equation reads $y_1'' = x y_1 + x^2 y_1'$, i.e. $y = y_1$ solves the original equation.

(b) The equation is of third order, so we need three unknowns:

$$y_1 = y, \qquad y_2 = y', \qquad y_3 = y'' \qquad (\text{book: } y,\ z = y',\ w = y'').$$

Then $y_1' = y_2$ and $y_2' = y_3$ by definition, and the equation $y''' = y'' - x^2 (y')^2$ gives $y_3' = y_3 - x^2 y_2^2$. The system is

$$\begin{cases} y_1' = y_2, \\ y_2' = y_3, \\ y_3' = y_3 - x^2 y_2^2, \end{cases} \qquad \text{i.e.} \qquad \begin{cases} \dfrac{dy}{dx} = z, \\ \dfrac{dz}{dx} = w, \\ \dfrac{dw}{dx} = w - x^2 z^2. \end{cases}$$

Equivalence is shown exactly as in (a): the first two equations force $y_2 = y_1'$ and $y_3 = y_2' = y_1''$, and then the third becomes $y_1''' = y_1'' - x^2 (y_1')^2$.

Answer. (a) $\dfrac{dy}{dx} = z$, $\dfrac{dz}{dx} = xy + x^2 z$; (b) $\dfrac{dy}{dx} = z$, $\dfrac{dz}{dx} = w$, $\dfrac{dw}{dx} = w - x^2 z^2$ (with $z = y'$, $w = y''$).

Problem 54-2 (Simmons)

Solution. Each second order equation needs one extra unknown, namely the corresponding velocity component. Put

$$v_x = \frac{dx}{dt}, \qquad v_y = \frac{dy}{dt}.$$

Then $\dfrac{dx}{dt} = v_x$ and $\dfrac{dy}{dt} = v_y$ by definition, and since $\dfrac{d^2x}{dt^2} = \dfrac{dv_x}{dt}$, $\dfrac{d^2y}{dt^2} = \dfrac{dv_y}{dt}$, the equations of motion (divided by $m \neq 0$) become

$$\frac{dv_x}{dt} = \frac{f(t,x,y)}{m}, \qquad \frac{dv_y}{dt} = \frac{g(t,x,y)}{m}.$$

The four unknowns $x,\ v_x,\ y,\ v_y$ (functions of t) therefore satisfy

$$\begin{cases} \dfrac{dx}{dt} = v_x, \\ \dfrac{dv_x}{dt} = \dfrac{f(t,x,y)}{m}, \\ \dfrac{dy}{dt} = v_y, \\ \dfrac{dv_y}{dt} = \dfrac{g(t,x,y)}{m}, \end{cases}$$

which has the form (1) with the independent variable t and the right sides depending on t, x, v_x, y, v_y. Equivalence: the first and third equations say that v_x, v_y are the derivatives of x, y; substituting this in the second and fourth gives back $mx'' = f$, $my'' = g$; and conversely a solution (x, y) of the original system gives, with $v_x = x'$, $v_y = y'$, a solution of the four-equation system.

Answer. $\dfrac{dx}{dt} = v_x,\ \dfrac{dv_x}{dt} = \dfrac{f(t,x,y)}{m},\ \dfrac{dy}{dt} = v_y,\ \dfrac{dv_y}{dt} = \dfrac{g(t,x,y)}{m}.$

Section 55 Linear Systems

Throughout, the homogeneous system is

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)x + b_1(t)y, \\ \dfrac{dy}{dt} = a_2(t)x + b_2(t)y, \end{cases} \tag{5}$$

and the nonhomogeneous system (2) is (5) with the extra terms $f_1(t)$, $f_2(t)$ added to the right sides. The coefficients and f_1, f_2 are continuous on $[a, b]$.

Problem 55-1 (Simmons)

Solution. *Theorem B.* If $x = x_1(t)$, $y = y_1(t)$ and $x = x_2(t)$, $y = y_2(t)$ are solutions of (5) on $[a, b]$, then $x = c_1x_1 + c_2x_2$, $y = c_1y_1 + c_2y_2$ is a solution of (5) on $[a, b]$ for all constants c_1, c_2.

Proof. By hypothesis, for all t in $[a, b]$,

$$x_1' = a_1x_1 + b_1y_1, \qquad y_1' = a_2x_1 + b_2y_1,$$
$$x_2' = a_1x_2 + b_1y_2, \qquad y_2' = a_2x_2 + b_2y_2.$$

Let $x = c_1x_1 + c_2x_2$ and $y = c_1y_1 + c_2y_2$. Differentiation is linear, so

$$\begin{aligned} x' &= c_1x_1' + c_2x_2' = c_1(a_1x_1 + b_1y_1) + c_2(a_1x_2 + b_1y_2) \\ &= a_1(c_1x_1 + c_2x_2) + b_1(c_1y_1 + c_2y_2) = a_1x + b_1y, \\ y' &= c_1y_1' + c_2y_2' = c_1(a_2x_1 + b_2y_1) + c_2(a_2x_2 + b_2y_2) \\ &= a_2(c_1x_1 + c_2x_2) + b_2(c_1y_1 + c_2y_2) = a_2x + b_2y. \end{aligned}$$

Thus both equations of (5) hold throughout $[a, b]$, and (x, y) is a solution. ■

Answer. Shown: $(c_1x_1 + c_2x_2,\ c_1y_1 + c_2y_2)$ satisfies both equations of (5) by linearity of the derivative.

Problem 55-2 (Simmons)

Solution. *Theorem F.* If (x_1, y_1), (x_2, y_2) are linearly independent solutions of (5) on $[a, b]$ and (x_p, y_p) is any particular solution of (2), then $x = c_1x_1 + c_2x_2 + x_p$, $y = c_1y_1 + c_2y_2 + y_p$ is the general solution of (2) on $[a, b]$.

The text reduces the proof to the following claim, which we now prove. *Claim:* if (x, y) is an arbitrary solution of (2), then $(X, Y) = (x - x_p,\ y - y_p)$ is a solution of (5).

Since (x, y) and (x_p, y_p) both solve (2),

$$x' = a_1x + b_1y + f_1, \qquad y' = a_2x + b_2y + f_2,$$
$$x_p' = a_1x_p + b_1y_p + f_1, \qquad y_p' = a_2x_p + b_2y_p + f_2.$$

Subtracting, the terms f_1 and f_2 cancel:

$$\begin{aligned} X' &= x' - x_p' = a_1(x - x_p) + b_1(y - y_p) = a_1X + b_1Y, \\ Y' &= y' - y_p' = a_2(x - x_p) + b_2(y - y_p) = a_2X + b_2Y. \end{aligned}$$

So (X, Y) solves (5), proving the claim.

Conclusion. (i) Every function pair $(c_1x_1 + c_2x_2 + x_p,\ c_1y_1 + c_2y_2 + y_p)$ solves (2): the combination $c_1(x_1, y_1) + c_2(x_2, y_2)$ solves (5) by Theorem B, and adding (x_p, y_p) adds exactly f_1, f_2 to the right sides, because (x_p, y_p) solves (2). (ii) Conversely, if (x, y) is any solution of (2), the claim shows $(x - x_p,\ y - y_p)$ solves (5). Since (x_1, y_1), (x_2, y_2) are linearly independent, Theorem E says that $c_1(x_1, y_1) + c_2(x_2, y_2)$ is the general solution of (5), so there are constants c_1, c_2 with $x - x_p = c_1x_1 + c_2x_2$ and $y - y_p = c_1y_1 + c_2y_2$, i.e. (x, y) has the form (14). Hence (14) contains every solution of (2) and is the general solution. ■

Answer. Shown: the difference of any two solutions of (2) solves (5), so every solution of (2) equals (x_p, y_p) plus a solution of (5), i.e. has the form (14).

Problem 55-3 (Simmons)

Solution. Equation (9) is $\dfrac{dW}{dt} = [a_1(t) + b_2(t)]W$, where for two solutions (x_1, y_1), (x_2, y_2) of (5)

$$W = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1y_2 - x_2y_1.$$

Differentiate with the product rule:

$$W' = x_1'y_2 + x_1y_2' - x_2'y_1 - x_2y_1'.$$

Substitute $x_1' = a_1x_1 + b_1y_1$, $y_2' = a_2x_2 + b_2y_2$, $x_2' = a_1x_2 + b_1y_2$, $y_1' = a_2x_1 + b_2y_1$:

$$\begin{aligned} W' &= (a_1x_1 + b_1y_1)y_2 + x_1(a_2x_2 + b_2y_2) - (a_1x_2 + b_1y_2)y_1 - x_2(a_2x_1 + b_2y_1) \\ &= a_1x_1y_2 + b_1y_1y_2 + a_2x_1x_2 + b_2x_1y_2 - a_1x_2y_1 - b_1y_1y_2 - a_2x_1x_2 - b_2x_2y_1. \end{aligned}$$

The terms $b_1y_1y_2$ and $a_2x_1x_2$ cancel, leaving

$$W' = a_1(x_1y_2 - x_2y_1) + b_2(x_1y_2 - x_2y_1) = (a_1 + b_2)W.$$

This is (9). (Solving this linear equation gives (10), $W = c\,e^{\int (a_1 + b_2)\,dt}$.)

Answer. Verified: $W' = (a_1 + b_2)W$.

Problem 55-4 (Simmons)

Solution. *The reduction.* With $y = x'$ we have $x' = y$ and $y' = x'' = -Px' - Qx = -Qx - Py$, which is $(**)$; so if $x_i(t)$ solves $(*)$, the corresponding solution of $(**)$ is $x = x_i(t)$, $y = y_i(t) = x_i'(t)$, $i = 1, 2$.

Wronskians. In the sense of Section 15, the Wronskian of x_1, x_2 is

$$W_{15}(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = x_1x_2' - x_2x_1'.$$

In the sense of Section 55, the Wronskian of the two solutions of $(**)$ is

$$W(t) = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1y_2 - x_2y_1.$$

Since $y_1 = x_1'$ and $y_2 = x_2'$,

$$W(t) = x_1x_2' - x_2x_1' = W_{15}(t)$$

for every t; in fact the two determinants are literally the same determinant.

Answer. Shown: since $y_i = x_i'$, $\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix}$.

Problem 55-5 (Simmons)

Solution.

(a) First solution: $x = y = e^{4t}$. Then $x' = 4e^{4t}$ and $y' = 4e^{4t}$, while $x + 3y = e^{4t} + 3e^{4t} = 4e^{4t}$ and $3x + y = 4e^{4t}$. Both equations hold.

Second solution: $x = e^{-2t}$, $y = -e^{-2t}$. Then $x' = -2e^{-2t}$, $y' = 2e^{-2t}$, while $x + 3y = e^{-2t} - 3e^{-2t} = -2e^{-2t}$ and $3x + y = 3e^{-2t} - e^{-2t} = 2e^{-2t}$. Both equations hold.

(b) *Way 1 (Wronskian, Theorem C).*

$$W(t) = \begin{vmatrix} e^{4t} & e^{-2t} \\ e^{4t} & -e^{-2t} \end{vmatrix} = -e^{4t}e^{-2t} - e^{-2t}e^{4t} = -2e^{2t},$$

which is nowhere 0, hence nonzero on every closed interval. (Check with (9): $a_1 + b_2 = 1 + 1 = 2$ and $W' = -4e^{2t} = 2W$.)

Way 2 (not constant multiples, Theorem E). If the solutions were linearly dependent, one of the first components would be a constant multiple of the other: $e^{4t} = ke^{-2t}$ or $e^{-2t} = ke^{4t}$ for all t in the interval. But $e^{4t}/e^{-2t} = e^{6t}$ is not constant on any interval (its derivative $6e^{6t}$ never vanishes), and neither is its reciprocal e^{-6t}. Hence neither solution is a constant multiple of the other: they are linearly independent.

By Theorem C (or E) the general solution is

$$x = c_1e^{4t} + c_2e^{-2t}, \qquad y = c_1e^{4t} - c_2e^{-2t}.$$

(c) Impose $x(0) = 5$, $y(0) = 1$:

$$c_1 + c_2 = 5, \qquad c_1 - c_2 = 1.$$

Adding gives $2c_1 = 6$, $c_1 = 3$, and then $c_2 = 2$. Therefore

$$x = 3e^{4t} + 2e^{-2t}, \qquad y = 3e^{4t} - 2e^{-2t}.$$

(Check: $x(0) = 5$, $y(0) = 1$.)

Answer. (b) $x = c_1e^{4t} + c_2e^{-2t}$, $y = c_1e^{4t} - c_2e^{-2t}$; (c) $x = 3e^{4t} + 2e^{-2t}$, $y = 3e^{4t} - 2e^{-2t}$.

Problem 55-6 (Simmons)

Solution.

(a) First: $x = 2e^{4t}$, $y = 3e^{4t}$ give $x' = 8e^{4t}$, $y' = 12e^{4t}$; and $x + 2y = 2e^{4t} + 6e^{4t} = 8e^{4t}$, $3x + 2y = 6e^{4t} + 6e^{4t} = 12e^{4t}$. Both equations hold.

Second: $x = e^{-t}$, $y = -e^{-t}$ give $x' = -e^{-t}$, $y' = e^{-t}$; and $x + 2y = e^{-t} - 2e^{-t} = -e^{-t}$, $3x + 2y = 3e^{-t} - 2e^{-t} = e^{-t}$. Both equations hold.

(b) *Way 1 (Wronskian).*

$$W(t) = \begin{vmatrix} 2e^{4t} & e^{-t} \\ 3e^{4t} & -e^{-t} \end{vmatrix} = -2e^{3t} - 3e^{3t} = -5e^{3t} \neq 0$$

for every t, so the solutions are independent on every closed interval (Theorem C).

Way 2. Linear dependence would require, e.g., $2e^{4t} = ke^{-t}$ for all t in the interval, or $e^{-t} = k \cdot 2e^{4t}$; but $2e^{4t}/e^{-t} = 2e^{5t}$ (and its reciprocal) is not constant on any interval (derivative $10e^{5t} \neq 0$). So neither solution is a constant multiple of the other, and they are linearly independent (Theorem E).

General solution of the homogeneous system:

$$x = 2c_1e^{4t} + c_2e^{-t}, \qquad y = 3c_1e^{4t} - c_2e^{-t}.$$

(c) For $x = 3t - 2$, $y = -2t + 3$ we have $x' = 3$, $y' = -2$. The right sides are

$$x + 2y + t - 1 = (3t - 2) + (-4t + 6) + t - 1 = (3 - 4 + 1)t + (-2 + 6 - 1) = 3,$$
$$3x + 2y - 5t - 2 = (9t - 6) + (-4t + 6) - 5t - 2 = (9 - 4 - 5)t + (-6 + 6 - 2) = -2.$$

They equal x' and y' respectively, so this is a particular solution. By Theorem F the general solution is (general solution of the homogeneous system) + (particular solution):

$$x = 2c_1e^{4t} + c_2e^{-t} + 3t - 2, \qquad y = 3c_1e^{4t} - c_2e^{-t} - 2t + 3.$$

Answer. (b) $x = 2c_1e^{4t} + c_2e^{-t}$, $y = 3c_1e^{4t} - c_2e^{-t}$; (c) $x = 2c_1e^{4t} + c_2e^{-t} + 3t - 2$, $y = 3c_1e^{4t} - c_2e^{-t} - 2t + 3$.

Problem 55-7 (Simmons)

Solution. The system is $x' = x + 2y$, $y' = 3x + 2y$.

(a) Differentiate the first equation: $x'' = x' + 2y'$. Substitute $y' = 3x + 2y$: $x'' = x' + 6x + 4y$. From the first equation, $2y = x' - x$, so $4y = 2x' - 2x$ and

$$x'' = x' + 6x + 2x' - 2x = 3x' + 4x, \qquad \text{i.e.} \qquad x'' - 3x' - 4x = 0.$$

The characteristic equation $m^2 - 3m - 4 = (m - 4)(m + 1) = 0$ has roots 4 and -1, so

$$x = Ae^{4t} + Be^{-t}.$$

The first equation gives y without further integration: $y = \frac{1}{2}(x' - x)$, and $x' = 4Ae^{4t} - Be^{-t}$, so

$$y = \tfrac{1}{2}\left(3Ae^{4t} - 2Be^{-t}\right) = \tfrac{3}{2}Ae^{4t} - Be^{-t}.$$

Putting $A = 2c_1$, $B = c_2$:

$$x = 2c_1e^{4t} + c_2e^{-t}, \qquad y = 3c_1e^{4t} - c_2e^{-t},$$

which is the general solution found in Problem 6(b). (Taking $c_1 = 1, c_2 = 0$ and $c_1 = 0, c_2 = 1$ gives the two given solutions $(2e^{4t}, 3e^{4t})$ and $(e^{-t}, -e^{-t})$.) Because y is determined by x, no extraneous solutions appear: with $y = \frac{1}{2}(x' - x)$ the first equation $x' = x + 2y$ holds identically, and the second, $y' = \frac{1}{2}(x'' - x') = \frac{1}{2}(2x' + 4x) = x' + 2x = 3x + 2y$, holds because $x'' = 3x' + 4x$.

(b) Differentiate the second equation: $y'' = 3x' + 2y'$. Substitute $x' = x + 2y$: $y'' = 3x + 6y + 2y'$. From the second equation $3x = y' - 2y$, hence

$$y'' = y' - 2y + 6y + 2y' = 3y' + 4y, \qquad \text{i.e.} \qquad y'' - 3y' - 4y = 0,$$

so again $y = Ae^{4t} + Be^{-t}$. Now the second equation gives $x = \frac{1}{3}(y' - 2y)$, with $y' = 4Ae^{4t} - Be^{-t}$:

$$x = \tfrac{1}{3}\left(2Ae^{4t} - 3Be^{-t}\right) = \tfrac{2}{3}Ae^{4t} - Be^{-t}.$$

Putting $A = 3c_1$, $B = -c_2$:

$$x = 2c_1e^{4t} + c_2e^{-t}, \qquad y = 3c_1e^{4t} - c_2e^{-t},$$

the same general solution. (Again no extraneous solutions: $x = \frac{1}{3}(y' - 2y)$ makes the second equation hold identically, and $x' = \frac{1}{3}(y'' - 2y') = \frac{1}{3}(y' + 4y) = x + 2y$ because $y'' = 3y' + 4y$.)

Answer. Both eliminations give $x'' - 3x' - 4x = 0$ (resp. $y'' - 3y' - 4y = 0$) and the solutions $x = 2c_1e^{4t} + c_2e^{-t}$, $y = 3c_1e^{4t} - c_2e^{-t}$.

Problem 55-8 (Simmons)

Solution. Here the second equation involves only y, so we solve it first (this is the elimination of Problem 7(b), in its simplest form). From $y' = y$ we get $y = c_2e^t$. Substituting in the first equation:

$$x' - x = c_2e^t.$$

This is a first order linear equation; multiplying by the integrating factor e^{-t} gives $(xe^{-t})' = c_2$, so $xe^{-t} = c_2t + c_1$ and $x = (c_1 + c_2t)e^t$.

Alternatively, differentiating the first equation and eliminating y as in Problem 7(a): $x'' = x' + y' = x' + y = x' + (x' - x)$, so $x'' - 2x' + x = 0$, with characteristic equation $(m-1)^2 = 0$, hence $x = (c_1 + c_2t)e^t$, and $y = x' - x = c_2e^t$, the same result.

Verification: $x' = (c_1 + c_2 + c_2t)e^t$ and $x + y = (c_1 + c_2t + c_2)e^t$ agree, and $y' = c_2e^t = y$. The two solutions $(e^t, 0)$ and (te^t, e^t) have Wronskian $\begin{vmatrix} e^t & te^t \\ 0 & e^t \end{vmatrix} = e^{2t} \neq 0$, so this is the general solution.

Answer. $x = c_1e^t + c_2te^t, \quad y = c_2e^t$.

Problem 55-9 (Simmons)

Solution.

(a) The equations are uncoupled. $x' = x$ gives $x = c_1e^t$ and $y' = y$ gives $y = c_2e^t$ with independent constants:

$$x = c_1e^t, \qquad y = c_2e^t.$$

(Wronskian of $(e^t, 0)$ and $(0, e^t)$ is $e^{2t} \neq 0$.)

(b) *What is to be shown.* The solutions of the system are exactly the pairs (c_1e^t, c_2e^t) (part (a) and Theorem E). In particular the first components of the solutions of the system are exactly the functions c_1e^t, a *one*-parameter family, and the system gives no relation at all between x and y. A second order equation for the single unknown x that is obtained from the system must be satisfied by every x-component c_1e^t; we show that every such equation also has solutions $\tilde{x}(t)$ for which no function $\tilde{y}(t)$ exists with $(\tilde{x}, \tilde{y})$ a solution of the system. (The same applies to an equation for y, by symmetry of the system under $x \leftrightarrow y$.)

Step 1: the equations actually obtained by differentiating. Differentiating $x' = x$ gives $x'' = x'$; using $x' = x$ once more one may also write $x'' = x$, or in general

$$x'' + px' + qx = 0 \qquad \text{with } p + q = -1, \tag{†}$$

because for every solution $x'' = x' = x$ gives $x'' + px' + qx = (1 + p + q)x = 0$. (The case $p = -1$, $q = 0$ is $x'' - x' = 0$, and $p = 0$, $q = -1$ is $x'' - x = 0$.) The only "other" equation is the one from $y' = y$, which is the same with y in place of x. Equation (†) has characteristic polynomial

$$m^2 + pm + q = m^2 + pm - 1 - p = (m - 1)(m + 1 + p),$$

with roots 1 and $-(1 + p)$.

- If $p \neq -2$, the roots are distinct and $\tilde{x} = e^{-(1+p)t}$ solves (†). If this were the first component of a solution of the system we would need $\tilde{x}' = \tilde{x}$, i.e. $-(1 + p)e^{-(1+p)t} = e^{-(1+p)t}$, i.e. $-(1 + p) = 1$, i.e. $p = -2$, which is excluded.
- If $p = -2$ (so $q = 1$), (†) is $x'' - 2x' + x = 0$, with solution $\tilde{x} = te^t$. Then $\tilde{x}' = (1 + t)e^t \neq te^t = \tilde{x}$ for every t.

In particular, for $x'' - x' = 0$ ($p = -1$): $\tilde{x} = a + be^t$ is the general solution, and $\tilde{x}' = \tilde{x}$ forces $be^t = a + be^t$, i.e. $a = 0$; so e.g. $\tilde{x} \equiv 1$ (which satisfies $1'' - 1' = 0$) is not the x-component of any solution of the system, since it would need $0 = \tilde{x}' = \tilde{x} = 1$. Likewise $\tilde{x} = e^{-t}$ solves $x'' - x = 0$ but $\tilde{x}' = -e^{-t} \neq \tilde{x}$.

Step 2: any second order equation whatever. Let $x'' = \Phi(t, x, x')$ be any second order equation for x with Φ continuous and locally Lipschitz in (x, x') (so that the existence and uniqueness theorem applies), and suppose every x-component c_1e^t of a solution of the system satisfies it. By the existence theorem there is a solution $\tilde{x}$ of it on some interval around $t = 0$ with $\tilde{x}(0) = 1$, $\tilde{x}'(0) = 0$ (the initial data $(\tilde{x}(0), \tilde{x}'(0))$ of a second order equation can be prescribed arbitrarily). If $\tilde{x}$ were the first component of a solution of the system, we would have $\tilde{x}'(0) = \tilde{x}(0)$, i.e. $0 = 1$, a contradiction. So this $\tilde{x}$ is a solution of the second order equation that is part of no solution of the system.

Conclusion. The solutions of any such second order equation form a two-parameter family (initial values $x(0)$, $x'(0)$ arbitrary), whereas the x-components of the solutions of the system form only the one-parameter family c_1e^t (the extra constant c_2 of the system lives in y and is invisible to x). Hence no second order equation obtained from the system is equivalent to it: passing from the system to such an equation enlarges the solution set (and discards y). In contrast, in Problem 7 the second unknown could be recovered from the first, $y = \frac{1}{2}(x' - x)$, which is why the second order equation was equivalent there. Thus higher order equations are equivalent to systems, but not conversely, and systems are more general.

Answer. (a) $x = c_1e^t$, $y = c_2e^t$. (b) Shown: every second order equation obtained from the system has a solution that is not the x-component of any solution of the system (e.g. $x'' - x' = 0$ has the solution $x = 1$, while system solutions need $x = c_1e^t$); the same holds for y.

Section 56 Homogeneous Linear Systems with Constant Coefficients

Throughout, the system is

$$\frac{dx}{dt} = a_1x + b_1y, \qquad \frac{dy}{dt} = a_2x + b_2y,$$

and we seek solutions $x = Ae^{mt}$, $y = Be^{mt}$. Substitution gives the algebraic system

$$(a_1 - m)A + b_1B = 0, \qquad a_2A + (b_2 - m)B = 0, \tag{3}$$

which has a nontrivial solution exactly when the auxiliary equation $m^2 - (a_1 + b_2)m + (a_1b_2 - a_2b_1) = 0$ holds (the text's equation (4)). In each part below, the auxiliary equation is the characteristic polynomial of the coefficient matrix: its roots are $m_{1,2}$, the pair (A, B) is read off from (3), and the three cases of the text (distinct real, complex, equal roots) are applied.

Problem 56-1 (Simmons)

Solution.

(a) Here $a_1 = -3$, $b_1 = 4$, $a_2 = -2$, $b_2 = 3$, so $a_1 + b_2 = 0$ and $a_1b_2 - a_2b_1 = -9 + 8 = -1$. The auxiliary equation is $m^2 - 1 = 0$, with distinct real roots $m = 1$ and $m = -1$.

$m = 1$: (3) reads $-4A + 4B = 0$ and $-2A + 2B = 0$, both equivalent to $A = B$; take $A = B = 1$. Solution: $x = e^t$, $y = e^t$.

$m = -1$: (3) reads $-2A + 4B = 0$ and $-2A + 4B = 0$, i.e. $A = 2B$; take $A = 2$, $B = 1$. Solution: $x = 2e^{-t}$, $y = e^{-t}$.

The two solutions are linearly independent (their Wronskian at $t = 0$ is $\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1 \neq 0$), so

$$x = 2c_1e^{-t} + c_2e^t, \qquad y = c_1e^{-t} + c_2e^t.$$

(b) Here $a_1 + b_2 = 4 + 2 = 6$ and $a_1b_2 - a_2b_1 = 8 + 10 = 18$, so the auxiliary equation is $m^2 - 6m + 18 = 0$, with roots $m = 3 \pm 3i$ (complex, $a = 3$, $b = 3$).

Take $m = 3 + 3i$. Equation (3) is $(1 - 3i)A - 2B = 0$ and $5A + (-1 - 3i)B = 0$. The first gives $B = \frac{1-3i}{2}A$; with $A = 2$ we get $B = 1 - 3i$. The second equation is then $10 + (-1 - 3i)(1 - 3i) = 10 + (-1 + 3i - 3i + 9i^2) = 10 - 10 = 0$, consistent. So the complex solution is

$$x = 2e^{(3+3i)t}, \qquad y = (1 - 3i)e^{(3+3i)t}.$$

By Euler's formula $e^{(3+3i)t} = e^{3t}(\cos 3t + i \sin 3t)$:

$$\begin{aligned} x &= e^{3t}(2\cos 3t) + i\,e^{3t}(2\sin 3t), \\ y &= (1 - 3i)e^{3t}(\cos 3t + i\sin 3t) = e^{3t}\big[(\cos 3t + 3\sin 3t) + i(\sin 3t - 3\cos 3t)\big]. \end{aligned}$$

(In the notation of the text, $A_1 = 2$, $A_2 = 0$, $B_1 = 1$, $B_2 = -3$.) The real and imaginary parts are real solutions, and they are linearly independent: their Wronskian at $t = 0$ is $\begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} = -6 \neq 0$. Hence

$$\begin{aligned} x &= e^{3t}\,(2c_1\cos 3t + 2c_2\sin 3t), \\ y &= e^{3t}\big[c_1(\cos 3t + 3\sin 3t) + c_2(\sin 3t - 3\cos 3t)\big]. \end{aligned}$$

(c) $a_1 + b_2 = 6$, $a_1b_2 - a_2b_1 = 5 + 4 = 9$, so $m^2 - 6m + 9 = (m-3)^2 = 0$: equal roots $m = 3$ (the system is not uncoupled, since $b_1 = 4 \neq 0$).

With $m = 3$, (3) reads $2A + 4B = 0$ and $-A - 2B = 0$, i.e. $A = -2B$; take $A = -2$, $B = 1$:

$$x = -2e^{3t}, \qquad y = e^{3t}.$$

For a second solution we try $x = (A_1 + A_2t)e^{3t}$, $y = (B_1 + B_2t)e^{3t}$. Substituting into $x' = 5x + 4y$, $y' = -x + y$ and dividing by e^{3t}:

$$\begin{aligned} A_2 + 3(A_1 + A_2t) &= 5(A_1 + A_2t) + 4(B_1 + B_2t), \\ B_2 + 3(B_1 + B_2t) &= -(A_1 + A_2t) + (B_1 + B_2t), \end{aligned}$$

that is,

$$\begin{aligned} (-2A_2 - 4B_2)t + (A_2 - 2A_1 - 4B_1) &= 0, \\ (A_2 + 2B_2)t + (B_2 + A_1 + 2B_1) &= 0. \end{aligned}$$

These must be identities in t. The coefficients of t give $A_2 = -2B_2$ (both equations say the same thing); this is the same relation as for (A, B), as Problem 4 shows in general. Choose $A_2 = 2$, $B_2 = -1$. The constant terms then give $2 - 2A_1 - 4B_1 = 0$ and $-1 + A_1 + 2B_1 = 0$, both equivalent to $A_1 + 2B_1 = 1$; take $A_1 = 1$, $B_1 = 0$. The second solution is

$$x = (1 + 2t)e^{3t}, \qquad y = -te^{3t}.$$

It is independent of the first (Wronskian at $t = 0$: $\begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$), so

$$x = -2c_1e^{3t} + c_2(1 + 2t)e^{3t}, \qquad y = c_1e^{3t} - c_2te^{3t}.$$

(d) $a_1 + b_2 = -2$, $a_1b_2 - a_2b_1 = -24 + 24 = 0$, so $m^2 + 2m = m(m+2) = 0$: distinct real roots $m = 0$, $m = -2$.

$m = 0$: (3) is $4A - 3B = 0$ and $8A - 6B = 0$ (equivalent); take $A = 3$, $B = 4$: $x = 3$, $y = 4$ (a constant solution).

$m = -2$: (3) is $6A - 3B = 0$ and $8A - 4B = 0$, i.e. $B = 2A$; take $A = 1$, $B = 2$: $x = e^{-2t}$, $y = 2e^{-2t}$.

Wronskian at $t = 0$: $\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 2 \neq 0$. Hence

$$x = 3c_1 + c_2e^{-2t}, \qquad y = 4c_1 + 2c_2e^{-2t}.$$

(e) The system is uncoupled: $a_1 = 2$, $b_2 = 3$, $b_1 = a_2 = 0$, and the auxiliary equation $m^2 - 5m + 6 = (m-2)(m-3) = 0$ has distinct roots 2, 3.

$m = 2$: (3) is $0 \cdot A + 0 \cdot B = 0$ and $0 \cdot A + B = 0$, so $B = 0$; take $A = 1$: $x = e^{2t}$, $y = 0$.

$m = 3$: (3) is $-A = 0$ and $0 = 0$, so $A = 0$; take $B = 1$: $x = 0$, $y = e^{3t}$.

These are clearly independent, and

$$x = c_1e^{2t}, \qquad y = c_2e^{3t}.$$

(Equivalently, each equation $x' = 2x$, $y' = 3y$ can be integrated separately.)

(f) $a_1 + b_2 = -6$, $a_1b_2 - a_2b_1 = 8 + 1 = 9$, so $m^2 + 6m + 9 = (m+3)^2 = 0$: equal roots $m = -3$ (not uncoupled since $b_1 = -1 \neq 0$).

With $m=-3$, (3) is $-A-B=0$ and $A+B=0$, so $B=-A$; take $A=1$, $B=-1$: $x=e^{-3t}$, $y=-e^{-3t}$.

Second solution $x=(A_1+A_2t)e^{-3t}$, $y=(B_1+B_2t)e^{-3t}$. Substituting in $x'=-4x-y$, $y'=x-2y$ and dividing by e^{-3t}:

$$A_2-3(A_1+A_2t)=-4(A_1+A_2t)-(B_1+B_2t),$$
$$B_2-3(B_1+B_2t)=(A_1+A_2t)-2(B_1+B_2t),$$

i.e.

$$(A_2+B_2)t+(A_2+A_1+B_1)=0,$$
$$(-A_2-B_2)t+(B_2-A_1-B_1)=0.$$

The coefficients of t require $B_2=-A_2$; take $A_2=-1$, $B_2=1$. The constant terms then require $A_1+B_1=-A_2=1$ (from the first) and $A_1+B_1=B_2=1$ (from the second), which agree; take $A_1=1$, $B_1=0$. The second solution is

$$x=(1-t)e^{-3t},\qquad y=te^{-3t}.$$

Wronskian at $t=0$: $\begin{vmatrix}1&1\\-1&0\end{vmatrix}=1\neq 0$. Hence

$$x=c_1e^{-3t}+c_2(1-t)e^{-3t},\qquad y=-c_1e^{-3t}+c_2te^{-3t}.$$

(g) $a_1+b_2=13$, $a_1b_2-a_2b_1=42-12=30$, so $m^2-13m+30=(m-10)(m-3)=0$: $m=10$, $m=3$.

$m=10$: (3) is $-3A+6B=0$, $2A-4B=0$, i.e. $A=2B$; take $A=2$, $B=1$: $x=2e^{10t}$, $y=e^{10t}$.

$m=3$: (3) is $4A+6B=0$, $2A+3B=0$, i.e. $2A=-3B$; take $A=3$, $B=-2$: $x=3e^{3t}$, $y=-2e^{3t}$.

Wronskian at $t=0$: $\begin{vmatrix}2&3\\1&-2\end{vmatrix}=-7\neq 0$. Hence

$$x=2c_1e^{10t}+3c_2e^{3t},\qquad y=c_1e^{10t}-2c_2e^{3t}.$$

(h) $a_1+b_2=6$, $a_1b_2-a_2b_1=5+8=13$, so $m^2-6m+13=0$, $m=3\pm 2i$ ($a=3$, $b=2$).

Take $m=3+2i$. Equation (3) is $(-2-2i)A-2B=0$ and $4A+(2-2i)B=0$. The first gives $B=-(1+i)A$; with $A=1$, $B=-1-i$. The second then reads $4-(2-2i)(1+i)=4-(2+2i-2i+2)=0$, consistent. The complex solution is

$$x=e^{3t}(\cos 2t+i\sin 2t),$$
$$y=(-1-i)e^{3t}(\cos 2t+i\sin 2t)=e^{3t}\big[(\sin 2t-\cos 2t)+i(-\sin 2t-\cos 2t)\big].$$

Its real and imaginary parts are solutions:

$$\begin{cases}x=e^{3t}\cos 2t\\ y=e^{3t}(\sin 2t-\cos 2t)\end{cases}\qquad\qquad \begin{cases}x=e^{3t}\sin 2t\\ y=-e^{3t}(\sin 2t+\cos 2t)\end{cases}$$

with Wronskian at $t=0$ equal to $\begin{vmatrix}1&0\\-1&-1\end{vmatrix}=-1\neq 0$. Hence

$$x=e^{3t}(c_1\cos 2t+c_2\sin 2t),$$
$$y=e^{3t}\big[c_1(\sin 2t-\cos 2t)-c_2(\sin 2t+\cos 2t)\big].$$

Answer. (a) $x = 2c_1e^{-t} + c_2e^{t}$, $y = c_1e^{-t} + c_2e^{t}$;
(b) $x = e^{3t}(2c_1\cos 3t + 2c_2\sin 3t)$, $y = e^{3t}[c_1(\cos 3t + 3\sin 3t) + c_2(\sin 3t - 3\cos 3t)]$;
(c) $x = -2c_1e^{3t} + c_2(1+2t)e^{3t}$, $y = c_1e^{3t} - c_2te^{3t}$;
(d) $x = 3c_1 + c_2e^{-2t}$, $y = 4c_1 + 2c_2e^{-2t}$;
(e) $x = c_1e^{2t}$, $y = c_2e^{3t}$;
(f) $x = c_1e^{-3t} + c_2(1-t)e^{-3t}$, $y = -c_1e^{-3t} + c_2te^{-3t}$;
(g) $x = 2c_1e^{10t} + 3c_2e^{3t}$, $y = c_1e^{10t} - 2c_2e^{3t}$;
(h) $x = e^{3t}(c_1\cos 2t + c_2\sin 2t)$, $y = e^{3t}[c_1(\sin 2t - \cos 2t) - c_2(\sin 2t + \cos 2t)]$.

Problem 56-2 (Simmons)

Solution. Solutions of the form (2) are $x = Ae^{mt}$, $y = Be^{mt}$ with $(A,B) \neq (0,0)$ and m a root of (4); real-valued ones require m, A, B real. The discriminant of (4) is

$$(a_1+b_2)^2 - 4(a_1b_2 - a_2b_1) = (a_1-b_2)^2 + 4a_2b_1.$$

Sufficiency. If $a_2b_1 > 0$ then the discriminant is $\geq 4a_2b_1 > 0$, so (4) has two distinct real roots $m_1 \neq m_2$. For each m_j the real homogeneous system (3) has zero determinant, hence a real nontrivial solution (A_j, B_j). The two solutions $x = A_je^{m_jt}$, $y = B_je^{m_jt}$ are linearly independent: their Wronskian at $t = 0$ is $A_1B_2 - A_2B_1$, and if this vanished then $(A_2, B_2) = k(A_1, B_1)$ with $k \neq 0$ (both vectors are nonzero), so $(A, B) = (A_1, B_1)$ would satisfy (3) for both $m = m_1$ and $m = m_2$; subtracting the two versions gives $(m_1 - m_2)A_1 = 0$ and $(m_1 - m_2)B_1 = 0$, so $A_1 = B_1 = 0$, a contradiction. Hence $W(0) \neq 0$, and the solutions are linearly independent.

Not necessary. The system $x' = x$, $y' = 2y$ has $a_2b_1 = 0 \not> 0$, yet $x = e^t$, $y = 0$ and $x = 0$, $y = e^{2t}$ are two real linearly independent solutions of the form (2). Even $a_2b_1 < 0$ is possible: for $x' = 3x + y$, $y' = -x$ we have $a_2b_1 = -1 < 0$, but the discriminant is $9 - 4 = 5 > 0$, so $m^2 - 3m + 1 = 0$ has two real roots $m = (3 \pm \sqrt{5})/2$, and the argument above (distinct real roots) gives two real linearly independent solutions of the form (2).

Answer. Shown: $a_2b_1 > 0$ forces a positive discriminant $(a_1 - b_2)^2 + 4a_2b_1$, hence two distinct real roots and two independent real solutions (2); but $x' = x$, $y' = 2y$ (where $a_2b_1 = 0$) and $x' = 3x + y$, $y' = -x$ (where $a_2b_1 < 0$) also have them.

Problem 56-3 (Simmons)

Solution. Write $c = \cos bt$, $s = \sin bt$. The solutions are

$$x_1 = e^{at}(A_1c - A_2s), \qquad y_1 = e^{at}(B_1c - B_2s),$$
$$x_2 = e^{at}(A_1s + A_2c), \qquad y_2 = e^{at}(B_1s + B_2c),$$

and $W = x_1y_2 - x_2y_1$. Then

$$\begin{aligned} We^{-2at} &= (A_1c - A_2s)(B_1s + B_2c) - (A_1s + A_2c)(B_1c - B_2s) \\ &= \big(A_1B_1cs + A_1B_2c^2 - A_2B_1s^2 - A_2B_2sc\big) - \big(A_1B_1sc - A_1B_2s^2 + A_2B_1c^2 - A_2B_2cs\big) \\ &= A_1B_2(c^2 + s^2) - A_2B_1(s^2 + c^2) = A_1B_2 - A_2B_1, \end{aligned}$$

so $W(t) = (A_1B_2 - A_2B_1)e^{2at}$.

Nonvanishing. Let $M = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$ (real) and let $\mu = a + ib$, $b \neq 0$, be a root of (4), with $(A_1 + iA_2,\ B_1 + iB_2)$ a nontrivial solution of (3), i.e. with $u = (A_1, B_1)^T$ and $v = (A_2, B_2)^T$ (real vectors, not both zero),

$$M(u + iv) = (a + ib)(u + iv).$$

Suppose $A_1B_2 - A_2B_1 = 0$. Then u and v are linearly dependent in $\mathbb{R}^2$, so there is a real vector $w \neq 0$ and real numbers α, β, not both 0, with $u = \alpha w$, $v = \beta w$ (if $u \neq 0$ take $w = u$; if $u = 0$ take $w = v$). Then $u + iv = (\alpha + i\beta)w$ with $\alpha + i\beta \neq 0$, and dividing the displayed equation by $\alpha + i\beta$ gives

$$Mw = (a + ib)w.$$

The left side is a real vector, while the right side has imaginary part $bw \neq 0$ (since $b \neq 0$, $w \neq 0$). This is a contradiction. Hence $A_1B_2 - A_2B_1 \neq 0$, so $W(t) \neq 0$ for all t and (15), (16) are linearly independent.

Answer. Shown: $W(t) = (A_1B_2 - A_2B_1)e^{2at}$, and $A_1B_2 - A_2B_1 \neq 0$ because otherwise $Mw = (a + ib)w$ with a real $w \neq 0$ and $b \neq 0$, which is impossible.

Problem 56-4 (Simmons)

Solution. Let m be the double root of (4), and substitute $x = (A_1 + A_2t)e^{mt}$, $y = (B_1 + B_2t)e^{mt}$ into (1). After dividing by e^{mt},

$$A_2 + m(A_1 + A_2t) = a_1(A_1 + A_2t) + b_1(B_1 + B_2t),$$
$$B_2 + m(B_1 + B_2t) = a_2(A_1 + A_2t) + b_2(B_1 + B_2t).$$

Collecting powers of t:

$$\big[(a_1 - m)A_2 + b_1B_2\big]t + \big[(a_1 - m)A_1 + b_1B_1 - A_2\big] = 0,$$
$$\big[a_2A_2 + (b_2 - m)B_2\big]t + \big[a_2A_1 + (b_2 - m)B_1 - B_2\big] = 0.$$

These are identities in t, so the coefficients of t vanish:

$$(a_1 - m)A_2 + b_1B_2 = 0, \qquad a_2A_2 + (b_2 - m)B_2 = 0,$$

which is exactly system (3) (with m the double root) for the unknowns A_2, B_2; this is the system satisfied by A, B.

Now we show that its solutions are all proportional. For a double root, $m = (a_1 + b_2)/2$, so $a_1 - m = \delta := (a_1 - b_2)/2$ and $b_2 - m = -\delta$, and the matrix of (3) is $N = \begin{pmatrix} \delta & b_1 \\ a_2 & -\delta \end{pmatrix}$, with $\det N = -\delta^2 - a_2b_1 = 0$ (the discriminant $(a_1 - b_2)^2 + 4a_2b_1$ vanishes). Unless $\delta = b_1 = a_2 = 0$ (the uncoupled case of the text's footnote, where $m = a_1 = b_2$ and A, B are unrestricted), $N \neq 0$, so N has rank 1 and its null space is a line. Therefore (A_2, B_2) and (A, B), both in that line, satisfy $(A_2, B_2) = k(A, B)$ for some constant k. If $k = 0$ then (19) is just (A_1e^{mt}, B_1e^{mt}), a solution of the form (18), which gives nothing new; so for a second, independent solution $k \neq 0$. Then $\frac{1}{k}$ times the solution (19) is again a solution, namely

$$x = \left(\tfrac{A_1}{k} + At\right)e^{mt}, \qquad y = \left(\tfrac{B_1}{k} + Bt\right)e^{mt},$$

and c_2 times (19) with arbitrary c_2 is the same family as c_2' times this one. Thus in (20) we may assume $A_2 = A$, $B_2 = B$ without loss of generality.

Answer. Shown: the t-coefficients give $(a_1 - m)A_2 + b_1 B_2 = 0$, $a_2 A_2 + (b_2 - m)B_2 = 0$, the same system as for A, B; its solution space is a line, so $(A_2, B_2) = k(A, B)$, $k \neq 0$, and after dividing by k we may take $A_2 = A$, $B_2 = B$.

Problem 56-5 (Simmons)

Solution.

(a) Put $x = v_1 x_1 + v_2 x_2$, $y = v_1 y_1 + v_2 y_2$. By the product rule,

$$x' = v_1' x_1 + v_2' x_2 + v_1 x_1' + v_2 x_2',$$
$$y' = v_1' y_1 + v_2' y_2 + v_1 y_1' + v_2 y_2'.$$

Since (x_j, y_j) solve (**), $x_j' = a_1 x_j + b_1 y_j$ and $y_j' = a_2 x_j + b_2 y_j$ for $j = 1, 2$, hence

$$v_1 x_1' + v_2 x_2' = a_1(v_1 x_1 + v_2 x_2) + b_1(v_1 y_1 + v_2 y_2) = a_1 x + b_1 y,$$
$$v_1 y_1' + v_2 y_2' = a_2 x + b_2 y.$$

Therefore

$$x' = a_1 x + b_1 y + (v_1' x_1 + v_2' x_2), \qquad y' = a_2 x + b_2 y + (v_1' y_1 + v_2' y_2),$$

and if $v_1' x_1 + v_2' x_2 = f_1$ and $v_1' y_1 + v_2' y_2 = f_2$, these are exactly the equations (*). So (x, y) is a solution of (*). (The linear system for v_1', v_2' is solvable because its determinant $x_1 y_2 - x_2 y_1$ is the Wronskian of the independent solutions, which is nonzero.)

(b) By Example 1 (equations (10), (11)), independent solutions of the homogeneous system $x' = x + y$, $y' = 4x - 2y$ are

$$x_1 = e^{-3t},\ y_1 = -4e^{-3t}; \qquad x_2 = e^{2t},\ y_2 = e^{2t}.$$

With $f_1 = -5t + 2$ and $f_2 = -8t - 8$, the equations for v_1', v_2' are

$$v_1' e^{-3t} + v_2' e^{2t} = -5t + 2,$$
$$-4v_1' e^{-3t} + v_2' e^{2t} = -8t - 8.$$

Subtracting the second from the first: $5v_1' e^{-3t} = (-5t + 2) - (-8t - 8) = 3t + 10$, so

$$v_1' = \tfrac{1}{5}(3t + 10)e^{3t}.$$

Then from the first equation

$$v_2' e^{2t} = -5t + 2 - \tfrac{1}{5}(3t + 10) = \tfrac{1}{5}(-25t + 10 - 3t - 10) = -\tfrac{28}{5}t, \qquad v_2' = -\tfrac{28}{5}\,te^{-2t}.$$

Integrate (any antiderivatives suffice, since we want one particular solution). By parts,

$$\int (3t + 10)e^{3t}\,dt = \tfrac{1}{3}(3t + 10)e^{3t} - \int e^{3t}\,dt = \tfrac{1}{3}(3t + 10)e^{3t} - \tfrac{1}{3}e^{3t} = (t + 3)e^{3t},$$

$$\int te^{-2t}\,dt = -\tfrac{1}{2}te^{-2t} + \tfrac{1}{2}\int e^{-2t}\,dt = -\tfrac{1}{2}te^{-2t} - \tfrac{1}{4}e^{-2t}.$$

Hence

$$v_1 = \tfrac{1}{5}(t + 3)e^{3t}, \qquad v_2 = -\tfrac{28}{5}\left(-\tfrac{1}{2}t - \tfrac{1}{4}\right)e^{-2t} = \tfrac{7}{5}(2t + 1)e^{-2t}.$$

The particular solution is

$$x = v_1 x_1 + v_2 x_2 = \tfrac{1}{5}(t + 3) + \tfrac{7}{5}(2t + 1) = \tfrac{1}{5}(15t + 10) = 3t + 2,$$
$$y = v_1 y_1 + v_2 y_2 = -\tfrac{4}{5}(t + 3) + \tfrac{7}{5}(2t + 1) = \tfrac{1}{5}(10t - 5) = 2t - 1.$$

Check. $x' = 3$ and $x + y - 5t + 2 = (3t + 2) + (2t - 1) - 5t + 2 = 3$; $y' = 2$ and $4x - 2y - 8t - 8 = 12t + 8 - 4t + 2 - 8t - 8 = 2$.

Answer. (a) shown: $x' - a_1 x - b_1 y = v_1' x_1 + v_2' x_2$ etc.; (b) $x = 3t + 2$, $y = 2t - 1$ (with $v_1 = \frac{1}{5}(t+3)e^{3t}$, $v_2 = \frac{7}{5}(2t+1)e^{-2t}$).

Section 57 Nonlinear Systems. Volterra's Prey-Predator Equations

The system under study is Volterra's prey-predator system

$$\frac{dx}{dt} = x(a - by), \qquad \frac{dy}{dt} = -y(c - dx), \tag{1}$$

with $a, b, c, d > 0$; x is the number of rabbits (prey) and y the number of foxes (predators), so $x, y > 0$ for the populations of interest.

Problem 57-1 (Simmons)

Solution. The first equation of (1) gives $x' = x(a - by)$. For $x \neq 0$ (we have $x > 0$), solve for y:

$$by = a - \frac{x'}{x}, \qquad y = \frac{ax - x'}{bx}. \tag{i}$$

Differentiate the first equation of (1) with respect to t (product rule):

$$x'' = x'(a - by) - bxy'.$$

Insert $y' = -y(c - dx)$ from the second equation:

$$x'' = x'(a - by) + bxy(c - dx).$$

Now eliminate y using (i): $a - by = x'/x$ and $by = (ax - x')/x$. Then

$$x'' = \frac{(x')^2}{x} + x(c - dx)\,\frac{ax - x'}{x} = \frac{(x')^2}{x} + (c - dx)(ax - x').$$

Multiplying by x,

$$\begin{aligned} xx'' &= (x')^2 + x(c - dx)(ax - x') \\ &= (x')^2 + x\left[acx - adx^2 - cx' + dxx'\right] \\ &= (x')^2 + (acx^2 - adx^3) + (dx^2 - cx)x'. \end{aligned}$$

That is,

$$x\,\frac{d^2x}{dt^2} = (dx^2 - cx)\frac{dx}{dt} + (acx^2 - adx^3) + \left(\frac{dx}{dt}\right)^2.$$

Answer. $x\,\dfrac{d^2x}{dt^2} = (dx^2 - cx)\dfrac{dx}{dt} + (acx^2 - adx^3) + \left(\dfrac{dx}{dt}\right)^2.$

Problem 57-2 (Simmons)

Solution. From (1), $y' = -y(c - dx) = y(dx - c)$. Differentiating with respect to t (product rule),

$$y'' = y'(dx - c) + y\,dx' = y(dx - c)^2 + d\,y\,x',$$

where we used $y' = y(dx - c)$ again. Here $y > 0$ (a population) and $d > 0$. The first term $y(dx - c)^2$ is ≥ 0 always, and the second term $d\,y\,x'$ is > 0 whenever $x' > 0$. Therefore

$$x' > 0 \implies y'' > 0.$$

Meaning. Wherever the rabbit curve $x(t)$ of Figure 70 is rising ($x' > 0$), the fox curve $y(t)$ is concave up ($y'' > 0$), i.e. it bends upward and lies above its tangent lines there.

Answer. $y'' = y(dx - c)^2 + dy\,x' > 0$ when $x' > 0$; the fox curve is concave up whenever the rabbit curve is rising.

Chapter 11

Nonlinear Equations

Section 58 Autonomous Systems. The Phase Plane and Its Phenomena

Throughout, a system $x' = F(x,y)$, $y' = G(x,y)$ with F, G continuous and having continuous first partial derivatives is called *autonomous*; the equation $x'' = f(x,x')$ is replaced by the equivalent system $x' = y$, $y' = f(x,y)$ (equations (4) and (5) of the text), and a *critical point* is a point where $F = G = 0$.

Problem 58-1 (Simmons)

Solution. The bob has mass m and hangs on a rigid massless rod (or string) of length a; x is the angle of deviation from the downward vertical, measured counterclockwise. The bob moves on a circle of radius a, so its arc-length coordinate is $s = ax$, its speed is $|ds/dt| = a|dx/dt|$, and its tangential acceleration is $d^2s/dt^2 = a\,d^2x/dt^2$.

Forces on the bob:

- the tension in the rod, directed along the rod (radially), has no tangential component;
- gravity mg (downward) has tangential component $-mg\sin x$ (it always pulls the bob back toward the lowest point, i.e. in the direction of decreasing x when $x > 0$);
- the damping force is proportional to the velocity of the bob and opposite to it: $-c\,\dfrac{ds}{dt} = -ca\,\dfrac{dx}{dt}$, where $c > 0$ is the constant of proportionality.

Newton's second law in the tangential direction, $m\,\dfrac{d^2s}{dt^2} = (\text{tangential force})$, gives

$$ma\,\frac{d^2x}{dt^2} = -mg\sin x - ca\,\frac{dx}{dt}.$$

Dividing by ma and moving everything to the left,

$$\frac{d^2x}{dt^2} + \frac{c}{m}\,\frac{dx}{dt} + \frac{g}{a}\sin x = 0,$$

which is equation (2). (For $c = 0$ it reduces to equation (1).)

Answer. Tangential Newton law $ma\,x'' = -mg\sin x - ca\,x'$, i.e. $x'' + \dfrac{c}{m}x' + \dfrac{g}{a}\sin x = 0$.

Problem 58-2 (Simmons)

Solution. Here (6) is the autonomous system $x' = F(x,y)$, $y' = G(x,y)$, where F, G are continuous with continuous first partial derivatives, so Theorem 54-A (existence and uniqueness) applies.

Put $c = t_1 - t_2$ and define

$$u(t) = x_1(t+c), \qquad v(t) = y_1(t+c).$$

Step 1: (u,v) is a solution of (6). By the chain rule, and because F and G do not contain t,

$$u'(t) = x_1'(t+c) = F\big(x_1(t+c), y_1(t+c)\big) = F\big(u(t),v(t)\big), \qquad v'(t) = G\big(u(t),v(t)\big).$$

(For a nonautonomous system this step would fail, because $t + c \neq t$ in the explicit t-dependence.)

Step 2: same value at $t = t_2$.

$$u(t_2) = x_1(t_2 + c) = x_1(t_1) = x_0 = x_2(t_2), \qquad v(t_2) = y_1(t_1) = y_0 = y_2(t_2).$$

Step 3: uniqueness. The solutions (u, v) and (x_2, y_2) of (6) take the same value (x_0, y_0) at the same time $t = t_2$. By the uniqueness part of Theorem 54-A they coincide on every interval containing t_2 on which both are defined; that is,

$$x_1(t + c) = x_2(t), \qquad y_1(t + c) = y_2(t).$$

Answer. Shown: with $c = t_1 - t_2$, the translate $\big(x_1(t + c), y_1(t + c)\big)$ solves the same autonomous system and passes through (x_0, y_0) at $t = t_2$, hence equals (x_2, y_2) by uniqueness. Consequently two paths with a common point coincide, so at most one path passes through each point.

Problem 58-3 (Simmons)

Solution. Call the first system (I) and the second system (II).

Critical points. $-F = -G = 0$ exactly when $F = G = 0$, so (I) and (II) have the same critical points.

Paths. Suppose $x = x(t)$, $y = y(t)$ solves (I). Put $\tilde{x}(t) = x(-t)$, $\tilde{y}(t) = y(-t)$. By the chain rule

$$\tilde{x}'(t) = -x'(-t) = -F\big(x(-t), y(-t)\big) = -F(\tilde{x}, \tilde{y}), \qquad \tilde{y}'(t) = -G(\tilde{x}, \tilde{y}),$$

so $(\tilde{x}, \tilde{y})$ solves (II). Conversely, if $(\tilde{x}, \tilde{y})$ solves (II), then $(\tilde{x}(-t), \tilde{y}(-t))$ solves (I) by the same computation. The two solutions $(x(t), y(t))$ and $(x(-t), y(-t))$ trace out exactly the same set of points of the plane, but in opposite directions (the point that is reached later on the first is reached earlier on the second). Equivalently, the differential equation of the paths $dy/dx = G/F$ is the same for (I) and (II), since the minus signs cancel; only the sign of the velocity vector $V = F\mathbf{i} + G\mathbf{j}$ changes.

Answer. The two systems have the same critical points and the same paths (curves), but the direction of every path is reversed (arrows reversed): $t \to -t$. In particular a node, spiral, or center for one is a node, spiral, or center for the other with the roles of $t \to \infty$ and $t \to -\infty$ interchanged (stability becomes instability).

Problem 58-4 (Simmons)

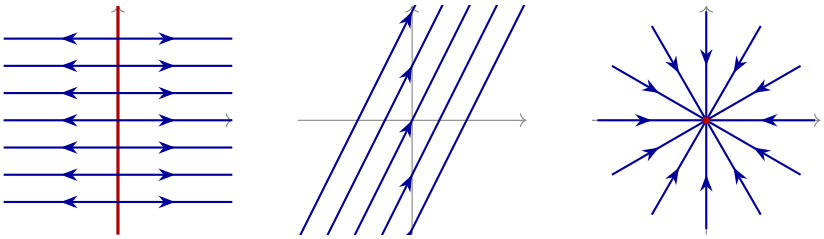
Solution.

(a) $F = G = 0$ at every point of the plane, so every point is a critical point (the critical points are not isolated). Solutions are $x = c_1$, $y = c_2$, constant, which define no path. The phase portrait consists only of the critical points: there are no paths.

(b) Critical points: $F = x = 0$ and $G = 0$ hold exactly for $x = 0$, i.e. the whole y-axis consists of critical points. The general solution is $x = c_1 e^t$, $y = c_2$. If $c_1 = 0$ this is a constant (a critical point on the y-axis). If $c_1 \neq 0$, the path is the horizontal line $y = c_2$ with x having the sign of c_1: the half-line $x > 0$ ($c_1 > 0$) or $x < 0$ ($c_1 < 0$), and $|x| = |c_1| e^t$ increases with t. Thus the paths are the horizontal open half-lines leaving the y-axis, directed away from it, to the right in the half-plane $x > 0$ and to the left in $x < 0$ (each half-line approaches, as $t \to -\infty$, the critical point $(0, c_2)$ on the y-axis).

(c) $F = 1 \neq 0$, so there are no critical points. The general solution is $x = t + c_1$, $y = 2t + c_2$. Eliminating t: $y - 2x = c_2 - 2c_1$ is constant, i.e. every path is a whole straight line $y = 2x + k$ of slope 2 (k arbitrary, each value of k giving one path), traversed upward and to the right as t increases (both x and y increase). The paths are the family of parallel lines of slope 2; all points move with the constant velocity $\mathbf{i} + 2\mathbf{j}$.

(d) $F = -x = 0$, $G = -y = 0$ only at the origin, the only critical point (isolated). The general solution is $x = c_1 e^{-t}$, $y = c_2 e^{-t}$. For $(c_1, c_2) \neq (0, 0)$ the point is $e^{-t}(c_1, c_2)$: it moves along the open ray from the origin through (c_1, c_2), toward the origin, approaching it as $t \to \infty$ without reaching it. Since $y/x = c_2/c_1$ is constant, each path approaches *and enters* the origin (with a definite direction; for $c_1 = 0$ the direction is vertical). Thus the paths are all the half-lines through the origin (all slopes, including the two halves of the y-axis) directed toward the origin. (This is a node, "a star".)



(b) (c) (d)

Answer. (a) every point is critical, no paths. (b) the y-axis is made of critical points; paths are horizontal half-lines directed away from the y-axis (right for $x > 0$, left for $x < 0$). (c) no critical points; paths are the lines $y = 2x + k$ directed up and to the right. (d) the origin is the only critical point; paths are all half-lines directed toward the origin.

Problem 58-5 (Simmons)

Solution. Equation (4) is $x'' = f(x, x')$, with equivalent system $x' = y$, $y' = f(x, y)$; its critical points are the points $(x_0, 0)$ with $f(x_0, 0) = 0$.

(1) $x'' + \frac{g}{a}\sin x = 0$ gives $x' = y$, $y' = -\frac{g}{a}\sin x$. Critical points: $y = 0$ and $\sin x = 0$ (since $g/a \neq 0$), i.e. $x = n\pi$:

$$(n\pi, 0), \qquad n = 0, \pm 1, \pm 2, \ldots$$

(2) $x'' + \frac{c}{m}x' + \frac{g}{a}\sin x = 0$ gives $x' = y$, $y' = -\frac{c}{m}y - \frac{g}{a}\sin x$. Setting $x' = y = 0$ and then $y' = -\frac{g}{a}\sin x = 0$ gives the same points $(n\pi, 0)$, $n = 0, \pm 1, \pm 2, \ldots$ (the damping term vanishes when $y = 0$).

(3) $x'' + \mu(x^2 - 1)x' + x = 0$ gives $x' = y$, $y' = -\mu(x^2 - 1)y - x$. From $y = 0$ the second equation becomes $-x = 0$, so $x = 0$. The only critical point is $(0, 0)$.

Answer. (1) and (2): $(n\pi, 0)$, $n = 0, \pm 1, \pm 2, \ldots$; (3): only $(0, 0)$.

Problem 58-6 (Simmons)

Solution.

(a) With $y = dx/dt$ the equation is equivalent to

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = -y + x^3 + x^2 - 2x.$$

Critical points: $y = 0$, and then $x^3 + x^2 - 2x = 0$. Factor: $x(x^2 + x - 2) = x(x+2)(x-1) = 0$, so $x = -2,\ 0,\ 1$. The critical points are $(-2,0)$, $(0,0)$, $(1,0)$.

(b) Setting $x - y = 0$ gives $y = x$. Then $y^2 - 5x + 6 = x^2 - 5x + 6 = (x-2)(x-3) = 0$, so $x = 2$ or $x = 3$. The critical points are $(2,2)$ and $(3,3)$.

Answer. (a) $(-2,0)$, $(0,0)$, $(1,0)$; (b) $(2,2)$, $(3,3)$.

Problem 58-7 (Simmons)

Solution. The first equation is separable (linear): $x = c_1 e^t$. Substituting into the second,

$$\frac{dy}{dt} = c_1 e^t + e^t = (c_1 + 1)e^t \implies y = (c_1 + 1)e^t + c_2.$$

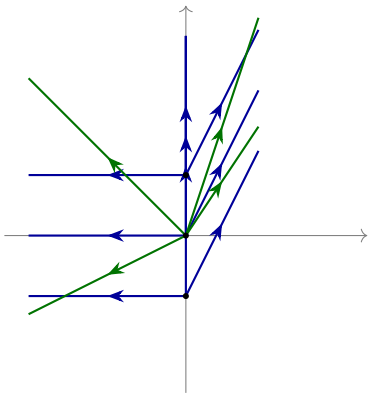
Thus all solutions are

$$x = c_1 e^t, \qquad y = c_1 e^t + e^t + c_2 \qquad (c_1, c_2 \text{ arbitrary constants}).$$

The curves. If $c_1 = 0$: $x = 0$, $y = e^t + c_2$ runs over the open vertical half-line $y > c_2$ of the y-axis (upward as t increases). If $c_1 \neq 0$: $e^t = x/c_1 > 0$, so

$$y = (c_1 + 1)\frac{x}{c_1} + c_2 = \Big(1 + \frac{1}{c_1}\Big)x + c_2, \qquad \frac{x}{c_1} > 0.$$

This is the open half-line starting at the point $(0, c_2)$ of the y-axis (not reached, since $e^t > 0$), lying in $x > 0$ if $c_1 > 0$ and in $x < 0$ if $c_1 < 0$, with slope $m = 1 + 1/c_1$. As t increases, $|x| = |c_1|e^t$ increases, so the point moves away from the y-axis. For $c_1 > 0$ the slope takes all values $m > 1$; for $c_1 < 0$ it takes all values $m < 1$ ($c_1 = -1$ gives $m = 0$, $c_1 = -\frac{1}{2}$ gives $m = -1$, $c_1 = -2$ gives $m = \frac{1}{2}$). Hence from each point $(0, c_2)$ of the y-axis there emanate a vertical ray upward, rays to the right of slope $m > 1$, and rays to the left of slope $m < 1$ (the slope $m = 1$ is never attained).



(Blue: from $(0,-1)$, $(0,0)$, $(0,1)$: the ray of slope 2 [$c_1 = 1$], the horizontal ray to the left [$c_1 = -1$], the vertical ray [$c_1 = 0$]. Green: further rays from $(0,0)$ with slopes 1.5, 3 [$c_1 = 2,\ \frac{1}{2}$] and $-1,\ \frac{1}{2}$ [$c_1 = -\frac{1}{2},\ -2$].)

Remark. Through the point (x_0, y_0) with $x_0 \neq 0$ there pass infinitely many of these curves, one for each c_1 with $c_1 x_0 > 0$ (take $c_2 = y_0 - (1 + 1/c_1)x_0$). For example $y = 1.5x$ (from $(0, 0)$) and $y = 3x - 1$ (from $(0, -1)$) cross at $x = 2/3$. This does not contradict uniqueness: the system is not autonomous (the right side contains t), so the curves are not paths of a phase portrait, and several of them can pass through one point (at different times).

Answer. $x = c_1 e^t$, $y = c_1 e^t + e^t + c_2$. The curves are open half-lines issuing from the points $(0, c_2)$ of the y-axis: the vertical ray ($c_1 = 0$) and rays with slope $1 + 1/c_1$ ($x/c_1 > 0$), directed away from the y-axis.

Section 59 Types of Critical Points. Stability

For an autonomous system $x' = F(x,y)$, $y' = G(x,y)$ the paths satisfy $\dfrac{dy}{dx} = \dfrac{G(x,y)}{F(x,y)}$ (equation (4) of the text) wherever $(F,G) \neq (0,0)$. The direction of increasing t is read from the signs of $F = dx/dt$ and $G = dy/dt$. Stability is meant in the sense of the text: the critical point O is *stable* if for each $R > 0$ there is $r \leq R$ such that every path inside the circle of radius r about O at some time t_0 stays inside the circle of radius R for all $t > t_0$; it is *asymptotically stable* if in addition paths starting near enough tend to O as $t \to \infty$; otherwise it is *unstable*.

Problem 59-1 (Simmons)

Solution.

(a) Here $F = y(x^2+1)$, $G = 2xy^2$.

(i) $F = 0$ forces $y = 0$ because $x^2 + 1 \geq 1 > 0$; and then $G = 0$ automatically. So the critical points are all points $(x, 0)$ of the x-axis (not isolated).

(ii) For $y \neq 0$,

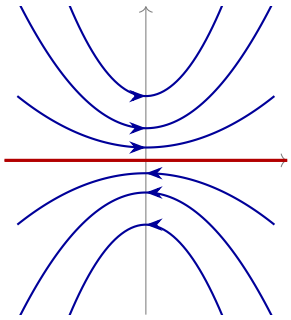
$$\frac{dy}{dx} = \frac{2xy^2}{y(x^2+1)} = \frac{2xy}{x^2+1}.$$

(iii) The equation is separable: $\dfrac{dy}{y} = \dfrac{2x\,dx}{x^2+1}$, so $\log|y| = \log(x^2+1) + k$, i.e.

$$y = c\,(x^2+1), \qquad c \neq 0$$

(the value $c = 0$ gives $y = 0$, the critical points). The paths are the parabolas $y = c(x^2+1)$.

(iv) Direction: $dx/dt = y(x^2+1)$ has the sign of y. So on the parabolas with $c > 0$ ($y > 0$) x increases: the motion is from left to right; on those with $c < 0$ the motion is from right to left. (Check with $dy/dt = 2xy^2$: for $c > 0$, y decreases while $x < 0$ and increases while $x > 0$, as it must on a parabola with its vertex at $x = 0$.) Each path moves away from the x-axis as $|x| \to \infty$; none enters a critical point.



(The red line is the line of critical points.)

(b) $F = y(x^2+1)$, $G = -x(x^2+1)$.

(i) $F = 0 \Rightarrow y = 0$, and $G = 0 \Rightarrow x = 0$ (again since $x^2 + 1 \neq 0$). The only critical point is $(0,0)$.

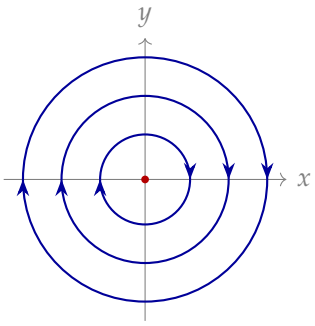
(ii) For $y \neq 0$: $\frac{dy}{dx} = \dfrac{-x(x^2+1)}{y(x^2+1)} = -\dfrac{x}{y}$ (the common factor x^2+1 cancels).

(iii) $y\,dy = -x\,dx$ gives $\frac{1}{2}y^2 = -\frac{1}{2}x^2 + k$, i.e.

$$x^2 + y^2 = c^2.$$

The paths are circles centered at the origin ($y = 0$ is covered, because the points $(\pm c, 0)$ are ordinary points of the circle, where the tangent is vertical: $F = 0$, $G \neq 0$).

(iv) Direction: at the point $(0, c)$ at the top of a circle, $dx/dt = c(0 + 1) = c > 0$, so the motion is to the right at the top, i.e. *clockwise*. Since the system has no other critical point, every circle is a closed path traversed clockwise (periodic solutions).



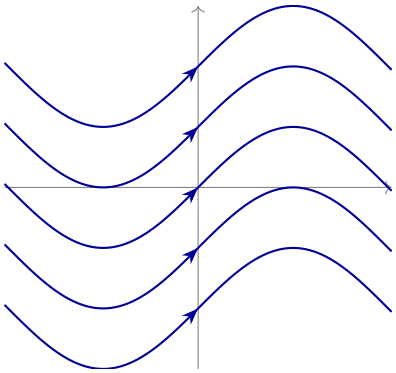
(c) $F = e^y$, $G = e^y \cos x$.

(i) $F = e^y > 0$ for all y, so F never vanishes: there are no critical points.

(ii) $\frac{dy}{dx} = \dfrac{e^y \cos x}{e^y} = \cos x$ (valid everywhere, since $F \neq 0$).

(iii) Integrating, $y = \sin x + c$. These are all the paths: translates of the sine curve in the y-direction.

(iv) $dx/dt = e^y > 0$, so x increases along every path: all paths are traversed from left to right.



(d) $F = -x$, $G = 2x^2y^2$.

(i) $F = 0$ iff $x = 0$, and then $G = 0$. The critical points are all points $(0, y)$ of the y-axis (not isolated).

(ii) For $x \neq 0$ and $y \neq 0$ (so that $F \neq 0$):

$$\frac{dy}{dx} = \frac{2x^2y^2}{-x} = -2xy^2.$$

(iii) Separate variables ($y \neq 0$): $\dfrac{dy}{y^2} = -2x\,dx$, so $-\dfrac{1}{y} = -x^2 + k$, i.e. $\dfrac{1}{y} = x^2 + c$,

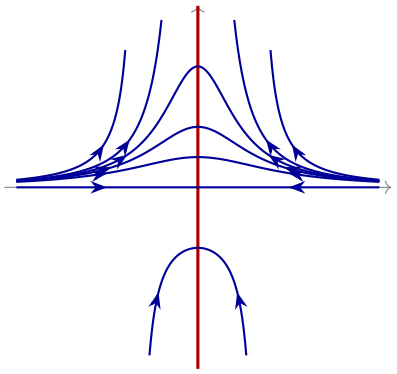
$$y = \frac{1}{x^2 + c},$$

together with the solution $y = 0$ lost in the division. For $x \neq 0$ the line $y = 0$ is not made of critical points (there $F = -x \neq 0$), so the two half-lines $y = 0$, $x > 0$ and $y = 0$, $x < 0$ are paths.

Shape of the paths (all with $x \neq 0$, because the y-axis consists of critical points and no path can meet it in finite time: $x(t) = x_0 e^{-t} \neq 0$):

- $c > 0$: for each sign of x, half of a bell-shaped curve that runs from far out along the x-axis ($y \to 0$ as $|x| \to \infty$) to the point $(0, 1/c)$ of the y-axis;
- $c = 0$: $y = 1/x^2$, which tends to $+\infty$ as $x \to 0$;
- $c < 0$, $c = -k^2$: $y = 1/(x^2 - k^2)$ has an outer piece $|x| > k$ with $y > 0$ (asymptotic to the vertical line $x = \pm k$ as $y \to +\infty$) and an inner piece $0 < |x| < k$ with $y < 0$ (tending to $-\infty$ as $|x| \to k$ and to $-1/k^2$ as $x \to 0$).

(iv) Direction: $dx/dt = -x$, so $x = x_0 e^{-t}$ decreases in absolute value: every path moves *toward the y-axis* (to the left on the right side, to the right on the left side). The paths $y = 0$, $x \neq 0$ move toward the origin.



(Shown: $c = 2, 1, \frac{1}{2}$; $c = 0$; $c = -1$ with its outer and inner pieces; the half-axes $y = 0$; red: the line of critical points.)

Answer. (a) (i) the x-axis; (ii) $dy/dx = 2xy/(x^2+1)$; (iii) $y = c(x^2+1)$. (b) (i) $(0,0)$; (ii) $dy/dx = -x/y$; (iii) $x^2 + y^2 = c^2$ (clockwise circles). (c) (i) none; (ii) $dy/dx = \cos x$; (iii) $y = \sin x + c$ (left to right). (d) (i) the y-axis; (ii) $dy/dx = -2xy^2$; (iii) $y = 1/(x^2 + c)$ and $y = 0$ (toward the y-axis).

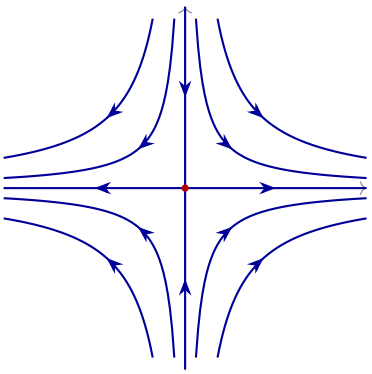
Problem 59-2 (Simmons)

Solution.

(a) (i) The equations are uncoupled: $x = c_1 e^t$, $y = c_2 e^{-t}$.

(ii) $\frac{dy}{dx} = \dfrac{-y}{x}$ (for $x \neq 0$).

(iii) $\dfrac{dy}{y} = -\dfrac{dx}{x}$ gives $\log|y| = -\log|x| + k$, so $xy = c$. (For $c = 0$ this gives the axes, which are paths, too: $x = c_1 e^t$, $y = 0$ and $x = 0$, $y = c_2 e^{-t}$.) The paths are the four half-axes and the hyperbolas $xy = c$. Directions: $x = c_1 e^t$ moves away from the y-axis, and $y = c_2 e^{-t}$ moves toward the x-axis. So on the x-axis the motion is outward from the origin, on the y-axis inward, and on each hyperbola $|x|$ grows while $|y|$ shrinks.

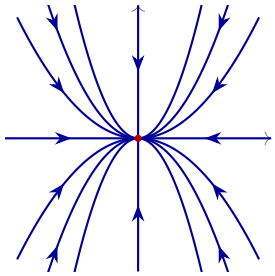


(iv) The origin is a saddle point and is *unstable*: take $R = 1$. For any $r > 0$ the path $x = \frac{r}{2}e^t$, $y = 0$ is inside the circle of radius r at $t = 0$, but $x \to \infty$, so it leaves the circle of radius 1.

(b) (i) $x = c_1 e^{-t}$, $y = c_2 e^{-2t}$.

(ii) $\frac{dy}{dx} = \dfrac{-2y}{-x} = \dfrac{2y}{x}$.

(iii) $\dfrac{dy}{y} = \dfrac{2\,dx}{x}$ gives $\log|y| = 2\log|x| + k$, so $y = cx^2$ (the value $c = 0$ is the x-axis, a path), together with the y-axis $x = 0$ (arising from $c_1 = 0$, which the equation $dy/dx = 2y/x$ does not cover). The paths are the half-axes and the parabolas $y = cx^2$ (halves with $x > 0$ and $x < 0$), all tangent to the x-axis at the origin. Since x and y both decay to 0, every path is directed toward the origin, and enters it (for $x \neq 0$, $y/x = (c_2/c_1)e^{-t} \to 0$; for $c_1 = 0$ the path enters vertically).



(iv) The origin is an (improper) node and is *asymptotically stable*. Indeed, given $R > 0$ take $r = R$: if $(x(t_0), y(t_0))$ lies inside the circle of radius r, then for $t > t_0$ $x(t) = x(t_0)e^{-(t-t_0)}$ and $y(t) = y(t_0)e^{-2(t-t_0)}$, so $x^2 + y^2 \le x(t_0)^2 + y(t_0)^2 < R^2$; the point stays inside the circle of radius R, and moreover $(x(t), y(t)) \to (0,0)$ as $t \to \infty$, for every initial point.

(c) (i) From $x' = 4y$, $y' = -x$ we get $x'' = 4y' = -4x$, so $x = A\cos 2t + B\sin 2t$ and $y = x'/4 = \frac{1}{2}(-A\sin 2t + B\cos 2t)$. With $A = 2c_1$, $B = 2c_2$:

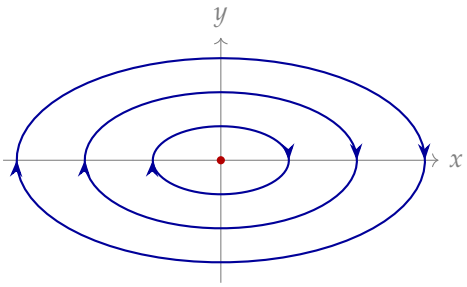
$$x = 2c_1\cos 2t + 2c_2\sin 2t, \qquad y = -c_1\sin 2t + c_2\cos 2t.$$

(ii) $\frac{dy}{dx} = -\dfrac{x}{4y}$ (for $y \neq 0$).

(iii) $4y\,dy = -x\,dx$ gives $2y^2 = -\frac{1}{2}x^2 + k$, i.e. $x^2 + 4y^2 = 4c^2$:

$$\frac{x^2}{4c^2} + \frac{y^2}{c^2} = 1,$$

a family of ellipses with semi-axes $2c$ (along x) and c (along y). From the general solution $x^2 + 4y^2 = 4(c_1\cos 2t + c_2\sin 2t)^2 + 4(-c_1\sin 2t + c_2\cos 2t)^2 = 4(c_1^2 + c_2^2)$, so $c^2 = c_1^2 + c_2^2$ and each ellipse is a closed path. Direction: where $y > 0$, $dx/dt = 4y > 0$, so the motion is clockwise.



(iv) The origin is a center and is *stable but not asymptotically stable*. Stability: given $R > 0$ take $r = R/2$. If $x(t_0)^2 + y(t_0)^2 < r^2$ then $x(t_0)^2 + 4y(t_0)^2 \le 4\big(x(t_0)^2 + y(t_0)^2\big) < 4r^2 = R^2$. Since $x^2 + 4y^2$ is constant along a path, for all $t > t_0$ we have $x^2 + y^2 \le x^2 + 4y^2 < R^2$, so the path stays inside the circle of radius R. Not asymptotically stable: along any path other than the origin, $x^2 + 4y^2 = 4(c_1^2 + c_2^2) > 0$ is constant, so the path does not tend to the origin however close it starts.

Answer. (a) $x = c_1 e^{t}$, $y = c_2 e^{-t}$; $dy/dx = -y/x$; $xy = c$; saddle, unstable. (b) $x = c_1 e^{-t}$, $y = c_2 e^{-2t}$; $dy/dx = 2y/x$; $y = cx^2$; asymptotically stable node. (c) $x = 2c_1 \cos 2t + 2c_2 \sin 2t$, $y = -c_1 \sin 2t + c_2 \cos 2t$; $dy/dx = -x/(4y)$; $x^2/(4c^2) + y^2/c^2 = 1$; stable but not asymptotically stable (center).

Problem 59-3 (Simmons)

Solution. With $y = dx/dt$ the equation is equivalent to the system

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = 2x^3.$$

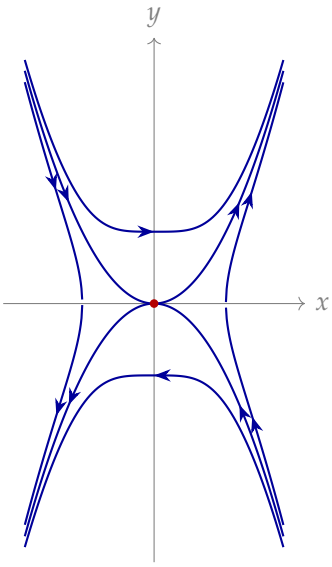
Critical points. $y = 0$ and $2x^3 = 0$ give $x = 0$: the origin is the only critical point, hence it is isolated.

Paths. For $y \neq 0$, $\frac{dy}{dx} = \dfrac{2x^3}{y}$, so $y\,\mathrm{d}y = 2x^3\,\mathrm{d}x$, $\frac{1}{2}y^2 = \frac{1}{2}x^4 + k$:

$$y^2 = x^4 + C.$$

- $C = 0$: $y = x^2$ and $y = -x^2$, four half-paths ending at the origin (two on each curve).
- $C > 0$: the two separate curves $y = \pm\sqrt{x^4 + C}$ (above and below the origin; they cross the y-axis at $y = \pm\sqrt{C}$).
- $C = -k^4 < 0$: $y^2 = x^4 - k^4$ requires $|x| \ge k$, two "C-shaped" curves, one in $x \ge k$ and one in $x \le -k$, with vertical tangent at $(\pm k, 0)$.

Directions. $dx/dt = y$: x increases in the upper half-plane and decreases in the lower. For $C = 0$, along $y = x^2$ we have $x' = x^2 > 0$: moving right on both sides; along $y = -x^2$, $x' = -x^2 < 0$: moving left. Thus the half-path $y = x^2$, $x < 0$ moves *into* the origin, and $y = x^2$, $x > 0$ moves out of it; similarly for $y = -x^2$ (in for $x > 0$, out for $x < 0$). On the C-shaped curves: at $(k, 0)$, $dy/dt = 2k^3 > 0$, so the path comes in along the lower branch (moving left, up toward $(k, 0)$) and leaves along the upper branch (moving right); at $(-k, 0)$, $dy/dt < 0$, so the path comes in along the upper branch (moving right) and leaves along the lower branch. The picture is that of a saddle, with the four "separatrices" $y = \pm x^2$.



Instability. Observe that $x(t) = \dfrac{x_0}{1 - x_0 t}$, $y(t) = x(t)^2$ is a solution: indeed $x' = \dfrac{x_0^2}{(1 - x_0 t)^2} = x^2 = y$ and $y' = 2xx' = 2x \cdot x^2 = 2x^3$. Its path is the half-parabola $y = x^2$, $x > 0$ (for $x_0 > 0$). Take $R = 1$ and let $r > 0$ be arbitrary. Choose x_0 with $0 < x_0 < \frac{1}{2}\min(1, r)$. At $t = 0$ the point (x_0, x_0^2) satisfies $x_0^2 + x_0^4 < 2x_0^2 < r^2$, so it lies inside the circle of radius r. But $x(t) \to +\infty$ as $t \to 1/x_0^-$, so the path leaves the circle of radius $R = 1$ (in finite time). Thus for $R = 1$ no r works: the origin is not stable, i.e. it is an *unstable* isolated critical point.

Answer. The paths are $y^2 = x^4 + C$ (saddle-type portrait with separatrices $y = \pm x^2$); the origin is the only critical point, so it is isolated, and it is unstable because the path $y = x^2$, $x = x_0/(1 - x_0 t)$, starting arbitrarily close to O, leaves every fixed circle.

Section 60 Critical Points and Stability for Linear Systems

Throughout, the linear system is

$$\frac{dx}{dt} = a_1x + b_1y, \qquad \frac{dy}{dt} = a_2x + b_2y \tag{1}$$

with $a_1b_2 - a_2b_1 \neq 0$, whose auxiliary equation is $m^2 + pm + q = 0$, where $p = -(a_1 + b_2)$ and $q = a_1b_2 - a_2b_1$ (equation (3) of the text). We use the classification of the text, in terms of the roots m_1, m_2 and of p, q and the discriminant $p^2 - 4q$:

- $q < 0$: roots real of opposite signs: *saddle point*, unstable;
- $q > 0$, $p^2 - 4q > 0$: roots real, distinct, of the same sign (that of $-p$): *node*; asymptotically stable if $p > 0$, unstable if $p < 0$;
- $q > 0$, $p^2 - 4q = 0$: real equal roots $-p/2$: (borderline) *node*; asymptotically stable if $p > 0$, unstable if $p < 0$;
- $p^2 - 4q < 0$, $p \neq 0$: roots $a \pm ib$ with $a = -p/2 \neq 0$: *spiral*; asymptotically stable if $p > 0$, unstable if $p < 0$;
- $p = 0$, $q > 0$: roots $\pm i\sqrt{q}$: *center*, stable but not asymptotically stable.

(Theorem A: stable iff both roots have nonpositive real parts; asymptotically stable iff both have negative real parts.)

Problem 60-1 (Simmons)

Solution. In each part we read off a_1, b_1, a_2, b_2, compute $p = -(a_1 + b_2)$, $q = a_1b_2 - a_2b_1$, and solve $m^2 + pm + q = 0$.

(a) $a_1 = 2, b_1 = 0, a_2 = 0, b_2 = 3$. Then $p = -5$, $q = 6$, and $m^2 - 5m + 6 = (m-2)(m-3) = 0$, so $m = 2, 3$. The roots are real, distinct and positive ($q > 0$, $p^2 - 4q = 1 > 0$): a node. Since both roots are positive ($p < 0$), paths move away from the origin: *unstable node*.

(b) $a_1 = -1, b_1 = -2, a_2 = 4, b_2 = -5$. Then $p = 6$, $q = (-1)(-5) - (4)(-2) = 5 + 8 = 13$, and $m^2 + 6m + 13 = 0$ gives $m = \dfrac{-6 \pm \sqrt{36 - 52}}{2} = -3 \pm 2i$. The roots are complex with real part $-3 < 0$: a spiral, with all paths approaching the origin: *asymptotically stable spiral*.

(c) $a_1 = -3, b_1 = 4, a_2 = -2, b_2 = 3$. Then $p = -(-3 + 3) = 0$, $q = (-3)(3) - (-2)(4) = -9 + 8 = -1$, and $m^2 - 1 = 0$, $m = \pm 1$. The roots are real with opposite signs ($q < 0$): *unstable saddle point*.

(d) $a_1 = 5, b_1 = 2, a_2 = -17, b_2 = -5$. Then $p = -(5 - 5) = 0$, $q = (5)(-5) - (-17)(2) = -25 + 34 = 9$, and $m^2 + 9 = 0$, $m = \pm 3i$. The roots are pure imaginary: a center; the paths are closed curves around the origin, so the point is *stable but not asymptotically stable* (a center).

(e) $a_1 = -4, b_1 = -1, a_2 = 1, b_2 = -2$. Then $p = 6$, $q = (-4)(-2) - (1)(-1) = 8 + 1 = 9$, and $m^2 + 6m + 9 = (m + 3)^2 = 0$, so $m_1 = m_2 = -3$. The roots are real and equal and negative (here $p^2 - 4q = 0$, and $a_1 \neq b_2$, so this is the subcase of Case D with a single line of half-line paths): a node (borderline), and since $m = -3 < 0$, *asymptotically stable node*.

(f) $a_1 = 4, b_1 = -3, a_2 = 8, b_2 = -6$. Then $a_1b_2 - a_2b_1 = -24 + 24 = 0$, so the basic hypothesis (2) fails. The critical points are the solutions of $4x - 3y = 0$, $8x - 6y = 0$; the second equation is twice the first, so they are all points of the line $4x = 3y$, which consists of infinitely many critical points, none of them isolated (cf. Problem 2). Thus $(0, 0)$ *is not an isolated critical point*, and the classification of the section does not apply.

(g) $a_1 = 4, b_1 = -2, a_2 = 5, b_2 = 2$. Then $p = -6$, $q = (4)(2) - (5)(-2) = 8 + 10 = 18$, and

$m^2 - 6m + 18 = 0$ gives $m = \dfrac{6 \pm \sqrt{36 - 72}}{2} = 3 \pm 3i$. The roots are complex with real part $3 > 0$: *unstable spiral*.

Answer. (a) unstable node; (b) asymptotically stable spiral; (c) unstable saddle point; (d) stable but not asymptotically stable center; (e) asymptotically stable node; (f) not isolated (a whole line $4x = 3y$ of critical points); (g) unstable spiral. This agrees with the book.

Problem 60-2 (Simmons)

Solution. The critical points are the solutions (x, y) of the linear homogeneous equations

$$a_1 x + b_1 y = 0, \qquad a_2 x + b_2 y = 0. \tag{$*$}$$

Step 1: there is a solution other than $(0,0)$.

- If $(a_1, b_1) \neq (0,0)$, put $(x_0, y_0) = (b_1, -a_1) \neq (0,0)$. Then $a_1 x_0 + b_1 y_0 = a_1 b_1 - b_1 a_1 = 0$ and $a_2 x_0 + b_2 y_0 = a_2 b_1 - b_2 a_1 = -(a_1 b_2 - a_2 b_1) = 0$.
- If $a_1 = b_1 = 0$ and $(a_2, b_2) \neq (0,0)$, put $(x_0, y_0) = (b_2, -a_2) \neq (0,0)$: the first equation of $(*)$ is $0 = 0$ and the second is $a_2 b_2 - b_2 a_2 = 0$.
- If $a_1 = b_1 = a_2 = b_2 = 0$, every point is a critical point; take $(x_0, y_0) = (1, 0)$.

Step 2: infinitely many critical points. Since $(*)$ is linear and homogeneous, (sx_0, sy_0) is a solution for every real s: $a_1(sx_0) + b_1(sy_0) = s(a_1 x_0 + b_1 y_0) = 0$, and likewise for the second equation. Because $(x_0, y_0) \neq (0,0)$, distinct values of s give distinct points, so there are infinitely many critical points.

Step 3: none is isolated. Let (ξ, η) be any critical point and $\varepsilon > 0$. The point $(\xi, \eta) + s(x_0, y_0)$ is also a critical point (the solutions of $(*)$ form a vector space: sums of solutions are solutions), it differs from (ξ, η) for $s \neq 0$, and its distance from (ξ, η) is $|s|\sqrt{x_0^2 + y_0^2} < \varepsilon$ for $|s|$ small. So every neighborhood of every critical point contains another critical point, i.e. no critical point is isolated.

Answer. Shown: when $a_1 b_2 - a_2 b_1 = 0$ the critical points fill a line (or the whole plane) through the origin, so there are infinitely many and none is isolated.

Problem 60-3 (Simmons)

Solution.

(a) Critical points satisfy $a_1 x + b_1 y = -c_1$, $a_2 x + b_2 y = -c_2$. Put $D = a_1 b_2 - a_2 b_1 \neq 0$. By Cramer's rule this linear system has exactly one solution,

$$x_0 = \frac{(-c_1)b_2 - b_1(-c_2)}{D} = \frac{b_1 c_2 - b_2 c_1}{D}, \qquad y_0 = \frac{a_1(-c_2) - a_2(-c_1)}{D} = \frac{a_2 c_1 - a_1 c_2}{D}.$$

There is exactly one critical point, so it is isolated.

(b) Let $\bar{x} = x - x_0$, $\bar{y} = y - y_0$. Since x_0, y_0 are constants, $d\bar{x}/dt = dx/dt$ and $d\bar{y}/dt = dy/dt$, and $x = \bar{x} + x_0$, $y = \bar{y} + y_0$. Thus

$$\frac{d\bar{x}}{dt} = a_1(\bar{x} + x_0) + b_1(\bar{y} + y_0) + c_1 = a_1\bar{x} + b_1\bar{y} + (a_1 x_0 + b_1 y_0 + c_1) = a_1\bar{x} + b_1\bar{y},$$
$$\frac{d\bar{y}}{dt} = a_2\bar{x} + b_2\bar{y} + (a_2 x_0 + b_2 y_0 + c_2) = a_2\bar{x} + b_2\bar{y},$$

because (x_0, y_0) is a critical point, i.e. $a_1x_0 + b_1y_0 + c_1 = 0 = a_2x_0 + b_2y_0 + c_2$. This is a system of the form (1), with the same coefficients, and the critical point (x_0, y_0) corresponds to the origin $(\bar{x}, \bar{y}) = (0, 0)$.

(c) Here $a_1 = 2$, $b_1 = -2$, $c_1 = 10$, $a_2 = 11$, $b_2 = -8$, $c_2 = 49$, and $D = -16 + 22 = 6 \neq 0$. Critical point: set $2x - 2y + 10 = 0$, $11x - 8y + 49 = 0$. The first gives $x = y - 5$; substituting, $11y - 55 - 8y + 49 = 3y - 6 = 0$, so $y = 2$, $x = -3$. (Check in the second: $11(-3) - 8(2) + 49 = -33 - 16 + 49 = 0$.) By the formulas of (a): $x_0 = \dfrac{(-2)(49) - (-8)(10)}{6} = \dfrac{-98 + 80}{6} = -3$, $y_0 = \dfrac{(11)(10) - (2)(49)}{6} = \dfrac{110 - 98}{6} = 2$.

With $\bar{x} = x + 3$, $\bar{y} = y - 2$:

$$\frac{d\bar{x}}{dt} = 2\bar{x} - 2\bar{y}, \qquad \frac{d\bar{y}}{dt} = 11\bar{x} - 8\bar{y},$$

since, e.g., $2(\bar{x} - 3) - 2(\bar{y} + 2) + 10 = 2\bar{x} - 2\bar{y}$ and $11(\bar{x} - 3) - 8(\bar{y} + 2) + 49 = 11\bar{x} - 8\bar{y}$. For this system $p = -(2 - 8) = 6$, $q = (2)(-8) - (11)(-2) = -16 + 22 = 6$, and $m^2 + 6m + 6 = 0$ gives $m = -3 \pm \sqrt{3}$, i.e. $m_1 \approx -4.732$, $m_2 \approx -1.268$. The roots are real, distinct and both negative ($p^2 - 4q = 12 > 0$, $q > 0$, $p > 0$): an asymptotically stable node.

Answer. (a) $(x_0, y_0) = \left(\dfrac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \dfrac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}\right)$; (b) the shift gives (1) with the same coefficients; (c) critical point $(-3, 2)$; transformed system $\bar{x}' = 2\bar{x} - 2\bar{y}$, $\bar{y}' = 11\bar{x} - 8\bar{y}$; *asymptotically stable node*. Agrees with the book.

Problem 60-4 (Simmons)

Solution.

(a) Here $a_1 = 0$, $b_1 = 1$, $a_2 = -a^2$, $b_2 = -2b$, and $a_1b_2 - a_2b_1 = 0 \cdot (-2b) - (-a^2)(1) = a^2 \neq 0$ (so $(0, 0)$ is the only critical point). The auxiliary equation $m^2 - (a_1 + b_2)m + (a_1b_2 - a_2b_1) = 0$ is

$$m^2 + 2bm + a^2 = 0, \qquad \text{so} \qquad p = 2b, \quad q = a^2.$$

(This is the same as the characteristic equation of the second-order equation, as it must be.) The roots are $m = -b \pm \sqrt{b^2 - a^2}$ and $p^2 - 4q = 4(b^2 - a^2)$. Always $q = a^2 > 0$ and $p = 2b \geq 0$.

(b) *(i)* $b = 0$. Then $p = 0$, $q = a^2 > 0$, roots $m = \pm ia$ (pure imaginary): a *center*, stable but not asymptotically stable. Physically there is no damping: the displacement x and velocity $y = dx/dt$ are periodic functions of t (period $2\pi/a$); the mass oscillates forever, with constant amplitude. The paths in the xy-plane are ellipses $a^2x^2 + y^2 = \text{const}$ (constant energy).

(ii) $0 < b < a$. Then $p > 0$, $p^2 - 4q = 4(b^2 - a^2) < 0$, roots $m = -b \pm i\sqrt{a^2 - b^2}$ with negative real part: an *asymptotically stable spiral*. The mass executes damped (underdamped) oscillations; $x \to 0$ and $dx/dt \to 0$ as $t \to \infty$, through oscillations of smaller and smaller amplitude (the mass passes through the equilibrium position infinitely often).

(iii) $b = a$. Then $p = 2a > 0$, $p^2 - 4q = 0$, double root $m = -a < 0$ (here $a_1 = 0 \neq b_2 = -2a$, so we are in the subcase with one distinguished direction): an *asymptotically stable node* (borderline). The motion is critically damped: the mass does not oscillate (it crosses the equilibrium at most once), and $x \to 0$, $dx/dt \to 0$ as $t \to \infty$ (indeed $x = (c_1 + c_2t)e^{-at}$).

(iv) $b > a$. Then $p > 0$, $p^2 - 4q > 0$, roots $m = -b \pm \sqrt{b^2 - a^2}$, real, distinct and both negative (their product is $a^2 > 0$ and their sum is $-2b < 0$): an *asymptotically stable node*. The motion is overdamped: the mass returns to equilibrium without oscillating, $x \to 0$, $dx/dt \to 0$.

Answer. (a) $m^2 + 2bm + a^2 = 0$; $p = 2b$, $q = a^2$. (b) (i) stable but not asymptotically stable center: undamped periodic oscillation; (ii) asymptotically stable spiral: damped oscillations; (iii) asymptotically stable (borderline) node: critical damping, no oscillation; (iv) asymptotically stable node: overdamping, no oscillation. Agrees with the book.

Problem 60-5 (Simmons)

Solution. Equation (14) is the differential equation of the paths of (1):

$$\frac{dy}{dx} = \frac{a_2x + b_2y}{a_1x + b_1y}.$$

Hypotheses of Case E. The roots of $m^2 + pm + q = 0$ are pure imaginary exactly when $p = 0$ and $q > 0$, i.e.

$$a_1 + b_2 = 0, \qquad a_1b_2 - a_2b_1 > 0.$$

Hence $b_2 = -a_1$ and $q = -a_1^2 - a_2b_1 > 0$, i.e.

$$a_1^2 + a_2b_1 < 0. \tag{1}$$

In particular $a_2b_1 < 0$, so $a_2 \neq 0$, $b_1 \neq 0$, and a_2 and b_1 have opposite signs.

Solution. Put (14) in differential form with $b_2 = -a_1$:

$$(a_2x - a_1y)\,dx - (a_1x + b_1y)\,dy = 0.$$

With $M = a_2x - a_1y$, $N = -(a_1x + b_1y)$ we have $\frac{\partial M}{\partial y} = -a_1 = \frac{\partial N}{\partial x}$, so the equation is exact. We seek f with $f_x = M$, $f_y = N$:

$$f = \int (a_2x - a_1y)\,dx = \tfrac{1}{2}a_2x^2 - a_1xy + h(y),$$
$$f_y = -a_1x + h'(y) = -a_1x - b_1y \;\Rightarrow\; h'(y) = -b_1y,\; h = -\tfrac{1}{2}b_1y^2.$$

Thus $f = \frac{1}{2}(a_2x^2 - 2a_1xy - b_1y^2)$ and the general solution, with a constant c (twice the constant of integration), is

$$a_2x^2 - 2a_1xy - b_1y^2 = c.$$

(Directly: along a solution of (1), $\frac{d}{dt}(a_2x^2 - 2a_1xy - b_1y^2) = (2a_2x - 2a_1y)(a_1x + b_1y) + (-2a_1x - 2b_1y)(a_2x - a_1y) = 0$, using $b_2 = -a_1$; so every path lies on such a curve. The division by $a_1x + b_1y$ in (14) is thus harmless.)

Ellipses. The equation has the form $Ax^2 + Bxy + Cy^2 = D$ with $A = a_2$, $B = -2a_1$, $C = -b_1$, $D = c$. The discriminant is

$$B^2 - 4AC = (-2a_1)^2 - 4(a_2)(-b_1) = 4a_1^2 + 4a_2b_1 = 4(a_1^2 + a_2b_1) < 0$$

by (1). So each real curve of the family is an ellipse (or the single point $(0,0)$ when $c = 0$). Moreover the quadratic form $Q(x,y) = a_2x^2 - 2a_1xy - b_1y^2$ is definite (Theorem B of Section 61: $B^2 - 4AC < 0$) with the sign of a_2. Hence $Q(x,y) = c$ has no real points if c has the opposite sign to a_2, reduces to the origin for $c = 0$, and for c of the same sign as a_2 is an ellipse centered at the origin; as c varies over these values we obtain a one-parameter family of ellipses, all centered at the origin, nested around it (they are the level sets of a definite quadratic form, which shrink to the origin as $c \to 0$). Every point $\neq (0,0)$ lies on exactly one of them.

Answer. $a_2x^2 - 2a_1xy - b_1y^2 = c$. Since $B^2 - 4AC = 4(a_1^2 + a_2b_1) = -4q < 0$ in Case E, these curves are ellipses surrounding the origin. Agrees with the book.

Section 61 Stability by Liapunov's Direct Method

For the autonomous system

$$\frac{dx}{dt} = F(x,y), \qquad \frac{dy}{dt} = G(x,y) \tag{1}$$

with isolated critical point at the origin, and a function $E(x,y)$ with continuous first partial derivatives, the rate of change of E along the paths is

$$\frac{dE}{dt} = \frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G. \tag{2}$$

Theorem A of the text: if E is positive definite and (2) is negative semidefinite (a *Liapunov function*) then $(0,0)$ is stable; if (2) is negative definite then $(0,0)$ is asymptotically stable. Theorem B: $E = ax^2 + bxy + cy^2$ is positive definite iff $a > 0$ and $b^2 - 4ac < 0$, and negative definite iff $a < 0$ and $b^2 - 4ac < 0$.

Problem 61-1 (Simmons)

Solution. Each is $ax^2 + bxy + cy^2$, so Theorem B applies.

(a) $a = 1, b = -1, c = -1$: $b^2 - 4ac = 1 + 4 = 5 > 0$. Not definite of either sign (indeed $E(1,0) = 1 > 0$ and $E(0,1) = -1 < 0$): *neither*.

(b) $a = 2, b = -3, c = 3$: $b^2 - 4ac = 9 - 24 = -15 < 0$ and $a = 2 > 0$: *positive definite*.

(c) $a = -2, b = 3, c = -1$: $b^2 - 4ac = 9 - 8 = 1 > 0$. Not definite ($E(1,0) = -2 < 0$, whereas $E(2,3) = -8 + 18 - 9 = 1 > 0$): *neither*.

(d) $a = -1, b = -4, c = -5$: $b^2 - 4ac = 16 - 20 = -4 < 0$ and $a = -1 < 0$: *negative definite*.

Answer. (a) neither; (b) positive definite; (c) neither; (d) negative definite. Agrees with the book.

Problem 61-2 (Simmons)

Solution. Let $E(x,y) = ax^3 + bx^2y + cxy^2 + dy^3$. Every term has degree 3, so

$$E(-x,-y) = -ax^3 - bx^2y - cxy^2 - dy^3 = -E(x,y) \quad \text{for all } (x,y).$$

Suppose E were positive definite, i.e. $E(x,y) > 0$ for all $(x,y) \neq (0,0)$ near the origin. Pick any such point (x_1,y_1) in the region (the region contains a disc about the origin, hence also $(-x_1,-y_1)$). Then $E(x_1,y_1) > 0$, but $E(-x_1,-y_1) = -E(x_1,y_1) < 0$, although $(-x_1,-y_1) \neq (0,0)$: a contradiction. Hence E is not positive definite. If E were negative definite, then $-E$ would be positive definite, and $-E$ is again a cubic form of the same kind, which was just excluded (the same argument: $E(x_1,y_1) < 0$ forces $E(-x_1,-y_1) > 0$). So E is neither.

Answer. Shown: since $E(-x,-y) = -E(x,y)$, E cannot have the same strict sign at a point and at its reflection through the origin, so it is neither positive nor negative definite.

Problem 61-3 (Simmons)

Solution. We look for a Liapunov function of the form $E = ax^{2m} + by^{2n}$ $(a, b > 0)$, which is positive definite, and use (2). The origin is a critical point in each case, and it is isolated: it is the only solution of $F = G = 0$ in a neighborhood (see below).

(a) $F = -3x^3 - y$, $G = x^5 - 2y^3$. Try $E = ax^6 + by^2$ (so that $E_x F = 6ax^5(-3x^3 - y)$ contains the term $-6ax^5y$, which can be cancelled by $E_y G = 2by(x^5 - 2y^3)$ which contains $+2bx^5y$):

$$\begin{aligned}\frac{dE}{dt} &= 6ax^5(-3x^3 - y) + 2by(x^5 - 2y^3) \\ &= -18ax^8 - 6ax^5y + 2bx^5y - 4by^4 = -18ax^8 + (2b - 6a)x^5y - 4by^4.\end{aligned}$$

Choose $b = 3a$ to kill the mixed term, and take $a = 1$, $b = 3$:

$$E(x, y) = x^6 + 3y^2, \qquad \frac{dE}{dt} = -18x^8 - 12y^4.$$

E is positive definite ($E(0,0) = 0$, $E > 0$ otherwise) and $dE/dt < 0$ for $(x, y) \neq (0, 0)$ (both terms are ≤ 0 and vanish simultaneously only at the origin): negative definite. By Theorem A, $(0, 0)$ is asymptotically stable. (It is an isolated critical point: $F = 0$ gives $y = -3x^3$, and then $G = x^5 - 2(-3x^3)^3 = x^5 + 54x^9 = x^5(1 + 54x^4)$, which vanishes only for $x = 0$.)

(b) $F = -2x + xy^3$, $G = -x^2y^2 - y^3$. Try $E = ax^2 + by^2$:

$$\begin{aligned}\frac{dE}{dt} &= 2ax(-2x + xy^3) + 2by(-x^2y^2 - y^3) \\ &= -4ax^2 + 2ax^2y^3 - 2bx^2y^3 - 2by^4 = -4ax^2 + 2(a - b)x^2y^3 - 2by^4.\end{aligned}$$

Take $a = b = 1$: $E = x^2 + y^2$ and

$$\frac{dE}{dt} = -4x^2 - 2y^4,$$

which is negative definite. E is positive definite, so by Theorem A the origin is asymptotically stable. (It is isolated: $G = -y^2(x^2 + y) = 0$ and $F = x(-2 + y^3) = 0$. If $x = 0$, $G = -y^3 = 0$ gives $y = 0$; if $y^3 = 2$, then $y \neq 0$ and $G = 0$ would need $x^2 = -y < 0$, impossible.)

Answer. (a) $E = x^6 + 3y^2$, $E' = -18x^8 - 12y^4 < 0$; (b) $E = x^2 + y^2$, $E' = -4x^2 - 2y^4 < 0$. In both cases E is positive definite and E' negative definite, so $(0, 0)$ is asymptotically stable.

Problem 61-4 (Simmons)

Solution. Recall that $(0, 0)$ is *stable* if for each $R > 0$ there is an $r > 0$ (with $r < R$) such that every path which is inside the circle C_r at some time t_0 stays inside C_R for all $t > t_0$; it is unstable if this fails. We assume as always that F, G are such that through every point of the plane region there passes a unique path (e.g. continuous first partial derivatives), and that a path stays defined as long as it stays in a bounded region.

Write $H(x, y) = E_x F + E_y G$, so that along any path $dE/dt = H$. Fix $R > 0$ so small that the closed disc $\overline{D}_R = \{x^2 + y^2 \leq R^2\}$ lies in the region of (a) and (d) (so $H > 0$ on $\overline{D}_R$ except at the origin). We show that *no* $r > 0$ works for this R.

Let $0 < r < R$ be arbitrary. By (c), the circle of radius $r/2$ (centered on the origin) contains a point (x_0, y_0) with $E(x_0, y_0) = E_0 > 0$; this point lies inside C_r and $(x_0, y_0) \neq (0, 0)$ since $E(0, 0) = 0$. Let $C = [x(t), y(t)]$ be the path with $(x(0), y(0)) = (x_0, y_0)$. We claim C leaves $\overline{D}_R$ for some $t > 0$, which shows instability.

Suppose instead that C stays in $\overline{D}_R$ for all $t \geq 0$. Let $E(t) = E(x(t), y(t))$. Then $E'(t) = H(x(t), y(t)) \geq 0$ (as $H \geq 0$ in $\overline{D}_R$), so $E(t) \geq E(0) = E_0 > 0$ for all $t \geq 0$. Hence the path lies in

$$K = \{(x, y) \in \overline{D}_R : E(x, y) \geq E_0\}.$$

K is closed and bounded (closed by continuity of E), hence compact, and $(0,0) \notin K$ since $E(0,0) = 0 < E_0$. The function H is continuous and positive at every point of K (because K avoids the origin, where (d) allows $H = 0$). So H attains a positive minimum $k > 0$ on K. Therefore, for $t \geq 0$,

$$E(t) = E_0 + \int_0^t H(x(s), y(s))\, ds \;\geq\; E_0 + kt \;\longrightarrow\; \infty \quad (t \to \infty).$$

But E is continuous on the compact set $\overline{D}_R$, hence bounded there, $E \leq M$, and $E(t) \leq M$ for all t: contradiction. Thus C must leave $\overline{D}_R$.

So for this R, whatever $r < R$ we choose, there is a path starting inside C_r which does not stay inside C_R for all $t > 0$. By definition, $(0,0)$ is unstable.

Answer. Shown: along a path started where $E > 0$ near the origin, E increases at a rate bounded below by a positive constant while the path stays in a small disc; since E is bounded there, the path must leave the disc, so $(0,0)$ is unstable.

Problem 61-5 (Simmons)

Solution. We use Problem 4. Here $F = 2xy + x^3$, $G = -x^2 + y^5$; $(0,0)$ is a critical point. Try $E = ax^2 + by^2$ $(a, b > 0)$:

$$\begin{aligned} E_x F + E_y G &= 2ax(2xy + x^3) + 2by(-x^2 + y^5) \\ &= 4ax^2y + 2ax^4 - 2bx^2y + 2by^6 = 2ax^4 + 2by^6 + (4a - 2b)x^2y. \end{aligned}$$

Choose $b = 2a$ to kill the mixed term, $a = 1$, $b = 2$:

$$E(x, y) = x^2 + 2y^2, \qquad E_x F + E_y G = 2x^4 + 4y^6.$$

Check the conditions of Problem 4: (a) E is a polynomial; (b) $E(0,0) = 0$; (c) $E > 0$ at every point other than the origin, so every circle centered on the origin contains points where $E > 0$; (d) $2x^4 + 4y^6 > 0$ for $(x, y) \neq (0,0)$ and $= 0$ at the origin, so it is positive definite. Hence $(0,0)$ is unstable.

Answer. With $E = x^2 + 2y^2$ one gets $E' = 2x^4 + 4y^6$, positive definite; by Problem 4, $(0,0)$ is unstable.

Problem 61-6 (Simmons)

Solution. We assume, as is implicit, that f (and g) are continuous, so that the integral has a continuous derivative. (The book writes dx/df in (c); dx/dt is meant.) To avoid the clash of x as limit and as variable we write $W(x) = \int_0^x f(s)\, ds$.

(a) $E(x, y) = \frac{1}{2}y^2 + W(x)$ is continuous with continuous partial derivatives $E_x = f(x)$, $E_y = y$, and $E(0,0) = 0$.

- If $x > 0$: $f(s) > 0$ for $0 < s \leq x$ and f is continuous, so $W(x) = \int_0^x f(s)\, ds > 0$.
- If $x < 0$: $W(x) = -\int_x^0 f(s)\, ds$, and $f(s) < 0$ for $x \leq s < 0$, so $\int_x^0 f(s)\, ds < 0$ and $W(x) > 0$.

Thus $W(x) > 0$ for $x \neq 0$ and $W(0) = 0$. Therefore, if $x \neq 0$, $E \geq W(x) > 0$; if $x = 0$ and $y \neq 0$, $E = \frac{1}{2}y^2 > 0$. So $E > 0$ for $(x, y) \neq (0, 0)$: positive definite.

(b) Put $y = dx/dt$; the equation is equivalent to the system

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = -f(x).$$

Its critical points: $y = 0$ and $f(x) = 0$; since $f(x) \neq 0$ for $x \neq 0$, the only one is $(0, 0)$, which is therefore isolated. Using E of (a),

$$\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G = f(x) \cdot y + y \cdot (-f(x)) = 0,$$

which is negative semidefinite (it is ≤ 0, and $= 0$ at the origin). Thus E is a Liapunov function and by Theorem A the critical point $(0, 0)$ is stable. (In fact E is the total energy, constant along paths, so the point is not asymptotically stable.)

(c) Equivalent system: $\frac{dx}{dt} = y$, $\frac{dy}{dt} = -g(x)y - f(x)$. Again the critical points satisfy $y = 0$, then $f(x) = 0$, hence only $(0, 0)$. With the same E,

$$\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G = f(x)y + y\big(-g(x)y - f(x)\big) = -g(x)y^2 \leq 0$$

in the neighborhood of the origin where $g \geq 0$. It is 0 at the origin and ≤ 0 elsewhere: negative semidefinite. Hence E is a Liapunov function and $(0, 0)$ is stable.

Answer. (a) $E > 0$ for $(x, y) \neq (0, 0)$ since $\int_0^x f > 0$ for $x \neq 0$; (b) $E' = 0$, so $(0, 0)$ is stable; (c) $E' = -g(x)y^2 \leq 0$, so $(0, 0)$ is stable.

Section 62 Simple Critical Points of Nonlinear Systems

Throughout, the system is

$$\frac{dx}{dt} = a_1x + b_1y + f(x,y), \qquad \frac{dy}{dt} = a_2x + b_2y + g(x,y), \tag{4}$$

with related linear system $x' = a_1x + b_1y$, $y' = a_2x + b_2y$ (3). The origin is a *simple* critical point if $\Delta = a_1b_2 - a_2b_1 \neq 0$ (5), f, g are continuous with continuous first partial derivatives, and $f/r \to 0$, $g/r \to 0$ as $r = \sqrt{x^2+y^2} \to 0$ (6). We write

$$p = -(a_1 + b_2), \qquad q = a_1b_2 - a_2b_1,$$

so that the characteristic equation of the linear system is $m^2 + pm + q = 0$. We use the following results quoted in the text: Theorem A (a simple critical point of (4) has the same type as that of (3) in the node, saddle and spiral cases of Section 60); Theorem B (if the origin is asymptotically stable for (3), i.e. $p > 0$, $q > 0$, it is asymptotically stable for (4)); Theorem 61-A (Liapunov's theorem), and Problem 61-4 (Chetaev-type instability criterion: if there is a function E, continuous with continuous first partial derivatives near the origin, with $E(0,0) = 0$, with $E > 0$ at some point inside every circle about the origin, and with $\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G$ positive definite, then the origin is unstable).

Problem 62-1 (Simmons)

Solution. First, $(0,0)$ *is* a critical point: f and g are continuous and f/r, $g/r \to 0$ force $f(0,0) = \lim_{r\to 0} r \cdot (f/r) = 0$, and likewise $g(0,0) = 0$, so both right sides of (4) vanish at the origin.

Suppose the origin were not isolated. Then there are points $(x_n, y_n) \neq (0,0)$, with $r_n = \sqrt{x_n^2 + y_n^2} \to 0$, at which both right sides of (4) vanish:

$$a_1x_n + b_1y_n + f(x_n,y_n) = 0, \qquad a_2x_n + b_2y_n + g(x_n,y_n) = 0.$$

Put $x_n = r_n\cos\theta_n$, $y_n = r_n\sin\theta_n$ and $\epsilon_{1,n} = f(x_n,y_n)/r_n$, $\epsilon_{2,n} = g(x_n,y_n)/r_n$; by (6), $\epsilon_{1,n} \to 0$ and $\epsilon_{2,n} \to 0$. Dividing the two equations by $r_n \neq 0$ gives, with $c = \cos\theta_n$, $s = \sin\theta_n$,

$$a_1c + b_1s = -\epsilon_{1,n}, \qquad a_2c + b_2s = -\epsilon_{2,n}.$$

By (5) the determinant $\Delta = a_1b_2 - a_2b_1 \neq 0$, so Cramer's rule solves this linear system for c, s:

$$\cos\theta_n = \frac{\epsilon_{2,n}b_1 - \epsilon_{1,n}b_2}{\Delta}, \qquad \sin\theta_n = \frac{a_2\epsilon_{1,n} - a_1\epsilon_{2,n}}{\Delta}.$$

(Indeed, $a_1 \cdot \frac{\epsilon_2 b_1 - \epsilon_1 b_2}{\Delta} + b_1 \cdot \frac{a_2\epsilon_1 - a_1\epsilon_2}{\Delta} = \frac{-\epsilon_1(a_1b_2 - a_2b_1)}{\Delta} = -\epsilon_1$, and similarly for the second equation.) Since $\epsilon_{1,n}, \epsilon_{2,n} \to 0$ and a_i, b_i, Δ are fixed constants with $\Delta \neq 0$, both $\cos\theta_n \to 0$ and $\sin\theta_n \to 0$. But then

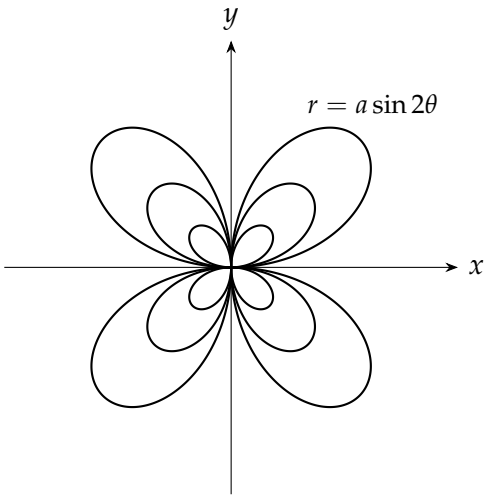
$$1 = \cos^2\theta_n + \sin^2\theta_n \longrightarrow 0,$$

a contradiction. Hence there is a disc about the origin containing no other critical point: $(0,0)$ is isolated.

Answer. Shown: if $\Delta \neq 0$ and $f/r, g/r \to 0$, then $\cos\theta_n, \sin\theta_n \to 0$ at any hypothetical nearby critical points, contradicting $\cos^2\theta + \sin^2\theta = 1$; so a simple critical point is isolated.

Problem 62-2 (Simmons)

Solution. **The sketch.** Since $\sin 2\theta$ has period π and changes sign at $\theta = 0, \pi/2, \pi, 3\pi/2$, the curve $r = a\sin 2\theta$ ($a > 0$) is a four-leaved rose. For $0 \le \theta \le \pi/2$, $r \ge 0$ and we get the leaf in the first quadrant (maximum length a at $\theta = \pi/4$); for $\pi/2 \le \theta \le \pi$, $r \le 0$, so the points lie in the fourth quadrant (the point (r,θ) with $r < 0$ is $(|r|, \theta + \pi)$); for $\pi \le \theta \le 3\pi/2$ we get the third quadrant, and for $3\pi/2 \le \theta \le 2\pi$ the second quadrant. All leaves pass through the origin, tangent there to the coordinate axes. Replacing a by $-a$ gives the same rose, because $-\sin 2\theta = \sin 2(\theta + \pi/2)$ (the rose is carried into itself by a rotation through $\pi/2$). So the family consists of similar roses of "size" $|a|$, all with leaves along the diagonals $y = \pm x$ (compare Figure 90).



(shown for $a = 1,\ 0.6,\ 0.3$).

The differential equation. Since $x = r\cos\theta$, $y = r\sin\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta = 2xy/r^2$, the equation $r = a\sin 2\theta$ becomes $r = 2axy/r^2$, i.e.

$$(x^2 + y^2)^{3/2} = 2a\,xy.$$

Differentiate implicitly with respect to x (with $y = y(x)$, $r^2 = x^2 + y^2$):

$$\tfrac{3}{2}(x^2 + y^2)^{1/2}(2x + 2yy') = 2a(y + xy') \quad\Longrightarrow\quad 3r(x + yy') = 2a(y + xy').$$

To eliminate the parameter a, use the equation of the family itself, $2a = r^3/(xy)$ (valid where $xy \neq 0$):

$$3r(x + yy') = \frac{r^3}{xy}(y + xy') \implies 3xy(x + yy') = (x^2 + y^2)(y + xy')$$

(after dividing by $r \neq 0$ and multiplying by xy). Expand:

$$3x^2y + 3xy^2y' = x^2y + y^3 + (x^3 + xy^2)y' \implies (3xy^2 - x^3 - xy^2)\,y' = y^3 + x^2y - 3x^2y,$$

that is, $(2xy^2 - x^3)y' = y^3 - 2x^2y$. Multiplying by -1,

$$\frac{dy}{dx} = \frac{2x^2y - y^3}{x^3 - 2xy^2}.$$

Check by the polar slope. For $r = a\sin 2\theta$, $dy/dx = \dfrac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$ with $r' = 2a\cos 2\theta$; at the sample points tested numerically this agrees with G/F (see the verification script).

Answer. The family $r = a\sin 2\theta$ is a four-leaved rose (leaves along the lines $y = \pm x$), and

$$\frac{dy}{dx} = \frac{G(x,y)}{F(x,y)} = \frac{2x^2y - y^3}{x^3 - 2xy^2}.$$

Problem 62-3 (Simmons)

Solution. Write $T = a_1 + b_2 = -p$. By hypothesis $T > 0$ and $q > 0$, so $Tq > 0$.

Step 1: the function E. Define

$$E(x,y) = \tfrac{1}{2}\left(\mathcal{A}x^2 + 2\mathcal{B}xy + \mathcal{C}y^2\right),$$

$$\mathcal{A} = \frac{a_2^2 + b_2^2 + q}{Tq}, \qquad \mathcal{B} = -\frac{a_1a_2 + b_1b_2}{Tq}, \qquad \mathcal{C} = \frac{a_1^2 + b_1^2 + q}{Tq}.$$

(These are the coefficients a, b, c of the proof of Theorem B with $D = pq = -Tq$ replaced by $-D = Tq$, i.e. with E replaced by $-E$.) Then $\partial E/\partial x = \mathcal{A}x + \mathcal{B}y$ and $\partial E/\partial y = \mathcal{B}x + \mathcal{C}y$, so

$$\begin{aligned} L &:= \frac{\partial E}{\partial x}(a_1x + b_1y) + \frac{\partial E}{\partial y}(a_2x + b_2y) \\ &= (\mathcal{A}a_1 + \mathcal{B}a_2)x^2 + (\mathcal{A}b_1 + \mathcal{B}a_1 + \mathcal{B}b_2 + \mathcal{C}a_2)xy + (\mathcal{B}b_1 + \mathcal{C}b_2)y^2. \end{aligned}$$

We compute the three coefficients, multiplying each by Tq and using $q = a_1b_2 - a_2b_1$:

Coefficient of x^2:

$$\begin{aligned} Tq(\mathcal{A}a_1 + \mathcal{B}a_2) &= a_1(a_2^2 + b_2^2 + q) - a_2(a_1a_2 + b_1b_2) = a_1b_2^2 - a_2b_1b_2 + a_1q \\ &= b_2(a_1b_2 - a_2b_1) + a_1q = (a_1 + b_2)q = Tq. \end{aligned}$$

Coefficient of y^2:

$$\begin{aligned} Tq(\mathcal{B}b_1 + \mathcal{C}b_2) &= -b_1(a_1a_2 + b_1b_2) + b_2(a_1^2 + b_1^2 + q) = a_1^2b_2 - a_1a_2b_1 + b_2q \\ &= a_1(a_1b_2 - a_2b_1) + b_2q = (a_1 + b_2)q = Tq. \end{aligned}$$

Coefficient of xy:

$$\begin{aligned} Tq(\mathcal{A}b_1 + \mathcal{B}a_1 + \mathcal{B}b_2 + \mathcal{C}a_2) &= b_1(a_2^2 + b_2^2 + q) - (a_1 + b_2)(a_1a_2 + b_1b_2) + a_2(a_1^2 + b_1^2 + q) \\ &= \left(a_2^2b_1 + b_1b_2^2 + b_1q\right) - \left(a_1^2a_2 + a_1b_1b_2 + a_1a_2b_2 + b_1b_2^2\right) \\ &\quad + \left(a_1^2a_2 + a_2b_1^2 + a_2q\right) \\ &= a_2^2b_1 - a_1b_1b_2 - a_1a_2b_2 + a_2b_1^2 + (a_2 + b_1)q \\ &= \left(a_2^2b_1 - a_1a_2b_2\right) + \left(a_2b_1^2 - a_1b_1b_2\right) + (a_2 + b_1)q \\ &= -a_2q - b_1q + (a_2 + b_1)q = 0. \end{aligned}$$

(In the second line, $b_1b_2^2$ and $a_1^2a_2$ cancel between the three brackets; in the third line we used $a_2^2b_1 - a_1a_2b_2 = -a_2(a_1b_2 - a_2b_1) = -a_2q$ and $a_2b_1^2 - a_1b_1b_2 = -b_1(a_1b_2 - a_2b_1) = -b_1q$.)

Dividing by Tq the three coefficients are $1,\ 0,\ 1$:

$$\boxed{\frac{\partial E}{\partial x}(a_1x + b_1y) + \frac{\partial E}{\partial y}(a_2x + b_2y) = x^2 + y^2}.$$

Step 2: E is positive definite. By Theorem 61-B (for $E = \frac{1}{2}(\mathcal{A}x^2 + 2\mathcal{B}xy + \mathcal{C}y^2)$) it is enough that $\mathcal{A} > 0$ and $\mathcal{AC} - \mathcal{B}^2 > 0$. First, $\mathcal{A} = (a_2^2 + b_2^2 + q)/(Tq) > 0$ since $q > 0$, $T > 0$. Next,

$$\begin{aligned}(Tq)^2(\mathcal{AC} - \mathcal{B}^2) &= (a_2^2 + b_2^2 + q)(a_1^2 + b_1^2 + q) - (a_1a_2 + b_1b_2)^2 \\ &= \big[(a_1^2 + b_1^2)(a_2^2 + b_2^2) - (a_1a_2 + b_1b_2)^2\big] + q(a_1^2 + b_1^2 + a_2^2 + b_2^2) + q^2 \\ &= q^2 + q(a_1^2 + b_1^2 + a_2^2 + b_2^2) + q^2 \;>\; 0,\end{aligned}$$

where we used Lagrange's identity $(a_1^2 + b_1^2)(a_2^2 + b_2^2) - (a_1a_2 + b_1b_2)^2 = (a_1b_2 - a_2b_1)^2 = q^2$. Hence E is positive definite. Obviously E is a polynomial, so it is continuous with continuous partial derivatives, $E(0,0) = 0$, and $E > 0$ at every point other than the origin, in particular inside every circle about the origin.

Step 3: the nonlinear system. Let $F = a_1x + b_1y + f$, $G = a_2x + b_2y + g$. By Step 1,

$$\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G = (x^2 + y^2) + (\mathcal{A}x + \mathcal{B}y)f + (\mathcal{B}x + \mathcal{C}y)g.$$

In polar coordinates this is

$$r^2 + r\big[(\mathcal{A}\cos\theta + \mathcal{B}\sin\theta)f + (\mathcal{B}\cos\theta + \mathcal{C}\sin\theta)g\big].$$

Let K be the largest of $|\mathcal{A}|, |\mathcal{B}|, |\mathcal{C}|$ ($K > 0$ since $\mathcal{A} > 0$). Then $|\mathcal{A}\cos\theta + \mathcal{B}\sin\theta| \le 2K$ and $|\mathcal{B}\cos\theta + \mathcal{C}\sin\theta| \le 2K$. By (6) there is $\delta > 0$ such that $|f| < r/(6K)$ and $|g| < r/(6K)$ for $0 < r < \delta$. For those r,

$$\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G > r^2 - r\Big(2K\frac{r}{6K} + 2K\frac{r}{6K}\Big) = r^2 - \tfrac{2}{3}r^2 = \tfrac{1}{3}r^2 > 0,$$

so $\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G$ is positive definite near the origin.

Conclusion. All hypotheses of Problem 61-4 hold, so $(0,0)$ is an unstable critical point of (4).

Answer. Shown: with $T = a_1 + b_2 > 0$, $q > 0$, the positive definite form $E = \frac{1}{2}(\mathcal{A}x^2 + 2\mathcal{B}xy + \mathcal{C}y^2)$ above satisfies $E_x(a_1x + b_1y) + E_y(a_2x + b_2y) = x^2 + y^2$, and then $E_xF + E_yG > \frac{1}{3}r^2$ near the origin, so $(0,0)$ is unstable by Problem 61-4.

Problem 62-4 (Simmons)

Solution. In both systems $(0,0)$ is a critical point, and the related linear system is $x' = -y$, $y' = x$, whose characteristic equation is $m^2 + 1 = 0$ ($p = 0$, $q = 1$): $m = \pm i$, a center. The nonlinear terms $\mp x^3$, $\mp y^3$ satisfy $|f|/r \le r^2 \to 0$, so $(0,0)$ is a simple critical point of each system, but since the linear system has a center, Theorems A and B and Problem 3 give no information. We use the function $E(x,y) = x^2 + y^2$, which is continuous with continuous first partials, $E(0,0) = 0$, and positive definite.

(First system) Here $F = -y - x^3$, $G = x - y^3$, and

$$\begin{aligned}\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G &= 2x(-y - x^3) + 2y(x - y^3) \\ &= -2xy - 2x^4 + 2xy - 2y^4 = -2(x^4 + y^4).\end{aligned}$$

This is negative definite ($x^4 + y^4 > 0$ for $(x,y) \ne (0,0)$). So E is a Liapunov function whose derivative along the paths is negative definite, and by Liapunov's Theorem 61-A, $(0,0)$ is *asymptotically stable*. (In polar form, $r\,r' = xx' + yy' = -(x^4 + y^4)$, i.e. $r' = -r^3(\cos^4\theta + \sin^4\theta) \le -\frac{1}{2}r^3$,

since $\cos^4\theta + \sin^4\theta = 1 - \frac{1}{2}\sin^2 2\theta \geq \frac{1}{2}$; r decreases to 0 but only like $t^{-1/2}$, i.e. algebraically rather than exponentially.)

(Second system) Here $F = -y + x^3$, $G = x + y^3$, and

$$\frac{\partial E}{\partial x}F + \frac{\partial E}{\partial y}G = 2x(-y + x^3) + 2y(x + y^3) = 2(x^4 + y^4),$$

which is positive definite. Moreover E satisfies all the other requirements of Problem 61-4: it is continuous with continuous partial derivatives, $E(0,0) = 0$, and $E > 0$ at every point $\neq (0,0)$, hence inside every circle about the origin. Therefore $(0,0)$ is *unstable*. (Polar form: $r' = r^3(\cos^4\theta + \sin^4\theta) \geq \frac{1}{2}r^3 > 0$, so r increases along every path.)

(Relation to Problem 3) The parenthetical remark in Problem 3 says that the information in Figure 86 — asymptotic stability, stability, instability — carries over from the linear system to the nonlinear system *for simple critical points* when the linear system is not on a borderline (that is, when $p \neq 0$, $q \neq 0$). The two systems here have the *same* linear part (a center, $p = 0$) — the borderline case which Theorem A, Theorem B and Problem 3 do not cover. The nonlinear terms, however small near the origin, decide the stability: $-x^3, -y^3$ add damping and make the origin asymptotically stable, whereas $+x^3, +y^3$ add "anti-damping" and make it unstable (while in a third example, $x' = -y$, $y' = x$ itself, the origin is stable but not asymptotically stable). Thus no conclusion about stability of the nonlinear system can be drawn when the related linear system has a center, in agreement with the answer given in the book to this section.

Answer. With $E = x^2 + y^2$: $\dot{E} = -2(x^4 + y^4) < 0$ for the first system (asymptotically stable) and $\dot{E} = +2(x^4 + y^4) > 0$ for the second (unstable). The linear part is the same center in both: the center is a borderline case not covered by Theorems A, B or Problem 3, so the nonlinear terms decide.

Problem 62-5 (Simmons)

Solution.

(a) Here $a_1 = 1,\ b_1 = 1,\ a_2 = -2,\ b_2 = 1,\ f = -2xy,\ g = 3y^2$.

Simple critical point. $\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} = 1 + 2 = 3 \neq 0$. The functions f, g are polynomials, hence continuous with continuous partials, and since $|x| \leq r$, $|y| \leq r$,

$$\frac{|f|}{r} = \frac{2|x||y|}{r} \leq \frac{2r^2}{r} = 2r \to 0, \qquad \frac{|g|}{r} = \frac{3y^2}{r} \leq 3r \to 0 \qquad (r \to 0).$$

So (6) holds and $(0,0)$ is a simple critical point.

Nature. The linear system $x' = x + y$, $y' = -2x + y$ has characteristic equation $\begin{vmatrix} 1-m & 1 \\ -2 & 1-m \end{vmatrix} = (1-m)^2 + 2 = 0$, i.e. $m^2 - 2m + 3 = 0$, with roots $m = 1 \pm i\sqrt{2}$. These are conjugate complex but not pure imaginary ($p = -2$, $q = 3$, $p^2 - 4q = -8 < 0$, $p \neq 0$): a spiral (the case of complex roots that are not pure imaginary). Since the real part is $+1 > 0$ ($p < 0$, $q > 0$), it is an unstable spiral. By Theorem A the same is true for the nonlinear system, and by Problem 3 it is unstable.

Result: $(0,0)$ is an *unstable spiral point*.

(b) Here $a_1 = -1,\ b_1 = -1,\ a_2 = -2,\ b_2 = -4,\ f = -3x^2y,\ g = y\sin x$.

Simple critical point. $\begin{vmatrix} -1 & -1 \\ -2 & -4 \end{vmatrix} = 4 - 2 = 2 \neq 0$. The functions f, g are continuously differentiable, and

$$\frac{|f|}{r} = \frac{3x^2|y|}{r} \leq \frac{3r^3}{r} = 3r^2 \to 0, \qquad \frac{|g|}{r} = \frac{|y||\sin x|}{r} \leq |\sin x| \leq |x| \leq r \to 0.$$

So $(0,0)$ is a simple critical point.

Nature. The linear system $x' = -x - y$, $y' = -2x - 4y$ has characteristic equation

$$\begin{vmatrix} -1-m & -1 \\ -2 & -4-m \end{vmatrix} = (m+1)(m+4) - 2 = m^2 + 5m + 2 = 0,$$

with roots

$$m = \frac{-5 \pm \sqrt{25-8}}{2} = \frac{-5 \pm \sqrt{17}}{2} \approx -0.438, \ -4.562.$$

These are real, distinct and both negative ($p = 5 > 0$, $q = 2 > 0$, $p^2 - 4q = 17 > 0$): a node, asymptotically stable. By Theorem A (type) and Theorem B (stability) the same holds for the nonlinear system.

Result: $(0,0)$ is an *asymptotically stable node*.

Answer. (a) unstable spiral point; (b) asymptotically stable node. (Agrees with the book.)

Problem 62-6 (Simmons)

Solution. Write the second equation as $y' = -x + \mu y - \mu x^2 y$. Then $a_1 = 0$, $b_1 = 1$, $a_2 = -1$, $b_2 = \mu$, $f = 0$, $g = -\mu x^2 y$.

Simple critical point. $\begin{vmatrix} 0 & 1 \\ -1 & \mu \end{vmatrix} = 1 \neq 0$, and $|g|/r = |\mu| x^2 |y|/r \leq |\mu| r^2 \to 0$, $f/r = 0$. So $(0,0)$ is a simple critical point for every real μ.

Linear system. $x' = y$, $y' = -x + \mu y$ has characteristic equation $\begin{vmatrix} -m & 1 \\ -1 & \mu - m \end{vmatrix} = -m(\mu - m) + 1 = m^2 - \mu m + 1 = 0$, so

$$p = -\mu, \qquad q = 1, \qquad m = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}.$$

- $\mu > 0$: $p < 0$ and $q > 0$. By Problem 3, $(0,0)$ is *unstable*. In more detail: if $0 < \mu < 2$ the roots are complex with real part $\mu/2 > 0$ (unstable spiral); if $\mu > 2$ they are real, distinct and positive (unstable node); if $\mu = 2$ the root $m = 1$ is double (borderline node for the linear system; Theorem A does not decide the type, but Problem 3 still gives instability).

- $\mu < 0$: $p > 0$ and $q > 0$. By Theorem B, $(0,0)$ is *asymptotically stable*: a stable spiral if $-2 < \mu < 0$ (real part $\mu/2 < 0$), a stable node if $\mu < -2$, and the borderline case $\mu = -2$ (double root $m = -1$) is still asymptotically stable by Theorem B.

(For $\mu = 0$ the equation is $x'' + x = 0$, the linear system itself, with a stable center, which is not asymptotically stable.)

Interpretation. For $\mu > 0$ (the physically relevant case) the origin repels small oscillations, which is why the van der Pol oscillator builds up to a self-sustained oscillation (Section 64).

Answer. $(0,0)$ is unstable if $\mu > 0$ (spiral for $0 < \mu < 2$, node for $\mu > 2$) and asymptotically stable if $\mu < 0$ (spiral for $-2 < \mu < 0$, node for $\mu < -2$). Agrees with the book.

Section 63 Nonlinear Mechanics. Conservative Systems

Throughout we consider $\dfrac{d^2x}{dt^2} + f(x) = 0$ (the case $m = 1$ of equation (2) of the text), with f continuous (and, as in the text, with continuous derivative where we linearize), equivalent to the system

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = -f(x). \tag{3}$$

Its paths satisfy $\frac{1}{2}y^2 + V(x) = E$ with $V(x) = \int_0^x f(x)\,\mathrm{d}x$ (the potential energy). Indeed, along a path $\frac{d}{dt}\left[\frac{1}{2}y^2 + V(x)\right] = y\,y' + f(x)\,x' = -yf(x) + f(x)\,y = 0$. Facts used from the text: the critical points are the points $(x_c, 0)$ with $f(x_c) = 0$ (that is, $V'(x_c) = 0$); the paths are symmetric about the x-axis; they cross the x-axis at right angles; the motion is to the right in the upper half-plane ($x' = y > 0$) and to the left in the lower half-plane; and the method of Figure 91, $y = \pm\sqrt{2[E - V(x)]}$. All paths in the sketches below are drawn with these rules. Moreover, in all sketches arrows give the direction of increasing t.

Problem 63-1 (Simmons)

Solution.

(Case $xf(x) > 0$) Shape of V. The condition $xf(x) > 0$ says $f > 0$ for $x > 0$ and $f < 0$ for $x < 0$. Hence

$$V(x) = \int_0^x f(s)\,\mathrm{d}s > 0 \quad (x \neq 0), \qquad V(0) = 0,$$

V is strictly increasing on $[0, \infty)$ (because $V' = f > 0$ there) and strictly decreasing on $(-\infty, 0]$. The only critical point of (3) is $(0, 0)$, since $f(x) = 0$ only for $x = 0$.

The level curves are closed curves around the origin. Let $L_+ = \lim_{x\to+\infty} V(x)$ and $L_- = \lim_{x\to-\infty} V(x)$ (each possibly $+\infty$), and $c^* = \min(L_+, L_-) > 0$. Take $0 < c < c^*$. By continuity and strict monotonicity, there is exactly one $x_+ = x_+(c) > 0$ with $V(x_+) = c$ and exactly one $x_- = x_-(c) < 0$ with $V(x_-) = c$, and $V(x) < c$ for $x_- < x < x_+$, $V(x) > c$ outside $[x_-, x_+]$. The path $\frac{1}{2}y^2 + V(x) = c$ requires $V(x) \le c$, so it is the curve

$$y = \pm\sqrt{2[c - V(x)]}, \qquad x_- \le x \le x_+.$$

For $x_- < x < x_+$ the radicand is positive, so the curve has an upper branch ($y > 0$) and a lower branch ($y < 0$), which join at the two points $(x_\pm, 0)$ on the x-axis. It is therefore a bounded simple closed curve, symmetric about the x-axis, which surrounds the origin (the origin lies strictly between the branches, since $c - V(0) = c > 0$).

It is a closed path. The curve does not contain the critical point $(0, 0)$ (there $E = 0 \neq c$), and any other point of it is not a critical point either. On the compact curve the velocity vector $(x', y') = (y, -f(x))$ never vanishes (it vanishes only if $y = 0$ and $f(x) = 0$, i.e. at $(0, 0)$), so its length has a positive lower bound, while the curve has finite length. Since E is constant along the motion, the point $(x(t), y(t))$ stays on the curve; it moves clockwise (to the right on the upper branch, to the left on the lower branch) with speed bounded below, hence goes all the way around and returns to its starting point in finite time. So the path is a closed curve: every solution with $0 < E < c^*$ is periodic. (Its period is $2\int_{x_-}^{x_+} \mathrm{d}x/\sqrt{2[c - V(x)]}$.)

Stability. Let $\varepsilon > 0$ be given. Put

$$c_0 = \min\{V(\varepsilon/2),\ V(-\varepsilon/2),\ \varepsilon^2/8\} > 0,$$

and, since $E(x, y) = \frac{1}{2}y^2 + V(x)$ is continuous with $E(0, 0) = 0$, choose $\delta > 0$ so that $E < c_0$ inside the disc of radius δ. A path starting in that disc has constant $E = E_0 < c_0$. Then

$V(x) \le E_0 < c_0 \le V(\pm\varepsilon/2)$ gives (by the monotonicity of V on each side of 0) $|x| < \varepsilon/2$, and $\frac{1}{2}y^2 \le E_0 < \varepsilon^2/8$ gives $|y| < \varepsilon/2$, so $r < \varepsilon/\sqrt{2} < \varepsilon$ for all t. Hence $(0,0)$ is *stable*.

Not asymptotically stable. A path starting at $(x_0, y_0) \neq (0,0)$ has $E_0 > 0$ and stays on the closed curve $E = E_0$ (when $E_0 < c^*$), which is a fixed positive distance from the origin and which it traverses periodically; so it does not tend to $(0,0)$ as $t \to \infty$. Hence the critical point is a stable center that is not asymptotically stable.

(Case $xf(x) < 0$) Now $f < 0$ for $x > 0$ and $f > 0$ for $x < 0$, so $V(x) = \int_0^x f < 0$ for $x \neq 0$, with V strictly decreasing on $[0,\infty)$ and strictly increasing on $(-\infty, 0]$ (a strict local *maximum* of V at 0). The only critical point is again $(0,0)$.

Instability. Take $W(x,y) = xy$. It is a polynomial, $W(0,0) = 0$, and $W > 0$ at points arbitrarily close to the origin (for example $x = y > 0$). Along the paths of (3),

$$\frac{\partial W}{\partial x}\, y + \frac{\partial W}{\partial y}\,(-f(x)) = y \cdot y + x \cdot (-f(x)) = y^2 - xf(x),$$

and this is positive for $(x,y) \neq (0,0)$ (if $y \neq 0$ the term $y^2 > 0$ and $-xf(x) \ge 0$; if $y = 0$ then $x \neq 0$ and $-xf(x) > 0$), i.e. it is positive definite. By Problem 61-4, $(0,0)$ is *unstable*.

Nature. The paths are the curves $\frac{1}{2}y^2 + V(x) = E$, that is $y^2 = 2[E - V(x)] = 2[E + |V(x)|]$:

- $E = 0$: $y = \pm\sqrt{2|V(x)|}$, two smooth curves (one above, one below the x-axis away from 0, both vanishing only at $x = 0$), which cross at the origin: four half-paths run into or out of the origin (the "separatrices");
- $E > 0$: $y^2 \ge 2E > 0$, so the paths never meet the x-axis: one path lies above the origin, one below, and they are asymptotic in shape to the separatrices;
- $E < 0$: $|V(x)| \ge |E|$ forces $|x| \ge x_*(E) > 0$, so the path has two separate parts, one to the right of the origin (turning at $(x_+, 0)$) and one to the left.

This is exactly the picture of a saddle point (for $f(x) = -x$ the curves $y^2 - x^2 = 2E$ are the familiar hyperbolas, with asymptotes $y = \pm x$). Thus $(0,0)$ is an *unstable saddle point*.

Answer. If $f(0) = 0$ and $xf(x) > 0$: the level curves $\frac{1}{2}y^2 + V(x) = E$ (for E small enough) are closed paths around the origin, so $(0,0)$ is a stable center, not asymptotically stable. If $xf(x) < 0$ for $x \neq 0$: $(0,0)$ is an unstable saddle point. (Agrees with the book.)

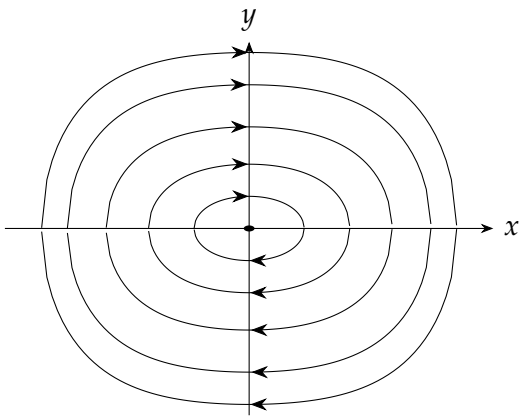
Problem 63-2 (Simmons)

Solution. Here $f(x) = kx + \alpha x^3$, so

$$V(x) = \int_0^x (ks + \alpha s^3)\, ds = \tfrac{1}{2}kx^2 + \tfrac{1}{4}\alpha x^4,$$

and the paths are $\frac{1}{2}y^2 + \frac{1}{2}kx^2 + \frac{1}{4}\alpha x^4 = E$, i.e. $y = \pm\sqrt{2E - kx^2 - \frac{1}{2}\alpha x^4}$. The critical points are the points $(x, 0)$ with $x(k + \alpha x^2) = 0$. We draw the sketches for $k = 1$ and $\alpha = \pm 1$; the qualitative picture is the same for any $k > 0$ and α of the same sign.

(Hard spring, $\alpha > 0$) Then $f(x) = x(k + \alpha x^2)$ vanishes only at $x = 0$, and $xf(x) = x^2(k + \alpha x^2) > 0$ for $x \neq 0$. Moreover $V(x) \to +\infty$ as $x \to \pm\infty$ (because $V \ge \frac{1}{4}\alpha x^4$), so $c^* = \infty$ in Problem 1: *every* path with $E > 0$ is a closed curve around the origin, and every motion is periodic. The origin is a stable center. For $k = \alpha = 1$ the path with energy E crosses the x-axis at $x = \pm x_E$, where $\frac{1}{2}x_E^2 + \frac{1}{4}x_E^4 = E$, i.e. $x_E^2 = -1 + \sqrt{1 + 4E}$, and crosses the y-axis at $y = \pm\sqrt{2E}$.



Hard spring ($k = \alpha = 1$): closed paths $E = 0.1, 0.4, 1, 2, 3$ around the center $(0, 0)$.

(Soft spring, $\alpha < 0$) Write $\alpha = -\beta$, $\beta > 0$. Now $f(x) = x(k - \beta x^2)$ vanishes at $x = 0$ and at $x = \pm a$, $a = \sqrt{k/\beta}$. Since $f'(x) = k - 3\beta x^2$, we have $f'(0) = k > 0$ and $f'(\pm a) = -2k < 0$. V has a local minimum at 0 (where $V = 0$) and local maxima at $\pm a$, where

$$V(\pm a) = \tfrac{1}{2}ka^2 - \tfrac{1}{4}\beta a^4 = \frac{k^2}{2\beta} - \frac{k^2}{4\beta} = \frac{k^2}{4\beta} =: E_s.$$

So $(0, 0)$ is a stable center (by Problem 1 applied near 0), and $(\pm a, 0)$ are unstable saddle points (by the case $xf(x) < 0$ of Problem 1, applied to $\xi = x \mp a$; the instability argument there, with $W = \xi y$, uses only a neighbourhood of the critical point, and near $x = \pm a$ we have $(x \mp a)f(x) < 0$ for $x \neq \pm a$, because $f = x(k - \beta x^2)$ changes sign there). For $k = 1$, $\beta = 1$: $a = 1$, $E_s = \frac{1}{4}$ and

$$y^2 = 2E - x^2 + \tfrac{1}{2}x^4.$$

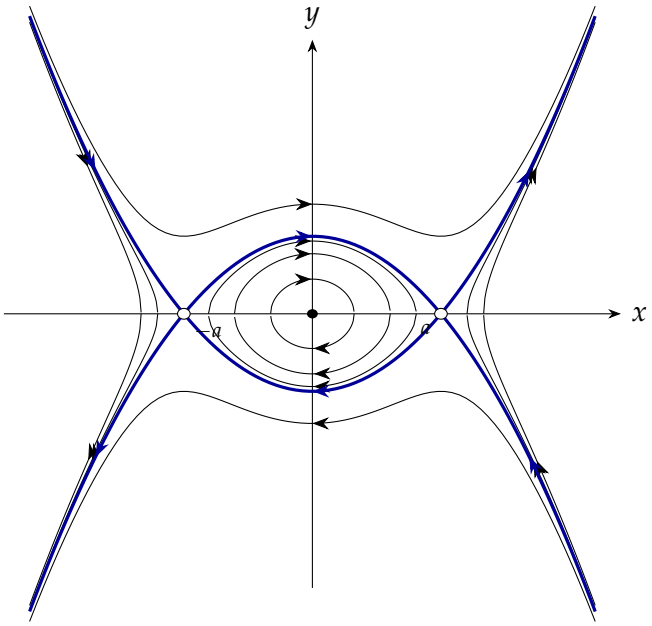
The energy levels give:

- $0 < E < \frac{1}{4}$, inner part: closed paths around the origin (inside $|x| < 1$) crossing the x-axis at $x = \pm x_i$, $x_i^2 = 1 - \sqrt{1 - 4E}$ — periodic oscillations;

- $E = \frac{1}{4}$: here $y^2 = \frac{1}{2} - x^2 + \frac{1}{2}x^4 = \frac{1}{2}(1 - x^2)^2$, so the *separatrices* are $y = \pm\dfrac{1 - x^2}{\sqrt{2}}$; the part with $|x| < 1$ is the boundary of the region of periodic motion, a loop through the two saddle points $(\pm 1, 0)$, and the parts with $|x| > 1$ enter or leave the saddles;

- $0 < E < \frac{1}{4}$, outer part: paths with $|x| \geq x_o$, $x_o^2 = 1 + \sqrt{1 - 4E} > 1$, which come in from infinity, turn at $(\pm x_o, 0)$ and go back out (the motion is not periodic: it escapes to infinity);

- $E > \frac{1}{4}$: the radicand $2E - x^2 + \frac{1}{2}x^4$ is positive for all x (its minimum is $2E - \frac{1}{2} > 0$), so the paths are two wavy curves, $y > 0$ moving right and $y < 0$ moving left, running off to $x = \pm\infty$.



Soft spring ($k=1$, $\alpha=-1$, $a=1$). Thick: separatrices $E=\frac{1}{4}$ through the saddles $(\pm1,0)$. Inside: closed paths ($E=0.05,\ 0.15,\ 0.22$); outside: $E=0.1,\ 0.2$ (turning at $|x|=x_o$) and $E=0.5$.

Answer. Paths: $\frac{1}{2}y^2+\frac{1}{2}kx^2+\frac{1}{4}\alpha x^4=E$. Hard spring ($\alpha>0$): the origin is a center and *all* paths are closed curves around it (all motions periodic). Soft spring ($\alpha<0$): the origin is a center surrounded by closed paths lying inside the separatrices $E=k^2/(4|\alpha|)$ through the saddle points $(\pm\sqrt{k/|\alpha|},0)$; all other paths are unbounded (see figures).

Problem 63-3 (Simmons)

Solution. Here $f(x)=-x+2x^3$, and the equivalent system is $x'=y$, $y'=x-2x^3$. Eliminating t,

$$\frac{dy}{dx}=\frac{x-2x^3}{y}\quad\Longrightarrow\quad y\,dy=(x-2x^3)\,dx\quad\Longrightarrow\quad \tfrac{1}{2}y^2=\tfrac{1}{2}x^2-\tfrac{1}{2}x^4+C.$$

Writing $2C=2E$ (E is the total energy, with $V(x)=\int_0^x(-s+2s^3)\,ds=-\frac{1}{2}x^2+\frac{1}{2}x^4$),

$$\boxed{y^2-x^2+x^4=2E}\qquad\text{i.e.}\qquad y=\pm\sqrt{2E+x^2-x^4}.$$

Critical points. $f(x)=x(2x^2-1)=0$ gives $x=0$ or $x=\pm\dfrac{1}{\sqrt{2}}=\pm\dfrac{\sqrt{2}}{2}$. So the critical points are $(0,0)$ and $\left(\pm\frac{\sqrt{2}}{2},0\right)$.

Nature. $f'(x)=-1+6x^2$.

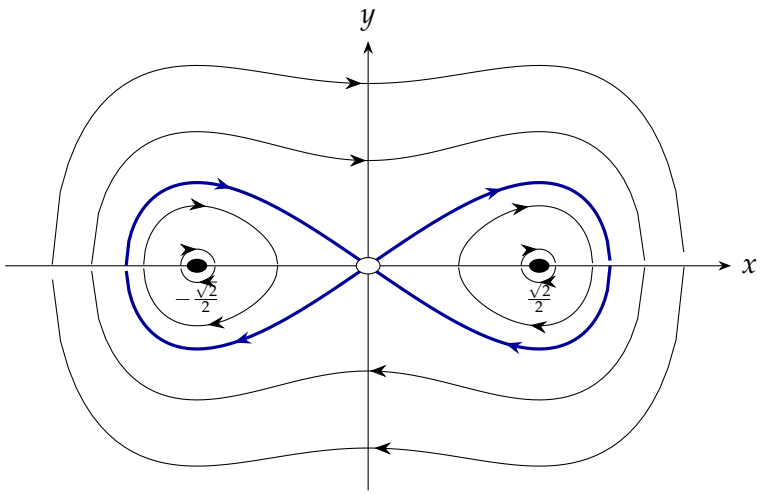
- At $(0,0)$: $f'(0)=-1$; the linearized system is $x'=y$, $y'=x$ with characteristic equation $m^2-1=0$, $m=\pm1$ real of opposite signs: a saddle point, and by Theorem 62-A (the critical point is simple: $\begin{vmatrix}0&1\\1&0\end{vmatrix}=-1\neq0$ and $|-2x^3|/r\le 2r^2\to0$) the nonlinear system has a saddle at the origin. It is unstable. (Alternatively use Problem 1: $xf(x)=-x^2(1-2x^2)<0$ for $0<|x|<1/\sqrt{2}$.)

- At $\left(\pm\frac{\sqrt{2}}{2},0\right)$: $f' = -1+3 = 2 > 0$. Put $\xi = x \mp \frac{\sqrt{2}}{2}$. Then $\xi' = y$, $y' = -f(\xi \pm \frac{\sqrt{2}}{2}) = -2\xi + (\text{terms of order } \xi^2)$, whose linearization is $m^2 + 2 = 0$, $m = \pm i\sqrt{2}$, a center. This is a borderline case, so Theorem 62-A does not decide it, but the system is conservative and $\xi\, f(\xi \pm \frac{\sqrt{2}}{2}) > 0$ for small $\xi \neq 0$ (for $f(x) = x(2x^2-1)$ changes sign from $-$ to $+$ at each of these points, V having a strict local minimum $-\frac{1}{8}$ there), so Problem 1 (applied in the variable ξ) shows that the closed level curves of E around these points are closed paths: each is a stable center (not asymptotically stable).

The paths ($V(x) = \frac{1}{2}x^2(x^2-1)$; $V(0) = 0$ is a local maximum, $V(\pm\frac{\sqrt{2}}{2}) = -\frac{1}{8}$ are the minima, $V(\pm 1) = 0$):

- $E = -\frac{1}{8}$: only the centers.
- $-\frac{1}{8} < E < 0$: two families of closed paths, one around each center, e.g. $y^2 = 2E + x^2 - x^4$ with x between the roots $x^2 = \frac{1\pm\sqrt{1+8E}}{2}$.
- $E = 0$: $y^2 = x^2(1-x^2)$, i.e. $y = \pm|x|\sqrt{1-x^2}$, $|x| \le 1$: the separatrix, a figure-eight (two loops meeting at the saddle $(0,0)$ and reaching $x = \pm 1$ on the axis). The loops enclose the two centers.
- $E > 0$: $y^2 = 2E + x^2 - x^4 > 0$ for $x^2 \le x_E^2 = \frac{1+\sqrt{1+8E}}{2}$: a single closed path enclosing all three critical points.

All paths are traversed clockwise (the upper halves to the right, the lower halves to the left).



Paths of $y^2 - x^2 + x^4 = 2E$: $E = -0.12$ (small loops), -0.06, 0 (thick, figure-eight through the saddle), 0.2, 0.6.

Answer. Paths: $y^2 - x^2 + x^4 = 2E$. Critical points: $(0,0)$, a (unstable) saddle point; $\left(-\frac{\sqrt{2}}{2},0\right)$ and $\left(\frac{\sqrt{2}}{2},0\right)$, centers (stable, not asymptotically stable). Agrees with the book.

Problem 63-4 (Simmons)

Solution. Let x_c be a point with $V'(x_c) = f(x_c) = 0$. Translate so that $x_c = 0$ and $V(0) = 0$ (replace x by $x - x_c$ and V by $V - V(x_c)$); then $V'(x) = f(x)$ and the paths are $y^2 = 2[E - V(x)]$.

We examine the three cases using the simplest model for each: $V = -\frac{1}{2}x^2$, $V = \frac{1}{2}x^2$ and $V = \frac{1}{3}x^3$, i.e. $f(x) = -x$, $f(x) = x$, $f(x) = x^2$. In general, near x_c the shape of V is the same as that of the model, and the phase portrait is obtained from the graph of V by the construction of Figure 91.

(i) *Maximum of V.* Here $V' = f$ changes sign from $+$ to $-$, so $xf(x) < 0$ near 0 ($f(x) \approx V''(0)x$ with $V''(0) < 0$). By Problem 1, $(0,0)$ is a *saddle point* (unstable). For the model $V = -\frac{1}{2}x^2$:

$$y^2 = 2E + x^2.$$

$E = 0$: $y = \pm x$ (the separatrices, crossing at the origin). $E > 0$: $y = \pm\sqrt{2E + x^2}$, hyperbola-like paths above and below. $E < 0$: $y = \pm\sqrt{x^2 - 2|E|}$ for $|x| \geq \sqrt{2|E|}$, hyperbola branches to the right and left.

(ii) *Minimum of V.* Here $V' = f$ changes sign from $-$ to $+$, so $xf(x) > 0$ near 0 ($V''(0) > 0$ in the model). By Problem 1, $(0,0)$ is a *center* (stable, not asymptotically stable): the level curves $\frac{1}{2}y^2 + V(x) = E$ are closed curves around the critical point (for the model, the circles $x^2 + y^2 = 2E$).

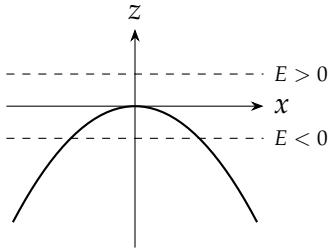
(iii) *Point of inflection of V.* Here $V' = f$ vanishes at 0 without changing sign ($f \geq 0$ on both sides in the model $f = x^2$, $V = \frac{1}{3}x^3$). Then

$$y^2 = 2E - \tfrac{2}{3}x^3.$$

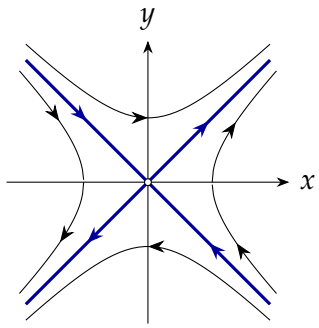
For $E = 0$: $y^2 = -\frac{2}{3}x^3$, i.e. $y = \pm\sqrt{\frac{2}{3}}(-x)^{3/2}$ for $x \leq 0$, a semicubical parabola with a *cusp* at the origin, where the two branches meet tangentially to the x-axis (the point $(x, y) = (-\gamma s^2, s^3)$, $\gamma = (3/2)^{1/3}$, traces it). The upper branch ($y > 0$) is traversed to the right, into the origin; the lower branch ($y < 0$) is traversed to the left, away from the origin. For $E > 0$: $y = \pm\sqrt{2E - \frac{2}{3}x^3}$ for $x \leq (3E)^{1/3}$, curves opening to the left that cross the x-axis at $(3E)^{1/3} > 0$; for $E < 0$: curves opening to the left that cross the x-axis at $-(3|E|)^{1/3} < 0$. No path is closed. The critical point is a *cusp*: the two branches of the $E = 0$ path end at it, and all other paths in its neighbourhood miss it.

The sketches. In each figure the left panel is the graph $z = V(x)$ with horizontal lines $z = E$, and the right panel is the phase portrait obtained from it by the construction of Figure 91 (arrows: direction of increasing t; the open dot is the critical point).

(i) Maximum of V: saddle point ($V = -\frac{1}{2}x^2$).

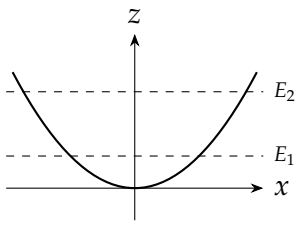


$z = V(x)$

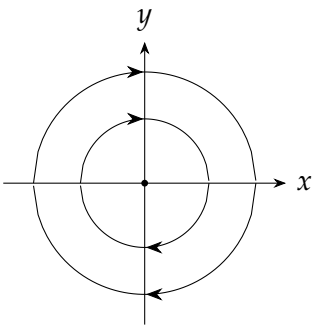


saddle: $y^2 = 2E + x^2$

(ii) Minimum of V: center ($V = \frac{1}{2}x^2$).

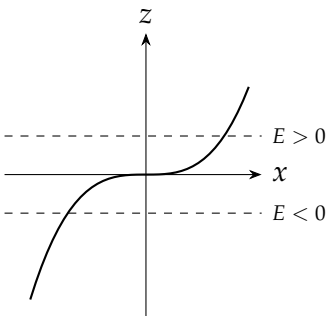


$z = V(x)$

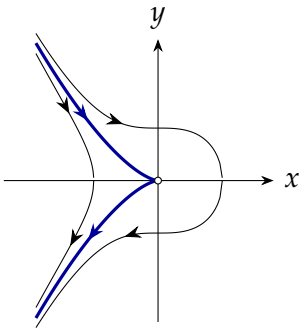


center: $x^2 + y^2 = 2E$

(iii) Point of inflection of V: cusp ($V = \frac{1}{3}x^3$).



$z = V(x)$



cusp: $y^2 = 2E - \frac{2}{3}x^3$

Answer. (i) Maximum of V: saddle point. (ii) Minimum of V: center. (iii) Point of inflection of V: cusp. (Agrees with the book.)

Section 64 Periodic Solutions. The Poincaré–Bendixson Theorem

Throughout, (1) denotes the system $x' = F(x,y)$, $y' = G(x,y)$ of the text. We use Theorems A–D of the section: A (a closed path surrounds a critical point), B (Bendixson: $F_x + G_y$ of one strict sign in a region excludes closed paths there), C (Poincaré–Bendixson), and D (Liénard).

Problem 64-1 (Simmons)

Solution.

(a) Let C be a closed path, $x = x(t)$, $y = y(t)$, with period T. Since C is a path it contains no critical point, so $\mathbf{V} = (x'(t), y'(t)) \neq \mathbf{0}$ at each point, and $\mathbf{V}$ is a continuous, nonvanishing tangent vector to C (it is the velocity vector of the parametrization, which is C^1 because F, G are continuous). Let $\mathbf{T}$ be the unit tangent vector to C pointing in the *counterclockwise* sense. Then at every point $\mathbf{V} = \lambda\mathbf{T}$ with $\lambda > 0$ if the path is traversed counterclockwise and $\lambda < 0$ if it is traversed clockwise; in either case the direction angle of $\mathbf{V}$ differs from that of $\mathbf{T}$ by a constant (0 or π) and so changes by exactly the same total amount as the direction angle of $\mathbf{T}$ when P goes once around C counterclockwise. It remains to show that the unit tangent of a simple closed C^1 curve turns through exactly $+2\pi$ in one counterclockwise circuit (Hopf's *theorem of turning tangents*, *Umlaufsatz*; see e.g. M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Sec. 5-7, or H. Hopf, Compositio Math. 2 (1935), 50–62, whose chord argument is used below). The book only quotes continuity of the index; here is a short self-contained proof of this fact, in the form it is needed. A closed path is a simple closed curve (if $(x(t_1), y(t_1)) = (x(t_2), y(t_2))$ with $0 \le t_1 < t_2 < t_1 + T$, then by uniqueness of solutions of the autonomous system (1) the solution would repeat after $t_2 - t_1 < T$, contradicting that T is the smallest period). Reparametrize C by arclength, $\gamma : [0, L] \to \mathbb{R}^2$, $\gamma(0) = \gamma(L)$, $\gamma'(0) = \gamma'(L)$, with γ injective on $[0, L)$ and γ' continuous and of length 1. Choose the starting point $\gamma(0)$ to be a lowest point of C (minimum of the y-coordinate). There $\gamma'(0) = \pm(1, 0)$ because y is minimal; the interior of C lies above, and traversing C counterclockwise means that $\mathbf{T}(0) = \gamma'(0) = (1, 0)$. On the closed triangle $\Omega = \{0 \le s \le t \le L\}$ define the unit vector

$$\Phi(s,t) = \frac{\gamma(t) - \gamma(s)}{|\gamma(t) - \gamma(s)|} \ (s < t,\ (s,t) \neq (0,L)), \qquad \Phi(s,s) = \gamma'(s), \qquad \Phi(0,L) = -\gamma'(0).$$

It is well defined (γ is injective on $[0, L)$) and continuous: on the diagonal because $(\gamma(t) - \gamma(s))/(t - s) \to \gamma'(s)$ uniformly (as γ' is continuous), and at the corner $(0, L)$ because $\gamma(t) - \gamma(s) = [\gamma(t) - \gamma(L)] + [\gamma(0) - \gamma(s)] \approx -\big((L - t) + s\big)\gamma'(0)$, using $\gamma(L) = \gamma(0)$ and $\gamma'(L) = \gamma'(0)$. Since Ω is simply connected, $\Phi = (\cos\varphi, \sin\varphi)$ with a continuous angle function φ on Ω, and the total change of φ along any path in Ω depends only on its endpoints. Compare the two routes from $(0,0)$ to (L,L). *Route 1* is the diagonal $s = t$, along which $\Phi = \mathbf{T}$, so the change of φ is the total turning $\Delta\theta_{\mathbf{T}}$ of the tangent. *Route 2* goes first along $s = 0$ from $(0,0)$ to $(0,L)$ and then along $t = L$ from $(0,L)$ to (L,L). On the first leg $\Phi(0,t)$ is the direction of the chord from the lowest point $\gamma(0)$ to $\gamma(t)$, whose y-component is ≥ 0; so Φ stays in the closed upper half circle, $\varphi \in [0, \pi]$ (taking $\varphi = 0$ at the start, where $\Phi = (1, 0)$), and it ends at $-\gamma'(0) = (-1, 0)$, i.e. at $\varphi = \pi$: the change is $+\pi$. On the second leg $\Phi(s, L)$ is the direction of $\gamma(0) - \gamma(s)$, with y-component ≤ 0, so Φ stays in the closed lower half circle, starting at $(-1, 0)$ and ending at $\Phi(L,L) = \gamma'(L) = (1, 0)$: the angle runs in $[\pi, 2\pi]$ from π to 2π, a change of $+\pi$. Hence the change along Route 2 is 2π, and so is the change along Route 1: $\Delta\theta_{\mathbf{T}} = 2\pi$. Since $\mathbf{V} = \lambda\mathbf{T}$ with λ of constant sign, $\Delta\theta = \Delta\theta_{\mathbf{T}} = 2\pi$ for the counterclockwise circuit, i.e. the index is $n = 1$.

(b) Now suppose C is a closed path and that there is *no critical point* of (1) inside C (and none on C, as C is a path). Then there is no critical point in the closed region bounded by C, so C can be shrunk continuously, staying inside the region, to a very small simple closed curve C_0 (say a

tiny circle) around some interior point P_0, without ever crossing a critical point. By the quoted principle, the index of C equals that of C_0.

Now compute the index of C_0. Since $\mathbf{V}(P_0) \neq \mathbf{0}$ and $\mathbf{V}$ is continuous, if C_0 is small enough then $\mathbf{V}$ stays within a small angle $\varepsilon < \pi/2$ of the fixed direction $\mathbf{V}(P_0)$ at all points of C_0. Hence the direction angle θ of $\mathbf{V}$ on C_0 can be chosen as a continuous function with values in an interval of length less than π around $\theta(P_0)$; after one circuit it returns to the same value, so $\Delta\theta = 0 = 2\pi \cdot 0$. Thus the index of C_0 is 0.

Therefore the index of C would be 0, contradicting part (a), which says it is 1. The assumption that no critical point lies inside C is therefore false: *every closed path of (1) surrounds at least one critical point*, which is Theorem A.

Answer. Shown: (a) a closed path has index 1 (its $\mathbf{V}$ is tangent to it); (b) if no critical point were enclosed the index would equal that of a small curve, namely 0, a contradiction. Hence a closed path must surround a critical point (Theorem A).

Problem 64-2 (Simmons)

Solution.

(a) Put $x = r\cos\theta$, $y = r\sin\theta$. From $x^2 + y^2 = r^2$ and $\theta = \tan^{-1}(y/x)$ we have (formulas (3) of the text)

$$x\frac{dx}{dt} + y\frac{dy}{dt} = r\frac{dr}{dt}, \qquad x\frac{dy}{dt} - y\frac{dx}{dt} = r^2\frac{d\theta}{dt}.$$

Multiply the first equation of the system by x, the second by y, and add:

$$r\frac{dr}{dt} = 4x^2 + 4xy - x^2r^2 - 4xy + 4y^2 - y^2r^2 = 4r^2 - r^2(x^2+y^2) = 4r^2 - r^4 = r^2(4 - r^2).$$

Multiply the second by x, the first by y, and subtract (second minus first):

$$r^2\frac{d\theta}{dt} = (-4x^2 + 4xy - xyr^2) - (4xy + 4y^2 - xyr^2) = -4x^2 - 4y^2 = -4r^2.$$

The only critical point is the origin (shown below), so for $r > 0$ we may divide by r and r^2:

$$\boxed{\frac{dr}{dt} = r(4 - r^2), \qquad \frac{d\theta}{dt} = -4}$$

Critical points: in polar form $\theta' = -4 \neq 0$, so $x' = y' = 0$ forces $r = 0$; hence $(0,0)$ is the only critical point.

(b) Let R be the ring $1 \le r \le 3$. It is bounded and contains no critical point (the origin is not in it). The vector field $\mathbf{V} = (F, G)$ has radial component $dr/dt = r(4 - r^2)$:

$$r = 1: \ \frac{dr}{dt} = 1 \cdot 3 = 3 > 0, \qquad r = 3: \ \frac{dr}{dt} = 3(4 - 9) = -15 < 0.$$

So on the inner circle $\mathbf{V}$ points into R and on the outer circle $\mathbf{V}$ points into R as well (inward). Hence every path that starts on the boundary of R enters R and stays there for all later t. By the Poincaré–Bendixson theorem (Theorem C), each such path is either closed or spirals toward a closed path, so R contains a closed path. In fact the closed path is $r = 2$, where $dr/dt = 0$.

(c) Solve the two equations of (a) separately. The θ-equation gives $\theta = -4(t + t_0)$. The r-equation is a Bernoulli (separable) equation; with $u = r^2$ we have $u' = 2rr' = 2u(4 - u)$, so

$$\frac{du}{u(4-u)} = 2\,\mathrm{d}t, \qquad \frac{1}{4}\left(\frac{1}{u} + \frac{1}{4-u}\right)du = 2\,\mathrm{d}t \implies \frac{1}{4}\log\left|\frac{u}{4-u}\right| = 2t + c_1,$$

that is, $\dfrac{u}{4-u} = Ae^{8t}$ ($A \neq 0$), so $u = \dfrac{4Ae^{8t}}{1+Ae^{8t}} = \dfrac{4}{1+ce^{-8t}}$ with $c = 1/A$. Therefore

$$r = \frac{2}{\sqrt{1+ce^{-8t}}}, \qquad \theta = -4(t+t_0).$$

(The constant solutions $u = 0$ and $u = 4$ are $r = 0$ and $r = 2$, i.e. $c = \infty$ and $c = 0$.) Returning to $x = r\cos\theta$, $y = r\sin\theta$, and using $\cos(-\alpha) = \cos\alpha$, $\sin(-\alpha) = -\sin\alpha$:

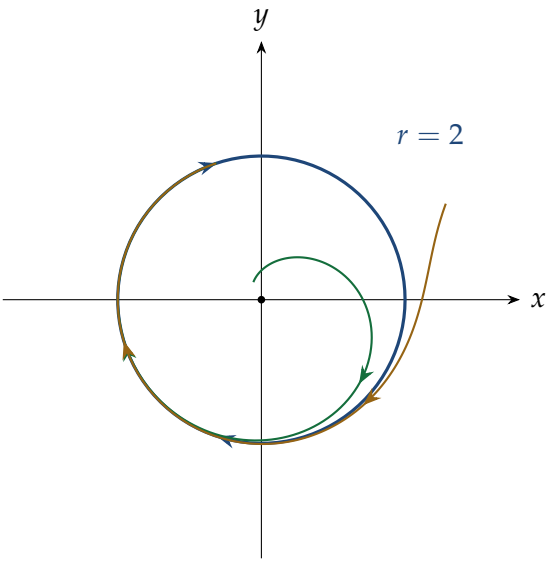
$$\boxed{x = \frac{2\cos 4(t+t_0)}{\sqrt{1+ce^{-8t}}}, \qquad y = \frac{-2\sin 4(t+t_0)}{\sqrt{1+ce^{-8t}}}}$$

The closed path of (b) is the case $c = 0$ (so $r \equiv 2$); taking $t_0 = 0$ gives the periodic solution

$$x = 2\cos 4t, \qquad y = -2\sin 4t,$$

which satisfies $x^2 + y^2 = 4$, has period $2\pi/4 = \pi/2$, and traces the circle $r = 2$ clockwise (as $\theta' = -4 < 0$).

(d) For $c > 0$ we have $r < 2$ and $r \to 2$ as $t \to \infty$ (also $r \to 0$ as $t \to -\infty$), so these paths spiral outward from the origin onto the circle. For $c < 0$ we have $r > 2$ and again $r \to 2$ as $t \to \infty$, so these paths spiral inward onto the circle from outside. All paths turn clockwise. The origin is an unstable spiral point: the linearization $x' = 4x + 4y$, $y' = -4x + 4y$ has eigenvalues $4 \pm 4i$.



(Thick circle: the closed path $r = 2$; the inner curve is the path with $c = 1$, the outer one has $c = -0.2$.)

Answer. (a) $\dfrac{dr}{dt} = r(4-r^2)$, $\dfrac{d\theta}{dt} = -4$. (b) $dr/dt > 0$ on $r = 1$, < 0 on $r = 3$, no critical point in the ring, so it holds a closed path (namely $r = 2$). (c) $x = \dfrac{2\cos 4(t+t_0)}{\sqrt{1+ce^{-8t}}}$, $y = \dfrac{-2\sin 4(t+t_0)}{\sqrt{1+ce^{-8t}}}$; periodic solution $x = 2\cos 4t$, $y = -2\sin 4t$. (d) clockwise circle $r = 2$ with paths spiraling onto it from inside and outside.

Problem 64-3 (Simmons)

Solution. Write $\rho = x^2 + y^2 = r^2$. As in Problem 2, multiply the first equation by x, the second by y, and add; then multiply the second by x, the first by y and subtract:

$$r\frac{dr}{dt} = 3x^2 - xy - x^2e^{r^2} + xy + 3y^2 - y^2e^{r^2} = r^2\left(3 - e^{r^2}\right),$$
$$r^2\frac{d\theta}{dt} = (x^2 + 3xy - xye^{r^2}) - (3xy - y^2 - xye^{r^2}) = x^2 + y^2 = r^2.$$

So for $r > 0$:

$$\frac{dr}{dt} = r\left(3 - e^{r^2}\right), \qquad \frac{d\theta}{dt} = 1.$$

Critical points. Since $\theta' = 1 \neq 0$ for $r > 0$, the only critical point is the origin. (Directly: $x' = y' = 0$ gives $(3 - e^{r^2})x = y$ and $x = (e^{r^2} - 3)y$; hence $y = -(e^{r^2} - 3)^2 y$, so $y\,[1 + (e^{r^2} - 3)^2] = 0$, $y = 0$, then $x = 0$.)

Poincaré–Bendixson. Take the ring R: $\frac{1}{2} \le r \le 2$. It is bounded and contains no critical point. On the inner circle $r = \frac{1}{2}$, $dr/dt = \frac{1}{2}(3 - e^{1/4}) > 0$ (since $e^{1/4} < 3$); on the outer circle $r = 2$, $dr/dt = 2(3 - e^4) < 0$ (since $e^4 > 3$). Thus $\mathbf{V}$ points into R everywhere on the boundary, so every path starting on the boundary enters R and remains in it for all later t. By the Poincaré–Bendixson theorem, R contains a closed path, hence the system has a periodic solution.

In fact it can be found explicitly: $dr/dt = 0$ at $e^{r^2} = 3$, i.e. $r = \sqrt{\log 3} \approx 1.048$, which lies in R. The circle $r = \sqrt{\log 3}$, $\theta = t + t_0$ is a closed path and

$$x = \sqrt{\log 3}\,\cos t, \qquad y = \sqrt{\log 3}\,\sin t$$

is a periodic solution of period 2π. (Check: $x' = -\sqrt{\log 3}\sin t = 3x - y - 3x$ because $e^{x^2+y^2} = e^{\log 3} = 3$, and $3x - y - 3x = -y$; similarly $y' = x$.)

Answer. Shown: in the ring $\frac{1}{2} \le r \le 2$ there are no critical points and $dr/dt > 0$ on $r = \frac{1}{2}$, $dr/dt < 0$ on $r = 2$, so by Poincaré–Bendixson there is a closed path; explicitly $x = \sqrt{\log 3}\cos t$, $y = \sqrt{\log 3}\sin t$ is a periodic solution.

Problem 64-4 (Simmons)

Solution. Each equation is written as an equivalent system with $y = dx/dt$.

(a) This is Liénard's equation $x'' + f(x)x' + g(x) = 0$ with $f(x) = 5x^4 - 9x^2$, $g(x) = x^5$. Check the hypotheses of Theorem D. (i) f and g are polynomials: continuous with continuous derivatives. (ii) g is odd, $g(x) = x^5 > 0$ for $x > 0$; f is even. (iii)

$$F(x) = \int_0^x (5s^4 - 9s^2)\,ds = x^5 - 3x^3 = x^3(x^2 - 3).$$

For $x > 0$ the factor x^3 is positive, so the sign of F is that of $x^2 - 3$: F has exactly one positive zero, $a = \sqrt{3}$, is negative for $0 < x < \sqrt{3}$, and positive for $x > \sqrt{3}$. Moreover $F(x) \to \infty$ as $x \to \infty$, and $F'(x) = f(x) = x^2(5x^2 - 9) \ge 0$ for $x \ge 3/\sqrt{5} \approx 1.342$, which is less than $\sqrt{3} \approx 1.732$; so F is positive and nondecreasing for $x > a$. All conditions hold, so the equation *has a (unique, stable) periodic solution*.

(b) The equivalent system is $x' = y$, $y' = -x^5 + (x^2 + 1)y$, so $F(x, y) = y$ and $G(x, y) = -x^5 + (x^2 + 1)y$. Then

$$\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0 + (x^2 + 1) = x^2 + 1 > 0 \quad \text{everywhere.}$$

By Theorem B (Bendixson) there are no closed paths in the whole phase plane: *no periodic solution*. (Liénard's theorem does not apply: here $f(x) = -(x^2+1) < 0$ for all x, so F has no positive zero.)

(c) The system is $x' = y$, $y' = y^2 + 1 + x^2$. A critical point needs $y = 0$ and $y^2 + 1 + x^2 = 1 + x^2 = 0$, which is impossible for real x. The system has *no critical points at all*, so it has no closed path by Theorem A (a closed path would have to surround a critical point): *no periodic solution*. (Theorem B is not conclusive here, since $F_x + G_y = 2y$ changes sign.)

(d) The system is $x' = y$, $y' = 3x^3 - y - y^5$. Here

$$\frac{\partial}{\partial x}(y) + \frac{\partial}{\partial y}\left(3x^3 - y - y^5\right) = -1 - 5y^4 < 0 \quad \text{everywhere,}$$

so by Theorem B there is *no periodic solution*.

(e) Here $f(x) = x^6 - x^2$, $g(x) = x$. (i) Both are polynomials. (ii) g is odd and $g > 0$ for $x > 0$; f is even. (iii)

$$F(x) = \int_0^x (s^6 - s^2)\,ds = \frac{x^7}{7} - \frac{x^3}{3} = x^3\left(\frac{x^4}{7} - \frac{1}{3}\right).$$

For $x > 0$ this vanishes only when $x^4 = 7/3$, i.e. at $a = (7/3)^{1/4} \approx 1.236$; it is negative for $0 < x < a$, positive for $x > a$, and $F(x) \to \infty$. Also $F'(x) = x^6 - x^2 = x^2(x^4 - 1) \geq 0$ for $x \geq 1$, and $a > 1$, so F is nondecreasing for $x > a$. By Liénard's theorem the equation *has a periodic solution* (unique, and approached by every other nontrivial path).

Answer. (a) periodic solution (Liénard); (b) none (Theorem B); (c) none (Theorem A); (d) none (Theorem B); (e) periodic solution (Liénard).

Problem 64-5 (Simmons)

Solution. The equation is $a\,x'' + b(x^2-1)x' + cx = 0$ with $a, b, c > 0$. Introduce a new independent variable $\tau = kt$ with a constant $k > 0$ to be chosen, and regard x as a function of τ. By the chain rule

$$\frac{dx}{dt} = k\frac{dx}{d\tau}, \qquad \frac{d^2x}{dt^2} = k^2\frac{d^2x}{d\tau^2}.$$

Substituting,

$$ak^2\frac{d^2x}{d\tau^2} + bk(x^2-1)\frac{dx}{d\tau} + cx = 0.$$

Divide by ak^2:

$$\frac{d^2x}{d\tau^2} + \frac{b}{ak}(x^2-1)\frac{dx}{d\tau} + \frac{c}{ak^2}\,x = 0.$$

Choose k so that the coefficient of x equals 1: $c/(ak^2) = 1$, i.e. $k = \sqrt{c/a} > 0$. Then $\dfrac{b}{ak} = \dfrac{b}{a\sqrt{c/a}} = \dfrac{b}{\sqrt{ac}}$, and the equation becomes

$$\frac{d^2x}{d\tau^2} + \mu(x^2-1)\frac{dx}{d\tau} + x = 0, \qquad \mu = \frac{b}{\sqrt{ac}} > 0, \quad \tau = \sqrt{\frac{c}{a}}\,t,$$

which is the van der Pol equation (11) of the text with a positive constant μ.

Answer. Shown: with $\tau = \sqrt{c/a}\,t$ the equation becomes $x_{\tau\tau} + \mu(x^2 - 1)x_\tau + x = 0$, where $\mu = b/\sqrt{ac} > 0$.

Note on Section 65. Section 65 (More about the van der Pol Equation) is purely expository (historical remarks on radar, Cartwright and Littlewood) and has no problem list of its own.

Chapter 12

The Calculus of Variations

Section 67 Euler's Differential Equation for an Extremal

Throughout, $y' = dy/dx$, and for $I = \int_{x_1}^{x_2} f(x, y, y')\,dx$ the extremals are the solutions of Euler's equation (9),

$$\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) - \frac{\partial f}{\partial y} = 0, \qquad \text{i.e.} \qquad f_{y'y'}\,y'' + f_{y'y}\,y' + (f_{y'x} - f_y) = 0 \quad \text{(equation (10))}.$$

We use the three special cases of the text: Case A (x and y missing from f: the extremals are straight lines, if $f_{y'y'} \neq 0$); Case B (y missing: $f_{y'} = c_1$); Case C (x missing: $f_{y'}\,y' - f = c_1$). In Case C recall the identity $\frac{d}{dx}(f_{y'}y' - f) = y'\left[\frac{d}{dx}f_{y'} - f_y\right] - f_x$: the first integral is obtained by multiplying Euler's equation by y', so it may admit spurious solutions with $y' \equiv 0$ (constants), which must be tested in Euler's equation itself. The symbol log is the natural logarithm. When Euler's equation is a second order equation which can be solved for y'' with a right side that is smooth in the region considered (i.e. $f_{y'y'} \neq 0$), the existence and uniqueness theorem shows that a two-parameter family of solutions that matches arbitrary initial data (x_1, y_1, y_1') contains *all* the extremals; we use this to confirm that no extremals are lost.

Problem 67-1 (Simmons)

Solution.

(a) Here $f = \sqrt{1+y'^2}/y$ (defined for $y \neq 0$; we take $y > 0$, the case $y < 0$ being identical) does not contain x, so Case C applies: $f_{y'}y' - f = c_1$. Since $f_{y'} = \dfrac{y'}{y\sqrt{1+y'^2}}$,

$$f_{y'}y' - f = \frac{y'^2}{y\sqrt{1+y'^2}} - \frac{1+y'^2}{y\sqrt{1+y'^2}} = -\frac{1}{y\sqrt{1+y'^2}} = c_1.$$

The left side never vanishes, so $c_1 \neq 0$, and with $k = 1/|c_1|$ we get

$$y\sqrt{1+y'^2} = k, \qquad \text{so} \qquad y'^2 = \frac{k^2 - y^2}{y^2}, \qquad y' = \pm\frac{\sqrt{k^2-y^2}}{y}.$$

Separating variables,

$$\frac{y\,dy}{\sqrt{k^2-y^2}} = \pm\,dx \quad\Longrightarrow\quad -\sqrt{k^2-y^2} = \pm(x - c_2) \quad\Longrightarrow\quad (x-c_2)^2 + y^2 = k^2.$$

So the extremals are *semicircles* $y = \sqrt{k^2 - (x-c_2)^2}$ with center $(c_2, 0)$ on the x-axis and radius k. (In the book's notation the radius is again called c_1.)

Checks that nothing is lost or spurious. Differentiating $(x-c_2)^2 + y^2 = k^2$ gives $yy' = -(x-c_2)$, hence $y^2(1+y'^2) = y^2 + (x-c_2)^2 = k^2$, which is the first integral; the semicircles do satisfy Euler's equation (see also the SymPy check). The constant $y \equiv k$, which satisfies the first integral ($y' = 0$), does *not* satisfy Euler's equation: with $y' = 0$, $\frac{d}{dx}f_{y'} = 0$ but $f_y = -\sqrt{1+y'^2}/y^2 = -1/k^2 \neq 0$. Finally, $f_{y'y'} = \dfrac{1}{y(1+y'^2)^{3/2}} \neq 0$, so Euler's equation is a regular second order equation for $y > 0$; given any (x_1, y_1, y_1') with $y_1 > 0$, the semicircle with center $c_2 = x_1 + y_1y_1'$ and radius $k = y_1\sqrt{1+y_1'^2}$ passes through (x_1, y_1) with slope y_1' (because $yy' = -(x-c_2)$ gives $y_1y_1' = c_2 - x_1$). By uniqueness the semicircles are all the extremals.

(b) Here $f = y^2 - y'^2$, so $f_{y'} = -2y'$ and $f_y = 2y$. Euler's equation is

$$\frac{d}{dx}(-2y') - 2y = 0 \quad\Longrightarrow\quad y'' + y = 0,$$

with general solution $y = A\cos x + B\sin x$. Put $c_1 = \sqrt{A^2+B^2}$ and choose c_2 with $\cos c_2 = B/c_1$, $\sin c_2 = -A/c_1$ (if $A = B = 0$ take $c_1 = 0$ and c_2 arbitrary). Then $A = -c_1\sin c_2$, $B = c_1\cos c_2$ and

$$y = c_1\sin x\cos c_2 - c_1\cos x\sin c_2 = c_1\sin(x - c_2).$$

Answer. (a) $(x-c_2)^2 + y^2 = c_1^2$ (semicircles centered on the x-axis); (b) $y = c_1\sin(x-c_2)$ (equivalently $y = A\cos x + B\sin x$).

Problem 67-2 (Simmons)

Solution. Here $f = xy' - y'^2$, so $f_{y'} = x - 2y'$ and $f_y = 0$. Euler's equation is

$$\frac{d}{dx}(x - 2y') - 0 = 0 \implies 1 - 2y'' = 0 \implies y'' = \tfrac{1}{2}.$$

Integrating twice, $y' = \frac{1}{2}x + c_1$ and $y = \frac{1}{4}x^2 + c_1x + c_2$. Boundary conditions:

$$y(0) = c_2 = 0,$$
$$y(4) = 4 + 4c_1 + c_2 = 3 \implies c_1 = -\tfrac{1}{4}.$$

Hence

$$y = \tfrac{1}{4}x^2 - \tfrac{1}{4}x = \tfrac{1}{4}(x^2 - x).$$

Check: $y(0) = 0$ and $y(4) = \frac{1}{4}(16 - 4) = 3$.

Remark. This stationary function gives a *maximum*. Let y_0 be the stationary function and η any function with $\eta(0) = \eta(4) = 0$. Then

$$I(y_0 + \alpha\eta) - I(y_0) = \alpha\int_0^4 (x - 2y_0')\eta'\,dx - \alpha^2\int_0^4 \eta'^2\,dx.$$

Integrating by parts, $\int_0^4 (x-2y_0')\eta'\,dx = \big[(x-2y_0')\eta\big]_0^4 - \int_0^4 (1-2y_0'')\eta\,dx = 0$, so the difference equals $-\alpha^2\int_0^4\eta'^2\,dx \le 0$.

Answer. $y = \dfrac{1}{4}(x^2 - x)$.

Problem 67-3 (Simmons)

Solution. Let $f = a(x)y'^2 + 2b(x)yy' + c(x)y^2$, where a, b, c are given functions of x (with a and b differentiable). Then

$$f_{y'} = 2a(x)y' + 2b(x)y, \qquad f_y = 2b(x)y' + 2c(x)y.$$

Differentiating $f_{y'}$ totally with respect to x (product rule; y and y' depend on x):

$$\frac{d}{dx}f_{y'} = 2ay'' + 2a'y' + 2by' + 2b'y.$$

Euler's equation $\frac{d}{dx}f_{y'} - f_y = 0$ therefore reads

$$2ay'' + 2a'y' + 2by' + 2b'y - 2by' - 2cy = 0.$$

The terms $+2by'$ and $-2by'$ cancel, and after division by 2,

$$a(x)\,y'' + a'(x)\,y' + \big[b'(x) - c(x)\big]y = 0, \qquad \text{i.e.} \qquad \big(ay'\big)' + \big(b' - c\big)y = 0.$$

This has the form $P(x)y'' + Q(x)y' + R(x)y = 0$ with $P = a$, $Q = a'$, $R = b' - c$: a homogeneous *linear* equation of order two (wherever $a(x) \neq 0$, which is exactly the condition $f_{y'y'} = 2a \neq 0$). It is already in self-adjoint (Sturm–Liouville) form.

Answer. Shown: Euler's equation is $a\,y'' + a'\,y' + (b' - c)\,y = 0$, a linear second order equation.

Problem 67-4 (Simmons)

Solution. Recall $x = r\cos\theta$, $y = r\sin\theta$, so that $r\cos(\theta - \theta_0) = x\cos\theta_0 + y\sin\theta_0$; thus $r\cos(\theta - c_2) = c_1$ is the equation of the straight line $x\cos c_2 + y\sin c_2 = c_1$, at distance $|c_1|$ from the origin, with normal direction c_2.

(a) With $r = r(\theta)$ and $r' = dr/d\theta$, we minimize $\int \sqrt{r'^2 + r^2}\,d\theta$. The integrand $f(r, r') = \sqrt{r'^2 + r^2}$ does not contain θ, so Case C applies (r plays the role of y, θ that of x): $f_{r'}r' - f = c$, where

$$f_{r'}r' - f = \frac{r'^2}{\sqrt{r'^2 + r^2}} - \sqrt{r'^2 + r^2} = -\frac{r^2}{\sqrt{r'^2 + r^2}} = c.$$

For $r > 0$ the left side is negative, so $c = -k$ with $k > 0$:

$$\frac{r^2}{\sqrt{r'^2 + r^2}} = k.$$

Substitute $u = 1/r$, so $r' = -u'/u^2$ and $r'^2 + r^2 = (u'^2 + u^2)/u^4$. Then $\dfrac{r^2}{\sqrt{r'^2 + r^2}} = \dfrac{u^{-2}\,u^2}{\sqrt{u'^2 + u^2}} = \dfrac{1}{\sqrt{u'^2 + u^2}}$, and the equation is

$$u'^2 + u^2 = \frac{1}{k^2}.$$

The functions $u = \dfrac{1}{k}\cos(\theta - \theta_0)$ satisfy this ($u'^2 + u^2 = k^{-2}(\sin^2 + \cos^2)$), i.e.

$$r = \frac{k}{\cos(\theta - \theta_0)} \quad \Longleftrightarrow \quad r\cos(\theta - \theta_0) = k.$$

To see that these are all the extremals: Euler's equation is regular for $r > 0$ because $f_{r'r'} = r^2/(r'^2 + r^2)^{3/2} \neq 0$. For initial data $(\theta_1, r_1 > 0, r_1')$ choose $\psi_1 = \tan^{-1}(r_1'/r_1) \in (-\pi/2, \pi/2)$, $k = r_1\cos\psi_1 > 0$ and $\theta_0 = \theta_1 - \psi_1$; then $r = k\sec(\theta - \theta_0)$ has $r(\theta_1) = r_1$ and $r'(\theta_1) = r_1\tan\psi_1 = r_1'$, so by uniqueness no other extremal exists. Hence the extremals are the lines

$$r\cos(\theta - c_2) = c_1 \qquad (c_1 = k > 0),$$

that is, $x\cos c_2 + y\sin c_2 = c_1$. (Lines through the origin cannot be written as $r = r(\theta)$ with $r > 0$ except as the rays $\theta = \text{const}$, which appear in part (b).)

(b) Now $\theta = \theta(r)$, $\theta' = d\theta/dr$, and $ds = \sqrt{1 + r^2\theta'^2}\,dr$, so $f(r, \theta') = \sqrt{1 + r^2\theta'^2}$ does not contain θ: Case B gives $f_{\theta'} = c$, i.e.

$$\frac{r^2\theta'}{\sqrt{1 + r^2\theta'^2}} = c.$$

If $c = 0$ then $\theta' = 0$, so $\theta =$ const: the rays through the origin (these are $r\cos(\theta - c_2) = 0$ with $c_2 = \theta \pm \pi/2$). If $c \neq 0$, squaring gives $r^4\theta'^2 = c^2(1 + r^2\theta'^2)$, i.e. $\theta'^2\, r^2(r^2 - c^2) = c^2$, which requires $r > |c|$, and θ' has the sign of c:

$$\theta' = \frac{c}{r\sqrt{r^2 - c^2}}.$$

Take first $c > 0$. Since $\frac{d}{dr}\cos^{-1}\frac{c}{r} = \dfrac{c/r^2}{\sqrt{1 - c^2/r^2}} = \dfrac{c}{r\sqrt{r^2 - c^2}}$ for $r > c > 0$, integration gives

$$\theta - c_2 = \cos^{-1}\frac{c}{r} \quad \Longrightarrow \quad r\cos(\theta - c_2) = c.$$

For $c < 0$ the same computation with $|c|$ gives $\theta - c_2 = -\cos^{-1}(|c|/r)$, i.e. again $r\cos(\theta - c_2) = |c|$. Renaming $|c|$ as c_1, the extremals are

$$r\cos(\theta - c_2) = c_1,$$

(with $c_1 = 0$ giving the rays through the origin; replacing c_2 by $c_2 + \pi$ changes the sign of c_1). This is the same family as in (a).

Answer. (a), (b): $r\cos(\theta - c_2) = c_1$, i.e. $x\cos c_2 + y\sin c_2 = c_1$, a straight line.

Problem 67-5 (Simmons)

Solution. *The length integral.* Along $\theta = \theta(\phi)$, with $\theta' = d\theta/d\phi$,

$$\begin{aligned} dx &= a\cos\phi\cos\theta\,d\phi - a\sin\phi\sin\theta\,d\theta,\\ dy &= a\cos\phi\sin\theta\,d\phi + a\sin\phi\cos\theta\,d\theta,\\ dz &= -a\sin\phi\,d\phi. \end{aligned}$$

In $dx^2 + dy^2$ the cross terms cancel, giving $a^2\cos^2\phi\,d\phi^2 + a^2\sin^2\phi\,d\theta^2$; adding $dz^2 = a^2\sin^2\phi\,d\phi^2$,

$$dx^2 + dy^2 + dz^2 = a^2\big(d\phi^2 + \sin^2\phi\,d\theta^2\big), \qquad L = a\int\sqrt{1 + \theta'^2\sin^2\phi}\;d\phi.$$

Euler's equation. Here ϕ is the independent variable and θ the unknown function; the integrand $f = \sqrt{1 + \theta'^2\sin^2\phi}$ does not contain θ, so Case B applies: $f_{\theta'} = c$, i.e.

$$\frac{\theta'\sin^2\phi}{\sqrt{1 + \theta'^2\sin^2\phi}} = c.$$

Squaring: $\theta'^2\sin^4\phi = c^2(1 + \theta'^2\sin^2\phi)$, i.e. $\theta'^2\sin^2\phi\,(\sin^2\phi - c^2) = c^2$. Hence $\sin^2\phi > c^2$ unless $c = 0$, and since θ' has the sign of c and $\sin\phi > 0$ on $0 < \phi < \pi$,

$$\theta' = \frac{c}{\sin\phi\sqrt{\sin^2\phi - c^2}}.$$

Case $c = 0$. Then $\theta' = 0$, so $\theta = \theta_0$ is constant: a meridian, which lies in the plane $x\sin\theta_0 - y\cos\theta_0 = 0$ through the center.

Case $0 < c^2 < 1$ (necessarily $c^2 < \sin^2\phi \le 1$). Write $\sin\phi\sqrt{\sin^2\phi - c^2} = \sin^2\phi\sqrt{1 - c^2/\sin^2\phi}$, and put $u = \cot\phi$, so that $du = -\,\mathrm{d}\phi/\sin^2\phi$ and $1/\sin^2\phi = 1 + u^2$. Then $1 - c^2/\sin^2\phi = 1 - c^2(1+u^2) = (1-c^2) - c^2u^2$, and

$$\theta = \int \frac{c\,\mathrm{d}\phi}{\sin^2\phi\sqrt{(1-c^2) - c^2\cot^2\phi}} = -\int \frac{c\,\mathrm{d}u}{\sqrt{(1-c^2) - c^2u^2}} = -\sin^{-1}\frac{c\,u}{\sqrt{1-c^2}} + \theta_0,$$

because $\frac{d}{du}\sin^{-1}\frac{cu}{s} = \frac{c}{\sqrt{s^2 - c^2u^2}}$ with $s = \sqrt{1-c^2}$. Hence $\sin(\theta_0 - \theta) = \dfrac{c\cot\phi}{\sqrt{1-c^2}}$, that is,

$$\cot\phi = K\sin(\theta_0 - \theta) = K\sin\theta_0\cos\theta - K\cos\theta_0\sin\theta, \qquad K = \frac{\sqrt{1-c^2}}{c}.$$

Back to rectangular coordinates. Multiply by $a\sin\phi$ and use $z = a\cos\phi$, $x = a\sin\phi\cos\theta$, $y = a\sin\phi\sin\theta$:

$$z = (K\sin\theta_0)\,x - (K\cos\theta_0)\,y, \qquad \text{i.e.} \qquad Ax + By - z = 0.$$

This is the equation of a plane through the origin, the center of the sphere. (The limit $c^2 \to 1$ gives $K = 0$ and the plane $z = 0$, the equator, which cannot be written as $\theta = F(\phi)$ because it has $\phi \equiv \pi/2$.) Thus every extremal of the form $\theta = F(\phi)$ lies on the sphere and on a plane through its center, so it is an arc of a great circle.

Answer. Shown: the extremals satisfy $\cot\phi = A\cos\theta + B\sin\theta$ (or $\theta = \text{const}$), i.e. $Ax + By - z = 0$, a plane through the center, so the geodesics are arcs of great circles.

Problem 67-6 (Simmons)

Solution. Write $k = \sqrt{1+a^2} > 1$, so $x = \dfrac{r}{k}\cos k\theta$, $y = \dfrac{r}{k}\sin k\theta$, $z = \dfrac{ar}{k}$.

The parametrization lies on the cone. $x^2 + y^2 = r^2/k^2$ and $z^2 = a^2r^2/k^2 = a^2(x^2+y^2)$, with $z \ge 0$ for $r \ge 0$. Also $x^2 + y^2 + z^2 = r^2(1+a^2)/k^2 = r^2$, so r is the distance from the apex (the slant distance along a generator), and a generator is a curve $\theta = \text{const}$.

Arc length. Differentiating,

$$\begin{aligned} dx &= \tfrac{1}{k}\cos k\theta\,\mathrm{d}r - r\sin k\theta\,\mathrm{d}\theta, \\ dy &= \tfrac{1}{k}\sin k\theta\,\mathrm{d}r + r\cos k\theta\,\mathrm{d}\theta, \\ dz &= \tfrac{a}{k}\,\mathrm{d}r. \end{aligned}$$

In $dx^2 + dy^2$ the cross terms $\mp\frac{r}{k}\sin k\theta\cos k\theta\,\mathrm{d}r\,\mathrm{d}\theta$ cancel, so $dx^2 + dy^2 = \frac{1}{k^2}dr^2 + r^2d\theta^2$, and

$$ds^2 = dx^2 + dy^2 + dz^2 = \frac{1+a^2}{k^2}dr^2 + r^2d\theta^2 = dr^2 + r^2d\theta^2.$$

Flattening. The cone minus the generator $\theta = 0$ is described by $0 < \theta < 2\pi/k$ (at $\theta = 2\pi/k$ the point returns to the generator $\theta = 0$), $r > 0$. In the plane with ordinary polar coordinates (r,θ) this is the sector $0 < \theta < 2\pi/k$, which is a genuine sector since $2\pi/k < 2\pi$. The formula $ds^2 = dr^2 + r^2d\theta^2$ says exactly that the length of every curve $r = r(\theta)$ (or $\theta = \theta(r)$) on the cone equals the length of the curve with the same polar equation in the plane sector, the arc element in the plane being $\sqrt{dr^2 + r^2d\theta^2}$. So cutting the cone along a generator and unrolling it onto a plane sector maps (r,θ) to the ordinary polar coordinates of the plane, and this map preserves

the lengths of all curves. (As a check, the circle $r = \text{const}$ on the cone has length $2\pi r/k$, which is the arc length $r \cdot \dfrac{2\pi}{k}$ of the sector.)

Geodesics. A geodesic is a curve on the cone which gives a stationary value to its length. For a curve $r = r(\theta)$ the length is $\int \sqrt{r'^2 + r^2}\,\mathrm{d}\theta$, which is exactly the integral minimized in Problem 4(a); its extremals are

$$r\cos(\theta - c_2) = c_1,$$

which, in the ordinary polar coordinates of the flattened cone, is the straight line $x_{\mathrm{p}}\cos c_2 + y_{\mathrm{p}}\sin c_2 = c_1$ ($x_{\mathrm{p}} = r\cos\theta$, $y_{\mathrm{p}} = r\sin\theta$ being the rectangular coordinates in the plane of the flattened cone). The remaining geodesics are the generators $\theta = \text{const}$ (Problem 4(b) with $c_1 = 0$), which are the rays from the origin, also straight lines. Hence every geodesic of the cone becomes a straight line when the cone is flattened.

Answer. Shown: $ds^2 = dr^2 + r^2 d\theta^2$, so (r, θ) are plane polar coordinates of the flattened cone, and the geodesics $r\cos(\theta - c_2) = c_1$ (and generators $\theta = \text{const}$) are straight lines.

Problem 67-7 (Simmons)

Solution. We take $g(z) > 0$. Along a curve $\theta = \theta(z)$ on the surface,

$$\begin{aligned} dx &= g'\cos\theta\,\mathrm{d}z - g\sin\theta\,\mathrm{d}\theta,\\ dy &= g'\sin\theta\,\mathrm{d}z + g\cos\theta\,\mathrm{d}\theta,\\ dz &= \mathrm{d}z, \end{aligned}$$

and in $dx^2 + dy^2$ the cross terms cancel: $dx^2 + dy^2 = g'^2 dz^2 + g^2 d\theta^2$. Hence

$$ds = \sqrt{(1 + g'^2) + g^2\theta'^2}\,dz, \qquad \theta' = \frac{d\theta}{dz},$$

and a geodesic is an extremal of $\displaystyle\int f\,\mathrm{d}z$ with $f(z,\theta') = \sqrt{1 + g'(z)^2 + g(z)^2\theta'^2}$. The function f does not contain θ, so Case B applies (independent variable z, unknown θ): $f_{\theta'} = c_1$,

$$\frac{g^2\theta'}{\sqrt{1 + g'^2 + g^2\theta'^2}} = c_1.$$

Squaring, $g^4\theta'^2 = c_1^2(1 + g'^2 + g^2\theta'^2)$, so

$$g^2\theta'^2(g^2 - c_1^2) = c_1^2(1 + g'^2), \qquad \theta' = \frac{c_1\sqrt{1 + g'^2}}{g\sqrt{g^2 - c_1^2}}.$$

(The sign of the square root is the sign of c_1, because the equation $f_{\theta'} = c_1$ shows that θ' and c_1 have the same sign; this requires $g^2 > c_1^2$ along the curve, and for $c_1 = 0$ we get $\theta' = 0$, the meridians.) Integrating with respect to z,

$$\theta = c_1\int \frac{\sqrt{1 + [g'(z)]^2}}{g(z)\sqrt{g(z)^2 - c_1^2}}\,dz + c_2.$$

Check of the algebra: with θ' as above, $1 + g'^2 + g^2\theta'^2 = (1 + g'^2)\left(1 + \dfrac{c_1^2}{g^2 - c_1^2}\right) = \dfrac{(1 + g'^2)g^2}{g^2 - c_1^2}$, so

$$f_{\theta'} = \frac{g^2 \cdot c_1\sqrt{1 + g'^2}/(g\sqrt{g^2 - c_1^2})}{g\sqrt{1 + g'^2}/\sqrt{g^2 - c_1^2}} = c_1.$$

Answer. Shown: $\theta = c_1 \displaystyle\int \frac{\sqrt{1+g'(z)^2}}{g(z)\sqrt{g(z)^2 - c_1^2}}\,dz + c_2$.

Problem 67-8 (Simmons)

Solution. For a right circular cylinder of radius R, $g(z) = R$ is constant, so $g' = 0$. The result of Problem 7 gives, with $|c_1| < R$ (needed for $g^2 > c_1^2$),

$$\theta = c_1 \int \frac{dz}{R\sqrt{R^2 - c_1^2}} + c_2 = mz + c_2, \qquad m = \frac{c_1}{R\sqrt{R^2 - c_1^2}},$$

since the integrand is a constant. Therefore the geodesic is the curve

$$x = R\cos(mz + c_2), \qquad y = R\sin(mz + c_2), \qquad z = z.$$

If $m \neq 0$ (i.e. $c_1 \neq 0$) this is a circular helix of radius R and pitch $2\pi/|m|$ about the z-axis: as z increases by $2\pi/|m|$ the point goes once around the cylinder. If $m = 0$ (i.e. $c_1 = 0$), then $\theta = c_2$ is constant and the curve $x = R\cos c_2$, $y = R\sin c_2$, z arbitrary, is a generator (a straight line parallel to the axis). As c_1 runs over $(-R, R)$, m takes every real value, so all helices occur.

Answer. Shown: $\theta = mz + c_2$ with $m = \dfrac{c_1}{R\sqrt{R^2 - c_1^2}}$; a helix if $m \neq 0$, a generator if $m = 0$.

Section 68 Isoperimetric Problems

Notation and tools (from the text). To find a function $y(x)$ giving a stationary value to $I = \int_{x_1}^{x_2} f(x,y,y')\,dx$ subject to $J = \int_{x_1}^{x_2} g(x,y,y')\,dx = c$ and fixed end values, we form $F = f + \lambda g$ with a constant Lagrange multiplier λ and solve Euler's equation (16), $\frac{d}{dx}F_{y'} - F_y = 0$, then fix the two constants of integration and λ from the two boundary conditions and the side condition $J = c$. For finite constraints in ordinary calculus we use $F(x,y,\dots,\lambda) = f + \lambda g$ and the equations (7). For curves in space the equations (24)–(26) (with $\lambda = \lambda(t)$) are used. As in Section 67, in Case C (x missing from F) the first integral $F_{y'}y' - F = \text{const}$ is obtained by multiplying Euler's equation by y', so constants $y \equiv \text{const}$ must be tested separately in Euler's equation. log is the natural logarithm.

Problem 68-1 (Simmons)

Solution. Formula (1) states that, for a closed curve $x = x(t)$, $y = y(t)$, $t_1 \le t \le t_2$, traversed once counterclockwise, the enclosed area is

$$A = \frac{1}{2}\int_{t_1}^{t_2}\Big(x\frac{dy}{dt} - y\frac{dx}{dt}\Big)dt = \frac{1}{2}\oint (x\,dy - y\,dx).$$

Let the curve be convex, with leftmost point Q at $x = a$ and rightmost point P at $x = b$, so that the curve consists of an upper arc $y = y_u(x)$ and a lower arc $y = y_l(x)$ ($a \le x \le b$), with $y_u \ge y_l$, which join at P and Q. Counterclockwise, the upper arc is traversed from P to Q (from right to left, x decreasing from b to a) and the lower arc from Q to P (from left to right). Therefore

$$\oint y\,dx = \int_P^Q y\,dx\Big|_{\text{upper}} + \int_Q^P y\,dx\Big|_{\text{lower}} = \int_b^a y_u\,dx + \int_a^b y_l\,dx = -\int_a^b (y_u - y_l)dx = -A,$$

because $\int_a^b (y_u - y_l)\,dx$ is the area between the two arcs, i.e. the area enclosed by the curve. So

$$A = -\oint y\,dx. \tag{i}$$

In the same way let R and S be the lowest and highest points ($y = c_R$ and $y = c_S$), and let $x = x_r(y)$, $x = x_l(y)$ be the right and left arcs ($x_r \ge x_l$). Counterclockwise the right arc is traversed upward from R to S and the left arc downward from S to R, so

$$\oint x\,dy = \int_{c_R}^{c_S} x_r\,dy + \int_{c_S}^{c_R} x_l\,dy = \int_{c_R}^{c_S} (x_r - x_l)\,dy = A. \tag{ii}$$

Adding (i) and (ii), $2A = \oint x\,dy - \oint y\,dx$, i.e.

$$A = \frac{1}{2}\oint (x\,dy - y\,dx) = \frac{1}{2}\int_{t_1}^{t_2}\Big(x\frac{dy}{dt} - y\frac{dx}{dt}\Big)dt,$$

which is formula (1). (Both (i) and (ii) are special cases of Green's theorem.)

Answer. Shown: for a closed convex curve traversed counterclockwise, $\oint y\,dx = -A$ and $\oint x\,dy = A$, hence $A = \frac{1}{2}\oint (x\,dy - y\,dx)$.

Problem 68-2 (Simmons)

Solution. As t increases from 0 to 2π the point $(a\cos t, a\sin t)$ traverses the circle once counterclockwise. We have $dx/dt = -a\sin t$ and $dy/dt = a\cos t$, so

$$x\frac{dy}{dt} - y\frac{dx}{dt} = a\cos t \cdot a\cos t - a\sin t \cdot (-a\sin t) = a^2(\cos^2 t + \sin^2 t) = a^2.$$

Hence

$$A = \frac{1}{2}\int_0^{2\pi} a^2\,\mathrm{d}t = \frac{1}{2}a^2 \cdot 2\pi = \pi a^2,$$

which is the area of a circle of radius a.

Answer. Formula (1) gives $A = \pi a^2$, as it should.

Problem 68-3 (Simmons)

Solution.

(a) We assume $(a,b,c) \neq (0,0,0)$. Let $F = x^2 + y^2 + z^2 + \lambda(ax + by + cz - d)$. The conditions (7) are

$$F_x = 2x + \lambda a = 0, \quad F_y = 2y + \lambda b = 0, \quad F_z = 2z + \lambda c = 0, \quad F_\lambda = ax + by + cz - d = 0.$$

The first three give $(x,y,z) = -\frac{\lambda}{2}(a,b,c)$. Substituting into the fourth, $-\frac{\lambda}{2}(a^2+b^2+c^2) = d$, so $-\frac{\lambda}{2} = \dfrac{d}{a^2+b^2+c^2}$ and

$$x = \frac{ad}{a^2+b^2+c^2}, \qquad y = \frac{bd}{a^2+b^2+c^2}, \qquad z = \frac{cd}{a^2+b^2+c^2}.$$

This is the only stationary point; since $w = x^2+y^2+z^2 \to \infty$ along the plane as the point goes to infinity, w has a minimum on the plane, and it must occur at the unique stationary point. The minimum distance is $\sqrt{w} = |d|/\sqrt{a^2+b^2+c^2}$ (for the plane $ax+by+cz=d$).

(b) Fix the perimeter $x + y + z = 2s$ (so s is a constant). The sides of a triangle satisfy the triangle inequalities $x < y+z$, $y < z+x$, $z < x+y$; since $y+z = 2s-x$, the first is equivalent to $x < s$, and similarly for the others. So the admissible triangles correspond to $u = s-x > 0$, $v = s-y > 0$, $w = s-z > 0$, with $u+v+w = 3s-2s = s$ (and then $x,y,z > 0$ automatically, as $x = s-u > 0$ etc. because $u < s$). Maximizing A is equivalent to maximizing $A^2/s = (s-x)(s-y)(s-z)$ (positive), and so to maximizing its logarithm. Let

$$F = \log(s-x) + \log(s-y) + \log(s-z) + \lambda(x+y+z-2s).$$

Then

$$F_x = -\frac{1}{s-x} + \lambda = 0, \qquad F_y = -\frac{1}{s-y} + \lambda = 0, \qquad F_z = -\frac{1}{s-z} + \lambda = 0,$$

so $s-x = s-y = s-z = 1/\lambda$, i.e. $x = y = z$, and by the constraint $x = y = z = \frac{2s}{3}$. To see that this point gives a maximum: on the closed triangle $u,v,w \geq 0$, $u+v+w = s$ (a compact set), the continuous function uvw is 0 on the boundary and positive inside, so its maximum is attained at an interior stationary point; there is only one, so it is the maximum. Thus the triangle of greatest area has $x = y = z$: it is equilateral, and its area is $A = \sqrt{s\,(s/3)^3} = \dfrac{s^2}{3\sqrt{3}}$.

(c) Let $s > 0$ and maximize $P = x_1 x_2 \cdots x_n$ over $x_i > 0$ with $x_1 + \cdots + x_n = s$. Since $P > 0$ it is equivalent to maximize $\log P = \sum \log x_i$. Let $F = \sum_{i=1}^n \log x_i + \lambda\big(\sum x_i - s\big)$. Then

$$\frac{\partial F}{\partial x_i} = \frac{1}{x_i} + \lambda = 0 \quad (i = 1,\ldots,n) \quad \Longrightarrow \quad x_1 = x_2 = \cdots = x_n = -\frac{1}{\lambda},$$

and the constraint gives $x_i = s/n$ for all i, with product $(s/n)^n = s^n/n^n$. This is the maximum: on the compact simplex $x_i \geq 0$, $\sum x_i = s$, the continuous function P vanishes on the boundary (some $x_i = 0$) and is positive inside, so its maximum is attained at an interior stationary point, and $(s/n, \dots, s/n)$ is the only one. Hence $x_1 \cdots x_n \leq s^n/n^n$ whenever $x_1 + \cdots + x_n = s$.

Now let $x_1, \dots, x_n$ be any positive numbers and put $s = x_1 + \cdots + x_n$. Then $x_1 x_2 \cdots x_n \leq (s/n)^n$, and taking the positive nth root,

$$\sqrt[n]{x_1 x_2 \cdots x_n} \leq \frac{s}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Answer. (a) $\left(\dfrac{ad}{a^2+b^2+c^2}, \dfrac{bd}{a^2+b^2+c^2}, \dfrac{cd}{a^2+b^2+c^2}\right)$; (b) $x = y = z$ (equilateral); (c) maximum product s^n/n^n at $x_i = s/n$, and hence $\sqrt[n]{x_1 \cdots x_n} \leq (x_1 + \cdots + x_n)/n$.

Problem 68-4 (Simmons)

Solution. Let the curve be $y = y(x)$, $0 \leq x \leq 1$, with $y(0) = y(1) = 0$ and $y \geq 0$. We minimize $I = \int_0^1 \sqrt{1 + y'^2}\, dx$ subject to the side condition $J = \int_0^1 y\, dx = A$ (the given area, $A > 0$). With $F = \sqrt{1 + y'^2} + \lambda y$, Euler's equation (16) is

$$\frac{d}{dx}\frac{y'}{\sqrt{1 + y'^2}} - \lambda = 0.$$

The variable y appears in F only through λy, so this integrates at once:

$$\frac{y'}{\sqrt{1 + y'^2}} = \lambda x + c_1 =: w.$$

Here $\lambda \neq 0$: if $\lambda = 0$, then y' is constant, so y is linear with $y(0) = y(1) = 0$, i.e. $y \equiv 0$, which encloses area $0 \neq A$. Solving for y': $y'^2 = w^2(1 + y'^2)$, $y'^2(1 - w^2) = w^2$, so

$$y' = \frac{w}{\sqrt{1 - w^2}} = \frac{\lambda x + c_1}{\sqrt{1 - (\lambda x + c_1)^2}}$$

(the sign of y' is that of w). Integrating, with $dw = \lambda\, dx$,

$$y = \frac{1}{\lambda}\int \frac{w\, dw}{\sqrt{1 - w^2}} = -\frac{1}{\lambda}\sqrt{1 - (\lambda x + c_1)^2} + c_2.$$

Hence $(y - c_2)^2 = \dfrac{1 - (\lambda x + c_1)^2}{\lambda^2}$, that is,

$$\left(x + \frac{c_1}{\lambda}\right)^2 + (y - c_2)^2 = \frac{1}{\lambda^2},$$

a circle of radius $1/|\lambda|$ centered at $(-c_1/\lambda, c_2)$. (Equivalently: by Example 1 of the text the curvature $y''/(1 + y'^2)^{3/2} = \lambda$ is constant.) Thus every stationary curve is an arc of a circle.

The constants are fixed by the end points and the area. The conditions $y(0) = y(1) = 0$ force the center to lie on the perpendicular bisector $x = \frac{1}{2}$ of the chord from $(0,0)$ to $(1,0)$, so the circle is

$$\left(x - \tfrac{1}{2}\right)^2 + (y - k)^2 = R^2, \qquad R^2 = \tfrac{1}{4} + k^2,$$

and the arc is the part above the x-axis. When it is a graph $y = y(x)$ (center on or below the x-axis, $k \le 0$, $y = k + \sqrt{R^2 - (x - \tfrac12)^2}$), the area beneath is the circular segment $A = R^2 \sin^{-1}\dfrac{1}{2R} - \tfrac12\sqrt{R^2 - \tfrac14}$, which decreases from $\pi/8$ (at $R = \tfrac12$, a semicircle) to 0 (as $R \to \infty$); so for each A with $0 < A \le \pi/8$ there is exactly one such arc, which is the required curve.

Answer. Shown: the stationary curves satisfy $\left(x + c_1/\lambda\right)^2 + (y - c_2)^2 = 1/\lambda^2$, a circle; the shortest curve is the arc of the circle through $(0,0)$ and $(1,0)$ with the prescribed area beneath it.

Problem 68-5 (Simmons)

Solution. Let the points be (x_1, y_1) and (x_2, y_2), with y measured upward, and let the chain be $y = y(x)$ with constant linear density ρ. A small piece of length ds has potential energy $\rho g y\,\mathrm{d}s$, so, up to the constant factor ρg, we must minimize

$$I = \int_{x_1}^{x_2} y\sqrt{1 + y'^2}\,\mathrm{d}x \quad \text{subject to} \quad J = \int_{x_1}^{x_2} \sqrt{1 + y'^2}\,\mathrm{d}x = L$$

(the given length L exceeds the distance between the two points), with $y(x_1) = y_1$, $y(x_2) = y_2$. Then $F = (y + \lambda)\sqrt{1 + y'^2}$ and Euler's equation is

$$\frac{d}{dx}\frac{(y + \lambda)y'}{\sqrt{1 + y'^2}} - \sqrt{1 + y'^2} = 0.$$

First integral. F does not contain x, so Case C applies:

$$F_{y'}y' - F = \frac{(y + \lambda)y'^2}{\sqrt{1 + y'^2}} - (y + \lambda)\sqrt{1 + y'^2} = -\frac{y + \lambda}{\sqrt{1 + y'^2}} = c.$$

Write $v = y + \lambda$. Then $v = k\sqrt{1 + v'^2}$ with $k = -c$. If $k = 0$ then $v \equiv 0$, which is not a solution of Euler's equation (it gives $0 - 1 \neq 0$); so $k \neq 0$, and v has the sign of k and $v^2 \ge k^2$. Then

$$v'^2 = \frac{v^2 - k^2}{k^2}, \qquad \frac{\mathrm{d}v}{\sqrt{v^2 - k^2}} = \pm\frac{\mathrm{d}x}{|k|}.$$

Set $v = k\cosh t$ (possible since $v/k \ge 1$): then $\sqrt{v^2 - k^2} = |k|\sinh t$ for $t \ge 0$ and $\mathrm{d}v/\sqrt{v^2 - k^2} = \operatorname{sgn}(k)\,\mathrm{d}t$, so $t = \pm(x - c_2)/|k|$, and since cosh is even,

$$y + \lambda = c_1 \cosh\frac{x - c_2}{c_1} \qquad (c_1 = k).$$

All extremals are of this form. Expanding Euler's equation, $\dfrac{y'^2}{\sqrt{1 + y'^2}} + \dfrac{(y + \lambda)y''}{(1 + y'^2)^{3/2}} - \sqrt{1 + y'^2} = 0$, i.e.

$$(y + \lambda)\,y'' = 1 + y'^2,$$

a regular second order equation wherever $y + \lambda \neq 0$. The function $v = c_1\cosh\frac{x - c_2}{c_1}$ satisfies $v'' = \cosh\frac{x - c_2}{c_1}/c_1$, so $vv'' = \cosh^2 = 1 + \sinh^2 = 1 + v'^2$. Given initial data $(x_0, v_0 \neq 0, p_0)$, take $t_0 = \sinh^{-1} p_0$, $c_1 = v_0/\cosh t_0$, $c_2 = x_0 - c_1 t_0$: then $v(x_0) = v_0$, $v'(x_0) = \sinh t_0 = p_0$. By uniqueness this two-parameter family contains every extremal.

The three constants c_1, c_2, λ are determined by the two boundary conditions and the length condition $J = L$. The chain hangs in the shape of a catenary.

Answer. The catenary $y + \lambda = c_1 \cosh \dfrac{x - c_2}{c_1}$.

Problem 68-6 (Simmons)

Solution. Take the origin at a point of the curve and the polar axis along the tangent there. We assume the curve is a smooth closed curve with nonzero curvature at the origin, lying on one side of its tangent there (as for any convex curve), so that it is described by $r = r(\theta)$ for $0 \le \theta \le \pi$ with $r(0) = r(\pi) = 0$ and $r > 0$ for $0 < \theta < \pi$. The enclosed area is $A = \frac{1}{2}\int_0^\pi r^2\,d\theta$ and the length is $L = \int_0^\pi \sqrt{r'^2 + r^2}\,d\theta$, a given constant. (Near the origin $r \approx 2\theta/\kappa$ for a curve of curvature $\kappa \neq 0$, so $r'(0) = 2/\kappa \neq 0$.)

Put $S = \sqrt{r'^2 + r^2}$ and

$$F = \tfrac{1}{2}r^2 + \lambda S, \qquad F_{r'} = \frac{\lambda r'}{S}, \qquad F_r = r + \frac{\lambda r}{S}.$$

F does not contain θ, so Case C: $F_{r'}r' - F = c$, i.e.

$$\frac{\lambda r'^2}{S} - \tfrac{1}{2}r^2 - \lambda S = -\tfrac{1}{2}r^2 - \lambda\frac{r^2}{S} = c.$$

At $\theta = 0$ we have $r = 0$ and $S = |r'(0)| \neq 0$, so the left side is 0 and $c = 0$. Hence

$$r^2\left(\tfrac{1}{2} + \frac{\lambda}{S}\right) = 0,$$

and for $0 < \theta < \pi$ (where $r > 0$) we get $S = -2\lambda$: $S = \sqrt{r'^2 + r^2}$ is constant (so $\lambda < 0$). Substituting $S = -2\lambda$ in Euler's equation $\frac{d}{d\theta}F_{r'} - F_r = 0$:

$$\frac{\lambda}{S}r'' - r - \frac{\lambda}{S}r = -\tfrac{1}{2}r'' - r + \tfrac{1}{2}r = 0 \quad\Longrightarrow\quad r'' + r = 0.$$

So $r = A_1\cos\theta + B_1\sin\theta$, and $r(0) = 0$ gives $A_1 = 0$: $r = B_1\sin\theta$, and $r > 0$ on $(0, \pi)$ means $B_1 > 0$. (Then $S = \sqrt{B_1^2\cos^2\theta + B_1^2\sin^2\theta} = B_1$, which is consistent with $B_1 = -2\lambda$.) Write $D = B_1$:

$$r = D\sin\theta \qquad\Longleftrightarrow\qquad x^2 + y^2 = Dy,$$

(using $r^2 = x^2 + y^2$, $r\sin\theta = y$), i.e. $x^2 + \left(y - \frac{D}{2}\right)^2 = \left(\frac{D}{2}\right)^2$: a circle of diameter D tangent to the polar axis at the origin. Its length is $L = \int_0^\pi D\,d\theta = \pi D$, so $D = L/\pi$, and the area is

$$A = \frac{1}{2}\int_0^\pi D^2\sin^2\theta\,d\theta = \frac{D^2}{2}\cdot\frac{\pi}{2} = \frac{\pi D^2}{4} = \frac{L^2}{4\pi}.$$

Thus of all closed curves of length L the circle encloses the largest area, $A \le L^2/(4\pi)$.

Answer. The maximizing curve is $r = (L/\pi)\sin\theta$, a circle of diameter L/π (radius $L/2\pi$), with area $A = L^2/4\pi$.

Problem 68-7 (Simmons)

Solution. The surface $G(x, y, z) = g(x, z) = 0$ is a cylinder whose generators are parallel to the y-axis, and $G_y = 0$. A geodesic $x = x(t)$, $y = y(t)$, $z = z(t)$ is a stationary curve of $\int f\,dt$ with

$f = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$ on the surface. By (24)–(26) of the text (equivalently, Euler's equations for $\int [f + \lambda(t) G]\, dt$) there is a function $\lambda(t)$ with

$$\frac{d}{dt}\frac{\partial f}{\partial \dot{x}} = \lambda G_x, \qquad \frac{d}{dt}\frac{\partial f}{\partial \dot{y}} = \lambda G_y, \qquad \frac{d}{dt}\frac{\partial f}{\partial \dot{z}} = \lambda G_z.$$

(In the text the derivation solves $G = 0$ for z and uses $G_z \neq 0$; at any point of a regular surface one of the derivatives G_x, G_z is nonzero and one may solve for that variable, and the resulting equations are the same three.) Since $G_y = g_y = 0$, the middle equation reads

$$\frac{d}{dt}\frac{\partial f}{\partial \dot{y}} = \frac{d}{dt}\frac{\dot{y}}{f} = 0 \qquad \Longrightarrow \qquad \frac{\dot{y}}{f} = \text{const.}$$

Now $f = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$ is the speed, so $\mathbf{T} = (\dot{x}, \dot{y}, \dot{z})/f$ is the unit tangent vector of the curve, and $\mathbf{T} \cdot \mathbf{j} = \dot{y}/f$, where $\mathbf{j} = (0, 1, 0)$ is the unit vector along the y-axis. If γ is the angle between the curve and the y-axis, then $\cos\gamma = \dot{y}/f$ is constant. (If $t = s$ is the arc length, $f = 1$ and the statement is simply $dy/ds = \text{const.}$)

Independent check. Let σ be arc length along the plane curve $g(x, z) = 0$ in the xz-plane, with $x = h(\sigma)$, $z = k(\sigma)$, $h'^2 + k'^2 = 1$. The cylinder is parametrized by (σ, y) via $(h(\sigma), y, k(\sigma))$ and $ds^2 = (h'^2 + k'^2)d\sigma^2 + dy^2 = d\sigma^2 + dy^2$. A curve $y = y(\sigma)$ has length $\int \sqrt{1 + y'^2}\, d\sigma$, whose extremals (Case A) are $y = m\sigma + b$, with $ds = \sqrt{1 + m^2}\, d\sigma$, so $dy/ds = m/\sqrt{1 + m^2}$ is constant, in agreement with the above.

Answer. Shown: $\dfrac{d}{dt}\left(\dot{y}/f\right) = 0$, so $\cos\gamma = \dot{y}/\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$ is constant along every geodesic: it makes a constant angle with the y-axis.

Chapter 13

The Existence and Uniqueness of Solutions

Section 69 The Method of Successive Approximations

Throughout, the initial value problem $y' = f(x,y)$, $y(x_0) = y_0$ is replaced by the equivalent integral equation $y(x) = y_0 + \int_{x_0}^{x} f[t, y(t)]\,dt$, and Picard's iteration is

$$y_n(x) = y_0 + \int_{x_0}^{x} f[t, y_{n-1}(t)]\,dt, \qquad n = 1,2,3,\ldots$$

starting from a given $y_0(x)$ (the constant y_0 unless stated otherwise).

Problem 69-1 (Simmons)

Solution. **Exact solution.** Separating variables ($y \neq 0$, which is the case near $x = 0$ since $y(0) = 1$): $\dfrac{dy}{y^2} = dx$, so $-\dfrac{1}{y} = x + c$. The condition $y(0) = 1$ gives $c = -1$, so

$$y = \frac{1}{1-x}, \qquad |x| < 1,$$

(on the interval $x < 1$ containing 0), with Taylor series $1 + x + x^2 + x^3 + x^4 + \cdots$ for $|x| < 1$.
Picard iterates. Here $f(t,y) = y^2$, so $y_n(x) = 1 + \int_0^x [y_{n-1}(t)]^2\,dt$.

$$\begin{aligned}
y_1(x) &= 1 + \int_0^x 1\,dt = 1 + x,\\
y_2(x) &= 1 + \int_0^x (1+t)^2\,dt = 1 + \int_0^x (1 + 2t + t^2)\,dt = 1 + x + x^2 + \tfrac{1}{3}x^3,\\
y_3(x) &= 1 + \int_0^x \left(1 + t + t^2 + \tfrac{1}{3}t^3\right)^2 dt.
\end{aligned}$$

Squaring the polynomial term by term,

$$\left(1 + t + t^2 + \tfrac{1}{3}t^3\right)^2 = 1 + 2t + 3t^2 + \tfrac{8}{3}t^3 + \tfrac{5}{3}t^4 + \tfrac{2}{3}t^5 + \tfrac{1}{9}t^6,$$

(coefficients: t^2: $2\cdot 1\cdot 1 + 1 = 3$; t^3: $2\cdot\frac{1}{3} + 2\cdot 1 = \frac{8}{3}$; t^4: $2\cdot\frac{1}{3} + 1 = \frac{5}{3}$; t^5: $2\cdot\frac{1}{3} = \frac{2}{3}$; t^6: $\frac{1}{9}$). Integrating,

$$y_3(x) = 1 + x + x^2 + x^3 + \tfrac{2}{3}x^4 + \tfrac{1}{3}x^5 + \tfrac{1}{9}x^6 + \tfrac{1}{63}x^7.$$

Comparison. The exact solution is $1 + x + x^2 + x^3 + x^4 + \cdots$. We see that y_1 agrees with it through the term in x, y_2 through the term in x^2, and y_3 through the term in x^3: in each case $y_n(x)$ matches the Taylor series through x^n and first differs in the coefficient of x^{n+1} (for y_3 the coefficient of x^4 is $\frac{2}{3}$ instead of 1). So the iterates approach $1/(1-x)$ near $x = 0$, improving by one more correct Taylor coefficient at every step.

Answer. Exact solution $y = \dfrac{1}{1-x}$ ($|x| < 1$). $y_1 = 1 + x$, $y_2 = 1 + x + x^2 + \frac{1}{3}x^3$, $y_3 = 1 + x + x^2 + x^3 + \frac{2}{3}x^4 + \frac{1}{3}x^5 + \frac{1}{9}x^6 + \frac{1}{63}x^7$; each y_n agrees with the Taylor series $1 + x + x^2 + \cdots$ of the exact solution through the term in x^n.

Problem 69-2 (Simmons)

Solution. **Exact solution.** The equation is linear and separable: $\dfrac{dy}{1+y} = 2x\,dx$ (the constant solution $y=-1$ is lost in the division, but it does not satisfy $y(0)=0$). Integrating, $\log|1+y| = x^2 + c$, so $1+y = Ce^{x^2}$; $y(0)=0$ gives $C=1$, hence

$$y = e^{x^2} - 1 = x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} + \cdots.$$

Picard iterates. $y_n(x) = \int_0^x 2t\,[1 + y_{n-1}(t)]\,dt$, $y_0 = 0$.

$$\begin{aligned}
y_1 &= \int_0^x 2t\,dt = x^2,\\
y_2 &= \int_0^x 2t(1+t^2)\,dt = \int_0^x (2t+2t^3)\,dt = x^2 + \frac{x^4}{2},\\
y_3 &= \int_0^x 2t\Big(1+t^2+\frac{t^4}{2}\Big)\,dt = x^2 + \frac{x^4}{2} + \frac{x^6}{2\cdot 3},\\
y_4 &= \int_0^x 2t\Big(1+t^2+\frac{t^4}{2}+\frac{t^6}{6}\Big)\,dt = x^2 + \frac{x^4}{2} + \frac{x^6}{2\cdot 3} + \frac{x^8}{2\cdot 3\cdot 4}.
\end{aligned}$$

In general, $y_n(x) = \sum_{k=1}^{n} \dfrac{x^{2k}}{k!}$. Indeed this holds for $n=1$, and if it holds for n then

$$y_{n+1}(x) = \int_0^x 2t\Big(1 + \sum_{k=1}^{n} \frac{t^{2k}}{k!}\Big)\,dt = \sum_{k=0}^{n} \frac{2}{k!}\cdot\frac{x^{2k+2}}{2k+2} = \sum_{k=0}^{n} \frac{x^{2k+2}}{(k+1)!} = \sum_{j=1}^{n+1} \frac{x^{2j}}{j!}.$$

Comparison. y_n is exactly the nth partial sum (after the constant term 1 is removed) of the series of e^{x^2}; it agrees with $e^{x^2}-1$ through the term x^{2n} (and so, in particular, through x^{2n+1}), and the first discrepancy is the omitted term $x^{2n+2}/(n+1)!$. Since $\sum_{k\ge 1} x^{2k}/k!$ converges to $e^{x^2}-1$ for every x, $y_n(x) \to e^{x^2}-1$ for all x.

Answer. $y = e^{x^2}-1$; $y_1 = x^2$, $y_2 = x^2 + \frac{x^4}{2}$, $y_3 = x^2 + \frac{x^4}{2} + \frac{x^6}{2\cdot 3}$, $y_4 = x^2 + \frac{x^4}{2} + \frac{x^6}{2\cdot 3} + \frac{x^8}{2\cdot 3\cdot 4}$; in general $y_n = \sum_{k=1}^{n} x^{2k}/k!$, the partial sums of the Taylor series of the exact solution.

Problem 69-3 (Simmons)

Solution. Problem (4) of the text is $y' = x + y$, $y(0) = 1$, with exact solution $y(x) = 2e^x - x - 1$ (check: $y' = 2e^x - 1 = x + y$ and $y(0) = 1$). The iteration is

$$y_n(x) = 1 + \int_0^x [t + y_{n-1}(t)]\,dt = 1 + \frac{x^2}{2} + \int_0^x y_{n-1}(t)\,dt. \tag{$*$}$$

Note that the starting functions below are not constants, but the recursion is the same.

(a) $y_0 = e^x$. Then

$$y_1 = 1 + \frac{x^2}{2} + \int_0^x e^t\,dt = 1 + \frac{x^2}{2} + e^x - 1 = e^x + \frac{x^2}{2!}.$$

We claim $y_n(x) = e^x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots + \dfrac{x^{n+1}}{(n+1)!}$. It holds for $n=1$ (and for $n=0$ with an empty sum), and if it holds for n, then by $(*)$

$$y_{n+1} = 1 + \frac{x^2}{2} + (e^x - 1) + \sum_{j=2}^{n+1} \frac{x^{j+1}}{(j+1)!} = e^x + \frac{x^2}{2!} + \sum_{i=3}^{n+2} \frac{x^i}{i!},$$

which is the claim for $n+1$. As $n \to \infty$ the sum tends to $\sum_{j\ge 2} x^j/j! = e^x - 1 - x$, so

$$y_n(x) \to e^x + (e^x - x - 1) = 2e^x - x - 1,$$

the exact solution.

(b) $y_0 = 1 + x$. Then $y_1 = 1 + \frac{x^2}{2} + \int_0^x (1+t)\,\mathrm{d}t = 1 + \frac{x^2}{2} + x + \frac{x^2}{2} = 1 + x + x^2$, and we claim

$$y_n(x) = 1 + x + 2\Big[\frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^{n+1}}{(n+1)!}\Big].$$

True for $n = 1$ ($2 \cdot x^2/2! = x^2$) and for $n = 0$ (empty bracket). If true for n, then by $(*)$

$$y_{n+1} = 1 + \frac{x^2}{2} + x + \frac{x^2}{2} + 2\sum_{j=2}^{n+1} \frac{x^{j+1}}{(j+1)!} = 1 + x + 2\Big[\frac{x^2}{2!} + \sum_{i=3}^{n+2} \frac{x^i}{i!}\Big],$$

using $x^2 = 2 \cdot x^2/2!$. Letting $n \to \infty$:

$$y_n(x) \to 1 + x + 2(e^x - 1 - x) = 2e^x - x - 1,$$

the exact solution again.

(c) $y_0 = \cos x$. By $(*)$:

$$\begin{aligned}
y_1 &= 1 + \frac{x^2}{2} + \int_0^x \cos t\,\mathrm{d}t = 1 + \frac{x^2}{2} + \sin x,\\
y_2 &= 1 + \frac{x^2}{2} + \int_0^x \Big(1 + \frac{t^2}{2} + \sin t\Big)\,\mathrm{d}t = 1 + \frac{x^2}{2} + x + \frac{x^3}{6} + 1 - \cos x\\
&= 2 - \cos x + x + \frac{x^2}{2} + \frac{x^3}{3!},\\
y_3 &= 1 + \frac{x^2}{2} + \int_0^x \Big(2 - \cos t + t + \frac{t^2}{2} + \frac{t^3}{6}\Big)\,\mathrm{d}t = 1 + \frac{x^2}{2} + 2x - \sin x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\\
&= 1 + 2x + x^2 - \sin x + \frac{x^3}{3!} + \frac{x^4}{4!},\\
y_4 &= 1 + \frac{x^2}{2} + \int_0^x \Big(1 + 2t + t^2 - \sin t + \frac{t^3}{6} + \frac{t^4}{24}\Big)\,\mathrm{d}t\\
&= 1 + \frac{x^2}{2} + x + x^2 + \frac{x^3}{3} + (\cos x - 1) + \frac{x^4}{24} + \frac{x^5}{120}\\
&= \cos x + x + \frac{3x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4!} + \frac{x^5}{5!}.
\end{aligned}$$

(These agree with the book's forms, e.g. $y_4 = \big(\cos x - 1 + \frac{x^2}{2!} - \frac{x^4}{4!}\big) + 1 + x + \big(x^2 + \frac{x^3}{3} + \frac{x^4}{3\cdot 4}\big) + \frac{x^5}{5!}$.)

Convergence. Let $y = 2e^x - x - 1$ be the exact solution and $d_n = y_n - y$. Since y satisfies $y = 1 + \frac{x^2}{2} + \int_0^x y$, subtracting from $(*)$ gives $d_n(x) = \int_0^x d_{n-1}(t)\,\mathrm{d}t$, hence $d_n(x) = \int_0^x \frac{(x-t)^{n-1}}{(n-1)!} d_0(t)\,\mathrm{d}t$ (iterated integration). Here $d_0 = \cos x - 2e^x + x + 1$ is continuous, so on any interval $|x| \le a$ it is bounded by some M, and then

$$|d_n(x)| \le M\frac{|x|^n}{n!} \le M\frac{a^n}{n!} \longrightarrow 0.$$

So $y_n \to 2e^x - x - 1$ uniformly on every bounded interval, for the starting function $\cos x$ too. (The same argument shows that *any* continuous starting function leads to the exact solution; parts (a), (b) are special cases.)

Answer. (a) $y_n = e^x + \frac{x^2}{2!} + \cdots + \frac{x^{n+1}}{(n+1)!} \to 2e^x - x - 1$; (b) $y_n = 1 + x + 2\left[\frac{x^2}{2!} + \cdots + \frac{x^{n+1}}{(n+1)!}\right] \to 2e^x - x - 1$; (c) $y_1 = 1 + \frac{x^2}{2} + \sin x$, $y_2 = 2 - \cos x + x + \frac{x^2}{2} + \frac{x^3}{3!}$, $y_3 = 1 + 2x + x^2 - \sin x + \frac{x^3}{3!} + \frac{x^4}{4!}$, $y_4 = \cos x + x + \frac{3x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4!} + \frac{x^5}{5!}$, and $y_n \to 2e^x - x - 1$.

Section 70 Picard's Theorem

We use Theorem A of the text (Picard's theorem): if $f(x,y)$ and $\partial f/\partial y$ are continuous on a closed rectangle R with sides parallel to the axes and (x_0,y_0) is an interior point of R, then $y' = f(x,y)$, $y(x_0) = y_0$ has one and only one solution on some interval $|x - x_0| \le h$. We also use the *Lipschitz condition*: f satisfies a Lipschitz condition (with respect to y) on a region if there is a constant K such that

$$|f(x,y_1) - f(x,y_2)| \le K|y_1 - y_2|$$

for all points (x,y_1), (x,y_2) of the region.

Problem 70-1 (Simmons)

Solution. **Why Theorem A applies.** $f = y^2$ and $f_y = 2y$ are polynomials, hence continuous on the whole plane, in particular on the closed rectangle $R = \{|x - x_0| \le 1,\ |y - y_0| \le 1\}$, and (x_0,y_0) is an interior point of R. By Theorem A there is an $h > 0$ and exactly one solution on $|x - x_0| \le h$.

The solution through $(0,0)$. The function $y \equiv 0$ satisfies $y' = 0 = y^2$ and $y(0) = 0$, and it is defined for all x. By the uniqueness part of Theorem A there is no other solution through $(0,0)$ (on any interval). So here the unique solution is valid for all x.

The solution through $(0,1)$. Suppose y is a solution on an interval $J \ni 0$ with $y(0) = 1$. It never vanishes on J: if $y(x_1) = 0$ for some $x_1 \in J$, then y and the zero function would be two solutions through $(x_1,0)$, so by uniqueness $y \equiv 0$ near x_1, and the set where $y = 0$ would be open and closed in J, making $y \equiv 0$, contradicting $y(0) = 1$. Hence on J we may divide by y^2:

$$\frac{d}{dx}\Big(-\frac{1}{y}\Big) = \frac{y'}{y^2} = 1 \quad\Longrightarrow\quad -\frac{1}{y} = x + c.$$

$y(0) = 1$ gives $c = -1$, so $y = \dfrac{1}{1-x}$ on J. This function is not defined at $x = 1$ and $y \to +\infty$ as $x \to 1^-$; therefore $1 \notin J$, i.e. every solution through $(0,1)$ lives on an interval contained in $(-\infty,1)$. There is no solution through $(0,1)$ valid for all x.

Conclusion. Both problems satisfy the hypotheses of Theorem A on the whole plane, but the solution through $(0,0)$ exists for all x and the solution through $(0,1)$ does not. (More generally, the solution through (x_0,y_0) is $y = y_0/[1 - y_0(x - x_0)]$, which is valid for all x only when $y_0 = 0$.) Hence Theorem A, which is a *local* theorem (it gives only an interval $|x - x_0| \le h$, where h may be small), cannot be used to infer that a solution is valid for all x, even when f and f_y are continuous everywhere.

Answer. Theorem A applies since y^2 and $2y$ are continuous everywhere. Through $(0,0)$: $y \equiv 0$, valid for all x. Through $(0,1)$: $y = 1/(1-x)$, which blows up at $x = 1$, so it is valid only for $x < 1$. Hence continuity of f, f_y on the whole plane does not give global validity.

Problem 70-2 (Simmons)

Solution.

(a) Suppose a constant K existed with $|\sqrt{y_1} - \sqrt{y_2}| \le K|y_1 - y_2|$ for all $y_1, y_2 \in [0,1]$ (and $|x| \le 1$). Take $y_2 = 0$ and $y_1 = u \in (0,1]$:

$$\sqrt{u} \le Ku \quad\Longrightarrow\quad \frac{1}{\sqrt{u}} \le K \qquad (0 < u \le 1).$$

But $1/\sqrt{u} \to \infty$ as $u \to 0^+$ (take $u = 1/(4K^2+4)$ if $K > 0$, giving $1/\sqrt{u} = 2\sqrt{K^2+1} > K$), a contradiction. So no Lipschitz condition holds.

(b) For $y_1, y_2 \in [c,d]$, $y_1 \neq y_2$, rationalize:

$$|\sqrt{y_1} - \sqrt{y_2}| = \frac{|y_1 - y_2|}{\sqrt{y_1} + \sqrt{y_2}} \leq \frac{|y_1 - y_2|}{2\sqrt{c}},$$

because $\sqrt{y_1} + \sqrt{y_2} \geq 2\sqrt{c}$. Thus the Lipschitz condition holds with $K = \dfrac{1}{2\sqrt{c}}$. (Equivalently, by the mean value theorem, $f_y = \frac{1}{2\sqrt{y}}$ is bounded by $\frac{1}{2\sqrt{c}}$ on $[c,d]$.)

Answer. (a) With $y_2 = 0$, $y_1 = u$, the quotient $|f(x,u) - f(x,0)|/u = 1/\sqrt{u}$ is unbounded as $u \to 0^+$; no Lipschitz constant exists. (b) Lipschitz with $K = 1/(2\sqrt{c})$.

Problem 70-3 (Simmons)

Solution. Lipschitz condition. By the inequality $\big||y_1| - |y_2|\big| \leq |y_1 - y_2|$ (reverse triangle inequality) and $x^2 \leq 1$,

$$|f(x,y_1) - f(x,y_2)| = x^2 \big||y_1| - |y_2|\big| \leq x^2 |y_1 - y_2| \leq |y_1 - y_2|,$$

so $K = 1$ works.

Failure of $\partial f/\partial y$. Fix $x \neq 0$ with $|x| \leq 1$ and look at the point $(x,0)$. For $h \neq 0$,

$$\frac{f(x,h) - f(x,0)}{h} = \frac{x^2|h|}{h} = \begin{cases} x^2, & h > 0, \\ -x^2, & h < 0. \end{cases}$$

The two one-sided limits as $h \to 0$ are $x^2 \neq -x^2$ (since $x^2 > 0$), so the limit does not exist: $\partial f/\partial y$ does not exist at any point $(x,0)$ with $0 < |x| \leq 1$, that is, along the whole segment $y = 0$ of the rectangle except its midpoint. (At $x = 0$, $f(0,y) = 0$, so $f_y = 0$ there; for $y \neq 0$, $f_y = x^2 \operatorname{sgn} y$.) Thus a Lipschitz condition can hold even where $\partial f/\partial y$ does not exist, so the hypothesis on $\partial f/\partial y$ in Theorem A is sufficient but not necessary for a Lipschitz condition.

Answer. Shown: $|f(x,y_1) - f(x,y_2)| \leq |y_1 - y_2|$ on the rectangle ($K = 1$), while at every point $(x,0)$, $0 < |x| \leq 1$, the one-sided derivatives in y are $\pm x^2$, so $\partial f/\partial y$ does not exist.

Problem 70-4 (Simmons)

Solution.

(a) Let $m = \max(|a|,|b|)$ and $N = \max(|c|,|d|)$, so $|x| \leq m$ and $|y| \leq N$ on the rectangle. Then

$$|f(x,y_1) - f(x,y_2)| = |x|\,|y_1^2 - y_2^2| = |x|\,|y_1 + y_2|\,|y_1 - y_2| \leq m \cdot 2N\,|y_1 - y_2|,$$

so $K = 2mN$ is a Lipschitz constant.

(b) (Here the strip is assumed to have $a < b$, or at least to contain some $x_1 \neq 0$; for the degenerate strip $a = b = 0$, $f \equiv 0$.) Choose $x_1 \in [a,b]$ with $x_1 \neq 0$. Suppose a constant K existed. Take $y_2 = y$ and $y_1 = y + 1$ with $y > 0$ arbitrary: $|f(x_1, y+1) - f(x_1, y)| = |x_1|\,|(y+1)^2 - y^2| = |x_1|(2y+1)$, while $|y_1 - y_2| = 1$. The Lipschitz inequality would require

$$|x_1|(2y+1) \leq K \qquad \text{for all } y > 0,$$

which is false for $y > K/(2|x_1|)$. Hence no Lipschitz condition holds on the strip.

Answer. (a) $|f(x,y_1)-f(x,y_2)| \le 2\max(|a|,|b|)\max(|c|,|d|)\,|y_1-y_2|$. (b) At fixed $x_1 \neq 0$, $|f(x_1,y+1)-f(x_1,y)| = |x_1|(2y+1)$ is unbounded in y, so no K exists.

Problem 70-5 (Simmons)

Solution.

(b) Let $m = \max(|a|,|b|)$. For any $x \in [a,b]$ and all real y_1, y_2,

$$|f(x,y_1)-f(x,y_2)| = |x|\,|y_1-y_2| \le m\,|y_1-y_2|,$$

so $K = m$ works on the strip.

(a) A rectangle $a \le x \le b$, $c \le y \le d$ is contained in the strip of part (b), so the same inequality (with $K = m$) holds on it.

(c) Suppose $|xy_1 - xy_2| \le K|y_1-y_2|$ for all points of the plane. Taking $y_1 = 1$, $y_2 = 0$ gives $|x| \le K$ for every real x, which is absurd (take $x = K+1$). So no Lipschitz condition holds on the entire plane.

Answer. (a), (b) $|f(x,y_1)-f(x,y_2)| = |x||y_1-y_2| \le \max(|a|,|b|)\,|y_1-y_2|$. (c) With $y_1 = 1$, $y_2 = 0$ we would need $|x| \le K$ for all x, impossible.

Problem 70-6 (Simmons)

Solution. Write $f(x,y) = y|y|$, i.e. $f = y^2$ for $y \ge 0$ and $f = -y^2$ for $y \le 0$. Clearly f is continuous on the plane.

(a) *Derivative of f in y.* For $y > 0$, $f_y = 2y = 2|y|$; for $y < 0$, $f = -y^2$ and $f_y = -2y = 2|y|$. At $y = 0$, from the definition,

$$f_y(x,0) = \lim_{h\to 0}\frac{h|h|-0}{h} = \lim_{h\to 0}|h| = 0 = 2|0|.$$

So $\partial f/\partial y = 2|y|$ for all (x,y), a function continuous on the whole plane. Hence for *every* point (x_0,y_0) both f and f_y are continuous on a closed rectangle $|x-x_0| \le 1$, $|y-y_0| \le 1$ having (x_0,y_0) in its interior, and Theorem A applies: the problem has a unique solution on some interval $|x-x_0| \le h$ for every (x_0,y_0).

(b) Since Theorem A already gives existence and uniqueness at every point, the problem has a unique solution near x_0 for every (x_0,y_0). We can also see this explicitly. Put $s = x - x_0$.

- $y_0 = 0$: the solution is $y \equiv 0$, and it is the only one by Theorem A.
- $y_0 > 0$: while $y > 0$, $y' = y^2$, so $-1/y = s + c$, and the initial condition gives $y = \dfrac{y_0}{1-y_0 s}$ for $s < 1/y_0$; it stays positive, so the assumption $y > 0$ is consistent.
- $y_0 < 0$: while $y < 0$, $y' = -y^2$, so $1/y = s + c$ and $y = \dfrac{y_0}{1+y_0 s}$ for $s < -1/y_0 = 1/|y_0|$; here $1 + y_0 s > 0$, so $y < 0$ throughout.

(Check for $y_0 < 0$: $y' = -y_0^2/(1+y_0 s)^2 = -y^2 = y|y|$.) Thus there is a unique solution through every point of the plane.

Answer. (a) All points (x_0, y_0) (because $f = y|y|$ and $f_y = 2|y|$ are continuous everywhere). (b) All points (x_0, y_0).

Remark. The book's answer for (a) is "all points with $y_0 \neq 0$". This is not what Theorem A gives: $f_y = 2|y|$ exists and is continuous at $y = 0$ too (the derivative at 0 is 0, computed above), so the hypotheses hold at the points with $y_0 = 0$ as well. The book's answer to (b) and to Problem 7 ("all points") agrees with ours. (The key's reasoning for (b), that $f = |y|$ satisfies a Lipschitz condition, together with its "$y_0 \neq 0$" in (a), is what one obtains for the equation $y' = |y|$, whose f_y fails to exist at $y = 0$; for the equation as printed, $y' = y|y|$, the answer to (a) is all points.)

Problem 70-7 (Simmons)

Solution. This is the question of Problem 6(a). With $f(x, y) = y|y|$, we showed there that f is continuous and $f_y = 2|y|$ exists and is continuous at every point of the plane (at $y = 0$ because $\lim_{h\to 0} h|h|/h = 0$). So for every (x_0, y_0), the closed rectangle $|x - x_0| \leq 1$, $|y - y_0| \leq 1$ has (x_0, y_0) in its interior and satisfies the hypotheses of Theorem A.

Answer. All points (x_0, y_0) of the plane.

Section 71 Systems. The Second Order Linear Equation

Section 71 has a single problem (printed at the end of the section, page 641 of the book). For the system $y' = f(x,y,z)$, $z' = g(x,y,z)$, $y(x_0) = y_0$, $z(x_0) = z_0$, Picard's iteration is

$$y_n(x) = y_0 + \int_{x_0}^{x} f[t, y_{n-1}(t), z_{n-1}(t)]\,dt, \qquad z_n(x) = z_0 + \int_{x_0}^{x} g[t, y_{n-1}(t), z_{n-1}(t)]\,dt,$$

where both new functions are computed from the *previous* pair.

Problem 71-1 (Simmons)

Solution. **Exact solution.** Differentiating the first equation, $y'' = z' = -y$, i.e. $y'' + y = 0$, so $y = A\cos x + B\sin x$ with $y(0) = A = 1$ and $y'(0) = z(0) = B = 0$. Thus $y = \cos x$ and $z = y' = -\sin x$. Check: $y' = -\sin x = z$, $z' = -\cos x = -y$, $y(0) = 1$, $z(0) = 0$.

Picard iterates. Start with $y_0(x) = 1$, $z_0(x) = 0$, and use $y_n = 1 + \int_0^x z_{n-1}(t)\,dt$, $z_n = -\int_0^x y_{n-1}(t)\,dt$.

$$y_1 = 1 + \int_0^x 0\,dt = 1, \qquad z_1 = -\int_0^x 1\,dt = -x,$$

$$y_2 = 1 + \int_0^x (-t)\,dt = 1 - \frac{x^2}{2!}, \qquad z_2 = -\int_0^x 1\,dt = -x,$$

$$y_3 = 1 + \int_0^x (-t)\,dt = 1 - \frac{x^2}{2!}, \qquad z_3 = -\int_0^x \left(1 - \frac{t^2}{2}\right) dt = -x + \frac{x^3}{3!},$$

$$y_4 = 1 + \int_0^x \left(-t + \frac{t^3}{6}\right) dt = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}, \qquad z_4 = -\int_0^x \left(1 - \frac{t^2}{2}\right) dt = -x + \frac{x^3}{3!}.$$

The pattern: for $n \geq 0$,

$$y_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j x^{2j}}{(2j)!}, \qquad z_n(x) = -\sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \frac{(-1)^j x^{2j+1}}{(2j+1)!}$$

(an empty sum for $z_0 = 0$). *Proof by induction.* The formulas hold for $n = 0$ ($y_0 = 1$, $z_0 = 0$). Assume them for $n-1$ (with $n \geq 1$). Then

$$z_n = -\int_0^x \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \frac{(-1)^j t^{2j}}{(2j)!}\,dt = -\sum_{j=0}^{\lfloor (n-1)/2 \rfloor} \frac{(-1)^j x^{2j+1}}{(2j+1)!},$$

which is the formula for z_n, and

$$y_n = 1 + \int_0^x \left(-\sum_{j=0}^{\lfloor (n-2)/2 \rfloor} \frac{(-1)^j t^{2j+1}}{(2j+1)!}\right) dt = 1 + \sum_{j=0}^{\lfloor n/2 \rfloor - 1} \frac{(-1)^{j+1} x^{2j+2}}{(2j+2)!} = \sum_{i=0}^{\lfloor n/2 \rfloor} \frac{(-1)^i x^{2i}}{(2i)!}$$

(with $i = j+1$ and $\lfloor (n-2)/2 \rfloor = \lfloor n/2 \rfloor - 1$), which is the formula for y_n.

Comparison. The iterates are the partial sums of the Taylor series

$$\cos x = \sum_{j \geq 0} \frac{(-1)^j x^{2j}}{(2j)!}, \qquad \sin x = \sum_{j \geq 0} \frac{(-1)^j x^{2j+1}}{(2j+1)!}.$$

Since both series converge for all x, $y_n(x) \to \cos x$ and $z_n(x) \to -\sin x$ for every x, which is the exact solution. (Each new iterate adds one more correct term, alternately to y and to z.)

Answer. $y_1 = 1, z_1 = -x; y_2 = 1 - \frac{x^2}{2!}, z_2 = -x; y_3 = 1 - \frac{x^2}{2!}, z_3 = -x + \frac{x^3}{3!}; y_4 = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$, $\ldots$; the iterates are partial sums of the Taylor series and converge to the exact solution $y = \cos x, z = -\sin x$.

Chapter 14

Numerical Methods

Section 73 The Method of Euler

Conventions. The Euler method for $y' = f(x,y)$, $y(x_0) = y_0$, with step size h is

$$x_k = x_0 + kh, \qquad y_{k+1} = y_k + h\,f(x_k, y_k), \qquad k = 0, 1, 2, \ldots$$

and the relative error is $E_k = |y(x_k) - y_k| / |y(x_k)|$ (expressed in percent; the printed definition (5) of the text has a misprinted "+" in place of "−"). All iterates in this section were computed with 40-digit arithmetic (no rounding between steps) and are displayed rounded to five decimals, as in the text; a hand calculation that rounds to five decimals at every step may therefore differ from our table in the last digit. The tables are checked in the script `verify/sec73.py`, which also reproduces the book's own Tables 1 and 2 of Section 73 (the benchmark problem $y' = x + y$, $y(0) = 1$).

Section 72. The introductory Section 72 contains a single (discussion) problem, which we answer first.

Problem 72-1 (Simmons)

Solution. This is an open-ended question; here are typical, accurate examples.

- *Projectile motion with air resistance* (physics, Section 5 of this book). With quadratic drag the equations $m\,\ddot{\mathbf{r}} = -mg\,\mathbf{e}_y - c\,|\dot{\mathbf{r}}|\,\dot{\mathbf{r}}$ have no closed-form solution; the range and the maximum height are found by stepping the system forward in time (Euler or Runge–Kutta). Without drag the same problem is solved exactly by parabolas, and comparing the two shows how much the drag matters.
- *The pendulum with large amplitude.* The equation $\ddot{\theta} + (g/L)\sin\theta = 0$ is linearized to $\ddot{\theta} + (g/L)\theta = 0$ only for small angles; for large angles the period is an elliptic integral and the motion itself is generated numerically.
- *Planetary and many-body motion* (Section 21). The two-body problem is solvable in closed form (Kepler's laws), but the three-body problem is not, and orbit predictions for spacecraft are numerical solutions of the equations of motion.
- *Chemical kinetics and epidemics.* Systems such as the SIR model $S' = -\beta SI$, $I' = \beta SI - \gamma I$, $R' = \gamma I$ or multi-species reaction networks are nonlinear and are solved numerically; "stiff" reaction networks (very different rate constants) need special methods.

In each instance the instructor's message is the one of this chapter: the solution is a table of approximate values $y_k \approx y(x_k)$, and one must know how large the error is and how it depends on the step size h.

Answer. Open-ended; examples: projectile with drag, large-angle pendulum, many-body motion, epidemic/chemical kinetics models.

Problem 73-1 (Simmons)

Solution. **Exact solution.** The equation is linear: $y' - 2y = 2x$. Multiplying by the integrating factor e^{-2x} gives $(e^{-2x}y)' = 2xe^{-2x}$. Integration by parts ($u = 2x$, $dv = e^{-2x}dx$) gives $\int 2xe^{-2x}\,dx = -xe^{-2x} - \frac{1}{2}e^{-2x} + C$, hence

$$y = Ce^{2x} - x - \tfrac{1}{2}, \qquad y(0) = 1 \;\Rightarrow\; C = \tfrac{3}{2},$$

$$y(x) = \tfrac{3}{2}e^{2x} - x - \tfrac{1}{2}, \qquad y(1) = \tfrac{3}{2}e^2 - \tfrac{3}{2} = 9.58358 \text{ (5 d.p.)}.$$

Euler formula. Here $f(x,y) = 2x + 2y$, so $y_{k+1} = y_k + h(2x_k + 2y_k) = (1+2h)y_k + 2hx_k$. This linear recursion can even be solved in closed form: trying $y_k = \alpha k + \beta$ for the particular solution gives $\alpha = -h$, $\beta = -\tfrac{1}{2}$, so with $y_0 = 1$

$$y_k = \tfrac{3}{2}(1+2h)^k - x_k - \tfrac{1}{2}, \qquad \text{in particular} \qquad y_N = \tfrac{3}{2}(1+2h)^{1/h} - \tfrac{3}{2} \;\; (x_N = 1).$$

(Compare with $y(x) = \tfrac{3}{2}e^{2x} - x - \tfrac{1}{2}$: Euler replaces e^{2x} by $(1+2h)^{x/h}$, which tends to e^{2x} as $h \to 0$, but from below, since $(1+2h)^{1/h} < e^2$.)

Step size $h = 0.1$ ($N = 10$ steps; $y_{k+1} = 1.2\,y_k + 0.2\,x_k$; the column f is the slope used in the step $y_{k+1} = y_k + 0.1f$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	2.00000	1.00000	0.00000	0.000
0.10	1.20000	2.60000	1.23210	0.03210	2.606
0.20	1.46000	3.32000	1.53774	0.07774	5.055
0.30	1.79200	4.18400	1.93318	0.14118	7.303
0.40	2.21040	5.22080	2.43831	0.22791	9.347
0.50	2.73248	6.46496	3.07742	0.34494	11.209
0.60	3.37898	7.95795	3.88018	0.50120	12.917
0.70	4.17477	9.74954	4.88280	0.70803	14.500
0.80	5.14973	11.89945	6.12955	0.97982	15.985
0.90	6.33967	14.47934	7.67447	1.33480	17.393
1.00	7.78760	–	9.58358	1.79598	18.740

Step size $h = 0.05$ ($N = 20$ steps; $y_{k+1} = 1.1\,y_k + 0.1\,x_k$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	2.00000	1.00000	0.00000	0.000
0.05	1.10000	2.30000	1.10776	0.00776	0.700
0.10	1.21500	2.63000	1.23210	0.01710	1.388
0.15	1.34650	2.99300	1.37479	0.02829	2.058
0.20	1.49615	3.39230	1.53774	0.04159	2.704
0.25	1.66576	3.83153	1.72308	0.05732	3.326
0.30	1.85734	4.31468	1.93318	0.07584	3.923
0.35	2.07308	4.84615	2.17063	0.09755	4.494
0.40	2.31538	5.43077	2.43831	0.12293	5.042
0.45	2.58692	6.07384	2.73940	0.15248	5.566
0.50	2.89061	6.78123	3.07742	0.18681	6.070

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.55	3.22968	7.55935	3.45625	0.22657	6.555
0.60	3.60764	8.41529	3.88018	0.27253	7.024
0.65	4.02841	9.35681	4.35395	0.32554	7.477
0.70	4.49625	10.39250	4.88280	0.38655	7.917
0.75	5.01587	11.53174	5.47253	0.45666	8.345
0.80	5.59246	12.78492	6.12955	0.53709	8.762
0.85	6.23171	14.16341	6.86092	0.62922	9.171
0.90	6.93988	15.67975	7.67447	0.73460	9.572
0.95	7.72386	17.34773	8.57884	0.85498	9.966
1.00	8.59125	–	9.58358	0.99233	10.355

Step size $h = 0.01$ ($N = 100$ steps, $y_{k+1} = 1.02\,y_k + 0.02\,x_k$; only every tenth iterate is displayed):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	2.00000	1.00000	0.00000	0.000
0.10	1.22849	2.65698	1.23210	0.00361	0.293
0.20	1.52892	3.45784	1.53774	0.00882	0.573
0.30	1.91704	4.43408	1.93318	0.01614	0.835
0.40	2.41206	5.62412	2.43831	0.02625	1.077
0.50	3.03738	7.07476	3.07742	0.04004	1.301
0.60	3.82155	8.84309	3.88018	0.05863	1.511
0.70	4.79934	10.99867	4.88280	0.08346	1.709
0.80	6.01316	13.62632	6.12955	0.11639	1.899
0.90	7.51470	16.82940	7.67447	0.15977	2.082
1.00	9.36697	–	9.58358	0.21661	2.260

Summary at $x = 1$.

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^1$
0.1	10	7.78760	9.58358	1.79598	18.74	17.960
0.05	20	8.59125	9.58358	0.99233	10.35	19.847
0.01	100	9.36697	9.58358	0.21661	2.26	21.661

Discussion. The Euler method performs poorly here: the estimates 7.788, 8.591, 9.367 underestimate $y(1) = 9.584$ by 18.7%, 10.4% and 2.3%. The error is roughly proportional to h (the last column, error$/h$, tends to $3e^2 = 22.17$: indeed $y(1) - y_N = \frac{3}{2}\left(e^2 - (1+2h)^{1/h}\right) \approx 3e^2 h$), exactly the behavior predicted by the text, so that an error of a few hundredths ($\approx 22h$) requires h of the order of 10^{-3}, i.e. about a thousand steps. The solution is convex and increasing fast ($y'' = 6e^{2x}$), and each tangent line lies below the curve, so all Euler values are too small.

Answer. $y_N(1) = 7.78760,\ 8.59125,\ 9.36697$ for $h = 0.1,\ 0.05,\ 0.01$; exact $y(1) = \frac{3}{2}e^2 - \frac{3}{2} = 9.58358$. Errors 1.796, 0.992, 0.217 (relative 18.7%, 10.4%, 2.3%), proportional to h; Euler is accurate only for very small h.

Problem 73-2 (Simmons)

Solution. Exact solution. Separating variables, $y\,dy = dx$, so $\frac{1}{2}y^2 = x + c$; $y(0) = 1$ gives $c = \frac{1}{2}$, i.e. $y^2 = 2x + 1$, and since $y(0) = 1 > 0$ (and y can never vanish because $y^2 = 2x + 1 \geq 1$ for $x \geq 0$)

$$y(x) = \sqrt{2x+1}, \qquad y(1) = \sqrt{3} = 1.73205.$$

Euler formula. $y_{k+1} = y_k + \dfrac{h}{y_k}$.

Step size $h = 0.1$ ($y_{k+1} = y_k + 0.1/y_k$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	1.00000	1.00000	0.00000	0.000
0.10	1.10000	0.90909	1.09545	−0.00455	0.416
0.20	1.19091	0.83969	1.18322	−0.00769	0.650
0.30	1.27488	0.78439	1.26491	−0.00997	0.788
0.40	1.35332	0.73892	1.34164	−0.01168	0.870
0.50	1.42721	0.70067	1.41421	−0.01300	0.919
0.60	1.49728	0.66788	1.48324	−0.01404	0.946
0.70	1.56406	0.63936	1.54919	−0.01487	0.960
0.80	1.62800	0.61425	1.61245	−0.01555	0.964
0.90	1.68943	0.59192	1.67332	−0.01611	0.962
1.00	1.74862	–	1.73205	−0.01657	0.956

Step size $h = 0.05$ ($y_{k+1} = y_k + 0.05/y_k$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	1.00000	1.00000	0.00000	0.000
0.05	1.05000	0.95238	1.04881	−0.00119	0.114
0.10	1.09762	0.91106	1.09545	−0.00217	0.198
0.15	1.14317	0.87476	1.14018	−0.00300	0.263
0.20	1.18691	0.84252	1.18322	−0.00369	0.312
0.25	1.22904	0.81365	1.22474	−0.00429	0.350
0.30	1.26972	0.78758	1.26491	−0.00481	0.380
0.35	1.30910	0.76389	1.30384	−0.00526	0.403
0.40	1.34729	0.74223	1.34164	−0.00565	0.421
0.45	1.38440	0.72233	1.37840	−0.00600	0.435
0.50	1.42052	0.70397	1.41421	−0.00631	0.446

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.55	1.45572	0.68695	1.44914	−0.00658	0.454
0.60	1.49007	0.67111	1.48324	−0.00683	0.460
0.65	1.52362	0.65633	1.51658	−0.00705	0.465
0.70	1.55644	0.64249	1.54919	−0.00724	0.468
0.75	1.58856	0.62950	1.58114	−0.00742	0.469
0.80	1.62004	0.61727	1.61245	−0.00759	0.470
0.85	1.65090	0.60573	1.64317	−0.00773	0.471
0.90	1.68119	0.59482	1.67332	−0.00787	0.470
0.95	1.71093	0.58448	1.70294	−0.00799	0.469
1.00	1.74015	–	1.73205	−0.00810	0.468

Step size $h = 0.01$ ($y_{k+1} = y_k + 0.01/y_k$, 100 steps, every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	1.00000	1.00000	0.00000	0.000
0.10	1.09586	0.91252	1.09545	−0.00042	0.038
0.20	1.18393	0.84464	1.18322	−0.00072	0.061
0.30	1.26585	0.78999	1.26491	−0.00094	0.074
0.40	1.34274	0.74474	1.34164	−0.00110	0.082
0.50	1.41545	0.70649	1.41421	−0.00123	0.087
0.60	1.48458	0.67359	1.48324	−0.00134	0.090
0.70	1.55061	0.64491	1.54919	−0.00142	0.092
0.80	1.61394	0.61960	1.61245	−0.00149	0.092
0.90	1.67487	0.59706	1.67332	−0.00155	0.092
1.00	1.73364	–	1.73205	−0.00159	0.092

Summary at $x = 1$.

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^1$
0.1	10	1.74862	1.73205	−0.01657	0.96	−0.166
0.05	20	1.74015	1.73205	−0.00810	0.47	−0.162
0.01	100	1.73364	1.73205	−0.00159	0.09	−0.159

Discussion. The Euler values 1.74862, 1.74015, 1.73364 *overestimate* $\sqrt{3}$, with relative errors 0.96%, 0.47%, 0.09%: the error again scales like h (error$/h \approx -0.16$). Overestimation is expected because $y'' = -1/y^3 < 0$: the solution is concave, so tangent lines lie above it. This problem is much milder than Problem 1 because the solution varies slowly ($|y''| \le 1$).

Answer. $y_N(1) = 1.74862,\ 1.74015,\ 1.73364$ for $h = 0.1,\ 0.05,\ 0.01$; exact $y(1) = \sqrt{3} = 1.73205$. Errors $-0.0166,\ -0.0081,\ -0.0016$ (0.96%, 0.47%, 0.09%); Euler does quite well, with error $\propto h$.

Problem 73-3 (Simmons)

Solution. Exact solution. Separating variables, $e^{-y}dy = dx$, so $-e^{-y} = x + c$; $y(0) = 0$ gives $c = -1$, so $e^{-y} = 1 - x$ and

$$y(x) = -\log(1 - x), \qquad 0 \le x < 1.$$

This solution tends to $+\infty$ as $x \to 1^-$: *the exact solution does not exist at* $x = 1$. (Equivalently, $y'' = e^{2y} = (1-x)^{-2}$ is unbounded.) So there is nothing to compare to at $x = 1$ itself; we compare the Euler iterates with $-\log(1-x)$ for $x < 1$ and we examine what the numbers at $x = 1$ mean.

Euler formula. $y_{k+1} = y_k + h\, e^{y_k}$.

Step size $h = 0.1$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10000	1.10517	0.10536	0.00536	5.088
0.20	0.21052	1.23432	0.22314	0.01263	5.658
0.30	0.33395	1.39647	0.35667	0.02273	6.372
0.40	0.47360	1.60576	0.51083	0.03723	7.288
0.50	0.63417	1.88546	0.69315	0.05898	8.508
0.60	0.82272	2.27668	0.91629	0.09357	10.212
0.70	1.05039	2.85875	1.20397	0.15359	12.757
0.80	1.33626	3.80479	1.60944	0.27318	16.973
0.90	1.71674	5.56635	2.30259	0.58585	25.443
1.00	2.27337	–	$+\infty$	–	–

Step size $h = 0.05$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.05	0.05000	1.05127	0.05129	0.00129	2.521
0.10	0.10256	1.10801	0.10536	0.00280	2.655
0.15	0.15796	1.17112	0.16252	0.00455	2.803
0.20	0.21652	1.24175	0.22314	0.00662	2.968
0.25	0.27861	1.32129	0.28768	0.00907	3.154
0.30	0.34467	1.41153	0.35667	0.01200	3.365
0.35	0.41525	1.51475	0.43078	0.01553	3.606
0.40	0.49099	1.63393	0.51083	0.01984	3.884
0.45	0.57268	1.77302	0.59784	0.02516	4.208
0.50	0.66133	1.93737	0.69315	0.03181	4.590

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.55	0.75820	2.13443	0.79851	0.04031	5.048
0.60	0.86492	2.37482	0.91629	0.05137	5.606
0.65	0.98366	2.67424	1.04982	0.06616	6.302
0.70	1.11738	3.05682	1.20397	0.08660	7.193
0.75	1.27022	3.56163	1.38629	0.11608	8.373
0.80	1.44830	4.25587	1.60944	0.16114	10.012
0.85	1.66109	5.26506	1.89712	0.23603	12.441
0.90	1.92434	6.85066	2.30259	0.37824	16.427
0.95	2.26688	9.64923	2.99573	0.72885	24.330
1.00	2.74934	–	$+\infty$	–	–

Step size $h = 0.01$ (100 steps, every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10478	1.11047	0.10536	0.00058	0.550
0.20	0.22176	1.24828	0.22314	0.00138	0.618
0.30	0.35416	1.42498	0.35667	0.00252	0.706
0.40	0.50663	1.65969	0.51083	0.00419	0.821
0.50	0.68634	1.98643	0.69315	0.00681	0.982
0.60	0.90510	2.47218	0.91629	0.01119	1.221
0.70	1.18454	3.26920	1.20397	0.01943	1.614
0.80	1.57121	4.81246	1.60944	0.03823	2.375
0.90	2.19941	9.01972	2.30259	0.10317	4.481
1.00	3.95946	–	$+\infty$	–	–

Summary at $x = 1$**.**

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^1$
0.1	10	2.27337	$+\infty$	–	–	–
0.05	20	2.74934	$+\infty$	–	–	–
0.01	100	3.95946	$+\infty$	–	–	–

Discussion. The method returns finite numbers, 2.27337, 2.74934, 3.95946, at $x = 1$, but they are meaningless as approximations of $y(1)$, because $y(1)$ does not exist. The warning sign is that the three values do not settle down as h shrinks: they keep growing, and in fact with $h = 10^{-3}$ and $h = 10^{-4}$ we obtain 5.8519 and 7.8527, an increase of about $\log 10 = 2.3$ per decade, as for $-\log(1 - x)$ evaluated at distance $\approx h$ from the singularity. For $x < 1$ the method behaves as before, underestimating the (convex, increasing) solution, with relative error growing as $x \to 1$

(for $h = 0.01$: 0.6% at $x = 0.1$, 1.0% at $x = 0.5$, 4.5% at $x = 0.9$). Euler's method cannot detect that a solution has blown up; the numerical solution just keeps going.

Answer. Euler gives $y_N(1) = 2.27337,\ 2.74934,\ 3.95946$ ($h = 0.1,\ 0.05,\ 0.01$), but the exact solution $y = -\log(1-x)$ blows up at $x = 1$, so $y(1)$ does not exist and the estimates keep growing as $h \to 0$ (5.85 at $h = 10^{-3}$, 7.85 at $h = 10^{-4}$); they are meaningless.

Problem 73-4 (Simmons)

Solution. **Exact solution.** The equation $y' - y = -\sin x$ is linear. With the integrating factor e^{-x}: $(e^{-x}y)' = -e^{-x}\sin x$. Since $\int e^{-x}\sin x\,dx = -\frac{1}{2}e^{-x}(\sin x + \cos x)$ (check by differentiating: $\frac{1}{2}e^{-x}(\sin x + \cos x) - \frac{1}{2}e^{-x}(\cos x - \sin x) = e^{-x}\sin x$),

$$e^{-x}y = \tfrac{1}{2}e^{-x}(\sin x + \cos x) + C, \qquad y = \tfrac{1}{2}(\sin x + \cos x) + Ce^{x},$$

and $y(0) = \frac{1}{2} + C = -1$ gives $C = -\frac{3}{2}$:

$$y(x) = \tfrac{1}{2}(\sin x + \cos x) - \tfrac{3}{2}e^{x}, \qquad y(1) = -3.38654.$$

Euler formula. $y_{k+1} = y_k + h(y_k - \sin x_k)$.

Step size $h = 0.1$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	−1.00000	−1.00000	−1.00000	0.00000	0.000
0.10	−1.10000	−1.19983	−1.11034	−0.01034	0.931
0.20	−1.21998	−1.41865	−1.24274	−0.02275	1.831
0.30	−1.36185	−1.65737	−1.39936	−0.03751	2.681
0.40	−1.52759	−1.91700	−1.58250	−0.05491	3.470
0.50	−1.71929	−2.19871	−1.79458	−0.07529	4.196
0.60	−1.93916	−2.50380	−2.03819	−0.09903	4.859
0.70	−2.18954	−2.83375	−2.31610	−0.12656	5.464
0.80	−2.47291	−3.19027	−2.63128	−0.15837	6.019
0.90	−2.79194	−3.57527	−2.98694	−0.19500	6.528
1.00	−3.14947	–	−3.38654	−0.23707	7.000

Step size $h = 0.05$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	−1.00000	−1.00000	−1.00000	0.00000	0.000
0.05	−1.05000	−1.09998	−1.05254	−0.00254	0.242
0.10	−1.10500	−1.20483	−1.11034	−0.00534	0.481
0.15	−1.16524	−1.31468	−1.17365	−0.00841	0.716
0.20	−1.23097	−1.42964	−1.24274	−0.01176	0.946
0.25	−1.30246	−1.54986	−1.31788	−0.01542	1.170
0.30	−1.37995	−1.67547	−1.39936	−0.01941	1.387
0.35	−1.46372	−1.80662	−1.48747	−0.02374	1.596
0.40	−1.55405	−1.94347	−1.58250	−0.02844	1.797
0.45	−1.65123	−2.08619	−1.68476	−0.03353	1.990
0.50	−1.75554	−2.23496	−1.79458	−0.03904	2.175

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.55	−1.86729	−2.38997	−1.91227	−0.04499	2.353
0.60	−1.98678	−2.55143	−2.03819	−0.05140	2.522
0.65	−2.11436	−2.71954	−2.17268	−0.05832	2.684
0.70	−2.25033	−2.89455	−2.31610	−0.06577	2.840
0.75	−2.39506	−3.07670	−2.46884	−0.07378	2.988
0.80	−2.54890	−3.26625	−2.63128	−0.08238	3.131
0.85	−2.71221	−3.46349	−2.80384	−0.09163	3.268
0.90	−2.88538	−3.66871	−2.98694	−0.10155	3.400
0.95	−3.06882	−3.88223	−3.18102	−0.11220	3.527
1.00	−3.26293	–	−3.38654	−0.12361	3.650

Step size $h = 0.01$ (100 steps, every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	−1.00000	−1.00000	−1.00000	0.00000	0.000
0.10	−1.10924	−1.20907	−1.11034	−0.00110	0.099
0.20	−1.24032	−1.43899	−1.24274	−0.00242	0.195
0.30	−1.39537	−1.69089	−1.39936	−0.00399	0.285
0.40	−1.57664	−1.96606	−1.58250	−0.00586	0.370
0.50	−1.78653	−2.26596	−1.79458	−0.00805	0.448
0.60	−2.02758	−2.59223	−2.03819	−0.01061	0.520
0.70	−2.30252	−2.94673	−2.31610	−0.01358	0.586
0.80	−2.61425	−3.33161	−2.63128	−0.01703	0.647
0.90	−2.96592	−3.74925	−2.98694	−0.02101	0.704
1.00	−3.36093	–	−3.38654	−0.02560	0.756

Summary at $x = 1$**.**

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^1$
0.1	10	−3.14947	−3.38654	−0.23707	7.00	−2.371
0.05	20	−3.26293	−3.38654	−0.12361	3.65	−2.472
0.01	100	−3.36093	−3.38654	−0.02560	0.76	−2.560

Discussion. The Euler values -3.14947, -3.26293, -3.36093 are too large (not negative enough) by 7.0%, 3.7%, 0.8%; the error is proportional to h (error$/h \approx -2.4$ to -2.6). The solution is concave ($y'' = -\frac{1}{2}(\sin x + \cos x) - \frac{3}{2}e^x < 0$), so tangent lines lie above it.

Answer. $y_N(1) = -3.14947,\ -3.26293,\ -3.36093$ for $h = 0.1,\ 0.05,\ 0.01$; exact $y(1) = \frac{1}{2}(\sin 1 + \cos 1) - \frac{3}{2}e = -3.38654$. Errors 7.0%, 3.7%, 0.8%, roughly proportional to h.

Problem 73-5 (Simmons)

Solution. Exact solution. Put $u = x + y - 1$. Then $u' = 1 + y' = 1 + u^2$ and $u(0) = -1$. Separating, $\dfrac{du}{1+u^2} = dx$, so $\tan^{-1} u = x + c$; $u(0) = -1$ gives $c = \tan^{-1}(-1) = -\pi/4$. Thus $u = \tan\left(x - \frac{\pi}{4}\right)$ and

$$y(x) = 1 - x + \tan\left(x - \tfrac{\pi}{4}\right), \qquad -\tfrac{\pi}{4} < x < \tfrac{3\pi}{4},$$

a solution which exists on all of $[0, 1]$ (it blows up only at $x = 3\pi/4 \approx 2.356$). Then $y(1) = \tan(1 - \pi/4) = 0.21796$.

Euler formula. $y_{k+1} = y_k + h(x_k + y_k - 1)^2$.
Step size $h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10000	0.64000	0.08237	−0.01763	21.402
0.20	0.16400	0.40450	0.13709	−0.02691	19.630
0.30	0.20445	0.24557	0.17251	−0.03194	18.516
0.40	0.22901	0.13764	0.19431	−0.03469	17.854
0.50	0.24277	0.06617	0.20659	−0.03618	17.512
0.60	0.24939	0.02268	0.21245	−0.03694	17.387
0.70	0.25166	0.00234	0.21439	−0.03726	17.380
0.80	0.25189	0.00269	0.21460	−0.03729	17.375
0.90	0.25216	0.02315	0.21511	−0.03705	17.225
1.00	0.25447	–	0.21796	−0.03652	16.753

Step size $h = 0.05$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.05	0.05000	0.81000	0.04531	−0.00469	10.342
0.10	0.09050	0.65529	0.08237	−0.00813	9.869
0.15	0.12326	0.52814	0.11258	−0.01068	9.486
0.20	0.14967	0.42293	0.13709	−0.01258	9.179
0.25	0.17082	0.33545	0.15681	−0.01401	8.934
0.30	0.18759	0.26256	0.17251	−0.01508	8.743
0.35	0.20072	0.20185	0.18483	−0.01589	8.597
0.40	0.21081	0.15147	0.19431	−0.01650	8.490
0.45	0.21838	0.10997	0.20143	−0.01695	8.416
0.50	0.22388	0.07624	0.20659	−0.01729	8.370

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.55	0.22770	0.04942	0.21016	−0.01754	8.346
0.60	0.23017	0.02884	0.21245	−0.01772	8.340
0.65	0.23161	0.01402	0.21377	−0.01784	8.346
0.70	0.23231	0.00458	0.21439	−0.01792	8.356
0.75	0.23254	0.00030	0.21459	−0.01795	8.366
0.80	0.23255	0.00106	0.21460	−0.01795	8.365
0.85	0.23261	0.00682	0.21469	−0.01791	8.344
0.90	0.23295	0.01768	0.21511	−0.01784	8.294
0.95	0.23383	0.03379	0.21610	−0.01773	8.203
1.00	0.23552	–	0.21796	−0.01756	8.058

Step size $h = 0.01$ (100 steps, every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.08390	0.66601	0.08237	−0.00153	1.861
0.20	0.13948	0.43628	0.13709	−0.00240	1.747
0.30	0.17540	0.27521	0.17251	−0.00289	1.676
0.40	0.19749	0.16201	0.19431	−0.00318	1.635
0.50	0.20993	0.08414	0.20659	−0.00334	1.618
0.60	0.21588	0.03390	0.21245	−0.00343	1.617
0.70	0.21787	0.00674	0.21439	−0.00348	1.623
0.80	0.21809	0.00033	0.21460	−0.00349	1.626
0.90	0.21857	0.01406	0.21511	−0.00347	1.613
1.00	0.22137	–	0.21796	−0.00341	1.565

Summary at $x = 1$.

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^1$
0.1	10	0.25447	0.21796	−0.03652	16.75	−0.365
0.05	20	0.23552	0.21796	−0.01756	8.06	−0.351
0.01	100	0.22137	0.21796	−0.00341	1.57	−0.341

Discussion. The estimates $0.25447,\ 0.23552,\ 0.22137$ overestimate $y(1) = 0.21796$ by 16.8%, 8.1%, 1.6%, again with error roughly proportional to h (error$/h \approx -0.34$ to -0.37). Euler overshoots from the very first step because the exact solution is concave for $x < \pi/4$ ($y'' = 2u(1+u^2) < 0$ for $u = \tan(x - \pi/4) < 0$); the resulting error (≈ -0.037 for $h = 0.1$) persists, being reduced only slightly by the mild convexity for $x > \pi/4$. The relative errors are large because $y(1)$ is small, so a moderate absolute error is a large percentage.

Answer. $y_N(1) = 0.25447,\ 0.23552,\ 0.22137$ for $h = 0.1,\ 0.05,\ 0.01$; exact $y(1) = \tan(1 - \frac{\pi}{4}) = 0.21796$. Relative errors 16.8%, 8.1%, 1.6%, proportional to h.

Problem 73-6 (Simmons)

Solution. **Exact solution.** $y = \int \sec^2 x\,dx = \tan x + c$, and $y(0) = 0$ gives $c = 0$: $y = \tan x$, which is a solution exactly on the interval $-\pi/2 < x < \pi/2$ (the maximal interval containing $x = 0$ on which $\tan x$ is defined and differentiable); $\tan x \to +\infty$ as $x \to \pi/2^-$.

Euler. $y_{k+1} = y_k + 0.1\sec^2(x_k)$, $x_k = 0.1k$, $y_0 = 0$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10000	1.01007	0.10033	0.00033	0.334
0.20	0.20101	1.04109	0.20271	0.00170	0.840
0.30	0.30512	1.09569	0.30934	0.00422	1.364
0.40	0.41468	1.17875	0.42279	0.00811	1.918
0.50	0.53256	1.29845	0.54630	0.01374	2.516
0.60	0.66240	1.46804	0.68414	0.02173	3.177
0.70	0.80921	1.70945	0.84229	0.03308	3.927
0.80	0.98015	2.06016	1.02964	0.04948	4.806
0.90	1.18617	2.58800	1.26016	0.07399	5.871
1.00	1.44497	3.42552	1.55741	0.11244	7.220

So $y_{10} = 1.44497$ is the Euler estimate at $x = 1$. Compared with $\tan 1 = 1.55741$ the error is 0.11244 (7.2%); Euler underestimates because $y'' = 2\sec^2 x \tan x > 0$ (convex solution).

The statement about $x = 1$. As we have just seen, the exact solution $\tan x$ is perfectly well defined at $x = 1$, because $1 < \pi/2 = 1.5708\ldots$, so the claim "no solution at $x = 1$" cannot be literally right for the equation as printed: the initial value problem *has* the solution $\tan 1 = 1.55741$ there, and Euler's value 1.44497 is a (poor but reasonable) approximation. What is true is that the solution *ceases to exist at* $x = \pi/2 \approx 1.5708$, where it has a vertical asymptote, and therefore the IVP has no solution at any $x \geq \pi/2$. (The exercise presumably meant a point beyond $\pi/2$, such as $x = 2$.) The danger of blind application is seen by marching Euler's method past $\pi/2$:

x_k	y_k	$f(x_k, y_k) = \sec^2 x_k$	$\tan x_k$
1.0	1.44497	3.42552	1.55741
1.1	1.78752	4.86028	1.96476
1.2	2.27355	7.61596	2.57215
1.3	3.03515	13.97514	3.60210
1.4	4.43266	34.61546	5.79788
1.5	7.89421	199.85004	14.10142
1.6	27.87921	1172.86630	−34.23253
1.7	145.16584	60.23768	−7.69660
1.8	151.18961	19.37204	−4.28626
1.9	153.12681	9.56790	−2.92710
2.0	154.08360	–	−2.18504

At $x = 1.6$ and $x = 2.0$ the method produces perfectly respectable-looking finite numbers (27.879 and 154.084), although the IVP has no solution there ($\tan x$ at these points, -34.23 and -2.185, belongs to a different branch and is not a continuation of the solution through $x = 0$). The numerical values grow monotonically through the singularity at $\pi/2$ (since $\sec^2 x > 0$ always), giving no warning whatsoever. Even before $x = \pi/2$ the relative error grows quickly (7.2% at $x = 1$, 44% at $x = 1.5$) as the solution blows up.

Answer. Euler ($h = 0.1$): $y_{10} = 1.44497$ at $x = 1$ (exact $\tan 1 = 1.55741$). The solution is $y = \tan x$, which exists only for $|x| < \pi/2 \approx 1.5708$; it does exist at $x = 1$, but beyond $\pi/2$ the IVP has no solution, yet Euler keeps producing finite numbers (e.g. 154.08 at $x = 2$).

Problem 73-7 (Simmons)

Solution. Geometric argument. Euler's polygon is built from segments that start at (x_k, y_k) with the slope $f(x_k, y_k)$ of the solution curve *through that point*; the segment stays on the curve only if the curve does not bend away from its tangent line. If the exact solution is a *straight line*, $y = ax + b$, then at each point of the line the tangent is the line itself, so starting on the line the Euler step ends again on the line: by induction every (x_k, y_k) lies on the exact solution, and $y_k = y(x_k)$ for all k. Analytically: the local error is $\frac{1}{2}y''(\xi)h^2$ (Section 73, formula for y_1), and $y'' \equiv 0$ for a line. Conversely, if $y'' \neq 0$ somewhere the first step started there has a nonzero local error of size $\frac{1}{2}y''h^2$, so (barring accidental cancellation of errors later) the method is exact only for solutions with $y'' = 0$, i.e. straight lines.

Dependence on h. None: for a straight-line solution the Euler values are exact for *every* step size $h > 0$ (even a single big step). For other solutions the accuracy does depend on h (error proportional to h); the smaller the curvature $|y''|$ is, the closer to exact the method is for a given h.

Example 1. $y' = 2$, $y(0) = 1$, exact solution $y = 1 + 2x$. Euler: $y_{k+1} = y_k + 2h$, so $y_k = 1 + 2kh = 1 + 2x_k$ exactly. For $h = 0.5$: $y_0, \ldots, y_4 = 1, 2, 3, 4, 5$ at $x = 0, 0.5, 1, 1.5, 2$, and these are the exact values $1 + 2x$; the same is true for $h = 0.25$ (or any other h).

Example 2. $y' = y/x$, $y(1) = 2$ on $x > 0$. The solutions are the lines through the origin ($y = cx$,

since $(y/x)' = 0$), and here $y = 2x$. Euler gives $y_{k+1} = y_k + h\dfrac{y_k}{x_k} = y_k\,\dfrac{x_k + h}{x_k} = y_k\dfrac{x_{k+1}}{x_k}$; if $y_k = 2x_k$ then $y_{k+1} = 2x_{k+1}$, so by induction $y_k = 2x_k$ exactly. For $h = 0.5$: $y_1 = 2 + 0.5 \cdot 2/1 = 3$ $(= 2 \cdot 1.5)$ and $y_2 = 3 + 0.5 \cdot 3/1.5 = 4$ $(= 2 \cdot 2)$; for $h = 0.25$: 2, 2.5, 3, 3.5, 4, again exact. Unlike Example 1, here the slope field is not constant: the slope y/x varies from point to point, but it is constant *along* each solution, which is what matters.

Answer. Euler is exact for solutions that are straight lines ($y'' \equiv 0$), independently of h. Examples: $y' = 2$, $y(0) = 1$ $(y = 1 + 2x)$ and $y' = y/x$, $y(1) = 2$ $(y = 2x)$.

Problem 73-8 (Simmons)

Solution. **Prediction.** Since $f(y) = y(1 - y^2) > 0$ for $0 < y < 1$, a solution starting at $y(0) = 0.1$ is increasing. It cannot cross the equilibrium $y = 1$ (uniqueness), so it stays below 1, is bounded, increasing, and therefore converges to a limit which must be an equilibrium: $y(x) \to 1$ as $x \to \infty$. It starts near the unstable equilibrium 0 and leaves it slowly, then rises along an S-shaped (sigmoid) curve, with inflection where $y^2 = 1/3$, and then creeps up to 1.

Exact solution (for comparison). Let $w = y^{-2}$. Then $w' = -2y^{-3}y' = -2y^{-3}(y - y^3) = -2w + 2$, whose solution is $w = 1 + Ce^{-2x}$; $w(0) = 100$ gives $C = 99$, so

$$y(x) = \left(1 + 99e^{-2x}\right)^{-1/2}, \qquad y(3) = 0.89608.$$

(Inflection at $1 + 99e^{-2x} = 3$, i.e. $x = \frac{1}{2}\log 49.5 = 1.95$.)

Euler. $y_{k+1} = y_k + 0.1\,y_k(1 - y_k^2)$, $y_0 = 0.1$, $k = 0, \ldots, 29$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.0	0.10000	0.09900	0.10000	0.00000	0.000
0.1	0.10990	0.10857	0.11039	0.00049	0.448
0.2	0.12076	0.11900	0.12184	0.00108	0.889
0.3	0.13266	0.13032	0.13443	0.00178	1.322
0.4	0.14569	0.14260	0.14828	0.00259	1.745
0.5	0.15995	0.15586	0.16347	0.00352	2.156
0.6	0.17553	0.17013	0.18013	0.00460	2.553
0.7	0.19255	0.18541	0.19837	0.00582	2.934
0.8	0.21109	0.20168	0.21828	0.00719	3.296
0.9	0.23126	0.21889	0.23998	0.00872	3.634
1.0	0.25315	0.23692	0.26354	0.01039	3.944
1.1	0.27684	0.25562	0.28904	0.01221	4.223
1.2	0.30240	0.27475	0.31653	0.01413	4.463
1.3	0.32987	0.29398	0.34600	0.01613	4.661
1.4	0.35927	0.31290	0.37742	0.01815	4.809
1.5	0.39056	0.33099	0.41069	0.02013	4.901

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
1.6	0.42366	0.34762	0.44564	0.02198	4.931
1.7	0.45842	0.36208	0.48202	0.02360	4.896
1.8	0.49463	0.37361	0.51952	0.02489	4.791
1.9	0.53199	0.38143	0.55774	0.02574	4.616
2.0	0.57014	0.38481	0.59621	0.02607	4.373
2.1	0.60862	0.38318	0.63442	0.02580	4.067
2.2	0.64693	0.37618	0.67184	0.02491	3.708
2.3	0.68455	0.36376	0.70797	0.02342	3.308
2.4	0.72093	0.34624	0.74232	0.02139	2.882
2.5	0.75555	0.32424	0.77451	0.01895	2.447
2.6	0.78798	0.29872	0.80422	0.01625	2.020
2.7	0.81785	0.27081	0.83127	0.01343	1.615
2.8	0.84493	0.24173	0.85558	0.01065	1.245
2.9	0.86910	0.21264	0.87715	0.00805	0.918
3.0	0.89036	–	0.89608	0.00571	0.638

What happens. The numerical solution does exactly what the true solution does, qualitatively: it increases monotonically from 0.1 and approaches 1 from below, reaching $y_{30} = 0.89036$ at $x = 3$ (exact: 0.89608, error 0.6%). It lags behind the exact curve, with the largest discrepancy, 0.0261, at $x = 2.0$ (shortly after the inflection point where the exact solution is steepest). The reason there is no pathology: the Euler map $y \mapsto y + h\,y(1 - y^2)$ has the same fixed points $0, \pm 1$ as the differential equation, and their stability is preserved when h is small: the multiplier at $y = 0$ is $1 + h = 1.1 > 1$ (unstable) and at $y = 1$ it is $1 + h(1 - 3) = 1 - 2h = 0.8$, of absolute value < 1 (stable). *Caution:* this holds only for $h < 1$. For larger steps the multiplier $1 - 2h$ has $|1 - 2h| \geq 1$, and the numerical solution misbehaves although the exact one is unchanged: for $h = 1.2$ the iterates settle into a two-cycle ($\approx 0.731, 1.140$) instead of converging to 1, and for $h = 2.1$ they blow up (after 12 steps they are of size 10^{25}). With $h = 0.1$ nothing of the sort happens.

Answer. Predicted: $y \uparrow 1$ monotonically (S-shaped curve, $y \to 1$). Euler with $h = 0.1$ reproduces this: $y_{30} = 0.89036$ at $x = 3$ versus exact 0.89608, rising monotonically toward 1 (stable for $h < 1$).

Section 74 Errors

Conventions. For the Euler method $y_{k+1} = y_k + hf(x_k, y_k)$ the *total discretization error* at x_N is $y(x_N) - y_N$ (we give it with its sign; its absolute value is the "total error"), and the relative error is $E_N = |y(x_N) - y_N| / |y(x_N)|$. The text's heuristic bound is

$$|y(x_n) - y_n| \le (x_n - x_0)\,\frac{Mh}{2}, \qquad M \ge \max_{x_0 \le x \le x_n} |y''(x)|,$$

obtained from the local errors $\tfrac{1}{2}y''(\xi)h^2$ with $n = (x_n - x_0)/h$ steps. In Problems 1–5 the exact solutions are those found in Section 73 (each is re-derived there); we compute the Euler iterates for $h = 0.2$ (5 steps) and $h = 0.1$ (10 steps), tabulate them against the exact solution and estimate the total error at $x = 1$ as $y(1) - y_N$, which we also compare with the bound above. (Values: 40-digit arithmetic, displayed to five decimals; see the script `verify/sec74.py`.)

Problem 74-1 (Simmons)

Solution. Exact solution (Section 73, Problem 1): $y = \tfrac{3}{2}e^{2x} - x - \tfrac{1}{2}$, $y(1) = \tfrac{3}{2}e^2 - \tfrac{3}{2} = 9.58358$. Euler: $y_{k+1} = y_k + h(2x_k + 2y_k)$.

$h = 0.2$ ($y_{k+1} = 1.4y_k + 0.4x_k$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	2.00000	1.00000	0.00000	0.000
0.20	1.40000	3.20000	1.53774	0.13774	8.957
0.40	2.04000	4.88000	2.43831	0.39831	16.336
0.60	3.01600	7.23200	3.88018	0.86418	22.272
0.80	4.46240	10.52480	6.12955	1.66715	27.199
1.00	6.56736	–	9.58358	3.01622	31.473

$h = 0.1$ ($y_{k+1} = 1.2y_k + 0.2x_k$):

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	2.00000	1.00000	0.00000	0.000
0.10	1.20000	2.60000	1.23210	0.03210	2.606
0.20	1.46000	3.32000	1.53774	0.07774	5.055
0.30	1.79200	4.18400	1.93318	0.14118	7.303
0.40	2.21040	5.22080	2.43831	0.22791	9.347
0.50	2.73248	6.46496	3.07742	0.34494	11.209
0.60	3.37898	7.95795	3.88018	0.50120	12.917
0.70	4.17477	9.74954	4.88280	0.70803	14.500
0.80	5.14973	11.89945	6.12955	0.97982	15.985
0.90	6.33967	14.47934	7.67447	1.33480	17.393
1.00	7.78760	–	9.58358	1.79598	18.740

The last column (relative error) and the column $y(x_k) - y_k$ give the total error at every node; at $x = 1$:

h	x_N	y_N	exact $y(x_N)$	total error $y(x_N) - y_N$	E_N (%)	bound $\frac{Mh}{2}(x_N - x_0)$	error/bound
0.2	1.0	6.56736	9.58358	3.01622	31.47	4.433	0.680
0.1	1.0	7.78760	9.58358	1.79598	18.74	2.217	0.810

Bound. $y'' = 2 + 2y' = 6e^{2x}$ (since $y' = 3e^{2x} - 1$), which is increasing, so $M = 6e^2 = 44.334$ on $[0, 1]$ and the bound is $\frac{Mh}{2} = 3e^2 h = 22.167\,h$. The actual errors, 3.016 and 1.796, are below the

bounds 4.433 and 2.217 (about 68% and 81% of the bound): the bound is fairly sharp here and, as predicted, halving h roughly halves the error (3.016 → 1.796).

Answer. Total discretization error at $x = 1$: $y(1) - y_N = 3.016$ for $h = 0.2$ (31.5%) and 1.796 for $h = 0.1$ (18.7%); bounds 4.433 and 2.217.

Problem 74-2 (Simmons)

Solution. Exact solution (Section 73, Problem 2): $y = \sqrt{2x+1}$, $y(1) = \sqrt{3} = 1.73205$. Euler: $y_{k+1} = y_k + h/y_k$.

$h = 0.2$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	1.00000	1.00000	0.00000	0.000
0.20	1.20000	0.83333	1.18322	−0.01678	1.419
0.40	1.36667	0.73171	1.34164	−0.02503	1.865
0.60	1.51301	0.66093	1.48324	−0.02977	2.007
0.80	1.64520	0.60783	1.61245	−0.03274	2.031
1.00	1.76676	–	1.73205	−0.03471	2.004

$h = 0.1$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	1.00000	1.00000	1.00000	0.00000	0.000
0.10	1.10000	0.90909	1.09545	−0.00455	0.416
0.20	1.19091	0.83969	1.18322	−0.00769	0.650
0.30	1.27488	0.78439	1.26491	−0.00997	0.788
0.40	1.35332	0.73892	1.34164	−0.01168	0.870
0.50	1.42721	0.70067	1.41421	−0.01300	0.919
0.60	1.49728	0.66788	1.48324	−0.01404	0.946
0.70	1.56406	0.63936	1.54919	−0.01487	0.960
0.80	1.62800	0.61425	1.61245	−0.01555	0.964
0.90	1.68943	0.59192	1.67332	−0.01611	0.962
1.00	1.74862	–	1.73205	−0.01657	0.956

At $x = 1$:

h	x_N	y_N	exact $y(x_N)$	total error $y(x_N) - y_N$	E_N (%)	bound $\frac{Mh}{2}(x_N - x_0)$	error/bound
0.2	1.0	1.76676	1.73205	−0.03471	2.00	0.100	0.347
0.1	1.0	1.74862	1.73205	−0.01657	0.96	0.050	0.331

Bound. $y'' = -y'/y^2 = -1/y^3 = -(2x+1)^{-3/2}$, whose absolute value decreases from 1 at $x = 0$; so $M = 1$ and the bound is $h/2$: 0.1 and 0.05. The (negative, because Euler overshoots a concave solution) errors −0.03471 and −0.01657 are about a third of the bound: here the bound is conservative because $|y''|$ is 1 only at $x = 0$ and is only 0.19 at $x = 1$.

Answer. $y(1) - y_N = -0.03471$ for $h = 0.2$ (2.0%) and −0.01657 for $h = 0.1$ (0.96%); bounds 0.1 and 0.05.

Problem 74-3 (Simmons)

Solution. Exact solution (Section 73, Problem 3): $y = -\log(1-x)$ for $0 \le x < 1$; it tends to $+\infty$ as $x \to 1^-$, so $y(1)$ *does not exist* and the total discretization error at $x = 1$ is not defined (any finite Euler value is infinitely far from a nonexistent value; the bound is useless too, because $y'' = e^{2y} = (1-x)^{-2}$ is unbounded on $[0,1)$). What can be done is to estimate the error on an interval $[0, x_N]$ with $x_N < 1$ common to both grids; we take $x_N = 0.8$ ($N = 4$ for $h = 0.2$, $N = 8$ for $h = 0.1$). Euler: $y_{k+1} = y_k + he^{y_k}$.

$h = 0.2$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.20	0.20000	1.22140	0.22314	0.02314	10.372
0.40	0.44428	1.55937	0.51083	0.06655	13.027
0.60	0.75615	2.13007	0.91629	0.16014	17.477
0.80	1.18217	3.26144	1.60944	0.42727	26.548
1.00	1.83446	–	$+\infty$	–	–

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10000	1.10517	0.10536	0.00536	5.088
0.20	0.21052	1.23432	0.22314	0.01263	5.658
0.30	0.33395	1.39647	0.35667	0.02273	6.372
0.40	0.47360	1.60576	0.51083	0.03723	7.288
0.50	0.63417	1.88546	0.69315	0.05898	8.508
0.60	0.82272	2.27668	0.91629	0.09357	10.212
0.70	1.05039	2.85875	1.20397	0.15359	12.757
0.80	1.33626	3.80479	1.60944	0.27318	16.973
0.90	1.71674	5.56635	2.30259	0.58585	25.443
1.00	2.27337	–	$+\infty$	–	–

At $x = 0.8$ (exact value $-\log 0.2 = \log 5 = 1.60944$):

h	x_N	y_N	exact $y(x_N)$	total error $y(x_N) - y_N$	E_N (%)	bound $\frac{Mh}{2}(x_N - x_0)$	error/bound
0.2	0.8	1.18217	1.60944	0.42727	26.55	2.000	0.214
0.1	0.8	1.33626	1.60944	0.27318	16.97	1.000	0.273

Bound on $[0, 0.8]$**.** $y'' = e^{2y} = (1-x)^{-2} \le 25$ there, so $M = 25$ and the bound is $0.8 \cdot 25h/2 = 10h$: 2.0 and 1.0; the actual errors 0.4273 and 0.2732 satisfy it. Note how the error blows up as $x \to 1$: at $x = 0.9$ the relative error for $h = 0.1$ is already 25.4% (table), because the solution grows without bound.

Answer. $y(1)$ does not exist ($y = -\log(1-x) \to \infty$), so the error at $x = 1$ is infinite/undefined and the bound ($M = \sup y'' = \infty$) is meaningless. On $[0, 0.8]$: errors 0.4273 ($h = 0.2$) and 0.2732 ($h = 0.1$), within the bounds 2.0 and 1.0.

Problem 74-4 (Simmons)

Solution. Exact solution (Section 73, Problem 4): $y = \frac{1}{2}(\sin x + \cos x) - \frac{3}{2}e^x$, $y(1) = -3.38654$. Euler: $y_{k+1} = y_k + h(y_k - \sin x_k)$.

$h = 0.2$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	−1.00000	−1.00000	−1.00000	0.00000	0.000
0.20	−1.20000	−1.39867	−1.24274	−0.04274	3.439
0.40	−1.47973	−1.86915	−1.58250	−0.10276	6.494
0.60	−1.85356	−2.41821	−2.03819	−0.18462	9.058
0.80	−2.33721	−3.05456	−2.63128	−0.29407	11.176
1.00	−2.94812	–	−3.38654	−0.43842	12.946

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	−1.00000	−1.00000	−1.00000	0.00000	0.000
0.10	−1.10000	−1.19983	−1.11034	−0.01034	0.931
0.20	−1.21998	−1.41865	−1.24274	−0.02275	1.831
0.30	−1.36185	−1.65737	−1.39936	−0.03751	2.681
0.40	−1.52759	−1.91700	−1.58250	−0.05491	3.470
0.50	−1.71929	−2.19871	−1.79458	−0.07529	4.196
0.60	−1.93916	−2.50380	−2.03819	−0.09903	4.859
0.70	−2.18954	−2.83375	−2.31610	−0.12656	5.464
0.80	−2.47291	−3.19027	−2.63128	−0.15837	6.019
0.90	−2.79194	−3.57527	−2.98694	−0.19500	6.528
1.00	−3.14947	–	−3.38654	−0.23707	7.000

At $x = 1$:

h	x_N	y_N	exact $y(x_N)$	total error $y(x_N) - y_N$	E_N (%)	bound $\frac{Mh}{2}(x_N - x_0)$	error/bound
0.2	1.0	−2.94812	−3.38654	−0.43842	12.95	0.477	0.919
0.1	1.0	−3.14947	−3.38654	−0.23707	7.00	0.238	0.994

Bound. $y'' = y' - \cos x = -\frac{3}{2}e^x - \frac{1}{2}(\sin x + \cos x)$, so $|y''| = \frac{3}{2}e^x + \frac{1}{2}(\sin x + \cos x)$, which is increasing on $[0, 1]$ (its derivative $\frac{3}{2}e^x + \frac{1}{2}(\cos x - \sin x) > 1.5 - 0.5 > 0$). Hence $M = \frac{3}{2}e + \frac{1}{2}(\sin 1 + \cos 1) = 4.7683$ and the bound is $2.3842\,h$: 0.4768 and 0.2384. The actual errors, −0.4384 and −0.2371, are 92% and 99% of the bound: the bound is almost attained, since $|y''|$ is near its maximum for most of the interval.

Answer. $y(1) - y_N = -0.4384$ for $h = 0.2$ (12.9%) and −0.2371 for $h = 0.1$ (7.0%); bounds 0.4768 and 0.2384.

Problem 74-5 (Simmons)

Solution. Exact solution (Section 73, Problem 5): $y = 1 - x + \tan(x - \pi/4)$, $y(1) = \tan(1 - \pi/4) = 0.21796$. Euler: $y_{k+1} = y_k + h(x_k + y_k - 1)^2$.

$h = 0.2$**:**

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.20	0.20000	0.36000	0.13709	−0.06291	45.891
0.40	0.27200	0.10758	0.19431	−0.07769	39.979
0.60	0.29352	0.01134	0.21245	−0.08107	38.159
0.80	0.29578	0.00917	0.21460	−0.08118	37.829
1.00	0.29762	–	0.21796	−0.07966	36.549

$h = 0.1$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.00	0.00000	1.00000	0.00000	0.00000	–
0.10	0.10000	0.64000	0.08237	−0.01763	21.402
0.20	0.16400	0.40450	0.13709	−0.02691	19.630
0.30	0.20445	0.24557	0.17251	−0.03194	18.516
0.40	0.22901	0.13764	0.19431	−0.03469	17.854
0.50	0.24277	0.06617	0.20659	−0.03618	17.512
0.60	0.24939	0.02268	0.21245	−0.03694	17.387
0.70	0.25166	0.00234	0.21439	−0.03726	17.380
0.80	0.25189	0.00269	0.21460	−0.03729	17.375
0.90	0.25216	0.02315	0.21511	−0.03705	17.225
1.00	0.25447	–	0.21796	−0.03652	16.753

At $x = 1$:

h	x_N	y_N	exact $y(x_N)$	total error $y(x_N) - y_N$	E_N (%)	bound $\frac{Mh}{2}(x_N - x_0)$	error/bound
0.2	1.0	0.29762	0.21796	−0.07966	36.55	0.400	0.199
0.1	1.0	0.25447	0.21796	−0.03652	16.75	0.200	0.183

Bound. With $u = x + y - 1 = \tan(x - \pi/4)$ we have $y' = u^2$ and $y'' = 2u\,u' = 2u(1 + u^2)$ (since $u' = 1 + u^2$). On $[0, 1]$, u increases from -1 to $\tan(1 - \pi/4) = 0.218$ and $g(u) = 2u + 2u^3$ is increasing, so $|y''| \le \max(|g(-1)|, |g(0.218)|) = \max(4, 0.45) = 4$; $M = 4$ (attained at $x = 0$), bound $2h$: 0.4 and 0.2. The actual errors, -0.0797 and -0.0365, are only about 20% of the bound, because $|y''|$ drops from 4 to about 0.45 along the interval.

Answer. $y(1) - y_N = -0.0797$ for $h = 0.2$ (36.6%) and -0.0365 for $h = 0.1$ (16.8%); bounds 0.4 and 0.2.

Problem 74-6 (Simmons)

Solution. **Exact solution.** Integrating, $y = -\dfrac{\cos 3\pi x}{3\pi} + c$, and $y(0) = 0$ gives $c = \dfrac{1}{3\pi}$:

$$y(x) = \frac{1 - \cos 3\pi x}{3\pi}, \qquad y(1) = \frac{1 - \cos 3\pi}{3\pi} = \frac{2}{3\pi} = 0.21221.$$

Since $y' = \sin 3\pi x$, y increases on $[0, \frac{1}{3}]$ (maximum $\frac{2}{3\pi}$ at $x = \frac{1}{3}$), decreases on $[\frac{1}{3}, \frac{2}{3}]$ back to 0, and increases again on $[\frac{2}{3}, 1]$ to $\frac{2}{3\pi}$ at $x = 1$.

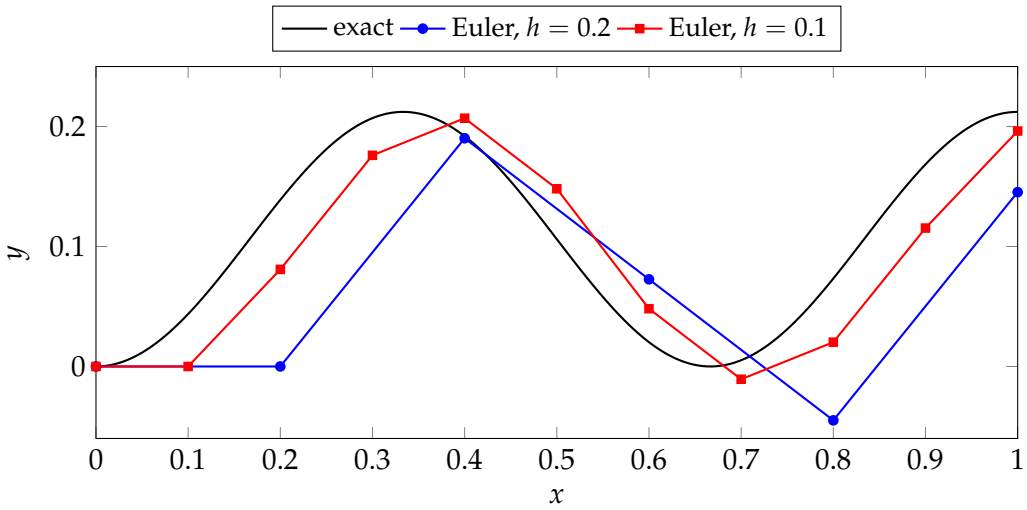
Euler iterates. $y_{k+1} = y_k + h\sin(3\pi x_k)$.

$h = 0.2$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.0	0.00000	0.00000	0.00000	0.00000	–
0.2	0.00000	0.95106	0.13889	0.13889	100.000
0.4	0.19021	−0.58779	0.19194	0.00173	0.902
0.6	0.07265	−0.58779	0.02026	−0.05239	258.540
0.8	−0.04490	0.95106	0.07332	0.11822	161.246
1.0	0.14531	–	0.21221	0.06690	31.525

$h = 0.1$:

x_k	y_k	$f(x_k, y_k)$	exact $y(x_k)$	$y(x_k) - y_k$	E_k (%)
0.0	0.00000	0.00000	0.00000	0.00000	–
0.1	0.00000	0.80902	0.04374	0.04374	100.000
0.2	0.08090	0.95106	0.13889	0.05799	41.752
0.3	0.17601	0.30902	0.20701	0.03101	14.978
0.4	0.20691	−0.58779	0.19194	−0.01497	7.797
0.5	0.14813	−1.00000	0.10610	−0.04203	39.610
0.6	0.04813	−0.58779	0.02026	−0.02787	137.518
0.7	−0.01065	0.30902	0.00519	0.01584	305.043
0.8	0.02025	0.95106	0.07332	0.05306	72.375
0.9	0.11536	0.80902	0.16847	0.05311	31.525
1.0	0.19626	–	0.21221	0.01595	7.514



Discussion of the sketch. For $h = 0.2$ there are only 3.33 steps per period (2/3) of the slope function, and the polygon does not resemble the solution: the first step uses the slope $\sin 0 = 0$, so $y_1 = 0$ while $y(0.2) = 0.13889$; the value at $x = 0.4$ (0.19021 against 0.19194) is close only by coincidence, because errors of opposite sign compensate; at $x = 0.6$ and $x = 0.8$ the values 0.07265 and -0.04490 are far from 0.02026 and 0.07332 (the polygon even becomes negative, although the exact solution is ≥ 0), and the final value is $y_5 = 0.14531$ (error 0.0669, 31.5% of $y(1)$). For $h = 0.1$ (6.7 steps per period) the polygon has the right shape (two humps) but is displaced: it lags on the rising parts, overshoots on the falling parts, and ends at $y_{10} = 0.19626$ (error 0.01595, 7.5%). (The large relative errors in the tables near $x = 0.6$–0.7 are due to the exact solution being close to 0 there.) Euler always uses the slope at the *left* end of the interval, which, for a rapidly oscillating slope, is a poor representative of the slope over the interval; the step size must be small compared with the period.

Step size guaranteeing a total error 0.01**.** Here $y'' = 3\pi \cos 3\pi x$, so $|y''| \leq M = 3\pi = 9.4248$ on $[0, 1]$ (attained at $x = 0, \frac{1}{3}, \frac{2}{3}, 1$), and the bound at $x = 1$ is

$$|y(1) - y_N| \leq (1-0)\frac{Mh}{2} = \frac{3\pi h}{2} \leq 0.01 \iff h \leq \frac{0.02}{3\pi} = 0.0021221.$$

So $h = 0.002$ ($N = 500$ steps) is sufficient (any $h \leq 0.00212$ would be).

Applying Euler with $h = 0.002$**.** The sum can be done in closed form: with $\theta = 3\pi h$ and $N\theta = 3\pi$, $\sum_{k=0}^{N-1} \sin k\theta = \dfrac{\sin(N\theta/2)\sin((N-1)\theta/2)}{\sin(\theta/2)} = \dfrac{(-1)\cdot(-\cos(\theta/2))}{\sin(\theta/2)} = \cot\dfrac{\theta}{2}$, so

$$y_N = h \cot \frac{3\pi h}{2} \qquad (Nh = 1).$$

(Check: $h = 0.2$ gives $0.2\cot 0.3\pi = 0.14531$ and $h = 0.1$ gives $0.1\cot 0.15\pi = 0.19626$, matching the tables.) For $h = 0.002$: $y_{500} = 0.002\cot(0.003\pi) = 0.21220$ against the exact value 0.21221; the error is 6.28×10^{-6}. Selected iterates:

x_k	y_k	exact $y(x_k)$	$y(x_k) - y_k$
0.10	0.042927	0.043737	0.0008103
0.20	0.137936	0.138891	0.0009552
0.30	0.206698	0.207014	0.0003151
0.40	0.192525	0.191943	−0.0005821
0.50	0.107100	0.106103	−0.0009969
0.60	0.020851	0.020264	−0.0005872
0.70	0.004884	0.005193	0.0003092
0.80	0.072362	0.073316	0.0009532
0.90	0.167655	0.168469	0.0008140
1.00	0.212200	0.212207	0.0000063

Why is the step so small? The guaranteed error 0.01 has been met by a factor of about 1600: the actual error, 6.3×10^{-6}, is far below 0.01. Two reasons. First, $M = 3\pi$ is large because the right-hand side oscillates fast, forcing a small h in the bound $3\pi h/2$. Second, and more importantly, the bound assumes the worst case in which all local errors $\frac{1}{2}y''(\xi_k)h^2$ have the same sign and add up; here $y'' = 3\pi\cos 3\pi x$ changes sign, so the local errors largely cancel (the leading term of the total error is $\frac{h}{2}\left(f(1) - f(0)\right) = \frac{h}{2}(\sin 3\pi - \sin 0) = 0$), leaving an error of order h^2: exactly, $\dfrac{2}{3\pi} - h\cot\dfrac{3\pi h}{2} \approx \dfrac{\pi}{2}h^2$ (using $\cot\alpha = \frac{1}{\alpha} - \frac{\alpha}{3} - \cdots$ and $\alpha = \frac{3\pi h}{2}$), which is 0.01 already for $h \approx 0.079$, about 37 times larger than the "guaranteed" step. The bound is a pessimistic worst case over all problems with $|y''| \le M$, and cannot know about the cancellation.

Answer. $y = \dfrac{1 - \cos 3\pi x}{3\pi}$. Euler: $y_5 = 0.14531$ ($h = 0.2$), $y_{10} = 0.19626$ ($h = 0.1$) versus $y(1) = 0.21221$. With $M = 3\pi$, $h \le 0.02/(3\pi) = 0.00212$, e.g. $h = 0.002$; Euler then gives 0.21220 (error 6.3×10^{-6}). The step is small because the worst-case bound $3\pi h/2$ ignores the sign cancellation of the local errors (actual error $\approx \frac{\pi}{2}h^2$).

Section 75 An Improvement to Euler

Conventions. The improved Euler (Heun) method for $y' = f(x,y)$, $y(x_0) = y_0$, step h, $x_k = x_0 + kh$, is

$$z_{k+1} = y_k + h f(x_k, y_k), \qquad y_{k+1} = y_k + \frac{h}{2}\Big[f(x_k,y_k) + f(x_{k+1}, z_{k+1})\Big].$$

In the tables the columns $f(x_k,y_k)$, z_{k+1}, $f(x_{k+1},z_{k+1})$ are the numbers entering the step from row k to row $k+1$. The relative error is $E_k = |y(x_k) - y_k|/|y(x_k)|$. The exact solutions are those of Section 73 (re-derived there). All numbers were computed with 40 digits and are shown to five decimals (the book's own Tables 1–3 of Section 75 are reproduced in `verify/sec75.py`). For each problem we show the complete iteration for $h = 0.1$ and $h = 0.05$ and every tenth iterate for $h = 0.01$, and compare the value at $x = 1$ with the Euler method of Section 73.

Problem 75-1 (Simmons)

Solution. Exact: $y = \frac{3}{2}e^{2x} - x - \frac{1}{2}$, $y(1) = 9.58358$. Here $f(x,y) = 2x + 2y$, so $z_{k+1} = y_k + h(2x_k + 2y_k)$ and $y_{k+1} = y_k + h\big[(x_k + y_k) + (x_{k+1} + z_{k+1})\big]$.

$h = 0.1$:

x_k	y_k	$f(x_k,y_k)$	z_{k+1}	$f(x_{k+1},z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	2.00000	1.20000	2.60000	1.00000	0.000
0.10	1.23000	2.66000	1.49600	3.39200	1.23210	0.171
0.20	1.53260	3.46520	1.87912	4.35824	1.53774	0.334
0.30	1.92377	4.44754	2.36853	5.53705	1.93318	0.487
0.40	2.42300	5.64600	2.98760	6.97520	2.43831	0.628
0.50	3.05406	7.10812	3.76487	8.72975	3.07742	0.759
0.60	3.84596	8.89191	4.73515	10.87029	3.88018	0.882
0.70	4.83407	11.06813	5.94088	13.48176	4.88280	0.998
0.80	6.06156	13.72312	7.43387	16.66775	6.12955	1.109
0.90	7.58110	16.96221	9.27733	20.55465	7.67447	1.217
1.00	9.45695	–	–	–	9.58358	1.321

$h = 0.05$:

x_k	y_k	$f(x_k,y_k)$	z_{k+1}	$f(x_{k+1},z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	2.00000	1.10000	2.30000	1.00000	0.000
0.05	1.10750	2.31500	1.22325	2.64650	1.10776	0.023
0.10	1.23154	2.66308	1.36469	3.02938	1.23210	0.046
0.15	1.37385	3.04770	1.52623	3.45247	1.37479	0.068
0.20	1.53635	3.47271	1.70999	3.91998	1.53774	0.090
0.25	1.72117	3.94234	1.91829	4.43657	1.72308	0.111
0.30	1.93064	4.46129	2.15371	5.00741	1.93318	0.131
0.35	2.16736	5.03472	2.41910	5.63819	2.17063	0.151
0.40	2.43418	5.66837	2.71760	6.33520	2.43831	0.169
0.45	2.73427	6.36855	3.05270	7.10540	2.73940	0.187
0.50	3.07112	7.14224	3.42823	7.95647	3.07742	0.205

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.55	3.44859	7.99718	3.84845	8.89690	3.45625	0.222
0.60	3.87094	8.94188	4.31803	9.93607	3.88018	0.238
0.65	4.34289	9.98578	4.84218	11.08436	4.35395	0.254
0.70	4.86964	11.13929	5.42661	12.35321	4.88280	0.269
0.75	5.45696	12.41391	6.07765	13.75530	5.47253	0.285
0.80	6.11119	13.82237	6.80230	15.30461	6.12955	0.300
0.85	6.83936	15.37872	7.60830	17.01659	6.86092	0.314
0.90	7.64924	17.09849	8.50417	18.90834	7.67447	0.329
0.95	8.54941	18.99883	9.49936	20.99871	8.57884	0.343
1.00	9.54935	–	–	–	9.58358	0.357

$h = 0.01$ (100 steps, every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	2.00000	1.02000	2.06000	1.00000	0.000
0.10	1.23208	2.66416	1.25872	2.73744	1.23210	0.002
0.20	1.53768	3.47536	1.57243	3.56486	1.53774	0.004
0.30	1.93307	4.46614	1.97773	4.57546	1.93318	0.006
0.40	2.43814	5.67627	2.49490	5.80980	2.43831	0.007
0.50	3.07715	7.15431	3.14870	7.31740	3.07742	0.009
0.60	3.87978	8.95957	3.96938	9.15876	3.88018	0.010
0.70	4.88224	11.16448	4.99389	11.40777	4.88280	0.011
0.80	6.12877	13.85754	6.26734	14.15469	6.12955	0.013
0.90	7.67340	17.14680	7.84487	17.50973	7.67447	0.014
1.00	9.58213	–	–	–	9.58358	0.015

Summary and comparison with Euler at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^2$
0.1	10	9.45695	9.58358	0.12664	1.32	12.664
0.05	20	9.54935	9.58358	0.03423	0.36	13.693
0.01	100	9.58213	9.58358	0.00146	0.02	14.557

h	Euler y_N	improved Euler y_N	exact $y(1)$	Euler error	improved Euler error
0.1	7.78760	9.456947	9.583584	1.79598	1.266×10^{-1}
0.05	8.59125	9.549352	9.583584	0.99233	3.423×10^{-2}
0.01	9.36697	9.582128	9.583584	0.21661	1.456×10^{-3}

Discussion. The improved Euler estimates 9.45695, 9.54935, 9.58213 have relative errors 1.32%, 0.36%, 0.015%, compared with 18.7%, 10.4%, 2.3% for Euler. The error is proportional to h^2: halving h divides it by about 3.7, and (error)$/h^2$ increases slowly (12.7, 13.7, 14.6) towards about 15. For the same step $h = 0.1$ the error is reduced by a factor of 14; Heun with $h = 0.1$ is better than Euler with $h = 0.01$ (error 0.1266 against 0.2166), at a cost of twice as many evaluations of f per step (20 against 100).

Answer. Improved Euler: $y_N(1) = 9.45695,\ 9.54935,\ 9.58213$ for $h = 0.1,\ 0.05,\ 0.01$ (errors 0.1266, 0.0342, 0.0015; 1.3%, 0.36%, 0.015%); exact $y(1) = 9.58358$; Euler gave 7.78760, 8.59125, 9.36697.

Problem 75-2 (Simmons)

Solution. Exact: $y = \sqrt{2x+1}$, $y(1) = \sqrt{3} = 1.73205$. Here $z_{k+1} = y_k + h/y_k$, $y_{k+1} = y_k + \frac{h}{2}\left[1/y_k + 1/z_{k+1}\right]$.

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	1.00000	1.10000	0.90909	1.00000	0.000
0.10	1.09545	0.91286	1.18674	0.84264	1.09545	0.001
0.20	1.18323	0.84514	1.26774	0.78880	1.18322	0.001
0.30	1.26493	0.79056	1.34398	0.74406	1.26491	0.001
0.40	1.34166	0.74535	1.41619	0.70612	1.34164	0.001
0.50	1.41423	0.70710	1.48494	0.67343	1.41421	0.001
0.60	1.48326	0.67419	1.55068	0.64488	1.48324	0.001
0.70	1.54921	0.64549	1.61376	0.61967	1.54919	0.001
0.80	1.61247	0.62017	1.67449	0.59720	1.61245	0.001
0.90	1.67334	0.59761	1.73310	0.57700	1.67332	0.001
1.00	1.73207	–	–	–	1.73205	0.001

$h = 0.05$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	1.00000	1.05000	0.95238	1.00000	0.000
0.05	1.04881	0.95346	1.09648	0.91201	1.04881	0.000
0.10	1.09545	0.91287	1.14109	0.87636	1.09545	0.000
0.15	1.14018	0.87706	1.18403	0.84457	1.14018	0.000
0.20	1.18322	0.84515	1.22548	0.81601	1.18322	0.000
0.25	1.22475	0.81650	1.26557	0.79016	1.22474	0.000
0.30	1.26491	0.79057	1.30444	0.76661	1.26491	0.000
0.35	1.30384	0.76696	1.34219	0.74505	1.30384	0.000
0.40	1.34164	0.74535	1.37891	0.72521	1.34164	0.000
0.45	1.37841	0.72548	1.41468	0.70687	1.37840	0.000
0.50	1.41422	0.70711	1.44957	0.68986	1.41421	0.000

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.55	1.44914	0.69006	1.48364	0.67402	1.44914	0.000
0.60	1.48324	0.67420	1.51695	0.65922	1.48324	0.000
0.65	1.51658	0.65938	1.54955	0.64535	1.51658	0.000
0.70	1.54920	0.64550	1.58147	0.63232	1.54919	0.000
0.75	1.58114	0.63245	1.61276	0.62005	1.58114	0.000
0.80	1.61245	0.62017	1.64346	0.60847	1.61245	0.000
0.85	1.64317	0.60858	1.67360	0.59751	1.64317	0.000
0.90	1.67332	0.59761	1.70320	0.58713	1.67332	0.000
0.95	1.70294	0.58722	1.73230	0.57727	1.70294	0.000
1.00	1.73205	–	–	–	1.73205	0.000

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	1.00000	1.00000	1.01000	0.99010	1.00000	0.000
0.10	1.09545	0.91287	1.10457	0.90533	1.09545	0.000
0.20	1.18322	0.84515	1.19167	0.83916	1.18322	0.000
0.30	1.26491	0.79057	1.27282	0.78566	1.26491	0.000
0.40	1.34164	0.74536	1.34909	0.74124	1.34164	0.000
0.50	1.41421	0.70711	1.42128	0.70359	1.41421	0.000
0.60	1.48324	0.67420	1.48998	0.67115	1.48324	0.000
0.70	1.54919	0.64550	1.55565	0.64282	1.54919	0.000
0.80	1.61245	0.62017	1.61865	0.61780	1.61245	0.000
0.90	1.67332	0.59761	1.67930	0.59549	1.67332	0.000
1.00	1.73205	–	–	–	1.73205	0.000

Summary and comparison with Euler at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^2$
0.1	10	1.73207	1.73205	-0.00002	0.00	-0.002
0.05	20	1.73205	1.73205	0.00000	0.00	-0.001
0.01	100	1.73205	1.73205	0.00000	0.00	0.000

h	Euler y_N	improved Euler y_N	exact $y(1)$	Euler error	improved Euler error
0.1	1.74862	1.732068	1.732051	-0.01657	-1.709×10^{-5}
0.05	1.74015	1.732053	1.732051	-0.00810	-2.074×10^{-6}
0.01	1.73364	1.732051	1.732051	-0.00159	-1.615×10^{-8}

Discussion. The improved Euler values 1.732068, 1.732053, 1.732051 already agree with $\sqrt{3} = 1.732051$ to 4–6 digits, errors -1.7×10^{-5}, -2.1×10^{-6}, -1.6×10^{-8}, versus -1.7×10^{-2}, -8.1×10^{-3}, -1.6×10^{-3} for Euler. Interestingly the error here decreases like h^3 rather than h^2 (ratios 8.2 and $128 \approx 5^3$): for $y' = 1/y$, $f_y y''/6 = D/12$ with $D = f_{xx} + 2f f_{xy} + f^2 f_{yy}$ (both equal $1/(6y^5)$), so the h^3 term of Heun's local error (see Problem 6 below) cancels identically.

Answer. Improved Euler: $y_N(1) = 1.732068,\ 1.732053,\ 1.732051$ ($h = 0.1,\ 0.05,\ 0.01$); exact $\sqrt{3} = 1.732051$; errors -1.7×10^{-5}, -2.1×10^{-6}, -1.6×10^{-8} (Euler: -0.0166, -0.0081, -0.0016).

Problem 75-3 (Simmons)

Solution. Exact (Section 73): $y = -\log(1 - x)$ for $x < 1$, which tends to $+\infty$ as $x \to 1^-$; *there is no exact solution at* $x = 1$. Here $z_{k+1} = y_k + h e^{y_k}$, $y_{k+1} = y_k + \frac{h}{2}\left[e^{y_k} + e^{z_{k+1}}\right]$.

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.10000	1.10517	0.00000	–
0.10	0.10526	1.11100	0.21636	1.24155	0.10536	0.097
0.20	0.22289	1.24968	0.34785	1.41602	0.22314	0.116
0.30	0.35617	1.42785	0.49896	1.64700	0.35667	0.141
0.40	0.50991	1.66515	0.67643	1.96684	0.51083	0.179
0.50	0.69151	1.99673	0.89119	2.43802	0.69315	0.236
0.60	0.91325	2.49241	1.16249	3.19789	0.91629	0.332
0.70	1.19777	3.31271	1.52904	4.61373	1.20397	0.516
0.80	1.59409	4.92383	2.08647	8.05644	1.60944	0.954
0.90	2.24310	9.42251	3.18535	24.17579	2.30259	2.583
1.00	3.92302	–	–	–	$+\infty$	–

$h = 0.05$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.05000	1.05127	0.00000	–
0.05	0.05128	1.05262	0.10391	1.10950	0.05129	0.022
0.10	0.10533	1.11108	0.16089	1.17455	0.10536	0.024
0.15	0.16248	1.17642	0.22130	1.24769	0.16252	0.027
0.20	0.22308	1.24992	0.28557	1.33053	0.22314	0.029
0.25	0.28759	1.33321	0.35425	1.42511	0.28768	0.032
0.30	0.35655	1.42839	0.42797	1.53414	0.35667	0.036
0.35	0.43061	1.53820	0.50752	1.66117	0.43078	0.040
0.40	0.51060	1.66628	0.59391	1.81105	0.51083	0.045
0.45	0.59753	1.81762	0.68841	1.99055	0.59784	0.052
0.50	0.69273	1.99917	0.79269	2.20933	0.69315	0.060

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.55	0.79795	2.22097	0.90899	2.48182	0.79851	0.070
0.60	0.91552	2.49806	1.04042	2.83040	0.91629	0.085
0.65	1.04873	2.85402	1.19143	3.29178	1.04982	0.104
0.70	1.20237	3.32800	1.36877	3.93052	1.20397	0.133
0.75	1.38383	3.99017	1.58334	4.87121	1.38629	0.177
0.80	1.60537	4.97970	1.85435	6.38757	1.60944	0.253
0.85	1.88955	6.61640	2.22037	9.21075	1.89712	0.399
0.90	2.28523	9.82794	2.77663	16.06474	2.30259	0.754
0.95	2.93255	18.77539	3.87132	48.00553	2.99573	2.109
1.00	4.60207	–	–	–	$+\infty$	–

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.01000	1.01005	0.00000	–
0.10	0.10536	1.11111	0.11647	1.12352	0.10536	0.001
0.20	0.22314	1.25000	0.23564	1.26572	0.22314	0.001
0.30	0.35667	1.42856	0.37096	1.44912	0.35667	0.001
0.40	0.51082	1.66665	0.52748	1.69466	0.51083	0.002
0.50	0.69313	1.99997	0.71313	2.04037	0.69315	0.002
0.60	0.91626	2.49992	0.94126	2.56321	0.91629	0.003
0.70	1.20391	3.33312	1.23724	3.44609	1.20397	0.005
0.80	1.60927	4.99917	1.65926	5.25544	1.60944	0.010
0.90	2.30184	9.99253	2.40176	11.04263	2.30259	0.032
1.00	6.20029	–	–	–	$+\infty$	–

Values at $x = 1$ (no exact value to compare with):

h	Euler y_N	improved Euler y_N	exact $y(1)$	Euler error	improved Euler error
0.1	2.27337	3.92302	$+\infty$	–	–
0.05	2.74934	4.60207	$+\infty$	–	–
0.01	3.95946	6.20029	$+\infty$	–	–

Discussion. For $x < 1$ the improved Euler method follows $-\log(1-x)$ much better than Euler (compare, e.g., $x = 0.9$ with $h = 0.1$: error about 2.6% versus 25% for Euler), but at $x = 1$ it is of no use: 3.92302, 4.60207, 6.20029 increase without bound as h decreases (with $h = 10^{-3}, 10^{-4}$ one gets 8.5004 and 10.8027, against 5.85 and 7.85 for Euler); the numerical solution tracks the blow-up of $-\log(1-x)$ ever more closely, and the value at the last mesh point $x_N = 1$ simply measures how close to the singularity the last step is (for $-\log(1-x)$ at distance h from the singularity this is $\log(1/h)$). Such a failure is not signalled by the method; one has to know that the solution ceases to exist (here by solving the equation exactly).

Answer. Improved Euler gives $3.92302,\ 4.60207,\ 6.20029$ at $x = 1$ ($h = 0.1,\ 0.05,\ 0.01$), but the exact solution $-\log(1-x)$ does not exist at $x = 1$ (blow-up), so these numbers have no meaning and grow without bound as $h \to 0$. (Euler: $2.27337,\ 2.74934,\ 3.95946$.)

Problem 75-4 (Simmons)

Solution. Exact: $y = \frac{1}{2}(\sin x + \cos x) - \frac{3}{2}e^{x}$, $y(1) = -3.38654$. Here $z_{k+1} = y_k + h(y_k - \sin x_k)$, $y_{k+1} = y_k + \frac{h}{2}\left[(y_k - \sin x_k) + (z_{k+1} - \sin x_{k+1})\right]$.

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−1.00000	−1.10000	−1.19983	−1.00000	0.000
0.10	−1.10999	−1.20983	−1.23097	−1.42964	−1.11034	0.031
0.20	−1.24197	−1.44063	−1.38603	−1.68155	−1.24274	0.062
0.30	−1.39807	−1.69359	−1.56743	−1.95685	−1.39936	0.092
0.40	−1.58060	−1.97001	−1.77760	−2.25702	−1.58250	0.120
0.50	−1.79195	−2.27137	−2.01909	−2.58373	−1.79458	0.147
0.60	−2.03470	−2.59935	−2.29464	−2.93886	−2.03819	0.171
0.70	−2.31161	−2.95583	−2.60720	−3.32455	−2.31610	0.194
0.80	−2.62563	−3.34299	−2.95993	−3.74326	−2.63128	0.215
0.90	−2.97995	−3.76327	−3.35627	−4.19774	−2.98694	0.234
1.00	−3.37800	–	–	–	−3.38654	0.252

$h = 0.05$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−1.00000	−1.05000	−1.09998	−1.00000	0.000
0.05	−1.05250	−1.10248	−1.10762	−1.20746	−1.05254	0.004
0.10	−1.11025	−1.21008	−1.17075	−1.32019	−1.11034	0.008
0.15	−1.17350	−1.32294	−1.23965	−1.43832	−1.17365	0.012
0.20	−1.24254	−1.44121	−1.31460	−1.56200	−1.24274	0.016
0.25	−1.31762	−1.56502	−1.39587	−1.69139	−1.31788	0.020
0.30	−1.39903	−1.69455	−1.48375	−1.82665	−1.39936	0.024
0.35	−1.48706	−1.82995	−1.57855	−1.96797	−1.48747	0.028
0.40	−1.58200	−1.97142	−1.68058	−2.11554	−1.58250	0.031
0.45	−1.68418	−2.11914	−1.79014	−2.26956	−1.68476	0.035
0.50	−1.79390	−2.27332	−1.90756	−2.43025	−1.79458	0.038

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.55	−1.91149	−2.43417	−2.03319	−2.59784	−1.91227	0.041
0.60	−2.03729	−2.60193	−2.16738	−2.77257	−2.03819	0.044
0.65	−2.17165	−2.77683	−2.31049	−2.95471	−2.17268	0.047
0.70	−2.31494	−2.95915	−2.46289	−3.14453	−2.31610	0.050
0.75	−2.46753	−3.14917	−2.62499	−3.34234	−2.46884	0.053
0.80	−2.62982	−3.34717	−2.79718	−3.54846	−2.63128	0.056
0.85	−2.80221	−3.55349	−2.97988	−3.76321	−2.80384	0.058
0.90	−2.98513	−3.76845	−3.17355	−3.98696	−2.98694	0.061
0.95	−3.17901	−3.99243	−3.37863	−4.22010	−3.18102	0.063
1.00	−3.38432	–	–	–	−3.38654	0.065

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−1.00000	−1.01000	−1.02000	−1.00000	0.000
0.10	−1.11033	−1.21017	−1.12244	−1.23221	−1.11034	0.000
0.20	−1.24273	−1.44140	−1.25714	−1.46560	−1.24274	0.001
0.30	−1.39935	−1.69487	−1.41629	−1.72135	−1.39936	0.001
0.40	−1.58248	−1.97190	−1.60220	−2.00081	−1.58250	0.001
0.50	−1.79455	−2.27398	−1.81729	−2.30547	−1.79458	0.002
0.60	−2.03815	−2.60279	−2.06418	−2.63705	−2.03819	0.002
0.70	−2.31605	−2.96027	−2.34565	−2.99749	−2.31610	0.002
0.80	−2.63122	−3.34858	−2.66471	−3.38899	−2.63128	0.002
0.90	−2.98686	−3.77019	−3.02456	−3.81407	−2.98694	0.002
1.00	−3.38645	–	–	–	−3.38654	0.003

Summary and comparison with Euler at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^2$
0.1	10	−3.37800	−3.38654	−0.00854	0.25	−0.854
0.05	20	−3.38432	−3.38654	−0.00221	0.07	−0.885
0.01	100	−3.38645	−3.38654	−0.00009	0.00	−0.910

h	Euler y_N	improved Euler y_N	exact $y(1)$	Euler error	improved Euler error
0.1	−3.14947	−3.377996	−3.386536	−0.23707	-8.540×10^{-3}
0.05	−3.26293	−3.384324	−3.386536	−0.12361	-2.212×10^{-3}
0.01	−3.36093	−3.386445	−3.386536	−0.02560	-9.103×10^{-5}

Discussion. Improved Euler: −3.37800, −3.38432, −3.38645 with relative errors 0.25%, 0.065%, 0.0027% (Euler: 7.0%, 3.7%, 0.76%); the error is proportional to h^2 (it divides by 3.9 when h is halved, and (error)$/h^2 \approx -0.9$).

Answer. Improved Euler: $y_N(1) = -3.37800,\ -3.38432,\ -3.38645$ ($h = 0.1,\ 0.05,\ 0.01$); exact −3.38654; errors $-0.0085,\ -0.0022,\ -0.00009$ (Euler: $-3.14947,\ -3.26293,\ -3.36093$).

Problem 75-5 (Simmons)

Solution. Exact: $y = 1 - x + \tan(x - \pi/4)$, $y(1) = \tan(1 - \pi/4) = 0.21796$. Here $z_{k+1} = y_k + h(x_k + y_k - 1)^2$, $y_{k+1} = y_k + \frac{h}{2}\left[(x_k + y_k - 1)^2 + (x_{k+1} + z_{k+1} - 1)^2\right]$.

$h = 0.1$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.10000	0.64000	0.00000	–
0.10	0.08200	0.66912	0.14891	0.42392	0.08237	0.451
0.20	0.13665	0.44003	0.18066	0.26972	0.13709	0.319
0.30	0.17214	0.27864	0.20000	0.16000	0.17251	0.214
0.40	0.19407	0.16478	0.21055	0.08378	0.19431	0.125
0.50	0.20650	0.08614	0.21511	0.03418	0.20659	0.045
0.60	0.21252	0.03515	0.21603	0.00705	0.21245	0.032
0.70	0.21463	0.00729	0.21535	0.00024	0.21439	0.108
0.80	0.21500	0.00023	0.21502	0.01323	0.21460	0.186
0.90	0.21567	0.01338	0.21701	0.04709	0.21511	0.264
1.00	0.21870	–	–	–	0.21796	0.340

$h = 0.05$**:**

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.05000	0.81000	0.00000	–
0.05	0.04525	0.81857	0.08618	0.66231	0.04531	0.141
0.10	0.08227	0.66868	0.11571	0.53919	0.08237	0.120
0.15	0.11247	0.54395	0.13967	0.43604	0.11258	0.103
0.20	0.13697	0.43961	0.15895	0.34934	0.13709	0.088
0.25	0.15669	0.35201	0.17429	0.27637	0.15681	0.074
0.30	0.17240	0.27836	0.18632	0.21500	0.17251	0.062
0.35	0.18474	0.21647	0.19556	0.16357	0.18483	0.050
0.40	0.19424	0.16464	0.20247	0.12078	0.19431	0.040
0.45	0.20137	0.12154	0.20745	0.08559	0.20143	0.030
0.50	0.20655	0.08611	0.21086	0.05719	0.20659	0.020

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.55	0.21013	0.05754	0.21301	0.03497	0.21016	0.011
0.60	0.21245	0.03518	0.21420	0.01844	0.21245	0.001
0.65	0.21379	0.01855	0.21471	0.00727	0.21377	0.008
0.70	0.21443	0.00732	0.21480	0.00124	0.21439	0.018
0.75	0.21465	0.00125	0.21471	0.00022	0.21459	0.027
0.80	0.21468	0.00022	0.21469	0.00419	0.21460	0.037
0.85	0.21479	0.00420	0.21500	0.01323	0.21469	0.047
0.90	0.21523	0.01328	0.21589	0.02752	0.21511	0.057
0.95	0.21625	0.02764	0.21763	0.04736	0.21610	0.066
1.00	0.21812	–	–	–	0.21796	0.076

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	$f(x_k, y_k)$	z_{k+1}	$f(x_{k+1}, z_{k+1})$	exact $y(x_k)$	E_k (%)
0.00	0.00000	1.00000	0.01000	0.96040	0.00000	–
0.10	0.08237	0.66852	0.08905	0.64152	0.08237	0.005
0.20	0.13708	0.43946	0.14148	0.42058	0.13709	0.004
0.30	0.17250	0.27825	0.17529	0.26493	0.17251	0.003
0.40	0.19431	0.16458	0.19596	0.15527	0.19431	0.002
0.50	0.20659	0.08609	0.20745	0.07983	0.20659	0.001
0.60	0.21245	0.03518	0.21280	0.03140	0.21245	0.000
0.70	0.21439	0.00733	0.21447	0.00571	0.21439	0.000
0.80	0.21461	0.00021	0.21461	0.00061	0.21460	0.001
0.90	0.21511	0.01325	0.21524	0.01569	0.21511	0.002
1.00	0.21796	–	–	–	0.21796	0.003

Summary and comparison with Euler at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	$(\text{error})/h^2$
0.1	10	0.21870	0.21796	-0.00074	0.34	-0.074
0.05	20	0.21812	0.21796	-0.00016	0.08	-0.066
0.01	100	0.21796	0.21796	-0.00001	0.00	-0.061

h	Euler y_N	improved Euler y_N	exact $y(1)$	Euler error	improved Euler error
0.1	0.25447	0.218698	0.217958	-0.03652	-7.402×10^{-4}
0.05	0.23552	0.218123	0.217958	-0.01756	-1.650×10^{-4}
0.01	0.22137	0.217964	0.217958	-0.00341	-6.136×10^{-6}

Discussion. Improved Euler: 0.21870, 0.21812, 0.21796 with relative errors 0.34%, 0.076%, 0.0028% (Euler: 16.8%, 8.1%, 1.6%): at $h = 0.1$ the error is reduced by a factor 49; the error behaves like h^2 (it divides by 4.5 when h is halved).

Answer. Improved Euler: $y_N(1) = 0.21870,\ 0.21812,\ 0.21796$ ($h = 0.1,\ 0.05,\ 0.01$); exact 0.21796; errors -7.4×10^{-4}, -1.7×10^{-4}, -6×10^{-6} (Euler: 0.25447, 0.23552, 0.22137).

Problem 75-6 (Simmons)

Solution. The three-term Taylor method is $y_{k+1} = y_k + hy'_k + \frac{h^2}{2}y''_k$ with $y' = f$ and $y'' = f_x + ff_y$, evaluated at (x_k, y_k).

Comparing the local errors. Write $D = f_{xx} + 2ff_{xy} + f^2 f_{yy}$. Expanding $f(x+h,\, y+hf)$ about (x, y) in Heun's formula gives

$$y^{+}_{\text{Heun}} = y + hf + \tfrac{h^2}{2}(f_x + ff_y) + \tfrac{h^3}{4}D + O(h^4),$$

while the exact solution has $y(x+h) = y + hf + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + O(h^4)$ with $y''' = D + f_y\, y''$ (differentiate $y'' = f_x + ff_y$ along the solution). Hence the local errors (exact minus approximate), starting from an exact value, are

$$\varepsilon_{\text{Taylor}} = \frac{h^3}{6}\,y''' = \frac{h^3}{6}\,(D + f_y y''), \qquad \varepsilon_{\text{Heun}} = h^3\Big(\frac{f_y y''}{6} - \frac{D}{12}\Big) + O(h^4).$$

(For $f = f(x)$ alone these are $\frac{h^3}{6}f''$ and $-\frac{h^3}{12}f''$; the latter is the value $-y'''h^3/12$ quoted in the text for Heun's method. The correct local error of the three-term Taylor formula is $y'''h^3/6$, the first neglected term of the Taylor series; the text's "$y'''h^3/12$" for the Taylor method seems to be a misprint.) Both methods have total error $O(h^2)$, so which is better depends on the sizes of these two coefficients, and the Taylor method is better when y''' is small while D is not.

Case 1: solutions with $y''' \equiv 0$ (quadratic polynomials). Then $\varepsilon_{\text{Taylor}} = 0$ and the Taylor method is *exact*, for every h, while Heun's $\varepsilon = h^3(f_y y''/6 - D/12)$ is in general not zero.

(A) $y' = 2y/x$, $y(1) = 1$ (solution $y = x^2$). Here $f_x = -2y/x^2$, $f_y = 2/x$, so $y'' = f_x + ff_y = 2y/x^2$ ($= 2$ on the solution). Taylor, $h = 0.5$: $y_1 = 1 + 0.5\cdot 2 + 0.125\cdot 2 = 2.25 = 1.5^2$ and $y_2 = 2.25 + 0.5\cdot 3 + 0.125\cdot 2 = 4 = 2^2$, exact. Heun, $h = 0.5$: $y_1 = 1 + 0.25(2 + 2\cdot 2/1.5) = 2.16667$, $y_2 = 3.79167$.

x_k	exact	Taylor y_k	improved Euler y_k	exact − improved Euler
1.00	1.00000	1.00000	1.00000	0.00000
1.50	2.25000	2.25000	2.16667	0.08333
2.00	4.00000	4.00000	3.79167	0.20833

With $h = 0.1$ the Taylor values are again exact ($y_{10} = 4.00000$), while Heun gives $y_{10} = 3.98668$:

x_k	exact	Taylor y_k	improved Euler y_k	exact − improved Euler
1.00	1.00000	1.00000	1.00000	0.00000
1.10	1.21000	1.21000	1.20909	0.00091
1.20	1.44000	1.44000	1.43809	0.00191
1.30	1.69000	1.69000	1.68698	0.00302
1.40	1.96000	1.96000	1.95579	0.00421
1.50	2.25000	2.25000	2.24450	0.00550
1.60	2.56000	2.56000	2.55312	0.00688
1.70	2.89000	2.89000	2.88165	0.00835
1.80	3.24000	3.24000	3.23008	0.00992
1.90	3.61000	3.61000	3.59842	0.01158
2.00	4.00000	4.00000	3.98668	0.01332

(B) $y' = y/x + x$, $y(1) = 2$ (solution $y = x^2 + x$, since $(y/x)' = 1$). Here $y'' = f_x + f f_y = (1 - y/x^2) + (y/x + x)/x = 2$ identically; Taylor reproduces $x^2 + x$ exactly for any h, while Heun, $h = 0.5$, gives 3.70833 at $x = 1.5$ and 5.91319 at $x = 2$ instead of 3.75 and 6.

x_k	exact	Taylor y_k	improved Euler y_k	exact − improved Euler
1.00	2.00000	2.00000	2.00000	0.00000
1.50	3.75000	3.75000	3.70833	0.04167
2.00	6.00000	6.00000	5.91319	0.08681

With $h = 0.1$, Heun gives 5.99536 at $x = 2$ (exact 6), Taylor 6.00000 (exact):

x_k	exact	Taylor y_k	improved Euler y_k	exact − improved Euler
1.00	2.00000	2.00000	2.00000	0.00000
1.10	2.31000	2.31000	2.30955	0.00045
1.20	2.64000	2.64000	2.63909	0.00091
1.30	2.99000	2.99000	2.98863	0.00137
1.40	3.36000	3.36000	3.35816	0.00184
1.50	3.75000	3.75000	3.74770	0.00230
1.60	4.16000	4.16000	4.15723	0.00277
1.70	4.59000	4.59000	4.58677	0.00323
1.80	5.04000	5.04000	5.03630	0.00370
1.90	5.51000	5.51000	5.50583	0.00417
2.00	6.00000	6.00000	5.99536	0.00464

Case 2: simple f, cheap derivatives, higher accuracy wanted. When f is simple (polynomial, exponential, trigonometric), so that f_x and f_y are available in closed form, the Taylor step needs only one evaluation of f (plus the cheap closed form of y''), whereas Heun needs two evaluations of f; and the Taylor method can be pushed to higher order simply by adding terms (Problem 8), which for such f gives accuracy that the two-stage Heun formula cannot reach. For the benchmark problem $y' = x + y$ the two methods give identical values (Table 3 of the text equals Table 2), because $D = 0$ and $y''' = y''$ make both local errors equal to $\frac{h^3}{6} y''$.

Answer. The Taylor method is exact (and Heun is not) for every equation whose solutions are quadratic polynomials ($y''' \equiv 0$), e.g. $y' = 2y/x$, $y(1) = 1$ ($y = x^2$) and $y' = y/x + x$, $y(1) = 2$ ($y = x^2 + x$). Local errors: Taylor $\frac{h^3}{6} y'''$, Heun $h^3(\frac{f_y y''}{6} - \frac{D}{12})$.

Problem 75-7 (Simmons)

Solution. **(i) The derivative $y'' = f_x + f f_y$ is unbounded or does not exist at some mesh point.** Take $y' = \sqrt{x}$, $y(0) = 0$, with solution $y = \frac{2}{3} x^{3/2}$. Then $y'' = f_x = \dfrac{1}{2\sqrt{x}}$ is infinite at the starting point $x_0 = 0$, so the three-term formula cannot even take its first step ($y_1 = 0 + 0 + \frac{h^2}{2} \cdot \infty$). Heun's method needs only values of f and runs without difficulty:

x_k	improved Euler y_k	exact $\frac{2}{3}x^{3/2}$	error
0.0	0.00000	0.00000	0.00000
0.1	0.01581	0.02108	0.00527
0.2	0.05398	0.05963	0.00565
0.3	0.10373	0.10954	0.00581
0.4	0.16274	0.16865	0.00592
0.5	0.22972	0.23570	0.00598
0.6	0.30380	0.30984	0.00604
0.7	0.38437	0.39044	0.00608
0.8	0.47092	0.47703	0.00611
0.9	0.56308	0.56921	0.00613
1.0	0.66051	0.66667	0.00616

(its error at $x = 1$ is 0.0062 for $h = 0.1$, 2.0×10^{-4} for $h = 0.01$; it is only $O(h^{3/2})$, not $O(h^2)$, because of the singularity, but it is usable). The same difficulty occurs when f is not differentiable (e.g. $f = |x - a|$ at a mesh point, or data with kinks).

(ii) The derivatives are laborious, expensive or unavailable. For $y' = e^{-x^2}\sin(xy)$ one finds already

$$y'' = f_x + f f_y = e^{-x^2}\left[y\cos xy - 2x\sin xy\right] + x\,e^{-2x^2}\sin xy\,\cos xy,$$

and the formulas for the higher derivatives needed for higher-order Taylor methods grow quickly (after expansion, y''' has 10 terms and $y^{(4)}$ has 21 terms); coding them by hand invites errors, and each step costs the evaluation of these long expressions. If $f(x, y)$ is available only numerically (a table of experimental data, or the output of another program), f_x and f_y do not exist in usable form, and numerical differentiation of noisy data destroys the accuracy; Heun's method needs only values of f.

(iii) The Taylor error constant is unfavorable. For $y' = f(x)$ the Taylor error is $\frac{h^3}{6}f''$ per step against $-\frac{h^3}{12}f''$ for Heun, twice as large with the opposite sign. Example: $y' = \cos 10x$, $y(0) = 0$ ($y = \frac{1}{10}\sin 10x$, $y(1) = \frac{1}{10}\sin 10 = -0.0544021$), with the Taylor step $y_{k+1} = y_k + h\cos 10x_k - 5h^2\sin 10x_k$:

h	Taylor y_N	Heun y_N	exact $y(1)$	Taylor error	Heun error	ratio
0.020	−0.0547033	−0.0542206	−0.0544021	3.012×10^{-4}	-1.815×10^{-4}	−1.66
0.010	−0.0544851	−0.0543568	−0.0544021	8.300×10^{-5}	-4.534×10^{-5}	−1.83
0.005	−0.0544238	−0.0543908	−0.0544021	2.171×10^{-5}	-1.133×10^{-5}	−1.92

The ratio of the errors tends to -2 as $h \to 0$ (the Taylor error is about twice the Heun error and has the opposite sign; for $f = f(x)$ the total errors are $\approx \frac{h^2}{6}[f'(1) - f'(0)]$ and $-\frac{h^2}{12}[f'(1) - f'(0)]$ respectively, with $f'(1) - f'(0) = -10\sin 10 = 5.44$), so the Taylor method is the less accurate of the two here, in addition to being more laborious.

Answer. Poor cases: f_x singular at a mesh point (e.g. $y' = \sqrt{x}$, $y(0) = 0$: Taylor cannot start); complicated or unavailable derivatives of f (e.g. $y' = e^{-x^2}\sin xy$, tabulated f); and cases with larger error constants ($y' = f(x)$: Taylor error $h^3 f''/6$ vs. Heun's $-h^3 f''/12$).

Problem 75-8 (Simmons)

Solution. **Derivation.** Taylor's theorem gives

$$y(x_k + h) = y(x_k) + hy'(x_k) + \tfrac{h^2}{2}y''(x_k) + \tfrac{h^3}{6}y'''(x_k) + \tfrac{h^4}{24}y^{(4)}(\xi).$$

With $y' = f$,

$$y'' = f_x + f f_y, \qquad y''' = \frac{d}{dx}\big(f_x + f f_y\big) = f_{xx} + 2 f f_{xy} + f^2 f_{yy} + f_y(f_x + f f_y)$$

(using $\frac{d}{dx}F(x,y(x)) = F_x + F_y f$ twice; e.g. $\frac{d}{dx}f_x = f_{xx} + f_{xy}f$, $\frac{d}{dx}(f f_y) = (f_x + f_y f)f_y + f(f_{xy} + f_{yy}f)$). The *four-term Taylor method* is

$$\boxed{y_{k+1} = y_k + hf + \frac{h^2}{2}(f_x + f f_y) + \frac{h^3}{6}\big[f_{xx} + 2 f f_{xy} + f^2 f_{yy} + f_y(f_x + f f_y)\big]_{(x_k, y_k)}}$$

with local error $\frac{h^4}{24}y^{(4)}(\xi)$ and total error proportional to h^3.

Benchmark problem $y' = x + y$, $y(0) = 1$ (exact $y = 2e^x - x - 1$). Here $f_x = f_y = 1$ and the second derivatives vanish, so $y'' = 1 + (x + y) = 1 + x + y$ and $y''' = y''$ (indeed $f_y(f_x + f f_y) = 1 + x + y$). Write $w_k = 1 + x_k + y_k$: then

$$y_{k+1} = y_k + h(x_k + y_k) + \Big(\frac{h^2}{2} + \frac{h^3}{6}\Big) w_k.$$

Explicitly, for $k = 0$: $w_0 = 2$, so $y_1 = 1 + 0.1 \cdot 1 + (0.005 + 0.000166667) \cdot 2 = 1.1103333$ (exact $y(0.1) = 1.1103418$). Table (the 3-term values, which reproduce the book's Table 3, are listed for comparison):

x_k	3-term y_k	4-term y_k	$w_k = 1 + x_k + y_k$ (4-term)	exact	err. 3-term	err. 4-term	E_k 4-term (%)
0.0	1.00000	1.00000	2.00000	1.00000	0.000000	0.000000	0.00000
0.1	1.11000	1.11033	2.21033	1.11034	0.000342	0.000009	0.00077
0.2	1.24205	1.24279	2.44279	1.24281	0.000756	0.000019	0.00151
0.3	1.39847	1.39969	2.69969	1.39972	0.001252	0.000031	0.00223
0.4	1.58180	1.58360	2.98360	1.58365	0.001845	0.000046	0.00290
0.5	1.79489	1.79738	3.29738	1.79744	0.002549	0.000063	0.00353
0.6	2.04086	2.04415	3.64415	2.04424	0.003380	0.000084	0.00411
0.7	2.32315	2.32740	4.02740	2.32751	0.004358	0.000108	0.00466
0.8	2.64558	2.65094	4.45094	2.65108	0.005504	0.000137	0.00517
0.9	3.01236	3.01904	4.91904	3.01921	0.006843	0.000170	0.00564
1.0	3.42816	3.43635	–	3.43656	0.008402	0.000209	0.00609

Closed form: since $w_{k+1} = w_k\,(1 + h + \frac{h^2}{2} + \frac{h^3}{6})$ one finds $y_k = 2\big(1 + h + \frac{h^2}{2} + \frac{h^3}{6}\big)^k - 1 - x_k$, so the 4-term value at $x = 1$ is $y_{10} = 2(1.1051667)^{10} - 2 = 3.43635$, while the three-term method has factor $1 + h + h^2/2$ and gives $y_{10} = 3.42816$.

Accuracy. At $x = 1$ the exact value is $y(1) = 2e - 2 = 3.43656$:

method ($h = 0.1$)	$y(1)$ estimate	error	relative error
three-term Taylor	3.42816	8.40×10^{-3}	0.244%
four-term Taylor	3.43635	2.09×10^{-4}	0.0061%

So a considerable accuracy is gained: the error is smaller by a factor of about 40. The error behaves as predicted, like h^3: for $h = 0.05$ and $h = 0.025$ the four-term errors at $x = 1$ are 2.72×10^{-5} and 3.47×10^{-6}, ratios 7.7 and $7.8 \approx 2^3$ (the three-term errors 8.40×10^{-3}, 2.18×10^{-3}, 5.56×10^{-4} shrink by $\approx 4 = 2^2$). The price is the extra derivative y''', which here is trivial but is complicated for general f.

Answer. $y_{k+1} = y_k + hf + \frac{h^2}{2}(f_x + f f_y) + \frac{h^3}{6}\big[f_{xx} + 2 f f_{xy} + f^2 f_{yy} + f_y(f_x + f f_y)\big]$. For $y' = x + y$: $y_{k+1} = y_k + h(x_k + y_k) + (\frac{h^2}{2} + \frac{h^3}{6})(1 + x_k + y_k)$, giving $y_{10} = 3.43635$ at $x = 1$ (exact

3.43656, error 2.1×10^{-4}) versus 3.42816 (error 8.4×10^{-3}) for the three-term method: about 40 times more accurate.

Section 76 Higher Order Methods

Conventions. The (classical fourth-order) Runge–Kutta method for $y' = f(x,y)$, $y(x_0) = y_0$, step h, $x_k = x_0 + kh$, is

$$m_1 = hf(x_k, y_k), \qquad m_2 = hf\left(x_k + \tfrac{h}{2},\ y_k + \tfrac{m_1}{2}\right),$$
$$m_3 = hf\left(x_k + \tfrac{h}{2},\ y_k + \tfrac{m_2}{2}\right), \qquad m_4 = hf(x_k + h,\ y_k + m_3),$$

$$y_{k+1} = y_k + \tfrac{1}{6}\,(m_1 + 2m_2 + 2m_3 + m_4).$$

In the tables the columns $m_1, \dots, m_4$ in row k are the numbers used for the step from row k to row $k+1$; $E_k = |y(x_k) - y_k| / |y(x_k)|$ in percent. Exact solutions are those of Section 73. Computations are done with 40 digits and displayed to five decimals (the tables of the text are reproduced in `verify/sec76.py`); for each problem we show the complete iteration for $h = 0.1$ and $h = 0.05$ and every tenth iterate for $h = 0.01$, and compare the value at $x = 1$ with those of the Euler method (Section 73) and the improved Euler method (Section 75).

Problem 76-1 (Simmons)

Solution. Exact: $y = \frac{3}{2}e^{2x} - x - \frac{1}{2}$, $y(1) = 9.58358$. Here $f(x,y) = 2x + 2y$.

$h = 0.1$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.20000	0.23000	0.23300	0.26660	1.00000	0.00000
0.10	1.23210	0.26642	0.30306	0.30673	0.34777	1.23210	0.00034
0.20	1.53773	0.34755	0.39230	0.39678	0.44690	1.53774	0.00066
0.30	1.93316	0.44663	0.50130	0.50676	0.56798	1.93318	0.00096
0.40	2.43828	0.56766	0.63442	0.64110	0.71588	2.43831	0.00124
0.50	3.07738	0.71548	0.79702	0.80518	0.89651	3.07742	0.00150
0.60	3.88011	0.89602	0.99562	1.00558	1.11714	3.88018	0.00174
0.70	4.88270	1.11654	1.23819	1.25036	1.38661	4.88280	0.00197
0.80	6.12941	1.38588	1.53447	1.54933	1.71575	6.12955	0.00219
0.90	7.67429	1.71486	1.89634	1.91449	2.11776	7.67447	0.00240
1.00	9.58333	–	–	–	–	9.58358	0.00261

$h = 0.05$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.10000	0.10750	0.10787	0.11579	1.00000	0.00000
0.05	1.10776	0.11578	0.12406	0.12448	0.13322	1.10776	0.00001
0.10	1.23210	0.13321	0.14237	0.14283	0.15249	1.23210	0.00002
0.15	1.37479	0.15248	0.16260	0.16311	0.17379	1.37479	0.00003
0.20	1.53774	0.17377	0.18496	0.18552	0.19733	1.53774	0.00004
0.25	1.72308	0.19731	0.20967	0.21029	0.22334	1.72308	0.00006
0.30	1.93318	0.22332	0.23698	0.23767	0.25208	1.93318	0.00007
0.35	2.17063	0.25206	0.26717	0.26792	0.28385	2.17063	0.00007
0.40	2.43831	0.28383	0.30052	0.30136	0.31897	2.43831	0.00008
0.45	2.73940	0.31894	0.33739	0.33831	0.35777	2.73940	0.00009
0.50	3.07742	0.35774	0.37813	0.37915	0.40066	3.07742	0.00010

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.55	3.45625	0.40062	0.42316	0.42428	0.44805	3.45625	0.00011
0.60	3.88017	0.44802	0.47292	0.47416	0.50043	3.88018	0.00012
0.65	4.35394	0.50039	0.52791	0.52929	0.55832	4.35395	0.00013
0.70	4.88279	0.55828	0.58869	0.59021	0.62230	4.88280	0.00013
0.75	5.47253	0.62225	0.65587	0.65755	0.69301	5.47253	0.00014
0.80	6.12954	0.69295	0.73010	0.73196	0.77115	6.12955	0.00015
0.85	6.86091	0.77109	0.81215	0.81420	0.85751	6.86092	0.00016
0.90	7.67446	0.85745	0.90282	0.90509	0.95295	7.67447	0.00016
0.95	8.57883	0.95288	1.00303	1.00553	1.05844	8.57884	0.00017
1.00	9.58357	–	–	–	–	9.58358	0.00018

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.02000	0.02030	0.02030	0.02061	1.00000	0.00000
0.10	1.23210	0.02664	0.02701	0.02701	0.02738	1.23210	0.00000
0.20	1.53774	0.03475	0.03520	0.03521	0.03566	1.53774	0.00000
0.30	1.93318	0.04466	0.04521	0.04522	0.04577	1.93318	0.00000
0.40	2.43831	0.05677	0.05743	0.05744	0.05812	2.43831	0.00000
0.50	3.07742	0.07155	0.07236	0.07237	0.07320	3.07742	0.00000
0.60	3.88018	0.08960	0.09060	0.09061	0.09162	3.88018	0.00000
0.70	4.88280	0.11166	0.11287	0.11288	0.11411	4.88280	0.00000
0.80	6.12955	0.13859	0.14008	0.14009	0.14159	6.12955	0.00000
0.90	7.67447	0.17149	0.17330	0.17332	0.17516	7.67447	0.00000
1.00	9.58358	–	–	–	–	9.58358	0.00000

Runge–Kutta at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^4$
0.1	10	9.58333386	9.58358415	2.503×10^{-4}	2.61×10^{-3}	2.503
0.05	20	9.58356715	9.58358415	1.700×10^{-5}	1.77×10^{-4}	2.720
0.01	100	9.58358412	9.58358415	2.907×10^{-8}	3.03×10^{-7}	2.907

Comparison of the three methods at $x = 1$:

h	Euler y_N	improved Euler y_N	Runge–Kutta y_N	exact $y(1)$
0.1	7.787605	9.456947	9.58333386	9.58358415
0.05	8.591250	9.549352	9.58356715	9.58358415
0.01	9.366969	9.582128	9.58358412	9.58358415

h	Euler error	improved Euler error	Runge–Kutta error
0.1	1.80×10^{0}	1.27×10^{-1}	2.50×10^{-4}
0.05	9.92×10^{-1}	3.42×10^{-2}	1.70×10^{-5}
0.01	2.17×10^{-1}	1.46×10^{-3}	2.91×10^{-8}

Discussion. Runge–Kutta gives 9.583334, 9.583567, 9.583584: relative errors 0.0026%, 0.00018%, 3×10^{-7}%, against 1.3%, 0.36%, 0.015% for improved Euler and 18.7%, 10.4%, 2.3% for Euler. The error is proportional to h^4: halving h divides it by 14.7 (≈ 16) and reducing h by 5 divides it by 585 ($\approx 5^4 = 625$). With $h = 0.1$ (40 evaluations of f) RK is more accurate than Heun with $h = 0.01$ (200 evaluations) by a factor of about 6 (error 2.5×10^{-4} against 1.5×10^{-3}), and than Euler with $h = 0.01$ (100 evaluations) by a factor of about 860.

Answer. RK: $y_N(1) = 9.583334,\ 9.583567,\ 9.583584$ ($h = 0.1,\ 0.05,\ 0.01$); exact 9.583584; errors 2.5×10^{-4}, 1.7×10^{-5}, 2.9×10^{-8} (Heun: $0.127,\ 0.034,\ 0.0015$; Euler: $1.80,\ 0.99,\ 0.22$).

Problem 76-2 (Simmons)

Solution. Exact: $y = \sqrt{2x+1}$, $y(1) = \sqrt{3} = 1.7320508076$. Here $f(x,y) = 1/y$.

$h = 0.1$:

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.10000	0.09524	0.09545	0.09129	1.00000	0.00000
0.10	1.09545	0.09129	0.08764	0.08778	0.08452	1.09545	0.00001
0.20	1.18322	0.08452	0.08160	0.08170	0.07906	1.18322	0.00002
0.30	1.26491	0.07906	0.07666	0.07673	0.07454	1.26491	0.00002
0.40	1.34164	0.07454	0.07252	0.07257	0.07071	1.34164	0.00002
0.50	1.41421	0.07071	0.06899	0.06903	0.06742	1.41421	0.00002
0.60	1.48324	0.06742	0.06592	0.06595	0.06455	1.48324	0.00002
0.70	1.54919	0.06455	0.06323	0.06326	0.06202	1.54919	0.00001
0.80	1.61245	0.06202	0.06085	0.06087	0.05976	1.61245	0.00001
0.90	1.67332	0.05976	0.05871	0.05873	0.05774	1.67332	0.00001
1.00	1.73205	–	–	–	–	1.73205	0.00001

$h = 0.05$:

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.05000	0.04878	0.04881	0.04767	1.00000	0.00000
0.05	1.04881	0.04767	0.04661	0.04664	0.04564	1.04881	0.00000
0.10	1.09545	0.04564	0.04471	0.04473	0.04385	1.09545	0.00000
0.15	1.14018	0.04385	0.04303	0.04304	0.04226	1.14018	0.00000
0.20	1.18322	0.04226	0.04152	0.04153	0.04082	1.18322	0.00000
0.25	1.22474	0.04082	0.04016	0.04017	0.03953	1.22474	0.00000
0.30	1.26491	0.03953	0.03892	0.03893	0.03835	1.26491	0.00000
0.35	1.30384	0.03835	0.03779	0.03780	0.03727	1.30384	0.00000
0.40	1.34164	0.03727	0.03676	0.03676	0.03627	1.34164	0.00000
0.45	1.37840	0.03627	0.03580	0.03581	0.03536	1.37840	0.00000
0.50	1.41421	0.03536	0.03492	0.03492	0.03450	1.41421	0.00000

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.55	1.44914	0.03450	0.03410	0.03410	0.03371	1.44914	0.00000
0.60	1.48324	0.03371	0.03333	0.03334	0.03297	1.48324	0.00000
0.65	1.51658	0.03297	0.03261	0.03262	0.03227	1.51658	0.00000
0.70	1.54919	0.03227	0.03194	0.03195	0.03162	1.54919	0.00000
0.75	1.58114	0.03162	0.03131	0.03131	0.03101	1.58114	0.00000
0.80	1.61245	0.03101	0.03071	0.03072	0.03043	1.61245	0.00000
0.85	1.64317	0.03043	0.03015	0.03015	0.02988	1.64317	0.00000
0.90	1.67332	0.02988	0.02962	0.02962	0.02936	1.67332	0.00000
0.95	1.70294	0.02936	0.02911	0.02911	0.02887	1.70294	0.00000
1.00	1.73205	–	–	–	–	1.73205	0.00000

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	1.00000	0.01000	0.00995	0.00995	0.00990	1.00000	0.00000
0.10	1.09545	0.00913	0.00909	0.00909	0.00905	1.09545	0.00000
0.20	1.18322	0.00845	0.00842	0.00842	0.00839	1.18322	0.00000
0.30	1.26491	0.00791	0.00788	0.00788	0.00786	1.26491	0.00000
0.40	1.34164	0.00745	0.00743	0.00743	0.00741	1.34164	0.00000
0.50	1.41421	0.00707	0.00705	0.00705	0.00704	1.41421	0.00000
0.60	1.48324	0.00674	0.00673	0.00673	0.00671	1.48324	0.00000
0.70	1.54919	0.00645	0.00644	0.00644	0.00643	1.54919	0.00000
0.80	1.61245	0.00620	0.00619	0.00619	0.00618	1.61245	0.00000
0.90	1.67332	0.00598	0.00597	0.00597	0.00595	1.67332	0.00000
1.00	1.73205	–	–	–	–	1.73205	0.00000

Runge–Kutta at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^4$
0.1	10	1.73205102	1.73205081	-2.130×10^{-7}	1.23×10^{-5}	-0.002
0.05	20	1.73205082	1.73205081	-1.273×10^{-8}	7.35×10^{-7}	-0.002
0.01	100	1.73205081	1.73205081	-1.953×10^{-11}	1.13×10^{-9}	-0.002

Comparison of the three methods at $x = 1$:

h	Euler y_N	improved Euler y_N	Runge–Kutta y_N	exact $y(1)$
0.1	1.748617	1.732068	1.73205102	1.73205081
0.05	1.740152	1.732053	1.73205082	1.73205081
0.01	1.733643	1.732051	1.73205081	1.73205081

h	Euler error	improved Euler error	Runge–Kutta error
0.1	-1.66×10^{-2}	-1.71×10^{-5}	-2.13×10^{-7}
0.05	-8.10×10^{-3}	-2.07×10^{-6}	-1.27×10^{-8}
0.01	-1.59×10^{-3}	-1.62×10^{-8}	-1.95×10^{-11}

Discussion. RK: 1.73205102, 1.73205082, 1.73205081, i.e. errors -2.1×10^{-7}, -1.3×10^{-8}, -2.0×10^{-11} (Heun: -1.7×10^{-5}, -2.1×10^{-6}, -1.6×10^{-8}; Euler: -1.7×10^{-2}, -8.1×10^{-3}, -1.6×10^{-3}). The error scales like h^4 (ratios 16.7 and 652).

Answer. RK: $y_N(1) = 1.73205102,\ 1.73205082,\ 1.73205081$ for $h = 0.1,\ 0.05,\ 0.01$; exact $\sqrt{3} = 1.73205081$ (errors -2.1×10^{-7}, -1.3×10^{-8}, -2×10^{-11}).

Problem 76-3 (Simmons)

Solution. Exact (Section 73): $y = -\log(1 - x)$ for $x < 1$, which tends to $+\infty$ as $x \to 1^-$; *there is no solution at* $x = 1$. Here $f(x, y) = e^y$.

$h = 0.1$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.10000	0.10513	0.10540	0.11112	0.00000	–
0.10	0.10536	0.11111	0.11746	0.11783	0.12501	0.10536	0.00003
0.20	0.22314	0.12500	0.13306	0.13360	0.14287	0.22314	0.00004
0.30	0.35668	0.14286	0.15343	0.15425	0.16668	0.35667	0.00006
0.40	0.51083	0.16667	0.18115	0.18247	0.20003	0.51083	0.00008
0.50	0.69315	0.20000	0.22103	0.22337	0.25006	0.69315	0.00013
0.60	0.91629	0.25000	0.28329	0.28804	0.33345	0.91629	0.00024
0.70	1.20398	0.33334	0.39379	0.40587	0.50021	1.20397	0.00049
0.80	1.60946	0.50001	0.64203	0.68928	0.99616	1.60944	0.00119
0.90	2.30259	1.00000	1.64872	2.28043	9.78086	2.30259	0.00004
1.00	5.40911	–	–	–	–	$+\infty$	–

$h = 0.05$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.05000	0.05127	0.05130	0.05263	0.00000	–
0.05	0.05129	0.05263	0.05404	0.05407	0.05556	0.05129	0.00000
0.10	0.10536	0.05556	0.05712	0.05717	0.05882	0.10536	0.00000
0.15	0.16252	0.05882	0.06058	0.06063	0.06250	0.16252	0.00000
0.20	0.22314	0.06250	0.06448	0.06455	0.06667	0.22314	0.00000
0.25	0.28768	0.06667	0.06893	0.06900	0.07143	0.28768	0.00000
0.30	0.35667	0.07143	0.07403	0.07412	0.07692	0.35667	0.00000
0.35	0.43078	0.07692	0.07994	0.08006	0.08333	0.43078	0.00001
0.40	0.51083	0.08333	0.08688	0.08703	0.09091	0.51083	0.00001
0.45	0.59784	0.09091	0.09514	0.09534	0.10000	0.59784	0.00001
0.50	0.69315	0.10000	0.10513	0.10540	0.11112	0.69315	0.00001

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.55	0.79851	0.11111	0.11746	0.11783	0.12501	0.79851	0.00001
0.60	0.91629	0.12500	0.13306	0.13360	0.14287	0.91629	0.00002
0.65	1.04982	0.14286	0.15343	0.15425	0.16668	1.04982	0.00003
0.70	1.20397	0.16667	0.18115	0.18247	0.20003	1.20397	0.00005
0.75	1.38630	0.20000	0.22103	0.22337	0.25006	1.38629	0.00008
0.80	1.60944	0.25000	0.28329	0.28804	0.33345	1.60944	0.00015
0.85	1.89713	0.33334	0.39379	0.40587	0.50021	1.89712	0.00032
0.90	2.30260	0.50001	0.64203	0.68928	0.99616	2.30259	0.00085
0.95	2.99573	1.00000	1.64873	2.28043	9.78090	2.99573	0.00005
1.00	6.10227	–	–	–	–	$+\infty$	–

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.01000	0.01005	0.01005	0.01010	0.00000	–
0.10	0.10536	0.01111	0.01117	0.01117	0.01124	0.10536	0.00000
0.20	0.22314	0.01250	0.01258	0.01258	0.01266	0.22314	0.00000
0.30	0.35667	0.01429	0.01439	0.01439	0.01449	0.35667	0.00000
0.40	0.51083	0.01667	0.01681	0.01681	0.01695	0.51083	0.00000
0.50	0.69315	0.02000	0.02020	0.02020	0.02041	0.69315	0.00000
0.60	0.91629	0.02500	0.02531	0.02532	0.02564	0.91629	0.00000
0.70	1.20397	0.03333	0.03389	0.03390	0.03448	1.20397	0.00000
0.80	1.60944	0.05000	0.05127	0.05130	0.05263	1.60944	0.00000
0.90	2.30259	0.10000	0.10513	0.10540	0.11112	2.30259	0.00000
1.00	7.71171	–	–	–	–	$+\infty$	–

Comparison of the three methods at $x = 1$ (no exact value exists):

h	Euler y_N	improved Euler y_N	Runge–Kutta y_N	exact $y(1)$
0.1	2.27337	3.92302	5.40911	$+\infty$
0.05	2.74934	4.60207	6.10227	$+\infty$
0.01	3.95946	6.20029	7.71171	$+\infty$

Discussion. For $x < 1$ Runge–Kutta follows $-\log(1-x)$ extremely well (e.g. for $h = 0.1$ at $x = 0.9$ the relative error is only 4×10^{-5}%, against 2.6% for improved Euler and 25% for Euler); but at $x = 1$ all three methods return finite numbers (5.40911, 6.10227, 7.71171 for RK) which are meaningless because $y(1)$ does not exist. The RK values still grow without bound as $h \to 0$ (10.0143 for $h = 10^{-3}$, 12.3169 for $h = 10^{-4}$) and keep climbing by about $2.3 = \log 10$ per decade, as for $-\log(1-x)$ evaluated at a point at distance $\approx h$ from the singularity; the more accurate the method, the better it follows the blow-up. No method of this type can reveal that the solution has ceased to exist.

Answer. RK: 5.40911, 6.10227, 7.71171 at $x = 1$ ($h = 0.1,\ 0.05,\ 0.01$), but the exact solution $-\log(1-x)$ blows up at $x = 1$; the numbers have no meaning and grow without bound as $h \to 0$ (Heun: 3.92302, 4.60207, 6.20029; Euler: 2.27337, 2.74934, 3.95946).

Problem 76-4 (Simmons)

Solution. Exact: $y = \frac{1}{2}(\sin x + \cos x) - \frac{3}{2}e^x$, $y(1) = -3.386536097$. Here $f(x,y) = y - \sin x$.

$h = 0.1$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−0.10000	−0.11000	−0.11050	−0.12103	−1.00000	0.00000
0.10	−1.11034	−0.12102	−0.13203	−0.13258	−0.14416	−1.11034	0.00001
0.20	−1.24274	−0.14414	−0.15622	−0.15683	−0.16951	−1.24274	0.00003
0.30	−1.39936	−0.16949	−0.18270	−0.18336	−0.19721	−1.39936	0.00004
0.40	−1.58250	−0.19719	−0.21161	−0.21233	−0.22742	−1.58250	0.00005
0.50	−1.79458	−0.22740	−0.24310	−0.24388	−0.26031	−1.79458	0.00007
0.60	−2.03819	−0.26028	−0.27735	−0.27820	−0.29606	−2.03819	0.00008
0.70	−2.31610	−0.29603	−0.31458	−0.31550	−0.33490	−2.31610	0.00009
0.80	−2.63128	−0.33486	−0.35500	−0.35601	−0.37706	−2.63128	0.00010
0.90	−2.98693	−0.37703	−0.39889	−0.39998	−0.42284	−2.98694	0.00011
1.00	−3.38653	–	–	–	–	−3.38654	0.00011

$h = 0.05$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−0.05000	−0.05250	−0.05256	−0.05513	−1.00000	0.00000
0.05	−1.05254	−0.05513	−0.05775	−0.05782	−0.06051	−1.05254	0.00000
0.10	−1.11034	−0.06051	−0.06326	−0.06333	−0.06616	−1.11034	0.00000
0.15	−1.17365	−0.06615	−0.06904	−0.06911	−0.07207	−1.17365	0.00000
0.20	−1.24274	−0.07207	−0.07509	−0.07517	−0.07827	−1.24274	0.00000
0.25	−1.31788	−0.07826	−0.08143	−0.08151	−0.08475	−1.31788	0.00000
0.30	−1.39936	−0.08474	−0.08805	−0.08813	−0.09152	−1.39936	0.00000
0.35	−1.48747	−0.09152	−0.09497	−0.09506	−0.09860	−1.48747	0.00000
0.40	−1.58250	−0.09860	−0.10221	−0.10230	−0.10599	−1.58250	0.00000
0.45	−1.68476	−0.10599	−0.10975	−0.10985	−0.11370	−1.68476	0.00000
0.50	−1.79458	−0.11370	−0.11763	−0.11773	−0.12175	−1.79458	0.00000

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.55	−1.91227	−0.12175	−0.12585	−0.12595	−0.13014	−1.91227	0.00000
0.60	−2.03819	−0.13014	−0.13442	−0.13452	−0.13890	−2.03819	0.00001
0.65	−2.17268	−0.13889	−0.14335	−0.14346	−0.14802	−2.17268	0.00001
0.70	−2.31610	−0.14802	−0.15266	−0.15278	−0.15753	−2.31610	0.00001
0.75	−2.46884	−0.15752	−0.16237	−0.16249	−0.16743	−2.46884	0.00001
0.80	−2.63128	−0.16743	−0.17248	−0.17260	−0.17776	−2.63128	0.00001
0.85	−2.80384	−0.17776	−0.18301	−0.18314	−0.18852	−2.80384	0.00001
0.90	−2.98694	−0.18851	−0.19399	−0.19413	−0.19972	−2.98694	0.00001
0.95	−3.18101	−0.19972	−0.20543	−0.20557	−0.21140	−3.18102	0.00001
1.00	−3.38654	–	–	–	–	−3.38654	0.00001

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	−1.00000	−0.01000	−0.01010	−0.01010	−0.01020	−1.00000	0.00000
0.10	−1.11034	−0.01210	−0.01221	−0.01221	−0.01232	−1.11034	0.00000
0.20	−1.24274	−0.01441	−0.01454	−0.01454	−0.01466	−1.24274	0.00000
0.30	−1.39936	−0.01695	−0.01708	−0.01708	−0.01722	−1.39936	0.00000
0.40	−1.58250	−0.01972	−0.01986	−0.01986	−0.02001	−1.58250	0.00000
0.50	−1.79458	−0.02274	−0.02290	−0.02290	−0.02306	−1.79458	0.00000
0.60	−2.03819	−0.02603	−0.02620	−0.02620	−0.02637	−2.03819	0.00000
0.70	−2.31610	−0.02960	−0.02979	−0.02979	−0.02998	−2.31610	0.00000
0.80	−2.63128	−0.03349	−0.03369	−0.03369	−0.03389	−2.63128	0.00000
0.90	−2.98694	−0.03770	−0.03792	−0.03792	−0.03814	−2.98694	0.00000
1.00	−3.38654	–	–	–	–	−3.38654	0.00000

Runge–Kutta at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^4$
0.1	10	−3.38653222	−3.38653610	-3.878×10^{-6}	1.15×10^{-4}	−0.039
0.05	20	−3.38653584	−3.38653610	-2.526×10^{-7}	7.46×10^{-6}	−0.040
0.01	100	−3.38653610	−3.38653610	-4.177×10^{-10}	1.23×10^{-8}	−0.042

Comparison of the three methods at $x = 1$:

h	Euler y_N	improved Euler y_N	Runge–Kutta y_N	exact $y(1)$
0.1	−3.149466	−3.377996	−3.38653222	−3.38653610
0.05	−3.262929	−3.384324	−3.38653584	−3.38653610
0.01	−3.360935	−3.386445	−3.38653610	−3.38653610

h	Euler error	improved Euler error	Runge–Kutta error
0.1	-2.37×10^{-1}	-8.54×10^{-3}	-3.88×10^{-6}
0.05	-1.24×10^{-1}	-2.21×10^{-3}	-2.53×10^{-7}
0.01	-2.56×10^{-2}	-9.10×10^{-5}	-4.18×10^{-10}

Discussion. RK gives -3.386532, -3.386536, -3.386536 with errors -3.9×10^{-6}, -2.5×10^{-7}, -4.2×10^{-10} (relative $1.1 \times 10^{-4}\%$, $7 \times 10^{-6}\%$, $10^{-8}\%$); Heun's errors are -8.5×10^{-3}, -2.2×10^{-3}, -9.1×10^{-5} and Euler's -0.24, -0.12, -0.026. The error ratios, 15.4 and 605, confirm the order h^4.

Answer. RK: $y_N(1) = -3.386532,\ -3.386536,\ -3.386536$ for $h = 0.1,\ 0.05,\ 0.01$; exact -3.386536 (errors -3.9×10^{-6}, -2.5×10^{-7}, -4×10^{-10}).

Problem 76-5 (Simmons)

Solution. Exact: $y = 1 - x + \tan(x - \pi/4)$, $y(1) = \tan(1 - \pi/4) = 0.2179580985$. Here $f(x,y) = (x+y-1)^2$.

$h = 0.1$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.10000	0.08100	0.08272	0.06679	0.00000	–
0.10	0.08237	0.06685	0.05391	0.05486	0.04393	0.08237	0.00116
0.20	0.13709	0.04394	0.03492	0.03546	0.02782	0.13709	0.00101
0.30	0.17251	0.02782	0.02149	0.02179	0.01646	0.17251	0.00089
0.40	0.19432	0.01646	0.01207	0.01223	0.00861	0.19431	0.00079
0.50	0.20659	0.00861	0.00572	0.00579	0.00352	0.20659	0.00071
0.60	0.21245	0.00352	0.00184	0.00187	0.00073	0.21245	0.00065
0.70	0.21439	0.00073	0.00012	0.00013	0.00002	0.21439	0.00059
0.80	0.21460	0.00002	0.00042	0.00042	0.00132	0.21460	0.00055
0.90	0.21511	0.00132	0.00275	0.00277	0.00475	0.21511	0.00052
1.00	0.21796	–	–	–	–	0.21796	0.00050

$h = 0.05$**:**

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.05000	0.04512	0.04536	0.04092	0.00000	–
0.05	0.04531	0.04092	0.03691	0.03709	0.03342	0.04531	0.00006
0.10	0.08237	0.03343	0.03010	0.03023	0.02719	0.08237	0.00005
0.15	0.11258	0.02719	0.02442	0.02451	0.02197	0.11258	0.00005
0.20	0.13709	0.02197	0.01965	0.01972	0.01759	0.13709	0.00005
0.25	0.15681	0.01759	0.01565	0.01570	0.01391	0.15681	0.00005
0.30	0.17251	0.01391	0.01228	0.01232	0.01082	0.17251	0.00004
0.35	0.18483	0.01082	0.00945	0.00948	0.00823	0.18483	0.00004
0.40	0.19431	0.00823	0.00709	0.00711	0.00608	0.19431	0.00004
0.45	0.20143	0.00607	0.00514	0.00515	0.00430	0.20143	0.00004
0.50	0.20659	0.00430	0.00354	0.00355	0.00288	0.20659	0.00003

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.55	0.21016	0.00288	0.00228	0.00228	0.00176	0.21016	0.00003
0.60	0.21245	0.00176	0.00131	0.00131	0.00093	0.21245	0.00003
0.65	0.21377	0.00093	0.00061	0.00062	0.00037	0.21377	0.00003
0.70	0.21439	0.00037	0.00018	0.00018	0.00006	0.21439	0.00003
0.75	0.21459	0.00006	0.00001	0.00001	0.00001	0.21459	0.00003
0.80	0.21460	0.00001	0.00008	0.00008	0.00021	0.21460	0.00003
0.85	0.21469	0.00021	0.00040	0.00040	0.00066	0.21469	0.00002
0.90	0.21511	0.00066	0.00099	0.00099	0.00138	0.21511	0.00002
0.95	0.21610	0.00138	0.00184	0.00184	0.00238	0.21610	0.00002
1.00	0.21796	–	–	–	–	0.21796	0.00002

$h = 0.01$ (every tenth iterate shown):

x_k	y_k	m_1	m_2	m_3	m_4	exact $y(x_k)$	E_k (%)
0.00	0.00000	0.01000	0.00980	0.00980	0.00961	0.00000	–
0.10	0.08237	0.00669	0.00655	0.00655	0.00642	0.08237	0.00000
0.20	0.13709	0.00439	0.00430	0.00430	0.00421	0.13709	0.00000
0.30	0.17251	0.00278	0.00272	0.00272	0.00265	0.17251	0.00000
0.40	0.19431	0.00165	0.00160	0.00160	0.00155	0.19431	0.00000
0.50	0.20659	0.00086	0.00083	0.00083	0.00080	0.20659	0.00000
0.60	0.21245	0.00035	0.00033	0.00033	0.00031	0.21245	0.00000
0.70	0.21439	0.00007	0.00006	0.00006	0.00006	0.21439	0.00000
0.80	0.21460	0.00000	0.00000	0.00000	0.00001	0.21460	0.00000
0.90	0.21511	0.00013	0.00014	0.00014	0.00016	0.21511	0.00000
1.00	0.21796	–	–	–	–	0.21796	0.00000

Runge–Kutta at $x = 1$:

h	N	$y_N \approx y(1)$	exact $y(1)$	$y(1) - y_N$	E_N (%)	(error)$/h^4$
0.1	10	0.21795918	0.21795810	-1.083×10^{-6}	4.97×10^{-4}	-0.011
0.05	20	0.21795815	0.21795810	-4.860×10^{-8}	2.23×10^{-5}	-0.008
0.01	100	0.21795810	0.21795810	-5.613×10^{-11}	2.58×10^{-8}	-0.006

Comparison of the three methods at $x = 1$:

h	Euler y_N	improved Euler y_N	Runge–Kutta y_N	exact $y(1)$
0.1	0.254474	0.218698	0.21795918	0.21795810
0.05	0.235521	0.218123	0.21795815	0.21795810
0.01	0.221369	0.217964	0.21795810	0.21795810

h	Euler error	improved Euler error	Runge–Kutta error
0.1	-3.65×10^{-2}	-7.40×10^{-4}	-1.08×10^{-6}
0.05	-1.76×10^{-2}	-1.65×10^{-4}	-4.86×10^{-8}
0.01	-3.41×10^{-3}	-6.14×10^{-6}	-5.61×10^{-11}

Discussion. RK gives 0.2179592, 0.2179581, 0.2179581, with errors -1.1×10^{-6}, -4.9×10^{-8}, -5.6×10^{-11} (Heun: -7.4×10^{-4}, -1.7×10^{-4}, -6.1×10^{-6}; Euler: -0.037, -0.018, -0.0034). The error ratios 22 and 866 are larger than 16 and 625 since the error is not yet in its asymptotic regime for $h = 0.1$ and 0.05.

Answer. RK: $y_N(1) = 0.2179592,\ 0.2179581,\ 0.2179581$ for $h = 0.1,\ 0.05,\ 0.01$; exact 0.2179581 (errors -1.1×10^{-6}, -4.9×10^{-8}, -6×10^{-11}).

Problem 76-6 (Simmons)

Solution. Yes. Any quadrature rule of order 4 or more can be used; for example Newton's $\frac{3}{8}$ rule, Boole's rule (five nodes, order 6 as a quadrature rule), and Gauss–Legendre rules (the 2-point rule has order 4 and leads to an *implicit* Runge–Kutta method). We work out Newton's $\frac{3}{8}$ rule.

The rule. With nodes $x_0,\ x_0 + \frac{h}{3},\ x_0 + \frac{2h}{3},\ x_1 = x_0 + h$,

$$\int_{x_0}^{x_1} f(x, y(x))\,\mathrm{d}x \approx \frac{h}{8}\Big[f(x_0, y_0) + 3f\big(x_0 + \tfrac{h}{3}, y(x_0 + \tfrac{h}{3})\big)$$

$$+ 3f\big(x_0 + \tfrac{2h}{3}, y(x_0 + \tfrac{2h}{3})\big) + f(x_1, y(x_1))\Big],$$

exact for polynomials of degree ≤ 3 (error $\frac{h^5}{6480} f^{(4)}(\xi)$ for $f = f(x)$, even smaller than Simpson's $\frac{h^5}{2880} f^{(4)}$ used by the classical method). As in the text, the unknown values $y(x_0 + h/3)$, $y(x_0 + 2h/3)$, $y(x_1)$ must be replaced by estimates. Put

$$k_1 = f(x_k, y_k), \qquad k_2 = f\big(x_k + \tfrac{h}{3},\ y_k + h a_{21} k_1\big),$$
$$k_3 = f\big(x_k + \tfrac{2h}{3},\ y_k + h(a_{31}k_1 + a_{32}k_2)\big), \qquad k_4 = f\big(x_k + h,\ y_k + h(a_{41}k_1 + a_{42}k_2 + a_{43}k_3)\big),$$

and $y_{k+1} = y_k + \frac{h}{8}(k_1 + 3k_2 + 3k_3 + k_4)$.

Determining the coefficients. Let $c = (0, \frac{1}{3}, \frac{2}{3}, 1)$ be the nodes and $b = (\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})$ the weights. Matching the Taylor series of the exact solution through h^4 (the standard order conditions for a 4-stage explicit method) requires: the *row sums* $\sum_j a_{ij} = c_i$, i.e. $a_{21} = \frac{1}{3}$, $a_{31} + a_{32} = \frac{2}{3}$, $a_{41} + a_{42} + a_{43} = 1$ (the estimate must be exact for $y' = 1$); the quadrature conditions $\sum_i b_i c_i^q = \frac{1}{q+1}$ for $q = 0, 1, 2, 3$ (which hold for the $\frac{3}{8}$ rule: $\sum b_i c_i = \frac{1}{8} + \frac{2}{8} + \frac{1}{8} = \frac{1}{2}$, $\sum b_i c_i^2 = \frac{3}{72} + \frac{12}{72} + \frac{9}{72} = \frac{1}{3}$, $\sum b_i c_i^3 = \frac{3}{216} + \frac{24}{216} + \frac{27}{216} = \frac{1}{4}$); and the four conditions

(i) $\sum_{i,j} b_i a_{ij} c_j = \frac{1}{6}$: $\frac{3}{8} \cdot \frac{1}{3}\, a_{32} + \frac{1}{8}\big(\frac{1}{3} a_{42} + \frac{2}{3} a_{43}\big) = \frac{1}{6}$, i.e. $3a_{32} + a_{42} + 2a_{43} = 4$;

(ii) $\sum_{i,j} b_i c_i a_{ij} c_j = \frac{1}{8}$: $\frac{3}{8} \cdot \frac{2}{3} \cdot \frac{1}{3}\, a_{32} + \frac{1}{8}\big(\frac{1}{3} a_{42} + \frac{2}{3} a_{43}\big) = \frac{1}{8}$, i.e. $2a_{32} + a_{42} + 2a_{43} = 3$;

(iii) $\sum_{i,j} b_i a_{ij} c_j^2 = \frac{1}{12}$: $\frac{3}{8} \cdot \frac{1}{9}\, a_{32} + \frac{1}{8}\big(\frac{1}{9} a_{42} + \frac{4}{9} a_{43}\big) = \frac{1}{12}$, i.e. $3a_{32} + a_{42} + 4a_{43} = 6$;

(iv) $\sum_{i,j,k} b_i a_{ij} a_{jk} c_k = \frac{1}{24}$: $\frac{1}{8}\, a_{43} a_{32} \cdot \frac{1}{3} = \frac{1}{24}$, i.e. $a_{32} a_{43} = 1$.

Subtracting (ii) from (i) gives $a_{32} = 1$; then $a_{32} a_{43} = 1$ gives $a_{43} = 1$; (ii) gives $a_{42} = 3 - 2 - 2 = -1$; (iii) is then satisfied ($3 - 1 + 4 = 6$). Finally $a_{31} = \frac{2}{3} - 1 = -\frac{1}{3}$ and $a_{41} = 1 + 1 - 1 = 1$. The algorithm is therefore (the $\frac{3}{8}$*-rule Runge–Kutta method*):

$$k_1 = f(x_k, y_k), \qquad k_2 = f\big(x_k + \tfrac{h}{3},\ y_k + \tfrac{h}{3} k_1\big),$$
$$k_3 = f\big(x_k + \tfrac{2h}{3},\ y_k + h(-\tfrac{1}{3}k_1 + k_2)\big), \qquad k_4 = f\big(x_k + h,\ y_k + h(k_1 - k_2 + k_3)\big),$$
$$y_{k+1} = y_k + \tfrac{h}{8}(k_1 + 3k_2 + 3k_3 + k_4).$$

It has local error $O(h^5)$ and total error $O(h^4)$, like the classical method. (The linear system has a unique solution, so for this choice of nodes and weights these are the only coefficients; the method is known in the literature as Kutta's $\frac{3}{8}$-rule method.)

Benchmark problem $y' = x + y$, $y(0) = 1$, exact $y = 2e^x - x - 1$, $y(1) = 3.43656$. With $h = 0.2$ (the columns $k_1, \dots, k_4$ belong to the step from row k to row $k + 1$):

x_k	y_k	k_1	k_2	k_3	k_4	exact	classical RK y_k	E_k (%)
0.0	1.00000	1.00000	1.13333	1.29333	1.43200	1.00000	1.00000	0.00000
0.2	1.24280	1.44280	1.60565	1.80108	1.97044	1.24281	1.24280	0.00044
0.4	1.58364	1.98364	2.18254	2.42124	2.62810	1.58365	1.58364	0.00085
0.6	2.04421	2.64421	2.88716	3.17870	3.43136	2.04424	2.04421	0.00121
0.8	2.65104	3.45104	3.74778	4.10386	4.41247	2.65108	2.65104	0.00152
1.0	3.43650	–	–	–	–	3.43656	3.43650	0.00179

and with $h = 0.1$:

x_k	y_k	k_1	k_2	k_3	k_4	exact	classical RK y_k	E_k (%)
0.0	1.00000	1.00000	1.06667	1.14000	1.20733	1.00000	1.00000	0.00000
0.1	1.11034	1.21034	1.28402	1.36507	1.43948	1.11034	1.11034	0.00002
0.2	1.24281	1.44281	1.52423	1.61380	1.69604	1.24281	1.24281	0.00003
0.3	1.39972	1.69972	1.78971	1.88870	1.97959	1.39972	1.39972	0.00004
0.4	1.58365	1.98365	2.08310	2.19250	2.29295	1.58365	1.58365	0.00006
0.5	1.79744	2.29744	2.40736	2.52826	2.63928	1.79744	1.79744	0.00007
0.6	2.04424	2.64424	2.76571	2.89933	3.02202	2.04424	2.04424	0.00008
0.7	2.32750	3.02750	3.16175	3.30943	3.44502	2.32751	2.32750	0.00009
0.8	2.65108	3.45108	3.59945	3.76265	3.91251	2.65108	2.65108	0.00010
0.9	3.01920	3.91920	4.08318	4.26355	4.42916	3.01921	3.01920	0.00011
1.0	3.43656	–	–	–	–	3.43656	3.43656	0.00012

Hence $y(1) \approx 3.43650$ for $h = 0.2$ (the book's Table 1 value for the classical method) and 3.43656 for $h = 0.1$, matching the exact value 3.43656 to five decimals (relative errors 0.0018% and 0.00012%). A refinement study:

h	3/8-rule RK	classical RK	exact $y(1)$	3/8-rule error	classical error
0.200	3.4365023	3.4365023	3.4365637	6.138×10^{-5}	6.138×10^{-5}
0.100	3.4365595	3.4365595	3.4365637	4.169×10^{-6}	4.169×10^{-6}
0.050	3.4365634	3.4365634	3.4365637	2.716×10^{-7}	2.716×10^{-7}
0.025	3.4365636	3.4365636	3.4365637	1.733×10^{-8}	1.733×10^{-8}

Halving h divides the error by $\approx$ 15–16 (order h^4). On this benchmark the two methods give *identical* results, in every digit: for $f = x + y$, linear in (x, y), every four-stage fourth-order method reproduces the same polynomial amplification factor $1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}$, so the benchmark cannot tell them apart. On nonlinear problems they differ slightly, in either direction:

problem	h	classical RK error at $x = 1$	3/8-rule error at $x = 1$	ratio
$y' = 1/y$	0.10	-2.130×10^{-7}	-1.543×10^{-7}	0.72
$y' = 1/y$	0.05	-1.273×10^{-8}	-8.833×10^{-9}	0.69
$y' = y - \sin x$	0.10	-3.878×10^{-6}	-3.792×10^{-6}	0.98
$y' = y - \sin x$	0.05	-2.526×10^{-7}	-2.467×10^{-7}	0.98
$y' = (x + y - 1)^2$	0.10	-1.083×10^{-6}	-4.098×10^{-6}	3.78
$y' = (x + y - 1)^2$	0.05	-4.860×10^{-8}	-2.105×10^{-7}	4.33

(for $y' = (x + y - 1)^2$ the classical method is 4 times better and for $y' = 1/y$ the $\frac{3}{8}$-rule method is 1.4 times better; for $y' = y - \sin x$, which is linear, they differ only through the inhomogeneity $-\sin x$). Both need four evaluations of f per step and have the same order, so the $\frac{3}{8}$ rule gives a method of the same quality as the classical Runge–Kutta method, not a better one; to get more accuracy one needs a higher-order method (e.g. Boole's rule leads to a method of order 5 or 6 with more stages).

Answer. Newton's $\frac{3}{8}$ rule gives the fourth-order method: $k_1 = f(x, y)$, $k_2 = f(x + \frac{h}{3}, y + \frac{h}{3}k_1)$, $k_3 = f(x + \frac{2h}{3}, y + h(-\frac{1}{3}k_1 + k_2))$, $k_4 = f(x + h, y + h(k_1 - k_2 + k_3))$, $y_+ = y + \frac{h}{8}(k_1 + 3k_2 + 3k_3 + k_4)$. On $y' = x + y$: $y(1) \approx 3.43650$ ($h = 0.2$) and 3.43656 ($h = 0.1$), exact 3.43656; error $\propto h^4$, same accuracy as classical RK.

Problem 76-7 (Simmons)

Solution. **Exact solution.** The homogeneous equation $t^2y'' - 3ty' + 3y = 0$ is an Euler (equidimensional) equation: $y = t^r$ gives $r(r-1) - 3r + 3 = r^2 - 4r + 3 = (r-1)(r-3) = 0$, so

$y_h = c_1 t + c_2 t^3$. A particular solution of the nonhomogeneous equation is the constant $y_p = \frac{1}{3}$ ($3y_p = 1$). Thus $y = \frac{1}{3} + c_1 t + c_2 t^3$ with $y(1) = \frac{1}{3} + c_1 + c_2 = 0$ and $y'(1) = c_1 + 3c_2 = 0$; so $c_1 = -3c_2$, $\frac{1}{3} - 2c_2 = 0$, $c_2 = \frac{1}{6}$, $c_1 = -\frac{1}{2}$:

$$y(t) = \tfrac{1}{3} - \tfrac{t}{2} + \tfrac{t^3}{6}, \qquad y'(t) = -\tfrac{1}{2} + \tfrac{t^2}{2}.$$

(Check: $y'' = t$, and $t^2 \cdot t - 3t(-\frac{1}{2} + \frac{t^2}{2}) + 3(\frac{1}{3} - \frac{t}{2} + \frac{t^3}{6}) = t^3 + \frac{3t}{2} - \frac{3t^3}{2} + 1 - \frac{3t}{2} + \frac{t^3}{2} = 1$.)

First order system. Put $v = y'$. For $t \neq 0$,

$$y' = v, \qquad v' = \frac{1 + 3tv - 3y}{t^2}, \qquad y(1) = 0,\ v(1) = 0.$$

The Runge–Kutta method for a system (equations (9), (10) of Section 77, with $f = v$ and $g = (1 + 3tv - 3y)/t^2$) is used with $h = 0.2$; m_i denote the increments of y and n_i those of v: $m_1 = hv_k$, $n_1 = hg(t_k, y_k, v_k)$; $m_2 = h(v_k + \frac{n_1}{2})$, $n_2 = hg(t_k + \frac{h}{2}, y_k + \frac{m_1}{2}, v_k + \frac{n_1}{2})$; $m_3 = h(v_k + \frac{n_2}{2})$, $n_3 = hg(t_k + \frac{h}{2}, y_k + \frac{m_2}{2}, v_k + \frac{n_2}{2})$; $m_4 = h(v_k + n_3)$, $n_4 = hg(t_k + h, y_k + m_3, v_k + n_3)$; $y_{k+1} = y_k + \frac{1}{6}(m_1 + 2m_2 + 2m_3 + m_4)$, $v_{k+1} = v_k + \frac{1}{6}(n_1 + 2n_2 + 2n_3 + n_4)$.

First step in detail ($t_0 = 1$, $y_0 = v_0 = 0$, $h = 0.2$): $m_1 = 0$, $n_1 = 0.2 \cdot (1 + 0 - 0)/1 = 0.2$; $m_2 = 0.2 \cdot 0.1 = 0.02$, $n_2 = 0.2\,(1 + 3 \cdot 1.1 \cdot 0.1 - 0)/1.21 = 0.219835$; $m_3 = 0.2 \cdot (0 + 0.109917) = 0.021983$, $n_3 = 0.2\,(1 + 3 \cdot 1.1 \cdot 0.109917 - 3 \cdot 0.01)/1.21 = 0.220285$; $m_4 = 0.2 \cdot 0.220285 = 0.044057$, $n_4 = 0.2\,(1 + 3 \cdot 1.2 \cdot 0.220285 - 3 \cdot 0.021983)/1.44 = 0.239872$. Then $y_1 = \frac{1}{6}(0 + 0.04 + 0.043967 + 0.044057) = 0.021337$ and $v_1 = \frac{1}{6}(0.2 + 0.439669 + 0.440571 + 0.239872) = 0.220019$ (exact: $y(1.2) = 0.021333$, $y'(1.2) = 0.22$).

Marching to the right ($t = 1 \to 2$, five steps): stage values

t_k	y_k	v_k	m_1	m_2	m_3	m_4
1.0	0.000000	0.000000	0.000000	0.020000	0.021983	0.044057
1.2	0.021337	0.220019	0.044004	0.068005	0.069993	0.096047
1.4	0.090678	0.480039	0.096008	0.124009	0.126000	0.156043
1.6	0.216023	0.780061	0.156012	0.188014	0.190007	0.224041
1.8	0.405372	1.120085	0.224017	0.260019	0.262014	0.300042
2.0	0.666727	1.500112	–	–	–	–

t_k	n_1	n_2	n_3	n_4
1.0	0.200000	0.219835	0.220285	0.239872
1.2	0.240008	0.259890	0.260218	0.279898
1.4	0.280013	0.299925	0.300174	0.319921
1.6	0.320017	0.339949	0.340145	0.359940
1.8	0.360021	0.379967	0.380124	0.399955
2.0	–	–	–	–

and comparison with the exact solution:

t_k	y_k	exact y	$y - y_k$	$v_k \approx y'$	exact y'	$y' - v_k$
1.0	0.000000	0.000000	0	0.000000	0.000000	0
1.2	0.021337	0.021333	-4.01×10^{-6}	0.220019	0.220000	-1.87×10^{-5}
1.4	0.090678	0.090667	-1.16×10^{-5}	0.480039	0.480000	-3.89×10^{-5}
1.6	0.216023	0.216000	-2.31×10^{-5}	0.780061	0.780000	-6.10×10^{-5}
1.8	0.405372	0.405333	-3.90×10^{-5}	1.120085	1.120000	-8.52×10^{-5}
2.0	0.666727	0.666667	-5.98×10^{-5}	1.500112	1.500000	-1.12×10^{-4}

The results agree with the exact values to about 10^{-5}–10^{-4} (at $t = 2$: $y_5 = 0.666727$ against 0.666667).

Marching to the left ($t = 1 \to 0.2$, four steps $h = -0.2$):

t_k	y_k	v_k	m_1	m_2	m_3	m_4
1.0	0.000000	0.000000	0.000000	0.020000	0.018025	0.035885
0.8	0.018656	−0.180063	0.036013	0.052009	0.050050	0.063808
0.6	0.069312	−0.320175	0.064035	0.076021	0.074103	0.083596
0.4	0.143959	−0.420518	0.084104	0.092041	0.090277	0.094687
0.2	0.234530	−0.482991	–	–	–	–

t_k	n_1	n_2	n_3	n_4
1.0	−0.200000	−0.180247	−0.179424	−0.161034
0.8	−0.159963	−0.140376	−0.138977	−0.122006
0.6	−0.119860	−0.100683	−0.097806	−0.085221
0.4	−0.079377	−0.061738	−0.052918	−0.066148
0.2	–	–	–	–

t_k	y_k	exact y	$y - y_k$	$v_k \approx y'$	exact y'	$y' - v_k$
1.0	0.000000	0.000000	0	0.000000	0.000000	0
0.8	0.018656	0.018667	1.10×10^{-5}	−0.180063	−0.180000	6.26×10^{-5}
0.6	0.069312	0.069333	2.12×10^{-5}	−0.320175	−0.320000	1.75×10^{-4}
0.4	0.143959	0.144000	4.13×10^{-5}	−0.420518	−0.420000	5.18×10^{-4}
0.2	0.234530	0.234667	1.36×10^{-4}	−0.482991	−0.480000	2.99×10^{-3}

The errors now grow as $t \to 0$: at $t = 0.2$, $y_4 = 0.234530$ against 0.234667 (relative error 0.06%), and $v_4 = -0.482991$ against -0.48 (0.6%). The reason is that the equation is equidimensional, so only the *relative* step h/t matters, and this grows (from 0.2 at $t = 1$ to 1 at $t = 0.2$) as $t \to 0$.

Is there a solution at $t = 0$? Yes. Written as $y'' = (1 + 3ty' - 3y)/t^2$ the equation is singular at $t = 0$, but in the form $t^2y'' - 3ty' + 3y = 1$ it makes sense there, and the polynomial $y = \frac{1}{3} - \frac{t}{2} + \frac{t^3}{6}$ (entire in t) satisfies it for every t, including $t = 0$: $0 - 0 + 3 \cdot \frac{1}{3} = 1$. So the solution through $(1, 0)$ continues to $t = 0$ with $y(0) = \frac{1}{3}$, $y'(0) = -\frac{1}{2}$. Note, however: *every* solution $y = \frac{1}{3} + c_1 t + c_2 t^3$ has $y(0) = \frac{1}{3}$ (so $y(0) \neq \frac{1}{3}$ is impossible), while c_1, c_2 are free: the uniqueness theorem fails at the singular point $t = 0$ (also c_2 may differ for $t < 0$ and $t > 0$ and still give a twice differentiable solution). Only the continuation from $t = 1$ by analyticity, i.e. by the same coefficients, picks the polynomial above.

Computing the solution at $t = 0$ by Runge–Kutta. The standard Runge–Kutta formulas cannot be applied at $t = 0$ (division by $t^2 = 0$ in m_4, n_4 of the last step; the $0/0$ is not defined). Possible remedies:

(a) March to a small $t = \delta > 0$ and finish with the information from the equation: since every solution has $y(0) = \frac{1}{3}$ the value at 0 is known and $y'(0)$ follows by extrapolation/Taylor series from the computed $y(\delta)$, $y'(\delta)$.

(b) *Change variables to remove the singularity.* Put $t = e^s$, $y(t) = Y(s)$. Then $ty' = \dot{Y}$, $t^2y'' = \ddot{Y} - \dot{Y}$, and the equation becomes the constant-coefficient problem $\ddot{Y} - 4\dot{Y} + 3Y = 1$, regular for all real s; $t \to 0^+$ corresponds to $s \to -\infty$, and equal steps in s are equal *relative* steps in t, which cures the loss of accuracy seen above (the relative step is now constant, about 22% of t). We apply RK to the system $\dot{Y} = V$, $\dot{V} = 4V - 3Y + 1$ with initial data $Y(0) = 0$, $V(0) = \dot{Y}(0) = ty'|_{t=1} = 0$ and the step $h = -0.25$ in s (24 steps down to $s = -6$, $t = e^{-6} = 0.00248$; every fourth iterate is shown):

s_k	$t = e^{s_k}$	RK y	exact $y(t)$	error
0.0	1.00000	0.0000000	0.0000000	0
-1.0	0.36788	0.1578081	0.1576915	-1.17×10^{-4}
-2.0	0.13534	0.2660858	0.2660788	-7.01×10^{-6}
-3.0	0.04979	0.3084583	0.3084604	2.06×10^{-6}
-4.0	0.01832	0.3241751	0.3241765	1.41×10^{-6}
-5.0	0.00674	0.3299637	0.3299644	6.72×10^{-7}
-6.0	0.00248	0.3320937	0.3320940	2.98×10^{-7}

At $t = e^{-6}$ the computed value is 0.3320937 against the exact 0.3320940 (error 3×10^{-7}), and the error decreases as $t \to 0$ instead of increasing as in the march in t. Both values approach $y(0) = \frac{1}{3}$ (the exact value at $t = 0$; we reach any $t > 0$ as close to 0 as we like, the value $t = 0$ itself being the limit $s \to -\infty$).

(c) Use a power-series start at $t = 0$: substituting $y = \sum a_n t^n$ gives $(n-1)(n-3)a_n = 0$ for $n \geq 1$ and $3a_0 = 1$, hence $a_0 = \frac{1}{3}$, a_1 and a_3 free, all other $a_n = 0$; the free constants are then fixed by matching the Runge–Kutta values computed at some $t > 0$ (e.g. at $t = 0.2$), which gives $a_1 = -\frac{1}{2}$, $a_3 = \frac{1}{6}$ and the value $y(0) = \frac{1}{3}$.

Answer. Exact: $y = \frac{1}{3} - \frac{t}{2} + \frac{t^3}{6}$. RK ($h = 0.2$) from $t = 1$ to 2 gives $y(2) \approx 0.666727$ (exact 0.666667), and to $t = 0.2$ gives 0.234530 (exact 0.234667). The solution exists at $t = 0$ ($y(0) = \frac{1}{3}$, $y'(0) = -\frac{1}{2}$) though the equation is singular there (and uniqueness fails); RK cannot be applied at $t = 0$ itself, but works well in the variable $s = \log t$ ($Y_{ss} - 4Y_s + 3Y = 1$), or by marching to small $t > 0$.

Section 77 Systems

Conventions. A second order equation $y'' = F(t, y, y')$ is rewritten, as in the text, as a first order system by putting $x = y'$ (or $v = y'$). For the system $x' = f(t, x, y)$, $y' = g(t, x, y)$, $x(t_0) = x_0$, $y(t_0) = y_0$ with step h, $t_k = t_0 + kh$: **Euler:**

$$x_{k+1} = x_k + hf(t_k, x_k, y_k), \qquad y_{k+1} = y_k + hg(t_k, x_k, y_k)$$

(both right sides use the old values x_k, y_k).

Runge–Kutta:

$$x_{k+1} = x_k + \tfrac{1}{6}(\mu_{k1} + 2\mu_{k2} + 2\mu_{k3} + \mu_{k4}), \qquad y_{k+1} = y_k + \tfrac{1}{6}(\nu_{k1} + 2\nu_{k2} + 2\nu_{k3} + \nu_{k4}),$$

where

$$\begin{aligned}
\mu_{k1} &= hf(t_k, x_k, y_k), & \nu_{k1} &= hg(t_k, x_k, y_k),\\
\mu_{k2} &= hf\big(t_k + \tfrac{h}{2},\, x_k + \tfrac{\mu_{k1}}{2},\, y_k + \tfrac{\nu_{k1}}{2}\big), & \nu_{k2} &= hg\big(t_k + \tfrac{h}{2},\, x_k + \tfrac{\mu_{k1}}{2},\, y_k + \tfrac{\nu_{k1}}{2}\big),\\
\mu_{k3} &= hf\big(t_k + \tfrac{h}{2},\, x_k + \tfrac{\mu_{k2}}{2},\, y_k + \tfrac{\nu_{k2}}{2}\big), & \nu_{k3} &= hg\big(t_k + \tfrac{h}{2},\, x_k + \tfrac{\mu_{k2}}{2},\, y_k + \tfrac{\nu_{k2}}{2}\big),\\
\mu_{k4} &= hf\big(t_k + h,\, x_k + \mu_{k3},\, y_k + \nu_{k3}\big), & \nu_{k4} &= hg\big(t_k + h,\, x_k + \mu_{k3},\, y_k + \nu_{k3}\big)
\end{aligned}$$

(the text's μ_{ki}, v_{ki}; in the tables of Problem 3 we write m_i, n_i). The tables of the text (Tables 1 and 2 of Section 77) are reproduced by `verify/sec77.py`. Numbers are computed with 40 digits and displayed rounded.

Problem 77-1 (Simmons)

Solution. **The statement contains misprints, and our interpretation.** Reading the printed text literally, (i) "$r = 0.4$" must be $t = 0.4$ (the points are $t = 0.2$ and $t = 0.4$, i.e. the first two steps of size $h = 0.2$); (ii) the conditions $y(0) = 0$, $y'(1) = 0$ are given at two different points, so as printed this is a *boundary* value problem, whereas the Euler method needs initial data $y(0)$, $y'(0)$ at $t = 0$. We therefore treat the problem as follows.

(a) The problem exactly as printed. The general solution of $y'' - y = 0$ is $y = A\cosh t + B\sinh t$. $y(0) = A = 0$, and $y'(1) = B\cosh 1 = 0$ forces $B = 0$ (as $\cosh 1 > 0$). Hence the unique solution is $y \equiv 0$, and $y'(0) = 0$. With the (now determined) initial data $y(0) = 0$, $y'(0) = 0$ one can apply Euler: with $v = y'$, the system is $y' = v$, $v' = y$ and

$$y_{k+1} = y_k + hv_k, \qquad v_{k+1} = v_k + hy_k, \qquad y_0 = v_0 = 0.$$

t_k	y_k	v_k	exact y	exact y'	$y - y_k$	$y' - v_k$
0.0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
0.2	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
0.4	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

So $y(0.2) \approx 0$, $y(0.4) \approx 0$, *exactly* equal to the exact values $y(0.2) = y(0.4) = 0$ (error 0): the zero solution is an equilibrium of the system, and Euler reproduces constant solutions exactly. The same happens if the misprint is "$y'(0) = 0$" instead of "$y'(1) = 0$": then $y \equiv 0$ again.

(b) A nontrivial variant. Because the problem as printed is trivial, the intended initial conditions were presumably $y(0) = 0$, $y'(0) = 1$ (in which case the exact solution is $y = \sinh t$). With $h = 0.2$, $y_0 = 0$, $v_0 = 1$:

$$\begin{aligned}
y_1 &= y_0 + 0.2v_0 = 0.2, & v_1 &= v_0 + 0.2y_0 = 1,\\
y_2 &= y_1 + 0.2v_1 = 0.4, & v_2 &= v_1 + 0.2y_1 = 1.04.
\end{aligned}$$

t_k	y_k	$v_k \approx y'(t_k)$	exact $\sinh t_k$	exact $\cosh t_k$	$\sinh t_k - y_k$	$\cosh t_k - v_k$
0.0	0.00000	1.00000	0.00000	1.00000	0.00000	0.00000
0.2	0.20000	1.00000	0.20134	1.02007	0.00134	0.02007
0.4	0.40000	1.04000	0.41075	1.08107	0.01075	0.04107

At $t = 0.2$: $y \approx 0.2$ against $\sinh 0.2 = 0.201336$ (error 0.001336, 0.66%); at $t = 0.4$: $y \approx 0.4$ against $\sinh 0.4 = 0.410752$ (error 0.010752, 2.6%). Euler underestimates since $y'' = y > 0$ (convex solution), and the error grows with t.

Answer. As printed (data $y(0) = 0$, $y'(1) = 0$ force $y \equiv 0$): Euler gives $y(0.2) = y(0.4) = 0$, exactly the exact values. For the likely intended data $y(0) = 0$, $y'(0) = 1$: $y(0.2) \approx 0.2$, $y(0.4) \approx 0.4$, against exact $\sinh 0.2 = 0.20134$, $\sinh 0.4 = 0.41075$ (errors 0.0013, 0.0108).

Problem 77-2 (Simmons)

Solution. **Interpretation.** As printed, the first equation reads $x' = y'$, in which y' appears on the right side; this is evidently a misprint for $x' = y$, so that the system is the first order form of $x'' = x(1 - x)$ (with the roles of the book's Section 77: $f = y$, $g = x(1 - x)$). We solve this. (If the printed system is read literally, then $x' = y'$ and $x(0) = y(0)$ give $x \equiv y$, hence $x' = x(1 - x)$ with $x(0) = 1$, an equilibrium: $x \equiv y \equiv 1$, which the Euler method reproduces exactly, so the literal reading is trivial and certainly not intended.)

Euler formulas ($h = 0.1$, $t_k = 0.1k$, $x_0 = y_0 = 1$):

$$x_{k+1} = x_k + 0.1\, y_k, \qquad y_{k+1} = y_k + 0.1\, x_k(1 - x_k).$$

t_k	x_k	y_k	$f = y_k$	$g = x_k(1-x_k)$	$x(t_k)$	$y(t_k)$	$x - x_k$	$y - y_k$
0.0	1.0000000	1.0000000	1.0000000	0.0000000	1.0000000	1.0000000	0.000000	0.000000
0.1	1.1000000	1.0000000	1.0000000	−0.1100000	1.0998251	0.9946717	−0.000175	−0.005328
0.2	1.2000000	0.9890000	0.9890000	−0.2400000	1.1985369	0.9774282	−0.001463	−0.011572
0.3	1.2989000	0.9650000	0.9650000	−0.3882412	1.2948561	0.9465569	−0.004044	−0.018443
0.4	1.3954000	0.9261759	0.9261759	−0.5517412	1.3873475	0.9006760	−0.008053	−0.025500
0.5	1.4880176	0.8710018	–	–	1.4744595	0.8388587	−0.013558	−0.032143

For example, $x_1 = 1 + 0.1 \cdot 1 = 1.1$, $y_1 = 1 + 0.1 \cdot 1 \cdot 0 = 1$; $x_2 = 1.1 + 0.1 = 1.2$, $y_2 = 1 + 0.1 \cdot 1.1 \cdot (-0.1) = 0.989$; $x_3 = 1.2 + 0.0989 = 1.2989$, $y_3 = 0.989 + 0.1 \cdot 1.2 \cdot (-0.2) = 0.965$; $x_4 = 1.2989 + 0.0965 = 1.3954$, $y_4 = 0.965 + 0.1 \cdot 1.2989 \cdot (-0.2989) = 0.926176$; $x_5 = 1.3954 + 0.0926176 = 1.4880176$, $y_5 = 0.926176 + 0.1 \cdot 1.3954 \cdot (-0.3954) = 0.8710018$.

Result: $x(0.5) \approx 1.48802$, $y(0.5) \approx 0.87100$.

Comparison. The system has no elementary solution, but it has the first integral $y^2 = x^2 - \frac{2}{3}x^3 + \frac{2}{3}$ (since $\frac{d}{dt}\left[y^2 - x^2 + \frac{2}{3}x^3\right] = 2y\,x(1 - x) - 2xy + 2x^2y = 0$ along solutions, using $x' = y$, $y' = x(1 - x)$; the constant is $1 - 1 + \frac{2}{3}$). A high-accuracy numerical solution (Taylor series method with 30 digits) gives $x(0.5) = 1.474459$, $y(0.5) = 0.838859$ (and these satisfy the first integral to 10^{-30}). The Euler errors are therefore $x(0.5) - x_5 = -0.01356$ (0.92%) and $y(0.5) - y_5 = -0.03214$ (3.8%): Euler overestimates both components.

Answer. (Reading the first equation as $x' = y$.) Euler, $h = 0.1$: $x(0.5) \approx 1.48802$, $y(0.5) \approx 0.87100$; accurate values $x(0.5) = 1.47446$, $y(0.5) = 0.83886$ (errors -0.0136, -0.0321).

Problem 77-3 (Simmons)

Solution. **System.** Put $v = y'$ (the book's x). Then

$$y' = v, \qquad v' = y(1-y)v - y, \qquad y(0) = 1,\ v(0) = 1,$$

i.e. in the book's notation $x = v$, $f(t,x,y) = y(1-y)x - y$ and $g(t,x,y) = x$. The Runge–Kutta formulas for the pair (y, v), with $F_1(y,v) = v$ and $F_2(y,v) = y(1-y)v - y$ (neither depends on t): $m_1 = hF_1(y_k, v_k)$, $n_1 = hF_2(y_k, v_k)$; $m_2 = hF_1(y_k + \frac{m_1}{2}, v_k + \frac{n_1}{2})$, $n_2 = hF_2(y_k + \frac{m_1}{2}, v_k + \frac{n_1}{2})$; m_3, n_3 the same with m_2, n_2; $m_4 = hF_1(y_k + m_3, v_k + n_3)$, $n_4 = hF_2(y_k + m_3, v_k + n_3)$; $y_{k+1} = y_k + \frac{1}{6}(m_1 + 2m_2 + 2m_3 + m_4)$, $v_{k+1} = v_k + \frac{1}{6}(n_1 + 2n_2 + 2n_3 + n_4)$.

First step with $h = 0.5$ in detail: $y_0 = v_0 = 1$. $m_1 = 0.5 \cdot 1 = 0.5$, $n_1 = 0.5(1 \cdot 0 \cdot 1 - 1) = -0.5$. Then $(y, v) = (1.25,\ 0.75)$: $m_2 = 0.5 \cdot 0.75 = 0.375$, $n_2 = 0.5(1.25 \cdot (-0.25) \cdot 0.75 - 1.25) = -0.7421875$. Then $(y, v) = (1.1875,\ 0.62890625)$: $m_3 = 0.5 \cdot 0.62890625 = 0.3144531$, $n_3 = 0.5(1.1875 \cdot (-0.1875) \cdot 0.62890625 - 1.1875) = -0.6637650$. Then $(y, v) = (1.3144531,\ 0.3362350)$: $m_4 = 0.5 \cdot 0.3362350 = 0.1681175$, $n_4 = 0.5(1.3144531 \cdot (-0.3144531) \cdot 0.3362350 - 1.3144531) = -0.7267152$. Hence $y_1 = 1 + \frac{1}{6}(0.5 + 2 \cdot 0.375 + 2 \cdot 0.3144531 + 0.1681175) = 1.3411706$ and $v_1 = 1 + \frac{1}{6}(-0.5 - 2 \cdot 0.7421875 - 2 \cdot 0.6637650 - 0.7267152) = 0.3268966$.

$h = 0.5$ (2 steps):

t_k	y_k	$x_k = y'_k$	m_1	m_2	m_3	m_4
0.0	1.000000	1.000000	0.500000	0.375000	0.314453	0.168118
0.5	1.341171	0.326897	0.163448	−0.022895	−0.010969	−0.166578
1.0	1.329361	−0.348562	–	–	–	–

t_k	n_1	n_2	n_3	n_4
0.0	−0.500000	−0.742188	−0.663765	−0.726715
0.5	−0.745374	−0.697671	−0.660052	−0.591934
1.0	–	–	–	–

$h = 0.2$ (5 steps):

t_k	y_k	$x_k = y'_k$	m_1	m_2	m_3	m_4
0.0	1.000000	1.000000	0.200000	0.180000	0.176020	0.152946
0.2	1.177498	0.763833	0.152767	0.126024	0.123677	0.095766
0.4	1.302154	0.477669	0.095534	0.065732	0.065431	0.036280
0.6	1.367843	0.180273	0.036055	0.006884	0.007969	−0.019608
0.8	1.375535	−0.098906	−0.019781	−0.046270	−0.044784	−0.069609
1.0	1.330286	−0.348693	–	–	–	–

t_k	n_1	n_2	n_3	n_4
0.0	−0.200000	−0.239800	−0.235268	−0.266864
0.2	−0.267428	−0.290894	−0.285002	−0.297764
0.4	−0.298019	−0.301034	−0.296268	−0.291755
0.6	−0.291710	−0.280855	−0.278314	−0.265024
0.8	−0.264889	−0.250024	−0.249137	−0.235512
1.0	–	–	–	–

$h = 0.1$ (10 steps):

t_k	y_k	$x_k = y_k'$	m_1	m_2	m_3	m_4
0.0	1.000000	1.000000	0.100000	0.095000	0.094501	0.089055
0.1	1.094676	0.890410	0.089041	0.083106	0.082686	0.076399
0.2	1.177513	0.763839	0.076384	0.069698	0.069392	0.062474
0.3	1.247020	0.624582	0.062458	0.055261	0.055084	0.047783
0.4	1.302175	0.477681	0.047768	0.040317	0.040266	0.032829
0.5	1.342469	0.328145	0.032815	0.025348	0.025402	0.018041
0.6	1.367861	0.180283	0.018028	0.010735	0.010865	0.003739
0.7	1.378690	0.037285	0.003729	−0.003262	−0.003089	−0.009882
0.8	1.375547	−0.098907	−0.009891	−0.016513	−0.016324	−0.022744
0.9	1.359162	−0.227519	−0.022752	−0.028992	−0.028811	−0.034863
1.0	1.330292	−0.348707	–	–	–	–

t_k	n_1	n_2	n_3	n_4
0.0	−0.100000	−0.109988	−0.109452	−0.118661
0.1	−0.118696	−0.127098	−0.126422	−0.133690
0.2	−0.133717	−0.139848	−0.139102	−0.143924
0.3	−0.143941	−0.147480	−0.146749	−0.149007
0.4	−0.149014	−0.150038	−0.149396	−0.149336
0.5	−0.149333	−0.148249	−0.147740	−0.145865
0.6	−0.145858	−0.143258	−0.142892	−0.139825
0.7	−0.139816	−0.136341	−0.136102	−0.132454
0.8	−0.132445	−0.128672	−0.128531	−0.124817
0.9	−0.124810	−0.121189	−0.121114	−0.117713
1.0	–	–	–	–

Values at $t = 1$. For comparison we also ran $h = 0.01$ and computed a reference solution with a 30-digit Taylor-series integrator:

h	steps	$y(1) \approx$	$y'(1) \approx$	$y(1) - y_N$	$y'(1) - v_N$
0.50	2	1.32936089	−0.34856226	9.31×10^{-4}	-1.46×10^{-4}
0.20	5	1.33028601	−0.34869299	5.81×10^{-6}	-1.51×10^{-5}
0.10	10	1.33029179	−0.34870703	2.63×10^{-8}	-1.08×10^{-6}
0.01	100	1.33029182	−0.34870810	-2.63×10^{-11}	-1.14×10^{-10}
reference		1.33029182	−0.34870810	–	–

Discussion. The three requested estimates of $y(1)$ are 1.32936, 1.33029, 1.33029 (digits: 1.329361, 1.330286, 1.330292), converging rapidly to the reference value 1.330292; for $y'(1)$: -0.348562, -0.348693, -0.348707 against -0.348708. The error in $y(1)$ drops by a factor 160 from $h = 0.5$ to 0.2 and by 221 from 0.2 to 0.1 (expected for order 4: $2.5^4 = 39$ and $2^4 = 16$ in the asymptotic regime; the error in y' falls by 10 and 14). With $h = 0.1$ the error in $y(1)$ is 2.6×10^{-8} and the error in $y'(1)$ is 1.1×10^{-6}.

Answer. RK: $y(1) \approx 1.329361$ $(h = 0.5)$, 1.330286 $(h = 0.2)$, 1.330292 $(h = 0.1)$; $y'(1) \approx -0.348562$, -0.348693, -0.348707. Reference: $y(1) = 1.3302918$, $y'(1) = -0.3487081$.

Problem 77-4 (Simmons)

Solution. Consider the system

$$x' = f(t, x, y, z), \qquad y' = g(t, x, y, z), \qquad z' = q(t, x, y, z),$$

$$x(t_0) = x_0, \qquad y(t_0) = y_0, \qquad z(t_0) = z_0.$$

Following the derivation of Section 73 for each component (replace each unknown on $[t_k, t_{k+1}]$ by its tangent line, i.e. integrate each right side with the value at the left endpoint t_k), with $t_k = t_0 + kh$:

$$\begin{aligned} x_{k+1} &= x_k + h\, f(t_k, x_k, y_k, z_k), \\ y_{k+1} &= y_k + h\, g(t_k, x_k, y_k, z_k), \\ z_{k+1} &= z_k + h\, q(t_k, x_k, y_k, z_k), \qquad k = 0, 1, 2, \ldots \end{aligned}$$

All three right sides are evaluated at the *old* values (t_k, x_k, y_k, z_k) (not using the already updated x_{k+1}, y_{k+1}). In vector form, with $\mathbf{u} = (x, y, z)^T$ and $\mathbf{F} = (f, g, q)^T$: $\mathbf{u}' = \mathbf{F}(t, \mathbf{u})$, $\mathbf{u}_{k+1} = \mathbf{u}_k + h\mathbf{F}(t_k, \mathbf{u}_k)$, which is the scalar formula $y_{k+1} = y_k + hf(x_k, y_k)$ of Section 73 with vectors; the same formula works for any number of equations, and the local error is again $O(h^2)$ per step, the total error proportional to h.

Example (a third order equation). $y''' = y$, $y(0) = y'(0) = y''(0) = 1$ (exact: $y = e^t$). Put $u = y$, $v = y'$, $w = y''$: $u' = v$, $v' = w$, $w' = u$. Euler with $h = 0.1$:

$$u_{k+1} = u_k + 0.1 v_k, \quad v_{k+1} = v_k + 0.1 w_k, \quad w_{k+1} = w_k + 0.1 u_k.$$

Starting from $(1, 1, 1)$ all three components remain equal, so $u_k = 1.1^k$:

t_k	y_k	v_k	w_k	e^{t_k}
0.0	1.00000	1.00000	1.00000	1.00000
0.1	1.10000	1.10000	1.10000	1.10517
0.2	1.21000	1.21000	1.21000	1.22140
0.3	1.33100	1.33100	1.33100	1.34986

(error $e^{0.3} - 1.331 = 0.0189$, consistent with an $O(h)$ error).

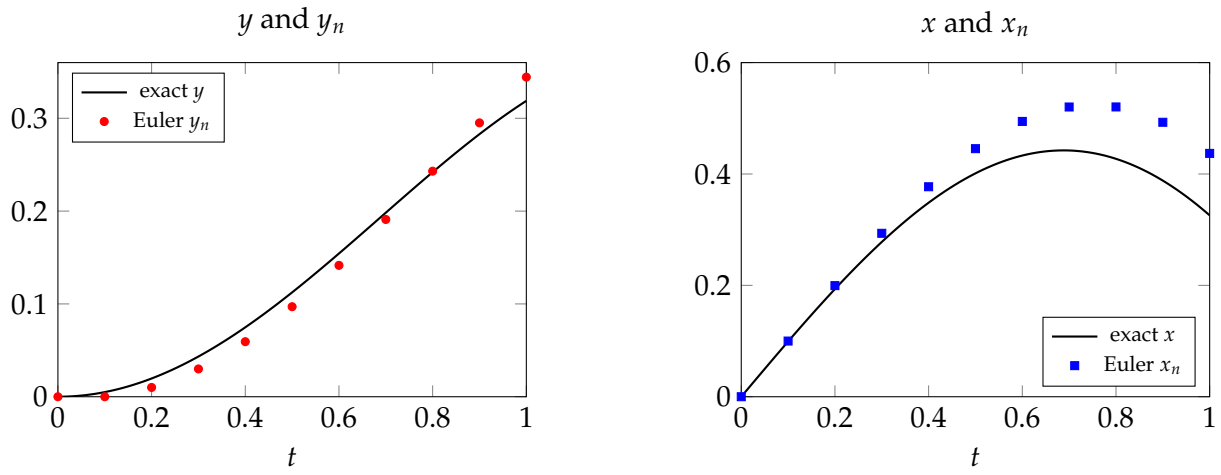
Answer. $x_{k+1} = x_k + hf(t_k, x_k, y_k, z_k)$, $y_{k+1} = y_k + hg(t_k, x_k, y_k, z_k)$, $z_{k+1} = z_k + hq(t_k, x_k, y_k, z_k)$ for $x' = f$, $y' = g$, $z' = q$ (all evaluated at the old values); in vector form $\mathbf{u}_{k+1} = \mathbf{u}_k + h\mathbf{F}(t_k, \mathbf{u}_k)$.

Problem 77-5 (Simmons)

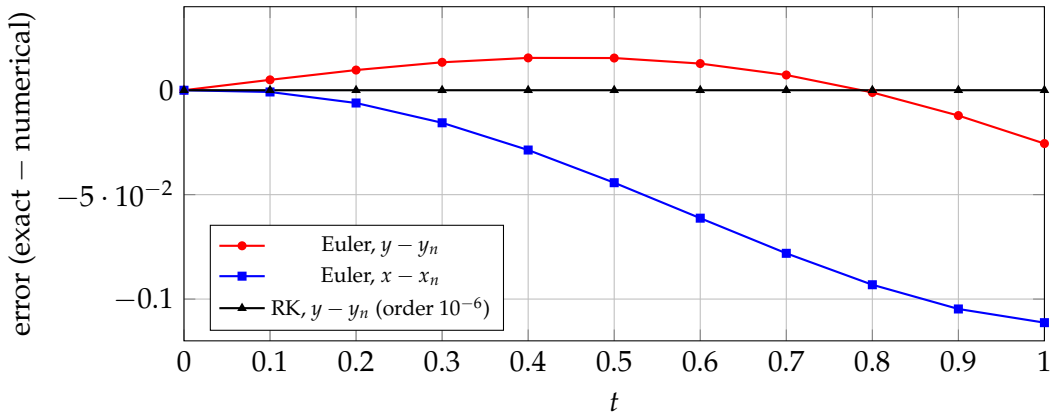
Solution. Setting. $y'' + 4y = \cos t$, $y(0) = y'(0) = 0$, with $x = y'$: $x' = -4y + \cos t$, $y' = x$; exact solution $y = \frac{1}{3}(\cos t - \cos 2t)$, $x = y' = \frac{1}{3}(2\sin 2t - \sin t)$. The Euler iteration with $h = 0.1$ (our recomputation of the text's Table 1, with the signed errors added; E_n is the relative error in percent) is:

t_n	x_n	y_n	exact x	exact y	$x - x_n$	$y - y_n$	E_n for x (%)	E_n for y (%)
0.0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	–	–
0.1	0.10000	0.00000	0.09917	0.00498	−0.00083	0.00498	0.84	100.00
0.2	0.19950	0.01000	0.19339	0.01967	−0.00611	0.00967	3.16	49.16
0.3	0.29351	0.02995	0.27792	0.04333	−0.01559	0.01338	5.61	30.88
0.4	0.37706	0.05930	0.34843	0.07478	−0.02863	0.01548	8.22	20.70
0.5	0.44545	0.09701	0.40117	0.11243	−0.04427	0.01542	11.04	13.72
0.6	0.49440	0.14155	0.43315	0.15433	−0.06126	0.01277	14.14	8.28
0.7	0.52032	0.19099	0.44223	0.19829	−0.07809	0.00730	17.66	3.68
0.8	0.52040	0.24302	0.42726	0.24197	−0.09314	−0.00105	21.80	0.44
0.9	0.49286	0.29506	0.38812	0.28294	−0.10474	−0.01213	26.99	4.29
1.0	0.43700	0.34435	0.32571	0.31882	−0.11129	−0.02553	34.17	8.01

The graphs. (Left: y_n against y; right: x_n against x.)



and the signed errors of the numerical solutions:



Explanation of the fluctuation for y. The relative error in y in Table 1 starts at 100%, decreases to 0.4% at $t = 0.8$, and then increases again (4.3%, 8.0%). The signed error $\varepsilon_n = y(t_n) - y_n$ explains it: it is positive and increasing at first (0.0050, 0.0097, 0.0134, 0.0155 at $t = 0.1, \dots, 0.4$), reaches its maximum ≈ 0.0155 near $t = 0.4$–0.5, then *decreases, passes through zero between* $t = 0.7$ *and* $t = 0.8$ ($\varepsilon = +0.0073$ at 0.7, -0.0011 at 0.8), and becomes negative (-0.0121 at 0.9, -0.0255 at 1.0). The relative error $|\varepsilon_n|/y(t_n)$ therefore passes through (almost) zero at the sign change and rises again afterwards: this is the fluctuation. The sign change has a clear cause. Subtracting the Euler formulas from the exact relations $y(t_{n+1}) = y(t_n) + hx(t_n) + \tau_n$, $x(t_{n+1}) = x(t_n) + h(-4y(t_n) + \cos t_n) + \sigma_n$ (with local errors $\tau_n = \frac{h^2}{2}y''(\xi_n) = \frac{h^2}{2}x'(\xi_n)$ and $\sigma_n = \frac{h^2}{2}x''(\xi_n')$) gives, for the signed errors $\delta_n = x(t_n) - x_n$ and ε_n,

$$\varepsilon_{n+1} = \varepsilon_n + h\,\delta_n + \tau_n,$$
$$\delta_{n+1} = \delta_n - 4h\,\varepsilon_n + \sigma_n.$$

Here $y'' = x' = \cos t - 4y$ is positive while $y < \frac{1}{4}\cos t$, which holds up to $t \approx 0.688$ (where $8c^2 - c - 4 = 0$ for $c = \cos t$, i.e. $c = \frac{1+\sqrt{129}}{16} = 0.7724$), and negative afterwards. So up to $t \approx 0.688$ the solution y is convex, the tangent-line steps fall below it ($\tau_n > 0$, Euler lags: the iterate $y_1 = 0$ misses the term $\frac{h^2}{2}y''(0) = 0.005$ altogether, which is the 100% error: the relative error is huge only because $y(t) \approx t^2/2$ itself is small), and ε_n grows. But $x'' = -4x - \sin t < 0$ for $t > 0$ ($x > 0$): x is concave, Euler overestimates it, $\sigma_n < 0$ and $\delta_n < 0$ for all $n \geq 1$ ($x_1 = 0.1 > 0.09917$), and the error in x enters the equation for ε through $h\delta_n < 0$. After $t \approx 0.688$ also $\tau_n < 0$, so both terms

push ε_n down, ε_n crosses zero and becomes negative: the numerical y_n lags below the exact y first and overshoots later.

Does the same happen for x_n and x? No. The signed error $\delta_n = x(t_n) - x_n$ is negative at every step and its absolute value increases monotonically (-0.0008, -0.0061, -0.0156, -0.0286, …, -0.1113); the relative error E_n for x increases monotonically from 0.84% at $t = 0.1$ to 34% at $t = 1.0$ (see the table). There is no sign change, since x is concave throughout ($\sigma_n < 0$ always). In the recursion $\delta_{n+1} = \delta_n - 4h\varepsilon_n + \sigma_n$ the term σ_n is always negative; after ε_n turns negative, the term $-4h\varepsilon_n$ becomes positive but is smaller than $|\sigma_n|$ (for example, at $t = 0.9$ it is $+0.0049$ against -0.0114), so δ_n keeps decreasing. Also, $x(t) \approx t$ is not small near $t = 0$, so there is no inflated early relative error as for y.

Why Runge–Kutta behaves differently. With the same $h = 0.1$ the Runge–Kutta iteration (our recomputation of Table 2 with signed errors) is:

t_n	x_n	y_n	exact x	exact y	$x - x_n$	$y - y_n$	E_n for x (%)	E_n for y (%)
0.0	0.00000	0.00000	0.00000	0.00000	0	0	–	–
0.1	0.09917	0.00498	0.09917	0.00498	1.24×10^{-6}	2.83×10^{-8}	0.0013	0.0006
0.2	0.19339	0.01967	0.19339	0.01967	2.41×10^{-6}	3.46×10^{-7}	0.0012	0.0018
0.3	0.27792	0.04333	0.27792	0.04333	3.32×10^{-6}	9.31×10^{-7}	0.0012	0.0021
0.4	0.34843	0.07478	0.34843	0.07478	3.81×10^{-6}	1.74×10^{-6}	0.0011	0.0023
0.5	0.40117	0.11242	0.40117	0.11243	3.75×10^{-6}	2.70×10^{-6}	0.0009	0.0024
0.6	0.43314	0.15432	0.43315	0.15433	3.05×10^{-6}	3.73×10^{-6}	0.0007	0.0024
0.7	0.44223	0.19829	0.44223	0.19829	1.66×10^{-6}	4.75×10^{-6}	0.0004	0.0024
0.8	0.42726	0.24196	0.42726	0.24197	-4.12×10^{-7}	5.63×10^{-6}	0.0001	0.0023
0.9	0.38813	0.28293	0.38812	0.28294	-3.12×10^{-6}	6.30×10^{-6}	0.0008	0.0022
1.0	0.32571	0.31881	0.32571	0.31882	-6.34×10^{-6}	6.64×10^{-6}	0.0019	0.0021

(i) The errors are $O(h^4)$ with small constants: the absolute error of y never exceeds 7×10^{-6} (compared with 0.026 for Euler), so even at $t = 0.1$ the relative error is only 6×10^{-4}%: the first Runge–Kutta step reproduces the Taylor expansion of the exact solution through h^4, including the leading term $y \approx t^2/2$ that Euler misses entirely ($y_1 = 0$). (ii) The error does not change sign: the local error of the fourth order method is proportional to the fifth derivative $y^{(5)} = \frac{1}{3}(32 \sin 2t - \sin t)$, which is positive for all $0 < t \leq 1$, so all local errors have the same sign and accumulate monotonically ($y - y_n = 2.8 \times 10^{-8}$, 3.5×10^{-7}, …, $6.6 \times 10^{-6} > 0$ in the table). Since y itself grows like t^2 at about the same rate as the accumulated error, the relative error stays nearly constant, 0.0006% to 0.0024%, and does not exhibit the dip that occurs in Euler's case when the errors from the early (convex) and late (concave) parts of the solution cancel. (The Runge–Kutta error of x does change sign near $t = 0.78$, but it is of size 10^{-6} and invisible in any comparison.)

Answer. Graphs: the Euler points y_n lie below the exact curve for $t \leq 0.7$ and above it for $t \geq 0.8$. The relative error in y falls from 100% to 0.4% at $t = 0.8$ because the signed error $y - y_n$ changes sign there (Euler underestimates the convex part $y'' > 0$, $t < 0.688$, and overestimates later, helped by the always-overestimated x); it then grows again. For x the error has one sign (x is concave) and grows steadily to 34%. RK: errors $\sim 10^{-6}$, with $y^{(5)} > 0$ so no sign change, and the first step reproduces the $t^2/2$ behavior, so the relative error is tiny and smooth.

Appendix A

Remarks on the Textbook and Its Answer Key

While working every problem we compared each result with the answer key printed in the book. Almost all answers agree. The items below are the places where our answer differs from the key, or where the printed statement of a problem appears to contain a misprint; each is explained in a remark inside the relevant solution. "Problem s–n" means problem n of Section s.

Misprints in problem statements or text

- **4–17:** "$x =$ when $t = t_1$" is read as "$x = x_1$ when $t = t_1$" (consistent with the key).
- **33–8:** "Figure 33" should be Figure 39 (the sawtooth of series (11)).
- **35–16:** "$\cos^3 x = \frac{3}{4}x + \frac{1}{4}\cos 3x$" should read $\frac{3}{4}\cos x + \frac{1}{4}\cos 3x$.
- **42–2(b):** the statement prints $f(\theta) = 0$, but the key's answer is the solution for $f(\theta) = \theta$; both are given.
- **42–3:** the Poisson kernel denominator "$2R2\cos$" should be $2Rr\cos(\theta - \varphi)$.
- **43–6(a):** the equation should read $Py'' + P'y' + Ry = 0$ (a factor y is missing on the last term).
- **47, formula (5):** $J_p' - \frac{p}{x}J_p = +J_{p+1}$ should have the sign $-J_{p+1}$ (as follows from formula (3), and as (6) requires).
- **49–5:** $\varepsilon \to \infty$ should be $\varepsilon \to 0$.
- **60–3:** "dx/dy" should be dx/dt; **61–6(c):** "dx/df" should be dx/dt.
- **Chapter 1 Miscellaneous 6:** the cross-reference "Problem 3–2" should be Problem 3–5. **Misc. 13** contains the printed typo "is is".
- **73, formula (5):** the relative error is $(y - y_n)/y$ with a minus sign; the plus sign printed is a misprint (the text's own Table 1 confirms this).
- **73–6:** the problem has a solution at $x = 1$ ($y = \tan x$, since $1 < \pi/2$), contrary to the text.
- **75–6:** the local error of the three-term Taylor method is $y'''h^3/6$, not $y'''h^3/12$.
- **77–1, 77–2:** as printed these contain misprints ("$y'(1) = 0$", "$r = 0.4$", "$x' = y'$"); the interpretation used is stated in each solution.

Differences from the answer key

- **4–7, 4–12, 4–19, 4–29:** the key's numbers (3531; 35.35%; 27.72 min; 3330, 3850, 10 510, 7010 years) differ slightly from the exact values (3535.5; 35.36%; 27.73; 3323, 3833, 10 502, 7016), because of intermediate rounding.
- **7–6(b):** the key gives $n = \frac{3}{4}$; $n = -\frac{1}{2}$ also separates the variables (both lead to the same solution).
- **10–9:** the key prints $\frac{8}{10}(10 - t)^4$; the coefficient must be $\frac{8}{10^4}$ for $x(0) = 2$.
- **11–3(b):** the key's family $y = \log(c_1 e^x + e^{-x}) + c_2$ is incomplete; the complete family is $y = \log(Ae^x + Be^{-x})$.
- **11–4 and 20–5:** the key's "about 89 minutes" and "about 90 minutes" are about 85 minutes for $R = 4000$ miles, $g = 32$ ft/s^2.
- **22–10:** the key's answer duplicates that of Problem 11 and does not satisfy the equation; the correct answer is $(c_1 + c_2 x)\cos ax + (c_3 + c_4 x)\sin ax$.
- **Chapter 2 Miscellaneous 46:** the key's relation should read $c_1^2 y = c_1 x + \log(c_1 x - 1) + c_2$. **Misc. 11:** the key gives only one of three families.
- **35–15:** the key's signs in the $\cos^2 x$ sine series are those of the $\sin^2 x$ series; the correct signs are $+$.
- **51–8:** the key's entry labelled "8(a)" is the answer to part (b).
- **70–6(a):** the key's "all points with $y_0 \neq 0$" is the answer for $y' = |y|$; for the printed equation $y' = y|y|$ the answer is all points.

Where the key differs only by an arbitrary constant, a different but equivalent closed form, or a different (equally valid) particular solution, no remark is made in this list.